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Weak Convergence and Empirical Processes

With Applications to Statistics

Second Edition

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A.W. van der Vaart • Jon A. Wellner

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Second Edition

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To Maryse

To Vera

Preface

This book tries to do three things. The first goal is to give an exposition of certain modes of stochastic convergence, in particular convergence in distribution. The classical theory of this subject was developed mostly in the 1950s and is well summarized in Billingsley (1968). During the last 15 years, the need for a more general theory allowing random elements that are not Borel measurable has become well established, particularly in developing the theory of empirical processes. Part 1 of the book, *Stochastic Convergence*, gives an exposition of such a theory following the ideas of J. Hoffmann-Jørgensen and R. M. Dudley.

A second goal is to use the weak convergence theory background developed in Part 1 to present an account of major components of the modern theory of empirical processes indexed by classes of sets or functions. The weak convergence theory developed in Part 1 is important for this, simply because the empirical processes studied in Part 2, *Empirical Processes*, are naturally viewed as taking values in non-separable Banach spaces, even in the most elementary cases, and are typically *not* Borel measurable. Much of the theory presented in Part 2 has previously been scattered in the journal literature and has, as a result, been accessible only to a relatively small number of specialists. In view of the importance of this theory for statistics, we hope that the presentation given here will make this theory more accessible to statisticians as well as to probabilists interested in statistical applications.

Our third goal is to illustrate the usefulness of modern weak convergence theory and modern empirical process theory for statistics by a wide variety of applications. On the one hand, as is also clear through the work of David Pollard, the theory of empirical processes provides a collection of extremely powerful tools for proving many of the main limit theorems of asymptotic statistics. On the other hand, the empirical distribution indexed by a collection of sets or functions is an object of independent statistical interest; for instance, as a measure of goodness of fit. The topics

included in Part 3 of the book, *Statistical Applications*, range from rates of convergence in semiparametric estimation, to the functional delta-method, bootstrap, permutation empirical processes, and the convolution theorem. We have not aimed at giving an exhaustive coverage of the statistical background or related literature in presenting these applications. The choices made reflect our personal interests and research efforts over the past few years, so the reader should understand that many equally interesting applications and further research directions are not covered here. For instance, we expect significant progress in semiparametric theory through the use of empirical processes in the next few years. Wellner (1992) reviews various applications of empirical process methods through 1992.

This project began with a joint effort to develop some results in the Hoffmann-Jørgensen theory (namely Prohorov's Theorem 1.3.9) needed to prove the convolution and asymptotic minimax theorem for estimators with values in nonseparable Banach spaces (see Chapter 3.12). That effort was successful, but it resulted in a manuscript that was too awkward for publication in a journal. The generality offered by the new weak convergence theory and the many applications of general empirical process theory in statistics led us to explore the area further. The result, several years later, is this book.

Along the way we have learned much from the main contributors to empirical process theory, from colleagues, and from friends. In particular, we owe thanks to Lucien Birgé, R. M. Dudley, Peter Gaenssler, Sara van de Geer, Richard Gill, Evarist Giné, Piet Groeneboom, Lucien Le Cam, Michel Ledoux, Pascal Massart, Susan Murphy, David Pollard, Michel Talagrand, and Joel Zinn, for advice, corrections, discussions, clarifications, inspiration, preprints, and help.

The authors presented preliminary versions of various parts of the manuscript in courses at the Free University, Amsterdam, and the University of Washington to patient and sharp-eyed groups of students, including Jian Huang, Marianne Jonker, Brad McNeney, Jinko Graham, Jens Praestgaard, and Anne Sheehy. We owe them as well as Franz Strobl, Klaus Ziegler, and Lutz Dümbgen thanks for picking up errors and points needing clarification.

The authors have been partially supported in their research efforts over the period of work on this book (1988–1995) by grants from the National Science Foundation, NIAID, NWO, NATO, CNRS, Stieltjes Institute, and Free University Amsterdam (which funded several visits of the second author to that fair city). Our thanks also go to our editor at Springer, Martin Gilchrist, and to the staff at Springer-Verlag for their efficient job in turning the $\text{\TeX}$ manuscript into a published book.

Amsterdam and Seattle
August 1995

Preface to the Second Edition

The three basic aims of this Second Edition remain the same as those of the First Edition (published in 1996). In the meantime the basic theory has continued to evolve and grow, while applications of the theory to problems in statistics, machine learning, biostatistics and econometrics have expanded dramatically. This Second Edition aims to capture developments in both directions, primarily by including a selection of the new tools in Parts 1 and 2 and by presenting further applications of the tools to problems in statistics and machine learning in Part 3.

In brief we have added six new chapters, some small and some extensive:

- Chapter 1.13 on weak convergence of conditional distributions.
- Chapter 1.14 on weak convergence relative to the Skorokhod topology.
- Chapter 1.15 on stable convergence.
- Chapter 2.15 on concentration Inequalities, presenting the bounded difference inequality and Talagrand's inequality.
- Chapter 3.5 on model selection.
- Chapter 3.13 on random empirical processes.

We also added significantly to existing chapters throughout the book. An incomplete list of major additions is:

- Chapter 1.12: new material on the bounded Lipschitz metric.
- Chapter 2.2: an exposition of generic chaining and majorizing measures.
- Chapter 2.5: inclusion of the converse of the random entropy Glivenko-Cantelli theorem (in support of Glivenko-Cantelli preservation theorems in Chapter 2.10).
- Chapter 2.7: new sections on Besov and Sobolev classes of functions, analytic functions, and Gaussian mixtures. The chapter has been renamed "Entropies of function classes".
- Chapter 2.10: new Glivenko-Cantelli preservation theorems.

- Chapter 2.14: this chapter now gives a unified treatment of bounds for the expected value of the supremum of the empirical process, in terms of the uniform entropy and bracketing integral. Concentration inequalities for the supremum have been moved to a separate chapter.
- Chapter 3.4: seven new sections have been added. These include sections on sieved and penalized maximum contrast estimators, as well as new sections on penalized least squares regression, lasso, classification, and support vector machines.
- Chapter A.2: additional inequalities on Gaussian processes have been added, including the isoperimetric inequality, and Borell's and Ehrhard's inequalities, and proofs for most results.

These additions are accompanied by the addition of 112 new problems, 184 referenced papers and books (giving totals of 322 problems and 489 references), and five figures (giving six figures in total). When typeset in the *Computer Modern* fonts used for the First Edition, the Second Edition would have been 198 pages longer (706 versus 508 pages). We shaved off some pages by using the compacter *Times* font.

We hope that these additions will give readers access to parts of the theory that have developed rapidly in the recent past, including important developments in the applications of the basic tools to problems in semi-parametric statistics, non-parametric statistics, nonparametric Bayesian methods, and machine learning. The caveats of the preface to the first edition remain: the choices made reflect our personal interests and research efforts over the 26 years since appearance of the first edition, so the reader should understand that many equally interesting applications and further research directions are not covered here.

We continue to owe thanks to a large number of colleagues and students who have helped by pointing out misprints, minor errors, and other deficiencies in the first edition. In particular we would like to thank: Moulinath Banerjee, Richard M. Dudley, Lutz Dümbgen, Bart Eggen, Zheng Fang, Stefan Franssen, the late Evarist Giné, the late Jørgen Hoffmann-Jørgensen, Geerten Koers, Michael Kosorok, Thomas Nagler, Lasse Vuursteen, Marten Wegkamp, and the late Joel Zinn. The authors welcome comments on errors that remain from the First Edition or on new errors in the Second Edition. We intend to maintain a list of errata on our website, as already available for the First Edition.

The authors have been partially supported in their research efforts over the period of work on this book (1996–2022) by grants from the European Research Council, National Science Foundation, NIAID, NWO, NATO, CNRS, Stieltjes Institute, the Simons Foundation, and TU Delft. Our thanks also go to our editor(s) at Springer, Allison DeVille, Laura Briskman, Neil Levine and Faith Su.

We have retained our personal format files, written in plain T_EX in the early 1990s but providing a L^aT_EX like environment, for this new edition. The second edition is typeset in *Times* fonts instead of the *Computer Modern* fonts used for the first edition. We thank Pascal van der Vaart for help with writing Python scripts that automate the compilation of the index files.

Delft and Seattle, December 2022

Reading Guide

This book consists of three “parts” and an appendix.

Part 1 is an exposition of (mostly) three modes of convergence of stochastic variables with values in metric spaces: convergence in distribution, convergence in probability, and almost sure convergence. A new aspect of this part, as compared to existing literature, is the fact that the stochastic variables are not assumed measurable with respect to the Borel σ -field. Part 1 is also useful in collecting a large number of results that have thus far not been available in book form in sufficient generality. In particular this concerns the systematic treatment of convergence in distribution in spaces of bounded functions equipped with the supremum metric. Most subjects treated in Part 1 are used in the two later parts of the book, but Chapters 1.2, 1.3, 1.5, 1.9, and 1.10 probably form the core of this part, and should be read in this order. Part 1 begins with a thorough introduction.

Part 2 is mostly concerned with the empirical measure and empirical process of a sample of observations, indexed by a class of functions. These are the maps

$$f \mapsto \mathbb{P}_n f := \frac{1}{n} \sum_{i=1}^n f(X_i)$$

and

$$f \mapsto \mathbb{G}_{n,P} f := \frac{1}{\sqrt{n}} \sum_{i=1}^n (f(X_i) - Pf),$$

whose domain is a class $\mathcal{F}$ of measurable functions. The main results in this part are contained in Chapters 2.4–2.5, and Chapters 2.14–2.15. The first pair of chapters concern the uniform law of large numbers (Glivenko-Cantelli theorem) and the uniform central limit theorem (Donsker theorem), respectively. The second pair of chapters gives bounds on the expectation and concentration of the supremum of the

empirical process. Chapters 2.6, 2.7, and 2.10 contain many examples of classes that satisfy the conditions of the theorems in these chapters. Chapter 2.1 is an introduction and Chapters 2.2 and 2.3 are necessary preparation for the main results of Part 2. Again, many of the results from the other chapters reappear later in the book but need not be studied in sequential order.

Part 3 consists of 15 chapters, which in principle can be read independently. This part shows the wide range of applications of the results obtained in the earlier parts in statistics, ranging from parametric and nonparametric estimation to the bootstrap, the functional delta-method, and Kolmogorov-Smirnov statistics. Every chapter assumes familiarity of the basic notation of Parts 1 and 2, but no section requires knowledge of more than a few sections of Part 2. Chapter 3.1 gives an overview of this part.

The material presented in the three parts is self-contained to a reasonable extent. The appendix covers a number of auxiliary subjects that are used to develop some of the material in the three main parts. Most results are presented with proof and some without, to serve as an easy reference.

Most of the chapters contain a number of “Problems and Complements” at the end. A number of these are real, textbook-style problems, but a lot of this material is meant as a supplement to the main text. Some problems present technical details, while other problems concern additional results of interest. We should warn the reader that the density of errors in these problem sections is probably higher than in the main text. Not all problems and complements have been triple-checked.

Bibliographical and historical information is placed in “Notes” sections, which appear at the end of each of the three parts and the appendix, organised by chapter.

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PART 1

Stochastic Convergence

1.1

Introduction

The first goal in this book is to give an exposition of the modern weak convergence theory suitable for the study of empirical processes.

Let $(\mathbb{D}, d)$ be a metric space, and let $\{P_n\}$ and P be Borel probability measures on $(\mathbb{D}, \mathcal{D})$, where $\mathcal{D}$ is the Borel σ -field on $\mathbb{D}$, the smallest σ -field containing all the open sets. Then the sequence P_n *converges weakly* to P , which we write as $P_n \rightsquigarrow P$, if and only if

$$(1.1.1) \quad \int_{\mathbb{D}} f dP_n \rightarrow \int_{\mathbb{D}} f dP, \quad \text{for all } f \in C_b(\mathbb{D}).$$

Here $C_b(\mathbb{D})$ denotes the set of all bounded, continuous, real functions on $\mathbb{D}$. Equivalently, if X_n and X are $\mathbb{D}$ -valued random variables with distributions P_n and P respectively, then $X_n \rightsquigarrow X$ if and only if

$$(1.1.2) \quad \mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X), \quad \text{for all } f \in C_b(\mathbb{D}).$$

These definitions yield the classical theory of *weak convergence* as treated in Billingsley (1968) and which has proved very useful in probability theory and statistics. The basic elements of this theory include the portmanteau theorem, continuous mapping theorems, Prohorov's theorem, tools for establishing tightness and uniform tightness, and weak convergence results for product spaces.

The classical theory requires that P_n is defined, for each n , on the Borel σ -field $\mathcal{D}$, or, equivalently, that X_n is a Borel measurable map for each n . If $(\Omega_n, \mathcal{A}_n, P_n)$ are the underlying probability spaces on which the maps X_n are defined, this means that $X_n^{-1}(D) \in \mathcal{A}_n$ for every Borel set D . This required measurability usually holds when $\mathbb{D}$ is a separable metric space such as $\mathbb{R}^k$ or $C[0, 1]$ with the supremum metric and sometimes even when $\mathbb{D}$ is nonseparable (for example, in the case of partial-sum processes).

However, this apparently modest requirement can and does easily fail when the metric space $\mathbb{D}$ is not separable. For example, this occurs when $\mathbb{D}$ is the Skorokhod space $D[0, 1]$ of all right-continuous functions on $[0, 1]$ with left limits endowed with the metric induced by the supremum norm or for the space $\ell^\infty(\mathcal{F})$ of all bounded functions from a set $\mathcal{F}$ to $\mathbb{R}$ equipped with the supremum norm $\|z\|_{\mathcal{F}} = \sup_{f \in \mathcal{F}} |z(f)|$.

The classical example of this difficulty comes from empirical process theory. Suppose that for each n the random variables $\xi_1, \dots, \xi_n$ are the independent, uniformly distributed variables, defined as the coordinate projections on the product probability space $([0, 1], \mathcal{B}, \lambda)^n$, where λ denotes Lebesgue measure on $[0, 1]$ and $\mathcal{B}$ the Borel σ -field. The empirical distribution function $\mathbb{F}_n$ is the random function

$$\mathbb{F}_n(t) = \frac{1}{n} \sum_{i=1}^n 1_{[0, t]}(\xi_i), \quad 0 \leq t \leq 1;$$

and the uniform empirical process X_n is

$$X_n(t) = \sqrt{n}(\mathbb{F}_n(t) - t), \quad 0 \leq t \leq 1.$$

Both $\mathbb{F}_n$ and X_n can be viewed as maps from $[0, 1]^n$ into $D[0, 1]$, but neither $\mathbb{F}_n$ nor X_n is a Borel measurable map if $D[0, 1]$ is endowed with the supremum norm. It turns out that the Borel σ -field $\mathcal{D}$ is so large in this case that the inclusion $X_n^{-1}(\mathcal{D}) \subset \mathcal{B}^n$ fails to hold (cf. Problem 1.7.3). This failure was pointed out by Chibisov (1965) and is nicely explained by Billingsley (1968); see Billingsley's Chapter 18, pages 150–153. Thus, the basic definitions (1.1.1) and (1.1.2) of weak convergence cannot be used for X_n viewed as a random function with values in $(D[0, 1], \|\cdot\|_\infty)$, even though this is a very natural and useful space in which to view the processes X_n , and even though the natural limiting process X , a Brownian bridge process on $[0, 1]$, can be taken to be a Borel measurable map (which takes its values in the separable subspace $C[0, 1]$).

Through the late 1960s, several different approaches were suggested to deal with this difficulty. Skorokhod (1956) and Billingsley (1968) endowed $D[0, 1]$ with the Skorokhod metric (or a modification) under which $D[0, 1]$ is separable (and complete) and the classical theory could be applied without difficulties. Dudley (1966, 1967a) developed an alternative weak convergence theory based on the smaller ball σ -field generated by the $\|\cdot\|_\infty$ -open balls in $D[0, 1]$. Pyke and Shorack (1968) proposed yet another definition of weak convergence requiring convergence of the integrals only for those functions $f \in C_b(\mathbb{D})$ for which $f(X_n)$ is a measurable map from $(\Omega_n, \mathcal{A}_n)$ into $\mathbb{R}$. While Dudley's proposition successfully handles the uniform empirical process and many simple extensions, it cannot handle the general empirical process and certain aspects of large sample theory in statistics.

The key idea put forward more recently by J. Hoffmann-Jørgensen is to drop the requirement of Borel measurability of each X_n , meanwhile upholding the requirement (1.1.2), where the expectations are now to be interpreted as *outer expectations* and the X_n may be arbitrary (possibly nonmeasurable) maps. Provided the limiting variable X is Borel measurable, a fruitful theory of weak convergence is

still possible. This chapter is a systematic exposition of this theory, and it also investigates the relationships with other modes of stochastic convergence such as convergence in probability and almost sure convergence, which can also be extended to nonmeasurable maps.

The first job is to define outer integrals and the basic techniques related to these integrals. If T is an arbitrary (not necessarily measurable) map from a probability space $(\Omega, \mathcal{A}, P)$ to the extended real line $\bar{\mathbb{R}}$, then the *outer integral* of T with respect to P is defined as

$$E^*T = \inf \left\{ EU : U \geq T, U: \Omega \rightarrow \bar{\mathbb{R}} \text{ measurable and } EU \text{ exists} \right\}.$$

It is useful to know that the infimum is achieved in a strong sense: if $E^*T < \infty$, then $E^*T = ET^*$, where T^* is a “smallest measurable function above T .”

In terms of outer integrals, weak convergence of arbitrary, possibly nonmeasurable, maps X_n from underlying probability spaces $(\Omega_n, \mathcal{A}_n, P_n)$ to a metric space $\mathbb{D}$ is defined as follows. The sequence X_n converges weakly to a Borel measurable map X , and we write $X_n \rightsquigarrow X$, if

$$(1.1.3) \quad E^*f(X_n) \rightarrow Ef(X), \quad \text{for every } f \in C_b(\mathbb{D}).$$

This part develops the weak convergence theory connected with this definition, more or less in parallel to the classical theory, including a portmanteau theorem, continuous mapping theorems, Prohorov’s theorem, tightness and basic tools for establishing tightness, and weak convergence on product spaces.

Because of several important applications of nets of processes in statistics, we have formulated the theory in terms of *nets* $\{X_\alpha\}_{\alpha \in A}$, with A a directed set, rather than just for sequences. Recall that a *directed set* A is a set equipped with a partial order $\leq$ with the further property that every pair α_1, α_2 has a “successor” α_3 (satisfying $\alpha_3 \geq \alpha_1$ and $\alpha_3 \geq \alpha_2$). For a sequence, the directed set is the set of natural numbers with the usual ordering, and all results for nets are valid for sequences in particular.

Once the basic elements of the theory are available, Part 1 develops further connections and relationships. Because spaces of uniformly bounded functions are important for empirical process theory, Chapter 1.5 develops conditions for weak convergence in these spaces. For example, Theorem 1.5.4 asserts that asymptotic tightness together with weak convergence of the finite-dimensional distributions suffices for weak convergence. This resembles the classical theorem for convergence in $C[0, 1]$, as in Billingsley (1968), but the present formulation is considerably more general and flexible.

The relationship between the present theory and measurability with respect to the ball σ -field is explored in Chapter 1.7. Chapter 1.8 develops the theory for the special case of a Hilbert space.

Just as it is useful to develop a theory of weak convergence for nonmeasurable maps, it is useful to extend the notions of “convergence in probability” and “convergence almost surely.” The appropriate analogues are developed and investigated in Chapter 1.9, including the corresponding continuous mapping theorems. Chapter 1.10 gives relationships and connections between these stronger convergence

concepts and weak convergence as defined in (1.1.3). In particular, this section contains an analogue of the Skorokhod-Dudley-Wichura almost sure representation theorem for nonmeasurable maps.

The main result of Chapter 1.11 is the “extended continuous mapping theorem,” which applies to a sequence of functions g_n rather than to a fixed continuous function g . Chapter 1.12 explores uniformity of the convergence (1.1.3) in f for appropriate subsets $\mathcal{F}$ of $C_b(\mathbb{D})$ and uses this to metrize weak convergence under the additional condition that the limit X is separable. Metrization allows to define and study weak convergence of conditional distributions, in probability or almost surely, in Chapter 1.13.

Many interesting applications of weak convergence theory involve random processes X_n (and a limit process X) with values in the Skorokhod spaces $D[0, 1]$ or $D[0, \infty)$. While the theory developed in Chapters 1.5–1.7 handles such applications when the limit process X is continuous (or has “fixed jumps” only), when X has jumps at random locations a weaker metric than the uniform metric is necessary. The Skorokhod topology provide such a metric and is useful when the jumps in two functions that occur at nearby locations should be identified. In Chapter 1.14 we review the theory of weak convergence in these spaces including criteria for tightness (both classical and more recently developed). We illustrate the theory with several examples, including convergence to Poisson type processes, to more general processes with independent increments, and to semimartingales.

Stable convergence, considered in Chapter 1.15, can be viewed as intermediate between weak convergence and convergence in probability. It is connected to weak convergence jointly with other variables, and also to convergence of conditional distributions. The weak convergence entailed in the central limit theorem is actually stable convergence.

Much of the theory of stochastic convergence for nonmeasurable maps follows the same lines as the classical theory. Some of the key differences between the present theory and the classical theory that emerge in the course of Part 1 are as follows:

- (i) The notion of (uniform) tightness of sequences needs modification.
- (ii) The “direct half” of Prohorov’s theorem (asymptotic tightness implies relative compactness) must be modified by the addition of the requirement of *asymptotic measurability*.
- (iii) Almost sure convergence or even convergence everywhere is meaningless without some (asymptotic) measurability.
- (iv) There is no general version of Fubini’s theorem (but see Lemmas 1.2.6 and 1.2.7).

In spite of these differences and slight difficulties resulting from sacrificing measurability of the converging random quantities, the weak convergence theory outlined in Part 1 parallels the classical theory to a remarkable degree and will prove valuable in dealing with empirical processes, which are studied in Part 2, and asymptotic statistical theory, which is treated in Part 3.

1.2

Outer Integrals and Measurable Majorants

Let $(\Omega, \mathcal{A}, P)$ be an arbitrary probability space and $T: \Omega \rightarrow \bar{\mathbb{R}}$ an arbitrary map. The *outer integral* of T with respect to P is defined as

$$E^*T = \inf \left\{ EU: U \geq T, U: \Omega \rightarrow \bar{\mathbb{R}} \text{ measurable and } EU \text{ exists} \right\}.$$

Here, as usual, EU is understood to exist if at least one of EU^+ or EU^- is finite. The *outer probability* of an arbitrary subset B of Ω is

$$P^*(B) = \inf \left\{ P(A): A \supset B, A \in \mathcal{A} \right\}.$$

Note that the functions U in the definition of outer integral are allowed to take the value ∞ , so that the infimum is never empty.

Inner integral and *inner probability* can be defined in a similar fashion — their definition should be obvious. Equivalently, they can be defined by $E_*T = -E^*(-T)$ and $P_*(B) = 1 - P^*(\Omega - B)$, respectively.

A very useful fact is that the infima in the definitions of the outer integral and probability are always achieved. This even happens for an essentially minimal U and A as in the definitions (provided $E^*T < \infty$), which will be denoted T^* and B^* . For outer integrals, this is contained in the following lemma; for outer probabilities, a proof is deferred to Lemma 1.2.3.

1.2.1 Lemma (Measurable cover function). *For any map $T: \Omega \rightarrow \bar{\mathbb{R}}$, there exists a measurable function $T^*: \Omega \rightarrow \bar{\mathbb{R}}$ with*

- (i) $T^* \geq T$;
- (ii) $T^* \leq U$ a.s., for every measurable $U: \Omega \rightarrow \bar{\mathbb{R}}$ with $U \geq T$ a.s.

For any T^ satisfying these requirements, it holds that $E^*T = ET^*$, provided ET^* exists. The latter is certainly true if $E^*T < \infty$.*

Proof. Choose a measurable sequence $U_m \geq T$ with $E \arctan U_m \downarrow E^* \arctan T$, and set

$$T^*(\omega) = \lim_{m \rightarrow \infty} \inf_{1 \leq k \leq m} U_k(\omega).$$

This defines a measurable function T^* taking values in the extended real line, with $T^* \geq T$, and by monotone convergence $E \arctan T^* = E^* \arctan T$. Every measurable $U \geq T$ satisfies $\arctan U \wedge T^* \geq \arctan T$, so that $E \arctan U \wedge T^* \geq E^* \arctan T = E \arctan T^*$. But the integrand on the left side is trivially pointwise smaller than the integrand on the right side. Since they have the same expectation, they must be equal: $\arctan U \wedge T^* = \arctan T^*$ a.s. This implies $T^* \leq U$ a.s.

If ET^* exists, then it is larger than E^*T by (i) and smaller by (ii). Hence $ET^* = E^*T$. If $E^*T < \infty$, then there exists a measurable $U \geq T$ with $EU^+ < \infty$. Then $E(T^*)^+ \leq EU^+$ and ET^* exists. ■

The function T^* is called a *minimal measurable majorant* of T , or also a measurable cover or envelope function. It is unique only up to P null sets but, somewhat abusing notation, we write T^* for any member of its equivalence class of a.s. equal functions. Its expectation does not always exist, so some care is needed when applying the identity $E^*T = ET^*$ (Problem 1.2.2). A *maximal measurable minorant* is defined by $T_* = -(-T)^*$ and satisfies the obvious relations. Some of the properties of these functions are collected in the following lemma. It is tedious, but useful. The same is true for the other results in this section; perhaps it is better to skip them at first reading.

1.2.2 Lemma. *The following statements are true a.s. for arbitrary maps $S, T: \Omega \rightarrow \mathbb{R}$, provided the statement is well-defined:*

- (i) $(S + T)^* \leq S^* + T^*$, with equality if S is measurable;
- (ii) $(S - T)^* \geq S^* - T^*$;
- (iii) $|S^* - T^*| \leq |S - T|^*$;
- (iv) If S is measurable, then $(ST)^* = S1_{S>0}T^* + S1_{S<0}T_*$;
- (v) $(ST)^* \leq S^*T^*1_{S^*>0, T^*>0} + S^*T_*1_{S^*<0, T_*>0} + S_*T^*1_{S_*>0, T^*<0} + S_*T_*1_{S_*<0, T_*<0}$;
- (vi) $(1_{T>c})^* = 1_{T^*>c}$ for any $c \in \mathbb{R}$;
- (vii) $|T|^* = T^* \vee (-T)^* = T^* \vee (-T_*) = |T^*| \vee |T_*|$;
- (viii) $(S \vee T)^* = S^* \vee T^*$;
- (ix) $(S \wedge T)^* \leq S^* \wedge T^*$ with equality if S is measurable.

Moreover, $P^*(T > c) = P(T^* > c)$ for any $c \in \mathbb{R}$.

Proof. Every inequality in this proof means inequality a.s. The inequality in (i) is trivial. If S is measurable and $U \geq S + T$ and is measurable, then $U - S \geq T$ and is measurable, so $U - S \geq T^*$, which shows that $(S + T)^* = S + T^*$. Statement (ii) is a rearrangement and relabelling of (i). Next, (iii) follows from $S^* - T^* \leq (S - T)^* \leq |S - T|^*$.

In (iv) it is trivial that the left side is smaller than the right side. Suppose $U \geq$

ST and is measurable. Then the following string of implications holds:

$$\begin{aligned}
 U1_{S>0} &\geq ST1_{S>0}, \\
 U/S1_{S>0} &\geq T1_{S>0}, \\
 U/S1_{S>0} + T^*1_{S\leq 0} &\geq T, \\
 U/S1_{S>0} + T^*1_{S\leq 0} &\geq T^*, \\
 U/S1_{S>0} &\geq T^*1_{S>0}, \\
 U1_{S>0} &\geq ST^*1_{S>0}.
 \end{aligned}$$

Furthermore, since the starting inequality can also be written $U \geq (-S)(-T)$, we also have

$$U1_{-S>0} \geq (-S)(-T)^*1_{-S>0}.$$

Adding the last two displayed inequalities to the trivial one $U1_{S=0} \geq 0$ yields $U \geq S1_{S>0}T^* + S1_{S<0}T^*$. This concludes the proof of (iv).

To obtain (v), first write $ST \leq S^*T1_{T>0} + S_*T1_{T<0}$. Next use (iv) repeatedly, together with a number of simple inequalities, such as $(T1_{T>0})^* \leq T^*1_{T^*>0}$.

In (vi) it is trivial that $(1_{T>c})^* \leq 1_{T^*>c}$. Suppose $U \geq 1_{T>c}$ and is measurable. Then $S = T^*1_{U\geq 1} + (T^* \wedge c)1_{U<1} \geq T$ and is measurable. Thus $S \geq T^*$, whence $T^* \leq c$ if $U < 1$. So $1_{T^*>c} \leq U$.

It is immediate in (vii) that the functions are nondecreasing from left to right. Furthermore, $|T^*| \leq |T|^*$ follows from (iii) applied with $S = 0$, while $|T_*| = |(-T)^*| \leq |-T|^*$ by a second application of (iii).

In (viii) it is clear that $S^* \vee T^*$ is a measurable majorant of $S \vee T$. This gives one-half of the equality. The other half follows from this: if $U \geq S$ and $U \geq T$ and is measurable, then U is greater than or equal to the measurable majorants of S and T , and hence $U \geq S^* \vee T^*$.

The first part of (ix) is trivial. To obtain the second part, first note that $(S \wedge T)^* \leq S \wedge T^*$. Next, if $U \geq S \wedge T$ and is measurable, then $U1_{U<S} \geq T1_{U<S}$. Hence $U1_{U<S} \geq (T1_{U<S})^* = T^*1_{U<S}$, so that $U \geq T^* \wedge S$.

To obtain the last assertion of the lemma, write $P(T^* > c) \geq P^*(T > c) \geq E^*1_{T>c} = E1_{T^*>c} = P(T^* > c)$, where the inequalities are immediate consequences of the definitions and the equality follows from (vi). ■

The following lemma shows that outer probabilities are special cases of outer integrals. Furthermore, just as for outer integrals, the infimum in their definition is achieved, by measurable sets that we denote by B^* .

1.2.3 Lemma. *For any subset B of Ω ,*

- (i) $P^*(B) = E^*1_B$; $P_*(B) = E(1_B)_*$;
- (ii) *there exists a measurable set $B^* \supset B$ with $P(B^*) = P^*(B)$; for any such B^* , it holds that $1_{B^*} = (1_B)^*$;*
- (iii) $(1_B)^* + (1_{\Omega-B})_* = 1$.

Proof. From the definitions it is immediate that $P^*(B) \geq E^*1_B$. Next with $A = \{1_B^* \geq 1\}$, one has $E^*1_B = E1_B^* \geq P(A) \geq P^*(B)$, where the inequalities are

direct consequences of the definitions. Combination yields that all the inequalities are in fact equalities. This shows the first part of (i) and (ii). The second part of (i) follows from $P_*(B) = 1 - P^*(\Omega - B) = 1 - E(1 - 1_B)^* = 1 - E(1 - (1_B)_*)$. The second part of (ii) follows from the trivial inequality $1_{B^*} \geq (1_B)^*$ if $B^* \supset B$ and $E1_{B^*} = P(B^*) = E(1_B)^*$. To obtain (iii), write $(1_{\Omega-B})_* = (1 - 1_B)_* = 1 - (1_B)^*$. ■

Though we didn't let this show up in the notation, the minimal cover function T^* depends on the underlying probability measure P . Suppose we have a set $\mathcal{P}$ of probability measures. Is it possible to find one function that is a minimal measurable cover function for every $P \in \mathcal{P}$ at the same time? If $\mathcal{P}$ is dominated (by a σ -finite measure), the answer is affirmative.

1.2.4 Lemma. *Let $\mathcal{P}$ be a dominated set of probability measures on $(\Omega, \mathcal{A})$. Then there exists a measurable function $T^*: \Omega \rightarrow \bar{\mathbb{R}}$ with*

- (i) $T^* \geq T$;
- (ii) $T^* \leq U$ P -a.s., for every measurable $U: \Omega \rightarrow \bar{\mathbb{R}}$ with $U \geq T$ P -a.s., for every $P \in \mathcal{P}$.

Proof. Let P_0 be a probability measure that dominates $\mathcal{P}$. Take T^* equal to the minimal measurable cover function that satisfies (i) and (ii) for $\mathcal{P} = \{P_0\}$. Given an arbitrary $P \ll P_0$, there is a decomposition $\Omega = \Omega^a + \Omega^\perp$ of Ω into disjoint, measurable sets with $P(\Omega^\perp) = 0$ and P_0 absolutely continuous with respect to P on Ω^a . (Take densities p and p_0 , and set $\Omega^a = \{p > 0, p_0 > 0\}$ and $\Omega^\perp = \{p = 0, \text{ or } p_0 = 0\}$.) If $U \geq T$, P -a.s., and is measurable, then $U1_{\Omega^a} \geq T1_{\Omega^a}$, P_0 -a.s., and is measurable. Hence $U1_{\Omega^a} \geq (T1_{\Omega^a})^{*P_0} = T^*1_{\Omega^a}$, P_0 -a.s. Since $\mathcal{P}$ is dominated by P_0 , this then also holds P -a.s., but then also $U \geq T^*$, P -a.s., because $P(\Omega^\perp) = 0$. ■

Consider what happens to a minimal measurable cover if a map $T: \Omega \rightarrow \mathbb{R}$ is composed with a measurable map $\phi: \tilde{\Omega} \rightarrow \Omega$ defined on some probability space to form

$$T \circ \phi: (\tilde{\Omega}, \tilde{\mathcal{A}}, \tilde{P}) \xrightarrow{\phi} (\Omega, \mathcal{A}, \tilde{P} \circ \phi^{-1}) \xrightarrow{T} \mathbb{R}.$$

Let T^* be the minimal measurable cover of T for $\tilde{P} \circ \phi^{-1}$. Since $T^* \circ \phi \geq T \circ \phi$ and is measurable, trivially $(T \circ \phi)^* \leq T^* \circ \phi$. The map ϕ is called *perfect*[†] if $(T \circ \phi)^* = T^* \circ \phi$, for every bounded $T: \Omega \rightarrow \mathbb{R}$. It is exactly the property that ensures that

$$E^*T \circ \phi = \int^* T d\tilde{P} \circ \phi^{-1}, \quad \text{for every bounded } T: \Omega \rightarrow \mathbb{R}.$$

[†] There is some clash of terminology here. A probability space $(\Omega, \mathcal{A}, P)$ is called *perfect* if for every Borel measurable map $\phi: \Omega \rightarrow \mathbb{R}$, the collection of sets $\{B: \phi^{-1}(B) \in \mathcal{A}\}$ is contained in the $P \circ \phi^{-1}$ -completion of the Borel sets. (So the biggest σ -field on $\mathbb{R}$ for which ϕ is measurable is contained in the Borel σ -field up to differences in $P \circ \phi^{-1}$ -null sets.) Dudley (1985) calls a ϕ for which the condition is satisfied *quasi-perfect*, so that a probability space is perfect if every Borel measurable real function on it is quasi-perfect. Any perfect function is quasi-perfect, but not the other way around.

Note that perfectness of ϕ depends on $(\tilde{\Omega}, \tilde{\mathcal{A}}, \tilde{P})$ as well as on ϕ and $(\Omega, \mathcal{A})$.

In particular, $P^*(\phi \in A) = (\tilde{P} \circ \phi^{-1})^*(A)$ for every set $A \subset \Omega$.

Perfect maps do exist. An example that will be encountered frequently results from the following. Suppose a map T is defined on a product probability space $(\Omega_1 \times \Omega_2, \mathcal{A}_1 \times \mathcal{A}_2, P_1 \times P_2)$, but really depends only on the first coordinate of $\omega = (\omega_1, \omega_2)$. Then T^* (for $P_1 \times P_2$) can be computed (for P_1) after just ignoring Ω_2 and thinking of T as a map on Ω_1 . More formally, suppose $T = T_1 \circ \pi_1$, where π_1 is the projection on the first coordinate. Then the next lemma shows that $T^* = T_1^* \circ \pi_1$.

1.2.5 Lemma. *A coordinate projection on a product probability space (with product measure) is perfect.*

Proof. Let $\pi_1: (\Omega_1 \times \Omega_2, \mathcal{A}_1 \times \mathcal{A}_2, P_1 \times P_2) \rightarrow \Omega_1$ be the projection on the first coordinate and $T: \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}$ be bounded, but arbitrary otherwise. Let T^* be the least measurable cover of T for $(P_1 \times P_2) \circ \pi_1^{-1} = P_1$. Trivially $(T \circ \pi_1)^* \leq T^* \circ \pi_1$. Suppose $U \geq T \circ \pi_1$, $P_1 \times P_2$ -a.s. and is measurable (where $U: \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}$). Then, by Fubini's theorem, for P_2 -almost all ω_2 : $U(\omega_1, \omega_2) \geq T(\omega_1)$ for all ω_1 . Since for ω_2 fixed, U is a measurable function of ω_1 , for P_2 -almost all ω_2 : $U(\omega_1, \omega_2) \geq T^*(\omega_1)$ for P_1 -almost all ω_1 . By Fubini's theorem, the jointly measurable set $\{(\omega_1, \omega_2): U < T^* \circ \pi_1\}$ is $P_1 \times P_2$ -null. ■

Now we must consider Fubini's theorem. Unfortunately, measurability plays an important role in this result, and there is no general Fubini's theorem for outer expectations. This is the main reason why certain arguments work only under measurability assumptions.

Fubini's theorem is valid as a string of inequalities: repeated outer expectations are always less than joint outer integrals. This can only be formulated in a somewhat awkward notation. Let T be a real-valued map defined on a product space $(\Omega_1 \times \Omega_2, \mathcal{A}_1 \times \mathcal{A}_2, P_1 \times P_2)$. Then write E^*T for the outer expectation as before, and write $E_1^*E_2^*T$ for the outer expectations taken in turn: define for every ω_1 :

$$(E_2^*T)(\omega_1) = \inf \int U(\omega_2) dP_2(\omega_2),$$

where the infimum is taken over all measurable functions $U: \Omega_2 \rightarrow \bar{\mathbb{R}}$ with $U(\omega_2) \geq T(\omega_1, \omega_2)$ for every ω_2 and such that $\int U dP_2$ exists. Next $E_1^*E_2^*T$ is the outer integral of the function $E_2^*T: \Omega_1 \rightarrow \bar{\mathbb{R}}$. Repeated inner expectations are defined analogously.

1.2.6 Lemma (Fubini's theorem). *Let T be defined on a product probability space. Then $E_*T \leq E_{1*}E_{2*}T \leq E_1^*E_2^*T \leq E^*T$.*

Proof. For the inequality on the far right, it may be assumed that $E^*T < \infty$, so that $E^*T = ET^*$ and $(T^*)^+$ is integrable. Since T^* is jointly measurable for the product σ -field, the map $\omega_2 \mapsto T^*(\omega_1, \omega_2)$ is a measurable majorant of $\omega_2 \mapsto T(\omega_1, \omega_2)$ for P_1 -almost all ω_1 . Hence $\int T^*(\omega_1, \omega_2) dP_2(\omega_2) \geq (E_2^*T)(\omega_1)$ for almost every ω_1 . Here the left side is defined as the difference

$$\int (T^*)^+(\omega_1, \omega_2) dP_2(\omega_2) - \int (T^*)^-(\omega_1, \omega_2) dP_2(\omega_2),$$

in which the first term is finite for almost every ω_1 and the second is possibly infinite. By Fubini's theorem, both terms are measurable functions of ω_1 . It follows that the difference is a measurable majorant of the map $\omega_1 \mapsto (E_2^*T)(\omega_1)$, and its integral with respect to P_1 is an upper bound for $E_1^*E_2^*T$, provided it exists. Since the first term of the difference is integrable, the integral of the difference exists and is the difference of the integrals. Next by Fubini's theorem, the double integrals in both terms can be replaced by a joint integral, which yields $E(T^*)^+ - E(T^*)^- \geq E_1^*E_2^*T$.

Finish the proof by considering $-T$. ■

If a function T on Euclidean space is continuous, then it is measurable and Fubini's theorem holds. In the case that T is continuous in only one argument but not necessarily measurable, it may still be possible to write the joint outer expectation as a repeated outer expectation (in one of the two possible orders). Consider a map T on a product probability space as before. Write T^{2*} for any map such that for every ω_1 the map $\omega_2 \mapsto T^{2*}(\omega_1, \omega_2)$ is a measurable cover function for $\omega_2 \mapsto T(\omega_1, \omega_2)$ (for P_2).

1.2.7 Lemma (Fubini's theorem). *Let $(\Omega_1, \mathcal{A}_1)$ be a separable metric space equipped with its Borel σ -field. Suppose the map $T: \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}$ is defined on a product probability space and satisfies $|T(\omega_1, \omega_2) - T(\omega'_1, \omega_2)| \leq d(\omega_1, \omega'_1) H(\omega_2)$, for a function H with $P_2^*H < \infty$ and for every sufficiently small $d(\omega_1, \omega'_1)$. Then $E^*T = E_1E_2^*T$ whenever the left side is finite.*

Proof. For every n , partition $\Omega_1 = \bigcup_{j=1}^{\infty} A_{n,j}$ into measurable sets of diameter at most $1/n$. Choose an arbitrary point $a_{n,j}$ from each set $A_{n,j}$ and define the discretization

$$T_n(\omega_1, \omega_2) = \sum_j 1_{A_{n,j}}(\omega_1) T(a_{n,j}, \omega_2).$$

Then

$$T_n^*(\omega_1, \omega_2) = \sum_j 1_{A_{n,j}}(\omega_1) T_n^{2*}(a_{n,j}, \omega_2).$$

Indeed, it is certainly true that T_n^* is jointly measurable and dominates T_n . If $U \geq T_n$ and is jointly measurable, then $U(\omega_1, \omega_2) \geq T(a_{n,j}, \omega_2)$ for every $\omega_1 \in A_{n,j}$ and every ω_2 , whence $U(\omega_1, \cdot) \geq T_n^{2*}(a_{n,j}, \cdot)$ almost surely for every $\omega_1 \in A_{n,j}$. In view of Fubini's theorem this shows that the jointly measurable set $\{(\omega_1, \omega_2): U(\omega_1, \omega_2) < T_n^*(\omega_1, \omega_2)\}$ is $P_1 \times P_2$ -null.

Next $|T_n^* - T^*| \leq |T_n - T|^* \leq 1/nH^*$. Take the expectation on the left and right, and let $n \rightarrow \infty$, to conclude that

$$E^*T = \lim E^*T_n = \lim \sum_j P_1(A_{n,j}) E_2 T_n^{2*}(a_{n,j}, \cdot).$$

Since the map $\omega_1 \mapsto E_2 T_n^{2*}(\omega_1, \cdot) = (E_2^*T_n)(\omega_1)$ is continuous, it follows that the right side of the preceding display is equal to $E_1E_2^*T$. ■

The preceding lemma applies in particular to countable Ω_1 with discrete topology, in which case the continuity condition is automatic. In this case we also have that any versions $T^{2*}(\omega_1, \cdot)$ (determined independently for different values of ω_1) automatically yield a measurable cover $T^*(\omega_1, \omega_2) = T^{2*}(\omega_1, \omega_2)$ for T . This follows since $T \leq T^{2*} \leq T^*$ always and for countable, discrete Ω_1 , any version T^{2*} is automatically jointly measurable.

We note that under the conditions of the preceding theorem, it need not be true that $E^*T = E_2^*E_1T$.

Problems and Complements

1. **(Essential infima)** Let $(\Omega, \mathcal{A}, P)$ be a probability space and $\mathcal{U}$ an arbitrary set of measurable functions $U: \Omega \rightarrow \mathbb{R}$. Show that there exists a measurable function $U_*: \Omega \rightarrow \mathbb{R}$ with
 - (i) $U_* \leq U$ for every $U \in \mathcal{U}$;
 - (ii) $U_* \geq V$ a.s., for every measurable V with $V \leq U$ a.s. for every $U \in \mathcal{U}$.
 Such a function U_* is called the *essential infimum* of $\mathcal{U}$. Show that for a map $T: \Omega \rightarrow \mathbb{R}$, the minimal measurable cover T^* is the essential infimum of the set of all measurable $U \geq T$.
2. **(E^*T is not always ET^*)** Let $(\Omega, \mathcal{A}, P)$ be $\mathbb{R}$ with the σ -field generated by $[0, \infty)$ and the Borel sets in $(-\infty, 0)$ and P equal to the Cauchy measure. The map $T: \Omega \rightarrow \mathbb{R}$ defined by $T(\omega) = \omega$ has $E^*T = \infty$, but ET^* does not exist. Orlic

3. **(Monotone convergence)** Let $T_n, T: \Omega \rightarrow \mathbb{R}$ be maps on a probability space with $T_n \uparrow T$ pointwise on a set of probability one. Then $T_n^* \uparrow T^*$ almost surely. If the maps are bounded from below, then $E^*T_n \uparrow E^*T$. For a decreasing sequence T_n , similar results are true for the lower starred versions, but not for the upper starred objects.

[Hint: Since $T_n^* \leq T^*$ for every n , certainly $\liminf T_n^* \leq \limsup T_n^* \leq T^*$. Conversely, $\liminf T_n^* \geq \liminf T_n = T$ and is a measurable function. If $E^*T_n < \infty$ for every n , then $E^*T_n = ET_n^* \uparrow ET^*$ by the monotone convergence theorem for measurable maps; consequently, ET^* exists and equals E^*T . If $E^*T_n = \infty$ for some n , then it is infinite for every larger n and also $E^*T = \infty$ from its definition.]

4. **(Dominated convergence)** The “naive” dominated convergence theorem fails for outer integrals; in fact, on $\Omega = [0, 1]$ with the uniform measure on the Borel sets, there exist maps T_n with $T_n \downarrow 0$ everywhere and $|T_n| \leq 1$ for every n , such that $E^*T_n = 1$ for every n . However, if almost sure convergence is strengthened to $|T_n - T|^* \xrightarrow{\text{a.s.}} 0$ and $|T_n| \leq S$ for every n and S with $E^*S < \infty$, then $E^*T_n \rightarrow E^*T$.

[Hint: If $T_n \rightarrow T$ almost surely and $|T_n| \leq S$ for every n , then $|T_n - T|^* \leq 2S^*$. If $|T_n - T|^* \xrightarrow{\text{a.s.}} 0$ and $E^*S < \infty$, then $E|T_n - T|^* \rightarrow 0$, which implies the result.]

5. Let $S, T: \Omega \rightarrow \mathbb{R}$ be arbitrary. Then $|S^* - T_*| \leq |S - T|^* + (S^* - S_*) \wedge (T^* - T_*)$. Also, $|S - T|^* \leq (S^* - T_*) \vee (T^* - S_*) \leq |S - T|^* + (S^* - S_*) \wedge (T^* - T_*)$.
6. It can happen that $(S \wedge T)^* < S^* \wedge T^*$ with probability 1.

[Hint: There exist subsets of $[0, 1]$ with $\lambda_*(B) = 0$ and $\lambda^*(B) = 1$.]

7. For arbitrary $T: \Omega \rightarrow \mathbb{R}$, one has $P(T^* \leq c) = P_*(T \leq c)$ for every c . It can happen that $1 = P^*(T \leq c) > P(T^* \leq c) = 0$.

8. Let $T: \Omega \rightarrow \mathbb{R}$ be an arbitrary map defined on a probability space. If $g: \mathbb{R} \rightarrow \mathbb{R}$ is nondecreasing and left-continuous on an interval $(a, b]$ that contains the range of both T and T^* , then $g(T)^* = g(T^*)$. If g is nonincreasing and right-continuous, then $g(T)_* = g(T_*)$. The continuity conditions can be replaced by the condition that g is one-to-one, is measurable, and has a measurable inverse.

[Hint: That the left side is smaller than the right side is immediate from $g(T^*) \geq g(T)$. Suppose $g(T^*) \geq U \geq g(T)$ almost surely for a measurable U . The “inverse” $g^{-1}(u) = \sup\{x: g(x) \leq u\}$ satisfies $g(x) \leq u$ if and only if $x \leq g^{-1}(u)$. Consequently, $g^{-1}(U) \geq T$ almost surely. Since it is also measurable, $g^{-1}(U) \geq T^*$. This implies $U \geq g(T^*)$ almost surely. The second statement follows from the first applied to the function $x \mapsto -g(-x)$.]

9. Let $T_i: \Omega_i \rightarrow \mathbb{R}$ be maps defined on probability spaces $(\Omega_i, \mathcal{A}_i, P_i)$ and define (T_1, T_2) on $\Omega_1 \times \Omega_2$ by $(\omega_1, \omega_2) \mapsto (T_1(\omega_1), T_2(\omega_2))$. Let $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ be coordinatewise nondecreasing and left-continuous on an interval $(a, \infty]$ that contains the range of (T_1, T_2) . Then $g(T_1, T_2)^* = g(T_1^*, T_2^*)$, where the first measurable cover is computed for $P_1 \times P_2$ on the product σ -field $\mathcal{A}_1 \times \mathcal{A}_2$ and the T_i^* on the right each for the corresponding P_i . In particular, for “independent” T_1 and T_2 as above:

- (i) $(T_1 + T_2)^* = T_1^* + T_2^*$;
- (ii) $(T_1 T_2)^* = T_1^* T_2^*$ if $T_1 \geq 0$ and $T_2 \geq 0$ everywhere.

[Hint: It is clear that the left side is smaller than the right side. If $U \geq g(T_1, T_2)$ almost surely and is measurable, then, by Fubini’s theorem, for P_1 -almost all ω_1 , one has $U(\omega_1, \omega_2) \geq g(T_1(\omega_1), T_2(\omega_2))$ for P_2 -almost all ω_2 . For every ω_1 for which this holds, it follows that $U(\omega_1, \omega_2) \geq g(T_1(\omega_1), T_2^*(\omega_2))$ for P_2 -almost all ω_2 by the previous exercise. Now reverse the roles of ω_1 and ω_2 .]

10. (*P-completion*) The *P-completion* of a probability space $(\Omega, \mathcal{A}, P)$ is the triplet $(\Omega, \tilde{\mathcal{A}}, \tilde{P})$, where $\tilde{\mathcal{A}}$ consists of all sets $A \cup N$ (and also $A - N$) with $A \in \mathcal{A}$ and N a subset of Ω with $P^*(N) = 0$, and $\tilde{P}(A \cup N) = P(A)$. An equivalent description is that $\tilde{\mathcal{A}}$ is the collection of all sets with equal inner and outer probabilities under P , and $\tilde{P}$ maps each of these sets in this common value of P^* and P_* . A completion is a probability space, and for every measurable map $\tilde{U}: (\Omega, \tilde{\mathcal{A}}) \rightarrow \mathbb{R}$, there is a measurable map $U: (\Omega, \mathcal{A}) \rightarrow \mathbb{R}$ with $P^*(U \neq \tilde{U}) = 0$.

[Hint: For the construction of U , first assume that $\tilde{U} = \sum a_i 1_{\tilde{A}_i}$ with $\tilde{A}_i \in \tilde{\mathcal{A}}$. Take sets $A_i \in \mathcal{A}$ with $\tilde{A}_i = A_i \cup N_i$ and $P^*(N_i) = 0$. Then $U = \sum a_i 1_{A_i}$ satisfies the requirements. Then a nonnegative $\tilde{U}$ can be approximated pointwise from below by a sequence $\tilde{U}_n$ of the form just considered. Construct U_n for every n , and set $U = \liminf U_n$. A general $\tilde{U}$ can be split into its positive and negative parts.]

11. A minimal measurable cover T^* of a map $T: (\Omega, \mathcal{A}, P) \rightarrow \bar{\mathbb{R}}$ is also (a version of) a minimal measurable cover for T as a map on the P -completion of $(\Omega, \mathcal{A}, P)$.

[Hint: If $\tilde{T}^{*c}$ is a minimal measurable cover for the completion, then there is measurable function $T^{*c}: (\Omega, \mathcal{A}) \rightarrow \mathbb{R}$ with $T^{*c} = \tilde{T}^{*c}$, almost surely. Check that T^{*c} is a version of T^* . So $T^* = T^{*c} = \tilde{T}^{*c}$, almost surely.]

12. Let $(\Omega_i, \tilde{\mathcal{A}}_i, \tilde{P}_i)$ be the completions of probability spaces $(\Omega_i, \mathcal{A}_i, P_i)$, and let the probability space $(\Omega_1 \times \Omega_2, \tilde{\mathcal{A}}_1 \times \tilde{\mathcal{A}}_2, \tilde{P}_1 \times \tilde{P}_2)$ be the completion of their product. Then $\tilde{\mathcal{A}}_1 \times \Omega_2 \subset \tilde{\mathcal{A}}_1 \times \tilde{\mathcal{A}}_2$ and $\tilde{P}_1 \times \tilde{P}_2(A_1 \times \Omega_2) = \tilde{P}_1(A_1)$. Moreover, a map $T: \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}$ of the form $T(\omega_1, \omega_2) = T_1(\omega_1)$ is measurable for $\tilde{\mathcal{A}}_1 \times \tilde{\mathcal{A}}_2$ if and only if T_1 is measurable for $\tilde{\mathcal{A}}_1$. Finally, $T^* = T_1^*$, where the first is computed for $P_1 \times P_2$ on $\mathcal{A}_1 \times \mathcal{A}_2$ and the second for P_1 on $\mathcal{A}_1$.

- 13. (Outer measure)** A real function μ^* defined on the collection of all subsets of a set Ω is called an *outer measure* if $\mu^*(\emptyset) = 0$, $\mu^*(B_1) \leq \mu^*(B_2)$ if $B_1 \subset B_2$ and $\mu^*(\bigcup_{n=1}^{\infty} B_n) \leq \sum_{n=1}^{\infty} \mu^*(B_n)$. A subset B of Ω is said to be μ^* -measurable if for every subset C one has $\mu^*(C) = \mu^*(C \cap B) + \mu^*(C \cap \Omega - B)$. (According to a fundamental result of measure theory, the μ^* -measurable sets form a σ -field and the restriction of μ^* to this σ -field is a complete measure.) The outer probability P^* defined in the present section is an outer measure and the P^* -measurable sets are exactly the sets with $P^*(B) = P_*(B)$; they are also the sets in the P -completion of $\mathcal{A}$.

[Hint: If B is P^* -measurable, then the outer measure rule applied to $C = B^*$ yields $P(B^*) = P(B^*) + P^*(B^* - B)$.]

- 14.** For any $\varepsilon \in [0, 1]$, there exists a set $B \subset [0, 1]$ with $\lambda_*(B) = 0$ and $\lambda^*(B) = \varepsilon$, where λ is Lebesgue measure. Hence there exist subsets A and B of $[0, 1]$ with $A \cap B = \emptyset$ and $\lambda^*(A) = \lambda^*(B) = 1$.

- 15.** For any sets A and B ,

- (i) $(A \cup B)^* = A^* \cup B^*$; $(A \cap B)_* = A_* \cap B_*$;
(ii) $(A \cap B)^* \subset A^* \cap B^*$; $(A \cup B)_* \supset A_* \cup B_*$.

For sets A and B with $A \cap B = \emptyset$,

- (iii) $P_*(A) + P_*(B) \leq P_*(A \cup B) \leq P^*(A \cup B) \leq P^*(A) + P^*(B)$.

The last two equalities in (iii) are always valid. The inclusions in (ii) cannot be replaced by equalities, in general, but they are equalities if one of the two sets is measurable.

- 16. (Trace measure)** Let $(\Omega, \mathcal{A}, P)$ be a probability space and B a subset of Ω , possibly not measurable. The equality $P_B(A \cap B) = P(A \cap B^*)$ defines a measure on the trace σ -field $\mathcal{A} \cap B$ with the property $P_B(C) = P^*(C)$ for every $C \in \mathcal{A} \cap B$. It is a probability measure if and only if $P^*(B) = 1$.

[Hint: If A_1 and A_2 are measurable sets with $A_1 \cap B = A_2 \cap B$, then their symmetric difference is measurable and disjoint with B , so it is also disjoint with (a version of) B^* . Thus $P(A_1 \cap B^*) = P(A_2 \cap B^*)$ and P_B is well defined. It is clearly a measure. Next use that $A \cap B^* = (A \cap B)^*$ for measurable A .]

- 17.** Let $(\Omega, \mathcal{A}, P)$ be a probability space and $B \notin \mathcal{A}$. For every constant p with $P_*(B) \leq p \leq P^*(B)$ there exists an extension $\tilde{P}$ of P to a σ -field that contains $\mathcal{A}$ and B with the further property that $\tilde{P}(B) = p$.

[Hint: Take the σ -field equal to the collection of sets of the form $(A_1 \cup B) \cup (A_2 \cap \Omega - B)$. For $p = P^*(B)$, define $\tilde{P}(A) = P_B(A \cap B) + P(A \cap (\Omega - B^*))$, where P_B is the trace of P on $\mathcal{A} \cap B$. For $p = P_*(B)$, define $\tilde{P}(A) = P(A \cap B_*) + P_{\Omega-B}(A \cap (\Omega - B))$. For general p , take a linear combination.]

- 18.** Let L be a finite Borel measure on a metric space $\mathbb{D}$, and let $\mathbb{D}_0 \subset \mathbb{D}$ be an arbitrary subset. If G_n is a sequence of relatively open subsets of $\mathbb{D}_0$ with $\limsup G_n = \emptyset$, then $L^*(G_n) \rightarrow 0$. “Relatively open” could just as well be “relatively Borel.”

[Hint: Consider the Borel measure $L_{\mathbb{D}_0}(B) = L(B \cap \mathbb{D}_0^*)$ on $\mathbb{D}_0$.]

- 19.** For $i = 1, 2$, let $\phi_i: (\Omega_i, \mathcal{U}_i, P_i) \rightarrow (\mathcal{X}, \mathcal{A})$ be measurable maps such that the induced measures $P_1 \circ \phi_1^{-1} = P_2 \circ \phi_2^{-1}$ are the same (on $\mathcal{A}$). Then it is not necessarily true that $E^*T \circ \phi_1 = E^*T \circ \phi_2$ for every $T: \mathcal{X} \rightarrow [0, 1]$.

[Hint: Take $\mathcal{X} = \Omega_1 = \Omega_2$ and both maps ϕ_i the identity. Take $\mathcal{A} = \mathcal{U}_1$ and $\mathcal{U}_2$ the smallest σ -field generated by $\mathcal{U}_1$ and a nonmeasurable set B . Take $T = 1_B$.]

1.3

Weak Convergence

In this section $\mathbb{D}$ and $\mathbb{E}$ are metric spaces with metrics d and e , respectively. The set of all continuous, bounded functions $f: \mathbb{D} \rightarrow \mathbb{R}$ is denoted $C_b(\mathbb{D})$.

The *Borel σ -field* on $\mathbb{D}$ is the smallest σ -field containing the open sets. A function between two topological spaces is continuous if and only if the inverse image of every open set is open. Hence a continuous function is Borel measurable. This is always true; for a metric space $\mathbb{D}$, the case we consider throughout, there is also a converse.

1.3.1 Lemma. *The Borel σ -field on a metric space $\mathbb{D}$ is the smallest σ -field making all elements of $C_b(\mathbb{D})$ measurable (with respect to the Borel sets on $\mathbb{R}$).*

Proof. A closed set F in $\mathbb{D}$ is the null set $\{x: f(x) = 0\}$ of the continuous, bounded function $x \mapsto f(x) = d(x, F) \wedge 1$. Hence it is contained in the σ -field generated by $C_b(\mathbb{D})$. Since the closed sets generate the Borel σ -field, all Borel sets must be contained in the σ -field generated by $C_b(\mathbb{D})$. The reverse inclusion was argued previously. ■

A *finite Borel measure* is simply a finite measure on the Borel sets.[‡] A Borel probability measure L is *tight* if for every $\varepsilon > 0$ there exists a compact set K with $L(K) \geq 1 - \varepsilon$. A Borel measurable map $X: \Omega \rightarrow \mathbb{D}$ is called *tight* if its law $\mathcal{L}(X) = P \circ X^{-1}$ is tight. This is equivalent to there being a σ -compact set (a countable union of compacts) that has probability 1 under L or X . If there is a separable, measurable set with probability 1, then L or X is called *separable*. Since a σ -compact set in a metric space is separable, separability is slightly weaker than

[‡] A general *Borel measure* is a measure μ on the Borel sets with $\mu(K) < \infty$ for every compact set K .

tightness. The following lemma shows that the two properties are the same if the metric space is complete. So the difference can be made to disappear by completing the space.^b Actually, every separable Borel probability measure on a metric space is *pre-tight*: for every $\varepsilon > 0$, there exists a totally bounded, measurable set with probability at least $1 - \varepsilon$.[‡] This concept is usually not considered, because it depends on the metric, while both separability and tightness depend on the topology only. Another topological property of a Borel probability measure is Polishness: L is *Polish* if it gives mass 1 to a Polish set.[†]

1.3.2 Lemma. *A Borel probability measure on a metric space is pre-tight if and only if it is separable. On a complete metric space separability, pre-tightness and tightness are equivalent. Any Polish Borel probability measure is tight.*

Proof. A pre-tight measure concentrates on a countable union of totally bounded sets; a totally bounded set is separable — so a pre-tight measure is certainly separable. Conversely, let $\mathbb{D}_0$ be the closure of a separable set with probability 1. Take a sequence x_m that is dense in $\mathbb{D}_0$. For every $\delta > 0$, the balls of radius δ around the x_m cover $\mathbb{D}_0$. Thus their union has mass 1; the union of some finitely many of them has mass at least $1 - \varepsilon$. Conclude that for every j there exist finitely many balls of radius $1/j$ whose union G_j has mass at least $1 - \varepsilon/2^j$. The intersection $\bigcap_{j=1}^{\infty} G_j$ is totally bounded and has mass at least $1 - \varepsilon$. This concludes the proof of the first assertion. The second is a consequence of the fact that a set is compact if and only if it is totally bounded and complete. ■

1.3.3 Definition (Weak convergence). Let $(\Omega_\alpha, \mathcal{A}_\alpha, P_\alpha)$ be a net of probability spaces and $X_\alpha: \Omega_\alpha \rightarrow \mathbb{D}$ arbitrary maps. The net X_α *converges weakly* to a Borel measure L if

$$\mathbb{E}^* f(X_\alpha) \rightarrow \int f dL, \quad \text{for every } f \in C_b(\mathbb{D}).$$

This is denoted by $X_\alpha \rightsquigarrow L$. If X has a Borel law L , we also say that X_α converges weakly to X and write $X_\alpha \rightsquigarrow X$. Throughout, it will be silently understood that the statements $X_\alpha \rightsquigarrow L$ or $X_\alpha \rightsquigarrow X$ include that L is a Borel measure and X Borel measurable, or in the latter case X can at least be chosen Borel measurable. The names “convergence in distribution”, “convergence in law,” and “weak star convergence” are sometimes used instead of “weak convergence.” In the present context, these terms are equivalent.

^b Every metric space $\mathbb{D}$ has a *completion*: a complete metric space of which a dense subset can be isometrically identified with $\mathbb{D}$.

[‡] A subset of a semimetric space is *totally bounded* if for every $\varepsilon > 0$ it can be covered with finitely many balls of radius ε . This is equivalent to the completion of the space being compact. Also, a subset of a semimetric space is compact if and only if it is totally bounded and complete. Every totally bounded space is separable.

[†] A topological space is called *Polish* if it is separable and there exists a metric that generates the topology for which the space is complete. Any separable, complete metric space is Polish; so is any open subset of a Polish space. A metric space is Polish if and only if it is a G_δ -subset of its completion (a countable intersection of open sets). Concrete examples of Polish spaces are $\mathbb{R}$, $(0,1)$, $[0,1)$, and $\mathbb{R} - \mathbb{Q}$.

A closely related concept is weak convergence of Borel measures. A net L_α of Borel measures on $\mathbb{D}$ is said to converge weakly to L if

$$\int f dL_\alpha \rightarrow \int f dL, \quad \text{for every } f \in C_b(\mathbb{D}).$$

This is denoted $L_\alpha \rightsquigarrow L$. In the special case that every X_α is Borel measurable, weak convergence is equivalent to weak convergence of their induced laws: $X_\alpha \rightsquigarrow L$ if and only if $P_\alpha \circ X_\alpha^{-1} \rightsquigarrow L$. In general, such a reduction to induced laws is impossible. Instead, the definition must be based on outer expectations.

The probability spaces $(\Omega_\alpha, \mathcal{A}_\alpha, P_\alpha)$ are of crucial importance because they determine the outer expectations. They may be different for each α , though in everything that is to follow it is not a loss of generality to take them all equal — the case of different spaces can be reduced to that of a single Ω through suitable “canonical representations” (see Problem 1.3.4).

In any case, we write P^* and E^* without an index α to denote “general” probability and expectation, unless this would cause confusion. This index α , of course, is understood to run through a directed set A . Some results are special to sequences; in this case α will be changed to n , and it is silently understood that the directed set is formed by the natural numbers.

The portmanteau theorem gives equivalent ways of describing weak convergence. Characterization (vi) has intuitive meaning: weak convergence means convergence of probabilities of certain (but not all) sets. The other characterizations are mainly useful as technical tools.

1.3.4 Theorem (Portmanteau). *The following statements are equivalent:*

- (i) $X_\alpha \rightsquigarrow L$;
- (ii) $\liminf P_*(X_\alpha \in G) \geq L(G)$ for every open G ;
- (iii) $\limsup P^*(X_\alpha \in F) \leq L(F)$ for every closed F ;
- (iv) $\liminf E_* f(X_\alpha) \geq \int f dL$ for every lower semicontinuous f that is bounded below;[‡]
- (v) $\limsup E^* f(X_\alpha) \leq \int f dL$ for every upper semicontinuous f that is bounded above;
- (vi) $\lim P^*(X_\alpha \in B) = \lim P_*(X_\alpha \in B) = L(B)$ for every Borel set B with $L(\partial B) = 0$;[‡]
- (vii) $\liminf E_* f(X_\alpha) \geq \int f dL$ for every bounded, Lipschitz continuous, nonnegative f .

Proof. The equivalence of (ii) and (iii) follows by taking complements; the equivalence of (iv) and (v) by replacing f by $-f$. The implication (i) $\Rightarrow$ (vii) is trivial.

(vii) $\Rightarrow$ (ii). Suppose (vii) holds. For every open G , there exists a sequence of Lipschitz continuous functions with $0 \leq f_m \uparrow 1_G$. For instance, $f_m(x) =$

[‡] A function f on $\mathbb{D}$ is called *upper semicontinuous* if $\{f \geq c\}$ is closed for every c , or equivalently $\limsup_{y \rightarrow y_0} f(y) \leq f(y_0)$ for every y_0 . A function f is *lower semicontinuous* if $-f$ is upper semicontinuous.

[‡] The set ∂B is the *boundary* of B , the closure minus the interior. A set B with $L(\partial B) = 0$ is often called an *L-continuity set*.

$m d(x, \mathbb{D} - G) \wedge 1$. For every fixed m , $\liminf P_*(X_\alpha \in G) \geq \liminf E_* f_m(X_\alpha) \geq \int f_m dL$. Letting $m \rightarrow \infty$ yields (ii).

(ii) $\Rightarrow$ (iv). Let (ii) hold and f be lower semicontinuous with $f \geq 0$. Define the sequence f_m by $f_m = \sum_{i=1}^{m^2} (1/m) 1_{G_i}$, where $G_i = \{x: f(x) > i/m\}$; this is f truncated to i/m if $i/m < f(x) \leq (i+1)/m \leq m$ and truncated to m if $f(x) > m$. Thus $f_m \leq f$ and $|f_m - f|(x) \leq 1/m$ whenever $f(x) \leq m$. Fix m . Since every G_i is open

$$\liminf E_* f(X_\alpha) \geq \liminf E_* f_m(X_\alpha) \geq \sum_{i=1}^{m^2} \frac{1}{m} P(X \in G_i) = \int f_m dL.$$

Letting $m \rightarrow \infty$ yields the assertion of (iv) for nonnegative, lower semicontinuous f . Add and subtract a constant to complete the proof.

Since a continuous function is both upper and lower semicontinuous, (v) implies (i). It remains to show the equivalence of (vi) and the others.

(ii) $\Rightarrow$ (vi). If (ii) and (iii) hold, then

$$L(\text{int } B) \leq \liminf P_*(X_\alpha \in \text{int } B) \leq \limsup P^*(X_\alpha \in \bar{B}) \leq L(\bar{B}).$$

When $L(\partial B) = 0$, all inequalities in the previous display are equalities. This yields (vi).

(vi) $\Rightarrow$ (iii). Suppose (vi) holds, and let F be closed. The boundaries ∂F^ε of the sets $F^\varepsilon = \{x: d(x, F) < \varepsilon\}$ for different values of $\varepsilon > 0$ are disjoint, so that at most countably many of them can have nonzero L -measure. Choose a sequence $\varepsilon_m \downarrow 0$ with $L(\partial F^{\varepsilon_m}) = 0$ for every m . For fixed m ,

$$\limsup P^*(X_\alpha \in F) \leq \limsup P^*(X_\alpha \in \overline{F^{\varepsilon_m}}) = L(\overline{F^{\varepsilon_m}}).$$

Letting $m \rightarrow \infty$ yields (iii). ■

1.3.5 Example. When $\mathbb{D} = \mathbb{R}^k$, the Borel σ -field is the usual σ -field generated by the cells $(a, b]$ and every Borel measure is tight. Every Borel measure L is uniquely determined by its cumulative distribution function $L(x) = L((-\infty, x])$. In addition to the characterizations of the portmanteau theorem, weak convergence $X_\alpha \rightsquigarrow L$ is equivalent to:

(viii) $\lim P^*(X_\alpha \leq x) = \lim P_*(X_\alpha \leq x) = L(x)$ for all continuity points x of L ; and, for measurable sequences X_α , also to pointwise convergence of *characteristic functions*:

$$(ix) \lim E e^{it' X_\alpha} = \int e^{it' x} dL(x) \text{ for every } t \in \mathbb{R}^k.$$

For measurable sequences X_α with nonnegative coordinates, a similar characterization is valid in terms of the *Laplace transform*:

$$(x) [\text{if } \mathbb{D} = [0, \infty)^k] \lim E e^{t' X_\alpha} = \int e^{t' x} dL(x) \text{ for every } t \leq 0.$$

Here the inequality $t \leq 0$, for $t \in \mathbb{R}^k$, is understood coordinatewise.

Proofs of (viii)–(x) can be based on Prohorov's theorem combined with the fact that a probability distribution on $\mathbb{R}^k$, $\mathbb{R}^k$ or $[0, \infty)^k$ is uniquely determined by its cumulative distribution function, characteristic function, or Laplace transform (see Problems 1.3.12–1.3.14).

Characterization (ix) implies that weak convergence of vectors in $\mathbb{R}^k$ is equivalent to weak convergence of all linear combinations in $\mathbb{R}$. This is known as the *Cramér-Wold device*.

Characterization (ix) can be strengthened to:

- (A) If $Ee^{it'X_\alpha} \rightarrow \psi(t)$ for every $t \in \mathbb{R}^k$, for a function $\psi: \mathbb{R}^k \rightarrow \mathbb{C}$ that is continuous at 0, then ψ is the characteristic function of a probability measure L on $\mathbb{R}^k$, and $X_\alpha \rightsquigarrow L$.

Similarly, characterization (x) can be strengthened to:

- (B) If $\mathbb{D} = [0, \infty)^k$ and $Ee^{t'X_\alpha} \rightarrow \psi(t)$ for every $t \leq 0$, for a function $\psi: (-\infty, 0]^k \rightarrow [0, \infty)$ such that $\lim_{t \uparrow 0} \psi(t) = 1$, then there exists a probability measure L on $[0, \infty)^k$ such that $\psi(t) = \int e^{t'x} dL(x)$ for every $t \leq 0$, and $X_\alpha \rightsquigarrow L$.

Finally the convergence of Laplace transforms is also sufficient for certain vectors that are not necessarily nonnegative:

- (C) If $Ee^{t'X_\alpha} \rightarrow \psi(t)$ for every t in a neighborhood G of $0 \in \mathbb{R}^k$ and some function $\psi: G \rightarrow \mathbb{R}$, then there exists a unique probability distribution L on $\mathbb{R}^k$ such that $\psi(t) = \int e^{t'x} dL(x)$ for every $t \in G$, and $X_\alpha \rightsquigarrow L$.

See Problems 1.3.12–1.3.14 for proofs.

It is immediate from the definition that $X_\alpha \rightsquigarrow X$ implies $g(X_\alpha) \rightsquigarrow g(X)$ for every continuous g . This is actually already true under only the condition that g is continuous almost surely under X . The continuous mapping theorem is not a very deep result, but it is what makes the concept of weak convergence successful.

1.3.6 Theorem (Continuous mapping). Let $g: \mathbb{D} \rightarrow \mathbb{E}$ be continuous at every point of a set $\mathbb{D}_0 \subset \mathbb{D}$. If $X_\alpha \rightsquigarrow X$ and X takes its values in $\mathbb{D}_0$, then $g(X_\alpha) \rightsquigarrow g(X)$.

Proof. The set D_g of all points at which g is discontinuous can be written as $D_g = \bigcup_{m=1}^{\infty} \bigcap_{k=1}^{\infty} G_k^m$, where G_k^m is the set of all x for which there are y and z within the open ball of radius $1/k$ around x with $e(g(y), g(z)) > 1/m$. Every G_k^m is open, so D_g is a Borel set. For every closed F , it holds that

$$\overline{g^{-1}(F)} \subset g^{-1}(F) \cup D_g.$$

Since g is continuous on the range of the limit variable, $g(X)$ can be chosen Borel measurable. By the portmanteau theorem, $\limsup P^*(g(X_\alpha) \in F) \leq \limsup P^*(X_\alpha \in \overline{g^{-1}(F)}) \leq P(X \in \overline{g^{-1}(F)})$. Since the set of discontinuities has probability zero under X , the last expression equals $P(g(X) \in F)$. Finally, apply the portmanteau theorem again. ■

Next to the continuous mapping theorem, Prohorov's theorem is the most important theorem on weak convergence. To formulate the result, two new concepts are needed.

1.3.7 Definition (Asymptotic measurability and tightness). The net of maps X_α is *asymptotically measurable* if and only if

$$E^*f(X_\alpha) - E_*f(X_\alpha) \rightarrow 0, \quad \text{for every } f \in C_b(\mathbb{D}).$$

The net X_α is *asymptotically tight* if for every $\varepsilon > 0$ there exists a compact set K such that

$$\liminf P_*(X_\alpha \in K^\delta) \geq 1 - \varepsilon, \quad \text{for every } \delta > 0.$$

Here $K^\delta = \{y \in \mathbb{D} : d(y, K) < \delta\}$ is the “ δ -enlargement” around K .

The δ in the definition of tightness may seem a bit overdone. It is not — asymptotic tightness as defined is essentially weaker than the same condition but with K instead of K^δ . This is caused by a second difference with the classical concept of uniform tightness[‡]: the (enlarged) compacts need to contain mass $1 - \varepsilon$ only in the limit.

On the other hand, nothing is gained in simple cases: for Borel measurable maps in a Polish space, asymptotic tightness and uniform tightness are the same (Problem 1.3.9). It may also be noted that, although K^δ is dependent on the metric, the property of asymptotic tightness depends on the topology only (Problem 1.3.6). One nice consequence of the present tightness concept is that weak convergence usually implies asymptotic measurability and tightness.

1.3.8 Lemma.

- (i) If $X_\alpha \rightsquigarrow X$, then X_α is asymptotically measurable.
- (ii) If $X_\alpha \rightsquigarrow X$, then X_α is asymptotically tight if and only if X is tight.

Proof. (i). This follows upon applying the definition of weak convergence to both f and $-f$.

(ii). Fix $\varepsilon > 0$. If X is tight, then there is a compact K with $P(X \in K) > 1 - \varepsilon$. By the portmanteau theorem, $\liminf P_*(X_\alpha \in K^\delta) \geq P(X \in K^\delta)$, which is larger than $1 - \varepsilon$ for every $\delta > 0$. Conversely, if X_α is tight, then there is a compact K with $\liminf P_*(X_\alpha \in K^\delta) \geq 1 - \varepsilon$. By the portmanteau theorem, $P(X \in \overline{K^\delta}) \geq 1 - \varepsilon$. Let $\delta \downarrow 0$. ■

The next version of Prohorov’s theorem may be considered a converse of the previous lemma. It comes in two parts, one for nets and one for sequences, neither one of which implies the other. The sequence case is the deeper of the two.

1.3.9 Theorem (Prohorov’s theorem).

- (i) If the net X_α is asymptotically tight and asymptotically measurable, then it has a subnet $X_{\alpha(\beta)}$ that converges in law to a tight Borel law.
- (ii) If the sequence X_n is asymptotically tight and asymptotically measurable, then it has a subsequence X_{n_j} that converges weakly to a tight Borel law.

[‡] A collection of Borel measurable maps X_α is called *uniformly tight* if, for every $\varepsilon > 0$, there is a compact K with $P(X_\alpha \in K) \geq 1 - \varepsilon$ for every α .

Proof. (i). Consider $(E^*f(X_\alpha))_{f \in C_b(\mathbb{D})}$ as a net in the product space

$$\prod_{f \in C_b(\mathbb{D})} [-\|f\|_\infty, \|f\|_\infty].$$

By Tychonov's theorem, this space is compact in the product topology. Hence the given net has a converging subnet. This is equivalent to the existence of constants $L(f) \in [-\|f\|_\infty, \|f\|_\infty]$ such that

$$E^*f(X_{\alpha(\beta)}) \rightarrow L(f), \quad \text{for every } f \in C_b(\mathbb{D}).$$

It suffices to show that the functional $L: C_b(\mathbb{D}) \rightarrow \mathbb{R}$ is representable by a Borel probability measure L in the sense that $L(f) = \int f dL$ for every $f \in C_b(\mathbb{D})$.

Because of the asymptotic measurability, the numbers $L(f)$ are also the limits of the corresponding inner expectations $E_*f(X_\alpha)$. Consequently,

$$\begin{aligned} L(f_1 + f_2) &\leq \lim \left(E^*f_1(X_{\alpha(\beta)}) + E^*f_2(X_{\alpha(\beta)}) \right) \\ &= L(f_1) + L(f_2) \\ &= \lim \left(E_*f_1(X_{\alpha(\beta)}) + E_*f_2(X_{\alpha(\beta)}) \right) \leq L(f_1 + f_2). \end{aligned}$$

Thus $L: C_b(\mathbb{D}) \rightarrow \mathbb{R}$ is additive. By a similar argument, it follows that $L(\lambda f) = \lambda L(f)$ for every $\lambda \in \mathbb{R}$. So L is even linear. Trivially, it is positive: if $f \geq 0$, then $L(f) \geq 0$.

Finally, L also has another property of an integral: if $f_m \downarrow 0$ pointwise, then $L(f_m) \downarrow 0$. Indeed, fix $\varepsilon > 0$. There is a compact K such that $\liminf P_*(X_\alpha \in K^\delta) \geq 1 - \varepsilon$ for every $\delta > 0$. By Dini's theorem, $f_m \downarrow 0$ uniformly on compacts. Hence for sufficiently large m , it holds that $|f_m(x)| \leq \varepsilon$ for every $x \in K$. Fix an m for which this holds. By an easy argument using the compactness of K , there exists a $\delta > 0$ such that $|f_m(x)| \leq 2\varepsilon$ for every $x \in K^\delta$. One has that $(1_{X_\alpha \in K^\delta})_* = 1_{A_\alpha}$ for a measurable set $A_\alpha \subset \{X_\alpha \in K^\delta\}$. Hence

$$L(f_m) = \lim E f_m(X_\alpha)^* ((1_{X_\alpha \in K^\delta})_* + (1_{X_\alpha \notin K^\delta})^*) \leq 2\varepsilon + \|f_1\|_\infty \varepsilon.$$

It has been shown that L has all the properties of an *abstract integral*. By the Daniell-Stone theorem[†], L is representable by a Borel probability measure L . This concludes the proof of (i).

(ii). The proof of (ii) is the same as for (i) except in the first part, where for (ii) it has to be shown that there is a subsequence, rather than a subnet, along which outer expectations converge. This is achieved in the following manner.

For $m \in \mathbb{N}$, let K_m be a compact set with $\liminf P_*(X_n \in K_m^\delta) \geq 1 - 1/m$ for every $\delta > 0$. Since K_m is compact, the space $C_b(K_m)$, and hence also its unit ball $\{f \in C_b(K_m): |f(x)| \leq 1 \text{ for every } x \in K_m\}$, is separable. According to Tietze's extension theorem[‡], every continuous function defined on a closed subset of a normal space with values in $[-1, 1]$ can be extended to a continuous function on

[†] Bauer (1981), Theorem 7.1.4, or Dudley (1989), Theorem 4.5.2.

[‡] Jameson (1974), Theorem 12.4.

the whole space with values in $[-1, 1]$. Hence every f in the unit ball of $C_b(K_m)$ can be extended to an f in the unit ball of $C_b(\mathbb{D})$. Combination of these facts yields that there exists a countable subset of the unit ball of $C_b(\mathbb{D})$ of which the restrictions to K_m are dense in the unit ball of $C_b(K_m)$.

Pick such a countable set for every m , and let $\mathcal{F}$ be the countably many functions obtained this way. For a fixed, bounded f , there clearly is a subsequence X_{n_j} such that $E^*f(X_{n_j})$ converges to some number. By a diagonalization argument, obtain a subsequence such that

$$E^*f(X_{n_j}) \rightarrow L(f), \quad \text{for every } f \in \mathcal{F},$$

for numbers $L(f) \in [-1, 1]$.

Next let $f \in C_b(\mathbb{D})$ take values in $[-1, 1]$, but be arbitrary otherwise. Fix $\varepsilon > 0$ and m . There exists $f_m \in \mathcal{F}$ with $|f(x) - f_m(x)| \leq \varepsilon$, for every $x \in K_m$. Then, as before, there exists $\delta > 0$ such that $|f(x) - f_m(x)| \leq 2\varepsilon$, for every $x \in K_m^\delta$. Then

$$\begin{aligned} & |E^*f(X_n) - E^*f_m(X_n)| \\ & \leq E|f(X_n) - f_m(X_n)|^* (1_{X_n \in K_m^\delta})_* + 2P^*(X_n \notin K_m^\delta) \leq 2\varepsilon + 2/m, \end{aligned}$$

for sufficiently large n . Conclude that the sequence $E^*f(X_{n_j})$ has the property that, for every $\eta > 0$, there is a converging subsequence of numbers that is eventually within distance η . This implies that $E^*f(X_{n_j})$ itself converges to a limit.

Complete the proof as under (i). ■

Of course, under the conditions of Prohorov's theorem — that X_α is asymptotically tight and measurable — the seemingly stronger conclusion is true that every subnet of X_α has a further subnet that converges to a tight Borel law. A net with this property is called *relatively compact*. Relatively compact sequences are defined analogously, with subnets replaced by subsequences. The converse of this stronger form of Prohorov's theorem is false: a relatively compact net or sequence is not necessarily asymptotically tight. However, if, in addition, all limit points concentrate on a fixed Polish space, then a relatively compact net is asymptotically tight (Problem 1.12.3). Thus for Borel measures on Polish spaces the concepts “relatively compact,” “asymptotically tight,” and “uniformly tight” are all equivalent.

The next theorem is trivial but important. In many applications there are several possible spaces $\mathbb{D}$ in which one could consider weak convergence. For instance, a net of positive real-valued maps can be considered as maps in $\mathbb{R}^+$, in $\mathbb{R}$, or even in $\mathbb{R}^2$. Less trivial examples arise with infinite-dimensional spaces. It wouldn't be nice if our choice of $\mathbb{D}$ influenced the conclusion too much. Fortunately, as long as the topology is not changed, this is not the case.

1.3.10 Theorem. *Let $\mathbb{D}_0 \subset \mathbb{D}$ be arbitrary metric spaces equipped with same metric, and let X and X_α take their values in $\mathbb{D}_0$. Then $X_\alpha \rightsquigarrow X$ as maps in $\mathbb{D}_0$ if and only if $X_\alpha \rightsquigarrow X$ as maps in $\mathbb{D}$.*

Proof. Because a set G_0 in $\mathbb{D}_0$ is open if and only if it is of the form $G \cap \mathbb{D}_0$ for an open set G in $\mathbb{D}$, this is an easy corollary of (ii) of the portmanteau theorem. ■

1.3.11 Example (Weak convergence of discrete measures). Let X and every X_α be maps taking their values in a countable subset $S \subset \mathbb{D}$ of isolated points. (Every $s \in S$ has an open neighborhood that contains no other points of S .) Then $X_\alpha \rightsquigarrow X$ if and only if X is Borel measurable and $P_*(X_\alpha = s) \rightarrow P(X = s)$ for every $s \in S$. In this case, both $P^*(X_\alpha \in B)$ and $P_*(X_\alpha \in B)$ converge to $P(X \in B)$ for every Borel set $B \subset \mathbb{D}$.

Indeed, in view of the previous theorem, it may be assumed without loss of generality that $S = \mathbb{D}$. Then every set $B \subset \mathbb{D}$ is open and closed at the same time and therefore has empty boundary. If $X_\alpha \rightsquigarrow X$, then outer and inner probabilities of every set converge by the portmanteau theorem. Conversely, suppose inner measures of one-point sets converge. Let G be arbitrary. For any finite subset K of G , one has $P_*(X_\alpha \in K) \geq \sum_{s \in K} P_*(X_\alpha = s) \rightarrow P(X \in K)$. Conclude that $\liminf P_*(X_\alpha \in G) \geq P(X \in G)$ for every G , in particular open ones, so that $X_\alpha \rightsquigarrow X$ by the portmanteau theorem.

The condition that the points of S are isolated cannot be omitted. It is crucial in the above that both X and every X_α take their values in S .

As a consequence of the previous theorem, a net that converges weakly to a separable limit is asymptotically tight when seen as maps into the completion of $\mathbb{D}$. There are no known examples of nonseparable Borel measures. Thus, there is no great loss of generality in considering only those weakly convergent nets that are asymptotically tight. (It is actually known that, starting from the usual axioms of mathematics, including the axiom of choice, it is impossible to construct nonseparable Borel measures — the axiom that nonseparable Borel measures do not exist can be added to the Zermelo-Fr nkel system without creating inconsistencies. It is apparently unknown whether nonseparable Borel measures can consistently exist.)

A question that has been ignored so far is whether weak limits are unique — they are. A Borel measure is uniquely defined by the expectations it gives to the elements of $C_b(\mathbb{D})$. Tight Borel measures are already determined by the expectations of relatively small subclasses of $C_b(\mathbb{D})$.

1.3.12 Lemma. Let L_1 and L_2 be finite Borel measures on $\mathbb{D}$.

(i) If $\int f dL_1 = \int f dL_2$ for every $f \in C_b(\mathbb{D})$, then $L_1 = L_2$.

Let L_1 and L_2 be tight Borel probability measures on $\mathbb{D}$.

(ii) If $\int f dL_1 = \int f dL_2$ for every f in a vector lattice^b $\mathcal{F} \subset C_b(\mathbb{D})$ that contains the constant functions and separates points of $\mathbb{D}$, then $L_1 = L_2$.

Proof. (i). For every open G , there exists a sequence of continuous functions with $0 \leq f_m \uparrow 1_G$ (compare the proof of the portmanteau theorem). By monotone convergence, $L_1(G) = L_2(G)$ for every open G . Since $L_1(\mathbb{D}) = L_2(\mathbb{D})$, the collection of Borel sets for which $L_1(B) = L_2(B)$ is a σ -field.

(ii). Fix $\varepsilon > 0$. Take a compact K such that $L_1(K) \wedge L_2(K) \geq 1 - \varepsilon$. According to a version of the Stone-Weierstrass theorem (Jameson (1974), p. 263), a vector

^b A vector lattice $\mathcal{F} \subset C_b(\mathbb{D})$ is a vector space that is closed under taking positive parts: if $f \in \mathcal{F}$, then $f^+ = f \vee 0 \in \mathcal{F}$. Then automatically $f \vee g \in \mathcal{F}$ and $f \wedge g \in \mathcal{F}$ for every $f, g \in \mathcal{F}$. A set of functions on $\mathbb{D}$ separates points of $\mathbb{D}$ if, for every pair $x \neq y \in \mathbb{D}$, there is $f \in \mathcal{F}$ with $f(x) \neq f(y)$.

lattice $\mathcal{F} \subset C_b(K)$ that contains the constant functions and separates points of K is uniformly dense in $C_b(K)$. Given $g \in C_b(\mathbb{D})$ with $0 \leq g \leq 1$, take $f \in \mathcal{F}$ with $|g(x) - f(x)| \leq \varepsilon$ for every $x \in K$. Then $|\int g dL_1 - \int g dL_2| \leq |\int (f \wedge 1)^+ dL_1 - \int (f \wedge 1)^+ dL_2| + 4\varepsilon$, which equals 4ε because $(f \wedge 1)^+ \in \mathcal{F}$. Conclude that $\int g dL_1 = \int g dL_2$. By adding and multiplying with scalars, obtain the same result for every $g \in C_b(\mathbb{D})$. ■

Requiring only asymptotic measurability of a net X_α , as opposed to Borel measurability of every X_α , extends the applicability of weak convergence considerably. However, asymptotic measurability is hard to establish directly. The most promising method seems to be to prove measurability into some smaller σ -field plus some additional property. For asymptotically tight nets, the situation is nice — asymptotic tightness plus only a little bit of measurability gives asymptotic measurability. The next lemma is an abstract version of this result. Interesting special cases are discussed in the next sections.

1.3.13 Lemma. *Let the net X_α be asymptotically tight, and suppose $E^*f(X_\alpha) - E_*f(X_\alpha) \rightarrow 0$ for every f in a subalgebra $\mathcal{F}$ of $C_b(\mathbb{D})$ that separates points of $\mathbb{D}$.[‡] Then the net X_α is asymptotically measurable.*

Proof. Fix $\varepsilon > 0$ and a compact K such that $\limsup P^*(X_\alpha \notin K^\delta) \leq \varepsilon$ for every $\delta > 0$. Assume without loss of generality that $\mathcal{F}$ contains the constant functions. By the Stone-Weierstrass theorem (Jameson (1974), p. 266), the restrictions of the functions in $\mathcal{F}$ to K are uniformly dense in $C_b(K)$. Hence given $f \in C_b(\mathbb{D})$, there exists $g \in \mathcal{F}$ with $|f(x) - g(x)| \leq \varepsilon/4$ for every $x \in K$. Using the compactness of K , it is easily seen that there is a $\delta > 0$ such that $|f(x) - g(x)| \leq \varepsilon/3$ for every $x \in K^\delta$. Let $\{X_\alpha \in K^\delta\}_*$ be a measurable set contained in $\{X_\alpha \in K^\delta\}$ and with the same inner measure. Then $P(\Omega_\alpha - \{X_\alpha \in K^\delta\}_*) = P^*(X_\alpha \notin K^\delta)$ and, for large α ,

$$\begin{aligned} P(|f(X_\alpha)^* - f(X_\alpha)_*| > \varepsilon) \\ &\leq P(|f(X_\alpha)^* - f(X_\alpha)_*| > \varepsilon \cap \{X_\alpha \in K^\delta\}_*) + 2\varepsilon \\ &\leq P(|g(X_\alpha)^* - g(X_\alpha)_*| > \varepsilon/3) + 2\varepsilon. \end{aligned}$$

Hence $f(X_\alpha)^* - f(X_\alpha)_* \rightarrow 0$ in probability. By dominated convergence, $E(f(X_\alpha)^* - f(X_\alpha)_*) \rightarrow 0$. ■

By the continuous mapping theorem convergence $X_\alpha \rightsquigarrow X$ in $\mathbb{D}$ implies convergence $g(X_\alpha) \rightsquigarrow g(X)$ for every continuous transformation $g: \mathbb{D} \rightarrow \mathbb{E}$ into another metric space $\mathbb{E}$. The following result reverses this and ascertains weak convergence through convergence of a collection of transformations.

[‡] An algebra $\mathcal{F} \subset C_b(\mathbb{D})$ is a vector space that is closed under taking products: if $f, g \in \mathcal{F}$, then $fg \in \mathcal{F}$.

1.3.14 Theorem (Characterization through maps). Let $\mathcal{G}$ be a countable collection of maps $g: \mathbb{D} \rightarrow \mathbb{E}_g$ into metric spaces $\mathbb{E}_g$ such that $g(x_n) \rightarrow g(x)$, for every $g \in \mathcal{G}$, implies that $x_n \rightarrow x$ in $\mathbb{D}$, for any sequence x_n and x in $\mathbb{D}$. Then for any net of maps X_α in $\mathbb{D}$ and Borel measurable map X in $\mathbb{D}$:

- (i) If $(g_1(X_\alpha), \dots, g_k(X_\alpha)) \rightsquigarrow (g_1(X), \dots, g_k(X))$ in $\times_{i=1}^k \mathbb{E}_{g_i}$, for every finite set $g_1, \dots, g_k$ in $\mathcal{G}$, then $X_\alpha \rightsquigarrow X$ in $\mathbb{D}$.
- (ii) If the net $g(X_\alpha)$ is asymptotically tight in $\mathbb{E}_g$, for every $g \in \mathcal{G}$, then the net X_α is asymptotically tight in $\mathbb{D}$.

If X takes its values in a subset $\mathbb{D}_0 \subset \mathbb{D}$, then (i) remains true if $\mathcal{G}$ separates the points of $\mathbb{D}$ and the condition on $\mathcal{G}$ holds only for limits $x \in \mathbb{D}_0$.

Proof. Enumerate $\mathcal{G}$ as $\{g_1, g_2, \dots\}$, and for $x, y \in \mathbb{D}$ and e_i the metric of $\mathbb{E}_{g_i}$, set

$$e(x, y) = \sum_{i=1}^{\infty} \frac{1}{2^i} (e_i(g_i(x), g_i(y)) \wedge 1).$$

Then e is a semimetric on $\mathbb{D}$ and $e(x_n, x) \rightarrow 0$ implies that $g_i(x_n) \rightarrow g_i(x)$, for every $i \in \mathbb{N}$, and hence by assumption $d(x_n, x) \rightarrow 0$. In particular $e(y, x) = 0$ implies that $d(y, x) = 0$ and hence $x = y$, so that e is a metric. Furthermore, it follows that the identity map $\text{id}: (\mathbb{D}, e) \rightarrow (\mathbb{D}, d)$ is continuous. In view of the continuous mapping theorem for (i) it now suffices to show that $X_\alpha \rightsquigarrow X$ in $(\mathbb{D}, e)$. Furthermore, since continuous images of compact sets are compact, for (ii) it suffices to show that the net X_α is asymptotically tight in $(\mathbb{D}, e)$.

The map $x \mapsto x' = (g(x))_{g \in \mathcal{G}}$ from $\mathbb{D}$ to $\mathbb{E}' := \times_{g \in \mathcal{G}} \mathbb{E}_g$ is an isometry for the metric e on $\mathbb{D}$ and $e'(x', y') = \sum_{i=1}^{\infty} 2^{-i} (e_i(x'_i, y'_i) \wedge 1)$ on $\mathbb{E}'$. Since the latter map generates the product topology, the convergence $X'_\alpha \rightsquigarrow X'$ of the images in $(\mathbb{E}', e')$ follows by Theorem 1.4.8. By the isometry this is equivalent to $X_\alpha \rightsquigarrow X$ in $(\mathbb{D}, e)$. Furthermore, the net X'_α is asymptotically tight in $\mathbb{E}'$ if and only if each net of coordinates $(X'_\alpha)_g = g(X_\alpha)$ is asymptotically tight in $\mathbb{E}_g$ (the extension of Lemma 1.4.3 to countable products).

If the class of functions $\mathcal{G}$ separates the points of $\mathbb{D}$, then e is still a metric on $\mathbb{D}$, and the map $x \mapsto (g(x))_{g \in \mathcal{G}}$ is an isometry between $(\mathbb{D}, e)$ and $(\mathbb{E}, e')$, as before. Hence $X_\alpha \rightsquigarrow X$ in $(\mathbb{D}, e)$, as before. Under the weaker assumption that $g(x_n) \rightarrow g(x)$, for every $g \in \mathcal{G}$, implies that $x_n \rightarrow x$ in $\mathbb{D}$, for any sequence x_n in $\mathbb{D}$ and $x \in \mathbb{D}_0$, the identity map $\text{id}: (\mathbb{D}, e) \rightarrow (\mathbb{D}, d)$ is continuous at every point in $\mathbb{D}_0$. This is sufficient for an application of the continuous mapping theorem to infer that $X_\alpha \rightsquigarrow X$ in $(\mathbb{D}, d)$. ■

For real-valued maps $g: \mathbb{D} \rightarrow \mathbb{R}$, the marginals $(g_1(X_\alpha), \dots, g_k(X_\alpha))$ are vectors in $\mathbb{R}^k$. By the Cramér-Wold device, see Example 1.3.5, the weak convergence of these vectors is equivalent to the convergence of all linear combinations $\sum_{i=1}^k t_i g_i(X_\alpha)$. Thus in this case the weak convergence can be reduced to convergence $g(X_\alpha) \rightsquigarrow g(X)$ in $\mathbb{R}$, for every g in the linear span of $\mathcal{G}$.

Problems and Complements

1. **(Regularity of Borel measures)** Every Borel probability measure L on a metric space is *outer regular*: for every Borel set B ,

$$L(B) = \sup_{\substack{F \subset B \\ F \text{ closed}}} L(F) = \inf_{\substack{G \supset B \\ G \text{ open}}} L(G).$$

A Borel probability measure L is called *inner regular* if, for every Borel set B ,

$$L(B) = \sup_{\substack{K \subset B \\ K \text{ compact}}} L(K).$$

A Borel probability measure on a metric space is inner regular if and only if it is tight. In particular, every Borel probability measure on a Polish space is inner regular. An inner regular measure is sometimes called a *Radon measure*. The first equation is also true with “ F closed” replaced by “ F totally bounded”. (More generally, every finite Borel measure on a Suslin space is regular.)

[Hint: The collection of all Borel sets B for which the first equation is valid can be seen to be a σ -field. It includes all open and closed sets.]

2. The metrics $d(x, y) = |x - y|$ and $e(x, y) = |\arctan x - \arctan y|$ generate the same topology on $\mathbb{R}$. However, $\mathbb{R}$ is complete for d , but incomplete for e . The completion of $\mathbb{R}$ under e is the extended real line $[-\infty, \infty]$. The real line is totally bounded for e , but not for d .
3. The *Baire σ -field* on a topological space $\mathbb{D}$ (not necessarily metrizable) is the smallest σ -field for which all continuous (bounded) functions $f: \mathbb{D} \rightarrow \mathbb{R}$ are measurable. It is the smallest σ -field containing all open F_σ -sets and/or closed G_δ -sets.

[Hint: Bauer (1981), page 206.]

4. **(Canonical representation)** Let $X_\alpha: \Omega_\alpha \rightarrow \mathbb{D}$ be an arbitrary collection of maps indexed by $\alpha \in A$ and defined on probability spaces $(\Omega_\alpha, \mathcal{A}_\alpha, P_\alpha)$. Let $(\Omega, \mathcal{A}, P)$ be the product of these probability spaces, and define $\tilde{X}_\alpha: \Omega \rightarrow \mathbb{D}$ by $\tilde{X}_\alpha = X_\alpha \circ \pi_\alpha$, where $\pi_\alpha: \Omega \rightarrow \Omega_\alpha$ is the projection on the α th coordinate. Then $E^*f(\tilde{X}_\alpha) = E^*f(X_\alpha)$ for every bounded $f: \mathbb{D} \rightarrow \mathbb{R}$. Consequently, whenever one is interested only in the “laws” of maps X_α , it is no loss of generality to assume that the maps are defined on a single probability space.

[Hint: Coordinate projections are perfect maps.]

5. Let d and e be metrics on the same set $\mathbb{D}$ with the following property: if $x_n \rightarrow x$ for e and a limit $x \in \mathbb{D}_0$, then $x_n \rightarrow x$ for d ; for every sequence x_n and a given subset $\mathbb{D}_0$ of $\mathbb{D}$. Then $X_\alpha \rightsquigarrow X$ for e , where X takes its values in $\mathbb{D}_0$, implies the same for d .

[Hint: Apply the continuous mapping theorem to the identity map from $(\mathbb{D}, e)$ to $(\mathbb{D}, d)$.]

6. A net X_α is asymptotically tight if and only if, for every $\varepsilon > 0$, there exists a compact set K with $\liminf P_*(X_\alpha \in G) \geq 1 - \varepsilon$ for every open $G \supset K$.

[Hint: For every open $G \supset K$ where K is compact, there is a $\delta > 0$ with $G \supset K^\delta \supset K$. If there were not such a δ , then there would be a sequence x_n with $d(x_n, K) \rightarrow 0$ and $x_n \notin G$ for every n . By compactness of K , a subsequence would converge; the limit would be both in K and not in G .]

7. Let X_α be maps in a metric space $\mathbb{D}$ and $g: \mathbb{D} \rightarrow \mathbb{E}$ a continuous map into a metric space $\mathbb{E}$.

- (i) If X_α is asymptotically tight, then $g(X_\alpha)$ is asymptotically tight.
- (ii) If X_α is asymptotically measurable, then $g(X_\alpha)$ is asymptotically measurable.

Is full continuity of g necessary?

[Hint: For a given metric e on $\mathbb{E}$, the function $x \mapsto e(g(x), g(K))$ is continuous and identically zero on K . For a compact K , there exists for every $\varepsilon > 0$ a $\delta > 0$ with $e(g(x), g(K)) < \varepsilon$ whenever $d(x, K) < \delta$.]

8. If $X_\alpha \rightsquigarrow X$ and X is separable, then X_α is asymptotically pre-tight: for every $\varepsilon > 0$, there exists a totally bounded measurable set K with $\liminf P_*(X_\alpha \in K^\delta) \geq 1 - \varepsilon$ for every $\delta > 0$. Asymptotic pre-tightness cannot replace asymptotic tightness in Prohorov's theorem: a net that is measurable and asymptotically pre-tight is not necessarily relatively compact.

[Hint: For a counterexample, let X_n be maps in the rationals $\mathbb{D} = \mathbb{Q}$ with $P(X_n = i/n) = 1/(n+1)$, for $i = 0, 1, \dots, n$.]

9. **(Asymptotic and uniform tightness of a sequence)** According to the "classical" definition, a sequence of Borel measurable maps X_n is *uniformly tight* if, for every $\varepsilon > 0$, there exists a compact K with $P(X_n \in K) \geq 1 - \varepsilon$ for every n . A sequence X_n is uniformly tight if and only if it is asymptotically tight and each element X_n is tight. In particular, for Borel measurable sequences in a Polish space, uniform tightness and asymptotic tightness are the same.

[Hint: Fix $\varepsilon > 0$. Take a compact K_0 with $\liminf P(X_n \in K_0^\delta) \geq 1 - \varepsilon$ for every $\delta > 0$. Choose $n_1 < n_2 < \dots$ such that $P(X_n \in K_0^{1/m}) \geq 1 - 2\varepsilon$ for $n \geq n_m$. For $n_m < n \leq n_{m+1}$, choose a compact $K_n \subset K_0^{1/m}$ with $P(X_n \in K_0^{1/m} - K_n) < \varepsilon$. Now $K = \bigcup_{n=0}^\infty K_n$ is compact and $P(X_n \in K) \geq 1 - 3\varepsilon$ for every n .]

10. A relatively compact sequence of Borel measures on a metric space is not necessarily asymptotically tight.

[Hint: For every rational q , let L_q be the measure on $\ell^\infty(\mathbb{Q})$ with $L_q\{0\} = L_q\{z_q\} = 1/2$, where 0 is the function that is identically zero and z_q is the function that is zero except at the point q , where it is 1. (So L_q is a two-point measure.) Take a sequence $\{q_n: n = 1, 2, \dots\}$ of rational numbers that reaches every rational number infinitely often, e.g. $0, 1, -1, 0, 1/2, -1/2, 1, -1, 3/2, -3/2, 2, -2, 0, 1/3, \dots$. Then the sequence L_{q_n} has all measures L_q as weak limit points, and it is relatively compact. If the sequence were asymptotically tight, then it would be uniformly tight (in the old sense), because every L_q is tight. But there is no compact K with $L_q(K) > 3/4$ for every q . Indeed, K would have to contain both 0 and z_q for every q , which is impossible, since $\|z_q - z_r\| = 1$ for every pair of distinct rationals q, r .]

11. Define X_n as $Z_n(2\log(n+1))^{-1/2}$ for i.i.d. standard normal variables Z_n . Then $(X_1, X_2, \dots)$ takes its values with probability 1 in the sequence space ℓ_∞ , but it does not induce a tight Borel law on this space.

[Hint: By Anderson's lemma, every tight, Borel measurable centered Gaussian variable X satisfies $P(\|X - x\| \leq \varepsilon) \leq P(\|X\| \leq \varepsilon)$ for every x . Conclude from this that $P(\|X\| \leq \varepsilon)$ is positive for every $\varepsilon > 0$. For the given process, this probability is zero for $\varepsilon < 1$. (Instead of Anderson's lemma, one may use the inequality $P(\|X - x\| \leq \varepsilon)^2 \leq P(\|X\| \leq \varepsilon\sqrt{2})$, which can be established easily by rewriting the left side as $P(\|X - x\| \leq \varepsilon, \|Y - x\| \leq \varepsilon)$ for an independent copy Y of X .])

12. If $\psi(t) = \mathbb{E}e^{itX}$ is the characteristic function of the random variable X , then, for every $\delta > 0$,

$$\mathbb{P}\left(|X| > \frac{2}{\delta}\right) \leq \frac{1}{\delta} \int_{-\delta}^{\delta} \operatorname{Re}(1 - \psi(t)) dt.$$

Deduce that pointwise convergence of a sequence of characteristic functions to a function that is continuous at zero implies asymptotic tightness of the corresponding random variables. Deduce (A) of Example 1.3.5.

[Hint: E.g. the proof of Theorem 2.13 in Van der Vaart (1998).]

13. Prove (B) in Example 1.3.5.

[Hint: Considered as maps into the compact metric space $[0, \infty]^k$, the net X_α is uniformly tight. If X is a weak limit point, then $\mathbb{E}e^{t'X_\alpha} \rightarrow \mathbb{E}e^{t'X}$, for every $t \leq 0$, where we interpret the values of the function $x \mapsto e^{t'x}$ as $\prod_{i:t_i < 0} e^{t_i x_i}$, and to be 0 if $x_i = \infty$ for some $i = 1, \dots, k$ and $t_i < 0$, so that it is continuous and bounded on $[0, \infty]^k$. Thus $\mathbb{E}e^{t'X} = \psi(t)$, for every $t \leq 0$. When evaluated at vectors t with nonpositive integer coordinates, this determines the mixed moments of the random vector (e^{-X_i}) . Since this vector takes its values in $[0, 1]^k$, its distribution is determined by its moments. Hence the distribution of X is determined by ψ , and the net X_α has only one limit point. Because by assumption $\mathbb{E}e^{t'X} = \psi(t) \rightarrow 1$ as $t \uparrow 0$, it follows that $\mathbb{P}(X \in [0, \infty)^k) = 1$. Thus $X_\alpha \rightsquigarrow X$ in $[0, \infty)^k$, by Theorem 1.3.10.]

14. Prove (C) in Example 1.3.5.

[Hint: Since $\mathbb{P}(X_{\alpha,i} > K) \leq e^{-sK} \mathbb{E}e^{sX_{\alpha,i}}$ for $s > 0$ and $\mathbb{P}(X_{\alpha,i} < -K) \leq e^{sK} \mathbb{E}e^{sX_{\alpha,i}}$ for $s < 0$, and $\mathbb{E}e^{sX_{\alpha,i}} = O(1)$ for some $s > 0$ and some $s < 0$ by assumption, the net X_α is asymptotically tight. Let X be a weak limit point. Fix vectors with $a < 0 < b$, coordinatewise, such that $(a, b) \subset G$. Since for $0 \leq t < b$ there exists a scalar $r > 1$ with $rt < b$, we have $\mathbb{E}(e^{t'X_\alpha})^r = O(1)$ by the assumption, and hence $\mathbb{E}e^{t'X_\alpha} \rightarrow \mathbb{E}e^{t'X} < \infty$ for every $0 \leq t < b$, by Example 1.11.4, whence $\mathbb{E}e^{t'X} = \psi(t)$. Similarly $\mathbb{E}e^{t'X} = \psi(t)$, for $a < t \leq 0$. In particular, the Laplace transform of X exists in the neighborhood (a, b) of $0 \in \mathbb{R}^k$. Then it has a unique extension to an analytic function on $\{z \in \mathbb{C}^k : \operatorname{Re} z \in (a, b)\}$, which contains the imaginary axes. Evaluating the function on these axes gives the characteristic function, and hence X is uniquely determined by ψ . (This is Cramér's condition for uniqueness of the moment problem.)]

1.4

Product Spaces

Let $\mathbb{D}$ and $\mathbb{E}$ be metric spaces with metrics d and e . Then the Cartesian product $\mathbb{D} \times \mathbb{E}$ is a metric space for any of the metrics

$$\begin{aligned}c((x_1, y_1), (x_2, y_2)) &= d(x_1, x_2) \vee e(y_1, y_2), \\c((x_1, y_1), (x_2, y_2)) &= \sqrt{d(x_1, x_2)^2 + e(y_1, y_2)^2}, \\c((x_1, y_1), (x_2, y_2)) &= d(x_1, x_2) + e(y_1, y_2).\end{aligned}$$

These generate the same topology, the *product topology*.

On the product space $\mathbb{D} \times \mathbb{E}$, there are two natural σ -fields: the product of the Borel σ -fields and the Borel σ -field for the product topology. In general, these are not the same. The reason is that a product topology is built up from *arbitrary* unions of (open) cylinders $G_1 \times G_2$, whereas the product σ -field is generated through *countable* set-theoretic operations on these sets. A separable metric space has a countable base for its topology. Not surprisingly, for separable spaces the two σ -fields are the same.

1.4.1 Lemma. *If $\mathbb{D}$ and $\mathbb{E}$ are separable, then the product of the Borel σ -fields equals the Borel σ -field for the product topology on $\mathbb{D} \times \mathbb{E}$.*

1.4.2 Lemma. *A separable Borel probability measure L on $\mathbb{D} \times \mathbb{E}$ is uniquely determined by the numbers $\int f(x)g(y) dL(x, y)$, where f and g range over the non-negative Lipschitz functions in $C_b(\mathbb{D})$ and $C_b(\mathbb{E})$, respectively.*

Proofs. Since $\mathbb{D}$ and $\mathbb{E}$ are separable (and metric), it is possible to find countable bases $\mathcal{G}_1$ and $\mathcal{G}_2$ for the open sets of their topologies. A set in $\mathbb{D} \times \mathbb{E}$ is open if and only if it is the union of sets of the form $G_1 \times G_2$ with $G_1 \in \mathcal{G}_1$ and $G_2 \in \mathcal{G}_2$. Since

there are only countably many sets of the latter type, a set is open in the product topology if and only if it is a countable union of sets of the form $G_1 \times G_2$, where G_1 and G_2 are open. Thus the Borel σ -field is generated by the sets $G_1 \times G_2$, where G_1 and G_2 are open. But so is the product of the Borel σ -fields.

For the proof of the second lemma, conclude first that the given integrals determine the probabilities $L(G_1 \times G_2)$ for every pair of open G_1 and G_2 , because the indicator of $G_1 \times G_2$ is the monotone limit from below of a sequence of functions of the form $f_m(x)g_m(y)$, with f_m and g_m Lipschitz continuous and $0 \leq f_m \leq 1_{G_1}$ and $0 \leq g_m \leq 1_{G_2}$. The open product sets generate the trace of the Borel σ -field on any separable subset of $\mathbb{D} \times \mathbb{E}$ by the first lemma. They also form an intersection stable collection of sets. So the trace of L on any separable subset is uniquely determined (by a monotone class theorem). ■

Given $X_\alpha: \Omega_\alpha \rightarrow \mathbb{D}$ and $Y_\alpha: \Omega_\alpha \rightarrow \mathbb{E}$, one can form the joint variable $(X_\alpha, Y_\alpha): \Omega_\alpha \rightarrow \mathbb{D} \times \mathbb{E}$. If both components are Borel measurable, then (X_α, Y_α) is measurable in the product of the Borel σ -fields, but not necessarily in the Borel σ -field of the product. This is an inconvenience, since we would want to use the second one in the weak convergence theory in the product space. The problem disappears if the metric spaces are separable. Furthermore, in the general case, asymptotic measurability is retained under tightness.

1.4.3 Lemma. *Nets $X_\alpha: \Omega_\alpha \rightarrow \mathbb{D}$ and $Y_\alpha: \Omega_\alpha \rightarrow \mathbb{E}$ are asymptotically tight if and only if the same is true for $(X_\alpha, Y_\alpha): \Omega_\alpha \rightarrow \mathbb{D} \times \mathbb{E}$.*

1.4.4 Lemma. *Asymptotically tight nets $X_\alpha: \Omega_\alpha \rightarrow \mathbb{D}$ and $Y_\alpha: \Omega_\alpha \rightarrow \mathbb{E}$ are asymptotically measurable if and only if the same is true for $(X_\alpha, Y_\alpha): \Omega_\alpha \rightarrow \mathbb{D} \times \mathbb{E}$.*

Proofs. A set of the form $K_1 \times K_2$ is compact if and only if K_1 and K_2 are compact. Second, if $K \subset \mathbb{D} \times \mathbb{E}$ is compact, then it is contained in $\pi_1 K \times \pi_2 K$, where $\pi_i K$ are the projections on the first and second coordinates and are compact. Third, for the first of the three mentioned product metrics, it holds that $(K_1 \times K_2)^\delta = K_1^\delta \times K_2^\delta$. It is now easy to see that asymptotic tightness of both marginals is equivalent to asymptotic tightness of the joint maps.

Asymptotic measurability of the joint maps trivially implies asymptotic measurability of the marginals (Problem 1.3.7).

Finally, assume (X_α, Y_α) is asymptotically tight and X_α and Y_α are asymptotically measurable. Let $f \in C_b(\mathbb{D})$ and $g \in C_b(\mathbb{E})$ take values in the interval $[-1, 1]$. Then

$$\begin{aligned} & (f(X_\alpha)g(Y_\alpha))^* - (f(X_\alpha)g(Y_\alpha))_* \\ & \leq (1 + f(X_\alpha))^* (1 + g(Y_\alpha))^* - (1 + f(X_\alpha))_* (1 + g(Y_\alpha))_* + 1 \\ & \quad - (1 + f(X_\alpha))_* (1 + g(Y_\alpha))_* + (1 + f(X_\alpha))^* + (1 + g(Y_\alpha))^* - 1. \end{aligned}$$

This converges to zero in probability, because each of the four terms in the bottom line asymptotically cancels the corresponding term in the second line of the display. By similar reasoning it follows that $h(X_\alpha, Y_\alpha)$ is asymptotically measurable for every h of the form $h(x, y) = \sum_{i=1}^m f_i(x)g_i(y)$ with $f_i \in C_b(\mathbb{D})$ and $g_i \in C_b(\mathbb{E})$. This

set of functions h forms an algebra and separates points of $\mathbb{D} \times \mathbb{E}$. Thus (X_α, Y_α) is asymptotically measurable by Lemma 1.3.13. ■

A useful corollary is that for weak convergence of vectors (X_α, Y_α) , it suffices to consider expressions of the type $E^*f(X_\alpha)g(Y_\alpha)$.

1.4.5 Corollary. *Let $(X_\alpha, Y_\alpha): \Omega_\alpha \rightarrow \mathbb{D} \times \mathbb{E}$ be an arbitrary net such that $E^*f(X_\alpha)g(Y_\alpha) \rightarrow \int f(x)g(y) dL(x, y)$ for all bounded, nonnegative Lipschitz functions $f: \mathbb{D} \rightarrow \mathbb{R}$ and $g: \mathbb{E} \rightarrow \mathbb{R}$ and a separable Borel measure L on $\mathbb{D} \times \mathbb{E}$. Then $(X_\alpha, Y_\alpha) \rightsquigarrow L$.*

Proof. Since g and f can be taken equal to 1, it follows that the nets X_α and Y_α converge marginally in distribution. Thus the nets X_α and Y_α are asymptotically tight and asymptotically measurable in the completions $\overline{\mathbb{D}}$ and $\overline{\mathbb{E}}$ of $\mathbb{D}$ and $\mathbb{E}$, respectively. The joint maps (X_α, Y_α) are asymptotically tight in $\overline{\mathbb{D}} \times \overline{\mathbb{E}}$. By Prohorov's theorem, every subnet has a further weakly converging subnet. Every weak limit point gives the same expectation to the function $(x, y) \mapsto f(x)g(y)$ as L . Thus L is the only limit point. ■

It is not hard to find an example of weakly convergent nets X_α and Y_α with tight limits for which the joint maps (X_α, Y_α) do not converge weakly. However, the previous result and Prohorov's theorem show that (X_α, Y_α) is at least relatively compact. Under special conditions there is only one limit point and the joint maps do converge weakly. The next two examples consider the case of independent coordinates and the case that one coordinate is asymptotically degenerate.

1.4.6 Example. Call X_α and Y_α *asymptotically independent* if

$$E^*f(X_\alpha)g(Y_\alpha) - E^*f(X_\alpha)E^*g(Y_\alpha) \rightarrow 0$$

for all bounded, nonnegative, Lipschitz functions f and g on $\mathbb{D}$ and $\mathbb{E}$, respectively. If X_α and Y_α are asymptotically independent and $X_\alpha \rightsquigarrow L_1$ and $Y_\alpha \rightsquigarrow L_2$ for separable L_i , then $(X_\alpha, Y_\alpha) \rightsquigarrow L_1 \otimes L_2$.

To see this, assume without loss of generality that $\mathbb{D}$ and $\mathbb{E}$ are complete, so that the marginals of (X_α, Y_α) are asymptotically tight. By Prohorov's theorem the net of joint variables (X_α, Y_α) is relatively compact. Every limit point L must have $\int f(x)g(y) dL(x, y) = \int f dL_1 \int g dL_2$. These numbers uniquely identify the (one) limit point.

1.4.7 Example (Slutsky's lemma). *If $X_\alpha \rightsquigarrow X$ and $Y_\alpha \rightsquigarrow c$ with X separable and c a constant, then $(X_\alpha, Y_\alpha) \rightsquigarrow (X, c)$.*

Again assume without loss of generality that X is tight. Then the net (X_α, Y_α) is asymptotically tight and asymptotically measurable. The limit points have marginals X and c and so are uniquely determined as (X, c) . (Now that the result has been obtained, it is easy to see that this is actually a special case of the previous example: since X and c are independent, X_α and Y_α are asymptotically independent.)

If X_α and Y_α take their values in a fixed topological vector space, the previous result can be combined with the continuous mapping theorem to obtain *Slutsky's lemma*: if $X_\alpha \rightsquigarrow X$ and $Y_\alpha \rightsquigarrow c$ with X separable and c a constant, then $X_\alpha + Y_\alpha \rightsquigarrow X + c$. Similarly, under the same conditions, $X_\alpha Y_\alpha \rightsquigarrow cX$ in the case Y_α are scalars; and provided $c \neq 0$, $X_\alpha/Y_\alpha \rightsquigarrow X/c$ also.

The results of this section easily extend to products of finitely many metric spaces. They can also be extended to countable products, with slightly more effort. If $\mathbb{D}_i$ is a metric space with metric d_i for every natural number i , then the Cartesian product $\mathbb{D}_1 \times \mathbb{D}_2 \times \cdots$ can be metrized by the metrics

$$d((x_1, x_2, \dots), (z_1, z_2, \dots)) = \sup_i \frac{1}{i} (d_i(x_i, z_i) \wedge 1),$$

$$d((x_1, x_2, \dots), (z_1, z_2, \dots)) = \sum_i 2^{-i} (d_i(x_i, z_i) \wedge 1).$$

These generate the same topology; the product topology. The results of this section hold for $\mathbb{D}_1 \times \mathbb{D}_2 \times \cdots$. In particular, coordinatewise asymptotic tightness is the same as joint asymptotic tightness, and given asymptotic tightness, coordinatewise asymptotic measurability is the same as joint asymptotic measurability.

For weak convergence, a countable product yields hardly anything new over finite products. A map $X: \Omega \rightarrow \mathbb{D}_1 \times \mathbb{D}_2 \times \cdots$ is separable if and only if every of its coordinates is, and weak convergence to a separable limit is equivalent to weak convergence of every finite set of coordinates. Write X_i for the i th coordinate of X : $X(\omega) = (X_1(\omega), X_2(\omega), \dots)$.

1.4.8 Theorem. *Let $X_\alpha: \Omega_\alpha \rightarrow \mathbb{D}_1 \times \mathbb{D}_2 \times \cdots$ be an arbitrary net and X separable. Then $X_\alpha \rightsquigarrow X$ if and only if $(X_{\alpha,1}, \dots, X_{\alpha,m}) \rightsquigarrow (X_1, \dots, X_m)$, for every $m \in \mathbb{N}$.*

Proof. If $X_\alpha \rightsquigarrow X$, then all marginals converge by the continuous mapping theorem. For the converse, assume without loss of generality that every $\mathbb{D}_i$ is complete. If $X_{\alpha,i} \rightsquigarrow X_i$ with X_i separable, then $X_{\alpha,i}$ is asymptotically tight and asymptotically measurable. By a straightforward argument, conclude that the vector X_α is asymptotically tight in the product space. Moreover, by a similar argument as for Lemma 1.4.4, obtain that $h(X_\alpha)$ is asymptotically measurable for every h in the linear span of the functions of the form $f_1(x_1)f_2(x_2)\cdots f_m(x_m)$, where f_i ranges over $C_b(\mathbb{D}_i)$ and $m \in \mathbb{N}$. This set of h forms a subalgebra of $C_b(\mathbb{D}_1 \times \mathbb{D}_2 \times \cdots)$ and separates points of the product. By Lemma 1.3.13, the net X_α is asymptotically measurable. Apply Prohorov's theorem to see that it is relatively compact. Every limit point L has the same finite-dimensional marginal distributions as X , so that the numbers $\int f_1(x_1)\cdots f_m(x_m) dL(x_1, x_2, \dots)$ are uniquely determined. These determine L completely. ■

1.5

Spaces of Bounded Functions

Let T be an arbitrary set. The space $\ell^\infty(T)$ is defined as the set of all uniformly bounded, real functions on T : all functions $z: T \rightarrow \mathbb{R}$ such that

$$\|z\|_T = \sup_{t \in T} |z(t)| < \infty.$$

It is a metric space with respect to the *uniform distance* $d(z_1, z_2) = \|z_1 - z_2\|_T$.

The space $\ell^\infty(T)$, or a suitable subspace of it, is a natural space for stochastic processes with bounded sample paths. A *stochastic process* is simply an indexed collection $\{X(t): t \in T\}$ of random variables defined on the same probability space: every $X(t): \Omega \rightarrow \mathbb{R}$ is a measurable map. If every *sample path* $t \mapsto X(t, \omega)$ is bounded, then a stochastic process yields a map $X: \Omega \rightarrow \ell^\infty(T)$. Sometimes the sample paths have additional properties, such as measurability or continuity, and it may be fruitful to consider X as a map into a subspace of $\ell^\infty(T)$. If in either case the uniform metric is used, this does not make a difference for weak convergence of a net; but for measurability it can. Here is one example of this situation; more examples are discussed in the next section.

1.5.1 Example (Continuous functions). Let T be a compact semimetric space; for instance, a compact interval in the real line, or the extended real line $[-\infty, \infty]$ with the metric $\rho(s, t) = |\arctan s - \arctan t|$. The set $C(T)$ of all continuous functions $z: T \rightarrow \mathbb{R}$ is a separable, complete subspace of $\ell^\infty(T)$. The Borel σ -field of $C(T)$ equals the σ -field generated by the coordinate projections $z \mapsto z(t)$, the *projection σ -field* (Problem 1.7.1). Thus a map $X: \Omega \rightarrow C(T)$ is Borel measurable if and only if it is a stochastic process.

In most cases a map $X: \Omega \rightarrow \ell^\infty(T)$ is a stochastic process. The small amount of measurability this gives may already be enough for asymptotic measurability. The

special role played by the *marginals* $(X(t_1), \dots, X(t_k))$, which are considered as maps into $\mathbb{R}^k$, is underlined by the following three results. Weak convergence in $\ell^\infty(T)$ can be characterized as asymptotic tightness plus convergence of marginals.

1.5.2 Lemma. *Let $X_\alpha: \Omega_\alpha \rightarrow \ell^\infty(T)$ be asymptotically tight. Then it is asymptotically measurable if and only if $X_\alpha(t)$ is asymptotically measurable for every $t \in T$.*

1.5.3 Lemma. *Let X and Y be tight Borel measurable maps into $\ell^\infty(T)$. Then X and Y are equal in Borel law if and only if all corresponding marginals of X and Y are equal in law.*

1.5.4 Theorem (Tightness plus marginal convergence). *Let $X_\alpha: \Omega_\alpha \rightarrow \ell^\infty(T)$ be arbitrary. Then X_α converges weakly to a tight limit if and only if X_α is asymptotically tight and the marginals $(X_\alpha(t_1), \dots, X_\alpha(t_k))$ converge weakly to a limit for every finite subset $t_1, \dots, t_k$ of T . If X_α is asymptotically tight and its marginals converge weakly to the marginals $(X(t_1), \dots, X(t_k))$ of a stochastic process X , then there is a version of X with uniformly bounded sample paths and $X_\alpha \rightsquigarrow X$.*

Proofs. For the proof of both lemmas, consider the collection $\mathcal{F}$ of all functions $f: \ell^\infty(T) \rightarrow \mathbb{R}$ of the form

$$f(z) = g(z(t_1), \dots, z(t_k)), \quad g \in C_b(\mathbb{R}^k), \quad t_1, \dots, t_k \in T, \quad k \in \mathbb{N}.$$

This forms an algebra and a vector lattice, contains the constant functions, and separates points of $\ell^\infty(T)$. Therefore, the lemmas are corollaries of Lemmas 1.3.13 and 1.3.12, respectively.

If X_α is asymptotically tight and the marginals converge, then X_α is asymptotically measurable by the first lemma. By Prohorov's theorem, X_α is relatively compact. To prove weak convergence, it suffices to show that all limit points are the same. This follows from marginal convergence and the second lemma. ■

Marginal convergence can be established by any of the well known methods for proving weak convergence on Euclidean space. Tightness can be given a more concrete form, either through finite approximation or (essentially) with the help of the Arzelà-Ascoli theorem. Finite approximation leads to the simpler of the two characterizations, but the second approach is perhaps of more interest, because it connects tightness to (asymptotic, uniform, equi-) continuity of the sample paths $t \mapsto X_\alpha(t)$.

The idea of finite approximation is that for any $\varepsilon > 0$ the index set T can be partitioned into finitely many subsets T_i such that (asymptotically) the variation of the sample paths $t \mapsto X_\alpha(t)$ is less than ε on every one of the sets T_i . More precisely, it is assumed that for every $\varepsilon, \eta > 0$, there exists a partition $T = \bigcup_{i=1}^k T_i$ such that

$$(1.5.5) \quad \limsup_{\alpha} \mathbf{P}^* \left(\sup_i \sup_{s, t \in T_i} |X_\alpha(s) - X_\alpha(t)| > \varepsilon \right) < \eta.$$

Clearly, under this condition the asymptotic behavior of the process can be described within error margin ε, η by the behavior of the marginal $(X_\alpha(t_1), \dots, X_\alpha(t_k))$ for arbitrary fixed points $t_i \in T_i$. If the process can thus be reduced to a finite set of coordinates for any $\varepsilon, \eta > 0$ and the nets of marginal distributions are tight, then the net X_α is asymptotically tight.

1.5.6 Theorem (Tightness through partitioning). *A net $X_\alpha: \Omega_\alpha \rightarrow \ell^\infty(T)$ is asymptotically tight if and only if $X_\alpha(t)$ is asymptotically tight in $\mathbb{R}$ for every t and, for all $\varepsilon, \eta > 0$, there exists a finite partition $T = \cup_{i=1}^k T_i$ such that (1.5.5) holds.*

Proof. The necessity of the conditions follows easily from the next theorem. For instance, take the partition equal to (disjointified) balls of radius δ for a semimetric on T as in the next theorem. We prove sufficiency.

For any partition, as in the condition of the theorem, the norm $\|X_\alpha\|_T$ is bounded by $\max_i |X_\alpha(t_i)| + \varepsilon$, with inner probability at least $1 - \eta$, if $t_i \in T_i$ for each i . Since a maximum of finitely many tight nets of real variables is tight, it follows that the net $\|X_\alpha\|_T$ is asymptotically tight in $\mathbb{R}$.

Fix $\zeta > 0$ and a sequence $\varepsilon_m \downarrow 0$. Take a constant M such that $\limsup P^*(\|X_\alpha\|_T > M) < \zeta$, and for each $\varepsilon = \varepsilon_m$ and $\eta = 2^{-m}\zeta$, take a partition $T = \cup_{i=1}^k T_i$ as in (1.5.5). For the moment m is fixed and we do not let it appear in the notation. Let $\{z_1, \dots, z_p\}$ be the set of all functions in $\ell^\infty(T)$ that are constant on each T_i and take on only the values $0, \pm\varepsilon_m, \dots, \pm \lfloor M/\varepsilon_m \rfloor \varepsilon_m$. (It is easy to express p in terms of the other constants, but it is only relevant that it is finite.) Let K_m be the union of the p closed balls of radius ε_m around the z_i . Then, by construction, the two conditions

$$\|X_\alpha\|_T \leq M \quad \text{and} \quad \sup_i \sup_{s, t \in T_i} |X_\alpha(s) - X_\alpha(t)| \leq \varepsilon_m$$

imply that $X_\alpha \in K_m$. This is true for each fixed m .

Let $K = \cap_{m=1}^\infty K_m$. Then K is closed and totally bounded (by construction of the K_m and because $\varepsilon_m \downarrow 0$) and hence compact. Furthermore, for every $\delta > 0$, there is an m with $K^\delta \supset \cap_{i=1}^m K_i$. If not, then there would be a sequence z_m not in K^δ , but with $z_m \in \cap_{i=1}^m K_i$ for every m . This would have a subsequence contained in one of the balls making up K_1 , a further subsequence eventually contained in one of the balls making up K_2 , and so on. The “diagonal” sequence, formed by taking the first of the first subsequence, the second of the second subsequence and so on, would eventually be contained in a ball of radius ε_m for every m ; hence Cauchy. Its limit would be in K , contradicting the fact that $d(z_m, K) \geq \delta$ for every m .

Conclude that if X_α is not in K^δ , then it is not in $\cap_{i=1}^m K_i$ for some fixed m . Then

$$\limsup P^*(X_\alpha \notin K^\delta) \leq \limsup P^*(X_\alpha \notin \bigcap_{i=1}^m K_i) \leq \zeta + \sum_{i=1}^m \zeta 2^{-m} < 2\zeta.$$

This concludes the proof of the theorem. ■

The second type of characterization of asymptotic tightness is deeper and relates the concept to asymptotic continuity of the sample paths. Suppose ρ is a semimetric on T . A net $X_\alpha: \Omega_\alpha \rightarrow \ell^\infty(T)$ is *asymptotically uniformly ρ -equicontinuous in probability* if for every $\varepsilon, \eta > 0$ there exists a $\delta > 0$ such that

$$\limsup_\alpha \mathbf{P}^* \left(\sup_{\rho(s,t) < \delta} |X_\alpha(s) - X_\alpha(t)| > \varepsilon \right) < \eta.$$

1.5.7 Theorem (Tightness through equicontinuity). *A net $X_\alpha: \Omega_\alpha \rightarrow \ell^\infty(T)$ is asymptotically tight if and only if $X_\alpha(t)$ is asymptotically tight in $\mathbb{R}$ for every t and there exists a semimetric ρ on T such that (T, ρ) is totally bounded and X_α is asymptotically uniformly ρ -equicontinuous in probability.*

1.5.8 Addendum. *If, moreover, $X_\alpha \rightsquigarrow X$, then almost all paths $t \mapsto X(t, \omega)$ are uniformly ρ -continuous; and the semimetric ρ can without loss of generality be taken equal to any semimetric ρ for which this is true and (T, ρ) is totally bounded.*

Proof. $\Leftarrow$. The sufficiency follows from the previous theorem. First take $\delta > 0$ sufficiently small so that the last displayed inequality is valid. Since T is totally bounded, it can be covered with finitely many balls of radius δ . Construct a partition of T by disjointifying these balls.

$\Rightarrow$. If X_α is asymptotically tight, then $g(X_\alpha)$ is asymptotically tight for every continuous map g ; in particular, for each coordinate projection.

Let $K_1 \subset K_2 \subset \cdots$ be compacts with $\liminf \mathbf{P}_*(X_\alpha \in K_m^\varepsilon) \geq 1 - 1/m$ for every $\varepsilon > 0$. For every fixed m , define a semimetric ρ_m on T by

$$\rho_m(s, t) = \sup_{z \in K_m} |z(s) - z(t)|, \quad s, t \in T.$$

Then (T, ρ_m) is totally bounded. Indeed, cover K_m by finitely many balls of (arbitrarily small) radius η , centered at $z_1, \dots, z_k$. Partition $\mathbb{R}^k$ into cubes of edge η , and for every cube pick at most one $t \in T$ such that $(z_1(t), \dots, z_k(t))$ is in the cube. Since $z_1, \dots, z_k$ are uniformly bounded, this gives finitely many points $t_1, \dots, t_p$. Now the balls $\{t: \rho_m(t, t_i) < 3\eta\}$ cover T : t is in the ball around t_i for which $(z_1(t), \dots, z_k(t))$ and $(z_1(t_i), \dots, z_k(t_i))$ fall in the same cube. This follows because $\rho_m(t, t_i)$ can be bounded by $2 \sup_{z \in K_m} \inf_i \|z - z_i\|_T + \sup_j |z_j(t_i) - z_j(t)|$.

Next set

$$\rho(s, t) = \sum_{m=1}^{\infty} 2^{-m} (\rho_m(s, t) \wedge 1).$$

Fix $\eta > 0$. Take a natural number m with $2^{-m} < \eta$. Cover T with finitely many ρ_m -balls of radius η . Let $t_1, \dots, t_p$ be their centers. Since $\rho_1 \leq \rho_2 \leq \cdots$, there is for every t a t_i with $\rho(t, t_i) \leq \sum_{k=1}^m 2^{-k} \rho_k(t, t_i) + 2^{-m} < 2\eta$. Thus (T, ρ) is totally bounded for ρ , too.

It is clear from the definitions that $|z(s) - z(t)| \leq \rho_m(s, t)$ for every $z \in K_m$ and that $\rho_m(s, t) \wedge 1 \leq 2^m \rho(s, t)$. Also, if $\|z_0 - z\|_T < \varepsilon$ for $z \in K_m$, then $|z_0(s) - z_0(t)| < 2\varepsilon + |z(s) - z(t)|$ for any pair s, t . Deduce that

$$K_m^\varepsilon \subset \left\{ z: \sup_{\rho(s,t) < 2^{-m}\varepsilon} |z(s) - z(t)| \leq 3\varepsilon \right\}.$$

Thus for given ε and m , and for $\delta < 2^{-m}\varepsilon$,

$$\liminf P_* \left(\sup_{\rho(s,t) < \delta} |X_\alpha(s) - X_\alpha(t)| \leq 3\varepsilon \right) \geq 1 - \frac{1}{m}.$$

Finally, we prove the addendum. If $X_\alpha \rightsquigarrow X$, then with notation as in the second part of the proof, $P(X \in K_m) \geq 1 - 1/m$; hence X concentrates on $\bigcup_{m=1}^\infty K_m$. The elements of K_m are uniformly ρ_m -equicontinuous and hence also uniformly ρ -continuous. This yields the first statement.

The set of uniformly continuous functions on a totally bounded, semimetric space is complete and separable, so a map X that takes its values in this set is tight. Next if $X_\alpha \rightsquigarrow X$ and X is tight, then X_α is asymptotically tight and the compacts for asymptotical tightness can be chosen equal to the compacts for tightness of X . If X has uniformly continuous paths, then the latter compacts can be chosen within the space of uniformly continuous functions. Since a compact is totally bounded, every one of the compacts is necessarily uniformly equicontinuous. Combination of these facts proves the second statement of the addendum. ■

Asymptotic uniform ρ -equicontinuity is a fairly complicated concept. Nevertheless, it is what must be shown for weak convergence. For particular problems, reasonable methods are available; for instance, methods based on the Markov property or the chaining method. In Part 2 we develop such methods for empirical processes.

At a general level, a few things can be said about which semimetric to use to establish asymptotic equicontinuity of a net X_α . Note that in principle the index set T need not have a semimetric on it when it comes to us. Furthermore, if it does, this semimetric may not be the right one to use. The next few results of this section are somewhat technical, but they lead to the important result that for Gaussian limits one can always use a certain variance metric.

A possible limit X of a net X_α can be identified from marginal convergence. Next, according to the addendum, we should look for a semimetric ρ that makes T totally bounded and the paths of X uniformly continuous. It can be shown that if any semimetric does this at all, the semimetric

$$\rho_0(s, t) = E \arctan |X(s) - X(t)|$$

should do the job (Problem 1.5.2). However, ρ_0 may not be the most convenient semimetric with which to work. Sometimes there is an obvious semimetric ρ to try. Alternatively, consider the family of semimetrics

$$\rho_p(s, t) = \left(E |X(s) - X(t)|^p \right)^{1/(p \vee 1)} \quad (0 < p < \infty).$$

None of these will always qualify; in fact, the expectations need not even be finite. Furthermore, these semimetrics are special in that they make the process $t \mapsto X(t)$ continuous in p th mean. It turns out that the latter is exactly what makes them work or not: if there is a semimetric ρ for which T is totally bounded, the paths of X are uniformly ρ -continuous, and also the process $t \mapsto X(t)$ is uniformly ρ -continuous in p th mean, then and only then can ρ_p be used without loss of generality when

showing tightness. Here uniform continuity in p th mean with respect to ρ means that $E|X(s_n) - X(t_n)|^p \rightarrow 0$ whenever $\rho(s_n, t_n) \rightarrow 0$.

1.5.9 Lemma. *Let X be a tight, Borel measurable map into $\ell^\infty(T)$. Then there is a semimetric on T for which almost all paths of X are uniformly continuous and T is totally bounded. Moreover, for any fixed $p > 0$, the following statements are equivalent:*

- (i) *for the semimetric ρ_p , the set T is totally bounded and almost all paths of X are uniformly ρ_p -continuous;*
- (ii) *for every semimetric ρ for which almost all paths of X are uniformly ρ -continuous, the map $t \mapsto X(t)$ is uniformly ρ -continuous in p th mean;*
- (iii) *there is a semimetric ρ for which T is totally bounded, the map $t \mapsto X(t)$ is uniformly ρ -continuous in p th mean, and almost all paths of X are uniformly ρ -continuous.*

In particular, if T is compact for a semimetric ρ such that the map $t \mapsto E|X(t)|^p$ and almost all sample paths are ρ -continuous, then (i) holds.

Proof. Since $X \rightsquigarrow X$ and is asymptotically tight, the first statement is a special case of the addendum to the previous theorem.

(i) $\Rightarrow$ (ii). Suppose that for sequences s_n and t_n in T , $\rho(s_n, t_n) \rightarrow 0$, but there is an ε with $E|X(s_n) - X(t_n)|^p \geq \varepsilon > 0$ for every n . By total boundedness of T under ρ_p , there exist subsequences $s_{n'}$ and $t_{n'}$ that converge with respect to ρ_p to limits s and t in the ρ_p -completion of T . These subsequences are Cauchy, which by definition of ρ_p is the same as $X(s_{n'})$ and $X(t_{n'})$ being Cauchy in the L_p -metric. Let $X(s)$ and $X(t)$ be their L_p -limits. Since almost all sample paths of X are uniformly ρ_p -continuous, $X(s_{n'})$ and $X(t_{n'})$ converge almost surely also; the almost sure limits must equal the L_p -limits $X(s)$ and $X(t)$. Because almost all sample paths of X are also uniformly continuous with respect to ρ (!) and $\rho(s_n, t_n) \rightarrow 0$ by assumption, $X(s_n) - X(t_n) \rightarrow 0$ almost surely. Conclude that $X(s) = X(t)$ almost surely, so that the two constructed subsequences have the same L_p -limits. It follows that $E|X(s_{n'}) - X(t_{n'})|^p \rightarrow 0$, contradicting the assumption.

(ii) $\Rightarrow$ (iii). By the first statement, there always exists a semimetric for which T is totally bounded and X is almost surely uniformly continuous. If (ii) holds, then $t \mapsto X(t)$ is uniformly continuous in p th mean for this semimetric. Hence we obtain a semimetric as in (iii).

(iii) $\Rightarrow$ (i). Since the map $t \mapsto X(t)$ is uniformly ρ -continuous in p th mean, there exists for every $\varepsilon > 0$ a $\delta > 0$ with $\rho_p(s, t) < \varepsilon$ whenever $\rho(s, t) < \delta$. Thus the total boundedness of T for ρ implies the same for ρ_p . Now assume for simplicity that all paths of X are uniformly ρ -continuous. Every uniformly continuous function defined on a subset of a semimetric space has a unique extension to a continuous function on the closure of its domain. Therefore all sample paths of X can be extended to continuous functions on $(\bar{T}, \rho)$, where $\bar{T}$ is the ρ -completion of T , which is compact.

If a path $t \mapsto X(t, \omega)$ is not uniformly ρ_p -continuous, then there exist $\varepsilon > 0$ and sequences s_n and t_n with $\rho_p(s_n, t_n) \rightarrow 0$ and $|X(s_n, \omega) - X(t_n, \omega)| \geq \varepsilon$ for every n . These have subsequences that converge with respect to ρ to limits s and t

in $\bar{T}$. Consequently, first $X(s_{n'}, \omega) - X(t_{n'}, \omega) \rightarrow X(s, \omega) - X(t, \omega)$. Second, the subsequences converge to s and t under ρ_p , too, so $\rho_p(s, t) = 0$. Conclude that the path $t \mapsto X(t, \omega)$ is uniformly ρ_p -continuous for every ω for which there do not exist $s, t \in \bar{T}$ with $\rho_p(s, t) = 0$, but $X(s, \omega) \neq X(t, \omega)$. Let N be the exceptional set of ω for which there do exist such s, t . Take a countable, ρ -dense subset A of $\{(s, t) \in \bar{T} \times \bar{T} : \rho_p(s, t) = 0\}$. Since $t \mapsto X(t, \omega)$ is ρ -continuous, N is also the set of all ω such that there exist $(s, t) \in A$ with $X(s, \omega) \neq X(t, \omega)$. From the definition of ρ_p , it is clear that for every fixed $(s, t) \in A$, it holds that $X(s, \omega) = X(t, \omega)$ for almost every ω . Conclude that N is a null set. Hence, almost all paths of X are uniformly ρ_p -continuous.

For the last remark of the lemma, it suffices to note that $X(t_n) \rightarrow X(t)$ almost surely and $E|X(t_n)|^p \rightarrow E|X(t)|^p$ implies that $X(t_n) \rightarrow X(t)$ in p th mean. ■

1.5.10 Example (Gaussian limit). A stochastic process X is called *Gaussian* if each of its finite-dimensional marginals $(X(t_1), \dots, X(t_k))$ has a multivariate normal distribution on Euclidean space.

Let X be a Gaussian process with “intrinsic” semimetrics ρ_p , and let X_α be a net of random elements with values in $\ell^\infty(T)$. Then there exists a version of X which is a tight Borel measurable map into $\ell^\infty(T)$, and X_α converges weakly to X if and only if for some p (and then for all p):

- the marginals of X_α converge weakly to the corresponding marginals of X ;
- X_α is asymptotically equicontinuous in probability with respect to ρ_p ;
- T is totally bounded for ρ_p .

The second moment metric ρ_2 is often the easiest with which to work.

The sufficiency of the three conditions is immediate from the preceding theorems and is not special to Gaussian processes. In fact, for the sufficiency, the semimetric ρ_p can be replaced by any other semimetric. The extra information is that for Gaussian processes one can always use any of the intrinsic semimetrics. This necessity is not immediate from the previous results but can be derived as follows. In view of the addendum to Theorem 1.5.7, it is certainly enough to show that

A Gaussian process X in $\ell^\infty(T)$ is tight if and only if (T, ρ_p) is totally bounded and almost all paths $t \mapsto X(t, \omega)$ are uniformly ρ_p -continuous for some p (and then for all p).

For any tight X , there is a semimetric ρ that makes T totally bounded and almost all paths of X uniformly continuous. The assertion would follow by (iii) of the previous lemma if the map $t \mapsto X(t)$ is uniformly ρ -continuous in p th mean for a Gaussian process. This is automatically the case.

If a Gaussian process in $\ell^\infty(T)$ has uniformly continuous paths with respect to a semimetric ρ that makes T totally bounded, then the map $t \mapsto X(t)$ is uniformly ρ -continuous in p th mean.

Indeed, if $\rho(s_n, t_n) \rightarrow 0$, then $X(s_n) - X(t_n) \rightarrow 0$ almost surely, hence weakly. Since all the random variables are normal, this can only happen if the mean and variance of $X(s_n) - X(t_n)$ converge to zero. (Use characteristic functions.) For a

normal variable, the p th absolute moment is a continuous function of the first two moments.

Problems and Complements

1. (**Arzelà–Ascoli**) Let (T, ρ) be a totally bounded, semimetric space and $K \subset \ell^\infty(T)$. Suppose that for some $\varepsilon, \delta > 0$ and every $z \in K$: $|z(s) - z(t)| < \varepsilon$ whenever $\rho(s, t) < \delta$. Moreover, suppose $\{z(t) : z \in K\}$ is bounded for every t . Then

- (i) K is uniformly bounded;
- (ii) K can be covered with finitely many balls of radius 2ε with centers in the set $UC(T, \rho)$ of uniformly continuous functions.

Deduce the Arzelà–Ascoli theorem: a uniformly bounded, uniformly ρ -equicontinuous subset of $\ell^\infty(T)$ is totally bounded.

[Hint: For (ii) choose a $\delta/2$ -net $t_1, \dots, t_p$ on T . Let d be the distance between the closest pair of t_i . Define functions

$$\begin{aligned}\beta_i(t) &= (1 - \delta^{-1} \rho(t, t_i))^+, \\ \alpha_i(t) &= 1 - (1 - d^{-1} \rho(t, t_i))^+, \\ w_i(t) &= \frac{\beta_i(t) \prod_{j \neq i} \alpha_j(t)}{\sum_i \beta_i(t) \prod_{j \neq i} \alpha_j(t)}.\end{aligned}$$

Then each w_i is defined and continuous on the ρ -completion of T , the w_i sum up to 1, $w_i(t_i) = 1$, and w_i can be nonzero only at t with $\rho(t, t_i) < \delta$ that do not equal one of the other t_j . (The α_i are just there to take care of the latter.) For any $u \in \mathbb{R}^p$, define a function $z_u(t) = \sum_i w_i(t) u_i$. This is uniformly continuous with $z_u(t_i) = u_i$ for every i . Let $S = \{0, \varepsilon, -\varepsilon, \dots, k\varepsilon, -k\varepsilon\}$ where k is chosen such that $k\varepsilon$ is larger than a uniform bound on K . Then $\{z_u : u \in S^p\}$ is a 2ε -net over K : a given z is in the 2ε -ball around z_u for which $|u_i - z(t_i)| < \varepsilon$ for every i . This is true because, for every fixed t , the number $z_u(t)$ is a convex combination of values u_i with $w_i(t) \neq 0$; hence i with $\rho(t, t_i) < \delta$. For every such i : $|z(t) - u_i| < |z(t) - z(t_i)| + \varepsilon < 2\varepsilon$.]

2. Let (T, ρ) be a totally bounded, semimetric space and X a map into $\ell^\infty(T)$ with uniformly ρ -continuous paths. Then T is totally bounded for the semimetric

$$\rho_0(s, t) = \mathbb{E} \arctan |X(s) - X(t)|.$$

Furthermore, almost all paths of X are uniformly continuous with respect to ρ_0 .

3. Let $X_\alpha: \Omega_\alpha \rightarrow \ell^\infty(S)$ and $Y_\alpha: \Omega_\alpha \rightarrow \ell^\infty(T)$ be asymptotically tight nets such that

$$(X_\alpha(s_1), \dots, X_\alpha(s_k), Y_\alpha(t_1), \dots, Y_\alpha(t_l)) \rightsquigarrow (X(s_1), \dots, X(s_k), Y(t_1), \dots, Y(t_l)),$$

for stochastic processes X and Y . Then there exist versions of X and Y with bounded sample paths and $(X_\alpha, Y_\alpha) \rightsquigarrow (X, Y)$ in $\ell^\infty(S) \times \ell^\infty(T)$.

[Hint: The net (X_α, Y_α) can be identified with a net in $\ell^\infty(S \cup T)$, where $S \cup T$ is the formal union of S and T , counting every element of S different from every element of T . This identification is an isometry.]

1.6

Spaces of Locally Bounded Functions

Let $T_1 \subset T_2 \subset \dots$ be arbitrary sets and $T = \bigcup_{i=1}^{\infty} T_i$. The space $\ell^\infty(T_1, T_2, \dots)$ is defined as the set of all functions $z: T \rightarrow \mathbb{R}$ that are uniformly bounded on every T_i (but not necessarily on T). This is a complete metric space with respect to the metric

$$d(z_1, z_2) = \sum_{i=1}^{\infty} \left(\|z_1 - z_2\|_{T_i} \wedge 1 \right) 2^{-i}.$$

A sequence converges in this metric if it converges uniformly on each T_i . In the case that T_i equals the interval $[-i, i] \subset \mathbb{R}^d$, the metric d induces the topology of uniform convergence on compacta.

The space $\ell^\infty(T_1, T_2, \dots)$ is of interest in applications, but its weak convergence theory is uneventful. Weak convergence of a net is equivalent to convergence of each of the restrictions to T_i .

1.6.1 Theorem. *Let $X_\alpha: \Omega_\alpha \rightarrow \ell^\infty(T_1, T_2, \dots)$ be arbitrary maps. Then the net X_α converges weakly to a tight limit if and only if every of the nets of restrictions $X_{\alpha|T_i}: \Omega_\alpha \rightarrow \ell^\infty(T_i)$ converges weakly ($i \in \mathbb{N}$) to a tight limit.*

Proof. The necessity of the weak convergence of all restrictions follows from the continuous mapping theorem. For the sufficiency part of the theorem, fix $\varepsilon > 0$ and take for each i a compact $K_i \subset \ell^\infty(T_i)$ such that

$$\limsup \mathbf{P}^*(X_{\alpha|T_i} \notin K_i^\delta) < \frac{\varepsilon}{2^i}, \quad \text{for every } \delta > 0.$$

Define K as the set of all $z: T \rightarrow \mathbb{R}$ such that $z|_{T_i - T_{i-1}}$ is contained in $(K_i)|_{T_i - T_{i-1}}$ for every i . A diagonal argument shows that K is compact in $\ell^\infty(T_1, T_2, \dots)$. Furthermore, if $z|_{T_i} \in K_i^\delta$ for every i , then $z \in K^\delta$. Conclude that

$$\limsup \mathbf{P}^*(X_\alpha \notin K^\delta) < \varepsilon, \quad \text{for every } \delta > 0.$$

It follows that the net X_α is asymptotically tight in $\ell^\infty(T_1, T_2, \dots)$.

Let $\mathcal{F}$ be the set of all continuous functions $f: \ell^\infty(T_1, T_2, \dots) \rightarrow \mathbb{R}$ of the form $f(z) = g(z(t_1), \dots, z(t_k))$ with $g \in C_b(\mathbb{R}^k)$, $t_1, \dots, t_k \in T$, and $k \in \mathbb{N}$. Then $\mathcal{F}$ is an algebra that separates points and contains the constant functions. Since the net $f(X_\alpha)$ is asymptotically measurable for every $f \in \mathcal{F}$, Lemma 1.3.13 implies that X_α is asymptotically measurable in $\ell^\infty(T_1, T_2, \dots)$.

Apply Prohorov's theorem to see that every subnet has a further weakly converging subnet with a tight limit. There is only one limit point, in view of the marginal convergence of the net and Lemma 1.3.12. ■

Given the preceding theorem, convergence in distribution in the space $\ell^\infty(T_1, T_2, \dots)$ can be proved by showing weak convergence of all marginals $(X_\alpha(t_1), \dots, X_\alpha(t_k))$ plus asymptotic equicontinuity on every T_i . For a given semimetric ρ on T , the latter takes the following form: for all $\varepsilon > 0$, $\eta > 0$, and i , there exists $\delta > 0$ such that

$$\limsup_{\alpha} \mathbf{P}^* \left(\sup_{\substack{\rho(s,t) < \delta \\ s, t \in T_i}} |X_\alpha(s) - X_\alpha(t)| > \varepsilon \right) < \eta.$$

Under this condition, the limit process necessarily has uniformly ρ -continuous sample paths on each T_i . In particular, it has continuous sample paths on T . For a Gaussian limit, this is necessarily the case for the standard deviation metric.

1.6.1 Convex processes

Stochastic processes with convex sample paths are of interest in statistical minimization problems.[†] For Euclidean index sets marginal convergence implies uniform weak convergence on sets in the interior of the domain. This is due to the fact that a convex function on Euclidean space is automatically Lipschitz.

1.6.2 Lemma (Convex processes). *Let X_α be a net of stochastic processes indexed by a convex, open subset C of $\mathbb{R}^k$ such that every sample path $t \mapsto X_\alpha(t)$ is convex (on C). If the net X_α converges marginally in distribution to a limit, then it converges in distribution to a tight limit in the space $\ell^\infty(K_1, K_2, \dots)$ for any sequence of compact sets $K_1 \subset K_2 \subset \dots \subset C$.*

Proof. It suffices to show that the processes converge in $\ell^\infty(K)$ for every compact $K \subset C$. Since C is open and K compact, there exists an $\varepsilon > 0$ such that $K^\varepsilon \subset K^{2\varepsilon} \subset C$. There exists a finite subset K_0 of $K^{2\varepsilon}$, which depends only on K and ε , such that the supremum norms over K^ε and K_0 of any convex function $z: K^{2\varepsilon} \rightarrow \mathbb{R}$ satisfy $\|z\|_{K^\varepsilon} \leq 3^k \|z\|_{K_0}$ (see Problem 1.6.4). Thus the assumed marginal tightness implies that the net $\|X_\alpha\|_{K^\varepsilon}$ is asymptotically tight. A bounded, convex function z on K^ε is automatically Lipschitz on K with Lipschitz constant $(2/\varepsilon)\|z\|_{K^\varepsilon}$ (see Problem 1.6.3). Conclude that the net X_α is asymptotically equicontinuous in $\ell^\infty(K)$. ■

[†] Centered criteria are typically not convex, rendering the next corollary perhaps more useful than the lemma.

1.6.3 Corollary (Convex processes). *Let X_α be a net of stochastic processes indexed by a convex, open subset $C \subset \mathbb{R}^k$ such that every sample path $t \mapsto X_\alpha(t)$ is convex (on C). If the net $X_\alpha(t)$ converges in probability to a limit $\mu(t)$ for every t , then $\sup_{t \in K} |X_\alpha(t) - \mu(t)|$ tends to zero in outer probability for every compact set $K \subset C$.*

Proof. This is Lemma 1.6.2 with a degenerate limit. ■

The problem section considers extensions to non-Euclidean index sets.

Problems and Complements

- (Convex processes on normed space)** Let X_α be a net of stochastic processes indexed by a convex, open subset C of a normed space such that every sample path $t \mapsto X_\alpha(t)$ is convex (on C). If the net $\sup_{t \in C} X_\alpha(t)$ is bounded in probability and the net X_α converges marginally in distribution to a limit, then it converges in distribution to a tight limit in the space $\ell^\infty(K)$ for any compact set $K \subset C$. (If the normed space is Euclidean space, then the boundedness in probability is automatic. See Lemma 1.6.2.)

[Hint: The assumptions imply the existence of a constant M and events A_α of probability (asymptotically) arbitrarily close to 1 on which $\sup_{t \in C} X_\alpha(t) \leq M$ and $X_\alpha(t_0) \geq -M$, for some $t_0 \in K$. Use Problem 1.6.5 to see that the net $\|X_\alpha\|_{K^\varepsilon}$ is asymptotically tight and next argue as in Lemma 1.6.2.]

- Let $x_\alpha: C \rightarrow \mathbb{R}$ be convex functions on a convex, open subset $C \subset \mathbb{R}^k$. If $x_\alpha(t)$ converges to a finite limit for every t , then x_α converges uniformly on compacta $K \subset C$.

[Hint: This is Corollary 1.6.3 with degenerate processes. See also Rockafellar (1970), Theorem 10.8.]

- (Convex functions are Lipschitz)** Let C be a convex subset of a normed space, $\varepsilon > 0$, and $f: C^\varepsilon \rightarrow \mathbb{R}$ a bounded and convex function, for $C^\varepsilon = \{x: \|x - C\| < \varepsilon\}$. Then $|f(x) - f(y)| \leq 2\varepsilon^{-1} \|f\|_{C^\varepsilon} \|x - y\|$, for every x, y in C , and $\|f\|_D$ the supremum norm over D .

[Hint: Take $x \neq y$ in C and define $z = y + \eta(y - x)/\|y - x\|$ for fixed $\eta < \varepsilon$. Then $z \in C^\varepsilon$ and y is a convex combination of x and z . By convexity, $f(y) \leq (1 - \lambda)f(x) + \lambda f(z)$, whence $f(y) - f(x) \leq \lambda(f(z) - f(x)) \leq 2\lambda\|f\|_{C^\varepsilon}$. Calculate λ .]

- (Convex functions are uniformly bounded)** Let $\varepsilon > 0$ and let D be a compact, convex subset of $\mathbb{R}^k$. Then there exists a finite set $D_0 \subset D^\varepsilon$ depending only on ε and D , such that $\|f\|_D = \sup_{x \in D} |f(x)|$ is bounded above by $3^k \|f\|_{D_0}$ for every convex function $f: D^\varepsilon \rightarrow \mathbb{R}$.

[Hint: Since we can cover D by finitely many cubes that are contained in D^ε , it suffices to bound the supremum norm of f over a given closed cube $C \subset D^\varepsilon$. By convexity, the maximum of f over C is bounded above by the maximum of $|f|$ over the corners. For a lower bound we can apply Problem 2.7.9 sequentially to every of the k coordinates separately to bound the infimum over a cube $\{x: a_i \leq x_i \leq b_i\}$ by $-3^k \max_{x \in S} |f(x)|$, for S the set of all points $S = (c_1, \dots, c_k)$ with $c_i \in \{a_i, b_i, (a_i + b_i)/2\}$, for every i .]

5. **(Convex functions are Lipschitz)** Let C be a bounded, convex subset of a normed space, $\varepsilon > 0$ and $\mathcal{F}$ a class of convex functions $f: C^\varepsilon \rightarrow \mathbb{R}$ that is uniformly bounded above on C^ε and such that $\{f(x): f \in \mathcal{F}\}$ is bounded below for at least one $x \in C$. Then the class of restrictions to C of the functions in $\mathcal{F}$ is uniformly bounded and uniformly Lipschitz.

[Hint: By assumption the constants $M = \sup_f \sup_{x \in C^\varepsilon} f(x)$ and $N = \inf_f f(x_0)$ satisfy $-\infty < N \leq M < \infty$, for some $x_0 \in C$. For $x \neq y$ in C and $\eta < \varepsilon$ set $z = y + \eta(y - x)/\|y - x\|$. Then $z \in C^\varepsilon$ and $y = \lambda z + (1 - \lambda)x$, for $\lambda = \|y - x\|/(\eta + \|y - x\|)$. By convexity $f(y) \leq \lambda f(z) + (1 - \lambda)f(x)$. Apply this first with $y = x_0$ to see that $f(x) \geq (1 - \lambda)^{-1}f(x_0) - \lambda(1 - \lambda)^{-1}f(z)$, which is greater than $N - \lambda(1 - \lambda)^{-1}(M - N) \geq N - \text{diam}(C)\eta^{-1}(M - N)$, by the definition of λ , for every $x \in C$. Thus the functions are uniformly bounded below on C . Apply the inequality a second time to conclude that $f(y) - f(x) \leq \lambda(f(z) - f(x))$. Insert the upper bound M on $f(z)$ and the lower bound on $f(x)$ just obtained to see that this is bounded above by $\lambda(M - N)(1 + \text{diam}(C)\eta^{-1})$. Finally use that $\lambda \leq \|y - x\|/\eta$ to conclude that f is Lipschitz with constant $\eta^{-1}(M - N)(1 + \text{diam}(C)\eta^{-1})$.]

6. Let $C \subset \mathbb{R}^k$ be bounded and convex, $\varepsilon > 0$, and $\mathcal{F}$ a class of convex functions $f: C^\varepsilon \rightarrow \mathbb{R}$ such that $\{f(x): f \in \mathcal{F}\}$ is bounded above for every $x \in C^\varepsilon$ and bounded below for at least one $x \in C$. Then $\mathcal{F}$ is uniformly bounded and uniformly Lipschitz on C .

[Hint: The function $x \mapsto \sup_{f \in \mathcal{F}} f(x)$ is finite and convex on C^ε , and hence continuous and bounded on compact subsets of C^ε , including the closure of $C^{\varepsilon/2}$. Use Problem 1.6.5.]

1.7

Ball Sigma-Field and Measurability of Suprema

The *ball σ -field* on $\mathbb{D}$ is the smallest σ -field containing all the open (and/or closed) balls in $\mathbb{D}$. In general, this is smaller than the Borel σ -field, although the two σ -fields are equal for separable spaces (Problems 1.7.3 and 1.7.4). For some nonseparable spaces, it is even fairly common that maps are ball measurable even though they are not Borel measurable. Thus one may wonder about the possibility of a weak convergence theory for ball measurable maps. It turns out that the set of ball measurable $f \in C_b(\mathbb{D})$ is rich enough to make this fruitful, but at the same time it is so rich that the theory is a special case of the theory that we have discussed so far.

However, establishing ball measurability is a good way to take care of the measurability part of weak convergence. Since maps of the form $x \mapsto d(x, s)$ generate the ball σ -field, a map X in $\mathbb{D}$ is ball measurable if and only if the map $\omega \mapsto d(X(\omega), s)$ is measurable for every $s \in \mathbb{D}$. A more common situation is that the maps $d(X(\omega), s)$ are measurable for a subset of s . This may yield enough measurability for weak convergence too.

1.7.1 Lemma. *Let $X_\alpha: \Omega_\alpha \rightarrow \mathbb{D}$ be asymptotically tight. Then it is asymptotically measurable if and only if $f(X_\alpha)$ is asymptotically measurable for every ball measurable $f \in C_b(\mathbb{D})$.*

1.7.2 Theorem. *Let X be separable and Borel measurable. Then*

- (i) $X_\alpha \rightsquigarrow X$ if and only if $E^* f(X_\alpha) \rightarrow Ef(X)$ for all ball measurable $f \in C_b(\mathbb{D})$;
- (ii) $X_\alpha \rightsquigarrow X$ if and only if there exists a sequence s_i with $P(X \in \overline{\{s_1, s_2, \dots\}}) = 1$ such that $(d(X_\alpha, s_1), d(X_\alpha, s_2), \dots) \rightsquigarrow (d(X, s_1), d(X, s_2), \dots)$ in $\mathbb{R}^\infty$.

Proofs. The set of all ball measurable $f \in C_b(\mathbb{D})$ forms an algebra and separates points of $\mathbb{D}$. So the lemma is a special case of Lemma 1.3.13.

The “only if” parts of the theorem are trivial. Furthermore, the “if” part of (ii) is stronger than that of (i). It suffices to show that $X_\alpha \rightsquigarrow X$ if the condition on the right side in (ii) holds.

Let $\mathcal{F}$ be the collection of functions of the form

$$f(x) = (1 - p d(x, s_i))^+, \quad p \in \mathbb{Q}^+, \quad i \in \mathbb{N}.$$

Then for any open G , with a supremum over the empty collection defined to be zero,

$$1_G(x) = \sup\{f(x) : 0 \leq f \leq 1_G, f \in \mathcal{F}\}, \quad \text{for every } x \in \overline{s_1, s_2, \dots}.$$

Place the countably many elements of $\mathcal{F}$ with $0 \leq f \leq 1_G$ in a sequence, and let f_m be the maximum of the first m functions. Then $0 \leq f_m \leq 1_G$ and $f_m \uparrow 1_G$ on a set of probability 1 under X . Hence first, for fixed m , $\liminf P_*(X_\alpha \in G) \geq \liminf E_* f_m(X_\alpha) = E f_m(X)$. Next, letting $m \rightarrow \infty$ completes the proof in view of the portmanteau theorem. ■

1.7.3 Example (Cadlag functions). The space $D[a, b]$ is the set of all *cadlag* functions on an interval $[a, b] \subset \mathbb{R}$: functions $z: [a, b] \rightarrow \mathbb{R}$ that are continuous from the right and have limits from the left everywhere. The ball σ -field for the uniform norm equals the σ -field generated by the coordinate projections (Problem 1.7.2). This means that a map $X: \Omega \rightarrow D[a, b]$ is ball measurable if and only if $X(t): \Omega \rightarrow \mathbb{R}$ is measurable for every $t \in [a, b]$, a rather weak requirement.

1.7.4 Example (Pointwise separable processes). The same idea applied to a more general situation leads to the following. Let $\mathcal{H}$ be a uniformly bounded class of functions $h: \mathcal{X} \rightarrow \mathbb{R}$ defined on some measurable space $(\mathcal{X}, \mathcal{B})$, with the property that there is a countable subcollection $\mathcal{H}_0 \subset \mathcal{H}$ such that every $h \in \mathcal{H}$ is the pointwise limit of a sequence h_m in $\mathcal{H}_0$: $h_m(x) \rightarrow h(x)$ for every x . Let $D(\mathcal{H})$ be the subspace of $\ell^\infty(\mathcal{H})$ of all z with the property

$$z(h_m) \rightarrow z(h), \quad \text{if } h_m \rightarrow h \text{ pointwise,}$$

for every sequence h_m in $\mathcal{H}_0$. Then the ball σ -field for the uniform norm is contained in the σ -field generated by the coordinate projections $z \mapsto z(h)$, $h \in \mathcal{H}$. Thus every stochastic process $X: \Omega \rightarrow D(\mathcal{H})$ is ball measurable. Remember that X is called a stochastic process if every coordinate $X(h)$ is a measurable map into $\mathbb{R}$.

A finite signed measure P on $(\mathcal{X}, \mathcal{B})$ induces an element $h \mapsto Ph = \int h dP$ of $D(\mathcal{H})$. So any stochastic process X for which every $X(\omega)$ has the character of a signed measure can be viewed a ball measurable map into $D(\mathcal{H})$. An example is the empirical measure $\mathbb{P}_n = n^{-1} \sum_{i=1}^n \delta_{X_i}$ of a sample $X_1, \dots, X_n$ of random elements in $(\mathcal{X}, \mathcal{B})$; another is the empirical process $\sqrt{n}(\mathbb{P}_n - P)$.[‡]

The condition on $\mathcal{H}$ is met, for instance, by the set of all indicator functions of closed or open ellipsoids, rectangles, and half-spaces in a Euclidean space.

[‡] Here δ_x denotes the *Dirac measure* at x , the discrete probability measure that gives mass 1 to the point x .

1.7.5 Example (Measurable processes with Suslin index set). Let T be a metric space equipped with its Borel σ -field. A map $X: \Omega \rightarrow \mathbb{R}^T$ is a *measurable stochastic process* if $X: \Omega \times T \rightarrow \mathbb{R}$ is jointly measurable for the product of $\mathcal{A}$ and the Borel sets of T . This is a little stronger than measurability of every coordinate $X(t)$.

If T is a Borel subset of a Polish space and $X: \Omega \rightarrow \ell^\infty(T)$ is a measurable process, then $d(X, z) = \|X - z\|_T$ is measurable for the P -completion of $(\Omega, \mathcal{A})$ for every measurable z . Consequently, every measurable process X with bounded sample paths is a ball measurable map into $\ell_m^\infty(T)$, the subspace of all measurable $z \in \ell^\infty(T)$, at least if $(\Omega, \mathcal{A}, P)$ is complete.

The claim follows from the theorem on measurable projections. The set $\{\omega: \|X(\omega) - z\|_T > c\}$ is the projection on Ω of the product measurable set $\{(\omega, t): |X(t, \omega) - z(t)| > c\}$. Projections are not always measurable, but under the stated conditions the present one is.^b

At closer inspection the argument actually yields more. Suppose $\mathbf{S}$ is a Suslin topological space[#] and let $\phi: \mathbf{S} \rightarrow T$ be an arbitrary map. Then any $X: \Omega \rightarrow \ell^\infty(T)$ for which $(s, \omega) \mapsto X(\phi(s), \omega)$ is jointly measurable has a measurable supremum $\sup_{t \in \phi(\mathbf{S})} |X(t)|$.

Problems and Complements

1. Let T be a compact semimetric space with dense subset S . The Borel σ -field on $C(T)$ is the smallest σ -field making all projections $z \mapsto z(s)$, $s \in S$, measurable.

[Hint: The closed ball around z_0 of radius r equals $\bigcup_{s \in S_0} \{z: |z(s) - z_0(s)| \leq r\}$, where S_0 is a countable dense subset of S . So every closed and consequently every open ball is projection measurable. For a separable metric space, the ball σ -field and Borel σ -field are the same.]

2. The ball σ -field on $D[a, b]$ with respect to the uniform metric equals the σ -field generated by the coordinate projections.

[Hint: The set $\{z: z(t) > c\}$ is the union of the balls of radius n around $z_{n,c} = c + (n + n^{-1})1_{[t, t+n^{-1}]}$.]

3. Equip $D[0, 1]$ with the uniform metric, and define $X: [0, 1] \rightarrow \mathbb{D}$ by $X(\omega) = 1_{[\omega, 1]}$ (the empirical process based on one observation ω). If $[0, 1]$ is equipped with the Borel σ -field, then X is ball measurable, but not Borel measurable.

[Hint: Let B_s be the open ball of radius $1/2$ in $\mathbb{D}$ around the function $1_{[s, 1]}$. Since an open ball is open, so is the union $G = \bigcup_{s \in S} B_s$, for any $S \subset [0, 1]$. Now $X(\omega) \in B_s$ if and only if $\omega = s$. Thus $\{X \in G\} = S$. The conclusion is that X is Borel measurable if and only if every subset of $[0, 1]$ is a Borel set.]

^b If $(\Omega, \mathcal{A})$ is a measurable space and Y a Polish space, then the projection on Ω of any product measurable subset of $\Omega \times Y$ (a set contained in the product σ -field of $\mathcal{A}$ and the Borel sets of Y) is *universally measurable*; that is, it is contained in the P -completion of $\mathcal{A}$ for every probability measure P on $(\Omega, \mathcal{A})$ (Cohn (1980), p. 281).

[#] A (Hausdorff) topological space is called *Suslin* if it is the continuous image of a Polish space. A subset of a Polish space is called *analytic* if it is Suslin for the relative topology. Every Borel subset of a Polish space is analytic/Suslin. See Cohn (1980), page 292.

4. (Properties of the ball σ -field)

- (i) For a separable, semimetric space, the ball σ -field equals the Borel σ -field.
- (ii) Every closed, separable set is ball measurable.
- (iii) If K is compact, then K^δ is ball measurable.
- (iv) Every separable probability measure on the ball σ -field has a unique extension to the Borel σ -field.

[Hint: If the sequence s_i is dense in a closed F , then $F^\delta = \cup_{i=1}^\infty B(s_i, \delta)$, where $B(s, \delta)$ is the ball of radius δ around s . Since a compact set is closed and separable, (iii) follows. For (ii) note that $F = \cap_{m=1}^\infty F^{1/m}$; (i) is a consequence of (ii). For (iv) let S be a closed, separable set with $L(\mathbb{D}_0) = 1$. For every Borel set B , the set $B \cap \mathbb{D}_0$ is a Borel set contained in $\mathbb{D}_0$, hence is contained in the ball σ -field on $\mathbb{D}_0$ by (i), but then also in the ball σ -field on $\mathbb{D}$. Define the extension by $\tilde{L}(B) = L(B \cap \mathbb{D}_0)$.]

- 5. (Completely regular points)** Pollard (1984) calls a point x in a metric space $\mathbb{D}$ equipped with a σ -field $\mathcal{A}$ *completely regular* if, for every open neighborhood G of x , there is a measurable, uniformly continuous $f: \mathbb{D} \rightarrow \mathbb{R}$ with $0 \leq f \leq 1_G$ and $f(x) = 1$. If X is an $\mathcal{A}$ -measurable map that takes all its values in a separable set of completely regular points, then it is Borel measurable. If, moreover, X_α are arbitrary maps with $E^* f(X_\alpha) \rightarrow E f(X)$ for every $\mathcal{A}$ -measurable, uniformly continuous f , then $X_\alpha \rightsquigarrow X$. Since every point is completely regular for the ball σ -field, this extends the results obtained in the section about the ball σ -field.

[Hint: Pollard (1984), page 88, Exercises [14]–[16].]

1.8

Hilbert Spaces

Let $\mathbb{H}$ be a (real) Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and complete orthonormal system $\{e_j: j \in J\}$. Thus $\langle e_i, e_j \rangle$ equals 0 or 1 if $i \neq j$ or $i = j$, respectively, and every $x \in \mathbb{H}$ can be written as

$$x = \sum_j \langle x, e_j \rangle e_j.$$

The sum contains at most countably many nonzero terms, because only countably many inner products $\langle x, e_j \rangle$ are nonzero for every x . The series converges unconditionally and $\|x\|^2 = \sum_j \langle x, e_j \rangle^2$. The orthogonal projection of x into the linear span of a subset $\{e_i: i \in I\}$ of the base equals $\sum_{i \in I} \langle x, e_i \rangle e_i$; the square distance of x to this linear span equals $\sum_{j \notin I} \langle x, e_j \rangle^2$.

Call a net $X_\alpha: \Omega_\alpha \rightarrow \mathbb{H}$ *asymptotically finite-dimensional* if, for all $\delta, \varepsilon > 0$, there exists a finite subset $\{e_i: i \in I\}$ of the orthonormal base such that

$$\limsup_\alpha \mathbf{P}^* \left(\sum_{j \notin I} \langle X_\alpha, e_j \rangle^2 > \delta \right) < \varepsilon.$$

Compact sets in a Banach space can be characterized by being bounded and being contained in the δ -shell of a finite-dimensional subspace, for every δ . This leads to the following characterization of asymptotic tightness of a sequence of random maps.

1.8.1 Lemma. *A net of random maps $X_\alpha: \Omega_\alpha \rightarrow \mathbb{H}$ is asymptotically tight if and only if it is asymptotically finite-dimensional and the nets $\langle X_\alpha, e_j \rangle$ are asymptotically tight for every j .*

Proof. The set $K = \left\{ \sum_{i \in I} a_i e_i : a \in \mathbb{R}^I, \max |a_i| \leq k \right\}$ is compact for every finite set I and constant k . For every fixed δ ,

$$P^*(X_\alpha \notin K^\delta) \leq P^*\left(\max_{i \in I} |\langle X_\alpha, e_i \rangle| > k\right) + P^*\left(\sum_{j \notin I} \langle X_\alpha, e_j \rangle^2 \geq \delta^2\right).$$

If the net X_α is asymptotically finite-dimensional, then the last probability can be made arbitrarily small by the choice of I . Next, the first probability on the right can be made arbitrarily small by the choice of k .

Conversely, suppose that the net X_α is asymptotically tight. Then every net $\langle X_\alpha, h \rangle$ is asymptotically tight by continuity of the inner product. Given $\varepsilon > 0$, let K be a compact set such that $\limsup_\alpha P^*(X_\alpha \notin K^\delta) < \varepsilon$, for every $\delta > 0$. Since K is compact, there exists a finite set I such that $K \subset \text{lin}(e_i : i \in I)^\delta$. Then every $x \in K^\delta$ is at a distance of at most 2δ from $\text{lin}(e_i : i \in I)$; the square of this distance equals $\sum_{j \notin I} \langle x, e_j \rangle^2$. Conclude that $\sum_{j \notin I} \langle X_\alpha, e_j \rangle^2 \geq 4\delta^2$ implies that $X_\alpha \notin K^\delta$. Hence the net X_α is asymptotically finite-dimensional. ■

1.8.2 Lemma. Let $X_\alpha : \Omega_\alpha \rightarrow \mathbb{H}$ be asymptotically tight. Then it is asymptotically measurable if and only if $\langle X_\alpha, e_j \rangle$ is asymptotically measurable for every j .

1.8.3 Lemma. Tight Borel measurable random elements X and Y in $\mathbb{H}$ are equal in distribution if and only if the random variables $\langle X, h \rangle$ and $\langle Y, h \rangle$ are equal in distribution for every $h \in \mathbb{H}$.

1.8.4 Theorem. A net of random maps $X_\alpha : \Omega_\alpha \rightarrow \mathbb{H}$ converges in distribution to a tight Borel measurable random variable X if and only if it is asymptotically finite-dimensional and the net $\langle X_\alpha, h \rangle$ converges in distribution to $\langle X, h \rangle$ for every $h \in \mathbb{H}$.

Proofs. For the lemmas, consider the collection $\mathcal{F}$ of all functions $f : \mathbb{H} \rightarrow \mathbb{R}$ of the form

$$f(x) = g(\langle x, e_{j_1} \rangle, \dots, \langle x, e_{j_k} \rangle), \quad g \in C_b(\mathbb{R}^k), \quad j_1, \dots, j_k \in J, \quad k \in \mathbb{N}.$$

This collection forms an algebra and a vector lattice and separates points of $\mathbb{H}$. Therefore, the lemmas are consequences of Lemmas 1.3.13 and 1.3.12, respectively.

The theorem follows from combination of the lemmas with Prohorov's theorem. ■

1.8.5 Example (Central limit theorem). If $X_1, X_2, \dots$ are i.i.d. Borel measurable random elements in a separable Hilbert space $\mathbb{H}$ with mean zero (i.e., $E\langle X_1, h \rangle = 0$ for every h), and $E\|X_1\|^2 < \infty$, then the sequence $n^{-1/2} \sum_{i=1}^n X_i$ converges in distribution to a Gaussian variable G . The distribution of G is determined by the distribution of its marginals $\langle G, h \rangle$, which are $N(0, E\langle X, h \rangle^2)$ distributed for every $h \in \mathbb{H}$.

This follows from the theorem. The sequence $\langle n^{-1/2} \sum_{i=1}^n X_i, h \rangle$ converges in distribution to a normal distribution by the central limit theorem for real-valued random variables. Second, if $e_1, e_2, \dots$ is an orthonormal base for $\mathbb{H}$, then

$$\mathbb{E} \sum_{j>J} \langle n^{-1/2} \sum_{i=1}^n X_i, e_j \rangle^2 = \mathbb{E} \sum_{j>J} \langle X_1, e_j \rangle^2 \rightarrow 0, \quad J \rightarrow \infty,$$

by dominated convergence, because the series on the right is bounded by $\|X_1\|^2$, by Bessel's inequality. Thus the sequence $n^{-1/2} \sum_{i=1}^n X_i$ is asymptotically finite-dimensional.

1.8.6 Example (Anderson–Darling statistic). Let $X_1, X_2, \dots$ be real-valued random variables with cumulative distribution function F . The random functions

$$Z_i(t) = \frac{1\{X_i \leq t\} - F(t)}{\sqrt{F(t)(1-F)(t)}}$$

are contained in $L_2(\mathbb{R}, \mu)$, and $\mathbb{E}\|Z_1\|_2^2 = \mu(\mathbb{R}) < \infty$ for every finite measure μ . Conclude that the sequence $n^{-1/2} \sum_{i=1}^n Z_i$ converges in distribution to a Gaussian variable G in $L_2(\mathbb{R}, \mu)$. One consequence is the weak convergence of the sequence of $L_2(\mu)$ -norms

$$\int \frac{n(\mathbb{F}_n - F)^2(t)}{F(t)(1-F)(t)} d\mu(t) \rightsquigarrow \int G^2(t) d\mu(t),$$

where $\mathbb{F}_n$ is the empirical distribution function of $X_1, \dots, X_n$. The choice $\mu = F$ yields the Anderson–Darling statistic.

Problems and Complements

1. If K is compact, then for every $\delta > 0$ there exists a finite set I with $K \subset \text{lin}(e_i; i \in I)^\delta$.

[Hint: If not then for every finite subset I there exists $x_I \in K$ at distance at least δ from $\text{lin}(e_i; i \in I)$. Direct the finite subsets by inclusion. The net has a converging subnet. Its limit point x has distance at least δ to every finite-dimensional space $\text{lin}(e_i; i \in I)$. Thus $\sum_{j \notin I} \langle x, e_j \rangle^2 \geq \delta^2$ for every finite set I .]

1.9

Almost Sure and in Probability Convergence

For nets of maps defined on a single, fixed probability space $(\Omega, \mathcal{A}, P)$, convergence almost surely and in probability are frequently used modes of stochastic convergence, stronger than weak convergence. In this section we consider their nonmeasurable extensions together with the concept of almost uniform convergence, which is equivalent to outer almost sure convergence for sequences, but stronger and more useful for general nets.

1.9.1 Definition. Let $X_\alpha, X: \Omega \rightarrow \mathbb{D}$ be arbitrary maps.

- (i) X_α *converges in outer probability* to X if $d(X_\alpha, X)^* \rightarrow 0$ in probability; this means that $P(d(X_\alpha, X)^* > \varepsilon) = P^*(d(X_\alpha, X) > \varepsilon) \rightarrow 0$, for every $\varepsilon > 0$, and is denoted by $X_\alpha \xrightarrow{P^*} X$.
- (ii) X_α *converges almost uniformly* to X if, for every $\varepsilon > 0$, there exists a measurable set A with $P(A) \geq 1 - \varepsilon$ and $d(X_\alpha, X) \rightarrow 0$ uniformly on A ; this is denoted $X_\alpha \xrightarrow{\text{au}} X$.
- (iii) X_α *converges outer almost surely* to X if $d(X_\alpha, X)^* \rightarrow 0$ almost surely for some versions of $d(X_\alpha, X)^*$; this is denoted $X_\alpha \xrightarrow{\text{as}^*} X$.
- (iv) X_α *converges almost surely* to X if $P_*(\lim d(X_\alpha, X) = 0) = 1$; this is denoted $X_\alpha \xrightarrow{\text{as}} X$.

The first three concepts in this list are the most important. The fourth is tricky — it does not behave as one might expect, in general. In fact, for nonmeasurable sequences, even convergence *everywhere* (which is stronger than (iv)) does not imply any of the other three forms of convergence. This is one of those phenomena that remind us that measurability, though often present, should not be taken too lightly. The counterexample given below isn't even very complicated, though contrived, perhaps.

For sequences the second and third modes of convergence are equivalent, but for general nets the third loses much of its value. First, there is the nuisance that for general nets outer almost sure convergence depends on the versions of the minimal measurable covers one uses. Second, even for measurable nets X_α , outer almost sure convergence is rather weak. For instance, it does not imply convergence in probability. For these reasons we consider (iii) and (iv) for sequences only. The Problems and Complements section gives more details about the general implications between the four forms of convergence. In the following text, we use the notation X_α for a general net and X_n for a sequence, without further mention.

For sequences things are partly as they should be. Outer almost sure convergence (iii) is stronger than convergence in outer probability. Equivalence of almost uniform and outer almost sure convergence, part (iii) of the following lemma, is known as *Egorov's theorem*.

1.9.2 Lemma. *Let X be Borel measurable. Then*

- (i) $X_n \xrightarrow{\text{as}^*} X$ implies $X_n \xrightarrow{P^*} X$;
- (ii) $X_n \xrightarrow{P^*} X$ if and only if every subsequence $X_{n'}$ has a further subsequence $X_{n''}$ with $X_{n''} \xrightarrow{\text{as}^*} X$;
- (iii) $X_n \xrightarrow{\text{as}^*} X$ if and only if $X_n \xrightarrow{\text{au}} X$.

1.9.3 Lemma. *Let X be Borel measurable. Then*

- (i) $X_\alpha \xrightarrow{\text{au}} X$ if and only if $\sup_{\beta \geq \alpha} d(X_\beta, X)^* \xrightarrow{P} 0$ if and only if $\sup_{\beta \geq \alpha} d(X_\beta, X) \xrightarrow{P^*} 0$;
- (ii) $X_\alpha \xrightarrow{\text{au}} X$ implies $X_\alpha \xrightarrow{P^*} X$.

Proofs. (iii). Suppose X_n converges outer almost surely to X . Fix $\varepsilon > 0$. Set $A_n^k = \{\sup_{m \geq n} d(X_m, X)^* > 1/k\}$. Then, for every fixed k , it holds that $P(A_n^k) \downarrow 0$ as $n \rightarrow \infty$. Choose n_k with $P(A_{n_k}^k) \leq \varepsilon/2^k$, and set $A = \Omega - \bigcup_{k=1}^\infty A_{n_k}^k$. Then $P(A) \geq 1 - \varepsilon$ and $d(X_n, X)^* \leq 1/k$, for $n \geq n_k$ and $\omega \in A$. Thus X_n converges to X almost uniformly. Conversely, suppose X_n converges almost uniformly to X . Fix $\varepsilon > 0$, and let A be as in the definition of almost uniform convergence. Fix $\eta > 0$. Then $d(X_n, X)^* 1_A = (d(X_n, X) 1_A)^* \leq \eta$ for sufficiently large n , since η is a measurable function and $\eta \geq d(X_n, X) 1_A$ for sufficiently large n . Thus $d(X_n, X)^* \rightarrow 0$ for almost all $\omega \in A$.

Next consider the second lemma first.

(i). It is easy to see that the first statement implies the second, which implies the third. For the converse, fix $\varepsilon > 0$. Take α_k such that $P^*(\sup_{\beta \geq \alpha_k} d(X_\beta, X) > 1/k) \leq \varepsilon/2^k$. Call the set within brackets A_k , and set $A = \Omega - \bigcup_{k=1}^\infty A_k$. Then $P(A) \geq 1 - \varepsilon$, and for every $\omega \in A$ and $\alpha \geq \alpha_k$, it holds that $d(X_\alpha, X) \leq 1/k$.

(ii). This is an immediate consequence of (i).

Finally, consider the first lemma again. Part (i) follows from a combination of the other statements. For (ii) suppose $X_n \xrightarrow{P^*} X$. Take $n_1 < n_2 < \dots$ such that $P(d(X_{n_j}, X)^* > 1/j) < 2^{-j}$. Then $P(d(X_{n_j}, X)^* > 1/j, \text{i.o.}) = 0$, by the Borel-Cantelli lemma. Thus $d(X_{n_j}, X) \leq 1/j$ eventually, for almost every ω . The converse is trivial in view of (i). ■

1.9.4 Counterexample. $X_n(\omega) \rightarrow 0$ for every ω does not imply $X_n \xrightarrow{\text{as}^*} 0$.

Let $(\Omega, \mathcal{A}, \lambda)$ be $[0,1]$ with Borel sets and Lebesgue measure. There exists a decreasing sequence of sets B_n , with $\bigcap_{n=1}^{\infty} B_n = \emptyset$, but $\lambda^*(B_n) = 1$ for every n . Thus $1_{B_n}(\omega) \rightarrow 0$ for every ω , but $|1_{B_n} - 0|^* = 1_{B_n^*} = 1$ for every n , so that 1_{B_n} certainly doesn't converge to zero almost uniformly.

One construction of the sequence B_n is as follows.[†] A set $H \subset \mathbb{R}$ is called a *Hamel base* for $\mathbb{R}$ over $\mathbb{Q}$ if every element of $\mathbb{R}$ can be expressed as a finite linear combination $\sum_{i=1}^k q_i h_i$, where $q_i \in \mathbb{Q}$, $h_i \in H$, and $k \in \mathbb{N}$. Such a base exists by Zorn's lemma (which is equivalent to the axiom of choice). Any such Hamel base H is an uncountable set. Cut one such Hamel base H into countably many disjoint, nonempty subsets (this is possible by the axiom of choice), and let H_n be the union of the first n cuts. Thus $H_n \uparrow$ and $H = \bigcup_{n=1}^{\infty} H_n$. Let C_n be the subspace of $\mathbb{R}$ spanned by H_n : all finite linear combinations $\sum_{i=1}^k q_i h_i$ with $q_i \in \mathbb{Q}$ and $h_i \in H_n$. Then $C_n \uparrow$ and $\mathbb{R} = \bigcup_{n=1}^{\infty} C_n$.

Let K be an arbitrary compact set. Then either $\lambda(K) = 0$ or $K - K$ contains an interval $(-\delta, \delta)$ for some $\delta > 0$. Indeed, if $\lambda(K) > 0$, then for $\delta > 0$ sufficiently small, $\lambda(K^\delta) < 2\lambda(K)$. If $(-\delta, \delta)$ is not contained in $K - K$, then there is an $|x| < \delta$ for which the sets K and $K + x$ are disjoint. But this is impossible, since it would imply $2\lambda(K) = \lambda(K \cup K + x) \leq \lambda(K^\delta)$.

Now suppose that K is a compact set contained in C_n . Then $K - K \subset C_n = C_n - C_n$. Since C_n does not contain an interval, it can be inferred that $\lambda(K) = 0$. Together with the inner regularity of Lebesgue measure, this implies $\lambda_*(C_n) = 0$. Finally, set $B_n = [0, 1] - C_n$.

There is a continuous mapping theorem for convergence in outer probability and almost uniform convergence under exactly the same conditions as for weak convergence.

1.9.5 Theorem (Continuous mapping). *Let $g: \mathbb{D} \rightarrow \mathbb{E}$ be continuous at every point of a Borel set $\mathbb{D}_0 \subset \mathbb{D}$. Let X be Borel measurable with $P(X \in \mathbb{D}_0) = 1$. Then*

- (i) $X_\alpha \xrightarrow{P^*} X$ implies that $g(X_\alpha) \xrightarrow{P^*} g(X)$;
- (ii) $X_\alpha \xrightarrow{\text{au}} X$ implies that $g(X_\alpha) \xrightarrow{\text{au}} g(X)$.

Proof. (i). Fix $\varepsilon > 0$. Let B_k be the set of all x for which there exist y and z within the open ball of radius $1/k$ around x with $e(g(y), g(z)) > \varepsilon$. Then B_k is open and the sequence B_k is decreasing. Moreover, $P(X \in B_k) \downarrow 0$, since every point in $\bigcap_{k=1}^{\infty} B_k$ is a point of discontinuity of g . Now, for every fixed k ,

$$\begin{aligned} P^*(e(g(X_\alpha), g(X)) > \varepsilon) &\leq P^*(X \in B_k \text{ or } d(X_\alpha, X) \geq 1/k) \\ &\rightarrow P^*(X \in B_k). \end{aligned}$$

Finally, let $k \rightarrow \infty$.

(ii). Add a supremum in the proof of (i) twice to obtain that the convergence $\sup_{\beta \geq \alpha} d(X_\beta, X) \xrightarrow{P^*} 0$ implies that $\sup_{\beta \geq \alpha} e(g(X_\beta), g(X)) \xrightarrow{P^*} 0$. This is equivalent to the statement of (ii). ■

[†] Cohn (1980).

It was shown by counterexample that convergence almost surely does not imply convergence outer almost surely or in outer probability. The problem is a possible lack of measurability. Since convergence almost surely is so easy to work with, it is of interest to know how much measurability is sufficient to remove the difference. A trivial but useful result is that $X_n \xrightarrow{\text{as}} X$ together with measurability of $d(X_n, X)$ implies $X_n \xrightarrow{\text{as}*} X$. The next theorem gives an exact answer to the problem, although, admittedly, it may often not be more useful than the trivial sufficient condition just mentioned.

What is needed is some sort of asymptotic measurability of X_n . Asymptotic measurability as introduced before is slightly too weak. Call X_n *strongly asymptotically measurable* if

$$f(X_n)^* - f(X_n)_* \xrightarrow{\text{as}} 0, \quad \text{for every } f \in C_b(\mathbb{D}).$$

This is stronger than (ordinary) asymptotic measurability: the latter requires convergence in probability rather than almost sure convergence. Of course, both are implied by measurability for every n . It can be shown that $X_n \xrightarrow{\text{as}} X$ together with asymptotic measurability implies $X_n \xrightarrow{P^*} X$ (Problem 1.9.1). For the stronger conclusion that $X_n \xrightarrow{\text{as}*} X$, strong asymptotic measurability is necessary. Perhaps the best part of the next theorem is (iv), which implies that almost sure convergence plus ball measurability implies almost uniform convergence.

1.9.6 Theorem. *Let X be Borel measurable and separable. Then the following statements are equivalent:*

- (i) $X_n \xrightarrow{\text{as}*} X$;
- (ii) $X_n \xrightarrow{\text{as}} X$ and $d(X_n, X)^* - d(X_n, X)_* \xrightarrow{\text{as}} 0$;
- (iii) $X_n \xrightarrow{\text{as}} X$ and X_n is strongly asymptotically measurable;
- (iv) $X_n \xrightarrow{\text{as}} X$ and $d(X_n, s)$ is strongly asymptotically measurable for every s in a set S with $P(X \in \overline{S}) = 1$.

In particular, if $X_n \xrightarrow{\text{as}} X$ and every X_n and X is ball measurable, then $X_n \xrightarrow{\text{as}} X$.*

Proof. The equivalence of (i) and (ii) is easy to see.

(i) $\Rightarrow$ (iii) $\Rightarrow$ (iv). If (i) holds, then trivially $X_n \xrightarrow{\text{as}} X$. Furthermore, by the continuous mapping theorem $f(X_n) \xrightarrow{\text{as}*} f(X)$, for every continuous f . Consequently

$$\begin{aligned} f(X_n)^* - f(X_n)_* &= (f(X_n) - f(X))^* - (f(X_n) - f(X))_* \\ &\leq 2|f(X_n) - f(X)|^* \xrightarrow{\text{as}} 0. \end{aligned}$$

Thus (iii) holds. Next, (iv) is immediate from the fact that the function $x \mapsto f(d(x, s))$ is contained in $C_b(\mathbb{D})$ for every bounded, continuous function f on $\mathbb{R}$ and $s \in \mathbb{D}$.

(iv) $\Rightarrow$ (i). Assume without loss of generality that S is countable and that $\mathbb{D}$ is complete. Let $f_1, f_2, \dots$ be the set of functions $x \mapsto (1 - m d(x, s))^+$, where $m \in \mathbb{N}$ and $s \in S$. Define a semimetric on $\mathbb{D}$ by

$$e(x, y) = \sup_{j \in \mathbb{N}} \frac{1}{j} |f_j(x) - f_j(y)|.$$

A sequence x_n in $\mathbb{D}$ converges to a point $x \in \overline{S}$ if and only if $f_j(x_n) \rightarrow f_j(x)$ for every j ; equivalently, if and only if $e(x_n, x) \rightarrow 0$. Conclude that, for every compact $K \subset \overline{S}$, there is for every $\varepsilon > 0$ a $\delta > 0$ such that, for every $x \in K$ and $y \in \mathbb{D}$, $e(y, x) < \delta$ implies $d(y, x) < \varepsilon$.

Fix j . If the first assertion of (iv) holds, then $f_j(X_n) \xrightarrow{\text{as}} f_j(X)$. Under the second assertion,

$$(f_j(X_n) - f_j(X))^* - (f_j(X_n) - f_j(X))_* = f_j(X_n)^* - f_j(X_n)_* \xrightarrow{\text{as}*} 0.$$

Combination yields that $|f_j(X_n) - f_j(X)| \xrightarrow{\text{as}*} 0$. By Egorov's theorem, there is a measurable set A_j with $P(\Omega - A_j) < \varepsilon/2^j$ and $f_j(X_n) - f_j(X) \rightarrow 0$ uniformly on A_j . The set $A = \bigcap_{j=1}^{\infty} A_j$ has $P(\Omega - A) < \varepsilon$ and $e(X_n, X) \rightarrow 0$ uniformly on A .

Take a compact K with $P(X \in K) \geq 1 - \varepsilon$. Then $B = A \cap \{X \in K\}$ has $P(\Omega - B) < 2\varepsilon$ and $d(X_n, X) \rightarrow 0$ uniformly on B . Thus $X_n \xrightarrow{\text{as}*} X$. ■

Problems and Complements

1. Let X be Borel measurable and separable. Then the following statements are equivalent:

- (i) $X_n \xrightarrow{\text{as}} X$ and $d(X_n, X)$ is asymptotically measurable;
- (ii) $X_n \xrightarrow{\text{as}} X$ and X_n is asymptotically measurable;
- (iii) $X_n \xrightarrow{\text{as}} X$ and $d(X_n, s)$ is asymptotically measurable for all s in a set S with $P(X \in \overline{S}) = 1$.

Furthermore, the statements imply

- (iv) $X_n \xrightarrow{P^*} X$.

It is possible to give this a fancier formulation by introducing the notion of *convergence in probability* (different from convergence in outer probability). Say that $X_n \xrightarrow{P} X$ if every subsequence $X_{n'}$ has a further subsequence with $X_{n''} \xrightarrow{\text{as}} X$. If, in (i)–(iii), $X_n \xrightarrow{\text{as}} X$ is replaced by $X_n \xrightarrow{P} X$, then (i)–(iv) are equivalent.

1.10

Weak, Almost Uniform and in Probability Convergence

Consider the relationships between the convergence concepts introduced in the previous section and weak convergence. First we shall be a bit formal and note that convergence in probability to a constant can be defined for maps with *different* domains $(\Omega_\alpha, \mathcal{A}_\alpha, P_\alpha)$ too, so that it is not covered by Definition 1.9.1 in the preceding section.

1.10.1 Definition. Let $X_\alpha: \Omega_\alpha \rightarrow \mathbb{D}$ be an arbitrary net of maps and $c \in \mathbb{D}$. Then X_α *converges in outer probability* to c if $P^*(d(X_\alpha, c) > \varepsilon) \rightarrow 0$, for every $\varepsilon > 0$. This is denoted $X_\alpha \xrightarrow{P^*} c$.

In general, convergence in outer probability is stronger than weak convergence, though they are equivalent if the limit is constant.

1.10.2 Lemma. Let X_α, Y_α be arbitrary maps and X Borel measurable.

- (i) If $X_\alpha \rightsquigarrow X$ and $d(X_\alpha, Y_\alpha) \xrightarrow{P^*} 0$, then $Y_\alpha \rightsquigarrow X$.
- (ii) If $X_\alpha \xrightarrow{P^*} X$, then $X_\alpha \rightsquigarrow X$.
- (iii) $X_\alpha \xrightarrow{P^*} c$ if and only if $X_\alpha \rightsquigarrow c$.

Proof. (i). Let F be closed. Then, for every fixed $\varepsilon > 0$, it holds that $\limsup P^*(Y_\alpha \in F) = \limsup P^*(Y_\alpha \in F \wedge d(X_\alpha, Y_\alpha)^* \leq \varepsilon) \leq \limsup P^*(X_\alpha \in \overline{F^\varepsilon}) \leq P(X \in \overline{F^\varepsilon})$. Letting $\varepsilon \downarrow 0$ completes the proof.

(ii). Clearly, $X \rightsquigarrow X$ and $d(X, X_\alpha) \xrightarrow{P^*} 0$. Apply (i).

(iii). One direction follows from (ii). For the other, note that $P^*(d(X_\alpha, c) \geq \varepsilon) = P^*(X_\alpha \notin B(c, \varepsilon))$, where $B(c, \varepsilon)$ is the ball of radius ε around c . By the portmanteau theorem, the limsup of this is smaller than or equal to $P(c \notin B(c, \varepsilon)) = 0$. ■

The second assertion of the previous lemma can certainly not be inverted: weakly convergent maps need not even be defined on the same probability space. However, according to the almost sure representation theorem, for every weakly convergent net there is an almost surely convergent net (defined on some probability space) that is the “same” as far as laws are concerned. The nonmeasurable version of this result sounds somewhat more complicated than the measurable version; so it is worth considering the case of measurable maps first. For every α , let $\mathcal{D}_\alpha$ be a σ -field on $\mathbb{D}$, not larger than the Borel σ -field. For convenience of notation, the limit variable will be written as X_∞ rather than X and a statement that is valid “for every α ” will be understood to apply to $\alpha = \infty$ too.

1.10.3 Theorem (a.s. representations). *Let $X_\alpha: \Omega_\alpha \rightarrow \mathbb{D}$ be $\mathcal{D}_\alpha$ -measurable maps. If $X_\alpha \rightsquigarrow X_\infty$ and X_∞ is separable, then there exist $\mathcal{D}_\alpha$ -measurable maps $\tilde{X}_\alpha: \tilde{\Omega} \rightarrow \mathbb{D}$ defined on some probability space $(\tilde{\Omega}, \tilde{\mathcal{A}}, \tilde{P})$ with*

(i) $\tilde{X}_\alpha \xrightarrow{\text{a.s.}} \tilde{X}_\infty$;

(ii) $\tilde{X}_\alpha$ and X_α are equal in law on $\mathcal{D}_\alpha$ for every α .

Usually this theorem will be applied with every $\mathcal{D}_\alpha$ equal to the Borel or ball σ -field. In any case, the smaller the $\mathcal{D}_\alpha$ are, the weaker the result. In the extreme case that the $\mathcal{D}_\alpha$ are the trivial σ -fields, the theorem is still true, but it yields “representations” $\tilde{X}_\alpha$ with no relationship to the original X_α whatsoever. Thus it is worthwhile to pursue a stronger formulation for nonmeasurable maps. The problem is to generalize the statement that every pair $\tilde{X}_\alpha$ and X_α are equal in law. In the initial formulation, equality in law will be interpreted in the sense that

$$E^* f(\tilde{X}_\alpha) = E^* f(X_\alpha), \quad \text{for every bounded } f: \mathbb{D} \rightarrow \mathbb{R}.$$

In particular, $P^*(\tilde{X}_\alpha \in B) = P^*(X_\alpha \in B)$ for every set B , and the same for inner probabilities. If $(\tilde{\Omega}, \tilde{\mathcal{A}}, \tilde{P})$ is complete (which may be assumed without loss of generality), this implies that the laws of $\tilde{X}_\alpha$ and X_α are the same on the maximal σ -field $\{B \subset \mathbb{D}: X_\alpha^{-1}(B) \in \mathcal{A}_\alpha\}$ for which they are defined, that is, for which X_α is measurable (Problem 1.2.10). However, in general, equality of the outer expectations says a lot more than just equality in law.

The following nonmeasurable representation theorem holds for sequences but not nets in general. Call a directed set *nontrivial* if it permits a net of strictly positive numbers δ_α with $\delta_\alpha \rightarrow 0$. Of course, the set of natural numbers with the usual ordering is nontrivial.[‡]

1.10.4 Theorem (a.s. representations). *Let $X_\alpha: \Omega_\alpha \rightarrow \mathbb{D}$ be an arbitrary net indexed by a nontrivial directed set, and let X_∞ be Borel measurable and separable. If $X_\alpha \rightsquigarrow X_\infty$, then there exists a probability space $(\tilde{\Omega}, \tilde{\mathcal{A}}, \tilde{P})$ and maps $\tilde{X}_\alpha: \tilde{\Omega} \rightarrow \mathbb{D}$ with*

[‡] There do exist directed sets that are trivial. For each of them there exists a net X_α for which the theorem fails (Problem 1.10.7). On the other hand, for such a directed set, convergence $X_\alpha \rightsquigarrow X_\infty$ to a separable limit X_∞ implies that X_α is eventually Borel measurable and equal in law to X_∞ (Problem 1.10.6). So only rather trivial cases have to be excluded from the present formulation of the almost sure representation theorem. The reason that these have to be excluded is that (ii) says things about the relation between $\tilde{X}_\alpha$ and X_α that go beyond their Borel laws.

- (i) $\tilde{X}_\alpha \xrightarrow{\text{au}} \tilde{X}_\infty$;
 (ii) $E^*f(\tilde{X}_\alpha) = E^*f(X_\alpha)$, for every bounded $f: \mathbb{D} \rightarrow \mathbb{R}$ and every α .

1.10.5 Addendum. In addition to (i) and (ii), $\tilde{X}_\alpha$ can be chosen according to the following diagram:

$$\begin{array}{ccc} \Omega_\alpha & \xrightarrow{X_\alpha} & \mathbb{D} \\ \phi_\alpha \uparrow & \nearrow & \\ \tilde{\Omega} & & \tilde{X}_\alpha = X_\alpha \circ \phi_\alpha \end{array}$$

with the maps ϕ_α measurable and perfect, and $P_\alpha = \tilde{P} \circ \phi_\alpha^{-1}$.

The perfectness of ϕ_α asserted by the addendum improves part (ii) of the theorem. However, the main interest of the addendum is the implication that every $\tilde{X}_\alpha$ can be forced to take no other values than the original X_α . Thus $\tilde{X}_\alpha$ and X_α also share other properties, which may not be directly expressible in terms of their “laws.”

1.10.6 Example (Empirical process). Consider the empirical process $\mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - P)$ indexed by a set of functions $\mathcal{H}$ on a measurable space $(\mathcal{X}, \mathcal{B})$. Under suitable conditions this converges weakly when seen as maps into $\ell^\infty(\mathcal{H})$. Of course, every realization $\mathbb{G}_n(\omega)$ can also be seen as a signed measure on $(\mathcal{X}, \mathcal{B})$. According to the addendum, it is possible to construct the representations $\tilde{\mathbb{G}}_n$ as maps into $\ell^\infty(\mathcal{H})$ in such a way that the double interpretation — as a bounded function and a signed measure — is retained.

Proofs. If the directed set is trivial, then $X_\alpha \rightsquigarrow X_\infty$ for a separable X can happen only if X_α is Borel measurable (for the completion of the underlying probability space) and equal in law to X_∞ eventually (Problem 1.10.6). In that case, the measurable version of the theorem is trivial. For nontrivial directed sets, the nonmeasurable version implies the measurable version. For $\tilde{X}_\alpha$ as in the addendum,

$$E^*f(\tilde{X}_\alpha) = \int^* f \circ X_\alpha \circ \phi_\alpha d\tilde{P} = \int^* f \circ X_\alpha d\tilde{P} \circ \phi_\alpha^{-1} = E^*f(X_\alpha),$$

for every bounded f , where perfectness of ϕ_α is used in the second equality. Thus it is enough to construct $\tilde{X}_\alpha$ as in the addendum that also satisfy (i). The construction consists of five steps. Call a set $B \subset \mathbb{D}$ a *continuity set* if $P(X_\infty \in \partial B) = 0$.

(i). For every $\varepsilon > 0$, there exists a partition of $\mathbb{D}$ into finitely many disjoint continuity sets

$$B_0^{(\varepsilon)}, B_1^{(\varepsilon)}, \dots, B_{k_\varepsilon}^{(\varepsilon)},$$

with the properties $P(X_\infty \in B_0^{(\varepsilon)}) < \varepsilon^2$ and $\text{diam } B_i^{(\varepsilon)} < \varepsilon$ for $i = 1, 2, \dots, k_\varepsilon$.

Indeed, let the sequence $s_1, s_2, \dots$ be dense in a set with probability 1 under X_∞ . Every open ball $B(s_i, r)$ is a discontinuity set for at most countably many values of r . Take $\varepsilon/3 < r_i < \varepsilon/2$ such that $B(s_i, r_i)$ is a continuity set. For $i \geq 1$ set $B_i^{(\varepsilon)} = B(s_i, r_i) - \bigcup_{j < i} B(s_j, r_j)$. Since the boundary of a finite intersection is contained in the union of the boundaries, every $B_i^{(\varepsilon)}$ is a continuity set. Moreover,

the union of all $B_i^{(\varepsilon)}$ has probability 1 under X_∞ . Take k_ε sufficiently large, and set $B_0^{(\varepsilon)} = \mathbb{D} - \cup_{i=1}^{k_\varepsilon} B_i^{(\varepsilon)}$.

(ii). There exists a net $\varepsilon_\alpha \rightarrow 0$ taking values $1/m$ with $m \in \mathbb{N}$ only, and

$$(1.10.7) \quad P_*(X_\alpha \in B_i^{(\varepsilon_\alpha)}) \geq (1 - \varepsilon_\alpha) P_\infty(X_\infty \in B_i^{(\varepsilon_\alpha)}), \quad i = 1, \dots, k_{\varepsilon_\alpha},$$

for all sufficiently large α , say $\alpha \geq \alpha_1$.

Indeed, for every fixed $\varepsilon > 0$ and every i , one has $P_*(X_\alpha \in B_i^{(\varepsilon)}) \rightarrow P_\infty(X_\infty \in B_i^{(\varepsilon)})$, by the portmanteau theorem. Let α_m be such that, for $\alpha \geq \alpha_m$ and $i = 1, \dots, k_{1/m}$,

$$(1.10.8) \quad P_*(X_\alpha \in B_i^{(1/m)}) \geq (1 - \frac{1}{m}) P_\infty(X_\infty \in B_i^{(1/m)}).$$

Assume without loss of generality that $\alpha_1 \leq \alpha_2 \leq \dots$. Set

$$\eta_\alpha = \inf \left\{ \frac{1}{m} : \alpha \geq \alpha_m, m \in \mathbb{N} \right\},$$

where the infimum over the empty set is 2. For $\alpha \geq \alpha_m$, it holds that $\eta_\alpha \leq 1/m$; hence $\eta_\alpha \rightarrow 0$. This net qualifies for ε_α in case it is never zero. In general, ε_α will be defined as η_α kept away from zero. For a fixed $\alpha \geq \alpha_1$, either $\eta_\alpha = 1/m$, in which case $\alpha \geq \alpha_m$, or $\eta_\alpha = 0$, in which case $\alpha \geq \alpha_m$ for every m . In the first case, (1.10.8) holds for $1/m = \eta_\alpha$; in the second, (1.10.8) holds for every $1/m$, $m \in \mathbb{N}$. Let δ_α be an arbitrary net with $\delta_\alpha \rightarrow 0$ and taking values $1/m$ with $m \in \mathbb{N}$ only. Set $\varepsilon_\alpha = \eta_\alpha$ if $\eta_\alpha > 0$ and $\varepsilon_\alpha = \delta_\alpha$ if $\eta_\alpha = 0$.

(iii). To simplify notation, assume that (1.5.5) holds for every α and is positive (throw away the beginning of the net). For $i = 1, \dots, k_{\varepsilon_\alpha}$, let A_i^α be a measurable set contained in $\{X_\alpha \in B_i^{(\varepsilon_\alpha)}\}$ with the same (inner) probability. Set $A_0^\alpha = \Omega_\alpha - \cup_{i=1}^{k_{\varepsilon_\alpha}} A_i^\alpha$. Define

$$\begin{aligned} \tilde{\Omega} &= \Omega_\infty \times \prod_{\alpha} \left[\Omega_\alpha \times \prod_{i=0}^{k_{\varepsilon_\alpha}} A_i^\alpha \right] \times [0, 1], \\ \tilde{\mathcal{A}} &= \mathcal{A}_\infty \times \prod_{\alpha} \left[\mathcal{A}_\alpha \times \prod_{i=0}^{k_{\varepsilon_\alpha}} \mathcal{A}_\alpha \cap A_i^\alpha \right] \times \mathcal{B}_o, \\ \tilde{P} &= P_\infty \times \prod_{\alpha} \left[\mu_\alpha \times \prod_{i=0}^{k_{\varepsilon_\alpha}} P_\alpha(\cdot \mid A_i^\alpha) \right] \times \lambda. \end{aligned}$$

Here $P_\alpha(\cdot \mid A_i^\alpha)$ is the conditional P_α -measure given A_i^α , and $\mathcal{B}_o$ and λ are the Borel sets and Lebesgue measure on $[0, 1]$, respectively. Furthermore, μ_α is the probability measure defined by

$$(1.10.9) \quad \mu_\alpha(A) = \varepsilon_\alpha^{-1} \sum_{i=0}^{k_{\varepsilon_\alpha}} P_\alpha(A \mid A_i^\alpha) \left[P_\alpha(A_i^\alpha) - (1 - \varepsilon_\alpha) P_\infty(X_\infty \in B_i^{(\varepsilon_\alpha)}) \right].$$

Write an element $\tilde{\omega}$ of $\tilde{\Omega}$ as

$$\tilde{\omega} = (\omega_\infty, \dots, \omega_\alpha, \omega_{\alpha 0}, \omega_{\alpha 1}, \dots, \omega_{\alpha k_{\varepsilon_\alpha}}, \dots, \xi).$$

Define

$$\begin{aligned} \phi_\infty(\tilde{\omega}) &= \omega_\infty, \\ \phi_\alpha(\tilde{\omega}) &= \begin{cases} \omega_\alpha, & \text{if } \xi > 1 - \varepsilon_\alpha, \\ \omega_{\alpha i}, & \text{if } \xi \leq 1 - \varepsilon_\alpha \text{ and } X_\infty(\omega_\infty) \in B_i^{(\varepsilon_\alpha)}. \end{cases} \end{aligned}$$

Thus if X_∞ falls in $B_i^{(\varepsilon_\alpha)}$, “choose” ϕ_α with probability $1 - \varepsilon_\alpha$ in A_i^α (so that $X_\alpha \circ \phi_\alpha \in B_i^{(\varepsilon_\alpha)}$, close to X_∞ for $i \geq 1$) according to the conditional distribution $P_\alpha(\cdot | A_i^\alpha)$. Next, the mass ε_α is used to correct this “discretization” so as to ensure that $\tilde{P} \circ \phi_\alpha^{-1} = P_\alpha$. If ϕ_α is chosen with probability ε_α , according to a law μ_α , then

$$\tilde{P}(\phi_\alpha \in A) = (1 - \varepsilon_\alpha) \sum_{i=0}^{k_{\varepsilon_\alpha}} \tilde{P}(\omega_{\alpha, i} \in A \wedge X_\infty(\omega_\infty) \in B_i^{(\varepsilon_\alpha)}) + \varepsilon_\alpha \mu_\alpha(A).$$

Elementary algebra shows that to make this equal to $P_\alpha(A)$, the measure μ_α must be defined by (1.10.9). Because of (1.10.7), this is possible.

(iv). It is clear from the construction that $d(\tilde{X}_\alpha, \tilde{X}_\infty) \leq \varepsilon_\alpha$, for every $\tilde{\omega}$, with $X_\infty(\omega_\infty) \notin B_0^{(\varepsilon_\alpha)}$ and $\xi \leq 1 - \varepsilon_\alpha$. Set $A_k = \cup_{m \geq k} \{\tilde{\omega}: X_\infty(\omega_\infty) \in B_0^{(1/m)} \text{ or } \xi > 1 - 1/k\}$. For every $\varepsilon > 0$, there exists a k such that $\tilde{P}(A_k) \leq \varepsilon$. For $\tilde{\omega} \in \tilde{\Omega} - A_k$ and α such that $\varepsilon_\alpha \leq 1/k$, it holds that $d(\tilde{X}_\alpha, \tilde{X}_\infty) \leq \varepsilon_\alpha$. Thus $\tilde{X}_\alpha$ converges to $\tilde{X}_\infty$ almost uniformly.

(v). Let $T: \Omega_\alpha \rightarrow \mathbb{R}$ be bounded, and let T^* be its minimal measurable cover for P_α . Write

$$T \circ \phi_\alpha = 1_{\pi_\xi \leq 1 - \varepsilon_\alpha} \sum_{i=0}^{k_{\varepsilon_\alpha}} T|_{A_i^\alpha} \circ \pi_{\alpha, i} 1_{X_\infty \circ \pi_\infty \in B_i^{(\varepsilon_\alpha)}} + 1_{\pi_\xi > 1 - \varepsilon_\alpha} T \circ \pi_\alpha,$$

where $\pi_\xi: \tilde{\Omega} \rightarrow [0, 1]$, $\pi_{\alpha, i}: \tilde{\Omega} \rightarrow A_i^\alpha$, and $\pi_\alpha: \tilde{\Omega} \rightarrow \Omega_\alpha$ are the coordinate projections. Then the minimal measurable cover of $T \circ \phi_\alpha$ for $\tilde{P}$ can be computed as

$$\begin{aligned} (T \circ \phi_\alpha)^* &= 1_{\pi_\xi \leq 1 - \varepsilon_\alpha} \sum_{i=0}^{k_{\varepsilon_\alpha}} (T|_{A_i^\alpha} \circ \pi_{\alpha, i})^* 1_{X_\infty \circ \pi_\infty \in B_i^{(\varepsilon_\alpha)}} \\ &\quad + 1_{\pi_\xi > 1 - \varepsilon_\alpha} (T \circ \pi_\alpha)^* \\ &= 1_{\pi_\xi \leq 1 - \varepsilon_\alpha} \sum_{i=0}^{k_{\varepsilon_\alpha}} (T|_{A_i^\alpha})^* P(\cdot | A_i^\alpha) \circ \pi_{\alpha, i} 1_{X_\infty \circ \pi_\infty \in B_i^{(\varepsilon_\alpha)}} \\ &\quad + 1_{\pi_\xi > 1 - \varepsilon_\alpha} T^{* \mu_\alpha} \circ \pi_\alpha, \end{aligned}$$

since coordinate projections are perfect. Because P_α and $P(\cdot | A_i^\alpha)$ or μ_α are equivalent on the spaces where they are defined, the measurable covers in the last formula can just as well be computed under P_α , whence $(T \circ \phi_\alpha)^* = T^* \circ \phi_\alpha$. ■

A typical use of the almost sure representation theorem is to extend convergence theorems for expectations of almost surely convergent nets to weakly convergent nets.

1.10.10 Example. Let $f, g: \mathbb{D} \rightarrow \mathbb{R}$ be continuous and satisfy $|f| \leq g$. If $X_n \rightsquigarrow X$ and $E^*g(X_n) \rightarrow Eg(X) < \infty$, then $E^*f(X_n) \rightarrow Ef(X)$.

To see this, let $\tilde{X}_n \xrightarrow{\text{as}^*} \tilde{X}$ be almost sure representations. By the continuous mapping theorem for outer almost sure convergence, $f(\tilde{X}_n) \xrightarrow{\text{as}^*} f(\tilde{X})$ and $g(\tilde{X}_n) \xrightarrow{\text{as}^*} g(\tilde{X})$. The latter gives $g(\tilde{X}_n)^* \xrightarrow{\text{as}} g(\tilde{X})$. Combined with convergence of expectations, $Eg(\tilde{X}_n)^* \rightarrow Eg(\tilde{X})$, it yields $E|g(\tilde{X}_n)^* - g(\tilde{X})| \rightarrow 0$, by a well known convergence lemma. Thus the sequence $g(\tilde{X}_n)^*$ is uniformly integrable. Since $0 \leq |f(\tilde{X}_n)|^* \leq g(\tilde{X}_n)^*$, the same is true for the sequence $f(\tilde{X}_n)^*$, whence $Ef(\tilde{X}_n)^* \rightarrow Ef(\tilde{X})$.

1.10.11 Example. A particular case of the previous example is the following. Let L_n be a sequence of Borel measures. For a continuous positive function g , consider the measures $M_n(B) = \int_B g dL_n$. If $L_n \rightsquigarrow L_\infty$ and $\int g dL_n \rightarrow \int g dL_\infty$, then $M_n \rightsquigarrow M_\infty$. A closer look reveals that g need be continuous almost everywhere under the limit L_∞ only.

Certain results are easier to prove for measurable maps. The following proposition can often be used to turn a result for measurable maps into a general one.

1.10.12 Proposition. Let $X_\alpha: \Omega_\alpha \rightarrow \mathbb{D}$ be arbitrary. If $X_\alpha \rightsquigarrow X$ and X is separable, then there exist Borel measurable $Y_\alpha: \Omega_\alpha \rightarrow \mathbb{D}$ with $d(X_\alpha, Y_\alpha) \xrightarrow{P^*} 0$.

Proof. As explained in the proof of the almost sure representation theorem, there exist for every $\varepsilon > 0$ finitely many X -continuity sets $B_0^{(\varepsilon)}, B_1^{(\varepsilon)}, \dots, B_{k_\varepsilon}^{(\varepsilon)}$ such that $P(X \in B_0^{(\varepsilon)}) < \varepsilon$ and $\text{diam } B_i^{(\varepsilon)} < \varepsilon$ for $i = 1, 2, \dots, k_\varepsilon$. Choose a point $x_i^{(\varepsilon)}$ from every $B_i^{(\varepsilon)}$, and define

$$Y_\alpha^{(\varepsilon)} = \begin{cases} x_i^{(\varepsilon)}, & \text{on } \{X_\alpha \in B_i^{(\varepsilon)}\}_*, \quad i = 1, \dots, k_\varepsilon, \\ x_0^{(\varepsilon)}, & \text{on } \Omega_\alpha - \bigcup_{i=1}^{k_\varepsilon} \{X_\alpha \in B_i^{(\varepsilon)}\}_*. \end{cases}$$

Then each $Y_\alpha^{(\varepsilon)}$ is measurable, and

$$\begin{aligned} P^*\left(d(Y_\alpha^{(\varepsilon)}, X_\alpha) > \varepsilon\right) &\leq \sum_{i=1}^{k_\varepsilon} P\left(\{X_\alpha \in B_i^{(\varepsilon)}\}^* - \{X_\alpha \in B_i^{(\varepsilon)}\}_*\right) \\ &\quad + P^*(X_\alpha \in B_0^{(\varepsilon)}) \rightarrow P(X \in B_0^{(\varepsilon)}) < \varepsilon. \end{aligned}$$

Assume for the moment that there exists a net $\delta_\alpha \rightarrow 0$ taking values $1/m$, $m \in \mathbb{N}$ only. Let $R(\alpha, \varepsilon)$ be the probability on the left side in the last displayed equation. For $m \in \mathbb{N}$, there exists α_m such that, for all $\alpha \geq \alpha_m$,

$$R\left(\alpha, \frac{1}{m}\right) < \frac{2}{m}.$$

Choose the α_m increasing: $\alpha_1 \leq \alpha_2 \leq \dots$. Set

$$\eta_\alpha = \inf\left\{\frac{1}{m}: \alpha \geq \alpha_m\right\},$$

where the infimum over the empty set is 2. Then $\eta_\alpha \rightarrow 0$. Moreover, for $\alpha \geq \alpha_1$, either $\eta_\alpha = 1/m$ for some m , in which case $\alpha \geq \alpha_m$, or $\eta_\alpha = 0$. In the first case, $R(\alpha, 1/m) < 2/m$; while in the second case, $R(\alpha, 1/m) < 2/m$, for every m . Define ε_α to be η_α if $\eta_\alpha > 0$ and δ_α if $\eta_\alpha = 0$. Then $R(\alpha, \varepsilon_\alpha) < 2\varepsilon_\alpha$, for every α . Consequently, $d(Y_\alpha^{(\varepsilon_\alpha)}, X_\alpha) \xrightarrow{P^*} 0$.

Finally, if there does not exist a net δ_α as assumed, then necessarily X_α is Borel measurable for the completion of $(\Omega_\alpha, \mathcal{A}_\alpha, P_\alpha)$ and is equal in law to X for all sufficiently large α . By a standard argument, there exists a Borel measurable map $Y_\alpha: (\Omega_\alpha, \mathcal{A}_\alpha) \rightarrow \mathbb{D}$ with $P_*(X_\alpha = Y_\alpha) = 1$ for such α (Problem 1.2.10). ■

Problems and Complements

1. Let X_n and X be maps into $\mathbb{R}$ with X Borel measurable.

- (i) $X_n \xrightarrow{\text{as}^*} X$ if and only if $X_n^* \xrightarrow{\text{as}} X$ and $X_{n*} \xrightarrow{\text{as}} X$.
- (ii) $X_n \rightsquigarrow X$ if and only if $X_n^* \rightsquigarrow X$ and $X_{n*} \rightsquigarrow X$.

[Hint: For (i) use $|X_n - X|^* = |X_n^* - X| \vee |X_{n*} - X|$. For (ii) suppose $X_n \rightsquigarrow X$. Let $\tilde{X}_n = X_n \circ \phi_n \xrightarrow{\text{as}^*} \tilde{X}$ be almost sure representations. By (i) $\tilde{X}_n^* \xrightarrow{\text{as}} \tilde{X}$, whence $\tilde{X}_n^* \rightsquigarrow \tilde{X}$. For perfect ϕ_n , one has $\tilde{X}_n^* = X_n^* \circ \phi_n$, and it follows that $\text{Ef}(\tilde{X}_n^*) = \text{Ef}(X_n^*)$, for every measurable f . So $X_n^* \rightsquigarrow X$. By considering the negatives of the functions, obtain $X_{n*} \rightsquigarrow X$. The converse follows from $P(X_n^* \leq x) \leq P^*(X_n \leq x) \leq P(X_{n*} \leq x)$, for every x .]

2. Let $(\Omega_n, \mathcal{A}_n, P_n)$ be the product of n copies of a probability space $(\Omega, \mathcal{A}, P)$. Let $X: \Omega \rightarrow \mathbb{R}$ be a fixed map, let $X_i = X$ for every i , and let $S_n: \Omega_n \rightarrow \mathbb{R}$ be defined by $S_n(\omega_1, \dots, \omega_n) = \sum_{i=1}^n X_i(\omega_i)$. Then the central limit theorem $S_n/\sqrt{n} \rightsquigarrow N(0, 1)$ can hold only if X is measurable for the P -completion of $\mathcal{A}$. Similarly, the law of large numbers $S_n/n \rightsquigarrow 0$ can hold only if X is P -completion measurable.

[Hint: If $S_n/\sqrt{n} \rightsquigarrow N(0, 1)$, then $(S_n/\sqrt{n})^*$ converges weakly to the same limit. Since $S_n^* = \sum_{i=1}^n X_i^*$, this implies that the central limit theorem holds for the measurable maps X_i^* . This implies $\text{EX}_i^* = 0$ (and $\text{E}(X_i^*)^2 = 1$). By an analogous argument, $\text{EX}_{i*} = 0$. Finally, $\text{E}(X_i^* - X_{i*}) = 0$ implies $X_i^* = X_{i*}$ almost surely.]

3. The following statements are *not* true:

- (i) X_α measurable and $X_\alpha \xrightarrow{\text{as}^*} 0$ imply $X_\alpha \xrightarrow{P^*} 0$ (for general nets);
- (ii) $X_\alpha \xrightarrow{\text{as}^*} 0$ implies that X_α is asymptotically measurable (for general nets);
- (iii) $X_n \xrightarrow{\text{as}} 0$ and X_n is asymptotically measurable imply $X_n \xrightarrow{\text{as}^*} X$.

[Hint: Let A be the collection of all finite subsets of $[0, 1]$, directed through inclusion: $\alpha_1 \leq \alpha_2$ if and only if $\alpha_1 \subset \alpha_2$. Let P be Lebesgue measure on the Borel sets of $[0, 1]$. For (i) define $X_\alpha: [0, 1] \rightarrow [0, 1]$ by $X_\alpha = 1_{[0, 1] - \alpha}$. Then X_α is Borel measurable, and $X_\alpha(\omega) \rightarrow 0$ for every ω . However, $P(X_\alpha = 1) = 1$ for every α , so X_α does not converge to zero in probability; in fact, $X_\alpha \xrightarrow{P^*} 1$. Note that the dominated convergence theorem fails too, since $\text{EX}_\alpha = 1$ for every α . For (ii) let B be a subset of $[0, 1]$ with $P_*(B) < P^*(B)$. Define $X_\alpha = 1_{[0, 1] - \alpha} 1_B$. Then $|X_\alpha|^* = 1_{[0, 1] - \alpha} (1_B)^* \xrightarrow{\text{as}^*} 0$, while $\text{E}(X_\alpha)^* - \text{E}(X_\alpha)_* = \text{E}(1_B)^* - \text{E}(1_B)_*$ does not converge to 0. Finally, for (iii), let I_n be the n th interval in the sequence $[0, 1]$, $[0, 1/2]$, $[1/2, 1]$, $[0, 1/4]$, $\dots$. For B_n as in Example 1.9.4, take $X_n = 1_{I_n} 1_{B_n}$. Then $d(X_n, 0)^* = X_n^* = 1_{I_n}$ converges to zero in probability and weakly, but not almost surely.]

4. For α running through a directed set, consider the statements

- (i) $X_\alpha \xrightarrow{\text{au}} X$;
- (ii) $\sup_{\beta \geq \alpha} d(X_\beta, X)^* \xrightarrow{P^*} 0$;
- (iii) $X_\alpha \xrightarrow{\text{as}^*} X$;
- (iv) $X_\alpha \xrightarrow{P^*} X$.

The following implications are always true:

$$\begin{array}{ccc} (i) & \longleftrightarrow & (ii) \\ \downarrow & \searrow & \downarrow \\ (iii) & & (iv) \end{array} .$$

There is a counterexample to every implication that is not indicated in the diagram.

5. There exist directed sets that cannot index a net of strictly positive numbers that converges to zero. Equivalently, there exist directed sets A such that, if $\{\delta_\alpha: \alpha \in A\}$ is a net of real numbers with $\delta_\alpha \rightarrow 0$, then $\delta_\alpha = 0$ eventually.

[Hint: A trivial example is obtained by taking A equal to a partially ordered set with a largest element; for instance, $A = \{1, 2, \dots, n\}$ with usual order, or $A = [0, 1]$ with reversed natural order. For a nontrivial example, take A equal to an uncountable, well-ordered set for which every section $\{\alpha: \alpha < \alpha_0\}$ is countable. Such sets exist provided one assumes the axiom of choice (and the well-ordering theorem), though it is hard to give a description of one. (See Munkres (1975), page 66.) If δ_α is indexed by such a set and converges to zero without being zero eventually, then $\varepsilon_\alpha = \sup_{\beta \geq \alpha} |\delta_\beta|$ is strictly positive and converges monotonically to zero. By the well-ordering, each set $S_i = \{\alpha: \varepsilon_\alpha \leq 1/i\}$ has a smallest element α_i . Since the net ε_α is decreasing, it follows that $S_i = \{\alpha: \alpha \geq \alpha_i\}$. Since each ε_α is positive, there is for each α an i with $\varepsilon_\alpha > 1/i$, that is, $\alpha < \alpha_i$. Then A is covered by the sets $\{\alpha < \alpha_i\}$, but these are only countably many sets, each with countably many elements.]

6. Let X_α be maps indexed by a directed set that is trivial defined on complete probability spaces.

- (i) If $X_\alpha \rightsquigarrow X$ and X is separable, then X_α is Borel measurable and equal in law to X eventually.
- (ii) If $X_\alpha \xrightarrow{\text{au}} X$, then $P_*(X_\alpha = X) = 1$ eventually.

[Hint: For (i) use metrization of weak convergence to reduce $X_\alpha \rightsquigarrow X$ to convergence of a set of numbers. For (ii) set $\delta_\alpha = P^*(d(X_\alpha, X) > 0)$. Then $\delta_\alpha \rightarrow 0$. Indeed, for every $\varepsilon > 0$ there is a set B with $\sup_{\omega \in B} d(X_\alpha, X) \rightarrow 0$ and $P(B) > 1 - \varepsilon$. Since the directed set is trivial, the supremum is zero for all sufficiently large α , whence $\delta_\alpha \leq \varepsilon$ eventually. Finally, $\delta_\alpha \rightarrow 0$ and trivialness imply $\delta_\alpha = 0$ eventually.]

7. Any directed set that is trivial indexes a weakly convergent net $X_\alpha \rightsquigarrow X_\infty$ with values in $\mathbb{R}$ for which there is no almost uniform representation: there is no net $\tilde{X}_\alpha$ and a $\tilde{X}_\infty$ with $\tilde{X}_\alpha \xrightarrow{\text{au}} \tilde{X}_\infty$ and $E^*f(X_\alpha) = E^*f(\tilde{X}_\alpha)$ for every α and bounded f .

[Hint: Take a Borel measure on $\mathbb{R}$ that can be extended in two different ways to a σ -field $\mathbb{D}$ that is strictly larger than the Borel sets. (Every L for which there is a (nonmeasurable) set B with $L^*(B) > L_*(B)$ does the job.) Let L_1 and L_2 be the extensions. Take $(\Omega_\infty, \mathcal{A}_\infty, P_\infty) = (\mathbb{R}, \mathbb{D}, L_1)$ and $(\Omega, \mathcal{A}_\alpha, P_\alpha) = (\mathbb{R}, \mathbb{D}, L_2)$ for every other α ; let both X_α and X_∞ be the identity map. The almost uniform representations would be $\mathbb{D}$ -measurable maps into $\mathbb{R}$; $\tilde{X}_\infty$ would have law L_1 , while the other $\tilde{X}_\alpha$ would have law L_2 . But according to Problem 1.10.6, also $\tilde{X}_\infty = \tilde{X}_\alpha$ almost surely.]

8. Let X_n and X be Borel measurable elements in a metric space. For a fixed Borel set, define Y_n to be X_n if $X \in B$ and to be X if $X \notin B$.
- (i) If $X_n \xrightarrow{P} X$, then $Y_n \xrightarrow{P} X$.
 - (ii) If $X_n \rightsquigarrow X$, then not necessarily $Y_n \rightsquigarrow X$.

[Hint: For (ii) take X distributed as $N(0, 1)$, $X_n = -X$, and $B = [0, \infty)$. Then Y_n is nonnegative with probability 1.]

1.11

Refinements

The continuous mapping theorems for the three modes of stochastic convergence considered so far can be refined to cover maps $g_n(X_n)$, rather than $g(X_n)$, for a fixed g . Then the g_n should have a property that might be called *asymptotic equicontinuity* almost everywhere under the limit measure.

For simplicity, it will be assumed that the limit measure is separable, though this is not necessary for (iii) and can be replaced by other conditions for (i) and (ii) (Problem 1.11.1).

1.11.1 Theorem (Extended continuous mapping). *Let $\mathbb{D}_n \subset \mathbb{D}$ and $g_n: \mathbb{D}_n \rightarrow \mathbb{E}$ satisfy the following statements: if $x_n \rightarrow x$ with $x_n \in \mathbb{D}_n$ for every n and $x \in \mathbb{D}_0$, then $g_n(x_n) \rightarrow g(x)$, where $\mathbb{D}_0 \subset \mathbb{D}$ and $g: \mathbb{D}_0 \rightarrow \mathbb{E}$. Let X_n be maps with values in $\mathbb{D}_n$, let X be Borel measurable and separable, and take values in $\mathbb{D}_0$. Then*

- (i) $X_n \rightsquigarrow X$ implies that $g_n(X_n) \rightsquigarrow g(X)$;
- (ii) $X_n \xrightarrow{P_*} X$ implies that $g_n(X_n) \xrightarrow{P_*} g(X)$;
- (iii) $X_n \xrightarrow{\text{as}^*} X$ implies that $g_n(X_n) \xrightarrow{\text{as}^*} g(X)$.

Proof. Assume the weakest of the three assumptions: the one in (i) that $X_n \rightsquigarrow X$. Let $\mathbb{D}_\infty$ be the set of all x for which there exists a sequence x_n with $x_n \in \mathbb{D}_n$ and $x_n \rightarrow x$. First, $P_*(X \in \mathbb{D}_\infty) = 1$; second, the restriction of g to $\mathbb{D}_0 \cap \mathbb{D}_\infty$ is continuous; and third, if some subsequence satisfies $x_{n'} \rightarrow x$ with $x_{n'} \in \mathbb{D}_{n'}$ for every n' and $x \in \mathbb{D}_0 \cap \mathbb{D}_\infty$, then $g_{n'}(x_{n'}) \rightarrow g(x)$.

To see the first, invoke the almost sure representation theorem. If $\tilde{X}_n \xrightarrow{\text{as}^*} \tilde{X}$ are representing versions, then the range of $\tilde{X}$ is contained in $\mathbb{D}_\infty$ up to a null set, so $P_*(X \in \mathbb{D}_\infty) = P_*(\tilde{X} \in \mathbb{D}_\infty) = 1$. For the third, let the subsequence $x_{n'}$ be given. Since its limit x is in $\mathbb{D}_\infty$, there is a sequence $y_n \rightarrow x$ with $y_n \in \mathbb{D}_n$ for every n . Fill out the subsequence $x_{n'}$ to a whole sequence by putting $x_n = y_n$ if $n \notin \{n'\}$.

Then, by assumption, $g_n(x_n) \rightarrow g(x)$ along the whole sequence, so also along the subsequence. To prove the second, let $x_m \rightarrow x$ in $\mathbb{D}_0 \cap \mathbb{D}_\infty$. For every m , there is a sequence $y_{m,n} \in \mathbb{D}_n$ with $y_{m,n} \rightarrow x_m$ as $n \rightarrow \infty$. Since $x_m \in \mathbb{D}_0$, then also $g_n(y_{m,n}) \rightarrow g(x_m)$. For every m , take n_m such that both $|y_{m,n_m} - x_m| < 1/m$ and $|g_{n_m}(y_{m,n_m}) - g(x_m)| < 1/m$ and such that n_m is increasing with m . Then $y_{m,n_m} \rightarrow x$, so $g_{n_m}(y_{m,n_m}) \rightarrow g(x)$. This implies $g(x_m) \rightarrow g(x)$.

For simplicity of notation, now write $\mathbb{D}_0$ for $\mathbb{D}_0 \cap \mathbb{D}_\infty$. The limit variable X may, without loss of generality, be assumed to take its values in $\mathbb{D}_0$. By the continuity property of g , the map $g(X)$ is then Borel measurable.

(i). Let F be closed. Then

$$\bigcap_{n=1}^{\infty} \overline{\bigcup_{m=n}^{\infty} g_m^{-1}(F)} \subset g^{-1}(F) \cup (\mathbb{D} - \mathbb{D}_0).$$

Indeed, suppose x is in the left side. Then for every n there is $m_n \geq n$ and $x_{m_n} \in g_{m_n}^{-1}(F)$, with $d(x_{m_n}, x) < 1/n$. Thus there exists $x_{m'_n} \in \mathbb{D}_{m'_n}$, with $m'_n \rightarrow \infty$ and $x_{m'_n} \rightarrow x$. Then either $g_{m'_n}(x_{m'_n}) \rightarrow g(x)$ or $x \notin \mathbb{D}_0$. Because F is closed, this implies $g(x) \in F$ or $x \notin \mathbb{D}_0$.

Now, for every fixed k , by the portmanteau theorem,

$$\begin{aligned} \limsup P^*(g_n(X_n) \in F) &\leq \limsup P^*\left(X_n \in \overline{\bigcup_{m=k}^{\infty} g_m^{-1}(F)}\right) \\ &\leq P\left(X \in \overline{\bigcup_{m=k}^{\infty} g_m^{-1}(F)}\right). \end{aligned}$$

As $k \rightarrow \infty$, the last probability converges to $P(X \in \bigcap_{k=1}^{\infty} \overline{\bigcup_{m=k}^{\infty} g_m^{-1}(F)})$, which is smaller than or equal to $P(g(X) \in F)$.

(ii). Fix $\varepsilon > 0$. Choose $\delta_n \downarrow 0$ with $P^*(d(X_n, X) \geq \delta_n) \rightarrow 0$. Let B_n be the set of all x for which there is $y \in \mathbb{D}_n$, with $d(y, x) < \delta_n$ and $e(g_n(y), g(x)) > \varepsilon$. Suppose $x \in B_n$ infinitely often. Then there is a sequence $x_{n_m} \in \mathbb{D}_{n_m}$, with $x_{n_m} \rightarrow x$ and $e(g_{n_m}(x_{n_m}), g(x)) > \varepsilon$, for every m . So $x \notin \mathbb{D}_0$. Conclude that $\limsup B_n \cap \mathbb{D}_0 = \emptyset$. From the continuity of g , it is not hard to see that $B_n \cap \mathbb{D}_0$ is relatively open in $\mathbb{D}_0$ and hence relatively Borel. Consequently, $P^*(X \in B_n) \rightarrow 0$ (Problem 1.2.18). Now

$$P^*(e(g_n(X_n), g(X)) > \varepsilon) \leq P^*(X \in B_n \text{ or } d(X_n, X) \geq \delta_n) \rightarrow 0.$$

(iii). It suffices to prove that $\sup_{m \geq n} e(g_m(X_m), g(X)) \xrightarrow{P^*} 0$. Choose $\delta_n \downarrow 0$ such that $P^*(\sup_{m \geq n} d(X_m, X) \geq \delta_n) \rightarrow 0$. Define B_n as the set of all x such that there exists $m \geq n$ and $y \in \mathbb{D}_m$ with $d(y, x) < \delta_n$ and $e(g_m(y), g(x)) > \varepsilon$. Finish the proof along the lines of the proof of (ii). ■

1.11.2 Counterexample. One might think that in part (iii) of the previous theorem it is enough that $g_n(X_n(\omega)) \rightarrow g(X(\omega))$ for every $\omega \in \Omega$. This is not so. Take $\Omega_n = [0, 1]$ with Borel sets and Lebesgue measure. Furthermore, set $X_n(\omega) = X(\omega) = \omega$, and let $g_n(\omega) = 1_{B_n}(\omega)$ as in Example 1.9.4. Then $g_n(X_n(\omega)) \rightarrow 0$ for every ω , but $|g_n(X_n)|^* = 1$ for every n , so that $g_n(X_n)$ definitely does not converge to zero almost uniformly.

It is not difficult to find examples of weakly convergent nets $X_\alpha \rightsquigarrow X$ and unbounded continuous functions f such that $E^*f(X_\alpha)$ does not converge to $Ef(X)$. However, in many situations, such convergence for unbounded functions actually does hold true. Call a net of real-valued maps X_α *asymptotically uniformly integrable* if

$$\lim_{M \rightarrow \infty} \limsup E^*|X_\alpha| \{ |X_\alpha| > M \} = 0.$$

1.11.3 Theorem (Convergence of means). *Let $f: \mathbb{D} \rightarrow \mathbb{R}$ be continuous at every point in a set $\mathbb{D}_0$. Let $X_\alpha \rightsquigarrow X$, where X takes its values in $\mathbb{D}_0$.*

- (i) *If $f(X_\alpha)$ is asymptotically uniformly integrable, then $E^*f(X_\alpha) \rightarrow Ef(X)$.*
- (ii) *If $\limsup E^*|f(X_\alpha)| \leq E|f(X)| < \infty$, then $E^*f(X_\alpha) \rightarrow Ef(X)$.*

Proof. For $M > 0$, write f^M for f truncated to $[-M, M]$, that is, $f^M(x) = (f(x) \vee (-M)) \wedge M$. By the continuous mapping theorem, $f^M(X_\alpha) \rightsquigarrow f^M(X)$ as maps into the interval $[-M, M]$, for every fixed M . Since the identity function is bounded and continuous on this interval, one has $E^*f^M(X_\alpha) \rightarrow Ef^M(X)$; similarly for $|f^M|$.

First consider (ii). If y^M is an arbitrary real number y truncated to $[-M, M]$, then $|y - y^M| = |y| - |y^M|$. Consequently, for every $M > 0$,

$$\begin{aligned} \limsup |E^*f(X_\alpha) - E^*f^M(X_\alpha)| &\leq \limsup E^* (|f(X_\alpha)| - |f^M(X_\alpha)|) \\ &\leq E|f(X)| - E|f^M(X)|. \end{aligned}$$

Fix M such that the right side is smaller than $\varepsilon > 0$ and also $|Ef^M(X) - Ef(X)| < \varepsilon$. Then the net $E^*f(X_\alpha)$ is eventually within distance ε of $Ef^M(X)$, and so within distance 2ε of $Ef(X)$.

For (i) note first that $|E^*|f|^M(X_\alpha) - E^*|f|(X_\alpha)|$ is bounded above by $2E^*|f(X_\alpha)| \{ |f(X_\alpha)| > M \}$. Fix some $\varepsilon > 0$. For any constant M such that $\limsup E^*|f(X_\alpha)| \{ |f(X_\alpha)| > M \} < \varepsilon$, the net $E^*|f|(X_\alpha)$ is eventually within distance 2ε of $E|f|^M(X)$. This happens for all sufficiently large M . By monotone convergence, $E|f|^M(X) \rightarrow E|f|(X)$ as $M \rightarrow \infty$. So $E^*|f|(X_\alpha) \rightarrow E|f|(X)$. The asymptotic uniform integrability implies that the limit is finite. Finally, apply (ii) to obtain (i). ■

1.11.4 Example (Moments). If $X_\alpha \rightsquigarrow X$ in $\mathbb{R}$ and $\limsup E^*|X_\alpha|^p < \infty$, then $E^*X_\alpha^q \rightarrow EX^q$ for every $q < p$ for which the statement makes sense. It suffices to note that X_α^q is asymptotically uniformly integrable, since $\limsup E^*|X_\alpha|^q \{ |X_\alpha| > M \} \leq M^{q-p} \limsup E^*|X_\alpha|^p < \infty$.

1.11.5 Example. If $X_\alpha \rightsquigarrow X$ in $\mathbb{R}^k$ and every X_α has a multivariate Gaussian distribution, then $E\|X_\alpha\|^p \rightarrow E\|X\|^p$ for every $p > 0$ and every norm on $\mathbb{R}^k$. The reason is that a Gaussian net can converge weakly only if the nets of means and covariances converge. Thus $\limsup E^*\|X_\alpha\|^p < \infty$, for every $p > 0$.

1.11.6 Example (Dominated weak convergence). If $|X_\alpha| \leq Z_\alpha$ for every α and Z_α is asymptotically uniformly integrable, then X_α is asymptotically uniformly integrable. In particular, the following dominated convergence theorem holds: if $X_\alpha \rightsquigarrow X$ in $\mathbb{R}$ and $|X_\alpha| \leq Z$ for every α and some Z with $E^*Z < \infty$, then $E^*X_\alpha \rightarrow EX$.

Problems and Complements

1. Separability of X in the refined continuous mapping theorem 1.11.1 can be omitted from the conditions for (iii). If every $\mathbb{D}_n$ contains $\mathbb{D}_0$, it is not necessary for (i) and (ii) either. Furthermore, suppose the condition on $g_n: \mathbb{D}_n \rightarrow \mathbb{E}$ is strengthened as follows: for every subsequence n' , if $x_{n'} \rightarrow x$ with $x_{n'} \in \mathbb{D}_{n'}$ and $x \in \mathbb{D}_0$, then $g_{n'}(x_{n'}) \rightarrow g(x)$. Then separability can be dropped for (ii) and replaced by the condition that $g(X)$ is Borel measurable for (i).

[Hint: The proof as given uses separability only to ensure that the set $\mathbb{D}_\infty$ has inner measure 1 under the limit. For (iii) this is always true. If $\mathbb{D}_n$ contains $\mathbb{D}_0$ for every n , then $\mathbb{D}_\infty$ trivially contains $\mathbb{D}_0$ and so always has inner measure 1. For (ii) the set $\mathbb{D}_\infty$ can be replaced by the set of all x for which there is a subsequence $x_{n'} \in \mathbb{D}_{n'}$ with $x_{n'} \rightarrow x$. Then the restriction of g to $\mathbb{D}_\infty \cap \mathbb{D}_0$ is still continuous. Furthermore, since there is a subsequence such that $X_{n'} \xrightarrow{\text{as}^*} X$, the inner measure of $\mathbb{D}_\infty$ under X is 1 and the proof applies as it stands. Finally, the proof of (i) does not use separability, but only the conditions as given.]

1.12

Uniformity and Metrization

In principle, weak convergence is the pointwise convergence of “operators” X_α or L_α on the space $C_b(\mathbb{D})$. However, there is automatically uniform convergence over certain subsets. These subsets can be fairly big: equicontinuity and boundedness suffice. On the other hand, there also exist small (countable) subsets such that pointwise convergence on such a subset is automatically uniform, and equivalent to pointwise convergence on the whole of $C_b(\mathbb{D})$, i.e. weak convergence. For separable $\mathbb{D}$, it is even possible to pick such a countable subset that works for every X_α at the same time.

1.12.1 Theorem (Uniformity). *Let $\mathcal{F} \subset C_b(\mathbb{D})$ be bounded and equicontinuous at every point x in a set $\mathbb{D}_0$. If $X_\alpha \rightsquigarrow X$ where the limit X is separable and takes its values in $\mathbb{D}_0$, then $E^*f(X_\alpha) \rightarrow Ef(X)$ uniformly in $f \in \mathcal{F}$.*

Proof. Add all Lipschitz functions with $|f(x) - f(y)| \leq d(x, y)$ that are bounded by 1 to $\mathcal{F}$. Then $\mathcal{F}$ is still bounded and equicontinuous on $\mathbb{D}_0$, and a new metric can be defined on $\mathbb{D}$ through

$$e(x, y) = \sup_{f \in \mathcal{F}} |f(x) - f(y)|.$$

With respect to this metric, the class $\mathcal{F}$ is uniformly equicontinuous on the whole of $\mathbb{D}$. Also, as a consequence of the original equicontinuity, if $x_n \rightarrow x \in \mathbb{D}_0$ with respect to d , then x_n converges with respect to e to the same limit. Thus global continuity of $f: \mathbb{D} \rightarrow \mathbb{R}$ with respect to e implies continuity of f with respect to d at every $x \in \mathbb{D}_0$. If f is also bounded, then $E^*f(X_\alpha) \rightarrow Ef(X)$ by the continuous mapping theorem. Thus $X_\alpha \rightsquigarrow X$ also for e . Conclude that it is no loss of generality to assume that the original set $\mathcal{F}$ is uniformly equicontinuous on the whole of $\mathbb{D}$.

Viewed as maps into the completion $\bar{\mathbb{D}}$ of $\mathbb{D}$, the X_α satisfy $X_\alpha \rightsquigarrow X$ and form an asymptotically tight net. Fix $\varepsilon > 0$. Let $K \subset \bar{\mathbb{D}}$ be a compact set with $\liminf P_*(X_\alpha \in K^\delta) \geq 1 - \varepsilon$ for every $\delta > 0$. Since it is uniformly continuous, every element of $\mathcal{F}$ can be extended to an element of $C_b(\bar{\mathbb{D}})$. The set of restrictions $\mathcal{F}_K$ to K is totally bounded in $C_b(K)$ by the Arzelà-Ascoli theorem. Take finitely many balls of radius $\varepsilon > 0$ in $C_b(K)$ that cover and intersect $\mathcal{F}_K$. Let $f_1, \dots, f_m$ be the centers of the balls. By Tietze's theorem^b, each f_i can be extended to an element of $C_b(\bar{\mathbb{D}})$ of the same norm. Abusing notation, call the extensions $f_1, \dots, f_m$ too. For sufficiently large α , it is certainly true that $|E^*f_i(X_\alpha) - Ef_i(X)| < \varepsilon$ for every i . For any $f \in \mathcal{F}$, there is an f_i and $\delta > 0$ with $|f(x) - f_i(x)| < \varepsilon$ for all $x \in K^\delta$. Then for sufficiently large α , one has $|E^*f(X_\alpha) - Ef(X)| < 3\varepsilon + 4\varepsilon M$ if $M - \varepsilon$ is bigger than the uniform bound on $\mathcal{F}$. ■

1.12.2 Theorem. *For every separable subset $\mathbb{D}_0 \subset \mathbb{D}$, there is a countable, uniformly equicontinuous, and bounded collection $\mathcal{F} \subset C_b(\mathbb{D})$ such that the following statements are equivalent for every Borel measurable X with $P(X \in \mathbb{D}_0) = 1$:*

- (i) $X_\alpha \rightsquigarrow X$;
- (ii) $E^*f(X_\alpha) \rightarrow Ef(X)$ for every $f \in \mathcal{F}$;
- (iii) $E^*f(X_\alpha) \rightarrow Ef(X)$ uniformly in $f \in \mathcal{F}$.

1.12.3 Addendum. *The collection $\mathcal{F}$ can be taken equal to the set of functions of the form*

$$f(x) = q (1 - p_1 d(x, s_1))^+ \vee \dots \vee (1 - p_k d(x, s_k))^+,$$

where $k \in \mathbb{N}$; $s_1, \dots, s_k$ range over a countable, dense subset of $\mathbb{D}_0$; $q \in \mathbb{Q}$; and each $p_i \in \mathbb{Q}^+$, with $|q| \leq 1$ and $|q|(p_1 \vee \dots \vee p_k) \leq 1$.

Proof. The collection $\mathcal{F}$ as suggested in the addendum is uniformly bounded, and every $f \in \mathcal{F}$ is Lipschitz continuous with Lipschitz constant smaller than 1. By Theorem 1.12.1, statement (i) implies (iii), which trivially implies (ii). Assume (ii), and let G be open. Let S be a countable, dense subset of $\mathbb{D}_0$. Then

$$1_G(y) = \sup \left\{ f(y) : 0 \leq f \leq 1_G, f(x) = (1 - p d(x, s))^+, p \in \mathbb{Q}^+, s \in S \right\},$$

for every $y \in \mathbb{D}_0$. Order the countably many functions in this supremum in a sequence, and let f_m be the maximum of the first m functions. Then $0 \leq f_m \uparrow 1_G$ on $\mathbb{D}_0$. Now, for fixed m , it holds that $\liminf P_*(X_\alpha \in G) \geq \liminf E_* f_m(X_\alpha) = Ef_m(X)$, because every f_m is a multiple of a function in $\mathcal{F}$. Letting $m \rightarrow \infty$ completes the proof. ■

A consequence of the previous results is that weak convergence to separable limits is “metrizable.” Let BL_1 be the set of all real functions on $\mathbb{D}$ with a Lipschitz

^b Jameson (1974), Theorem 12.4.

norm bounded by 1: all f with $\|f\|_\infty \leq 1$ and $|f(x) - f(y)| \leq d(x, y)$, for every x, y . Then $X_\alpha \rightsquigarrow X$, where X is Borel measurable and separable, if and only if

$$\sup_{f \in BL_1} |E^* f(X_\alpha) - E f(X)| \rightarrow 0.$$

In particular, weak convergence of separable Borel measures on a metric space is metrizable; for instance, by the metric

$$d_{BL}(L_1, L_2) = \sup_{f \in BL_1} \left| \int f dL_1 - \int f dL_2 \right|.$$

This is called the *bounded Lipschitz metric*.[#]

1.12.4 Theorem (Metrizability). *Weak convergence of separable Borel probability measures on a metric space $\mathbb{D}$ corresponds to a topology that is metrizable by the bounded Lipschitz metric. The set of all separable Borel probability measures is complete under this metric if and only if $\mathbb{D}$ is complete. This set is separable for the weak topology if and only if $\mathbb{D}$ is separable.*

Proof. That the bounded Lipschitz metric generates the topology of weak convergence on the set of separable Borel measures follows from the preceding theorem, as is argued in the preceding. We only need to consider separability and completeness.

It can be shown that $d_{BL}(\delta_x, \delta_y) = d(x, y) \wedge 2$ (see Problem 1.12.5). This shows that $\mathbb{D}$ equipped with d is homeomorphic to the set $\{\delta_x : x \in \mathbb{D}\}$, viewed as subset of the set $\mathcal{M}_1^+$ of separable Borel probability measures on $\mathbb{D}$, equipped with the weak topology (or bounded Lipschitz metric). Separability of the latter set implies separability of the subset of Dirac measures and hence of $\mathbb{D}$.

For the converse, it suffices to show that the set of all empirical measures $m^{-1} \sum_{i=1}^m \delta_{x_i}$, for $m \in \mathbb{N}$, on finite sets of points x_i in a given countable dense subset of $\mathbb{D}$ is dense in $\mathcal{M}_1^+$. Because $\delta_{x_n} \rightsquigarrow \delta_x$ if $d(x_n, x) \rightarrow 0$, this set of measures has the same weak closure as the set of empirical measures on arbitrary points in $\mathbb{D}$. For a given Borel probability measure P and $X_1, X_1, \dots$ an i.i.d. sample from P , we have $n^{-1} \sum_{i=1}^n f(X_i) \rightarrow Pf$, almost surely, for every bounded, measurable function $f: \mathbb{D} \rightarrow \mathbb{R}$, by the law of large numbers. Because the union of the null sets attached to a countable set of such f is still a null set, it follows by Theorem 1.12.2, that $n^{-1} \sum_{i=1}^n \delta_{X_i} \rightsquigarrow P$, almost surely. Hence P is certainly contained in the closure of all empirical measures.

Finally consider completeness. A Cauchy sequence is always totally bounded. Thus a Cauchy sequence in $\mathcal{M}_1^+$ is totally bounded for the bounded Lipschitz metric. For a complete metric space $\mathbb{D}$ such a sequence is relatively compact for the weak topology (see Problem 1.12.1), meaning that every subsequence has a further weakly converging subsequence. A Cauchy sequence with a converging subsequence always converges itself. Thus completeness of $\mathbb{D}$ implies completeness of $\mathcal{M}_1^+$.

[#] The class BL_1 is the unit ball in the space $BL(\mathbb{D})$ of bounded Lipschitz functions on $\mathbb{D}$ if this is equipped with the norm $\|f\|_{BL}$ equal to the maximum of $\|f\|_\infty$ and $\inf\{c: |f(x) - f(y)| \leq c d(x, y) \text{ for every } x, y\}$. (Abusively the same notation is used for the norm on measures in (1.12.5).) Taking a sum instead of a maximum in the definition of $\|f\|_{BL}$ leads to a slightly different, but equivalent, bounded Lipschitz metric on the set of measures.

Conversely, consider a Cauchy sequence x_n in $\mathbb{D}$. Then the corresponding sequence of Dirac measures δ_{x_n} is Cauchy in $\mathcal{M}_1^+$, and hence $\delta_{x_n} \rightsquigarrow P$, for some separable Borel measure P . The Cauchy property of x_n shows that for every $m \in \mathbb{N}$, there exists n_m such that $d(x_n, x_{n_m}) < 1/m$, for $n \geq n_m$. By the Portmanteau theorem it follows that $P(F_m) \geq \limsup_n \delta_{x_n}(F_m) \geq 1$, for F_m the closed ball of radius $1/m$ around x_{n_m} . Then the set $F = \bigcap_m F_m$ has probability one under P also, and hence cannot be empty. Since its diameter is zero, the set must be a single point x . Then $x_n \rightarrow x$, since $d(x_n, x) \leq 2/m$, for $n \geq n_m$. ■

The set of all Borel probability measures on the metric space $\mathbb{D}$ can be embedded into the linear space $\mathcal{M}$ of all (finite) signed Borel measures on $\mathbb{D}$: the set of all differences $L_1 - L_2$ of nonnegative multiples L_1 and L_2 of Borel probability measures. On $\mathcal{M}$ we can define the norm

$$(1.12.5) \quad \|L\|_{BL} = \sup_{f \in BL_1} \left| \int f dL \right|.$$

Clearly, the bounded Lipschitz metric on probability measures is generated by this norm: $d_{BL}(L_1, L_2) = \|L_1 - L_2\|_{BL}$. The following lemma shows that, for a Polish space $\mathbb{D}$, the dual space of the normed space $(\mathcal{M}, \|\cdot\|_{BL})$ can be identified with the set $BL(\mathbb{D})$ of all bounded, Lipschitz functions $f: \mathbb{D} \rightarrow \mathbb{R}$ through the identification $f: L \mapsto \int f dL$. Consequently, a sequence of separable probability measures L_n converges relative to the weak topology generated on the normed space $(\mathcal{M}, \|\cdot\|_{BL})$ by its dual space (in the sense of functional analysis: $\phi(L_n) \rightarrow \phi(L)$ for every ϕ from the dual space) if and only if the sequence converges weakly in the sense of Definition 1.3.3: $\int f dL_n \rightarrow \int f dL$, for every $f \in C_b(\mathbb{D})$.

This justifies the name *weak convergence* in terms of the terminology of functional analysis, although this name was commonly adopted in the stochastics literature before this connection was noted.

1.12.6 Lemma (Dual space). *Let $\mathcal{M}$ be the linear space of signed Borel measures on a Polish space $\mathbb{D}$. Any linear map $\phi: \mathcal{M} \rightarrow \mathbb{R}$ that is continuous relative to $\|\cdot\|_{BL}$ can be represented as $\phi(L) = \int f dL$, for a bounded, Lipschitz function $f: \mathbb{D} \rightarrow \mathbb{R}$, and then*

$$\sup_{\|L\|_{BL} \leq 1} |\phi(L)| = \sup_x |f(x)| \vee \sup_{x \neq y} \frac{|f(x) - f(y)|}{d(x, y)}.$$

Consequently, the space $BL(\mathbb{D})$, equipped with the norm on the right side, is the dual space of $(\mathcal{M}, \|\cdot\|_{BL})$ and the correspondence $\phi \leftrightarrow f$ is an isometry.

Proof. Given ϕ define the function f by $f(x) = \phi(\delta_x)$. Then $\phi(L) = \int f dL$, for every Dirac measure L , and hence by linearity for all finitely discrete signed measures. For a Polish sample space, the finitely discrete measures are weakly dense in the set of all signed measures (see the proof of Theorem 1.12.4). Hence by continuity of ϕ we obtain that $\phi(L) = \int f dL$, for every signed measure L , provided f can be shown to be bounded and continuous.

Now $|f(x)| = |\phi(\delta_x)| \leq \|\phi\| \|\delta_x\|_{BL}$, for every $x \in \mathbb{D}$, and similarly $|f(x) - f(y)| \leq \|\phi\| \|\delta_x - \delta_y\|_{BL}$, for every $x, y \in \mathbb{D}$. It is immediate from the definition

(1.12.5) of $\|\cdot\|_{BL}$, that $\|\delta_x\|_{BL} \leq 1$, for every x , and that $\|\delta_x - \delta_y\|_{BL} \leq d(x, y)$, for every x and y . Thus $f \in BL(\mathbb{D})$ and $\|f\|_{BL} \leq \|\phi\|$.

Conversely, it is immediate again from definition (1.12.5) that $|\int g dL| \leq \|L\|_{BL}$, for every $g \in BL_1$ and $L \in \mathcal{M}$. Applied with $g = f/\|f\|_{BL}$, this gives $|\int f dL| \leq \|f\|_{BL}\|L\|_{BL}$, and hence $\|\phi\| \leq \|f\|_{BL}$. ■

The connection to functional analysis can alternatively be made through viewing $\mathcal{M}$ as a subset of the dual space of the Banach space $(C_b(\mathbb{D}), \|\cdot\|_\infty)$ of all bounded, continuous functions with the supremum norm, through the identification $L: f \mapsto \int f dL$, for $L \in \mathcal{M}$. The space $C_b(\mathbb{D})$ can be canonically embedded in its second dual, as usual, and weak convergence of a sequence of measures in $\mathcal{M}$ becomes *weak star convergence* of these measures viewed as elements of the dual space. This explains that the names “weak convergence” and “weak star” convergence both appear in the stochastics literature, depending on the point of view, and refer to the *same* property.

Because in general the dual space of $C_b(\mathbb{D})$ is larger than $\mathcal{M}$, the weak star interpretation is perhaps of less interest. An exception is when $\mathbb{D}$ is compact, in which case $\mathcal{M}$ is exactly the dual of $C_b(\mathbb{D})$, by the Riesz representation theorem.[†]

Problems and Complements

- 1. (Uniform tightness and weak compactness of a set of Borel measures)** Call a set $\mathcal{L}$ of Borel probability measures on a metric space *uniformly tight* if, for every $\varepsilon > 0$, there exists a compact K with $L(K) \geq 1 - \varepsilon$ for every $L \in \mathcal{L}$. The following statements are equivalent for a collection of separable Borel measures on a complete metric space:

- (i) $\mathcal{L}$ is totally bounded for the bounded Lipschitz metric;
- (ii) $\mathcal{L}$ is uniformly tight;
- (iii) $\mathcal{L}$ is relatively compact.

In (iii) *relatively compact* may be taken to mean either that every net in $\mathcal{L}$ has a weakly convergent subnet or that every sequence in $\mathcal{L}$ has a weakly convergent subsequence.

[Hint: Prohorov’s theorem implies that the second statement implies the third. Relative compactness implies total boundedness for any semimetric space. It suffices to show that (i) implies (ii). For any Borel set B , the function $f(x) = (1 - \varepsilon^{-1}d(x, B))^+$ has Lipschitz constant ε^{-1} and is sandwiched between 1_B and 1_{B^c} . Deduce that, for any probability measures L and L' , $L'(B^c) \geq L(B) - \varepsilon^{-1}d_{BL}(L, L')$. Fix $\delta > 0$. Take a finite set $\mathcal{L}_0 \subset \mathcal{L}$ of measures with $\mathcal{L} \subset \mathcal{L}_0^{\delta^2}$. For every $L_0 \in \mathcal{L}_0$, take a compact with probability at least $1 - \delta$. Let F be the union of the compacts, and take a finite set G with $F \subset G^\delta$. Then, for every $L \in \mathcal{L}$, $L(G^{2\delta}) \geq L_0(G^\delta) - \delta^{-1}d_{BL}(L, L_0)$, which is at least $1 - 2\delta$ for some L_0 . Take K equal to the closure of $\cap_{m=1}^\infty G^{\varepsilon 2^{-m}}$. Then K is totally bounded and complete, hence compact, with $L(K) \geq 1 - \varepsilon$.]

- 2.** A collection of separable Borel measures $\mathcal{L}$ on a complete metric space is compact for the weak topology if and only if it is uniformly tight and contains the limit of every converging sequence in $\mathcal{L}$.

[†] See Dudley (1989), Theorem 7.4.1.

3. **(Converse of Prohorov's theorem)** Let the net X_α be relatively compact with all limit points concentrating on a fixed Polish subset $\mathbb{D}_0 \subset \mathbb{D}$. Then X_α is asymptotically tight.

[Hint: Use “metrizability” of weak convergence to a separable limit to see that the set of all limit points is compact for the weak topology. (The proof is the same as for the proposition that the set of limit points of a relatively compact net in a metric space is compact.) Next use Problem 1.12.1.]

4. **(Lipschitz functions)** For a Lipschitz function $f: \mathbb{D} \rightarrow \mathbb{R}$, define $\|f\|_l = \sup\{|f(x) - f(y)|/d(x, y) : d(x, y) \neq 0\}$. This is equal to the smallest c for which $|f(x) - f(y)| \leq c d(x, y)$, for every x, y . The expressions $\|f\|_\infty + \|f\|_l$ and $\|f\|_\infty \vee \|f\|_l$ define norms on the set of all Lipschitz functions on $\mathbb{D}$. They are equivalent and make the Lipschitz functions into a Banach space.

5. The bounded Lipschitz distance on the set of Dirac measures satisfies $d_{BL}(\delta_x, \delta_y) = d(x, y) \wedge 2$, for every x, y .

[Hint: The bounded Lipschitz distance is $d_{BL}(\delta_x, \delta_y) = \sup_{f \in BL_1} |f(x) - f(y)|$. It is immediate from the definition of BL_1 that this is bounded above by $d(x, y) \wedge 2$. For the converse, consider the function $f(z) = (d(x, z) - 1) \wedge 1$.]

1.13

Conditional Weak Convergence

The metrizability of weak convergence makes it possible to give a natural definition of weak convergence in probability of conditional distributions. We shall use this extensively when discussing the bootstrap in Chapter 3.7 and Section 3.10.3, and the multiplier central limit theorems in Chapter 2.9.

A sequence of Borel measurable maps $X_n: \Omega_n \rightarrow \mathbb{D}$ from given probability spaces $(\Omega_n, \mathcal{A}_n, P_n)$ into a metric space $\mathbb{D}$ *converges in conditional distribution* given some sub σ -fields $\mathcal{B}_n \subset \mathcal{A}_n$ if the bounded Lipschitz distance between the conditional law of X_n given $\mathcal{B}_n$ and a Borel probability measure L tends to zero, where the convergence can be in outer probability or almost uniformly. More precisely, the sequence X_n tends to L conditionally given $\mathcal{B}_n$ in outer probability if

$$d_{BL}(\mathcal{L}(X_n | \mathcal{B}_n), L) = \sup_{f \in BL_1} \left| E(f(X_n) | \mathcal{B}_n) - \int f dL \right| \xrightarrow{P^*} 0.$$

For almost uniform convergence we replace the convergence in this display appropriately. Thus metrization puts this notion in the ordinary framework of stochastic convergence.

While using a metric is desirable for a definition, the notion is actually equivalent to pointwise (or “functionwise”) convergence of conditional expectations.

1.13.1 Theorem. *For a sequence of Borel measurable maps $X_n: \Omega_n \rightarrow \mathbb{D}$ on probability spaces $(\Omega_n, \mathcal{A}_n, P_n)$ with values in a metric space $\mathbb{D}$, sub σ -fields $\mathcal{B}_n \subset \mathcal{A}_n$ and a separable Borel measure L , the following statements are equivalent:*

- (i) $d_{BL}(\mathcal{L}(X_n | \mathcal{B}_n), L) \xrightarrow{P^*} 0$;
 - (ii) $E(f(X_n) | \mathcal{B}_n) \xrightarrow{P} \int f dL$, for every bounded Lipschitz $f: \mathbb{D} \rightarrow \mathbb{R}$.
- If $\mathbb{D} = \mathbb{R}^k$ these statements are implied by any of the following:*
- (iii) $P(X_n \leq x | \mathcal{B}_n) \xrightarrow{P} L(x)$, for every continuity point x of L ;

- (iv) $E(e^{it'X_n} | \mathcal{B}_n) \xrightarrow{P} \int e^{it'x} dL(x)$, for every $t \in \mathbb{R}^k$;
- (v) $E(e^{t'X_n} | \mathcal{B}_n) \xrightarrow{P} \int e^{t'x} dL(x) < \infty$, for every t in a neighborhood of 0;
- (vi) $X_n \geq 0$ for every n , and $E(e^{t'X_n} | \mathcal{B}_n) \xrightarrow{P} \int e^{t'x} dL(x) < \infty$, for every $t \leq 0$.

Proof. (i) $\Rightarrow$ (ii). This is immediate from the definition of d_{BL} .

(ii) $\Rightarrow$ (i). Because we can work within the completion of $\mathbb{D}$, it is no loss of generality to assume that L is tight. By (ii) and the dominated convergence theorem $E f(X_n) \rightarrow \int f dL$, for every bounded Lipschitz f , whence $X_n \rightsquigarrow L$, by the portmanteau theorem. In particular, the sequence X_n is asymptotically tight. Given $\varepsilon > 0$, let $K \subset \mathbb{D}$ be a compact set such that $L(K) > 1 - \varepsilon$, whence $\limsup P(X_n \notin K^\delta) < \varepsilon$, for every $\delta > 0$. The set of restrictions of the functions in BL_1 to K is totally bounded in $C(K)$ by the Arzelà-Ascoli theorem. Therefore, for every $\eta > 0$ it can be covered by finitely many uniform balls of radius η , where the centers of these balls can be chosen inside the set without loss of generality. These functions can be extended to uniformly Lipschitz functions on the full space of the same norm[‡]. Thus for every $\eta > 0$ there exist finitely many elements $f_1, \dots, f_m$ of BL_1 , with $\min_i \sup_{x \in K} |f(x) - f_i(x)| < \eta$, for every $f \in BL_1$. By the compactness of K and continuity of $f - f_i$, it is then automatically true that $\min_i \sup_{x \in K^\delta} |f(x) - f_i(x)| < \eta$, for some $\delta > 0$, which can be chosen independently of $f \in BL_1$ (see Problem 1.13.3). Thus for every $f \in BL_1$, there exists i such that

$$|E(f(X_n) - f_i(X_n) | \mathcal{B}_n)| \leq E(\eta 1_{X_n \in K^\delta} + 21_{X_n \notin K^\delta} | \mathcal{B}_n) \leq \eta + 2S_n,$$

where $S_n = P(X_n \notin K^\delta | \mathcal{B}_n)$ satisfies $\limsup ES_n < \varepsilon$ and does not depend on $f \in BL_1$. Similarly, we have that $|\int (f - f_i) dL| \leq \eta + 2\varepsilon$. By the triangle inequality we conclude that for every $\varepsilon, \eta > 0$ there exists a finite set $f_1, \dots, f_m$ in BL_1 such that

$$d_{BL}(\mathcal{L}(X_n | \mathcal{B}_n), L) \leq \max_i \left| E(f_i(X_n) | \mathcal{B}_n) - \int f_i dL \right| + 2\eta + 2S_n + 2\varepsilon.$$

As $n \rightarrow \infty$ the expectation of the latter bound is bounded above by $2\eta + 4\varepsilon$. We conclude that $d_{BL}(\mathcal{L}(X_n | \mathcal{B}_n), L) \rightarrow 0$, in outer probability.

To show that either of (iii), (iv), (v) or (vi) implies (i), by Lemma 1.9.2 it suffices to prove that every subsequence of $\{n\}$ has a further subsequence along which $d_{BL}(\mathcal{L}(X_n | \mathcal{B}_n), L) \rightarrow 0$, almost surely. This reduces the random laws to deterministic sequences, for which the convergence in the bounded Lipschitz metric is equivalent to ordinary weak convergence $\mathcal{L}(X_n | \mathcal{B}_n) \rightsquigarrow L$, pointwise on the underlying probability space.

(iii), (v), or (vi) $\Rightarrow$ (i). In each case the assumed convergence in probability, for a given x or t , implies existence of a subsequence along which the same convergence is true almost surely. By a diagonalization scheme there exists a subsequence that works simultaneously for every x or t in a countable set. The portmanteau theorem (viii), (C) or (B) in Example 1.3.5 can be refined to see that this is enough to conclude that $\mathcal{L}(X_n | \mathcal{B}_n) \rightsquigarrow L$, almost surely. Indeed, for (viii) in Example 1.3.5

[‡] Dudley (1989), Theorem 6.1.1.

this follows by the monotonicity of a cumulative distribution function, for (C) because the argument based on analytic continuation works for a countable dense set, and for (B) because only vectors t with nonpositive integer coordinates are needed to determine the distribution by moments.

(iv) $\Rightarrow$ (i). By assumption the process $Z_n(t) = E(e^{it'X_n} | \mathcal{B}_n)$ converges marginally to the degenerate process $Z(t) = \int e^{it'x} dL(x)$. The continuity modulus $\sup_{|s-t| < \delta} |Z_n(s) - Z_n(t)|$ is bounded above by $E[\delta \|X_n\| \wedge 2 | \mathcal{B}_n]$, and hence $P(\sup_{|s-t| < \delta} |Z_n(s) - Z_n(t)| > \varepsilon) \leq \varepsilon^{-1} E[\delta \|X_n\| \wedge 2]$, by Markov's inequality. Because assumption (iv) implies that $Ee^{it'X_n} \rightarrow \int e^{it'x} dL(x)$, for every $t \in \mathbb{R}^k$, Example 1.3.5(ix) gives $X_n \rightsquigarrow L$, and hence $E[\delta \|X_n\| \wedge 2] \rightarrow 0$ as $n \rightarrow \infty$ followed by $\delta \downarrow 0$, by the dominated convergence theorem (see Example 1.11.6). By Theorem 1.5.7 the process Z_n converges in distribution in $\ell^\infty[-T, T]$, for every $T > 0$, to the degenerate limit Z . Thus by Lemma 1.10.2(iii) the assumed convergence $Z_n(t) \xrightarrow{P} Z(t)$, for every $t \in \mathbb{R}^k$, can be strengthened to convergence in probability uniformly on compacta. This implies that for fixed $T \in \mathbb{N}$ every subsequence has a further subsequence such that $Z_n(t) \rightarrow Z(t)$ for every $\|t\| \leq T$, almost surely. A diagonalization argument gives a subsequence with this property for every T , whence $E(e^{it'X_n} | \mathcal{B}_n) \rightarrow \int e^{it'x} dL(x)$, for every $t \in \mathbb{R}^k$. Example 1.3.5(ix) implies that $\mathcal{L}(X_n | \mathcal{B}_n) \rightsquigarrow L$, almost surely, along the subsequence. ■

If the metric space $\mathbb{D}$ is Polish, then the conditional distributions can be identified with Markov kernels L_n given by $L_n(B | \omega) = P(X_n \in B | \mathcal{B}_n)(\omega)$. Therefore, for maps X_n defined on a common probability space $(\Omega, \mathcal{A}, P)$, the notion of *weak convergence of Markov kernels* applies (see Problem 1.15.3): $L_n \xrightarrow{w} L$ if, for every $f \in C_b(\mathbb{D})$ and bounded random variable $Z: \Omega \rightarrow \mathbb{R}$,

$$\int \int f(y) L_n(dy | \omega) Z(\omega) dP(\omega) \rightarrow \int \int f(y) L(dy | \omega) Z(\omega) dP(\omega).$$

This is a weaker notion than the one in the preceding theorem, although the latter one is also less general in assuming the limit to be degenerate.

1.13.2 Lemma. Suppose that $X_n: (\Omega, \mathcal{A}, P) \rightarrow \mathbb{D}$ are measurable maps from a common probability space into a Polish space $\mathbb{D}$. If X_n converges weakly in probability in conditional distribution given sub σ -fields $\mathcal{B}_n \subset \mathcal{A}$ to a measure L , then the conditional distributions $L_n = \mathcal{L}(X_n | \mathcal{B}_n)$ converge weakly in the sense of Markov kernels to the degenerate Markov kernel $(\omega, B) \mapsto L(B)$.

Proof. Conditional weak convergence implies convergence in probability of the variables $E(f(X_n) | \mathcal{B}_n)$ to the numbers $\int f dL$, for every bounded Lipschitz function $f: \mathbb{D} \rightarrow \mathbb{R}$, and hence $EE(f(X_n) | \mathcal{B}_n)Z \rightarrow \int f dL EZ$, for every bounded measurable random variable Z , by the dominated convergence theorem. The same convergence for general $f \in C_b(\mathbb{D})$ next follows by approximation, as in the portmanteau theorem (or see Problem 1.15.3). ■

As an illustration we consider “Hoeffding’s trick”. Consider sequences of Borel measurable maps $X_n, Y_n: \Omega_n \rightarrow \mathbb{D}$ from given probability spaces

$(\Omega_n, \mathcal{A}_n, P_n)$ into a metric space $\mathbb{D}$ such that for every n the variables X_n and Y_n are independent given some sub σ -field $\mathcal{B}_n \subset \mathcal{A}_n$. To ease notation we denote this by $X_n \perp\!\!\!\perp Y_n | \mathcal{B}_n$, and take this to mean that, for every bounded, measurable functions $f, g: \mathbb{D} \rightarrow \mathbb{R}$,

$$(1.13.3) \quad E(f(X_n)g(Y_n) | \mathcal{B}_n) = E(f(X_n) | \mathcal{B}_n) E(g(Y_n) | \mathcal{B}_n), \quad \text{a.s..}$$

Furthermore, assume that given $\mathcal{B}_n$ the sequence X_n converges in distribution, to a Borel probability distribution L , in probability. Again to ease notation we denote this by $X_n | \mathcal{B}_n \rightsquigarrow L$, in probability. That the limit distribution L is nonrandom is an essential part of the assumptions of the next result.

We write $X_n \perp\!\!\!\perp Y_n$ to denote independence between X_n and Y_n , and $X_n | \mathcal{B}_n \sim Y_n | \mathcal{B}_n$ to denote that X_n and Y_n are equal in conditional distribution given $\mathcal{B}_n$, meaning that $P(X_n \in A | \mathcal{B}_n) = P(Y_n \in A | \mathcal{B}_n)$, for every Borel set $A \subset \mathbb{D}$, almost surely.

1.13.4 Lemma (Hoeffding's trick). *Suppose that $X_n, Y_n: \Omega_n \rightarrow \mathbb{D}$ are sequences of Borel measurable maps in a metric space $\mathbb{D}$ such that $X_n \perp\!\!\!\perp Y_n | \mathcal{B}_n$, for a sub σ -field $\mathcal{B}_n$, for every n . Let L, L_1 and L_2 be separable Borel probability measures on $\mathbb{D}$.*

- (i) *If $X_n | \mathcal{B}_n \rightsquigarrow L_1$ and $Y_n | \mathcal{B}_n \rightsquigarrow L_2$ in $\mathbb{D}$, in probability, then $(X_n, Y_n) \rightsquigarrow L_1 \times L_2$ in $\mathbb{D} \times \mathbb{D}$.*
 - (ii) *If $X_n | \mathcal{B}_n \sim Y_n | \mathcal{B}_n$, for every n and $(X_n, Y_n) \rightsquigarrow L_1 \times L_2$ in $\mathbb{D} \times \mathbb{D}$, then $L_1 = L_2$ and $X_n | \mathcal{B}_n \rightsquigarrow L_1$ and $Y_n | \mathcal{B}_n \rightsquigarrow L_2$, in probability.*
- In particular, if $X_n \perp\!\!\!\perp Y_n | \mathcal{B}_n$ and $X_n | \mathcal{B}_n \sim Y_n | \mathcal{B}_n$, then $X_n | \mathcal{B}_n \rightsquigarrow L$ in $\mathbb{D}$, in probability, if and only if $(X_n, Y_n) \rightsquigarrow L \times L$ in $\mathbb{D} \times \mathbb{D}$.*

Proof. The assumption of conditional independence gives that (1.13.3) holds, for any bounded measurable functions f, g .

(i). By assumption the two conditional expectations on the right side of (1.13.3) tend in probability to $\int f dL_1$ and $\int g dL_2$, for any bounded Lipschitz functions f and g , respectively. The expectation of their product then tends to $\int f dL_1 \int g dL_2$, by the dominated convergence theorem. By (1.13.3) this expectation is same as $E(f(X_n)g(Y_n))$. This implies assertion (i), in view of Corollary 1.4.5.

(ii). The assumption that $X_n | \mathcal{B}_n \sim Y_n | \mathcal{B}_n$ gives that $E(g(X_n) | \mathcal{B}_n) = E(g(Y_n) | \mathcal{B}_n)$, almost surely, for every indicator $g = 1_A$ of a set $A \in \mathcal{A}_n$, and then, by approximation, for every bounded measurable function g . Thus (1.13.3) remains valid if (only) in its right side Y_n is replaced by X_n . Next taking expectation, we see that $EE(f(X_n) | \mathcal{B}_n)E(g(X_n) | \mathcal{B}_n) = Ef(X_n)g(Y_n) \rightarrow \int f dL_1 \int g dL_2$, by the assumption that $(X_n, Y_n) \rightsquigarrow L_1 \times L_2$, for every bounded Lipschitz functions f and g . Since the left side is symmetric in f and g , so must be the limit, which implies $L_1 = L_2$. The convergence with $f = g$ gives that $E[E(f(X_n) | \mathcal{B}_n)]^2 \rightarrow (\int f dL)^2$, for every bounded Lipschitz f and $L = L_1 = L_2$. With f replaced by $f - \int f dL$, the right side vanishes, and we conclude that $E(f(X_n) | \mathcal{B}_n) \rightarrow \int f dL$ in quadratic mean. Thus $X_n | \mathcal{B}_n \rightsquigarrow L$, by Theorem 1.13.1. ■

1.13.5 Example (Random projection). Given an $(n \times p)$ -matrix Z_n of rank p , the matrix $Z_n(Z_n^t Z_n)^{-1/2}$ has the same column space, but has p orthonormal columns. The orthogonal projection of a vector $X_n \in \mathbb{R}^n$ onto this column space has coordinates $(Z_n^t Z_n)^{-1/2} Z_n^t X_n$ relative to the latter orthonormal basis. Consider the distribution of this set of coordinates in the case that the np entries of Z_n are i.i.d. standard normal variables independent of a vector X_n with some arbitrary distribution on $\mathbb{R}^n$, in the setting that $n \rightarrow \infty$ and p is fixed. (Not much is lost by taking $p = 1$, which gives the length of the projection of X_n in a random direction Z_n .)

We shall show that $(Z_n^t Z_n)^{-1/2} Z_n^t X_n | Z_n \rightsquigarrow L$ if and only if both $n^{-1} X_n^t X_n$ converges weakly and $n^{-1} X_n^t \tilde{X}_n \xrightarrow{P} 0$, for independent copies $\tilde{X}_n$ of X_n . The limit L is then equal to the law of RW , for $\mathbb{R}^2$ the weak limit of $n^{-1} X_n^t X_n$ and W an independent p -dimensional standard normal vector.

For $\tilde{X}_n$ defined to be an independent copy of X_n that is also independent of Z_n , the random vectors $Y_n := (Z_n^t Z_n)^{-1/2} Z_n^t X_n$ and $\tilde{Y}_n := (Z_n^t Z_n)^{-1/2} Z_n^t \tilde{X}_n$ trivially satisfy $Y_n \perp\!\!\!\perp \tilde{Y}_n | Z_n$ and $Y_n | Z_n \sim \tilde{Y}_n | Z_n$. Therefore by Hoeffding's trick, the conditional weak convergence $Y_n | Z_n \rightsquigarrow L$ is equivalent to the unconditional joint convergence $(Y_n, \tilde{Y}_n) \rightsquigarrow L \times L$.

By the law of large numbers the sequence $n^{-1} Z_n^t Z_n$ tends in probability to the identity matrix I_p . Thus by Slutsky's lemma the sequence $(Y_n, \tilde{Y}_n)$ converges in distribution if and only if the sequence $n^{-1/2}(Z_n^t X_n, Z_n^t \tilde{X}_n)$ converges in distribution, with identical weak limits.

Given $(X_n, \tilde{X}_n)$, the latter $2p$ -vector is multivariate Gaussian with mean zero and covariance matrix, for I_p the $(p \times p)$ identity matrix,

$$\frac{1}{n} \begin{pmatrix} X_n^t X_n I_p & X_n^t \tilde{X}_n I_p \\ X_n^t \tilde{X}_n I_p & \tilde{X}_n^t \tilde{X}_n I_p \end{pmatrix}.$$

Therefore this vector converges in distribution if and only if the sequence $n^{-1}(X_n^t X_n, X_n^t \tilde{X}_n, \tilde{X}_n^t \tilde{X}_n)$ converges in distribution (in $\mathbb{R}^3$), where the limit is a product law if and only if the second coordinates $n^{-1} X_n^t \tilde{X}_n$, tend to zero (see Problem 1.13.2).

Problems and Complements

1. The assumption of equal conditional distributions in (ii) of Lemma 1.13.4 can be relaxed while still maintaining some converse of (i), although it appears that some extra condition is necessary. In the case of variables not depending on n and a σ -field $\mathcal{A}$ generated by a variable T , we assume that $X \perp\!\!\!\perp Y | T$, and assertions (i)–(ii) become:

- (i) if $X \perp\!\!\!\perp T$ and $Y \perp\!\!\!\perp T$, then $X \perp\!\!\!\perp Y$,
- (ii) if $X \perp\!\!\!\perp Y$ and $X | T \sim Y | T$, then $X \perp\!\!\!\perp T$ and $Y \perp\!\!\!\perp T$.

The latter assertion can be strengthened to:

- (ii') if $X \perp\!\!\!\perp Y$ and there exists a randomization $h(Y, U)$ of Y such that $X | T \sim h(Y, U) | T$, then $X \perp\!\!\!\perp T$ and $Y \perp\!\!\!\perp T$.

A "randomization" is a measurable map applied to Y and an independent uniform variable U . The existence of such a randomization is also necessary for the conclusion $X \perp\!\!\!\perp T$ and $Y \perp\!\!\!\perp T$, since in that case it suffices to construct h such that $X \sim h(Y, U)$, which is always possible.

[Hint: Since $E \operatorname{cov}(f(X), g(Y, U) | T) = 0$, we can argue that $0 = \operatorname{cov}(f(X), g(Y, U)) = 0$ in (ii') is equal to $\operatorname{cov}(E(f(X) | T), E(g(Y, U) | T))$. Choose $g(Y, U) = f(h(Y, U))$ to obtain that $E(g(Y, U) | T) = E(f(X) | T)$, and hence $\operatorname{var}(E(f(X) | T)) = 0$.]

2. Let Y_n be random p -vectors and Γ_n random nonnegative-definite $(p \times p)$ -matrices such that $Y_n | \Gamma_n \sim N_p(0, \Gamma_n)$.

- (i) If the sequence Γ_n converges in distribution, then so does the sequence Y_n .
- (ii) If the sequence Y_n converges in distribution, then the sequence Γ_n is tight and the sequence $s^t \Gamma_n s$ converges in distribution for every $s \in \mathbb{R}^p$.

Let $Y_n = (Y_{n1}, Y_{n2})$ be random $2p$ -vectors and correspondingly $\Gamma_n = (\Gamma_{n,i,j})_{i,j=1,2}$ partitioned random $(2p \times 2p)$ -matrices such that $Y_n | \Gamma_n \sim N_{2p}(0, \Gamma_n)$.

- (iii) If $\Gamma_{n,1,1}$ and $\Gamma_{n,2,2}$ are independent and the sequence Y_n converges in distribution to a vector $Y = (Y_1, Y_2)$ with independent components, then $\Gamma_{n,1,2} \xrightarrow{p} 0$.

[Hint: (i). Represent Y_n as $Y_n = \Gamma_n^{1/2} Z$, for Z standard normal and independent of Γ_n , and apply the continuous mapping theorem. (ii). The characteristic function of Y_n is $s \mapsto E e^{-s^t \Gamma_n s / 2}$. If this converges pointwise to the characteristic function ψ , then the Laplace transform of the variable $s^t \Gamma_n s$ tends to $u \mapsto \psi(\sqrt{us})$, for $u > 0$, which tends to $\psi(0) = 1$, as $u \downarrow 0$. For s equal to the unit vectors, this gives the convergence of the diagonal elements of Γ_n . Nonnegative-definiteness and symmetry give that $(\Gamma_n)_{k,l}^2 \leq (\Gamma_n)_{k,k}(\Gamma_n)_{l,l}$, which implies tightness of the off-diagonal elements. (iii). By (ii) the sequence Γ_n is tight. It suffices to show that every weak limit point Γ has $\Gamma_{1,2} = 0$. Since $Y_n \sim \Gamma_n^{1/2} Z \rightsquigarrow \Gamma^{1/2} Z$, it follows that $Y \sim \Gamma^{1/2} Z$, for the limit point Γ and an independent standard normal Z . Thus the characteristic function of Y at $s = (s_1, s_2)$ is equal to $E e^{i s_1^t Y_1 + i s_2^t Y_2} = E e^{-s^t \Gamma s / 2}$. This factorizes as $E e^{i s_1^t Y_1} E e^{i s_2^t Y_2}$, by independence of Y_1 and Y_2 . Because $Y_i \sim (\Gamma^{1/2} Z)_i$, the product is equal to $E e^{-s_1^t \Gamma_{1,1} s_1 / 2} E e^{-s_2^t \Gamma_{2,2} s_2 / 2} = E e^{-s_1^t \Gamma_{1,1} s_1 / 2 - s_2^t \Gamma_{2,2} s_2 / 2}$, because $\Gamma_{1,1}$ and $\Gamma_{2,2}$ inherit the assumed independence of the matrices $\Gamma_{n,1,1}$ and $\Gamma_{n,2,2}$. We conclude that $E e^{-A-B} = E e^{-A}$ for $A = s_1^t \Gamma_{1,1} s_1 + s_2^t \Gamma_{2,2} s_2$ and $B = 2s_1^t \Gamma_{1,2} s_2$, for every (s_1, s_2) . Applied to $(s_1, -s_2)$, we see that $E e^{-A+B} = E e^{-A}$ also. This implies that $E e^{-A} (e^{-B} + e^B - 2) = 0$, and hence $B = 0$, since $e^x + e^{-x} - 2 > 0$, for every $x \neq 0$.]

3. Given a compact subset $K \subset \mathbb{D}$ of a metric space, finitely many continuous functions $f_i: \mathbb{D} \rightarrow \mathbb{R}$, and $\eta > 0$, suppose that for every (bounded Lipschitz) function $f \in BL_1(\mathbb{D})$ there exists f_i with $|f(x) - f_i(x)| < \eta$ for every $x \in K$. Then there exists $\delta > 0$ such that the latter inequality is true for every $x \in K^\delta$.

[Hint: Were this not the case, then for every n there would exist $g_n \in BL_1(\mathbb{D})$ such that $\min_i \sup_{x \in K^{1/n}} |g_n(x) - f_i(x)| \geq \eta$, which means that for all i there exists $x_{n,i} \in K^{1/n}$ with $|g_n(x_{n,i}) - f_i(x_i)| > \eta - 1/n$. By the compactness of K , there exists a subsequence along which $x_{n,i} \rightarrow x_i$ for a point $x_i \in K$, for every i , and in view of the Arzelà-Ascoli theorem it can be assured that $g_n \rightarrow g$ uniformly on K along this subsequence, for some $g: K \rightarrow \mathbb{R}$. The function g will be bounded and Lipschitz on K and hence can be extended to a function $g \in BL_1(\mathbb{D})$. Then $g_n(x_{n,i}) = g_n(x_i) + o(1) \rightarrow g(x_i)$ and $f_i(x_{n,i}) \rightarrow f_i(x_i)$, so that $|g(x_i) - f_i(x_i)| \geq \eta$, for every i , contradicting the supposition.]

1.14

Skorokhod Space

The *Skorokhod space* $D[0, \infty)$ is the set of all functions $z: [0, \infty) \rightarrow \mathbb{R}$ that are *cad-lag*: functions that are continuous from the right throughout the domain and possess left limits $z(t-):= \lim_{s \uparrow t} z(s)$ at every $t > 0$. For given $T > 0$ the Skorokhod space $D[0, T]$ consists of the restrictions of these functions to $[0, T]$. Every cad-lag function is bounded on compacts (see Problem 1.14.1) and hence the spaces $D[0, \infty)$ and $D[0, T]$ are subsets of the spaces $\ell^\infty([0, 1], [0, 2], \dots)$ and $\ell^\infty[0, T]$ of locally bounded and bounded functions, respectively, considered in Chapters 1.5 and 1.6. Because left and right limits are retained under uniform convergence, the spaces are closed subsets and hence Banach spaces under the uniform norm. For many purposes it is best to treat them as such and apply the theory for weak convergence relative to the uniform norm developed in Chapters 1.5 and 1.6, which is conceptually simpler and yields stronger results than obtained in the present chapter. This is true in particular for the classical empirical process. However, for some applications, specifically when the limit process has jumps at a random set of locations, the uniform norm is too strong. In this chapter we discuss the weaker “Skorokhod topology” (also known as J_1 -topology) that is suitable to these cases.

The topology on $D[0, \infty)$ can be extended from the topologies on the spaces $D[0, T]$, roughly by the same procedure as used to construct the uniform topology of locally bounded functions in Chapter 1.6. Because possible discontinuities near the end point T of a compact interval play different roles in $D[0, \infty)$ and $D[0, T]$, we describe this extension in detail in Section 1.14.2. In Section 1.14.1 we first concentrate on the topology for a compact interval, which we take to be $[0, 1]$ without loss of generality.

1.14.1 Compact Domain

The *Skorokhod metric* on $D[0, 1]$ is defined, for $x, y \in D[0, 1]$, as

$$d(x, y) = \inf_{\lambda \in \Lambda} \left[\sup_{t \in [0, 1]} |x(t) - y(\lambda(t))| + \sup_{t \in [0, 1]} |t - \lambda(t)| \right],$$

where the infimum is taken over the set Λ of all strictly increasing bijections $\lambda: [0, 1] \rightarrow [0, 1]$. Clearly $z_n \rightarrow z$ relative to this metric if and only if there exist $\lambda_n \in \Lambda$ such that both $z_n \circ \lambda_n \rightarrow z$ and $\lambda_n \rightarrow \text{id}$ uniformly on $[0, 1]$, for $\text{id}: [0, 1] \rightarrow [0, 1]$ the identity map. Since the inverse λ^{-1} of an element of Λ is itself contained in Λ and $\lambda_n^{-1} \rightarrow \text{id}$ uniformly if and only if $\lambda_n \rightarrow \text{id}$ uniformly, an equivalent statement is the existence of $\lambda_n \in \Lambda$ such that both $z_n - z \circ \lambda_n \rightarrow 0$ and $\lambda_n \rightarrow \text{id}$ uniformly.

The fact that $\text{id} \in \Lambda$ shows that the Skorokhod metric is bounded above by the uniform distance, whence the *Skorokhod topology* is weaker than the uniform topology. The weakening is considerable, but only when functions with discontinuities are involved. Whereas the uniform distance is harsh when comparing such functions, the possibility of “warping the time axis”, incorporated in the Skorokhod metric, allows to match up jumps at slightly different locations and may make distances much smaller. For instance, the uniform distance between $1_{[0, 1/2]}$ and $1_{[0, 1/2 + 1/n]}$ is 1 for every n , whereas the Skorokhod distance between these functions tends to zero. However, a sequence of functions that converges in the Skorokhod topology to a *continuous* limit, will automatically converge to the same limit relative to the uniform norm (see Problem 1.14.3). In particular, the relative Skorokhod topology induced on the set of continuous functions is identical to the uniform topology, and convergence in distribution relative to the Skorokhod distance to a process with continuous sample paths implies convergence in distribution relative to the uniform distance (as may be shown using the almost sure representation theorem, Theorem 1.10.4). We conclude that there is no gain in using the Skorokhod topology whenever the limit variable has continuous sample paths.

The space $D[0, 1]$ is separable under d (see Problem 1.14.6), but not complete (Problem 1.14.7). However, the Skorokhod topology is Polish; Problem 1.14.8 gives an alternative metric that generates the same topology under which the space is complete. Another desirable property is that the Borel σ -field has a simple description.

1.14.1 Lemma. *The Borel σ -field on $D[0, 1]$ is equal to the σ -field generated by the coordinate projections.*

Proof. It can be checked that for any given $\varepsilon > 0$, $t \in [0, 1 - \varepsilon]$ and bounded, Lipschitz function $f: \mathbb{R} \rightarrow \mathbb{R}$, the map $g_\varepsilon: D[0, 1] \rightarrow \mathbb{R}$ defined by $g_\varepsilon(z) = \varepsilon^{-1} \int_t^{t+\varepsilon} f(z(s)) ds$ is continuous with respect to the Skorokhod topology and hence Borel measurable. Furthermore for given $t < 1$ and f as before, $g_\varepsilon(z) \rightarrow f(z(t)) = f(\pi_t z)$, as $\varepsilon \downarrow 0$, by the dominated convergence theorem, for every right-continuous z . Since measurability is retained under pointwise limits, the map $z \mapsto f(\pi_t z)$ is Borel measurable for every bounded Lipschitz f . This shows that $z \mapsto \pi_t z$ is Borel measurable, for every $t < 1$. From the fact that $\lambda(1) = 1$ for every

$\lambda \in \Lambda$, it is seen that the coordinate projection π_1 is continuous, and hence Borel measurable. It follows that the projection σ -field is contained in the Borel σ -field.

To show the converse define for $\alpha \in \mathbb{R}^{k+1}$ and a fixed partition $0 = t_0 < t_1 < \dots < t_k = 1$ a function z_α by $z_\alpha(t) = \sum_i \alpha_i 1_{[t_{i-1}, t_i)}(t) + \alpha_k 1_1(t)$. Then $d(z_\alpha, z_{\alpha'}) \leq \|\alpha - \alpha'\|_\infty$, by the fact that the Skorokhod metric is bounded above by the uniform distance. Hence by the triangle inequality the map $\alpha \mapsto d(z_0, z_\alpha)$ from $\mathbb{R}^{k+1}$ to $\mathbb{R}$ is (Lipschitz) continuous, for every $z_0 \in D[0, 1]$ and fixed partition $0 = t_0 < t_1 < \dots < t_k = 1$. Therefore, the composed map $z \mapsto (z(t_0), \dots, z(t_k)) \mapsto d(z_0, \sum_i z(t_{i-1})1_{[t_{i-1}, t_i)} + z(t_k)1_1)$ is measurable relative to the projection σ -field. For a sequence of partitions with mesh widths tending to zero these maps tend pointwise to $z \mapsto d(z_0, z)$. (For a precise proof, see for instance (1.14.7) and Problem 1.14.11.) Hence the latter map is projection measurable, so that the Skorokhod ball σ -field is contained in the projection σ -field. By separability of the Skorokhod topology, the Skorokhod Borel and ball σ -fields are the same. ■

As a consequence of the preceding lemma, a map $X: (\Omega, \mathcal{A}, P) \rightarrow D[0, 1]$ is Borel measurable relative to the Skorokhod topology if and only if its coordinates $X(t)$, for $t \in [0, 1]$, are Borel measurable maps in $\mathbb{R}$.

Although measurable, the coordinate projections are not continuous (see Problem 1.14.4). For this reason weak convergence of a sequence of random elements relative to the Skorokhod topology does not necessarily imply weak convergence of the finite-dimensional marginals. However, it turns out that, for a given limit, marginal convergence takes place at all except at most countably many points. With this adaptation weak convergence is equivalent to tightness and marginal convergence, just as for processes in $\ell^\infty[0, 1]$.

1.14.2 Theorem (Tightness plus marginal convergence). *A net of maps X_α in $D[0, 1]$ converges weakly to a limit relative to the Skorokhod topology if and only if the net is asymptotically tight and there exists a dense set $S \subset [0, 1]$ that includes 1 such that the net $(X_\alpha(t_1), \dots, X_\alpha(t_k))$ converges in distribution in $\mathbb{R}^k$ for every $t_1, \dots, t_k \in S$.*

1.14.3 Addendum. *If $X_\alpha \rightsquigarrow X$, then $S - \{1\}$ can without loss of generality be taken equal to the set of all t such that $X(t) = X(t-)$ almost surely. For any measurable map X into $D[0, 1]$, the complement of this set is at most countable.*

Proof. Because $D[0, 1]$ is Polish, every random element with values in $D[0, 1]$ is tight and hence every weakly converging net is asymptotically tight. Conversely, by Prohorov's theorem every asymptotically tight net converges weakly provided all weak limit points are the same. Thus it suffices to show that weak convergence implies marginal convergence at all points in some dense set S that contains 1, and, for the converse, that the limit points of an asymptotically tight net are uniquely determined by these marginal weak limits.

For the second we can use that the Skorokhod Borel σ -field is the projection σ -field, and the fact that cadlag processes are determined by their values at any dense

subset of $[0, 1]$ that contains 1. For t in the set S as in the addendum, marginal convergence follows by the continuous mapping theorem, as the almost sure equality $X(t) = X(t-)$ implies the a.s. continuity of the coordinate projection π_t , and π_1 is continuous everywhere (see Problem 1.14.4). It suffices to show that the set S in the addendum is dense, which certainly follows if it has an at most countable complement.

For a given measurable map X into $D[0, 1]$, consider the set $\bar{S}$ of all $t \in (0, 1]$ such that $X(t) \neq X(t-)$ with positive probability. This set is the union over $\delta, \varepsilon > 0$ of the sets $\{t: P(J_\varepsilon(t)) \geq \delta\}$, for $J_\varepsilon(t) = \{|X(t) - X(t-)| > \varepsilon\}$. Since we can count off over sequences $\delta_m \downarrow 0$ and $\varepsilon_m \downarrow 0$, the set $\bar{S}$ is certainly at most countable if each of these sets is finite. If it were true that $P(J_\varepsilon(t_n)) \geq \delta$ for an infinite sequence of distinct points t_n and given $\delta, \varepsilon > 0$, then $\limsup J_\varepsilon(t_n)$ would also have probability at least δ , and hence be nonempty. This implies that at least one realization of X would have infinitely many jumps of size at least ε , contradicting the assumption that X takes its values in $D[0, 1]$ (see Problem 1.14.1). ■

We present two types of characterizations of asymptotic tightness relative to the Skorokhod topology. The first is based on a modulus of continuity, and can be viewed a relaxation of the characterization of tightness relative to the uniform norm, as given in Theorem 1.5.7. The second exploits the structure of the time set $[0, 1]$ and is in terms of stopping times.

For a given element $z \in D[0, 1]$, define a *Skorokhod modulus* by

$$(1.14.4) \quad w(z, \delta) = \inf_{\substack{0=t_0 < t_1 < \dots < t_k=1 \\ \min_{1 \leq i \leq k} (t_i - t_{i-1}) > \delta}} \sup_{1 \leq i \leq k} \sup_{t \in [t_{i-1}, t_i]} |z(t) - z(t_{i-1})|.$$

The first infimum is taken over all partitions $0 = t_0 < t_1 < \dots < t_k = 1$ of mesh width bigger than δ . That these partitions may depend on z is important and distinguishes the modulus from similar devices in Chapter 1.5. Jump points of z will be obvious candidates for the points in a partition, as the jump sizes at the partitioning points do not count to the size of the modulus. However, the number of points in a partition is limited by the requirement that the distance between them must be at least δ .

1.14.5 Theorem (Tightness through modulus). *A net of arbitrary maps X_α in $D[0, 1]$ is asymptotically tight relative to the Skorokhod topology if and only if the net $X_\alpha(t)$ is asymptotically tight for every t in a set that is dense in $[0, 1]$ and includes 1, and for every $\varepsilon, \eta > 0$ there exists $\delta > 0$ such that*

$$(1.14.6) \quad \limsup_{\alpha} P^*(w(X_\alpha, \delta) > \varepsilon) < \eta.$$

In this case the net $\sup_{t \in [0, 1]} |X_\alpha(t)|$ is asymptotically tight in $\mathbb{R}$.

Proof. First consider the sufficiency of “dense marginal tightness” and (1.14.6) for asymptotic tightness.

Let $0 = s_0 < s_1 < \dots < s_l = 1$ be an arbitrary partition consisting (besides possibly $s_0 = 0$) of points t in the given dense set where $X_\alpha(t)$ is asymptotically

tight and such that $\max_j(s_j - s_{j-1}) \leq \delta$, for given $\delta > 0$. For $z \in D[0, 1]$, define the discretization $\Pi z = \sum_{j=1}^l z(s_{j-1})1_{[s_{j-1}, s_j)} + z(1)1_{\{1\}}$. We first show that

$$(1.14.7) \quad \begin{aligned} d(z, \Pi z) &\leq 2w(z, \delta) + \delta, \\ \|z\|_\infty &\leq \sup_{0 < j \leq l} |z(s_j)| + 2w(z, \delta). \end{aligned}$$

For any $\varepsilon > w(z, \delta)$, there exists a partition $0 = t_0 < t_1 < \dots < t_{k-1} < t_k = 1$ with $\min_i(t_i - t_{i-1}) > \delta$ and $|z(t) - z(t_{i-1})| < \varepsilon$, for every $t \in [t_{i-1}, t_i]$ and every $1 \leq i \leq k$. For $1 \leq i < k$ let s'_i be the smallest point s_j in the partition (s_j) with $s_j \geq t_i$. Complete this into a partition of $[0, 1]$ by defining $s'_0 = 0$ and $s'_k = 1$. Then $0 < s'_1 < \dots < s'_{k-1} < 1$, because $\min_i(t_i - t_{i-1}) > \delta$ and $\max_j(s_j - s_{j-1}) \leq \delta$. Let $\lambda \in \Lambda$ be the strictly increasing bijection of $[0, 1]$ that is linear on each interval $[s'_{i-1}, s'_i]$ and satisfies $\lambda(s'_i) = t_i$ for $1 \leq i < k$. Then $|\lambda(s'_i) - s'_i| = |t_i - s'_i| \leq \delta$, by construction of the grid (s_j) , and hence $\|\lambda - \text{id}\|_\infty \leq \delta$, by piecewise linearity. If $s'_{i-1} \leq s < s'_i$, then $t_{i-1} \leq \lambda(s) < t_i$, and hence $|z \circ \lambda(s) - z(t_{i-1})| < \varepsilon$. This implies that $|z \circ \lambda(s) - z(s_{j-1})| < 2\varepsilon$, for every $s_{j-1} \in [t_{i-1}, t_i]$ and $s \in [s_{j-1}, s_j]$, since $[s_{j-1}, s_j] \subset [s'_{i-1}, s'_i]$ for such s_{j-1} . Hence $\|z \circ \lambda - \Pi z\|_\infty < 2\varepsilon$, whence $d(z, \Pi z) \leq 2\varepsilon + \delta$ and the first claim in (1.14.7) follows. The second claim is clear from the triangle inequality and the fact that every interval $[t_{i-1}, t_i]$ of any partition used in the definition of $w(z, \delta)$ contains at least one point s_j with $j > 0$.

By the second inequality in (1.14.7) and (1.14.6), the variable $\|X_\alpha\|_\infty$ is bounded by $\max_{j>0} |X_\alpha(s_j)| + 2\varepsilon$, with inner probability at least $1 - \eta$, eventually. As the net of the latter variables is asymptotically tight by assumption, it follows that the net $\|X_\alpha\|_\infty$ is asymptotically tight.

Fix $\zeta > 0$ and a sequence $\varepsilon_m \downarrow 0$. Take a constant M such that $\limsup P^*(\|X_\alpha\|_\infty > M) < \zeta$, and for each $\varepsilon = \varepsilon_m$ and $\eta = 2^{-m}\zeta$, let $\delta = \delta_m$ satisfy (1.14.6). Without loss of generality we may choose $\delta_m \downarrow 0$, as $m \rightarrow \infty$. Take a partition $0 = s_0 < s_1 < \dots < s_l = 1$ as constructed in the preceding paragraph, for this δ_m . For the moment m is fixed and we do not let it appear in the notation. Let $z_1, \dots, z_p$ be the set of all functions in $D[0, 1]$ that are constant on each interval $[s_{j-1}, s_j]$ and take on only the values $0, \pm\varepsilon_m, \dots, \pm\lfloor M/\varepsilon_m \rfloor \varepsilon_m$. Let K_m be the union of the p closed d -balls of radius $3\varepsilon_m + \delta_m$ around the z_i . Then, for each fixed m , the two conditions

$$w(X_\alpha, \delta_m) \leq \varepsilon_m \quad \text{and} \quad \|X_\alpha\|_\infty \leq M$$

imply that $X_\alpha \in K_m$. Indeed, the discretization ΠX_α is a piecewise constant function, taking values in $[-M, M]$ and hence it is at uniform distance at most ε_m of one of the functions z_j . Then $d(X_\alpha, z_j) \leq d(X_\alpha, \Pi X_\alpha) + \|\Pi X_\alpha - z_j\|_\infty \leq 2\varepsilon_m + \delta_m + \varepsilon_m$, by (1.14.7).

Define $K = \cap_m K_m$. We claim that for every $\delta > 0$, there exists m such that $\cap_{i=1}^m K_i \subset K^\delta$. If not, then for every m there would exist z_m with $z_m \in \cap_{i=1}^m K_i$ and $z_m \notin K^\delta$. The sequence z_m would have a subsequence contained in one of the balls making up K_1 , a further subsequence eventually contained in one of the balls making up K_2 , and so on. The sequence formed by taking the first of the first subsequence, the second of the second subsequence, and so on, would eventually be

contained in a ball of radius $3\varepsilon_m + \delta_m$, which tends to zero, and hence be a Cauchy sequence. Its limit in the completion of $D[0, 1]$ would belong to the completion of K , contradicting the fact that $d(z_m, K) \geq \delta$, for every m . Conclude that if X_α is not in K^δ , then it is not in $\cap_{i=1}^m K_i$ for some m . By construction of the sets K_m , this implies that $\limsup P^*(X_\alpha \notin K^\delta) \leq \eta + \sum_{i=1}^m 2^{-m}\eta < 2\eta$, for every $\delta > 0$. (This part of the proof is the same as in the last paragraph of the proof of Theorem 1.5.6.)

By construction the set K is closed and totally bounded for the Skorokhod metric d , but it is not clear that K is d -compact, as the Skorokhod metric is not complete. To remedy this we employ the metric d_0 given in Problem 1.14.8, under which $D[0, 1]$ is complete. We first replace K_m by $K'_m = K_m \cap \{z: w(z, \delta_m) \leq \varepsilon_m\}$ for every m , and K by $K' = K \cap \cap_m \{z: w(z, \delta_m) \leq \varepsilon_m\}$, which still has $\limsup P^*(X_\alpha \notin (K')^\delta) < 2\eta$, in view of the preceding display. Since the set K is d -totally bounded, it can for every fixed m be covered with finitely many d -balls of radius δ_m^2 . The intersection of such a ball with K' is contained in a d_0 -ball of radius $5\delta_m + 2w(z, \delta_m) \leq 5\delta_m + 2\varepsilon_m$ (see Problem 1.14.9). Hence K' is covered by finitely many d_0 -balls of radius $5\delta_m + 2\varepsilon_m$, for every m , and therefore totally bounded and precompact relative to d_0 . This concludes the proof of sufficiency of the given conditions for asymptotic tightness.

For the proof of the necessity of (1.14.6), we may assume without loss of generality that $X_\alpha \rightsquigarrow X$, for some X , in view of Prohorov's theorem. Then the upper semicontinuity of the modulus $z \mapsto w(z, \delta)$ (see Problem 1.14.11) and the portmanteau theorem give that $\limsup P^*(w(X_\alpha, \delta) \geq \varepsilon) \leq P(w(X, \delta) \geq \varepsilon)$, for every fixed $\delta, \varepsilon > 0$. Because $w(z, \delta) \rightarrow 0$ as $\delta \rightarrow 0$, for every $z \in D[0, 1]$ (see Problem 1.14.11), the variables $w(X, \delta)$ tend to zero almost surely and in probability as $\delta \rightarrow 0$, and hence the right side tends to zero, for every $\varepsilon > 0$.

The necessity of the asymptotic tightness of marginals is clear if $X_\alpha \rightsquigarrow X$ for some X , because in that case $\pi_t X_\alpha \rightsquigarrow \pi_t X$ by the continuous mapping theorem, for every t such that the limit X is continuous almost surely at t , which includes all except at most countably many t . If X_α is asymptotically tight, then every subset has a further converging subnet, by Prohorov's theorem. The same argument is then valid provided the weak limit points share a dense set of continuity points. This is true, because they all concentrate on the σ -compact set from the definition of asymptotic tightness, and functions in a compact set possess at most countably many discontinuity points (see Problem 1.14.10). ■

For a sequence of variables X_n , condition (1.14.6) is equivalent to convergence $w(X_n, \delta_n) \rightarrow 0$ in probability, for every sequence $\delta_n \downarrow 0$ (see Problem 2.1.5 or 2.1.6).

As noted previously, the partition in the definition of the Skorokhod modulus $w(z, \delta)$ is allowed to depend on the function z . For a random modulus $w(X_n, \delta)$ the partition may be different for every sample path. The next theorem exploits the idea of random partitions directly. Given a fixed $\varepsilon > 0$ and a process X_n , define $\sigma_{n,0}^\varepsilon = \tau_{n,0}^\varepsilon = 0$ and, for $i = 1, 2, \dots$, with the infimum and supremum over the empty set equal to 1 and 0, respectively, and restricted to $t \in [0, 1]$,

$$\begin{aligned}\tau_{n,i}^\varepsilon &= \inf\{t \geq \tau_{n,i-1}^\varepsilon : |X_n(t) - X_n(\tau_{n,i-1}^\varepsilon)| > \varepsilon\}, \\ \sigma_{n,i}^\varepsilon &= \sup\{t \leq \tau_{n,i}^\varepsilon : |X_n(t) - X_n(\tau_{n,i}^\varepsilon)| \geq \varepsilon\}.\end{aligned}$$

The variable $\tau_{n,i}^\varepsilon$ looks forward in time and records the first time t in $[\tau_{n,i-1}^\varepsilon, 1]$ that $X_n(t)$ is outside the vertical band at height $[X_n(\tau_{n,i-1}^\varepsilon) - \varepsilon, X_n(\tau_{n,i-1}^\varepsilon) + \varepsilon]$. The variable $\sigma_{n,i}^\varepsilon$ looks backward in time from $\tau_{n,i}^\varepsilon$ and records the “first” time t when $X_n(t)$ is outside the band $(X_n(\tau_{n,i}^\varepsilon) - \varepsilon, X_n(\tau_{n,i}^\varepsilon) + \varepsilon)$.

Variability in the Skorokhod sense can be measured by the size of the intervals between these random times: long intervals indicate less variability. Specifically, the sequence X_n can be shown to be asymptotically tight in $D[0, 1]$ if and only if it is marginally asymptotically tight and both $X_n(1 - \delta_n) - X_n(1-) \xrightarrow{P} 0$ and

$$\sup_i P(\tau_{n,i}^\varepsilon - \sigma_{n,i-1}^\varepsilon < \delta_n, \tau_{n,i}^\varepsilon < 1) \rightarrow 0,$$

as $n \rightarrow \infty$, for every $\delta_n \downarrow 0$ and every fixed $\varepsilon > 0$ (see the proof of the next theorem and Problem 1.14.15).

The variables $\tau_{n,i}^\varepsilon$ are stopping times, whereas the $\sigma_{n,i}^\varepsilon$ are general random times, relative to the processes X_n . The next theorem formulates a sufficient criterion for asymptotic tightness in terms of general stopping and random times.

In the following theorem an *X-stopping time* is a nonnegative random variable τ defined on the same probability space as X such that for every $t \geq 0$ the event $\{\tau \leq t\}$ is contained in the σ -field generated by the variables $X(s)$, for $s \leq t$, possibly up to a null set.^b We call it finitely discrete if it assumes only a finite set of values within every given finite time interval. By a *random time* we understand any nonnegative random variable defined on the given probability space. In view of the cadlag sample paths (which make the process “progressively measurable”), for a random element X in $D[0, 1]$ the variable $X(\tau)$ is measurable for every given random time τ taking values in $[0, 1]$.

1.14.8 Theorem (Tightness through stopping). *A sequence of Borel measurable maps X_n in $D[0, 1]$ is asymptotically tight relative to the Skorokhod topology if it satisfies the following three conditions:*

- (i) *the sequence $X_n(t)$ is asymptotically tight for every t in a set that is dense in $[0, 1]$ and includes 1,*
- (ii) *$X_n(\delta_n) - X_n(0) \xrightarrow{P} 0$ and $X_n(1 - \delta_n) - X_n(1-) \xrightarrow{P} 0$, for every $\delta_n \downarrow 0$,*
- (iii) *for any finitely discrete X_n -stopping times $\tau_n \leq 1$, any deterministic numbers $\delta_n \downarrow 0$, and any random times v_n with $0 \leq v_n \leq \delta_n \wedge \tau_n$,*

$$|X_n((\tau_n + \delta_n) \wedge 1) - X_n(\tau_n)| \wedge |X_n(\tau_n) - X_n(\tau_n - v_n)| \xrightarrow{P} 0.$$

Proof. We first show that (1.14.6) is satisfied if $X_n(1 - \delta_n) - X_n(1-) \xrightarrow{P} 0$ and $P(\min_{i: \tau_{n,i}^\varepsilon < 1} (\tau_{n,i}^\varepsilon - \sigma_{n,i-1}^\varepsilon) < \delta_n) \rightarrow 0$, as $n \rightarrow \infty$, for any $\delta_n \downarrow 0$, for the stopping times $\tau_{n,i}^\varepsilon$ and random times $\sigma_{n,i}^\varepsilon$ defined in the discussion preceding the theorem (which depend on $\varepsilon > 0$), and every $\varepsilon > 0$. We drop the superscript ε from the notation.

For the moment also leave off the index n . Right-continuity of X and the definition of τ_i imply that $|X(\tau_i) - X(\tau_{i-1})| \geq \varepsilon$, if $\tau_i < 1$. Together with the definition

^b By standard theory the variables $\tau_{n,i}^\varepsilon$ defined previously are stopping times relative to the “usual filtration” (i.e. the completed, right-continuous version) of X_n , and not X_n -stopping times in general.

of σ_i this shows that $\sigma_i \geq \tau_{i-1}$ for $i \geq 1$, whence $\sigma_{i-1} \leq \tau_{i-1} \leq \sigma_i \leq \tau_i$, for all $1 \leq i \leq k := \max\{i: \tau_i < 1\}$. Furthermore, the definitions imply, for $i = 1, \dots, k+1$,

$$\sup_{\sigma_{i-1} < t \leq \tau_{i-1}} |X(t) - X(\tau_{i-1})| \leq \varepsilon, \quad \sup_{\tau_{i-1} \leq t < \tau_i} |X(t) - X(\tau_{i-1})| \leq \varepsilon.$$

By right-continuity of X , the interval $(\sigma_{i-1}, \tau_{i-1}]$ in the first supremum may be augmented with σ_{i-1} . (The half-open interval is empty if $\sigma_{i-1} = \tau_{i-1}$; then only the second inequality in the display will be used.) By combining the two inequalities, we see that the modulus $\sup\{|X(s) - X(t)|: s, t \in [\sigma_{i-1}, \tau_i]\}$ is bounded above by 2ε , for every $i \leq k+1$. The intervals $[\sigma_{i-1}, \tau_i]$ may overlap and not form a partition. We replace them by the intervals $[s_{i-1}, s_i]$, where $s_i = (\sigma_i + \tau_i)/2$, for $0 \leq i \leq k$ and $s_{k+1} = 1$. Since $\sigma_{i-1} \leq s_{i-1} \leq \tau_{i-1} \leq \sigma_i \leq s_i \leq \tau_i$, for every $1 \leq i \leq k$, the interval $[s_{i-1}, s_i]$ is contained in $[\sigma_{i-1}, \tau_i]$, for $i \leq k+1$. Hence the modulus of X over every interval $[s_{i-1}, s_i]$, for $i \leq k+1$, is bounded above by 2ε .

If $\min_{1 \leq i \leq k} (\tau_i - \sigma_{i-1}) > 2\delta$, then $s_i - s_{i-1} = (\tau_i - \sigma_{i-1})/2 + (\sigma_i - \tau_{i-1})/2 > \delta$, for $1 \leq i \leq k$. Hence $0 = s_0 < s_1 < \dots < s_k < s_{k+1} = 1$ is a partition as in the definition of $w(X, \delta)$, except that the final interval $[s_k, 1)$ may have length shorter than δ . If $1 - s_k > \delta/2$, then we still obtain the bound $w(X, \delta/2) \leq 2\varepsilon$. In the other case, that $1 - s_k \leq \delta/2$, we delete s_k from the partition, and obtain a partition $0 = s_0 < s_1 < \dots < s_{k-1} < 1$ with minimal spacing bigger than δ . Since $s_{k-1} < 1 - \delta < s_k < 1$ and the moduli of X over both $[s_{k-1}, s_k]$ and $[s_k, 1)$ are bounded above by 2ε , we obtain $|X(s_{k-1}) - X(1 - \delta)| \leq 2\varepsilon$ and $|X(s_k) - X(1 -)| \leq 2\varepsilon$ and hence $|X(s_{k-1}) - X(s_k)| \leq 4\varepsilon + |X(1 - \delta) - X(1 -)|$, by the triangle inequality. This shows that the modulus over the “new” interval $[s_{k-1}, 1)$ is bounded above by $8\varepsilon + |X(1 - \delta) - X(1 -)|$.

Thus we conclude that $\min_{1 \leq i \leq k_n} (\tau_{n,i} - \sigma_{n,i-1}) > 2\delta_n$, for $k_n = \max\{i: \tau_{n,i} < 1\}$, implies the bound $w(X_n, \delta_n/2) \leq 8\varepsilon + |X_n(1 - \delta) - X_n(1 -)|$. Together with the assumption that $X_n(1 - \delta_n) - X_n(1 -) \xrightarrow{P} 0$, this shows that (1.14.6) is satisfied if $P(\min_{i: \tau_{n,i} < 1} (\tau_{n,i} - \sigma_{n,i-1}) < \delta_n) \rightarrow 0$, as $n \rightarrow \infty$, for every $\delta_n \downarrow 0$, for every $\varepsilon > 0$. In the remainder of the proof we verify that (ii) and (iii) imply the latter condition.

For $1 \leq i \leq k_n := \max\{i: \tau_{n,i} < 1\}$, set $\Delta_{n,i} = (\tau_{n,i} - \tau_{n,i-1}) \wedge \delta_n$, and for $i \geq 2$ define

$$V_{n,i} = \sup_{0 \leq s \leq \delta_n \wedge \tau_{n,i-1}} |X_n(\tau_{n,i-1}) - X_n(\tau_{n,i-1} - s)|.$$

Then $\tau_{n,i-1} + \Delta_{n,i} = \tau_{n,i} \wedge (\tau_{n,i-1} + \delta_n)$ is the minimum of two stopping times and hence a stopping time. If $\tau_{n,i} - \tau_{n,i-1} < \delta_n$ and $1 \leq i \leq k_n$, then $\tau_{n,i-1} + \Delta_{n,i} = \tau_{n,i}$ and hence $|X_n(\tau_{n,i-1} + \Delta_{n,i}) - X_n(\tau_{n,i-1})| \geq \varepsilon$, by the definition of $\tau_{n,i}$. If $\tau_{n,i-1} - \sigma_{n,i-1} < \delta_n$ and $i \geq 2$, then $V_{n,i} \geq \varepsilon$, by the definition of $\sigma_{n,i-1}$.

If $\tau_{n,i} - \sigma_{n,i-1} < \delta_n$, then both $\tau_{n,i} - \tau_{n,i-1} < \delta_n$ and $\tau_{n,i-1} - \sigma_{n,i-1} < \delta_n$,

and hence combining the preceding gives, for every $i \geq 2$,

$$\begin{aligned} & P(\tau_{n,i} - \sigma_{n,i-1} < \delta_n, \tau_{n,i} < 1) \\ & \leq P\left(|X_n(\tau_{n,i-1} + \Delta_{n,i}) - X_n(\tau_{n,i-1})| \geq \varepsilon, V_{n,i} \geq \varepsilon, \tau_{n,i} < 1\right) \\ & \leq 4 \max_{\rho} \sup_{0 \leq s \leq 2\delta_n} P\left(|X_n(\rho + s) - X_n(\rho)| \geq \frac{\varepsilon}{2}, V_{n,i} \geq \varepsilon, \rho < 1\right), \end{aligned}$$

by Lemma 1.14.9 below, applied with $A = \{V_{n,i} \geq \varepsilon, \tau_{n,i} < 1\}$ and the sample paths of X_n extended as being constant beyond $[0, 1]$, and the maximum taken over all $\rho \in \{\tau_{n,i-1}, \tau_{n,i-1} + \Delta_{n,i}\}$. If we define $W_n(\rho) = \sup_{0 \leq s \leq 2\delta_n \wedge \rho} |X_n(\rho) - X_n(\rho - s)|$, then clearly $V_{n,i} \leq W_n(\rho)$, if $\rho = \tau_{n,i-1}$, as δ_n in $V_{n,i}$ is replaced by $2\delta_n$. Furthermore, since $\Delta_{n,i} \leq \delta_n$ and $|X_n(\tau_{n,i-1}) - X_n(\tau_{n,i-1} - s)| \leq |X_n(\rho) - X_n(\tau_{n,i-1})| + |X_n(\rho) - X_n(\tau_{n,i-1} - s)|$ by the triangle inequality, we have that $V_{n,i} \leq 2W_n(\rho)$ if $\rho = \tau_{n,i-1} + \Delta_{n,i}$. It follows that the preceding display is bounded above by

$$\begin{aligned} & 4 \max_{\rho} \sup_{0 \leq s \leq 2\delta_n} P\left(|X_n(\rho + s) - X_n(\rho)| > \frac{\varepsilon}{2}, \right. \\ & \quad \left. \sup_{0 \leq v \leq 2\delta_n \wedge \rho} |X_n(\rho) - X_n(\rho - v)| > \frac{\varepsilon}{2}, \rho < 1\right). \end{aligned}$$

For the given stopping times ρ , and $m \in \mathbb{N}$, the discretization ρ_m defined by $\rho_m = j/m$ if $(j-1)/m \leq \rho < j/m$, for $j = 1, \dots, m$, and $\rho_m = 1$ if $\rho \geq 1$, is a finitely discrete stopping time relative to the natural filtration of X_n with $\rho_m \downarrow \rho$ on the event $\{\rho < 1\}$, as $m \rightarrow \infty$. By the right-continuity of the process X_n , the preceding display is bounded above by the same expression but with the maximum replaced by the supremum over all ρ_m attached to the given pair of stopping times ρ , where the restriction $\rho < 1$ in the probability is deleted (cf. Problem 1.14.17.) Hence it is certainly bounded above by the expression with the supremum extended to all finitely discrete stopping times.

Write the supremum inside the probability as $|X_n(\rho) - X_n(\rho - v_n)|$, for a random time $v_n \leq 2\delta_n \wedge \rho$, and conclude that the preceding display tends to zero as $n \rightarrow \infty$ by assumption (iii) of the theorem.

This concludes the proof that $P(\tau_{n,i} - \sigma_{n,i-1} < \delta_n, \tau_{n,i} < 1) \rightarrow 0$, uniformly in $i \geq 2$. If $i = 1$, then $\sigma_{n,i-1} = \tau_{n,i-1} = 0$ and we cannot use the variables $V_{n,i}$ in the preceding, but must show that $P(\tau_{n,1} < \delta_n) \rightarrow 0$. If $\tau_{n,1} < \delta_n < 1$, then $\tau_{n,1}$ is equal to the stopping time $\Delta_{n,1} := \tau_{n,1} \wedge \delta_n$ and $|X_n(\Delta_{n,1}) - X_n(0)| \geq \varepsilon$, by the definition of $\tau_{n,1}$. By Lemma 1.14.9 with $\tau = \Delta_{n,1}$, $\sigma = 0$ and $A = \{\tau_{n,1} < \delta_n\}$ (and X extended beyond $[0, 1]$ as being constant), the probability of the event $\{|X_n(\Delta_{n,1}) - X_n(0)| \geq \varepsilon\} \cap A$ is bounded above by

$$\begin{aligned} & 4 \sup_{0 \leq s \leq 2\delta_n} P\left(|X_n(\Delta_{n,1} + s) - X_n(\Delta_{n,1})| \geq \frac{\varepsilon}{2}, \tau_{n,1} < \delta_n\right) \\ & \quad + 4 \sup_{0 \leq s \leq 2\delta_n} P\left(|X_n(s) - X_n(0)| \geq \frac{\varepsilon}{2}\right). \end{aligned}$$

The second supremum tends to zero by the first half of assumption (ii). The probability in the first supremum becomes bigger if $\tau_{n,1} < \delta_n$ is replaced by

$|X_n(\Delta_{n,1}) - X_n(0)| \geq \varepsilon$. We can approximate the stopping times $\Delta_{n,1}$ from above by finitely discrete stopping times, as before, and the resulting supremum is then seen to tend to zero in view of assumption (iii).

The final step of the proof is to strengthen the convergence $\sup_{i \geq 1} P(\tau_{n,i} - \sigma_{n,i-1} < \delta_n, \tau_{n,i} < 1) \rightarrow 0$ to the desired claim on the minimum value $\min_{i: \tau_{n,i} < 1} (\tau_{n,i} - \sigma_{n,i-1})$. Because the convergence is true for any $\delta_n \rightarrow 0$, we can first infer that for every $\eta > 0$, there exists δ such that $\limsup_n \sup_{i \geq 1} P(\tau_{n,i} - \sigma_{n,i-1} < 2\delta, \tau_{n,i} < 1) < \eta$. Next we choose k equal to the smallest integer bigger than $2/\delta$, and apply the inequality given in Lemma 1.14.10 to see that, still with $k_n = \max\{i: \tau_{n,i} < 1\}$,

$$P(k_n > k) = P(s_{n,k} < 1, k_n > k) \leq 2 \max_{i \leq k} P(s_{n,i} - s_{n,i-1} < \delta, k_n > k).$$

This is smaller than η , eventually, as $s_{n,i} - s_{n,i-1} \geq (\tau_{n,i} - \sigma_{n,i-1})/2$ and $k_n > k$ implies $\tau_{n,i} < 1$ for $i \leq k$. We combine this with the bound

$$P\left(\min_{i: \tau_{n,i} < 1} (\tau_{n,i} - \sigma_{n,i-1}) < \delta_n, k_n \leq k\right) \leq k \max_i P(\tau_{n,i} - \sigma_{n,i-1} < \delta_n, \tau_{n,i} < 1).$$

Since the right side tends to zero as $n \rightarrow \infty$, for any fixed k , it follows that $\limsup_n P(\min_{i: \tau_{n,i} < 1} (\tau_{n,i} - \sigma_{n,i-1}) < \delta_n) < \eta$, for every $\eta > 0$, whence it is zero. ■

1.14.9 Lemma. For any $\delta, \varepsilon > 0$ and random times σ, τ with $0 \leq \sigma \leq \tau \leq \sigma + \delta$, and any random element X in $D[0, \infty)$ and events A ,

$$P(|X(\tau) - X(\sigma)| \geq \varepsilon, A) \leq 4 \max_{\rho \in \{\sigma, \tau\}} \sup_{0 \leq s \leq 2\delta} P(|X(\rho + s) - X(\rho)| \geq \frac{\varepsilon}{2}, A).$$

Proof. By the triangle inequality, for any $s \geq 0$, the left side of the lemma is bounded above by

$$E \left[1_A 1_{|X(\sigma+s) - X(\sigma)| \geq \varepsilon/2} + 1_A 1_{|X(\sigma+s) - X(\tau)| \geq \varepsilon/2} \right].$$

Integrate over $s \in [0, \delta]$ and apply Fubini's theorem to see that the left side of the lemma is bounded above by

$$\begin{aligned} & \frac{1}{\delta} E \int_0^\delta (1_A 1_{|X(\sigma+s) - X(\sigma)| \geq \varepsilon/2} + 1_A 1_{|X(\sigma+s) - X(\tau)| \geq \varepsilon/2}) ds \\ & \leq \frac{1}{\delta} E \int_0^{2\delta} (1_A 1_{|X(\sigma+s) - X(\sigma)| \geq \varepsilon/2} + 1_A 1_{|X(\tau+s) - X(\tau)| \geq \varepsilon/2}) ds, \end{aligned}$$

by the change of variable $s \rightarrow \tau - \sigma + s$ in the second integral, followed by enlarging the range of integration. (The assumption that $\sigma \leq \tau \leq \sigma + \delta$ implies that $(\delta - \tau + \sigma, 2\delta - \tau + \sigma) \subset (0, 2\delta)$.) The two integrands have the same form, but refer to the stopping times σ and τ , respectively. We exchange the expectation and integral and bound the integrals by the supremum of the integrands times the integration range. ■

1.14.10 Lemma. *For any increasing sequence of random variables $0 = S_0 < S_1 < S_2 < \dots$, any event A , and any $k \in \mathbb{N}$ and $\delta > 0$ with $k\delta \geq 2$,*

$$P(S_k < 1, A) \leq 2 \max_{1 \leq i \leq k} P(S_i - S_{i-1} < \delta, S_k < 1, A).$$

Proof. We have $k\delta \leq \delta \sum_{i=1}^k 1_{S_i - S_{i-1} < \delta} + \sum_{i=1}^k (S_i - S_{i-1})$. The second sum telescopes out to S_k . Rearrange the inequality to see that $(k\delta - S_k) \leq \delta \sum_{i=1}^k 1_{S_i - S_{i-1} < \delta}$. On the event $S_k < 1$ the left side is bigger than $k\delta - 1 \geq k\delta/2$, by assumption. Integrate the resulting inequality over the intersection of this event with A . ■

The displayed condition in Theorem 1.14.8 is clearly implied by *Aldous's condition*: for any X_n -stopping times τ_n and any deterministic numbers $\delta_n \downarrow 0$,[#]

$$(1.14.11) \quad X_n(\tau_n + \delta_n) - X_n(\tau_n) \xrightarrow{P} 0.$$

Aldous's condition also implies the condition that $X_n(\delta_n) - X_n(0) \xrightarrow{P} 0$ in (ii) of the theorem, since the stopping times τ_n can be taken equal to 0. The second condition in (ii) remains.

We discuss applications of Aldous's condition within the context of processes with an unbounded time set in the next section.

1.14.2 Unbounded Domain

A Skorokhod topology can be defined on the space $D[0, T]$, for any $T > 0$, in complete analogy with the definition for $D[0, 1]$. An inconvenient property of the resulting spaces is that the restriction map from $D[0, T]$ to $D[0, S]$, for $S < T$, is continuous only at functions that are continuous at S (see Problem 1.14.18). This is because the Skorokhod topology on the bigger domain allows deformations of the full time set $[0, T]$ that may include a deformation just past the point S , whereas the transformations used in the definition of $D[0, S]$ are restricted to map the point S onto itself.

The discontinuity of the restriction map suggests that it would not be “correct” to define weak convergence in $D[0, \infty)$ as being equivalent to weak convergence of the restrictions in every space $D[0, T]$. On the other hand, the continuity of the restriction map at functions that are continuous at the end point, makes it natural that a definition of weak convergence in $D[0, \infty)$ should imply weak convergence of the restrictions to intervals with end point a continuity point of the limit process. It was seen in Theorem 1.14.2 that these continuity points exclude at most a countable set, so that they determine the full processes. We might therefore define convergence $X_n \rightsquigarrow X$ in $D[0, \infty)$ by the convergence $X_{n|[0, T]} \rightsquigarrow X_{|[0, T]}$ of the restricted processes in $D[0, T]$, for every T such that almost all sample paths of X are continuous at T .

[#] If needed, extend X_n beyond $[0, 1]$ as being constant.

However, as a definition of weak convergence in the context of the present chapter this would be flawed in that it does not refer to a metric on $D[0, \infty)$. One possibility to remedy this is the metric, for $x, y \in D[0, \infty)$,

$$d_\infty(x, y) = \int_0^\infty e^{-u} [d_u(x|_{[0, u]}, y|_{[0, u]}) \wedge 1] du,$$

where d_u is the Skorokhod metric on $D[0, u]$, defined for $u > 0$ in analogy to the metric on $D[0, 1]$ in Section 1.14.1, by

$$d_u(x, y) = \inf_{\lambda \in \Lambda_u} \left[\sup_{t \in [0, u]} |x(t) - y(\lambda(t))| + \sup_{t \in [0, u]} |t - \lambda(t)| \right],$$

where the infimum is taken over the set Λ_u of all strictly increasing bijections of $[0, u]$. The following lemma shows that $z_n \rightarrow z$ relative to this metric if and only if $z_n|_{[0, T]} \rightarrow z|_{[0, T]}$ in the Skorokhod topology on $D[0, T]$, for all continuity points T of z . The lemma also gives a characterization of convergence in terms of a sequence of deformations of the time set $[0, \infty)$, which is a natural analogue for convergence in $D[0, 1]$, and yet another in terms of a tapering device, which will be technically useful.

For $T > 1$ the “tapering function” $g_T: [0, \infty) \rightarrow \mathbb{R}$ is defined to be the continuous function that is equal to 1 on $[0, T - 1]$, equal to 0 on $[T, \infty)$, and linear on $[T - 1, T]$. The product $z g_T$ is identical to z on $[0, T - 1]$, but is close to zero near T , which ensures that for convergence of a sequence $z_n g_T$, the values of z_n near T are less important.

1.14.12 Lemma. *For any sequence z_n and element z in $D[0, \infty)$, the following statements are equivalent:*

- (i) $d_\infty(z_n, z) \rightarrow 0$.
- (ii) $z_n|_{[0, T]} \rightarrow z|_{[0, T]}$ in the Skorokhod topology on $D[0, T]$, for all continuity points T of z .
- (iii) there exists a sequence of strictly increasing bijections $\lambda_n: [0, \infty) \rightarrow [0, \infty)$ such that $z_n \circ \lambda_n \rightarrow z$ uniformly on compacta and $\lambda_n \rightarrow \text{id}$ uniformly on $[0, \infty)$.
- (iv) $(z_n g_T)|_{[0, T]} \rightarrow (z g_T)|_{[0, T]}$ in the Skorokhod topology on $D[0, T]$, for every T in a sequence $1 \leq T_i \uparrow \infty$ (and then for every $T \geq 1$).

Proof. (ii) $\Rightarrow$ (i). Convergence of $z_n|_{[0, u]} \rightarrow z|_{[0, u]}$ in the Skorokhod topology on $D[0, u]$ is by definition the same as convergence $d_u(z_n|_{[0, u]}, z|_{[0, u]}) \rightarrow 0$. Since any z has at most countably many points of discontinuity, convergence for every continuity point u implies that $d_\infty(z_n, z) \rightarrow 0$, by the dominated convergence theorem.

(i) $\Rightarrow$ (iii). Let $\underline{d}_\infty \leq d_\infty$ be the discrepancy obtained from d_∞ by replacing in the definition of every d_u the set Λ_u of all strictly increasing bijections of $[0, u]$ by the set of all strictly increasing bijections of $[0, \infty)$ (restricted to $[0, u]$). The corresponding discrepancies $\underline{d}_u \leq d_u$ (which are not semimetrics, as they lack symmetry) are nondecreasing in u , and hence convergence of the integral $d_\infty(z, z_n)$ to 0 implies that $\underline{d}_u(z, z_n) \rightarrow 0$, for every $u > 0$. Conclude that for every $u \in \mathbb{N}$ there exist strictly increasing bijections $\lambda_{n, u}$ of $[0, \infty)$ with $\varepsilon_{n, u} := \sup_{t \leq u} |z -$

$z_n \circ \lambda_{n,u}|(t) + \sup_{t \leq u} |\text{id} - \lambda_{n,u}|(t) \rightarrow 0$, as $n \rightarrow \infty$. Then there exists $u_n \rightarrow \infty$ such that $\varepsilon_{n,u_n} \rightarrow 0$, as $n \rightarrow \infty$. Now define λ_n to be equal to λ_{n,u_n} on $[0, u_n]$ and to be equal to $\text{id} + \lambda_{n,u_n}(u_n) - u_n$ on $[u_n, \infty)$. Then $\sup_{t > 0} |\text{id} - \lambda_n|(t) \leq \sup_{t \leq u_n} |\text{id} - \lambda_{n,u_n}|(t) \leq \varepsilon_{n,u_n} \rightarrow 0$, and, if $u \leq u_n$, then $\sup_{t \leq u} |z - z_n \circ \lambda_n|(t) \leq \varepsilon_{n,u_n} \rightarrow 0$.

(iii) $\Rightarrow$ (iv). Let λ_n be strictly increasing bijections of $[0, \infty)$ with $\lambda_n \rightarrow \text{id}$, uniformly, and $z_n \circ \lambda_n \rightarrow z$ uniformly on compacta. Then $\lambda_n(u - \varepsilon) \rightarrow u - \varepsilon < u$, for every $\varepsilon > 0$ and fixed u , whence there exists $\varepsilon_n \rightarrow 0$ with $\lambda_n(u - \varepsilon_n) < u$, for every n . Define $\lambda_{n,u}$ to be equal to λ_n on $[0, u - \varepsilon_n]$, to satisfy $\lambda_{n,u}(u) = u$, and to be continuous and linear on $[u - \varepsilon_n, u]$. Then $\lambda_{n,u} \in \Lambda_u$ satisfies $\lambda_{n,u} \rightarrow \text{id}$, uniformly on $[0, u]$, and trivially $z_n \circ \lambda_{n,u} = z_n \circ \lambda_n \rightarrow z$, uniformly on $[0, u - \varepsilon_n]$. Together with the fact that g_u is bounded and Lipschitz, this implies the uniform convergence of $(z_n g_u) \circ \lambda_{n,u} \rightarrow z g_u$ on $[0, u - \varepsilon_n]$. On $[u - \varepsilon_n, u]$ the functions $(z_n g_u) \circ \lambda_{n,u}$ and $z g_u$ are bounded above in absolute value by the uniform norms $\|z_n\|_u$ and $\|z\|_u$ of z_n and z over $[0, u]$ times the maxima of $g_u \circ \lambda_{n,u}$ and g_u over $[u - \varepsilon_n, u]$. The first are uniformly bounded by the convergence of $z_n \circ \lambda_n$ on compacta, and the latter tend to zero. Thus $(z_n g_u)|_{[0,u]} \rightarrow (z g_u)|_{[0,u]}$ in $D[0, u]$.

(iv) $\Rightarrow$ (ii). For every $S \leq T - 1$, the restriction of $z g_T$ to an interval $[0, S]$ is equal to $z|_{[0,S]}$. Thus the assertion follows from the continuity of the restriction map for every continuity point S of z (see Problem 1.14.18). ■

1.14.13 Theorem (Tightness from compacta). *A net X_α of random elements in $D[0, \infty)$ equipped with the metric d_∞ converges weakly to a random element X in $D[0, \infty)$ if and only if the net of restrictions $X_\alpha|_{[0,T]}$ converges weakly in $D[0, T]$ to $X|_{[0,T]}$, for every T such that almost all sample paths of X are continuous at T , or equivalently, for every T in some sequence $T_1 \leq T_2 \leq \dots \uparrow \infty$.*

Proof. Because the restriction map $z \mapsto z|_{[0,T]}$ is $d_\infty - d_T$ continuous at every z that is continuous at T (see Problem 1.14.18), weak convergence $X_\alpha \rightsquigarrow X$ implies weak convergence of the restricted processes, at continuity points, by the continuous mapping theorem.

To prove the converse, define g_T as in Lemma 1.14.12, so that $z_n \rightarrow z$ in $D[0, \infty)$ is implied by $(z_n g_T)|_{[0,T]} \rightarrow (z g_T)|_{[0,T]}$ in $D[0, T]$, for every T in a sequence (T_i) that increases to infinity. Then, by Theorem 1.3.14, applied with the set $\mathcal{G}$ of maps $z \mapsto (z g_T)|_{[0,T]}$, it suffices to show that $((X_\alpha g_{T_1})|_{[0,T_1]}, \dots, (X_\alpha g_{T_k})|_{[0,T_k]}) \rightsquigarrow ((X g_{T_1})|_{[0,T_1]}, \dots, (X g_{T_k})|_{[0,T_k]})$ in the space $\times_{i=1}^k D[0, T_i]$, for every $k \in \mathbb{N}$.

Choose every T in the sequence (T_i) to be a continuity point of X , so that $X_\alpha|_{[0,T]} \rightsquigarrow X|_{[0,T]}$, by assumption. By the triangle inequality and the fact that g_T is bounded and Lipschitz, it can be checked that $d_T(x g_T, y g_T) \leq d_T(x, y)(1 + \|x\|_T)$, for $\|\cdot\|_T$ the supremum norm on $[0, T]$. Thus the map $z \mapsto (z g_T)|_{[0,T]}$ from $D[0, T]$ in itself is continuous, and hence $(X_\alpha g_T)|_{[0,T]} \rightsquigarrow (X g_T)|_{[0,T]}$ in $D[0, T]$, by the continuous mapping theorem. The joint convergence of finitely many of these nets follows similarly. ■

The preceding theorem is easy to remember, and shows that, as for the space of locally bounded functions in Chapter 1.6, convergence in the space $D[0, \infty)$ does

not add much to convergence on multiple compact domains. However, the constraints imposed at the end points of the intervals $[0, T]$ by convergence in $D[0, T]$ (e.g. (ii) in Theorem 1.14.8) are unnatural when considering the full time scale $[0, \infty)$, even if they become automatic if the limit is continuous at T . For this reason in the following we also develop the theory for $D[0, \infty)$ without reference to the spaces of restrictions.

From the separability and Polishness of $D[0, T]$, it can be inferred that $D[0, \infty)$ is also separable and Polish (see Problem 1.14.24).

The following results are also direct generalizations of the corresponding results for $D[0, 1]$. Theorem 1.14.15 only differs from Theorem 1.14.2 in that the point 1 has lost its special role as the end point of the time set.

1.14.14 Lemma. *The Borel σ -field on $D[0, \infty)$ is equal to the σ -field generated by the coordinate projections.*

1.14.15 Theorem (Tightness plus marginal convergence). *A net of maps X_α in $D[0, \infty)$ converges weakly to a limit relative to the Skorokhod topology if and only if X_α is asymptotically tight and there exists a dense set $S \subset [0, \infty)$ such that the net $(X_\alpha(t_1), \dots, X_\alpha(t_k))$ converges in distribution in $\mathbb{R}^k$ for every $t_1, \dots, t_k \in S$.*

1.14.16 Addendum. *If $X_\alpha \rightsquigarrow X$, then S can without loss of generality be taken equal to the set of all t such that $X(t) = X(t-)$ almost surely. For any measurable map X into $D[0, \infty)$, the complement of this set is at most countable.*

Asymptotic tightness in $D[0, \infty)$ can be characterized in terms of moduli over compact subsets $[0, T] \subset [0, \infty)$ which are almost the same as the moduli for $D[0, T]$, the difference being a relaxation of the requirement near the end points T . For a given element $z \in D[0, \infty)$ and $T > 0$, define the “infinite-time” Skorokhod modulus by

$$w_{\infty, T}(z, \delta) = \inf_{\substack{0=t_0 < t_1 < \dots < t_{k-1} < T \leq t_k \\ \min_{1 \leq i \leq k} (t_i - t_{i-1}) > \delta}} \sup_{1 \leq i \leq k} \sup_{t \in [t_{i-1}, t_i]} |z(t) - z(t_{i-1})|.$$

The first infimum is taken over all partitions $0 = t_0 < t_1 < \dots < t_k$ with $t_k \geq T$ in which all spacings $t_i - t_{i-1}$ are bigger than δ . That t_k is allowed to be bigger than T makes it easier to maintain this minimal spacing, and differentiates the modulus from the modulus for a function in $D[0, T]$.

1.14.17 Theorem. *A net of arbitrary maps X_α in $D[0, \infty)$ is asymptotically tight relative to the Skorokhod topology if and only if the net $X_\alpha(t)$ is asymptotically tight for every t in a set that is dense in $[0, \infty)$, and for every $T, \varepsilon, \eta > 0$ there exists $\delta > 0$ such that*

$$\limsup_{\alpha} P^*(w_{\infty, T}(X_\alpha, \delta) > \varepsilon) < \eta.$$

In this case the net $\sup_{t \in [0, T]} |X_\alpha(t)|$ is asymptotically tight in $\mathbb{R}$, for every $T > 0$.

Proof. For $T \geq 1$, define a function g_T , as in Lemma 1.14.12, and for $z \in D[0, \infty)$ identify zg_T with its restriction $(zg_T)|_{[0, T]} \in D[0, T]$. By (iv) of Lemma 1.14.12

and Theorem 1.3.14, for asymptotic tightness of X_α , it suffices to prove that the net $X_{\alpha g_T}$ is asymptotically tight in $D[0, T]$, for every $T \in \mathbb{N}$.

Define $w_T(z, \delta)$ to be the Skorokhod modulus of a function $z \in D[0, T]$, in analogy to the modulus $w(z, \delta)$ of a function $z \in D[0, 1]$, defined in (1.14.4). We show below that, for any $z \in D[0, \infty)$ and $T \geq 1$ and $\delta \in (0, 1)$, and any partition $s_0 < s_1 < \cdots < s_{l-1} < T \leq s_l$ with $\max_j(s_j - s_{j-1}) < \delta$,

$$\begin{aligned} w_T(zg_T, \delta) &\leq 4\delta\|z\|_T + w_{\infty, T}(z, 2\delta), \\ \|z\|_T &\leq 2w_{\infty, T+1}(z, \delta) + \max_{j: 0 < j \leq l} |z(s_j)|, \end{aligned}$$

where $\|z\|_T$ is the uniform norm on $[0, T]$. The second inequality shows the asymptotic tightness of $\|X_\alpha\|_T$ under the conditions of the theorem. The asymptotic tightness of the net $X_{\alpha g_T}$ in $D[0, T]$, for every $T \geq 1$ is next clear from the first inequality and Theorem 1.14.5, extended from $D[0, 1]$ to $D[0, T]$.

To prove the first inequality, we use that $|(zg_T)(t) - (zg_T)(s)| \leq |z(t) - z(s)| + \|z\|_T|t - s|$, for any $0 \leq s < t < \infty$, and $|zg_T|(t) \leq \|z\|_T(T - t)$, for $t \in [T - 1, T]$. The first follows by the triangle inequality and the fact that g_T is bounded and Lipschitz, the second is clear from the definition of g_T .

If $\varepsilon > w_{\infty, T}(z, \delta)$, then there exists a partition $0 = t_1 < t_2 < \cdots < t_{k-1} < T \leq t_k$ with $\sup_{t \in [t_{i-1}, t_i]} |z(t) - z(t_{i-1})| < \varepsilon$ and $t_i - t_{i-1} > \delta$, for every $1 \leq i \leq k$. Since the modulus does not increase if intervals are subdivided into smaller ones, we may assume that also $t_i - t_{i-1} \leq 2\delta$, for every $1 \leq i \leq k$. If $T - t_{k-1} > \delta/2$, then we can use the same partition to bound $w_T(zg_T, \delta/2)$, and obtain by the inequality in the preceding paragraph that this is bounded above by $\varepsilon + \|z\|_T 2\delta$. If the partition has $T - t_{k-1} \leq \delta/2$, then we delete t_{k-1} and obtain the partition $0 = t_1 < t_2 < \cdots < t_{k-2} < T$, which has all intervals of length bigger than δ . By the inequality of the preceding paragraph we have $|(zg_T)(t) - (zg_T)(t_{k-2})| \leq \varepsilon + \|z\|_T 2\delta$, for every $t \in [t_{k-2}, t_{k-1})$. Since $T - \delta$ is contained in the latter interval, the triangle inequality gives that $|(zg_T)(t) - (zg_T)(t_{k-2})| \leq \varepsilon + \|z\|_T 2\delta + |(zg_T)(t) - (zg_T)(T - \delta)|$, for every $t \in [t_{k-1}, t_k)$. By the second inequality of the preceding paragraph this is further bounded by $\varepsilon + \|z\|_T 2\delta + \|z\|_T(\delta/2 + \delta)$, for $t \in [t_{k-1}, T]$.

The second inequality is a consequence of the fact that every interval in a partition $0 = t_0 < t_1 < \cdots < t_{k-1} < T + 1 \leq t_k$ for $w_{\infty, T+1}(z, \delta)$ that intersects $[0, T]$ contains at least one point s_j , for $j > 0$, since $\max_j(s_j - s_{j-1}) < \delta < \min_i(t_i - t_{i-1})$.

If $X_\alpha \rightsquigarrow X$ in $D[0, \infty)$ for some X , then the net of restrictions satisfies $X_{\alpha|_{[0, T]}} \rightsquigarrow X|_{[0, T]}$ in $D[0, T]$, for every T at which X is continuous almost surely, by the continuous mapping theorem and Lemma 1.14.12. Then the necessity of the asymptotic tightness of marginals and the condition on the modulus can be deduced from Theorem 1.14.2 and the fact that $w_{T, \infty}(X_\alpha, \delta) \leq w_T(X_\alpha, \delta)$, for w_T the modulus for $D[0, T]$. (Alternatively use Problem 1.14.13, but note that the modulus $w_{T, \infty}$ is not upper semi-continuous.) If X_α is asymptotically tight, then the necessity of the condition on the modulus can be seen by applying the preceding argument to subnets, in view of Prohorov's theorem. For the proof of necessity of marginal tightness, it can be argued that the set of weak limit points share a dense set of continuity points, as in Theorem 1.14.2. ■

1.14.18 Theorem. *A sequence of Borel measurable maps X_n in $D[0, \infty)$ is asymptotically tight relative to the Skorokhod topology if the sequence $X_n(t)$ is asymptotically tight for every t in a set that is dense in $[0, \infty)$, and $X_n(\delta_n) - X_n(0) \xrightarrow{P} 0$ for every $\delta_n \downarrow 0$, and for any finitely discrete, uniformly bounded sequence of X_n -stopping times τ_n , any deterministic numbers $\delta_n \downarrow 0$, and any random times v_n with $0 \leq v_n \leq \delta_n \wedge \tau_n$,*

$$|X_n(\tau_n + \delta_n) - X_n(\tau_n)| \wedge |X_n(\tau_n) - X_n(\tau_n - v_n)| \xrightarrow{P} 0.$$

Proof. This follows the same lines as the proof of Theorem 1.14.8, except that the special treatment of the partitions and stopping times near 1 is now unnecessary. ■

In comparison to Theorem 1.14.8, the preceding theorem does not require tightness of the sequence $X_n(1)$, nor the convergence to zero in probability of the sequences $X_n(1 - \delta_n) - X_n(1-)$. The latter two conditions are both necessary for weak convergence of the restrictions of the processes in $D[0, 1]$, and also sufficient given convergence in $D[0, \infty)$ (see Problem 1.14.23).

As in the case of Theorem 1.14.8, the main condition of Theorem 1.14.18, as well as the condition that $X_n(\delta_n) - X_n(0) \xrightarrow{P} 0$, are implied by Aldous's condition.

1.14.19 Corollary (Aldous's condition). *A sequence of Borel measurable maps X_n in $D[0, \infty)$ is asymptotically tight relative to the Skorokhod topology if the sequence $X_n(t)$ is asymptotically tight for every t in a set that is dense in $[0, \infty)$, and the sequence X_n satisfies Aldous's condition (1.14.11) for any finitely discrete, uniformly bounded sequence of X_n -stopping times τ_n and any deterministic numbers $\delta_n \downarrow 0$.*

1.14.2.1 Conditional Distribution Bounds

The following corollary gives a sufficient condition for asymptotic tightness in terms of the conditional distributions of increments given the past.

1.14.20 Corollary. *Given random elements X_n in $D[0, \infty)$ let $Z_n(\delta, \varepsilon, T)$ be a random variable such that for given $\delta, \varepsilon, T > 0$ and every points $0 < t_1 < \dots < t_k \leq T$,*

$$P(|X_n(t_k + \delta) - X_n(t_k)| > \varepsilon | X_n(t_1), \dots, X_n(t_k)) \leq Z_n(\delta, \varepsilon, T).$$

If $Z_n(\delta_n, \varepsilon, T) \xrightarrow{P} 0$ as $n \rightarrow \infty$ for every $\delta_n \downarrow 0$ and every fixed (ε, T) , and the sequence $X_n(t)$ is asymptotically tight for every t in a dense subset of $[0, \infty)$, then the sequence X_n is asymptotically tight in $D[0, \infty)$.

Proof. In view of Corollary 1.14.19, it suffices to show that Aldous's condition is satisfied. By the martingale convergence theorem and the assumption, $P(|X_n(t + \delta_n) - X_n(t)| > \varepsilon | \mathcal{F}_{n,t}) \leq Z_n(\delta_n, \varepsilon, T)$, almost surely, for $\mathcal{F}_{n,t}$ the

σ -field generated by $X_n(s)$, for $s \leq t$, and $t \leq T$. For any finitely discrete stopping times τ_n with values in $[0, T]$,

$$\begin{aligned} \mathbb{P}(|X_n(\tau_n + \delta_n) - X_n(\tau_n)| > \varepsilon) &= \sum_t \mathbb{P}(|X_n(t + \delta_n) - X_n(t)| > \varepsilon, \tau_n = t) \\ &\leq \sum_t \mathbb{E}[1_{\tau_n=t}(Z_n(\delta_n, \varepsilon, T) \wedge 1)], \end{aligned}$$

since the event $\{\tau_n = t\}$ is contained in $\mathcal{F}_{n,t}$. The right side is equal to $\mathbb{E}Z_n(\delta_n, \varepsilon, T) \wedge 1$ and tends to zero as $n \rightarrow \infty$, for every $\delta_n \downarrow 0$, by dominated convergence and the assumption. ■

The preceding corollary is particularly interesting for Markov processes, for which the condition reduces to a bound involving the transition probabilities of the process. An especially simple case is that of independent increment processes, as in the following example.

1.14.21 Example (Independent increments). Any sequence of processes X_n in $D[0, \infty)$ with independent increments so that, for every bounded (deterministic) sequence $t_n \geq 0$ and every nonnegative sequence $\delta_n \downarrow 0$,

$$X_n(t_n + \delta_n) - X_n(t_n) \xrightarrow{\mathbb{P}} 0,$$

satisfies Aldous's condition (1.14.11) and hence converges in distribution in $D[0, \infty)$ if its marginal distributions on a dense set in $[0, \infty)$ converge in distribution.

This result is immediate from the preceding corollary, where $Z_n(\delta, \varepsilon, T)$ can be taken equal to $\sup_{t \leq T} \mathbb{P}(|X_n(t + \delta) - X_n(t)| > \varepsilon)$. The displayed condition implies that $Z_n(\delta_n, \varepsilon, T)$ tends to zero in probability.

1.14.2.2 Mixed Moment Bounds

To a certain extent all criteria for tightness so far depend on more than marginal distributions. The following theorem gives a sufficient condition in terms of the joint distribution of sets of three variables. The condition appears to be much stronger than necessary, but it is useful in many examples, in view of its simplicity.

1.14.22 Theorem (Tightness under increment bounds). Let X_n be random elements in $D[0, \infty)$ such that for every $T > 0$ there are constants $p, r > 0$ and a continuous, strictly increasing map $F: [0, \infty) \rightarrow [0, \infty)$ such that for $0 \leq s < t < u \leq T$ and every n ,

$$\mathbb{E} \left(|X_n(u) - X_n(t)| \wedge |X_n(t) - X_n(s)| \wedge 1 \right)^p \leq |F(u) - F(s)|^{1+r}.$$

Then, for every $T > 0$, as $n \rightarrow \infty$ followed by $\delta \rightarrow 0$,

$$\sup_{t \in [0, T]} \sup_{\substack{0 \leq s < t < u \\ u-s < \delta}} |X_n(u) - X_n(t)| \wedge |X_n(t) - X_n(s)| \xrightarrow{\mathbb{P}} 0.$$

In particular, the sequence X_n is asymptotically tight in $D[0, \infty)$.

Proof. The asymptotic tightness follows from Theorem 1.14.18, since the first assertion implies the condition of the latter theorem (even for $\tau_n = t$ equal to any random time, not necessarily a discrete stopping time). The first assertion is proved using a chaining argument. It suffices to prove the assertion for F equal to the identity, because the general case can then be obtained by an application to the processes $X_n \circ F^{-1}$.

First consider the case that $p \geq 1$. For the moment drop the index n from the notation. For fixed $m = 1, 2, \dots$ and $T > 0$ define variables, with $t_{j,m} = j2^{-m}$,

$$\Delta_m = \sum_{j=1}^{3T2^m} |X(t_{j+1,m}) - X(t_{j,m})|^p \wedge |X(t_{j,m}) - X(t_{j-1,m})|^p \wedge 1.$$

Fix $j_0 \in \mathbb{N}$ and $k \in \mathbb{N}$ with $j_0 2^{-k} \leq 2T$, and define a set of ordered triplets of points in the grid of mesh width 2^{-m} by

$$\mathcal{S}_m = \{(s_1, s_2, s_3) \in 2^{-m}\mathbb{Z}^3: (j_0 - 1)2^{-k} \leq s_1 < s_2 < s_3 \leq (j_0 + 1)2^{-k}\}.$$

We first show by induction on m that for all $m \geq k$,

$$(1.14.23) \quad \max_{(s_1, s_2, s_3) \in \mathcal{S}_m} |X(s_3) - X(s_2)| \wedge |X(s_2) - X(s_1)| \wedge 1 \leq 3 \sum_{l=k}^m \Delta_l^{1/p}.$$

For $m = k$ the set $\mathcal{S}_k$ consists of only one triplet and necessarily $s_2 = s_3 - 2^{-k} = s_1 + 2^{-k}$. Then inequality (1.14.23) is clear from the definition of Δ_k . Given that the claim is true for m , consider $(s_1, s_2, s_3) \in \mathcal{S}_{m+1}$ and relate this to a triplet in $(s'_1, s'_2, s'_3) \in \mathcal{S}_m$ as follows. If s_i , which by its definition belongs to $2^{-m-1}\mathbb{Z}$, is also contained in the coarser grid $2^{-m}\mathbb{Z}$, then set $s'_i = s_i$. Otherwise both $s_i - 2^{-m-1}$ and $s_i + 2^{-m-1}$ belong to $2^{-m}\mathbb{Z}$; then set s'_i equal to the one of these points that minimizes the increment $\varepsilon_i := |X(s_i) - X(s'_i)|$ across the levels, so that the increment is equal to

$$|X(s_i) - X(s_i - 2^{-m-1})| \wedge |X(s_i) - X(s_i + 2^{-m-1})|.$$

Clearly the increment ε_i vanishes in the first case that s_i belongs to $2^{-m}\mathbb{Z}$; in all cases $\varepsilon_i \wedge 1 \leq \Delta_{m+1}^{1/p}$, by the definition of Δ_{m+1} . By two applications of the triangle inequality we see that, for $i = 1, 2$,

$$|X(s_{i+1}) - X(s_i)| \leq |X(s'_{i+1}) - X(s'_i)| + \varepsilon_{i+1} + \varepsilon_i.$$

Consequently, the minimum of the left side over $i = 1, 2$ is bounded above by $\min_{i \in \{1, 2\}} |X(s'_{i+1}) - X(s'_i)| + \varepsilon_1 + \varepsilon_2 + \varepsilon_3$.

The points s'_1 and s'_3 can be strictly smaller and bigger than s_1 or s_3 only if the latter points do not belong to $2^{-m}\mathbb{Z}$, which is not the case if s_1 or s_3 is equal to one of the boundary points $(j_0 \pm 1)2^{-k}$, as these belong to a coarser grid if $m \geq k$. Hence the new triplets (s'_1, s'_2, s'_3) maintain the boundaries on the coordinates in the definition of the sets $\mathcal{S}_m$. However, it is possible that the strict inequalities

$s_1 < s_2 < s_3$ are not inherited by (s'_1, s'_2, s'_3) , so that the latter triplet does not belong to $\mathcal{S}_m$. In this case the triplet (s'_1, s'_2, s'_3) is not included in the range of the maximum in (1.14.23), but the minimum of the two increments is zero and hence the triplet could be added without consequence. We conclude that the terms in the maximum (1.14.23) for $m + 1$ are bounded by the terms in the maximum for m plus $(\varepsilon_3 + \varepsilon_2 + \varepsilon_1) \wedge 1$. Since the latter sum is bounded by $\Delta_{m+1}^{1/p}$, the proof of the induction step, and hence of (1.14.23) is complete.

The sets $\mathcal{S}_m$ depend on j_0 . The triplets in the union of the sets $\mathcal{S}_m$, for $j_0 \in \mathbb{N}$ with $j_0 2^{-k} \leq 2T$, can approximate any triplet $s < t < u$ with $u - s < 2^{-k}$ and $t \in [0, T]$ from the right, as $m \rightarrow \infty$. The right side of (1.14.23) does not depend on j_0 . Take the maximum of the left side of (1.14.23) over $j_0 \in \mathbb{N}$ with $j_0 2^{-k} \leq 2T$, and next let $m \rightarrow \infty$ to see that, by right-continuity of the sample paths of X ,

$$\sup_{t \in [0, T]} \sup_{\substack{0 \leq s < t < u \\ u - s < 2^{-k}}} |X(u) - X(t)| \wedge |X(t) - X(s)| \wedge 1 \leq \sum_{l=k}^{\infty} \Delta_l^{1/p}.$$

Since $E\Delta_l^{1/p} \leq (E\Delta_l)^{1/p} \leq (2T2^{-r(l-1)})^{1/p}$, by assumption, the right side tends to zero in mean as $k \rightarrow \infty$. This is true uniformly in n , which has been suppressed from the notation, and concludes the proof for $p \geq 1$.

For $p \in (0, 1)$, we use the same approach, but we redefine Δ_m as a sum of minimal increments without the power p , and remove p (or rather $1/p$) also from the induction statement (1.14.23). Finally we take the left side of the preceding display to the power p , bound this by $(\sum_{l=k}^{\infty} \Delta_l)^p \leq \sum_{l=k}^{\infty} \Delta_l^p$, and take the expectation. ■

Since $(x \wedge y \wedge 1) \leq x^\alpha y^{1-\alpha}$, for every $x, y \geq 0$ and $\alpha \in [0, 1]$, the p th moment of the minimum of the increments in the theorem can be replaced by a mixed moment of order p . Furthermore, the moment condition (on the minimum) is equivalent to a probability bound on the increments (see Problem 1.14.26).

1.14.24 Example (Central limit theorem). Consider i.i.d. stochastic processes $X, X_1, X_2, \dots$ with values in $D[0, 1]$, zero means and finite marginal second moments, such that for given continuous, strictly increasing functions $F, G: [0, 1] \rightarrow [0, \infty)$ and some $r > 0$, and every $0 \leq s < t < u \leq 1$,

$$\begin{aligned} E(X(t) - X(s))^2 &\leq |F(t) - F(s)|^{1/2+r}, \\ E(|X(u) - X(t)| |X(t) - X(s)|)^2 &\leq |G(u) - G(s)|^{1+r}. \end{aligned}$$

Then $n^{-1/2} \sum_{i=1}^n X_i \rightsquigarrow S$ in $D[0, 1]$, for a zero-mean Gaussian process S with continuous sample paths and the same covariance function as X .

Since the limit process is continuous, the convergence is also in the stronger sense of weak convergence in $\ell^\infty[0, 1]$, so that the Skorokhod topology could be avoided in this example. However, the following proof is based on the mixed moment condition of the preceding theorem. To accommodate that the theorem refers to processes in $D[0, \infty)$ and not $D[0, 1]$, we first extend all processes to $[0, \infty)$ by setting $X(t) = X(1)$, for $t \geq 1$. Next we obtain weak convergence in $D[0, \infty)$,

which implies weak convergence in $D[0, 1]$ of the restrictions, since the limit process is continuous.

Marginal weak convergence is clear from the multivariate central limit theorem, while asymptotic tightness can be verified as follows. The fourth power of the minimum in Theorem 1.14.22 can be bounded by the product of the squares of the two increments, and the square of the increments of $S_n = n^{-1/2} \sum_{i=1}^n X_i$ can be written as double sums. Therefore, the moment in Theorem 1.14.22 with $p = 4$ is bounded by

$$\begin{aligned} & \mathbb{E} \left(|S_n(u) - S_n(t)| \wedge |S_n(t) - S_n(s)| \right)^4 \\ & \leq \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \sum_{l=1}^n \mathbb{E} [[X_{i,u} - X_{i,t}][X_{j,u} - X_{j,t}][X_{k,t} - X_{k,s}][X_{l,t} - X_{l,s}]]. \end{aligned}$$

All terms in the four-fold sum in which one of i, j, k, l appears exactly once give contribution zero, by the independence of the X_i and the assumption that they have mean equal to zero. Only the quadruplets with $i = j = k = l$ and the quadruplets that form two unequal pairs ($i = j \neq k = l$ or $i = k \neq j = l$ or $i = l \neq j = k$) contribute to the sum. It follows that the preceding display is bounded above by

$$\begin{aligned} & \frac{1}{n^2} \left[n \mathbb{E} [X_u - X_t]^2 [X_t - X_s]^2 + n(n-1) \mathbb{E} [X_u - X_t]^2 \mathbb{E} [X_t - X_s]^2 \right. \\ & \quad \left. + 2n(n-1) (\mathbb{E} [X_u - X_t][X_t - X_s])^2 \right]. \end{aligned}$$

By the Cauchy-Schwarz inequality the square expectation in the third term is bounded by the product of the expectations in the second term, which is bounded above by $|F(u) - F(t)|^{1/2+r} |F(t) - F(s)|^{1/2+r} \leq |F(u) - F(s)|^{1+r} F(1)^r$. Therefore, under the given conditions the right side of the display is bounded above by $K [G(u) - G(s) + F(u) - F(s)]^{1+r}$, for $K \geq 2\sqrt{6}F(1)^r$. Thus Theorem 1.14.22 applies with $p = 4$ and the function F in the theorem taken equal to $(F + G)K^{1/(1+r)}$.

By the first condition the process X and hence the limit process S is continuous in second mean relative to the Euclidean distance. A Gaussian mean-zero process with sample paths in $D[0, T]$ that is continuous in probability is automatically continuous almost surely. Indeed, the jump process $t \mapsto S(t) - S(t-)$ is necessarily a deterministic process (see Itô and Nisio (1968)), almost surely, and hence must vanish, as it is also zero-mean Gaussian. Thus the limit process S has continuous sample paths.

1.14.2.3 Counting Processes

For nondecreasing processes that increase only by jumps of size 1, asymptotic tightness is automatic, if the limit is of the same type. We show this by giving a direct proof of convergence based on Theorem 1.3.14.

1.14.25 Theorem. Suppose that $X_1, X_2, \dots$ and X are random elements in $D[0, \infty)$ that have nondecreasing sample paths that increase only by jumps of size 1. If $(X_n(t_1), \dots, X_n(t_k)) \rightsquigarrow (X(t_1), \dots, X(t_k))$ for every finite subset $t_1, \dots, t_k$ of a dense subset of $[0, \infty)$, then $X_n \rightsquigarrow X$ in $D[0, \infty)$.

Proof. Let $\mathbb{D}_0$ be the set of all $z \in D[0, \infty)$ that are nondecreasing and increase only by jumps of size 1, equipped with the Skorokhod metric d_∞ . Since every X_n and X take their values in $\mathbb{D}_0$, weak convergence $X_n \rightsquigarrow X$ in $D[0, \infty)$ and $\mathbb{D}_0$ are equivalent, by Theorem 1.3.10. We shall prove weak convergence in $\mathbb{D}_0$.

We claim that for any sequence $z_n \in \mathbb{D}_0$ and element $z \in \mathbb{D}_0$, the convergence $z_n(t) \rightarrow z(t)$ for every t in a dense, countable subset of $[0, \infty)$ implies that $d_\infty(z_n, z) \rightarrow 0$. Then the theorem is a consequence of Theorem 1.3.14, applied with $\mathcal{G}$ the set of coordinate projections $z \mapsto z(t)$, for t in the given countable dense subset.

To prove the claim, start by noting that $z_n(t) \rightarrow z(t)$ implies that $z_n(t) = z(t)$ eventually, by the fact that all functions take their values in the discrete set $\mathbb{N}_0$. This is also true simultaneously in t belonging to a given finite set.

Because $z \in \mathbb{D}_0$ can have only finitely many jumps in a finite interval, we can list the jump points of z in a given interval $[0, T]$ as $t_1, \dots, t_k$. Choose $T > 0$ that is itself not a jump point of z , and select points $u_i < t_i < v_i$ and $u_{k+1} > T$ from the given dense set on which z_n converges to z such that $0 < u_1 < v_1 < u_2 < \dots < v_k < T < u_{k+1}$ and such that z does not jump on $[T, u_{k+1}]$. Then for large enough n we have $z_n(t) = z(t)$ for every $t \in \{u_1, v_1, \dots, u_k, v_k, u_{k+1}\}$, and then necessarily $z_n(t) = z(t)$ on every interval $[v_i, u_{i+1}]$, for $i = 1, \dots, k$, as z is constant on these intervals and z_n is nondecreasing. Furthermore, z_n must have exactly one jump in each interval $[u_i, v_i]$, say at $t_{n,i}$. Let $\lambda_n: [0, T] \rightarrow [0, T]$ be the strictly increasing bijection with $\lambda_n(t_{n,i}) = t_i$, for $i = 1, \dots, k$, that is piecewise linear with knots only at the points $t_{n,i}$. Then $z_n \circ \lambda_n = z$ and $\sup_{t \leq T} |\lambda_n(t) - t| \leq \max_i |t_i - t_{n,i}| \leq \max_i (v_i - u_i)$. The last quantity can be made arbitrarily small by choice of the points u_i, v_i . We conclude that $d_T(z_n, z) \rightarrow 0$, for every continuity point T of z , and finish with an appeal to Lemma 1.14.12. ■

1.14.26 Example (Extreme value limit). Let $X_1, X_2, \dots$ be i.i.d. nonnegative random variables with a continuous distribution function F_0 that satisfies, for some $\gamma > 0$,

$$\lim_{t \downarrow 0} \frac{F_0(ct)}{F_0(t)} = c^\gamma, \quad \text{for all } c > 0.$$

Fix a sequence $a_n \rightarrow 0$ so that $nF_0(a_n) \rightarrow 1$, and define processes $\mathbb{N}_n$ by

$$\mathbb{N}_n(t) = \sum_{i=1}^n 1\{X_i \leq a_n t\}.$$

We shall show that the sequence $\mathbb{N}_n$ converges in distribution in $D[0, \infty)$ to the process $t \mapsto \mathbb{N}(t^\gamma)$, for a standard Poisson process $\mathbb{N}$ on $[0, \infty)$.

A function F_0 such that the limit on the left side of the preceding display exists is said to be *regularly varying* at 0, and the limit then necessarily has the form of

the right side of the equation. It is always possible to take $a_n = F_0^{-1}(1/n)$. It is also known that the condition is equivalent to the convergence $nF_0(a_nt) \rightarrow t^\gamma$, for $t > 0$, and is necessary and sufficient for the existence of a sequence a_n such that the sequence $a_n^{-1} \min(X_1, \dots, X_n)$ converges in distribution to a nondegenerate limit. The distribution function of the limit variable then necessarily has the form $1 - e^{-t^\gamma}$, for $t \geq 0$. (See Gnedenko (1943) or De Haan and Ferreira (2006) for these claims.)

The processes $\mathbb{N}_n$ and their claimed limit are counting processes, whence by Theorem 1.14.25, the sequence converges in distribution in $D[0, \infty)$ as soon as it converges marginally. A single marginal $\mathbb{N}_n(t)$ is binomially distributed with parameters n and $F_0(a_nt)$, and hence the convergence $\mathbb{N}_n(t) \rightsquigarrow \mathbb{N}(t^\gamma)$ follows by the classical Poisson convergence of binomial distributions, since $nF_0(a_nt) \rightarrow t^\gamma$ by assumption. This may be extended to the convergence of finite-dimensional marginals in various ways. A direct approach is to note that, for any $0 < t_1 < \dots < t_{k-1} < t_k$, the vector $Y_{n,k} := (\mathbb{N}_n(t_1), \mathbb{N}_n(t_2) - \mathbb{N}_n(t_1), \dots, \mathbb{N}_n(t_k) - \mathbb{N}_n(t_{k-1}))$ is conditionally multinomially distributed given $\mathbb{N}_n(t_k) = m$, with parameters m and $(p_{n,1}, \dots, p_{n,k})$, where, for $i = 1, 2, \dots, k$,

$$p_{n,i} = \frac{F_0(a_nt_i) - F_0(a_nt_{i-1})}{F_0(a_nt_k)} \rightarrow \left(\frac{t_i}{t_k}\right)^\gamma - \left(\frac{t_{i-1}}{t_k}\right)^\gamma.$$

Combined with the Poisson convergence of $\mathbb{N}_n(t_k)$, this shows that the probability of the event $\{Y_{n,k} = (y_1, \dots, y_k)\}$ satisfies, for $m = \sum_{i=1}^k y_i$,

$$\frac{m!}{\prod_{i=1}^k y_i!} \prod_{i=1}^k p_{n,i}^{y_i} \mathbb{P}(\mathbb{N}_n(t_k) = m) \rightarrow \prod_{i=1}^k \frac{(t_i^\gamma - t_{i-1}^\gamma)^{y_i}}{y_i!} e^{-(t_i^\gamma - t_{i-1}^\gamma)}.$$

The right side can be recognized as the distribution of the independent, Poisson increments of the process $\mathbb{N}(t^\gamma)$.

1.14.2.4 Semi-Martingales

A *locally square-integrable martingale* M is a random element of $D[0, \infty)$ for which there exists an increasing sequence of stopping times $0 \leq T_n \uparrow \infty$ such that the stopped process M^{T_n} , defined by $M^{T_n}(t) = M(T_n \wedge t)$, is a martingale consisting of square-integrable variables, for every n . A square-integrable martingale is an obvious example, as the stopping times can be taken equal to ∞ . The martingale property may be understood relative to any filtration $(\mathcal{F}_t)_{t \geq 0}$ to which M is adapted and which satisfies the “usual conditions” of right-continuity and completeness, for instance the completed, right-continuous version of the natural filtration of M .

The *predictable quadratic variation* of a locally square-integrable martingale M is the unique predictable process $t \mapsto \langle M \rangle(t)$ such that the process $M^2 - \langle M \rangle$ is a local martingale. The latter property can be seen to imply that the predictable quadratic variation *Lenglart-dominates* M^2 , meaning that $\mathbb{E}M^2(\tau) \leq \mathbb{E}\langle M \rangle(\tau)$, for every bounded stopping time τ (see Problem 1.14.27). *Lenglart’s inequality* (see e.g. Chapter I, Lemma 3.30 in Jacod and Shiryaev (1987)) then gives that, for every stopping time τ and every $\varepsilon, \eta > 0$,

$$\mathbb{P}\left(\sup_{t \in [0, \tau]} M^2(t) \geq \varepsilon\right) \leq \frac{\eta}{\varepsilon} + \mathbb{P}(\langle M \rangle(\tau) \geq \eta).$$

This is a useful device to verify the tightness of sequences of martingales, provided that the predictable quadratic variation processes can be suitably controlled. The latter should be easier than handling the original processes, as the predictable variation is nondecreasing and often “more continuous” than the original martingale.

1.14.27 Theorem (Tightness through predictable variation). *Suppose that X_n is a locally square-integrable martingale with $X_n(0) = 0$, with respect to some usual filtration $\mathcal{F}_{n,t}$, for every n . If the sequence of predictable quadratic variation processes $\langle X_n \rangle$ is asymptotically tight in $D[0, \infty)$ with all weak limit points having continuous sample paths, then the sequence X_n is asymptotically tight in $D[0, \infty)$.*

Proof. For a given finitely discrete stopping time τ_n , the process $M_n = X_n - X_n^{\tau_n} = \int_0^\cdot 1_{(\tau_n, \infty)} dX_n$ is a martingale, with predictable quadratic variation process $\langle M_n \rangle(t) = \langle X_n \rangle(t) - \langle X_n \rangle(\tau_n \wedge t)$. For numbers $\delta_n \downarrow 0$, we have $X_n(\tau_n + \delta_n) - X_n(\tau_n) = M_n(\tau_n + \delta_n)$. Therefore, by Lenglart’s inequality applied with the stopping time $\tau_n + \delta_n$,

$$\mathbf{P}\left(|X_n(\tau_n + \delta_n) - X_n(\tau_n)| \geq \varepsilon\right) \leq \frac{\eta}{\varepsilon^2} + \mathbf{P}\left(\langle X_n \rangle(\tau_n + \delta_n) - \langle X_n \rangle(\tau_n) \geq \eta\right).$$

The variable $\langle X_n \rangle(\tau_n + \delta_n) - \langle X_n \rangle(\tau_n)$ is bounded above by the modulus $\sup_{|s-t| < \delta_n} |\langle X_n \rangle(s) - \langle X_n \rangle(t)|$, where s and t may also be restricted to a bounded interval that contains the range of $\tau_n + \delta_n$. Since weak convergence in $D[0, \infty)$ to a limit process with continuous sample paths is equivalent to weak convergence in $\ell^\infty[0, T]$ of the restrictions of the processes, for every $T > 0$, the modulus tends to zero in probability, by Theorem 1.5.7 and its addendum.

This implies that the sequence X_n satisfies Aldous’s condition (1.14.11). Using Lenglart’s inequality in its standard form, we also see that the sequence $\sup_{t \leq T} X_n^2(t)$ is asymptotically tight if the sequence $\langle X_n \rangle(T)$ is asymptotically tight. In particular, the sequence $X_n(t)$ is uniformly tight for every t , if the sequence of predictable variation processes is asymptotically tight in $D[0, \infty)$.

Thus the theorem follows from Corollary 1.14.19. ■

1.14.28 Example (Gaussian limit). Suppose that X_n is a locally square-integrable martingale with $X_n(0) = 0$, for every n .

If $\langle X_n \rangle(t) \xrightarrow{\mathbf{P}} C(t)$, for every $t > 0$ and a continuous, deterministic function C , then the sequence X_n is asymptotically tight in $D[0, \infty)$.

This follows from the preceding theorem, because the pointwise convergence to the continuous function C automatically implies uniform convergence on compacta, which implies that the sequence of quadratic variation processes tends in distribution in $D[0, \infty)$ to the deterministic process C .

If, moreover, $\sup_{s \leq t} |\Delta X_n|(s) \xrightarrow{\mathbf{P}} 0$, then the sequence X_n tends in distribution to a limit in $D[0, \infty)$. The limit process is a continuous, Gaussian martingale with variance function C . Thus $X_n \rightsquigarrow X$ for a stochastic process X with normally distributed, independent increments and $\text{var } X(t) = C(t)$.

For a proof, see Jacod and Shiryaev (1987), Section VIII.3b, in particular Theorem 3.12, and Chapter VIII. The pair of conditions $\sup_{s \leq t} |\Delta X_n|(s) \xrightarrow{\mathbf{P}} 0$ and

$\langle X_n \rangle(t) \xrightarrow{P} C(t)$, for every $t > 0$ and a continuous, deterministic function C , can also be proved to be equivalent to convergence $[X_n](t) \xrightarrow{P} C(t)$, for every $t > 0$, of the (ordinary) quadratic variation processes of the X_n . Without the assumption that C is continuous and deterministic, the situation is much more complicated.

The form of the limit process is explained by Lévy's theorem, which says that a martingale with continuous sample paths and deterministic quadratic variation is necessarily a deterministically rescaled Brownian motion (and/or the fact that a Gaussian martingale necessarily has independent increments, as the martingale property implies zero correlations).

A *semimartingale* is a random element in $D[0, \infty)$ that can be decomposed as $X = X_0 + M + A$, for a local martingale M and an adapted process A of locally bounded variation, both 0 at 0. Without loss of generality the local martingale can be taken to be locally square-integrable, as in the preceding theorem. Thus under appropriate conditions on the processes A the preceding theorem extends to semimartingales. One difficulty in this approach is that addition is not continuous in the Skorokhod topology, so that tightness cannot easily be obtained from conditions on M and A separately. (Addition is continuous at pairs of functions of which one is continuous, see Problem 1.14.5, so this route is not always closed.)

The following theorem employs a more concrete decomposition of a semimartingale, which separates out its jumps. The *jump measure* of a semimartingale X is the random measure μ^X on $[0, \infty) \times \mathbb{R}$ that puts point masses of size 1 at every point $(t, \Delta X(t))$. Here t is the time of a jump in the sample path $t \mapsto X(t)$ and $\Delta X(t) = X(t) - X(t-)$ is the size of the jump. The *compensator* of this jump measure is the unique predictable random measure ν^X such that the process $H * (\mu^X - \nu^X)$, defined by $H * (\mu^X - \nu^X)(t) = \int_{[0, t]} \int H(s, y) (\mu^X - \nu^X)(ds, dy)$, is a local martingale, for every suitably integrable predictable process H . Given a uniformly bounded function $h: \mathbb{R} \rightarrow \mathbb{R}$ that is equal to the identity in a neighborhood of 0, any semimartingale X can be decomposed as (called “canonical representation” in Jacod and Shiryaev (1987), Chapter II, 2.34)

$$X = X(0) + X^c + h * (\mu^X - \nu^X) + \bar{h} * \mu^X + B^X,$$

where X^c is the continuous martingale part of X , the function $\bar{h}$ is defined by $h + \bar{h} = \text{id}$, and B^X is a predictable process of locally bounded variation. In the formula the functions h and $\bar{h}$ are identified with the predictable processes $(s, y) \mapsto h(y)$ and $(s, y) \mapsto \bar{h}(y)$, which are constant in time, and the processes $h * (\mu^X - \nu^X)$ and $\bar{h} * \mu^X$ are understood correspondingly. The idea of the decomposition is that the process $X - \bar{h} * \mu^X$, obtained by removing the process of “big jumps” $\bar{h} * \mu^X(t) = \sum_{0 < s \leq t} \bar{h}(\Delta X(s))$ from X , is more manageable and allows a decomposition in a martingale part $X^c + h * (\mu^X - \nu^X)$ and a remaining predictable, bounded variation part B^X . The function h is fixed in this decomposition and the process B^X will depend on it; the classical choice is $h(y) = y1_{|y| \leq 1}$.[†]

[†] For nice semimartingales X , the function h will be integrable relative to μ^X and ν^X in the usual sense and $h * (\mu^X - \nu^X)$ is the difference of the two integrals, but in general the latter integral must be understood in a martingale sense (see Jacod and Shiryaev (1987), Chapter II, §1d) and cannot be split into two parts.

The processes $(B^X, \langle X^c \rangle, \nu^X)$ are known as the *characteristic triplet* of the semimartingale X . The following theorem gives a sufficient condition for tightness of a sequence of semimartingales in terms of characteristic triplets. The first main condition limits the intensity of big jumps. A second main condition involves tightness (and asymptotic continuity) of the nondecreasing, predictable processes A^X , defined by

$$A^X(t) = \int_{(0,t]} |dB^X(s)| + \langle X^c \rangle(t) + \int_{(0,t]} \int (y^2 \wedge 1) \nu^X(ds, dy).$$

1.14.29 Theorem. *Suppose that X_n is a semimartingale with $X_n(0) = 0$, for every n . If the associated sequence of predictable processes A^{X_n} is asymptotically tight in $D[0, \infty)$ with all weak limit points having continuous sample paths and $\nu^{X_n}([0, T] \times \{y: |y| > K\}) \xrightarrow{P} 0$, as $n \rightarrow \infty$, followed by $K \rightarrow \infty$, for every fixed $T > 0$, then the sequence X_n is asymptotically tight in $D[0, \infty)$.*

Proof. Abbreviate μ^{X_n} to μ_n , etc. If the increments of a sequence of processes Y_n are bounded in absolute value by the increments of a sequence of processes Z_n , then $w_T(Y_n, \delta) \leq w_T(Z_n, \delta)$, for every $T, \delta > 0$. If, moreover, both processes are 0 at 0, then the marginals of Y_n are bounded in absolute value by the marginals of Z_n , and hence the sequence Y_n is asymptotically tight if the sequence Z_n is. Thus the assumption on A_n implies that the processes B_n , $\langle X_n^c \rangle$, and $g * \nu^X$, are asymptotically tight, for any function g with $g(y) \lesssim y^2 \wedge 1$. Because the weak limit points of A_n are continuous by assumption, we may apply this argument with the modulus for the uniform norm on compacta, and conclude that the weak limit points of each of these processes must have continuous sample paths as well.

For given $K > 0$, let $h_K: \mathbb{R} \rightarrow \mathbb{R}$ be a uniformly bounded function that is equal to the identity on $[-K, K]$, and let $\bar{h}_K = \text{id} - h_K$. Using h_K instead of h in the decomposition of X_n , gives a predictable process $B_{n,K}$ that relates to the given process B_n as $B_{n,K} - B_n = (h_K - h) * \nu_n$. Since $|h_K - h|(y) \lesssim y^2 \wedge 1$, the preceding observation shows that the sequence $|B_{n,K} - B_n|$ is asymptotically tight, whence the sequence $B_{n,K}$ is asymptotically tight, where in both cases we may understand this to be true relative to the uniform norm on compacta.

The process $\tilde{G}_{n,K}(t) := \nu_n([0, t] \times \{y: |y| > K\})$ is the compensator of the nondecreasing process $G_{n,K}(t) := \sum_{0 < s \leq t} 1_{|\Delta X_n(s)| > K}$, which implies that $\tilde{G}_{n,K}$ Lengart-dominates $G_{n,K}$. Lengart's inequality gives that, for every $T, K, \eta > 0$,

$$P\left(\sup_{0 < t \leq T} |\Delta X_n(t)| > K\right) = P(G_{n,K}(T) > 1) \leq \eta + P(\tilde{G}_{n,K}(T) > \eta).$$

By assumption the probability on the far right tends to zero as $n \rightarrow \infty$ followed by $K \rightarrow \infty$, for every $T, \eta > 0$. We conclude that the probability on the left tends to zero, as $n \rightarrow \infty$ followed by $K \rightarrow \infty$, for every $T > 0$. Since $\bar{h}_K = \text{id} - h_K$ vanishes on $[-K, K]$, we have that $\bar{h}_K * \mu_n(t) = \sum_{0 < s \leq t} \bar{h}_K(\Delta X(s)) \neq 0$, for some $t \leq T$, implies that $\sup_{0 < t \leq T} |\Delta X_n(t)| > K$. It follows that $\sup_{0 < t \leq T} |\bar{h}_K * \mu_n(t)| \xrightarrow{P} 0$, as $n \rightarrow \infty$ followed by $K \rightarrow \infty$.

The process $M_{n,K} = X_n^c + h_K * (\mu_n - \nu_n)$ is a locally square-integrable martingale, with predictable quadratic variation process equal to the process (the “modified second characteristic”, see Jacod and Shiryaev (1987), Chapter II, 2.17)

$$\langle M_{n,K} \rangle = \langle X_n^c \rangle + h_K^2 * \nu_n - \sum_{0 < s \leq \cdot} (\Delta B_n(s))^2.$$

The jumps of B_n are bounded in absolute value by the uniform bound on h (see Jacod and Shiryaev (1987), Chapter I, 4.24; B_n is the predictable process of locally bounded variation in the decomposition of the special semimartingale $X_n - \bar{h} * \mu^X$). It follows that the increments of $\langle M_{n,K} \rangle$ are bounded above by the increments of a multiple of A_n , and the sequence of processes $\langle M_{n,K} \rangle$ is asymptotically tight with continuous limits points, for every $K > 0$. By Theorem 1.14.27 the sequence $M_{n,K}$ is asymptotically tight in $D[0, \infty)$, for every K .

We conclude that, for every $K > 0$ the process X_n can be decomposed as $X_n = M_{n,K} + B_{n,K} + \bar{h}_K * \mu_n$, where $M_{n,K}$ is asymptotically tight, $B_{n,K}$ is asymptotically tight with continuous limit points, for every fixed K , and $\bar{h}_K * \mu_n$ tends uniformly to zero in probability as $n \rightarrow \infty$ followed by $K \rightarrow \infty$. By Prohorov’s theorem any subsequence of n has a further subsequence along which the sequence $(M_{n,K}, B_{n,K}, \bar{h}_K * \mu_n)$ converges in distribution. By a diagonalization procedure this subsequence can be chosen so that this is true for every $K \in \mathbb{N}$ simultaneously. By a second array procedure, we see that there is a sequence $K_n \rightarrow \infty$ such that the sequence $(M_{n,K_n}, B_{n,K_n}, \bar{h}_{K_n} * \mu_n)$ converges in distribution, along the subsequence, where the second and third coordinates of the limit are a continuous process and 0, respectively. Because the map $(z_1, z_2, z_3) \mapsto z_1 + z_2 + z_3$ is continuous at (z_1, z_2, z_3) such that z_2 and z_3 are continuous, we conclude that $X_n = M_{n,K_n} + B_{n,K_n} + \bar{h}_{K_n} * \mu_n$ converges in distribution along the subsequence, and hence the sequence X_n is asymptotically tight. ■

Problems and Complements

1. Every element of $D[0, 1]$ is uniformly bounded and has at most finitely many jumps of size bigger than a prescribed $\varepsilon > 0$.

[Hint: Every infinite sequence t_n has a subsequence that decreases or increases to a limit point; the values of an element of $D[0, 1]$ at this subsequence must converge.]

2. The Skorokhod metric is a true metric: it is symmetric, satisfies the triangle inequality, and $d(x, y) = 0$ if and only if $x = y$.

[Hint: The uniform norms of $x \circ \lambda - y$ and $x - y \circ \lambda^{-1}$ are equal.]

3. If a net $z_\alpha \in D[0, 1]$ satisfies $d(z_\alpha, z) \rightarrow 0$ relative to the Skorokhod metric for a limit $z \in C[0, 1]$, then $z_\alpha \rightarrow z$ uniformly.

[Hint: The uniform continuity of z implies that $z \circ \lambda_\alpha \rightarrow z$ uniformly if $\lambda_\alpha \rightarrow \text{id}$, uniformly.]

4. Let $\pi_t: D[0, 1] \rightarrow D[0, 1]$ be the coordinate projection $z \mapsto z(t)$.
 (i) For $t \in (0, 1)$ the map π_t is continuous at z if and only if z is continuous at t .
 (ii) The maps π_0 and π_1 are continuous everywhere.

[Hint: For the proof of “only if” in (i) note that if z is discontinuous at t , then $z_n = z \circ \lambda_n$, for $\lambda_n(t) = t - 1/n$ and $\lambda_n \in \Lambda$ linear on $[0, t]$ and $[t, 1]$, satisfies $z_n \rightarrow z$, but $z_n(t)$ does not converge to $z(t)$. Assertion (ii) is a consequence of the requirements $\lambda(0) = 0$ and $\lambda(1) = 1$ on $\lambda \in \Lambda$.]

5. The addition map $(z_1, z_2) \mapsto z_1 + z_2$ from $D[0, 1] \times D[0, 1] \rightarrow D[0, 1]$ is Borel measurable, but not continuous relative to the Skorokhod topology. However, it is continuous at every (z_1, z_2) such that at least one of z_1 or z_2 is continuous.

[Hint: Measurability follows from the characterisation of the Borel σ -field as the projection σ -field. If z is continuous and $z_n \rightarrow z$ in $D[0, 1]$, then $\|z - z_n \circ \lambda_n\|_\infty \rightarrow 0$, for any $\lambda_n \in \Lambda$ with $\|\lambda_n - \text{id}\|_\infty \rightarrow 0$, and so if one of the functions z_1 or z_2 is continuous, the deformations λ_n in the definition of the Skorokhod metric can be chosen based solely on the discontinuous function. For a counterexample, take $z_{1,n} = 1_{[1/2-1/n, 1]}$ and $z_{2,n} = -1_{[1/2+1/n, 1]}$. Then $z_{1,n} + z_{2,n} = 1_{[1/2-1/n, 1/2+1/n]}$ does not have a limit.]

6. The space $D[0, 1]$ is separable under the Skorokhod topology.

[Hint: The set of all step functions with steps of rational size taken at the rationals is dense.]

7. The space $D[0, 1]$ is not complete under the Skorokhod metric d .

[Hint: The sequence of functions $z_n = 1_{[t, t+1/2^n]}$, for some arbitrary $t \in [0, 1)$, is Cauchy, but does not have a limit. To see the first in the case that $t = 0$, note that $z_{n+1} \circ \lambda_n = z_n$, for λ_n the strictly monotone transformation that is linear on $[0, 2^{-n}]$ and on $[2^{-n}, 1]$ and satisfies $\lambda_n(2^{-n}) = 2^{-n-1}$, so that $d(z_{n-1}, z_n) \leq \|\lambda_n - \text{id}\|_\infty = 2^{-n-1}$ and hence $d(z_n, z_m) \leq 2^{-m+1}$, for $n \geq m$. To see the second, also when $t = 0$, note first that a limit z would have to vanish on $[\varepsilon, 1]$, for every fixed $\varepsilon > 0$, as all z_n for large enough n vanish on intervals $[\eta, 1]$, for $\eta > 0$. Thus z would have to be identically zero, by right-continuity, but $d(z_n, 0) = 1$, for every n .]

8. The space $D[0, 1]$ is Polish under the Skorokhod topology. For Λ_0 the set of all strictly increasing, Lipschitz bijections $\lambda: [0, 1] \rightarrow [0, 1]$ and $x, y \in D[0, 1]$, set

$$d_0(x, y) = \inf_{\lambda \in \Lambda_0} \left[\sup_{t \in [0, 1]} |x(t) - y(\lambda(t))| + \sup_{0 \leq s < t \leq 1} \left| \log \frac{\lambda(t) - \lambda(s)}{t - s} \right| \right].$$

Then d_0 is a metric that generates the Skorokhod topology and $D[0, 1]$ is complete under d_0 .

[Hint: Problems 1.14.9 and 1.14.11 show that d_0 is not stronger than d . The converse follows from the inequality $\|\lambda - \text{id}\| \leq e^{J(\lambda)} - 1$, where $\|\cdot\|$ is the uniform norm on $[0, 1]$ and $J(\lambda)$ is the second supremum in the definition of d_0 . The inequality follows from $|\lambda(t) - t| = t|(\lambda(t) - \lambda(0))/(t - 0) - 1|$ combined with the inequality $|u - 1| \leq e^{| \log u |} - 1$, for $u > 0$.]

To prove completeness, suppose that $z_n \in D[0, 1]$ is a sequence with $d_0(z_n, z_{n+1}) < 2^{-n}$, for every n . Thus there exist $\lambda_n \in \Lambda_0$ with $\|z_n \circ \lambda_n^{-1} - z_{n+1}\| < 2^{-n}$ and $J(\lambda_n) < 2^{-n}$. It follows that $\|\lambda_{n+j} \circ \lambda_{n+j-1} \circ \cdots \circ \lambda_n - \lambda_{n+j-1} \circ \cdots \circ \lambda_n\| = \|\lambda_{n+j} - \text{id}\| \leq e^{J(\lambda_{n+j})} - 1 < 2 \cdot 2^{-n-j}$, since $e^u - 1 \leq 2u$, for $0 \leq u \leq 1/2$, and hence the functions $\lambda_{n+j} \circ \lambda_{n+j-1} \circ \cdots \circ \lambda_n$, for $j = 1, 2, \dots$ and fixed n , form a Cauchy sequence relative to

the uniform norm. If μ_n is the limit, then $J(\mu_n) \leq \limsup_{j \rightarrow \infty} J(\lambda_{n+j} \circ \lambda_{n+j-1} \circ \dots \circ \lambda_n) \leq \sum_{j=0}^{\infty} J(\lambda_{n+j}) \leq 2^{-n+1}$, by sublinearity $J(\lambda_1 \circ \lambda_2) \leq J(\lambda_1) + J(\lambda_2)$ of J . Thus $\mu_n \in \Lambda_0$ and $J(\mu_n) \rightarrow 0$, as $n \rightarrow \infty$. By the definition of μ_n we have $\mu_{n+1} \circ \lambda_n = \mu_n$. Therefore $z_n \circ \mu_n^{-1} = z_n \circ \lambda_n^{-1} \circ \mu_{n+1}^{-1}$ and hence $\|z_n \circ \mu_n^{-1} - z_{n+1} \circ \mu_{n+1}^{-1}\| = \|z_n \circ \lambda_n^{-1} - z_{n+1}\| \leq 2^{-n}$. Thus $z_n \circ \mu_n^{-1}$ is also a Cauchy sequence relative to the uniform norm, and possesses a uniform limit z . It follows that $d_0(z_n, z) \rightarrow 0$. (Cf. Billingsley (1999), Theorems 12.1 and 12.2.)]

9. If $x, y \in D[0, 1]$ satisfy $d(x, y) < \delta^2 \leq 1/16$, then $d_0(x, y) \leq 5\delta + 2w(x, \delta)$.

[Hint: Fix x, y with $d(x, y) < \delta^2$ and $w(x, \delta) < \varepsilon$. So there exists $\lambda \in \Lambda$ with $\|x - y \circ \lambda\|_{\infty} \vee \|\lambda - \text{id}\|_{\infty} < \delta^2$ and a partition (t_i) with $\min_i(t_i - t_{i-1}) > \delta$ and $|x(t) - x(t_{i-1})| < \varepsilon$, for every $t \in [t_{i-1}, t_i]$ and i . Define $\mu(t_i) = \lambda(t_i)$ for every i and extend linearly to $\mu \in \Lambda$. Since $\lambda^{-1} \circ \mu$ is a strictly increasing bijection of $[0, 1]$ and $\lambda^{-1} \circ \mu(t_i) = t_i$, for every i , it maps every interval $[t_{i-1}, t_i]$ onto itself. Then $\|x - x \circ \lambda^{-1} \circ \mu\|_{\infty} < 2\varepsilon$ and $\|x \circ \lambda^{-1} \circ \mu - y \circ \mu\|_{\infty} < \delta^2$ and hence $\|x - y \circ \mu\|_{\infty} < 2\varepsilon + \delta^2$. Furthermore

$$|\mu(t_i) - \mu(t_{i-1}) - (t_i - t_{i-1})| = |\lambda(t_i) - \lambda(t_{i-1}) - (t_i - t_{i-1})| < 2\delta^2 < 2\delta(t_i - t_{i-1}).$$

By piecewise linearity of μ this extends to the increment over any interval $[s, t]$, and hence $|\mu(t) - \mu(s)/(t - s) - 1| < 2\delta$. Then $|\log(\mu(t) - \mu(s))/(t - s)| < 4\delta$, if $\delta \leq 1/4$, and hence $d_0(x, y) < 2\varepsilon + \delta^2 + 4\delta$.]

10. The set $\cup_{z \in K} \{t \in [0, T]: \Delta z(t) \neq 0\}$ of discontinuity points of the functions in a compact set $K \subset D[0, T]$ for the Skorokhod topology, is at most countable. The same is true for a compact subset of $D[0, \infty)$.

[Hint: It suffices to show that $\cup_{z \in K} \{t \in [0, T]: |\Delta z(t)| > \varepsilon\}$ is finite for every $\varepsilon > 0$. If it were not for some $\varepsilon > 0$, then there would exist infinite sequences $t_n \in [0, T]$ and $z_n \in K$ with $|\Delta z_n(t_n)| > \varepsilon$ for every n . By compactness there would be a converging subsequence $z_{n_j} \rightarrow z$ and hence $\lambda_{n_j} \in \Lambda$ such that $\Delta z_{n_j}(t_{n_j}) - \Delta(z \circ \lambda_{n_j})(t_{n_j}) \rightarrow 0$, but this would imply that z has infinitely many jumps bigger than $\varepsilon/2$. For a compact set $K \subset D[0, \infty)$, it is similarly seen that $\cup_{z \in K} \{t \in [0, T]: |\Delta z(t)| > \varepsilon\}$ is finite for every $\varepsilon, T > 0$, using the λ_n given in Lemma 1.14.12(iii).]

11. The Skorokhod modulus $w(z, \delta)$ defined in (1.14.4) satisfies:

- (i) If $z \in D[0, 1]$, then $w(z, \delta) \rightarrow 0$, as $\delta \rightarrow 0$.
- (ii) If $d(z_{\alpha}, z) \rightarrow 0$, then $\limsup w(z_{\alpha}, \delta) \leq w(z, \delta)$, for any $\delta > 0$.

[Hint: For (i) consider the sequence defined by $t_0 = 0$ and $t_i = \inf\{t > t_{i-1}: |z(t) - z(t_{i-1})| \geq \varepsilon\}$, for a given $\varepsilon > 0$. Then $|z(t) - z(t_{i-1})| \leq \varepsilon \leq |z(t_i) - z(t_{i-1})|$, for every $t \in [t_{i-1}, t_i]$ and hence $0 = t_0 < t_1 < \dots$. If t_i were defined for infinitely many i , with $t_i < 1$, then $t_i \uparrow t$ and $z(t_i) \rightarrow z(t-)$, for some $t \leq 1$, but this contradicts the definition of the t_i . Thus we find a finite partition $0 = t_0 < t_1 < \dots < t_k = 1$, with $|z(t) - z(t_{i-1})| \leq \varepsilon$, for every $t \in [t_{i-1}, t_i]$ and i , and $w(z, \delta) \leq \varepsilon$, for δ smaller than the mesh width of this partition. For (ii) transform a partition (t_i) used for $w(z, \delta)$ into the partition $s_{i,\alpha} = \lambda_{\alpha}(t_i)$, for a sequence $\lambda_{\alpha} \in \Lambda$ such that $z_{\alpha} \circ \lambda_{\alpha} \rightarrow z$ and $\lambda_{\alpha} \rightarrow \text{id}$, uniformly. Since $\min_i(t_i - t_{i-1}) > \delta$, we have $\min_i(s_{i,\alpha} - s_{i-1,\alpha}) > \delta$ eventually, and the modulus of z_{α} over $[s_{i-1,\alpha}, s_{i,\alpha})$ will be bounded above by the modulus of z over $[t_{i-1}, t_i)$ plus $o(1)$.]

12. The Skorokhod moduli for $D[0, \infty)$ and $D[0, T]$ satisfy $w_{T,\infty}(z, \delta) \leq w_T(z, \delta)$, for every $T, \delta > 0$ and cadlag $z: [0, \infty) \rightarrow \mathbb{R}$.

[Hint: A partition as in the definition of $w_T(z, \delta)$ is also eligible as a partition as in the definition of $w_{T,\infty}(z, \delta)$.]

13. The Skorokhod modulus $w_{T,\infty}(z, \delta)$ satisfies, for every $T > 0$:

- (i) If $z \in D[0, \infty)$, then $w_{T,\infty}(z, \delta) \rightarrow 0$, as $\delta \rightarrow 0$.
(ii) If $d_\infty(z_n, z) \rightarrow 0$, then $\limsup w_{T,\infty}(z_n, \delta) \leq w_{T',\infty}(z, \delta)$, for any $\delta > 0$ and $T' > T$. Statement (ii) cannot be strengthened to $\limsup w_{T,\infty}(z_n, \delta) \leq w_{T,\infty}(z, \delta)$.

[Hint: For (i) combine the preceding exercise with Problem 1.14.11. For (ii) take λ_n such that $z_n \circ \lambda_n - z \rightarrow 0$ uniformly on compacta and $\lambda_n - \text{id} \rightarrow 0$ uniformly on $[0, \infty)$. Given a partition $0 = t_0 < t_1 < \dots < t_{k-1} < T' \leq t_k$ with $\min_i (t_i - t_{i-1}) > \delta$ as in the definition of $w_{T'}(z, \delta)$, consider the points $t_{n,i} = \lambda_n(t_i)$. These satisfy $\min_i (t_{n,i} - t_{n,i-1}) > \delta$ eventually, and $t_{n,k} > T$ eventually, as $t_{n,k} \rightarrow t_k \geq T' > T$. Therefore there exists $l \leq k$ such that $0 = t_{n,0} < t_{n,1} < \dots < t_{n,l-1} < T \leq t_{n,l}$ is a partition as in the definition of $w_{T,\infty}(z_n, \delta)$. The variation of z_n over the partition is $\max_{i \leq l} \sup_{t \in [t_{i-1}, t_i]} |z_n \circ \lambda_n(t) - z_n \circ \lambda_n(t_{i-1})|$, which is bounded above by $\max_i \sup_{t \in [t_{i-1}, t_i]} |z(t) - z(t_{i-1})| + o(1)$. As a counterexample to (ii) with $T' = T$, consider $z_n = 1_{[T-1/n, \infty)} \rightarrow z = 1_{[T, \infty)}$, for which $w_{T,\infty}(z_n, \delta) = 1$, for every n and $w_{T,\infty}(z, \delta) = 0$. (It is also true that $w_T(z_n, \delta) = 1$, for every n , and $w_T(z, \delta) = 0$, but z_n does not converge to z in $D[0, T]$.)]

14. For $z \in D[0, 1]$ and $\delta > 0$, define $w_<(z, \delta)$ as $w(z, \delta)$, but with the infimum taken over all partitions $0 < t_1 < \dots < t_{k-1} < t_k = 1$ with $\max_{1 \leq i < k} (t_i - t_{i-1}) > \delta$. Then $w(z, \delta/2) \leq 3w_<(z, \delta) + |z(1 - \delta) - z(1 -)|$.

[Hint: We need to show that for every partition (t_i) of mesh width $> \delta$ as in the definition of $w_<$ there is a partition as in the definition of w , but with mesh width $> \delta/2$, such that the maximum of the moduli over the intervals in the partition is bounded by the right side. If the partition has $t_k - t_{k-1} > \delta/2$, then the same partition applies. If not, then delete t_{k-1} , and bound the modulus over $[t_{k-2}, 1)$ by the maximum of the moduli over $[t_{k-2}, t_{k-1}]$ and $[t_{k-1}, t_k]$ plus $|z(t_{k-1}) - z(t_{k-2})|$. The two moduli are bounded above by $w_<(z, \delta)$, while $|z(t_{k-1}) - z(t_{k-2})|$ is bounded above by $2w_<(z, \delta) + |z(1 - \delta) - z(1 -)|$, since $1 - \delta \in [t_{k-2}, t_k]$.]

15. The moduli $w(X_n, \delta_n)$ of a sequence of Borel measurable maps X_n in $D[0, 1]$ tend to zero in probability for every $\delta_n \downarrow 0$ if and only if both $P(\min\{\tau_{n,i}^\varepsilon - \sigma_{n,i-1}^\varepsilon: \tau_{n,i}^\varepsilon < 1\} \leq \delta_n) \rightarrow 0$, for every $\varepsilon > 0$, and $X_n(1 - \delta_n) - X_n(1 -) \xrightarrow{P} 0$ for every $\delta_n \downarrow 0$. (Cf. Ethier and Kurtz (1986), Proposition 3.8.3, for maps in $D[0, \infty)$.)

[Hint: That convergence of the minima and the variables $|X_n(1 - \delta_n) - X_n(1 -)|$ imply convergence of the moduli is proved within the proof of Theorem 1.14.8. For the other direction, see next problem for the increment $X_n(1 - \delta_n) - X_n(1 -)$. The necessity of the condition on the random times $\tau_{n,i}^\varepsilon - \sigma_{n,i-1}^\varepsilon$ follows from the following statement: if there exists i such that $\tau_{n,i}^\varepsilon - \sigma_{n,i-1}^\varepsilon \leq \delta_n$ and $\tau_{n,i}^\varepsilon < 1$, then $w(X_n, \delta_n) > \varepsilon/2$. To see the validity of this statement, drop the indices n and ε and consider any partition $0 = t_0 < t_1 < \dots < t_k = 1$ with $\min_i (t_j - t_{j-1}) > \delta$. If $t_{j-1} \leq \tau_i < t_j$, then $\sigma_{i-1} > t_{j-1}$ or $\tau_i < t_j$, or both, as otherwise $t_j - t_{j-1} \leq \tau_i - \sigma_{i-1} \leq \delta$. Then $\sup\{|X(s) - X(t)|: s, t \in [t_{j-1}, t_j]\}$ is bounded below by $\sup\{|X(\tau_{i-1}) - X(\sigma_{i-1} - s)|: 0 \leq s < \eta\}$, for some $\eta > 0$, or $|X(\tau_i) - X(\tau_{i-1})|$, or both, and these are both at least ε , by the definitions of σ_{i-1} and τ_i (in the cases that $\sigma_{i-1} > t_{j-1} \geq 0$ or $\tau_i < 1$). Finally the symmetric modulus is bounded above by $2 \sup\{|X(t) - X(t_{j-1})|: t \in [t_{j-1}, t_j]\}$, by the triangle inequality, which is bounded above by $2w(X, \delta) + \eta$, for some partition.]

16. Conditions (i) and (ii) of Theorem 1.14.8 are necessary for asymptotic tightness of X_n .

[Hint: The variables $|X_n(\delta_n) - X_n(0)|$ and $|X_n(1 - \delta_n) - X_n(1-)|$ are bounded above by $2w(X_n, \delta_n)$, because the intervals $[0, \delta_n]$ and $[1 - \delta_n, 1]$ are contained in the first and last intervals of any partition of minimal mesh width bigger than δ_n .]

17. If $z: [0, \infty) \rightarrow \mathbb{R}$ is cadlag and $\delta_n \downarrow 0$, then $\sup_{a \leq t \leq b} z(t + \delta_n) \rightarrow \sup_{a \leq t \leq b} z(t)$, for any $0 \leq a \leq b$. It is not necessarily true that $\sup_{a \leq t \leq b} |z(t + \delta_n) - z(t)| \rightarrow 0$.

[Hint: By right-continuity $\sup_{b \leq t \leq b + \delta_n} z(t) \rightarrow z(b)$ and hence $\sup_{a + \delta_n \leq t \leq b + \delta_n} z(t) = \sup_{a + \delta_n \leq t \leq b} z(t) + o(1)$. Similarly $\sup_{a \leq t \leq a + \delta_n} z(t) \rightarrow z(a) = z(a + \delta_n) + o(1)$.]

18. The restriction map $z \mapsto z|_{[0, S]}$ from $D[0, T]$ to $D[0, S]$, for $S < T$, is continuous at $z \in D[0, T]$ if and only if z is continuous at S .

[Hint: For simplicity consider $T = 2$ and $S = 1$ and suppose $z_n \rightarrow z$ in $D[0, 2]$. Then there are increasing bijections $\lambda_n: [0, 2] \rightarrow [0, 2]$ with $\lambda_n \rightarrow \text{id}$ and $z_n - z \circ \lambda_n \rightarrow 0$ uniformly on $[0, 2]$. Take points $t_n \rightarrow 1$ from below such that $\lambda_n(t_n) < 1$ (e.g. $t_n = (1 - 1/n) \wedge \lambda_n^{-1}(1 - 1/n)$), and define $\lambda_{n,1}$ to be λ_n on $[0, t_n]$, and to be linear and continuous on $[t_n, 1]$ with $\lambda_{n,1}(1) = 1$. Then $z_n - z \circ \lambda_{n,1} \rightarrow 0$ uniformly on $[0, t_n]$, and uniformly on $[t_n, 1]$ if z is left-continuous at 1, and then $z_n \rightarrow z$ in $D[0, 1]$. Conversely, if $z(1-) \neq z(1)$, then $z_n = z \circ \lambda_n$, for $\lambda_n(t) = (1 - 1/n)t$ on $[0, 1]$ and $\lambda_n(t) = 2 + (1 + 1/n)(t - 2)$ on $[1, 2]$, is an example of a sequence with $z_n \rightarrow z$ in $D[0, 2]$, but not in $D[0, 1]$, as $z_n(1) = z(1 - 1/n) \rightarrow z(1-)$.]

19. The sequence $1_{[0, 1 - 1/n]}$ tends to $1_{[0, 1]}$ in $D[0, 2]$ and $D[0, \infty)$, but does not converge in $D[0, 1]$.

[Hint: For $\lambda_n \in \Lambda$, we have $1_{[0, 1 - 1/n]} \circ \lambda_n = 1_{[0, t_n]}$, for $t_n = \lambda_n^{-1}(1 - 1/n) \in (0, 1)$. The sequence $1_{[0, t_n]}$ has a uniform limit only if $t_n = t$ is constant, eventually, but then $\lambda_n(t) = 1 - 1/n \rightarrow 1$ and also $\lambda_n(t) \rightarrow t$, but $t = t_n$ cannot be 1.]

20. If $z_n \rightarrow z$ in $D[0, 1]$ or $D[0, \infty)$, then $z_n(\delta_n) \rightarrow z(0)$, for every nonnegative $\delta_n \downarrow 0$, and hence $\sup_{0 \leq t < \delta_n} |z_n(t) - z_n(0)| \rightarrow 0$.

[Hint: If $\lambda_n \in \Lambda$ with $\lambda_n - \text{id} \rightarrow 0$ and $z_n - z \circ \lambda_n \rightarrow 0$ uniformly on $[0, 1]$, then $\sup_{0 \leq t < \delta_n} |z_n - z \circ \lambda_n|(t) \rightarrow 0$ and $\sup_{0 \leq t < \delta_n} |z \circ \lambda_n(t) - z(0)| \rightarrow 0$, since $\lambda_n(\delta_n) \rightarrow 0$.]

21. If $z_n \rightarrow z$ in $D[0, 1]$, then $z_n(1 - \delta_n) \rightarrow z(1-)$, for every strictly positive $\delta_n \downarrow 0$, and $z_n(1-) \rightarrow z(1-)$ and $z_n(1) \rightarrow z(1)$. (The convergence $z_n(1 - \delta_n) \rightarrow z(1-)$, for every strictly positive $\delta_n \downarrow 0$, is equivalent to $\sup_{1 - \delta_n < t < 1} |z_n(t) - z(1-)| \rightarrow 0$, for every such δ_n .) Consequently “ z_n may have no jumps right before 1 and must have a jump of almost equal size as z at 1.”

[Hint: If $\lambda_n \in \Lambda$ with $\lambda_n - \text{id} \rightarrow 0$ and $z_n - z \circ \lambda_n \rightarrow 0$ uniformly on $[0, 1]$, then $\sup_{1 - \delta_n < t < 1} |z_n - z \circ \lambda_n|(t) \rightarrow 0$ and $\sup_{1 - \delta_n < t < 1} |z \circ \lambda_n(t) - z(1-)| \rightarrow 0$, since $\lambda_n(1 - \delta_n) \rightarrow 1$, for every n . The second assertion follows from the first. The third is immediate from the fact that $\lambda(1) = 1$, for every $\lambda \in \Lambda$.]

22. If $z_n \rightarrow z$ in $D[0, \infty)$, then $z_n|_{[0, 1]} \rightarrow z|_{[0, 1]}$ in $D[0, 1]$ if and only if $z_n(1) \rightarrow z(1)$ and $z_n(1 - \delta_n) \rightarrow z(1-)$, for every strictly positive $\delta_n \downarrow 0$. (The last condition is equivalent to $\sup_{1 - \delta_n < t < 1} |z_n(t) - z(t)| \rightarrow 0$, and also to $\sup_{1 - \delta_n < t < 1} |z_n(t) - z_n(1-)| \rightarrow 0$, for every strictly positive $\delta_n \downarrow 0$.) A sufficient condition is that z is continuous at 1.

[Hint: The necessity is clear from Problem 1.14.21. For the sufficiency, take strictly increasing bijections $\lambda_n: [0, \infty) \rightarrow [0, \infty)$ with $\lambda_n \rightarrow \text{id}$ and $z_n - z \circ \lambda_n \rightarrow 0$, uniformly on

compacta. Then $\lambda_n(1 - \varepsilon) \rightarrow 1 - \varepsilon < 1$, for every $\varepsilon > 0$, and hence there exist strictly positive $\varepsilon_n \rightarrow 0$ with $\lambda_n(1 - \varepsilon_n) < 1$, for every n . Define $\lambda_{n,1}$ to be λ_n on $[0, 1 - \varepsilon_n]$, and to be continuous and linear on $[1 - \varepsilon_n, 1]$ with $\lambda_{n,1}(1) = 1$. Then $z_n - z \circ \lambda_{n,1} = z_n - z \circ \lambda_n \rightarrow 0$, uniformly on $[0, 1 - \varepsilon_n]$ by construction, and $z_n(1) - z \circ \lambda_{n,1}(1) = z_n(1) - z(1) \rightarrow 0$, by assumption. Furthermore $\sup_{1 - \varepsilon_n < t < 1} |z_n - z \circ \lambda_{n,1}|(t) = \sup_{1 - \varepsilon_n < t < 1} |z_n(t) - z(1)| + o(1)$, since $\lambda_{n,1}(1 - \varepsilon_n) \rightarrow 1$ and $\lambda_{n,1}(t) < 1$, for $t < 1$. This also tends to zero by assumption, and hence $z_n - z \circ \lambda_{n,1} \rightarrow 0$, uniformly on $[0, 1]$. If z is continuous at 1, then $z_n - z \circ \lambda_n \rightarrow 0$, implies that $z_n(t) \rightarrow z(1)$, uniformly in a shrinking neighborhood $(\lambda_n(1 - \varepsilon_n), \lambda_n(1 + \eta_n))$ of 1, which implies the conditions.]

23. If $X_n \rightsquigarrow X$ in $D[0, \infty)$, then $X_n|_{[0,1]} \rightsquigarrow X|_{[0,1]}$ if and only if $X_n(1) \rightsquigarrow X(1)$ and $X_n(1 - \delta_n) - X_n(1) \xrightarrow{P} 0$, for every strictly positive $\delta_n \downarrow 0$. A sufficient condition is that almost all sample paths of X are continuous at 1.

[Hint: Invoke almost sure representations $\tilde{X}_n \rightarrow \tilde{X}$, almost surely, for the convergence in $D[0, \infty)$, and Problem 1.14.22.]

24. The space $D[0, \infty)$ is separable and Polish.

[Hint: One possible metric is $d_{0,\infty}(x, y) = \sum_{u \in \mathbb{N}} e^{-u} [d_{0,u}(xg_u, yg_u) \wedge 1]$, where $d_{0,u}$ is the generalization of d_0 as in Problem 1.14.8 to $D[0, u]$. For a $d_{0,\infty}$ -Cauchy sequence z_n the sequence of restrictions $(z_n g_u)|_{[0,u]}$ is Cauchy in $D[0, u]$ for every $u \in \mathbb{N}$ and hence has a converging subsequence. Diagonalization gives a subsequence that works for every u . The limits are compatible as the restrictions of the functions $z_n g_u$ to $[0, u - 1]$ are equal to the restrictions of the z_n itself.]

25. For any increasing sequence of random variables $0 = S_0 < S_1 < S_2 < \dots$ such that $\sup_i S_i > 1$, and any $k \in \mathbb{N}$ and $\delta > 0$,

$$\begin{aligned} \mathbb{P}\left(\min_{i: S_i \leq 1} (S_i - S_{i-1}) < \delta\right) &\leq k \sup_{i \in \mathbb{N}} \mathbb{P}(S_i - S_{i-1} < \delta, S_i \leq 1) \\ &+ e \int_0^\infty k e^{-kt} \sup_{i \in \mathbb{N}} \mathbb{P}(S_i - S_{i-1} < t, S_i \leq 1) dt. \end{aligned}$$

[Hint: For $K = \max(i: S_i \leq 1)$, we can bound the left side by $\sum_{i=1}^k \mathbb{P}(S_t - S_{i-1} < \delta, S_i \leq 1) + \mathbb{P}(K > k)$. The sum is bounded by the first term on the right side. Because $\sum_{i=1}^k (S_i - S_{i-1}) = S_k \leq 1$ on the event $K > k$, the probability $\mathbb{P}(K > k)$ is bounded by

$$e \mathbb{E} \left[1_{K > k} \prod_{i=1}^k e^{-(S_i - S_{i-1})} \right] \leq e \prod_{i=1}^k [\mathbb{E} 1_{S_i \leq 1} e^{-k(S_i - S_{i-1})}]^{1/k},$$

by Hölder's inequality. Here $\mathbb{E} 1_{S_i \leq 1} e^{-k(S_i - S_{i-1})} = \mathbb{E} 1_{S_i \leq 1} \int_{S_i - S_{i-1}}^\infty k e^{-kt} dt$, which is bounded by the integral in the second term on the right, in view of Fubini's theorem.]

26. Suppose that, for some $q \geq 0$ and every $x \in (0, 1]$,

$$\mathbb{P}(|X_n(u) - X_n(t)| \wedge |X_n(t) - X_n(s)| > x) \leq x^{-q} |F(u) - F(s)|^{1+r}.$$

Then the condition of Theorem 1.14.22 is satisfied for $p > q$.

27. If $M = (M_t; t \geq 0)$ is a square-integrable local martingale, then $\mathbb{E} M_\tau^2 \leq \mathbb{E} \langle M \rangle_\tau$, for every stopping time τ .

[Hint: By assumption there exist stopping times $T_n \uparrow \infty$ such that M^{T_n} is a square-integrable martingale, for every n . By replacing T_n by $T_n \wedge n$, we can ensure that M^{T_n} is

L_2 -bounded. Then $(M^{T_n})^2 - \langle M^{T_n} \rangle = (M^{T_n})^2 - \langle M \rangle^{T_n}$ is a uniformly integrable martingale and hence $EM_{T_n \wedge \tau}^2 = E\langle M \rangle_{T_n \wedge \tau}$, by the optional stopping theorem. Let $n \rightarrow \infty$ and apply Fatou's lemma to the left side and the monotone convergence theorem to the right side to obtain that $EM_\tau^2 \leq E\langle M \rangle_\tau$.]

1.15

Stable Convergence

A sequence of arbitrary maps $X_n: \Omega \rightarrow \mathbb{D}$ defined on a (single) probability space $(\Omega, \mathcal{A}, P)$ with values in a metric space $\mathbb{D}$ *converges stably* if there exists a probability measure L on the product space $\mathbb{D} \times \Omega$ such that, for every $f \in C_b(\mathbb{D})$ and every bounded measurable map $Z: \Omega \rightarrow \mathbb{R}$,

$$(1.15.1) \quad E^* f(X_n) Z \rightarrow \int f(x) Z(\omega) dL(x, \omega).$$

More generally X_n is said to converge stably relative to a sub σ -field $\mathcal{A}_0 \subset \mathcal{A}$, if (1.15.1) is true for every bounded $\mathcal{A}_0$ -measurable map $Z: \Omega \rightarrow \mathbb{R}$. In the latter case the distribution L need be defined (and is identified there) on the sub σ -field $\mathcal{B} \times \mathcal{A}_0$ only, for $\mathcal{B}$ the Borel σ -field on $\mathbb{D}$.

In the special case that the limit law L is a product measure, the stable convergence is said to be *mixing*.

Taking $Z = 1$ in the definition, we see that stable convergence implies weak convergence $X_n \rightsquigarrow L_1$, for L_1 the marginal distribution of L on $\mathbb{D}$. Thus stable convergence is stronger than weak convergence. For the trivial σ -field $\mathcal{A}_0 = \{\emptyset, \Omega\}$ they are equivalent. On the other hand, the strongest form of stable convergence is weaker than convergence in probability.

1.15.2 Lemma. *If the sequence of arbitrary maps $X_n: \Omega \rightarrow \mathbb{D}$ converges in outer probability to a random element $X: \Omega \rightarrow \mathbb{D}$, then the sequence X_n converges $\mathcal{A}$ -stably to the measure L defined by $L(B \times A) = E(1_B(X)1_A)$.*

Proof. The continuous mapping theorem gives that $f(X_n)Z \rightarrow f(X)Z$ in outer probability, for every $f \in C_b(\mathbb{D})$ and bounded measurable map $Z: \Omega \rightarrow \mathbb{R}$. Consequently $E^* f(X_n)Z \rightarrow E f(X)Z$, by the dominated convergence theorem. The

given measure L satisfies $\int 1_B(x) 1_A(\omega) dL(x, \omega) = L(B \times A) = E 1_B(X) 1_A$, for all measurable sets A and B . This extends to the equality $\int f(x) Z(\omega) dL(x, \omega) = E f(X) Z$, for every $f \in C_b(\mathbb{D})$ and bounded measurable map $Z: \Omega \rightarrow \mathbb{R}$. Hence (1.15.1) is satisfied. ■

The limit measure L in the lemma can be described intuitively as the “law of $(X(\omega), \omega)$ if ω has law P ”. In the definition of stable convergence there is no limit variable X , but a probability distribution L on the product space $\mathbb{D} \times \Omega$. We note below that (for notational convenience) a limit variable X can always be defined, but in general this must involve extra randomness, not obtainable from a deterministic map $\omega \mapsto X(\omega)$ on Ω .

Taking f equal to 1 and $Z = 1_A$ in the definition of stable convergence, we see that $E 1_A \rightarrow \int 1_A(\omega) dL(x, \omega)$ and hence the marginal of L on Ω (the law of the variable ω in the preceding paragraph) is the original measure P . If $L(\mathbb{D}_0 \times \Omega) = 1$ for a complete separable subset $\mathbb{D}_0 \subset \mathbb{D}$, then we can disintegrate the measure L on $(\mathbb{D} \times \Omega, \mathcal{B} \times \mathcal{A}_0)$ as

$$dL(x, \omega) = L^{\mathcal{A}_0}(dx|\omega) P(d\omega),$$

where $L^{\mathcal{A}_0}(\cdot|\omega)$ is the conditional distribution of the first argument x given the second argument ω in the distribution L . The superscript $\mathcal{A}_0$ indicates that the disintegration is dependent on the σ -field: the map $\omega \mapsto L^{\mathcal{A}_0}(B|\omega)$ is measurable relative to $\mathcal{A}_0$, for every measurable subset $B \subset \mathbb{D}$.

Stable convergence can be recast as convergence to this conditional distribution, in the sense of weak convergence in $L_1(\Omega, \mathcal{A}_0, P)$ in the functional analytic sense[‡]: $U_n \xrightarrow{w} U$ if and only if $EU_n Z \rightarrow EU Z$, for every $Z \in L_\infty(\Omega, \mathcal{A}_0, P)$.

1.15.3 Lemma. *An arbitrary sequence of maps $X_n: \Omega \rightarrow \mathbb{D}$ defined on a probability space $(\Omega, \mathcal{A}, P)$ converges $\mathcal{A}_0$ -stably to L if and only if $E(f(X_n)^*|\mathcal{A}_0) \xrightarrow{w} Uf$ in $L_1(\Omega, \mathcal{A}_0, P)$, where $Uf(\omega) = \int f(x) L^{\mathcal{A}_0}(dx|\omega)$, for every $f \in C_b(\mathbb{D})$.*

Proof. The weak convergence $\xrightarrow{w}$ means that $EE(f(X_n)^*|\mathcal{A}_0)Z \rightarrow E(Uf)Z$, for every bounded $\mathcal{A}_0$ -measurable Z , and every $f \in C_b(\mathbb{D})$. The left side is $E^*f(X_n)Z$, while the limit can be written as $E(Uf)Z = \int f(x) Z(\omega) dL(x, \omega)$, by the disintegration noted preceding the lemma. ■

If the maps $X_n: \Omega \rightarrow \mathbb{D}$ are Borel measurable and tight, then the conditional expectations in the lemma can be written as $E(f(X_n)|\mathcal{A}_0) = \int f dP^{X_n|\mathcal{A}_0}$, for a regular version $P^{X_n|\mathcal{A}_0}$ of the conditional distribution of X_n given $\mathcal{A}_0$. The lemma then shows that stable convergence is equivalent to convergence of the conditional distributions $P^{X_n|\mathcal{A}_0}$ to the conditional measure $L^{\mathcal{A}_0}$, in a weak sense, defined as follows.

Say that a sequence of Markov kernels $(\omega, B) \mapsto L_n(B|\omega)$ from $(\Omega, \mathcal{A}_0)$ into $\mathbb{D}$ converges weakly to a Markov kernel L , notation $L_n \xrightarrow{w} L$, if and only if, for every $f \in C_b(\mathbb{D})$ and every bounded measurable $Z: \Omega \rightarrow \mathbb{R}$,

$$\int \int f(x) L_n(dx|\omega) Z(\omega) dP(\omega) \rightarrow \int \int f(x) L(dx|\omega) Z(\omega) dP(\omega).$$

[‡] So not the convergence in distribution that is understood to be “weak convergence” in most of this manuscript.

Because the maps $\omega \mapsto \int f(x) L_n(dx|\omega)$ and $\omega \mapsto \int f(x) L(dx|\omega)$ are assumed $\mathcal{A}_0$ -measurable, requiring convergence only for every bounded $\mathcal{A}_0$ -measurable Z gives an equivalent definition.

For notational convenience the limit $L^{\mathcal{A}_0}$ may be written as the conditional distribution of a random variable. This works best if the measure L is defined on the full σ -field $\mathbb{B} \times \mathcal{A}$ on $\mathbb{D} \times \Omega$. Define a variable X on the probability space $(\mathbb{D} \times \Omega, \mathcal{B} \times \mathcal{A}, L)$ by $X(x, \omega) = x$. Then, for every $f \in C_b(\mathbb{D})$ and $A \in \mathcal{A}_0$,

$$\mathbb{E} \int f dP^{X|\mathbb{D} \times \mathcal{A}_0} 1_A = \mathbb{E} \mathbb{E}(f(X) | \mathbb{D} \times \mathcal{A}_0) 1_A = \mathbb{E} f(X) 1_A = \int f(x) 1_A(\omega) dL(x, \omega).$$

Thus the conditional law $P^{X|\mathbb{D} \times \mathcal{A}_0}$ of X given the σ -field $\mathbb{D} \times \mathcal{A}_0$ is a version of the disintegration $L^{\mathcal{A}_0}(\cdot|\omega)$.

With this notation, stable convergence obtains the following pleasing formulation.

1.15.4 Theorem (Stable and conditional). *A sequence of Borel measurable maps $X_n: \Omega \rightarrow \mathbb{D}$ defined on a probability space $(\Omega, \mathcal{A}, P)$ with values in a Polish space $\mathbb{D}$ converges $\mathcal{A}_0$ -stably to a measure L if and only if $P^{X_n|\mathcal{A}_0} \xrightarrow{w} P^{X|\mathbb{D} \times \mathcal{A}_0}$, for $X: (\mathbb{D} \times \Omega, \mathcal{B} \times \mathcal{A}, L) \rightarrow \mathbb{D}$ defined by $X(x, \omega) = x$.*

Proof. We can write $\mathbb{E} f(X_n) Z = \mathbb{E} \int f dP^{X_n|\mathcal{A}_0} Z$, for every $\mathcal{A}_0$ -measurable Z , and the display preceding the theorem shows that $\int f(x) Z(\omega) dL(x, \omega) = \mathbb{E} f(X) Z = \mathbb{E} \int f dP^{X|\mathbb{D} \times \mathcal{A}_0} Z$, for every $\mathcal{A}_0$ -measurable Z . Because all variables $\int f dP^{X_n|\mathcal{A}_0}$ and $\int f dP^{X|\mathbb{D} \times \mathcal{A}_0}$ are $\mathcal{A}_0$ -measurable, convergence of these variables in the spaces $L_1(\Omega, \mathcal{A}_0, P)$ or $L_1(\Omega, \mathcal{A}, P)$ are the same. ■

The convergence in the theorem is often referred as “stable convergence of X_n to X .” This terminology can be misleading in that the structure of X as a map on $\mathbb{D} \times \Omega$ matters, or at least the Markov kernel $P^{X|\mathbb{D} \times \mathcal{A}_0}$ that X induces on $\mathbb{D}$. This conditional distribution cannot be recovered from X alone, let alone from its law, or the weak convergence $X_n \rightsquigarrow X$.

In the case that $\mathcal{A}_0 = \mathcal{A}$, the conditional distributions $P^{X_n|\mathcal{A}}$ are the Dirac measures δ_{X_n} . In that case, still the stable limit $P^{X|\mathbb{D} \times \mathcal{A}}$ may be a true random distribution. In fact, the limit is degenerate if and only if the convergence is in probability (see Problem 1.15.1).

Weak convergence in the sense of Chapter 1.3 depends on the distributions of the maps X_n only, and their definitions on the underlying probability spaces are irrelevant, apart from the determination of the outer expectations when the maps are nonmeasurable. In contrast, stable convergence involves the definitions of the variables on the (single) underlying probability space. The following theorem shows that it can nevertheless be described in terms of weak convergence, of joint variables.

Say that the limit L in stable convergence is *tight* if its induced marginal law $B \mapsto \int 1_B(x) dL(x, \omega)$ is tight on $\mathbb{D}$.

1.15.5 Theorem (Stable and weak). For arbitrary maps $X_n: \Omega \rightarrow \mathbb{D}$ defined on a probability space $(\Omega, \mathcal{A}, P)$, the following statements are equivalent:

- (i) the sequence X_n converges $\mathcal{A}_0$ -stably to a tight limit;
- (ii) for every $\mathcal{A}_0$ -Borel measurable map $Y: \Omega \rightarrow \mathbb{E}$ with values in some Polish space $\mathbb{E}$, the sequence (X_n, Y) converges in distribution to a tight limit in the space $\mathbb{D} \times \mathbb{E}$;
- (iii) the sequence X_n is asymptotically tight and the sequence $E^*f(X_n)1_A$ converges to a limit for every $f \in C_b(\mathbb{D})$ and every set A in an intersection stable generator of $\mathcal{A}_0$ that includes Ω .

For L the limit law in (i), the limit distribution in (ii) is the law of $(x, Y(\omega))$ if (x, ω) is distributed according to L . It is also the distribution of a pair (x, y) generated by drawing $\omega \sim P$ and next drawing $x \sim L(\cdot | \omega)$ and setting $y = Y(\omega)$.

Proof. If $g \in C_b(\mathbb{E})$ and Y is an $\mathcal{A}_0$ -measurable map in $\mathbb{E}$, then $Z = g(Y)$ is a bounded, $\mathcal{A}_0$ -measurable map on Ω . Hence if (i) holds and X_n converges $\mathcal{A}_0$ -stably to L , then $E^*f(X_n)g(Y) \rightarrow \int f(x)g(Y(\omega)) dL(x, \omega)$. By Corollary 1.4.5 the sequence (X_n, Y) tends in distribution to the law L as indicated in the final statement of the theorem. Thus (i) implies (ii). Statement (ii) implies (iii), because the convergence in distribution of the sequence $(X_n, 1_A)$ implies the convergence of $E^*f(X_n)1_A$, the map $(x, z) \mapsto f(x)z$ being bounded and continuous on $\mathbb{D} \times \{0, 1\}$.

To prove that (iii) implies (i), we must show the existence of a measure L on $(\mathbb{D} \times \Omega, \mathcal{B} \times \mathcal{A}_0)$ so that $E^*f(X_n)Z$ tends to $\int f(x)Z(\omega) dL(x, \omega)$, for every $f \in C_b(\mathbb{D})$ and bounded $\mathcal{A}_0$ -measurable $Z: \Omega \rightarrow \mathbb{R}$. In fact, it is enough to show this for every Lipschitz f (see Problem 1.15.2), and by the assumed asymptotic tightness of X_n and Prohorov's theorem it is not a loss of generality to assume that X_n converges weakly to a limit. In view of Proposition 1.10.12, there exist Borel measurable maps $\tilde{X}_n$ with values in a separable subset of $\mathbb{D}$ such that $d(X_n, \tilde{X}_n) \rightarrow 0$, in outer probability. Because for these maps both $E^*f(X_n)Z - Ef(\tilde{X}_n)Z$ and $E^*f(X_n)1_A - Ef(\tilde{X}_n)1_A$ tend to zero, for every bounded measurable Z , measurable A and, bounded Lipschitz f , it is sufficient to prove that (iii) implies (i) for $\tilde{X}_n$ instead of X_n . Thus it is also no loss of generality to assume that the maps X_n are Borel measurable.

If $L_n = P^{X_n | \mathcal{A}_0}$, then $\int L_n(\cdot | \omega) dP(\omega) = P(X_n \in \cdot)$ is the marginal law of X_n . The assumption that the sequence X_n is tight implies that the sequence of Markov kernels L_n is relatively compact relative to the weak topology (see Problem 1.15.4). Thus every subsequence of n has a further subsequence along which $\iint f(x) L_n(dx | \omega) Z(\omega) dP(\omega) \rightarrow \iint f(x) L(dx | \omega) Z(\omega) dP(\omega)$, for some Markov kernel L and every $f \in C_b(\mathbb{D})$ and bounded, measurable Z . By the assumption that the sequence $Ef(X_n)1_A = \int_A \int f(x) L_n(dx | \omega) dP(\omega)$ converges to a limit, every limit point L possesses the same value $\int_A \int f(x) L(dx | \omega) dP(\omega)$, for every A in a stable generator of $\mathcal{A}_0$ and every bounded Lipschitz function f . This determines the full measure $A \mapsto \int_A \int f(x) L(dx | \omega) dP(\omega)$ on the σ -field $\mathcal{A}_0$, for every bounded Lipschitz f , and hence the Markov kernel L . Thus the limit point L is unique, and $Ef(X_n)Z = \iint f(x) L_n(dx | \omega) Z(\omega) dP(\omega) \rightarrow \int f(x)Z(\omega) dL(x, \omega)$ along the whole sequence, for the measure $dL(x, \omega) = L(dx | \omega) dP(\omega)$. ■

Part (iii) of the theorem implies that convergence in distribution $X_n \rightsquigarrow X$ can be upgraded to stable convergence, including existence of the limit measure L , as soon as every sequence $E^*f(X_n)1_A$ converges, for every separate A . The latter can be interpreted as convergence in distribution $X_n \rightsquigarrow X$ also under every conditional law $P_A(\cdot) = P(\cdot \cap A)/P(A)$ on Ω (for $P(A) > 0$). The convergence of $E^*f(X_n)Z$, for every Z , in the definition of stable convergence can likewise be interpreted as convergence $X_n \rightsquigarrow X$ under every law Q such that $dQ = Z dP$, for a bounded measurable Z . By approximation, this can be seen to extend to every Q that is absolutely continuous with respect to P . This gives another interpretation to stable convergence: weak convergence under multiple probability measures.

Assertion (ii) of the preceding theorem shows that stable convergence can be described through weak convergence and hence the concept does not really add to the toolkit of weak convergence. One reason that it is still interesting is that well known limit theorems turn out to be stable. Another reason is that stable convergence is a sufficient strengthening of weak convergence to yield extensions of *Slutsky's lemma* (Example 1.4.7) and the Delta-method to random centering and scaling (see Theorem 3.10.17), although also these results could be satisfactorily handled by just weak convergence.

1.15.6 Lemma (Stable Slutsky). *If the sequence of random variables X_n converges $\mathcal{A}_0$ -stably to X , and Y_n are random variables defined on the same probability space such that $d(Y_n, Y)$ tends to zero in probability for an $\mathcal{A}_0$ -measurable random variable Y , then $(X_n, Y_n) \rightsquigarrow (X, Y)$.*

Proof. By Lemma 1.15.5(ii), the sequence (X_n, Y) converges in distribution. Also by assumption $d((X_n, Y_n), (X_n, Y)) = d(Y_n, Y)$ tends to zero in probability. The result follows from Lemma 1.10.2(i). ■

Slutsky's lemma stated in this form is appealing, but it is only true because stable convergence of X_n to X , involves more than X alone. The joint distribution of the limit vector (X, Y) cannot be derived from the distributions of X and Y separately, in general, but involves the definitions of these variables on the underlying (extended) probability space $\mathbb{D} \times \Omega$. For a degenerate (i.e. constant) Y , this joint distribution is obvious, but then we fall back to the ordinary Slutsky's lemma. For a degenerate X (i.e. with a Dirac measure as conditional law), the convergence of X_n to X is in probability, and the lemma is also clear. In general, the lemma interpolates between these situations.

1.15.7 Example. If the sequence of random variables X_n converges $\mathcal{A}_0$ -mixing to X , and Y_n are random variables defined on the same probability space such that $d(Y_n, Y)$ tends to zero in probability for an $\mathcal{A}_0$ -measurable random variable Y , then $(X_n, Y_n) \rightsquigarrow (X, Y)$, for X and Y independent variables.

1.15.8 Example (Stable CLT). Let $X_1, X_2, \dots$ be i.i.d. random variables with mean zero and finite second moment, defined on some probability space $(\Omega, \mathcal{A}, P)$. Then the central limit theorem gives that $n^{-1/2} \sum_{i=1}^n X_i \rightsquigarrow N(0, 1)$ in the space $\mathbb{D} = \mathbb{R}$.

This convergence is stable relative to the sub σ -field $\mathcal{A}_0 = \sigma(X_1, X_2, \dots)$ generated by $X_1, X_2, \dots$, where the stable limit distribution is the distribution $L = N(0, 1) \times P$ on $\mathbb{R} \times \Omega$. Since this is a product measure, the convergence is mixing.

To see this, note that for every fixed m and variable $Y: \Omega \rightarrow \mathbb{R}$ that is a measurable function of $X_1, \dots, X_m$, the sequence $(n^{-1/2} \sum_{i=m+1}^n X_i, Y)$ tends in distribution to $(\tilde{S}, \tilde{Y})$, for $\tilde{S}$ and $\tilde{Y}$ independent variables, distributed as $N(0, 1)$ and Y , respectively. By Slutsky's lemma, the sequence $(n^{-1/2} \sum_{i=1}^n X_i, Y)$ tends in distribution to the same limit. In particular (iii) of Theorem 1.15.5 is satisfied for every measurable set $A \in \sigma(X_1, \dots, X_m)$, for every m , and hence every set A in the algebra $\cup_m \sigma(X_1, \dots, X_m)$, which generates $\mathcal{A}_0$.

Theorem 3.6.2 generalizes the present one-dimensional stable central limit theorem to general empirical processes, by roughly the same proof.

Problems and Complements

1. If X_n converges $\mathcal{A}_0$ -stably to X and X is $\mathcal{A}_0$ -measurable, then X_n tends in probability to X .

[Hint: The sequence (X_n, X) converges in distribution to (X, X) .]

2. If (1.15.1) holds for every Lipschitz $f: \mathbb{D} \rightarrow [0, 1]$ and measurable $Z: \Omega \rightarrow [0, 1]$, then it holds for every $f \in C_b(\mathbb{D})$ and bounded, measurable Z .

[Hint: First conclude that $X_n \rightsquigarrow X$, so that $E^*f(X_n) \rightarrow Ef(X)$, for every $f \in C_b(\mathbb{D})$. Every such nonnegative f can be approximated from below by a sequence of nonnegative Lipschitz functions. Conclude that $\liminf E^*f(X_n)Z \geq \int f(x)Z(\omega) dL(x, \omega)$, for every nonnegative f and Z . Apply this to $E^*f(X_n)(M - Z)$ and next to $f + M$, for a large constant M , to find that $E^*f(X_n)Z \rightarrow \int f(x)Z(\omega) dL(x, \omega)$.]

3. **(Weak convergence of Markov kernels)** For a sequence of Markov kernels $(\omega, B) \mapsto L_n(B|\omega)$ from $(\Omega, \mathcal{A})$ into $\mathbb{D}$, the following are equivalent definitions of convergence $L_n \xrightarrow{w} L$:

- (i) $\iint f(x) L_n(dx|\omega) Z(\omega) dP(\omega) \rightarrow \iint f(x) L(dx|\omega) Z(\omega) dP(\omega)$, for every $f \in C_b(\mathbb{D})$ and every bounded measurable $Z: \Omega \rightarrow \mathbb{R}$;
- (ii) $\int f(x) L_n(dx|\cdot) \xrightarrow{w} \int f(x) L(dx|\cdot)$ in $L_1(\Omega, \mathcal{A}, P)$, for every $f \in C_b(\mathbb{D})$;
- (iii) $\int L_n(\cdot|\omega) dQ(\omega) \rightsquigarrow \int L(\cdot|\omega) dQ(\omega)$, for every probability measure Q on $(\Omega, \mathcal{A})$ that is absolutely continuous with respect to P ;
- (iv) $\iint f(x) L_n(dx|\omega) Z(\omega) dP(\omega) \rightarrow \iint f(x) L(dx|\omega) Z(\omega) dP(\omega)$, for every Lipschitz $f: \mathbb{D} \rightarrow [0, 1]$ and every bounded measurable $Z: \Omega \rightarrow \mathbb{R}$.

[Hint: For the equivalence of (i) and (iii), see e.g. Häusler and Luschgy (2015), Theorem 2.3.]

4. **(Tightness of Markov kernels)** Every sequence of Markov kernels $(\omega, B) \mapsto L_n(B|\omega)$ from $(\Omega, \mathcal{A})$ into $\mathbb{D}$ such that the sequence of measures $\int L_n(\cdot|\omega) dP(\omega)$ is tight in $\mathbb{D}$ has a subsequence such that $L_n \xrightarrow{w} L$, for some Markov kernel L , in the sense of the preceding problem.

[Hint: See e.g. Häusler and Luschgy (2015), Theorem 2.7. Alternatively, compactify $\mathbb{D}$, use that the set of Markov kernels with values in a compact metric space (the set of randomized procedures in statistical decision theory) is compact for the weak topology,

and show that the limit points must concentrate on the original space if the tightness assumption holds.]

- 5. (Stable continuous mapping)** Let $g: \mathbb{D} \rightarrow \mathbb{E}$ be continuous at every point of a measurable set $\mathbb{D}_0 \subset \mathbb{D}$. If X_n converges $\mathcal{A}_0$ -stably to L , and $L(\mathbb{D}_0 \times \Omega) = 1$, then $g(X_n)$ converges $\mathcal{A}_0$ -stably to the measure M with $M(B \times A) = L(g^{-1}(B) \times A)$.

[Hint: The assumption implies that (X_n, Y) converges in distribution in $\mathbb{D} \times \mathbb{E}$, for every $\mathcal{A}_0$ -measurable $Y: \Omega \rightarrow \mathbb{E}$. It suffices to show that $(g(X_n), Y)$ converges likewise. This follows from the ordinary continuous mapping theorem. The limit distributions are given by $P(X \in B, Y \in C) = \int L(B|\omega) 1_C(Y(\omega)) dP(\omega)$ and the induced distribution $P(g(X) \in B, Y \in C)$.]

- 6. (Random time)** If the sequence of stochastic processes $(X_n(t): t > 0)$ converges $\mathcal{A}_0$ -stably in $\ell^\infty([0, 1], [0, 2], \dots)$ to a process with continuous sample paths, and $v_n: \Omega \rightarrow [0, \infty)$ are maps so that $v_n \xrightarrow{P} v$, for an $\mathcal{A}_0$ -measurable variable v , then the sequence of variables $X_n(v_n)$ converges in distribution.

[Hint: The sequence (X_n, v) converges in distribution. By Slutsky's lemma, the sequence (X_n, v_n) has the same limit. The map $g: \ell^\infty([0, 1], [0, 2], \dots) \times [0, \infty) \rightarrow \mathbb{R}$ given by $g(z, r) = z(r)$ is continuous at every (z, r) such that z is continuous at r . Hence $X_n(v_n) = g(X_n, v_n)$ converges in distribution, by the continuous mapping theorem.]

- 7. (Random sample size)** If the sequence of stochastic processes $(n^{-\alpha} X_{N_n}: t > 0)$ converges $\mathcal{A}_0$ -stably in $\ell^\infty([0, 1], [0, 2], \dots)$ to a process with continuous sample paths, for some $\alpha > 0$, and $N_n: \Omega \rightarrow [0, \infty)$ are maps so that $N_n/c_n \xrightarrow{P} v$, for some $c_n \rightarrow \infty$ and an $\mathcal{A}_0$ -measurable variable v with $P(v > 0) > 0$, then $N_n^{-\alpha} X_{N_n}$ converges in distribution under the conditional measure given $\{v > 0\}$.

[Hint: Assume without loss of generality that the c_n take values in the natural numbers. Then the processes $t \mapsto c_n^{-\alpha} X_{c_n t}$ are a subsequence of the given sequence and hence converge stably. Because v is $\mathcal{A}_0$ -measurable, the sequence $(c_n^{-\alpha} X_{c_n}, v)$ converges in distribution, also under the conditional measure given $v > 0$. So does the sequence $(c_n^{-\alpha} X_{c_n}, N_n/c_n)$, by Slutsky's lemma. The map $g: \ell^\infty([0, 1], [0, 2], \dots) \times [0, \infty) \rightarrow \mathbb{R}$ given by $g(z, r) = r^{-\alpha} z(r)$ is continuous at every (z, r) such that $r > 0$ and such that z is continuous at r . Hence $N_n^{-\alpha} X_{N_n} = g(c_n^{-\alpha} X_{c_n}, N_n/c_n)$ converges in distribution.]

1

Notes

1.2. Outer integrals and measurability are well known in probability theory. The minimal measurable cover is also the essential infimum of the collection of functions in the definition of E^*T ; see, e.g., Chow and Teicher (1978), page 190. Measurable cover or envelope functions were studied by Blumberg (1935) and then again independently by Eames and May (1967). Many parts of Lemma 1.2.2 are contained in May (1973), Dudley and Philipp (1983), Dudley (1984), and Dudley (1985). Perfect functions were introduced by Hoffmann-Jørgensen (1984, 1991).

1.3. Convergence in distribution of random variables in Euclidean spaces is an old concept. The study of weak convergence on abstract spaces started in the late 1940s and early 1950s with the study of the empirical process and the partial sum process by Doob (1949) and Donsker (1951, 1952). Prohorov (1956), Le Cam (1957), and Skorokhod (1956) gave a general theory for separable metric spaces. Billingsley (1968) and Parthasarathy (1967) have become standard references. The measurability problems were first ignored and then were solved through the introduction of “Skorokhod’s” topology on $D[0, 1]$. The more elegant solution to work with measurability in the ball σ -field was introduced by Dudley (1966). His stronger version of Prohorov’s theorem and characterization of tightness for nonseparable spaces was crucial for a more general development of weak convergence of the empirical process. Apparently, it went largely unnoticed until Gaenssler (1983) and Pollard (1984) showed the importance of the theory.

The more general approach to drop measurability altogether and work with outer integrals, which we follow here, is due to Hoffmann-Jørgensen (1984), Chapter 7; see Hoffmann-Jørgensen (1991). Hoffmann-Jørgensen’s approach was developed further in a series of papers by Andersen (1985) and Andersen and Do-

brić (1987, 1988). We have restricted our treatment to the weak convergence of measures on metric spaces. For various aspects of weak convergence theory on general topological spaces (which is the framework frequently chosen by physicists and probabilists studying interacting particle systems), see Le Cam (1957), Varadarajan (1961), Smolyanov and Fomin (1976), and Mitoma (1983).

1.4. The results in this section are classical in the case of measurable random elements, but were apparently first proved in the nonmeasurable setting by Van der Vaart and Wellner (1989).

1.5. Theorem 1.5.7 was given by Andersen and Dobrić (1987); see their Theorem 2.12, page 167. They built on results of Dudley (1984, 1985) and Hoffmann-Jørgensen (1984). Lemma 1.5.9 originated with Dudley (1966, 1973). The first published proof is apparently that of Andersen and Dobrić (1987), Theorem 3.2, page 169.

1.6. For the function space $C[0, \infty)$, results similar to those in this section were obtained by Stone (1963), Whitt (1970), Lindvall (1973), and Kim and Pollard (1990). Lemma 1.6.2 was contained in the first edition of this book as Problem 1.6.1, and is also contained in Arcones (1998). For Corollary 1.6.3, see also Andersen and Gill (1982), Theorem II.1. The applicability of the lemma is restricted by the fact that many processes must be centered before they converge, which destroys their convexity, a situation discussed in Arcones (1998).

1.7. The failure of measurability of the uniform empirical process as a process in $D[0, 1]$ with the uniform metric and Borel σ -field (Problem 1.7.3) was pointed out by Chibisov (1965); see Billingsley (1968), Section 18, pages 150–153, for a discussion. Dudley (1966, 1967a) initiated the study of weak convergence based on the ball σ -field. This theory was studied further by Wichura (1968), Gaenssler (1983), and Pollard (1984). Dudley (1978a, 1984) applied the theory of analytic sets to establish measurability properties in empirical process theory and introduced the concept of a class of functions that is “image-admissible-Suslin via a Suslin measurable space.” For a general introduction to the theory of analytic sets, including a proof of the projection theorem and references to the literature, see Cohn (1980), Chapter 8. For a thorough treatment, see also Hoffmann-Jørgensen (1970).

1.8. The basic result of this section goes back at least to Prohorov (1956); see his Theorem 1.13, page 171. For related results and further development, see Parthasarathy (1967), Chapter 6. For an application to tests of independence for directional data, see Jupp and Spurr (1985).

The characterization of tightness as approximate finite-dimensionality plus boundedness can be extended to general Banach spaces. See, for instance, Araujo and Giné (1980).

1.9. This section, including most of Lemma 1.9.3 and the Counterexample 1.9.4, is based on Dudley (1985). The continuous mapping theorem 1.9.5 extends the corresponding theorem in the classical theory. For a study of continuous functions for

limit theorems in $C[0, 1]$, $D[0, 1]$, $C[0, \infty)$, and $D[0, \infty)$, see Whitt (1980). Theorem 1.9.6 is new.

1.10. A precursor of the “Skorokhod construction”, Theorem 1.10.3, is due to Hammersley (1952). Hammersley showed that if a sequence of random variables converges in distribution, then there is a construction of a sequence of random variables and a limit random variable with the same distributions on a common probability space for which convergence in probability holds. Constructions of almost surely convergent versions of processes that converge weakly apparently began with Skorokhod (1956) for processes with values in a complete separable metric space. Dudley (1968) removed completeness as a hypothesis, and Wichura (1970) proved a Skorokhod type result without separability. The first constructions were based on embedding in Brownian motion. The potential uses of such constructions in statistics were made clear by Pyke (1969, 1970). Billingsley (1971) gave a nice proof of Skorokhod’s theorem for complete and separable spaces. Theorem 1.10.4 extends these results within the framework of the Hoffmann-Jørgensen weak convergence theory, with almost sure convergence replaced by outer almost sure convergence. For sequences, the theorem is due to Dudley (1985), who shows that $\tilde{\Omega}$ can be taken equal to the product of the Ω_α and one copy of $[0, 1]$, which is more economical than the space that is constructed here. Proposition 1.10.12 is due to Le Cam (1989). Exercise 1.10.2 is from Dudley (1984); see Theorem 3.3.1, page 25. Dudley and Philipp (1983) use a strong approximation approach to convergence of general empirical processes in order to avoid measurability difficulties.

1.11. The extended continuous mapping theorem 1.11.1 originated with Prohorov (1956) and continued with H. Rubin [see Anderson (1963), Billingsley (1968), page 34, and Topsøe (1967a, 1967b)]. The current extension to nonmeasurable random elements in Theorem 1.11.1 is used by Wellner (1989) in connection with the delta-method for Hadamard differentiable functions.

1.12. The metrization of convergence of laws began with Lévy; he provided a metric for convergence of laws on the real line; see Gnedenko and Kolmogorov (1954), Chapter 2, for a treatment of the Lévy metric. Prohorov (1956) defined a generalization, now called the Prohorov metric, and showed that it metrized weak convergence of distributions on complete, separable metric spaces. See Dudley (1989), Section 11.3, for a summary of this and other metrics. Uniformity of weak convergence over classes of functions was investigated by Ranga Rao (1962) and Topsøe (1967a). Fortet and Mourier (1953) introduced the bounded Lipschitz metric d_{BL} , and Dudley (1966, 1968) studied it further. Its extension to the Hoffmann-Jørgensen theory was studied in Dudley (1990) and Van der Vaart and Wellner (1989).

1.13. Theorem 1.13.1(v) was obtained in Ray and Van der Vaart (2020). Hoeffding’s trick originates in Hoeffding (1952). Dümbgen and Conte-Zerial (2013) used it to sharpen a result of Diaconis and Freedman (1984).

1.14. The Skorokhod topology was introduced by Skorokhod (1956). Reviews of the theory can be found at many places, including Gikhman and Skorokhod (2004),

Section VI-5. The presentation in Billingsley (1968, 1999) is classical and particularly well worth studying. The present chapter is also inspired by the reviews by Pollard (1981), Chapter VI, Jacod and Shiryaev (1987), Chapter VI, and especially by Ethier and Kurtz (1986), Sections 3.5–3.9. We follow Billingsley (1968, 1999) in first presenting $D[0, 1]$, and then extending to $D[0, \infty)$. Our choice of distance d_∞ releases the restrictions on functions in $D[0, u]$ imposed by the metrics d_u near u by taking a continuous average over u . The alternative, employed by Billingsley (1968, 1999), to multiply the functions first by the tapering functions g_u before taking the d_u -distance works with both continuous and discrete averages. The metric used by Ethier and Kurtz (1986), based on stopping the functions, also employs a continuous average. See Whitt (2002) for a treatment of other topologies, going under the names J_2 and M_1 and M_2 .

Aldous's condition was developed in Aldous (1978) and Rebolledo (1979), with further contributions by Jacod, Memin and Metivier (1983) and Ethier and Kurtz (1986). The extended version using also backward times in Theorem 1.14.18 is taken from Ethier and Kurtz (1986), who had input from Kurtz (1975). Aldous's condition is only useful for convergence to quasi-left continuous limit processes. Theorem 1.14.27 is taken from Jacod and Shiryaev (1987), and 1.14.29 is a simplified version of a theorem in Jacod and Shiryaev (1987), who discuss numerous refinements, suitable to more general, not necessarily quasi-left continuous semimartingales. Alternative approaches based on stopping times are developed in Aldous (1989). Theorem 1.14.22 is based on Kolmogorov's classical chaining scheme and in this form is due to Čentsov (1956). The power $1 + r > 1$ in its condition can be relaxed to a first power times a logarithmic term, as shown in Bloznelis and Paulauskas (1993), essentially with the same proof, and further refinement is given in Genest, Ghoudi and Rémillard (1996). Example 1.14.24 is due to Hahn (1978), and can similarly be refined. Theorem 1.14.25 is modelled after Billingsley (1999), Theorem 12.6.

An alternative “modulus of continuity” is

$$\sup_{t_1 < t_2: t_2 - t_1 < \delta} \sup_{t \in [t_1, t_2]} |z(t) - z(t_1)| \wedge |z(t) - z(t_2)|.$$

This is more like the usual continuity modulus, but it has the disadvantage that for use on $D[0, 1]$, additional control of the functions near the endpoints is necessary to characterize compactness (see Billingsley (1999), Theorem 12.4). Yet another modulus over a given interval $[a, c]$ is

$$\Delta(z, [a, c]) = \inf_{b \in (a, c)} \left[\sup_{a \leq t < b} |z(t) - z(a)| \vee \sup_{b \leq t \leq c} |z(t) - z(c)| \right].$$

This is the maximum of the uniform moduli over the intervals $[a, b)$ and $[b, c]$, where the split point $b \in (a, c)$ is chosen to minimize this quantity, and a possible jump at b is not counted. It allows to characterize compactness using deterministic partitions (see Pollard (1981), Theorem VI.10), the random locations of jumps being accommodated by the choice of the points b inside the modulus, again with additional control of the functions near the endpoints being needed for compactness in the case of a finite time set.

1.15. Stable convergence was introduced in Rényi (1963), as a generalization of mixing convergence, which goes back to at least Rényi (1950, 1958) and Rényi and Révész (1958), but was present in earlier papers. It was employed in Hall and Heyde (1980) for the martingale central limit theorem, and by Jacod and Shiryaev (1987) for martingale central limit theorems related to stochastic processes. The book Häusler and Luschgy (2015) gives an overview of many aspects of stable convergence.



PART 2

Empirical Processes

2.1

Introduction

This part is concerned with convergence of a particular type of random map: the empirical process. The *empirical measure* $\mathbb{P}_n$ of a sample of random elements $X_1, \dots, X_n$ in a measurable space $(\mathcal{X}, \mathcal{A})$ is the discrete random measure given by $\mathbb{P}_n(C) = n^{-1} \#(1 \leq i \leq n: X_i \in C)$. Alternatively (if points are measurable), it can be described as the random measure that puts mass $1/n$ at each observation. We shall frequently write the empirical measure as the linear combination $\mathbb{P}_n = n^{-1} \sum_{i=1}^n \delta_{X_i}$ of the Dirac measures at the observations.

Given a collection $\mathcal{F}$ of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$, the empirical measure induces a map from $\mathcal{F}$ to $\mathbb{R}$ given by

$$f \mapsto \mathbb{P}_n f.$$

Here, we use the abbreviation $Qf = \int f dQ$ for a given measurable function f and signed measure Q . Let P be the common distribution of the X_i . The centered and scaled version of the given map is the $\mathcal{F}$ -indexed *empirical process* $\mathbb{G}_n$ given by

$$f \mapsto \mathbb{G}_n f = \sqrt{n}(\mathbb{P}_n - P)f = \frac{1}{\sqrt{n}} \sum_{i=1}^n (f(X_i) - Pf).$$

Frequently the signed measure $\mathbb{G}_n = n^{-1/2} \sum_{i=1}^n (\delta_{X_i} - P)$ will be identified with the empirical process.

For a given function f , it follows from the law of large numbers and the central limit theorem that

$$\mathbb{P}_n f \xrightarrow{\text{as}} Pf,$$

$$\mathbb{G}_n f \rightsquigarrow N(0, P(f - Pf)^2),$$

provided Pf exists and $Pf^2 < \infty$, respectively. This part is concerned with making these two statements uniform in f varying over a class $\mathcal{F}$.

Classical empirical process theory concerns the special cases when the sample space $\mathcal{X}$ is the unit interval in $[0, 1]$, the real line $\mathbb{R}$, or $\mathbb{R}^d$ and the indexing collection $\mathcal{F}$ is taken to be the set of indicators of left half-lines in $\mathbb{R}$ or lower-left orthants in $\mathbb{R}^d$. In this part we are concerned with empirical measures and processes indexed by these classes of functions $\mathcal{F}$, but also many others, including indicators of half-spaces, balls, ellipsoids, and other sets in $\mathbb{R}^d$, classes of smooth functions from $\mathbb{R}^d$ to $\mathbb{R}$, and monotone functions.

With the notation $\|Q\|_{\mathcal{F}} = \sup\{|Qf|: f \in \mathcal{F}\}$, the uniform version of the law of large numbers becomes

$$\|\mathbb{P}_n - P\|_{\mathcal{F}} \rightarrow 0,$$

where the convergence is in outer probability or is outer almost surely. A class $\mathcal{F}$ for which this is true is called a *Glivenko-Cantelli class*, or also *P-Glivenko-Cantelli class* to bring out the dependence on the underlying measure P .

In order to discuss a uniform version of the central limit theorem, it is assumed that

$$\sup_{f \in \mathcal{F}} |f(x) - Pf| < \infty, \quad \text{for every } x.$$

Under this condition the empirical process $\{\mathbb{G}_n f: f \in \mathcal{F}\}$ can be viewed as a map into $\ell^\infty(\mathcal{F})$. Consequently, it makes sense to investigate conditions under which

$$(2.1.1) \quad \mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - P) \rightsquigarrow \mathbb{G}, \quad \text{in } \ell^\infty(\mathcal{F}),$$

where the limit $\mathbb{G}$ is a tight Borel measurable element in $\ell^\infty(\mathcal{F})$. A class $\mathcal{F}$ for which this is the true is called a *Donsker class*, or *P-Donsker class* to be more complete.

The nature of the limit process $\mathbb{G}$ follows from consideration of its marginal distributions. The marginals $\mathbb{G}_n f$ converge if and only if the functions f are square-integrable. In that case the multivariate central limit theorem yields that for any finite set $f_1, \dots, f_k$ of functions

$$(\mathbb{G}_n f_1, \dots, \mathbb{G}_n f_k) \rightsquigarrow N_k(0, \Sigma),$$

where the $k \times k$ -matrix Σ has (i, j) th element $P(f_i - Pf_i)(f_j - Pf_j)$. Since convergence in $\ell^\infty(\mathcal{F})$ implies marginal convergence, it follows that the limit process $\{\mathbb{G}f: f \in \mathcal{F}\}$ must be a zero-mean Gaussian process with covariance function

$$(2.1.2) \quad E\mathbb{G}f_1\mathbb{G}f_2 = P(f_1 - Pf_1)(f_2 - Pf_2) = Pf_1f_2 - Pf_1Pf_2.$$

According to Lemma 1.5.3, this and tightness completely determine the distribution of $\mathbb{G}$ in $\ell^\infty(\mathcal{F})$. It is called the *P-Brownian bridge*.

An application of Slutsky's lemma shows that every Donsker class is a Glivenko-Cantelli class in probability. In fact, this is also true with "in probability" replaced by "almost surely." Conversely, not every Glivenko-Cantelli class is Donsker, but the collection of all Donsker classes contains many members of interest. Many classes are also *universally Donsker*, which is defined as *P-Donsker* for every probability measure P on the sample space.

2.1.3 Example (Empirical distribution function). Let $X_1, \dots, X_n$ be i.i.d. random elements in $\mathbb{R}^d$, and let $\mathcal{F}$ be the collection of all indicator functions of lower rectangles $\{1\{(-\infty, t]\} : t \in \mathbb{R}^d\}$. The empirical measure indexed by $\mathcal{F}$ can be identified with the *empirical distribution function*

$$t \mapsto \mathbb{P}_n 1\{(-\infty, t]\} = \frac{1}{n} \sum_{i=1}^n 1\{X_i \leq t\}.$$

In this case, it is natural to identify $f = 1\{(-\infty, t]\}$ with $t \in \bar{\mathbb{R}}^d$ and the space $\ell^\infty(\mathcal{F})$ with the space $\ell^\infty(\bar{\mathbb{R}}^d)$. Of course, the sample paths of the empirical distribution function are all contained in a much smaller space (such as a $D[-\infty, \infty]$), but as long as this space is equipped with the supremum metric, this is irrelevant for the central limit theorem.

It is known from classical results (for the line by Donsker) that the class of lower rectangles is Donsker for any underlying law P of $X_1, \dots, X_n$. These classical results are a simple consequence of the results of this part. Moreover, with no additional effort, analogous results are obtained for the empirical process indexed by the collection of closed balls, rectangles, half-spaces, and ellipsoids, as well as many collections of functions.

2.1.4 Example (Empirical process indexed by sets). Let $\mathcal{C}$ be a collection of measurable sets in the sample space $(\mathcal{X}, \mathcal{A})$, and take the class of functions $\mathcal{F}$ equal to the set of indicator functions of sets in $\mathcal{C}$. This leads to the empirical distribution indexed by sets

$$C \mapsto \mathbb{P}_n(C) = \frac{1}{n} \#(X_i \in C).$$

In this case, it is convenient to make the identification $C \leftrightarrow 1\{C\}$, both in notation and in terminology. Hence $\mathcal{C}$ is called a Glivenko-Cantelli class if $\|\mathbb{P}_n - P\|_{\mathcal{C}}$ converges to zero in outer probability or outer almost surely, and $\mathcal{C}$ is a Donsker class if $\sqrt{n}(\mathbb{P}_n - P)$ converges weakly to a tight limit in $\ell^\infty(\mathcal{C})$.

An unfortunate technicality of the preceding definitions of Glivenko-Cantelli and Donsker classes is that they depend on the underlying probability space on which $X_1, X_2, \dots$ are defined, because this determines the outer expectations. In this book, unless specified otherwise, this technicality is resolved by always assuming that $X_1, X_2, \dots$ are defined as coordinate projections: the underlying probability space is the product space $(\mathcal{X}^\infty, \mathcal{A}^\infty, P^\infty)$, and X_i is the projection onto the i th coordinate.[†]

2.1.1 Overview

Whether a given class $\mathcal{F}$ is a Glivenko-Cantelli or Donsker class depends on the size of the class. A finite class of square integrable functions is always Donsker, while at the other extreme the class of all square integrable, uniformly bounded functions is

[†] Since the empirical measure depends only on the first n coordinates, and the projection of $\mathcal{X}^\infty$ onto $\mathcal{X}^n$ is a perfect map, any outer expectation $E^* h(\mathbb{P}_n)$ may be evaluated equivalently, assuming that $X_1, \dots, X_n$ are defined on $\mathcal{X}^n$ or on $\mathcal{X}^\infty$.

almost never Donsker. A relatively simple way to measure the size of a class $\mathcal{F}$ is to use *entropy numbers*. The ε -entropy of $\mathcal{F}$ is essentially the logarithm of the number of “balls” or “brackets” of size ε needed to cover $\mathcal{F}$. From this informal definition, it is already clear that the entropy numbers increase as ε decreases to zero. Simple sufficient conditions for a class to be Glivenko-Cantelli or Donsker can be given in terms of the rate of increase as ε tends to zero. The main results on Glivenko-Cantelli classes are given in Chapter 2.4, and the main results on Donsker classes are given in Chapter 2.5. These chapters are the core of Part 2.

For an informal description of the main results, we first define entropy with and without bracketing. Let $(\mathcal{F}, \|\cdot\|)$ be a subset of a (semi)normed space of real functions $f: \mathcal{X} \rightarrow \mathbb{R}$ on some set.

2.1.5 Definition (Covering numbers). The *covering number* $N(\varepsilon, \mathcal{F}, \|\cdot\|)$ is the minimal number of balls $\{g: \|g - f\| < \varepsilon\}$ of radius ε needed to cover the set $\mathcal{F}$. The centers of the balls need not belong to $\mathcal{F}$, but they should have finite norms. The *entropy* (without bracketing) is the logarithm of the covering number.

2.1.6 Definition (Bracketing numbers). Given two functions l and u , the *bracket* $[l, u]$ is the set of all functions f with $l \leq f \leq u$. An ε -bracket is a bracket $[l, u]$ with $\|u - l\| < \varepsilon$. The *bracketing number* $N_{[]}(\varepsilon, \mathcal{F}, \|\cdot\|)$ is the minimum number of ε -brackets needed to cover $\mathcal{F}$. The *entropy with bracketing* is the logarithm of the bracketing number. In the definition of the bracketing number, the upper and lower bounds u and l of the brackets need not belong to $\mathcal{F}$ themselves but are assumed to have finite norms.

The seminorms that will be of interest later, such as the $L_r(Q)$ -seminorms, all possess the (Riesz) property: for a pair of functions, if $|f| \leq |g|$, then $\|f\| \leq \|g\|$. For such norms, if f is in the 2ε -bracket $[l, u]$, then it is in the ball of radius ε around $(l + u)/2$. Thus covering and bracketing numbers are related by

$$N(\varepsilon, \mathcal{F}, \|\cdot\|) \leq N_{[]} (2\varepsilon, \mathcal{F}, \|\cdot\|).$$

In general, there is no converse inequality, so that bracketing numbers may be bigger than covering numbers. (A notable exception is the uniform norm, for which the previous inequality is an identity.) However, sufficient conditions for the theorems of interest in terms of bracketing numbers can often be stated using only a single norm on $\mathcal{F}$, while sufficient conditions in terms of entropy numbers usually involve many norms. This is intuitively obvious since brackets control a function $f(x)$ pointwise in its argument x , rather than in norm. As a consequence, sufficient conditions in terms of bracketing numbers or covering numbers are unrelated in general. Both are of interest.

We shall write $N(\varepsilon, \mathcal{F}, L_r(Q))$ for covering numbers relative to the $L_r(Q)$ -

seminorm[‡]

$$\begin{aligned}\|f\|_{Q,r} &= (f|f|^r dQ)^{1/r}, \\ \|f\|_{Q,\infty} &= Q - \text{esssup } f.\end{aligned}$$

This is similar for bracketing numbers. An *envelope function* of a class $\mathcal{F}$ is any function $x \mapsto F(x)$ such that $|f(x)| \leq F(x)$, for every x and f . The minimal envelope function is $x \mapsto \sup_f |f(x)|$. It will usually be assumed that this function is finite for every x . The *uniform entropy numbers* (relative to L_r) are defined as

$$\sup_Q \log N(\varepsilon \|F\|_{Q,r}, \mathcal{F}, L_r(Q)),$$

where the supremum is over all probability measures Q on $(\mathcal{X}, \mathcal{A})$, with $0 < QF^r < \infty$.^b

In Chapter 2.4, two Glivenko-Cantelli theorems are given, based on entropy with and without bracketing, respectively. The first theorem asserts that $\mathcal{F}$ is P -Glivenko-Cantelli if

$$N_{[]}(\varepsilon, \mathcal{F}, L_1(P)) < \infty, \quad \text{for every } \varepsilon > 0.$$

The proof of this theorem is elementary and extends the usual proof of the classical Glivenko-Cantelli theorem for the empirical distribution function. The second theorem is more complicated but has a simple corollary involving the uniform covering numbers. A class $\mathcal{F}$ is P -Glivenko-Cantelli if it is P -measurable with envelope F such that $P^*F < \infty$ and satisfies

$$\sup_Q N(\varepsilon \|F\|_{Q,1}, \mathcal{F}, L_1(Q)) < \infty, \quad \text{for every } \varepsilon > 0.$$

The full statement of the second theorem yields necessary and sufficient conditions for a P -measurable class $\mathcal{F}$ to be P -Glivenko-Cantelli in terms of the random covering numbers $N(\varepsilon, \mathcal{F}_M, L_1(\mathbb{P}_n))$ where $\mathcal{F}_M = \{f1_{[F \leq M]} : f \in \mathcal{F}\}$. This characterization of the Glivenko-Cantelli theorem will be used in Chapter 2.10 to establish useful Glivenko-Cantelli preservation theorems. Chapter 2.4 concludes with the statement of a result characterizing universal Glivenko-Cantelli classes. As mentioned already, the second Glivenko-Cantelli theorem requires the assumption that $\mathcal{F}$ is a “ P -measurable class,” a concept that is defined in Definition 2.3.3. The presence of this measurability hypothesis is a consequence of the way uniform entropy is used to control suprema via “randomization by Rademacher random variables” (or “symmetrization”) together with the lack of a general version of Fubini’s theorem for outer integrals. This same type of additional measurability hypothesis also enters in the corresponding Donsker theorems with uniform entropy hypotheses. Without these hypotheses, the theorems are false, but on the other hand it requires

[‡] The set $\mathcal{L}_r(Q)$ is the set of all measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$ with $\|f\|_{Q,r} < \infty$. This latter seminorm becomes a norm on the set $L_r(Q)$ of equivalence classes of almost surely equal functions. Our class $\mathcal{F}$ consists of functions and not of equivalence classes, as otherwise $\mathbb{P}_n f$ is not well defined, null sets being problematic if we work with multiple measures Q or many functions f . Hence $\mathcal{F}$ can be a subset of $\mathcal{L}_r(Q)$ and should not be defined only as set of equivalence classes in $L_r(Q)$. As usual we shall often neglect this fine point.

^b Alternatively, the supremum could be taken over all discrete probability measures. This yields the same results.

some effort to construct a class for which the hypotheses fail. Thus, in applications, this type of measurability will hardly ever play a role.

The method of symmetrization is discussed in Chapter 2.3 together with measurability conditions. This section is a necessary preparation for the uniform entropy Glivenko-Cantelli and Donsker theorems in Chapters 2.4 and 2.5. The theorems that are proved with bracketing do not use symmetrization and do not need measurability hypotheses.

One of the two Donsker theorems proved in Chapter 2.5 uses bracketing entropy. A slightly simpler version of this theorem asserts that a class $\mathcal{F}$ of functions is P -Donsker under an integrability condition on the $L_2(P)$ -entropy with bracketing. A class $\mathcal{F}$ of measurable functions is P -Donsker if

$$\int_0^\infty \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L_2(P))} d\varepsilon < \infty.$$

The other Donsker theorem in this chapter asserts that a class $\mathcal{F}$ of functions is P -Donsker under a similar condition on uniform entropy numbers. If $\mathcal{F}$ is class of functions such that

$$(2.1.7) \quad \int_0^\infty \sup_Q \sqrt{\log N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q))} d\varepsilon < \infty,$$

then $\mathcal{F}$ is P -Donsker for every probability measure P such that $P^*F^2 < \infty$ for some envelope function F and for which certain measurability hypotheses are satisfied. The integral conditions both measure the rate at which the entropy grows as ε decreases to zero. Note that the entropies are zero for large ε (if $\mathcal{F}$ is bounded), so that convergence of the integrals at ∞ is automatic.

Once the main Glivenko-Cantelli theorems and Donsker theorems are in hand, the following question arises: how can we verify the hypotheses on uniform entropy numbers and bracketing numbers? Tools and techniques for this purpose are developed in Chapter 2.6 for uniform entropy numbers and in Chapter 2.7 for bracketing numbers.

One of the starting points for controlling uniform covering numbers is the notion of a *Vapnik-Červonenkis class of sets*, or simply *VC-class*. Say that a collection $\mathcal{C}$ of subsets of the sample space $\mathcal{X}$ picks out a certain subset of the finite set $\{x_1, \dots, x_n\} \subset \mathcal{X}$ if it can be written as $\{x_1, \dots, x_n\} \cap C$ for some $C \in \mathcal{C}$. The collection $\mathcal{C}$ is said to *shatter* $\{x_1, \dots, x_n\}$ if $\mathcal{C}$ picks out each of its 2^n subsets. The *VC-dimension* $V(\mathcal{C})$ of $\mathcal{C}$ is the largest n for which some set of size n is shattered by $\mathcal{C}$.[‡] A collection $\mathcal{C}$ of measurable sets is called a *VC-class* if its dimension $V(\mathcal{C})$ is finite. Thus, by definition, a VC-class of sets picks out strictly less than 2^n subsets from any set of $n > V(\mathcal{C})$ elements. The surprising fact is that such a class can necessarily pick out only a polynomial number $O(n^{V(\mathcal{C})})$ of subsets, well below the $2^n - 1$ that the definition appears to allow. This result is a consequence of a combinatorial result, known as Sauer's lemma, which states that the number

[‡] In the first edition of this book the notation $V(\mathcal{C})$ was used for the *VC-index*, which is defined as the smallest n such that no set of n points is shattered. This index is the present dimension plus 1.

$\Delta_n(\mathcal{C}, x_1, \dots, x_n)$ of subsets picked out by a VC-class $\mathcal{C}$ satisfies

$$\max_{x_1, \dots, x_n} \Delta_n(\mathcal{C}, x_1, \dots, x_n) \leq \sum_{j=0}^{V(\mathcal{C})} \binom{n}{j} \leq \left(\frac{ne}{V(\mathcal{C})} \right)^{V(\mathcal{C})}.$$

Some thought shows that the number of subsets picked out by a collection $\mathcal{C}$ is closely related to the covering numbers of the class of indicator functions $\{1_C: C \in \mathcal{C}\}$ in $L_1(Q)$ for discrete, empirical type measures Q . By a clever argument, Sauer's lemma can be used to bound the uniform covering (or entropy) numbers for this class. Theorem 2.6.4 asserts that there exists a universal constant K such that, for any VC-class $\mathcal{C}$ of sets,

$$N(\varepsilon, \mathcal{C}, L_r(Q)) \leq KV(\mathcal{C})(4e)^{V(\mathcal{C})} \left(\frac{1}{\varepsilon} \right)^{rV(\mathcal{C})},$$

for any probability measure Q and $r \geq 1$ and $0 < \varepsilon < 1$. Consequently, VC-classes are examples of *polynomial classes* in the sense that their covering numbers are bounded by a polynomial in $1/\varepsilon$. The upper bound shows that VC-classes satisfy the sufficient conditions for the Glivenko-Cantelli theorem and Donsker theorem discussed previously (with much to spare) provided they possess certain measurability properties.

It is possible to obtain similar results for classes of functions. A collection of real-valued, measurable functions $\mathcal{F}$ on a sample space $\mathcal{X}$ is called a *VC-subgraph class* (or simply a VC-class of functions) if the collection of all subgraphs of the functions in $\mathcal{F}$ forms a VC-class of sets in $\mathcal{X} \times \mathbb{R}$. Just as for sets, the covering numbers of a VC-subgraph class $\mathcal{F}$ grow polynomially in $1/\varepsilon$, so that suitably measurable VC-subgraph classes are Glivenko-Cantelli and Donsker under moment conditions on the envelope function.

Among the many examples of VC-classes are the collections of left half-lines and lower-left orthants, with which classical empirical process theory is concerned. Methods to construct a great variety of VC-classes are discussed in Sections 2.6.5 and 2.6.6.

Section 2.6.7 introduces the related notion of fat-shattering dimension. This notion of “dimension” is defined at a fixed level ϵ and is always smaller than a natural related VC-dimension. In spite of this, it is possible to bound $L_2(Q)$ covering numbers in terms of the fat-shattering dimension. Statements of the available bounds are provided in this section.

The application of the other main Glivenko-Cantelli and Donsker theorem requires estimates on the bracketing numbers of classes of functions. These are given in Chapter 2.7 for classes of smooth functions, including Hölder, Sobolev, Besov spaces and analytic functions, sets with smooth boundaries, convex sets and functions, monotone functions, functions smoothly depending on a parameter, and Gaussian mixtures. Coupled with the results of Chapters 2.4 and 2.5, these estimates yield many more interesting Donsker classes.

Given a basic collection of Glivenko-Cantelli and Donsker classes and estimates of their entropies, new classes with similar properties can be constructed in

several ways. For instance, the *symmetric convex hull* $\text{sconv } \mathcal{F}$ of a class $\mathcal{F}$ is the collection of all functions of the form $\sum_{i=1}^m \alpha_i f_i$, with each $f_i \in \mathcal{F}$ and $\sum_{i=1}^m |\alpha_i| \leq 1$. In Chapter 2.10 it is shown that the convex hull of a Donsker class is Donsker. In addition to this, Theorem 2.6.9 gives a useful bound for the entropy of a convex hull in the case that the $L_r(Q)$ covering numbers with $r > 1$ for $\mathcal{F}$ are polynomial: if

$$N(\varepsilon \|F\|_{Q,r}, \mathcal{F}, L_r(Q)) \leq C \left(\frac{1}{\varepsilon}\right)^V,$$

then the convex hull of $\mathcal{F}$ satisfies

$$\log N(\varepsilon \|F\|_{Q,r}, \text{sconv } \mathcal{F}, L_r(Q)) \leq \begin{cases} K \left(\frac{1}{\varepsilon}\right)^{rV/((r-1)V+r)}, & \text{if } 1 < r < 2, \\ K \left(\frac{1}{\varepsilon}\right)^{2V/(V+2)} & \text{if } 2 \leq r < \infty. \end{cases}$$

An important feature of this bound is that the power $2V/(V+2)$ of $1/\varepsilon$ in the second case is strictly less than 2 for any V , and in the first case the power $rV/((r-1)V+r)$ is strictly less than 2 if $V < 2r/(2-r)$; so that the convex hull of a polynomial class satisfies the uniform entropy condition (2.1.7). This bound is true in particular for *VC-hull classes*, which are defined as the sequential closures of the convex hulls of VC-classes.

Other operations preserving the Glivenko-Cantelli or Donsker properties are considered in Chapter 2.10. A short summary is that given Glivenko-Cantelli (or Donsker) classes $\mathcal{F}_1, \dots, \mathcal{F}_k$ and a fixed function ϕ on $\mathbb{R}^k$, the new class

$$\mathcal{H} = \{\phi(f_1, \dots, f_k): f_j \in \mathcal{F}_j, j = 1, \dots, k\}$$

is Glivenko-Cantelli (or Donsker) if ϕ is continuous (or Lipschitz) respectively, subject to having a suitably integrable envelope. For example, the classes of pairwise sums $\mathcal{F} + \mathcal{G}$, pairwise infima $\mathcal{F} \wedge \mathcal{G}$, and pairwise suprema $\mathcal{F} \vee \mathcal{G}$, as well as the union $\mathcal{F} \cup \mathcal{G}$, are Glivenko-Cantelli (or Donsker) classes if both $\mathcal{F}$ and $\mathcal{G}$ are Glivenko-Cantelli (or Donsker). Similarly, pairwise products $\mathcal{F}\mathcal{G}$ and quotients $\mathcal{F}/\mathcal{G}$ are Glivenko-Cantelli (or Donsker) if $\mathcal{F}$ and $\mathcal{G}$ are Glivenko-Cantelli (or Donsker) and bounded above, or bounded away from, zero. The latter boundedness assumptions can be traded against stronger conditions on the entropies of the classes. Such “permanence properties” applied to the basic Glivenko-Cantelli or Donsker classes resulting from Chapters 2.6 and 2.7 provide a very effective method to verify the Glivenko-Cantelli (or Donsker) property in, for instance, statistical applications.

The remainder of Part 2 explores refinements, extensions, special classes $\mathcal{F}$, uniformity in the underlying distribution, multiplier processes, partial-sum processes, the non-i.i.d. situation, and moment and tail bounds for the supremum of the empirical process.

The uniformity of weak convergence of $\mathbb{G}_n$ to $\mathbb{G}$ with respect to the underlying probability distribution P generating the data is addressed in Chapter 2.8. Again, the main results use either uniform entropy or bracketing entropy.

The viewpoint in Chapter 2.9 on multiplier central limit theorems is that, for a Donsker class $\mathcal{F}$,

$$\mathbb{G}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n (\delta_{X_i} - P) \equiv \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i \rightsquigarrow \mathbb{G},$$

in $\ell^\infty(\mathcal{F})$. Then we pose the following question: for what sequences of i.i.d., real-valued random variables $\xi_1, \dots, \xi_n$ independent from the original data does it follow that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \rightsquigarrow \mathbb{G}?$$

In fact, these two displays turn out to be equivalent if the ξ_i have zero mean, have variance 1, and satisfy the $L_{2,1}$ -condition

$$\|\xi_1\|_{2,1} = \int_0^\infty \sqrt{\mathbb{P}(|\xi_1| > t)} dt < \infty.$$

Chapter 2.9 also presents conditional versions of this multiplier central limit theorem. These are basic for part 3's development of limit theorems for the bootstrap empirical process.

Chapter 2.11 gives extensions of the Donsker theorem for sums of independent, but not identically distributed, processes $\sum_{i=1}^n Z_{ni}$ indexed by an arbitrary collection $\mathcal{F}$. One example, which occurs naturally in statistical contexts and is covered by this situation, is the empirical process indexed by a collection $\mathcal{F}$ of functions based on observations $X_{n1}, \dots, X_{nn}$ that are independent but not identically distributed. Once again we present two main central limit theorems, based on entropy with and without bracketing. Even if specified to the i.i.d. situation, the theorems in Chapter 2.11 are more general than the theorems obtained before. For instance, we use a random entropy condition, rather than a uniform entropy condition, and also consider bracketing using majorizing measures.

Two types of *partial-sum processes* are studied in Chapter 2.12. The first type is the *sequential empirical process*

$$\mathbb{Z}_n(s, f) = \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor ns \rfloor} (f(X_i) - Pf) = \sqrt{\frac{\lfloor ns \rfloor}{n}} \mathbb{G}_{\lfloor ns \rfloor}(f),$$

which records the “history” of the empirical processes $\mathbb{G}_n$ as sampling progresses. The second type of partial-sum processes studied are those generated by independent random variables located at the points of a lattice. These processes, which can be viewed as closely related to the multiplier processes studied in Chapter 2.9, since they are empirical processes with random masses (the multipliers) at (degenerately) random points, have some special features and deserve special attention.

Chapter 2.13 gives Donsker theorems for several special types of classes $\mathcal{F}$, namely, sequences and elliptical classes.

Chapter 2.14 gives a detailed study of moment bounds, tail bounds, and exponential bounds for suprema of the empirical process. The first moment $\mathbb{E}^* \|\mathbb{G}_n\|_{\mathcal{F}}$ can be viewed as expressing the magnitude of the empirical process. It depends on the size of the individual functions, as can in some cases be measured through the envelope function, but more significantly on the size of the class $\mathcal{F}$. As before the latter can be measured both by uniform entropy and bracketing entropy. We present inequalities in terms of the corresponding entropy integrals, where the integral is taken over the interval $(0, \delta)$ if the individual functions have norm bounded by δ . These bounds play an important role in Chapters 3.2–3.4 on the limit theory of

M -estimators. Next to the first moment the chapter also addresses tail probabilities $P^*(\|\mathbb{G}_n\|_{\mathcal{F}} > t)$ of the supremum, under various entropy conditions.

Although Chapter 2.15 is also concerned with the supremum $\|\mathbb{G}_n\|_{\mathcal{F}}$ of the empirical process, the point of view is very different. Rather than the size of this variable, the results in this chapter concern its deviation from its mean. As it turns out, bounds for probabilities of the type

$$P^*(\|\mathbb{G}_n\|_{\mathcal{F}} - E^*\|\mathbb{G}_n\|_{\mathcal{F}} > t)$$

depend hardly on the structure of the class $\mathcal{F}$, unlike bounds for the mean $E^*\|\mathbb{G}_n\|_{\mathcal{F}}$ itself. The main results here are the bounded difference inequality and Talagrand's inequality, which could roughly be compared to the Hoeffding and Bernstein inequalities for univariate averages. Several of the concentration inequalities in Chapter 2.15 are sharp in that they reduce to these classical bounds in the case when the class $\mathcal{F}$ consists of just a single function. The bounds show that the supremum is concentrated around its mean, with sub-Gaussian or sub-exponential deviations. The exponential decrease is important for the model selection theory presented in Chapter 3.5. Chapter 2.15 concludes Part 2.

2.1.2 Asymptotic Equicontinuity

A class of measurable functions is called *pre-Gaussian* if the (tight) limit process $\mathbb{G}$ in the uniform central limit theorem (2.1.1) exists. By Kolmogorov's extension theorem, there always exists a zero-mean Gaussian process $\{\mathbb{G}f: f \in \mathcal{F}\}$ with covariance function given by (2.1.2). A more precise description of pre-Gaussianity is that there exists a version of this Gaussian process that is a tight, Borel measurable map from some probability space into $\ell^\infty(\mathcal{F})$. Usually the name "Brownian bridge" introduced earlier refers to this tight process with its special sample path properties, rather than to the general stochastic process $\mathbb{G}$.

It is desirable to have a more concrete description of the tightness property of a Brownian bridge and hence of the notion of pre-Gaussianity. In Chapter 1.5 it was seen that tightness of a random map into $\ell^\infty(\mathcal{F})$ is closely connected to continuity of its sample paths. Define a seminorm ρ_P by

$$\rho_P(f) = (P(f - Pf)^2)^{1/2}.$$

Then, by Example 1.5.10, a class $\mathcal{F}$ is pre-Gaussian if and only if

- $\mathcal{F}$ is totally bounded for ρ_P ,
- there exists a version of $\mathbb{G}$ with uniformly ρ_P -continuous sample paths $f \mapsto \mathbb{G}(f)$.

Actually, this example shows that instead of the centered L_2 -norm ρ_P , any centered L_r -norm can be used as well. However, the L_2 -norm appears to be the most convenient.[†]

[†] It may be noted that the seminorm ρ_P is the $L_2(P)$ -norm of f centered at its expectation (in fact, the standard deviation of $f(X_i)$). The centering is natural in view of the fact that the empirical process is invariant under the addition of constants to $\mathcal{F}$: $\mathbb{G}_n f = \mathbb{G}_n(f - c)$ for every c . Under the condition that $\sup_{f \in \mathcal{F}} |Pf| < \infty$, the seminorm ρ_P can be replaced by the slightly simpler $L_2(P)$ -seminorm without loss of generality (Problem 2.1.2).

While pre-Gaussianity of a class $\mathcal{F}$ is necessary for the uniform central limit theorem, it is not sufficient. A Donsker class $\mathcal{F}$ satisfies the stronger condition that the sequence $\mathbb{G}_n$ is asymptotically tight. By Theorem 1.5.7, the latter entails replacing the condition that the sample paths of the limit process are continuous by the condition that the empirical process is asymptotically continuous: for every $\varepsilon > 0$,

$$(2.1.8) \quad \lim_{\delta \downarrow 0} \limsup_{n \rightarrow \infty} \mathbb{P}^* \left(\sup_{\rho_P(f-g) < \delta} |\mathbb{G}_n(f-g)| > \varepsilon \right) = 0.$$

Much of Part 2 is concerned with this equicontinuity condition. With the notation[‡] $\mathcal{F}_\delta = \{f - g: f, g \in \mathcal{F}, \rho_P(f - g) < \delta\}$, it is equivalent to the following statement: for every decreasing sequence $\delta_n \downarrow 0$,

$$\|\mathbb{G}_n\|_{\mathcal{F}_{\delta_n}} \xrightarrow{\mathbb{P}^*} 0$$

(Problem 2.1.5). In view of Theorem 1.5.7, a class $\mathcal{F}$ is Donsker if and only if $\mathcal{F}$ is totally bounded in $\mathcal{L}_2(P)$ and satisfies the asymptotic equicontinuity condition as in the preceding displays.

2.1.3 Maximal Inequalities

It follows that both the law of large numbers and the central limit theorem are concerned with showing that a supremum of real-valued variables converges to zero. To a certain extent, the derivation of these results is done in parallel, as both need to make use of *maximal inequalities* that bound probabilities involving suprema of random variables. Chapter 2.2 develops such inequalities in an abstract setting, using Orlicz norms. If ψ is a nondecreasing, convex function with $\psi(0) = 0$, then the Orlicz norm $\|X\|_\psi$ of a random variable X is defined by

$$\|X\|_\psi = \inf \left\{ C > 0: \mathbb{E} \psi \left(\frac{|X|}{C} \right) \leq 1 \right\}.$$

The functions ψ of primary interest are $\psi(x) = x^p$ and $\psi(x) = \exp(x^p) - 1$ (for $p \geq 1$ or adapted versions if $0 < p < 1$). For $\psi(x) = x^p$, the Orlicz norm $\|X\|_\psi$ is just the usual L_p -norm $\|X\|_p$. The key Lemma 2.2.2 provides a bound for the Orlicz norm of a finite maximum $\max_{1 \leq i \leq m} X_i$ in terms of the Orlicz norms of the variables X_i and the inverse function $\psi^{-1}(m)$. Suppose that $\limsup_{x, y \rightarrow \infty} \psi(x)\psi(y)/\psi(xy) < \infty$. Then there is a constant K depending only on ψ so that, for arbitrary random variables $X_1, \dots, X_m$,

$$\left\| \max_{1 \leq i \leq m} X_i \right\|_\psi \leq K \psi^{-1}(m) \max_{1 \leq i \leq m} \|X_i\|_\psi.$$

The interesting feature of this inequality is that a bound on $\max_i \|X_i\|_\psi$ for a rapidly growing function ψ , such as $\psi_2(x) = \exp(x^2) - 1$, yields a bound on the rate of growth with m of the Orlicz norm of $\max X_i$ governed by the slowly growing function $\psi^{-1}(m)$, such as $\psi_2^{-1}(m) = \sqrt{\log(m+1)}$. Since Gaussian random variables

[‡] In most of this part, the notation $\mathcal{F}_\delta$ means $\mathcal{F}_\delta = \{f - g: f, g \in \mathcal{F}, \rho(f - g) < \delta\}$, for ρ equal to either the $L_2(P)$ -semimetric or ρ_P .

and sums of independent, bounded random variables enjoy finite ψ_2 -Orlicz norms, the particular function $\psi_2^{-1}(m)$ figures prominently in the developments in later chapters.

Bounds on finite suprema can be extended to general maximal inequalities with the help of the *chaining method*, which Kolmogorov pioneered. The idea is to relate maxima over an infinite set to maxima of increments over an increasing sequence of finite sets. Here covering numbers are introduced to measure the size of the approximating finite sets. For a semimetric space (T, d) , the *covering number* $N(\varepsilon, T, d)$ is defined as the minimal number of balls of radius ε needed to cover T . Given a separable stochastic process $\{X_t: t \in T\}$ and a fixed function ψ , define a semimetric by $d(s, t) = \|X_s - X_t\|_\psi$ for every s, t . Then, for any $\eta, \delta > 0$ and a constant K ,

$$\left\| \sup_{d(s,t) \leq \delta} |X_s - X_t| \right\|_\psi \leq K \left[\int_0^\eta \psi^{-1}(N(\varepsilon, d)) d\varepsilon + \delta \psi^{-1}(N^2(\eta, d)) \right].$$

Thus the ψ -norm of a supremum is bounded by an integral that involves the inverse ψ^{-1} and the covering numbers of the index set T with respect to the *intrinsic semimetric* d .

In particular, a process $\{X_t: t \in T\}$ is said to be *sub-Gaussian* if the ψ_2 -norms of the increments $X_s - X_t$ are finite. This can be shown to be true if and only if, for some semimetric d ,

$$P(|X_s - X_t| > x) \leq 2e^{-\frac{1}{2}x^2/d^2(s,t)}, \quad \text{for every } x > 0.$$

Then the preceding inequality can be simplified to

$$E \sup_{d(s,t) \leq \delta} |X_s - X_t| \leq K \int_0^\delta \sqrt{\log N(\varepsilon, T, d)} d\varepsilon.$$

This inequality explains the appearance of the square root of the logarithm of the entropy in the sufficient integral conditions for a class to be Donsker, which were discussed in the preceding chapter.

While maximal inequalities based on covering numbers suffice for almost all statistical applications, they are known to not be sharp in general. This suboptimality is especially well understood for Gaussian maximal inequalities for which the classical bounds based on covering numbers fail, but bounds based on generic chaining and majorizing measures succeed in providing sharp bounds. Sections 2.2.5 and 2.2.5 present an introduction to these alternative methods and the resulting improvements.

* 2.1.4 Central Limit Theorem in Banach Spaces

The convergence in distribution in $\ell^\infty(\mathcal{F})$ of the empirical process $\mathbb{G}_n = n^{-1/2} \sum_{i=1}^n (\delta_{X_i} - P)$ can be considered a central limit theorem for the i.i.d. random elements $\delta_{X_1} - P, \dots, \delta_{X_n} - P$ in the Banach space $\ell^\infty(\mathcal{F})$. In this section it is first shown that, conversely, any central limit theorem in a Banach space can be

* This section may be skipped at first reading.

stated in terms of empirical processes. Next it is argued that this observation is only moderately useful.

Suppose $Z_1, \dots, Z_n$ are i.i.d. random maps with values in a Banach space $\mathbf{X}$. Let $\mathcal{F}$ be a subset of the dual space of $\mathbf{X}$ such that $\sup_f |f(x)| = \|x\|$ for every x and such that $f(Z_i)$ is measurable for each f . A mean of Z_i is an element $\mathbf{E}Z_i$ of $\mathbf{X}$ such that $f(\mathbf{E}Z_i) = \mathbf{E}f(Z_i)$, for every $f \in \mathcal{F}$. Every given element $x \in \mathbf{X}$ can be identified with a map

$$x^{**}: f \rightarrow f(x),$$

from $\mathcal{F}$ to $\mathbb{R}$. By assumption, this gives an isometric identification $x \leftrightarrow x^{**}$ of $\mathbf{X}$ with a subset of $\ell^\infty(\mathcal{F})$. Thus random maps $Z_1, \dots, Z_n$ in $\mathbf{X}$ satisfy the central limit theorem if and only if the random elements $Z_1^{**}, \dots, Z_n^{**}$ satisfy the central limit theorem in $\ell^\infty(\mathcal{F})$. Since $\sum_{i=1}^n (Z_i - \mathbf{E}Z_i)^{**}(f) = \sum_{i=1}^n (f(Z_i) - \mathbf{E}f(Z_i))$, this appears to be the case if and only if $\mathcal{F}$ is a Donsker class of functions. However, for an accurate statement, it is necessary to describe the measurability structure more precisely.

2.1.9 Example. In case of a separable Banach space it is usually assumed that $Z_1, \dots, Z_n$ are Borel measurable maps. Then “i.i.d.” can be understood in the usual manner. If $\mathcal{F}$ is taken equal to the unit ball of the dual space, then the equality $\sup_f |f(x)| = \|x\|$ is valid by the Hahn-Banach theorem, and the σ -field generated by $\mathcal{F}$ is precisely the Borel σ -field. It follows that the Z_i are Borel measurable if and only if every $f(Z_i)$ is measurable.

The conclusion is that the sequence $n^{-1/2} \sum_{i=1}^n (Z_i - \mathbf{E}Z_i)$ converges in distribution if and only if the unit ball of the dual space is Donsker with respect to the common Borel law of the Z_i .

2.1.10 Example. Suppose $Z_1, \dots, Z_n$ are i.i.d. stochastic processes with bounded sample paths, indexed by an arbitrary set $\tilde{\mathcal{F}}$ (not necessarily a class of functions). The notion “i.i.d.” should include that the finite-dimensional marginals $(Z_i(\tilde{f}_1), \dots, Z_i(\tilde{f}_k))$ are i.i.d. vectors. In general, this does not determine the “distribution” of the processes in $\ell^\infty(\tilde{\mathcal{F}})$ adequately. Assume in addition that each Z_i is defined on a product probability space $(\mathcal{X}^n, \mathcal{A}^n, P^n)$ as $Z_i(x_1, \dots, x_n)\tilde{f} = Z(\tilde{f}, x_i)$, for some fixed stochastic process $\{Z(\tilde{f}, x): \tilde{f} \in \tilde{\mathcal{F}}\}$. Then $\sum_{i=1}^n (Z_i(\tilde{f}) - \mathbf{E}Z_i(\tilde{f})) = \sum_{i=1}^n (f(x_i) - Pf)$, for the function f defined by $f(x) = Z(\tilde{f}, x)$. That Z is a stochastic process means precisely that each function $x \mapsto f(x)$ is measurable. The processes $Z_1, Z_2, \dots$ satisfy the central limit theorem in $\ell^\infty(\tilde{\mathcal{F}})$ if and only if the class of functions $\{f: \tilde{f} \in \tilde{\mathcal{F}}\}$ is P -Donsker.

Thus the central limit theorem in an arbitrary Banach space can be phrased as a result about empirical processes. Even though this conclusion is formally correct, the methods (traditionally) used for proving empirical central limit theorems yield unsatisfactory descriptions of the central limit theorem in special Banach spaces. For instance, the conditions for the central limit theorem in L_p -space are simple and well known. If $(S, \mathcal{S}, \mu)$ is a σ -finite measure space and $Z_1, \dots, Z_n$ are i.i.d., zero-mean Borel measurable maps into $L_p(S, \mathcal{S}, \mu)$, then the sequence $n^{-1/2} \sum_{i=1}^n Z_i$

converges weakly if and only if $P(\|Z_1\|_p > t) = o(t^{-2})$ as $t \rightarrow \infty$ and

$$\int_S (EZ_1^2(s))^{p/2} d\mu(s) < \infty.$$

(In case $p = 2$, this can be simplified to the single requirement $E\|Z_1\|_2^2 < \infty$. For this case, the theorem is given in Chapter 1.8.) In terms of empirical processes, this becomes as follows.

2.1.11 Proposition. *Let $(S, \mathcal{S}, \mu)$ be a σ -finite measure space, let $1 \leq p < \infty$, and let P be a Borel probability measure on $L_p(S, \mathcal{S}, \mu)$. Then the unit ball of the dual space of $L_p(S, \mathcal{S}, \mu)$ is P -Donsker if and only if $\int_S (\int z(\omega, s)^2 dP(\omega))^{p/2} d\mu(s) < \infty$ and $P(\|z\|_p > t) = o(t^{-2})$ as $t \rightarrow \infty$.*

This proposition is proved in texts on probability in Banach spaces.^b The methods developed in this part do not yield this result. In fact, already the formulation of the proposition is unnecessarily complicated: it is awkward to push this central limit theorem into an empirical mold.

On the other hand, Part 2 gives many interesting results on the central limit theorem in Banach spaces of the type $\ell^\infty(\mathcal{F})$, where $\mathcal{F}$ is a collection of measurable functions. For this large class of Banach spaces, sufficient conditions for the central limit theorem go beyond the Banach space structure, but they can be stated in terms of the structure of the function class $\mathcal{F}$.

Problems and Complements

1. **(Total boundedness in $L_2(P)$)** If $\mathcal{F}$ is totally bounded in $L_2(P)$, then it is totally bounded for the seminorm ρ_P . If $\mathcal{F}$ is totally bounded for ρ_P and $\|P\|_{\mathcal{F}} = \sup\{|Pf| : f \in \mathcal{F}\}$ is finite, then it is totally bounded in $L_2(P)$.

[Hint: Suppose $f_1, \dots, f_m$ form an ε -net for ρ_P . If $\sup\{|Pf| : f \in \mathcal{F}\}$ is finite, then the numbers Pf with f ranging over some ρ_P -ball of radius ε are contained in an interval. Choose for every f_i a finite collection $f_{i,j}$ such that, for every f with $\rho_P(f - f_i) < \varepsilon$, there is an $f_{i,j}$ with $|Pf - Pf_{i,j}| < \varepsilon$. Then the $f_{i,j}$ form a $\sqrt{5}\varepsilon$ -net for $\mathcal{F}$ in $L_2(P)$.]

2. **(Using $\|\cdot\|_{P,2}$ instead of ρ_P)** Suppose $\sup\{|Pf| : f \in \mathcal{F}\}$ is finite. Then a class $\mathcal{F}$ of measurable functions is P -Donsker if and only if $\mathcal{F}$ is totally bounded in $L_2(P)$ and the empirical process is asymptotically equicontinuous in probability for the $L_2(P)$ -semimetric.

[Hint: Use the previous problem and the next problem.]

3. Suppose ρ is a semimetric on a class of measurable functions $\mathcal{F}$ that is uniformly stronger than ρ_P in the sense that $\rho_P(f - g) \leq \phi(\rho(f, g))$, for a function ϕ with $\phi(\varepsilon) \rightarrow 0$ as $\varepsilon \downarrow 0$. Suppose $\mathcal{F}$ is totally bounded under ρ . Then $\mathcal{F}$ is P -Donsker if and only if the empirical process indexed by $\mathcal{F}$ is asymptotically equicontinuous in probability with respect to ρ .

[Hint: Use the addendum to Theorem 1.5.7.]

^b See Ledoux and Talagrand (1991), Theorem 10.10.

4. (**Brownian motion**) If $\mathcal{F}$ is P -pre-Gaussian and $\|P\|_{\mathcal{F}} < \infty$, then a P -Brownian bridge has uniformly continuous sample paths with respect to the $L_2(P)$ -semimetric. The process $\mathbb{Z}(f) = \mathbb{G}(f) + \xi Pf$ for a standard normal variable ξ independent of $\mathbb{G}$ has a version that is a Borel measurable, tight map in $\ell^\infty(\mathcal{F})$. This process, which is Gaussian with mean zero and covariance function Pfg , is called a *Brownian motion*.
5. If $a_n: (0, 1] \rightarrow [0, \infty)$ is a sequence of nondecreasing functions (hence $a_n(\delta)$ decreases if δ decreases, for every fixed n), then there exist numbers $\delta_n \downarrow 0$ such that $\limsup_{n \rightarrow \infty} a_n(\delta_n) = \lim_{\delta \rightarrow 0} \limsup_{n \rightarrow \infty} a_n(\delta)$. Deduce that $\lim_{\delta \rightarrow 0} \limsup_{n \rightarrow \infty} a_n(\delta) = 0$ if and only if $a_n(\delta_n) \rightarrow 0$ for every $\delta_n \rightarrow 0$.
6. For arbitrary functions $a_n: (0, 1] \rightarrow [0, \infty)$ consider:
- (i) $\lim_{n \rightarrow \infty} a_n(\delta_n) = 0$, for every $\delta_n \downarrow 0$.
 - (ii) $\lim_{\delta \downarrow 0} \limsup_{n \rightarrow \infty} a_n(\delta) = 0$.
 - (iii) $\lim_{n \rightarrow \infty} a_n(\delta_n) = 0$, for some $\delta_n \downarrow 0$.

Then (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii), but both inverses are false in general. If the functions a_n are nondecreasing, then (i) and (ii) are equivalent, but still (iii) does not imply (ii).

[**Hint:** Define $a(\delta) := \limsup_{n \rightarrow \infty} a_n(\delta)$.

(i) $\Rightarrow$ (ii). If the double limit in (ii) is positive, then there exist $\varepsilon > 0$ and $\eta_m \downarrow 0$ such that $a(\eta_m) > \varepsilon$, for every m , whence there exist $n_1 < n_2 < \dots$ with $a_{n_m}(\eta_m) > \varepsilon$, for every m . Define $\delta_n = \eta_m$ if $n \in [n_m, n_{m+1})$, and arbitrary for $n < n_1$. Then $\delta_n \downarrow 0$ and $a_{n_m}(\delta_{n_m}) > \varepsilon$, for every m , which contradicts that $a_n(\delta_n) \rightarrow 0$.

(ii) $\Rightarrow$ (iii). Set $A_n(\delta) = \sup_{m \geq n} a_m(\delta)$ and fix $\varepsilon > 0$ and $\eta_m \downarrow 0$. For every m there exists n_m with $|A_n(\eta_m) - a(\eta_m)| < \varepsilon$, for $n \geq n_m$. Without loss of generality take $n_1 < n_2 < \dots$ and set $\delta_n = \eta_m$ for $n \in [n_m, n_{m+1})$. Then $\delta_n \downarrow 0$ and $a_n(\delta_n) \leq A_n(\delta_n) \leq a(\delta_n) + \varepsilon$, for all $n \geq n_1$, so $a_n(\delta_n) \rightarrow 0$.

(ii) $\Rightarrow$ (i) if the a_n are decreasing. If $a_n(\delta_n)$ does not tend to zero for a given $\delta_n \downarrow 0$, then there exist $\varepsilon > 0$ and $n_1 < n_2 < \dots$ such that $a_{n_m}(\delta_{n_m}) > \varepsilon$, for every m . If a_n is increasing, then $a_{n_m}(\delta_{n_j}) > \varepsilon$, for every $j \leq m$. Thus $a(\delta_{n_j}) \geq \liminf_{m \rightarrow \infty} a_{n_m}(\delta_{n_j}) \geq \varepsilon$, for every j , which contradicts $a(\delta) \rightarrow 0$.

For counterexamples consider the array $g(n, m) = 1_{n > m}$, which has the properties that $m \mapsto g(n, m)$ is decreasing, and $g(n, m) \rightarrow 1$ if $n \rightarrow \infty$ followed by $m \rightarrow \infty$, while $g(n, n) = 0$, for every n . Set $a_n(\delta) = g(n, m)$, for $1/(m+1) < \delta \leq 1/m$, so that $a_n(1/m) = g(n, m)$. This function has $a_n(\delta) \rightarrow 1$ as $n \rightarrow \infty$ for every δ and hence does not satisfy (ii), but $a_n(\delta_n) = 0 \rightarrow 0$, for $\delta_n = 1/n$, so that a_n does satisfy (iii). Second, set $a_n(\delta) = 1 - g(n, m)$, for $1/(m+1) < \delta \leq 1/m$. Then $a_n(\delta) \rightarrow 0$ for every δ and hence a_n satisfies (ii), but $a_n(\delta_n) = 1$, for $\delta_n = 1/n$.]

7. The packing numbers of a ball $B(0, R) = \{x: \|x\| \leq R\}$ of radius R in $\mathbb{R}^d$ satisfy, for any norm,

$$D(\varepsilon, B(0, R), \|\cdot\|) \leq \left(\frac{3R}{\varepsilon}\right)^d, \quad 0 < \varepsilon \leq R.$$

[**Hint:** If $x_1, \dots, x_m$ is an ε -separated subset in $B(0, R)$, then the balls of radii $\varepsilon/2$ around the x_i are disjoint and contained in $B(0, R + \varepsilon/2)$. Comparison of the volume of their union and the volume of $B(0, R + \varepsilon/2)$ shows that $m(\varepsilon/2)^d \leq (R + \varepsilon/2)^d$.]

2.2

Maximal Inequalities

In this chapter we derive maximal inequalities for general stochastic processes. These inequalities will be used to establish the asymptotic equicontinuity of the empirical process, but have much wider applicability. Thus we temporarily leave the empirical process framework. After obtaining bounds for finite maxima in terms of Orlicz norms, we derive general maximal inequalities in terms of covering numbers and entropy, with special attention for sub-Gaussian processes and processes of mixed sub-Gaussian and sub-exponential tails. These are the main results needed for the treatment of empirical processes in the remainder of the manuscript. We close the chapter with a discussion of maximal inequalities in terms of generic chaining and majorizing measures, which will be useful in some later results.

2.2.1 Orlicz Norms

A *Young function* is a convex function $\psi: [0, \infty) \rightarrow [0, \infty)$ with $\psi(0) = 0$ that is not identically zero. For a given Young function and random variable X , the *Orlicz norm* $\|X\|_\psi$ is defined as

$$\|X\|_\psi = \inf \left\{ C > 0: \mathbb{E} \psi \left(\frac{|X|}{C} \right) \leq 1 \right\}.$$

(Here the infimum over the empty set is ∞ .) Using Jensen's inequality, it is not difficult to check that this indeed defines a norm (on the set of random variables for which $\|X\|_\psi$ is finite). The best known examples of Orlicz norms are those corresponding to the functions $x \mapsto x^p$ for $p \geq 1$: the corresponding Orlicz norms are simply the L_p -norms

$$\|X\|_p = (\mathbb{E}|X|^p)^{1/p}.$$

Orlicz norms that will be of special interest are the ones given by the functions $\psi_p(x) = e^{x^p} - 1$ for $p \geq 1$, which give much more weight to the tails of X . The

bound $x^p \leq \psi_p(x)$ for all nonnegative x implies that $\|X\|_p \leq \|X\|_{\psi_p}$ for each p . It is not true that the exponential Orlicz norms are all bigger than all L_p -norms. However, there exists a constant C_α that depends on α only so that

$$\begin{aligned}\|X\|_{\psi_p} &\leq \|X\|_{\psi_q} (\log 2)^{1/q-1/p}, & p \leq q, \\ \|X\|_p &\leq C_\alpha p^{1/\alpha} \|X\|_{\psi_\alpha}.\end{aligned}$$

(See Problems 2.2.4 and 2.2.5.) Since for the present purposes fixed constants in inequalities are irrelevant, this means that a bound on an exponential Orlicz norm always gives a better result than a bound on an L_p -norm. The second inequality in the display can be reversed in the sense that $\|X\|_{\psi_\alpha}$ is bounded by a multiple of the supremum of $\|X\|_p/p^{1/\alpha}$ over p , so that an exponential Orlicz norm may be viewed as equivalent to uniform control over L_p norms (see Problem 2.2.4).

Any Orlicz norm can be used to obtain an estimate of the tail of a distribution. By Markov's inequality,

$$\mathbf{P}(|X| > x) \leq \mathbf{P}(\psi(|X|/\|X\|_\psi) \geq \psi(x/\|X\|_\psi)) \leq \frac{1}{\psi(x/\|X\|_\psi)}.$$

For $\psi_p(x) = e^{x^p} - 1$, this leads to tail estimates $\exp(-Cx^p)$ for any random variable with a finite ψ_p -norm. Conversely, an exponential tail bound of this type shows that $\|X\|_{\psi_p}$ is finite.

2.2.1 Lemma. *Let X be a random variable with $\mathbf{P}(|X| > x) \leq Ke^{-Cx^p}$ for every $x > 0$, for constants K and C , and for $p \geq 1$. Then its Orlicz norm satisfies $\|X\|_{\psi_p} \leq ((1+K)/C)^{1/p}$.*

Proof. By Fubini's theorem

$$\mathbf{E}(e^{D|X|^p} - 1) = \mathbf{E} \int_0^{|X|^p} De^{Ds} ds = \int_0^\infty \mathbf{P}(|X| > s^{1/p}) De^{Ds} ds.$$

Now insert the inequality on the tails of $|X|$ and obtain the explicit upper bound $KD/(C-D)$. This is less than or equal to 1 for $D^{-1/p}$ greater than or equal to $((1+K)/C)^{1/p}$. ■

Orlicz norms are convenient to express the size of a maximum of finitely many random variables in the number of variables. In the special case of the L_p -norm, this is easily seen directly. Using the fact that $\max |X_i|^p \leq \sum |X_i|^p$, we obtain

$$\left\| \max_{1 \leq i \leq m} X_i \right\|_p = \left(\mathbf{E} \max_{1 \leq i \leq m} |X_i|^p \right)^{1/p} \leq m^{1/p} \max_{1 \leq i \leq m} \|X_i\|_p.$$

A similar inequality is valid for many Orlicz norms, in particular the exponential ones. Here, in the general case, the factor $m^{1/p}$ becomes $\psi^{-1}(m)$, where ψ^{-1} is the inverse function of ψ .

2.2.2 Lemma (Finite maximum). *Let ψ be a convex, nondecreasing, nonzero function with $\psi(0) = 0$ and $\limsup_{x,y \rightarrow \infty} \psi(x)\psi(y)/\psi(cxy) < \infty$ for some constant c . Then, for any random variables $X_1, \dots, X_m$, and a constant K depending only on ψ ,*

$$\left\| \max_{1 \leq i \leq m} X_i \right\|_{\psi} \leq K \psi^{-1}(m) \max_i \|X_i\|_{\psi}.$$

Proof. For simplicity of notation assume first that $\psi(x)\psi(y) \leq \psi(cxy)$ for all $x, y \geq 1$. In that case $\psi(x/y) \leq \psi(cx)/\psi(y)$ for all $x \geq y \geq 1$. Thus, for $y \geq 1$ and any C ,

$$\begin{aligned} \max \psi \left(\frac{|X_i|}{Cy} \right) &\leq \max \left[\frac{\psi(c|X_i|/C)}{\psi(y)} + \psi \left(\frac{|X_i|}{Cy} \right) 1_{\left\{ \frac{|X_i|}{Cy} < 1 \right\}} \right] \\ &\leq \sum \frac{\psi(c|X_i|/C)}{\psi(y)} + \psi(1). \end{aligned}$$

Set $C = c \max \|X_i\|_{\psi}$, and take expectations to get

$$E\psi \left(\frac{\max |X_i|}{Cy} \right) \leq \frac{m}{\psi(y)} + \psi(1).$$

When $\psi(1) \leq 1/2$, this is less than or equal to 1 for $y = \psi^{-1}(2m)$, which is greater than 1 under the same condition. Thus

$$\left\| \max |X_i| \right\|_{\psi} \leq \psi^{-1}(2m) c \max \|X_i\|_{\psi}.$$

By the convexity of ψ and the fact that $\psi(0) = 0$, it follows that $\psi^{-1}(2m) \leq 2\psi^{-1}(m)$. The proof is complete for every special ψ that meets the conditions made previously.

For a general ψ , there are constants $\sigma \leq 1$ and $\tau > 0$ such that $\phi(x) = \sigma\psi(\tau x)$ satisfies the conditions of the previous paragraph. Apply the inequality to ϕ , and observe that $\|X\|_{\psi} \leq \|X\|_{\phi}/(\sigma\tau) \leq \|X\|_{\psi}/\sigma$ (Problem 2.2.3). ■

For the present purposes, the value of the constant K in the previous lemma is uninteresting. (For the L_p -norms, it can be taken equal to 1.) The important conclusion is that the inverse of the ψ -function determines the size of the ψ -norm of a maximum in comparison to the ψ -norms of the individual terms. The ψ -norm grows slowest for rapidly increasing ψ . For $\psi(x) = e^{x^p} - 1$, the growth is at most logarithmic, because

$$\psi_p^{-1}(m) = (\log(1 + m))^{1/p}.$$

If ψ dominates the identity function, then $E \max_i |X_i| \leq \left\| \max_i |X_i| \right\|_{\psi}$ and hence the growth of $\psi^{-1}(m)$ can also be viewed as a bound on the growth of the distribution of the maximum.

2.2.2 Covering Numbers and Entropy

Lemma 2.2.1 is useless in the case of a maximum over infinitely many variables. However, such a case can be handled via repeated application of the lemma via a method known as *chaining*. Every random variable in the supremum is written as a sum of “little links,” and the bound depends on the number and sizes of the little links needed. For a stochastic process $\{X_t: t \in T\}$, the number of links depends on the entropy of the index set for the semimetric

$$d(s, t) = \|X_s - X_t\|_\psi.$$

The general definition of *metric entropy* is as follows.

2.2.3 Definition (Covering numbers). Let (T, d) be an arbitrary semimetric space. Then the *covering number* $N(\varepsilon, d)$ is the minimal number of balls of radius ε needed to cover T . Call a collection of points ε -*separated* if the distance between each pair of points is strictly larger than ε . The *packing number* $D(\varepsilon, d)$ is the maximum number of ε -separated points in T . The corresponding *entropy numbers* are the logarithms of the covering and packing numbers, respectively.

For the present purposes, both covering and packing numbers can be used. In all arguments one can be replaced by the other through the inequalities

$$N(\varepsilon, d) \leq D(\varepsilon, d) \leq N(\varepsilon/2, d).$$

Clearly, covering and packing numbers become bigger as $\varepsilon \downarrow 0$. By definition, the semimetric space T is totally bounded if and only if the covering and packing numbers are finite for every $\varepsilon > 0$. The upper bound in the following maximal inequality depends on the rate at which $D(\varepsilon, d)$ grows as $\varepsilon \downarrow 0$, as measured through an integral criterion.

2.2.4 Theorem (Maximal inequality). Let ψ be a convex, nondecreasing, nonzero function with $\psi(0) = 0$ and $\limsup_{x, y \rightarrow \infty} \psi(x)\psi(y)/\psi(cxy) < \infty$, for some constant c . Let $\{X_t: t \in T\}$ be a separable stochastic process[#] with

$$\|X_s - X_t\|_\psi \leq C d(s, t), \quad \text{for every } s, t,$$

for some semimetric d on T and a constant C . Then, for any $\eta, \delta > 0$, and a constant K depending on ψ and C only,

$$\left\| \sup_{d(s, t) \leq \delta} |X_s - X_t| \right\|_\psi \leq K \left[\int_0^\eta \psi^{-1}(D(\varepsilon, d)) d\varepsilon + \delta \psi^{-1}(D^2(\eta, d)) \right].$$

[#] *Separable* may be understood in the sense that $\sup_{d(s, t) < \delta} |X_s - X_t|$ remains almost surely the same if the index set T is replaced by a suitable countable subset.

2.2.5 Corollary. *The constant K can be chosen such that, for $\text{diam } T$ the diameter of T ,*

$$\left\| \sup_{s,t} |X_s - X_t| \right\|_{\psi} \leq K \int_0^{\text{diam } T} \psi^{-1}(D(\varepsilon, d)) d\varepsilon.$$

Proof. Assume without loss of generality that the packing numbers and the associated “covering integral” are finite. Construct nested sets $T_0 \subset T_1 \subset T_2 \subset \cdots \subset T$ such that every T_j is a maximal set of points such that $d(s, t) > \eta 2^{-j}$ for every $s, t \in T_j$ where “maximal” means that no point can be added without destroying the validity of the inequality. By the definition of packing numbers, the number of points in T_j is less than or equal to $D(\eta 2^{-j}, d)$.

“Link” every point $t_{j+1} \in T_{j+1}$ to a unique $t_j \in T_j$ such that $d(t_j, t_{j+1}) \leq \eta 2^{-j}$. Thus obtain for every t_{k+1} a chain $t_{k+1}, t_k, \dots, t_0$ that connects it to a point in T_0 . For arbitrary points s_{k+1}, t_{k+1} in T_{k+1} , the difference in increments along their chains can be bounded by

$$\begin{aligned} |(X_{s_{k+1}} - X_{s_0}) - (X_{t_{k+1}} - X_{t_0})| &= \left| \sum_{j=0}^k (X_{s_{j+1}} - X_{s_j}) - \sum_{j=0}^k (X_{t_{j+1}} - X_{t_j}) \right| \\ &\leq 2 \sum_{j=0}^k \max |X_u - X_v|, \end{aligned}$$

where for fixed j the maximum is taken over all links (u, v) from T_{j+1} to T_j . Thus the j th maximum is taken over at most $\#T_{j+1}$ links, with each link having a ψ -norm $\|X_u - X_v\|_{\psi}$ bounded by $C d(u, v) \leq C \eta 2^{-j}$. It follows with the help of Lemma 2.2.2 that, for a constant depending only on ψ and C ,

$$\begin{aligned} \left\| \max_{s,t \in T_{k+1}} |(X_s - X_{s_0}) - (X_t - X_{t_0})| \right\|_{\psi} &\leq K \sum_{j=0}^k \psi^{-1}(D(\eta 2^{-j-1}, d)) \eta 2^{-j} \\ (2.2.6) \qquad \qquad \qquad &\leq 4K \int_0^{\eta} \psi^{-1}(D(\varepsilon, d)) d\varepsilon. \end{aligned}$$

In this bound, s_0 and t_0 are the endpoints of the chains starting at s and t , respectively.

The maximum of the increments $|X_{s_{k+1}} - X_{t_{k+1}}|$ can be bounded by the maximum on the left side of (2.2.6) plus the maximum of the discrepancies $|X_{s_0} - X_{t_0}|$ at the end of the chains. The maximum of the latter discrepancies will be analyzed by a seemingly circular argument. For every pair of endpoints s_0, t_0 of chains starting at two points in T_{k+1} within distance δ of each other, choose exactly one pair s_{k+1}, t_{k+1} in T_{k+1} , with $d(s_{k+1}, t_{k+1}) < \delta$, whose chains end at s_0, t_0 . By definition of T_0 , this gives at most $D^2(\eta, d)$ pairs. By the triangle inequality,

$$|X_{s_0} - X_{t_0}| \leq |(X_{s_0} - X_{s_{k+1}}) - (X_{t_0} - X_{t_{k+1}})| + |X_{s_{k+1}} - X_{t_{k+1}}|.$$

Take the maximum over all pairs of endpoints s_0, t_0 as above. Then the corresponding maximum over the first term on the right in the last display is bounded by the

maximum in the left side of (2.2.6). Its ψ -norm can be bounded by the right side of this equation. Combine this with (2.2.6) to find that

$$\left\| \max_{\substack{s, t \in T_{k+1} \\ d(s, t) < \delta}} |X_s - X_t| \right\|_{\psi} \leq 8K \int_0^{\eta} \psi^{-1}(D(\varepsilon, d)) d\varepsilon + \left\| \max |X_{s_{k+1}} - X_{t_{k+1}}| \right\|_{\psi}.$$

Here the maximum on the right is taken over the pairs s_{k+1}, t_{k+1} in T_{k+1} uniquely attached to the pairs s_0, t_0 as above. Thus the maximum is over at most $D^2(\eta, d)$ terms, each of whose ψ -norm is bounded by δ . Its ψ -norm is bounded by $K \psi^{-1}(D^2(\eta, d)) \delta$.

Thus the upper bound given by the theorem is a bound for the maximum of increments over T_{k+1} . Let k tend to infinity to conclude the proof.

The corollary follows immediately from the previous proof, after noting that, for η equal to the diameter of T , the set T_0 consists of exactly one point. In that case $s_0 = t_0$ for every pair s, t , and the increments at the end of the chains are zero. The corollary also follows from the theorem upon taking $\eta = \delta = \text{diam } T$ and noting that $D(\eta, d) = 1$, so that the second term in the maximal inequality can also be written $\delta \psi^{-1}(D(\eta, d))$. Since the function $\varepsilon \mapsto \psi^{-1}(D(\varepsilon, d))$ is decreasing, this term can be absorbed into the integral, perhaps at the cost of increasing the constant K . ■

Though the theorem gives a bound on the continuity modulus of the process, if a bound on the maximum of the process is needed, this follows from, for any t_0 ,

$$\left\| \sup_t |X_t| \right\|_{\psi} \leq \|X_{t_0}\|_{\psi} + \left\| \sup_t |X_t - X_{t_0}| \right\|_{\psi}.$$

To state the maximal inequality in terms of the increments appears natural. The increment bound shows that the process X is continuous in ψ -norm whenever the covering integral $\int_0^{\eta} \psi^{-1}(D(\varepsilon, d)) d\varepsilon$ converges for some $\eta > 0$. (In that case, the right side in Theorem 2.2.4 can be made arbitrarily small by choosing first η small and next δ .) It is a small step to deduce the continuity of almost all sample paths from this inequality (Problem 2.2.17).

2.2.7 Example (Tightness under increment bounds). In Section 1.14.2.2 we derived a classical, simple sufficient condition for tightness of a sequence of stochastic processes indexed by the real numbers. The proof of Theorem 1.14.22 is based on direct chaining arguments. Similar results can be deduced from the general maximal inequality obtained previously. In this example we give a version of the latter theorem for tightness relative to the uniform norm.

Let $\{X_n(t) : t \in [0, 1]\}$ be a sequence of separable stochastic processes with bounded sample paths and increments satisfying

$$\mathbb{E}|X_n(s) - X_n(t)|^p \leq K|s - t|^{1+r},$$

for constants $p, K, r > 0$ independent of n . Assume that the sequences of marginals $(X_n(t_1), \dots, X_n(t_k))$ converge weakly to the corresponding marginals

of a stochastic process $\{X(t): t \in [0, 1]\}$. Then there exists a version of X with continuous sample paths and $X_n \rightsquigarrow X$ in $\ell^\infty[0, 1]$. (Hence also in $D[0, 1]$ or $C[0, 1]$, provided every X_n has all its sample paths in these spaces.)

To prove this for $p > 1$, apply Theorem 2.2.4 with $\psi(x) = x^p$ and $d(s, t) = |s - t|^\alpha$, where $\alpha = ((1+r)/p) \wedge 1$. A d -ball of radius ε around some t is simply the interval $[t - \varepsilon^{1/\alpha}, t + \varepsilon^{1/\alpha}]$. Thus the index set $[0, 1]$ can be covered with $N(\varepsilon, d) = (1/2)\varepsilon^{-1/\alpha}$ balls of radius ε . Since $\psi^{-1}(x) = x^{1/p}$, the covering integral can be bounded by a multiple of

$$\int_0^\eta \psi^{-1}(N(\varepsilon, d)) d\varepsilon \leq \int_0^\eta \varepsilon^{-1/(p\alpha)} d\varepsilon.$$

For $p > 1$, the integral converges. Since $\|X_n(s) - X_n(t)\|_p \leq K^{1/p}|s - t|^\alpha$, the general maximal inequality and Markov's inequality give

$$\mathbf{P}\left(\sup_{|s-t|<\delta} |X_n(s) - X_n(t)| > x\right) \leq \frac{C}{x} \left[\int_0^\eta \varepsilon^{-1/(p\alpha)} d\varepsilon + \delta \eta^{-2/(p\alpha)} \right],$$

for some constant C independent of n . By choosing first η and next δ , one can make the right side arbitrarily small. This verifies the asymptotic equicontinuity of X_n . Its weak convergence and the continuity of the limit process follow from Theorem 1.5.7.

For $p \leq 1$, use the inequality $|(x^+)^{1/q} - (y^+)^{1/q}| \leq |x - y|^{1/q}$ (valid for all reals x, y , and $q \geq 1$), with $q = 2/p$, to derive that

$$\mathbf{E}|X_n^+(s)^{p/2} - X_n^+(t)^{p/2}|^2 \leq K|s - t|^{1+r}, \quad \text{for every } s, t.$$

By the previous argument, it follows that $(X_n^+)^{p/2} \rightsquigarrow (X^+)^{p/2}$, where $(X^+)^{p/2}$ is a process with continuous sample paths. Since the map $x \mapsto x^q$ from $\ell^\infty[0, 1]$ to $\ell^\infty[0, 1]$ is continuous at every $x \in C[0, 1]$ (Problem 2.2.15), it follows that $X_n^+ \rightsquigarrow X^+$. By a similar argument, $X_n^- \rightsquigarrow X^-$. Together this yields $X_n \rightsquigarrow X$.

2.2.3 Sub-Gaussian Inequalities

A standard normal variable X has tails of the order $x^{-1} \exp(-x^2/2)$ and satisfies $\mathbf{P}(|X| > x) \leq 2 \exp(-x^2/2)$ for every $x \geq 0$. By direct calculation one finds a ψ_2 -norm $\|X\|_{\psi_2} = \sqrt{8/3}$. In this subsection we study random variables satisfying similar tail bounds.

Hoeffding's inequality asserts a "sub-Gaussian" tail bound for random variables of the form $X = \sum_i Y_i$ with $Y_1, \dots, Y_n$ i.i.d. with zero means and bounded range (see Proposition A.6.1). The following special case of Hoeffding's inequality will be needed.

2.2.8 Lemma (Hoeffding's inequality). Let $a_1, \dots, a_n$ be constants and $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher variables; i.e., random variables with $P(\varepsilon_i = 1) = P(\varepsilon_i = -1) = 1/2$. Then, for $\|a\|$ the Euclidean norm of $a \in \mathbb{R}^n$,

$$P\left(\left|\sum_{i=1}^n \varepsilon_i a_i\right| > x\right) \leq 2e^{-\frac{1}{2}x^2/\|a\|^2}.$$

Consequently, $\|\sum_i \varepsilon_i a_i\|_{\psi_2} \leq \sqrt{6}\|a\|$.

Proof. For any λ and Rademacher variable ε , one has $Ee^{\lambda\varepsilon} = (e^\lambda + e^{-\lambda})/2 \leq e^{\lambda^2/2}$, where the last inequality follows after writing out the power series. Thus by Markov's inequality, for any $\lambda > 0$,

$$P\left(\sum_{i=1}^n a_i \varepsilon_i > x\right) \leq e^{-\lambda x} Ee^{\lambda \sum_{i=1}^n a_i \varepsilon_i} \leq e^{(\lambda^2/2)\|a\|^2 - \lambda x}.$$

The best upper bound is obtained for $\lambda = x/\|a\|^2$ and is the exponential in the probability bound of the lemma. Combination with a similar bound for the lower tail yields the probability bound.

The bound on the ψ -norm is a consequence of the probability bound in view of Lemma 2.2.1. ■

A stochastic process is called *sub-Gaussian* with respect to the semimetric d on its index set if

$$P(|X_s - X_t| > x) \leq 2e^{-\frac{1}{2}x^2/d^2(s,t)}, \quad \text{for every } s, t \in T, x > 0.$$

The constants 2 and 1/2 are of no special importance (see Problem 2.2.14).

Any Gaussian process is sub-Gaussian for the standard deviation semimetric $d(s, t) = \sigma(X_s - X_t)$. Another example is the *Rademacher process*

$$X_a = \sum_{i=1}^n a_i \varepsilon_i, \quad a \in \mathbb{R}^n,$$

for independent Rademacher variables $\varepsilon_1, \dots, \varepsilon_n$. By Hoeffding's inequality, this is sub-Gaussian for the Euclidean distance $d(a, b) = \|a - b\|$. A further example is the empirical process $\{\mathbb{G}_n f : f \in \mathcal{F}\}$ relative to a multiple of the metric $d(f, g) = \|f - g\|_{P, \psi_2}$, induced by the ψ_2 -Orlicz norm on the class $\mathcal{F}$ of functions (and hence also relative to the stronger uniform metric). This may be derived from Proposition 2.14.23, see (2.14.24). Although useful, this metric is too strong in many cases. More practical maximal inequalities for the empirical process can be derived using symmetrization (see Chapter 2.3) or truncation in conjunction with the more natural L_2 -distance (see Chapter 2.14).

Sub-Gaussian processes satisfy the increment bound $\|X_s - X_t\|_{\psi_2} \leq \sqrt{6}d(s, t)$. Since the inverse of the ψ_2 -function is essentially the square root of the logarithm, the general maximal inequality leads for sub-Gaussian processes to a bound in terms of an entropy integral. (Remember that entropy is defined as the logarithm of packing numbers.) Furthermore, because of the special properties of the logarithm, the statement can be slightly simplified.

2.2.9 Corollary. Let $\{X_t: t \in T\}$ be a separable sub-Gaussian process. Then for every $\delta > 0$,

$$\mathbb{E} \sup_{d(s,t) \leq \delta} |X_s - X_t| \leq K \int_0^\delta \sqrt{\log D(\varepsilon, d)} d\varepsilon,$$

for a universal constant K . In particular, for any t_0 ,

$$\mathbb{E} \sup_t |X_t| \leq \mathbb{E} |X_{t_0}| + K \int_0^\infty \sqrt{\log D(\varepsilon, d)} d\varepsilon.$$

Proof. Apply the general maximal inequality with $\psi_2(x) = e^{x^2} - 1$ and $\eta = \delta$. Since $\psi_2^{-1}(m) = \sqrt{\log(1+m)}$, we have $\psi_2^{-1}(D^2(\delta, d)) \leq \sqrt{2} \psi_2^{-1}(D(\delta, d))$. Thus the second term in the maximal inequality can first be replaced by $\sqrt{2} \delta \psi_2^{-1}(D(\eta, d))$ and next be incorporated in the first (the covering integral) at the cost of increasing the constant. We obtain

$$\left\| \sup_{d(s,t) \leq \delta} |X_s - X_t| \right\|_{\psi_2} \leq K \int_0^\delta \sqrt{\log(1 + D(\varepsilon, d))} d\varepsilon.$$

Here $D(\varepsilon, d) \geq 2$ for every ε that is strictly less than the diameter of T . Since $\log(1+m) \leq 2 \log m$ for $m \geq 2$, the 1 inside the logarithm can be removed at the cost of increasing K . ■

2.2.4 Mixed Sub-Gaussian-Exponential Inequalities

Since many random variables have heavier than normal tails, the results of the previous section are not always useful. A sum of independent variables with mean zero and finite variances is, for large n , approximately normally distributed. For finite n , its lower tails are typically still sub-Gaussian, but further out the distribution may differ from normal. The simplest case of *Bernstein's inequality* shows that the tails beyond nv/M of a sum of bounded variables, with variance v and uniform bound M , are sub-exponential. A more general form of the inequality is as follows.

2.2.10 Lemma (Bernstein's inequality). If $Y_1, \dots, Y_n$ are independent random variables such that $n^{-1} \sum_{i=1}^n \mathbb{E} |Y_i|^m \leq m! M^{m-2} v/2$, for every $m \geq 2$, for some constants M and v , then, for every $x > 0$,

$$\mathbb{P}(|Y_1 - \mathbb{E} Y_1 + \dots + Y_n - \mathbb{E} Y_n| > x) \leq 2e^{-\frac{1}{2} \frac{x^2}{nv + Mx}}.$$

Proof. See Bennett (1962), pages 37–38, or Shorack and Wellner (1986), page 855, or in a slightly more general, one-sided form, Boucheron, Lugosi and Massart (2013), Corollary 2.11. ■

Because $e^x - 1 - x \geq x^m/m!$, for $m \geq 2$, the condition on the moments is implied by

$$\frac{1}{n} \sum_{i=1}^n \mathbb{E} \left(e^{|Y_i|/M} - 1 - \frac{|Y_i|}{M} \right) \leq \frac{v}{2M^2}.$$

Conversely, if the Y_i satisfy the moment conditions of the lemma, then the preceding display is valid with M replaced by $2M$ and v by $2v$. Thus for applications where constants in Bernstein's inequality are unimportant, the preceding display is "equivalent" to the moment condition of the lemma.

2.2.11 Example. If $Y_1, \dots, Y_n$ are i.i.d. mean zero variables with bounded range, then the conditions of the lemma are satisfied with $v = \text{var } Y_1$ and $M = \|Y_1\|_\infty/3$.

This is immediate from the estimate $\mathbb{E}|Y_1|^m \leq \|Y_1\|_\infty^{m-2} \mathbb{E}Y_1^2$, for every $m \geq 2$, and the fact that $m!/2 \geq 3^{m-2}$.

2.2.12 Example. If $Y_1, \dots, Y_n$ are i.i.d. mean zero variables with finite ψ_1 -norm, then the conditions of the lemma are satisfied with $v = \|Y_1\|_{\psi_1}^2$ and $M = \|Y_1\|_{\psi_1}$.

This follows because $\mathbb{E}\psi_1(|Y_1|/\|Y_1\|_{\psi_1}) \leq 1$, by the definition of the ψ_1 -norm, where $\psi_1(x) = e^x - 1 \geq 2x^m/m!$, for every $m \geq 2$ (see Problem 2.2.24), and obviously $v/M^2 \geq 1$, for the given choices.

Bernstein's inequality gives upper bounds of the general type, for $x > 0$,

$$\mathbb{P}(|X| > x) \leq 2e^{-\frac{x^2}{a+bx}}.$$

The upper bound is of sub-Gaussian type $e^{-x^2/a}$ for small x , while for larger x it is of exponential type $e^{-x/b}$. If the exact constants do not matter, then an easy further analysis is to note that the exponent in the display satisfies $h(x) \leq x^2/(a+bx) \leq 2h(x)$, for the function $h(x) = (x^2/2a)1_{x \leq a/b} + (x/2b)1_{x > a/b}$. Thus the left side of the display is bounded above by $2e^{-h(x)}$, which splits in separate bounds $2e^{-x^2/(2a)}$ and $2e^{-x/(2b)}$ on the intervals $x \leq a/b$ and $x > a/b$, respectively, (and can also be written as the bound $2e^{-x^2/(2a)} \vee e^{-x/(2b)}$ on the full half line). The function h is monotone and continuous, with inverse $h^{-1}(y) = \sqrt{2ay}1_{y \leq a/b^2} + 2by1_{y > a/b^2}$. The trivial upper bound $h^{-1}(y) \leq \sqrt{2ay} + 2by$, allows to "invert" the bound in the preceding display to, for $y > 0$,

$$\mathbb{P}(|X| > \sqrt{2ay} + 2by) \leq 2e^{-y}.$$

Conversely, this bound implies the original bound, but with the constants a and b replaced by $4\sqrt{2a}$ and $4b$. Thus essentially, up to constants, the bounds in the two preceding displays are equivalent, and are equivalent to separate sub-Gaussian and sub-exponential tail bounds on the intervals $x \leq a/b$ and $x > a/b$.

The separate sub-Gaussian and sub-exponential bounds suggest that a maximum of variables that satisfy a Bernstein-type bound can be controlled using a combination of the ψ_1 - and ψ_2 -Orlicz norms. This is true. The maximum also satisfies a mixed sub-Gaussian and sub-exponential tail bound.

2.2.13 Lemma. Let $X_1, \dots, X_m$ be arbitrary random variables that satisfy the tail bound, for fixed $a, b > 0$ and all $x > 0$,

$$\mathbb{P}(|X_i| > x) \leq 2e^{-\frac{x^2}{a+bx}},$$

Then, for a universal constant K ,

$$\left\| \max_{1 \leq i \leq m} |X_i| \right\|_{\psi_1} \leq K \left(\sqrt{a} \sqrt{\log(1+m)} + b \log(1+m) \right).$$

The constant K can be chosen so that, for $y > 0$,

$$\mathbb{P} \left(\max_{1 \leq i \leq m} |X_i| > K \sqrt{y a} \sqrt{\log(1+m)} + K y b \log(1+m) \right) \leq 2e^{-y}.$$

Proof. The condition implies the upper bound $2 \exp(-x^2/(2a))$ on $\mathbb{P}(|X_i| > x)$, for every $x \leq a/b$, and the upper bound $2 \exp(-x/(2b))$, for all other positive x . Consequently, the same upper bounds hold for all $x > 0$ for the probabilities $\mathbb{P}(|X_i| 1\{|X_i| \leq a/b\} > x)$ and $\mathbb{P}(|X_i| 1\{|X_i| > a/b\} > x)$, respectively. By Lemma 2.2.1, this implies that the Orlicz norms $\|X_i 1\{|X_i| \leq a/b\}\|_{\psi_2}$ and $\|X_i 1\{|X_i| > a/b\}\|_{\psi_1}$ are up to constants bounded by $\sqrt{a}$ and b , respectively. Next

$$\left\| \max_i |X_i| \right\|_{\psi_1} \leq \left\| \max_i |X_i| 1\{|X_i| \leq a/b\} \right\|_{\psi_1} + \left\| \max_i |X_i| 1\{|X_i| > a/b\} \right\|_{\psi_1}.$$

Since the ψ_p -norms are up to constants nondecreasing in p , the first ψ_1 -norm on the right can be replaced by a ψ_2 -norm. Apply Lemma 2.2.2 to find the Orlicz norm bound as stated.

The separate bounds $\sqrt{a \log(1+m)}$ and $b \log(1+m)$ on the ψ_2 - and ψ_1 -Orlicz norms of the variables $\max_i |X_i| 1\{|X_i| \leq a/b\}$ and $\max_i |X_i| 1\{|X_i| > a/b\}$ in the right side of the preceding display, imply sub-Gaussian and sub-exponential bounds on the tails of the variables $M 1\{M \leq a/b\}$ and $M 1\{M > a/b\}$, for $M = \max_i |X_i|$, which are bounded above by the two variables. The sum (and hence twice the maximum) of these two tail bounds at x dominates the tail probabilities $\mathbb{P}(M > 2x)$. This can next be inverted in the bound as given, by the arguments preceding the lemma. ■

2.2.5 Generic Chaining

Maximal inequalities based on covering numbers appear to suffice for most statistical applications, but are known to be suboptimal in general. The suboptimality is particularly well documented for Gaussian maximal inequalities, for which the alternate techniques generic chaining and majorizing measures give sharp bounds. In this section we discuss generic chaining within the context of Theorem 2.2.4.

As in the latter theorem, consider a separable stochastic process $\{X_t: t \in T\}$ indexed by a semimetric space (T, d) , with increments satisfying the bound $\|X_s - X_t\|_{\psi} \lesssim d(s, t)$, for a given Young function ψ . Unlike Theorem 2.2.4, the following theorem does not impose further conditions on ψ , but instead formulates its bound in terms of another Young function ϕ . (This can always be taken to be the identity, which then results in a bound on the mean of the supremum.)

The bound is expressed in terms of a sequence $T_0, T_1, T_2, \dots$ of subsets of T , not necessarily nested, which are only limited by their cardinalities. The set T_0 must consist of a single point, while the cardinality of T_j must satisfy $|T_j| \leq \psi(4^j)$, for every $j \geq 1$. Such a sequence of subsets is called ψ -admissible.[†]

[†] The base 4 in the powers 4^j is arbitrary and could be replaced by any other number bigger than 1. See Problems 2.2.18–2.2.19.

2.2.14 Theorem (Maximal inequality). *Let ϕ and ψ be convex, nondecreasing, nonzero functions with $\phi(0) = \psi(0) = 0$ and $\phi(y/x) \leq \psi(y)/\psi(x)$, for every $y \geq cx$ and $x \geq 1$, for some $c > 0$. Let $\{X_t: t \in T\}$ be a separable stochastic process with*

$$\|X_s - X_t\|_\psi \leq C d(s, t), \quad \text{for every } s, t,$$

for some semimetric d on T and a constant C . Then, for any sequence $T_0, T_1, T_2, \dots$ of nonempty subsets of T such that $|T_0| = 1$ and $|T_j| \leq \psi(4^j)$ for every $j \geq 1$,

$$\left\| \sup_{s, t \in T} |X_s - X_t| \right\|_\phi \leq K \sup_{t \in T} \sum_{j=0}^{\infty} d(t, T_j) 4^j + K \sum_{j=0}^{\infty} \frac{1}{\psi(4^{j+1})} \sum_{t \in T_{j+1}} d(t, T_j) 4^j,$$

for a constant K depending on ϕ, ψ, c and C only. Furthermore, in the special case that $\psi(x) = e^{x^p} - 1$, for some $p \geq 1$, this can be improved to

$$\left\| \sup_{s, t \in T} |X_s - X_t| \right\|_\psi \leq K \sup_{t \in T} \sum_{j=0}^{\infty} d(t, T_j) 4^j.$$

Proof. By redefining the metric d , we can take the constant C equal to 1. By the assumed separability of X and the monotone convergence theorem, it suffices to prove the bounds for a finite set T . As the bounds decrease if the sets T_j are enlarged (while remaining admissible), it is then no loss of generality to assume that $T = T_{k+1}$, for some k . For $t \in T$ and $j \geq 0$, let $\pi_j t$ be a point in T_j closest to t , so that $d(t, T_j) = d(t, \pi_j t)$, and for every $t = t_{k+1} \in T_{k+1}$ form the chain $t_{k+1}, t_k, \dots, t_0$ with $t_j = \pi_j t_{j+1}$, for every j . By telescoping and the triangle inequality, for every $x > 0$,

$$|X_t - X_{t_0}| \leq \sum_{j=0}^k |X_{t_{j+1}} - X_{t_j}| 1_{\Omega_{t_{j+1}, x}} + x \sum_{j=0}^k d(t_{j+1}, t_j) 4^{j+1},$$

for $\Omega_{u, x}$ the event that $|X_u - X_{\pi_j u}| > x d(u, \pi_j u) 4^{j+1}$. We show at the end of the proof that

$$A_k := \sup_{t \in T_{k+1}} \sum_{j=0}^k d(t_{j+1}, t_j) 4^{j+1} \leq 4 \sup_{t \in T} \sum_{j=0}^{\infty} d(t, T_j) 4^{j+1} =: A.$$

The right side A is the first term in the general bound given by the theorem, and the only term in the bound for the exponential ψ . Therefore it suffices to complement the bound $A_k \lesssim A$ by a bound on the first term on the right in the second last display. We derive this bound first for the exponential ψ and next for general ψ .

For $\psi(x) = e^{x^p} - 1$ the preceding decomposition gives that

$$\begin{aligned} \mathbb{P}\left(\sup_{t \in T} |X_t - X_{t_0}| > xA\right) &\leq \mathbb{P}\left(\sum_{j=0}^k |X_{t_{j+1}} - X_{t_j}| 1_{\Omega_{t_{j+1}, x}} > 0\right) \\ &\leq \sum_{j=0}^k \sum_{u \in T_{j+1}} \mathbb{P}(\Omega_{u, x}) \leq \sum_{j=0}^k |T_{j+1}| \frac{1}{\psi(x 4^{j+1})} \leq \sum_{j=0}^k \frac{\psi(4^{j+1})}{\psi(x 4^{j+1})} \leq 2e^{-x^p}, \end{aligned}$$

for $x^p \geq 2$, where we used Markov's inequality and the assumption that $E\psi(|X_u - X_{\pi_j u}|/d(u, \pi_j u)) \leq 1$, for every u . Lemma 2.2.1 gives that $\|\sup_{t \in T} |X_t - X_{t_0}|\|_{\psi} \lesssim A$, which completes the proof for the exponential function ψ .

For a general Young function ψ we choose $x = c$ in the decomposition of the first paragraph, and apply the assumption $\phi(y/x) \leq \psi(y)/\psi(x)$ with $y = |X_{t_{j+1}} - X_{t_j}|/d(t_{j+1}, t_j)$ and $x = 4^{j+1}$, on the event $\Omega_{t_{j+1}, c}$, where the restrictions $y \geq cx$ and $x \geq 1$ are valid, to see that

$$\begin{aligned} \sum_{j=0}^k |X_{t_{j+1}} - X_{t_j}| 1_{\Omega_{t_{j+1}, c}} &\leq \sum_{j=0}^k \phi^{-1} \left(\frac{\psi \left(\frac{|X_{t_{j+1}} - X_{t_j}|}{d(t_{j+1}, t_j)} \right)}{\psi(4^{j+1})} \right) d(t_{j+1}, t_j) 4^{j+1} \\ &\leq \phi^{-1} \left(\sum_{j=0}^k \frac{\psi \left(\frac{|X_{t_{j+1}} - X_{t_j}|}{d(t_{j+1}, t_j)} \right) d(t_{j+1}, t_j) 4^{j+1}}{\psi(4^{j+1}) A} \right) A, \end{aligned}$$

by concavity of ϕ^{-1} and $\phi^{-1}(0) = 0$, since $\sum_{j=0}^k d(t_{j+1}, t_j) 4^{j+1} / A \leq A_k / A \leq 1$, by the definition of A_k . As the map $s \mapsto \phi^{-1}(y/s)s$ is increasing, for any $y \geq 0$, by concavity of ϕ^{-1} with $\phi^{-1}(0) = 0$, the right side of the display does not become smaller if A is replaced by $A \vee \tilde{A}$, where we take $\tilde{A} = \sum_{j=0}^k \psi(4^{j+1})^{-1} \sum_{t \in T_{j+1}} d(t, T_j) 4^{j+1}$, which is bigger than four times the second infinite series in the upper bound given by the theorem. By monotonicity of ϕ we conclude that

$$\begin{aligned} \phi \left(\frac{\sum_{j=0}^k |X_{t_{j+1}} - X_{t_j}| 1_{\Omega_{t_{j+1}, c}}}{A \vee \tilde{A}} \right) &\leq \sum_{j=0}^k \frac{\psi \left(\frac{|X_{t_{j+1}} - X_{t_j}|}{d(t_{j+1}, t_j)} \right) d(t_{j+1}, t_j) 4^{j+1}}{\psi(4^{j+1}) A \vee \tilde{A}} \\ &\leq \sum_{j=0}^k \sum_{u \in T_{j+1}} \frac{\psi \left(\frac{|X_u - X_{\pi_j u}|}{d(u, \pi_j u)} \right) d(u, \pi_j u) 4^{j+1}}{\psi(4^{j+1}) A \vee \tilde{A}}. \end{aligned}$$

The right side is independent of $t \in T_{k+1}$, and hence by the assumption on the increments of X , the expectation of the supremum over $t \in T_{k+1}$ of the left side is bounded above by

$$\sum_{j=0}^k \sum_{u \in T_{j+1}} \frac{d(u, \pi_j u) 4^{j+1}}{\psi(4^{j+1}) A \vee \tilde{A}} \leq \frac{\tilde{A}}{A \vee \tilde{A}} \leq 1.$$

It follows that the ϕ -Orlicz norm of $\sum_{j=0}^k |X_{t_{j+1}} - X_{t_j}| 1_{\Omega_{t_{j+1}, c}}$ is bounded above by $A \vee \tilde{A}$. This concludes the proof of the theorem for a general Young function ψ .

Finally, we prove that $A_k \leq A$, for every k . By the triangle inequality $d(t, t_i) \leq d(t, t_{i+1}) + d(t_{i+1}, t_i) \leq d(t, t_{i+1}) + d(t_{i+1}, \pi_i t)$, since $\pi_i t \in T_i$, while $t_i = \pi_i t_{i+1}$ is a point in T_i closest to t_{i+1} . Applying the triangle inequality to $d(t_{i+1}, \pi_i t)$, we find that $d(t, t_i) \leq d(t, \pi_i t) + 2d(t, t_{i+1})$. Using that $d(t, \pi_i t) = d(t, T_i)$ and iterating the inequality for $i = j, j+1, \dots, k$, we find $d(t, t_j) \leq \sum_{i=j}^k d(t, T_i) 2^{i-j}$, where the remainder term $d(t, t_{k+1}) 2^{k+1-j}$ has vanished, as $t = t_{k+1}$ by construction. We conclude that

$$\sum_{j=0}^k d(t, t_j) 4^j \leq \sum_{j=0}^k \sum_{i=j}^k d(t, T_i) 2^{i-j} 4^j \leq 2 \sum_{i=0}^k d(t, T_i) 4^i.$$

By another application of the triangle inequality $\sum_{j=0}^k d(t_{j+1}, t_j)4^j$ is not bigger than twice the left side, and hence the right side, of this display, for any $t \in T_{k+1}$. This completes the proof that $A_k \leq A$. ■

The condition that $\phi(y/x) \leq \psi(y)/\psi(x)$, for $y \geq x \geq 1$, is satisfied for ϕ the identity and every Young function ψ (see Problem 2.2.23). Thus the theorem gives a bound on the mean $E \sup_{s,t} |X_s - X_t|$ of any stochastic process whose increments are bounded in terms of an arbitrary Young function ψ . If $\psi(x)\psi(y) \leq \psi(cxy)$ for every $x, y \geq 1$ and some $c > 0$, which is the assumption in Theorem 2.2.4 up to multiplicative constants and scaling, then the assumption of the theorem is also satisfied for $\phi(x) = \psi(x/c)$, and the theorem provides a bound on the ψ -norm of the supremum.

The first term in the bound in the theorem involves a supremum over t taken outside the infinite sum. This expression increases if the supremum is moved inside the sum. Since $\psi(4^{j+1}) \geq |T_{j+1}|$, the second term in the bound is smaller than a sum of averages over the sets T_{j+1} and increases if these are replaced by suprema. It follows that both terms are bounded above by

$$\sum_{j=0}^{\infty} \sup_{t \in T} d(t, T_j) 4^j.$$

The infimum of this expression over all ψ -admissible sequences $T_0, T_1, T_2, \dots$ is essentially the entropy bound of Theorem 2.2.4, as the optimal choice of T_j would roughly be a minimal ε_j -net, for ε_j determined so that $|T_j| \asymp \psi(4^j)$.

2.2.15 Lemma. *Let ψ be a convex, nondecreasing, positive function with $\psi(0) = 0$ and $\psi(4) \geq 1$. Then there exists a ψ -admissible sequence of sets T_j such that, for a constant K that depends on ψ only,*

$$\sum_{j=0}^{\infty} \sup_{t \in T} d(t, T_j) 4^j \leq K \int_0^{\text{diam}(T)} \psi^{-1}(N(\varepsilon, T, d)) d\varepsilon.$$

Proof. The function $\varepsilon \mapsto N(\varepsilon, T, d)$ is decreasing to 1 at $\varepsilon = \text{diam}(T)$. Given $j \in \mathbb{N}$, let ε_j be the smallest nonnegative number such that $N(\varepsilon, T, d) \leq \psi(4^j)$, for every $\varepsilon > \varepsilon_j$. If $\varepsilon_j > 0$, then $N(2\varepsilon_j, T, d) \leq \psi(4^j)$, and we choose T_j a minimal $2\varepsilon_j$ -net over T . If $\varepsilon_j = 0$, then $N(\varepsilon, T, d) \leq \psi(4^j)$, for every $\varepsilon > 0$, and we choose $T_j = T$. Let T_0 be an arbitrary one-point set. Then $T_0, T_1, \dots$, is an admissible sequence, and $\sum_{j=0}^{\infty} \sup_t d(t, T_j) 4^j \leq \text{diam}(T) + 3 \sum_{j=1}^{\infty} \varepsilon_j 4^j$, where we can truncate the sum at $k = \sup\{j : \varepsilon_j > 0\}$, if this is finite.

Since $N(\varepsilon, T, d) \geq \psi(4^j)$, for $\varepsilon < \varepsilon_j$ and $1 \leq j \leq k$, we have

$$\sum_{j=1}^{\infty} (\varepsilon_j - \varepsilon_{j+1}) 4^j \leq \sum_{j=1}^k \int_{\varepsilon_{j+1}}^{\varepsilon_j} \psi^{-1}(N(\varepsilon, T, d)) d\varepsilon \leq \int_0^{\varepsilon_1} \psi^{-1}(N(\varepsilon, T, d)) d\varepsilon,$$

where $\varepsilon_1 \leq \text{diam}(T)$, since $N(\text{diam}(T), T, d) = 1$ and $\psi(4) \geq 1$. Because $\sum_{j=1}^{\infty} \varepsilon_{j+1} 4^j \leq (1/4) \sum_{j=2}^{\infty} \varepsilon_j 4^j$, the left side of the display is no

smaller than $(3/4)$ times the series $\sum_{j=1}^{\infty} \varepsilon_j 4^j$. Thus the latter series is bounded above by $4/3$ times the entropy integral. Since $N(\varepsilon, T, d) \geq 1$, for any $\varepsilon > 0$, and hence $\psi^{-1}(N(\varepsilon, T, d)) \geq \psi^{-1}(1)$, we have that $\text{diam}(T) \leq \int_0^{\text{diam}(T)} \psi^{-1}(N(\varepsilon, T, d)) / \psi^{-1}(1) d\varepsilon$. Combination with the preceding paragraph concludes the proof. ■

In combination with Theorem 2.2.14, the preceding lemma gives an alternative proof of Corollary 2.2.5.

The following theorem generalizes Theorem 2.2.14 to a bound on the modulus of the process X instead of the supremum of its increments. This theorem can be shown to imply Theorem 2.2.4, in the same way as Theorem 2.2.14 implies Corollary 2.2.5, through the preceding corollary.

2.2.16 Theorem (Maximal inequality). *Under the conditions Theorem 2.2.14, for every $j_0 \geq 0$ and $\delta > 0$,*

$$\left\| \sup_{d(s,t) < \delta} |X_s - X_t| \right\|_{\phi} \leq K \sup_{t \in T} \sum_{j=j_0}^{\infty} d(t, T_j) 4^j + K \sum_{j=j_0}^{\infty} \frac{1}{\psi(4^{j+1})} \sum_{t \in T_{j+1}} d(t, T_j) 4^j + K \delta \psi^{-1}(|T_{j_0}|^2),$$

for a constant K depending on ϕ , ψ , c and C only. In the special case that $\psi(x) = e^{x^p} - 1$, for some $p \geq 1$, the second series on the right may be omitted.

Proof. As in the proof of Theorem 2.2.14, assume without loss of generality that $T = T_{k+1}$, and take any pair $s_{k+1}, t_{k+1} \in T$ with $d(s_{k+1}, t_{k+1}) < \delta$ and form chains $s_{k+1}, s_k, \dots, s_0$ and $t_{k+1}, t_k, \dots, t_0$, connecting sequentially every element $u \in T_{j+1}$ in the chain to a closest point $\pi_j u \in T_j$, for $j = k, k-1, \dots, 0$. By telescoping and the triangle inequality,

$$\begin{aligned} & |(X_{t_{k+1}} - X_{t_{j_0}}) - (X_{s_{k+1}} - X_{s_{j_0}})| \\ & \leq \sum_{j=j_0}^k (|X_{t_{j+1}} - X_{t_j}| 1_{\Omega_{t_{j+1}}} + |X_{s_{j+1}} - X_{s_j}| 1_{\Omega_{s_{j+1}}}) \\ & \quad + c \sum_{j=j_0}^k (d(t_{j+1}, t_j) + d(s_{j+1}, s_j)) 4^{j+1}, \end{aligned}$$

for Ω_u the event that $|X_u - X_{\pi_j u}| > cd(u, \pi_j u) 4^{j+1}$. By the argument in the last paragraph of the proof of Theorem 2.2.14, the second sum on the right is bounded above by a multiple of $A := \sup_{t \in T} \sum_{j=j_0}^k d(t, T_j) 4^{j+1}$. Also by the arguments in the same proof, the ϕ -Orlicz norms of the variables $\sup_t \sum_{j=j_0}^k |X_{t_{j+1}} - X_{t_j}| 1_{\Omega_{t_{j+1}}}$ and $\sup_s \sum_{j=j_0}^k |X_{s_{j+1}} - X_{s_j}| 1_{\Omega_{s_{j+1}}}$ are bounded above by a multiple of $A \vee \tilde{A}$, for $\tilde{A} = \sum_{j=j_0}^k \psi(4^{j+1})^{-1} \sum_{u \in T_{j+1}} d(u, T_j) 4^{j+1}$. It follows that the Orlicz norm of the variable on the left side of the preceding display is bounded similarly. Consequently, by the triangle inequality,

$$\left\| \sup_{d(s,t) < \delta} |X_t - X_s| \right\|_{\phi} \lesssim A \vee \tilde{A} + \left\| \sup_{d(s,t) < \delta} |X_{t_{j_0}} - X_{s_{j_0}}| \right\|_{\phi}.$$

The supremum in the term on the far right is taken over all pairs s_{j_0} and t_{j_0} of end points in T_{j_0} of chains starting at s_{k+1} and t_{k+1} such that $d(s_{k+1}, t_{k+1}) < \delta$. Select exactly one such chain for each endpoint s_{j_0} or t_{j_0} , thus giving at most $|T_{j_0}|^2$ pairs of starting points (s_{k+1}, t_{k+1}) . By the triangle inequality

$$|X_{t_{j_0}} - X_{s_{j_0}}| \leq |X_{t_{k+1}} - X_{s_{k+1}}| + |(X_{t_{k+1}} - X_{t_{j_0}}) - (X_{s_{k+1}} - X_{s_{j_0}})|.$$

The supremum over all pairs (s_{k+1}, t_{k+1}) of the second term on the right is bounded by the supremum that was already analyzed (which was over *all* pairs with $d(s_{k+1}, t_{k+1}) < \delta$). Since $\|X_{t_{k+1}} - X_{s_{k+1}}\|_\psi \lesssim d(s_{k+1}, t_{k+1}) < \delta$ and there are $|T_{j_0}|^2$ variables $X_{t_{k+1}} - X_{s_{k+1}}$, the ϕ -Orlicz norm of their maximum is bounded above by a multiple of $\psi^{-1}(|T_{j_0}|^2)\delta$, by an extension of Lemma 2.2.2 (see Problem 2.2.27). ■

The sequence of subsets $T_0, T_1, T_2, \dots$ may be replaced by a sequence of partitions of T . A nested sequence $T = T_0, T = \cup_k T_{1,k}, T = \cup_k T_{2,k}, \dots$ of partitions of T is called ψ -admissible if the number of sets in the j th partition is bounded above by $\psi(4^j)$, for every $j \geq 1$, and is equal to 1 for $j = 0$. For given $t \in T$ denote the set $T_{j,k}$ in the j th partition that contains t by $T_j(t)$, and let $T_{j,k}$ contain $m_{j,k}$ sets $T_{j+1,l}$. Then a *generic chaining bound* is defined as the infimum over all ψ -admissible partitions of the numbers

$$\sup_{t \in T} \sum_{j=0}^{\infty} \text{diam}(T_j(t)) 4^j + \sum_{j=0}^{\infty} \frac{1}{\psi(4^{j+1})} \sum_k m_{j,k} \text{diam}(T_{j,k}) 4^j.$$

If for every $j \geq 0$ a set $T_j \subset T$ of points is constructed by choosing exactly one point from each set $T_{j,k}$ in the partition $T = \cup_k T_{j,k}$, then the sequence of sets $T_0, T_1, T_2, \dots$ is clearly ψ -admissible, and the two series in the preceding display dominate the two series in Theorem 2.2.14.

In the case of the exponential Orlicz norms $\psi_p(x) = e^{x^p} - 1$, only the first of the two series is relevant, and the procedure of the preceding paragraph can also be reversed, in the sense that given a sequence of admissible subsets there exists a sequence of admissible partitions such that the sum in the preceding display is dominated by a multiple of the generic chaining bound of Theorem 2.2.14.

2.2.17 Lemma. *Given a sequence of subsets $T_j \subset T$ with $|T_j| \leq \psi_p(4^j)$ and a number $1 < r \leq 4$, there exists a nested sequence of partitions $T = \cup_k T_{j,k}$, each consisting of at most $\psi_p(r^j)$ sets, such that, for a constant C that depends on p and r only,*

$$\sup_{t \in T} \sum_{j=0}^{\infty} \text{diam}(T_j(t)) r^j \leq C \sup_{t \in T} \sum_{j=0}^{\infty} d(t, T_j) 4^j.$$

Proof. By relabeling, we can construct a (different) sequence of subsets T_j such that $|T_j| \leq \psi_p(r^j)$ and such that $\sup_{t \in T} \sum_{j=0}^{\infty} d(t, T_j) r^j$ is bounded above by a multiple of the series on the right (see Problems 2.2.18-2.2.19). Next for given $j \geq 1$, form

a Voronoi partition $T = \cup_k Q_{j,k}$, by assigning every $t \in T$ to a closest point $t_{j,k}$ in T_j . Split every set in this partition further into the sets

$$R_{j,k,l} = \{t \in Q_{j,k} : r^{-l-1} \text{diam}(T) < d(t, t_{j,k}) \leq r^{-l} \text{diam}(T)\}, \quad l = 0, 1, \dots, 2j,$$

$$\bar{R}_{j,k} = \{t \in Q_{j,k} : d(t, t_{j,k}) \leq r^{-2j-1} \text{diam}(T)\}.$$

Let $R_j(t)$ be the partitioning set in the j th partition $T = \cup_k (\cup_l R_{j,k,l} \cup \bar{R}_{j,k})$ that contains t . If $T_j(t) = R_{j,k,l}$, for some (k, l) , then $d(t, t_{j,k}) > r^{-l-1} \text{diam}(T)$ and $\text{diam}(T_j(t)) = \text{diam}(R_{j,k,l}) \leq 2r^{-l} \text{diam}(T) \leq 2rd(t, t_{j,k}) = 2rd(t, T_j)$. If $T_j(t) = \bar{R}_{j,k}$, for some k , then $\text{diam}(T_j(t)) \leq r^{-2j} \text{diam}(T)$. Therefore,

$$\sum_{j=1}^{\infty} \text{diam}(T_j(t)) r^j \leq 2r \sum_{j=1}^{\infty} d(t, T_j) r^j + \sum_{j=1}^{\infty} \text{diam}(T) r^{-j}.$$

The first series on the right is bounded by the right side of the lemma, while the second is bounded by $\text{diam}(T)/(r-1)$, which is bounded above by a multiple of the term for $j=0$ of the series on the right of the lemma.

Rename $\bar{R}_{j,k} = R_{j,k,l}$ for $l = 2j+1$, and consider the nested sequence of partitions obtained by intersecting all sets in the sequence of partitions $T = \cup_{k,l} R_{1,k,l} = \cup_{k,l} R_{2,k,l} = \dots$ up to a given j : the j th partition $T = \cup_k T_{j,k}$ consists of all sets $R_{1,k_1,l_1} \cap \dots \cap R_{j,k_j,l_j}$, for freely ranging $k_1, l_1, \dots, k_j, l_j$. Because it is a refinement, the corresponding series $\sup_{t \in T} \sum_{j=0}^{\infty} \text{diam}(T_j(t)) r^j$ is bounded by the series attached to the partitions $T = \cup_{k,l} R_{j,k,l}$ and hence is bounded by the right side of the lemma in view of the preceding paragraph. The number of sets in the j th partition is bounded above by $\prod_{i=1}^j (|T_i|(2i+2)) \leq \prod_{i=1}^j (e^{r^{p^i}} - 1) 3^j j! \leq e^{cr^{pj}} - 1$, for some constant $c = c_{p,r}$.

We conclude the proof by defining the final partition as the $(j-k)$ st partition for $j \geq k$ and as the partition in a single set for $j < k$, where k is chosen so that $cr^{-pk} \leq 1$, so that $e^{cr^{p(j-k)}} - 1 \leq \psi_p(r^j)$, for $j \geq k$. The infinite series corresponding to these shifted partitions is bounded above by $\sum_{j < k} \text{diam}(T) r^j + \sum_{j \geq k}^{\infty} \text{diam}(T_{j-k}(t)) r^j$, which is bounded above by a multiple of the original infinite series. ■

In the exponential case it is customary to choose $r = 2^{1/p}$ in the preceding lemma, giving a *generic chaining bound* in the form

$$\gamma_p(T, d) = \inf_{t \in T} \sup_{j=0}^{\infty} \text{diam}(T_j(t)) 2^{j/p},$$

where the infimum is taken over all sequences of nested partitions $T = \cup_k T_{j,k}$ consisting of at most $e^{2^j} - 1$ sets. (It is also customary to strengthen the last requirement to at most 2^{2^j} sets. Up to a multiplicative constant this is equivalent, as is seen by shifting the index of the partitions, as in the last paragraph of the preceding proof.)

A zero-mean Gaussian process X satisfies the increment bound $\|X_s - X_t\|_{\psi_2} \lesssim d(s, t)$ for $\psi_2(x) = e^{x^2} - 1$ and d the standard deviation metric. The relevant generic chaining number is $\gamma_2(T, d)$. In this case the generic chaining upper bound is known to be sharp up to multiplicative constants.

2.2.18 Proposition. *There exist universal constants $c, C > 0$ such that, for any zero-mean, separable Gaussian process X ,*

$$c \gamma_2(T, \rho_X) \leq \mathbb{E} \sup_{s, t \in T} |X_s - X_t| \leq C \gamma_2(T, \rho_X),$$

where ρ_X is the standard deviation semimetric $\rho_X(s, t) = \sigma(X_s - X_t)$.

Proof. See Talagrand (2014), Chapter 2. ■

The proposition underlines the superiority of generic chaining bounds over entropy bounds, as relatively simple examples of Gaussian processes show that the entropy bounds can be too large.[‡]

The bound $\|X_s - X_t\|_\psi \leq d(s, t)$ on the Orlicz norm of the increments of the process X , used in the preceding theorems, is a pretty way of controlling the tail probabilities $\mathbb{P}(|X_s - X_t| > x d(s, t))$. Tail probabilities of some processes can be bounded using multiple metrics, and this is more easily formalized directly. The following theorem gives an example of particular interest.

2.2.19 Theorem (Maximal inequality). *Let $\{X_t: t \in T\}$ be a separable stochastic process such that, for metrics d_1 and d_2 , and every $s, t \in T$ and every $x > 0$,*

$$\mathbb{P}(|X_s - X_t| > x d_1(s, t) + \sqrt{x} d_2(s, t)) \leq 2e^{-x}.$$

Then for any sequences $T_0^p, T_1^p, T_2^p, \dots$ of nonempty subsets with $|T_0^p| = 1$ and $|T_j^p| \leq \psi_p(4^j)$, for $p = 1, 2$, and a universal constant K ,

$$\mathbb{E} \sup_{s, t} |X_s - X_t| \leq K \sup_{t \in T} \sum_{j=0}^{\infty} d_1(t, T_j^1) 4^j + K \sup_{t \in T} \sum_{j=0}^{\infty} d_2(t, T_j^2) 4^j.$$

The two suprema on the right may be replaced by $\gamma_1(T, d_1)$ and $\gamma_2(T, d_2)$, and the constant K can be chosen so that, for every $x > 0$ and $\text{diam}(T, d_p)$ the diameter of T for the metric d_p , for $p = 1, 2$,

$$\mathbb{P}\left(\sup_{s, t} |X_s - X_t| > K \sum_{p=1}^2 (\gamma_p(T, d_p) + x^{1/p} \text{diam}(T, d_p))\right) \leq 2e^{-x}.$$

Proof. By Lemma 2.2.17, applied separately for $p = 1$ and $p = 2$, with $r = 2$ and $r = \sqrt{2}$ respectively, we can form two nested sequences of partitions $T = \cup_k T_{j,k}^p$, both with j th partition consisting of at most e^{2^j} sets, such that the series $\sum_{j=0}^{\infty} \text{diam}(T_j^p(t), d_p) 2^{1/p}$, for $p = 1, 2$, are bounded by multiples of the first and second series in the theorem. By the definitions of $\gamma_1(T, d_1)$ and $\gamma_2(T, d_2)$ as infima of such sums, these partitions can also be chosen so that the two series are bounded above by multiples of the latter numbers. For $j \in \mathbb{N}$, take an arbitrary point from every intersection $T_{j,k}^1 \cap T_{j,l}^2$ of sets from the two partitions at level j , and define T_j to be the set of these points. Set T_0 equal to an arbitrary point t_0 , and for $j \geq 0$ and

[‡] For instance, see Talagrand (2014), pages 51–55, for the case of linear Gaussian processes indexed by ellipsoids.

$t \in T$, let $\pi_j t \in T_j$ be the point from T_j that belongs to the same partitioning set $T_j(t)$ as t . Assume without loss of generality that T is finite, and that there exists k such that $T_{k+1} = T$.

By telescoping and the triangle inequality, for any $j_0 \geq 0$,

$$|X_t - X_{\pi_{j_0} t}| \leq \sum_{j=j_0}^k |X_{\pi_{j+1} t} - X_{\pi_j t}| 1_{\Omega_j} + \sum_{j=j_0}^k \sum_{p=1}^2 x^{1/p} 2^{j/p} d_p(\pi_{j+1} t, \pi_j t),$$

for Ω_j the event that $|X_v - X_u| > \sum_{p=1}^2 x^{1/p} 2^{j/p} d_p(v, u)$ for some $(u, v) \in T_j \times T_{j+1}$. Because the sequence of partitions is nested, the points $\pi_{j+1} t$ and $\pi_j t$ are contained in the same partitioning set $T_j(t) = T_j^1(t) \cap T_j^2(t)$ at level j , whence $d_p(\pi_{j+1} t, \pi_j t) \leq \text{diam}(T_j(t), d_p)$, for $p = 1, 2$. It follows that the double sum in the preceding display is (for any j_0) bounded above by a multiple of $x\gamma_1(T, d_1) + \sqrt{x}\gamma_2(T, d_2)$. We conclude that

$$\begin{aligned} \mathbb{P}\left(\sup_t |X_t - X_{\pi_{j_0} t}| \gtrsim \sum_{p=1}^2 x^{1/p} \gamma_p(T, d_p)\right) &\leq \mathbb{P}\left(\sum_{j=j_0}^k |X_{\pi_{j+1} t} - X_{\pi_j t}| 1_{\Omega_j} > 0\right) \\ &\leq \sum_{j=j_0}^k \mathbb{P}(\Omega_j) \leq \sum_{j=j_0}^k |T_{j+1}|^2 e^{-x2^j} \lesssim e^{-2^{j_0} x/2}, \end{aligned}$$

for $x \geq 8$, since $|T_j| \leq e^{2^{j+1}}$.

By construction $\pi_0 t = t_0$, for every t , so that for $j_0 = 0$, the variable in the left side is $\sup_t |X_t - X_{t_0}|$. By the triangle inequality the supremum $\sup_{s,t} |X_s - X_t|$ is bounded by twice this variable. We can integrate the bound to deduce that the mean of the supremum is bounded by a multiple of $\gamma_1(T, d_1) + \gamma_2(T, d_2)$.

For a proof of the probability bound, we first conclude for $2^{j_0} = m$ and a general j_0 , that $\|\sup_t |X_t - X_{\pi_{j_0} t}|\|_m \lesssim \gamma_1(T, d_1) + \gamma_2(T, d_2)$, where the multiplicative constant is independent of m (see Problem 2.2.25). Next we apply Lemma 2.2.13 with the L_m -norm to the $|T_{j_0}| \leq e^{2^m}$ variables $X_{\pi_{j_0} t} - X_{t_0}$ to see that $\|\max_t |X_{\pi_{j_0} t} - X_{t_0}|\|_m \leq e^2 \max_t \|X_{\pi_{j_0} t} - X_{t_0}\|_m$. Here $\|X_s - X_t\|_m \lesssim m d_1(s, t) + \sqrt{m} d_2(s, t)$, by the assumed tail bound (see Problem 2.2.25). Combining these bounds with the triangle inequality, we see that there exists a constant K with

$$\|\sup_t |X_t - X_{t_0}|\|_m \leq K \sum_{p=1}^2 (\gamma_p(T, d_p) + m^{1/p} \text{diam}(T_{j_0}(t), d_p)).$$

Therefore, by Markov's inequality, applied with the m th moment, for any $x > 0$,

$$\begin{aligned} \mathbb{P}\left(\sup_t |X_t - X_{t_0}| > e^2 K \sum_{p=1}^2 (\gamma_p(T, d_p) + x^{1/p} \text{diam}(T_{j_0}(t), d_p))\right) \\ \leq e^{-2m} \left(\frac{\sum_{p=1}^2 (\gamma_p(T, d_p) + m^{1/p} \text{diam}(T_{j_0}(t), d_p))}{\sum_{p=1}^2 (\gamma_p(T, d_p) + x^{1/p} \text{diam}(T_{j_0}(t), d_p))} \right)^m. \end{aligned}$$

For $m < x \leq 2m$, this is further bounded by e^{-x} . Since m can be chosen $m = 2^{j_0}$ for every $j_0 \in \mathbb{N}_0$, this gives the tail bound e^{-x} for every $x > 1$. Replacing the constant K by $2K$ gives the bound $e^{-2x} \leq e^{-x}$ for every $x > 1/2$. Since $2e^{-x} \geq 1$ for $x \leq 1/2$, the claimed tail bound $2e^{-x}$ is valid for every $x > 0$. ■

The bound $\|X_s - X_t\|_{\psi_p} \leq d(s, t)$ for the ψ_p -Orlicz norm implies a tail bound of the form $P(|X_s - X_t| \gtrsim x^{1/p} d(s, t)) \leq 2e^{-x}$. Thus the process in the preceding theorem can be viewed as being controlled by a combination of the ψ_1 - and ψ_2 -Orlicz norms. Up to constants its assumed tail bound is equivalent to, for every $x > 0$,

$$P(|X_s - X_t| > x) \leq 2 \exp\left(-\frac{x^2}{d_2^2(s, t) + x d_1(s, t)}\right).$$

Bernstein's inequality gives bounds of this type for X a sum of independent variables. The last assertion of the theorem shows that the supremum $\sup_{s,t} |X_s - X_t|$ then satisfies a tail bound of the same form.

2.2.6 Majorizing Measures

Majorizing measures offer a different approach that also gives sharp bounds in certain cases. A *majorizing measure* μ is any probability measure on T for which the integral in the following theorem is finite, which then gives a nontrivial bound on the supremum of the stochastic process X .^b

Let $B_d(t, \varepsilon)$ be the closed ball of radius ε around t relative to the semimetric d , and let $D(t, T) = \sup_{s \in T} d(t, s)$.

2.2.20 Theorem (Majorizing measure). *Let ψ be a convex, nondecreasing function with $\psi(0) = 0$ and $\psi(4) \geq 1$. Then, for any Borel probability measure μ on the metric space (T, d) , there exists a ψ -admissible sequence of sets T_j such that, for a constant K that depends on ψ only,*

$$\begin{aligned} \sup_{t \in T} \sum_{j=0}^{\infty} d(t, T_j) 4^j + \sum_{j=0}^{\infty} \frac{1}{\psi(4^{j+1})} \sum_{t \in T_{j+1}} d(t, T_j) 4^j \\ \leq K \sup_{t \in T} \int_0^{D(t, T)} \psi^{-1}\left(\frac{1}{\mu(B_d(t, \varepsilon))}\right) d\varepsilon. \end{aligned}$$

Proof. For the given probability measure μ and $j \geq 1$, define

$$r_j(t) = \inf\left\{\varepsilon > 0: \mu(B_d(t, \varepsilon)) \geq \frac{1}{\psi(4^j)}\right\}.$$

Next for every fixed $j \geq 1$ construct a set $T_j \subset T$ recursively, by first setting $t_1 = \operatorname{argmin}_{t \in T} r_j(t)$, and next given $t_1, \dots, t_{k-1}$ setting

$$t_k = \operatorname{argmin}_{t \in T - \cup_{l < k} A_j(t_l)} r_j(t),$$

^b In the remainder of this book majorizing measures are used in one proof, which can also be read independently from the present section.

where the sets $A_j(t_l)$ are defined by $A_j(t_l) = \{t \in T: d(t, t_l)/2 < r_j(t) + r_j(t_l)\}$, for $l \geq 1$, and the recursions continue until the sets $A(t_l)$ have exhausted T . The first point t_1 exists, since $\psi(4^j) \geq 1$, by assumption. By construction $d(t_k, t_l)/2 \geq r_j(t_k) + r_j(t_l)$, for $k > l$, whence the balls $B_d(t_l, r_j(t_l))$ for different $t_l \in T_j$ are disjoint (easily), and hence $1 = \mu(T) \geq \sum_l \mu(B_d(t_l, r_j(t_l))) \geq |T_j|/\psi(4^j)$, by the definition of r_j . This shows that the recursive procedure ends after finitely many steps and yields a set of cardinality $|T_j| \leq \psi(4^j)$. By construction $d(t, t_l) < 2r_j(t) + 2r_j(t_l) \leq 4r_j(t)$ if t_l is the first element of T_j with $t \in A_j(t_l)$, whence $d(t, T_j) \leq 4r_j(t)$, for every $t \in T$. If we set $r_0(t) = D(t, T)$, then this is also true for $j = 0$ and an arbitrary one-point set T_0 . Since $r_{j+1}(t) \leq r_j(t)$ and $\mu(B_d(t, \varepsilon)) < 1/\psi(4^j)$ for $\varepsilon \in [r_{j+1}(t), r_j(t))$, by the definition of $r_j(t)$, and hence $\psi^{-1}(1/\mu(B_d(t, \varepsilon))) \geq 4^j$, we have

$$\begin{aligned} \int_0^{D(t, T)} \psi^{-1}\left(\frac{1}{\mu(B_d(t, \varepsilon))}\right) d\varepsilon &\geq \sum_{j=0}^{\infty} 4^j (r_j(t) - r_{j+1}(t)) \\ &\geq \frac{3}{4} \sum_{j=0}^{\infty} r_j(t) 4^j \geq \frac{3}{4} \sum_{j=0}^{\infty} d(t, T_j) 4^{j-1}. \end{aligned}$$

This concludes the proof that the majorizing measure integral dominates the first generic chaining number.

Next consider the second generic chaining number of the sequence $T_0, T_1, \dots$. We claim that $r_{j+1}(t) \leq 2r_{j+1}(s)$, for any $t \in T_{j+1}$ and $s \in T$ with $d(s, t) \leq r_{j+1}(t)$. Indeed, let t_l be the first element of T_{j+1} such that $s \in A_{j+1}(t_l)$. If t_l appeared after t , then $r_{j+1}(t) \leq r_{j+1}(t_l) \leq r_{j+1}(s)$, by the definitions of t and t_l as minimizers of r_j . On the other hand, if t_l appeared before t in the recursions, then $t \notin A_{j+1}(t_l)$ as it was later selected for T_{j+1} and hence

$$r_{j+1}(t) + r_{j+1}(t_l) \leq \frac{d(t, t_l)}{2} \leq \frac{d(t, s)}{2} + \frac{d(s, t_l)}{2} < \frac{r_{j+1}(t)}{2} + r_{j+1}(s) + r_{j+1}(t_l),$$

which implies the claim $r_{j+1}(t) \leq 2r_{j+1}(s)$, after rearranging.

Since $B_d(t, r_j(s) + d(s, t)) \supset B_d(s, r_j(s))$ by the triangle inequality, the definitions of $r_j(s)$ and $r_j(t)$ give that $r_j(t) \leq r_j(s) + d(s, t)$. Combining this with the result of the preceding paragraph, we conclude that $r_j(t) \leq r_j(s) + 2r_{j+1}(s)$, whenever $t \in T_{j+1}$ and $d(s, t) \leq r_{j+1}(t)$. Consequently, for every $t \in T_{j+1}$, since $d(t, T_j) \leq 4r_j(t)$,

$$\frac{d(t, T_j)}{\psi(4^{j+1})} \leq 4r_j(t)\mu(B_d(t, r_{j+1}(t))) \leq \int_{B_d(t, r_{j+1}(t))} (r_j(s) + 2r_{j+1}(s)) d\mu(s).$$

Since the balls $B_d(t, r_{j+1}(t))$, for $t \in T_{j+1}$ are disjoint, the sum of the left side over $t \in T_{j+1}$ is bounded above by $\int_T (r_j(s) + 2r_{j+1}(s)) d\mu(s)$. Multiply by 4^j and sum over j to find that

$$\sum_{j=0}^{\infty} \sum_{t \in T_{j+1}} \frac{d(t, T_j) 4^j}{\psi(4^{j+1})} \leq \int_T \sum_{j=0}^{\infty} (r_j(s) + 2r_{j+1}(s)) 4^j d\mu(s).$$

The integrand is bounded above by $(3/2) \sum_{j=0}^{\infty} r_j(s) 4^j$, which was already seen to be bounded above by the majorizing measure integral at the end of the first part of the proof. ■

The left side of the theorem is the right side of Theorem 2.2.14, and hence by the latter theorem the majorizing measure integral is up to a multiplicative constant also an upper bound on expectation $E \sup_{s,t} |X_s - X_t|$, for any stochastic process X satisfying the increment bound $\|X_s - X_t\|_{\psi} \leq d(s, t)$, for $s, t \in T$. This may also be proved directly by a chaining argument: for any probability measure μ and every $\delta, \eta > 0$,[#]

$$\begin{aligned} E \sup_t |X_t - X_{t_0}| &\lesssim \sup_t \int_0^{\infty} \sqrt{\log \frac{1}{\mu(B(t, \varepsilon))}} d\varepsilon, \\ E \sup_{\rho(s, t) < \delta} |X_s - X_t| &\lesssim \sup_t \int_0^{\eta} \sqrt{\log \frac{1}{\mu(B(t, \varepsilon))}} d\varepsilon + \delta \sqrt{\log N(\eta, T, \rho)}. \end{aligned}$$

Compared to the entropy integral appearing in Theorem 2.2.4, the majorizing measure integral replaces the packing numbers $D(\varepsilon, T, d)$ by the inverses of the probabilities of the balls $B_d(t, \varepsilon)$. The latter are local to the point t , which is compensated by taking the supremum over $t \in T$, outside the integral, much as in the generic chaining bound. This local nature gives the majorizing measures more flexibility in measuring the complexity of the index set T . Indeed, the infimum of the majorizing measure integrals over all probability measures is smaller than a multiple of the entropy integral. In particular, for the exponential Young functions the majorizing measure integral of a convex combination of uniform measures on a sequence of ε -nets dominates (and is comparable to) the entropy integral.

In the case of the exponential Young functions $\psi(x) = e^{x^p} - 1$, the infimum of the majorizing measure integrals over all majorizing measures can be shown to be equivalent to $\gamma_p(T, d)$, up to multiplicative constants.[†] In particular, both generic chaining and majorizing measures give sharp bounds for suprema of Gaussian processes.

Besides boundedness, these devices do also characterize the continuity of Gaussian processes. For majorizing measures this takes the form as in the following proposition.

2.2.21 Proposition. *Let $\{X_t : t \in T\}$ be a separable zero-mean Gaussian process with standard deviation semimetric $\rho_X(s, t) = \sigma(X_s - X_t)$. Then almost all sample paths $t \mapsto X_t$ are uniformly ρ_X -continuous and bounded if and only if*

$$(2.2.22) \quad \limsup_{\delta \downarrow 0} \sup_{t \in T} \int_0^{\delta} \sqrt{\log \frac{1}{\mu(B_{\rho_X}(t, \varepsilon))}} d\varepsilon = 0,$$

for some Borel probability measure μ on (T, ρ_X) .

[#] Ledoux and Talagrand (1991), pages 320–321, and (11.15) on page 317.

[†] See Talagrand (2001).

The sufficiency of the continuity at zero (2.2.22) of the majorizing measure integral for sample path continuity follows from the maximal inequality stated preceding the proposition. In the other direction the proposition is harder to prove.[‡]

The equivalence of generic chaining numbers and majorizing measures integrals extends to more general Young functions that satisfy a mild regularity condition. Furthermore, for metrics d that are p -concave, i.e. metrics such that d^p is a metric, for some $p > 1$, these numbers also give sharp bounds on expected suprema of stochastic processes. Consider the quantities

$$\begin{aligned} S(T, d, \psi) &= \sup_{X \in \mathcal{D}, \psi} \mathbb{E} \sup_{s, t} |X_s - X_t|, \\ \bar{M}(T, d, \psi) &= \sup_{\mu} \int_0^{D(t, T)} \int_0^{\psi^{-1}\left(\frac{1}{\mu(B_d(t, \varepsilon))}\right)} d\varepsilon d\mu(t), \\ M(T, d, \psi) &= \inf_{\mu} \sup_{t \in T} \int_0^{D(t, T)} \psi^{-1}\left(\frac{1}{\mu(B_d(t, \varepsilon))}\right) d\varepsilon, \\ A(T, d, \psi) &= \inf_{(T_j) \in \psi} \left[\sup_{t \in T} \sum_{j=0}^{\infty} d(t, T_j) 4^j + \sum_{j=0}^{\infty} \frac{1}{\psi(4^{j+1})} \sum_{t \in T_{j+1}} d(t, T_j) 4^j \right]. \end{aligned}$$

Here the infimum in the definition of $S(T, d, \psi)$ is taken over all separable stochastic processes X that satisfy the increment bound $\|X_s - X_t\|_{\psi} \leq d(s, t)$, and the infimum in $A(T, d, \psi)$ is taken over all ψ -admissible sequences $T_0, T_1, \dots$ of subsets of T .

The first of these quantities may be smaller than the other three. However, the following theorem shows that the metric d may be replaced by a larger metric, giving a larger class of processes X in the definition of S and hence increasing S , so that all four quantities become equivalent, and are also equivalent to the last three quantities for the original metric d .

2.2.23 Proposition. *Let (T, d) be a compact metric space, and let ψ be a convex, nondecreasing, nonzero function with $\psi(0) = 0$ satisfying $\psi(2x) \geq 2C\psi(x)$, for every $x \geq 0$ and some constant $C > 1$. If $A(T, d, \psi)$ is finite, then there exists a metric ρ on T with $d \leq \rho$, such that all of $S(T, \rho, \psi)$, $\bar{M}(T, \rho, \psi)$, $M(T, \rho, \psi)$, $A(T, \rho, \psi)$, $M(T, d, \psi)$, $\bar{M}(T, d, \psi)$ and $A(T, d, \psi)$ are equivalent up to multiplicative constants. Furthermore, if d^p is a metric for some $p > 1$, then these quantities are also equivalent to $S(T, d, \psi)$. Finally, for the exponential ψ functions, these assertions are true with $A(T, d, \psi)$ defined as equal to only the first generic chaining series.*

Proof. The following three inequalities are true for any metric d : $S(T, d, \psi) \lesssim A(T, d, \psi)$ (by Theorem 2.2.14); $A(T, d, \psi) \lesssim M(T, d, \psi)$ (by Theorem 2.2.20); and $M(T, d, \psi) \leq 2\bar{M}(T, d, \psi)$ (by a convexity argument, see Problem 2.2.21). Furthermore, Bednorz (2010) shows that $\bar{M}(T, d, \psi) \lesssim S(T, d, \psi)$, for any p -concave metric d . It follows that the four quantities are equivalent up to multiplicative constants when d is p -concave.

[‡] See Talagrand (1992), for a direct, relatively simple proof.

The condition on ψ implies (1.2) in Bednorz (2012), who constructs an *ultra-metric* $\rho \geq d$ on T for which $A(T, \rho, \psi) \lesssim A(T, d, \psi)$. (By definition, an ultra-metric satisfies $d(s, t) \leq d(s, u) \vee d(t, u)$, for every s, t, u ; in particular, it is p -concave for every $p > 1$.) By the preceding paragraph the four quantities with d replaced by ρ are then equivalent. Since $\rho \geq d$, the four quantities with d are bounded above by the corresponding quantity with ρ . Then $A(T, \rho, \psi)$ and $A(T, d, \psi)$ are equivalent, and equivalent to all quantities with ρ . Furthermore, the quantities $A(T, d, \psi) \lesssim M(T, d, \psi) \leq 2\bar{M}(T, d, \psi)$ are bounded above by the corresponding quantities with ρ instead of d , which are equivalent to $A(T, \rho, \psi)$. This concludes the proof of the first assertion of the proposition.

The final assertion follows by the same arguments. ■

Problems and Complements

1. A standard normal variable X possesses norm $\|X\|_{\psi_2} = \sqrt{8/3}$ for $\psi_2(x) = \exp(x^2) - 1$.
2. The constant random variable $X = 1$ possesses norm $\|X\|_{\psi_p} = (\log 2)^{-1/p}$ for $\psi_p(x) = e^{x^p} - 1$.
3. For any constant $C \geq 1$, convex, increasing ψ and random variable X , one has $\|X\|_{C\psi} \leq C \|X\|_{\psi}$. For $C \leq 1$ the reverse inequality holds.
[Hint: By convexity of ψ , one has $E\psi(|X|/C\|X\|_{\psi}) \leq 1/C$, for $C \geq 1$.]

4. **(Comparing L_p - and ψ_α -norms)** For any $\alpha \geq 1$ there exist constants $0 < c_\alpha < C_\alpha < \infty$ such that, for any random variable X ,

$$c_\alpha \|X\|_{\psi_\alpha} \leq \sup_{p \geq 1} \frac{\|X\|_p}{p^{1/\alpha}} \leq C_\alpha \|X\|_{\psi_\alpha}.$$

In particular, $\|X\|_1 \leq \|X\|_2 \leq \|X\|_{\psi_2}$.

[Hint: For the lower bound set $D = \sup_{p \geq 1} \|X\|_p / p^{1/\alpha}$, so that $E|X|^{\alpha j} \leq D^{\alpha j} (\alpha j)^j$, for $j \geq 1$. Expand the exponential to conclude that $E\psi_\alpha(X/C) \leq \sum_{j=1}^{\infty} (D/C)^{\alpha j} (\alpha j)^j / j!$. Further bound the sum by using Stirling's formula $j^j / j! \leq e^j / (2\pi j)^{1/2}$, and reduce it to a number smaller than 1 by choosing C a sufficiently large multiple of D .

For the upper bound set $D = \|X\|_{\psi_\alpha}$ and use Markov's inequality to see that $P(|X| > x) \leq 2e^{-(x/D)^\alpha}$, for $x \geq x_\alpha := D(\log 2)^{1/\alpha}$. Conclude that

$$E|X|^p \leq x_\alpha^p + \int_{x_\alpha}^{\infty} px^{p-1} 2e^{-(x/D)^\alpha} dx \leq x_\alpha^p + 2D^p \Gamma(p/\alpha + 1),$$

by a change of variables. The second term on the far right is bounded above by a multiple of $D^p (p/(\alpha e))^{p/\alpha + 1/2}$, by Stirling's formula. Conclude that $D^{-1} \|X\|_p / p^{1/\alpha} \lesssim (\log 2/p)^{1/\alpha} + (p/\alpha)^{1/(2p)} (\alpha e)^{-1/\alpha}$. The right side is decreasing in p , for fixed α , and hence is bounded in $p \geq 1$.]

5. **(Comparing ψ_p -norms)** For any X , the numbers $(\log 2)^{1/p} \|X\|_{\psi_p}$ for $\psi_p(x) = \exp(x^p) - 1$ are nondecreasing in $p \geq 1$.

[Hint: The function ϕ for which $\psi_p(x(\log 2)^{1/p}) = \phi(\psi_q((\log 2)^{1/q} x))$ is concave and satisfies $\phi(1) = 1$ for $q \geq p$. Use Jensen's inequality.]

- 6. (Orlicz-norms-monotone convergence)** Let ψ be a convex, nondecreasing, nonzero function on $[0, \infty)$ with $\psi(0) = 0$. If $0 \leq X_n \uparrow X$ almost surely, then $\|X_n\|_\psi \uparrow \|X\|_\psi$.

[Hint: By the monotone convergence theorem, $\lim E\psi(X_n/r\|X\|_\psi) > 1$, for any $r < 1$.]

- 7.** The infimum in the definition of an Orlicz norm is attained (at $\|X\|_\psi$).

- 8.** Let ψ be any function as in the definition of an Orlicz norm. Then for any random variables $X_1, \dots, X_m$, one has $E \max |X_i| \leq \psi^{-1}(m) \max \|X_i\|_\psi$.

[Hint: For any C , one has $E \max |X_i|/C \leq \psi^{-1}(E \max \psi(|X_i|/C))$ by Jensen's inequality. Bound the maximum by a sum and take $C = \max \|X_i\|_\psi$.]

- 9.** If $\|X\|_\psi < \infty$, then $\|X\|_{\sigma\psi} < \infty$ for every $\sigma > 0$. Markov's inequality yields the bound $P(|X| > t) \leq 1/\sigma \cdot 1/\psi(t/\|X\|_{\sigma\psi})$ for every t . Is there an optimal value of σ ?

- 10.** The condition on the quotient $\psi(x)\psi(y)/\psi(cxy)$ in Lemma 2.2.2 is not automatic. For instance, the function $\psi(x) = (x+1)(\log(x+1)-1) + 1$ is convex and increasing for $x \geq 0$ but does not satisfy the condition.

- 11.** The function $\psi_p(x) = \exp(x^p) - 1$ satisfies the hypothesis of Lemma 2.2.2 for every $0 < p \leq 2$.

- 12.** Let ψ be a convex, positive function, with $\psi(0) = 0$ such that $\log \psi$ is convex. Then there is a constant σ such that $\phi(x) = \sigma\psi(x)$ satisfies both $\phi(1) < 1$ and $\phi(x)\phi(y) \leq \phi(xy)$, for all $x, y \geq 1$.

[Hint: Define $\phi(x) = \psi(x)/2\psi(1)$ for $x \geq 1$. Then $\phi(1) = 1/2$ and $g(x, y) = \phi(x)\phi(y)/\phi(xy)$ satisfies $g(1, 1) = 1/2$ and is decreasing in both x and y for $x, y \geq 1$ if $\log \psi$ is convex. The derivative of g with respect to x is given by $\psi(x)\psi(y)/(2\psi(1)\psi(xy))(\psi'/\psi(x) - y\psi'/\psi(xy))$, which is less than or equal to 0 since $y \geq 1$ and $\log \psi$ is convex. By symmetry, the derivative with respect to y is also bounded above by zero.]

- 13.** The functions ψ_p , with $1 \leq p \leq 2$, and more generally the function $\psi(x) = \exp(h(x))$, with $h(x)$ convex ($h(x) = x(\log(x) - 1) + 1$), satisfy the conditions in the preceding problem. The functions ψ_p , for $0 < p \leq 1$, do not satisfy these conditions.

- 14.** For given constants $K \geq 2$ and $C > 0$, and $D = C \log 2 / \log K$, we have $1 \wedge (Ke^{-Cx}) \leq 2e^{-Dx}$, for every $x > 0$.

[Hint: The functions $x \mapsto Ke^{-Cx}$ and $x \mapsto 2e^{-Dx}$ decrease from $K \geq 2$ and 2, respectively, to 1 on the interval $[0, \log K/C]$. Their graphs cross at the endpoint of this interval.]

- 15.** Let T be a compact, semimetric space and $q > 0$. Then the map $x \mapsto x^q$ from the nonnegative functions in $\ell^\infty(T)$ to $\ell^\infty(T)$ is continuous at every continuous x .

[Hint: It suffices to show that $\|x_n - x\|_\infty \rightarrow 0$ implies that $x_n^q(t_n) - x^q(t_n) \rightarrow 0$ for every sequence t_n . By compactness of T , the sequence t_n may be assumed convergent. Since x is continuous, $x(t_n) \rightarrow x(t)$. Thus $x_n(t_n) \rightarrow x(t)$, whence $x_n^q(t_n) \rightarrow x^q(t)$.]

- 16.** Suppose the metric space T has a finite diameter and $\log N(\varepsilon, T, d) \leq h(\varepsilon)$, for all sufficiently small $\varepsilon > 0$ and a fixed, positive, continuous function h . Then there exists a constant K with $\log N(\varepsilon, T, d) \leq Kh(\varepsilon)$ for all $\varepsilon > 0$.

[Hint: For sufficiently large ε , the entropy numbers are zero, and for sufficiently small ε , the inequality is valid with $K = 1$. The function $\log N(\varepsilon, T, d)/h(\varepsilon)$ is bounded on intervals that are bounded away from zero and infinity.]

17. Let X be a stochastic process with $\sup_{\rho(s,t) < \delta} |X(s) - X(t)| \xrightarrow{P^*} 0$ as $\delta \downarrow 0$. Then almost all sample paths of X are uniformly continuous.

[Hint: Given a decreasing sequence of numbers $\varepsilon_k \rightarrow 0$, there are numbers $\delta_k > 0$ such that $P^*(\sup_{\rho(s,t) < \delta_k} |X(s) - X(t)| > \varepsilon_k) \leq 2^{-k}$ for every k . By the Borel-Cantelli lemma, $|X(s) - X(t)| \leq \varepsilon_k$ whenever $\rho(s,t) < \delta_k$, for all sufficiently large k almost surely.]

18. Given a sequence $T_0, T_1, \dots$ of subsets with $|T_0| = 1$ and $|T_j| \leq \psi(4^j)$, for $j \geq 1$, and $1 < r < 4$, there exists a sequence $S_0, S_1, \dots$ of subsets with $|S_0| = 1$ and $|S_j| \leq \psi(r^j)$, for $j \geq 1$, such that $\sup_t \sum_{j=0}^{\infty} d(t, S_j) r^j \leq (16r/(r-1) + 1) \sup_t \sum_{j=0}^{\infty} d(t, T_j) 4^j$.

[Hint: For $j \geq 0$, define an integer k_j by $r^{k_j} \leq 4^j < r^{k_j+1}$. Because $r < 4$, we have $k_0 = 0 < k_1 < k_2 < \dots$. Define $S_i = T_j$ for $k_j < i \leq k_{j+1}$, whence $k_j + 1 \leq i$. Then $|S_i| = |T_j| \leq \psi(4^j) \leq \psi(r^{k_j+1}) \leq \psi(r^i)$, and $\sum_{i=1}^{\infty} d(t, S_i) r^i = \sum_{j=0}^{\infty} \sum_{k_j < i \leq k_{j+1}} d(t, T_j) r^i = \sum_{j=0}^{\infty} d(t, T_j) r^{k_j+1} (r^{k_{j+1}-k_j} - 1)/(r-1)$, where $r^{k_j} \leq 4^j$ and $r^{k_{j+1}-k_j} \leq 16$.]

19. Given a sequence $S_0, S_1, \dots$ of subsets with $|S_0| = 1$ and $|S_j| \leq \psi(r^j)$, for $j \geq 1$, and $1 < r < 4$, there exists a sequence $T_0, T_1, \dots$ of subsets with $|T_0| = 1$ and $|T_j| \leq \psi(4^j)$, for $j \geq 1$, such that $\sup_t \sum_{j=0}^{\infty} d(t, T_j) 4^j \leq r \sup_t \sum_{j=0}^{\infty} d(t, S_j) r^j$.

[Hint: With the integers k_j as defined in the hint to the preceding problem, define $T_j = S_{k_j}$. Then $|T_j| \leq \psi(r^{k_j}) \leq \psi(4^j)$ and $\sum_{j=0}^{\infty} d(t, T_j) 4^j \leq \sum_{j=0}^{\infty} d(t, S_{k_j}) r^{k_j+1}$.]

20. Given a sequence $T_0, T_1, \dots$ of subsets with $|T_0| = 1$ and $|T_j| \leq \psi(cr^j)$, for $j \geq 1$, for some $c > 0$ and $r > 1$ and ψ with $\psi(r) \geq 1$, there exists a sequence $S_0, S_1, \dots$ of subsets with $|S_0| = 1$ and $|S_j| \leq \psi(r^j)$, for $j \geq 1$, such that $\sup_t \sum_{j=0}^{\infty} d(t, S_j) r^j \leq C \sup_t \sum_{j=0}^{\infty} d(t, T_j) r^j$, for some $C > 0$.

[Hint: If $c \leq 1$, take $S_j = T_j$; otherwise take $S_j = T_{j-k}$, for $j \geq k$, and $S_j = T_0$, for $j < k$, for k such that $cr^{-k} \leq 1$. Then $|S_j| = |T_{j-k}| \leq \psi(cr^{j-k}) \leq \psi(r^j)$, for $j \geq k$. Furthermore $\sum_{j=0}^{k-1} d(t, S_j) r^j \leq \text{diam}(T)(r^k - 1)/(r-1)$ and $\sum_{j=k}^{\infty} d(t, S_j) r^j \leq \sum_{j=0}^{\infty} d(t, T_j) r^j r^k$. Because $\text{diam}(T) \leq 2 \sup_t d(t, T_0) \leq \sup_t \sum_{j=0}^{\infty} d(t, T_j) r^j$, the constant $C = 2(r^k - 1)/(r-1) + r^k$ works.]

21. For any compact metric space (T, d) and Young function ψ , we have $M(T, d, \psi) \leq 2\bar{M}(T, d, \psi)$. (See Section 2.2.5 for this notation.)

[Hint: Suppose first that T is finite and that the map $x \mapsto \psi^{-1}(1/x)$ is convex. Suppose $\bar{M} = \bar{M}(T, d, \psi) < \infty$ and set $f_{\mu}(t) = \int \psi^{-1}(1/\mu(B_d(t, \varepsilon))) d\varepsilon$. The set $\mathcal{F}$ of functions $f: T \rightarrow \mathbb{R}$ such that $f \geq f_{\mu}$, pointwise, for some μ , can be identified with a closed, convex subset of $\mathbb{R}^T$. If the constant function $\bar{M}$ were not contained in $\mathcal{F}$, then the separating hyperplane theorem would give a vector $v \neq 0$ such that $v^T \bar{M} < v^T f$, for every $f \in \mathcal{F}$. Since $f + u1_t \in \mathcal{F}$, for every $u \geq 0$ if $f \in \mathcal{F}$, the vector v cannot have negative coordinates, and hence can be assumed to be a probability vector. Then $\bar{M} < v^T f_v$, which contradicts the definition of $\bar{M}$. Thus $\bar{M} \in \mathcal{F}$, which implies that $\bar{M} \geq f_{\mu}$, for some μ . If T is not finite, then this argument gives for every finite $S \subset T$ a measure μ_S with $\bar{M} \geq f_{\mu_S}(t)$, for every $t \in S$. By Prohorov's theorem the net μ_S has a weak limit point μ and by the Portmanteau theorem this satisfies $f_{\mu}(t) \leq \liminf_S f_{\mu_S}(t)$, for every $t \in T$. Finally if $x \mapsto \psi^{-1}(1/x)$ is not convex, then there exists a convex function ξ with $\xi(x) \leq \psi^{-1}(1/x) \leq 2\xi(x)$.]

22. (Maximum inequality normal variables) If $X_1, \dots, X_n$ are i.i.d. standard normal variables, then $E \max_{1 \leq i \leq n} |X_i| / \sqrt{2 \log n} \rightarrow 1$, as $n \rightarrow \infty$. Furthermore,

$$(\pi \log 2)^{-1/2} \sqrt{\log n} \leq E \max_{1 \leq i \leq n} X_i \leq E \max_{1 \leq i \leq n} |X_i| \leq \sqrt{2 \log(2n)}.$$

[Hint: Lemma 2.2.2 gives the slightly worse upper bound $\sqrt{8/3 \log(n+1)}$, which is valid without independence. The preciser bound under independence can be obtained by exact calculation, as can the asymptotic result. The lower bound with its exact constant (which is sharp at $n = 2$) can be found in Fernique (1997); see the proof of Corollary A.2.6.]

- 23. (Young function)** A convex function $\psi: [0, \infty) \rightarrow [0, \infty)$ with $\psi(0) = 0$ is automatically nondecreasing and has the further property that $x \mapsto \psi(x)/x$ is nondecreasing.

[Hint: A point $0 < x \leq y$ is the convex combination $(1 - x/y)0 + (x/y)y$.]

- 24.** For every $x > 0$:

- (i) $e^x - 1 \geq 2x^m/m!$, for every $m \geq 2$.
- (ii) $e^x - 1 - x \geq 2x^m/m!$, for every $m \geq 3$.

[Hint: For $x \leq m$, we have that $x^{m-1}/(m-1)! \geq x^m/m!$, while for $x \geq m$, we have that $x^{m+1}/(m+1)! + x^{m+2}/(m+2)! \geq x^m/m!$.]

- 25.** If $P(|X| > x\gamma_1 + \sqrt{x}\gamma_2) \lesssim e^{-mx}$, for $x > x_0 \geq 0$ and $m \in \mathbb{N}$, then $\|X\|_m \lesssim (x_0 \vee 1)(\gamma_1 + \gamma_2)$, for a universal multiplicative constant. Consequently, if $P(|X| > xb + \sqrt{xa}) \lesssim e^{-x}$, for $x > 0$, then $\|X\|_m \lesssim bm + \sqrt{am}$, for every $m \in \mathbb{N}$.

[Hint: The inequality implies $P(Z > x) \lesssim e^{-mx}$, for $x > 1 \vee x_0$, for the variable $Z = |X|/(\gamma_1 + \gamma_2)$. Therefore $EZ^m = \int_0^\infty mx^{m-1}P(Z > x)dx \lesssim (1 \vee x_0)^m + \int_0^\infty mx^{m-1}e^{-mx}dx$ and hence $\|Z\|_m \lesssim 1 \vee x_0 + m^{1/m-1}\Gamma(m)^{1/m}$. This tends to $1 \vee x_0 + e^{-1}$, as $m \rightarrow \infty$, by Stirling's approximation $\Gamma(m) \sim \sqrt{2\pi} m^{m-1/2} e^{-m}$.]

- 26. (Maximal inequality Rademacher)** For $a \in \mathbb{R}^n$ and independent Rademacher variables $\varepsilon_1, \dots, \varepsilon_n$, let $X_a = \sum_{i=1}^n a_i \varepsilon_i$. Then $E \max_{a \in A} |X_a| \leq \sqrt{2 \log(2|A|)} \max_{a \in A} \|a\|$, for any (finite) set $A \subset \mathbb{R}^n$.

[Hint: In view of Lemma 2.2.8, this is essentially Lemma 2.2.2 with explicit constants. By scaling we can assume that $\|a\| \leq 1$, for every $a \in A$. By Jensen's inequality $e^{\lambda E \max_a |X_a|} \leq E e^{\lambda \max_a |X_a|} \leq \sum_a E(e^{\lambda X_a} + e^{-\lambda X_a})$, for any $\lambda > 0$. As seen in the proof of Lemma 2.2.8, the right side is bounded above by $2|A|e^{\lambda^2/2}$. Take logarithms and choose $\lambda = \sqrt{2 \log(2|A|)}$.]

- 27.** Let $\phi, \psi: [0, \infty) \rightarrow [0, \infty)$ be convex, nondecreasing, nonzero functions with $\phi(0) = \psi(0) = 0$ and $\phi(y/x) \leq \psi(y)/\psi(x)$, for every $y \geq cx$ and $x \geq 1$, for some $c > 0$. Then $\|\max_{1 \leq i \leq m} |X_i|\|_\phi \leq K \psi^{-1}(m) \max_i \|X_i\|_\psi$, for any random variables $X_1, \dots, X_m$ and a constant K that depends on ϕ, ψ, c only.

[Hint: Adapt the proof of Lemma 2.2.2.]

2.3

Symmetrization and Measurability

One of the two main approaches toward deriving Glivenko-Cantelli and Donsker theorems is based on the principle of comparing the empirical process to a “symmetrized” empirical process. In this chapter we derive the main symmetrization theorem, as well as a number of technical complements, which may be skipped at first reading.

2.3.1 Symmetrization

Let $\varepsilon_1, \dots, \varepsilon_n$ be i.i.d. Rademacher random variables. Instead of the empirical process

$$f \mapsto (\mathbb{P}_n - P)f = \frac{1}{n} \sum_{i=1}^n (f(X_i) - Pf),$$

consider the symmetrized process

$$f \mapsto \mathbb{P}_n^o f = \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i),$$

where $\varepsilon_1, \dots, \varepsilon_n$ are independent of $(X_1, \dots, X_n)$. Both processes have mean function zero (because $E(\varepsilon_i f(X_i) | X_i) = 0$ by symmetry of ε_i). It turns out that the law of large numbers or the central limit theorem for one of these processes holds if and only if the corresponding result is true for the other process. One main approach to proving empirical limit theorems is to pass from $\mathbb{P}_n - P$ to $\mathbb{P}_n^o$ and next apply arguments conditionally on the original X s. The idea is that, for fixed $X_1, \dots, X_n$, the symmetrized empirical measure is a Rademacher process, hence a sub-Gaussian process, to which Corollary 2.2.9 can be applied. in a Bayesian framework,

Thus we need to bound maxima and moduli of the process $\mathbb{P}_n - P$ by those of the symmetrized process. To formulate such bounds, we must be careful about

the possible nonmeasurability of suprema of the type $\|\mathbb{P}_n - P\|_{\mathcal{F}}$. The result will be formulated in terms of outer expectation, but it does not hold for every choice of an underlying probability space on which $X_1, \dots, X_n$ are defined. Throughout this part, if outer expectations are involved, it is assumed that $X_1, \dots, X_n$ are the coordinate projections on the product space $(\mathcal{X}^n, \mathcal{A}^n, P^n)$, and the outer expectations of functions $(X_1, \dots, X_n) \mapsto h(X_1, \dots, X_n)$ are computed for P^n . Thus “independent” is understood in terms of a product probability space. If auxiliary variables, independent of the X s, are involved, as in the next lemma, we use a similar convention. In that case, the underlying probability space is assumed to be of the form $(\mathcal{X}^n, \mathcal{A}^n, P^n) \times (\mathcal{Z}, \mathcal{C}, Q)$ with $X_1, \dots, X_n$ equal to the coordinate projections on the first n coordinates and the additional variables depending only on the $(n+1)$ st coordinate.

The following lemma will be used mostly with the choice $\Phi(x) = x$.

2.3.1 Lemma (Symmetrization). *For every nondecreasing, convex function $\Phi: \mathbb{R} \rightarrow \mathbb{R}$ and class of measurable functions $\mathcal{F}$,*

$$E^* \Phi(\|\mathbb{P}_n - P\|_{\mathcal{F}}) \leq E^* \Phi\left(2\|\mathbb{P}_n^o\|_{\mathcal{F}}\right),$$

where the outer expectations are computed as indicated in the preceding paragraph.

Proof. Let $Y_1, \dots, Y_n$ be independent copies of $X_1, \dots, X_n$, defined formally as the coordinate projections on the last n coordinates in the product space $(\mathcal{X}^n, \mathcal{A}^n, P^n) \times (\mathcal{Z}, \mathcal{C}, Q) \times (\mathcal{X}^n, \mathcal{A}^n, P^n)$. The outer expectations in the statement of the lemma are unaffected by this enlargement of the underlying probability space, because coordinate projections are perfect maps. For fixed values $X_1, \dots, X_n$,

$$\|\mathbb{P}_n - P\|_{\mathcal{F}} = \sup_{f \in \mathcal{F}} \frac{1}{n} \left| \sum_{i=1}^n [f(X_i) - Ef(Y_i)] \right| \leq E_Y^* \sup_{f \in \mathcal{F}} \frac{1}{n} \left| \sum_{i=1}^n [f(X_i) - f(Y_i)] \right|,$$

where E_Y^* is the outer expectation with respect to $Y_1, \dots, Y_n$ computed for P^n for given, fixed values of $X_1, \dots, X_n$. Combination with Jensen's inequality yields

$$\Phi(\|\mathbb{P}_n - P\|_{\mathcal{F}}) \leq E_Y \Phi \left(\left\| \frac{1}{n} \sum_{i=1}^n [f(X_i) - f(Y_i)] \right\|_{\mathcal{F}}^{*Y} \right),$$

where $*Y$ denotes the minimal measurable majorant of the supremum with respect to $Y_1, \dots, Y_n$, still with $X_1, \dots, X_n$ fixed. Because Φ is nondecreasing and continuous, the $*Y$ inside Φ can be moved to E_Y^* (Problem 1.2.8). Next take the expectation with respect to $X_1, \dots, X_n$ to get

$$E^* \Phi(\|\mathbb{P}_n - P\|_{\mathcal{F}}) \leq E_X^* E_Y^* \Phi \left(\frac{1}{n} \left\| \sum_{i=1}^n [f(X_i) - f(Y_i)] \right\|_{\mathcal{F}} \right).$$

Here the repeated outer expectation can be bounded above by the joint outer expectation E^* by Lemma 1.2.6.

Adding a minus sign in front of a term $[f(X_i) - f(Y_i)]$ has the effect of exchanging X_i and Y_i . By construction of the underlying probability space as a product space, the outer expectation of any function $f(X_1, \dots, X_n, Y_1, \dots, Y_n)$ remains unchanged under permutations of its $2n$ arguments. Hence the expression

$$\mathbb{E}^* \Phi \left(\left\| \frac{1}{n} \sum_{i=1}^n e_i [f(X_i) - f(Y_i)] \right\|_{\mathcal{F}} \right)$$

is the same for any n -tuple $(e_1, \dots, e_n) \in \{-1, 1\}^n$. Deduce that

$$\mathbb{E}^* \Phi (\|\mathbb{P}_n - P\|_{\mathcal{F}}) \leq \mathbb{E}_{\varepsilon} \mathbb{E}_{X,Y}^* \Phi \left(\left\| \frac{1}{n} \sum_{i=1}^n \varepsilon_i [f(X_i) - f(Y_i)] \right\|_{\mathcal{F}} \right).$$

Use the triangle inequality to separate the contributions of the X s and the Y s and next use the convexity of Φ to bound the previous expression by

$$\frac{1}{2} \mathbb{E}_{\varepsilon} \mathbb{E}_{X,Y}^* \Phi \left(2 \left\| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right\|_{\mathcal{F}} \right) + \frac{1}{2} \mathbb{E}_{\varepsilon} \mathbb{E}_{X,Y}^* \Phi \left(2 \left\| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(Y_i) \right\|_{\mathcal{F}} \right).$$

By perfectness of coordinate projections, the expectation $\mathbb{E}_{X,Y}^*$ is the same as $\mathbb{E}_X^*$ and $\mathbb{E}_Y^*$ in the two terms, respectively. Finally, replace the repeated outer expectations by a joint outer expectation. ■

The symmetrization lemma is valid for any class $\mathcal{F}$. In the proofs of Glivenko-Cantelli and Donsker theorems, it will be applied not only to the original set of functions of interest, but also to several classes constructed from such a set $\mathcal{F}$ (such as the class $\mathcal{F}_{\delta}$ of small differences). The next step in these proofs is to apply a maximal inequality to the right side of the lemma, conditionally on $X_1, \dots, X_n$. At that point we need to write the joint outer expectation as the repeated expectation $\mathbb{E}_X^* \mathbb{E}_{\varepsilon}$, where the indices X and ε mean expectation over X and ε , respectively, conditionally on the remaining variables. Unfortunately, Fubini's theorem is not valid for outer expectations. To overcome this problem, it is assumed that the integrand in the right side of the lemma is jointly measurable in $(X_1, \dots, X_n, \varepsilon_1, \dots, \varepsilon_n)$. Since the Rademacher variables are discrete, this is the case if and only if the maps

$$(2.3.2) \quad (X_1, \dots, X_n) \mapsto \left\| \sum_{i=1}^n e_i f(X_i) \right\|_{\mathcal{F}}$$

are measurable for every n -tuple $(e_1, \dots, e_n) \in \{-1, 1\}^n$. For the intended application of Fubini's theorem, it suffices that this is the case for the completion of $(\mathcal{X}^n, \mathcal{A}^n, P^n)$.

2.3.3 Definition (Measurable class). A class $\mathcal{F}$ of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$ on a probability space $(\mathcal{X}, \mathcal{A}, P)$ is called a *P-measurable class* if the function (2.3.2) is measurable on the completion of $(\mathcal{X}^n, \mathcal{A}^n, P^n)$ for every n and every vector $(e_1, \dots, e_n) \in \mathbb{R}^n$.

In the following, one cannot dispense with measurability conditions of some form. Examples show that the law of large numbers or the central limit theorem

may fail only because of the violation of measurability conditions such as the one just introduced. On the other hand, the measurability conditions in the present form are not necessary, but they are suggested by the methods of proof. A trivial, but nevertheless useful, device to relax measurability requirements a little is to replace the class $\mathcal{F}$ by a class $\mathcal{G}$ for which measurability can be checked. If $\|\mathbb{P}_n - P\|_{\mathcal{F}} = \|\mathbb{P}_n - P\|_{\mathcal{G}}$ inner almost surely, then a law of large numbers for $\mathcal{G}$ clearly implies one for $\mathcal{F}$. Similarly, if $\|\mathbb{P}_n - P\|_{\mathcal{F}_\delta} = \|\mathbb{P}_n - P\|_{\mathcal{G}_\delta}$ inner almost surely for every $\delta > 0$, then asymptotic equicontinuity of the $\mathcal{F}$ -indexed empirical process follows from the same property for $\mathcal{G}$. Furthermore, if in both cases $\mathcal{G}$ is chosen to be a subclass of $\mathcal{F}$, then the other conditions for the uniform law or the central limit theorem typically carry over from $\mathcal{F}$ onto $\mathcal{G}$. Rather than formalizing this principle into a concept such as “nearly measurable class,” we will state the conditions directly in terms of $\mathcal{F}$.

2.3.4 Example (Pointwise measurable classes). Suppose $\mathcal{F}$ contains a countable subset $\mathcal{G}$ such that for every $f \in \mathcal{F}$ there exists a sequence g_m in $\mathcal{G}$ with $g_m(x) \rightarrow f(x)$ for every x . Then $\mathcal{F}$ is P -measurable for every P .

This claim is immediate from the fact that $\|\sum e_i f(X_i)\|_{\mathcal{F}} = \|\sum e_i f(X_i)\|_{\mathcal{G}}$.

Some examples of this situation are the collection of indicators of cells in Euclidean space, the collection of indicators of balls, and collections of functions that are separable for the supremum norm.

In Section 2.3.3 it will be seen that every separable collection $\mathcal{F} \subset L_2(P)$ has a “version” that is “almost” pointwise separable. (The name *pointwise separable class* will be reserved for these versions.)

2.3.5 Example (Measurable Suslin). Suppose $\mathcal{F}$ is a Suslin topological space (with respect to an arbitrary topology) and the map $(x, f) \mapsto f(x)$ is jointly measurable on $\mathcal{X} \times \mathcal{F}$ for the product σ -field of $\mathcal{A}$ and the Borel σ -field. [Recall that every Borel subset (even analytic subset) of a Polish space is Suslin for the relative topology.] Then $\mathcal{F}$ is P -measurable for every probability measure P .

This follows from Example 1.7.5 upon noting that the conditions imply that the process $(x_1, \dots, x_n, f) \mapsto \sum_{i=1}^n e_i f(x_i)$ is jointly measurable on $\mathcal{X}^n \times \mathcal{F}$.

This “measurable Suslin condition” replaces the original problem of showing P -measurability of $\mathcal{F}$ by the problem of finding a suitable topology on $\mathcal{F}$ for which the map $(x, f) \mapsto f(x)$ is measurable. This is sometimes easy – for instance, if the class $\mathcal{F}$ is indexed by a parameter in a nice manner – but sometimes as hard as the original problem.

* 2.3.2 More Symmetrization

This section continues with additional results on symmetrization. These are not needed for the main results of this part (Chapters 2.2, 2.4, and 2.5), and the reader may wish to skip the remainder of this chapter at first reading.

For future reference it is convenient to generalize the notation. Instead of the empirical distribution, consider sums $\sum_{i=1}^n Z_i$ of independent stochastic processes

* This section may be skipped at first reading.

$\{Z_i(f): f \in \mathcal{F}\}$. Again the processes need not possess any further measurability properties besides measurability of all marginals $Z_i(f)$. However, for computing outer expectations, it will be understood that the underlying probability space is a product space $\prod_{i=1}^n (\mathcal{X}_i, \mathcal{A}_i, P_i) \times (\mathcal{Z}, \mathcal{C}, Q)$ and each Z_i is a function of the i th coordinate of $(x, z) = (x_1, \dots, x_n, z)$ only. For i.i.d. stochastic processes, it is understood in addition that the spaces $(\mathcal{X}_i, \mathcal{A}_i, P_i)$ as well as the maps $Z_i(f)$ defined on them are identical. Additional (independent) Rademacher or other variables are understood to be functions of the $(n+1)$ st coordinate z only. The empirical distribution corresponds to taking $Z_i(f) = f(X_i) - Pf$.

First consider “desymmetrization.” The inequality of the preceding lemma can only be reversed (up to constants) for classes $\mathcal{F}$ that are centered (Problem 2.3.1). The following lemma shows that the maxima of the processes $\mathbb{P}_n - P$ and $n^{-1} \sum \varepsilon_i (\delta_{X_i} - P)$ are fully comparable.

2.3.6 Lemma. *Let $Z_1, \dots, Z_n$ be independent stochastic processes with mean zero. Then*

$$\mathbb{E}^* \Phi \left(\frac{1}{2} \left\| \sum_{i=1}^n \varepsilon_i Z_i \right\|_{\mathcal{F}} \right) \leq \mathbb{E}^* \Phi \left(\left\| \sum_{i=1}^n Z_i \right\|_{\mathcal{F}} \right) \leq \mathbb{E}^* \Phi \left(2 \left\| \sum_{i=1}^n \varepsilon_i (Z_i - \mu_i) \right\|_{\mathcal{F}} \right),$$

for every nondecreasing, convex $\Phi: \mathbb{R} \rightarrow \mathbb{R}$ and arbitrary functions $\mu_i: \mathcal{F} \rightarrow \mathbb{R}$.

Proof. The inequality on the right follows by similar arguments as in the proof of Lemma 2.3.1. For the inequality on the left, let $Y_1, \dots, Y_n$ be an independent copy of $Z_1, \dots, Z_n$ (suitably defined on the probability space $\prod_{i=1}^n (\mathcal{X}_i, \mathcal{A}_i, P_i) \times (\mathcal{Z}, \mathcal{C}, Q) \times \prod_{i=1}^n (\mathcal{X}_i, \mathcal{A}_i, P_i)$ and depending on the last n coordinates exactly as $Z_1, \dots, Z_n$ depend on the first n coordinates). Since $\mathbb{E} Y_i(f) = 0$, the left side of the lemma is an average of expressions of the type

$$\mathbb{E}_Z^* \Phi \left(\left\| \frac{1}{2} \sum_{i=1}^n e_i [Z_i(f) - \mathbb{E} Y_i(f)] \right\|_{\mathcal{F}} \right),$$

where $(e_1, \dots, e_n)$ ranges over $\{-1, 1\}^n$. (cf. Lemma 1.2.7). By Jensen’s inequality, this expression is bounded above by

$$\mathbb{E}_{Z,Y}^* \Phi \left(\left\| \frac{1}{2} \sum_{i=1}^n e_i [Z_i(f) - Y_i(f)] \right\|_{\mathcal{F}} \right) = \mathbb{E}_{Z,Y}^* \Phi \left(\left\| \frac{1}{2} \sum_{i=1}^n [Z_i(f) - Y_i(f)] \right\|_{\mathcal{F}} \right).$$

Finally, apply the triangle inequality and convexity of Φ . ■

The most important choice for Φ in the preceding lemmas is $\Phi(x) = x$. The requirement that Φ be convex rules out $\Phi(x) = 1\{x > a\}$. Nevertheless, there is an analogous symmetrization inequality for probabilities.

2.3.7 Lemma (Symmetrization for probabilities). *For arbitrary independent stochastic processes $Z_1, \dots, Z_n$ and arbitrary functions $\mu_1, \dots, \mu_n: \mathcal{F} \mapsto \mathbb{R}$,*

$$\beta_n(x) \mathbf{P}^* \left(\left\| \sum_{i=1}^n Z_i \right\|_{\mathcal{F}} > x \right) \leq 2\mathbf{P}^* \left(4 \left\| \sum_{i=1}^n \varepsilon_i (Z_i - \mu_i) \right\|_{\mathcal{F}} > x \right),$$

for every $x > 0$ and $\beta_n(x) \leq \inf_f \mathbf{P} \left(\left| \sum_{i=1}^n Z_i(f) \right| < x/2 \right)$. In particular, this is true for i.i.d. mean-zero processes, and $\beta_n(x) = 1 - (4n/x^2) \sup_f \text{var } Z_1(f)$. Here the outer expectations are computed as indicated previously.

Proof. Let $Y_1, \dots, Y_n$ be an independent copy of $Z_1, \dots, Z_n$, suitably defined on a product space as previously. If $\left\| \sum_{i=1}^n Z_i \right\|_{\mathcal{F}} > x$, then there is certainly some f for which $\left| \sum_{i=1}^n Z_i(f) \right| > x$. Fix a realization, $Z_1, \dots, Z_n$ and f for which both are the case. For this fixed realization,

$$\begin{aligned} \beta &\leq \mathbf{P}_Y^* \left(\left| \sum_{i=1}^n Y_i(f) \right| < \frac{x}{2} \right) \leq \mathbf{P}_Y^* \left(\left| \sum_{i=1}^n Y_i(f) - \sum_{i=1}^n Z_i(f) \right| > \frac{x}{2} \right) \\ &\leq \mathbf{P}_Y^* \left(\left\| \sum_{i=1}^n (Y_i - Z_i) \right\|_{\mathcal{F}} > \frac{x}{2} \right). \end{aligned}$$

The far left and far right sides do not depend on the particular f , and the inequality between them is valid on the set $\left\{ \left\| \sum_{i=1}^n Z_i \right\|_{\mathcal{F}} > x \right\}$. Integrate the two sides out with respect to $Z_1, \dots, Z_n$ over this set to obtain

$$\beta \mathbf{P}^* \left(\left\| \sum_{i=1}^n Z_i \right\|_{\mathcal{F}} > x \right) \leq \mathbf{P}_Z^* \mathbf{P}_Y^* \left(\left\| \sum_{i=1}^n (Y_i - Z_i) \right\|_{\mathcal{F}} > \frac{x}{2} \right).$$

By symmetry, the right side equals

$$\mathbf{E}_\varepsilon \mathbf{P}_Z^* \mathbf{P}_Y^* \left(\left\| \sum_{i=1}^n \varepsilon_i (Y_i - Z_i) \right\|_{\mathcal{F}} > \frac{x}{2} \right).$$

In view of the triangle inequality, this expression is bounded above by twice $\mathbf{P}^* \left(\left\| \sum_{i=1}^n \varepsilon_i (Y_i - \mu_i) \right\|_{\mathcal{F}} > x/4 \right)$.

I.i.d. processes $Z_1, \dots, Z_n$ with mean zero satisfy the condition for the given β in view of Chebyshev's inequality. ■

Since they only add in signs, Rademacher variables appear to yield an efficient method of symmetrization. Nevertheless, in most arguments they can be replaced by other variables; for instance, standard normal ones. Then the form of the symmetrization inequalities may become more complicated. A discussion is deferred to Chapter 2.9, where it is shown in general that the sequences of processes $n^{-1/2} \sum_{i=1}^n \xi_i Z_i$ and $n^{-1/2} \sum_{i=1}^n Z_i$ possess the same asymptotic behavior for most choices of ξ_i .

Inequalities in terms of the means of suprema of processes are easier to handle than inequalities in terms of probabilities. However, so far the main condition (2.1.8)

for the empirical central limit theorem has been stated in terms of probabilities. In view of Markov's inequality, the condition

$$(2.3.8) \quad E^* \sqrt{n} \|\mathbb{P}_n - P\|_{\mathcal{F}_{\delta_n}} \rightarrow 0, \quad \text{for every } \delta_n \rightarrow 0,$$

is sufficient for asymptotic equicontinuity (2.1.8). In view of Lemma 2.3.6, this condition is equivalent to the analogous condition in terms of the symmetrized empirical measure $\mathbb{P}_n^o$. It is therefore of interest that there is no loss of generality in passing from probabilities (2.1.8) to expectations.

For the proof we need a preparatory lemma.

2.3.9 Lemma. *Let $Z_1, Z_2, \dots$ be i.i.d stochastic processes such that $n^{-1/2} \sum_{i=1}^n Z_i$ converges weakly in $\ell^\infty(\mathcal{F})$ to a tight Gaussian process. Then*

$$\lim_{x \rightarrow \infty} x^2 \sup_n P^* \left(\frac{1}{\sqrt{n}} \left\| \sum_{i=1}^n Z_i \right\|_{\mathcal{F}} > x \right) = 0.$$

In particular, the random variable $\|Z_1\|_{\mathcal{F}}^$ possesses a weak second moment.*

Proof. Let $Y_1, \dots, Y_n$ be an independent copy of $Z_1, \dots, Z_n$, for the benefit of the outer expectations defined as coordinate projections on an extra factor of a product probability space, as usual. Marginal convergence to a Gaussian process implies that $E Z_i(f) = 0$ and $\text{var } Z_i(f) < \infty$ for every f . Furthermore, the existence of a tight limit ensures that $\mathcal{F}$ is totally bounded for the semimetric $\rho(f, g) = \sigma(Z(f) - Z(g))$. In particular, $\alpha^2 = \sup_{f \in \mathcal{F}} \text{var } Z(f)$ is finite. By an intermediate step in the proof of the symmetrization Lemma 2.3.7,

$$\left(1 - \frac{4\alpha^2}{x^2}\right) P \left(\frac{1}{\sqrt{n}} \left\| \sum_{i=1}^n Z_i \right\|_{\mathcal{F}}^* > x \right) \leq P \left(\frac{1}{\sqrt{n}} \left\| \sum_{i=1}^n (Z_i - Y_i) \right\|_{\mathcal{F}}^* > \frac{x}{2} \right).$$

It suffices to prove the lemma for $Z_i - Y_i$ instead of Z_i . For convenience of notation, suppose that the original Z_i are symmetric.

Fix $\varepsilon > 0$. The norm $\|\mathbb{G}\|_{\mathcal{F}}$ of the Gaussian limit variable $\mathbb{G}$ has moments of all orders (see Proposition A.2.3). Thus there exists x such that $P(\|\mathbb{G}\|_{\mathcal{F}} \geq x) \leq \varepsilon/x^2 \leq 1/8$. By the portmanteau theorem, there exists N such that, for all $n \geq N$,

$$(2.3.10) \quad P^* \left(\left\| \sum_{i=1}^n Z_i \right\|_{\mathcal{F}} > x\sqrt{n} \right) \leq 2P(\|\mathbb{G}\|_{\mathcal{F}} \geq x) \leq \frac{2\varepsilon}{x^2}.$$

By Lévy's inequalities A.1.2,

$$P \left(\max_{1 \leq i \leq n} \|Z_i\|_{\mathcal{F}}^* > x\sqrt{n} \right) \leq 2P \left(\left\| \sum_{i=1}^n Z_i \right\|_{\mathcal{F}}^* > x\sqrt{n} \right) \leq \frac{4\varepsilon}{x^2}.$$

In view of Problem 2.3.2, for every $n \geq N$,

$$x^2 n P^* (\|Z_1\|_{\mathcal{F}} > x\sqrt{n}) \leq 8\varepsilon.$$

This immediately implies the second assertion of the lemma.

To obtain the first assertion, apply the preceding argument to the processes $Z_i = m^{-1/2} \sum_{j=1}^m Z_{i,j}$ for each $1 \leq i \leq n$, where $Z_{1,1}, \dots, Z_{n,m}$ are i.i.d. copies of Z_1 . The sequence $n^{-1/2} \sum_{i=1}^n Z_i = (nm)^{-1/2} \sum_{i=1}^n \sum_{j=1}^m Z_{i,j}$ converges to the same limit $\mathbb{G}$ for each m as $n \rightarrow \infty$. Since the convergence is “accelerated,” there exists N such that (2.3.10) holds for $n \geq N$ and all m , where x can be chosen dependent on ε and $\mathbb{G}$ only, as before. In other words, there exists x and N such that for $n \geq N$,

$$x^2 n \mathbf{P}^* \left(\frac{1}{\sqrt{m}} \left\| \sum_{j=1}^m Z_{1,j} \right\|_{\mathcal{F}} > x \sqrt{n} \right) \leq 8\varepsilon,$$

for every m . This implies the lemma. ■

If a distribution on the real line has tails of the order $o(x^{-2})$, then all its moments of order $0 < r < 2$ exist. One interesting consequence of the preceding lemma is that the sequence of moments $\mathbf{E}^* \|n^{-1/2} \sum_{i=1}^n Z_i\|_{\mathcal{F}}^r$ is uniformly bounded for every $0 < r < 2$. Given the weak convergence, this gives that the sequence $\mathbf{E}^* \|n^{-1/2} \sum_{i=1}^n Z_i\|_{\mathcal{F}}^r$ converges to the corresponding moment of the limit variable.

2.3.11 Lemma. *Let $Z_1, Z_2, \dots$ be i.i.d. stochastic processes, linear in f . Set $\rho_Z(f, g) = \sigma(Z_1(f) - Z_1(g))$ and $\mathcal{F}_\delta = \{f - g : \rho_Z(f, g) < \delta\}$. Then the following statements are equivalent:*

- (i) $n^{-1/2} \sum_{i=1}^n Z_i$ converges weakly to a tight limit in $\ell^\infty(\mathcal{F})$;
 - (ii) $(\mathcal{F}, \rho_Z)$ is totally bounded and $\|n^{-1/2} \sum_{i=1}^n Z_i\|_{\mathcal{F}_{\delta_n}} \xrightarrow{\mathbf{P}^*} 0$ for every $\delta_n \downarrow 0$;
 - (iii) $(\mathcal{F}, \rho_Z)$ is totally bounded and $\mathbf{E}^* \|n^{-1/2} \sum_{i=1}^n Z_i\|_{\mathcal{F}_{\delta_n}}^r \rightarrow 0$ for every $\delta_n \downarrow 0$.
- These conditions imply that the sequence $\mathbf{E}^* \|n^{-1/2} \sum_{i=1}^n Z_i\|_{\mathcal{F}}^r$ converges to $\mathbf{E} \|\mathbb{G}\|_{\mathcal{F}}^r$ for every $0 < r < 2$.

Proof. The equivalence of (i) and (ii) is clear from the general results on weak convergence in $\ell^\infty(\mathcal{F})$ obtained in Chapter 1.5. Condition (iii) implies (ii) by Markov’s inequality.

Suppose (i) and (ii) hold. Then $\mathbf{P}^*(\|Z_1\|_{\mathcal{F}} > x) = o(x^{-2})$ as x tends to infinity by Lemma 2.3.9. This implies that

$$\mathbf{E}^* \max_{1 \leq i \leq n} \frac{\|Z_i\|_{\mathcal{F}}}{\sqrt{n}} \rightarrow 0$$

(Problem 2.3.3). In view of the triangle inequality, the same is true with $\mathcal{F}$ replaced by $\mathcal{F}_{\delta_n}$. Convergence to zero in probability of $\|n^{-1/2} \sum_{i=1}^n Z_i\|_{\mathcal{F}_{\delta_n}}$ implies pointwise convergence to zero of the sequence of their quantile functions. Apply Hoffmann-Jørgensen’s inequality (Proposition A.1.5) to obtain (iii).

The last assertion follows from the remark after Lemma 2.3.9. ■

To recover the empirical process, set $Z_i(f) = f(X_i) - Pf$.

2.3.12 Corollary. *Let $\mathcal{F}$ be a class of measurable functions. Then the following are equivalent:*

- (i) $\mathcal{F}$ is P -Donsker;
- (ii) $(\mathcal{F}, \rho_P)$ is totally bounded and (2.1.8) holds;
- (iii) $(\mathcal{F}, \rho_P)$ is totally bounded and (2.3.8) holds.

2.3.13 Corollary. Every P -Donsker class $\mathcal{F}$ satisfies $P(\|f - Pf\|_{\mathcal{F}}^* > x) = o(x^{-2})$ as x tends to infinity. Consequently, if $\|Pf\|_{\mathcal{F}} < \infty$, then $\mathcal{F}$ possesses an envelope function F with $P(F > x) = o(x^{-2})$.

* 2.3.3 Separable Versions

A stochastic process $\{\mathbb{G}(f, \omega): f \in \mathcal{F}\}$ indexed by a given separable, semimetric space $(\mathcal{F}, \rho)$ is called *separable* if there exists a null set N and a countable subset $\mathcal{G} \subset \mathcal{F}$ such that, for all $\omega \notin N$ and $f \in \mathcal{F}$, there exists a sequence g_m in $\mathcal{G}$ with $g_m \rightarrow f$ and $\mathbb{G}(g_m, \omega) \rightarrow \mathbb{G}(f, \omega)$. A stochastic process $\{\tilde{\mathbb{G}}(f): f \in \mathcal{F}\}$ is a *separable version* of a given process $\{\mathbb{G}(f): f \in \mathcal{F}\}$ if $\tilde{\mathbb{G}}$ is separable and $\tilde{\mathbb{G}}(f) = \mathbb{G}(f)$ almost surely for every f . (The two processes must be defined on the same probability space, and the exceptional null sets may depend on f .)

It is well known that every stochastic process possesses a separable version (possibly taking values in the extended real line). By its definition, a separable process has excellent measurability properties. In this subsection it is shown that an empirical process possesses a separable version that is itself an empirical process. This fact is used in Chapter 2.10 (and there only) to remove unnecessary measurability conditions from two preservation theorems.

For any separable process $\tilde{\mathbb{G}}$ with countable “separant” $\mathcal{G}$,

$$\sup_{\substack{\rho(f,g) < \delta \\ f,g \in \mathcal{F}}} |\tilde{\mathbb{G}}(f) - \tilde{\mathbb{G}}(g)| = \sup_{\substack{\rho(f,g) < \delta \\ f,g \in \mathcal{G}}} |\tilde{\mathbb{G}}(f) - \tilde{\mathbb{G}}(g)|, \quad \text{a.s.}$$

If $\tilde{\mathbb{G}}$ is a version of $\mathbb{G}$, then the countability of $\mathcal{G}$ implies that the right side changes at most on a null set if $\tilde{\mathbb{G}}$ is replaced by $\mathbb{G}$. Next the expression increases if $\mathcal{G}$ is replaced by $\mathcal{F}$. Thus, for every separable version $\tilde{\mathbb{G}}$ of a process $\mathbb{G}$,

$$\sup_{\substack{\rho(f,g) < \delta \\ f,g \in \mathcal{F}}} |\tilde{\mathbb{G}}(f) - \tilde{\mathbb{G}}(g)| \leq \sup_{\substack{\rho(f,g) < \delta \\ f,g \in \mathcal{F}}} |\mathbb{G}(f) - \mathbb{G}(g)|, \quad \text{a.s.}$$

It follows that the modulus of continuity of a process decreases when replacing it by a separable version. Consequently, asymptotic equicontinuity of a sequence of processes implies the same for any sequence of separable versions. Since the marginal distributions of a process do not change when passing to a separable version, this yields the following lemma.

2.3.14 Lemma. Let $\tilde{\mathbb{G}}_n$ be separable versions of a sequence of stochastic processes $\mathbb{G}_n$ with sample paths in $\ell^\infty(\mathcal{F})$. Then the sequence $\mathbb{G}_n$ converges in distribution to a tight limit in $\ell^\infty(\mathcal{F})$ if and only if the sequence $\tilde{\mathbb{G}}_n$ converges to a tight limit and $\mathbb{G}_n - \tilde{\mathbb{G}}_n$ converges to zero in outer probability in $\ell^\infty(\mathcal{F})$.

Empirical processes allow a special type of separable version, constructed from a separable version of the index class $\mathcal{F}$. Let $\mathcal{F}$ be a class of square integrable measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$ on a probability space $(\mathcal{X}, \mathcal{A}, P)$ equipped with a

* This section may be skipped at first reading.

semi-metric ρ . Call $\mathcal{F}$ a *pointwise separable class* relative to ρ if there exists a countable subset $\mathcal{G} \subset \mathcal{F}$ and for each n a P^n -null set $N_n \subset \mathcal{X}^n$ such that, for all $(x_1, \dots, x_n) \notin N_n$ and $f \in \mathcal{F}$, there exists a sequence g_m in $\mathcal{G}$ such that $\rho(g_m, f) \rightarrow 0$ and $(g_m(x_1), \dots, g_m(x_n)) \rightarrow (f(x_1), \dots, f(x_n))$. A class $\tilde{\mathcal{F}}$ of measurable functions is a *pointwise separable version* of $\mathcal{F}$ if $\tilde{\mathcal{F}}$ is pointwise separable and there exists a bijection $f \leftrightarrow \tilde{f}$ between $\mathcal{F}$ and $\tilde{\mathcal{F}}$ such that $f = \tilde{f}$ almost surely.

Let $\mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - P)$ be the empirical process indexed by the class of functions $\mathcal{F}$. If $\tilde{\mathcal{F}}$ is a pointwise separable version of $\mathcal{F}$, then

$$\tilde{\mathbb{G}}_n(f) = \mathbb{G}_n(\tilde{f})$$

defines a separable version of $\mathbb{G}_n$, for every n . This separable version is itself an empirical process, and its index class is obtained by changing each element of the original index class on a null set. Furthermore, the difference process $\mathbb{G}_n - \tilde{\mathbb{G}}_n$ is the empirical process indexed by the class $\{f - \tilde{f} : f \in \mathcal{F}\}$. Each of the functions in this class is zero almost surely. It turns out that the empirical process indexed by a zero class converges in probability to zero if and only if the same is true for the class of absolute values. This yields the following theorem.

2.3.15 Theorem. *Let $\tilde{\mathcal{F}}$ be a pointwise separable version relative to the $L_2(P)$ -norm of a class of square integrable measurable functions $\mathcal{F}$. Then $\mathcal{F}$ is Donsker if and only if $\tilde{\mathcal{F}}$ is Donsker and $\sup_{f \in \mathcal{F}} \sqrt{n} \mathbb{P}_n |f - \tilde{f}| \rightarrow 0$ in outer probability.*

Proof. By the preceding lemma, $\mathcal{F}$ is Donsker if and only if $\tilde{\mathcal{F}}$ is Donsker and $\sqrt{n} \|\mathbb{P}_n(f - \tilde{f})\|_{\mathcal{F}} \rightarrow 0$ in outer probability. It suffices to show that for a class of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$ such that $f = 0$ almost surely, for every f , the two statements $\sqrt{n} \|\mathbb{P}_n f\|_{\mathcal{F}} \xrightarrow{P_*} 0$ and $\sqrt{n} \|\mathbb{P}_n |f|\|_{\mathcal{F}} \xrightarrow{P_*} 0$ are equivalent. It is clear that the second statement implies the first.

For a proof in the other direction, fix $\varepsilon > 0$ and define $U = \{x \in \mathcal{X}^n : \|\mathbb{P}_n\|_{\mathcal{F}} \leq \varepsilon\}_*$. It will be shown that there exists a measurable set $U' \subset U$ of the same measure as U such that, for all $x \in U'$ and every subset $I \subset \{1, 2, \dots, n\}$,

$$\frac{1}{n} \left| \sum_{i \in I} f(x_i) \right| \leq \varepsilon.$$

Then it follows that $P_*(\|\mathbb{P}_n\|_{|\mathcal{F}|} \leq 2\varepsilon) \geq P_*(\|\mathbb{P}_n\|_{\mathcal{F}} \leq \varepsilon)$, and the desired result follows.

For $I \subset \{1, 2, \dots, n\}$, identify $x \in \mathcal{X}^n$ with (α, β) , where $\alpha \in \mathcal{X}^I$ are the coordinates x_i with $i \in I$ and $\beta \in \mathcal{X}^{n-I}$ are the remaining coordinates. Let $U_\alpha \subset \mathcal{X}^n$ be the “vertical” section $\{\alpha\} \times \{\beta : (\alpha, \beta) \in U\}$ of U at “width” α , and let πU_α be the projection of U_α on $\mathcal{X}^{n-I}$ (so that $U_\alpha = \{\alpha\} \times \pi U_\alpha$). Let V_I be the union of all sections U_α with projection πU_α equal to a null set (under P^{n-I}). By Fubini’s theorem, $P^n(V_I) = 0$. Set

$$U' = U - \bigcup_I V_I,$$

where I ranges over all nonempty, proper subsets of $\{1, \dots, n\}$.

Fix I and f . Each $x \in U'$ can be identified with a pair (α, β) as before. By construction $P^{n-I}(\pi U_\alpha) > 0$ and by assumption $(f(\beta'_1), \dots, f(\beta'_{n-I})) = 0$ almost surely under P^{n-I} . Conclude that there must be a $\beta' \in \pi U_\alpha$ with $(f(\beta'_1), \dots, f(\beta'_{n-I})) = 0$. Thus

$$\frac{1}{n} \left| \sum_{i \in I} f(x_i) \right| = \frac{1}{n} \left| \sum_{i \in I} f(\alpha_i) + \sum_{i \notin I} f(\beta'_i) \right| \leq \varepsilon,$$

because $(\alpha, \beta') \in U$. This concludes the proof. ■

The preceding theorem shows that the difference between $\mathbb{G}_n$ and $\tilde{\mathbb{G}}_n$ must be small due to the fact that the differences $f - \tilde{f}$ are small: the difference cannot be small through the cancellation of positive terms by negative terms.

There is an analogous, but simpler result for Glivenko-Cantelli classes.

2.3.16 Theorem. *Let $\tilde{\mathcal{F}}$ be a pointwise separable version relative to the $L_1(P)$ -norm of a class of integrable measurable functions $\mathcal{F}$. Then $\mathcal{F}$ is weak P -Glivenko-Cantelli if and only if $\tilde{\mathcal{F}}$ is weak P -Glivenko-Cantelli and $\sup_{f \in \mathcal{F}} \mathbb{P}_n |f - \tilde{f}| \rightarrow 0$ in outer probability.*

Proof. Because the process $f \mapsto \mathbb{P}_n \tilde{f}$ is a separable version of $f \mapsto \mathbb{P}_n f$, there exists a countable separant $\mathcal{G} \subset \mathcal{F}$ such that, almost surely,

$$\sup_{f \in \mathcal{F}} |\mathbb{P}_n \tilde{f} - P \tilde{f}| = \sup_{f \in \mathcal{G}} |\mathbb{P}_n \tilde{f} - P \tilde{f}| = \sup_{f \in \mathcal{G}} |\mathbb{P}_n f - P f| \leq \sup_{f \in \mathcal{F}} |\mathbb{P}_n f - P f|.$$

It follows that the class $\tilde{\mathcal{F}}$ is Glivenko-Cantelli if $\mathcal{F}$ is Glivenko-Cantelli. We next conclude that $\mathcal{F}$ is Glivenko-Cantelli if and only if the class $\tilde{\mathcal{F}}$ is Glivenko-Cantelli and $\sup_{f \in \mathcal{F}} |\mathbb{P}_n(f - \tilde{f})|$ tends to zero in outer probability.

The functions $f - \tilde{f}$ are zero almost surely. The argument in the proof of Theorem 2.3.15 shows that the last mentioned convergence is equivalent to convergence in outer probability to zero of the sequence $\sup_{f \in \mathcal{F}} \mathbb{P}_n |f - \tilde{f}|$. ■

A pointwise separable version of a given class of functions can be constructed from a special type of lifting of $L_\infty(\mathcal{X}, \mathcal{A}, P)$. A *lifting* is a map $\rho: L_\infty(\mathcal{X}, \mathcal{A}, P) \rightarrow \mathcal{L}_\infty(\mathcal{X}, \mathcal{A}, P)$ that assigns to each equivalence class f of essentially bounded functions a representative ρf in such a way that

- (i) $\rho 1 = 1$;
- (ii) ρ is positive: $\rho f(x) \geq 0$ for all x if $f \geq 0$ almost surely;
- (iii) ρ is linear;
- (iv) ρ is multiplicative: $\rho(fg) = \rho f \rho g$.

Any lifting automatically possesses the further property:

- (vi) $\rho h(f_1, \dots, f_n) \geq h(\rho f_1, \dots, \rho f_n)$ for every lower semicontinuous $h: \mathbb{R}^n \rightarrow \mathbb{R}$ and $f_1, \dots, f_n$ in $\mathcal{L}_\infty(\mathcal{X}, \mathcal{A}, P)$ (Problem 2.3.8).

It is well known that every complete probability space $(\mathcal{X}, \mathcal{A}, P)$ admits a lifting ρ of the space $L_\infty(\mathcal{X}, \mathcal{A}, P)$. For the construction of a separable version of the empirical process, a lifting with a further property is needed. This exists by the following proposition.

2.3.17 Proposition (Consistent lifting). *Any complete probability space $(\mathcal{X}, \mathcal{A}, P)$ admits a lifting of the space $L_\infty(\mathcal{X}, \mathcal{A}, P)$ such that for each n the map ρ_n given by the diagram*

$$\left((x_1, \dots, x_n) \mapsto f(x_i) \right) \xrightarrow{\rho_n} \left((x_1, \dots, x_n) \mapsto \rho f(x_i) \right)$$

is the restriction [to the set of functions of the type $(x_1, \dots, x_n) \mapsto f(x_i)$, where f ranges over $L_\infty(\mathcal{X}, \mathcal{A}, P)$] of a lifting of $L_\infty(\mathcal{X}^n, \mathcal{A}^n, P^n)$ for each $1 \leq i \leq n$.

This proposition is proved by Talagrand (1982), who calls a lifting ρ with the given property a *consistent lifting*. There also exist liftings that are not consistent.

Suppose $\mathcal{F}$ is a class of measurable, uniformly bounded functions. The following theorem shows that a pointwise separable version $\tilde{\mathcal{F}}$ of $\mathcal{F}$ can be defined by

$$\tilde{f}(x) = \rho f(x)$$

for a consistent lifting ρ of $L_\infty(\mathcal{X}, \mathcal{A}, P)$. (Interpret ρf as the image of the class of f under ρ .) This separable version inherits the nice properties of a lifting: the map $f \mapsto \tilde{f}$ is positive, linear, and multiplicative.

If $\mathcal{F}$ contains unbounded functions, then a pointwise separable version can be defined by

$$\tilde{f}(x) = \Phi^{-1} \circ \rho(\Phi \circ f)(x),$$

for a consistent lifting as before and a fixed, strictly monotone bijection Φ of $\mathbb{R}$ onto a compact interval. This separable modification $f \mapsto \tilde{f}$ lacks linearity and multiplicativity but is positive.

2.3.18 Theorem (Separable modification). *Let $(\mathcal{X}, \mathcal{A}, P)$ be a complete probability space and $\mathcal{F}$ be a class of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$. If $\mathcal{F}$ is separable in $L_r(P)$, then there exists a pointwise separable version of $\mathcal{F}$ relative to the $L_r(P)$ – norm.*

Proof. Define $\tilde{f}$ as indicated, using a consistent lifting, so that the map ρ_n in the preceding proposition is a lifting. Then $f = \tilde{f}$ almost surely, because $\rho\Phi \circ f$ is a representative of the class of $\Phi \circ f$.

Fix n and open sets $U \subset \mathcal{F}$ and $G \subset \mathbb{R}^n$. Define

$$A_f = \left\{ (x_1, \dots, x_n) \in \mathcal{X}^n : (\rho\Phi \circ f(x_1), \dots, \rho\Phi \circ f(x_n)) \in G \right\}.$$

For $D \subset \mathcal{F}$ set $A_D = \bigcup_{f \in D} A_f$. There exists a countable subset $D \subset U$ for which $P^n(A_D)$ is maximal over all countable subsets D of U . Any maximizing D satisfies $1_{A_f} \leq 1_{A_D}$ almost surely for all $f \in U$. By positivity of the lifting ρ_n , it follows that

$$\rho_n 1_{A_f}(x) \leq \rho_n 1_{A_D}(x), \quad \text{for every } x.$$

Set $N_n = \{x \in \mathcal{X}^n : 1_{A_D}(x) = 0, \rho_n 1_{A_D}(x) = 1\}$. By multiplicativity the image of an indicator function under a lifting is an indicator function. It follows that N_n can also be described as the set of x such that $1_{A_D}(x) < \rho_n 1_{A_D}(x)$. Let $(\Phi \circ f)_i$

denote the function $(x_1, \dots, x_n) \mapsto \Phi \circ f(x_i)$. Since G is open, property (vi) of the lifting ρ_n gives

$$\rho_n 1_{A_f} = \rho_n 1_G((\Phi \circ f)_1, \dots, (\Phi \circ f)_n) \geq 1_G(\rho_n(\Phi \circ f)_1, \dots, \rho_n(\Phi \circ f)_n) = 1_{A_f}.$$

Combination of the last two displayed equations yields that $\rho_n 1_{A_D}(x) \geq 1_{A_f}(x)$ for every x . If $x \in A_f$ and $x \notin N_n$, then it follows that $x \in A_D$.

It has been proved that, for every open $U \subset \mathcal{F}$ and open $G \subset \mathbb{R}^n$, there exists a countable set $D_{U,G} \subset U$ and a null set $N_{n,U,G} \subset \mathcal{X}^n$ such that

$$\begin{aligned} x \notin N_{n,U,G} \wedge (\rho\Phi \circ f(x_1), \dots, \rho\Phi \circ f(x_n)) \in G \wedge f \in U \\ \Rightarrow \exists g \in D_{U,G}: (\rho\Phi \circ g(x_1), \dots, \rho\Phi \circ g(x_n)) \in G. \end{aligned}$$

Repeat the construction for every U and G in countable bases for the open sets in $\mathcal{F}$ and $\mathbb{R}^n$, respectively. Let N_n and $\mathcal{G}$ be the union of all null sets $N_{n,U,G}$ and countable $D_{U,G}$. Then for every $x \notin N_n$ and $f \in \mathcal{F}$, there exists a sequence g_m in $\mathcal{G}$ with $(\rho\Phi \circ g_m(x_1), \dots, \rho\Phi \circ g_m(x_n)) \rightarrow (\rho\Phi \circ f(x_1), \dots, \rho\Phi \circ f(x_n))$ and $g_m \rightarrow f$ in $L_r(P)$. Thus the class of functions $\{\rho(\Phi \circ f): f \in \mathcal{F}\}$ is pointwise separable. ■

Problems and Complements

1. There is no universal constant K such that $E|\sum \varepsilon_i X_i| \leq KE|\sum (X_i - EX_i)|$ for any i.i.d. random variables $X_1, \dots, X_n$.

[Hint: Take $n = 1$ and $X_1 \sim N(\mu, 1)$.]

2. For independent random variables $\xi_1, \dots, \xi_n$,

$$P(\max_i |\xi_i| > x) \geq \frac{\sum_i P(|\xi_i| > x)}{1 + \sum_i P(|\xi_i| > x)}.$$

In particular, if the left side is less than $1/2$, then $2P(\max_i |\xi_i| > x) \geq \sum_i P(|\xi_i| > x)$.

[Hint: For $x \geq 0$, one has $1 - x \leq \exp(-x)$ and $1 - e^{-x} \geq x/(1+x)$.]

3. Let $X_{n1}, \dots, X_{nn}$ be an arbitrary array of real random variables.

(i) If $\sup_{x > \varepsilon\sqrt{n}} n^{-1} \sum_{i=1}^n x^2 P(|X_{ni}| > x) \rightarrow 0$ as $n \rightarrow \infty$ for every $\varepsilon > 0$, then $E \max_{1 \leq i \leq n} |X_{ni}| / \sqrt{n} \rightarrow 0$.

(ii) If the array is row-wise i.i.d., then $\max_{1 \leq i \leq n} |X_{ni}| = o_P(\sqrt{n})$ if and only if $P(|X_{ni}| > \varepsilon\sqrt{n}) = o(n^{-1})$ for every $\varepsilon > 0$.

If the variables in the triangular array are i.i.d., then all assertions are equivalent to $P(|X_1| > x) = o(x^{-2})$.

[Hint: $E \max |X_{ni}| \{ |X_{ni}| > \varepsilon\sqrt{n} \}$ is bounded by $\sum \int_0^\infty P(|X_{ni}| 1\{|X_{ni}| > \varepsilon\sqrt{n}\} > t) dt$. This integral splits into two pieces. The integral over the second part $[\varepsilon\sqrt{n}, \infty)$ can be bounded by $\sup_{x > \varepsilon\sqrt{n}} \sum x^2 P(|X_{ni}| > x)$ times $\int_{\varepsilon\sqrt{n}}^\infty x^{-2} dx$.]

4. Let $\xi_1, \dots, \xi_n$ be i.i.d. random variables.
- (i) Show that the following are equivalent.
 - (a) $P(|\xi_1| > x) = o(x^{-1})$.
 - (b) $\max_{1 \leq i \leq n} |\xi_i|/n$ converges to zero in probability.
 - (c) $\max_{1 \leq i \leq n} |\xi_i|^r/n^r$ converges to zero in mean for every $r < 1$.
 - (ii) Show that the following are equivalent:
 - (a) $E|\xi_1| < \infty$.
 - (b) $\max_{1 \leq i \leq n} |\xi_i|/n$ converges to zero almost surely.
 - (c) $\max_{1 \leq i \leq n} |\xi_i|/n$ converges to zero in mean.
 - (iii) Let $r > 1$. Use (i) and (ii) to show the following:
 - (a) Show that $P(|\xi_1| > x) = o(x^{-r})$ if and only if $\max_{1 \leq i \leq n} |\xi_i|/n^{1/r}$ converges to zero in probability if and only if $\max_{1 \leq i \leq n} |\xi_i|/n^{1/r}$ converges to zero in mean.
 - (b) Show that $E|\xi_1|^r < \infty$ if and only if $\max_{1 \leq i \leq n} |\xi_i|^r/n$ converges to zero almost surely if and only if $\max_{1 \leq i \leq n} |\xi_i|^r/n$ converges to zero in mean.
5. If $\{\xi_i\}$ is a sequence of independent, positive random variables with $E|\xi_i|^r < \infty$, for all i and some $r > 0$, then, for $t_\lambda = \inf\{t > 0: \sum_i P(\xi_i > t) \leq \lambda\}$ and any $\lambda > 0$,

$$\frac{\lambda t_\lambda^r}{1 + \lambda} + \frac{1}{1 + \lambda} \sum_i \int_{t_\lambda}^\infty P(\xi_i > t) dt^r \leq E \max_i |\xi_i|^r \leq t_\lambda^r + \sum_i \int_{t_\lambda}^\infty P(\xi_i > t) dt^r.$$

[Hint: Use Problem 2.3.2.]

6. If $X_1, X_2, \dots$ are i.i.d. random variables with $E|X_1| < \infty$, then $E \sup_{n \geq 1} (|X_n|/n)^r < \infty$, for every $r < 1$.

[Hint: Use Problem 2.3.5.]

7. For i.i.d. random variables $X_1, X_2, \dots$, the expectation $E \sup_{n \geq 1} [X_n/n]$ is finite if and only if $E[|X_1|(1 \vee \log |X_1|)]$ is finite.

[Hint: Use Problem 2.3.5.]

8. Any lifting ρ of $L_\infty(\mathcal{X}, \mathcal{A}, P)$ has the following additional properties:
- (i) $\|\rho f\|_\infty \leq \|f\|_\infty$;
 - (ii) $\rho h(f_1, \dots, f_n) = h(\rho f_1, \dots, \rho f_n)$ for every continuous function $h: \mathbb{R}^n \rightarrow \mathbb{R}$ and $f_1, \dots, f_n$ in $L_\infty(\mathcal{X}, \mathcal{A}, P)$;
 - (iii) $\rho h(f_1, \dots, f_n) \geq h(\rho f_1, \dots, \rho f_n)$ for every lower semicontinuous function $h: \mathbb{R}^n \rightarrow \mathbb{R}$ and $f_1, \dots, f_n$ in $L_\infty(\mathcal{X}, \mathcal{A}, P)$;
 - (iv) $\rho 1_A = 1_{\tilde{A}}$ for some measurable set $\tilde{A}$.

[Hint: For polynomials h , property (ii) is a consequence of linearity and multiplicativity of a lifting. By the Stone-Weierstrass theorem, every continuous h can be uniformly approximated by polynomials on any given compact subset of $\mathbb{R}^n$. Thus the general form of (ii) can be reduced to polynomials in view of (i).

Property (iii) is a consequence of (ii), because every lower semicontinuous function can be approximated pointwise from below by a sequence of continuous functions.]

2.4

Glivenko-Cantelli Theorems

In this chapter we prove two types of Glivenko-Cantelli theorems. The first theorem is the simplest and is based on entropy with bracketing. Its proof relies on finite approximation and the law of large numbers for real variables. The second theorem uses random L_1 -entropy numbers and is proved through symmetrization followed by a maximal inequality.

Recall Definition 2.1.6 of the bracketing numbers of a class $\mathcal{F}$ of functions.

2.4.1 Theorem (Glivenko-Cantelli). *Let $\mathcal{F}$ be a class of measurable functions with $N_{[]}(\varepsilon, \mathcal{F}, L_1(P)) < \infty$ for every $\varepsilon > 0$. Then $\mathcal{F}$ is Glivenko-Cantelli.*

Proof. Fix $\varepsilon > 0$. Choose finitely many ε -brackets $[l_i, u_i]$ whose union contains $\mathcal{F}$ and such that $P(u_i - l_i) < \varepsilon$ for every i . Then, for every $f \in \mathcal{F}$, there is a bracket such that

$$(\mathbb{P}_n - P)f \leq (\mathbb{P}_n - P)u_i + P(u_i - f) \leq (\mathbb{P}_n - P)u_i + \varepsilon.$$

Consequently,

$$\sup_{f \in \mathcal{F}} (\mathbb{P}_n - P)f \leq \max_i (\mathbb{P}_n - P)u_i + \varepsilon.$$

The right side converges almost surely to ε by the strong law of large numbers for real variables. Combination with a similar argument for $\inf_{f \in \mathcal{F}} (\mathbb{P}_n - P)f$ yields that $\limsup \|\mathbb{P}_n - P\|_{\mathcal{F}}^* \leq \varepsilon$ almost surely, for every $\varepsilon > 0$. Take a sequence $\varepsilon_m \downarrow 0$ to see that the limsup must actually be zero almost surely. ■

2.4.2 Example (Cells). The previous proof generalizes a well known proof of the Glivenko-Cantelli theorem for the empirical distribution function on the

real line. Indeed, the set of indicator functions of cells $(-\infty, c]$ possesses finite bracketing numbers for any underlying distribution. Indeed, the brackets $[1\{(-\infty, t_i]\}, 1\{(-\infty, t_{i+1}]\})$ for a grid of points $-\infty = t_0 < t_1 < \dots < t_m = \infty$ with the property $P(t_i, t_{i+1}) < \varepsilon$ for each i , give a suitable covering. Bracketing numbers of many other classes of functions are discussed in Chapter 2.7.

Both the statement and the proof of the following theorem are more complicated than the previous bracketing theorem. However, its sufficiency condition for the Glivenko-Cantelli property can be checked for many classes of functions by elegant combinatorial arguments, as discussed in a later chapter. Furthermore, its random entropy condition is necessary.

2.4.3 Theorem (Glivenko-Cantelli). *Let $\mathcal{F}$ be a P -measurable class of measurable functions with envelope F such that $P^*F < \infty$. Let $\mathcal{F}_M$ be the class of functions $f1\{F \leq M\}$ when f ranges over $\mathcal{F}$. Then $\|\mathbb{P}_n - P\|_{\mathcal{F}}^* \rightarrow 0$ almost surely if and only if $n^{-1} \log N(\varepsilon, \mathcal{F}_M, L_1(\mathbb{P}_n)) \rightarrow 0$ in outer probability, for every ε and $M > 0$. In that case both convergences are also in outer mean. In particular, the class $\mathcal{F}$ is Glivenko-Cantelli if $\log N(\varepsilon, \mathcal{F}, L_1(\mathbb{P}_n)) = o_P^*(n)$ for every $\varepsilon > 0$.*

Proof. The last assertion of the theorem is a consequence of the inequality $N(\varepsilon, \mathcal{F}_M, L_r(Q)) \leq N(\varepsilon, \mathcal{F}, L_r(Q))$, which is valid for every $\varepsilon > 0$, every measure Q and every $r > 0$.

We first show the sufficiency of the entropy condition. By the symmetrization Lemma 2.3.1, measurability of the class $\mathcal{F}$, and Fubini's theorem,

$$\begin{aligned} E^* \|\mathbb{P}_n - P\|_{\mathcal{F}} &\leq 2E_X E_\varepsilon \left\| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right\|_{\mathcal{F}} \\ &\leq 2E_X E_\varepsilon \left\| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right\|_{\mathcal{F}_M} + 2P^*F\{F > M\}, \end{aligned}$$

by the triangle inequality, for every $M > 0$. For sufficiently large M , the last term is arbitrarily small. To prove convergence in mean, it suffices to show that the first term converges to zero for fixed M . Fix $X_1, \dots, X_n$. If $\mathcal{G}$ is an ε -net in $L_1(\mathbb{P}_n)$ over $\mathcal{F}_M$, then

$$(2.4.4) \quad E_\varepsilon \left\| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right\|_{\mathcal{F}_M} \leq E_\varepsilon \left\| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right\|_{\mathcal{G}} + \varepsilon.$$

The cardinality of $\mathcal{G}$ can be chosen equal to $N(\varepsilon, \mathcal{F}_M, L_1(\mathbb{P}_n))$. Bound the L_1 -norm on the right by the Orlicz-norm for $\psi_2(x) = \exp(x^2) - 1$, and use the maximal inequality Lemma 2.2.2 to find that the last expression does not exceed a multiple of

$$\sqrt{1 + \log N(\varepsilon, \mathcal{F}_M, L_1(\mathbb{P}_n))} \sup_{f \in \mathcal{G}} \left\| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right\|_{\psi_2|X} + \varepsilon,$$

where the Orlicz norms $\|\cdot\|_{\psi_2|X}$ are taken over $\varepsilon_1, \dots, \varepsilon_n$ with $X_1, \dots, X_n$ fixed. By Hoeffding's inequality, they can be bounded by $\sqrt{6/n} (\mathbb{P}_n f^2)^{1/2}$, which is less than $\sqrt{6/n} M$. Thus the last displayed expression is bounded by

$$\sqrt{1 + \log N(\varepsilon, \mathcal{F}_M, L_1(\mathbb{P}_n))} \sqrt{\frac{6}{n}} M + \varepsilon \xrightarrow{\mathbb{P}_n^*} \varepsilon,$$

if the entropy condition holds, for any $\varepsilon > 0$ and $M > 0$. It has been shown that the left side of (2.4.4) converges to zero in probability. Since it is bounded by M , its expectation with respect to $X_1, \dots, X_n$ converges to zero by the dominated convergence theorem.

This concludes the proof that $\|\mathbb{P}_n - P\|_{\mathcal{F}}^* \rightarrow 0$ in mean. That it also converges almost surely follows from the fact that the sequence $\|\mathbb{P}_n - P\|_{\mathcal{F}}^*$ is a reverse martingale with respect to a suitable filtration. See Lemma 2.4.6 below.

The proof of the converse result relies on the multiplier inequality (with Gaussian multipliers) given by Lemma 2.9.1 together with Sudakov's inequality. We shall actually show the (seemingly) stronger assertion that convergence to zero of $\|\mathbb{P}_n - P\|_{\mathcal{F}}^*$ in probability implies that, for every $\varepsilon > 0$ and $M > 0$,

$$(2.4.5) \quad \mathbb{E}^* \log N(\varepsilon, \mathcal{F}_M, L_2(\mathbb{P}_n)) = o(n).$$

The assumed outer integrability of the envelope function F implies (see Problem 2.3.4) that $\mathbb{E} \max_{k \leq n} F^*(X_k)/n \rightarrow 0$. Combination with the Hoffmann-Jørgensen inequality for moments, Proposition A.1.5, allows to strengthen the convergence in probability of $\|\mathbb{P}_n - P\|_{\mathcal{F}}^*$ to convergence in mean. Next, by the desymmetrization inequality, Lemma 2.3.6,

$$\frac{1}{2} \mathbb{E}^* \left\| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f(X_i) - Pf) \right\|_{\mathcal{F}} \leq \mathbb{E}^* \|\mathbb{P}_n - P\|_{\mathcal{F}} \rightarrow 0.$$

Because $\mathbb{E} \|n^{-1} \sum_{i=1}^n \varepsilon_i Pf\|_{\mathcal{F}} \leq \mathbb{E} \|n^{-1} \sum_{i=1}^n \varepsilon_i\| \|P\|_{\mathcal{F}}$ tends to zero in mean by the law of large numbers, we also obtain convergence to zero of the left side without the term Pf . Next by the multiplier inequalities, Lemma 2.9.1, we see that also $\mathbb{E}^* \|n^{-1} \sum_{i=1}^n \xi_i f(X_i)\|_{\mathcal{F}} \rightarrow 0$ for i.i.d. standard normal multipliers $\xi_1, \dots, \xi_n$ independent from $X_1, \dots, X_n$. (Apply the inequality for fixed n_0 and let first $n \rightarrow \infty$ and next $n_0 \rightarrow \infty$.) Given $X_1, \dots, X_n$ the process $n^{-1/2} \sum_{i=1}^n \xi_i f(X_i)$ is mean-zero Gaussian, with natural pseudo-metric given by the $L_2(\mathbb{P}_n)$ -norm. By Sudakov's inequality, Proposition A.2.6,

$$\frac{1}{\sqrt{n}} \sup_{\varepsilon > 0} \varepsilon \sqrt{\log N(\varepsilon, \mathcal{F}, L_2(\mathbb{P}_n))} \leq 3\mathbb{E}_{\xi} \left\| \frac{1}{n} \sum_{i=1}^n \xi_i f(X_i) \right\|_{\mathcal{F}}.$$

Taking the outer expectation across this inequality, we see that the sequence $n^{-1/2} \sqrt{\log N(\varepsilon, \mathcal{F}, L_2(\mathbb{P}_n))}$ tends to zero in outer expectation, for every $\varepsilon > 0$. Because $N(\varepsilon, \mathcal{F}_M, L_2(\mathbb{P}_n))$ is bounded above both by $N(\varepsilon, \mathcal{F}_M, L_{\infty}(\mathbb{P}_n)) \leq (2M/\varepsilon)^n$ and $N(\varepsilon, \mathcal{F}, L_2(\mathbb{P}_n))$, for every ε and $M > 0$, it follows that

$$\frac{1}{n} \log N(\varepsilon, \mathcal{F}_M, L_2(\mathbb{P}_n)) \leq \sqrt{\frac{\log((2M/\varepsilon)^n)}{n}} \sqrt{\frac{\log N(\varepsilon, \mathcal{F}, L_2(\mathbb{P}_n))}{n}}.$$

The first term is bounded for fixed $\varepsilon, M > 0$, and the second was seen to converge to zero in mean. This concludes the proof of (2.4.5). ■

If $\mathcal{F}$ has a measurable envelope with $PF < \infty$, then $\mathbb{P}_n F = O(1)$ almost surely and the random entropy condition of the preceding theorem is implied by

$$\log N(\varepsilon \|F\|_{\mathbb{P}_{n,1}}, \mathcal{F}, L_1(\mathbb{P}_n)) = o_P^*(n).$$

In Chapter 2.6 it is shown that the entropy in the left side is uniformly bounded by a constant for so-called Vapnik-Červonenkis classes $\mathcal{F}$. Hence any appropriately measurable Vapnik-Červonenkis class is Glivenko-Cantelli, provided its envelope function is integrable.

2.4.6 Lemma. *Let $\mathcal{F}$ be a class of measurable functions with envelope F such that $P^*F < \infty$. Define a filtration by letting $\mathcal{S}_n$ be the σ -field generated by all measurable functions $h: \mathcal{X}^\infty \rightarrow \mathbb{R}$ that are permutation-symmetric in their first n arguments. Then*

$$\mathbb{E}(\|\mathbb{P}_n - P\|_{\mathcal{F}}^* | \mathcal{S}_{n+1}) \geq \|\mathbb{P}_{n+1} - P\|_{\mathcal{F}}^*, \quad \text{a.s.}$$

Furthermore, there exist versions of the measurable cover functions $\|\mathbb{P}_n - P\|_{\mathcal{F}}^$ that are adapted to the filtration. Any such versions form a reverse submartingale and converge almost surely to a constant.*

Proof. Assume without loss of generality that $Pf = 0$ for every f . The function $h: \mathcal{X}^n \rightarrow \mathbb{R}$ given by $h(x_1, \dots, x_n) = \|n^{-1} \sum_{i=1}^n f(x_i)\|_{\mathcal{F}}$ is permutation-symmetric. Its measurable cover h^* (for P^n) can be chosen permutation-symmetric as well (Problem 2.4.4). Then $h^*(X_1, \dots, X_n)$ is a version of $\|\mathbb{P}_n\|_{\mathcal{F}}^*$ and is $\mathcal{S}_n$ measurable.

Let $\mathbb{P}_n^i$ be the empirical measure of $X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_{n+1}$. Then $\mathbb{P}_{n+1}f = (n+1)^{-1} \sum_{i=1}^{n+1} \mathbb{P}_n^i f$ for every f , whence

$$\|\mathbb{P}_{n+1}\|_{\mathcal{F}}^* \leq \frac{1}{n+1} \sum_{i=1}^{n+1} \|\mathbb{P}_n^i\|_{\mathcal{F}}^*, \quad \text{a.s.}$$

This is true for any version of the measurable covers. Choose the left side $\mathcal{S}_{n+1}$ -measurable and take conditional expectations with respect to $\mathcal{S}_{n+1}$ to arrive at

$$\|\mathbb{P}_{n+1}\|_{\mathcal{F}}^* \leq \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbb{E}(\|\mathbb{P}_n^i\|_{\mathcal{F}}^* | \mathcal{S}_{n+1}), \quad \text{a.s..}$$

Each term in the sum on the right side changes on at most a null set if $\|\mathbb{P}_n^i\|_{\mathcal{F}}^*$ is replaced by another version. For h^* given in the first paragraph of the proof, choose the version $h^*(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_{n+1})$ for $\|\mathbb{P}_n^i\|_{\mathcal{F}}^*$. Since the function h^* is permutation-symmetric, it follows by symmetry that the conditional expectations $\mathbb{E}(\|\mathbb{P}_n^i\|_{\mathcal{F}}^* | \mathcal{S}_{n+1})$ do not depend on i (almost surely). Thus the right side of the display equals $\mathbb{E}(\|\mathbb{P}_n^{n+1}\|_{\mathcal{F}}^* | \mathcal{S}_{n+1})$. ■

While bracketing numbers need not be finite for validity of the Glivenko-Cantelli property under a given underlying distribution P , this changes when the property is imposed for every P on the sample space. A class $\mathcal{F}$ of functions that is P -Glivenko-Cantelli for every probability measure P on $(\mathcal{X}, \mathcal{A})$ is called *universally Glivenko-Cantelli*. Under some regularity conditions a uniformly bounded, universal Glivenko-Cantelli class has finite bracketing numbers.

2.4.7 Theorem (Universal Glivenko-Cantelli). *Suppose that the sample space $(\mathcal{X}, \mathcal{A})$ is a Polish space with its Borel σ -field and that the class $\mathcal{F}$ of measurable functions is uniformly bounded and contains a countable subset that is dense for the topology of pointwise convergence on $\mathcal{X}$. Then the following assertions are equivalent:*

- (i) $\mathcal{F}$ is a universal Glivenko-Cantelli class.
 - (ii) $N_{[]}(\varepsilon, \mathcal{F}, L_1(P)) < \infty$, for every $\varepsilon > 0$ and probability measure P on $(\mathcal{X}, \mathcal{A})$.
 - (iii) $N(\varepsilon, \mathcal{F}, L_1(P)) < \infty$, for every $\varepsilon > 0$ and probability measure P on $(\mathcal{X}, \mathcal{A})$.
- Furthermore, under these conditions $\|\mathbb{P}_n - P\|_{\mathcal{F}}$ is a measurable random variable.

Proof. See Van Handel (2013). ■

Problems and Complements

1. **(Necessity of integrability of the envelope)** Suppose that for a class $\mathcal{F}$ of measurable functions the empirical measure of an i.i.d. sample satisfies $\|\mathbb{P}_n - P\|_{\mathcal{F}} \xrightarrow{\text{a.s.}} 0$. Then $P^* \|f - P\|_{\mathcal{F}} < \infty$. Consequently, if $\|P\|_{\mathcal{F}} < \infty$, then $P^* F < \infty$ for an envelope function F .

[**Hint:** Since $(1/n)\|f(X_n) - Pf\|_{\mathcal{F}}$ is bounded above by $\|\mathbb{P}_n - P\|_{\mathcal{F}} + (1 - 1/n)\|\mathbb{P}_{n-1} - P\|_{\mathcal{F}}$, one has that $P(\|f(X_n) - Pf\|_{\mathcal{F}} \geq n, \text{i.o.}) = 0$. By the Borel-Cantelli lemma, it follows that the series $\sum P(\|f(X_n) - Pf\|_{\mathcal{F}} \geq n)$ converges. This series is an upper bound for the given first moment, because the X_n are identically distributed.]

2. The $L_r(Q)$ -entropy numbers of the class $\mathcal{F}_M = \{f1\{F \leq M\} : f \in \mathcal{F}\}$ are smaller than those of $\mathcal{F}$ for any probability measure Q and for numbers $M > 0$ and $r \geq 1$.
3. **(Stability of the Glivenko-Cantelli property)** If $\mathcal{F}$, $\mathcal{F}_1$, and $\mathcal{F}_2$ are Glivenko-Cantelli classes of functions, then
 - (i) $\{a_1 f_1 + a_2 f_2 : f_i \in \mathcal{F}_i, |a_i| \leq 1\}$ is Glivenko-Cantelli;
 - (ii) $\mathcal{F}_1 \cup \mathcal{F}_2$ is Glivenko-Cantelli;
 - (iii) the class of all functions that are both the pointwise limit and the $L_1(P)$ -limit of a sequence in $\mathcal{F}$ is Glivenko-Cantelli.

4. Let $h: \mathcal{X}^n \rightarrow \mathbb{R}$ be permutation-symmetric. Then there exists a version h^* of the measurable cover for P^n on $\mathcal{A}^n$ which is permutation-symmetric.

[**Hint:** Take an arbitrary measurable cover $\tilde{h} \geq h$, and set $h^* = \min_{\sigma} \tilde{h} \circ \sigma$.]

5. If the underlying probability space is not the product space $(\mathcal{X}^\infty, \mathcal{A}^\infty, P^\infty)$, then $\|\mathbb{P}_n - P\|_{\mathcal{F}}$ may fail to be a reverse martingale, even if $\|\mathbb{P}_n - P\|_{\mathcal{F}}$ is measurable for each n .

[Hint: There exists a (nonmeasurable) set $A \subset [0, 1]$ with inner and outer Lebesgue measures $\lambda_*(A) = 0$ and $\lambda^*(A) = 1$, respectively. Take $(\mathcal{X}_n, \mathcal{A}_n, P_n)$ equal to $(A, \mathcal{B} \cap A, \lambda_A)$ for odd values of n and $(A^c, \mathcal{B} \cap A^c, \lambda_{A^c})$ for even values of n , where λ_A is the trace measure defined in Problem 1.2.16. Define X_n to be the embedding of $\mathcal{X}_n$ into the unit interval, viewed as a map on the infinite product $\prod \mathcal{X}_i$, and let $\mathcal{F}$ be equal to the set of indicators of all finite subsets of A^c . Then $\|\mathbb{P}_n - P\|_{\mathcal{F}}$ equals $(n-1)/2n$, for odd values of n , and $1/2$ for n even.]

6. If the filtration $\mathcal{S}_n$ in Lemma 2.4.6 is replaced by the filtration consisting of the σ -fields generated by the variables $\mathbb{P}_k f$ for $k \geq n$ and f ranging over $\mathcal{F}$, then $\|\mathbb{P}_n - P\|_{\mathcal{F}}^*$ may fail to be a reverse submartingale.

[Hint: Let $(\mathcal{X}, \mathcal{A}, P)$ be the unit interval in $\mathbb{R}$ with Borel sets and Lebesgue measure and $X_1, X_2, \dots$ the coordinate projections of the infinite product space, as usual. For the collection $\mathcal{F} = \{1_x : x \in [1/2, 1]\}$, the variable $\|\mathbb{P}_n - P\|_{\mathcal{F}}$ is measurable with respect to $\mathcal{A}^\infty$, but not $\mathcal{A}_n$ -measurable for any n . It is also not measurable with respect to the P^∞ -completion of $\mathcal{A}_n$ even though $\mathcal{F}$ is image-admissible Suslin.]

7. The σ -field $\mathcal{S}_n$ in Lemma 2.4.6 equals the σ -field generated by the variables $\mathbb{P}_k f$ with k ranging over all integers greater than or equal to n and f ranging over the set of all measurable, integrable $f: \mathcal{X} \rightarrow \mathbb{R}$.

8. The sequence of averages $\bar{Y}_n$ of the first n elements of a sequence of i.i.d. random variables $Y_1, Y_2, \dots$ with finite first absolute moment is a reverse martingale with respect to its natural filtration: $E(\bar{Y}_n | \bar{Y}_{n+1}, \bar{Y}_{n+2}, \dots) = \bar{Y}_{n+1}$.

[Hint: The conditional expectation equals $n^{-1} \sum_{i=1}^n E(Y_i | \bar{Y}_{n+1}, Y_{n+2}, \dots)$, and $E(Y_i | \bar{Y}_{n+1})$ is constant in $1 \leq i \leq n+1$ by symmetry.]

9. **(Block-bracketing)** Given a class $\mathcal{F}$ of functions $f: \mathcal{X} \rightarrow \mathbb{R}$ with integrable envelope, define $\mathcal{F}^{\oplus m}$ to be the set of functions $(x_1, \dots, x_m) \mapsto \sum_{i=1}^m f(x_i)$ on $\mathcal{X}^m$. Suppose that for every $\varepsilon > 0$ there exists m such that $N_{[]}(\varepsilon m, \mathcal{F}^{\oplus m}, L_1(P^m)) < \infty$. Then $\mathcal{F}$ is Glivenko-Cantelli.

[Hint: For every $\varepsilon > 0$ and f , there exists m and a bracket $[l_m, u_m]$ in $L_1(P^m)$ such that $m^{-1} P^m u_m - m^{-1} P^m l_m < \varepsilon$ and

$$\frac{1}{nm} \sum_{i=1}^n l_m(Y_{i,m}) \leq \mathbb{P}_{nm} f \leq \frac{1}{nm} \sum_{i=1}^n u_m(Y_{i,m}),$$

$$\frac{1}{m} P^m l_m \leq P f \leq \frac{1}{m} P^m u_m,$$

where $Y_{1,m}, Y_{2,m}, \dots$ are the blocks $[X_1, \dots, X_m], [X_{m+1}, \dots, X_{2m}], \dots$. Conclude that for every $\varepsilon > 0$ there exists m such that $\limsup_{n \rightarrow \infty} \|\mathbb{P}_{nm} - P\|_{\mathcal{F}}^* < \varepsilon$. It suffices to “close the gaps.”]

10. Theorem 2.4.3 remains valid if the $L_1(\mathbb{P}_n)$ -norm is replaced by the $L_p(\mathbb{P}_n)$ -norm, for any $0 < p < \infty$.

[Hint: For $0 < r < s < \infty$ and $f, g \in \mathcal{F}_M$ we have $\|f - g\|_{\mathbb{P}_{n,s}}^{s/r} \leq (2M)^{s/r-1} \|f - g\|_{\mathbb{P}_{n,r}} \leq \|f - g\|_{\mathbb{P}_{n,s}}$. Use this to compare the entropy numbers $N(\varepsilon, \mathcal{F}_M, L_r(\mathbb{P}_n))$.]

11. A P -Glivenko Cantelli class $\mathcal{F}$ with $\|P\|_{\mathcal{F}} < \infty$ is totally bounded in $L_1(P)$. If $\mathcal{F}$ has an envelope that is contained in $L_r(P)$, then $\mathcal{F}$ is also totally bounded in $L_r(P)$, for every $r \in (1, \infty)$.

[Hint: By Problem 2.4.1 $\mathcal{F}$ possesses an integrable envelope. It is no loss of generality to assume that this is finite everywhere. The class $\mathcal{F} - \mathcal{F}$ is Glivenko-Cantelli. By a result of Dudley (1998) (and also by Theorem 2.10.5, but this uses the present problem to resolve measurability issues), the classes $(\mathcal{F} - \mathcal{F})^+$ and $(\mathcal{F} - \mathcal{F})^-$ are Glivenko-Cantelli, whence the class $|\mathcal{F} - \mathcal{F}|$ of absolute differences is Glivenko-Cantelli. Thus there exists a sequence of finitely discrete probability measures P_n such that $L_n := \sup_{f, g \in \mathcal{F}} |(P_n - P)(f - g)| \rightarrow 0$. For given $\varepsilon > 0$ there exists n_0 such that $L_{n_0} < \varepsilon$. Because restricted to the support of P_{n_0} the functions $f \in \mathcal{F}$ are equivalent to a cube in $\mathbb{R}^{n_0}$, there exists a finite ε -net $f_1, \dots, f_N$ over $\mathcal{F}$ relative to the $L_1(P_{n_0})$ -norm. This is a 2ε -net over $\mathcal{F}$ relative to $L_1(P)$ -norm.]

2.5

Donsker Theorems

In this chapter we present the two main empirical central limit theorems. The first is based on uniform entropy, and its proof relies on symmetrization. The second is based on bracketing entropy.

2.5.1 Uniform Entropy

In this section weak convergence of the empirical process will be established under the condition that the envelope function F be square integrable, combined with the uniform entropy bound

$$(2.5.1) \quad \int_0^\infty \sup_Q \sqrt{\log N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q))} d\varepsilon < \infty.$$

Here the supremum is taken over all finitely discrete probability measures Q on $(\mathcal{X}, \mathcal{A})$ with $\|F\|_{Q,2}^2 = \int F^2 dQ > 0$. These conditions are by no means necessary, but they suffice for many examples. Finiteness of the previous integral will be referred to as the *uniform entropy condition*.

2.5.2 Theorem (Donsker). *Let $\mathcal{F}$ be a class of measurable functions that satisfies the uniform entropy bound (2.5.1). Let the classes $\mathcal{F}_\delta = \{f - g: f, g \in \mathcal{F}, \|f - g\|_{P,2} < \delta\}$ and $\mathcal{F}_\infty^2$ be P -measurable for every $\delta > 0$. If $P^*F^2 < \infty$, then $\mathcal{F}$ is P -Donsker.*

Proof. Let $\delta_n \downarrow 0$ be arbitrary. By Markov's inequality and the symmetrization Lemma 2.3.1,

$$P^*(\|\mathbb{G}_n\|_{\mathcal{F}_{\delta_n}} > x) \leq \frac{2}{x} E^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i f(X_i) \right\|_{\mathcal{F}_{\delta_n}}.$$

Since the supremum in the right-hand side is measurable by assumption, Fubini's theorem applies and the outer expectation can be calculated as $E_X E_\varepsilon$. Fix $X_1, \dots, X_n$. By Hoeffding's inequality, Lemma 2.2.8, the stochastic process $f \mapsto n^{-1/2} \sum_{i=1}^n \varepsilon_i f(X_i)$ is sub-Gaussian for the $L_2(\mathbb{P}_n)$ -seminorm

$$\|f\|_n = \sqrt{\frac{1}{n} \sum_{i=1}^n f^2(X_i)}.$$

Use the second part of the maximal inequality Corollary 2.2.9 to find that

$$(2.5.3) \quad E_\varepsilon \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i f(X_i) \right\|_{\mathcal{F}_{\delta_n}} \lesssim \int_0^\infty \sqrt{\log N(\varepsilon, \mathcal{F}_{\delta_n}, L_2(\mathbb{P}_n))} d\varepsilon.$$

For large values of ε the set $\mathcal{F}_{\delta_n}$ fits in a single ball of radius ε around the origin, in which case the integrand is zero. This is certainly the case for values of ε larger than θ_n , where

$$\theta_n^2 = \sup_{f \in \mathcal{F}_{\delta_n}} \|f\|_n^2 = \left\| \frac{1}{n} \sum_{i=1}^n f^2(X_i) \right\|_{\mathcal{F}_{\delta_n}}.$$

Furthermore, covering numbers of the class $\mathcal{F}_\delta$ are bounded by covering numbers of $\mathcal{F}_\infty = \{f - g : g \in \mathcal{F}\}$. The latter satisfy $N(\varepsilon, \mathcal{F}_\infty, L_2(Q)) \leq N^2(\varepsilon/2, \mathcal{F}, L_2(Q))$ for every measure Q .

Limit the integral in (2.5.3) to the interval $(0, \theta_n)$, make a change of variables, and bound the integrand to obtain the bound

$$\int_0^{\theta_n / \|F\|_n} \sup_Q \sqrt{\log N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q))} d\varepsilon \|F\|_n.$$

Here the supremum is taken over all discrete probability measures. The integrand is integrable by assumption. Furthermore, $\|F\|_n$ is bounded below by $\|F_*\|_n$, which converges almost surely to its expectation, which may be assumed positive. Use the Cauchy-Schwarz inequality and the dominated convergence theorem to see that the expectation (with respect to $X_1, \dots, X_n$) of this integral converges to zero provided $\theta_n \xrightarrow{P^*} 0$. This would conclude the proof of asymptotic equicontinuity.

Since $\sup\{Pf^2 : f \in \mathcal{F}_{\delta_n}\} \rightarrow 0$ and $\mathcal{F}_{\delta_n} \subset \mathcal{F}_\infty$, it is certainly enough to prove that

$$\|\mathbb{P}_n f^2 - P f^2\|_{\mathcal{F}_\infty} \xrightarrow{P^*} 0.$$

This is a uniform law of large numbers for the class $\mathcal{F}_\infty^2$. This class has integrable envelope $(2F)^2$ and is measurable by assumption. For any pair f, g of functions in $\mathcal{F}_\infty$,

$$\mathbb{P}_n |f^2 - g^2| \leq \mathbb{P}_n |f - g| 4F \leq \|f - g\|_n \|4F\|_n.$$

It follows that the covering number $N(\varepsilon \|2F\|_n^2, \mathcal{F}_\infty^2, L_1(\mathbb{P}_n))$ is bounded by the covering number $N(\varepsilon \|F\|_n, \mathcal{F}_\infty, L_2(\mathbb{P}_n))$. By assumption, the latter number is bounded by a fixed number, so its logarithm is certainly $o_P^*(n)$, as required for the uniform law of large numbers, Theorem 2.4.3. This concludes the proof of asymptotic equicontinuity.

Finally, we show that $\mathcal{F}$ is totally bounded in $L_2(P)$. By the result of the last paragraph, there exists a sequence of discrete measures P_n with $\|(P_n - P)f^2\|_{\mathcal{F}_\infty}$ converging to zero. Take n sufficiently large so that the latter supremum is bounded by ε^2 . By assumption, $N(\varepsilon, \mathcal{F}, L_2(P_n))$ is finite. Any ε -net for $\mathcal{F}$ in $L_2(P_n)$ is a $\sqrt{2}\varepsilon$ -net in $L_2(P)$. ■

2.5.4 Example (Cells). The set $\mathcal{F}$ of all indicator functions $1\{(-\infty, t]\}$ of cells in $\mathbb{R}$ satisfies

$$N(\varepsilon, \mathcal{F}, L_2(Q)) \leq N_{[]}(\varepsilon^2, \mathcal{F}, L_1(Q)) \leq \frac{2}{\varepsilon^2},$$

for any probability measure Q and $\varepsilon \leq 1$. Since $\int_0^1 \log(1/\varepsilon) d\varepsilon < \infty$, the class of cells in $\mathbb{R}$ is Donsker. The covering numbers of the class of cells $(-\infty, t]$ in higher dimensions satisfy a similar bound, but with a higher power of $(1/\varepsilon)$. Thus the class of all cells $(-\infty, t]$ in $\mathbb{R}^k$ is Donsker for any dimension.

In the next chapter these classes of sets are shown to be examples of the large collection of VC-classes, which are all Donsker provided they are suitably measurable.

2.5.2 Bracketing

The second main empirical central limit theorem uses bracketing entropy rather than uniform entropy. The simplest version of this theorem uses $L_2(P)$ -brackets, and requires finiteness of the corresponding *bracketing integral*

$$(2.5.5) \quad \int_0^\infty \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L_2(P))} d\varepsilon < \infty.$$

2.5.6 Theorem (Donsker). Any class $\mathcal{F}$ of measurable functions satisfying (2.5.5) is P -Donsker.

Unlike the uniform entropy condition, the bracketing integral involves only the true underlying measure P . However, part of this gain is offset by the fact that bracketing numbers can be larger than covering numbers. As a result, the two sufficient conditions for a class to be Donsker are not comparable. Examples of classes of functions that satisfy the bracketing condition are given in Chapter 2.7.

We shall prove a slightly stronger version of the theorem. From a given finite set of $L_2(P)$ -brackets it is not difficult to construct an envelope function F that is square-integrable, and hence the existence of such an envelope is implicit in the assumption of the preceding theorem. It is known that the latter is not necessary for a class $\mathcal{F}$ to be Donsker, though the envelope must possess a weak second moment (meaning that $P^*(F > x) = o(x^{-2})$ as $x \rightarrow \infty$). Similarly, the $L_2(P)$ -norm used to measure the size of the brackets can be replaced by a weaker norm, which makes the bracketing numbers smaller and the convergence of the integral easier. The result below measures the size of the brackets by their $L_{2,\infty}$ -norm given by

$$\|f\|_{P,2,\infty} = \sup_{x>0} \left(x^2 P(|f| > x) \right)^{1/2}.$$

(Actually this is not really a norm, because it does not satisfy the triangle inequality. It can be shown that there exists a norm that is equivalent to this “norm” up to a constant 2, but this is irrelevant for the present purposes.) Note that $\|f\|_{P,2,\infty} \leq \|f\|_{P,2}$, so that the bracketing numbers relative to $L_{2,\infty}(P)$ are smaller.

The proofs of the preceding theorem and its strengthening below are based on a chaining argument. Unlike in the proof of the uniform entropy theorem, this will be applied to the original summands, without an initial symmetrization. Because the summands are not appropriately bounded, Hoeffding’s inequality and the sub-Gaussian maximal inequality do not apply. Instead, Bernstein’s inequality is used in the form

$$P(|\mathbb{G}_n f| > x) \leq 2e^{-\frac{1}{2} \frac{x^2}{Pf^2 + 1/3\|f\|_\infty x/\sqrt{n}}}.$$

This is valid for every square integrable, uniformly bounded function f . Therefore, Lemma 2.2.13 implies that for a finite set $\mathcal{F}$ of cardinality $|\mathcal{F}| \geq 2$,

$$(2.5.7) \quad E\|\mathbb{G}_n\|_{\mathcal{F}} \lesssim \max_f \frac{\|f\|_\infty}{\sqrt{n}} \log |\mathcal{F}| + \max_f \|f\|_{P,2} \sqrt{\log |\mathcal{F}|}.$$

The chaining argument is set up in such a way that the two terms on the right are of comparable magnitude. This is the case if $\|f\|_\infty \asymp \sqrt{n}\|f\|_{P,2}/\sqrt{\log |\mathcal{F}|}$; the inequality is applied after truncating the functions f at this level.

2.5.8 Theorem (Donsker). *Let $\mathcal{F}$ be a class of measurable functions such that*

$$\int_0^\infty \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L_{2,\infty}(P))} d\varepsilon + \int_0^\infty \sqrt{\log N(\varepsilon, \mathcal{F}, L_2(P))} d\varepsilon < \infty.$$

Moreover, assume that the envelope function F of $\mathcal{F}$ possesses a weak second moment. Then $\mathcal{F}$ is P -Donsker.

Proofs. It suffices to prove the second theorem. For each natural number q , there exists a partition $\mathcal{F} = \cup_{i=1}^{N_q} \mathcal{F}_{qi}$ of $\mathcal{F}$ into N_q disjoint subsets such that

$$\begin{aligned} \sum 2^{-q} \sqrt{\log N_q} &< \infty, \\ \left\| \left(\sup_{f,g \in \mathcal{F}_{qi}} |f - g| \right)^* \right\|_{P,2,\infty} &< 2^{-q}, \\ \sup_{f,g \in \mathcal{F}_{qi}} \|f - g\|_{P,2} &< 2^{-q}. \end{aligned}$$

To see this, first cover $\mathcal{F}$ separately with minimal numbers of $L_2(P)$ -balls and $L_{2,\infty}(P)$ -brackets of size 2^{-q} , disjointify, and take the intersection of the two partitions. The total number of sets will be $N_q = N_q^1 N_q^2$ if N_q^i are the number of sets in the two separate partitions. The logarithm turns the product into a sum, and the first condition is satisfied if it is satisfied for both N_q^i .

The sequence of partitions can, without loss of generality, be chosen as successive refinements. Indeed, first construct a sequence of partitions $\mathcal{F} = \cup_{i=1}^{N_q} \tilde{\mathcal{F}}_{qi}$ possibly without this property. Next take the partition at stage q to consist of all intersections of the form $\cap_{p=1}^q \tilde{\mathcal{F}}_{p,i_p}$. This gives partitions into $N_q = \tilde{N}_1 \cdots \tilde{N}_q$ sets.

Using the inequality $(\log \prod \bar{N}_p)^{1/2} \leq \sum (\log \bar{N}_p)^{1/2}$ and rearranging sums, it is seen that the first of the three displayed conditions is still satisfied.

Choose for each q a fixed element f_{qi} from each partitioning set $\mathcal{F}_{qi}$, and set

$$\begin{aligned}\pi_q f &= f_{qi} && \text{if } f \in \mathcal{F}_{qi}, \\ \Delta_q f &= \sup_{f, g \in \mathcal{F}_{qi}} |f - g|^*, && \text{if } f \in \mathcal{F}_{qi}.\end{aligned}$$

Note that $\pi_q f$ and $\Delta_q f$ run through a set of N_q functions if f runs through $\mathcal{F}$. In view of Theorem 1.5.6, it suffices to show that the sequence $\|\mathbb{G}_n(f - \pi_{q_0} f)\|_{\mathcal{F}}$ converges in probability to zero as $n \rightarrow \infty$ followed by $q_0 \rightarrow \infty$.

Define for each fixed n and (large) $q \geq q_0$ numbers and indicator functions;

$$\begin{aligned}a_q &= 2^{-q} / \sqrt{\log N_{q+1}}, \\ A_{q-1} f &= 1 \{ \Delta_{q_0} f \leq \sqrt{n} a_{q_0}, \dots, \Delta_{q-1} f \leq \sqrt{n} a_{q-1} \}, \\ B_q f &= 1 \{ \Delta_{q_0} f \leq \sqrt{n} a_{q_0}, \dots, \Delta_{q-1} f \leq \sqrt{n} a_{q-1}, \Delta_q f > \sqrt{n} a_q \}, \\ B_{q_0} f &= 1 \{ \Delta_{q_0} f > \sqrt{n} a_{q_0} \}.\end{aligned}$$

Note that $A_q f$ and $B_q f$ are constant in f on each of the partitioning sets $\mathcal{F}_{qi}$ at level q , because the partitions are nested. Now decompose, pointwise in x (which is suppressed in the notation),

$$f - \pi_{q_0} f = (f - \pi_{q_0} f) B_{q_0} f + \sum_{q_0+1}^{\infty} (f - \pi_q f) B_q f + \sum_{q_0+1}^{\infty} (\pi_q f - \pi_{q-1} f) A_{q-1} f.$$

The idea here is to write the left side as the sum of $f - \pi_{q_1} f$ and $\sum_{q_0+1}^{q_1} (\pi_q f - \pi_{q-1} f)$ for the largest $q_1 = q_1(f, x)$ such that each of the “links” $\pi_q f - \pi_{q-1} f$ in the “chain” is bounded in absolute value by $\sqrt{n} a_q$ (note that $|\pi_q f - \pi_{q-1} f| \leq \Delta_{q-1} f$). For a rigorous derivation, note that either all $B_q f$ are zero or there is a unique q_1 with $B_{q_1} f = 1$. In the first case, the first two terms in the decomposition are zero and the third term is an infinite series (all $A_q f$ equal 1) whose q th partial sum telescopes out to $\pi_q f - \pi_{q_0} f$ and converges to $f - \pi_{q_0} f$ by the definition of the $A_q f$. In the second case, $A_{q-1} f = 1$ if and only if $q \leq q_1$ and the decomposition is as mentioned, apart from the separate treatment of the case that $q_1 = q_0$, when already the first link fails the test.

Next apply the empirical process $\mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - P)$ to each of the three terms separately, and take suprema over $f \in \mathcal{F}$. It will be shown that the resulting three variables converge to zero in probability as $n \rightarrow \infty$ followed by $q_0 \rightarrow \infty$.

First, since $|f - \pi_{q_0} f| B_{q_0} f \leq 2F 1\{2F > \sqrt{n} a_{q_0}\}$, one has

$$E^* \|\mathbb{G}_n(f - \pi_{q_0} f) B_{q_0} f\|_{\mathcal{F}} \leq 4\sqrt{n} P^* F \{2F > \sqrt{n} a_{q_0}\}.$$

The right side converges to zero as $n \rightarrow \infty$, for each fixed q_0 , by the assumption that F has a weak second moment (Problem 2.5.6).

Second, since the partitions are nested, $\Delta_q f B_q f \leq \Delta_{q-1} f B_q f$, whence by the inequality of Problem 2.5.5,

$$\sqrt{n} a_q P \Delta_q f B_q f \leq 2 \|\Delta_q f\|_{P, 2, \infty}^2 \leq 2 \cdot 2^{-2q}.$$

Since $\Delta_{q-1}f B_q f$ is bounded by $\sqrt{n}a_{q-1}$ for $q > q_0$, we obtain

$$P(\Delta_q f B_q f)^2 \leq \sqrt{n}a_{q-1} P\Delta_q f \{\Delta_q f > \sqrt{n}a_q\} \leq 2 \frac{a_{q-1}}{a_q} 2^{-2q}.$$

Apply the triangle inequality and inequality (2.5.7) to find

$$\begin{aligned} E^* \left\| \sum_{q_0+1}^{\infty} \mathbb{G}_n(f - \pi_q f) B_q f \right\|_{\mathcal{F}} &\leq \sum_{q_0+1}^{\infty} E^* \left\| \mathbb{G}_n \Delta_q f B_q f \right\|_{\mathcal{F}} + \sum_{q_0+1}^{\infty} 2\sqrt{n} \left\| P\Delta_q f B_q f \right\|_{\mathcal{F}} \\ &\lesssim \sum_{q_0+1}^{\infty} \left[a_{q-1} \log N_q + \sqrt{\frac{a_{q-1}}{a_q}} 2^{-q} \sqrt{\log N_q} + \frac{4}{a_q} 2^{-2q} \right]. \end{aligned}$$

Since a_q is decreasing, the quotient a_{q-1}/a_q can be replaced by its square. Then in view of the definition of a_q , the series on the right can be bounded by a multiple of $\sum_{q_0+1}^{\infty} 2^{-q} \sqrt{\log N_q}$. This upper bound is independent of n and converges to zero as $q_0 \rightarrow \infty$.

Third, there are at most N_q functions $\pi_q f - \pi_{q-1} f$ and at most N_{q-1} functions $A_{q-1} f$. Since the partitions are nested, the function $|\pi_q f - \pi_{q-1} f| A_{q-1} f$ is bounded by $\Delta_{q-1} f A_{q-1} f \leq \sqrt{n} a_{q-1}$. The $L_2(P)$ -norm of $|\pi_q f - \pi_{q-1} f|$ is bounded by 2^{-q+1} . Apply inequality (2.5.7) to find

$$E^* \left\| \sum_{q_0+1}^{\infty} \mathbb{G}_n(\pi_q f - \pi_{q-1} f) A_{q-1} f \right\|_{\mathcal{F}} \lesssim \sum_{q_0+1}^{\infty} \left[a_{q-1} \log N_q + 2^{-q} \sqrt{\log N_q} \right].$$

Again this upper bound is independent of n and converges to zero as $q_0 \rightarrow \infty$. ■

2.5.9 Example (Cells). The classical empirical central limit theorem was shown to be a special case of the uniform entropy central limit theorem in the preceding subsection. This classical theorem is a special case of the bracketing central limit theorem as well. This follows since $N_{[]}(\sqrt{2}\varepsilon, \mathcal{F}, L_2(P)) \leq N_{[]}(\varepsilon^2, \mathcal{F}, L_1(P))$ for every class of functions $f: \mathcal{X} \rightarrow [0, 1]$. The L_1 -bracketing numbers were bounded above by a power of $1/\varepsilon$ in the preceding subsection.

Problems and Complements

- (Taking the supremum over all Q)** The supremum in (2.5.1) can be replaced by the supremum over all probability measures Q , such that $0 < QF^2 < \infty$, without changing the condition. In fact, for any probability measure P and $r > 0$ such that $0 < PF^r < \infty$,

$$D(2\varepsilon \|F\|_{P,r}, \mathcal{F}, L_r(P)) \leq \sup_Q D(\varepsilon \|F\|_{Q,r}, \mathcal{F}, L_r(Q))$$

if the supremum on the right is taken over all discrete probability measures Q .

[Hint: If the left side is equal to m , then there exist functions $f_1, \dots, f_m$ such that $P|f_i - f_j|^r > 2^r \varepsilon^r PF^r$ for $i \neq j$. By the strong law of large numbers, $\mathbb{P}_n|f_i - f_j|^r \rightarrow P|f_i - f_j|^r$ almost surely. Also $\mathbb{P}_n F^r \rightarrow PF^r$ almost surely. Thus there exist n and ω such that $\mathbb{P}_n(\omega)|f_i - f_j|^r > 2^r \varepsilon^r PF^r$ and $\mathbb{P}_n(\omega)F^r < 2^r PF^r$.]

2. Can the constant 2 in the preceding problem be replaced by 1? Is a similar inequality valid for covering numbers, and what is the best constant?
3. The presence of the factor $\|F\|_{Q,2}$ in the uniform entropy condition (2.5.1) is helpful if it is larger than 1 but is detrimental otherwise. However, given any class $\mathcal{F}$ with a square integrable envelope, there is always a square integrable envelope F with $\|F\|_{Q,2} \geq 1$ for all Q .
4. If $\mathcal{F}$ is compact, then the function $\varepsilon \mapsto N(\varepsilon, \mathcal{F}, L_2(P))$ is left continuous.
[Hint: The function $f \mapsto \inf_i \|f - f_i\|$ attains its maximum for every finite set $f_1, \dots, f_p$.]
5. For each positive random variable X , one has the inequalities $\|X\|_{2,\infty}^2 \leq \sup_{t>0} t \mathbb{E}X\{X > t\} \leq 2\|X\|_{2,\infty}^2$.
[Hint: The first inequality follows from Markov's inequality. For the second, write the expectation in the expression in the middle as $\int_0^\infty P(X1\{X > t\} > x) dx$. Next, split the integral in the part from zero to t and its complement. On the first part, the integrand is constant; on the second, it can be bounded by x^{-2} times the $L_{2,\infty}$ -norm.]
6. Each random variable with $P(|X| > t) = o(t^{-2})$ as $t \rightarrow \infty$ (a “weak second moment”) satisfies $\mathbb{E}|X|\{ |X| > t \} = o(t^{-1})$ as $t \rightarrow \infty$.
7. For any random variable X , one has $\|X\|_1 \leq 3\|X\|_{2,\infty}$.

2.6

Uniform Entropy Numbers

In Section 2.5.1 the empirical process was shown to converge weakly for indexing sets $\mathcal{F}$ satisfying a uniform entropy condition. In particular, if

$$\sup_Q \log N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q)) \leq K \left(\frac{1}{\varepsilon} \right)^{2-\delta},$$

for some $\delta > 0$, then the entropy integral (2.5.1) converges and $\mathcal{F}$ is a Donsker class for any probability measure P such that $P^*F^2 < \infty$, provided measurability conditions are met. Many classes of functions satisfy this condition and often even the much stronger condition

$$\sup_Q N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q)) \leq K \left(\frac{1}{\varepsilon} \right)^V, \quad 0 < \varepsilon < 1,$$

for some number V . In this chapter this is shown for classes satisfying certain combinatorial conditions. For classes of sets, these were first studied by Vapnik and Červonenkis, whence the name VC-classes. In the second part of this chapter, VC-classes of functions are defined in terms of VC-classes of sets. The remainder of this chapter considers operations on classes that preserve entropy properties, such as taking convex hulls.

2.6.1 VC-Classes of Sets

Let $\mathcal{C}$ be a collection of subsets of a set $\mathcal{X}$. An arbitrary set of n points $\{x_1, \dots, x_n\}$ possesses 2^n subsets. Say that $\mathcal{C}$ *picks out* a certain subset from $\{x_1, \dots, x_n\}$ if this can be formed as a set of the form $C \cap \{x_1, \dots, x_n\}$ for a C in $\mathcal{C}$. The collection $\mathcal{C}$ is said to *shatter* $\{x_1, \dots, x_n\}$ if each of its 2^n subsets can be picked out in this manner. The *VC-dimension* $V(\mathcal{C})$ of the class $\mathcal{C}$ is the largest n such that some set

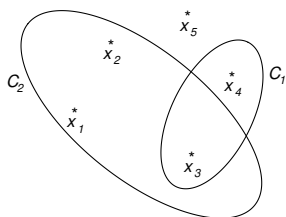


Figure 2.6.1. The sets C_1 and C_2 pick out the sets of points $\{x_3, x_4\}$ and $\{x_1, x_2, x_3\}$, respectively.

of size n is shattered by $\mathcal{C}$.^b Clearly, the more refined $\mathcal{C}$ is, the larger is its index. The index is more formally defined through

$$\Delta_n(\mathcal{C}, x_1, \dots, x_n) = \#\{C \cap \{x_1, \dots, x_n\} : C \in \mathcal{C}\},$$

$$V(\mathcal{C}) = \sup\left\{n : \max_{x_1, \dots, x_n} \Delta_n(\mathcal{C}, x_1, \dots, x_n) = 2^n\right\}.$$

Thus the dimension is infinite if $\mathcal{C}$ shatters sets of arbitrarily large size.

A collection of measurable sets $\mathcal{C}$ is called a *VC-class* if its dimension is finite. The main result of this section is the remarkable fact that the covering numbers of any VC-class grow polynomially in $1/\varepsilon$ as $\varepsilon \rightarrow 0$, of order dependent on the dimension of the class. Since the envelope of a class of sets is certainly square integrable, it follows that a VC-class of sets is Donsker for any underlying probability measure, provided measurability conditions are met.

2.6.1 Example (Cells). The collection of all cells of the form $(-\infty, c]$ in $\mathbb{R}$ shatters no two-point set $\{x_1, x_2\}$, because it fails to pick out the largest of the two points. Hence its VC-dimension equals 1. The collection of all cells $(a, b]$ in $\mathbb{R}$ shatters every two-point set but cannot pick out the subset consisting of the smallest and largest points of any set of three points. Thus its VC-dimension equals 2. With more effort, it can be seen that the indices of the same type of sets in $\mathbb{R}^d$ are d and $2d$, respectively.

Techniques to show that many other collections of sets are VC are discussed in Sections 2.6.5 and 2.6.6.

The following combinatorial result is needed: the number of subsets shattered by a class $\mathcal{C}$ is at least the number of subsets picked out by $\mathcal{C}$. In particular, if the class $\mathcal{C}$ shatters no set of $V + 1$ points, then the number of subsets picked out by $\mathcal{C}$ is at most $\sum_{j=0}^V \binom{n}{j}$, the number of subsets of size $\leq V$.

2.6.2 Lemma. *Let $\{x_1, \dots, x_n\}$ be arbitrary points. Then the total number of subsets $\Delta_n(\mathcal{C}, x_1, \dots, x_n)$ picked out by $\mathcal{C}$ is bounded above by the number of subsets of $\{x_1, \dots, x_n\}$ shattered by $\mathcal{C}$.*

^b In the first edition of this book the notation $V(\mathcal{C})$ was used for the *VC-index*, which is defined as the smallest n such that no set of n points is shattered. This index is the present dimension plus 1.

Proof. Assume without loss of generality that every C is a subset of the given set of points, so that each C picks out itself and $\Delta_n(\mathcal{C}, x_1, \dots, x_n)$ is the cardinality of $\mathcal{C}$.

Call $\mathcal{C}$ *hereditary* if it has the property that $B \in \mathcal{C}$ whenever $B \subset C$, for a set $C \in \mathcal{C}$. Each of the sets in a hereditary collection of sets is shattered, whence a hereditary collection shatters at least $|\mathcal{C}|$ sets and the assertion of the lemma is certainly true for hereditary collections. It will be shown that a general $\mathcal{C}$ can be transformed into a hereditary collection, without changing its cardinality and without increasing the number of its shattered subsets.

Given $1 \leq i \leq n$ and $C \in \mathcal{C}$, define the set $T_i(C)$ to be $C - \{x_i\}$ if $C - \{x_i\}$ is not contained in $\mathcal{C}$ and to be C otherwise. Thus x_i is deleted from C if this creates a “new” set; otherwise it is retained. If $x_i \notin C$, then C is left unchanged.

Since the map T_i is one-to-one, the collections $\mathcal{C}$ and $T_i(\mathcal{C})$ have the same cardinality. Furthermore, every subset $A \subset \{x_1, \dots, x_n\}$ that is shattered by $T_i(\mathcal{C})$ is shattered by $\mathcal{C}$. If $x_i \notin A$, this is clear from the fact that the collection of sets $C \cap A$ does not change if each C is replaced by its transformation $T_i(C)$. Second, if $x_i \in A$ and A is shattered by $T_i(\mathcal{C})$, then for every $B \subset A$, there is $C \in \mathcal{C}$ with $B \cup \{x_i\} = T_i(C) \cap A$. This implies $x_i \in T_i(C)$, whence $T_i(C) = C$, whence $C - \{x_i\} \in \mathcal{C}$. Thus both $B \cup \{x_i\}$ and $B - \{x_i\} = (C - \{x_i\}) \cap A$ are picked out by $\mathcal{C}$. One of them equals B .

It has been shown that the assertion of the lemma is true for $\mathcal{C}$ if it is true for $T_i(\mathcal{C})$. The same is true for the operator $T_1 \circ T_2 \circ \dots \circ T_n$ playing the role of T_i . Apply this operator repeatedly, until the collection of sets does not change any more. This happens after at most $\sum_C |C|$ steps, because $\sum_C |T_i(C)| < \sum_C |C|$ whenever the collections $T_i(\mathcal{C})$ and $\mathcal{C}$ are different. The collection $\mathcal{D}$ obtained in this manner has the property that $D - \{x_i\}$ is contained in $\mathcal{D}$ for every $D \in \mathcal{D}$ and every x_i . Consequently, $\mathcal{D}$ is hereditary. ■

2.6.3 Corollary. *For a VC-class of sets of dimension $V(\mathcal{C})$, one has*

$$\max_{x_1, \dots, x_n} \Delta_n(\mathcal{C}, x_1, \dots, x_n) \leq \sum_{j=0}^{V(\mathcal{C})} \binom{n}{j}.$$

Consequently, the numbers on the left side grow polynomially of order at most $O(n^{V(\mathcal{C})})$ as $n \rightarrow \infty$.

Proof. All sets shattered by a VC-class are among the sets of size at most $V(\mathcal{C})$. The number of shattered sets gives an upper bound on Δ_n by the preceding lemma. ■

2.6.4 Theorem (VC sets covering). *There exists a universal constant K such that for any VC-class $\mathcal{C}$ of sets, any probability measure Q , any $r \geq 1$, and $0 < \varepsilon < 1$,*

$$N(\varepsilon, \mathcal{C}, L_r(Q)) \leq KV(\mathcal{C})(4e)^{V(\mathcal{C})} \left(\frac{1}{\varepsilon}\right)^{rV(\mathcal{C})}.$$

Proof. In view of the equality $\|1_C - 1_D\|_{Q,r} = Q^{1/r}(C \triangle D)$ for any pair of sets C and D , the upper bound for a general r is an easy consequence of the bound for

$r = 1$. The proof for $r = 1$ is long. Problem 2.6.4 gives a short proof of a slightly weaker result.

By Problem 2.6.3, it suffices to consider the case that Q is an empirical type measure: a measure on a finite set of points $y_1, \dots, y_k$ such that $Q\{y_i\} = l_i/n$ for integers $l_1, \dots, l_k$ that add up to n . Then it is no loss of generality to assume that each set in the collection $\mathcal{C}$ is a subset of these points.

Let $x_1, \dots, x_n$ be the set of points $y_1, \dots, y_k$ with each y_i occurring l_i times. For each subset C of $y_1, \dots, y_k$, form a subset $\tilde{C}$ of $x_1, \dots, x_n$ by taking all x_i that are copies of some y_i in C . More precisely, let $\phi: \{1, \dots, n\} \rightarrow \{y_1, \dots, y_k\}$ be an arbitrary, fixed map such that $\#\{j: \phi(j) = y_i\} = l_i$ for every i . Let $\tilde{C} = \{j: \phi(j) \in C\}$. Now identify $x_1, \dots, x_n$ with $1, \dots, n$.

If a set $\{x_j: j \in J\}$ is shattered by the collection $\tilde{\mathcal{C}}$ of sets $\tilde{C}$, then every x_j in this set must correspond to a different y_i (i.e., the map ϕ must be one-to-one on $\{x_j: j \in J\}$), and the subset $\{\phi(x_j): j \in J\}$ of $\{y_1, \dots, y_k\}$ must be shattered by $\mathcal{C}$. This shows that the collection $\tilde{\mathcal{C}}$ is VC of the same dimension as $\mathcal{C}$. By construction, $Q(C \triangle D) = \tilde{Q}(\tilde{C} \triangle \tilde{D})$ for $\tilde{Q}$ equal to the discrete uniform measure on $x_1, \dots, x_n$. Thus the $L_1(Q)$ -distance on $\mathcal{C}$ corresponds to the $L_1(\tilde{Q})$ -distance on $\tilde{\mathcal{C}}$. Conclude that $N(\varepsilon, \tilde{\mathcal{C}}, L_1(\tilde{Q})) = N(\varepsilon, \mathcal{C}, L_1(Q))$, and it suffices to prove the upper bound for $\tilde{\mathcal{C}}$ and $\tilde{Q}$. For simplicity of notation, assume that Q is the discrete uniform measure on a set of points $x_1, \dots, x_n$.

Each set C can be represented by an n -vector of ones and zeros indicating whether or not the points x_i are contained in the set. Thus the collection $\mathcal{C}$ is identified with a subset $\mathcal{Z}$ of the vertices of the n -dimensional hypercube $[0, 1]^n$. Alternatively, $\mathcal{C}$ can be identified with an $(n \times \#\mathcal{C})$ -matrix of zeros and ones, the columns representing the individual sets C . For a subset I of rows, let $\mathcal{Z}_I$ be the matrix obtained by first deleting the other rows and next dropping duplicate columns. Alternatively, $\mathcal{Z}_I$ is the projection of the point set $\mathcal{Z}$ onto $[0, 1]^I$. In these terms $\mathcal{Z}_I$ corresponds to the collection of subsets $C \cap \{x_i: i \in I\}$ of the set of points $\{x_i: i \in I\}$, and this set is shattered if the matrix $\mathcal{Z}_I$ consists of all $2^{|I|}$ possible columns or if, as a subset of the I -dimensional hypercube, $\mathcal{Z}_I$ contains all vertices. By assumption, this is possible only for $I \leq V(\mathcal{C})$. Abbreviate $V = V(\mathcal{C})$.

The normalized *Hamming metric* on the point set $\mathcal{Z}$ is defined by

$$d(w, z) = \frac{1}{n} \sum_{j=1}^n |w_j - z_j|, \quad z, w \in \mathcal{Z}.$$

This corresponds exactly to the $L_1(Q)$ -metric on the collection $\mathcal{C}$: if the sets C and D correspond to the vertices w and z , then $Q(C \triangle D) = d(w, z)$ for the discrete uniform measure Q .

Fix a maximal ε -separated collection of sets $C \in \mathcal{C}$. For simplicity of notation assume that $\mathcal{C}$ (or equivalently, $\mathcal{Z}$) itself is ε -separated. We shall bound its cardinality, which constitutes a bound on the packing number $D(\varepsilon, \mathcal{C}, L_1(Q))$.

Let Z be a random variable with the discrete uniform distribution on the point set $\mathcal{Z}$ in $[0, 1]^n$. The coordinates of $Z = (Z_1, \dots, Z_n)$ are (correlated, non-i.i.d.) Bernoulli variables taking values in $\{0, 1\}$. Fix an integer $V \leq m < n$. Given a subset I of $\{1, 2, \dots, n\}$ of size $m+1$, apply Lemma 2.6.6 (below) to the projection

Z_I of Z onto $\mathcal{Z}_I$ to conclude that

$$\sum_{i \in I} \mathbb{E} \operatorname{var}(Z_i | Z_{I - \{i\}}) \leq V.$$

Take the sum over all $\binom{n}{m+1}$ of such subsets I on both left and right. The double sum on the left can be rearranged. Instead of leaving out one element at a time from every possible subset of size $m+1$, we may add missing elements to every possible set of size m . Conclude that

$$(2.6.5) \quad \sum_J \mathbb{E} \sum_{i \notin J} \operatorname{var}(Z_i | Z_J) \leq \binom{n}{m+1} V,$$

where the first sum is over all subsets J of size m .

Conditionally on $Z_J = s$, the vector Z is uniformly distributed over the set of columns z in $\mathcal{Z}$ for which $z_J = s$. Suppose there are N_s of such columns. Let W and $\tilde{W}$ be independent random vectors defined on a common probability space distributed uniformly over these columns. By the ε -separation of $\mathcal{Z}$, the vectors W and $\tilde{W}$ are at Hamming distance $d(W, \tilde{W})$ at least ε whenever they are unequal. The latter happens with probability $1 - 1/N_s$. Since $\operatorname{var} W_i = \mathbb{E}(W_i - \tilde{W}_i)^2/2$,

$$\sum_{i \notin J} \operatorname{var}(Z_i | Z_J = s) = \frac{1}{2} \sum \mathbb{E}(W_i - \tilde{W}_i)^2 = \frac{1}{2} n \mathbb{E} d(W, \tilde{W}) \geq \frac{1}{2} n \varepsilon \left(1 - \frac{1}{N_s}\right).$$

If we integrate the left side with respect to s for the distribution of Z_J and next sum over J , then we obtain the left side of (2.6.5). Since $\mathbb{P}(Z_J = s) = N_s/\#\mathcal{Z}$, the resulting expression is bounded below by, with $\overline{\#\mathcal{Z}_J}$ an average over J ,

$$\sum_J \sum_{s \in \mathcal{Z}_J} \frac{N_s}{\#\mathcal{Z}} \frac{1}{2} \varepsilon n \left(1 - \frac{1}{N_s}\right) = \binom{n}{m} \frac{1}{2} \varepsilon n \left(1 - \frac{\overline{\#\mathcal{Z}_J}}{\#\mathcal{Z}}\right).$$

This is bounded above by the right side of (2.6.5). Rearrange and simplify the resulting inequality to obtain

$$\#\mathcal{Z} \leq \frac{\overline{\#\mathcal{Z}_J} n \varepsilon (m+1)}{n \varepsilon m + n \varepsilon - 2nV + 2mV} \lesssim \frac{\overline{\#\mathcal{Z}_J} \varepsilon m}{\varepsilon m - 2V}.$$

The number of points in $\mathcal{Z}_J$ is equal to the number of subsets picked out by $\mathcal{C}$ from the points $\{x_i; i \in J\}$. By Sauer's lemma, this is bounded by $\sum_{j=0}^V \binom{m}{j}$, which is smaller than $(em/V)^V$ for $m \geq V$. Thus we obtain

$$\#\mathcal{Z} \leq \left(\frac{e}{V}\right)^V \frac{m^{V+1} \varepsilon}{m \varepsilon - 2V},$$

for every integer $V \leq m < n$. The optimal unrestricted choice of $m \in (0, \infty)$ is $m = 2(V+1)/\varepsilon$, for which the upper bound takes the form

$$\left(\frac{e}{V}\right)^V \frac{1}{2} \left(\frac{2(V+1)}{\varepsilon}\right)^{V+1} \leq (V+1) e \left(\frac{2e}{\varepsilon}\right)^V.$$

Since the upper bound evaluated at $m = 2(V+1)/\varepsilon + 1$ differs from this only up to a universally bounded constant, the discretization of m causes no problem.

Furthermore, for the optimal choice of m , the inequality $m > V$ is implied by $\varepsilon < 2$, while for $m = 2(V + 1)/\varepsilon \geq n$, we can use the trivial bound $\#\mathcal{Z} \leq n \leq m$, which is certainly bounded by the right side of the preceding display for $V \geq 1$. For $V = 0$, the collection $\mathcal{C}$ consists of at most one set, and the theorem is trivial. ■

2.6.6 Lemma. *Let Z be an arbitrary random vector taking values in a set $\mathcal{Z} \subset \{0, 1\}^n$ that corresponds to a VC-class $\mathcal{C}$ of subsets of a set of points $\{x_1, \dots, x_n\}$. Then*

$$\sum_{i=1}^n \mathbb{E} \text{var}(Z_i | Z_j, j \neq i) \leq V(\mathcal{C}).$$

Proof. If the values of all coordinates, except the i th coordinate, of Z are given, then the vector Z can have at most two values. If there is only one possible value, then the conditional variance of Z_i is zero. If there are two possible values, then we can call these v and w , where $v_i = 0$ and $w_i = 1$ and the other coordinates of v and w agree. Write $p(z)$ for $\mathbb{P}(Z = z)$. Then Z_i is 1 or 0 with conditional probabilities $p := p(w)/(p(v) + p(w))$ and $1 - p$, respectively, and the conditional variance of Z_i is $p(1 - p)$. The marginal probability that $(Z_j: j \neq i)$ is equal to $(v_j: j \neq i)$ is $p(v) + p(w)$.

Form a graph by taking the points of $\mathcal{Z}$ as the nodes and connecting any two nodes that are at Hamming distance $1/n$. This is a subgraph of the set of edges of the n -dimensional unit cube. Let $\mathcal{E}_i$ and $\mathcal{E}$ be the set of all edges that cross the i th dimension and all edges in the graph, respectively. If $\mathcal{E}_i$ is empty, then all variances in the i th term of the lemma are zero. In the other case, the edges $\{v, w\}$ in $\mathcal{E}$ concern nodes v and w as in the preceding paragraph. Then, with an empty sum understood as zero,

$$\begin{aligned} \sum_{i=1}^n \mathbb{E} \text{var}(Z_i | Z_j, j \neq i) &= \sum_{i=1}^n \sum_{\{v, w\} \in \mathcal{E}_i} (p(v) + p(w)) \frac{p(v)}{p(v) + p(w)} \frac{p(w)}{p(v) + p(w)} \\ &\leq \sum_{\{v, w\} \in \mathcal{E}} p(v) \wedge p(w). \end{aligned}$$

Suppose that the edges $\mathcal{E}$ can be directed in such a way that at each node (point in $\mathcal{Z}$) the number of arrows that are directed away is at most $V(\mathcal{C})$. Then the last sum can be rearranged as a double sum, the outer sum carried out over the nodes, and the inner sum over the arrows directed away from the particular node. Thus the sum is bounded by $\sum_{z \in \mathcal{Z}} p(z) \# \text{arrows}(z)$, which is bounded by $V(\mathcal{C})$.

It may be shown that the edges can always be directed in the given manner (Problem 2.6.5), but it is more direct to proceed in a different manner. Recall from the proof of Lemma 2.6.2 that a class $\mathcal{C}$ is hereditary if it contains every subset of every set contained in $\mathcal{C}$. Since each set in such a class $\mathcal{C}$ is shattered, each set in a hereditary VC-class $\mathcal{C}$ has at most $V(\mathcal{C})$ points. For such a class the edges of the graph can be directed as desired by decreasing number of elements (or downward in the cube): direct the edge between v and w in the direction $w \rightarrow v$ if the (one) coordinate of w by which w differs from v equals 1. For each given w this creates

at most $V(\mathcal{C})$ arrows as w has at most so many coordinates 1. This concludes the proof in the case that $\mathcal{C}$ is hereditary.

It was shown in the proof of Lemma 2.6.2 that an arbitrary collection $\mathcal{C}$ can be transformed into a hereditary collection by repeated application of certain operators $T_i: \mathcal{C} \rightarrow \mathcal{C}$, which are one-to-one and do not increase the VC-dimension. The identification of sets in $\mathcal{C}$ with points in $\mathcal{Z}$ allows to view the operators also as maps $T_i: \mathcal{Z} \rightarrow \mathcal{Z}$, and they can be described as pushing a point $z \in \mathcal{Z}$ that is at height 1 in the i th coordinate (i.e. $z_i = 1$) down to height 0 (i.e. changing z_i to 0) if the resulting point was not already in $\mathcal{Z}$, and leaving all other points invariant. We can form a new graph by taking the set of nodes equal to $T_i(\mathcal{Z})$ and connecting any two nodes at Hamming distance $1/n$. The operator T_i induces a one-to-one map from the set of vertices $\mathcal{E} = \mathcal{E}(\mathcal{Z})$ in the original graph into (but not necessarily onto) the set of vertices $\mathcal{E}(T_i(\mathcal{Z}))$ of the new graph as follows. The nodes v and w of an edge $\{v, w\}$ in $\mathcal{E}$ that crosses the i th dimension (i.e. v and w differ in their i th coordinates and no other ones) are left invariant by T_i ; we map such an edge onto itself. We do the same for an edge at height 0 for the i th coordinate. For an edge $\{v, w\} \in \mathcal{E}$ at height 1 (i.e. $v_i = w_i = 1$) there are three possibilities. If T_i pushes both v and w down, say to v^0 and w^0 , then the latter two nodes were not in $\mathcal{Z}$ and hence nor was $\{v^0, w^0\}$; we map $\{v, w\}$ to the latter edge. If T_i pushes exactly one of v and w down, say v to v^0 , then v^0 was not in $\mathcal{Z}$, but w^0 was in $\mathcal{Z}$, whence $\{v^0, w^0\}$ was not in $\mathcal{E}$ but is in $\mathcal{E}(T_i(\mathcal{Z}))$; we map $\{v, w\}$ to $\{v^0, w^0\}$. Third if T_i pushes none of v and w down, then we map $\{v, w\}$ to itself. Thus we have defined a one-to-one map from $\mathcal{E}$ into $\mathcal{E}(T_i(\mathcal{Z}))$. We also define a probability measure p_i on $T_i(\mathcal{Z})$ by shifting mass downward in the following manner: first we give every $z \in T_i(\mathcal{Z})$ the mass of its pre-image in $\mathcal{Z}$; next for every edge $\{z^0, z^1\}$ in $\mathcal{E}(T_i(\mathcal{Z}))$ that crosses the i th dimension we place the biggest of the two masses $p(z^0)$ and $p(z^1)$ at height 0 (i.e. at z^0) and the smallest at height 1 (i.e. at z^1). We claim that

$$\sum_{\{v, w\} \in \mathcal{E}(\mathcal{Z})} p(v) \wedge p(w) \leq \sum_{\{v, w\} \in \mathcal{E}(T_i(\mathcal{Z}))} p_i(v) \wedge p_i(w).$$

Indeed, any edge $\{v, w\} \in \mathcal{E}(\mathcal{Z})$ in the sum on the left that crosses the i th dimension reappears in the sum on the right, with the same value $p(v) \wedge p(w)$, which is invariant under redistributing the masses between the planes. Any such edge in $\mathcal{E}(\mathcal{Z})$ at height 1 that is pushed down by T_i can be matched to its image in the sum on the right, which has no smaller value attached as the bigger masses are redistributed to the bottom plane. Any such edge at height 1 that is *not* pushed down forms a pair with an edge in $\mathcal{E}(\mathcal{Z})$ at height 0; they reappear as a pair in the sum on the right and together contribute no smaller value in view of the inequality $(a_0 \wedge b_0) + (a_1 \wedge b_1) \leq (a_0 \vee a_1) \wedge (b_0 \vee b_1) + (a_0 \wedge a_1) \wedge (b_0 \wedge b_1)$ for any numbers a_i, b_i . Finally the remaining edges in $\mathcal{E}(\mathcal{Z})$ are at height 0 and do not have a paired edge above them; they reappear in the sum on the right with possibly a bigger value.

Thus by repeated application of the downward shift operation, the original graph and probability measure are changed into a hereditary graph and another probability measure, meanwhile increasing the target function. The lemma follows. ■

2.6.2 VC-Classs of Functions

The *subgraph* of a function $f: \mathcal{X} \rightarrow \mathbb{R}$ is the subset of $\mathcal{X} \times \mathbb{R}$ given by[‡]

$$\{(x, t): t < f(x)\}.$$

A collection $\mathcal{F}$ of measurable functions on a sample space is called a *VC-subgraph class*, or just a *VC-class*, if the collection of all subgraphs of the functions in $\mathcal{F}$ forms a VC-class of sets (in $\mathcal{X} \times \mathbb{R}$). Just as for sets, the covering numbers of VC-classes of functions grow at a polynomial rate.

Let $V(\mathcal{F})$ be the VC-dimension of the set of subgraphs of functions in $\mathcal{F}$.

2.6.7 Theorem (VC functions covering). *For a VC-class of functions with measurable envelope function F and $r \geq 1$, one has for any probability measure Q with $\|F\|_{Q,r} > 0$,*

$$N(\varepsilon \|F\|_{Q,r}, \mathcal{F}, L_r(Q)) \leq KV(\mathcal{F})(16e)^{V(\mathcal{F})} \left(\frac{1}{\varepsilon}\right)^{rV(\mathcal{F})},$$

for a universal constant K and $0 < \varepsilon < 1$.

Proof. Let $\mathcal{C}$ be the set of all subgraphs C_f of functions f in $\mathcal{F}$. By Fubini's theorem, $Q|f - g| = Q \times \lambda(C_f \triangle C_g)$ if λ is Lebesgue measure on the real line. Renormalize $Q \times \lambda$ to a probability measure on the set $\{(x, t): |t| \leq F(x)\}$ by defining $P = (Q \times \lambda)/(2QF)$. Then, by the result for sets in the previous section,

$$N(\varepsilon 2QF, \mathcal{F}, L_1(Q)) = N(\varepsilon, \mathcal{C}, L_1(P)) \leq KV(\mathcal{F}) \left(\frac{4e}{\varepsilon}\right)^{V(\mathcal{F})},$$

for a universal constant K .

This concludes the proof for $r = 1$. For $r > 1$, note that

$$Q|f - g|^r \leq Q|f - g|(2F)^{r-1} = 2^{r-1} R|f - g| QF^{r-1},$$

for the probability measure R with density F^{r-1}/QF^{r-1} with respect to Q . Thus the $L_r(Q)$ -distance is bounded by the distance $2(QF^{r-1})^{1/r} \|f - g\|_{R,1}^{1/r}$. Elementary manipulations yield

$$N(\varepsilon 2\|F\|_{Q,r}, \mathcal{F}, L_r(Q)) \leq N(\varepsilon^r RF, \mathcal{F}, L_1(R)).$$

This can be bounded by a constant times $1/\varepsilon$ raised to the power $rV(\mathcal{F})$ by the result of the previous paragraph. ■

The preceding theorem shows that a VC-class satisfies the uniform entropy condition (2.5.1), with much to spare. Thus Theorem 2.5.2 shows that a suitably measurable VC-class is P -Donsker for any underlying measure P for which the envelope is square integrable. The latter condition on the envelope can be relaxed a little to the minimal condition that the envelope has a weak second moment.

[‡] The form of this definition is different from the definitions given by other authors. However, it can be shown to lead to the same concept. See Problems 2.6.10 and 2.6.11.

2.6.8 Theorem. Let $\mathcal{F}$ be a pointwise separable, P -pre-Gaussian VC-class of functions with envelope function F such that $P^*(F > x) = o(x^{-2})$ as $x \rightarrow \infty$. Then $\mathcal{F}$ is P -Donsker.

Proof. Under the stronger condition that the envelope has a finite second moment, this follows from Theorem 2.5.2. (Then the assumption that $\mathcal{F}$ is pre-Gaussian is automatically satisfied.) For the refinement, see Alexander (1987c). ■

2.6.3 Convex Hulls and VC-Hull Classes

The *symmetric convex hull* $\text{sconv } \mathcal{F}$ of a class of functions is defined as the sets of functions $\sum_{i=1}^m \alpha_i f_i$, with $\sum_{i=1}^m |\alpha_i| \leq 1$ and each f_i contained in $\mathcal{F}$. The *convex hull* $\text{conv } \mathcal{F}$ is the subset of such functions $\sum_{i=1}^m \alpha_i f_i$, where $\alpha_i \geq 0$, for every i .

A set of measurable functions is called a *VC-hull class* if it is in the pointwise sequential closure of the symmetric convex hull of a VC-class of functions. More formally, a collection of functions $\mathcal{F}$ is VC-hull if there exists a VC-class $\mathcal{G}$ of functions such that every $f \in \mathcal{F}$ is the pointwise limit of a sequence of functions f_m contained in $\text{sconv } \mathcal{G}$. If the class $\mathcal{G}$ can be taken equal to a class of indicator functions, then the class $\mathcal{F}$ is called a *VC-hull class for sets*.

In Chapter 2.10 it is shown that the sequentially closed symmetric convex hull of a Donsker class is Donsker. Since a suitably measurable VC-class of functions is Donsker, provided its envelope has a weak second moment, many VC-hull classes can be seen to be Donsker by this general result. In most cases the same conclusion can also be obtained from the stronger result that any VC-hull class satisfies the uniform entropy condition. This is shown in the following, which gives an upper bound on the entropy.

Even though VC-hull classes are small enough to have a finite uniform entropy integral, they can be considerably larger than VC-classes. Their entropy numbers (logarithms of the covering numbers), rather than their covering numbers, behave polynomially in $1/\varepsilon$. More precisely, the L_r -entropy numbers of the convex hull of any polynomial class are of lower order than $(1/\varepsilon)^W$, where $W < 2$ if $r \geq 2$ (and also for smaller r depending on the polynomial order). The bound $W < 2$ is just enough to ensure that the class satisfies the uniform entropy condition (2.5.1).

2.6.9 Theorem (Convex hull entropy). Let Q be a probability measure on $(\mathcal{X}, \mathcal{A})$, and let $\mathcal{F}$ be a class of measurable functions with measurable square integrable envelope F such that $QF^r < \infty$ for some $1 < r < \infty$ and

$$N(\varepsilon \|F\|_{Q,r}, \mathcal{F}, L_r(Q)) \leq C \left(\frac{1}{\varepsilon}\right)^V, \quad 0 < \varepsilon < 1.$$

Then there exists a constant K that depends on r , C and V only such that

$$\log N(\varepsilon \|F\|_{Q,r}, \overline{\text{conv}} \mathcal{F}, L_r(Q)) \leq \begin{cases} K \left(\frac{1}{\varepsilon}\right)^{rV/((r-1)V+r)} & \text{if } 1 < r \leq 2, \\ K \left(\frac{1}{\varepsilon}\right)^{2V/(V+2)} & \text{if } 2 \leq r < \infty. \end{cases}$$

Proof. Every point in the convex hull of $\mathcal{F}$ is within distance ε of the convex hull of an ε -net over $\mathcal{F}$. Therefore, to prove the assertion of the theorem for a fixed ε , it is no loss of generality to assume that $\mathcal{F}$ is finite.

Set $r_0^{-1} = \min\{1 - 1/r, 1/2\}$, $W = r_0^{-1} + 1/V$, and $L = C^{1/V} \|F\|_{Q,r}$. Then by assumption $\mathcal{F}$ can be covered by n balls of radius at most $Ln^{-1/V}$ for every natural number n . (Note that the assumption is trivially true for $1 \leq \varepsilon \leq C^{1/V}$.) Form sets $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots \subset \mathcal{F}$ such that for each n the set $\mathcal{F}_n$ is a maximal, $Ln^{-1/V}$ -separated net over $\mathcal{F}$. Thus $\mathcal{F}_n$ contains at most n points. It will be shown by induction that there exist constants C_k and D_k depending only on C and V such that $\sup_k C_k \vee D_k < \infty$ and, for $q \geq 2V(1 + 1/r_0)$,

$$(2.6.10) \quad \log N(C_k Ln^{-W}, \text{conv } \mathcal{F}_{nk^q}, L_r(Q)) \leq D_k n, \quad n, k \geq 1.$$

This would imply the theorem. The proof of (2.6.10) consists of a nested induction argument. The outer layer is induction on k . The case $k = 1$ is proved for each n by induction on n .

Assume $k = 1$. For $n \leq n_0$ and fixed n_0 , the statement is trivially true for sufficiently large C_1 . It suffices to choose $C_1 Ln_0^{-W} \geq \|F\|_{Q,r}$, in which case the left side of (2.6.10) vanishes for every $n \leq n_0$. For general n , fix $m = n/d$ for large enough d to be chosen. Each $f \in \mathcal{F}_n - \mathcal{F}_m$ is within distance at most $Lm^{-1/V}$ of some element $\Pi_m f$ of $\mathcal{F}_m$. Thus each element of $\text{conv } \mathcal{F}_n$ can be written as

$$\sum_{f \in \mathcal{F}_n} \lambda_f f = \sum_{f \in \mathcal{F}_m} \mu_f f + \sum_{f \in \mathcal{F}_n - \mathcal{F}_m} \lambda_f (f - \Pi_m f),$$

where $\mu_f \geq 0$ and $\sum \mu_f = \sum \lambda_f = 1$. If $\mathcal{G}_n$ is the set of functions 0 and $f - \Pi_m f$ with f ranging over $\mathcal{F}_n - \mathcal{F}_m$, then it follows that $\text{conv } \mathcal{F}_n \subset \text{conv } \mathcal{F}_m + \text{conv } \mathcal{G}_n$ for a set $\mathcal{G}_n$ containing at most n elements, each of norm smaller than $Lm^{-1/V}$, whence $\text{diam } \mathcal{G}_n \leq 2Lm^{-1/V}$. Apply Lemma 2.6.11 (below) to $\mathcal{G}_n$ with ε defined by $m^{-1/V} \varepsilon = (1/4)C_1 n^{-W}$ to find a $(1/2)C_1 Ln^{-W}$ -net over $\text{conv } \mathcal{G}_n$ consisting of at most

$$(e + en\varepsilon^{r_0}/b_r)^{b_r/\varepsilon^{r_0}} \leq \left(e + \frac{e C_1^{r_0}}{b_r 4^{r_0} d^{r_0/V}} \right)^{nb_r 4^{r_0} C_1^{-r_0} d^{r_0/V}}$$

elements. Apply the induction hypothesis to $\mathcal{F}_m$ to find a $C_1 Lm^{-W}$ -net over the set $\text{conv } \mathcal{F}_m$ consisting of at most e^m elements. This defines a partition of $\text{conv } \mathcal{F}_m$ into m -dimensional sets of diameter at most $2C_1 Lm^{-W}$. Such a set can be isometrically identified with a subset of a ball of radius $C_1 Lm^{-W}$ in $\mathbb{R}^m$. Thus each of these sets can be partitioned in

$$\left(\frac{3C_1 Lm^{-W}}{(1/2)C_1 Ln^{-W}} \right)^m = (6d^W)^{n/d}$$

sets of diameter $(1/2)C_1 Ln^{-W}$ (Problem 2.1.7). Take a function from each of these sets, and form all sums $f + g$ of a function f attached to $\text{conv } \mathcal{F}_m$ and a function

g attached to $\text{conv } \mathcal{G}_n$ by the preceding procedure. These form a $C_1 L n^{-W}$ -net over $\text{conv } \mathcal{F}_n$ of cardinality bounded by

$$e^{n/d} (6d^W)^{n/d} \left(e + \frac{e C_1^{r_0}}{b_r 4^{r_0} d^{r_0/V}} \right)^{n b_r 4^{r_0} C_1^{-r_0} d^{r_0/V}}$$

This is bounded by e^n for suitable choices of C_1 and d depending on V and r only. This concludes the proof of (2.6.10) for $k = 1$ and every n .

The argument continues by induction on k . By a similar construction as before, $\text{conv } \mathcal{F}_{nk^q} \subset \text{conv } \mathcal{F}_{n(k-1)^q} + \text{conv } \mathcal{G}_{n,k}$ for a set $\mathcal{G}_{n,k}$ containing at most $n k^q$ elements, each of norm smaller than $L(n(k-1)^q)^{-1/V}$, so that $\text{diam } \mathcal{G}_{n,k} \leq 2L n^{-1/V} k^{-q/V} 2^{q/V}$. Apply Lemma 2.6.11 to $\mathcal{G}_{n,k}$ with $\varepsilon = 2^{-1} k^{q/V-2} 2^{-q/V} n^{-1/r_0}$ to find an $L k^{-2} n^{-W}$ -net over $\text{conv } \mathcal{G}_{n,k}$ consisting of at most

$$\left(e + e n k^q \varepsilon^{r_0} / b_r \right)^{b_r / \varepsilon^{r_0}} = \left(e + \frac{e k^{q+r_0 q/V-2r_0}}{2^{q r_0/V+r_0} b_r} \right)^{n b_r 2^{q r_0/V+r_0} k^{2r_0-r_0 q/V}}$$

elements. Apply the induction hypothesis to obtain a $C_{k-1} L n^{-W}$ -net over the set $\text{conv } \mathcal{F}_{n(k-1)^q}$ with respect to $L_r(Q)$ consisting of at most $e^{D_{k-1} n}$ elements. Combine the nets as before to obtain a $C_k L n^{-W}$ -net over $\text{conv } \mathcal{F}_{nk^q}$ consisting of at most $e^{D_k n}$ elements, for

$$C_k = C_{k-1} + \frac{1}{k^2},$$

$$D_k = D_{k-1} + \frac{1 + \log \left(1 + k^{q+r_0 q/V-2r_0} / (2^{q r_0/V+r_0} b_r) \right)}{k^{r_0 q/V-2r_0} / (2^{q r_0/V+r_0} b_r)}.$$

For $(q/V - 2)r_0 \geq 2$, or equivalently $q \geq 2V(1 + 1/r_0)$, the resulting sequences C_k and D_k are bounded. ■

Since the symmetric convex hull $\text{sconv } \mathcal{F}$ of a class $\mathcal{F}$ is contained in the convex hull of the class $\mathcal{F} \cup -\mathcal{F} \cup \{0\}$, the bound of the preceding theorem is valid for $\text{sconv } \mathcal{F}$ as well. It suffices to note that the covering numbers of the class $\mathcal{F} \cup -\mathcal{F} \cup \{0\}$ are at most twice the covering numbers of $\mathcal{F}$ plus 1.

An important ingredient in the proof of the preceding theorem is the following lemma, which gives a useful bound on the covering numbers of the convex hull of an arbitrary finite set of small diameter.

2.6.11 Lemma. *Let $\mathcal{F}$ be an arbitrary set of n measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$ of finite $L_r(Q)$ -diameter $\text{diam } \mathcal{F}$. Then for every $r > 1$ there exists a constant b_r such that for $r_0^{-1} = \min(1 - 1/r, 1/2)$ and for every $\varepsilon > 0$,*

$$N(\varepsilon \text{diam } \mathcal{F}, \text{conv } \mathcal{F}, L_r(Q)) \leq \left(e + e n \varepsilon^{r_0} / b_r \right)^{b_r / \varepsilon^{r_0}}.$$

Proof. Write $\mathcal{F} = \{f_1, \dots, f_n\}$. For given λ in the n -dimensional unit simplex and natural number k , let $Y_1, \dots, Y_k$ be i.i.d. random elements with $P(Y_i = f_j) = \lambda_j$

for $j = 1, \dots, n$. Then $EY_i = \sum \lambda_j f_j$ and by Marcinkiewicz' inequality, Proposition A.1.9,

$$E \left| \sum_{i=1}^k (Y_i - EY_i) \right|^r \leq B_r E \left(\sum_{i=1}^k (Y_i - EY_i)^2 \right)^{r/2}.$$

For $r \leq 2$ we bound the right side further by $B_r E \sum_{i=1}^k |Y_i - EY_i|^r$ using subadditivity of the map $y \mapsto y^{r/2}$, while for $r \geq 2$ we use Hölder's inequality to bound it by this same expression times $k^{r/2-1}$. Next we integrate over Q , divide by k^r , and use Fubini's theorem to see that

$$E \|\bar{Y}_k - EY_1\|_{Q,r}^r \leq \frac{B_r(1 \vee k^{r/2-1})}{k^r} E \sum_{i=1}^k \int |Y_i - EY_1|^r dQ \leq \frac{B_r}{k^{r/r_0}} (\text{diam } \mathcal{F})^r,$$

since $r/r_0 = r - 1$ if $r \leq 2$ and $r/r_0 = r/2$ if $r \geq 2$. The inequality shows that at least one realization of $\bar{Y}_k$ must have $L_r(Q)$ -distance at most $B_r^{1/r} k^{-1/r_0} \text{diam } \mathcal{F}$ to the convex combination $\sum \lambda_j f_j$. Every realization is an average of the form $k^{-1} \sum_{i=1}^k f_{i_k}$ (allowing for multiple use of an $f_i \in \mathcal{F}$). There are at most $\binom{n+k-1}{k}$ of such averages. Conclude that

$$N(B_r^{1/r} k^{-1/r_0} \text{diam } \mathcal{F}, \text{conv } \mathcal{F}, L_r(Q)) \leq \binom{n+k-1}{k} \leq e^k \left(1 + \frac{n}{k}\right)^k.$$

The last inequality can be proved by using Stirling's inequality with bound. For $\varepsilon \geq 1$, the assertion of the lemma is trivial. For $\varepsilon < 1$, conclude the proof by choosing the smallest k such that $B_r^{1/r} k^{-1/r_0} \leq \varepsilon$. ■

2.6.12 Corollary. *For any VC-hull class $\mathcal{F}$ of measurable functions and probability measure Q ,*

$$\log N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q)) \leq K \left(\frac{1}{\varepsilon} \right)^{2V_m(\mathcal{F})/(2V_m(\mathcal{F})+1)},$$

for a constant K that depends only the VC-dimension $V_m(\mathcal{F})$ of the VC-subgraph class connected with $\mathcal{F}$ and for F an envelope function of this VC-subgraph class.

The entropy of the convex hull of a much bigger class of polynomial entropy $(1/\varepsilon)^V$ can be similarly bounded. Remarkably, taking the convex hull does not increase the order of the entropy in the case of a very big class with $V > 2$, while the entropy increases by a logarithmic factor only if $V = 2$. For $V < 2$ the entropy increases more substantially, but the entropy integral of the convex hull still converges if $V < 2/3$.

2.6.13 Theorem (Convex hull entropy). *Let Q be a probability measure on $(\mathcal{X}, \mathcal{A})$, and let $\mathcal{F}$ be a class of measurable functions with measurable envelope function F such that $QF^2 < \infty$ and, for some positive constants C, V ,*

$$\log N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q)) \leq C \left(\frac{1}{\varepsilon} \right)^V, \quad 0 < \varepsilon < 1.$$

Then there exists a constant K that depends on C and V only such that

$$\log N(\varepsilon \|F\|_{Q,2}, \text{sconv}(\mathcal{F}), L_2(Q)) \leq \begin{cases} K(\frac{1}{\varepsilon})^2 (\log \frac{1}{\varepsilon})^{1-2/V}, & \text{if } 0 < V < 2, \\ K(\frac{1}{\varepsilon})^2 (\log \frac{1}{\varepsilon})^2, & \text{if } V = 2, \\ K(\frac{1}{\varepsilon})^V, & \text{if } V > 2. \end{cases}$$

Proof. For a proof for the case $V \neq 2$, see Carl, Kyrezi and Pajor (1999); for the case $V = 2$, see Gao (2001). ■

2.6.4 VC-Major Classes

A class of measurable functions is called a *VC-major class* if the sets $\{x: f(x) > t\}$ with f ranging over $\mathcal{F}$ and t over $\mathbb{R}$ form a VC-class of sets.

Because any function $f: \mathcal{X} \rightarrow [0, 1]$ is the uniform limit of the sequence of discretizations $f_m = \sum_{i=1}^m m^{-1} 1\{f > i/m\}$, bounded VC-major classes are scalar multiples of VC-hull classes. Hence Corollary 2.6.12 allows to obtain a bound on their entropy, from which it follows that suitably measurable bounded VC-hull classes are universally Donsker.

Both results can be improved considerably by direct arguments.

2.6.14 Lemma (Bounded VC-major). *If $\mathcal{F}$ is a VC-major class of functions with envelope $F \leq 1$, then for every $r \geq 1$ there is a constant $A > 0$ such that, for every probability measure Q , and every $\varepsilon \in (0, 1/2)$,*

$$\log N(\varepsilon \|F\|_{Q,r}, \mathcal{F}, L_r(Q)) \leq \frac{A}{\varepsilon} \log\left(\frac{1}{\varepsilon}\right) \log\left(\frac{1}{\varepsilon \|F\|_{Q,r}}\right).$$

Proof. It is no loss of generality to assume that the functions f are nonnegative. For fixed $\varepsilon \in (0, 1/2)$ define a grid of points $t_0 = 1 > t_1 > t_2 \cdots > t_m > 0$ by $t_j = (1 + \varepsilon)^{-j}$, where we choose m the smallest integer such that $t_m \leq \varepsilon \|F\|_{Q,r}$. Furthermore, for $f \in \mathcal{F}$ set

$$f_\varepsilon = \sum_{j=1}^m t_j 1\{t_j < f \leq t_{j-1}\}.$$

Then $0 \leq f_\varepsilon \leq f$, and for a given x , either $f(x) \in (t_j, t_{j-1}]$ for some $1 \leq j \leq m$ or $f(x) \leq t_m$. In the first case $0 \leq f(x) - f_\varepsilon(x) \leq t_{j-1} - t_j = \varepsilon t_j \leq \varepsilon f(x) \leq \varepsilon F(x)$, while in the second case $0 \leq f(x) - f_\varepsilon(x) \leq f(x) \leq \varepsilon \|F\|_{Q,r}$. This implies that $Q|f - f_\varepsilon|^r \leq 2\varepsilon^r QF^r$. We conclude that the entropy of the class $\mathcal{F}$ at $3\varepsilon \|F\|_{Q,r}$ is bounded by the entropy of the class of functions $\mathcal{F}_\varepsilon = \{f_\varepsilon: f \in \mathcal{F}\}$ at ε .

In fact, the class $\mathcal{F}_\varepsilon$ is a VC-subgraph class. The subgraphs can be decomposed as

$$\{(x, t): f_\varepsilon(x) \geq t\} = \cup_{j=1}^m \{x: f(x) > t_{j-1}\} \times (t_j, t_{j-1}].$$

Since $\mathcal{F}$ is a VC-major class, the collection of sets $\{x: f(x) > t_{j-1}\}$, when f ranges over $\mathcal{F}$, is a VC-class of subsets of $\mathcal{X}$, which (trivially) implies that for every j the class $\mathcal{C}_j$ of sets $\{x: f(x) > t_{j-1}\} \times (t_j, t_{j-1}]$, when f ranges over $\mathcal{F}$, is a

VC-class of subsets of the set $\mathcal{X} \times (t_j, t_{j-1}]$. Because the latter sets are disjoint, the collection $\sqcup_j \mathcal{C}_j$ is a VC-class of subsets of $\mathcal{X} \times (0, 1]$, of VC dimension the sum of the dimensions of the classes $\mathcal{C}_j$, i.e. mV for V bounded by the dimension of the VC-class attached to the VC-major class $\mathcal{F}$ (cf. Problem 2.6.12).

Thus the class $\mathcal{F}_\varepsilon$ is VC-subgraph of dimension bounded by mV . Since F is also an envelope function for $\mathcal{F}_\varepsilon$, it follows from Theorem 2.6.7 that

$$N(\varepsilon \|F\|_{Q,r}, \mathcal{F}_\varepsilon, L_r(Q)) \leq \left(\frac{A}{\varepsilon}\right)^{rVm}.$$

Here by its definition $m = m(\varepsilon)$ satisfies $(1 + \varepsilon)^{m-1} \geq \varepsilon \|F\|_{Q,r}$, from which it follows that $m \leq 1 + 2 \log(1/(\varepsilon \|F\|_{Q,r}))/\varepsilon$. The proof is concluded by inserting this bound. ■

2.6.15 Theorem. *Let $\mathcal{F}$ be a pointwise separable, VC-major class of measurable functions with envelope F such that $\int \sqrt{P^*(F > x)} dx < \infty$. Then $\mathcal{F}$ is P -Donsker.*

Proof. See Dudley and Koltchinskii (1994). ■

2.6.5 Examples

There are many examples of VC classes. The lemmas in this section give some basic methods for generating VC-graph classes.

2.6.16 Lemma (Finite-dimensional vector space). *Any finite-dimensional vector space $\mathcal{F}$ of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$ is VC-subgraph of dimension smaller than or equal to $\dim(\mathcal{F}) + 1$.*

Proof. Take any collection of $n = \dim(\mathcal{F}) + 2$ points $(x_1, t_1), \dots, (x_n, t_n)$ in $\mathcal{X} \times \mathbb{R}$. By assumption, the vectors

$$(f(x_1) - t_1, \dots, f(x_n) - t_n)', \quad f \in \mathcal{F},$$

are contained in a $\dim(\mathcal{F}) + 1 = (n - 1)$ -dimensional subspace of $\mathbb{R}^n$. Any vector $a \neq 0$ that is orthogonal to this subspace satisfies

$$\sum_{a_i > 0} a_i (f(x_i) - t_i) = \sum_{a_i < 0} (-a_i) (f(x_i) - t_i), \quad \text{for every } f \in \mathcal{F}.$$

Define the sum over an empty set as zero. There exists such a vector a with at least one strictly positive coordinate. For this vector, the set $\{(x_i, t_i): a_i > 0\}$ cannot be of the form $\{(x_i, t_i): t_i < f(x_i)\}$ for some f , because then the left side of the equation would be strictly positive and the right side nonpositive for this f . Conclude that the subgraphs of $\mathcal{F}$ do not pick out the set $\{(x_i, t_i): a_i > 0\}$. Hence the subgraphs shatter no set of n points. ■

2.6.17 Lemma (Monotone translates). *The set of all translates $\{\psi(x - h): h \in \mathbb{R}\}$ of a fixed monotone function $\psi: \mathbb{R} \rightarrow \mathbb{R}$ is VC of dimension 1.*

Proof. By the monotonicity, the subgraphs are linearly ordered by inclusion: if ψ is nondecreasing, then the subgraph of $x \mapsto \psi(x - h_1)$ is contained in the subgraph of $x \mapsto \psi(x - h_2)$ if $h_1 \geq h_2$. Any collection of sets with this property has VC-dimension 1. ■

2.6.18 Lemma. *Let $F: \mathcal{X} \rightarrow \mathbb{R}$ be a fixed, nonnegative function. Then the class of functions $x \mapsto t1\{F(x) \geq t\}$ with t ranging over $\mathbb{R}$ is VC of dimension $V(\mathcal{F}) \leq 2$.*

Proof. Consider three arbitrary points (x_1, t_1) , (x_2, t_2) , and (x_3, t_3) in $\mathcal{X} \times [0, \infty)$ such that $F(x_1) \leq F(x_2) \leq F(x_3)$. Suppose the three-point set is shattered by $\mathcal{F}$. Since $\mathcal{F}$ selects the single point (x_1, t_1) , there exists s_1 with

$$(t_1 < s_1 \leq F(x_1)) \wedge (s_1 \leq t_2 \text{ or } s_1 > F(x_2)) \wedge (s_1 \leq t_3 \text{ or } s_1 > F(x_3)).$$

Because $F(x_i)$ is increasing in i , this can be reduced to

$$(t_1 < s_1 \leq F(x_1)) \wedge (s_1 \leq t_2) \wedge (s_1 \leq t_3).$$

Since $\mathcal{F}$ selects the single point (x_2, t_2) , there exists s_2 with

$$(s_2 \leq t_1 \text{ or } s_2 > F(x_1)) \wedge (t_2 < s_2 \leq F(x_2)) \wedge (s_2 \leq t_3 \text{ or } s_2 > F(x_3)).$$

In the first part of this logical statement, $s_2 \leq t_1$ is impossible because $t_1 < s_1 \leq t_2 < s_2$ in view of the first two parts of the preceding statement and the middle part of the last statement. Thus the last statement can be reduced to

$$(s_2 > F(x_1)) \wedge (t_2 < s_2 \leq F(x_2)) \wedge (s_2 \leq t_3).$$

This implies that $t_3 > F(x_1)$. Then $\mathcal{F}$ cannot select the set $\{(x_1, t_1), (x_3, t_3)\}$. ■

2.6.6 Permanence Properties

The VC-property is retained under certain operations, such as taking suprema. This allows to build new VC classes, or classes with another VC-property, from more basic classes.

2.6.19 Lemma. *Let $\mathcal{C}$ and $\mathcal{D}$ be VC-classes of sets in a set $\mathcal{X}$ and $\phi: \mathcal{X} \rightarrow \mathcal{Y}$ and $\psi: \mathcal{Z} \rightarrow \mathcal{X}$ fixed functions. Then*

- (i) $\mathcal{C}^c = \{C^c: C \in \mathcal{C}\}$ is VC;
- (ii) $\mathcal{C} \cap \mathcal{D} = \{C \cap D: C \in \mathcal{C}, D \in \mathcal{D}\}$ is VC;
- (iii) $\mathcal{C} \sqcup \mathcal{D} = \{C \cup D: C \in \mathcal{C}, D \in \mathcal{D}\}$ is VC;
- (iv) $\phi(\mathcal{C})$ is VC if ϕ is one-to-one;
- (v) $\psi^{-1}(\mathcal{C})$ is VC;
- (vi) the sequential closure of $\mathcal{C}$ for pointwise convergence of indicator functions is VC.

For VC-classes $\mathcal{C}$ and $\mathcal{D}$ in sets $\mathcal{X}$ and $\mathcal{Y}$,
(vii) $\mathcal{C} \times \mathcal{D}$ is VC in $\mathcal{X} \times \mathcal{Y}$.

Proof. The set $\mathcal{C}^c$ picks out the points of a given set $x_1, \dots, x_n$ that $\mathcal{C}$ does not pick out. Thus if $\mathcal{C}$ shatters a given set of points, so does $\mathcal{C}^c$. This proves (i) (and shows that the dimensions of $\mathcal{C}$ and $\mathcal{C}^c$ are equal). From n points $\mathcal{C}$ can pick out $O(n^{V(\mathcal{C})})$ subsets. From each of these subsets, $\mathcal{D}$ can pick out at most $O(n^{V(\mathcal{D})})$ further subsets. Thus $\mathcal{C} \cap \mathcal{D}$ can pick out $O(n^{V(\mathcal{C})+V(\mathcal{D})})$ subsets. For large n , this is certainly smaller than 2^n . This proves (ii). Next (iii) follows from a combination of (i) and (ii), since $\mathcal{C} \cup \mathcal{D} = (\mathcal{C}^c \cap \mathcal{D}^c)^c$. For (iv) note first that if $\phi(\mathcal{C})$ shatters $\{y_1, \dots, y_n\}$, then each y_i must be in the range of ϕ and there exist $x_1, \dots, x_n$ such that ϕ is a bijection between $x_1, \dots, x_n$ and $y_1, \dots, y_n$. Thus $\mathcal{C}$ must shatter $x_1, \dots, x_n$. For (v) the argument is analogous: if $\psi^{-1}(\mathcal{C})$ shatters $z_1, \dots, z_n$, then all $\psi(z_i)$ must be different and the restriction of ψ to $z_1, \dots, z_n$ is a bijection on its range.

To prove (vi) take any set of points $x_1, \dots, x_n$ and any set $\bar{\mathcal{C}}$ from the sequential closure. If $\bar{\mathcal{C}}$ is the pointwise limit of a net $\mathcal{C}_\alpha$, then for sufficiently large α the equality $1\{\bar{\mathcal{C}}\}(x_i) = 1\{\mathcal{C}_\alpha\}(x_i)$ is valid for each i . For such α the set $\mathcal{C}_\alpha$ picks out the same subset as $\bar{\mathcal{C}}$.

For (vii) note first that $\mathcal{C} \times \mathcal{Y}$ and $\mathcal{X} \times \mathcal{D}$ are VC-classes. Then by (ii) so is their intersection $\mathcal{C} \times \mathcal{D}$. ■

2.6.20 Lemma. Let $\mathcal{F}$ and $\mathcal{G}$ be VC-subgraph classes of functions on a set $\mathcal{X}$ and $g: \mathcal{X} \rightarrow \mathbb{R}$, $\phi: \mathbb{R} \rightarrow \mathbb{R}$, and $\psi: \mathcal{Z} \rightarrow \mathcal{X}$ fixed functions. Then

- (i) $\mathcal{F} \wedge \mathcal{G} = \{f \wedge g: f \in \mathcal{F}, g \in \mathcal{G}\}$ is VC-subgraph;
- (ii) $\mathcal{F} \vee \mathcal{G}$ is VC-subgraph;
- (iii) $\{\mathcal{F} > 0\} = \{\{f > 0\}: f \in \mathcal{F}\}$ is VC;
- (iv) $-\mathcal{F}$ is VC;
- (v) $\mathcal{F} + g = \{f + g: f \in \mathcal{F}\}$ is VC-subgraph;
- (vi) $\mathcal{F} \cdot g = \{fg: f \in \mathcal{F}\}$ is VC-subgraph;
- (vii) $\mathcal{F} \circ \psi = \{f(\psi): f \in \mathcal{F}\}$ is VC-subgraph;
- (viii) $\phi \circ \mathcal{F}$ is VC-subgraph for monotone ϕ .

Proof. The subgraphs of $f \wedge g$ and $f \vee g$ are the intersection and union of the subgraphs of f and g , respectively. Hence (i) and (ii) are consequences of the preceding lemma. For (iii) note that the sets $\{f > 0\}$ are one-to-one images of the intersections of the (open) subgraphs with the set $\mathcal{X} \times \{0\}$. Thus the class $\{\mathcal{F} > 0\}$ is VC by (ii) and (iv) of the preceding lemma.

The subgraphs of the class $-\mathcal{F}$ are the images of the open supergraphs of $\mathcal{F}$ under the map $(x, t) \mapsto (x, -t)$. The open supergraphs are the complements of the closed subgraphs, which are VC by Problem 2.6.10. Now (iv) follows from the previous lemma. For (v) it suffices to note that the subgraphs of the class $\mathcal{F} + g$ shatter a given set of points $(x_1, t_1), \dots, (x_n, t_n)$ if and only if the subgraphs of $\mathcal{F}$ shatter the set $(x_i, t_i - g(x_i))$. The subgraph of the function fg is the union of the

sets

$$\begin{aligned} C^+ &= \{(x, t) : t < f(x)g(x), g(x) > 0\}, \\ C^- &= \{(x, t) : t < f(x)g(x), g(x) < 0\}, \\ C^0 &= \{(x, t) : t < 0, g(x) = 0\}. \end{aligned}$$

It suffices to show that these sets are VC in $(\mathcal{X} \cap \{g > 0\}) \times \mathbb{R}$, $(\mathcal{X} \cap \{g < 0\}) \times \mathbb{R}$, and $(\mathcal{X} \cap \{g = 0\}) \times \mathbb{R}$, respectively (Problem 2.6.12). Now, for instance, $\{i : (x_i, t_i) \in C^-\}$ is the set of indices of the points $(x_i, t_i/g(x_i))$ picked out by the open supergraphs of $\mathcal{F}$. These are the complements of the closed subgraphs and hence form a VC class.

The subgraphs of the class $\mathcal{F} \circ \psi$ are the inverse images of the subgraphs of functions in $\mathcal{F}$ under the map $(z, t) \mapsto (\psi(z), t)$. Thus (v) of the previous lemma implies (vii).

For (viii) suppose that the subgraphs of the class $\phi \circ \mathcal{F}$ shatter the set of points $(x_1, t_1), \dots, (x_n, t_n)$. Choose $f_1, \dots, f_m$ from $\mathcal{F}$ such that the subgraphs of the functions $\phi \circ f_j$ pick out all $m = 2^n$ subsets. For each fixed i , define $s_i = \max\{f_j(x_i) : \phi(f_j(x_i)) \leq t_i\}$. Then $s_i < f_j(x_i)$ if and only if $t_i < \phi \circ f_j(x_i)$, for every i and j , and the subgraphs of $f_1, \dots, f_m$ shatter the points (x_i, s_i) . ■

2.6.21 Lemma. *If $\mathcal{F}$ is VC-major, then the class of functions $h \circ f$, with h ranging over the monotone functions $h : \mathbb{R} \rightarrow \mathbb{R}$ and f over $\mathcal{F}$, is VC-major.*

2.6.22 Lemma. *If $\mathcal{F}$ and $\mathcal{G}$ are VC-hull classes for sets (in particular, uniformly bounded VC-major classes), then $\mathcal{F}\mathcal{G}$ is a VC-hull class for sets.*

Proofs. The set $\{h \circ f > t\}$ can be rewritten as $\{f \cdot > h^{-1}(t)\}$ for a suitable inverse h and $\cdot >$ meaning either $>$ or $\geq$ depending on t . Now the sets $\{f \geq t\}$ with f ranging over $\mathcal{F}$ and t over $\mathbb{R}$ form a VC-class, since they are in the sequential pointwise closure of the same sets defined with strict inequality.

If $\mathcal{F}$ and $\mathcal{G}$ are VC-hull classes for sets, then any function fg can be approximated by a product of two symmetric convex combinations of indicator functions of sets from VC-classes. Since the set of pairwise intersections of sets from two VC-classes is VC, the products of the approximations are symmetric convex combinations of indicators of sets from a VC-class. ■

2.6.23 Example. The set of all monotone functions $f : \mathbb{R} \rightarrow [0, 1]$ is Donsker for every probability measure.

2.6.24 Example. For every strictly increasing function ϕ and arbitrary functions $F \geq 0$ and G , the set of functions $x \mapsto \phi(t)G(x)1\{F(x) \geq t\}$ (with t ranging over $\mathbb{R}$) is VC of dimension $V(\mathcal{F}) \leq 3$.

Indeed, for $G \equiv 1$ this class is contained in the class of functions $x \mapsto s1\{\phi \circ F(x) \geq s\}$. The general case follows from Lemma 2.6.20.

Consequently, the class of functions $x \mapsto t^{k-1}F(x)1\{F(x) \geq t\}$ is Donsker for every measurable function F with $\int F^{2k} dP < \infty$ by the uniform entropy central limit theorem.

2.6.7 Fat Shattering

The VC-dimension $V(\mathcal{F})$ of a class $\mathcal{F}$ of functions $f: \mathcal{X} \rightarrow \mathbb{R}$ is the largest number n for which there exist points $x_1, \dots, x_n$ in $\mathcal{X}$ and $t_1, \dots, t_n$ in $\mathbb{R}$ such that the set of points $(x_1, t_1), \dots, (x_n, t_n)$ in $\mathcal{X} \times \mathbb{R}$ is shattered by the subgraphs of $\mathcal{F}$, i.e. for every subset $I \subset \{1, 2, \dots, n\}$ there exists $f \in \mathcal{F}$ with

$$\begin{cases} f(x_i) > t_i, & \text{if } i \in I, \\ f(x_i) \leq t_i, & \text{if } i \notin I. \end{cases}$$

This notion is purely combinatorial, without quantification of the position of the points (x_i, t_i) , in contrast to the following, which is a “dimension at ε ”.

The *fat-shattering dimension* $V(\varepsilon, \mathcal{F})$ of a class of functions $\mathcal{F}$ at $\varepsilon > 0$ is defined as the largest n such that there exist $x_1, \dots, x_n$ and $t_1, \dots, t_n$ such that for every subset $I \subset \{1, 2, \dots, n\}$ there exists $f \in \mathcal{F}$ with

$$\begin{cases} f(x_i) \geq t_i + \varepsilon, & \text{if } i \in I, \\ f(x_i) \leq t_i, & \text{if } i \notin I. \end{cases}$$

The two cases of inequalities arise if the point (x_i, t_i) is “ ε below” or “above” the graph of f , respectively. We could describe this by saying that the subgraph of f picks out the points $\{(x_i, t_i): i \in I\}$ with *margin* ε . Clearly if this is true for some $\varepsilon > 0$, then the subgraph also picks out these points in the ordinary sense, whence it follows that $V(\varepsilon, \mathcal{F}) \leq V(\mathcal{F})$, for every $\varepsilon > 0$.

The following theorem shows that these smaller numbers can nevertheless be used to bound entropy numbers.

2.6.25 Theorem (Fat shattering). *There exist universal constants D and c such that, for any class $\mathcal{F}$ of functions $f: \mathcal{X} \rightarrow [0, 1]$, any probability measure Q and $0 < \varepsilon < 1$,*

$$N(\varepsilon, \mathcal{F}, L_2(Q)) \leq \left(\frac{2}{\varepsilon}\right)^{DV(c\varepsilon, \mathcal{F})}.$$

Proof. See Mendelson and Vershynin (2003). ■

A comparison of the bound with Theorem 2.6.7, which gives a bound of the order $(1/\varepsilon)^{2V(\mathcal{F})}$ on the covering number, is impossible without further knowledge of the constant D . However, if the fat-shattering dimension is considerably smaller than the VC-dimension, then this may compensate even a large constant D , and the bound of the preceding theorem may be preferable. The following example illustrates that this may be the case. In fact, the example shows that the VC-dimension can be infinite while the fat-shattering dimension is finite for every $\varepsilon > 0$.

2.6.26 Example (Monotone functions). The class $\mathcal{F}$ of monotone functions $f: \mathbb{R} \rightarrow [0, 1]$ is not VC. For instance, its subgraphs can pick out every subset from a set of points $(x_1, t_1), \dots, (x_n, t_n)$ with $x_1 < x_2 < \dots < x_n$ and $0 < t_1 < t_2 < \dots < t_n < 1$. Here choosing the x_i to be increasing is no loss of generality, but choosing the points t_i also increasing is crucial. Because strict increase, without quantification, suffices, the points easily fit within the range $[0, 1]$ of the functions

f . The latter changes if we require fat-shattering. If the points must be picked out with margin ε , then the sequence $t_1 < \dots < t_n$ must increase by steps bigger than ε and only of the order $1/\varepsilon$ of them fit the interval $[0, 1]$. This explains that

$$V(\varepsilon, \mathcal{F}) \leq 1 + \frac{1}{\varepsilon}.$$

The preceding theorem therefore gives the bound of the order $(1/\varepsilon) \log(1/\varepsilon)$ on the entropy of $\mathcal{F}$. This bound is not sharp (see Theorem 2.7.9 and the discussion below), but reasonable, whereas Theorem 2.6.7 is not applicable.

For a precise proof of the bound on the fat-shattering numbers, take any sequence $x_1 < \dots < x_n$ and assume that the subgraphs of $\mathcal{F}$ shatter the points (x_i, t_i) at margin ε , for some $t_1, \dots, t_n$. In particular, there exist two functions f_o and f_e that pick out the odd and even numbered points at margin ε :

$$\begin{cases} f_o(x_i) \geq t_i + \varepsilon, & i \text{ odd}, \\ f_o(x_i) \leq t_i, & i \text{ even}, \end{cases} \quad \begin{cases} f_e(x_i) \geq t_i + \varepsilon, & i \text{ even}, \\ f_e(x_i) \leq t_i, & i \text{ odd}. \end{cases}$$

Then $f_o(x_i) - f_e(x_i) \geq \varepsilon$ for odd i and $f_o(x_i) - f_e(x_i) \leq -\varepsilon$ for even i , and hence the variation of $f_o - f_e$ over $x_1, \dots, x_n$ is at least $(n-1)2\varepsilon$. Since f_o and f_e are monotone with range $[0, 1]$, this variation is at most 2, whence $n-1 \leq 1/\varepsilon$.

The preceding example illustrates the potential advantage of fat-shattering, but also that Theorem 2.6.25 is not sharp in general. An improved bound is available if the fat-shattering dimension satisfies an additional condition.

2.6.27 Theorem (Fat shattering). *For any class $\mathcal{F}$ of functions $f: \mathcal{X} \rightarrow [0, 1]$ with fat-shattering dimension such that $V(d\varepsilon, \mathcal{F}) \leq V(\varepsilon, \mathcal{F})/2$ for some $d > 1$ and every $\varepsilon > 0$, there exists a constant D that depends on d only and a universal constant $c > 0$ such that, for any probability measure Q and $0 < \varepsilon < 1$,*

$$\log N(\varepsilon, \mathcal{F}, L_2(Q)) \leq DV(c\varepsilon, \mathcal{F}).$$

This theorem removes the factor $\log(1/\varepsilon)$ from the bound on the entropy implied by Theorem 2.6.25. The additional condition $V(d\varepsilon, \mathcal{F}) \leq V(\varepsilon, \mathcal{F})/2$ that makes this possible is less innocuous than it may seem at first. For instance, it cannot be satisfied by a nontrivial VC-class, as then a bound $\log N(\varepsilon, \mathcal{F}, L_2(Q)) \leq KV(\mathcal{F})$, free of ε , would result.

The following theorem shows that in the uniform entropy integral the fat-shattering dimension can replace the uniform entropy. In particular, the fat-shattering dimension characterizes the uniform Donsker property of uniformly bounded classes of functions.

2.6.28 Theorem. *There exist positive constants $c < C$ such that, for any class $\mathcal{F}$ of functions $f: \mathcal{X} \rightarrow [0, 1]$,*

$$c \int_0^1 \sqrt{V(\varepsilon, \mathcal{F})} d\varepsilon \leq \int_0^1 \sup_Q \sqrt{\log N(\varepsilon, \mathcal{F}, L_2(Q))} d\varepsilon \leq C \int_0^1 \sqrt{V(\varepsilon, \mathcal{F})} d\varepsilon.$$

Proofs. For proofs of the preceding two theorems, see Rudelson and Vershynin (2006). ■

Problems and Complements

1. Every collection of sets that is pre-Gaussian for every underlying measure is VC.

[Hint: Dudley (1984), 11.4.1.]

2. Unlike the case of sets, the VC-subgraph property does not characterize universal Donsker classes of functions. There exist bounded universal Donsker classes that do not even satisfy the uniform entropy condition.

[Hint: Dudley (1987).]

3. Let $\mathcal{F}$ be a class of measurable functions such that $D(\varepsilon, \mathcal{F}, L_r(Q)) \leq g(\varepsilon)$ for every empirical type probability measure Q and a fixed function $g(\varepsilon)$. Then the bound is valid for every probability measure.

[Hint: If $D(\varepsilon, \mathcal{F}, L_r(P)) = m$, then there are functions $f_1, \dots, f_m$ such that $P|f_i - f_j|^r > \varepsilon^r$ for every $i \neq j$. By the strong law of large numbers, $\mathbb{P}_n|f_i - f_j|^r \rightarrow P|f_i - f_j|^r$ almost surely, for every (i, j) . Thus there exists ω and n such that $\mathbb{P}_n(\omega)|f_i - f_j|^r > \varepsilon^r$, for every $i \neq j$.]

4. There exists a short proof of the fact that for any VC-class of sets $\mathcal{C}$ and $\delta > 0$,

$$N(\varepsilon, \mathcal{C}, L_r(Q)) \leq K \left(\frac{1}{\varepsilon} \right)^{r(V(\mathcal{C}) + \delta)},$$

for any probability measure Q , $r \geq 1$, $0 < \varepsilon < 1$, and a constant K depending on $V(\mathcal{C})$ and δ only.

[Hint: Take any subcollection of sets $C_1, \dots, C_m$ from $\mathcal{C}$ such that $Q(C_i \triangle C_j) > \varepsilon$ for every pair $i \neq j$. Generate a sample $X_1, \dots, X_n$ from Q . Two sets C_i and C_j pick out the same subset from a realization of the sample if and only if no X_k falls in the symmetric difference $C_i \triangle C_j$. If every symmetric difference contains a point of the sample, then all C_i pick out a different subset from the sample. In that case $\mathcal{C}$ picks out at least m subsets from $X_1, \dots, X_n$. The probability that this event does not occur is bounded by

$$\begin{aligned} \sum_{i < j} Q(X_k \notin C_i \triangle C_j \text{ for every } k) &\leq \binom{m}{2} (1 - Q(C_i \triangle C_j))^n \\ &\leq \binom{m}{2} (1 - \varepsilon)^n. \end{aligned}$$

For sufficiently large n , the last expression is strictly less than 1. For such n there exists a set of n points from which $\mathcal{C}$ picks out at least m subsets. In other words, for $n > -\log \binom{m}{2} / \log(1 - \varepsilon)$, one has the first inequality in

$$m \leq \max_{x_1, \dots, x_n} \Delta_n(\mathcal{C}, x_1, \dots, x_n) \leq K n^{V(\mathcal{C})}.$$

The last inequality follows from Corollary 2.6.3 and the constant K depends only on $V(\mathcal{C})$. Since $-\log(1 - \varepsilon) > \varepsilon$, we can take $n = 3(\log m)/\varepsilon$ and obtain

$$m \leq K \left(\frac{3 \log m}{\varepsilon} \right)^{V(\mathcal{C})}.$$

Since $\log m$ is bounded by a constant times m^δ , it follows that $m^{1-\delta}$ is bounded by a constant times $(3/\varepsilon)^{V(\mathcal{C})}$.]

5. The edges of the graph with nodes $\mathcal{Z} \subset \{0, 1\}^n$ representing a VC-class $\mathcal{C}$ of subsets of a set of points $\{x_1, \dots, x_n\}$ can be directed in such a manner that at most $V(\mathcal{C})$ arrows are positively incident with each node.

[Hint: The number of edges of a hereditary collection $\mathcal{C}$ can be enumerated by listing for each node v the edges pointing inward: edges $\{v, w\}$ such that w has a zero in the (one) position in which it differs from v . Since every subset of a hereditary class has at most $V(\mathcal{C})$ points, it follows that $\#\mathcal{C}/\#\mathcal{Z} \leq V(\mathcal{C})$. By repeated application of the operators T_i , an arbitrary collection of sets can be transformed into a hereditary class. The operations do not decrease the quotient $\#\mathcal{C}/\#\mathcal{Z}$: the number of edges may increase, while the number of nodes remains the same. Conclude that for any VC-collection of sets, $\#\mathcal{C}/\#\mathcal{Z} \leq V(\mathcal{C})$. Since a subcollection of a given VC-class is VC of no greater dimension, it follows that $\#\mathcal{C}' \leq \#\mathcal{Z}'V(\mathcal{C})$ for every subgraph $(\mathcal{Z}', \mathcal{C}')$ as well.

Apply Hall's marriage lemma [e.g. Dudley (1989), Theorem 11.6.1, page 318], representing directing an edge $\{v, w\}$ as marrying the edge to one of the nodes v and w . In fact, let the eligible partners for $\{v, w\}$ be the collection of $V(\mathcal{C})$ copies of v and $V(\mathcal{C})$ copies of w . Then the total number of partners for a given set of edges is at least $V(\mathcal{C})$ times the number of edges. By the marriage lemma, successful marriage is possible.]

6. For every pair of integers $S, r \geq 1$, and $n = Sr$, there exists a subset $\mathcal{Z} \subset \{0, 1\}^n$ such that, for all $1 \leq k \leq n$ and $S = V(\mathcal{Z})$,

$$D(k/n, \mathcal{Z}, d) \geq \left(\frac{n}{2e(k+S)} \right)^S.$$

[Hint: Start with the subset $\mathcal{W} = \{(0, 0, \dots, 0), (1, 0, \dots, 0), (1, 1, \dots, 0), \dots, (1, 1, 1, \dots, 1)\}$ of $\{0, 1\}^r$, and let $\mathcal{Z} = \mathcal{W}^S$ be the set of all vectors in $\{0, 1\}^n$ obtained by concatenating S vectors from $\mathcal{W}$.]

7. Every VC-class $\mathcal{C}$ of sets satisfies $\Delta_n(\mathcal{C}, x_1, \dots, x_n) \leq (ne/V(\mathcal{C}))^{V(\mathcal{C})}$ for $n \geq V(\mathcal{C})$.

[Hint: Consider a random variable Y with a binomial distribution with parameters n and $1/2$. Bound $P(Y \leq k)$ by Er^{Y-k} for a simple (not optimal) choice of r .]

8. Let Q be a finitely discrete probability measure supported on $x_1, \dots, x_n$. Then for any collection of sets, one has $\Delta_n(\mathcal{C}, x_1, \dots, x_n) = N(\varepsilon, \mathcal{C}, L_r(Q))$ for all $\varepsilon \leq \inf Q\{x\}^{1/r}$.

[Hint: The L_r -distance equals $Q^{1/r}(C \triangle D)$ and can be less than a sufficiently small ε only if $C \triangle D$ does not contain a support point.]

9. If a collection of sets $\mathcal{C}$ is a VC-class, then the collection of indicators of sets in $\mathcal{C}$ is a VC-subgraph class of the same index.

10. (**Open and closed subgraphs**) For a set $\mathcal{F}$ of measurable functions, define “closed” and “open” subgraphs by $\{(x, t): t \leq f(x)\}$ and $\{(x, t): t < f(x)\}$, respectively. Then the collection of “closed” subgraphs has the same VC-dimension as the collection of “open” subgraphs. Consequently, “closed” and “open” are equivalent in the definition of a VC-subgraph class.

[Hint: Suppose the “closed” subgraphs shatter the set $(x_1, t_1), \dots, (x_n, t_n)$. Choose $f_1, \dots, f_m$ whose “closed” subgraphs pick out all $m = 2^n$ subsets. Set $2\varepsilon = \inf\{t_i - f_j(x_i): t_i - f_j(x_i) > 0\}$. Then the “open” subgraphs shatter the set $(x_1, t_1 - \varepsilon), \dots, (x_n, t_n - \varepsilon)$. The converse can be argued in a similar manner.]

- 11. (Between graphs)** For a set $\mathcal{F}$ of measurable functions, define the “between” graphs as the sets $\{(x, t): 0 \leq t \leq f(x) \text{ or } f(x) \leq t \leq 0\}$. The “between” graphs form a VC-class of sets if and only if $\mathcal{F}$ is a VC-subgraph class.

[Hint: The “closed” subgraphs intersected with the set $\{t \geq 0\}$ yields a VC-class of sets which is the positive half of the “between” graphs. The “open” subgraphs intersected with the set $\{t \leq 0\}$ and next complemented within $\{t \leq 0\}$ give the lower parts of the “between” graphs.]

- 12.** If $\mathcal{X}$ is the union of finitely many disjoint sets $\mathcal{X}_i$, and $\mathcal{C}_i$ is a VC-class of subsets of $\mathcal{X}_i$ for each $i, i = 1, \dots, m$, then $\cup_{i=1}^m \mathcal{C}_i$ is a VC-class in $\cup_{i=1}^m \mathcal{X}_i$ of dimension $\sum_{i=1}^m V(\mathcal{C}_i)$.

- 13.** If $\mathcal{F}$ is a VC-major class, then the sets $\{x: f(x) \geq t\}$, with f ranging over $\mathcal{F}$ and t over $\mathbb{R}$, form a VC-class.

[Hint: The set $\{x: f(x) \geq t\}$ is the intersection of the sets $\{x: f(x) \geq t - n^{-1}\}$. Use Lemma 2.6.19(vi).]

- 14.** The collection of all *half-spaces* in $\mathbb{R}^d$ is a VC-class of dimension $d + 1$. A half-space is a set of the form $\{x \in \mathbb{R}^d: \langle x, u \rangle \leq c\}$ for fixed $u \in \mathbb{R}^d$ and $c \in \mathbb{R}$. The collection of all closed balls in $\mathbb{R}^d$ is a VC-class of dimension $d + 1$.

[Hint: Use Lemma 2.6.16. See Dudley (1979).]

- 15.** The class of all closed convex subsets in $\mathbb{R}^d$ is not VC for $d \geq 2$. The same is true for the collection of all open convex sets.

[Hint: Any set of n points on the rim of the unit ball is shattered by the closed convex sets. The closed convex sets are in the sequential closure of the open convex sets.]

- 16.** The set of all open polygons with extreme points on the rim of the unit circle in $\mathbb{R}^2$ is not VC.

[Hint: Form a regular polygon with its n extreme points on the rim of the unit circle. Choose n points in the n gaps between the polygon and the unit circle.]

- 17.** If $\mathcal{C}$ is a VC-class of subsets of $\mathcal{X}$ and $\phi: \mathcal{X} \rightarrow \mathcal{Y}$ is an arbitrary map, then $\phi(\mathcal{C})$ need not be VC.

[Hint: Take $\mathcal{X} = \mathcal{Y} = \mathbb{N}$ with $\phi(2n-1) = \phi(2n) = n$. Let $\mathcal{C}$ be the collection consisting of the empty set, all singletons of odd integers, and all finite subsets of even integers with more than two elements: $\mathcal{C} = \{\emptyset, \{1\}, \{3\}, \dots, \{2, 4\}, \{2, 6\}, \dots, \{2, 4, 6\}, \dots\}$. Show that no subset of two points in $\mathcal{X}$ is shattered, while $\phi(\mathcal{C})$ consists of all finite subsets of $\mathbb{N}$.]

- 18.** The set of all monotone functions $f: \mathbb{R} \rightarrow [0, 1]$ is VC-hull but not VC-subgraph.

[Hint: Any set of points (x_i, t_i) with both x_i and t_i strictly increasing in i is shattered.]

- 19.** For a VC-subgraph class $\mathcal{F}$, the class $\{f - Pf: f \in \mathcal{F}\}$ is not necessarily VC-subgraph.

[Hint: For any countable collection of functions $g_n: \mathcal{X} \rightarrow [0, 1]$, the subgraphs of the collection $g_n + n$ form a linearly ordered set. Hence the functions $g_n + n$ form a VC-subgraph class.]

20. The class of functions of the form $x \mapsto c1_{(a,b]}(x)$ with a, b , and $c > 0$ ranging over $\mathbb{R}$ is VC of dimension 2.

[Hint: Any f whose subgraph picks out the subset $\{(x_1, t_1), (x_3, t_3)\}$ from three points (x_i, t_i) with $x_1 \leq x_2 \leq x_3$ also picks out (x_2, t_2) unless $t_2 > t_1 \vee t_3$. If a set of four points with nondecreasing x_i is shattered, it follows that $t_2 > t_1 \vee t_3$, but also $t_3 > t_2 \vee t_4$, $t_2 > t_1 \vee t_4$, and $t_3 > t_1 \vee t_4$.]

21. The “Box-Cox family of transformations” $\mathcal{F} = \{f_\lambda: (0, \infty) \rightarrow \mathbb{R}: \lambda \in \mathbb{R} - \{0\}\}$, with $f_\lambda(x) = (x^\lambda - 1)/\lambda$, is a VC-subgraph class.

[Hint: The “between” graph class of sets (as defined in Exercise 2.6.11) is the class of subsets $\mathcal{C} = \{C_\lambda: \lambda \neq 0\}$ of $\mathbb{R}^2$, where $C_\lambda = \{(x, t): 0 \leq t \leq (x^\lambda - 1)/\lambda\}$. Examine the “dual class” of subsets of $\mathbb{R}$ given by $\mathcal{D} = \{D_{(x,t)}: (x, t) \in (0, \infty) \times (0, \infty)\}$, where

$$D_{(x,t)} = \{\lambda \neq 0: (x, t) \in C_\lambda\} = \{\lambda \neq 0: 0 \leq t \leq (x^\lambda - 1)/\lambda\}.$$

Show that $\mathcal{D}$ is a VC-class of sets, and apply Assouad (1983), page 246, Proposition 2.12.]

22. The VC-dimension of a collection Θ of functions $\theta: \mathcal{X} \rightarrow \{0, 1\}$ on some arbitrary set $\mathcal{X}$ is equal to the VC-dimension of the collection $\{C_\theta: \theta \in \Theta\}$ of subsets $C_\theta = \{(x, y): y = \theta(x)\}$ of $\mathcal{X} \times \{0, 1\}$.

23. For VC-classes $\mathcal{C}_1, \dots, \mathcal{C}_m$ of subsets of a set $\mathcal{X}$, of VC-dimensions $V_1, \dots, V_m$, the VC-dimension of the collection of all intersections $C_1 \cap \dots \cap C_m$ and of the collection of all unions $C_1 \cup \dots \cup C_m$ of sets $C_i \in \mathcal{C}_i$ is bounded above by

$$c_1 m \bar{V} (\log(c_2 m) - \text{Ent}(\bar{V})/\bar{V}) \leq c_1 m \bar{V} \log(c_2 m),$$

where $\bar{V} = (V_1, \dots, V_m)$, $\bar{V} = m^{-1} \sum V_i$, $c_1 = e/((e-1)\log 2) \approx 2.28231$ and $c_2 = e/\log 2 \approx 3.92165$. The same is true for the collection of all Cartesian products $C_1 \times \dots \times C_m$ of sets C_i belonging to classes $\mathcal{C}_i$ of subsets of sets $\mathcal{X}_i$.

[Hint: Van der Vaart and Wellner (2009)]

24. If a class $\mathcal{F}$ of functions is closed under scalar multiplication (if $f \in \mathcal{F}$, then $cf \in \mathcal{F}$ for every $c \in \mathbb{R}$), then $V(\varepsilon, \mathcal{F}) = V(\mathcal{F})$, for every $\varepsilon > 0$.

2.7

Entropies of Function Classes

While the VC-theory gives control over the uniform entropy numbers of many interesting classes through simple combinatorial arguments, results on bracketing numbers can be found in approximation theory. This chapter gives bracketing numbers of the most important classes. In some cases, the bracketing numbers are actually uniform in the underlying measure.

2.7.1 Hölder Classes

Hölder spaces provide the simplest method for defining smoothness of functions. In recent years they have been superseded by the Besov spaces, which provide a more refined scale of regularity levels. In this section we develop the classical results for Hölder spaces, deferring a treatment of Besov spaces (which imply the results obtained in the present section) to the next section.

For given $\alpha > 0$ we study the class of all functions on a bounded set $\mathcal{X}$ in $\mathbb{R}^d$ that possess uniformly bounded partial derivatives up to order $\underline{\alpha}$ (the greatest integer strictly smaller than α) and whose highest partial derivatives are Lipschitz of order $\alpha - \underline{\alpha}$. A simple example is the set of Lipschitz functions of some order $0 < \alpha \leq 1$ (for which $\underline{\alpha} = 0$ even if $\alpha = 1$!).

For a more precise description, define for any vector $k = (k_1, \dots, k_d) \in \mathbb{N}_0^d$ of d nonnegative integers the differential operator

$$D^k = \frac{\partial^k}{\partial x_1^{k_1} \dots \partial x_d^{k_d}},$$

where $k_{\cdot} = \sum k_i$. Then for a function $f: \mathcal{X} \rightarrow \mathbb{R}$, let

$$\|f\|_{\alpha} = \max_{0 \leq k_{\cdot} \leq \underline{\alpha}} \sup_x |D^k f(x)| + \max_{k_{\cdot} = \alpha} \sup_{x, y} \frac{|D^k f(x) - D^k f(y)|}{\|x - y\|^{\alpha - \underline{\alpha}}},$$

where the suprema are taken over all x, y in the interior of $\mathcal{X}$ with $x \neq y$. The Hölder space $C^\alpha(\mathcal{X})$ of order α is the space of all continuous functions $f: \mathcal{X} \rightarrow \mathbb{R}$ for which the norm $\|f\|_\alpha$ is well defined and finite.^{†‡b} Let $C^\alpha(\mathcal{X})_M$ be the subset of all functions with $\|f\|_\alpha \leq M$, so that $C^\alpha(\mathcal{X})_1$ is the unit ball of the space.

Bounds on the entropy numbers of the classes $C^\alpha(\mathcal{X})_1$ with respect to the supremum norm $\|\cdot\|_\infty$ were among the first known after the introduction of the concept of covering numbers. These readily yield bounds on the $L_r(Q)$ -bracketing numbers for the present classes of functions, as well as for the class of subgraphs corresponding to them. (These sets with “smooth boundaries” are considered in Section 2.7.8.)

2.7.1 Theorem (Hölder space). *Let $\mathcal{X}$ be a bounded, convex subset of $\mathbb{R}^d$ with nonempty interior. There exists a constant K depending only on α and d such that*

$$\log N(\varepsilon, C^\alpha(\mathcal{X})_1, \|\cdot\|_\infty) \leq K \lambda(\mathcal{X}^1) \left(\frac{1}{\varepsilon}\right)^{d/\alpha},$$

for every $\varepsilon > 0$, where $\lambda(\mathcal{X}^1)$ is the Lebesgue measure of the set $\{x: \|x - \mathcal{X}\| < 1\}$.

Proof. Write $\lesssim$ for “less than equal a constant times,” where the constant depends on α and d only. Let β denote the greatest integer strictly smaller than α .

Since the functions in $C^\alpha(\mathcal{X})_1$ are continuous on $\mathcal{X}$ by assumption, it may be assumed without loss of generality that $\mathcal{X}$ is open, so that Taylor’s theorem applies everywhere on $\mathcal{X}$.

Fix $\delta = \varepsilon^{1/\alpha} \leq 1$, and form a δ -net for $\mathcal{X}$ of points $x_1, \dots, x_m$ contained in $\mathcal{X}$. The number m of points can be taken to satisfy $m \lesssim \lambda(\mathcal{X}^1)/\delta^d$. For each vector $k = (k_1, \dots, k_d)$ with $k_i \leq \beta$, form for each f the vector

$$A_k f = \left(\left\lfloor \frac{D^k f(x_1)}{\delta^{\alpha-k}} \right\rfloor, \dots, \left\lfloor \frac{D^k f(x_m)}{\delta^{\alpha-k}} \right\rfloor \right).$$

Then the vector $\delta^{\alpha-k} A_k f$ consists of the values $D^k f(x_i)$ discretized on a grid of mesh-width $\delta^{\alpha-k}$.

If a given pair of functions f and g satisfy $A_k f = A_k g$ for each k with $k_i \leq \beta$, then $\|f - g\|_\infty \lesssim \varepsilon$. Indeed, for each x there exists an x_i with $\|x - x_i\| \leq \delta$. By Taylor’s theorem,

$$(f - g)(x) = \sum_{k_i \leq \beta} D^k (f - g)(x_i) \frac{(x - x_i)^k}{k!} + R,$$

[†] Note the inconsistency in notation. In the present section the notation $\|\cdot\|_\alpha$ is used only with $\alpha < \infty$ to define the norm of the Hölder space $C^\alpha(\mathcal{X})$. The notation $\|\cdot\|_\infty$ always denotes the supremum norm of $\ell^\infty(\mathcal{X})$. Clearer notations are $\|\cdot\|_{C^\alpha(\mathcal{X})}$ or $\|\cdot\|_{C^\alpha(\mathcal{X})}$, but we shall here and in the following opt for brevity.

[‡] The space $C^1(\mathcal{X})$ is identical to the space $BL(\mathcal{X})$ encountered elsewhere in the manuscript, and is also the space $Lip^1_\infty(\mathcal{X})$ introduced in the next section.

^b Because finiteness of $\|f\|_\alpha$ implies that $f: \mathcal{X} \rightarrow \mathbb{R}$ is uniformly continuous, the extension to a continuous function on the closure $\overline{\mathcal{X}}$ of the domain comes for free. For integer α the notation $C^\alpha(\mathcal{X})$ is sometimes used for the set of functions $f: \mathcal{X} \rightarrow \mathbb{R}$ that have partial derivatives up to order α , which is a bigger space than here defined if $\mathcal{X}$ is open, as differentiability everywhere in an open domain does not imply good behaviour near the boundary (see e.g. Adams and Fournier (2003), page 10, who write $C^{\alpha, \alpha-\alpha}(\overline{\mathcal{X}})$ for our present $C^\alpha(\mathcal{X})$).

where $|R| \lesssim \|x - x_i\|^\alpha$, because the highest-order partial derivatives are uniformly bounded. In the multidimensional case, the notation $h^k/k!$ is short for $\prod_{i=1}^d h_i^{k_i}/k_i!$. Thus

$$|f - g|(x) \lesssim \sum_{k, \leq \beta} \delta^{\alpha-k} \cdot \frac{\delta^k}{k!} + \delta^\alpha \leq \delta^\alpha (e^d + 1).$$

It follows that there exists a constant C depending on α and d only such that the covering number $N(C\varepsilon, C^\alpha(\mathcal{X})_1, \|\cdot\|_\infty)$ is bounded by the number of different matrices

$$Af = \begin{pmatrix} A_{0,0,\dots,0}f \\ A_{1,0,\dots,0}f \\ \vdots \\ A_{0,0,\dots,\beta}f \end{pmatrix},$$

when f ranges over $C^\alpha(\mathcal{X})_1$. Each row in the matrix Af is one of the vectors $A_k f$ for k with $k, \leq \beta$. Hence the number of rows is certainly smaller than $(\beta + 1)^d$. By definition of each $A_k f$ and the fact that $|D^k f(x_i)| \leq 1$ for each i , the number of possible values of each element in the row $A_k f$ is bounded by $2/\delta^{\alpha-k} + 2$, which does not exceed $2\delta^{-\alpha} + 2$. Thus each column of the matrix can have at most $(2\delta^{-\alpha} + 2)^{(\beta+1)^d}$ different values.

Assume without loss of generality that $x_1, \dots, x_m$ have been chosen and ordered in such a way that for each $j > 1$ there is an index $i < j$ such that $\|x_i - x_j\| < 2\delta$. Then use the crude bound obtained previously for the first column only. For each later column, indexed by x_j , there exists a previous x_i with $\|x_i - x_j\| < 2\delta$. By Taylor's theorem,

$$D^k f(x_j) = \sum_{k, +l, \leq \beta} D^{k+l} f(x_i) \frac{(x_i - x_j)^l}{l!} + R,$$

where $|R| \lesssim \|x_i - x_j\|^{\alpha-k}$. Thus with $B_k f = \delta^{\alpha-k} \cdot A_k f$,

$$\begin{aligned} & \left| D^k f(x_j) - \sum_{k, +l, \leq \beta} B_{k+l} f(x_i) \frac{(x_i - x_j)^l}{l!} \right| \\ & \lesssim \sum_{k, +l, \leq \beta} |B_{k+l} f(x_i) - D^{k+l} f(x_i)| \frac{\|x_i - x_j\|^l}{l!} + \delta^{\alpha-k}. \\ & \lesssim \sum_{k, +l, \leq \beta} \delta^{\alpha-k-l} \cdot \frac{\delta^l}{l!} + \delta^{\alpha-k} \lesssim \delta^{\alpha-k}. \end{aligned}$$

Thus given the values in the i th column of Af , the values $D^k f(x_j)$ range over an interval of length proportional to $\delta^{\alpha-k}$. It follows that the values in the j th column of Af range over integers in an interval of length proportional to $\delta^{k, -\alpha} \delta^{\alpha-k} = 1$. Consequently, there exists a constant C depending only on α and d such that

$$\#Af \leq (2\delta^{-\alpha} + 2)^{(\beta+1)^d} C^{m-1}.$$

The theorem follows upon replacing δ by $\varepsilon^{1/\alpha}$ and m by its upper bound $\lambda(\mathcal{X}^1) \varepsilon^{-d/\alpha}$, respectively, taking logarithms, and bounding $\log(1/\varepsilon)$ by a constant times $(1/\varepsilon)^{d/\alpha}$. ■

2.7.2 Corollary. *Let $\mathcal{X}$ be a bounded, convex subset of $\mathbb{R}^d$ with nonempty interior. There exists a constant K depending only on α , $\text{diam } \mathcal{X}$, and d such that*

$$\log N_{[]}(\varepsilon, C^\alpha(\mathcal{X})_1, L_r(Q)) \leq K \left(\frac{1}{\varepsilon} \right)^{d/\alpha},$$

for every $r \geq 1$, $\varepsilon > 0$, and probability measure Q on $\mathbb{R}^d$.

Proof. Let $f_1, \dots, f_p$ be the centers of $\|\cdot\|_\infty$ -balls of radius ε that cover $C^\alpha(\mathcal{X})_1$. Then the brackets $[f_i - \varepsilon, f_i + \varepsilon]$ cover $C^\alpha(\mathcal{X})_1$. Each bracket has $L_r(Q)$ -size at most 2ε . By the previous theorem, $\log p$ can be chosen smaller than the given polynomial in $1/\varepsilon$. ■

The corollary, together with either the bracketing central limit theorem or the uniform entropy theorem, implies that $C^\alpha([0, 1]^d)_1$ is universally Donsker for $\alpha > d/2$. For instance, on the unit interval in the line, uniformly bounded and uniformly Lipschitz of order $> 1/2$ suffices and in the unit square it suffices that the partial derivatives exist and satisfy a Lipschitz condition.

The previous results are restricted to bounded subsets of Euclidean space. Under appropriate conditions on the tails of the underlying distributions they can be extended to classes of functions on the whole of Euclidean space. The general conclusion is that under a weak tail condition, the same amount of smoothness suffices for the class to be Donsker. For instance, for any distribution on the line with a finite $2 + \delta$ moment, the class of all uniformly bounded and uniformly Lipschitz functions of order $\alpha > 1/2$ has a finite bracketing integral and hence is Donsker. A more general result is an easy corollary to the first theorem of this section also.

2.7.3 Corollary (Hölder space–unbounded domain). *Let $\mathbb{R}^d = \cup_{j=1}^\infty \mathcal{X}_j$ be a partition of $\mathbb{R}^d$ into bounded, convex sets with nonempty interior, and let $\mathcal{F}$ be a class of functions $f: \mathbb{R}^d \rightarrow \mathbb{R}$ such that the restrictions $\mathcal{F}|_{\mathcal{X}_j}$ belong to $C^\alpha(\mathcal{X}_j)_{M_j}$ for every j . Then there exists a constant K depending only on α , V , r , and d such that*

$$\log N_{[]}(\varepsilon, \mathcal{F}, L_r(Q)) \leq K \left(\frac{1}{\varepsilon} \right)^V \left(\sum_{j=1}^\infty \lambda(\mathcal{X}_j^1)^{\frac{r}{V+r}} M_j^{\frac{Vr}{V+r}} Q(\mathcal{X}_j)^{\frac{V}{V+r}} \right)^{\frac{V+r}{r}},$$

for every $\varepsilon > 0$, $V \geq d/\alpha$, and probability measure Q .

Proof. Let a_j be any sequence of numbers in $(0, \infty]$. For each $j \in \mathbb{N}$, take an εa_j -net $f_{j,1}, \dots, f_{j,p_j}$ for $C_{M_j}^\alpha(\mathcal{X}_j)$ for the uniform norm on $\mathcal{X}_j$. By Theorem 2.7.1, p_j can be chosen to satisfy

$$\log p_j \leq K \lambda(\mathcal{X}_j^1) \left(\frac{M_j}{\varepsilon a_j} \right)^{d/\alpha},$$

for a constant K depending on d and α only. It is clear that, for $\varepsilon a_j > M_j$, the value p_j can be chosen equal to 1. Now form the brackets

$$\left[\sum_{j=1}^\infty (f_{j,i_j} - \varepsilon a_j) 1\{\mathcal{X}_j\}, \sum_{j=1}^\infty (f_{j,i_j} + \varepsilon a_j) 1\{\mathcal{X}_j\} \right],$$

where the sequence $i_1, i_2, \dots$ ranges over all possible values: for each j the integer i_j ranges over $\{1, 2, \dots, p_j\}$. The number of brackets is bounded by $\prod p_j$, and each bracket has $L_r(Q)$ -size equal to $2\varepsilon(\sum a_j^r Q(\mathcal{X}_j))^{1/r}$. It follows that

$$\begin{aligned} \log N_{[]} \left(2\varepsilon \left(\sum a_j^r Q(\mathcal{X}_j) \right)^{1/r}, \mathcal{F}, L_r(Q) \right) &\lesssim K \left(\frac{1}{\varepsilon} \right)^{d/\alpha} \sum_{\substack{j=1 \\ a_j \varepsilon \leq M_j}}^{\infty} \lambda(\mathcal{X}_j^1) \left(\frac{M_j}{a_j} \right)^{d/\alpha} \\ &\leq K \left(\frac{1}{\varepsilon} \right)^V \sum_{j=1}^{\infty} \lambda(\mathcal{X}_j^1) \left(\frac{M_j}{a_j} \right)^V, \end{aligned}$$

for every $V \geq d/\alpha$. The choice $a_j^{V+r} = \lambda(\mathcal{X}_j^1) M_j^V / Q(\mathcal{X}_j)$ reduces both series in this expression to essentially the series in the statement of the corollary. Simplify to obtain the result. ■

The preceding upper bound may be applied to obtain a simple sufficient condition for classes of smooth functions on the whole space to have a finite bracketing integral. As an example, consider the class $C^\alpha(\mathbb{R})_1$. If $\alpha > 1/2$ and $\sum_{j=1}^{\infty} P(\mathcal{X}_j)^s < \infty$ for a partition of $\mathbb{R}$ into intervals of fixed length and some $s < 1/2$, then for sufficiently small $\delta > 0$,

$$\log N_{[]}(\varepsilon, C^\alpha(\mathbb{R})_1, L_2(P)) \leq K \left(\frac{1}{\varepsilon} \right)^{2-\delta}$$

(for a constant K depending on P , α , and δ). Consequently, the class $C^\alpha(\mathbb{R})_1$ has a finite bracketing integral and is P -Donsker. Convergence of the series is implied by a tail condition: the series converges for some $s < 1/2$ if $\int |x|^{2+\delta} dP(x) < \infty$ for some $\delta > 0$.

The bound given by the preceding corollary can be proved to be the best in terms of a power of $(1/\varepsilon)$, but it is not sharp in terms of lower-order terms. As a consequence, the best conditions for the class $\mathcal{F}$ as in the corollary to be Glivenko-Cantelli or Donsker cannot be obtained from the corollary. The class is known to be Glivenko-Cantelli or Donsker if and only if $\sum_{j=1}^{\infty} M_j P(\mathcal{X}_j) < \infty$ or $\sum_{j=1}^{\infty} M_j P^{1/2}(\mathcal{X}_j) < \infty$, respectively. See Section 2.10.5 and the Notes at the end of Part 2 for further details.

2.7.2 Sobolev and Besov Spaces

Next to Hölder spaces, *Sobolev spaces* form another classical scale of function classes that capture smoothness. Sobolev spaces find extensive application in the theory of differential equations. In this section we treat them along with *Besov spaces*, which provide a still more refined, and more complicated, sense of smoothness. While Sobolev spaces are indexed by two parameters, a smoothness index α and a parameter that defines the way smoothness is measured, Besov spaces are indexed by three parameters, one of which is a smoothness index. For the limiting values of the auxiliary parameters, the spaces are slightly bigger than the Hölder spaces of corresponding α , but other values give more refined ways of measuring

smoothness. Sobolev and Besov spaces can be defined for smoothness $\alpha \in \mathbb{R}$, but we restrict to nonnegative levels.

Both Sobolev and Besov spaces can be defined in terms of continuity moduli, or alternatively, by expansions on suitable bases. These definitions can be roughly characterized as being in the space domain or in the frequency domain. Actually *wavelet* expansions for Besov spaces combine the two descriptions, resulting in a “multi-resolution analysis”. In this section we start with definitions of the classes in the space domain, and next describe wavelet expansions. The entropy estimates are based on an analysis of the coefficients in the wavelet expansion.

The main theorem of this section gives a bound on the covering numbers of a Besov ball relative to the L_τ -norm, for given $1 \leq \tau \leq \infty$. Sobolev balls can be embedded in Besov balls, with implied bounds on their covering numbers. Certain combinations of smoothness and dimension allow to choose $\tau = \infty$, resulting in uniform balls and hence bracketing numbers. The entropy estimates possess the same order $(1/\varepsilon)^{d/\alpha}$ as for the Hölder spaces $C^\alpha(\mathcal{X})$, but are more general in allowing a variety of combinations of smoothness levels.

Throughout the section we consider functions $f: \mathcal{X} \rightarrow \mathbb{R}$ on a given subset $\mathcal{X} \subset \mathbb{R}^d$. To simplify the exposition, we always assume that this is a *Lipschitz domain*. Here a *domain* is simply an open set, and “Lipschitz” refers to the boundary $\partial\mathcal{X}$ of the domain, which should be sufficiently regular. For instance, every bounded open rectangle or bounded open convex set is a suitable domain, and so is the full space $\mathcal{X} = \mathbb{R}^d$. Formally a bounded domain is a *Lipschitz domain* if its boundary can be covered by finitely many balls centered at boundary points such that the intersection $B \cap \mathcal{X}$ of every of these balls B with the domain is equal to the intersection $B \cap S$ for S a rotation of the supergraph of a Lipschitz function on $\mathbb{R}^{d-1}$ (see Figure 2.7.1).[‡]

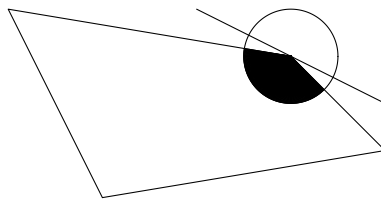


Figure 2.7.1. A quadrangle is a Lipschitz domain in $\mathbb{R}^2$. The shaded area is the rotated supergraph of a suitable function on $\mathbb{R}$ intersected with a ball centered at a boundary point. The boundary can be covered with finitely balls of this type. At the center of the ball shown, the boundary is not smooth, but it is Lipschitz. At all but four boundary points the function defining the supergraph can be chosen infinitely smooth, in fact constant.

[‡] See Adams and Fournier (2003), Chapter 4, or Triebel (2008), Definition 3.4(iii), where also several alternative conditions are discussed. For an unbounded domain, with unbounded boundary, the definition must be more complicated. A general open set $\mathcal{X}$ is called a *strong local Lipschitz domain* if its boundary can be covered by countably many open sets B such that any k sets have empty intersection, for some given k , any two points within distance ε of the boundary $\partial\mathcal{X}$ that are less than ε apart belong to the inner ε -shell of one of the sets B (the points in B at distance more than ε of its boundary), for some fixed $\varepsilon > 0$, and the intersection $B \cap \mathcal{X}$ is equal to intersection $B \cap S$ for S a rotated supergraph of a Lipschitz function. See Adams and Fournier (2003), Section 4.9 on page 83.

The scale of *Sobolev spaces* is indexed by two parameters, a level of regularity $\alpha \geq 0$, and a parameter $p \in [1, \infty]$ indicating the norm by which the regularity is measured. The Sobolev spaces indexed by $p = 2$ are the simplest examples; they are Hilbert spaces.

For a function f and vector $k = (k_1, \dots, k_d) \in \mathbb{N}_0^d$, let $D^k f$ denote the k th partial derivative, and set $k_\cdot = \sum_i k_i$, as in Section 2.7.1. Presently we understand the derivative in the weak sense. The function $f: \mathcal{X} \rightarrow \mathbb{R}$ is said to possess a *weak derivative* of order k if there exists a locally integrable function on $\mathcal{X}$, denoted $D^k f$, such that $\int (D^k f) \phi d\lambda = (-1)^{|k|} \int f(D^k \phi) d\lambda$, for every infinitely often differentiable function ϕ with compact support in $\mathcal{X}$. By partial integration it is seen that the weak derivatives of a sufficiently smooth function f coincide with the ordinary derivatives. The extension makes the Sobolev spaces, as defined below, into Banach spaces, and is also convenient when manipulating functions from Sobolev spaces, for instance when finding solutions to partial differential equations, but is not important in the following.

For $\alpha \in \mathbb{N}_0$ and $1 \leq p \leq \infty$, and a function $f: \mathcal{X} \rightarrow \mathbb{R}$, let

$$\|f\|_{\alpha|p} := \max_{0 \leq k_\cdot \leq \alpha} \|D^k f\|_{\lambda,p}.$$

where $\|\cdot\|_{\lambda,p}$ denotes the L_p -norm relative to the Lebesgue measure λ on $\mathcal{X}$. The *Sobolev space* $W_p^\alpha(\mathcal{X})$ is defined to be the subset of $L_p(\mathcal{X}, \lambda)$ of functions for which the norm $\|f\|_{\alpha|p}$ is finite. It can be shown that for $1 \leq p < \infty$, this is the same as the completion of the set of functions $\{f \in C^\alpha(\mathcal{X}): \|f\|_{\alpha|p} < \infty\}$ for the norm $\|\cdot\|_{\alpha|p}$.[†] These Sobolev spaces of integer order can be extended to arbitrary $\alpha > 0$ via interpolation (as $[L_p(\mathcal{X}), W_p^r(\mathcal{X})]_{\alpha/r}$ for r the smallest integer greater than α), but we shall not discuss this operation here.[‡] An alternative extension employs the difference functions $(x, y) \mapsto f(x) - f(y)$ on $\mathcal{X} \times \mathcal{X}$. For $0 < \alpha < 1$ and $1 \leq p < \infty$, the norm of $W_p^\alpha(\mathcal{X})$ is defined as

$$\|f\|_{\alpha|p} = \|f\|_{\lambda,p} + \left(\int_{\mathcal{X} \times \mathcal{X}} \frac{|f(x) - f(y)|^p}{\|x - y\|^{p\alpha+d}} dx dy \right)^{1/p}.$$

Next for a noninteger $\alpha > 1$ the space $W_p^\alpha(\mathcal{X})$ is defined to be the set of functions $f \in W_p^\alpha(\mathcal{X})$ with $D^k f \in W_p^{\alpha-\alpha|p}(\mathcal{X})$, for every k with $k_\cdot = \underline{\alpha}$, and with norm the maximum of $\|f\|_{\underline{\alpha}|p}$ and all norms $\|D^k f\|_{\alpha-\alpha|p}$.^b

The Sobolev spaces $W_2^\alpha(\mathcal{X})$ are commonly denoted by $H^\alpha(\mathcal{X})$, where the letter H can be motivated by the fact that they are Hilbert spaces for the inner product[‡]

$$\langle f, g \rangle_{\alpha|2} = \sum_{0 \leq k_\cdot \leq \alpha} \int_{\mathcal{X}} (D^k f)(D^k g) d\lambda.$$

[†] See e.g. Adams and Fournier (2003), page 60 and Theorem 3.17 on page 67. We assume a Lipschitz domain. See the same reference for other results on approximation by smooth functions. For instance, the set of restrictions to $\mathcal{X}$ of the collection $C_c^\infty(\mathbb{R}^d)$ of infinitely often differentiable functions $f: \mathbb{R}^d \rightarrow \mathbb{R}$ with compact support is dense in $W_p^\alpha(\mathcal{X})$, for $1 \leq p < \infty$ (see Theorem 3.22), which can be strengthened to functions with compact support within $\mathcal{X}$ if and only if $\mathcal{X}^c$ is $(\alpha, p/(p-1))$ -polar (see Theorem 3.37).

[‡] See e.g. Adams and Fournier (2003), page 250, Section 7.57 ff, or Bergh and Löfström (1976).

^b See Di Nezza, Paltucci and Valdinoci (2012) for an “elementary” exposition of this construction.

[‡] However, the notation $H_p^\alpha(\mathcal{X})$ instead of $W_p^\alpha(\mathcal{X})$ is also common. See e.g. Section 3.2 and Theorem 3.17 in Adams and Fournier (2003), or (1.19) ff. in Triebel (2008).

The spaces $H^\alpha(\mathbb{R}^d)$ for the full domain $\mathcal{X} = \mathbb{R}^d$ can also be described through the Fourier transforms of the functions. If $\hat{f}(t) := (2\pi)^{-d/2} \int e^{-it'x} f(x) dx$ and for $k \in \mathbb{N}_0^d$ as before $(\sigma_k \hat{f})(t) = (1 + \|t\|^k) \hat{f}(t)$, then an equivalent norm is[†]

$$\|f\|_{\alpha|2} \asymp \|\sigma_\alpha \hat{f}\|_{\lambda,2}.$$

This definition immediately generalizes to noninteger smoothness levels α , but is restricted to $\mathcal{X} = \mathbb{R}^d$. Sobolev spaces for general Lipschitz domains can next be described as the restrictions $f|_{\mathcal{X}}$ of the functions $f: \mathbb{R}^d \rightarrow \mathbb{R}$ in the global Sobolev spaces.[‡]

The space $W_p^1(a, b)$ for an interval $(a, b) \subset \mathbb{R}$ can be identified with the set of absolutely continuous functions $f: (a, b) \rightarrow \mathbb{R}$ in $L_p(a, b)$ with derivative in $L_p(a, b)$.^b For a bounded Lipschitz domain $\mathcal{X}$ and $p > d$, the functions in the space $W_p^1(\mathcal{X})$ are differentiable almost everywhere; the space $W_\infty^1(\mathcal{X})$ consists precisely of the functions that can be almost everywhere identified with a Lipschitz function $f: \mathcal{X} \rightarrow \mathbb{R}$.[#]

The scale of *Besov spaces* is indexed by three parameters, a level of regularity α , and two parameters p, q indicating norms by which the regularity is measured. Besov spaces are defined in terms of differences (“discretized derivatives”), rather than derivatives. The *directional difference operator* is defined recursively for $r \in \mathbb{N}$ and a “direction” $h \in \mathbb{R}^d$ by

$$\Delta_h f(x) = f(x + h) - f(x), \quad \Delta_h^r f(x) = \Delta_h \Delta_h^{r-1} f(x).$$

The last expression is well defined if $x, x + h$, or $x, x + h, \dots, x + rh$ are contained in $\mathcal{X}$, respectively. For x that do not satisfy this restriction, the difference operator $\Delta_h^r f(x)$ is defined to be 0.

For $\alpha > 0$ and $1 \leq p, q \leq \infty$, and fixed $r \in \mathbb{N}$ with $r > \alpha$, define,

$$\|f\|_{\alpha|p,q} = \|f\|_{\lambda,p} + \left[\int_0^\infty \left(\frac{1}{t^\alpha} \sup_{\|h\| \leq t} \|\Delta_h^r f\|_{\lambda,p} \right)^q \frac{dt}{t} \right]^{1/q}, \quad (q < \infty),$$

$$\|f\|_{\alpha|p,\infty} = \|f\|_{\lambda,p} + \sup_{t>0} \frac{1}{t^\alpha} \sup_{\|h\| \leq t} \|\Delta_h^r f\|_{\lambda,p}.$$

The corresponding *Besov space* $B_{p,q}^\alpha(\mathcal{X})$ is the collection of all $f \in L_p(\mathcal{X}, \lambda)$ such that $\|f\|_{\alpha|p,q} < \infty$. The norm $\|\cdot\|_{\lambda,p}$ is the L_p -norm relative to the Lebesgue measure on the domain $\mathcal{X}$. The value of r must be strictly larger than α , also if α is an integer (e.g. $r = 2$ if $\alpha = 1$). With this restriction, the norms for various values of r are equivalent, which is why the parameter is not referenced in the notation.[†]

[†] See Adams and Fournier (2003), page 252, Section 7.62, or Evans (2010), pages 282–283.

[‡] See the discussion of restriction and extension operators in the footnotes on the next pages. The norm is defined as the infimum of the norms of all extensions.

^b Brezis (2011), page 204. A function f is *absolutely continuous* if its increments can be written as the integral of an integrable function: $f(y) - f(x) = \int_x^y f'(t) dt$, for every $a \leq x < y \leq b$ and some locally integrable function f' .

[#] Evans (2010), pages 279–280.

[†] In Bergh and Löfström (1976), page 144, Theorem 6.2.5, it is shown that the scaled r th difference $t^{-\alpha} \Delta_h^r f$ may be replaced by the sum over the scaled lower order differences $t^{s-\alpha} \Delta_h^{r-s} \partial^s f / \partial x_i^s$ of partial derivatives, as long as $r > \alpha$ and $0 \leq s < \alpha$.

Because the function $h \mapsto \|\Delta_h^r f\|_{\lambda,p}$ is bounded above by a multiple of $\|f\|_{\lambda,p}$ and $\int_1^\infty t^{-\alpha q-1} dt = (\alpha q)^{-1} < \infty$, the integral in the definition of $\|f\|_{\alpha|p,q}$ is mostly determined by the behavior of the *modulus of continuity* $h \mapsto \|\Delta_h^r f\|_{\lambda,p}$ near zero, and its upper bound ∞ can be replaced by 1 without consequence. Because $\int_0^1 (1/t) dt = \infty$, with near convergence, in a very rough sense finiteness of the norm $\|f\|_{\alpha|p,q}$ requires that the modulus $\|\Delta_h^r f\|_{\lambda,p}$ is of smaller order than $\|h\|^{-\alpha}$, as $h \downarrow 0$. The parameter q determines how this order is measured. Higher values of q make the requirement less stringent (the norms $\|f\|_{\alpha|p,q}$ are up to constants decreasing in q), so that larger q lead to bigger Besov spaces (see Problem 2.7.11). A further intuitive understanding of the role of this parameter appears to be a delicate matter. For our purpose of bounding covering numbers, it suffices to consider the value $q = \infty$, which defines the biggest class of functions.

The use of the directional difference operator, and differences of order $r > \alpha$, make the definition of Besov spaces a bit complicated. It would be natural to use the same definition, but with first differences of derivatives. For instance, for functions on the line one might replace $t^{-\alpha} \|\Delta_h^r f\|_{\lambda,p}$ by $t^{-(\alpha-\underline{\alpha})} \|\Delta_h D^{\underline{\alpha}} f\|_{\lambda,p}$, for $\underline{\alpha}$ the largest integer smaller than α . This would result, in general, in spaces that are slightly smaller. Similarly using exactly α differences if α is an integer (i.e. $r = \alpha$) would give a smaller space (see Problem 2.7.12).

In particular, the Besov space $B_{p,\infty}^\alpha(\mathcal{X})$ contains the *Lipschitz space* $Lip_p^\alpha(\mathcal{X})$ of all functions $f: \mathcal{X} \rightarrow \mathbb{R}$ with finite α -Lipschitz norm[‡]

$$\|f\|_{\alpha;p} = \|f\|_{\lambda,p} + \sup_{t>0} \frac{1}{t^{\alpha-\underline{\alpha}}} \sup_{\|h\| \leq t} \max_{k=\underline{\alpha}} \|\Delta_h(D^k f)\|_{\lambda,p}.$$

The Lipschitz spaces are defined through a first order difference of the derivatives of order the biggest integer $\underline{\alpha}$ strictly smaller than α . These spaces are a natural generalization of the Hölder spaces considered in Section 2.7.1, replacing the uniform norm by the L_p -norm, and reduce to the latter spaces if $p = \infty$ (i.e. the norm $\|\cdot\|_{\alpha;\infty}$ is identical to the norm $\|\cdot\|_\alpha$ in Section 2.7.1). For $0 < \alpha \notin \mathbb{N}$ and $p \geq 1$ the spaces $Lip_p^\alpha(\mathcal{X})$ and $B_{p,\infty}^\alpha(\mathcal{X})$ coincide, but for integer-valued α the Besov space is slightly larger.^b

The space $B_{\infty,\infty}^\alpha(\mathcal{X})$ is also known as the *Hölder-Zygmund space* (or just *Zygmund space*). It is slightly bigger than the Hölder space $C^\alpha(\mathcal{X}) = Lip_\infty^\alpha(\mathcal{X})$ if α is an integer and coincides with it for noninteger α .[‡]

[‡] We included the norm $\|f\|_{\lambda,p}$, but one can also consider the spaces equipped with the *Lipschitz seminorm*, which is defined as just the second term on the right, still under the assumption that $f \in L_p(\mathcal{X}, \lambda)$. Our notation $\|\cdot\|_{\alpha;p}$ using the subscript $\alpha; p$ is all too brief, but used only in this section.

^b DeVore and Lorentz (1993), pages 51–52, where $B_{p,\infty}^\alpha$ is denoted by $Lip^*(\alpha, L_p)$. For instance, for $\alpha = 1$, the Besov space is defined in terms of second or higher order differences, whereas the Lipschitz space uses first order differences. We have $\|\Delta_h^2 f\|_\infty \leq 2\|\Delta_h f\|_\infty$, but this cannot be reversed (see Problem 2.7.17).

[‡] Triebel (1992), Theorem 1.2.2.

The Besov spaces “bracket” the Sobolev spaces:[†]

$$\begin{aligned} B_{p,p}^\alpha(\mathcal{X}) &\hookrightarrow W_p^\alpha(\mathcal{X}) \hookrightarrow B_{p,2}^\alpha(\mathcal{X}), & \text{if } 1 < p \leq 2, \\ B_{p,2}^\alpha(\mathcal{X}) &\hookrightarrow W_p^\alpha(\mathcal{X}) \hookrightarrow B_{p,p}^\alpha(\mathcal{X}), & \text{if } 2 \leq p < \infty. \end{aligned}$$

The arrows in this display denote embeddings: the set of functions on the left is included in the set of functions on the right, and the inclusion map (identity) is continuous relative to the norms of the two spaces. Thus the unit ball of each of the spaces on the left is included into a multiple of the unit ball of the spaces on the right. In particular, for $p = 2$ the inclusions clearly become identities. This shows that the Sobolev spaces $H^\alpha(\mathcal{X}) = W_2^\alpha(\mathcal{X})$ are exactly equal to the Besov spaces $B_{2,2}^\alpha(\mathcal{X})$, where the norms $\|\cdot\|_{\alpha|2}$ and $\|\cdot\|_{\alpha|2,2}$ defined in the preceding are equivalent.

We also have the following embeddings: for $\alpha > 0$ and $1 \leq p, q, p', q' \leq \infty$,[‡]

$$\begin{aligned} B_{p,q}^\alpha(\mathcal{X}) &\hookrightarrow B_{p,q'}^\alpha(\mathcal{X}), & \text{if } 1 \leq q \leq q' \leq \infty, \\ B_{p,q}^\alpha(\mathcal{X}) &\hookrightarrow B_{p',q}^{\alpha'}(\mathcal{X}), & \text{if } \alpha - d/p \geq \alpha' - d/p' \text{ and } p \leq p', \\ B_{p,q}^\alpha(\mathcal{X}) &\hookrightarrow C_b(\mathcal{X}), & \text{if } \alpha > d/p \text{ and } p < \infty, \\ B_{p,q}^\alpha(\mathcal{X}) &\hookrightarrow L_\tau(\mathcal{X}, \lambda), & \text{if } \alpha > d(1/p - 1/\tau)_+ \text{ and } \mathcal{X} \text{ is bounded.} \end{aligned}$$

These are examples or extensions of *Sobolev embedding theorems*. The first embedding was already noted in the preceding. The second embedding is easy in the case that $p = p'$, when it merely says that higher order smoothness classes are embedded in lower order smoothness classes, if the smoothness is measured in a single manner. In its stated, more general form the embedding shows that higher order (α) smoothness measured in a weaker sense (L_p) implies lower order (α') smoothness in a stronger sense ($L_{p'}$ for $p' \leq pd/(d - p(\alpha - \alpha'))$). For instance, a function on the line ($d = 1$) whose derivative is contained in L_2 ($\alpha = 1, p = 2$) belongs to $C^{1/2}$ ($\alpha' = 1/2, p' = \infty$), an assertion that can easily be proved directly using the Cauchy-Schwarz inequality. The third embedding can be viewed as a limiting case of this principle ($p' = \infty, \alpha' = 0$), and shows that functions in Besov spaces are continuous if the smoothness level is large enough relative to p . Only for $p = \infty$ any possible smoothness level $\alpha > 0$ qualifies to give continuity. The fourth embedding (restricted to bounded domains) shows that for embedding in the weaker L_τ -space, the restriction on the smoothness parameter can be relaxed. The theorem

[†] See Bergh and L fstr m (1976), page 152, Theorem 6.4.4. We assume that $\mathcal{X}$ is a bounded Lipschitz domain or strong local Lipschitz domain. The functions on the domain $\mathcal{X}$ can then be extended to a function with domain $\mathbb{R}^d$ of equivalent norm (Triebel (2008), pages 103–104, Theorem 4.4 and Corollary 4.12, or Cohen, Dahmen, Daubechies and DeVore (2001), page 203) and the restriction maps are continuous as well. The result follows then from the corresponding results for the domain $\mathcal{X} = \mathbb{R}^d$. The embeddings may also be proved using the characterizations of the norms in terms of wavelet coefficients given below (see Problem 2.7.14).

[‡] We assume that $\mathcal{X}$ is a bounded Lipschitz domain or strong local Lipschitz domain, which allows to reduce to the case that $\mathcal{X} = \mathbb{R}^d$, as in the preceding footnote. For the first assertion, see Problem 2.7.11, or Bergh and L fstr m (1976), page 142, Theorem 6.2.4. For the second inclusions, see Bergh and L fstr m (1976), page 153, Theorem 6.5.1. For the third inclusions, see Adams and Fournier (2003), page 231, Theorem 7.34 and pages 228 ff, Sections 7.30 and 7.31. The fourth assertion is implicit in Theorem 2.7.4 below. The embeddings may also be derived using the characterizations of the norms in terms of wavelet coefficients described below (see Problem 2.7.15).

below gives an upper bound on the entropy of the unit ball of the Besov space in L_τ and hence shows that the embedding is compact.^b

Together the preceding displays show that $W_p^\alpha(\mathcal{X}) \hookrightarrow B_{p,p\vee 2}^\alpha(\mathcal{X}) \hookrightarrow C_b(\mathcal{X})$ if $\alpha > d/p$ and $p < \infty$, which is a classical Sobolev embedding theorem, showing that the functions in a Sobolev space of sufficiently high order of smoothness (relative to dimension and norm) are continuous (and bounded). This may be improved to embedding in $C^{\alpha-d/p}(\mathcal{X})$ if $\alpha - d/p > 0$ and is not an integer. It suffices to note that $B_{p,q}^\alpha(\mathcal{X}) \hookrightarrow B_{\infty,q}^{\alpha-p/d}(\mathcal{X}) \hookrightarrow B_{\infty,\infty}^{\alpha-p/d}(\mathcal{X})$, by applying the second and first embeddings in the last display. As noted previously $B_{\infty,\infty}^\alpha(\mathcal{X}) = C^\alpha(\mathcal{X})$ if α is not an integer.[†]

In view of these inclusions an entropy bound on the Besov spaces automatically yields a bound on the Sobolev, Lipschitz, and Hölder spaces. The following theorem describes the entropy relative to the $L_\tau(\mathcal{X}, \lambda)$ -norm, for a bounded, Lipschitz domain $\mathcal{X} \subset \mathbb{R}^d$, and given $\tau \in (0, \infty]$, for combinations of parameters such that $\alpha > d(1/p - 1/\tau)_+$. This requires a larger smoothness level for larger τ or smaller p , with the full range $\alpha > 0$ only available if $\tau \leq p$. The order of the entropy depends on α and d only, and not on τ , p or q .[‡]

2.7.4 Theorem (Besov space). *Let $1 \leq \tau \leq \infty$, $1 \leq p, q \leq \infty$ and a bounded Lipschitz domain $\mathcal{X}$ be given. If $\alpha > d(1/p - 1/\tau)_+$, then the unit ball $B_{p,q}^\alpha(\mathcal{X})_1$ of the Besov space $B_{p,q}^\alpha(\mathcal{X})$ satisfies*

$$\log N\left(\varepsilon, B_{p,q}^\alpha(\mathcal{X})_1, L_\tau(\mathcal{X}, \lambda)\right) \lesssim \left(\frac{1}{\varepsilon}\right)^{d/\alpha}.$$

Proof. Because the Besov norms are decreasing in q , it suffices to prove the theorem for $q = \infty$, which gives the largest unit ball. We represent the functions f in a wavelet basis normalised in L_τ , as explained below, with coefficients $\beta_{j,k}^v|_\tau = \beta_{j,k}^v 2^{jd(1/2-1/\tau)}$. Then the $L_\tau(\mathcal{X}, \lambda)$ -norm is bounded above by the norm $\|\cdot\|_\tau$ in Section 2.7.2.1 applied to the array of coefficients $\beta = (\beta_{j,k}^v|_\tau)$, and the inclusion $f \in B_{p,\infty}^\alpha(\mathcal{X})_1$ translates into $\beta \in \mathbb{B}_p^\delta$ for the set of coefficients defined in Section 2.7.2.1 for $\delta = \alpha - d/p + d/\tau$. The theorem then is a corollary to Lemma 2.7.8 applied to the coefficients $\beta_{j,k}^v|_\tau$. The details of the representation are described in the following three pages. ■

For functions $f: \mathbb{R} \rightarrow \mathbb{R}$ a suitable wavelet expansion can be written in terms of a pair of functions ϕ and ψ , called *father wavelet* and *mother wavelet*. These functions can be chosen such that the set of translates $x \mapsto \phi_k(x) = \phi(x - k)$, for $k \in \mathbb{Z}$, of the father, together with the translated dilations $x \mapsto \psi_{j,k}(x) =$

^b For a bounded domain, the second of the four embeddings is also compact. See Triebel (2006), Theorem 1.107.

[‡] At the boundary value $\alpha = d(1/p - 1/\tau)$, the validity of the inclusion depends on the value of q . See DeVore and Lorentz (1993), page 385, Theorem 8.1. For instance, the embedding of $B_{p,q}^\alpha(\mathcal{X})$ in $C_b(\mathcal{X})$ is still true if $\alpha = d/p$ and $q = 1$ (Adams and Fournier (2003), Theorem 7.34).

[†] Also see Adams and Fournier (2003), page 2.33, Theorem 7.37, who prove an embedding in a smaller space of continuous functions, keeping the parameter q .

[‡] This order is sharp. See Cohen, Dahmen, Daubechies and DeVore (2001), Theorem 1.1.

$2^{j/2}\psi(2^j x - k)$, for $j, k \in \mathbb{Z}$, of the mother form an orthonormal basis of $L_2(\mathbb{R}, \lambda)$. Thus any function $f \in L_2(\mathbb{R}, \lambda)$ can be written as

$$f = \sum_{k \in \mathbb{Z}} \beta_{0,k} \phi_k + \sum_{j=1}^{\infty} \sum_{k \in \mathbb{Z}} \beta_{j,k} \psi_{j,k}.$$

In addition the father and mother wavelet can be chosen to have compact support.

To simplify notation we write $\phi_k = \psi_{0,k}$, so that we obtain an orthonormal basis $\psi_{j,k}$ of $L_2(\mathbb{R}, \lambda)$ indexed by the pairs $(j, k) \in \mathbb{N}_0 \times \mathbb{Z}$. Then any $f \in L_2(\mathbb{R}, \lambda)$ can be expanded as

$$f = \sum_{j=0}^{\infty} \sum_{k \in \mathbb{Z}} \beta_{j,k} \psi_{j,k},$$

where $\beta_{j,k} = \int f \psi_{j,k} d\lambda$ are the Fourier coefficients of f relative to the basis $\psi_{j,k}$. Now it can be shown that for sufficiently regular father and mother wavelets^b, the Besov norms defined earlier (for $\mathcal{X} = \mathbb{R}$) are equivalent to the corresponding coefficient norms given by

$$\|f\|_{\alpha|p,q} = \left(\sum_{j=0}^{\infty} \left[2^{j(\alpha+1/2)} \left(\sum_{k \in \mathbb{Z}} |\beta_{j,k}|^p \right)^{1/p} 2^{-j/p} \right]^q \right)^{1/q}, \quad q < \infty,$$

$$\|f\|_{\alpha|p,\infty} = \sup_{j=0,\dots,\infty} 2^{j(\alpha+1/2)} \left(\sum_{k \in \mathbb{Z}} |\beta_{j,k}|^p \right)^{1/p} 2^{-j/p}.$$

For $p = \infty$, the expression $(\sum_{k \in \mathbb{Z}} |\beta_{j,k}|^p)^{1/p}$ must be understood to be the maximum norm $\sup_k |\beta_{j,k}|$ of the coefficients at level j .

The Besov space $B_{p,q}^{\alpha}(\mathbb{R})$ consists exactly of all functions f whose wavelet coefficients renders the expression on the right finite, and its unit ball is the set of functions for which the expression is bounded above by a given constant. (We abuse notation by writing $\|\cdot\|_{\alpha|p,q}$ for both the function and coefficient norm, which are equivalent, but not identical.)

For functions $f: \mathbb{R}^d \rightarrow \mathbb{R}$ the wavelet expansion is slightly more complicated. A function $f \in L_2(\mathbb{R}^d)$ can be written

$$(2.7.5) \quad f = \sum_{k \in \mathbb{Z}^d} \sum_{v \in \{0,1\}^d} \beta_{0,k}^v \psi_{0,j}^v + \sum_{j=1}^{\infty} \sum_{k \in \mathbb{Z}^d} \sum_{v \in \{0,1\}^d - \{0\}} \beta_{j,k}^v \psi_{j,k}^v,$$

for functions $\psi_{j,k}^v$ which are orthogonal for different indices (j, k, v) , and are scaled and translated versions of the set of 2^d “base wavelets” $\psi_{0,0}^v$:

$$\psi_{j,k}^v(x) = 2^{jd/2} \psi_{0,0}^v(2^j x - k).$$

Such a higher-dimensional wavelet basis can be obtained as tensor products $\psi_{0,0}^v = \phi^{v_1} \times \dots \times \phi^{v_d}$ of a given father wavelet ϕ^0 and mother wavelet ϕ^1 in one dimension.

^b Essentially, besides being compactly supported, the wavelets must be sufficiently many times differentiable and have sufficiently many vanishing moments. See Meyer (1990, 1992), Daubechies (1988, 1990), or Triebel (2008), Sections 1.1. and 1.2.

If it is understood that a sum over v ranges over $\{0, 1\}^d$ if nested within $j = 0$ and over $\{0, 1\}^d - \{0\}$ if nested within $j \in \mathbb{N}$, then the two sums can be collapsed again to a single sum over $j \in \mathbb{N}_0$. The corresponding Besov norm then takes the form

$$\|f\|_{\alpha|p,q} = \left(\sum_{j=0}^{\infty} \left[2^{j(\alpha+d/2)} \left(\sum_{k \in \mathbb{Z}^d} \sum_v |\beta_{j,k}^v|^p \right)^{1/p} 2^{-jd/p} \right]^q \right)^{1/q}, \quad q < \infty,$$

$$\|f\|_{\alpha|p,\infty} = \sup_{j \in \mathbb{N}_0} 2^{j(\alpha+d/2)} \left(\sum_{k \in \mathbb{Z}^d} \sum_v |\beta_{j,k}^v|^p \right)^{1/p} 2^{-jd/p}.$$

For suitable base wavelets these norms are equivalent to the Besov norms defined previously (for the domain $\mathcal{X} = \mathbb{R}^d$).[‡]

In order to measure the size of a function in $L_\tau(\mathbb{R}^d, \lambda)$, it is useful to reexpress the expansion in terms of the “renormalized wavelet basis”. The base functions $\psi_{j,k}^v$ have been scaled so that their L_2 -norms are equal to the L_2 -norm of the base wavelet $\psi_{0,0}^v$, independent of the value of (j, k) . To stabilize the L_τ -norms the appropriate scaling factors are $2^{jd/\tau}$ rather than $2^{jd/2}$. For a given value of τ the function $2^{jd(1/\tau-1/2)}\psi_{j,k}^v$ has L_τ -norm independent of (j, k) , and yields a wavelet basis “normalized in L_τ ”. If we define

$$\beta_{j,k|\tau}^v = 2^{jd(1/2-1/\tau)} \beta_{j,k}^v, \quad \psi_{j,k|\tau}^v = 2^{jd(1/\tau-1/2)} \psi_{j,k}^v,$$

then the product $\beta_{j,k|\tau}^v \psi_{j,k|\tau}^v$ is independent of τ , and the function f can be reexpressed as

$$f = \sum_{j=0}^{\infty} \sum_{k \in \mathbb{Z}^d} \sum_v \beta_{j,k|\tau}^v \psi_{j,k|\tau}^v.$$

Under this normalization the L_τ -norm of the projection on level j is equivalent to the ℓ_τ -norm of the coefficients at that level:

$$\left\| \sum_{k \in \mathbb{Z}^d} \sum_v \beta_{j,k|\tau}^v \psi_{j,k|\tau}^v \right\|_{\lambda,\tau} \asymp \left(\sum_{k \in \mathbb{Z}^d} \sum_v |\beta_{j,k|\tau}^v|^\tau \right)^{1/\tau}.$$

This is a consequence of the localization of the basis functions; if the supports of the functions $\psi_{j,k|\tau}$ were completely disjoint, then the two norms would be exactly equal.[†] For $\tau = \infty$, the display is true as well, with the norm on the right understood to be the sup norm $\sup_{k,v} |\beta_{j,k|\tau}^v|$ of the coefficients. The equivalence constants do not depend on j . By the triangle inequality we conclude that the L_τ -norm of the function f satisfies

$$(2.7.6) \quad \|f\|_{\lambda,\tau} \lesssim \sum_{j=0}^{\infty} \left(\sum_{k \in \mathbb{Z}^d} \sum_v |\beta_{j,k|\tau}^v|^\tau \right)^{1/\tau}, \quad \tau < \infty,$$

$$\|f\|_\infty \lesssim \sum_{j=0}^{\infty} \sup_{k \in \mathbb{Z}^d} \max_v |\beta_{j,k|\tau}^v|.$$

[‡] See Meyer (1990, 1992), Daubechies (1988, 1990), Cohen, Dahmen, Daubechies and DeVore (2001), or Triebel (2008), Theorem 1.20.

[†] For general wavelets the equivalence can be found in Meyer (1990, 1992), Daubechies (1988, 1990), or Cohen, Dahmen, Daubechies and DeVore (2001), formula (2.8). It can also be derived from the characterization of the L_τ -norm in the following paragraph. See Problem 2.7.13.

Thus the L_τ -norm of a function can be bounded in terms of the “corrected” wavelet coefficients $\beta_{j,k|\tau}^v$, or equivalently in terms of its L_2 -wavelet coefficients $\beta_{j,k}^v$, with added correction factors $2^{jd(1/2-1/\tau)}$ if $\tau \neq 2$.

For $\tau = 2$ the functions $\psi_{j,k|2} = \psi_{j,k}$ are orthonormal and hence the inequalities in the preceding two displays are actually identities. For general τ the inequalities are valid up to constants only. More importantly, there is no inequality reversing the bounds (2.7.6), not even up to constants: the equivalence of norms is valid within the resolution levels only. A characterization of the L_τ -norm in terms of wavelet coefficients takes a more complicated form, involving the indicator functions of cubes $C_{j,k} = \{x: \|x - 2^{-j}k\|_\infty < 2^{-j}\}$ containing the supports of the wavelets. In fact, for a suitable wavelet basis the Sobolev norm of $W_\tau^\alpha(\mathbb{R}^d)$ is equivalent to[‡]

$$(2.7.7) \quad \|f\|_{\alpha|\tau} = \left\| \left(\sum_{j=0}^{\infty} 2^{2j(\alpha+d/2)} \sum_{k \in \mathbb{Z}^d} \sum_v |\beta_{j,k}^v|^2 1_{C_{j,k}} \right)^{1/2} \right\|_{\lambda, \tau}.$$

For $\alpha = 0$, the Sobolev space $W_\tau^0(\mathcal{X})$ reduces to the space $L_\tau(\mathcal{X}, \lambda)$, and hence the norm in the display reduces to $\|f\|_{\lambda, \tau}$, up to equivalence.

The inherent translations in their definition make wavelets a little awkward for approximating functions on domains $\mathcal{X}$ other than the full space $\mathbb{R}^d$. Periodic functions on intervals or cubes can be approximated by periodized wavelets. Wavelet approximations on general domains require breaking the regularity of the scheme of dilated basis functions located on a regular grid, particularly near the boundary of the domain.^b A simplistic approach, which will do for bounding covering numbers, is to extend a given function $f: \mathcal{X} \rightarrow \mathbb{R}$ to a function on the full space $\mathbb{R}^d$, next apply a wavelet approximation to the extension and finally restrict this approximation to the domain $\mathcal{X}$.

For this to work, the extensions should not be essentially more complex than the original functions. This *extension problem* has been well studied, and relies on the regularity of the boundary of the domain. It turns out that for a bounded Lipschitz domain $\mathcal{X}$, every function $f \in B_{p,q}^\alpha(\mathcal{X})$ can be extended to a function $\tilde{f} \in B_{p,q}^\alpha(\mathbb{R}^d)$ of essentially the same Besov norm: $\|\tilde{f}\|_{\alpha|p,q} \lesssim \|f\|_{\alpha|p,q}$, where the two Besov norms are understood to refer to the domains $\mathbb{R}^d$ and $\mathcal{X}$, respectively.[#]

It follows, that for a given bounded Lipschitz domain $\mathcal{X}$, the extensions of the functions in the unit ball $B_{p,q}^\alpha(\mathcal{X})_1$ are contained in a multiple of the unit ball of the space $B_{p,q}^\alpha(\mathbb{R}^d)$, whence their L_τ -covering numbers as in Theorem 2.7.4 are bounded by the $L_{\bar{\tau}}$ -covering numbers of this multiple, for $\bar{\tau}$ the restriction of Lebesgue measure to $\mathcal{X}$. The functions in $B_{p,q}^\alpha(\mathbb{R}^d)_1$ can be expanded in a suitable

[‡] See Meyer (1992), Section 6.2, or Triebel (2008), Theorem 1.20. The Sobolev space $W_p^\alpha(\mathbb{R}^d)$ is equivalent to the *Triebel-Lizorkin space* $F_{p,2}^\alpha(\mathbb{R}^d)$; see Triebel (2008), formulas (1.12)–(1.13) and (1.18)–(1.19). Our coefficients $\beta_{j,k}^v$ correspond to the coefficients $\lambda_k^{j,v} 2^{-jd/2}$ in Triebel (2008) (with $d \rightarrow n$ and $v \rightarrow G$; see his (1.94) and (1.99).

^b See Cohen, Dahmen and DeVore (2000), or Chapter 2 in Triebel (2008), where “*u*-wavelet systems” are introduced.

[#] In fact, there exists a continuous linear map $\text{Ext}: B_{p,q}^\alpha(\mathcal{X}) \rightarrow B_{p,q}^\alpha(\mathbb{R}^d)$ with $\text{Ext}(f)|_{\mathcal{X}} = f$ for every f , which can be taken the same for every $1 \leq p, q \leq \infty$ and every α in any given bounded range in $(0, \infty)$. Moreover, this map is also continuous as a map $\text{Ext}: W_p^\alpha(\mathcal{X}) \rightarrow W_p^\alpha(\mathbb{R}^d)$ for $1 \leq p < \infty$ and α in the same bounded range. See Rychkov (1999), or Triebel (2008), Theorem 4.4 and Corollary 4.5. The extension already exists for domains less regular than bounded Lipschitz domains.

wavelet basis, as previously. Terms of the expansion that refer to a base function whose support does not intersect $\mathcal{X}$, give zero contribution to the $L_{\bar{\tau}}$ -norm and hence do not play a role. Thus if we use compactly supported wavelets, then at each level j only of the order $2^{jd}2^d \lesssim 2^{jd}$ terms of the expansion are important. Consequently, for the proof of Theorem 2.7.4, it suffices to consider coefficients $\beta_{j,k}^v$ with $j \in \mathbb{N}_0$ and (k, v) ranging over a set of cardinality of the order 2^{jd} . The necessary calculations are given in Section 2.7.2.1, stripped of any interpretation in terms of function spaces. Because the parameter v loses its geometric meaning in this representation, it can without loss of generality be subsumed in the parameter k .

2.7.2.1 Besov Bodies

Let $\mathcal{B}_p^\delta$ be the collection of all doubly indexed sequences $\beta = (\beta_{j,k})$, where $j = 0, 1, \dots$ and $k = 1, 2, \dots, K_j$ for some $K_j \lesssim 2^{jd}$, with

$$\sup_{j \in \mathbb{N}_0} 2^{j\delta} \left(\sum_{k \in K_j} |\beta_{j,k}|^p \right)^{1/p} \leq 1.$$

We equip this with the norm

$$\begin{aligned} \|\beta\|_\tau &= \sum_{j=0}^{\infty} \left(\sum_{k=1}^{K_j} |\beta_{j,k}|^\tau \right)^{1/\tau}, \quad \text{if } \tau < \infty, \\ \|\beta\|_\infty &= \sum_{j=0}^{\infty} \max_{k=1, \dots, K_j} |\beta_{j,k}|. \end{aligned}$$

For β wavelet coefficients $(\beta_{j,k}^v)$ normalized in L_τ and $\delta = \alpha - d(1/p - 1/\tau)$, this corresponds to functions f with Besov norm $\|f\|_{\alpha|p,\infty}$ bounded by a constant, equipped with a norm that is stronger than the L_τ -norm, in view of (2.7.6).

2.7.8 Lemma. *If $\delta > d(1/\tau - 1/p)_+$, then, with $s = \delta + d(1/p - 1/\tau)$,*

$$\log N(\varepsilon, \mathcal{B}_p^\delta, \|\cdot\|_\tau) \lesssim \left(\frac{1}{\varepsilon} \right)^{d/s}.$$

Proof. For given j let $\beta_{j,\cdot}$ denote the vector $(\beta_{j,k})_{k=1, \dots, K_j}$ in $\mathbb{R}^{K_j}$, and let $\|\beta_{j,\cdot}\|_\tau$ be its τ -norm $(\sum_k |\beta_{j,k}|^\tau)^{1/\tau}$. These norms are decreasing in τ , while, in the other direction, for $p > \tau$ we have $\|\beta_{j,\cdot}\|_\tau \lesssim K_j^{1/\tau - 1/p} \|\beta_{j,\cdot}\|_p$, by Hölder's inequality. The latter comparison and the fact that $K_j \lesssim 2^{jd}$ shows that, for $p > \tau$ and some constant C ,

$$\mathcal{B}_p^\delta \subset C \mathcal{B}_\tau^{\delta + d(1/p - 1/\tau)}.$$

Therefore, the lemma for $\tau = p$ and $\delta > 0$ with upper bound $(1/\varepsilon)^{d/\delta}$ implies the lemma for $p > \tau$ and $\delta > d(1/\tau - 1/p)$ as given. Thus we may assume that $p \leq \tau$ and then prove the bound for arbitrary $\delta > 0$. It suffices to prove the bound for small $\varepsilon > 0$, as the left side vanishes for ε larger than the diameter of $\mathcal{B}_p^\delta$.

For a given sequence of numbers (λ_j) and $\beta \in \mathcal{B}_p^\delta$, let $\tilde{\beta}$ be defined by shrinking for each j each of the coordinates $\beta_{j,k}$ of β to the grid with mesh width λ_j . Thus $\tilde{\beta}_{j,k} = \text{sign}(\beta_{j,k}) \lfloor |\beta_{j,k}|/\lambda_j \rfloor \lambda_j$ for each (j,k) . If $\lambda_j = 0$, it will be understood that each $\beta_{j,k}$ is shrunk to 0. The proof proceeds by determining (λ_j) such that $\|\beta - \tilde{\beta}\|_\tau \leq C\varepsilon$ for some constant C , in which case the number of different arrays $\tilde{\beta}$ is an upper bound on the $C\varepsilon$ -covering number of $\mathcal{B}_p^\delta$. For defining the numbers (λ_j) and counting the number of different $\tilde{\beta}$, we differentiate between three cases:

- (a) $p \leq \tau < \infty$. We choose $\lambda_j = \lambda \asymp \varepsilon^{(s+d/\tau)/s}$, independent of j .
- (b) $p < \tau = \infty$. For j_ε the solution of the equation $2^{-j_\varepsilon(\delta+d/p)} = \varepsilon$, we choose $\lambda_j = \varepsilon/(j - j_\varepsilon)^2$ for $j \neq j_\varepsilon$ and $\lambda_{j_\varepsilon} = \varepsilon$ if $j_\varepsilon \in \mathbb{N}$.
- (c) $p = \tau = \infty$. For j_ε the solution of the equation $2^{-j_\varepsilon\delta} = \varepsilon$, we choose $\lambda_j = \varepsilon/(j - j_\varepsilon)^2$ for $j < j_\varepsilon$ and $\lambda_j = 0$ otherwise.

In all three cases clearly $|\beta_{j,k} - \tilde{\beta}_{j,k}| \leq \lambda_j$ for every (j,k) such that $\lambda_j > 0$. This implies that $\|\beta_{j,\cdot} - \tilde{\beta}_{j,\cdot}\|_\tau \leq \lambda_j K_j^{1/\tau} \lesssim \lambda_j 2^{jd/\tau}$ for $\tau < \infty$ and $\|\beta_{j,\cdot} - \tilde{\beta}_{j,\cdot}\|_\infty \leq \lambda_j$ whenever $\lambda_j > 0$.

In case (b) the last bound shows that $\|\beta - \tilde{\beta}\|_\infty \leq \sum_j \lambda_j$, which is bounded by a multiple of ε . In case (c) the same bound implies that $\|\beta - \tilde{\beta}\|_\infty \leq \sum_{j \leq j_\varepsilon} \lambda_j + \sum_{j > j_\varepsilon} |\beta_{j,k}|$, which is also bounded by a multiple of ε , since the assumption that $\beta \in \mathcal{B}_\infty^\delta$ implies that $|\beta_{j,k}| \leq 2^{-j\delta}$ for every (j,k) and $2^{-j_\varepsilon\delta} = \varepsilon$ by assumption.

In case (a), where $p \leq \tau < \infty$, we have next to the trivial bound $\|\beta_{j,\cdot} - \tilde{\beta}_{j,\cdot}\|_\tau \lesssim \lambda_j 2^{jd/\tau}$ also that $\sum_k |\beta_{j,k} - \tilde{\beta}_{j,k}|^\tau \leq \lambda_j^{\tau-p} \sum_k |\beta_{j,k} - \tilde{\beta}_{j,k}|^p$ and hence

$$\begin{aligned} \|\beta_{j,\cdot} - \tilde{\beta}_{j,\cdot}\|_\tau &\leq \lambda_j^{1-p/\tau} \|\beta_{j,\cdot} - \tilde{\beta}_{j,\cdot}\|_p^{p/\tau} \\ &\leq \lambda_j^{1-p/\tau} 2^{p/\tau} \|\beta_{j,\cdot}\|_p^{p/\tau} \lesssim \lambda_j^{1-p/\tau} 2^{-j\delta p/\tau}, \end{aligned}$$

since $\beta \in \mathcal{B}_p^\delta$. Combination with the trivial bound yields

$$\|\beta - \tilde{\beta}\|_\tau \lesssim \sum_{j=0}^{\infty} \left(\lambda_j^{1-p/\tau} 2^{-j\delta p/\tau} \wedge \lambda_j 2^{jd/\tau} \right) \lesssim \lambda^{s/(s+d/\tau)},$$

in view of Problem 2.7.4 and the definition of s . The choice of λ shows that this is a multiple of ε .

Next we proceed to count the number of $\tilde{\beta}$ in each of the three cases. For each fixed j we represent a typical vector $\tilde{\beta}_{j,\cdot}$ by

- (i) the set $(k_1, k_2, \dots, k_{L_j})$ of indices k of the nonzero coordinates $\tilde{\beta}_{j,k}$.
- (ii) the vector of signs $(\delta_1, \delta_2, \dots, \delta_{L_j})$ of these nonzero coordinates $\tilde{\beta}_{j,k}$.
- (iii) the absolute truncation levels $(n_1, n_2, \dots, n_{L_j})$ of these nonzero coordinates $\tilde{\beta}_{j,k}$ (i.e. $n_i = \lfloor |\beta_{j,k_i}|/\lambda_j \rfloor$ for every i .)

The number of different $\tilde{\beta}$ is bounded by the product over j of the product of the numbers of different entities of the three types, and is a bound on the $C\varepsilon$ -covering number of $\mathcal{B}_p^\delta$.

The number of different subsets as in (i) is bounded by, if $L_j > 0$,

$$\sum_{l=0}^{L_j} \binom{K_j}{l} \leq \left(\frac{eK_j}{L_j} \right)^{L_j} \lesssim \left(C \frac{2^{jd}}{L_j} \right)^{L_j}.$$

(Cf. Problem 2.7.7.) Each $\tilde{\beta}_{j,\cdot}$, with positive coordinates gives rise to 2^{L_j} vectors $\tilde{\beta}_{j,\cdot}$ with arbitrary signs, whence the number of sign vectors as in (ii) is bounded by 2^{L_j} .

The number L_j of nonzero coefficients $\tilde{\beta}_{j,k}$ trivially satisfies $L_j \leq K_j \lesssim 2^{jd}$ in all three cases (a), (b), (c). In cases (a) and (b), where $p < \infty$, the assumption that $\beta \in B_p^\delta$ implies that $\sum_k |\beta_{j,k}|^p \leq 2^{-j\delta p}$. Because $|\beta_{j,k}| \geq \lambda_j$ for each nonzero coordinate $\tilde{\beta}_{j,k}$, it follows that in cases (a) and (b) also $L_j \leq 2^{-j\delta p}/\lambda_j^p$ for each j . In case (c) we have $L_j = 0$ for $j \geq j_\varepsilon$ by construction.

In cases (a) and (b), the number of different sets of levels $(n_1, \dots, n_{L_j})$ as in (iii) is bounded by

$$\begin{aligned} & \#\left\{(n_1, \dots, n_{L_j}): \sum_k |\lambda_j n_k|^p \leq 2^{-j\delta p}\right\} \\ & \leq 4^{L_j} \max\left\{n_1^2 n_2^2 \cdots n_{L_j}^2: \sum_k n_k^p \leq \frac{2^{-j\delta p}}{\lambda_j^p}\right\}, \end{aligned}$$

in view of Problem 2.7.6. Here we can bound

$$\log n_1^2 n_2^2 \cdots n_{L_j}^2 = \frac{2}{p} \sum_k \log n_k^p \lesssim L_j \log\left(\frac{1}{L_j} \sum_k n_k^p\right) \leq L_j \log\left(\frac{2^{-j\delta p}/\lambda_j^p}{L_j}\right),$$

where we used Jensen's inequality in the second last step, and the restriction on $(n_1, \dots, n_{L_j})$ in the last step. It follows that for $p < \infty$ the logarithm of the number of different arrays $\tilde{\beta}$ is bounded by a multiple of

$$\sum_{j=0}^{\infty} \left[L_j \log\left(\frac{C2^{jd}}{L_j}\right) + L_j \log 2 + L_j \log 4 + L_j \log\left(\frac{2^{-j\delta p}/\lambda_j^p}{L_j}\right) \right].$$

Let j_ε be the solution to the equation $2^{-j_\varepsilon(d+\delta p)} = \lambda^p$, for $\lambda^{s/(s+d/\tau)} = \varepsilon$. In case (b), where $\tau = \infty$, the latter equation is understood to mean $\lambda = \varepsilon$, and $\lambda = \lambda_j(j - j_\varepsilon)^2$ for λ_j defined previously, while $\lambda_j = \lambda$ in case (a). For $A > 0$, the function $L \mapsto L \log(A/L)$ is increasing on $(0, A/e]$, whence $L \log(A/L) \leq L \log((A \vee L_0)e/L) \leq L_0 \log_+(A/L_0) + L_0$, for $L \leq L_0$. We apply this to the individual terms of the series, with the bound $L_j \leq K_j \leq C2^{jd}$ for the terms with $j \leq j_\varepsilon$ and the bound $L_j \leq 2^{-j\delta p}/\lambda_j^p$ if $j > j_\varepsilon$. Thus the preceding display is bounded above by a constant times

$$\begin{aligned} & \sum_{j \leq j_\varepsilon} 2^{jd} \left[1 + \log_+\left(\frac{2^{-j\delta p}/\lambda_j^p}{C2^{jd}}\right) \right] + \sum_{j > j_\varepsilon} \frac{2^{-j\delta p}}{\lambda_j^p} \left[\log_+\left(\frac{C2^{jd}}{2^{-j\delta p}/\lambda_j^p}\right) + 1 \right] \\ & = 2^{j_\varepsilon d} \sum_{j \leq j_\varepsilon} 2^{(j-j_\varepsilon)d} \left[1 + \log_+\left(\frac{\lambda^p/\lambda_j^p}{C2^{(j-j_\varepsilon)(d+\delta p)}}\right) \right] \\ & \quad + \frac{2^{-j_\varepsilon \delta p}}{\lambda^p} \sum_{j > j_\varepsilon} 2^{(j_\varepsilon-j)\delta p} \frac{\lambda^p}{\lambda_j^p} \left[\log_+\left(\frac{C2^{(j-j_\varepsilon)(d+\delta p)}}{\lambda^p/\lambda_j^p}\right) + 1 \right]. \end{aligned}$$

The leading factors $2^{j_\varepsilon d}$ and $2^{-j_\varepsilon \delta p}/\lambda^p$ in front of the two sums are both equal to the desired upper bound $(1/\varepsilon)^{d/s}$, whence it suffices to show that the sums are

bounded. In case (a) $\lambda_j = \lambda$, and the sums can be seen to be bounded by making the change of summation index $j - j_\varepsilon \rightarrow j$. In case (b) $\lambda/\lambda_j = (j - j_\varepsilon)^2$, and the situation is the same, apart from extra factors $(j - j_\varepsilon)^{2p}$.

Finally, in case (c) the arrays in $\mathbb{B}_p^\delta$ satisfy $\max_k |\beta_{j,k}| \leq 2^{-j\delta}$ and we count in (iii) the vectors $(n_1, \dots, n_{L_j}) \in \mathbb{N}^{L_j}$ with $\max_k n_k \leq 2^{-j\delta}/\lambda_j$, the number being understood to be 0 if $j > j_\varepsilon$ when $L_j = 0$. This gives an upper bound on the number of different $\tilde{\beta}$ equal to a multiple of

$$\begin{aligned} & \sum_{j \leq j_\varepsilon} \left[L_j \log \left(\frac{C 2^{j\delta}}{L_j} \right) + L_j \log 2 + L_j \log 4 + 2L_j \log \left(\frac{2^{-j\delta}}{\lambda_j} \right) \right] \\ & \lesssim \sum_{j \leq j_\varepsilon} 2^{jd} \left[1 + \log_+ \left(\frac{2^{-j\delta}}{\lambda_j} \right) \right] \lesssim 2^{j_\varepsilon d} \sum_{j \leq j_\varepsilon} 2^{(j-j_\varepsilon)d} \log_+ \left(\frac{(j_\varepsilon - j)^2}{2^{j\delta} \varepsilon} \right). \end{aligned}$$

The leading factor $2^{j_\varepsilon d}$ is equal to $(1/\varepsilon)^{d/\delta}$, which is the upper bound given in the lemma in this case, and the sum can be seen to be bounded after substituting $\varepsilon = 2^{-j_\varepsilon \delta}$ and changing summation index. ■

2.7.3 Monotone Functions

The class of all uniformly bounded, monotone functions on the real line is Donsker. This may be proved in many ways; for instance, by verifying that the class possesses a finite bracketing integral. The following theorem shows that the bracketing entropy is of the order $1/\varepsilon$ uniformly in the underlying measure.

2.7.9 Theorem (Monotone functions). *The class $\mathcal{F}$ of monotone functions $f: \mathbb{R} \rightarrow [0, 1]$ satisfies*

$$\log N_{[]}(\varepsilon, \mathcal{F}, L_r(Q)) \leq K \left(\frac{1}{\varepsilon} \right),$$

for every probability measure Q , every $r \geq 1$, and a constant K that depends on r only.

Proof. It suffices to establish the bound for the class of monotonically increasing functions. It also suffices to prove the bound for Q equal to the uniform measure λ on $[0, 1]$. To see the latter, note first that if $Q^{-1}(u) = \inf\{x: Q(x) \geq u\}$ denotes the quantile function of Q , then the class $\mathcal{F} \circ Q^{-1}$ consists of functions $f \circ Q^{-1}: [0, 1] \rightarrow [0, 1]$ that are monotone. Since $Q^{-1} \circ Q(x) \leq x$ for every x and $u \leq Q \circ Q^{-1}(u)$ for every u , an ε -bracket $[l, u]$ for $f \circ Q^{-1}$ for λ yields a function $l \circ Q$ with the properties $l \circ Q \leq f \circ Q^{-1} \circ Q \leq f$ and

$$\|f - l \circ Q\|_{Q,r} = \|f \circ Q^{-1} - l \circ Q \circ Q^{-1}\|_{\lambda,r} \leq \|f \circ Q^{-1} - l\|_{\lambda,r} < \varepsilon.$$

Thus $l \circ Q$ is a “lower bracket.” If Q is strictly increasing, then $u \circ Q$ can be used as an upper bracket. More generally, we can repeat the preceding construction with $\bar{Q}^{-1}(u) = \sup\{x: Q(x) \leq u\}$ and obtain upper brackets $\bar{u} \circ \bar{Q}$. A set of full brackets can next be formed by taking every pair $[l \circ Q, \bar{u} \circ \bar{Q}]$ of functions so constructed for some function $f \in \mathcal{F}$.

Call a function g a left bracket for f if $g \leq f$ and $\|f - g\|_{\lambda, r} \leq \varepsilon$. It suffices to construct a set of left brackets of the given cardinality. Right brackets may next be constructed from left brackets for the class of functions $h(x) = 1 - f(1 - x)$.

Fix ε and set $c = (1/2)^{1/r}$. Fix a function f . Let $\mathcal{P}_0$ be the partition $0 = x_0 < x_1 = 1$ of the unit interval. Given a partition $\mathcal{P}_i$ of the form $0 = x_0 < x_1 < \dots < x_n = 1$, define $\varepsilon_i = \varepsilon_i(f)$ by

$$\varepsilon_i = \max_j (f(x_j) - f(x_{j-1}))(x_j - x_{j-1})^{1/r}.$$

Form a partition $\mathcal{P}_{i+1}$ by dividing the intervals $[x_{j-1}, x_j]$ in $\mathcal{P}_i$ for which

$$(f(x_j) - f(x_{j-1}))(x_j - x_{j-1})^{1/r} \geq c\varepsilon_i$$

into two halves of equal length. It is clear that $\varepsilon_0 \leq 1$ and that $\varepsilon_{i+1} \leq c\varepsilon_i \leq 2\varepsilon_{i+1}$. Let $n_i = n_i(f)$ be the number of intervals in $\mathcal{P}_i$ and $s_i = n_{i+1} - n_i$ the number of members of $\mathcal{P}_i$ that are divided to obtain $\mathcal{P}_{i+1}$. Then by the definitions of s_i and ε_i ,

$$\begin{aligned} s_i (c\varepsilon_i)^{r/(r+1)} &\leq \sum_j (f(x_j) - f(x_{j-1}))^{r/(r+1)} (x_j - x_{j-1})^{1/(r+1)} \\ &\leq \left(\sum_j (f(x_j) - f(x_{j-1})) \right)^{r/(r+1)} \left(\sum_j (x_j - x_{j-1}) \right)^{1/(r+1)}, \end{aligned}$$

by Hölder's inequality. This is bounded by $(f(1) - f(0))^{r/(r+1)} 1 \leq 1$. Consequently, the sum of the numbers of intervals up to the i th partition satisfies

$$\begin{aligned} \sum_{j=1}^i n_j &= i + \sum_{j=1}^i j s_{i-j} \leq 2 \sum_{j=1}^i j (c\varepsilon_{i-j})^{-r/(r+1)} \\ (2.7.10) \quad &\lesssim \sum_{j=1}^i j c^{rj/(r+1)} \varepsilon_i^{-r/(r+1)} \lesssim \varepsilon_i^{-r/(r+1)}, \end{aligned}$$

where $\lesssim$ denotes smaller than up to a constant that depends on r only.

Each function f generates a sequence of partitions $\mathcal{P}_0 \subset \mathcal{P}_1 \subset \dots$. Call two functions equivalent at stage i if their partitions up to the i th are the same. For each i this yields a partitioning of the class $\mathcal{F}$ in equivalence classes, which can for increasing i be visualized as a tree structure with the different equivalence classes at stage i as the nodes at level i . Continue the partitioning of a certain branch until the first level k such that $\varepsilon_k(f)^r \leq \varepsilon^{r+1}$ for every f in that branch. This defines a partitioning of the class $\mathcal{F}$ into finitely many subsets, each corresponding to some sequence of partitions $\mathcal{P}_0 \subset \dots \subset \mathcal{P}_k$ of the unit interval. For each subset and i , define

$$\tilde{\varepsilon}_i = \sup_f \varepsilon_i(f),$$

where the supremum is taken over all functions f in the subset. While $\tilde{\varepsilon}_i = \tilde{\varepsilon}_i(f)$ may be thought of as depending on f , it really depends only on the subset of f in the final partitioning, unlike ε_i . Since the numbers $n_1 \leq n_2 \leq \dots \leq n_i$ also depend on

the sequence $\mathcal{P}_0 \subset \cdots \subset \mathcal{P}_i$ only, inequality (2.7.10) remains valid if ε_i is replaced by $\tilde{\varepsilon}_i$.

For a fixed function f , define a left-bracketing function f_i , which is constant on each interval $[x_{j-1}, x_j]$ in the partition $\mathcal{P}_i$, recursively as follows. First $f_0 = 0$. Next given f_{i-1} , define f_i on the interval $[x_{j-1}, x_j]$ by

$$f_i(x_{j-1}) = f_{i-1}(x_{j-1}) + l_j \frac{\tilde{\varepsilon}_i}{(x_j - x_{j-1})^{1/r}},$$

where $l_j \geq 0$ is the largest integer such that $f_i \leq f$. Thus to construct f_i , the left bracket f_{i-1} is raised at x_{j-1} by as many steps of size $\tilde{\varepsilon}_i(x_j - x_{j-1})^{-1/r}$ as possible.

We claim that the set of functions f_k , when f ranges over $\mathcal{F}$ and $k = k(f)$ is the final level of partitioning for f , constitutes a set of left brackets of size proportional to ε of the required cardinality.

First, it is immediate from the construction of f_i that, for each of the intervals $[x_{j-1}, x_j]$ in the i th partition,

$$(f(x_{j-1}) - f_i(x_{j-1}))(x_j - x_{j-1})^{1/r} \leq \tilde{\varepsilon}_i.$$

Combining this with the definition of ε_i and the monotonicity of f , we obtain

$$(2.7.11) \quad (f(x) - f_i(x_{j-1}))(x_j - x_{j-1})^{1/r} \leq \varepsilon_i + \tilde{\varepsilon}_i \leq 2\tilde{\varepsilon}_i, \quad x \in [x_{j-1}, x_j].$$

Consequently,

$$\|f - f_i\|_{\lambda, r}^r \leq n_i (2\tilde{\varepsilon}_i)^r \lesssim \tilde{\varepsilon}_i^{r^2/(r+1)},$$

since $n_i \lesssim \tilde{\varepsilon}_i^{-r/(r+1)}$ by (2.7.10). For $i = k(f)$, this is bounded up to a constant by ε^r . Thus the brackets have the correct size.

Second, we count the number of functions f_k obtained when f ranges over $\mathcal{F}$. In view of the definition of $k = k(f)$, we have that $\varepsilon_{k-1}(g)^r > \varepsilon^{r+1}$, for some function g which is equivalent to f at stage $k-1$. This implies that $\varepsilon_{k-1}(g)^{-r/(r+1)} < 1/\varepsilon$. Combining this with (2.7.10), we find $\sum_{j=1}^{k-1} n_j \lesssim 1/\varepsilon$ and trivially $n_k(f) \leq 2n_{k-1} \lesssim 1/\varepsilon$. (Note that the numbers n_j for $j \leq k-1$ are the same for f and g .) The number of sequences $1 = n_0 \leq n_1 \leq \cdots \leq n_k \leq C/\varepsilon$ is equal to the number of ways we can choose k integers from the set $\{2, 3, \dots, \lfloor C/\varepsilon \rfloor\}$. It does not exceed $2^{C/\varepsilon}$. Given a sequence of this type, the number of ways to obtain $\mathcal{P}_{i+1}$ from $\mathcal{P}_i$ is $\binom{n_i}{s_i}$, which is bounded by 2^{n_i} . We conclude that the total number of different final partitions $\mathcal{P}_0 \subset \cdots \subset \mathcal{P}_k$ generated when f ranges over $\mathcal{F}$ is bounded by

$$2^{C/\varepsilon} 2^{n_0} 2^{n_1} \cdots 2^{n_{k-1}} \leq 2^{2C/\varepsilon}.$$

Given a sequence of partitions $\mathcal{P}_0 \subset \cdots \subset \mathcal{P}_k$, let $\mathcal{F}_i$ be the set of all left brackets f_i constructed on the partition $\mathcal{P}_i$ when f ranges over the subset of $\mathcal{F}$ corresponding to this sequence of partitions. Then $\mathcal{F}_0 = \{0\}$, and since $\tilde{\varepsilon}_i$ is fixed for the given sequence of partitions,

$$\#\mathcal{F}_i \leq \prod_{j=1}^{n_i} m_j \#\mathcal{F}_{i-1},$$

where m_j is the number of nonnegative integers that can occur in the definition of a function $f_i \in \mathcal{F}_i$ from a function $f_{i-1} \in \mathcal{F}_{i-1}$. Let the interval $[x_{j-1}, x_j)$ occur in the i th partition. By (2.7.11) we have that $f(x_{j-1}) - f_{i-1}(x_{j-1}) \leq 2\tilde{\varepsilon}_{i-1}(x_{j,i} - x_{j-1,i})^{-1/r}$, where $[x_{j-1,i}, x_{j,i})$ is the interval in the partition $\mathcal{P}_{i-1}$ that contains $[x_{j-1}, x_j)$. It follows that

$$0 \leq m_j \leq \frac{2\tilde{\varepsilon}_{i-1}(x_{j,i} - x_{j-1,i})^{-1/r}}{\tilde{\varepsilon}_i(x_j - x_{j-1})^{-1/r}} + 2 \lesssim \frac{2}{c} \cdot 1 + 2 =: L.$$

Conclude that for a given sequence of partitions $\mathcal{P}_0 \subset \cdots \subset \mathcal{P}_k$, the total number of left brackets f_k is bounded by

$$L^{n_k} L^{n_{k-1}} \cdots L^{n_1} \leq L^{2C/\varepsilon}.$$

Multiply this with the upper bound on the number of partitions to conclude that the total number of left brackets does not exceed $(2L)^{2C/\varepsilon}$. ■

A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is of bounded variation if it is the distribution function $f(x) = \mu(-\infty, x]$ of a signed measure on $\mathbb{R}$. Each such function can be written as the difference of two monotone functions, corresponding to the positive and negative parts of the signed measure. The space $BV(\mathbb{R})$ is the collection of all functions of bounded variation, and it can be normed with the total variation norm on the corresponding signed measures. It follows from Theorem 2.7.9 that the $L_r(Q)$ -bracketing entropy of the unit ball of $BV(\mathbb{R})$ is of the order $(1/\varepsilon)$, uniformly in Q . Similar bounds are valid for the bounded variation functions $f: [a, b] \rightarrow \mathbb{R}$ on a subinterval of $\mathbb{R}$.

It is instructive to relate the bound on the entropy of the class $BV_1[a, b]$ of functions $f: [a, b] \rightarrow \mathbb{R}$ of variation bounded by 1 to the entropy bounds on Besov spaces. Because, for $h > 0$,

$$\int_a^{b-h} |f(x+h) - f(x)| dx = \int_a^{b-h} \left| \int_{(x, x+h]} df(x) \right| dx \leq h \int_a^b |df(s)|,$$

the set $BV_1[a, b]$ is contained in the unit ball of the Besov space $B_{1,\infty}^1[a, b]$. In view of Theorem 2.7.4 the $L_r(\lambda)$ -entropy of the latter unit ball is of the order $(1/\varepsilon)$ provided that $1 > 1/(1 - 1/\tau)_+$, i.e. $\tau < \infty$. Thus this theorem gives the correct bound on the entropy of the set of monotone functions, albeit not the *bracketing entropy*. (The restriction to Lebesgue measure on a compact interval is not a real disadvantage of this approach, as the second paragraph of the proof of Theorem 2.7.9 shows that the general setting can be easily reduced to this case.)

2.7.4 Convex Sets and Functions

For a pair of subsets C and D of a metric space, the *Hausdorff distance* is defined as

$$h(C, D) = \sup_{x \in C} d(x, D) \vee \sup_{x \in D} d(x, C).$$

Restricted to the closed subsets, this defines a metric (which may be infinite). The following lemma gives the entropy of the collection of all compact, convex subsets of a fixed, bounded subset of $\mathbb{R}^d$ with respect to the Hausdorff metric.

2.7.12 Lemma (Convex sets). *For the class $\mathcal{C}$ of all compact, convex subsets of a fixed, bounded subset of $\mathbb{R}^d$, with $d \geq 2$, one has*

$$K_1 \left(\frac{1}{\varepsilon} \right)^{(d-1)/2} \leq \log N(\varepsilon, \mathcal{C}, h) \leq K_2 \left(\frac{1}{\varepsilon} \right)^{(d-1)/2},$$

for constants K_i that depend on d and the bounded set only.

Proof. See Bronštein (1976), Dudley (1984), or Dudley (2014), Theorem 8.25. ■

As a consequence we obtain $L_r(Q)$ -bracketing numbers for Lebesgue absolutely continuous probability measures Q . Combination with the bracketing central limit theorem shows that the closed convex sets of the unit ball in two dimensions form a Donsker class for, for instance, the uniform measure. For any dimension, this class is Glivenko-Cantelli.

For an extension to unbounded convex sets, see Section 2.10.5.

2.7.13 Corollary. *For the class $\mathcal{C}$ of all compact, convex subsets of a fixed, bounded subset of $\mathbb{R}^d$ with $d \geq 2$, one has the entropy bound*

$$\log N_{[]}(\varepsilon, \mathcal{C}, L_r(Q)) \leq K \left(\frac{1}{\varepsilon} \right)^{(d-1)r/2},$$

for every Lebesgue absolutely continuous probability measure Q with bounded density and a constant K that depends on the bounded set, Q , and d only.

Proof. For a given set C , let ${}_{\varepsilon}C = \{x: d(x, C^c) > \varepsilon\}$ be the points that are at least a distance ε inside C , and let C^{ε} be the points within distance ε of C . Then there exists a constant K_0 depending only on $\mathcal{C}$ such that the Lebesgue measure λ satisfies

$$\lambda(C^{\varepsilon} - {}_{\varepsilon}C) \leq K_0 \varepsilon,$$

for every $0 < \varepsilon < 1$.[†]

If $h(C, D) < \varepsilon$, then it must be that ${}_{\varepsilon}C \subset D \subset C^{\varepsilon}$. Thus if $C_1, \dots, C_p$ are the centers of Hausdorff balls of radius ε that cover $\mathcal{C}$, then the pairs $[{}_{\varepsilon}C_i, C_i^{\varepsilon}]$ form a class of brackets that cover $\mathcal{C}$. Their sizes in $L_r(Q)$ are bounded by $Q^{1/r}(C^{\varepsilon} - {}_{\varepsilon}C)$, which is bounded by $\|q\|_{\infty}^{1/r} (K_0 \varepsilon)^{1/r}$. Now the corollary follows from the previous lemma. ■

Next consider the set of all convex functions $f: C \rightarrow \mathbb{R}$ defined on a compact, convex subset of $\mathbb{R}^d$, which we assume throughout to have nonempty interior. We consider the class of such functions that are uniformly bounded in L_p , for some p . The entropy of this class turns out to depend on the shape of C . The nicest case is that C is the convex hull of finitely many points, when the L_r -entropy is of order $(1/\varepsilon)^{d/2}$ for any $r < p$. This covers for instance the unit interval in $\mathbb{R}$, or a rectangle in $\mathbb{R}^d$. However, the constant in the bound increases with the number of extreme points of C , and for general convex sets, such as the Euclidean unit ball, the entropy is $(1/\varepsilon)^{d/2}$ only if $r < d/(d-1+d/p)$ and deteriorates otherwise.

[†] This is not trivial, although it is intuitively clear.

2.7.14 Theorem (Convex functions). *Let $\mathcal{F}_p$ be the class of all convex functions $f: C \rightarrow [0, 1]$ defined on a compact, convex subset $C \subset \mathbb{R}^d$ with $\|f\|_p \leq 1$, for some fixed $p \in [1, \infty]$. If C is the closed convex hull of m points, then, for $1 \leq r < p$,*

$$\log N(\varepsilon, \mathcal{F}_p, L_r(\lambda)) \leq K_d m^{\lceil d/2 \rceil} \lambda(C)^{d/(2r) - d/(2p)} \left(\frac{1}{\varepsilon}\right)^{d/2},$$

for a constant K_d that depends on the dimension d only. Furthermore, if C is an arbitrary compact, convex subset $\mathbb{R}^d$, then, for a constant K that depends on d, p, r and the diameter of C ,

$$\log N(\varepsilon, \mathcal{F}_p, L_r(\lambda)) \leq \begin{cases} K \left(\frac{1}{\varepsilon}\right)^{d/2}, & \text{if } r < \frac{d}{d-1+d/p}, \\ K \left(\frac{1}{\varepsilon}\right)^{d/2} \left(\log \frac{1}{\varepsilon}\right)^{1 + \frac{(p-r)d}{2pr}}, & \text{if } r = \frac{d}{d-1+d/p}, \\ K \left(\frac{1}{\varepsilon}\right)^{\frac{(d-1)pr}{2(p-r)}}, & \text{if } r > \frac{d}{d-1+d/p}. \end{cases}$$

In the last two formulas $(p-r)/p$ and $p/(p-r)$ are read as 1 if $p = \infty$. In both cases, if $p = \infty$, then the same bound is valid for the bracketing entropy.

Proof. See Gao and Wellner (2017). ■

The entropy in the preceding theorem is relative to the L_r -norm for the Lebesgue measure on C . The same bounds are valid for the $\|q\|_\infty^{1/r}$ ε -entropy relative to the $L_r(Q)$ -norm, for any measure Q with a uniformly bounded Lebesgue density q on C .

The theorem implies that the bracketing entropy integral of the classes $\mathcal{F}_\infty$ is finite for $d \leq 3$ in the case that C is a closed, convex polytope, and for $d \leq 2$ in the case of a general compact, convex domain. (In the latter case the cut-point for r is $d/(d-1)$; the bound is $(1/\varepsilon)$ times a logarithmic factor if $d = 2$ and $(1/\varepsilon)^2$ if $d = 3$, both times taken in the regime where the bound is worse than $(1/\varepsilon)^{d/2}$.)

The theorem does not provide bounds for the uniform norm. A bound for $r = \infty$ is available only if the class of functions is further restricted to uniformly Lipschitz functions. It can then be derived from the entropy of the class of their supergraphs, which are convex sets in $\mathbb{R}^{d+1}$, for the Hausdorff metric. The assumption that the functions are Lipschitz is unpleasant, though it may be noted that every function f that is convex on C^η for some $\eta > 0$ is automatically Lipschitz on C , with Lipschitz constant $2 \sup\{|f(x)|: x \in C^\eta\}/\eta$. (See Problem 1.6.3.) The proof of the following result is much shorter than the proof of Theorem 2.7.14, although the latter also uses the device that a convex function is automatically Lipschitz in the η -interior of its domain.

2.7.15 Corollary (Convex Lipschitz functions). *Let $\mathcal{F}$ be the class of all convex functions $f: C \rightarrow [0, 1]$ defined on a compact, convex subset $C \subset \mathbb{R}^d$ such that $|f(x) - f(y)| \leq L\|x - y\|$ for every x, y . Then*

$$\log N(\varepsilon, \mathcal{F}, \|\cdot\|_\infty) \leq K_{d,C} (1 + L)^{d/2} \left(\frac{1}{\varepsilon}\right)^{d/2},$$

for a constant $K_{d,C}$ that depends on the dimension d and C only.

Proof. Let C_f be the supergraph $\{(x, t): f(x) \leq t\}$ of a function f . For every pair of points x and y and functions f and g ,

$$|f(x) - g(x)| \leq |f(x) - g(y)| + L\|y - x\|.$$

Fix a point x such that $f(x) < g(x)$. Then the boundary of the supergraph of f is below the supergraph of g at x , and the projection (closest point) of the point $(x, f(x))$ on the supergraph of g has the form $(y, g(y))$ for some point y . The distance $d((x, f(x)), (y, g(y)))$ between the two points in $\mathbb{R}^{d+1}$ is bounded above by the Hausdorff distance between the two supergraphs. On the other hand, the distance between the two points is bounded below by a multiple of

$$\|x - y\| + |f(x) - g(y)| \geq (1 + L)^{-1} |f(x) - g(x)|.$$

Conclude that $\|f - g\|_\infty \lesssim (1 + L) h(C_f, C_g)$. Finally, apply Lemma 2.7.12 to the set of supergraphs intersected with the set $C \times [0, 1]$. ■

2.7.5 Analytic Functions

The ε -entropy of the unit ball in Hölder or Besov scales grows polynomially in $1/\varepsilon$ of degree d/α , which decreases to zero if the regularity index α increases indefinitely. Thus the entropy of classes of functions that are *infinitely* smooth is smaller than any polynomial in $1/\varepsilon$. The following theorem illustrates that it may be logarithmic in $1/\varepsilon$.

2.7.16 Theorem (Analytic functions). *The class $\mathcal{A}_A[0, 1]^d$ of all functions $f: [0, 1]^d \rightarrow \mathbb{R}$ that can be extended to an analytic function on the set $G = \{z \in \mathbb{C}^d: \|z - [0, 1]^d\|_\infty < A\}$ with $\sup_{z \in G} |f(z)| \leq 1$ satisfies, for $\varepsilon < 1/2$ and a constant c_d that depends on d only,*

$$\log N(\varepsilon, \mathcal{A}_A[0, 1]^d, \|\cdot\|_\infty) \leq c_d \left(\frac{1}{A}\right)^d \left(\log \frac{1}{\varepsilon}\right)^{1+d}.$$

Proof. Let $\{x_1, \dots, x_m\}$ be an $A/3$ -net in $[0, 1]^d$ for the maximum norm, and let $[0, 1]^d = \cup_i B_i$ be a partition in sets obtained by assigning every $x \in [0, 1]^d$ to a closest $x_i \in \{x_1, \dots, x_m\}$. Consider the piecewise polynomials $P = \sum_{i=1}^m P_{i, a_i} 1_{B_i}$, for

$$P_{i, a_i}(x) = \sum_{k: |k| \leq n} a_{i, k} (x - x_i)^k.$$

Here the sum ranges over all multi-index vectors $k = (k_1, \dots, k_d) \in \mathbb{N}_0^d$ with $|k| = k_1 + \dots + k_d \leq n$, and for $y = (y_1, \dots, y_d) \in \mathbb{R}^d$ the notation y^k is short for $y_1^{k_1} y_2^{k_2} \dots y_d^{k_d}$. We obtain a finite set of functions by discretizing the coefficients $a_{i, k}$ for each (i, k) over an $\varepsilon/A^{k \cdot}$ -net over the interval $[-1/A^{k \cdot}, 1/A^{k \cdot}]$. The log cardinality of this set is bounded by

$$\log \left(\prod_i \prod_{k: |k| \leq n} \#a_{i, k} \right) \leq m \log \left(\prod_{k: |k| \leq n} \frac{2/A^{k \cdot}}{\varepsilon/A^{k \cdot}} \right) \leq m(n+1)^d \log \left(\frac{2}{\varepsilon} \right).$$

We can choose $m \leq (3/A)^d$. The proof is complete once it is shown that the resulting set of functions is an ε -net for n of the order $\log(1/\varepsilon)$.

By the Cauchy formula (d applications of the formula in one dimension suffice), for $C_1, \dots, C_d$ circles of radius $\bar{A} < A$ in the complex plane around the coordinates $x_{i1}, \dots, x_{id}$ of x_i , and with D^k the partial derivative of order k and $k! = k_1!k_2! \cdots k_d!$,

$$\left| \frac{D^k f(x_i)}{k!} \right| = \left| \frac{1}{(2\pi i)^d} \oint_{C_1} \cdots \oint_{C_d} \frac{f(z)}{(z - x_i)^{k+1}} dz_1 \cdots dz_d \right| \leq \frac{1}{\bar{A}^k}.$$

Consequently, for any $z \in B_i$, appropriately chosen a_i , and $\bar{A} = 2A/3$,

$$\begin{aligned} \left| \sum_{k > n} \frac{D^k f(x_i)}{k!} (z - x_i)^k \right| &\leq \sum_{k > n} \frac{1}{\bar{A}^k} (A/3)^k \leq \sum_{l=n+1}^{\infty} \frac{(l+1)^d}{2^l} \lesssim \left(\frac{3}{4}\right)^n, \\ \left| \sum_{k \leq n} \frac{D^k f(x_i)}{k!} (z - x_i)^k - P_{i,a_i}(z) \right| &\leq \sum_{k \leq n} \frac{\varepsilon}{A^k} (A/3)^k \leq \sum_{l=0}^n \frac{(l+1)^d}{2^l} \varepsilon \lesssim \varepsilon. \end{aligned}$$

By analyticity, the function f is equal to its infinite Taylor series on every open ball within G and hence every set B_i . We conclude that the piecewise polynomials form a $c\varepsilon$ -net for n sufficiently large that $(3/4)^n$ is smaller than ε . ■

2.7.6 Parametrized Classes

A simple, but useful, application of bracketing is to classes of functions $x \mapsto f_t(x)$ that are Lipschitz in the index parameter $t \in T$. Suppose that

$$|f_s(x) - f_t(x)| \leq d(s, t) F(x),$$

for some metric d on the index set, function F on the sample space, and every x . Then $(\text{diam } T)F$ is an envelope function for the class $\{f_t - f_{t_0} : t \in T\}$ for any fixed t_0 . The bracketing numbers of this class are bounded by the covering numbers of T .

2.7.17 Theorem. *Let $\mathcal{F} = \{f_t : t \in T\}$ be a class of functions satisfying the preceding display for every s and t and some fixed function F . Then, for any norm $\|\cdot\|$,*

$$N_{[]} (2\varepsilon \|F\|, \mathcal{F}, \|\cdot\|) \leq N(\varepsilon, T, d).$$

Proof. Let $t_1, \dots, t_p$ be an ε -net for d for T . Then the brackets $[f_{t_i} - \varepsilon F, f_{t_i} + \varepsilon F]$ cover $\mathcal{F}$. They are of size $2\varepsilon \|F\|$. ■

2.7.7 Gaussian Mixtures

For $\sigma > 0$ and a probability distribution F on $\mathbb{R}$ let $p_{F,\sigma}: \mathbb{R} \rightarrow \mathbb{R}$ be the mixture density

$$p_{F,\sigma}(x) = \int \frac{1}{\sigma} \phi\left(\frac{x-z}{\sigma}\right) dF(z),$$

for ϕ the density of the standard normal distribution. In this section we estimate the bracketing entropy of classes of normal mixtures of this form. We consider these under the L_1 -norm relative to the Lebesgue measure on $\mathbb{R}$ and the uniform norm on $\mathbb{R}$.

A first theorem concerns mixtures with mixing distribution F restricted to an interval $[-a, a]$.

2.7.18 Theorem (Gaussian mixtures). *For $0 < b_1 < b_2 < \infty$ with $b_1 < 1$ and $a \geq e$, let $\mathcal{P}_{a,b_1,b_2}$ be the set of all normal mixtures $p_{F,\sigma}$ with F ranging over the probability distributions on $[-a, a]$ and $\sigma \in [b_1, b_2]$. Then there exists a universal constant K such that for d equal to the $L_1(\lambda)$ -norm or the uniform norm, and $0 < \varepsilon < 1/2$,*

$$\log N_{[]}(\varepsilon, \mathcal{P}_{a,b_1,b_2}, d) \leq K \frac{a}{b_1} \left(\log \frac{ab_2}{\varepsilon b_1} \right)^2.$$

For fixed b_1, b_2 and a the entropy is of the order $(\log(1/\varepsilon))^2$. This is bigger, but not much bigger than the entropy of finite-dimensional classes. On the other hand, the entropy bound increases proportionally to the minimal scale b_1 of the mixture components if this approaches zero.

The set of mixtures with an arbitrary mixing distribution on $\mathbb{R}$ is not totally bounded relative to the L_1 -metric (even if the scale is restricted to 1) and hence the preceding theorem can only be extended to mixing distributions with possibly noncompact support if the mixing distribution is restricted in some other way. We consider mixing distributions with tails bounded by a given decreasing function $A: (0, \infty) \rightarrow [0, 1]$.

Let $\tilde{A}$ be the function defined by $\tilde{A}(a) = A(a) + \int_a^\infty A d\lambda$, and let $\tilde{A}^{-1}$ be its inverse.

2.7.19 Theorem (Gaussian mixtures). *For $0 < b_1 < 1 < b_2 < \infty$ and A a given decreasing function, let $\mathcal{P}_{A,b_1,b_2}$ be the set of all normal mixtures $p_{F,\sigma}$ with F ranging over the probability distributions on $\mathbb{R}$ with $F[-a, a]^c \leq A(a)$ for all $a > 0$ and $\sigma \in [b_1, b_2]$. Then there exist universal constants c and K such that, for $0 < \varepsilon < 1/4 \wedge \tilde{A}(e)$,*

$$\log N_{[]} (c\varepsilon, \mathcal{P}_{A,b_1,b_2}, \|\cdot\|_1) \leq K \frac{\tilde{A}^{-1}(\varepsilon b_1/b_2)}{b_1} \left(\log \frac{\tilde{A}^{-1}(\varepsilon b_1/b_2) b_2}{\varepsilon b_1} \right)^2.$$

2.7.20 Example. If the mixing distributions have sub-Gaussian tails, then we can apply the preceding theorem with A equal to $1 - \Phi(a) \leq \phi(a)/a$, whence $\tilde{A}$ is bounded by a multiple of the same function. Then $\tilde{A}^{-1}(\varepsilon)$ is bounded by a multiple of $(\log(1/\varepsilon))^{1/2}$ and, for fixed b_1, b_2 , the bracketing entropy is bounded by a multiple of $(\log(1/\varepsilon))^{5/2}$. As long as the tails of the mixing distributions are bounded

by a function of the form $A(a) = \exp(-da^\delta)$, the entropy of the set of mixtures increases at most through a power of $\log(1/\varepsilon)$. On the other hand, polynomially decreasing tails incur an additional factor of ε^{-k} in the entropy bounds, so that the entropy bound is dominated by the tails in that case.

The proofs of the preceding theorems are based on approximation by discrete mixtures with finite support.

2.7.21 Lemma. *There exist universal constants C and D such that for any $0 < \varepsilon < 1/2$, any $a, \sigma > 0$ and any probability measure F on $[-a, a]$ there exists a discrete probability measure F' on the support of F with fewer than $D(a\sigma^{-1} \vee 1) \log(1/\varepsilon)$ support points such that $\|p_{F,\sigma} - p_{F',\sigma}\|_\infty \leq C\varepsilon/\sigma$.*

Proof. We first prove the lemma for $a = \sigma = 1$. For $\varepsilon < 1/2$ we have $M := \sqrt{8 \log(1/\varepsilon)} > 2$ and hence, for any probability measures F and F' on $[-1, 1]$,

$$\sup_{|x| \geq M} |p_{F,1}(x) - p_{F',1}(x)| \leq 2\phi(M-1) \leq 2\phi(M/2) \leq \varepsilon.$$

The remainder in the Taylor expansion of order $N-1$ of the function $x \mapsto e^{-x}$ on $[0, \infty)$ is bounded above by $|x|^N/N! \leq (e|x|)^N/N^N$. Consequently, there exists a polynomial P_{2N-2} of degree $2N-2$ such that

$$|\phi(x) - P_{2N-2}(x)| \leq \frac{(ex^2/2)^N}{N^N}.$$

If F and F' are probability measures on $[-1, 1]$ with the same moments of orders $1, \dots, 2N-2$, then $\int P_{2N-2}(x-z) d(F-F')(z) = 0$, and hence

$$\sup_{|x| \leq M} |p_{F,1}(x) - p_{F',1}(x)| \leq 2 \sup_{|x| \leq M, |z| \leq 1} |\phi(x-z) - P_{2N-2}(x-z)|.$$

The argument $x-z$ in the right side is bounded in absolute value by $M+1 \leq 3M/2 = \sqrt{18 \log(1/\varepsilon)}$. Therefore, there exists a universal constant c such that the left side of the preceding display is bounded above by a multiple of

$$\frac{(c \log(1/\varepsilon))^N}{N^N} = \exp[-N(\log N - \log(c \log(1/\varepsilon)))].$$

This is bounded by ε for N equal to a sufficiently large multiple of $\log(1/\varepsilon)$.

We can match $2N-2$ moments with a discrete distribution F' with $2N-1$ support points in the support of F . (See Problem 2.7.19.) This completes the proof of the lemma if $a = \sigma = 1$.

We partition a general interval $[-a, a]$ into $k = \lfloor 2a/\sigma \rfloor$ disjoint, consecutive subintervals $I_1, \dots, I_k$ of length σ and a final interval I_{k+1} of length l_{k+1} smaller than σ . We may write $F = \sum_{i=1}^{k+1} F(I_i)F_i$, where each F_i is a probability measure concentrated on I_i , and then $p_{F,\sigma} = \sum_{i=1}^{k+1} F(I_i)p_{F_i,\sigma}$. For ease of notation, let Z_i be a random variable distributed according to F_i , and for a_i the left end point of I_i , let G_i be the law of $W_i = (Z_i - a_i)/\sigma$. Thus G_i is a probability measure on the interval $[0, 1]$ for $i = 1, \dots, k$ and on the interval $[0, l_{k+1}/\sigma] \subset [0, 1]$ for $i = k+1$.

By the first part of the proof there exist probability measures G'_i on $[0, 1]$ for $i = 1, \dots, k$ and on $[0, l_{k+1}/\sigma]$ for $i = k + 1$, with $N_i \lesssim \log(1/\varepsilon)$ support points such that $\|p_{G_{i,1}} - p_{G'_{i,1}}\|_\infty \lesssim \varepsilon$. Let F'_i be the law of $a_i + \sigma W'_i$ if W'_i has law G'_i and set $F' = \sum_{i=1}^{k+1} F(I_i)F'_i$. The identity $p_{F_i,\sigma}(x) = \sigma^{-1}p_{G_{i,1}}((x - a_i)/\sigma)$, and similar identities for F'_i and G'_i show that $\|p_{F_i,\sigma} - p_{F'_i,\sigma}\|_\infty = \sigma^{-1}\|p_{G_{i,1}} - p_{G'_{i,1}}\|_\infty$. Combining this with the inequality $\|p_{F,\sigma} - p_{F',\sigma}\|_\infty \leq \sum_{i=1}^{k+1} F(I_i)\|p_{F_i,\sigma} - p_{F'_i,\sigma}\|_\infty$, we conclude that $p_{F',\sigma}$ has the required distance to $p_{F,\sigma}$. The number of support points of F' is bounded by the number of intervals $k + 1$ times the maximum number of support points of an F'_i , and hence is bounded as claimed. ■

Proof of Theorems 2.7.18 and 2.7.19. For given positive numbers α and β fix a minimal α -net Σ over the interval $[b_1, b_2]$, and let $\mathbb{F}$ be the set of discrete probability distributions on $[-a, a]$ with at most $N = D(ab_1^{-1} \vee 1) \log(1/\beta)$ support points, for the constant D of Lemma 2.7.21. For every $\sigma \in [b_1, b_2]$ there exists $\sigma' \in \Sigma$ with $|\sigma - \sigma'| \leq \alpha$, whence, with ϕ_σ the normal density with scale σ ,

$$\|p_{F,\sigma} - p_{F,\sigma'}\|_\infty \leq \|\phi_\sigma - \phi_{\sigma'}\|_\infty \lesssim \frac{|\sigma - \sigma'|}{(\sigma \wedge \sigma')^2} \lesssim \frac{\alpha}{b_1^2}.$$

By Lemma 2.7.21 there exists for every given probability measure F on $[-a, a]$ an element $F' \in \mathbb{F}$ (possibly depending on σ') such that $\|p_{F,\sigma'} - p_{F',\sigma'}\|_\infty \lesssim \beta/b_1$. Hence

$$\|p_{F,\sigma} - p_{F',\sigma'}\|_\infty \lesssim \frac{\alpha}{b_1^2} + \frac{\beta}{b_1}.$$

Thus the set $\mathcal{P}$ of mixtures $p_{F,\sigma}$ with F ranging over $\mathbb{F}$ and σ over Σ is an ε -net over $\mathcal{P}_{a,b_1,b_2}$ for the uniform norm, if the expression at the right of the last display is made to be less than ε .

We next construct a finite net over $\mathcal{P}$ by restricting the support points and weights of the mixing distributions to suitable grids. For a given $\gamma > 0$ let $\mathcal{S}$ be a γ -net over the N -dimensional simplex for the ℓ_1 -norm, and let $\mathcal{P}'$ be the set of all mixtures $p_{F,\sigma} \in \mathcal{P}$ such that the (maximally) N support points of F are among the points $\{0, \pm\delta, \pm 2\delta, \dots\} \cap [-a, a]$ with weights belonging to $\mathcal{S}$, for given $\delta > 0$. We may “project” $p_{F,\sigma} \in \mathcal{P}$ into $p_{F',\sigma} \in \mathcal{P}'$ by first moving each point mass of F to a closest point in the grid $\{0, \pm\delta, \pm 2\delta, \dots\}$, and next changing the vector of sizes of the point masses to a closest vector in $\mathcal{S}$. If $F = \sum_j p_j \delta_{z_j}$ and $F' = \sum_j p'_j \delta_{z'_j}$ with z'_j the projection of z_j on the grid, then

$$|p_{F,\sigma}(x) - p_{F',\sigma}(x)| \leq \sum_j \left(p_j |\phi_\sigma(x - z_j) - \phi_\sigma(x - z'_j)| + |p_j - p'_j| \phi_\sigma(x - z'_j) \right).$$

We conclude that $\|p_{F,\sigma} - p_{F',\sigma}\|_\infty \lesssim \delta \|\phi'_\sigma\|_\infty + \gamma \|\phi_\sigma\|_\infty \lesssim \delta/b_1^2 + \gamma/b_1$.

There are at most $(b_2 - b_1)/\alpha \vee 1$ possible choices of $\sigma \in \Sigma$. Each z'_j can assume at most $2a/\delta + 1$ different values, for $j = 1, \dots, N$. The cardinality of a minimal γ -net $\mathcal{S}$ over the N -dimensional unit simplex is bounded by $(5/\gamma)^N$ for

$\gamma \leq 1$ (cf. Problem 2.1.7). Therefore,

$$\begin{aligned} \log \# \mathcal{P}' &\lesssim \log \left[\left(\frac{b_2 - b_1}{\alpha} \vee 1 \right) \left(\frac{2a}{\delta} + 1 \right)^N \left(\frac{5}{\gamma \wedge 1} \right)^N \right] \\ &\lesssim \log \left(\frac{b_2 - b_1}{\alpha} \vee 1 \right) + \left(\frac{a}{b_1} \vee 1 \right) \log \frac{1}{\beta} \left[\log \left(\frac{2a}{\delta} + 1 \right) + \log \left(\frac{5}{\gamma \wedge 1} \right) \right]. \end{aligned}$$

This number is a bound on the ε -entropy of $\mathcal{P}_{a,b_1,b_2}$ for the uniform norm at ε equal to a multiple of $\alpha/b_1^2 + (\beta/b_1) + (\delta/b_1^2) + (\gamma/b_1)$. We choose $\alpha = \delta = b_1^2 \varepsilon$ and $\beta = \gamma = b_1 \varepsilon$ to conclude the proof of Theorem 2.7.18 for the uniform norm.

To extend this to a bound on the L_1 -bracketing numbers we first note that

$$H(x) = \frac{1}{b_1} \phi \left(\frac{x}{2b_2} \right) 1_{|x| > 2a} + \frac{1}{b_1} \phi(0) 1_{|x| \leq 2a}$$

is an envelope function for $\mathcal{P}_{a,b_1,b_2}$. Therefore, given an η -net $p_1, \dots, p_M$ over $\mathcal{P}_{a,b_1,b_2}$ for the uniform norm, the brackets $[l_i, u_i]$ defined by $l_i = (p_i - \eta) \vee 0$ and $u_i = (p_i + \eta) \wedge H$ cover $\mathcal{P}_{a,b_1,b_2}$. Because $u_i - l_i \leq (2\eta) \wedge H$ the size in L_1 of the bracket $[l_i, u_i]$ is bounded by, for any $B > 2a$,

$$\int (u_i - l_i) d\lambda \leq 2\eta B + \int_{|x| > B} H(x) dx \lesssim \eta B + \frac{b_2}{b_1} \phi \left(\frac{B}{2b_2} \right).$$

We choose $\eta = (c\varepsilon/ab_2)/\sqrt{\log(b_2/b_1\varepsilon)}$ and $B = dab_2\sqrt{\log(b_2/b_1\varepsilon)}$ to reduce the right side to $cd\varepsilon + (b_2/b_1)(b_1\varepsilon/b_2)^{d^2a^2/8}$. We can choose a universal constant d so that the second term is bounded by an arbitrarily small multiple of ε , for every $\varepsilon < 1/2$, $a > e$ and $b_1 < 1 < b_2$, and next choose $c < 1$ so that the first term $cd\varepsilon$ is an arbitrarily small multiple of ε as well. By the first part of the proof the number of brackets M is bounded above by

$$\frac{a}{b_1} \left(\log \frac{ab_2}{\eta b_1} \right)^2 \leq \frac{a}{b_1} \left(\log \frac{ab_2}{\varepsilon b_1} + \log \frac{ab_2}{c} + \frac{1}{2} \log \log \frac{b_2}{b_1 \varepsilon} \right)^2.$$

Under the assumed conditions $\varepsilon < 1/2$, $a > e$ and $b_1 < 1 < b_2$, the second and third terms within brackets are bounded by a multiple of the first. This concludes the proof of Theorem 2.7.18.

Consider any F that satisfies the tail bound assumed in Theorem 2.7.18. By partial integration, for any $x > a > 0$,

$$\int_a^\infty \phi_\sigma(x-z) dF(z) = (1-F)(a) \phi_\sigma(x-a) + \int_a^\infty \phi'_\sigma(z-x)(1-F)(z) dz.$$

Because the function ϕ'_σ is negative on the positive half line, the integral on the right is bounded by its restriction to the interval (a, x) . If $x/2 > a$, then it can be further bounded by

$$\int_a^{x/2} \phi'_\sigma(z-x) dz A(a) + \int_{x/2}^x |\phi'_\sigma(z-x)| dz A(x/2) \leq \phi_\sigma \left(-\frac{x}{2} \right) A(a) + \frac{A(x/2)}{\sigma}.$$

If $x/2 < a < x$, then the second integral in this expression alone is a valid upper bound. For $x < a$ we have $\phi_\sigma(x-z) \leq \phi_\sigma(x-a)$ for every $z > a$ and hence

$\int_a^\infty \phi_\sigma(x-z) dF(z) \leq \phi_\sigma(x-a)F[a, \infty) \leq \phi_\sigma(x-a)A(a)$. Together with analogous bounds for the left tail, we obtain that $\int_{|z|>a} \phi_\sigma(x-z) dF(z)$ is bounded above by, for every $x \in \mathbb{R}$,

$$H(x) := \frac{b_2}{b_1} \left(\phi_{b_2}(x-a) + \phi_{b_2}(x/2) + \phi_{b_2}(x+a) \right) A(a) + \frac{A(x/2)}{b_1} 1_{|x| \geq 2a}.$$

If F_a is the renormalized restriction to $[-a, a]$ of a probability measure F , then $F[-a, a]p_{F_a, \sigma} \leq p_{F, \sigma} \leq F[-a, a]p_{F_a, \sigma} + H$. Consequently, given ε -brackets $[l_i, u_i]$ that cover $\mathcal{P}_{a, b_1, b_2}$, there exists, for every (F, σ) as in the definition of $\mathcal{P}_{A, b_1, b_2}$, a bracket $[l_i, u_i]$ with $l_i(1 - A(a)) \leq p_{F, \sigma} \leq u_i + H$. Thus the brackets $[l_i(1 - A(a)), u_i + H]$ cover $\mathcal{P}_{A, b_1, b_2}$. The size in L_1 of such a bracket is bounded by $\|u_i - l_i\|_1 + \|l_i\|_1 A(a) + \|H\|_1 \lesssim \varepsilon + \tilde{A}(a)b_2/b_1$, as $b_1 < 1 < b_2$ by assumption. Now we choose a such that $\tilde{A}(a) \leq \varepsilon b_1/b_2$ and apply Theorem 2.7.18 to bound the number of brackets $[l_i, u_i]$. ■

2.7.8 Sets with Smooth Boundaries

Subgraphs of smooth functions $f: [0, 1]^{d-1} \rightarrow [0, 1]$ are subsets of $[0, 1]^d$ with smooth boundaries. Their bracketing entropy can easily be obtained from an entropy bound on the functions themselves.

2.7.22 Corollary (Hölder subgraphs). *Let $\mathcal{C}_{\alpha, d}$ be the collection of subgraphs of $C^\alpha([0, 1]^{d-1})_1$. There exists a constant K depending only on α and d such that*

$$\log N_{[]}(\varepsilon, \mathcal{C}_{\alpha, d}, L_r(Q)) \leq K \|q\|_\infty^{d/\alpha} \left(\frac{1}{\varepsilon} \right)^{(d-1)r/\alpha},$$

for every $r \geq 1$, $\varepsilon > 0$, and probability measure Q with bounded Lebesgue density q on $\mathbb{R}^d$.

Proof. Let $f_1, \dots, f_p$ be the centers of $\|\cdot\|_\infty$ -balls of radius ε that cover $C^\alpha([0, 1]^{d-1})_1$. For each i , define the sets C_i and D_i as the subgraphs of $f_i - \varepsilon$ and $f_i + \varepsilon$, respectively. Then the pairs $[C_i, D_i]$ form brackets that cover $\mathcal{C}_{\alpha, d-1}$. (More precisely, their indicator functions bracket the set of indicator functions of the sets in $\mathcal{C}_{\alpha, d-1}$.) Their $L_1(Q)$ -size equals

$$Q(C_i \triangle D_i) = \int_{[0, 1]^{(d-1)} \times \mathbb{R}} 1\{f_i(x) - \varepsilon \leq t < f_i(x) + \varepsilon\} dQ(t, x) \leq 2\varepsilon \|q\|_\infty.$$

The $L_r(Q)$ -size of the brackets is the $(1/r)$ th power of this. It follows that the bracketing number $N_{[]}((2\varepsilon \|q\|_\infty)^{1/r}, \mathcal{C}_{\alpha, d}, L_r(Q))$ is bounded by p . Finally, apply the previous theorem and make a change of variable. ■

It follows that the subgraphs of the functions $C^\alpha([0, 1]^{d-1})_1$ are P -Donsker for Lebesgue-dominated measures P with bounded density, provided $\alpha > d - 1$. For instance, for the sets cut out in the plane by functions $f: [0, 1] \rightarrow [0, 1]$, a uniform Lipschitz condition of any order on the derivatives suffices. The restriction

to smooth measures P is natural, as smoothness of the graphs is of little help if the probability mass is distributed in an erratic manner.

Subgraphs of smooth functions provide a simple way to define sets with smooth boundaries, but the asymmetry in the arguments is unattractive. A more natural definition of such sets is to map the unit sphere $S^{d-1} = \{x \in \mathbb{R}^d: \|x\| = 1\}$ smoothly into $\mathbb{R}^d$ and define a corresponding set as the “interior” of the range of the map, the latter giving the “boundary” of the set. For instance, the boundaries of smooth sets in $\mathbb{R}^2$ are defined as images of the unit circle under smooth maps. Unfortunately, it is somewhat complicated to make this definition precise.

To start we need to define smooth maps $f: S^{d-1} \rightarrow \mathbb{R}^d$. To this end we can cover S^{d-1} by finitely many sets (an “atlas”) V_i such that each V_i is the image $\phi_i(U^{d-1})$ of the open unit ball U^{d-1} in $\mathbb{R}^d$ under a C^∞ -isomorphism $\phi: U^{d-1} \rightarrow V_i$. These maps can be constructed to be restrictions of C^∞ -isomorphisms of a neighborhood of the closure of U^{d-1} into S^{d-1} . For given $\alpha, M > 0$ we then define $\mathcal{G}_{\alpha, M, d}$ to be the collection of all maps $f: S^{d-1} \rightarrow \mathbb{R}^d$ such that each of the maps $f_j \circ \phi_i: U^{d-1} \rightarrow \mathbb{R}$, where $f_j: S^{d-1} \rightarrow \mathbb{R}$ is a coordinate function of $f = (f_1, \dots, f_d)$, is contained in the Hölder ball $C_M^\alpha(U^{d-1})$.

From each function f in this class we now define a set $I(f) \subset \mathbb{R}^d$ as the set of points $x \in \mathbb{R}^d$ that do not belong to the range of f and such that within the collection of maps $g: S^{d-1} \rightarrow \mathbb{R}^d - \{x\}$ the map f is not homotopic to any (constant) map g (which maps S^{d-1} onto a point $y \neq x$). (Two functions $f, g: S^{d-1} \rightarrow \mathbb{R}^d - \{x\}$ are called *homotopic* if there exists a continuous map $F: [0, 1] \times S^{d-1} \rightarrow \mathbb{R}^d - \{x\}$ such that $F(0, \cdot) = f$ and $F(1, \cdot) = g$.) We interpret $I(f)$ as the interior of the set with boundary the range $f(S^{d-1})$ of f . (A point x is on the “outside” of the “closed curve” $f(S^{d-1})$ if and only if $f(S^{d-1})$ can be continuously contracted to a point without the contracted curves passing through x .) Finally we define $\mathcal{K}_{\alpha, M, d}$ as the collection of all sets $I(f) \cup f(S^{d-1})$ as f ranges over $\mathcal{G}_{\alpha, M, d}$.

The entropy of these collections of sets are of the same order as the entropies of the sets of subgraphs of Hölder functions considered in the preceding corollary.

2.7.23 Theorem (Smooth sets). *Let $d \geq 2$, $M \geq 1$ and $\alpha \geq 1$, let h be the Hausdorff metric, and let Q be a probability measure with bounded Lebesgue density q . There exists a constants K, K' depending only on α, M, d, r , and $\|q\|_\infty$ such that, for every $r \geq 1$ and $\varepsilon > 0$,*

$$\begin{aligned} \log N(\varepsilon, \mathcal{K}_{\alpha, M, d}, h) &\leq K \left(\frac{1}{\varepsilon} \right)^{(d-1)/\alpha}, \\ \log N_{[]}(\varepsilon, \mathcal{K}_{\alpha, M, d}, L_r(Q)) &\leq K' \left(\frac{1}{\varepsilon} \right)^{(d-1)r/\alpha}. \end{aligned}$$

Proof. See Sun and Pyke (1982) or Dudley (1999), Theorem 8.2.16. ■

Problems and Complements

1. **(Lower bound)** There exists a constant K depending only on d and α such that $\log N(\varepsilon, C^\alpha([0, 1]^d)_1, \|\cdot\|_\infty) \geq K(1/\varepsilon)^{d/\alpha}$ for every $\varepsilon > 0$.

[Hint: See Kolmogorov and Tikhomirov (1961).]

2. Let $\|\cdot\|$ be a norm on a vector space of real-valued functions. Then given a subclass $\mathcal{F}$ of functions, the entropy numbers of the class $M\mathcal{F} = \{Mf: f \in \mathcal{F}\}$ satisfy $N(\varepsilon, M\mathcal{F}, \|\cdot\|) = N(\varepsilon/M, \mathcal{F}, \|\cdot\|)$. The same relationship holds for bracketing numbers.
3. Let $\mathcal{F}$ be a class of measurable functions $x \mapsto f(x, r)$ indexed by $0 \leq r \leq 1$ such that $r \mapsto f(x, r)$ is monotone for each x . If the envelope function of $\mathcal{F}$ is square integrable, then the bracketing numbers of $\mathcal{F}$ are polynomial.

4. For any $a, b > 0$ there exists a constant K depending only on (a, b) such that, for every $c > 0$,

$$\sum_{j=0}^{\infty} (c2^{-aj} \wedge 2^{bj}) \leq Kc^{b/(a+b)}.$$

5. For any $a, b > 0$ there exists a constant K depending only on (a, b) such that, for every $c > 0$,

$$\sum_{j=0}^{\infty} (c2^{-aj} \wedge 2^{bj}) \left[\log \left(\frac{c2^{-aj} \vee 2^{bj}}{c2^{-aj} \wedge 2^{bj}} \right) + 1 \right] \leq Kc^{b/(a+b)}.$$

[Hint: The solution j_0 to the equation $c2^{-aj} = 2^{bj}$ satisfies $c2^{-aj_0} = 2^{bj_0} = c^{b/(a+b)}$. For the nearest integer these equalities hold up to a constant depending only on a, b . The sum without the “+1” in the logarithm is bounded by

$$\begin{aligned} & \sum_{j \leq j_0} 2^{bj} \log(c2^{-(a+b)j}) + \sum_{j > j_0} c2^{-aj} \log(2^{(a+b)j}/c) \\ & \lesssim c^{b/(a+b)} \left[\sum_{j \leq j_0} 2^{-b(j_0-j)} \log(2^{(a+b)(j_0-j)}) + \sum_{j > j_0} c2^{-a(j-j_0)} \log(2^{(a+b)(j-j_0)}) \right]. \end{aligned}$$

The series on the right are bounded by $(a+b) \log 2$ times $\sum_{j \geq 0} j2^{-bj}$ and $\sum_{j \geq 0} j2^{-aj}$, respectively.]

6. For any subset $C \subset \mathbb{N}^l$,

$$\#C \leq 4^l \sup_{(n_1, \dots, n_l) \in C} n_1^2 n_2^2 \cdots n_l^2.$$

[Hint: For given $(n_1, \dots, n_l)$ form a binary string as follows: write each n_k in its binary representation, duplicate every bit in this representation (e.g. 011 becomes 001111), and concatenate the strings with separator 01. This gives a string of length bounded by

$$2 \sum_{k=1}^l (2 \log n_k + 1) = 2(2 \log n_1 \cdots n_l) + 2l.$$

For N the supremum of these numbers over all $(n_1, \dots, n_l)$ in C , we can extend this string to a string of length N by padding it with strings 01 at the end. This recipe yields a one-to-one map of C into $\{0, 1\}^N$. Hence the number of elements of C is bounded by 2^N .]

7. The number of subsets of size smaller than l from a set of size n is bounded above by $(en/l)^l$.

[Hint: The number of subsets can be written as $2^n P(X \leq l)$ for a binomial $(n, \frac{1}{2})$ -distributed random variable X . By Markov's inequality this is bounded for any $r \leq 1$ by $2^n E r^{X-l} = 2^n r^{-l} (\frac{1}{2} + \frac{1}{2} r)^n$. Choose $r = l/n$ and use that $(1 + l/n)^n \leq e^l$.]

8. For any $c > 1$ and $\varepsilon > 0$,

$$(c-1) \sum_{j \in \mathbb{N}_0: c^j \varepsilon \leq 1} c^j \varepsilon \log\left(\frac{1}{\varepsilon c^j}\right) \leq c + c \log(1/c).$$

[Hint: The left side is bounded by the integral $\int_0^{c/\varepsilon} \varepsilon \log(1/\varepsilon x) dx = c + c \log(1/c)$, as can be seen by discretization on the grid points c^j .]

9. If $f: \{x_\lambda: 0 \leq \lambda \leq 1\} \rightarrow \mathbb{R}$ is convex, for $x_\lambda = (1-\lambda)x_0 + \lambda x_1$ and given x_0 and x_1 in a vector space, then $\inf_{0 \leq \lambda \leq 1} f(x_\lambda) \geq -3 \max(|f(x_0)|, |f(x_1)|, |f(x_{1/2})|)$.

[Hint: For $\lambda \in (0, 1/2)$ we can write $x_{1/2} = \mu x_\lambda + (1-\mu)x_1$, for $\mu(1-\lambda) = 1/2$. By convexity $f(x_{1/2}) \leq \mu f(x_\lambda) + (1-\mu)f(x_1)$, so that $f(x_\lambda) \geq \mu^{-1}f(x_{1/2}) - \mu^{-1}(1-\mu)f(x_1) \geq -M(2/\mu-1)$, if M is the given maximum value. Now $2/\mu-1 = 3-4\lambda \leq 3$. (Cf. Arcones (1998).)]

10. The ℓ_q -norms $\|\gamma\|_q = (\sum_j |\gamma_j|^q)^{1/q}$ are decreasing in q with limit the ℓ_∞ -norm $\|\gamma\|_\infty = \sup_j |\gamma_j|$, as $q \rightarrow \infty$. This is true both for the norms on finite sequences $\mathbb{R}^d$ and for the infinite sequence spaces.

[Hint: Combine $\|\gamma\|_\infty \leq \|\gamma\|_p$ with $\|\gamma\|_q \leq \|\gamma\|_p^{p/q} \|\gamma\|_\infty^{(1-p/q)}$, for $p \leq q$.]

11. For a given bounded, nondecreasing function $g: [0, \infty) \rightarrow [0, \infty)$ and $\alpha, q > 0$, the expression $[\int_0^\infty (t^{-\alpha} g(t))^q t^{-1} dt]^{1/q}$ is bounded below and above by $(\log 2)^{1/q} 2^{-\alpha}$ and $(\log 2)^{1/q} 2^\alpha$ times the ℓ_q -norm of the sequence $(2^{-j\alpha} g(2^j))_{j \in \mathbb{Z}}$. The ℓ_q -norms are decreasing in q , with limit the ℓ_∞ -norm as $q \rightarrow \infty$. So are the numbers $(\log 2)^{1/q}$, with limit 1 as $q \rightarrow \infty$.

12. **(Differences and derivatives)** If $\mathcal{X}$ is convex, then, for integers $r \geq k \geq 1$, and any $f: \mathcal{X} \rightarrow \mathbb{R}$ that is sufficiently many times continuously differentiable (0 or 1 or k times),

(i) $\|\Delta_h f\|_p \leq 2\|f\|_p$ and $\|\Delta_h f\|_p \leq \|h\| \|\nabla f\|_p$.

(ii) $\|\Delta_h^r f\|_p \leq 2^{r-k} \|\Delta_h^k f\|_p$ and $\|\Delta_h^r f\|_p \leq 2^{r-k} \|h\|^{r-k} \|\nabla^k f\|_p$.

Here $|\nabla^k f|(x)$ is a number such that $|\sum_{l: |l|=k} \binom{k}{l} D^l f(x) h^l| \leq |\nabla^k f|(x) \|h\|^k$. In particular, it can be taken the Euclidean norm of the gradient $\nabla f(x)$ if $k = 1$ and the Euclidean matrix norm of the second derivative $D^2 f(x)$ if $k = 2$. (For a partial converse of (i) if $p > 1$, see Evans (2010), Theorem 3 on page 277.)

[Hint: Because $|\Delta_h f(x)|^p = |\int_0^1 \nabla f(x+th) \cdot h dt|^p \leq \int_0^1 |\nabla f(x+th)|^p \|h\|^p dt$, Fubini's theorem gives that $\|\Delta_h f\|_p \leq \|h\| (\int_0^1 \int_{\mathcal{X}_h} |\nabla f(x+th)|^p dx dt)^{1/p}$, for $\mathcal{X}_h$ the set of points x such that both $x, x+h \in \mathcal{X}$. Change coordinates and note that $\mathcal{X}_h + th \subset \mathcal{X}$, by the assumed convexity of $\mathcal{X}$, to see that the inner integral is bounded above by $\int_{\mathcal{X}} |\nabla f(x)|^p dx$, for every $0 \leq t \leq 1$. The first part of assertion (ii) follows upon repeatedly applying (i). For the second part write the k th difference in the form $\Delta_h^k f(x) = \int_0^1 \cdots \int_0^1 \nabla^k f(x+t_1 h + \cdots + t_k h)(h, \dots, h) dt_1 \cdots dt_k$, where $\nabla^k f(x)(h, \dots, h) = \sum_{l: |l|=k} \binom{k}{l} D^l f(x) h^l$, and argue similarly.]

13. For the wavelets as in Section 2.7.2 and every given $j \in \mathbb{N}_0$ and every $1 \leq \tau \leq \infty$, the norms $\left\| \sum_{k \in \mathbb{Z}^d} \sum_v \beta_{j,k}^v |\psi_{j,k}|^\tau \right\|_\tau$ and $\left(\sum_{k \in \mathbb{Z}^d} \sum_v |\beta_{j,k}^v|^\tau \right)^{1/\tau}$ are equivalent, where the constants do not depend on j .

[**Hint:** By (2.7.7) with $\alpha = 0$, the first norm is equivalent to $\left\| (2^{jd} \sum_{k,v} |\beta_{j,k}^v|^2 1_{C_{j,k}})^{1/2} \right\|_\tau$. For every x and fixed j , there are at most K cubes $C_{j,k}$ that contain x , for a given K that depends only on the basis. Then trivially $K^{-1} \sum_{k,v} |\beta_{j,k}^v|^\tau 1_{C_{j,k}} \leq \left(\sum_{k,v} |\beta_{j,k}^v|^2 1_{C_{j,k}} \right)^{\tau/2} \leq (K^{\tau/2-1} \vee 1) \sum_{k,v} |\beta_{j,k}^v|^\tau 1_{C_{j,k}}$, by Hölder's inequality for $\tau \geq 2$ and concavity otherwise. Integrate, use that the volume of $C_{j,k}$ is of the order 2^{-jd} and that $\beta_{j,k}^v = 2^{jd(1/\tau-1/2)} \beta_{j,k|\tau}^v$.]

14. (**Besov and Lizorkin embedding**) Given the wavelet coefficients $\beta_{j,k}^v$ of a function f as in Section 2.7.2, cubes $C_{j,k} \subset \mathbb{R}^d$ of volume 2^{jd} such that for every j every point in $\mathbb{R}^d$ is covered by at most K cubes, for a constant K independent of j , and $j \in \mathbb{N}_0$, set $f_j = 2^{j(\alpha+d/2)} \sum_{k,v} |\beta_{j,k}^v|^\tau 1_{C_{j,k}}$, and for $1 \leq p, q \leq \infty$, consider the norms

$$\|f\|_{\alpha|p,q} = \left(\sum_j \|f_j\|_{\lambda,p}^q \right)^{1/q}, \quad \|f\|_{\alpha;p,q} = \left\| \left(\sum_j f_j^q \right)^{1/q} \right\|_{\lambda,p}.$$

The first is equivalent to the Besov norm, as in Section 2.7.2, the second is equivalent to the norm of the *Triebel-Lizorkin space*. The special case $\|f\|_{\alpha;p,2}$ with $q = 2$ is equivalent to the Sobolev norm $\|f\|_{\alpha|p}$, as in Section 2.7.2. We have

- (i) $\|f\|_{\alpha|p,q} \lesssim \|f\|_{\alpha;p,q}$, for $p \leq q$;
- (ii) $\|f\|_{\alpha;p,q} \lesssim \|f\|_{\alpha|p,q}$, for $q \leq p$.

[**Hint:** For the comparison of norms (i)–(ii) apply Minkowski's inequality as in Problem 2.7.16, with μ and ν the Lebesgue measure λ and counting measure on $\mathbb{Z}$ for (i), and vice versa for (ii). To see the equivalence of the given norms to the Besov or Sobolev norms given in Section 2.7.2, use that the function f_j is pointwise equivalent to the function $2^{j(\alpha+d/2)} \left(\sum_{k,v} |\beta_{j,k}^v|^\tau 1_{C_{j,k}} \right)^{1/\tau}$, for every τ , as seen in Problem 2.7.13.]

15. (**Besov embedding**) Given the notation as in Problem 2.7.14, we have

- (i) $\|f\|_{\alpha|p,q'} \lesssim \|f\|_{\alpha|p,q}$, if $q \leq q'$;
- (ii) $\|f\|_{\alpha'|p',q} \lesssim \|f\|_{\alpha|p,q}$, if $0 < p \leq p'$ and $\alpha - d/p \geq \alpha' - d/p'$.

[**Hint:** The ℓ_q -norms are decreasing in q and so are the numbers $2^{-j(\alpha+d/2)} 2^{jd/p} \|f_j\|_p \asymp \left(\sum_{k,v} |\beta_{j,k}^v|^p \right)^{1/p}$, up to equivalence constants.]

16. (**Minkowski's inequality**) If μ and ν are σ -finite measures on measurable spaces $(\mathcal{X}, \mathcal{A})$ and $(\mathcal{Y}, \mathcal{B})$ and $f: \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ is measurable, then, for $1 \leq p \leq q$,

$$\left(\int \left[\int |f(x,y)|^p \mu(x) \right]^{q/p} d\nu(y) \right)^{1/q} \leq \left(\int \left[\int |f(x,y)|^q d\nu(y) \right]^{p/q} d\mu(x) \right)^{1/p}.$$

[**Hint:** Take the p th power on both sides and set $g = |f|^p$ and $r = q/p$ to reduce this to $\| \int g(x, \cdot) d\mu(x) \|_{\nu,r} \leq \int \| \int g(x, \cdot) d\nu(y) \|_{\mu,r} d\mu(x)$, where the norms are relative to the functions of the argument $\cdot$. This is a generalization of *Minkowski's inequality* for the L_r -norm (the triangle inequality for the norm) from two functions to possibly infinitely many functions. It can be derived by finite approximation, or see Folland (1999), Theorem 6.19.]

17. The function $x \mapsto x \log x$ belongs to $B_{\infty,\infty}^1([0, 1])$, but not to $C^1([0, 1])$.

18. For sufficiently regular domains, for instance bounded convex domains, the maximum over the index k in the definitions of the Hölder and Sobolev norms with $\alpha \in \mathbb{N}$ may be reduced to the maximum over k with $k \in \{0, \alpha\}$.

[Hint: For an interval in the line one may use the Landau-Kolmogorov inequalities. In higher-dimensional bounded connected domains, one may repeatedly employ *Poincaré's inequality* $\|f - \int_{\mathcal{X}} f d\lambda\|_p \lesssim \|\nabla f\|_p$, valid for $1 \leq p \leq \infty$ (Evans (2010), Theorem 1 on page 275), to bound lower order derivatives minus a polynomial by higher order derivatives, given bounds of the type $\|D^k f - \sum_{l \geq k, |l| < \alpha} c_l x^{l-k} / (l-k)!\|_p \lesssim \max_{|l|=\alpha} \|D^l f\|_p$, and next bound the polynomial coefficients c_l using that the norm of the polynomial $\sum_{|l| < \alpha} c_l x^l / l!$, obtained for $k = 0$, is bounded above by a multiple of $\|f\|_p + \max_{|l|=\alpha} \|D^l f\|_p$, by the triangle inequality. See Problem 3.4.8 for an example.]

19. **(Moment matching)** Let $\psi_1, \dots, \psi_N: K \rightarrow \mathbb{R}$ be continuous functions on a compact metric space K . For any probability measure F on K , there exists a discrete probability measure F' on K with at most $N + 1$ support points such that $F\psi_j = F'\psi_j$ for every $j = 1, \dots, N$.

[Hint: By the geometric form of Jensen's inequality $(F\psi_1, \dots, F\psi_N)$ is contained in the convex hull of the (compact) set $C = \{(\psi_1(x), \dots, \psi_N(x)): x \in K\}$. Any element of this convex hull can be written as a convex combination of at most $N + 1$ elements of C (see Rudin (1973), page 73).]

2.8

Uniformity in the Underlying Distribution

The previous chapters present empirical laws of large numbers and central limit theorems for observations from a fixed underlying distribution P . Many of the sufficient conditions given there are actually satisfied by very large classes of underlying measures; typically, the only limitation is finiteness of some appropriate moment of the envelope function. For instance, classes satisfying the uniform entropy condition are, up to measurability, Glivenko-Cantelli or Donsker for all P with $P^*F < \infty$ or $P^*F^2 < \infty$, respectively. In particular, many bounded classes of functions are universally Donsker: Donsker for every probability measure on the sample space.

In this chapter we note that even stronger results are typically true. For example, not only does the central limit theorem hold for all underlying measures, or very large classes of measures, it is also valid uniformly in the underlying measure, when this ranges over large classes of measures. Essentially, the main empirical limit theorems are valid uniformly in the underlying measure if the conditions hold in a uniform sense, which appears to be the case frequently. This type of uniformity is certainly of interest in statistical applications.

2.8.1 Glivenko-Cantelli Theorems

We start with a uniform Glivenko-Cantelli theorem. Convergence almost surely to zero of a sequence of random variables X_n is equivalent to convergence in probability of $\sup_{m \geq n} |X_m|$ to zero. The latter characterization can be used to define “almost sure convergence uniformly in the underlying measure.” A class $\mathcal{F}$ of measurable functions on a measurable space $(\mathcal{X}, \mathcal{A})$ is said to be *Glivenko-Cantelli uniformly* in $P \in \mathcal{P}$ for a given class $\mathcal{P}$ of probability measures on $(\mathcal{X}, \mathcal{A})$ if

$$\sup_{P \in \mathcal{P}} P_P^* \left(\sup_{m \geq n} \|\mathbb{P}_m - P\|_{\mathcal{F}} > \varepsilon \right) \rightarrow 0,$$

for every $\varepsilon > 0$ as $n \rightarrow \infty$.

The set $\mathcal{Q}_n$ in the statement of the following theorem is the collection of all possible realizations of empirical measures of n observations.

2.8.1 Theorem (Uniform Glivenko-Cantelli). *Let $\mathcal{F}$ be a P -measurable class of functions on a measurable space for every probability measure P in a class $\mathcal{P}$. Suppose that, for some measurable envelope function F ,*

$$\begin{aligned} \lim_{M \rightarrow \infty} \sup_{P \in \mathcal{P}} PF\{F > M\} &= 0, \\ \sup_{Q \in \mathcal{Q}_n} \log N(\varepsilon \|F\|_{Q,1}, \mathcal{F}, L_1(Q)) &= o(n), \quad \text{for every } \varepsilon > 0, \end{aligned}$$

where the supremum is taken over the set $\mathcal{Q}_n$ of all discrete probability measures with atoms of size integer multiples of $1/n$. Then $\mathcal{F}$ is Glivenko-Cantelli uniformly in $P \in \mathcal{P}$.

Proof. For fixed M , let $\mathcal{F}_M$ be the class of all functions $f1\{f \leq M\}$ as f ranges over $\mathcal{F}$. Then

$$\|\mathbb{P}_n - P\|_{\mathcal{F}} \leq \|(\mathbb{P}_n - P)f1\{F \leq M\}\|_{\mathcal{F}} + \mathbb{P}_n F 1\{F > M\} + PF\{F > M\}.$$

By the uniform strong law for real random variables (Proposition A.5.1), the second term on the right converges uniformly in P almost surely to $PF\{F > M\}$. By the first condition of the theorem, this quantity can be made arbitrarily small uniformly in P by choosing large M . Conclude that it suffices to show that the first term on the right, which equals $\|\mathbb{P}_n - P\|_{\mathcal{F}_M}$, converges almost surely to zero uniformly in P for every fixed M .

The class $\mathcal{F}_M$ satisfies the entropy condition of the theorem relative to the envelope function M . This follows from the inequality

$$\sup_{Q \in \mathcal{Q}_n} N(\varepsilon M, \mathcal{F}_M, L_1(Q)) \leq \sup_{n \leq k \leq 2n} \sup_{Q \in \mathcal{Q}_k} N(\varepsilon \|F\|_{Q,1}, \mathcal{F}, L_1(Q)),$$

which may be proved as follows. Suppose that k out of the n support points $x_1, \dots, x_n$ of a given measure $Q \in \mathcal{Q}_n$ (with multiple appearance of a given point permitted) satisfy $F(x_i) \leq M$. If $Q_k \in \mathcal{Q}_k$ is the discrete measure on these points, then $Q|f\{F \leq M\}| = (k/n)Q_k|f|$ for any function f . Since $Q_k F \leq M$, it follows that an $\varepsilon Q_k F$ -net for $\mathcal{F}$ in $L_1(Q_k)$ yields an $\varepsilon(k/n)M$ net for $\mathcal{F}_M$ in $L_1(Q)$. Thus $N(\varepsilon M, \mathcal{F}_M, L_1(Q)) \leq N(\varepsilon \|F\|_{Q_k,1}, \mathcal{F}, L_1(Q_k))$. Since $\mathcal{Q}_k \subset \mathcal{Q}_{2k} \subset \dots$, the right side is bounded by the right side of the preceding display. This completes the proof that $\mathcal{F}_M$ satisfies the entropy condition of the theorem relative to the envelope function M .

Fix $\eta > 0$ and values $X_1, \dots, X_n$, and take a minimal ηM -net $\mathcal{F}_{nX}$ in $\mathcal{F}_M$ for the $L_1(\mathbb{P}_n)$ semimetric. It has just been proved that the cardinality $N(\eta M, \mathcal{F}_M, L_1(\mathbb{P}_n))$ of $\mathcal{F}_{nX}$ is bounded by a deterministic sequence $N_n(\eta)$ (uniformly in $X_1, \dots, X_n$) satisfying $\log N_n(\eta) = o(n)$. Let $\mathbb{P}_n^o$ be the symmetrized empirical measure as defined in Chapter 2.3. Then

$$\|\mathbb{P}_n^o\|_{\mathcal{F}_M} \leq \|\mathbb{P}_n^o\|_{\mathcal{F}_{nX}} + \eta M.$$

The measurability of $\mathcal{F}$ implies the measurability of $\mathcal{F}_M$, so that the probability $P_P(\|\mathbb{P}_n^o\|_{\mathcal{F}_M} > \varepsilon)$ can be written as $E_{P,X}P_\varepsilon(\|\mathbb{P}_n^o\|_{\mathcal{F}_M} > \varepsilon)$. By Hoeffding's inequality, Lemma 2.2.8 applied for fixed $X_1, \dots, X_n$,

$$E_{P,X}P_\varepsilon\left(\|\mathbb{P}_n^o f\|_{\mathcal{F}_{n,X}} > \varepsilon - \eta M\right) \leq E_{P,X}N_n(\eta)2e^{-\frac{1}{2}n(\varepsilon - \eta M)^2/M^2}.$$

The integrand on the right side is independent of $X_1, \dots, X_n$. Since $N_n(\eta) = \exp o(n)$, the integrand is bounded by $2 \exp(-n\varepsilon^2/4M^2)$ for sufficiently large n and small η . Conclude that $\sum_{m \geq n} P(\|\mathbb{P}_m^o\|_{\mathcal{F}_M} > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$. In view of the symmetrization Lemma 2.3.7 for probabilities, the same is true for the empirical measure replacing the symmetrized empirical. This concludes the proof that $\|\mathbb{P}_n - P\|_{\mathcal{F}_M}$ converges to zero outer almost surely uniformly in P . ■

Vapnik-Červonenkis classes of sets or functions satisfy

$$\sup_Q \log N(\varepsilon \|F\|_{Q,1}, \mathcal{F}, L_1(Q)) < \infty,$$

for the supremum taken over all probability measures Q . This trivially implies the entropy condition of the previous theorem. It follows that a suitably measurable VC-class is uniformly Glivenko-Cantelli over any class of underlying measures for which its envelope function is uniformly integrable. In particular, a uniformly bounded VC-class is uniformly Glivenko-Cantelli over all probability measures, provided it satisfies the measurability condition.

2.8.2 Donsker Theorems

Next consider uniform in P central limit theorems. Let $\mathcal{F}$ be a class of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$ that is Donsker for every P in a set $\mathcal{P}$ of probability measures on $(\mathcal{X}, \mathcal{A})$. Thus the empirical process $\mathbb{G}_{n,P} = \sqrt{n}(\mathbb{P}_n - P)$ converges weakly in $\ell^\infty(\mathcal{F})$ to a tight, Borel measurable version of the Brownian bridge $\mathbb{G}_P$. According to Chapter 1.12, this is equivalent to

$$\sup_{h \in BL_1} |E_P^* h(\mathbb{G}_{n,P}) - E h(\mathbb{G}_P)| \rightarrow 0.$$

(Here BL_1 is the set of all functions $h: \ell^\infty(\mathcal{F}) \rightarrow \mathbb{R}$ which are uniformly bounded by 1 and satisfy $|h(z_1) - h(z_2)| \leq \|z_1 - z_2\|_{\mathcal{F}}$.) We shall call $\mathcal{F}$ *Donsker uniformly* in $P \in \mathcal{P}$ if this convergence is uniform in P .

The weak convergence $\mathbb{G}_{n,P} \rightsquigarrow \mathbb{G}_P$ under a fixed P is equivalent to asymptotic equicontinuity of the sequence $\mathbb{G}_{n,P}$ and total boundedness of $\mathcal{F}$ under the seminorm $\rho_P(f) = \|f - Pf\|_{P,2}$. It is to be expected that uniform versions of these two conditions are sufficient for the uniform Donsker property. The sequence $\mathbb{G}_{n,P}$ is called *asymptotically equicontinuous uniformly* in $P \in \mathcal{P}$ if, for every $\varepsilon > 0$,

$$\lim_{\delta \downarrow 0} \limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}} P_P^* \left(\sup_{\rho_P(f,g) < \delta} |\mathbb{G}_{n,P}(f) - \mathbb{G}_{n,P}(g)| > \varepsilon \right) = 0.$$

Furthermore, the class $\mathcal{F}$ is ρ_P -totally bounded uniformly in $P \in \mathcal{P}$ if its covering numbers satisfy $\sup_{P \in \mathcal{P}} N(\varepsilon, \mathcal{F}, \rho_P) < \infty$.

These uniform versions of the two conditions actually imply more than the uniform Donsker property. In the situation for a fixed P , the asymptotic equicontinuity of the sequence $\mathbb{G}_{n,P}$ has its counterpart in the continuity of the sample paths of the limit process $\mathbb{G}_P$. Requiring the asymptotic equicontinuity uniformly in P yields, apart from uniform weak convergence, also uniformity in the continuity of the limit process. The latter is expressed in the following. The class $\mathcal{F}$ is *pre-Gaussian uniformly* in $P \in \mathcal{P}$ if the Brownian bridges satisfy the two conditions

$$\sup_{P \in \mathcal{P}} E \|\mathbb{G}_P\|_{\mathcal{F}} < \infty; \quad \lim_{\delta \downarrow 0} \sup_{P \in \mathcal{P}} E \sup_{\rho_P(f,g) < \delta} |\mathbb{G}_P(f) - \mathbb{G}_P(g)| = 0.$$

(Some authors include the requirement of uniform pre-Gaussianity in the concept of a uniform Donsker class.)

Throughout this section it is assumed that the class $\mathcal{F}$ possesses a measurable envelope function F with the property

$$\lim_{M \rightarrow \infty} \sup_{P \in \mathcal{P}} P F^2 \{F > M\} = 0.$$

Such an envelope function is called square integrable uniformly in $P \in \mathcal{P}$.

2.8.2 Theorem (Uniform Donsker). *Let $\mathcal{F}$ be a class of measurable functions with envelope function F that is square integrable uniformly in $P \in \mathcal{P}$. Then the following statements are equivalent:*

- (i) $\mathcal{F}$ is Donsker and pre-Gaussian, both uniformly in $P \in \mathcal{P}$;
- (ii) the sequence $\mathbb{G}_{n,P}$ is asymptotically ρ_P -equicontinuous uniformly in $P \in \mathcal{P}$ and $\sup_{P \in \mathcal{P}} N(\varepsilon, \mathcal{F}, \rho_P) < \infty$ for every $\varepsilon > 0$.

Furthermore, uniform pre-Gaussianity implies uniform total boundedness of $\mathcal{F}$.

Proof. The last assertion is a consequence of Sudakov's minorization inequality A.2.6 for Gaussian processes, which gives the upper bound $3E\|\mathbb{G}_P\|_{\mathcal{F}}$ for $\varepsilon \sqrt{\log N(\varepsilon, \mathcal{F}, \rho_P)}$ for every $\varepsilon > 0$ and P .

(i) $\Rightarrow$ (ii). For each fixed $\delta > 0$ and P , the truncated modulus function

$$z \mapsto h_P(z) = \sup_{\rho_P(f,g) < \delta} |z(f) - z(g)| \wedge 1$$

is contained in $2BL_1$. If $\mathcal{F}$ is uniformly Donsker, then $E_P^* h_P(\mathbb{G}_{n,P}) \rightarrow E h_P(\mathbb{G}_P)$ uniformly in P . Given uniform pre-Gaussianity, the expressions $E h_P(\mathbb{G}_P)$ can be made uniformly small by choice of a sufficiently small δ . Thus $\limsup E_P^* h_P(\mathbb{G}_{n,P})$ is uniformly small for sufficiently small δ . This implies that the sequence $\mathbb{G}_{n,P}$ is uniformly asymptotically ρ_P -equicontinuous.

(ii) $\Rightarrow$ (i). Fix an arbitrary finite subset $\mathcal{G}$ of $\mathcal{F}$ and $P \in \mathcal{P}$. The sequence of random vectors $\{\mathbb{G}_{n,P}(g) : g \in \mathcal{G}\}$ converges weakly to the Gaussian vector $\{\mathbb{G}_P(g) : g \in \mathcal{G}\}$. By the portmanteau theorem,

$$\begin{aligned} & P \left(\sup_{\substack{\rho_P(f,g) < \delta \\ f,g \in \mathcal{G}}} |\mathbb{G}_P(f) - \mathbb{G}_P(g)| > \varepsilon \right) \\ & \leq \liminf_{n \rightarrow \infty} \sup_{P \in \mathcal{P}} P_P^* \left(\sup_{\substack{\rho_P(f,g) < \delta \\ f,g \in \mathcal{G}}} |\mathbb{G}_{n,P}(f) - \mathbb{G}_{n,P}(g)| > \varepsilon \right), \end{aligned}$$

for every $\delta > 0$. On the right side $\mathcal{G}$ can be replaced by $\mathcal{F}$. By assumption the resulting expression can be made arbitrarily small by choice of δ . Conclude that, for every $\varepsilon, \eta > 0$, there exists $\delta > 0$ such that the left side of the display is smaller than η for every P . Let $\mathcal{G}$ increase to a ρ_P -countable dense subset of $\mathcal{F}$, and use the continuity of the sample paths of $\mathbb{G}_P$ to see that the same statement is true (with the same ε, η , and δ) with $\mathcal{G}$ replaced by $\mathcal{F}$.

This gives a probability form of the second requirement of uniform pre-Gaussianity. The expectation form follows with the help of Borell's inequality A.2.1, which allows one to bound the moment $E\|\mathbb{G}_P\|_{\mathcal{F}_\delta}$ by a universal constant times the median of $\|\mathbb{G}_P\|_{\mathcal{F}_\delta}$ (Problem A.2.2).

Fix $\delta > 0$. By the uniform total boundedness, there exists for each P a δ -net $\mathcal{G}_P$ for ρ_P over $\mathcal{F}$ such that the cardinalities of the $\mathcal{G}_P$ are uniformly bounded by a constant k . Then

$$\|\mathbb{G}_P\|_{\mathcal{F}} \leq \|\mathbb{G}_P\|_{\mathcal{G}_P} + \sup_{\rho_P(f,g) < \delta} |\mathbb{G}_P(f) - \mathbb{G}_P(g)|.$$

By the result of the preceding paragraph, the expectation of the second term on the right can be made arbitrarily small uniformly in P by choice of δ ; it is certainly finite. The first term on the right is a maximum over at most k Gaussian random variables, each having mean zero and variance not exceeding $\sup_{P \in \mathcal{P}} PF^2 < \infty$. Thus $E\|\mathbb{G}_P\|_{\mathcal{G}_P}$ is uniformly bounded. This concludes the proof of uniform pre-Gaussianity.

For each P let $\Pi_P: \mathcal{F} \rightarrow \mathcal{G}_P$ map each f into a ρ_P -closest element of $\mathcal{G}_P$. Write $z \circ \Pi_P$ for the discretized element of $\ell^\infty(\mathcal{F})$ taking the value $z(\Pi_P f)$ at each $f \in \mathcal{F}$. By the uniform central limit theorem for $\mathbb{R}^k$, the random vectors $\{\mathbb{G}_{n,P}(f): f \in \mathcal{G}_P\}$ in $\mathbb{R}^k$ (with zero coordinates added if $|\mathcal{G}_P| < k$) converge weakly to the random vectors $\{\mathbb{G}_P(f): f \in \mathcal{G}_P\}$ uniformly in $P \in \mathcal{P}$ (Proposition A.5.2). This implies that

$$\sup_{h \in BL_1} |E_P^* h(\mathbb{G}_{n,P} \circ \Pi_P) - Eh(\mathbb{G}_P \circ \Pi_P)| \rightarrow 0.$$

Next, since every $h \in BL_1$ satisfies the inequality $|h(z_1) - h(z_2)| \leq 2 \wedge \|z_1 - z_2\|_{\mathcal{F}}$, we have, for every $\varepsilon > 0$,

$$\sup_{h \in BL_1} |E_P^* h(\mathbb{G}_{n,P} \circ \Pi_P) - E^* h(\mathbb{G}_{n,P})| \leq \varepsilon + 2P_P^*(\|\mathbb{G}_{n,P} \circ \Pi_P - \mathbb{G}_{n,P}\|_{\mathcal{F}} > \varepsilon).$$

By construction of $\mathcal{G}_P$, the random variable $\|\mathbb{G}_{n,P} \circ \Pi_P - \mathbb{G}_{n,P}\|_{\mathcal{F}}$ is bounded by the modulus of continuity $\|\mathbb{G}_{n,P}\|_{\mathcal{F}_\delta}$ of the process $\mathbb{G}_{n,P}$. In view of the uniform asymptotic equicontinuity, $\limsup P_P^*(\|\mathbb{G}_{n,P} \circ \Pi_P - \mathbb{G}_{n,P}\|_{\mathcal{F}} > \varepsilon)$ can be made arbitrarily small by choice of δ , uniformly in P , for every fixed ε . Thus the limsup of the left side of the last display can be made arbitrarily small by choice of δ .

Using the uniform pre-Gaussianity, we can obtain the analogous result for the limit processes:

$$\sup_{h \in BL_1} |Eh(\mathbb{G}_P \circ \Pi_P) - Eh(\mathbb{G}_P)| \rightarrow 0, \quad \delta \downarrow 0.$$

Combination of the last three displayed equations shows that $E_P^* h(\mathbb{G}_{n,P})$ converges to $Eh(\mathbb{G}_P)$ uniformly in $h \in BL_1$ and P . ■

In the situation that $\mathcal{P}$ consists of all probability measures on the underlying measurable space, there is an interesting addendum to the previous theorem. In that case, the uniform pre-Gaussianity implies the uniform Donsker property (under suitable measurability conditions on $\mathcal{F}$). We do not reproduce the proof of this result here. (See the Notes for further comments and references.) Instead we derive the uniform versions of the two main empirical limit theorems: the central limit theorem under the uniform entropy condition and the central limit theorem with bracketing.

The uniform entropy condition (2.5.1) implies the uniform Donsker property.

2.8.3 Theorem (Uniform Donsker). *Let $\mathcal{F}$ be a class of measurable functions with measurable envelope function F such that $\mathcal{F}_{\delta,P} = \{f - g: f, g \in \mathcal{F}, \|f - g\|_{P,2} < \delta\}$ and $\mathcal{F}_\infty^2 = \{(f - g)^2: f, g \in \mathcal{F}\}$ are P -measurable for every $\delta > 0$ and $P \in \mathcal{P}$. Furthermore, suppose that, as $M \rightarrow \infty$,*

$$\sup_{P \in \mathcal{P}} P F^2 \{F > M\} \rightarrow 0, \\ \int_0^\infty \sup_Q \sqrt{\log N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q))} d\varepsilon < \infty,$$

where Q ranges over all finitely discrete probability measures. Then $\mathcal{F}$ is Donsker and pre-Gaussian uniformly in $P \in \mathcal{P}$.

Proof. It may be assumed that the envelope function satisfies $F \geq 1$, because replacing a given F by $F + 1$ does not affect the uniform integrability and makes the uniform entropy condition weaker. Given an arbitrary sequence $\delta_n \downarrow 0$, set

$$\theta_{n,P}^2 = \left\| \frac{1}{n} \sum_{i=1}^n f^2(X_i) \right\|_{\mathcal{F}_{\delta_n,P}}.$$

Exactly as in the proof of Theorem 2.5.2, it can be shown that for some universal constant K ,

$$\begin{aligned} & P_P^* \left(\|\mathbb{G}_{n,P}\|_{\mathcal{F}_{\delta_n,P}} > \varepsilon \right)^2 \\ & \leq \frac{K}{\varepsilon^2} E_P^* \int_0^{\theta_{n,P}/\|F\|_n} \sup_Q \sqrt{\log N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q))} d\varepsilon^2 E_P \|F\|_n^2. \end{aligned}$$

Here $\|F\|_n \geq 1$ and $E_P \|F\|_n^2 = P F^2$ is uniformly bounded in P . Conclude that the right side is up to a constant uniformly bounded by

$$\int_0^\eta \sup_Q \sqrt{\log N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q))} d\varepsilon^2 + \sup_{P \in \mathcal{P}} P_P^* (\theta_{n,P} > \eta),$$

for every $\eta > 0$. To establish uniform asymptotic equicontinuity, it suffices to show that the second term converges to zero for all η . This follows as in the proof of Theorem 2.5.2, except that presently we use the uniform law of large numbers, Theorem 2.8.1.

By assumption, $\sup_Q N(\varepsilon, \mathcal{F}, \rho_Q)$ is finite for every $\varepsilon > 0$ if Q ranges over all finitely discrete measures. That the supremum is still finite if Q also ranges over $\mathcal{P}$ can be proved as in the proof of Theorem 2.5.2. ■

The uniform entropy condition requires the supremum over all probability measures of the root entropy to be integrable. In terms of entropy with bracketing, it suffices to consider the supremum over the class of interest.

2.8.4 Theorem (Uniform Donsker). *Let $\mathcal{F}$ be a class of measurable functions such that*

$$\lim_{M \rightarrow \infty} \sup_{P \in \mathcal{P}} P F^2 \{F > M\} = 0,$$

$$\int_0^\infty \sup_{P \in \mathcal{P}} \sqrt{\log N_{[]}(\varepsilon \|F\|_{P,2}, \mathcal{F}, L_2(P))} d\varepsilon < \infty.$$

Then $\mathcal{F}$ is Donsker and pre-Gaussian uniformly in $P \in \mathcal{P}$.

Proof. The first condition implies that $\sup_P \|F\|_{P,2} < \infty$. Finiteness of the integral implies finiteness of the integrand. Therefore,

$$\sup_P N_{[]}(\varepsilon \|F\|_{P,2}, \mathcal{F}, L_2(P)) < \infty,$$

and $\mathcal{F}$ is totally bounded uniformly in P . Uniform asymptotic equicontinuity follows by making the proof of Theorem 2.5.6 uniform in P . Actually, the proof of this theorem contains the essence of the proof of the maximal inequality given by Theorem 2.14.16, which implies that

$$E_P^* \|\mathbb{G}_{n,P}\|_{\mathcal{F}_{\delta,P}} \lesssim \int_0^\delta \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L_2(P))} d\varepsilon + \frac{P F^2 \{F > \sqrt{nb}(\delta, P)\}}{b(\delta, P)}.$$

Here

$$b(\delta, P) = \delta / \sqrt{1 + \log N_{[]}(\delta, \mathcal{F}, L_2(P))}.$$

The right side of the first display converges to zero uniformly in P as $n \rightarrow \infty$ followed by $\delta \downarrow 0$. ■

2.8.3 Central Limit Theorem Under Sequences

As an application of the uniform central limit theorem, consider the empirical process based on triangular arrays. For each n , let $X_{n1}, \dots, X_{nn}$ be i.i.d. according to a probability measure P_n , and set

$$\mathbb{P}_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_{ni}}.$$

If the sequence of underlying measures P_n converges to a measure P_0 in a suitable sense, then we may hope to derive that the sequence $\mathbb{G}_{n,P_n} = \sqrt{n}(\mathbb{P}_n - P_n)$ converges in distribution to $\mathbb{G}_{P_0}$ in $\ell^\infty(\mathcal{F})$. According to the general results on weak convergence to Gaussian processes, this is equivalent to marginal convergence plus $\mathcal{F}$ being totally bounded and the sequence $\mathbb{G}_{n,P_n}$ being asymptotically uniformly

equicontinuous, both with respect to ρ_{P_0} . Suppose that the ρ_{P_n} semimetrics converge uniformly to ρ_{P_0} in the sense that

$$(2.8.5) \quad \sup_{f, g \in \mathcal{F}} |\rho_{P_n}(f, g) - \rho_{P_0}(f, g)| \rightarrow 0.$$

Then asymptotic equicontinuity for ρ_{P_0} follows from asymptotic equicontinuity “for the sequence ρ_{P_n} ,” defined as

$$\lim_{\delta \downarrow 0} \limsup_{n \rightarrow \infty} \mathbf{P}_{P_n}^* \left(\sup_{\rho_{P_n}(f, g) < \delta} |\mathbb{G}_{n, P_n}(f) - \mathbb{G}_{n, P_n}(g)| > \varepsilon \right) = 0.$$

In particular, it suffices that $\mathcal{F}$ be asymptotically equicontinuous uniformly in the set $\{P_m\}$. This is in turn implied by $\mathcal{F}$ being Donsker and pre-Gaussian uniformly in $\{P_m\}$.

For marginal convergence, we can apply the Lindeberg central limit theorem. Pointwise convergence of ρ_{P_n} to ρ_{P_0} means exactly that the covariance functions of the $\mathbb{G}_{n, P_n}$ converge to the desired limit. The Lindeberg condition for the sequence $\mathbb{G}_{n, P_n} f$ is certainly implied by

$$(2.8.6) \quad \limsup_{n \rightarrow \infty} P_n F^2 \{F \geq \varepsilon \sqrt{n}\} = 0, \quad \text{for every } \varepsilon > 0.$$

One possible conclusion is that $\mathcal{F}$ being uniformly Donsker and pre-Gaussian in the sequence $\{P_m\}$ together with (2.8.5) and (2.8.6) is sufficient for the convergence $\mathbb{G}_{n, P_n} \rightsquigarrow \mathbb{G}_{P_0}$. It is of interest that under the same conditions the Gaussian limit distributions are continuous in the underlying measure.

2.8.7 Lemma. *Let $\mathcal{F}$ be Donsker and pre-Gaussian uniformly in the sequence $\{P_m\}$, and let (2.8.5) and (2.8.6) hold. Then $\mathbb{G}_{n, P_n} \rightsquigarrow \mathbb{G}_{P_0}$ in $\ell^\infty(\mathcal{F})$.*

2.8.8 Lemma. *Let $\mathcal{F}$ be pre-Gaussian uniformly in the sequence $\{P_m\}$, and let (2.8.5) hold. Then $\mathbb{G}_{P_n} \rightsquigarrow \mathbb{G}_{P_0}$ in $\ell^\infty(\mathcal{F})$.*

Proofs. The first lemma was argued earlier (under slightly weaker conditions). (Inspection of the proof shows that the implication (i) $\Rightarrow$ (ii) of Theorem 2.8.2 is valid without uniform integrability of the envelope.)

For the second lemma, first note that pointwise convergence of the covariance functions and zero-mean normality of each $\mathbb{G}_P$ imply marginal convergence. The uniform pre-Gaussianity gives

$$\lim_{\delta \downarrow 0} \limsup_{n \rightarrow \infty} \mathbf{P}_{P_n} \left(\sup_{\rho_{P_n}(f, g) < \delta} |\mathbb{G}_{P_n}(f) - \mathbb{G}_{P_n}(g)| > \varepsilon \right) = 0.$$

By the uniform convergence of ρ_{P_n} to ρ_{P_0} , this is then also true with ρ_{P_n} replaced by ρ_{P_0} . The uniform pre-Gaussianity implies uniform total boundedness of $\mathcal{F}$ by Theorem 2.8.3, which together with (2.8.5) implies total boundedness for ρ_{P_0} . Finally, apply Theorems 1.5.4 and 1.5.7. ■

The uniform Donsker and pre-Gaussian property in the first lemma could be established by the uniform entropy condition or by a bracketing condition. For easy reference we formulate two easily applicable Donsker theorems for sequences. They are proved as the uniform Donsker theorems 2.8.3 and 2.8.4. Alternatively, compare Theorems 2.11.1 and 2.11.9 or Examples 2.14.15 and 2.14.20.

2.8.9 Theorem. Let $\mathcal{F}$ be a class of measurable functions with a measurable envelope function F such that $\mathcal{F}_{\delta, P_n} = \{f - g: f, g \in \mathcal{F}, \|f - g\|_{P_n, 2} < \delta\}$ and $\mathcal{F}_{\infty}^2 = \{(f - g)^2: f, g \in \mathcal{F}\}$ are P_n -measurable for every $\delta > 0$ and n . Furthermore, suppose that $\mathcal{F}$ satisfies the uniform entropy condition, that $P_n F^2 = O(1)$ and that (2.8.5) and (2.8.6) hold. Then $\mathbb{G}_{n, P_n} \rightsquigarrow \mathbb{G}_{P_0}$ in $\ell^\infty(\mathcal{F})$.

2.8.10 Theorem. Let $\mathcal{F}$ be a class of measurable functions, with a measurable envelope function F , that is totally bounded for ρ_{P_0} , satisfies (2.8.5) and (2.8.6), and is such that

$$\int_0^{\delta_n} \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L_2(P_n))} d\varepsilon \rightarrow 0, \quad \text{for every } \delta_n \downarrow 0.$$

Then $\mathbb{G}_{n, P_n} \rightsquigarrow \mathbb{G}_{P_0}$ in $\ell^\infty(\mathcal{F})$.

Problems and Complements

1. Suppose $\rho_{P_n} \rightarrow \rho_{P_0}$ uniformly on $\mathcal{F} \times \mathcal{F}$. Then $\mathcal{F}$ is totally bounded for ρ_{P_0} if and only if, for every $\varepsilon > 0$, there exists an N such that $\sup_{n \geq N} N(\varepsilon, \mathcal{F}, \rho_{P_n}) < \infty$.

2.9

Multiplier

Central Limit Theorems

With the notation $Z_i = \delta_{X_i} - P$, the empirical central limit theorem can be written

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i \rightsquigarrow \mathbb{G},$$

in $\ell^\infty(\mathcal{F})$, where $\mathbb{G}$ is a (tight) Brownian bridge. Given i.i.d. real-valued random variables $\xi_1, \dots, \xi_n$, which are independent of $Z_1, \dots, Z_n$, the *multiplier central limit theorem* asserts that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \rightsquigarrow \mathbb{G}.$$

If the ξ_i have mean zero, have variance 1, and satisfy a moment condition, then the multiplier central limit theorem is true if and only if $\mathcal{F}$ is Donsker: in that case, the two displays are equivalent.

A more refined and deeper result is the *conditional multiplier central limit theorem*, according to which

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \rightsquigarrow \mathbb{G},$$

given almost every sequence $Z_1, Z_2, \dots$. This more interesting assertion turns out to be true under only slightly stronger conditions. Next to this almost sure version, we also discuss a version in probability of the conditional central limit theorem.

The techniques developed in this chapter have applications to statistical problems, especially to the study of various ways of bootstrapping the empirical process (see Chapter 3.7). The results are also used in the proofs of Chapter 2.10.

It is unnecessary to make measurability assumptions on $\mathcal{F}$. However, as usual, outer expectations are understood relative to a product space. Throughout the section let $X_1, X_2, \dots$ be defined as the coordinate projections on the “first” ∞ coordinates in the probability space $(\mathcal{X}^\infty \times \mathcal{Z}, \mathcal{A}^\infty \times \mathcal{C}, P^\infty \times Q)$, and let $\xi_1, \xi_2, \dots$ depend on the last coordinate only.

The (unconditional) multiplier central limit theorem is a corollary of a symmetrization inequality, which complements the symmetrization inequalities for Rademacher variables obtained in Chapter 2.3. For a random variable ξ and $p \geq 1$, set

$$\|\xi\|_{p,1} = \int_0^\infty (P(|\xi| > x))^{1/p} dx.$$

In spite of the notation, this is not a norm (but there exists a norm that is equivalent to $\|\cdot\|_{p,1}$). Finiteness of $\|\xi\|_{p,1}$ requires slightly more than a finite p th moment, but it is implied by a finite $p + \varepsilon$ absolute moment (Problem 2.9.1).

In Chapter 2.3 it is shown that the norms of a given process $\sum_{i=1}^n Z_i$ and its symmetrized version $\sum_{i=1}^n \varepsilon_i Z_i$ are comparable in magnitude. The following lemma extends this comparison to the norm of a general multiplier process $\sum_{i=1}^n \xi_i Z_i$.

2.9.1 Lemma (Multiplier inequalities). *Let $Z_1, \dots, Z_n$ be i.i.d. stochastic processes with $E^* \|Z_i\|_{\mathcal{F}} < \infty$ independent of the Rademacher variables $\varepsilon_1, \dots, \varepsilon_n$. Then for every i.i.d. sample $\xi_1, \dots, \xi_n$ of mean-zero random variables independent of $Z_1, \dots, Z_n$, and any $1 \leq n_0 \leq n$,*

$$\begin{aligned} \frac{1}{2} \|\xi\|_1 E^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i Z_i \right\|_{\mathcal{F}} &\leq E^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}} \\ &\leq 2(n_0 - 1) E^* \|Z_1\|_{\mathcal{F}} E \max_{1 \leq i \leq n} \frac{|\xi_i|}{\sqrt{n}} \\ &\quad + 2\sqrt{2} \|\xi\|_{2,1} \max_{n_0 \leq k \leq n} E^* \left\| \frac{1}{\sqrt{k}} \sum_{i=n_0}^k \varepsilon_i Z_i \right\|_{\mathcal{F}}. \end{aligned}$$

For symmetrically distributed variables ξ_i , the constants $1/2$, 2 , and $2\sqrt{2}$ can all be replaced by 1 .

Proof. Define $\varepsilon_1, \dots, \varepsilon_n$ as independent of $\xi_1, \dots, \xi_n$ (on “their own” factor of a product probability space). If the ξ_i are symmetrically distributed, then the variables $\varepsilon_i |\xi_i|$ possess the same distribution as the ξ_i . In that case, the inequality on the left follows from

$$E^* \left\| \sum_{i=1}^n \varepsilon_i E \xi_i |Z_i| \right\|_{\mathcal{F}} \leq E^* \left\| \sum_{i=1}^n \varepsilon_i |\xi_i| Z_i \right\|_{\mathcal{F}}.$$

For the general case, let $\eta_1, \dots, \eta_n$ be an independent copy of $\xi_1, \dots, \xi_n$. Then $\|\xi_i\|_1 = E|\xi_i - E\eta_i| \leq \|\xi_i - \eta_i\|_1$, so that $\|\xi_i\|_1$ can be replaced by $\|\xi_i - \eta_i\|_1$ in the left-hand side. Next, apply the inequality for symmetric variables to the variables $\xi_i - \eta_i$, and then use the triangle inequality to see that

$$E^* \left\| \sum_{i=1}^n (\xi_i - \eta_i) Z_i \right\|_{\mathcal{F}} \leq 2E^* \left\| \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}}.$$

This concludes the proof of the inequality on the left.

Again assume that the ξ_i are symmetrically distributed. Let $\tilde{\xi}_1 \geq \dots \geq \tilde{\xi}_n$ be the (reversed) order statistics of $|\xi_1|, \dots, |\xi_n|$. By the definition of $Z_1, \dots, Z_n$ as fixed functions of the coordinates on the product space $(\mathcal{X}^n, \mathcal{B}^n)$, it follows that for any fixed $\xi_1, \dots, \xi_n$,

$$\mathbb{E}_\varepsilon \mathbb{E}_Z^* \left\| \sum_{i=1}^n \varepsilon_i |\xi_i| Z_i \right\|_{\mathcal{F}} = \mathbb{E}_\varepsilon \mathbb{E}_Z^* \left\| \sum_{i=1}^n \varepsilon_i \tilde{\xi}_i Z_i \right\|_{\mathcal{F}}.$$

In view of Lemma 1.2.7, the joint outer expectation $\mathbb{E}^*$ can be replaced by $\mathbb{E}_\xi \mathbb{E}_Z^*$. Thus

$$\begin{aligned} \mathbb{E}^* \left\| \sum_{i=1}^n \varepsilon_i \tilde{\xi}_i Z_i \right\|_{\mathcal{F}} &= \mathbb{E}_{\xi, \varepsilon} \mathbb{E}_Z^* \left\| \sum_{i=1}^n \varepsilon_i |\xi_i| Z_i \right\|_{\mathcal{F}} = \mathbb{E}_{\xi, \varepsilon} \mathbb{E}_Z^* \left\| \sum_{i=1}^n \varepsilon_i \tilde{\xi}_i Z_i \right\|_{\mathcal{F}} \\ &\leq (n_0 - 1) \mathbb{E} \tilde{\xi}_1 \mathbb{E}^* \|Z_1\|_{\mathcal{F}} + \mathbb{E}^* \left\| \sum_{i=n_0}^n \varepsilon_i \tilde{\xi}_i Z_i \right\|_{\mathcal{F}}. \end{aligned}$$

Substitute $\tilde{\xi}_i = \sum_{k=i}^n (\tilde{\xi}_k - \tilde{\xi}_{k+1})$ in the second term (with $\tilde{\xi}_{n+1} = 0$) and change the order of summation in the resulting double sum to see that it is equal to

$$\begin{aligned} \mathbb{E}^* \left\| \sum_{i=n_0}^n \varepsilon_i \tilde{\xi}_i Z_i \right\|_{\mathcal{F}} &= \mathbb{E}^* \left\| \sum_{k=n_0}^n (\tilde{\xi}_k - \tilde{\xi}_{k+1}) \sum_{i=n_0}^k \varepsilon_i Z_i \right\|_{\mathcal{F}} \\ &\leq \mathbb{E} \sum_{k=n_0}^n \sqrt{k} (\tilde{\xi}_k - \tilde{\xi}_{k+1}) \max_{n_0 \leq k \leq n} \mathbb{E}^* \left\| \frac{1}{\sqrt{k}} \sum_{i=n_0}^k \varepsilon_i Z_i \right\|_{\mathcal{F}}. \end{aligned}$$

For $\tilde{\xi}_{k+1} < t \leq \tilde{\xi}_k$, one has $k = \#\{i: |\xi_i| \geq t\}$. Thus the first expectation in the product can be written as

$$\mathbb{E} \sum_{k=n_0}^n \int_{\tilde{\xi}_{k+1}}^{\tilde{\xi}_k} \sqrt{k} dt \leq \int_0^\infty \mathbb{E} \sqrt{\#\{i: |\xi_i| \geq t\}} dt \leq \int_0^\infty \sqrt{n \mathbb{P}(|\xi_i| \geq t)} dt,$$

by Jensen's inequality. This concludes the proof of the upper bound for symmetric variables.

For possibly asymmetric variables, first note that

$$\mathbb{E}^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}} \leq \mathbb{E}^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n (\xi_i - \eta_i) Z_i \right\|_{\mathcal{F}}.$$

Next apply the bound for symmetric variables to the variables $\xi_i - \eta_i$, and then use the triangle inequality and the “corrected” triangle inequality $\|\xi - \eta\|_{2,1} \leq 2\sqrt{2} \|\xi\|_{2,1}$ (Problem 2.9.2). ■

For $n_0 = 1$, the preceding lemma gives the inequalities

$$\begin{aligned} \frac{1}{2} \|\xi\|_1 E^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i Z_i \right\|_{\mathcal{F}} &\leq E^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}} \\ &\leq 2\sqrt{2} \|\xi\|_{2,1} \max_{1 \leq k \leq n} E^* \left\| \frac{1}{\sqrt{k}} \sum_{i=1}^k \varepsilon_i Z_i \right\|_{\mathcal{F}}. \end{aligned}$$

For variables ξ_i with range $[-1, 1]$, the maximum in the right side may be replaced by only its n th term (Problem 2.9.3), but this appears to be untrue in general. This simpler inequality obtained for $n_0 = 1$ is too weak to yield the following theorem.

2.9.2 Theorem (Unconditional multiplier CLT). *Let $\mathcal{F}$ be a class of measurable functions. Let $\xi_1, \dots, \xi_n$ be i.i.d. random variables with mean zero, variance 1, and $\|\xi\|_{2,1} < \infty$, independent of $X_1, \dots, X_n$. Then the sequence $n^{-1/2} \sum_{i=1}^n \xi_i (\delta_{X_i} - P)$ converges to a tight limit process in $\ell^\infty(\mathcal{F})$ if and only if $\mathcal{F}$ is Donsker. In that case, the limit process is a P -Brownian bridge.*

Proof. Since both the empirical processes $n^{-1/2} \sum_{i=1}^n (\delta_{X_i} - P)$ and the multiplier processes $n^{-1/2} \sum_{i=1}^n \xi_i (\delta_{X_i} - P)$ do not change if indexed by the class of functions $\{f - Pf : f \in \mathcal{F}\}$ instead of $\mathcal{F}$, it may be assumed without loss of generality that $Pf = 0$ for every f . Marginal convergence of both sequences of processes is equivalent to $\mathcal{F} \subset \mathcal{L}_2(P)$. It suffices to show that the asymptotic equicontinuity conditions for the empirical and the multiplier processes are equivalent.

If $\mathcal{F}$ is Donsker, then $P^*(F > x) = o(x^{-2})$ as $x \rightarrow \infty$ by Lemma 2.3.9. By the same lemma convergence of the multiplier processes to a tight limit implies that $P^*(|\xi F| > x) = o(x^{-2})$. In particular, $P^*F < \infty$ in both cases.

Since the assumption $\|\xi\|_{2,1} < \infty$ implies the existence of a second moment, we have $E \max_{1 \leq i \leq n} |\xi_i| / \sqrt{n} \rightarrow 0$. Combination with the multiplier inequalities, Lemma 2.9.1, gives

$$\begin{aligned} \frac{1}{2} \|\xi\|_1 \limsup_{n \rightarrow \infty} E^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i Z_i \right\|_{\mathcal{F}_\delta} &\leq \limsup_{n \rightarrow \infty} E^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}_\delta} \\ &\leq 2\sqrt{2} \|\xi\|_{2,1} \sup_{n_0 \leq k} E^* \left\| \frac{1}{\sqrt{k}} \sum_{i=n_0}^k \varepsilon_i Z_i \right\|_{\mathcal{F}_\delta}, \end{aligned}$$

for every n_0 and $\delta > 0$. The inequality remains valid if the lower bound n_0 in the sum is replaced by 1. Next, by Lemma 2.3.6, the Rademacher variables can be deleted in this statement at the cost of changing the constants. Conclude that $E^* \left\| n^{-1/2} \sum_{i=1}^n Z_i \right\|_{\mathcal{F}_{\delta n}} \rightarrow 0$ if and only if $E^* \left\| n^{-1/2} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}_{\delta n}} \rightarrow 0$. These are the mean versions of the asymptotic equicontinuity conditions. By Lemma 2.3.11, these are equivalent to the probability versions. ■

Under the conditions of the preceding theorem, the sequence

$$\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n (\delta_{X_i} - P), \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i (\delta_{X_i} - P) \right)$$

is jointly asymptotically tight. Since the two coordinates are uncorrelated and the joint marginals converge to multivariate normal distributions, the sequence converges jointly to a vector of two independent Brownian bridges.

2.9.3 Corollary. *Under the conditions of the preceding theorem, the sequence of processes $(n^{-1/2} \sum_{i=1}^n (\delta_{X_i} - P), n^{-1/2} \sum_{i=1}^n \xi_i (\delta_{X_i} - P))$ converges in $\ell^\infty(\mathcal{F}) \times \ell^\infty(\mathcal{F})$ in distribution to a vector $(\mathbb{G}, \mathbb{G}')$ of independent (tight) Brownian bridges $\mathbb{G}$ and $\mathbb{G}'$.*

An additional corollary applies when the multipliers ξ_i do not have mean zero. Let $E\xi_i = \mu$ and $\text{var } \xi_i = \sigma^2$. Simple algebra gives

$$\begin{aligned} & \frac{1}{\sqrt{n}} \sum_{i=1}^n (\xi_i \delta_{X_i} - \mu P) \\ &= \frac{\mu}{\sqrt{n}} \sum_{i=1}^n (\delta_{X_i} - P) + \frac{\sigma}{\sqrt{n}} \sum_{i=1}^n \left(\frac{\xi_i - \mu}{\sigma} \right) (\delta_{X_i} - P) + \sqrt{n}(\bar{\xi}_n - \mu)P. \end{aligned}$$

The first two terms on the right converge weakly if and only if $\mathcal{F}$ is Donsker; the third is in $\ell^\infty(\mathcal{F})$ if and only if $\|P\|_{\mathcal{F}} < \infty$; in that case, it converges.

2.9.4 Corollary. *Let $\mathcal{F}$ be Donsker with $\|P\|_{\mathcal{F}} < \infty$. Let $\xi_1, \dots, \xi_n$ be i.i.d. random variables with mean μ , variance σ^2 , and $\|\xi_i\|_{2,1} < \infty$, independent of $X_1, \dots, X_n$. Then*

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (\xi_i \delta_{X_i} - \mu P) \rightsquigarrow \mu \mathbb{G} + \sigma \mathbb{G}' + \sigma Z P,$$

where $\mathbb{G}$ and $\mathbb{G}'$ are independent (tight) Brownian bridges and are independent of the random variable Z , which is standard normally distributed. The limit process $\mu \mathbb{G} + \sigma \mathbb{G}' + \sigma Z P$ is a Gaussian process with zero mean and covariance function $(\sigma^2 + \mu^2)Pfg - \mu^2 P f P g$.

Next consider conditional multiplier central limit theorems. For finite $\mathcal{F}$, the almost sure version is a simple consequence of the Lindeberg theorem.

2.9.5 Lemma. *Let $Z_1, Z_2, \dots$ be i.i.d. Euclidean random vectors with $E Z_i = 0$ and $E \|Z_i\|^2 < \infty$ independent of the i.i.d. sequence $\xi_1, \xi_2, \dots$, with $E \xi_i = 0$ and $E \xi_i^2 = 1$. Then, conditionally on $Z_1, Z_2, \dots$,*

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \rightsquigarrow N(0, \text{cov } Z_1),$$

for almost every sequence $Z_1, Z_2, \dots$.

Proof. According to the Lindeberg central limit theorem, the statement is true for every sequence $Z_1, Z_2, \dots$ such that, for every $\varepsilon > 0$,

$$\frac{1}{n} \sum_{i=1}^n Z_i Z_i' \rightarrow \text{cov } Z_1; \quad \frac{1}{n} \sum_{i=1}^n \|Z_i\|^2 E \xi_i^2 \{ |\xi_i| \|Z_i\| > \varepsilon \sqrt{n} \} \rightarrow 0.$$

The first condition is true for almost all sequences by the law of large numbers. Furthermore, a finite second moment, $E\|Z_i\|^2 < \infty$, implies $\max_{1 \leq i \leq n} \|Z_i\|/\sqrt{n} \rightarrow 0$, for almost all sequences. For the intersection of the two sets of sequences, both conditions are satisfied. ■

The preceding lemma takes care of marginal convergence in the conditional multiplier central limit theorems. For convergence in $\ell^\infty(\mathcal{F})$, it now suffices to check asymptotic equicontinuity.

Conditional weak convergence in probability must be formulated in terms of a metric on “conditional laws”. Chapter 1.13 formalizes conditional weak convergence using the bounded Lipschitz metric, as introduced in Chapter 1.12, but is restricted to Borel measurable maps, for which proper conditional laws exist. Presently we do not wish to assume measurability, and shall instead utilize a bounded Lipschitz distance based on outer expectations. In Chapter 1.12 it is shown that weak convergence, $G_n \rightsquigarrow G$, of a sequence of random elements G_n in $\ell^\infty(\mathcal{F})$ to a separable limit G is equivalent to, for BL_1 the set of all functions $h: \ell^\infty(\mathcal{F}) \rightarrow [-1, 1]$ such that $|h(z_1) - h(z_2)| \leq \|z_1 - z_2\|_{\mathcal{F}}$ for every z_1, z_2 ,

$$\sup_{h \in BL_1} |E^* h(G_n) - Eh(G)| \rightarrow 0.$$

To formulate conditional weak convergence, we replace the outer expectation $E^* h(G_n)$ in this expression by a conditional expectation. For a multiplier process $\mathbb{G}'_n = n^{-1/2} \sum_{i=1}^n \xi_i (\delta_{X_i} - P)$, the latter conditional expectation will be written as $E_\xi h(\mathbb{G}'_n)$, where the expectation is understood to be taken relative the multiplier variables $\xi_1, \dots, \xi_n$, for fixed values of the observations $X_1, \dots, X_n$.

2.9.6 Theorem (Conditional multiplier CLT in probability). *Let $\mathcal{F}$ be a class of measurable functions. Let $\xi_1, \dots, \xi_n$ be i.i.d. random variables with mean zero, variance 1, and $\|\xi\|_{2,1} < \infty$, independent of $X_1, \dots, X_n$. Let $\mathbb{G}'_n = n^{-1/2} \sum_{i=1}^n \xi_i (\delta_{X_i} - P)$. Then the following assertions are equivalent:*

- (i) $\mathcal{F}$ is Donsker;
- (ii) $\sup_{h \in BL_1} |E_\xi h(\mathbb{G}'_n) - Eh(\mathbb{G})| \rightarrow 0$ in outer probability, and the sequence $\mathbb{G}'_n$ is asymptotically measurable.

Proof. (i) $\Rightarrow$ (ii). For a Donsker class $\mathcal{F}$, the sequence $\mathbb{G}'_n$ converges in distribution to a Brownian bridge process by the unconditional multiplier theorem, Theorem 2.9.2. Thus, it is asymptotically measurable.

A Donsker class is totally bounded for ρ_P . For each fixed $\delta > 0$ and $f \in \mathcal{F}$, let $\Pi_\delta f$ denote a closest element in a given, finite δ -net for $\mathcal{F}$. By uniform continuity of the limit process $\mathbb{G}$, we have $\mathbb{G} \circ \Pi_\delta \rightarrow \mathbb{G}$ almost surely, as $\delta \downarrow 0$. Hence, it is certainly true that

$$\sup_{h \in BL_1} |Eh(\mathbb{G} \circ \Pi_\delta) - Eh(\mathbb{G})| \rightarrow 0.$$

Second, by the preceding lemma, as $n \rightarrow \infty$ and for every fixed $\delta > 0$,

$$\sup_{h \in BL_1} |E_\xi h(\mathbb{G}'_n \circ \Pi_\delta) - Eh(\mathbb{G} \circ \Pi_\delta)| \rightarrow 0,$$

for almost every sequence $X_1, X_2, \dots$ (To see this, define $A: \mathbb{R}^p \rightarrow \ell^\infty(\mathcal{F})$ by $Ay(f) = y_i$ if $\Pi_\delta f = f_i$. Then $h(\mathbb{G} \circ \Pi_\delta) = g(\mathbb{G}(f_1), \dots, \mathbb{G}(f_p))$, for the function g defined by $g(y) = h(Ay)$. If h is bounded Lipschitz on $\ell^\infty(\mathcal{F})$, then g is bounded Lipschitz on $\mathbb{R}^p$ with a smaller bounded Lipschitz norm.) Since $BL_1(\mathbb{R}^p)$ is separable for the topology of uniform convergence on compacta, the supremum in the preceding display can be replaced by a countable supremum. It follows that the variable in the display is measurable, because $h(\mathbb{G}'_n \circ \Pi_\delta)$ is measurable. Third,

$$\begin{aligned} \sup_{h \in BL_1} |E_\xi h(\mathbb{G}'_n \circ \Pi_\delta) - E_\xi h(\mathbb{G}'_n)| &\leq \sup_{h \in BL_1} E_\xi |h(\mathbb{G}'_n \circ \Pi_\delta) - h(\mathbb{G}'_n)| \\ &\leq E_\xi \|\mathbb{G}'_n \circ \Pi_\delta - \mathbb{G}'_n\|_{\mathcal{F}}^* \leq E_\xi \|\mathbb{G}'_n\|_{\mathcal{F}_\delta}^*, \end{aligned}$$

where $\mathcal{F}_\delta$ is the class $\{f - g: \rho_P(f - g) < \delta\}$. Thus, the outer expectation of the left side is bounded above by $E^* \|\mathbb{G}'_n\|_{\mathcal{F}_\delta}^*$. As in the proof of Theorem 2.9.2, we can use the multiplier inequalities of Lemma 2.9.1 to see that the latter expression converges to zero as $n \rightarrow \infty$ followed by $\delta \rightarrow 0$. Combine this with the previous displays to obtain one-half of the theorem.

(ii) $\Rightarrow$ (i). Letting $h(\mathbb{G}'_n)^*$ and $h(\mathbb{G}'_n)_*$ denote measurable majorants and minorants with respect to $(\xi_1, \dots, \xi_n, X_1, \dots, X_n)$ jointly, we have, by the triangle inequality and Fubini's theorem,

$$|E^* h(\mathbb{G}'_n) - Eh(\mathbb{G})| \leq |E_X E_\xi h(\mathbb{G}'_n)^* - E_X^* E_\xi h(\mathbb{G}'_n)| + E_X^* |E_\xi h(\mathbb{G}'_n) - Eh(\mathbb{G})|.$$

If (ii) holds, then, by dominated convergence, the second term on the right side converges to zero for every $h \in BL_1$. The first term on the right is bounded above by $E_X E_\xi h(\mathbb{G}'_n)^* - E_X E_\xi h(\mathbb{G}'_n)_*$, which converges to zero if $\mathbb{G}'_n$ is asymptotically measurable. Thus, (ii) implies that $\mathbb{G}'_n \rightsquigarrow \mathbb{G}$ unconditionally. Then $\mathcal{F}$ is Donsker by the converse part of the unconditional multiplier theorem, Theorem 2.9.2. ■

The functions $(\xi_1, \dots, \xi_n) \mapsto h(\mathbb{G}'_n)$ are (Lipschitz) continuous, and hence Borel measurable, for every given sequence $X_1, X_2, \dots$ (under the assumption that $\|f(x) - Pf\|_{\mathcal{F}} < \infty$ for every x , which is implicitly made to ensure that our processes take their values in $\ell^\infty(\mathcal{F})$). Therefore, the expectations $E_\xi h(\mathbb{G}'_n)$ in the statement of the preceding theorem make sense (without asterisks “*”), even though they may be nonmeasurable functions of $(X_1, \dots, X_n)$. The second assumption in (ii) implies that these functions are asymptotically measurable and appears to be necessary for the “easy” implication, (ii) $\Rightarrow$ (i).

The almost sure version of the conditional central limit theorem asserts weak convergence of the sequence $\mathbb{G}'_n = n^{-1/2} \sum_{i=1}^n \xi_i (\delta_{X_i} - P)$ given almost every sequence $X_1, X_2, \dots$. It was seen in Example 1.9.4 that almost sure convergence without proper measurability may not mean much. Thus, it is more attractive to define almost sure conditional convergence by strengthening part (ii) of the preceding theorem. It will be shown that

$$\sup_{h \in BL_1} |E_\xi h(\mathbb{G}'_n) - Eh(\mathbb{G})| \xrightarrow{\text{as}} 0.$$

This implies that $E_\xi h(\mathbb{G}'_n) \rightarrow Eh(\mathbb{G})$ for every $h \in BL_1$, for almost every sequence $X_1, X_2, \dots$. By the portmanteau theorem, this is then also true for every continuous,

bounded h . Thus, the sequence $\mathbb{G}'_n$ converges in distribution to $\mathbb{G}$ given almost every sequence $X_1, X_2, \dots$ also in a more naive sense of almost sure conditional convergence.

2.9.7 Theorem (Conditional multiplier CLT almost surely). *Let $\mathcal{F}$ be a class of measurable functions. Let $\xi_1, \dots, \xi_n$ be i.i.d. random variables with mean zero, variance 1, and $\|\xi\|_{2,1} < \infty$, independent of $X_1, \dots, X_n$. Define $\mathbb{G}'_n = n^{-1/2} \sum_{i=1}^n \xi_i (\delta_{X_i} - P)$. Then the following assertions are equivalent:*

- (i) $\mathcal{F}$ is Donsker and $P^* \|f - Pf\|_{\mathcal{F}}^2 < \infty$;
- (ii) $\sup_{h \in BL_1} |E_{\xi} h(\mathbb{G}'_n) - Eh(\mathbb{G})| \rightarrow 0$ outer almost surely, and the sequence $E_{\xi} h(\mathbb{G}'_n)^* - E_{\xi} h(\mathbb{G}'_n)_*$ converges almost surely to zero for every $h \in BL_1$. Here $h(\mathbb{G}'_n)^*$ and $h(\mathbb{G}'_n)_*$ denote measurable majorants and minorants with respect to $(\xi_1, \dots, \xi_n, X_1, \dots, X_n)$ jointly.

Proof. (i) $\Rightarrow$ (ii). For the first assertion of (ii), the proof of the in probability conditional central limit theorem applies, except that it must be argued that $E_{\xi} \|\mathbb{G}'_n\|_{\mathcal{F}_{\delta}}^*$ converges to zero outer almost surely, as $n \rightarrow \infty$ followed by $\delta \downarrow 0$. By Corollary 2.9.9,

$$\limsup_{n \rightarrow \infty} E_{\xi} \|\mathbb{G}'_n\|_{\mathcal{F}_{\delta}}^* \leq 6\sqrt{2} \limsup_{n \rightarrow \infty} E^* \|\mathbb{G}'_n\|_{\mathcal{F}_{\delta}}, \quad \text{a.s.}$$

The right-hand side decreases to zero as $\delta \downarrow 0$, because $\mathbb{G}'_n \rightsquigarrow \mathbb{G}$ unconditionally, by Theorem 2.9.2.

To see that the sequence $E_{\xi} h(\mathbb{G}'_n)$ is strongly asymptotically measurable, obtain first by the same proof, but with a star added, that

$$|E_{\xi} h(\mathbb{G}'_n)^* - Eh(\mathbb{G})| \xrightarrow{\text{as}} 0.$$

The same proof also shows that this is true with a lower star. Thus, the sequence $E_{\xi} h(\mathbb{G}'_n)^* - E_{\xi} h(\mathbb{G}'_n)_*$ converges to zero almost surely.

(ii) $\Rightarrow$ (i). The class $\mathcal{F}$ is Donsker in view of the in probability conditional central limit theorem. We need to prove the condition on its envelope function. As in the proof of the portmanteau theorem, there exists a sequence of Lipschitz functions $h_m: \mathbb{R} \rightarrow \mathbb{R}$ such that $1 \geq h_m(\|z\|_{\mathcal{F}}) \downarrow 1\{\|z\|_{\mathcal{F}} \geq t\}$ pointwise. By Lemma 1.2.2(vi),

$$P_{\xi}(\|\mathbb{G}'_n\|_{\mathcal{F}}^* > t) = E_{\xi} 1\{\|\mathbb{G}'_n\|_{\mathcal{F}} > t\}^* \leq E_{\xi} h_m(\|\mathbb{G}'_n\|_{\mathcal{F}})^*.$$

Under assumption (ii), the right side converges to $Eh_m(\|\mathbb{G}\|_{\mathcal{F}})$ almost surely as $n \rightarrow \infty$, for every fixed m . Conclude that

$$\limsup_{n \rightarrow \infty} P_{\xi}(\|\mathbb{G}'_n\|_{\mathcal{F}}^* > t) \leq P(\|\mathbb{G}\|_{\mathcal{F}} \geq t), \quad \text{a.s.}$$

Set $Z_i = \delta_{X_i} - P$. By the triangle inequality, $\|\xi_n Z_n\|_{\mathcal{F}}^* \leq \sqrt{n} \|\mathbb{G}'_n\|_{\mathcal{F}}^* + \sqrt{n-1} \|\mathbb{G}'_{n-1}\|_{\mathcal{F}}^*$. Thus, for sufficiently large t ,

$$\limsup_{n \rightarrow \infty} P_{\xi}(\|\xi_n Z_n\|_{\mathcal{F}}^* > t\sqrt{n}) < 1, \quad \text{a.s.}$$

Since ξ_n is distributed as ξ_1 , it follows that $\limsup \|Z_n\|_{\mathcal{F}}^* / \sqrt{n} < \infty$ almost surely. This implies $E^* \|Z_1\|_{\mathcal{F}}^2 < \infty$ (Problem 2.9.4). ■

The proof of the following lemma is based on isoperimetric inequalities for product measures, which are treated in Appendix A.4. The Problems section gives alternative proofs of similar results using more conventional methods.

2.9.8 Lemma. *Let $Z_1, Z_2, \dots$ be i.i.d. stochastic processes such that $E^* \|Z_1\|_{\mathcal{F}}^2 < \infty$. Let $\xi_1, \xi_2, \dots$ be i.i.d. random variables with mean zero, independent of $Z_1, Z_2, \dots$. Then there exists a constant K such that, for every $t > 0$,*

$$\sum_n \mathbb{P} \left(E_{\xi} \left\| \sum_{i=1}^{2^n} \xi_i Z_i \right\|_{\mathcal{F}}^* > 6E^* \left\| \sum_{i=1}^{2^n} \xi_i Z_i \right\|_{\mathcal{F}} + t2^{n/2} \right) \leq K \frac{E^* \|Z_1\|_{\mathcal{F}}^2}{t^2}.$$

Here the first star on the left denotes a measurable majorant with respect to the variables $(\xi_1, \dots, \xi_n, Z_1, \dots, Z_n)$ jointly.

2.9.9 Corollary. *In the situation of the preceding lemma,*

$$\begin{aligned} \limsup_{n \rightarrow \infty} E_{\xi} \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}}^* &\leq 6\sqrt{2} \limsup_{n \rightarrow \infty} E^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}}, \quad \text{a.s.,} \\ E \sup_n E_{\xi} \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}}^* &\leq C \sup_n E^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}} + C \sqrt{E^* \|Z_1\|_{\mathcal{F}}^2}, \end{aligned}$$

for a universal constant C .

Proofs. Abbreviate $S_n = E_{\xi} \left\| \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}}^*$. Fix n , and let $Z_{[1]}, \dots, Z_{[2^n]}$ be the processes $Z_1, \dots, Z_{2^n}$ ordered by decreasing norms $\|Z_{[i]}\|_{\mathcal{F}}^*$. Define the control number $f(A, A, A, Z_1, \dots, Z_{2^n})$ as in Appendix A.4 for the event $A = \{S_{2^n} \leq 2ES_{2^n}\}$. Then on the event $f(A, A, A, \tilde{Z}) < k$,

$$S_{2^n} \leq \sum_{i=1}^{k-1} \|Z_{[i]}\|_{\mathcal{F}}^* + 6ES_{2^n}.$$

(See the explanation after the proof of Proposition A.4.1.) By Markov's inequality, the event A has probability at least $1/2$. Combining this with Proposition A.4.1 yields, for every $k \geq 3$,

$$\begin{aligned} \mathbb{P}(S_{2^n} > 6ES_{2^n} + t2^{n/2}) &\leq \left(\frac{2}{3}\right)^k + \mathbb{P}\left(\sum_{i=1}^{k-1} \|Z_{[i]}\|_{\mathcal{F}}^* > t2^{n/2}\right) \\ &\leq \left(\frac{2}{3}\right)^k + \mathbb{P}\left(\|Z_{[1]}\|_{\mathcal{F}}^* > \frac{t}{2} 2^{n/2}\right) + \mathbb{P}\left(k\|Z_{[2]}\|_{\mathcal{F}}^* > \frac{t}{2} 2^{n/2}\right) \\ &\lesssim \left(\frac{1}{k}\right)^8 + \gamma_n\left(\frac{t}{2}\right) + \gamma_n^2\left(\frac{t}{2k}\right) \wedge 1. \end{aligned}$$

Here $\gamma_n(t) = 2^n \mathbb{P}(\|Z_1\|_{\mathcal{F}}^* > t2^{n/2})$, and the square arises from the binomial inequality of Problem 2.9.6. Let $\beta_n(t) = \sum_{l=0}^{\infty} \gamma_{n-l}(t)2^{-l}$ be the convolution of the sequences $\gamma_n(t)$ and $(1/2)^n$. Then $\beta_n(t) \geq \gamma_{n-l}(t)(1/2)^l$ for any natural number l , whence $\gamma_n(t2^{-l}) = 2^{2l}\gamma_{n-2l}(t) \leq 2^{4l}\beta_n(t)$. This readily yields that

$\gamma_n(t/k) \lesssim k^4 \beta_n(t)$ for every number $k \geq 3$. Use this inequality with the choice $k = \lfloor \beta_n(t)^{-1/8} \rfloor \vee 3$ to bound the last line of the preceding display up to a constant by

$$\left(\frac{1}{k}\right)^8 + \beta_n(t) + k^8 \beta_n^2(t) \wedge 1 \lesssim \beta_n(t).$$

The proof of the lemma is complete upon noting that $\sum \beta_n(t) \leq 2 \sum_{n=-\infty}^{\infty} \gamma_n(t)$, which in turn is bounded by a multiple of $E^* \|Z_1\|_{\mathcal{F}}^2/t^2$.

Since the random variables S_n are nondecreasing in n by Jensen's inequality, we obtain

$$\sup_{n \geq m} \frac{S_n}{\sqrt{n}} \leq \sqrt{2} \sup_{n \geq 2 \log m} \frac{S_{2^n}}{2^{n/2}}.$$

By the Borel-Cantelli lemma and Lemma 2.9.8 $\limsup (S_{2^n} - 6ES_{2^n})/2^{n/2} \leq 0$ almost surely. The first assertion of the corollary follows. Next

$$\begin{aligned} E \sup_n \frac{S_{2^n} - 6ES_{2^n}}{2^{n/2}} &\leq \int_0^\infty \sum_n P(S_{2^n} - 6ES_{2^n} > t2^{n/2}) dt \\ &\lesssim \int_0^\infty \left(\frac{E^* \|Z_1\|_{\mathcal{F}}^2}{t^2} \wedge 1 \right) dt. \end{aligned}$$

This is bounded by two times the square root of $E^* \|Z_1\|_{\mathcal{F}}^2$. The second assertion now follows similarly from the monotonicity of the variables S_n . ■

While the multiplier inequalities of Lemma 2.9.1 suffice for proving multiplier limit theorems (as seen in the first part of this chapter), they are not always tight, as they focus on the sufficiency of finiteness of $\|\xi\|_{2,1} < \infty$ for the central limit theorem. The following alternative inequality gives better finite sample behavior and allows for different norms.

It is phrased in terms of upper bounds $\psi_{\mathcal{F}}(k)$ for Rademacher processes for $1 \leq k \leq n$, of the form

$$(2.9.10) \quad E^* \left\| \sum_{i=1}^k \varepsilon_i Z_i \right\|_{\mathcal{F}} \leq \psi_{\mathcal{F}}(k).$$

2.9.11 Lemma (Multiplier inequalities). *For given n let $Z_1, \dots, Z_n$ be i.i.d. mean zero stochastic processes satisfying (2.9.10), for every $1 \leq k \leq n$, and a concave, nondecreasing function $\psi_{\mathcal{F}}: [0, \infty) \rightarrow [0, \infty)$. Then for any random variables $\xi_1, \dots, \xi_n$ independent of $Z_1, \dots, Z_n$,*

$$E^* \left\| \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}} \leq 2 \int_0^\infty \psi_{\mathcal{F}} \left(\sum_{i=1}^n P(|\xi_i| > t) \right) dt.$$

If the variables ξ_i are independent with zero mean, the same inequality is true for general exchangeable stochastic processes $Z_1, \dots, Z_n$, not necessarily with zero mean, but with the constant 2 replaced by 4.

Proof. Suppose first that the vector $(\xi_1, \dots, \xi_n)$ is identically distributed to the vector $(\varepsilon_1 |\xi_1|, \dots, \varepsilon_n |\xi_n|)$, for independent Rademacher variables ε_i , independent of

$\xi_1, \dots, \xi_n, Z_1, \dots, Z_n$, suitably defined on a product space. The left side of the lemma is then equal to $E^* \left\| \sum_{i=1}^n \varepsilon_i |\xi_i| Z_i \right\|_{\mathcal{F}}$. As in the proof of Lemma 2.9.1 (with $n_0 = 1$), this can be bounded by

$$E^* \left\| \sum_{k=1}^n (\tilde{\xi}_k - \tilde{\xi}_{k+1}) \sum_{i=1}^k \varepsilon_i Z_i \right\|_{\mathcal{F}} \leq E \sum_{k=1}^n (\tilde{\xi}_k - \tilde{\xi}_{k+1}) \psi_{\mathcal{F}}(k),$$

by (2.9.10), where the $\tilde{\xi}_k$ are the reverse order statistics of the $|\xi_k|$ (with $\tilde{\xi}_{n+1} = 0$). Because $k = \#(i: |\xi_i| \geq t)$, for $\tilde{\xi}_{k+1} < t \leq \tilde{\xi}_k$, the preceding display is bounded above by

$$E \sum_{k=1}^n \int_{\tilde{\xi}_{k+1}}^{\tilde{\xi}_k} \psi_{\mathcal{F}}(k) dt \leq \int_0^\infty E \psi_{\mathcal{F}}(\#(i: |\xi_i| \geq t)) dt.$$

By Jensen's inequality applied to the concave function $\psi_{\mathcal{F}}$, this is smaller than the right side of the lemma, without the factor 2.

For general variables ξ_i we can first apply Lemma 2.3.6 conditionally given $\xi_1, \dots, \xi_n$ to the independent mean zero processes $\xi_i Z_i$ to bound the left side of the lemma by $2E^* \left\| \sum_{i=1}^n \varepsilon'_i \xi_i Z_i \right\|_{\mathcal{F}}$ for independent Rademacher variables ε'_i . Next we apply the preceding with the variables $\varepsilon'_i \xi_i$ in the place of the variables ξ_i .

If the variables ξ_i have zero mean, then we can also bound the left side of the lemma by $E^* \left\| \sum_{i=1}^n (\xi_i - \xi'_i) Z_i \right\|_{\mathcal{F}}$, for an independent copy (ξ'_i) of (ξ_i) , by Jensen's inequality. If the variables ξ_i are independent, then the symmetric variables $\xi_i - \xi'_i$ satisfy the condition of the first paragraph, and this then gives the bound $\int_0^\infty \psi_{\mathcal{F}}(\sum_{i=1}^n P(|\xi_i - \xi'_i| > t)) dt$. This is bounded by twice the right side of the lemma, since $P(|\xi_i - \xi'_i| > t) \leq 2P(|\xi_i| > t/2)$ and $\psi_{\mathcal{F}}(2t) \leq 2\psi_{\mathcal{F}}(t)$, by concavity and nonnegativity of $\psi_{\mathcal{F}}$. ■

If $E^* \left\| \sum_{i=1}^k \varepsilon_i Z_i \right\|_{\mathcal{F}} \leq c k^{1/\gamma}$ for some constants $c > 0$ and $\gamma \geq 1$ and every k , then (2.9.10) is satisfied with $\psi_{\mathcal{F}}(t) = c t^{1/p}$, for any $1 \leq p \leq \gamma$. Then in the case that the variables ξ_i are i.i.d., the lemma yields the assertion

$$E^* \left\| \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}} \leq 2c n^{1/p} \|\xi_1\|_{p,1}.$$

The sharpest result in terms of growth in n is obtained by choosing the maximal $1 \leq p \leq \gamma$ such that $\|\xi_1\|_{p,1} < \infty$.

For example, if $\gamma = 2$, then the growth rate is $n^{1/2}$ when $\|\xi_1\|_{2,1} < \infty$ (and higher moments give no improvement), and $n^{1/p}$ when $\|\xi_1\|_{p,1} < \infty$ for $1 \leq p < 2$. This shows that boundedness in uniform mean of the scaled processes $k^{-1/2} \sum_{i=1}^k Z_i$ implies the boundedness of the multiplier processes $n^{-1/2} \sum_{i=1}^n \xi_i Z_i$ if $\|\xi_1\|_{2,1} < \infty$, but may not be sufficient for boundedness of the latter processes if the multipliers ξ_i possess only lower order moments. In the latter case the lemma gives the upper bound $n^{1/p-1/2}$.

Because the constants in the inequality are universal, it can be applied to classes $\mathcal{F}$ that change with n . Then the order γ as in the preceding paragraph may be obtained from an entropy bound. See Problem 2.9.12 for an example.

Problems and Complements

1. For any random variable ξ and $r > 2$, one has $(1/2)\|\xi\|_2 \leq \|\xi\|_{2,1} \leq r/(r-2)\|\xi\|_r$.
2. For any pair of random variables ξ and η , one has $\|\xi + \eta\|_{2,1}^2 \leq 4\|\xi\|_{2,1}^2 + 4\|\eta\|_{2,1}^2$.
3. In the situation of Lemma 2.9.1, if the variables ξ_i are symmetric and possess bounded range contained in $[-M, M]$, then $E^* \left\| \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}} \leq M E^* \left\| \sum_{i=1}^n \varepsilon_i Z_i \right\|_{\mathcal{F}}$.
[Hint: Use the contraction principle, Proposition A.1.11.]

4. If $X_1, X_2, \dots$ are i.i.d. random variables with $\limsup |X_n|/\sqrt{n} < \infty$ almost surely, then $EX_1^2 < \infty$.

[Hint: By the Kolmogorov 0-1 law, $\limsup |X_n|/\sqrt{n}$ is constant. Thus, there exists a constant M with $P(\limsup\{|X_n| > M\sqrt{n}\}) = 1$. By the Borel-Cantelli lemma, $\sum P(|X_n|^2 > nM^2) < \infty$.]

5. Let $\mathcal{F}$ be Donsker with $\|P\|_{\mathcal{F}} < \infty$. Let $\xi_1, \dots, \xi_n$ be i.i.d. random variables with mean μ , variance σ^2 , and $\|\xi_i\|_{2,1} < \infty$, independent of $X_1, \dots, X_n$. Let N_n be Poisson-distributed with mean n and independent of the ξ_i and X_i . Then the sequence $n^{-1/2} \sum_{i=1}^{N_n} \xi_i \delta_{X_i} - \sqrt{n} \mu P$ converges in distribution in $\ell^\infty(\mathcal{F})$ to $(\sigma^2 + \mu^2)^{1/2}$ times a Brownian motion process.
6. If $Z_{[1]}, \dots, Z_{[n]}$ are the reversed order statistics of an i.i.d. sample $Z_1, \dots, Z_n$, then $P(Z_{[k]} > x) \leq \binom{n}{k} P(Z_1 > x)^k$ for every x .
[Hint: Note that $P(Z_{[k]} > x) \leq \binom{n}{k} P(Z_1 > x, \dots, Z_k > x)$.]

7. (Alternative proof of Corollary 2.9.9) Let $Z_1, Z_2, \dots$ be i.i.d. stochastic processes with sample paths in $\ell^\infty(\mathcal{F})$ and $E^* \|Z_1\|_{\mathcal{F}}^2 < \infty$. Let $\xi_1, \xi_2, \dots$ be i.i.d. random variables with mean zero, independent of $Z_1, Z_2, \dots$. Then there exists a constant K with

$$\limsup_{n \rightarrow \infty} E_\xi \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}}^* \leq K \limsup_{n \rightarrow \infty} E^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}}, \quad \text{a.s.}$$

Here the star on the left side denotes a measurable majorant with respect to the variables $(\xi_1, \dots, \xi_n, Z_1, \dots, Z_n)$ jointly.

[Hint: Without loss of generality, assume that $E^* \|n^{-1/2} \sum_{i=1}^n \xi_i Z_i\|_{\mathcal{F}} \leq 1/2$, for every n , and $E|\xi_1| = 1$. Since $E\xi = 0$, the random variables $E_\xi \left\| \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}}^*$ are nondecreasing in n by Jensen's inequality. Therefore, it suffices to consider the limsup along the subsequence indexed by 2^n . Since $E^* \|Z_1\|_{\mathcal{F}}^2 < \infty$,

$$\sum_{n=1}^{\infty} P\left(\max_{1 \leq i \leq 2^n} \|Z_i\|_{\mathcal{F}}^* > 2^{n/2}\right) \leq \sum_{n=1}^{\infty} 2^n P\left(\|Z_1\|_{\mathcal{F}}^* > 2^{n/2}\right) < \infty.$$

It follows by the Borel-Cantelli lemma that each variable $\|Z_i\|_{\mathcal{F}}^*$, for $1 \leq i \leq 2^n$, is bounded by $2^{n/2}$ eventually, almost surely. Consequently, if $\tilde{Z}_i = Z_i 1\{\|Z_i\|_{\mathcal{F}}^* \leq 2^{n/2}\}$, then the variables $E_\xi \left\| \sum_{i=1}^{2^n} \xi_i Z_i \right\|_{\mathcal{F}}^*$ and $E_\xi \left\| \sum_{i=1}^{2^n} \xi_i \tilde{Z}_i \right\|_{\mathcal{F}}^*$ are equal eventually, almost surely. (We abuse notation, since $\tilde{Z}_i$ depends on n .) It suffices to show that the random variable $\limsup 2^{-n/2} E_\xi \left\| \sum_{i=1}^{2^n} \xi_i \tilde{Z}_i \right\|_{\mathcal{F}}^*$ is bounded by some fixed number almost surely. By the Borel-Cantelli lemma, this is certainly the case if

$$\sum_{n=1}^{\infty} P\left(E_\xi \left\| \sum_{i=1}^{2^n} \xi_i \tilde{Z}_i \right\|_{\mathcal{F}}^* > 5 \cdot 2^{n/2}\right) < \infty.$$

The probabilities in this series can be replaced by their squares at the cost of decreasing 5 to 2. Indeed, the random variables $S_{k,l} = E_\xi \left\| \sum_{i=k}^l \xi_i \tilde{Z}_i \right\|_{\mathcal{F}}^*$ are monotonely decreasing in k and increasing in l . Let T be the first l such that $S_{1,l} > 2 \cdot 2^{n/2}$. Then $T \leq 2^n$ if and only if $S_{1,2^n} > 2 \cdot 2^{n/2}$. Furthermore, if $T = k$ and $S_{1,2^n} > 5 \cdot 2^{n/2}$, then $S_{k+1,2^n} > 2 \cdot 2^{n/2}$, because $S_{k,k} = \|\tilde{Z}_k\|_{\mathcal{F}}^* \leq 2^{n/2}$. Thus

$$\begin{aligned} P(S_{1,2^n} > 5 \cdot 2^{n/2}) &\leq \sum_{k=1}^{2^n} P(T = k, S_{1,2^n} > 5 \cdot 2^{n/2}) \leq \sum_{k=1}^{2^n} P(T = k, S_{k+1,2^n} > 2 \cdot 2^{n/2}) \\ &\leq \sum_{k=1}^{2^n} P(T = k) \max_{1 \leq k \leq 2^n} P(S_{k,2^n} > 2 \cdot 2^{n/2}) \leq P(S_{1,2^n} > 2 \cdot 2^{n/2})^2. \end{aligned}$$

In view of the contraction principle stated in Proposition A.1.11, we have the inequality $E^* \left\| \sum_{i=1}^{2^n} \xi_i \tilde{Z}_i \right\|_{\mathcal{F}} \leq E^* \left\| \sum_{i=1}^{2^n} \xi_i Z_i \right\|_{\mathcal{F}} \leq 2^{n/2}$. Conclude that it suffices to prove that

$$(2.9.12) \quad \sum_{n=1}^{\infty} P \left(E_\xi \left\| \sum_{i=1}^{2^n} \xi_i \tilde{Z}_i \right\|_{\mathcal{F}}^* - E^* \left\| \sum_{i=1}^{2^n} \xi_i \tilde{Z}_i \right\|_{\mathcal{F}} > 2^{n/2} \right)^2 < \infty.$$

Let E^i denote conditional expectation with respect to the σ -field generated by $Z_1, \dots, Z_i$. More precisely, let E^0 denote unconditional expectation, and if Z_i is the projection on the i th coordinate of the product measurable space $(\mathcal{X}^\infty \times \mathcal{Z}, \mathcal{A}^\infty \times \mathcal{C})$, let E^i denote conditional expectation given $\mathcal{A}^i \times \mathcal{X}^{\infty-i} \times \mathcal{C}$. Set

$$D_i = (E^i - E^{i-1}) \left(E_\xi \left\| \sum_{i=1}^{2^n} \xi_i \tilde{Z}_i \right\|_{\mathcal{F}}^* \right) = (E^i - E^{i-1})(C_i),$$

where

$$C_i = E_\xi \left\| \sum_{j=1}^{2^n} \xi_j \tilde{Z}_j \right\|_{\mathcal{F}}^* - E_\xi \left\| \sum_{j=1, j \neq i}^{2^n} \xi_j \tilde{Z}_j \right\|_{\mathcal{F}}^*.$$

The random variables inside the probabilities in (2.9.12) can be written as the (telescoping) sums $\sum_{i=1}^{2^n} D_i$. By the triangle inequality, $|D_i| \leq 2C_i \leq 2\|\tilde{Z}_i\|_{\mathcal{F}}^*$. Furthermore, in view of Problem 2.9.9, the expectation of C_i can be bounded by $EC_i \leq 2^{-n} E^* \left\| \sum_{j=1}^{2^n} \xi_j \tilde{Z}_j \right\|_{\mathcal{F}} \leq 2^{-n/2}$. The random variables $D_1, \dots, D_{2^n}$ are uncorrelated and have mean zero. By Chebyshev's inequality, (2.9.12) is bounded by

$$\sum_{n=1}^{\infty} \left(2^{-n} \sum_{i=1}^{2^n} ED_i^2 \right)^2 \leq \sum_{n=1}^{\infty} \max_{1 \leq i \leq 2^n} E|D_i|^3 E|D_i| \lesssim \sum_{n=1}^{\infty} 2^{-n/2} E^* \|\tilde{Z}_1\|_{\mathcal{F}}^3.$$

Substitute $E^* \|\tilde{Z}_1\|_{\mathcal{F}}^3 = \sum_{k=-\infty}^n E^* \|Z_1\|_{\mathcal{F}}^3 \{2^{(k-1)/2} < \|Z_1\|_{\mathcal{F}}^* \leq 2^{k/2}\}$; next bound one factor of $\|Z_1\|_{\mathcal{F}}^3$ by $2^{k/2}$; and, finally, change the order of summation to bound the previous display by

$$\sum_{k=-\infty}^{\infty} \sum_{n=k}^{\infty} E^* \|Z_1\|_{\mathcal{F}}^2 \{2^{(k-1)/2} < \|Z_1\|_{\mathcal{F}}^* \leq 2^{k/2}\} 2^{k/2-n/2}.$$

This equals $E^* \|Z_1\|_{\mathcal{F}}^2 \sqrt{2}/(\sqrt{2}-1)$. The proof is complete.]

8. (Alternative proof of Corollary 2.9.9) Let $Z_1, Z_2, \dots$ be i.i.d. stochastic processes with sample paths in $\ell^\infty(\mathcal{F})$. Let $\xi_1, \xi_2, \dots$ be i.i.d. random variables with mean zero, inde-

pendent of $Z_1, Z_2, \dots$. Suppose $M = \sup_n E^* \left\| n^{-1/2} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}} < \infty$. Then there exists a universal constant K with

$$P \left(\sup_{n \geq 1} E_{\xi}^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}}^* > \sqrt{2}(2M + 3t) \right) \leq K \left(\frac{2}{t^2} + \frac{M}{t^3} \right) E^* \|Z_1\|_{\mathcal{F}}^2,$$

for every $t > 0$. Here the star on the left side denotes a measurable majorant with respect to $(\xi_1, \dots, \xi_n, Z_1, \dots, Z_n)$ jointly.

[Hint: The supremum over all $n \geq 1$ inside the probability can be replaced by $\sqrt{2}$ times the supremum over the numbers 2^n . Set $\tilde{Z}_i = Z_i 1\{\|Z_i\|_{\mathcal{F}}^* \leq t 2^{n/2}\}$. The given probability is bounded by

$$\begin{aligned} & \sum_n P \left(\max_{1 \leq i \leq 2^n} \|Z_i\|_{\mathcal{F}}^* > t 2^{n/2} \right) + \sum_n P \left(E_{\xi}^* \left\| \sum_{i=1}^{2^n} \xi_i \tilde{Z}_i \right\|_{\mathcal{F}}^* > 2^{n/2} (2M + 3t) \right) \\ & \leq \sum_n 2^n P \left(\|Z_1\|_{\mathcal{F}}^* > t 2^{n/2} \right) + \sum_n P \left(E_{\xi}^* \left\| \sum_{i=1}^{2^n} \xi_i \tilde{Z}_i \right\|_{\mathcal{F}}^* > 2^{n/2} (M + t) \right)^2. \end{aligned}$$

Here the square probability is obtained by the same argument as before, but with T equal to the first l such that $S_{1,l} > 2^{n/2} (M + t)$. The proof can be finished as before.]

9. Given i.i.d. stochastic processes $Z_1, Z_2, \dots$ with sample paths in $\ell^\infty(\mathcal{F})$, define $S_n = \sum_{i=1}^n Z_i$. Then $E^* \|S_n\|_{\mathcal{F}} - E^* \|S_{n-1}\|_{\mathcal{F}} \leq n^{-1} E^* \|S_n\|_{\mathcal{F}}$.

[Hint: The inequality is equivalent to $E^* \|S_n/n\|_{\mathcal{F}} \leq E^* \|S_{n-1}/(n-1)\|_{\mathcal{F}}$. The sequence $\|S_n/n\|_{\mathcal{F}}^*$ forms a reversed submartingale with respect to the filtration $\mathcal{S}_n$ generated by the symmetric σ -fields, as in Lemma 2.4.6.]

10. If $\varphi: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is concave with $\varphi(t) \rightarrow \infty$ as $t \rightarrow \infty$ and $\psi: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is the least concave majorant of $\{(n, a_n \varphi(n))\}_{n=1}^\infty$, then $\psi(t)/\varphi(t) \rightarrow 0$ as $t \rightarrow \infty$.

11. (Alternative proof of Theorem 2.9.6) If $\mathcal{F}$ is P -Donsker and $\|\xi_1\|_{2,1} < \infty$, then the sequence $n^{-1/2} \sum_{i=1}^n \xi_i (\delta_{X_i} - P)$ converges in distribution to a tight limit in $\ell^\infty(\mathcal{F})$.

[Hint: Use Lemma 2.9.11 and Exercise 2.9.10 rather than Lemma 2.9.1 to control the modulus of continuity of the process.]

12. Let $\mathcal{F}$ be a uniformly bounded class of functions with either a uniform entropy integral or a bracketing integral of the order $\varepsilon^{1-\alpha/2}$, for some $0 < \alpha < 2$. Then for i.i.d. ξ_i with $\|\xi_i\|_{p,1} < \infty$ and $p \geq 1$, Lemma 2.9.11 gives the bound

$$E \sup_{f \in \mathcal{F}: Pf^2 < n^{-2/(2+\alpha)}} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i (f(X_i) - Pf) \right| \lesssim \begin{cases} n^{1/p-1/2}, & \text{if } p \leq 1 + 2/\alpha, \\ n^{(\alpha-2)/(4+2\alpha)}, & \text{if } p \geq 1 + 2/\alpha. \end{cases}$$

The right side is of smaller order than $\min_{1 \leq n_0 \leq n} (n_0 \cdot n^{1/p-1/2} + n_0^{(\alpha-2)/(4+2\alpha)}) = n^{(\alpha-2)(1/2-1/p)/(6+\alpha)}$, which is the bound obtainable from Lemma 2.9.1. This illustrates the improvement of Lemma 2.9.11 over Lemma 2.9.1. (The value $\delta_n = n^{-1/(2+\alpha)}$ satisfies the entropy integral equation $J(\delta_n, \mathcal{F}) \asymp \sqrt{n} \delta_n^2$ arising in Chapter 3.4.)

[Hint: For $\mathcal{F}_n = \{f \in \mathcal{F}: Pf^2 < n^{-2/(2+\alpha)}\}$ and $k \leq n$, we have $E \|\mathbb{G}_k\|_{\mathcal{F}_n} \leq E \|\mathbb{G}_k\|_{\mathcal{F}_k} \lesssim J(k^{-1/(2+\alpha)}, \mathcal{F})$. Therefore (2.9.10) is satisfied with $\psi(k, \mathcal{F})$ a multiple of $k^{\alpha/(2+\alpha)}$.]

2.10

Permanence under Transformation

In this chapter we consider a number of operations that preserve the Glivenko-Cantelli and Donsker properties and allow the formation of many new Donsker classes from given examples. For instance, unions, convex hulls, and certain closures of Donsker classes are Donsker.

The main results of this chapter concern pointwise transformations of sets of Glivenko-Cantelli classes and Donsker classes. For given classes $\mathcal{F}_1, \dots, \mathcal{F}_k$ of real functions defined on some set $\mathcal{X}$ and a fixed map $\phi: \mathbb{R}^k \rightarrow \mathbb{R}$, let $\phi \circ (\mathcal{F}_1, \dots, \mathcal{F}_k)$ denote the class of all functions

$$x \mapsto \phi \circ f(x) := \phi(f_1(x), \dots, f_k(x)),$$

as $f = (f_1, \dots, f_k)$ ranges over $\mathcal{F}_1 \times \dots \times \mathcal{F}_k$. In Section 2.10.2 we show that if $\mathcal{F}_1, \dots, \mathcal{F}_k$ are Glivenko-Cantelli, then $\phi \circ (\mathcal{F}_1, \dots, \mathcal{F}_k)$ is Glivenko-Cantelli as soon as ϕ is continuous. In Section 2.10.3 we show that the Donsker property is similarly preserved if ϕ is Lipschitz.

Section 2.10.4 covers the preservation of the uniform-entropy condition, and in Section 2.10.5 new Glivenko-Cantelli and Donsker classes are formed through union of sample spaces.

2.10.1 Closures and Convex Hulls

Given a class $\mathcal{F}$ of measurable functions, let $\tilde{\mathcal{F}}_r$ denote the set of all $f: \mathcal{X} \rightarrow \mathbb{R}$ for which there exists a sequence f_m in $\mathcal{F}$ with $f_m \rightarrow f$ both pointwise and in $L_r(P)$. Let $\text{sconv}_r \mathcal{F}$ denote the set of convex combinations $\sum_{i=1}^{\infty} \lambda_i f_i$ of functions f_i in $\mathcal{F}$ where $\sum |\lambda_i| \leq 1$ and the series converges both pointwise and in $L_r(P)$.[‡]

[‡] Pointwise convergence means convergence for every argument, not convergence almost surely.

2.10.1 Theorem.

- (i) If $\mathcal{F}$ is Glivenko-Cantelli and $\mathcal{G} \subset \mathcal{F}$, then $\mathcal{G}$ is Glivenko-Cantelli.
- (ii) If $\mathcal{F}$ is Donsker and $\mathcal{G} \subset \mathcal{F}$, then $\mathcal{G}$ is Donsker.

2.10.2 Theorem.

- (i) If $\mathcal{F}$ is Glivenko-Cantelli, then $\tilde{\mathcal{F}}_1$ is Glivenko-Cantelli.
- (ii) If $\mathcal{F}$ is Donsker, then $\tilde{\mathcal{F}}_2$ is Donsker.

2.10.3 Theorem.

- (i) If $\mathcal{F}$ is Glivenko-Cantelli, then $\text{sconv}_1 \mathcal{F}$ is Glivenko-Cantelli.
- (ii) If $\mathcal{F}$ is Donsker, then $\text{sconv}_2 \mathcal{F}$ is Donsker.

Proofs. The preservation of the Glivenko-Cantelli property is immediate from the definition, in all three cases. In every case the supremum over the new class is equal to the supremum over the original class. For the Donsker property the first two results are consequences of the characterization of weak convergence in $\ell^\infty(\mathcal{F})$ as marginal convergence plus asymptotic equicontinuity. In both cases the modulus of continuity does not increase when passing from $\mathcal{F}$ to the new class.

For the Donsker property in the third result, let $\{\psi_i\}$ be an orthonormal base of a subspace of $L_2(P)$ that contains $\mathcal{F}$, and let $Z_1, Z_2, \dots$ be i.i.d. standard normally distributed random variables. Since $\mathcal{F}$ is pre-Gaussian, the series $\sum_{i=1}^\infty (P(f - Pf)\psi_i)Z_i$ is uniformly convergent in $\ell^\infty(\mathcal{F})$ and represents a tight Brownian bridge process (Problem 2.10.1). Thus the sequence of partial sums $\mathbb{G}_k = \sum_{i=1}^k (P(f - Pf)\psi_i)Z_i$ satisfies

$$\|\mathbb{G}_k - \mathbb{G}_l\|_{\text{sconv } \mathcal{F}} = \|\mathbb{G}_k - \mathbb{G}_l\|_{\mathcal{F}} \rightarrow 0, \quad \text{a.s.,}$$

as $k, l \rightarrow \infty$. It follows that the sequence of partial sums forms a Cauchy sequence in $\ell^\infty(\text{sconv } \mathcal{F})$ almost surely and hence converges almost surely to a limit $\mathbb{G}$. Since each of the partial sums is linear and ρ_P -uniformly continuous on $\text{sconv } \mathcal{F}$, so is the limit $\mathbb{G}$, with probability 1. This proves that the symmetric convex hull of $\mathcal{F}$ is pre-Gaussian.

According to the almost sure representation theorem, Theorem 1.10.4 and its addendum, there exists a probability space $(\tilde{\Omega}, \mathcal{U}, P)$ and perfect maps $\phi_n: \tilde{\Omega} \rightarrow \mathcal{X}^n$ and $\phi: \tilde{\Omega} \rightarrow \Omega$, for Ω the space where $\mathbb{G}(\omega) = (\mathbb{G}(f, \omega): f \in \mathcal{F})$ is defined, such that

$$\left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(f(\phi_n(\tilde{\omega})_i) - Pf \right) - \mathbb{G}(f, \phi(\tilde{\omega})) \right\|_{\mathcal{F}} \xrightarrow{\text{as}^*} 0.$$

The norm does not increase if $\mathcal{F}$ is replaced by $\text{sconv } \mathcal{F}$. Thus there exists a version of the empirical process that converges outer almost surely in $\ell^\infty(\text{sconv } \mathcal{F})$ to a process with uniformly continuous sample paths. This implies that $\text{sconv } \mathcal{F}$ is Donsker. (Note that by perfectness of ϕ_n , $\text{Eh}^*(\{\mathbb{G}_n f: f \in \text{sconv } \mathcal{F}\}) = (P^n)^* h(\{\mathbb{G}_n f: f \in \text{sconv } \mathcal{F}\})$ for every function h on $\ell^\infty(\text{sconv } \mathcal{F})$.) ■

2.10.4 Example. The class of all functions $x \mapsto \mu(-\infty, x]$ with μ ranging over all signed measures on $\mathbb{R}^k$ with total variation bounded by 1 is universally Donsker.

This can be deduced by applying the preceding results several times. The given class is in the convex hull of the set of cumulative distribution functions of probability measures. In view of the classical Glivenko-Cantelli theorem, any cumulative distribution function is the uniform limit of a sequence of finitely discrete cumulative distribution functions. The class of functions $x \mapsto \sum_i p_i 1\{t_i \leq x\}$, with $p_i \geq 0$ and $\sum p_i = 1$, is in the convex hull of the class of indicator functions of cells $[t, \infty)$. This is Vapnik-Červonenkis and suitably measurable, hence universally Donsker.

2.10.2 Transformations of Glivenko-Cantelli Classes

For given classes $\mathcal{F}_1, \dots, \mathcal{F}_k$ of functions and given $\phi: \mathbb{R}^k \rightarrow \mathbb{R}$ define the class of functions $\phi \circ (\mathcal{F}_1, \dots, \mathcal{F}_k)$ as indicated in the introduction of the chapter.

2.10.5 Theorem (Permanence Glivenko-Cantelli). *Let $\mathcal{F}_1, \dots, \mathcal{F}_k$ be P -Glivenko-Cantelli classes with integrable envelopes. If $\phi: \mathbb{R}^k \rightarrow \mathbb{R}$ is continuous, then the class $\phi \circ (\mathcal{F}_1, \dots, \mathcal{F}_k)$ is P -Glivenko-Cantelli provided that it has an integrable envelope function.*

Proof. We first assume that the classes of functions $\mathcal{F}_i$ are appropriately measurable. Let $F_1, \dots, F_k$ and H be integrable envelopes for the classes $\mathcal{F}_1, \dots, \mathcal{F}_k$ and $\mathcal{H} := \phi \circ (\mathcal{F}_1, \dots, \mathcal{F}_k)$, respectively, and set $F = F_1 \vee \dots \vee F_k$.

For $M \in (0, \infty)$ define $\mathcal{H}_M$ to be the class of functions $\phi(f)1_{F \leq M}$, where $f = (f_1, \dots, f_k)$ ranges over $\mathcal{F}_1 \times \dots \times \mathcal{F}_k$. Then

$$\|(\mathbb{P}_n - P)\phi(f)\|_{\mathcal{F}} \leq (\mathbb{P}_n + P)H1_{[F > M]} + \|(\mathbb{P}_n - P)h\|_{\mathcal{H}_M}.$$

Because the expectation of the first term on the right converges to 0 as $M \rightarrow \infty$, it suffices to show that the class $\mathcal{H}_M$ is P -Glivenko-Cantelli for every fixed M . By Lemma 2.10.6 below, applied to the function $\phi: [-M, M]^k \rightarrow \mathbb{R}$ and $\|\cdot\|$ the L_1 -norm on $\mathbb{R}^k$, there exists for every $\varepsilon > 0$ a $\delta > 0$, depending only on ε , ϕ and M , such that for any pairs $(f_j, g_j) \in \mathcal{F}_j$, the inequality

$$\mathbb{P}_n |f_j - g_j| 1_{[F_j \leq M]} \leq \frac{\delta}{k}, \quad j = 1, \dots, k$$

implies that

$$\mathbb{P}_n |\phi(f_1, \dots, f_k) - \phi(g_1, \dots, g_k)| 1_{F \leq M} \leq \varepsilon.$$

We conclude that

$$N(\varepsilon, \mathcal{H}_M, L_1(\mathbb{P}_n)) \leq \prod_{j=1}^k N\left(\frac{\delta}{k}, \mathcal{F}_j 1_{F_j \leq M}, L_1(\mathbb{P}_n)\right).$$

By the converse part of Theorem 2.4.3 the outer expectation of the logarithm of each of the terms in the product on the right is of order $o(n)$. It follows that $E^* \log N(\varepsilon, \mathcal{H}_M, L_1(\mathbb{P}_n)) = o(n)$, for every $\varepsilon > 0$ and $M > 0$. By Theorem 2.4.3 in the other direction the class $\phi(\mathcal{F})$ is P -Glivenko-Cantelli.

The preceding argument concludes the proof for classes $\mathcal{F}_1, \dots, \mathcal{F}_k$ that together with $\phi \circ (\mathcal{F}_1, \dots, \mathcal{F}_k)$ satisfy the measurability assumptions of Theorem 2.4.3. We extend the theorem to general Glivenko-Cantelli classes using separable versions of these classes, as defined in Section 2.3.3. As shown in the preceding argument, it is not a loss of generality to assume that the classes $\mathcal{F}_i$ are uniformly bounded. Furthermore, it suffices to show that the class $\phi \circ (\mathcal{F}_1, \dots, \mathcal{F}_k)$ is weak Glivenko-Cantelli.

Because a Glivenko-Cantelli class with integrable envelope is totally bounded in $L_1(P)$ (cf. Problem 2.4.11), it is separable as a subset of $L_1(P)$. Therefore, by Theorem 2.3.16 every class $\mathcal{F}_i$ admits a separable modification $\tilde{\mathcal{F}}_i$ relative to the $L_1(P)$ -norm, and by Theorem 2.3.16 the class $\mathcal{F}_i$ is Glivenko-Cantelli if and only if $\tilde{\mathcal{F}}_i$ is Glivenko-Cantelli, and $\sup_{f \in \mathcal{F}_i} \mathbb{P}_n |f - \tilde{f}| \rightarrow 0$ in outer probability. The classes $\tilde{\mathcal{F}}_i$ possess enough measurability to make the preceding argument work. Thus we conclude that the class $\phi \circ (\tilde{\mathcal{F}}_1, \dots, \tilde{\mathcal{F}}_k)$ is Glivenko-Cantelli. Finally we invoke Lemma 2.10.6 to see that convergence to zero of $\sup_{f \in \mathcal{F}_i} \mathbb{P}_n |f - \tilde{f}| \rightarrow 0$ for $j = 1, \dots, k$, implies convergence to zero of $\sup_{f \in \mathcal{F}} \mathbb{P}_n |\phi \circ (f_1, \dots, f_k) - \phi(\tilde{f}_1, \dots, \tilde{f}_k)|$. The theorem follows in the general case by application of Theorem 2.3.16 in the other direction. ■

The following lemma is the crux of the proof of Theorem 2.10.5. Let $\|\cdot\|$ be any norm on $\mathbb{R}^k$.

2.10.6 Lemma. *Suppose that $K \subset \mathbb{R}^k$ is compact and $\phi: K \rightarrow \mathbb{R}$ is continuous. Then for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all n and for all $a_1, \dots, a_n, b_1, \dots, b_n \in K$ the inequality $n^{-1} \sum_{i=1}^n \|a_i - b_i\| < \delta$ implies that $n^{-1} \sum_{i=1}^n |\phi(a_i) - \phi(b_i)| < \varepsilon$.*

Proof. Letting U_n be a random variable with the uniform distribution on $\{1, \dots, n\}$, and setting $X_n = a_{U_n}$ and $Y_n = b_{U_n}$, we can represent the two discrepancies in the lemma as $n^{-1} \sum_{i=1}^n \|a_i - b_i\| = \mathbb{E} \|X_n - Y_n\|$ and $n^{-1} \sum_{i=1}^n |\phi(a_i) - \phi(b_i)| = \mathbb{E} |\phi(X_n) - \phi(Y_n)|$. Therefore, it suffices to show that for every $\varepsilon > 0$ there exists $\delta > 0$ such that for random vectors (X, Y) taking values in $K \times K$,

$$\mathbb{E} \|X - Y\| < \delta \quad \text{implies} \quad \mathbb{E} |\phi(X) - \phi(Y)| < \varepsilon.$$

If this were not the case, then for some $\varepsilon > 0$ and for all $m = 1, 2, \dots$ there would exist (X_m, Y_m) such that $\mathbb{E} \|X_m - Y_m\| < m^{-1}$ and $\mathbb{E} |\phi(X_m) - \phi(Y_m)| \geq \varepsilon$. By Prohorov's theorem there would exist a subsequence $(X_{m'}, Y_{m'})$ that converges in distribution to a limit (X, Y) . Then, by dominated convergence, both $\mathbb{E} \|X - Y\| = \lim_{m' \rightarrow \infty} \mathbb{E} \|X_{m'} - Y_{m'}\| = 0$, and also $\mathbb{E} |\phi(X) - \phi(Y)| = \lim_{m' \rightarrow \infty} \mathbb{E} |\phi(X_{m'}) - \phi(Y_{m'})| \geq \varepsilon > 0$. This yields a contradiction, and hence the implication in the preceding display is valid. ■

2.10.3 Transformations of Donsker Classes

For given classes $\mathcal{F}_1, \dots, \mathcal{F}_k$ of functions and given $\phi: \mathbb{R}^k \rightarrow \mathbb{R}$ define the class of functions $\phi \circ (\mathcal{F}_1, \dots, \mathcal{F}_k)$ as indicated in the introduction of the chapter. It will be shown that the Donsker property is preserved if ϕ satisfies

$$(2.10.7) \quad |\phi \circ f(x) - \phi \circ g(x)|^2 \leq \sum_{l=1}^k (f_l(x) - g_l(x))^2,$$

for every $f, g \in \mathcal{F}_1 \times \dots \times \mathcal{F}_k$ and x . This is certainly the case for Lipschitz functions ϕ .

2.10.8 Theorem (Permanence Donsker). *Let $\mathcal{F}_1, \dots, \mathcal{F}_k$ be Donsker classes with $\|P\|_{\mathcal{F}_i} < \infty$ for each i . Let $\phi: \mathbb{R}^k \rightarrow \mathbb{R}$ satisfy (2.10.7). Then the class $\phi \circ (\mathcal{F}_1, \dots, \mathcal{F}_k)$ is Donsker, provided $\phi \circ (f_1, \dots, f_k)$ is square integrable for at least one $(f_1, \dots, f_k)$.*

2.10.9 Example. If $\mathcal{F}$ and $\mathcal{G}$ are Donsker classes and $\|P\|_{\mathcal{F} \cup \mathcal{G}} < \infty$, then the pairwise infima $\mathcal{F} \wedge \mathcal{G}$, the pairwise suprema $\mathcal{F} \vee \mathcal{G}$, and pairwise sums $\mathcal{F} + \mathcal{G}$ are Donsker classes.

Since $\mathcal{F} \cup \mathcal{G}$ is contained in the sum of $\mathcal{F} \cup \{0\}$ and $\mathcal{G} \cup \{0\}$, the union of $\mathcal{F}$ and $\mathcal{G}$ is Donsker as well.

2.10.10 Example. If $\mathcal{F}$ and $\mathcal{G}$ are uniformly bounded Donsker classes, then the pairwise products $\mathcal{F} \cdot \mathcal{G}$ form a Donsker class. The function $\phi(f, g) = fg$ is Lipschitz on bounded subsets of $\mathbb{R}^2$, but not on the whole plane. The condition that $\mathcal{F}$ and $\mathcal{G}$ be uniformly bounded cannot be omitted in general. (In fact, the envelope function of the product of two Donsker classes need not be weak L_2 .)

2.10.11 Example. If $\mathcal{F}$ is Donsker with $\|P\|_{\mathcal{F}} < \infty$ and $f \geq \delta$ for some constant $\delta > 0$ for every $f \in \mathcal{F}$, then $1/\mathcal{F} = \{1/f: f \in \mathcal{F}\}$ is Donsker.

2.10.12 Example. If $\mathcal{F}$ is a Donsker class with $\|P\|_{\mathcal{F}} < \infty$ and g , a uniformly bounded, measurable function, then $\mathcal{F} \cdot g$ is Donsker. Although the function $\phi(f, g) = fg$ is not Lipschitz on the domain of interest, we have

$$\left| \phi(f_1(x), g(x)) - \phi(f_2(x), g(x)) \right| \leq \|g\|_{\infty} |f_1(x) - f_2(x)|,$$

for all x . Thus, condition (2.10.7) is satisfied and the theorem applies.

2.10.13 Example. If $\mathcal{F}$ is Donsker with integrable envelope function F , then the class $\mathcal{F}_{\leq M} = \{f1\{F \leq M'\}: f \in \mathcal{F}, M' \leq M\}$ is Donsker. Indeed, the class of sets $\{F \leq M\}$ is linearly ordered by inclusion, so that it is VC. Furthermore, the function $\phi(f, g) = fg$ is Lipschitz on the set $\{(f, 0): f \in \mathbb{R}\} \cup \{(f, 1): |f| \leq M\}$.

Consequently, the class $\mathcal{F}^M = \{f1\{F > M\}: f \in \mathcal{F}\}$ is Donsker because it is contained in $\mathcal{F} - \mathcal{F}_{\leq M}$.

Many transformations of interest are Lipschitz, but not uniformly so. Consider a map $\phi: \mathbb{R}^k \rightarrow \mathbb{R}$ such that

$$(2.10.14) \quad |\phi \circ f(x) - \phi \circ g(x)|^2 \leq \sum_{l=1}^k L_{\alpha,l}^2(x) (f_l(x) - g_l(x))^2,$$

for given measurable functions $L_{\alpha,1}, \dots, L_{\alpha,k}$ and every $f, g \in \mathcal{F}_1 \times \dots \times \mathcal{F}_k$ and x . Then the class $\phi \circ (\mathcal{F}_1, \dots, \mathcal{F}_k)$ is Donsker if each of the classes $L_{\alpha,i}\mathcal{F}_i$, consisting of the functions $x \mapsto L_{\alpha,i}(x)f(x)$ with f ranging over $\mathcal{F}_i$, is Donsker. By Example 2.10.12, the class $g\mathcal{F}$ is Donsker whenever g is bounded and $\mathcal{F}$ is Donsker, but this is not useful to generalize the preceding theorem. Typically, it is possible to relax the requirement that g is bounded at the cost of more stringent conditions on $\mathcal{F}$. For instance, in the next subsection it is shown that $g\mathcal{F}$ is Donsker for every combination of a Donsker class $\mathcal{F}$ that satisfies the uniform entropy condition and for a function g such that $Pg^2F^2 < \infty$.

2.10.15 Corollary. *Let $\phi: \mathbb{R}^k \rightarrow \mathbb{R}$ satisfy (2.10.14). Let each of the classes $L_{\alpha,i}\mathcal{F}_i$ be Donsker with $\|P\|_{L_{\alpha,i}\mathcal{F}_i} < \infty$. Then the class $\phi \circ (\mathcal{F}_1, \dots, \mathcal{F}_k)$ is Donsker, provided $\phi \circ (f_1, \dots, f_k)$ is square integrable for at least one $(f_1, \dots, f_k)$.*

Proof. Without loss of generality, assume that each of the functions $L_{\alpha,i}$ is positive. For $f \in L_{\alpha}\mathcal{F} = L_{\alpha,1}\mathcal{F}_1 \times \dots \times L_{\alpha,k}\mathcal{F}_k$, define $\psi(f) = \phi(f/L_{\alpha})$. Then ψ is uniformly Lipschitz in the sense of (2.10.7). By the preceding theorem, $\psi(L_{\alpha}\mathcal{F}) = \phi(\mathcal{F})$ is Donsker. ■

The main tool in the proof of Theorem 2.10.8 is “Gaussianization.” Let $\xi_1, \xi_2, \dots$ be i.i.d. random variables with a standard normal distribution independent of $X_1, X_2, \dots$. By Chapter 2.9, the (conditional as well as unconditional) asymptotic behavior of the empirical process is related to the behavior of the processes

$$\mathbb{Z}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i \delta_{X_i}.$$

Given fixed values $X_1, \dots, X_n$, the process $\{\mathbb{Z}_n(f): f \in L_2(P)\}$ is Gaussian with zero mean and standard deviation metric

$$\sigma_{\xi}(\mathbb{Z}_n(f) - \mathbb{Z}_n(g)) = \left(\frac{1}{n} \sum_{i=1}^n (f(X_i) - g(X_i))^2 \right)^{1/2}$$

equal to the $L_2(\mathbb{P}_n)$ semimetric. This observation permits the use of several comparison principles for Gaussian processes, including Slepian’s lemma, Proposition A.2.5

Three lemmas precede the proof of Theorem 2.10.8. The first is a comparison principle of independent interest.

2.10.16 Lemma. Let $\mathcal{F}$ be a Donsker class with $\|P\|_{\mathcal{F}} < \infty$. Then the class $\mathcal{F}^2 = \{f^2: f \in \mathcal{F}\}$ is Glivenko-Cantelli in probability: $\|\mathbb{P}_n - P\|_{\mathcal{F}^2}^* \xrightarrow{P} 0$. If, in addition, $P^*F^2 < \infty$ for some envelope function F , then $\mathcal{F}^2$ is also Glivenko-Cantelli almost surely and in mean.

Proof. Let $\xi_1, \xi_2, \dots$ be i.i.d. random variables with a standard normal distribution independent of $X_1, X_2, \dots$. Given fixed values $X_1, \dots, X_n$ the stochastic process $\mathbb{Z}_n(f) = n^{-1/2} \sum_{i=1}^n \xi_i f(X_i)$ is a Gaussian process with zero mean and variance $\text{var}_{\xi}(\mathbb{Z}_n(f)) = \mathbb{P}_n f^2$. Thus

$$\|\mathbb{P}_n f^2\|_{\mathcal{F}}^{1/2} = \sqrt{\frac{\pi}{2}} \left\| \mathbb{E}_{\xi} |\mathbb{Z}_n(f)| \right\|_{\mathcal{F}} \leq \sqrt{\frac{\pi}{2}} \mathbb{E}_{\xi} \|\mathbb{Z}_n\|_{\mathcal{F}}.$$

Since $\mathcal{F}$ is Donsker and $\|P\|_{\mathcal{F}} < \infty$, the sequence $\mathbb{Z}_n$ converges (unconditionally) to a tight Gaussian process $\mathbb{Z}$ by the unconditional multiplier theorem, Theorem 2.9.2. By Lemma 2.3.11, this implies that $\mathbb{E}^* \|\mathbb{Z}_n\|_{\mathcal{F}} = O(1)$. Conclude that for every $\varepsilon > 0$ there exist constants M and N such that with inner probability at least $1 - \varepsilon$: $\mathbb{P}_n f^2 \leq M$ for every $f \in \mathcal{F}$ and every $n \geq N$. Since $\mathcal{F}$ is bounded in $L_2(P)$, one can also ensure that $P f^2 \leq M$ for every $f \in \mathcal{F}$.

For fixed M and any $\sigma^2, \tau^2 \leq M$, the bounded Lipschitz distance between the normal distributions $N(0, \sigma^2)$ and $N(0, \tau^2)$ is bounded below by $K |\sigma^2 - \tau^2|$ for a constant K depending on M only (Problem 2.10.2). Conclude that with inner probability at least $1 - \varepsilon$ and $n \geq N$,

$$K \left| \mathbb{P}_n f^2 - P f^2 \right| \leq \sup_{h \in BL_1(\mathbb{R})} \left| \mathbb{E}_{\xi} h(\mathbb{Z}_n(f)) - \mathbb{E} h(\mathbb{Z}(f)) \right|.$$

The right side is bounded by the bounded Lipschitz distance between the processes $\mathbb{Z}_n$ and $\mathbb{Z}$. Conclude that with inner probability at least $1 - \varepsilon$,

$$K \left\| \mathbb{P}_n f^2 - P f^2 \right\|_{\mathcal{F}} \leq \sup_{h \in BL_1} \left| \mathbb{E}_{\xi} h(\mathbb{Z}_n) - \mathbb{E} h(\mathbb{Z}) \right|.$$

The right side converges to zero in outer probability by the conditional multiplier theorem, Theorem 2.9.6.

Under the condition $P^*F^2 < \infty$, the submartingale argument in the proof of Theorem 2.4.3 (based on Lemma 2.4.6) applies and shows that the convergence in probability can be strengthened to convergence in mean and almost surely. ■

2.10.17 Lemma. If $\mathcal{F}$ is pre-Gaussian, then $\varepsilon^2 \log N(\varepsilon, \mathcal{F}, \rho_P) \rightarrow 0$ as $\varepsilon \rightarrow 0$. If, in addition, $\|P\|_{\mathcal{F}} < \infty$, then also $\varepsilon^2 \log N(\varepsilon, \mathcal{F}, L_2(P)) \rightarrow 0$.

Proof. For any partition $\mathcal{F} = \cup_{i=1}^m \mathcal{F}_i$ and $\varepsilon > 0$, one has $N(\varepsilon, \mathcal{F}, \rho_P) \leq \sum_{i=1}^m N(\varepsilon, \mathcal{F}_i, \rho_P)$. Bound the sum by m times the maximum of its terms, and take logarithms to obtain

$$\varepsilon \sqrt{\log N(\varepsilon, \mathcal{F}, \rho_P)} \leq \varepsilon \sqrt{\log m} + \varepsilon \max_{i \leq m} \sqrt{\log N(\varepsilon, \mathcal{F}_i, \rho_P)}.$$

For fixed $\delta > 0$, choose a partition into $m = N(\delta, \mathcal{F}, \rho_P)$ sets of diameter at most δ . Take an arbitrary element f_i from each partitioning set. Let $\mathbb{G}$ be a separable Brownian bridge process. According to Sudakov's inequality A.2.6, $\varepsilon \sqrt{\log N(\varepsilon, \mathcal{G}, \rho_P)} \leq 3E^* \|\mathbb{G}\|_{\mathcal{G}}$ for any set of functions $\mathcal{G}$ and ε . Since $N(\varepsilon, \mathcal{F}_i, \rho_P) = N(\varepsilon, \mathcal{F}_i - f_i, \rho_P)$, the preceding display can be bounded by

$$\frac{3\varepsilon}{\delta} E^* \|\mathbb{G}\|_{\mathcal{F}} + 3 \max_{i \leq m} E^* \|\mathbb{G}\|_{\mathcal{F}_i - f_i} \leq \frac{3\varepsilon}{\delta} E^* \|\mathbb{G}\|_{\mathcal{F}} + 3E^* \|\mathbb{G}\|_{\mathcal{F}_\delta},$$

where $\mathcal{F}_\delta = \{f - g: \rho_P(f - g) < \delta, f, g \in \mathcal{F}\}$. Since $\mathcal{F}$ is pre-Gaussian, $\mathbb{G}$ can be chosen bounded and uniformly continuous. This implies that $E^* \|\mathbb{G}\|_{\mathcal{F}} < \infty$ and $E^* \|\mathbb{G}\|_{\mathcal{F}_\delta} \rightarrow 0$ as $\delta \rightarrow 0$. Thus, the first term on the right is finite and converges to zero as $\varepsilon \downarrow 0$. The second term can be made arbitrarily small by choosing δ sufficiently small.

The second assertion of the lemma is proved in a similar way, now using a Brownian motion process $\mathbb{G} + \xi P$ (where ξ is standard normally distributed and is independent of $\mathbb{G}$) instead of $\mathbb{G}$ (cf. Problem 2.1.4). ■

2.10.18 Lemma. *Let $Z_1, \dots, Z_m$ be separable, mean-zero Gaussian processes indexed by arbitrary sets T_i . Then*

$$E \max_{1 \leq i \leq m} \|Z_i\|_{T_i} \leq K \left(\max_{1 \leq i \leq m} E \|Z_i\|_{T_i} + \sqrt{\log m} \max_{1 \leq i \leq m} \|\sigma(Z_{i,t})\|_{T_i} \right),$$

for a universal constant K .

Proof. Without loss of generality, assume that $m \geq 2$. Set $\sigma_i = \|\sigma(Z_{i,t})\|_{T_i}$ and $Y_i = \|Z_{i,t}\|_{T_i}$. By Borell's inequality A.2.1,

$$(2.10.19) \quad P(|Y_i - M_i| > x) \leq e^{-\frac{1}{2}x^2/\sigma_i^2},$$

where M_i is a median of the random variable Y_i . By Lemma 2.2.1, the variable $Y_i - M_i$ has Orlicz norm $\|Y_i - M_i\|_{\psi_2}$ bounded by a multiple of σ_i . In view of the triangle inequality,

$$E \max_i \|Z_i\|_{T_i} \leq E \max_i |Y_i - M_i| + \max_i M_i \lesssim \sqrt{\log m} \max_i \sigma_i + \max_i M_i,$$

by Lemma 2.2.2. Integration of (2.10.19) (or another application of the inequality $\|Y_i - M_i\|_{\psi_2} \lesssim \sigma_i$) yields $E|Y_i - M_i| \lesssim \sigma_i$. The lemma follows upon using this inequality to bound $\max_i M_i$ in the last displayed equation. ■

Proof of Theorem 2.10.8. The square integrability of $\phi \circ (f_1, \dots, f_k)$ for one element of $\mathcal{F}_1 \times \dots \times \mathcal{F}_k$ and the Lipschitz condition imply square integrability of every function of this type. This ensures marginal convergence.

Let $\xi_1, \xi_2, \dots, \xi_n$ be i.i.d. random variables with a standard normal distribution, independent of $X_1, \dots, X_n$. Given fixed values $X_1, \dots, X_n$, the process

$$H_n(f) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i \phi \circ f(X_i), \quad f = (f_1, \dots, f_k),$$

is zero-mean Gaussian with “intrinsic” semimetric $\sigma_\xi(H_n(f) - H_n(g))$ equal to the $L_2(\mathbb{P}_n)$ semimetric $\|\phi \circ f - \phi \circ g\|_n$. Because each $\mathcal{F}_l$ is Donsker and each $\|P\|_{\mathcal{F}_l} < \infty$, the set $\mathcal{F} = \mathcal{F}_1 \times \cdots \times \mathcal{F}_k$ is totally bounded in the product space $L_2(P)^k$. For given $\delta > 0$, let $\mathcal{F} = \cup_{j=1}^m \mathcal{H}_j$ be a partition of $\mathcal{F}$ into sets of diameter less than δ with respect to the supremum product metric. The number of sets in the partition can be chosen as $m \leq \prod_{l=1}^k N(\delta, \mathcal{F}_l, L_2(P))$. Choose an arbitrary element h_j from each $\mathcal{H}_j$. Given the same $X_1, \dots, X_n$ as before, the processes $\{H_n(f) - H_n(h_j) : f \in \mathcal{H}_j\}$ possess the same intrinsic semimetric as H_n . By the preceding lemma,

$$(2.10.20) \quad \begin{aligned} E_\xi \sup_j \|H_n - H_n(h_j)\|_{\mathcal{H}_j} &\lesssim \max_{j \leq m} E_\xi \|H_n - H_n(h_j)\|_{\mathcal{H}_j} \\ &\quad + \sqrt{\log m} \max_{j \leq m} \sup_{f \in \mathcal{H}_j} \|\phi \circ f - \phi \circ h_j\|_n. \end{aligned}$$

For each l , let $\xi_{l1}, \dots, \xi_{ln}$ be i.i.d. standard normal variables, chosen independently for different l . Define $G_n(f) = \sum_{l=1}^k n^{-1/2} \sum_{i=1}^n \xi_{li} f_l(X_i)$. In view of (2.10.7),

$$\text{var}_\xi(H_n(f) - H_n(g)) \leq \sum_{l=1}^k \frac{1}{n} \sum_{i=1}^n (f_l - g_l)^2(X_i) = \text{var}_\xi(G_n(f) - G_n(g)).$$

By Slepian’s lemma A.2.5, the expectation $E_\xi \|H_n - H_n(h_j)\|_{\mathcal{H}_j}$ is bounded above by two times the same expression, but with G_n instead of H_n . Make this substitution in the right side of (2.10.20), and conclude that the left side of (2.10.20) is bounded up to a constant by

$$E_\xi \sup_{e_P(f,h) < \delta} |G_n(f) - G_n(h)| + \sqrt{\log m} \sup_{e_P(f,h) < \delta} \|f - h\|_n.$$

(Here e_P and $\|f - g\|_n$ denote product metrics corresponding to $\|\cdot\|_{P,2}$ and $\|\cdot\|_n$.) Since each of the classes $\mathcal{F}_l$ is Donsker and satisfies $\|P\|_{\mathcal{F}_l} < \infty$, the sequence G_n is asymptotically equicontinuous with respect to e_P . (See Theorem 2.9.2 and Problem 2.1.2.) Hence the outer expectation of the first term converges to zero as $n \rightarrow \infty$ followed by $\delta \downarrow 0$. For the second term, note that the classes $\mathcal{F}_l - \mathcal{F}_l$ are Donsker by Theorem 2.10.3, so that their squares $(\mathcal{F}_l - \mathcal{F}_l)^2$ are Glivenko-Cantelli in probability, by Lemma 2.10.16. Thus, as $n \rightarrow \infty$,

$$\begin{aligned} \sqrt{\log m} \sup_{e_P(f,h) < \delta} \|f - h\|_n &\xrightarrow{P^*} \sqrt{\log m} \sup_{e_P(f,h) < \delta} e_P(f, h) \\ &\leq \sum_{l=1}^k \sqrt{\log N(\delta, \mathcal{F}_l, L_2(P))} \delta. \end{aligned}$$

This converges to zero as $\delta \downarrow 0$, by Lemma 2.10.17. Combination of the preceding displays gives that the left side of (2.10.20) converges to zero in probability as $n \rightarrow \infty$ and next as $\delta \downarrow 0$.

Suppose that the processes H_n are separable, so that the variables $\|H_n - H_n(h_j)\|_{\mathcal{H}_j}$ are measurable. Then it can be concluded that for all positive numbers ε and η there exists a finite partition $\mathcal{F} = \cup_{j=1}^m \mathcal{H}_j$ such that

$$\limsup_{n \rightarrow \infty} P\left(\sup_j \|H_n - H_n(h_j)\|_{\mathcal{H}_j} > \varepsilon\right) < \eta.$$

This implies that the sequence H_n is asymptotically tight in $\ell^\infty(\mathcal{F})$, by Theorem 1.5.6.

Define a map $T: \phi \circ \mathcal{F} \rightarrow \mathcal{F}$ by assigning to each function $g \in \phi \circ \mathcal{F}$ a unique (but arbitrary) vector $f = Tg \in \mathcal{F}$ such that $\phi(f) = g$. The map $z \mapsto z \circ T$ from $\ell^\infty(\mathcal{F})$ to $\ell^\infty(\phi \circ \mathcal{F})$ is (Lipschitz) continuous. Hence, by the continuous mapping theorem, the sequence of multiplier empirical processes $\mathbb{G}'_n = H_n \circ T$ converges weakly to a tight limit in $\ell^\infty(\phi \circ \mathcal{F})$.

Since $\|P\|_{\phi \circ \mathcal{F}} < \infty$, the sequence of processes $n^{-1/2} \sum_{i=1}^n \xi_i (\delta_{X_i} - P)$ converges weakly to a tight limit as well. By Theorem 2.9.2, the same is true for the sequence of empirical processes $\mathbb{G}_n$.

Finally, we remove the assumption that the processes H_n are separable. According to Chapter 2.3.3, there exist pointwise-separable versions $\tilde{\mathcal{F}}_l$ of the classes $\mathcal{F}_l$. Almost every value $\tilde{f}_l(x)$ is the limit of a sequence $\tilde{g}_{m,l}(x)$ with each $\tilde{g}_{m,l}$ contained in a countable “separant” of $\tilde{\mathcal{F}}_l$. Each element $\tilde{g}_l$ of the separant is almost surely equal to an element g_l of the original class $\mathcal{F}_l$. It follows that, perhaps possibly on a null set of x , the function ϕ is defined at each vector $\tilde{f}(x)$ or can be extended to this vector by continuity. The extended function satisfies (2.10.7) on $\tilde{\mathcal{F}} \cup \mathcal{F}$ except perhaps for x in a fixed null set. Then the class of functions $\phi \circ \tilde{\mathcal{F}}$ is (essentially) well defined and is a pointwise-separable version of $\phi \circ \mathcal{F}$. By the preceding argument, $\phi \circ \tilde{\mathcal{F}}$ is Donsker. By (2.10.7),

$$\sqrt{n} \|\mathbb{P}_n \phi(f) - \mathbb{P}_n \phi(\tilde{f})\|_{\mathcal{F}} \lesssim \sum_{l=1}^k \sqrt{n} \|\mathbb{P}_n |f_l - \tilde{f}_l|\|_{\mathcal{F}_l}.$$

According to Theorem 2.3.15, this converges to zero in probability. Thus $\phi \circ \mathcal{F}$ is Donsker also. ■

2.10.4 Uniform Entropy under Transformation

The main theorem of the preceding subsection combines the minimal condition that the classes $\mathcal{F}_1, \dots, \mathcal{F}_k$ are Donsker with the strong condition that ϕ is uniformly Lipschitz. It may be expected that stronger assumptions on the classes $\mathcal{F}_i$ combined with weaker conditions on ϕ will yield the same conclusion that the class $\phi(\mathcal{F}_1, \dots, \mathcal{F}_k)$ is Donsker. Obviously, a concrete formulation of this principle is useful only if it is sufficiently simple, because otherwise it is preferable to deal with the class $\phi(\mathcal{F}_1, \dots, \mathcal{F}_k)$ directly. Uniform entropy, integral-type conditions allow such a simple statement.

Let $\mathcal{F}_1, \dots, \mathcal{F}_k$ be classes of measurable functions and $\phi: \mathbb{R}^k \rightarrow \mathbb{R}$ a given map that is Lipschitz of orders $\alpha_1, \dots, \alpha_k \in (0, 1]$ in the sense that

$$(2.10.21) \quad |\phi \circ f(x) - \phi \circ g(x)|^2 \leq \sum_{i=1}^k L_{\alpha,i}^2(x) |f_i - g_i|^{2\alpha_i}(x),$$

for all $f = (f_1, \dots, f_k)$ and $g = (g_1, \dots, g_k)$ in $\mathcal{F} = \mathcal{F}_1 \times \dots \times \mathcal{F}_k$ and for every x . In the special case that $k = 1$, we can set

$$L_\alpha = \sup_{f,g} \frac{|\phi(f) - \phi(g)|}{|f - g|^\alpha}.$$

This relaxes condition (2.10.7) in two ways: the functions $L_{\alpha,i}$ need not be bounded, and a lower-order Lipschitz condition suffices. In the following theorem, the first weakening of (2.10.7) is compensated by a moment condition on L_α ; the second by a strengthening of the entropy condition on the $\mathcal{F}_i$.

If $\mathcal{F}_i$ has envelope function F_i , then for every pair f and f_0 in $\mathcal{F}$,

$$|\phi \circ f - \phi \circ f_0|^2 \leq 4 \sum_{i=1}^k L_{\alpha,i}^2 F_i^{2\alpha_i}.$$

Thus, the function $2(\sum_{i=1}^k L_{\alpha,i}^2 F_i^{2\alpha_i})^{1/2}$, which will be denoted $2L_\alpha \cdot F^\alpha$, is an envelope function of the class $\phi(\mathcal{F}) - \phi(f_0)$. We use this function to standardize the entropy numbers of the class $\phi \circ \mathcal{F}$.

2.10.22 Theorem (Permanence uniform entropy). *Let $\mathcal{F}_1, \dots, \mathcal{F}_k$ be classes of measurable functions with measurable envelopes F_i , and let $\phi: \mathbb{R}^k \rightarrow \mathbb{R}$ be a map that satisfies (2.10.21). Then, for every $\delta > 0$,*

$$\begin{aligned} & \int_0^\delta \sup_Q \sqrt{\log N(\varepsilon \|L_\alpha \cdot F^\alpha\|_{Q,2}, \phi(\mathcal{F}), L_2(Q))} d\varepsilon \\ & \leq \sum_{i=1}^k \int_0^{\delta^{1/\alpha_i}} \sup_Q \sqrt{\log N(\varepsilon \|F_i\|_{Q,2\alpha_i}, \mathcal{F}_i, L_{2\alpha_i}(Q))} \frac{d\varepsilon}{\varepsilon^{1-\alpha_i}}. \end{aligned}$$

Here the supremum is taken over all finitely discrete probability measures Q . Consequently, if the right side is finite and $P^*(L_\alpha \cdot F^\alpha)^2 < \infty$, then the class $\phi(\mathcal{F})$ is Donsker provided its members are square integrable and the class is suitably measurable.

Proof. For a given finitely discrete probability measure Q , define measures R_i by $dR_i = L_{\alpha,i}^2 dQ$. Then

$$\|\phi \circ f - \phi \circ g\|_{Q,2}^2 \leq \sum R_i |f_i - g_i|^{2\alpha_i}.$$

For each i , construct a minimal $\varepsilon^{1/\alpha_i} \|F_i\|_{R_i,2\alpha_i}$ -net $\mathcal{G}_i$ for $\mathcal{F}_i$ with respect to the $L_{2\alpha_i}(R_i)$ -norm. Then the set of points $\phi(g_1, \dots, g_k)$ with $(g_1, \dots, g_k)$ ranging over all possible combinations of g_i from $\mathcal{G}_i$ forms a net for $\phi(\mathcal{F})$ of $L_2(Q)$ -size bounded by the square root of

$$\sum \varepsilon^2 \|F_i\|_{R_i,2\alpha_i}^{2\alpha_i} = \varepsilon^2 Q \sum L_{\alpha,i}^2 F_i^{2\alpha_i}.$$

This equals $\varepsilon^2 \|L_\alpha \cdot F^\alpha\|_{Q,2}^2$. Conclude that

$$N(\varepsilon \|L_\alpha \cdot F^\alpha\|_{Q,2}, \phi(\mathcal{F}), L_2(Q)) \leq \prod_{i=1}^k N(\varepsilon^{1/\alpha_i} \|F_i\|_{R_i,2\alpha_i}, \mathcal{F}_i, L_{2\alpha_i}(R_i)).$$

Since expressions of the type $N(\varepsilon \|F\|_{cR,r}, \mathcal{F}, L_r(cR))$ are constant in c , the bound is also valid with R_i replaced by the probability measure $R_i/R_i 1$.

If Q ranges over all finitely discrete measures, then each of the R_i runs through finitely discrete measures. Substitute the bound of the preceding display in the left side of the theorem, and next make a change of variables $\varepsilon^{1/\alpha_i} \mapsto \varepsilon$ to conclude the proof. ■

2.10.23 Example. If $\mathcal{F}$ satisfies the uniform entropy condition and $P^*F^{2k} < \infty$ for $k > 1$, then $\mathcal{F}^k$ is Donsker if suitably measurable. In comparison with Theorem 2.10.8, this result has traded a stronger entropy condition for a weaker (almost optimal) condition on the envelope function: Theorem 2.10.8 shows that $\mathcal{F}^k$ is Donsker if $\mathcal{F}$ is Donsker and uniformly bounded.

The present result follows since $|f^k - g^k| \leq |f - g|(2F)^{k-1}$, so that (2.10.21) applies with $\alpha = 1$ and $L = (2F)^{k-1}$.

2.10.24 Example. Let the functions in $\mathcal{F}$ be nonnegative with lower envelope function $\underline{F}$, and consider the class $\sqrt{\mathcal{F}}$. If $\mathcal{F}$ satisfies the uniform entropy condition and $P^*F^2/\underline{F} < \infty$, then $\sqrt{\mathcal{F}}$ is Donsker if suitably measurable. This follows since $|\sqrt{f} - \sqrt{g}| \leq |f - g|/\sqrt{\underline{F}}$.

The conclusion that $\sqrt{\mathcal{F}}$ is Donsker can also be obtained under other combinations of moment and entropy conditions. First, Theorem 2.10.8 shows that $\sqrt{\mathcal{F}}$ is Donsker if $\mathcal{F}$ is Donsker and $\underline{F} \geq \delta > 0$. This combines a stronger assumption on the lower envelope with a weaker entropy condition. Second, the present theorem can be applied with a different Lipschitz order. For $1/2 \leq \alpha \leq 1$,

$$\frac{|\sqrt{f} - \sqrt{g}|}{|f - g|^\alpha} = \frac{|\sqrt{f} - \sqrt{g}|^{1-\alpha}}{|\sqrt{f} + \sqrt{g}|^\alpha} \leq (2\sqrt{\underline{F}})^{1-2\alpha}.$$

Thus, a suitably measurable class $\sqrt{\mathcal{F}}$ is Donsker if, for some $\alpha \geq 1/2$, the upper and lower envelopes of $\mathcal{F}$ satisfy $P^*F^{2\alpha}\underline{F}^{1-2\alpha} < \infty$ and, in addition, $\mathcal{F}$ satisfies

$$\int_0^\infty \sup_Q \sqrt{\log N(\varepsilon \|F\|_{Q, 2\alpha}, \mathcal{F}, L_{2\alpha}(Q))} \frac{d\varepsilon}{\varepsilon^{1-\alpha}} < \infty.$$

It appears that finiteness of the integral is more restrictive than a finite uniform L_2 -entropy integral (though the supremum in the integrand is increasing in α). For polynomial classes $\mathcal{F}$ and more generally classes with uniform L_1 -entropy of the order $(1/\varepsilon)^r$ with $r < 1$, the integral is certainly finite for $\alpha = 1/2$. For such classes, the class $\sqrt{\mathcal{F}}$ is Donsker provided its envelope $\sqrt{F}$ is square integrable.

2.10.25 Example. Let $\mathcal{F}$ and $\mathcal{G}$ satisfy the uniform entropy condition and be suitably measurable. Then $\mathcal{FG}$ is Donsker provided the envelopes F and G satisfy $P^*F^2G^2 < \infty$.

This follows from the theorem with $L_\alpha = (G, F)$ and Lipschitz orders $\alpha = (1, 1)$.

2.10.5 Partitioning the Sample Space

Let $\mathcal{X} = \bigcup_{j=1}^\infty \mathcal{X}_j$ be a partition of $\mathcal{X}$ into measurable sets, and let $\mathcal{F}_j$ be the class of functions $f1_{\mathcal{X}_j}$ when f ranges over $\mathcal{F}$. If the class $\mathcal{F}$ is Glivenko-Cantelli or Donsker, then each class $\mathcal{F}_j$ is Glivenko-Cantelli or Donsker. This follows from Theorems 2.10.5 and 2.10.8, because a restriction is a contraction in the sense that $|(f1_{\mathcal{X}_j})(x) - (g1_{\mathcal{X}_j})(x)| \leq |f(x) - g(x)|$ for every x . Consider the converse of these statements. While the sum of infinitely many Glivenko-Cantelli or Donsker classes

need not be Donsker, it is clear that if each $\mathcal{F}_j$ is Glivenko-Cantelli or Donsker and the classes $\mathcal{F}_j$ become suitably small as $j \rightarrow \infty$, then $\mathcal{F}$ is Glivenko-Cantelli or Donsker.

The following theorems make this precise. For the Glivenko-Cantelli property “suitably small” can be simply taken to be the (almost necessary) condition that $\mathcal{F}$ possesses an integrable envelope function.

2.10.26 Theorem (Partitions). *If the restriction $\mathcal{F}_j$ of $\mathcal{F}$ to $\mathcal{X}_j$ is P -Glivenko-Cantelli for each j and $\mathcal{F}$ has an integrable envelope function, then $\mathcal{F}$ is itself P -Glivenko-Cantelli.*

Proof. Since $f = \sum_{j=1}^{\infty} f 1_{\mathcal{X}_j}$, it follows that

$$E^* \|\mathbb{P}_n - P\|_{\mathcal{F}} \leq \sum_{j=1}^{\infty} E^* \|\mathbb{P}_n - P\|_{\mathcal{F}_j}.$$

Each term of the series on the right tends to zero by the hypothesis that the classes $\mathcal{F}_j$ are P -Glivenko-Cantelli. Furthermore, the j th terms is dominated by $E^* \mathbb{P}_n(F 1_{\mathcal{X}_j}) + P(F 1_{\mathcal{X}_j}) \leq 2P(F 1_{\mathcal{X}_j})$, where $\sum_{j=1}^{\infty} P(F 1_{\mathcal{X}_j}) = P(F) < \infty$. Thus the right side of the display tends to zero by the dominated convergence theorem. ■

For the Donsker property we impose a stronger, but still simple condition.

2.10.27 Theorem. *For each j , let the class $\mathcal{F}_j$, the restriction of $\mathcal{F}$ to $\mathcal{X}_j$, be Donsker and satisfy*

$$E^* \|\mathbb{G}_n\|_{\mathcal{F}_j} \leq C c_j,$$

for a constant C not depending on j or n . If $\sum_{j=1}^{\infty} c_j < \infty$ and $P^ F < \infty$ for some envelope function, then the class $\mathcal{F}$ is Donsker.*

Proof. We can assume without loss of generality that the class $\mathcal{F}$ contains the constant function 1.

Since $\mathcal{F}_j$ is Donsker, the sequence of empirical processes indexed by $\mathcal{F}_j$ converges in distribution in $\ell^\infty(\mathcal{F}_j)$ to a tight Brownian bridge $\mathbb{H}_j$ for each j . This implies that $E^* \|\mathbb{G}_n\|_{\mathcal{F}_j} \rightarrow E \|\mathbb{H}_j\|_{\mathcal{F}_j}$. Thus, $E \|\mathbb{H}_j\|_{\mathcal{F}_j} \leq C c_j$. Let Z_j be a standard normal variable independent of $\mathbb{H}_j$, constructed on the same probability space. Since $\sup_{f \in \mathcal{F}_j} |Pf| < \infty$, the process $f \mapsto \mathbb{Z}_j(f) = \mathbb{H}_j(f) + Z_j Pf$ is well defined on $\mathcal{F}_j$ and takes its values in $\ell^\infty(\mathcal{F}_j)$. We can construct these processes for different j as independent random elements on a single probability space. Then the series $\mathbb{Z}(f) = \sum_{j=1}^{\infty} \mathbb{Z}_j(f 1_{\mathcal{X}_j})$ converges in second mean for every f , and satisfies $E \mathbb{Z}(f) \mathbb{Z}(g) = Pfg$, for every f and g . Thus, the series defines a version of a Brownian motion process. Since each of the processes $\{\mathbb{Z}_j(f 1_{\mathcal{X}_j}) : f \in \mathcal{F}\}$ has bounded and uniformly continuous sample paths with respect to the $L_2(P)$ -seminorm, so

have the partial sums $\mathbb{Z}_{\leq k} = \{\sum_{j=1}^k \mathbb{Z}_j(f1_{\mathcal{X}_j}) : f \in \mathcal{F}\}$. Furthermore,

$$\begin{aligned} \mathbb{E} \sup_{f \in \mathcal{F}} \left| \sum_{j>k} \mathbb{Z}_j(f1_{\mathcal{X}_j}) \right| &\leq \sum_{j>k} (\mathbb{E} \|\mathbb{H}_j\|_{\mathcal{F}_j} + \mathbb{E} |Z_j| P^* F 1_{\mathcal{X}_j}) \\ &\leq C \sum_{j>k} c_j + \sqrt{\frac{2}{\pi}} P^* F 1_{\cup_{j>k} \mathcal{X}_j}. \end{aligned}$$

This converges to zero as $k \rightarrow \infty$. Thus the series $\mathbb{Z}(f) = \sum_{j=1}^{\infty} \mathbb{Z}_j(f1_{\mathcal{X}_j})$ converges in mean in the space $\ell^\infty(\mathcal{F})$. By the Ito-Nisio theorem, it also converges almost surely. Conclude that almost all sample paths of the process $\mathbb{Z}$ are bounded and uniformly continuous. Since $\mathbb{E} \|\mathbb{Z}\|_{\mathcal{F}} < \infty$, the class $\mathcal{F}$ is totally bounded in $L_2(P)$, by Sudakov's inequality. Hence $\mathbb{Z}$ is a tight version of a Brownian motion process indexed by $\mathcal{F}$. The process $\mathbb{G}(f) = \mathbb{Z}(f) - \mathbb{Z}(1)Pf$ defines a tight Brownian bridge process indexed by $\mathcal{F}$. We have proved that $\mathcal{F}$ is pre-Gaussian.

For each k set $\mathbb{G}_{n,\leq k}(f) = \mathbb{G}_n(f1_{\cup_{j\leq k} \mathcal{X}_j})$. The continuity modulus of the process $\{\mathbb{G}_{n,\leq k}(f); f \in \mathcal{F}\}$ is bounded by the continuity modulus of the empirical process indexed by the sum class $\sum_{j\leq k} \mathcal{F}_j$. Since the sum of finitely many Donsker classes is Donsker we can conclude that the sequence $(\mathbb{G}_{n,\leq k})_{n=1}^{\infty}$ is asymptotically tight in $\ell^\infty(\mathcal{F})$. Considering the marginal distributions, we can conclude that the sequence converges in distribution to

$$\mathbb{G}_{\leq k}(f) = \mathbb{Z}_{\leq k}(f) - d_k \mathbb{Z}_{\leq k}(1) P f 1_{\cup_{j\leq k} \mathcal{X}_j},$$

as $n \rightarrow \infty$ for each fixed k , for d_k a solution to the equation $d^2 P(\cup_{j\leq k} \mathcal{X}_j) - 2d = -1$.

As $k \rightarrow \infty$, the sequence $\mathbb{G}_{\leq k}$ converges almost surely, whence in distribution to $\mathbb{G}$ in $\ell^\infty(\mathcal{F})$. Since weak convergence to a tight limit is metrizable, and the array $\mathbb{G}_{n,\leq k}$ converges along every row to limits that converge to $\mathbb{G}$, there exist integers $k_n \rightarrow \infty$ such that the sequence $\mathbb{G}_{n,\leq k_n}$ converges in distribution to $\mathbb{G}$. We also have that the sequence $\mathbb{G}_n - \mathbb{G}_{n,\leq k_n}$ converges in outer probability to zero, since

$$\mathbb{E}^* \|\mathbb{G}_n - \mathbb{G}_{n,\leq k_n}\|_{\mathcal{F}} \leq C \sum_{j>k_n} c_j \rightarrow 0.$$

An application of Slutsky's lemma completes the proof. ■

Methods to obtain maximal inequalities for the L_1 -norm are discussed in Section 2.14.1. We give three applications of the preceding theorem.

2.10.28 Example (Smooth functions). Let $\mathbb{R}^d = \cup_{j=1}^{\infty} \mathcal{X}_j$ be a partition of $\mathbb{R}^d$ into uniformly bounded, convex sets with nonempty interior. Consider the class $\mathcal{F}$ of functions such that the class $\mathcal{F}_j$ of restrictions is contained in $C_{M_j}^\alpha(\mathcal{X}_j)$ for each j for given constants M_j .

If $\alpha > d/2$ and $\sum_{j=1}^{\infty} M_j P(\mathcal{X}_j)^{1/2} < \infty$, then the class $\mathcal{F}$ is P -Donsker. This follows by combining Theorem 2.10.27 with the (last) maximal inequality in Theorem 2.14.16, and the entropy estimate Theorem 2.7.1.

2.10.29 Example (Closed convex sets). Let $\mathbb{R}^2 = \bigcup_{j=1}^{\infty} \mathcal{X}_j$ be a partition into squares of fixed size. Let $\mathcal{C}$ be the class of closed convex subsets of $\mathbb{R}^2$.

Then $\mathcal{C}$ is P -Donsker for every probability measure P with a density p such that $\sum \|p\|_{\mathcal{X}_j}^{1/2} < \infty$. In particular, this is the case if $[(1 + |x|)(1 + |y|)]^{2+\delta} p(x, y)$ is bounded for some $\delta > 0$.

2.10.30 Example (Monotone functions). Consider the class $\mathcal{F}$ of all nondecreasing functions $f: \mathbb{R} \rightarrow \mathbb{R}$, such that $0 \leq f \leq F$, for a given nondecreasing function F . This class is Donsker provided $\|F\|_{2,1} < \infty$.

To see this, first reduce the problem to the case of uniform $[0, 1]$ observations by the quantile transformation. If P^{-1} is the quantile function corresponding to the underlying measure P , then the class $\mathcal{F} \circ P^{-1}$ is Donsker with respect to the uniform measure if and only if $\mathcal{F}$ is P -Donsker. The class $\mathcal{G} = \mathcal{F} \circ P^{-1}$ consists of monotone functions $g: [0, 1] \rightarrow \mathbb{R}$, with $0 \leq g \leq F \circ P^{-1}$. Furthermore,

$$\int_0^1 \frac{F \circ P^{-1}}{\sqrt{1-u}} du \leq \int \frac{F(x)}{\sqrt{1-P(-\infty, x]}} dP(x).$$

By partial integration the latter can be shown to be finite if and only if $\|F\|_{2,1}$ is finite. For simplicity of notation assume now that P is the uniform measure.

Use the theorem with the partition into the intervals $\mathcal{X}_j = (x_j, x_{j+1}]$, with $x_j = 1 - 2^{-j}$ for each integer $j \geq 0$. In view of Theorem 2.7.9 and the last inequality in Theorem 2.14.16 the mean $E^* \|\mathbb{G}_n\|_{\mathcal{F}}$ is bounded above by the $L_2(P)$ -norm of the envelope function F_j restricted to the interval $(x_j, x_{j+1}]$. In view of the monotonicity of the envelope function, the condition of the theorem involves the series

$$\sum_{j=0}^{\infty} \|F_j\|_{P,2} \leq \sum_{j=1}^{\infty} \frac{F(1 - 2^{-j})}{\sqrt{1 - (1 - 2^{-j})}} 2^{-j} \leq 2 \int_0^1 \frac{F(u)}{\sqrt{1-u}} du.$$

The result follows by the quantile transformation.

Problems and Complements

1. Let $\{\psi_i\}$ be an orthonormal base of a subspace of $L_2(P)$ that contains $\mathcal{F}$ and the constant functions, and let $Z_1, Z_2, \dots$ be i.i.d. standard normally distributed random variables. If $\mathcal{F}$ is P -pre-Gaussian, then the series $\sum_{i=1}^{\infty} Z_i (P(f - Pf)\psi_i)$ is almost surely convergent in $\ell^\infty(\mathcal{F})$ and represents a tight Brownian bridge process.

[Hint: The series converges in second mean for every single f , and the limit process (series) $\{\mathbb{G}(f): f \in L_2(P)\}$ is a well-defined stochastic process that is continuous in second mean with respect to ρ_P . Since $\mathcal{F}$ is separable, the latter implies that $\{\mathbb{G}(f): f \in \mathcal{F}\}$ is a separable process. A separable version of a uniformly continuous process is automatically uniformly continuous. Thus, $\{\mathbb{G}(f): f \in \mathcal{F}\}$ defines a tight Brownian bridge.

Let $\mathbb{G}_n$ be the n th partial sum of the series, and set $\mathcal{F}_\delta = \{f - g: f, g \in \mathcal{F}, \rho_P(f - g) < \delta\}$. By Jensen's inequality, $E\|\mathbb{G}_n\|_{\mathcal{F}_\delta} \leq E\|\mathbb{G}\|_{\mathcal{F}_\delta}$. Since $\mathcal{F}$ is pre-Gaussian, the

right side converges to zero as $\delta \downarrow 0$. Hence the sequence $\mathbb{G}_n$ is uniformly tight in $\ell^\infty(\mathcal{F})$. Since $\mathbb{G}_n - \mathbb{G} \rightsquigarrow 0$ marginally, it follows that $\mathbb{G}_n - \mathbb{G} \rightsquigarrow 0$ in $\ell^\infty(\mathcal{F})$. By the Itô-Nisio theorem A.1.3, it converges almost surely also.]

2. The bounded Lipschitz distance between the normal distributions $N(0, \sigma^2)$ and $N(0, \tau^2)$ on the line is bounded below by $|\sigma - \tau| g(1 \wedge \tau^{-1} \wedge \sigma^{-1})$ for $g(t) = t^2 \phi(t)$ and ϕ the standard normal density.

[Hint: For $0 \leq c \leq 1$, the function $h(x) = (c - |x|) \vee 0$ is contained in $BL_1(\mathbb{R})$. For $\sigma < \tau$, the integral $\left| \int (h(x\sigma) - h(x\tau)) \phi(x) dx \right|$ is bounded below by $\phi(c/\tau)$ times $\int_{-c/\tau}^{c/\tau} (h(x\sigma) - h(x\tau)) dx = (\tau - \sigma)(c/\tau)^2$. Take $c = \tau \wedge 1$.]

3. For any class $\mathcal{F}$ of functions with envelope function F and $0 < r < s < \infty$,

$$\sup_Q N(2\varepsilon \|F\|_{Q,s}, \mathcal{F}, L_s(Q)) \leq \sup_Q N(2\varepsilon^{r/s} \|F\|_{Q,r}, \mathcal{F}, L_r(Q))$$

if the supremum is taken over all finitely discrete measures.

[Hint: If $dR = (2F)^{s-r} dQ$, then $\|f - g\|_{Q,s} \leq \|f - g\|_{R,r}^{r/s}$.]

4. For any class $\mathcal{F}$ of functions with strictly positive envelope function F , the expression $\sup_Q N(\varepsilon \|F\|_{Q,r}, \mathcal{F}, L_r(Q))$, where the supremum is taken over all finitely discrete measures, is nondecreasing in $0 < r < \infty$. (Here $L_r(Q)$ is equipped with $(Q|f|^r)^{1/r}$ even though this is not a norm for $r < 1$.)

[Hint: Given $r < s$ and Q define $dR = F^{r-s} dQ$. Then $Q|f - g|^r \leq \|f - g\|_{R,s}^r \|F\|_{R,s}^{s-r}$, by Hölder's inequality, with $(p, q) = (s/r, s/(s-r))$.]

5. The expression $N(\varepsilon \|aF\|_{bQ,r}, a\mathcal{F}, L_r(bQ))$ is constant in a and b .
6. If $\mathcal{F}$ is P -pre-Gaussian and $A: L_2(P) \rightarrow L_2(P)$ is continuous, then the class $\{Af: f \in \mathcal{F}\}$ is P -pre-Gaussian.
7. If $\mathcal{F}$ is P -Donsker and $A: \mathcal{L}_2(P) \rightarrow \mathcal{L}_2(P)$ is an orthogonal projection, then the class $\{Af: f \in \mathcal{F}\}$ is not necessarily P -Donsker.
8. If both $\mathcal{F}$ and $\mathcal{G}$ are Donsker, then the class $\mathcal{F} \cdot \mathcal{G}$ of products $x \mapsto f(x)g(x)$ is Glivenko-Cantelli in probability.

2.11

Central Limit Theorem for Processes

So far we have focused on limit theorems and inequalities for the empirical process of independent and identically distributed random variables. Most of the methods of proof apply more generally. In this chapter we indicate some extensions to the case of independent but not identically distributed processes.

We consider the general situation of sums of independent stochastic processes $\{Z_{ni}(f): f \in \mathcal{F}\}$ with bounded sample paths indexed by an arbitrary set $\mathcal{F}$.

2.11.1 Random Entropy

For each n , let $Z_{n1}, \dots, Z_{nm_n}$ be independent stochastic processes indexed by a common (arbitrary) semimetric space $(\mathcal{F}, \rho)$. As usual, for computation of outer expectations, the independence is understood in the sense that the processes are defined on a product probability space $\prod_{i=1}^{m_n} (\mathcal{X}_{ni}, \mathcal{A}_{ni}, P_{ni})$ with each $Z_{ni}(f) = Z_{ni}(f, x)$ depending only on the i th coordinate of $x = (x_1, \dots, x_{m_n})$. In the first theorem it is assumed that every one of the maps

$$\begin{aligned} (x_1, \dots, x_{m_n}) &\mapsto \sup_{\rho(f,g) < \delta} \left| \sum_{i=1}^{m_n} e_i(Z_{ni}(f) - Z_{ni}(g)) \right|, \\ (x_1, \dots, x_{m_n}) &\mapsto \sup_{\rho(f,g) < \delta} \left| \sum_{i=1}^{m_n} e_i(Z_{ni}(f) - Z_{ni}(g))^2 \right|, \end{aligned}$$

is measurable, for every $\delta > 0$, every vector $(e_1, \dots, e_{m_n}) \in \{-1, 0, 1\}^{m_n}$, and every natural number n . (Measurability for the completion of the probability spaces $\prod_{i=1}^{m_n} (\mathcal{X}_{ni}, \mathcal{A}_{ni}, P_{ni})$ suffices.)

Define a random semimetric by

$$d_n^2(f, g) = \sum_{i=1}^{m_n} \left(Z_{ni}(f) - Z_{ni}(g) \right)^2.$$

The simplest result on marginal convergence of the processes $\sum_{i=1}^{m_n} Z_{ni}$ is the Lindeberg central limit theorem. The following theorem gives a uniform central limit theorem under a Lindeberg condition on norms combined with a condition on entropies with respect to the random semimetric d_n , a “random entropy condition.”

2.11.1 Theorem (Random entropy CLT). *For each n , let $Z_{n1}, \dots, Z_{n, m_n}$ be independent stochastic processes indexed by a totally bounded semimetric space $(\mathcal{F}, \rho)$. Assume that the sums $\sum_{i=1}^{m_n} e_i Z_{ni}$ are measurable as indicated and that*

$$(2.11.2) \quad \begin{aligned} & \sum_{i=1}^{m_n} \mathbb{E}^* \|Z_{ni}\|_{\mathcal{F}}^2 \{ \|Z_{ni}\|_{\mathcal{F}} > \eta \} \rightarrow 0, \quad \text{for every } \eta > 0, \\ & \sup_{\rho(f, g) < \delta_n} \sum_{i=1}^{m_n} \mathbb{E} (Z_{ni}(f) - Z_{ni}(g))^2 \rightarrow 0, \quad \text{for every } \delta_n \downarrow 0, \\ & \int_0^{\delta_n} \sqrt{\log N(\varepsilon, \mathcal{F}, d_n)} d\varepsilon \xrightarrow{P^*} 0, \quad \text{for every } \delta_n \downarrow 0. \end{aligned}$$

Then the sequence $\sum_{i=1}^{m_n} (Z_{ni} - \mathbb{E} Z_{ni})$ is asymptotically ρ -equicontinuous. It converges in distribution in $\ell^\infty(\mathcal{F})$ provided the sequence of covariance functions converges pointwise on $\mathcal{F} \times \mathcal{F}$.

Proof. The Lindeberg condition for norms implies the Lindeberg condition for marginals. Together with the assumption that the covariance function converges, this gives marginal weak convergence (to a Gaussian process).

Set $Z_{ni}^o = Z_{ni} - \mathbb{E} Z_{ni}$, and let $\delta_n \downarrow 0$ be arbitrary. For fixed t and sufficiently large n Chebyshev's inequality and the second condition give the bound $P(|\sum_{i=1}^{m_n} Z_{ni}^o(f) - Z_{ni}^o(g)| > t/2) \leq 1/2$ for every pair f, g with $\rho(f, g) < \delta_n$. By the symmetrization lemma, Lemma 2.3.7, for sufficiently large n ,

$$\begin{aligned} & P^* \left(\sup_{\rho(f, g) < \delta_n} \left| \sum_{i=1}^{m_n} Z_{ni}^o(f) - Z_{ni}^o(g) \right| > t \right) \\ & \leq 4P \left(\sup_{\rho(f, g) < \delta_n} \left| \sum_{i=1}^{m_n} \varepsilon_i (Z_{ni}(f) - Z_{ni}(g)) \right| > \frac{t}{4} \right). \end{aligned}$$

For fixed values of the processes $Z_{n1}, \dots, Z_{nm_n}$, define the subset $A_n \subset \mathbb{R}^{m_n}$ as the set of all vectors $(Z_{n1}(f) - Z_{n1}(g), \dots, Z_{nm_n}(f) - Z_{nm_n}(g))$ when the pairs (f, g) range over the set $\{(f, g) \in \mathcal{F} \times \mathcal{F} : \rho(f, g) < \delta_n\}$. By Hoeffding's inequality, Lemma 2.2.8, the stochastic process $\{\sum \varepsilon_i a_i : a \in A_n\}$ is sub-Gaussian for the Euclidean metric on A_n . By Corollary 2.2.9,

$$\mathbb{E}_\varepsilon \sup_{\rho(f, g) < \delta_n} \left| \sum_{i=1}^{m_n} \varepsilon_i (Z_{ni}(f) - Z_{ni}(g)) \right| \lesssim \int_0^\infty \sqrt{\log N(\varepsilon, A_n, \|\cdot\|)} d\varepsilon,$$

where $\|\cdot\|$ is the Euclidean norm. The integrand can be bounded using the inequality $N(\varepsilon, A_n, \|\cdot\|) \leq N^2(\varepsilon/2, \mathcal{F}, d_n)$. Furthermore, if

$$\theta_n^2 = \sup_{a \in A_n} \sum_{i=1}^{m_n} a_i^2 = \left\| \sum_{i=1}^{m_n} a_i^2 \right\|_{A_n},$$

then, for $\varepsilon > \theta_n$, the set A_n fits in the ball of radius ε around the origin and the integrand vanishes. Conclude from this and the entropy condition (2.11.2) that the integral converges to zero in outer probability if $\theta_n \rightarrow 0$ in probability. Under the measurability assumption, this implies the asymptotic equicontinuity of the sequence $\sum_{i=1}^{m_n} Z_{ni}^o$.

By the Lindeberg condition, there exists a sequence of numbers $\eta_n \downarrow 0$ such that $E^* \left\| \sum a_i^2 \{ \|Z_{ni}\|_{\mathcal{F}}^* > \eta_n \} \right\|_{A_n}$ converges to zero. Thus, for showing that $\theta_n \xrightarrow{P^*} 0$, it is not a loss of generality to assume that each Z_{ni} satisfies $\|Z_{ni}\|_{\mathcal{F}} \leq \eta_n$. Fix $Z_{n1}, \dots, Z_{nm_n}$, and take an ε -net B_n for A_n for the Euclidean norm. For every $a \in A_n$, there exists $b \in B_n$ with

$$|\sum \varepsilon_i a_i^2| = |\sum \varepsilon_i (a_i - b_i)^2 + 2 \sum \varepsilon_i (a_i - b_i) b_i + \sum \varepsilon_i b_i^2| \leq \varepsilon^2 + 2\varepsilon \|b\| + |\sum \varepsilon_i b_i^2|.$$

By Hoeffding's inequality, Lemma 2.2.8, the variable $\sum \varepsilon_i b_i^2$ has Orlicz norm for ψ_2 bounded by a multiple of $(\sum b_i^4)^{1/2} \lesssim \eta_n (\sum b_i^2)^{1/2}$. Apply Lemma 2.2.2 to the third term on the right, and replace suprema over B_n by suprema over A_n to obtain

$$E_\varepsilon \left\| \sum \varepsilon_i a_i^2 \right\|_{A_n} \lesssim \varepsilon^2 + 2\varepsilon \left\| \sum a_i^2 \right\|_{A_n}^{1/2} + \sqrt{1 + \log |B_n|} \eta_n \left\| \sum a_i^2 \right\|_{A_n}^{1/2}.$$

The size of the ε -net can be chosen $|B_n| \leq N^2(\varepsilon/2, \mathcal{F}, d_n)$. By the entropy condition (2.11.2), this variable is bounded in probability (Problem 2.11.1). Conclude that for some constant K ,

$$\begin{aligned} P \left(\left\| \sum \varepsilon_i a_i^2 \right\|_{A_n} > t \right) \\ \leq P^* (|B_n| > M) + \frac{K}{t} \left[\varepsilon^2 + (\varepsilon + \eta_n \sqrt{\log M}) E \left\| \sum a_i^2 \right\|_{A_n}^{1/2} \right]. \end{aligned}$$

For M and t sufficiently large, the right side is smaller than $1 - v_1$ for the constant v_1 in Hoffmann-Jørgensen's Proposition A.1.5. More precisely, this can be achieved for M such that $P^* (|B_n| > M) \leq (1 - v_1)/2$ and t equal to the numerator of the second term (i.e. K times the term in square brackets) divided by $(1 - v_1)/2$. Then t is bigger than the v_1 -quantile of the variable $\left\| \sum \varepsilon_i a_i^2 \right\|_{A_n}$. Thus, Proposition A.1.5 yields

$$\begin{aligned} E \left\| \sum \varepsilon_i a_i^2 \right\|_{A_n} &\lesssim E \left\| \max a_i^2 \right\|_{A_n} + t \\ &\lesssim \eta_n^2 + \varepsilon^2 + (\varepsilon + \eta_n \sqrt{\log M}) \left(E \left\| \sum a_i^2 \right\|_{A_n} \right)^{1/2}. \end{aligned}$$

By the second assumption of the theorem, $\left\| \sum E a_i^2 \right\|_{A_n} \rightarrow 0$. Combine this with Lemma 2.3.6 to see that

$$E \left\| \sum a_i^2 \right\|_{A_n} \leq E \left\| \sum \varepsilon_i a_i^2 \right\|_{A_n} + o(1) \leq \zeta + \zeta (E \left\| \sum a_i^2 \right\|_{A_n})^{1/2},$$

for $\zeta > \varepsilon^2 \vee \varepsilon$ and sufficiently large n . The inequality $c \leq \zeta + \zeta \sqrt{c}$ for a nonnegative number c implies that $c \leq (\zeta + \sqrt{\zeta^2 + 4\zeta})^2$. Apply this to $c = E \left\| \sum a_i^2 \right\|_{A_n}$ and conclude that $E \left\| \sum a_i^2 \right\|_{A_n} \rightarrow 0$ as n tends to infinity. ■

2.11.3 Example (I.i.d. observations). The empirical process of a sample $X_1, \dots, X_n$ studied in the earlier sections of this part can be recovered by setting $Z_{ni}(f) = n^{-1/2}f(X_i)$. In this situation, $\mathcal{F}$ is a class of measurable functions on the sample space.

Theorem 2.11.1 implies that the class $\mathcal{F}$ is Donsker if it is suitably measurable, is totally bounded in $L_2(P)$, possesses a square-integrable envelope, and satisfies for every sequence $\delta_n \downarrow 0$

$$\int_0^{\delta_n} \sqrt{\log N(\varepsilon, \mathcal{F}, L_2(\mathbb{P}_n))} d\varepsilon \xrightarrow{P} 0.$$

The latter random-entropy condition is certainly satisfied if $\mathcal{F}$ satisfies the uniform entropy condition (2.5.1) and the envelope function is square-integrable. Thus, Theorem 2.11.1 is a generalization of Theorem 2.5.2.

2.11.4 Example (Truncation). The Lindeberg condition on the norms is certainly not necessary for the central limit theorem. In combination with truncation, the preceding theorem applies to more general processes. Consider stochastic processes $Z_{n1}, \dots, Z_{nm_n}$ such that

$$\sum_{i=1}^{m_n} P^*(\|Z_{ni}\|_{\mathcal{F}} > \eta) \rightarrow 0, \quad \text{for every } \eta > 0.$$

Then the truncated processes $Z_{ni,\eta}(f) = Z_{ni}(f)1\{\|Z_{ni}\|_{\mathcal{F}} \leq \eta\}$ satisfy

$$\sum_{i=1}^{m_n} Z_{ni} - \sum_{i=1}^{m_n} Z_{ni,\eta} \xrightarrow{P^*} 0, \quad \text{in } \ell^\infty(\mathcal{F}).$$

Since this is true for every $\eta > 0$, it is also true for every sequence $\eta_n \downarrow 0$ that converges to zero sufficiently slowly. The processes Z_{ni,η_n} certainly satisfy the Lindeberg condition. If they (or the centered processes $Z_{ni,\eta_n} - \mathbb{E}Z_{ni,\eta_n}$) also satisfy the other conditions of the theorem, then the sequence $\sum_{i=1}^{m_n} (Z_{ni} - \mathbb{E}Z_{ni,\eta_n})$ converges weakly in $\ell^\infty(\mathcal{F})$. The random semimetrics d_n decrease by the truncation. Hence the conditions for the truncated processes are weaker than those for the original processes.

2.11.1.1 Measurelike Processes

The preceding theorem is valid for arbitrary index sets $\mathcal{F}$. Consider the special case that $\mathcal{F}$ is a set of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$ on a measurable space $(\mathcal{X}, \mathcal{A})$ that satisfies a uniform entropy condition:

$$(2.11.5) \quad \int_0^\infty \sup_{Q \in \mathcal{Q}} \sqrt{\log N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q))} d\varepsilon < \infty.$$

Then the preceding theorem readily yields a central limit theorem for processes with increments that are bounded by a (random) L_2 -metric on $\mathcal{F}$. Call the processes Z_{ni} *measurelike* with respect to (random) measures μ_{ni} if

$$(Z_{ni}(f) - Z_{ni}(g))^2 \leq \int (f - g)^2 d\mu_{ni}, \quad \text{every } f, g \in \mathcal{F}.$$

For measurelike processes, the random semimetric d_n is bounded above by the $L_2(\sum \mu_{ni})$ -semimetric, and the entropy condition (2.11.2) can be related to the uniform entropy condition.

2.11.6 Lemma. *Let $\mathcal{F}$ be a class of measurable functions with envelope function F . Let $Z_{n1}, \dots, Z_{nm_n}$ be measurelike processes indexed by $\mathcal{F}$. If $\mathcal{F}$ satisfies the uniform entropy condition (2.11.5) for a set $\mathcal{Q}$ that contains the measures $\sum_{i=1}^{m_n} \mu_{ni}$ and $\sum_{i=1}^{m_n} \mu_{ni} F^2 = O_P^*(1)$, then the entropy condition (2.11.2) is satisfied.*

Proof. Set $\mu_n = \sum_{i=1}^{m_n} \mu_{ni}$. Since d_n is bounded by the $L_2(\mu_n)$ -semimetric, we have

$$\int_0^{\delta_n} \sqrt{\log N(\varepsilon, \mathcal{F}, d_n)} d\varepsilon \leq \int_0^{\delta_n / \|F\|_{\mu_n}} \sqrt{\log N(\varepsilon \|F\|_{\mu_n}, \mathcal{F}, L_2(\mu_n))} d\varepsilon \|F\|_{\mu_n},$$

on the set where $\|F\|_{\mu_n}^2 = \mu_n F^2$ is finite. Abbreviate

$$J(\delta) = \int_0^\delta \sup_Q \sqrt{\log N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q))} d\varepsilon.$$

On the set where $\|F\|_{\mu_n} > \eta$, the right side of the next-to-last displayed equation is bounded by $J(\delta_n/\eta) O_P(1)$. This converges to zero in probability for every $\eta > 0$. On the set where $\|F\|_{\mu_n} \leq \eta$, we have the bound $J(\infty)\eta$. This can be made arbitrarily small by the choice of η . Thus, the entropy condition (2.11.2) is satisfied. ■

2.11.7 Example. Suppose the independent processes $Z_{n1}, \dots, Z_{nm_n}$ are measurelike. Let $\mathcal{F}$ satisfy the uniform entropy condition. Assume that, for some probability measure P with $P^* F^2 < \infty$,

$$\begin{aligned} \mathbf{E}^* \sum_{i=1}^{m_n} \mu_{ni} F^2 \{ \mu_{ni} F^2 > \eta \} &\rightarrow 0, & \text{for every } \eta > 0, \\ \sup_{\|f-g\|_{P,2} < \delta_n} \mathbf{E}^* \sum_{i=1}^{m_n} \mu_{ni} (f-g)^2 &\rightarrow 0, & \text{for every } \delta_n \downarrow 0, \\ \sum_{i=1}^{m_n} \mu_{ni} F^2 &= O_P^*(1). \end{aligned}$$

Then the preceding lemma verifies the entropy condition (2.11.2), while the first two conditions of the display ensure the other conditions of Theorem 2.11.1. Note that $\|Z_{ni}\|_{\mathcal{F}} \leq |Z_{ni}(f)| + 2(\mu_{ni} F^2)^{1/2}$, for any f .

Thus, the sequence $\sum_{i=1}^{m_n} (Z_{ni} - \mathbf{E} Z_{ni})$ converges in $\ell^\infty(\mathcal{F})$ provided that the sequence of covariance functions converges pointwise and that measurability conditions are met.

2.11.8 Example (Weighted empirical processes). For each n , let $X_{n1}, \dots, X_{nm_n}$ be independent random elements in a measurable space $(\mathcal{X}, \mathcal{A})$. Let X_{ni} have law P_{ni} and let $P_{ni}f$ exist for each element f of a class $\mathcal{F}$ of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$. Given a triangular array of constants c_{ni} , consider the weighted empirical process $\mathbb{G}_n(f) = \sum_{i=1}^{m_n} c_{ni}(f(X_{ni}) - P_{ni}f)$. Suppose $\mathcal{F}$ satisfies the uniform entropy condition and that

$$\max_{1 \leq i \leq m_n} |c_{ni}| \rightarrow 0,$$

$$\sum_{i=1}^{m_n} c_{ni}^2 P_{ni} \leq P,$$

for a probability measure P with $P^*F^2 < \infty$.

Then under measurability conditions, the sequence $\mathbb{G}_n$ converges weakly to a Gaussian process in $\ell^\infty(\mathcal{F})$, provided the sequence converges marginally. Furthermore, there is a version of the limit process with uniformly continuous sample paths with respect to the $L_2(P)$ -semimetric.

This follows from the preceding example applied to the processes $Z_{ni} = c_{ni}\delta_{X_{ni}}$, which are measurelike for the measures $\mu_{ni} = c_{ni}^2\delta_{X_{ni}}$.

2.11.2 Bracketing

For each n , let $Z_{n1}, \dots, Z_{nm_n}$ be independent stochastic processes indexed by a common index set $\mathcal{F}$. In this section we aim at generalizations of the bracketing central limit theorem, Theorem 2.5.6, in two directions: we prove a version for the non-i.i.d. case, and we also weaken the entropy conditions to conditions involving the existence of either majorizing measures or certain Gaussian processes.

For the first generalization, define, for every n , the bracketing number $N_{[]}(\varepsilon, \mathcal{F}, L_2^n)$ as the minimal number of sets N_ε in a partition $\mathcal{F} = \cup_{j=1}^{N_\varepsilon} \mathcal{F}_{\varepsilon j}^n$ of the index set into sets $\mathcal{F}_{\varepsilon j}^n$ such that, for every partitioning set $\mathcal{F}_{\varepsilon j}^n$

$$\sum_{i=1}^{m_n} \mathbb{E}^* \sup_{f, g \in \mathcal{F}_{\varepsilon j}^n} |Z_{ni}(f) - Z_{ni}(g)|^2 \leq \varepsilon^2.$$

Note that the partitions are allowed to depend on n .

2.11.9 Theorem (Bracketing CLT). For each natural number n , let $Z_{n1}, \dots, Z_{n, m_n}$ be independent stochastic processes with finite second moments indexed by a totally bounded semimetric space $(\mathcal{F}, \rho)$. Suppose

$$\sum_{i=1}^{m_n} \mathbb{E}^* \|Z_{ni}\|_{\mathcal{F}} \{ \|Z_{ni}\|_{\mathcal{F}} > \eta \} \rightarrow 0, \quad \text{for every } \eta > 0,$$

$$\sup_{\rho(f, g) < \delta_n} \sum_{i=1}^{m_n} \mathbb{E} (Z_{ni}(f) - Z_{ni}(g))^2 \rightarrow 0, \quad \text{for every } \delta_n \downarrow 0,$$

$$\int_0^{\delta_n} \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L_2^n)} d\varepsilon \rightarrow 0, \quad \text{for every } \delta_n \downarrow 0.$$

Then the sequence $\sum_{i=1}^{m_n} (Z_{ni} - \mathbb{E}Z_{ni})$ is asymptotically tight in $\ell^\infty(\mathcal{F})$ and converges in distribution provided it converges marginally. If the partitions can be chosen independent of n , then the middle of the displayed conditions is unnecessary.

A central limit theorem for the empirical process of i.i.d. observations can be recovered from the preceding theorem by setting $Z_{ni}(f) = n^{-1/2}f(X_i)$. Then the bracketing numbers in the theorem may be reduced to $N_{[]}(\varepsilon, \mathcal{F}, L_2(P))$, and we recover the bracketing central limit theorem, Theorem 2.5.6. The extension Theorem 2.5.8 of the latter theorem, which uses a combination of $L_2(P)$ - and $L_{2,\infty}(P)$ -entropies, also has a version for processes: the preceding theorem remains true if $N_{[]}(\varepsilon, \mathcal{F}, L_2^n)$ is replaced by the minimal number of sets N_ε in a partition $\mathcal{F} = \cup_{j=1}^{N_\varepsilon} \mathcal{F}_{\varepsilon j}^n$ of the index set in sets $\mathcal{F}_{\varepsilon j}^n$ such that, for every j and every n ,

$$\sup_{f,g \in \mathcal{F}_{\varepsilon j}^n} \sum_{i=1}^{m_n} \mathbb{E} (Z_{ni}(f) - Z_{ni}(g))^2 \leq \varepsilon^2,$$

$$\sup_{t>0} \sum_{i=1}^{m_n} t^2 \mathbb{P}^* \left(\sup_{f,g \in \mathcal{F}_{\varepsilon j}^n} |Z_{ni}(f) - Z_{ni}(g)| > t \right) \leq \varepsilon^2.$$

These bracketing numbers appear to be rather hard to work with.

Another refinement of the theorem is to replace its entropy assumption by a majorizing measure condition. This entails the existence of partitions $\mathcal{F} = \cup_j \mathcal{F}_{\varepsilon j}^n$ as before and discrete probability measures μ_n on $\mathcal{F}$ such that

$$(2.11.10) \quad \sup_f \int_0^{\delta_n} \sqrt{\log \frac{1}{\mu_n(\mathcal{F}_\varepsilon^n f)}} d\varepsilon \rightarrow 0, \quad \text{for every } \delta_n \downarrow 0.$$

Here $\mathcal{F}_\varepsilon^n f$ is defined as the partitioning set at level ε to which f belongs. This majorizing measure condition is a weakening of finiteness of the entropy integral. Entropies correspond to certain uniform majorizing measures; the majorizing measure criterion permits one to measure the size of $\mathcal{F}$ in a nonuniform way.^b

Majorizing measures are also hard to work with. The resulting central limit theorem can be expressed more elegantly in terms of the existence of Gaussian semimetrics. For simplicity, we utilize (only) a single semimetric, although it may be fruitful to allow the semimetric to depend on n . Call a semimetric ρ *Gaussian* if it is the standard deviation semimetric

$$\rho(f, g) = \left(\mathbb{E} (G(f) - G(g))^2 \right)^{1/2}$$

of a tight, zero-mean, Gaussian random element G in $\ell^\infty(\mathcal{F})$. The connection with majorizing measures is the characterization of continuity of Gaussian processes by majorizing measures. See Section 2.2.6 for a discussion. In this chapter this deep characterization is used only in the proof of the following theorem to translate its condition into the majorizing measure condition (2.11.10). The proof itself is next based on the existence of the majorizing measure.

^b See the Problems and Section 2.2.6.

Call a semimetric ρ *Gaussian-dominated* if it is bounded above (on $\mathcal{F} \times \mathcal{F}$) by a Gaussian semimetric. Any semimetric ρ such that

$$\int_0^\infty \sqrt{\log N(\varepsilon, \mathcal{F}, \rho)} d\varepsilon < \infty$$

is Gaussian-dominated (see Problem 2.11.4).

2.11.11 Theorem (CLT by Gaussian hypotheses). *For each natural number n , let $Z_{n1}, \dots, Z_{n, m_n}$ be independent stochastic processes indexed by an arbitrary index set $\mathcal{F}$. Suppose that there exists a Gaussian-dominated semimetric ρ on $\mathcal{F}$ such that*

$$\sum_{i=1}^{m_n} \mathbb{E}^* \|Z_{ni}\|_{\mathcal{F}} \{ \|Z_{ni}\|_{\mathcal{F}} > \eta \} \rightarrow 0, \quad \text{for every } \eta > 0,$$

$$\sum_{i=1}^{m_n} \mathbb{E} (Z_{ni}(f) - Z_{ni}(g))^2 \leq \rho^2(f, g), \quad \text{for every } f, g,$$

$$\sup_{t>0} \sum_{i=1}^{m_n} t^2 \mathbb{P}^* \left(\sup_{f, g \in B(\varepsilon)} |Z_{ni}(f) - Z_{ni}(g)| > t \right) \leq \varepsilon^2,$$

for every ρ -ball $B(\varepsilon) \subset \mathcal{F}$ of radius less than ε and for every n . Then the sequence $\sum_{i=1}^{m_n} (Z_{ni} - \mathbb{E} Z_{ni})$ is asymptotically tight in $\ell^\infty(\mathcal{F})$. It converges in distribution provided it converges marginally.

The specialization of the preceding theorem to the case of i.i.d. observations is of interest because of the use of a Gaussian semimetric instead of entropy integrals. For instance, the theorem contains the classical Chibisov-O'Reilly theorem on the weighted empirical distribution function on the real line (see Example 2.11.15).

To recover the i.i.d. case, let $X_1, \dots, X_n$ be a random sample in a measurable space $(\mathcal{X}, \mathcal{A})$, and set $Z_{ni}(f) = f(X_i)/\sqrt{n}$, for a class $\mathcal{F}$ of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$.

Recall that $\|f\|_{P,2,\infty} = \sup_{t>0} tP(|f| > t)^{1/2}$ is the weak L_2 -pseudonorm.

2.11.12 Corollary. *Let $\mathcal{F}$ be a pre-Gaussian class of measurable functions whose envelope function possesses a weak second moment. Suppose there exists a Gaussian-dominated semimetric ρ on $\mathcal{F}$ such that, for every $\varepsilon > 0$,*

$$\left\| \sup_{f, g \in B(\varepsilon)} |f - g| \right\|_{P,2,\infty} \leq \varepsilon,$$

for every ρ -ball $B(\varepsilon) \subset \mathcal{F}$ of radius ε . Then $\mathcal{F}$ is P -Donsker.

Proof. Since $\mathcal{F}$ is P -pre-Gaussian and $\|P\|_{\mathcal{F}} \leq P^*F < \infty$, there exists a tight version of Brownian motion. This means that the $L_2(P)$ semimetric e_P is Gaussian. The semimetric $d = \sqrt{e_P^2 + \rho^2}$ is the standard deviation metric of the sum of Brownian motion and an independent copy of the process corresponding to ρ . Thus, this semimetric is Gaussian-dominated also. The conditions of the preceding theorem are satisfied for d playing the role of ρ . ■

2.11.13 Example (Jain–Marcus theorem). For each $n \in \mathbb{N}$, let $Z_{n1}, \dots, Z_{n,m_n}$ be independent stochastic processes indexed by an arbitrary index set $\mathcal{F}$ such that

$$|Z_{ni}(f) - Z_{ni}(g)| \leq M_{ni} \rho(f, g), \quad \text{for every } f, g,$$

for independent random variables $M_{n1}, \dots, M_{n,m_n}$ and a semimetric ρ such that

$$\int_0^\infty \sqrt{\log N(\varepsilon, \mathcal{F}, \rho)} d\varepsilon < \infty,$$

$$\sum_{i=1}^{m_n} \mathbb{E} M_{ni}^2 = O(1).$$

If the triangular array of norms $\|Z_{ni}\|_{\mathcal{F}}$ satisfies the Lindeberg condition, then the sequence $\sum_{i=1}^{m_n} (Z_{ni} - \mathbb{E} Z_{ni})$ converges in distribution in $\ell^\infty(\mathcal{F})$ to a tight Gaussian process provided the sequence of covariance functions converges pointwise on $\mathcal{F} \times \mathcal{F}$.

This follows from Theorems 2.11.9 and 2.11.11. The entropy condition implies that ρ is Gaussian-dominated.

In view of the preceding discussion, the conditions may be relaxed in two ways: the L_2 -norm can be partly replaced by the weak L_2 -norm, and the entropy criterion can be replaced by the majorizing measure criterion.

2.11.14 Example. Let $Z, Z_1, Z_2, \dots$ be i.i.d. stochastic processes indexed by the unit interval $\mathcal{F} = [0, 1] \subset \mathbb{R}$, with $\|Z\|_{\mathcal{F}} \leq 1$, such that

$$\mathbb{E}|Z(f) - Z(g)| \leq K|f - g|,$$

for some constant K . Then the sequence $n^{-1/2} \sum_{i=1}^n (Z_i - \mathbb{E} Z_i)$ converges in distribution to a tight Gaussian process in $\ell^\infty[0, 1]$. Note that the condition bounds the mean of the increments, not the increments themselves as required by the Jain–Marcus theorem. On the other hand, the index set must be a compact interval in the real line and the processes uniformly bounded.

The result can be deduced from Theorem 2.11.11. For every interval $[a, a + \varepsilon]$,

$$\sup_{a < f \leq g < a + \varepsilon} |Z(f) - Z(g)| \leq \lim_{n \rightarrow \infty} \sum_{k=1}^{2^n} |Z(a + \varepsilon k 2^{-n}) - Z(a + \varepsilon(k-1)2^{-n})|.$$

(The process is separable, with any dense subset of $[0, 1]$ as the separant, because it is continuous in probability; the sums on the right side are increasing in n and bound dyadic increments.) The right-hand side has mean bounded by $K\varepsilon$. Since the processes are bounded by 1

$$\mathbb{E} \sup_{a < f \leq g < a + \varepsilon} |Z(f) - Z(g)|^2 \leq 2K\varepsilon.$$

Every set of diameter ε for the semimetric $\rho(f, g) = |f - g|^{1/2}$ is contained in an interval $[a, a + \varepsilon^2]$. Conclude that the condition of Theorem 2.11.11 is up to a constant satisfied for this semimetric. This semimetric is Gaussian-dominated, because it has a finite entropy integral.

2.11.15 Example (Weighted empirical distribution function). Let $X_1, X_2, \dots$ be i.i.d. random variables with the uniform distribution P on $[0, 1] \subset \mathbb{R}$. For a fixed function $q: (0, 1/2] \rightarrow \mathbb{R}^+$, consider the class of functions

$$\mathcal{F} = \left\{ \frac{1_{(0,t]}}{q(t)} : 0 < t \leq \frac{1}{2} \right\}.$$

For simplicity, assume that the function $1/q$ is decreasing. The empirical process indexed by this class is the classical weighted empirical process (restricted to $0 < t \leq 1/2$ for convenience). According to the Chibisov-O'Reilly theorem[‡], the following statements are equivalent:

- (i) $\mathcal{F}$ is Donsker;
- (ii) $\mathcal{F}$ is pre-Gaussian and $1/q(t) = o(1/\sqrt{t})$;
- (iii) $\int_0^{1/2} t^{-1} \exp(-\varepsilon q^2(t)/t) dt < \infty$, for every $\varepsilon > 0$.

The equivalence of (ii) and (iii) can be argued from properties of Brownian motion. Here we deduce that (ii) implies (i) from Theorem 2.11.11.

In view of the second condition of (ii), the envelope function $F = (1/q)1_{(0,1/2]}$ has a weak second moment and $\|F\|_{\mathcal{F}} = \sup\{t/q(t) : 0 < t \leq 1/2\}$ is finite. Hence the pre-Gaussianity implies the existence of a tight version of Brownian motion; its standard deviation metric is Gaussian. Since for $s < t$,

$$\frac{1_{(0,s]}}{q(s)} - \frac{1_{(0,t]}}{q(t)} = \frac{1_{(s,t]}}{q(t)} + \left(\frac{1}{q(t)} - \frac{1}{q(s)} \right) 1_{(0,s]},$$

the square of this metric equals

$$\rho^2(s, t) = \frac{t-s}{q^2(t)} + \left| \frac{1}{q(t)} - \frac{1}{q(s)} \right|^2 s, \quad s \leq t.$$

If $\rho(s, t) < \varepsilon$, then both terms on the right are bounded by ε^2 . Conclude that for every fixed s ,

$$\sup_{\substack{s < t \\ \rho(s, t) < \varepsilon}} \left| \frac{1_{(0,s]}}{q(s)} - \frac{1_{(0,t]}}{q(t)} \right| \leq \sup_{t > s} \frac{\varepsilon 1_{(s,t]}}{\sqrt{t-s}} + \frac{\varepsilon 1_{(0,s]}}{\sqrt{s}} = \frac{\varepsilon 1_{(s,1]}}{\sqrt{\cdot-s}} + \frac{\varepsilon 1_{(0,s]}}{\sqrt{s}}.$$

The $L_{2,\infty}(P)$ -norm of the function $x \mapsto 1_{(s,1]}/\sqrt{x-s}$ equals 1 for every s . It follows that the $L_{2,\infty}(P)$ -norm of the left side of the preceding display is bounded by a multiple of ε .

2.11.16 Example (Monotone processes). Let Z be a stochastic process indexed by an interval $[a, b] \subset \mathbb{R}$, whose sample paths $t \mapsto Z(t)$ are cadlag and nondecreasing. If $\mathbb{E}Z^2(a) < \infty$ and $\mathbb{E}Z^2(b) < \infty$, then Z satisfies the central limit theorem in $\ell^\infty[a, b]$. More precisely, if $Z_1, Z_2, \dots$ are i.i.d. copies of Z , then the sequence $n^{-1/2} \sum_{i=1}^n (Z_i - \mathbb{E}Z_i)$ converges in $\ell^\infty[a, b]$ to a tight Gaussian process.

We shall derive this from Theorem 2.11.9. The condition that $Z(a)$ and $Z(b)$ have finite second moments are necessary for the convergence of the processes evaluated at a and b , respectively.

[‡] Shorack and Wellner (1986), page 462.

Since we can replace $Z(t)$ by $Z(t) - Z(a)$, we may assume that $Z(a) = 0$. The envelope function of Z is $\|Z\| = Z(b)$ and is square integrable by assumption. It suffices to verify the entropy condition of Theorem 2.11.9, where we shall choose the partitions independent of n . Define $F(t) = EZ(t)Z(b)$. This function is right-continuous with left limits, and is nondecreasing from $F(a) = 0$ to $F(b) = EZ^2(b)$. Given $\varepsilon > 0$ choose a partition $a = t_0 < t_1 < \cdots < t_N = b$ such that $F(t_i-) - F(t_{i-1}) < \varepsilon^2$ for every i . Then, for every i ,

$$E \sup_{t_{i-1} \leq s, t < t_i} |Z(s) - Z(t)|^2 \leq E(Z(t_i-) - Z(t_{i-1}))Z(b) < \varepsilon^2.$$

The number of points in the partition can be chosen smaller than a constant times $1/\varepsilon^2$. (Make sure that the points where F jumps more than ε^2 are among $t_0, \dots, t_N$.) Thus the entropy condition of Theorem 2.11.9 is satisfied easily for the partition $[a, b] = \cup(t_{i-1}, t_i] \cup \{a\}$.

The proofs of Theorems 2.11.9 and 2.11.11 are based on Bernstein's inequality combined with a chaining argument that utilizes majorizing measures. The following lemma is used to control the finite suprema over the links at a fixed level in the chain.

2.11.17 Lemma. *Let X be an arbitrary random variable such that*

$$P(|X| > x) \leq 2e^{-\frac{1}{2} \frac{x^2}{b+ax}},$$

for every $x > 0$. Then there exists a universal constant K such that

$$E|X|1_A \leq K \left(a \log \frac{1}{\mu} + \sqrt{b} \sqrt{\log \frac{1}{\mu}} \right) (\mu + P(A)),$$

for every measurable set A and every constant $0 < \mu < e^{-1}$.

Proof. The condition implies the upper bound $2 \exp(-x^2/(4b))$ on $P(|X| > x)$, for every $x \leq b/a$, and the upper bound $2 \exp(-x/(4a))$, for all other positive x . Consequently, the same upper bounds hold for *all* $x > 0$ for the probabilities $P(|X|1_{\{|X| \leq b/a\}} > x)$ and $P(|X|1_{\{|X| > b/a\}} > x)$, respectively. By Lemma 2.2.1, this implies that the Orlicz norms $\|X1_{\{|X| \leq b/a\}}\|_{\psi_2}$ and $\|X1_{\{|X| > b/a\}}\|_{\psi_1}$ are up to constants bounded by $\sqrt{b}$ and a , respectively. For any random variable Y , Jensen's inequality yields for any convex, increasing, non-negative function ψ ,

$$\begin{aligned} E|Y|1_A &\leq \psi^{-1} \left(E\psi \left(\frac{|Y|}{\|Y\|_{\psi}} \right) \frac{1_A}{P(A)} \right) \|Y\|_{\psi} P(A) \\ &\leq \psi^{-1} \left(\frac{1}{P(A)} \right) \|Y\|_{\psi} P(A). \end{aligned}$$

Apply this to the random variables $|X|1_{\{|X| \leq b/a\}}$ and $|X|1_{\{|X| > b/a\}}$ separately and use the triangle inequality to find that

$$E|X|1_A \lesssim \psi_2^{-1} \left(\frac{1}{P(A)} \right) \sqrt{b} P(A) + \psi_1^{-1} \left(\frac{1}{P(A)} \right) a P(A).$$

Finally, apply the inequality $p \log^k(1 + 1/p) \lesssim (p + \mu) \log^k(1/\mu)$, which is valid for small $\mu > 0$ and $k = 1/2, 1$. ■

Proof of Theorems 2.11.9 and 2.11.11. There exists a sequence of numbers $\eta_n \downarrow 0$ such that $\sum E^* \|Z_{ni}\|_{\mathcal{F}} \{ \|Z_{ni}\|_{\mathcal{F}} > \eta_n \} \rightarrow 0$. Therefore, it is no loss of generality to assume that $\|Z_{ni}\|_{\mathcal{F}} \leq \eta_n$ for every i and n . Otherwise, replace Z_{ni} by $Z_{ni} 1\{\|Z_{ni}\|_{\mathcal{F}}^* \leq \eta_n\}$; the conditions of the theorems remain valid, and the difference between the centered sums and the centered truncated sums converges to zero.

Under the conditions of both theorems, there exists for every n a sequence of nested partitions $\mathcal{F} = \cup_j \mathcal{F}_{qj}^n$ and discrete subprobability measures μ_n on $\mathcal{F}$ such that, for every j and n ,

$$(2.11.18) \quad \lim_{q_0 \rightarrow \infty} \limsup_{n \rightarrow \infty} \sup_f \sum_{q > q_0} 2^{-q} \sqrt{\log \frac{1}{\mu_n(\mathcal{F}_q^n f)}} = 0,$$

$$\sup_{f, g \in \mathcal{F}_{qj}^n} \sum_{i=1}^{m_n} E(Z_{ni}(f) - Z_{ni}(g))^2 \leq 2^{-2q},$$

$$\sup_{t > 0} \sum_{i=1}^{m_n} t^2 P^* \left(\sup_{f, g \in \mathcal{F}_{qj}^n} |Z_{ni}(f) - Z_{ni}(g)| > t \right) \leq 2^{-2q}.$$

Here $\mathcal{F}_q^n f$ is the set in the q th partition to which f belongs. Under the conditions of Theorem 2.11.9, this is true with $1/\mu_n(\mathcal{F}_q^n f)$ replaced by the number N_q^n of sets in the q th partition. The measures μ_n can then be constructed as $\mu_n = \sum_q 2^{-q} \mu_{n,q}$, where $\mu_{n,q}(\mathcal{F}_q^n f) = (N_q^n)^{-1}$, for every f and q . (Alternatively, the expression $1/\mu_n(\mathcal{F}_q^n f)$ can be replaced by N_q^n throughout the proof.) Under the conditions of Theorem 2.11.11, a single measure $\mu = \mu_n$ can be derived from a majorizing measure corresponding to the Gaussian semimetric ρ . The existence of this measure is indicated in Proposition 2.2.21. The construction of a sequence of partitions (in sets of ρ -diameter at most 2^{-q}) is carried out in Lemma 2.11.21. In the following, it is not a loss of generality to assume that $\mu_n(\mathcal{F}_q^n f) \leq 1/4$ for every q and f . Most of the argument is carried out for a fixed n ; this index will be suppressed in the notation.

The remainder of the proof is similar to the proof of Theorem 2.5.6, except that Lemma 2.11.17 is substituted for Lemma 2.2.13 in the chaining argument, which now uses majorizing measures. Choose an element f_{qj} from each partitioning set $\mathcal{F}_{qj}$ and define

$$\pi_q f = f_{qj},$$

$$(\Delta_q f)_{ni} = \sup_{f, g \in \mathcal{F}_{qj}} |Z_{ni}(f) - Z_{ni}(g)|, \quad \text{if } f \in \mathcal{F}_{qj},$$

$$a_q f = 2^{-q} / \sqrt{\log \frac{1}{\mu(\mathcal{F}_{q+1} f)}}.$$

Next, for $q > q_0$, define indicator functions

$$\begin{aligned} (A_{q-1}f)_{ni} &= 1\{(\Delta_{q_0}f)_{ni} \leq a_{q_0}f, \dots, (\Delta_{q-1}f)_{ni} \leq a_{q-1}f\}, \\ (B_qf)_{ni} &= 1\{(\Delta_{q_0}f)_{ni} \leq a_{q_0}f, \dots, (\Delta_{q-1}f)_{ni} \leq a_{q-1}f, (\Delta_qf)_{ni} > a_qf\}, \\ (B_{q_0}f)_{ni} &= 1\{(\Delta_{q_0}f)_{ni} > a_{q_0}f\}. \end{aligned}$$

The indicator functions A_qf and B_qf are constant in f on each of the partitioning sets $\mathcal{F}_{qj}$ at level q , because the partitions are nested. Now decompose:

$$\begin{aligned} (2.11.19) \quad Z_{ni}(f) - Z_{ni}(\pi_{q_0}f) &= (Z_{ni}(f) - Z_{ni}(\pi_{q_0}f))(B_{q_0}f)_{ni} \\ &+ \sum_{q_0+1}^{\infty} (Z_{ni}(f) - Z_{ni}(\pi_qf))(B_qf)_{ni} \\ &+ \sum_{q_0+1}^{\infty} (Z_{ni}(\pi_qf) - Z_{ni}(\pi_{q-1}f))(A_{q-1}f)_{ni}. \end{aligned}$$

Center each term at zero expectation; take the sum over i ; and, finally, take the supremum over f for all three terms on the right separately. It will be shown that the resulting expressions converge to zero in mean as $n \rightarrow \infty$ followed by $q_0 \rightarrow \infty$, whence the centered processes Z_{ni}^o satisfy

$$(2.11.20) \quad \lim_{q_0 \rightarrow \infty} \limsup_{n \rightarrow \infty} E^* \left\| \sum_{i=1}^{m_n} (Z_{ni}^o(f) - Z_{ni}^o(\pi_{q_0}^n f)) \right\|_{\mathcal{F}} = 0.$$

In view of Theorem 1.5.6, this gives a complete proof in the case that the partitions do not depend on n . The case that the partitions depend on n requires an additional argument, given at the end of this proof.

Since $(\Delta_qf)_{ni} \leq 2\eta_n$, the first term in (2.11.19) is zero as soon as $2\eta_n \leq \inf_f a_{q_0}f$. For every fixed q_0 , this is true for sufficiently large n , because, by assumption (2.11.18), $\inf_f a_qf$ (which depends on n) is bounded away from zero as $n \rightarrow \infty$ for every fixed q_0 .

By the nesting of the partitions, $(\Delta_qf)_{ni} \leq (\Delta_{q-1}f)_{ni}$, which is bounded by $a_{q-1}f$ on the set where $(B_nqf)_{ni} = 1$. It follows that

$$\begin{aligned} |Z_{ni}(f) - Z_{ni}(\pi_qf)| (B_qf)_{ni} &\leq (\Delta_qf)_{ni} (B_qf)_{ni} \leq a_{q-1}f, \\ \text{var} \left(\sum_{i=1}^{m_n} (\Delta_qf)_{ni} (B_qf)_{ni} \right) &\leq \sum_{i=1}^{m_n} a_{q-1}f E(\Delta_qf)_{ni} \{(\Delta_qf)_{ni} > a_qf\} \\ &\leq 2 \frac{a_{q-1}f}{a_qf} 2^{-2q}, \end{aligned}$$

by Problem 2.5.5. Combination with Bernstein's inequality 2.2.10 shows that the random variable $\sum_{i=1}^{m_n} (\Delta_qf)_{ni} (B_qf)_{ni}$ centered at its expectation satisfies the condition of Lemma 2.11.17 with $b = 2(a_{q-1}f/a_qf) 2^{-2q}$ and $a = 2a_{q-1}f$. Thus, the

lemma implies that, for every f and measurable set A ,

$$\begin{aligned} & \mathbb{E} \left| \sum_{i=1}^{m_n} (\Delta_q f)_{ni} (B_q f)_{ni} - \mathbb{E}(\Delta_q f)_{ni} (B_q f)_{ni} \right| 1_A \\ & \lesssim \left(a_{q-1} f \log \frac{1}{\mu(\mathcal{F}_q f)} + \sqrt{\frac{a_{q-1} f}{a_q f}} 2^{-q} \sqrt{\log \frac{1}{\mu(\mathcal{F}_q f)}} \right) (\mathbb{P}(A) + \mu(\mathcal{F}_q f)) \\ & \lesssim 2^{-q} \sqrt{\log \frac{1}{\mu(\mathcal{F}_q f)}} (\mathbb{P}(A) + \mu(\mathcal{F}_q f)), \end{aligned}$$

since $\mu(\mathcal{F}_{q+1} f) \leq \mu(\mathcal{F}_q f)$. For each q , let $\Omega = \cup_j \Omega_{qj}$ be a partition of the underlying probability space such that the maximum $\left\| \sum_{i=1}^{m_n} (\Delta_q f)_{ni} (B_q f)_{ni} - \mathbb{E}(\Delta_q f)_{ni} (B_q f)_{ni} \right\|_{\mathcal{F}}$ is achieved at f_{qj} on the set Ω_{qj} . For every q , there are as many sets Ω_{qj} as there are sets $\mathcal{F}_{qj}$ in the q th partition. Then

$$\begin{aligned} & \mathbb{E} \left\| \sum_{q>q_0} \sum_{i=1}^{m_n} (\Delta_q f)_{ni} (B_q f)_{ni} - \mathbb{E}(\Delta_q f)_{ni} (B_q f)_{ni} \right\|_{\mathcal{F}} \\ & \leq \sum_{q>q_0} \sum_j \mathbb{E} \left| \sum_{i=1}^{m_n} (\Delta_q f_{qj})_{ni} (B_q f_{qj})_{ni} - \mathbb{E}(\Delta_q f)_{ni} (B_q f_{qj})_{ni} \right| 1_{\Omega_{qj}} \\ & \leq \sum_j \sup_f \sum_{q>q_0} 2^{-q} \sqrt{\log \frac{1}{\mu(\mathcal{F}_q f)}} (\mathbb{P}(\Omega_{qj}) + \mu(\mathcal{F}_q f_{qj})) \\ & \leq 2 \sup_f \sum_{q>q_0} 2^{-q} \sqrt{\log \frac{1}{\mu(\mathcal{F}_q f)}}. \end{aligned}$$

This converges to zero as $n \rightarrow \infty$ followed by $q_0 \rightarrow \infty$ and takes care of the second term resulting from the decomposition (2.11.19), apart from the centering. The centering is bounded by

$$\begin{aligned} \sum_{q>q_0} \sum_{i=1}^{m_n} \mathbb{E}(\Delta_q f)_{ni} (B_q f)_{ni} & \leq \sum_{q>q_0} \sum_{i=1}^{m_n} \mathbb{E}(\Delta_q f)_{ni} \{ (\Delta_q f)_{ni} > a_q f \} \\ & \leq \sup_f \sup_{t>0} \sum_{q>q_0} \frac{2^{-2q}}{a_q f} \lesssim \sup_f \sum_{q>q_0} 2^{-q} \sqrt{\log \frac{1}{\mu(\mathcal{F}_q f)}}. \end{aligned}$$

This concludes the proof that the second term in the decomposition resulting from (2.11.19) converges to zero as $n \rightarrow \infty$ followed by $q_0 \rightarrow \infty$.

The third term can be handled in a similar manner. The proof of (2.11.20) is complete.

Finally, if the partitions depend on n , then the discrepancies at the ends of the chains must be analyzed separately. If the q_0 th partition consists of $N_{q_0}^n$ sets, then the combination of Bernstein's inequality and Lemma 2.2.13 yields

$$\mathbb{E} \sup_{\rho(f,g)<\delta_n} \left| \sum_{i=1}^{m_n} Z_{ni}^o(\pi_{q_0}^n f) - Z_{ni}^o(\pi_{q_0}^n g) \right| \lesssim \log N_{q_0}^n \eta_n + \sqrt{\log N_{q_0}^n} (2^{-q_0} + \delta_n).$$

The entropy condition of Theorem 2.11.9 implies that $2^{-q_0} \log N_{q_0}^n \rightarrow 0$ as $n \rightarrow \infty$ followed by $q_0 \rightarrow \infty$. Hence the expression in the display converges to zero as $n \rightarrow \infty$ followed by $q_0 \rightarrow \infty$. Combine this with (2.11.20) to see that the sequence $\sum_{i=1}^{m_n} Z_{ni}^o$ is asymptotically equicontinuous with respect to ρ . Finally, Theorem 1.5.7 shows that the sequence is asymptotically tight. ■

2.11.21 Lemma. *Let (T, d) be a semimetric space and μ be a Borel probability measure on T such that*

$$(2.11.22) \quad \limsup_{\delta \downarrow 0} \sup_{t \in T} \int_0^\delta \sqrt{\log \frac{1}{\mu(B(t, \varepsilon))}} d\varepsilon = 0.$$

Then there exists a probability measure m on T and a sequence of nested, measurable partitions $T = \cup_i T_{qi}$ such that $\text{diam } T_{qi} \leq 2^{-q}$ for every i and

$$\lim_{q_0 \rightarrow \infty} \sup_{t \in T} \sum_{q > q_0} 2^{-q} \sqrt{\log \frac{1}{m(T_q t)}} = 0,$$

where $T_q t$ is the partitioning set T_{qi} at level q to which t belongs.

Proof. The set T is totally bounded: the existence of infinitely many disjoint balls $B(t_i, \delta)$ of fixed radius $\delta > 0$ would require that $\mu(B(t_i, \delta)) \rightarrow 0$ as $i \rightarrow \infty$. However,

$$\sup_t \delta \sqrt{\log \frac{1}{\mu(B(t, \delta))}} \leq \sup_t \int_0^\delta \sqrt{\log \frac{1}{\mu(B(t, \varepsilon))}} d\varepsilon.$$

The right-hand side is finite by assumption.

Fix $\delta > 0$. Find a point t_1 that (nearly) maximizes $\mu(B(t, \delta))$ over $t \in T$. Precisely, let $\mu(B(t_1, \delta)) \geq r \sup_t \mu(B(t, \delta))$ for some fixed $0 < r \leq 1$. Set $m\{t_1\} = \mu(B(t_1, \delta))$, and let $T_1 = B(t_1, 2\delta)$ be the ball of radius 2δ around t_1 . For every $t \in T_1$ (even for every $t \in T$),

$$r \mu(B(t, \delta)) \leq \mu(B(t_1, \delta)) = m\{t_1\} \leq m(T_1).$$

Next, find a t_2 that nearly maximizes $\mu(B(t, \delta))$ over $t \in T - T_1$. Set $m\{t_2\} = \mu(B(t_2, \delta))$, and let $T_2 = B(t_2, 2\delta) - T_1$. For every $t \in T_2$

$$r \mu(B(t, \delta)) \leq \mu(B(t_2, \delta)) = m\{t_2\} \leq m(T_2).$$

Continue this process, constructing disjoint sets $T_1, T_2, \dots, T_I$ until the set of t with $\rho(t, t_i) \geq 2\delta$ for every i is empty. Then $T = \cup_i T_i$ is a finite partition of T in sets of diameter at most 2δ and $m(T_i) \geq r \mu(B(t_i, \delta))$ for every $t \in T_i$.

Apply this construction for $\delta = 2^{-q-1}$ and every natural number q . This yields a sequence of partitions $T = \cup_i \bar{T}_{qi}$ and measures $\bar{m}_q$ such that

$$\sum_{q > q_0} 2^{-q} \sqrt{\log \frac{1}{\bar{m}_q(\bar{T}_q t)}} \leq \sum_{q > q_0} 2^{-q} \sqrt{\log \frac{1}{r \mu(B(t, 2^{-q-1}))}}.$$

The right side is bounded by a multiple of

$$\int_0^{2^{-q_0}} \sqrt{\log 1/\mu(B(t, \varepsilon))} d\varepsilon$$

if $\mu(B(t, \varepsilon))$ is bounded away from 1, which may be assumed.

The present sequence of partitions need not be nested. Replace the q th partition by the partition in the sets $\cap_{p=1}^q \bar{T}_{p, i_p}$, where $i_1, \dots, i_q$ range over all possible values. For each set in the q th partition, define

$$m_q\left(\cap_{p=1}^q \bar{T}_{p, i_p}\right) = \prod_{p=1}^q \bar{m}_p(\bar{T}_{p, i_p}).$$

(The exact location of the mass is irrelevant.) Next, set $m = \sum_q 2^{-q} m_q$. Then m is a subprobability measure and

$$\sum_{q > q_0} 2^{-q} \sqrt{\log \frac{1}{m(T_q t)}} \leq \sum_{q > q_0} 2^{-q} \sqrt{\log 2^q + \sum_{p=1}^q \log \frac{1}{\bar{m}_p(\bar{T}_p t)}}.$$

This converges to zero uniformly in t as $q_0 \rightarrow \infty$. ■

2.11.3 Changing Classes of Functions

Let $X_1, X_2, \dots$ be a sequence of independent random elements with common law P on a measurable space $(\mathcal{X}, \mathcal{A})$, and let $x \mapsto f_{n,t}(x)$ be functions from $\mathcal{X}$ to $\mathbb{R}$ indexed by $n \in \mathbb{N}$ and a fixed, totally bounded semimetric space (T, ρ) . We wish to derive conditions for the stochastic processes

$$\left\{ n^{-1/2} \sum_{i=1}^n (f_{n,t}(X_i) - P f_{n,t}) : t \in T \right\}$$

to converge in distribution in the space $\ell^\infty(T)$. These are the empirical processes $\{\mathbb{G}_n f_{n,t} : t \in T\}$ indexed by classes of functions $\mathcal{F}_n = \{f_{n,t} : t \in T\}$ changing with n .

This situation fits in the general set-up of this section upon setting $Z_{ni}(t) = f_{n,t}(X_i)/\sqrt{n}$.

Given envelope functions F_n , assume that

$$(2.11.23) \quad \begin{aligned} P^* F_n^2 &= O(1), \\ P^* F_n^2 \{F_n > \eta \sqrt{n}\} &\rightarrow 0, & \text{for every } \eta > 0, \\ \sup_{\rho(s,t) < \delta_n} P(f_{n,s} - f_{n,t})^2 &\rightarrow 0, & \text{for every } \delta_n \downarrow 0. \end{aligned}$$

Then the central limit theorem holds under an entropy condition. The following theorems impose a uniform entropy condition and a bracketing entropy condition, respectively.

2.11.24 Theorem. For each n , let $\mathcal{F}_n = \{f_{n,t}: t \in T\}$ be a class of measurable functions indexed by a totally bounded semimetric space (T, ρ) such that the classes $\mathcal{F}_{n,\delta} = \{f_{n,s} - f_{n,t}: \rho(s, t) < \delta\}$ and $\mathcal{F}_{n,\delta}^2$ are P -measurable for every $\delta > 0$. Suppose (2.11.23) holds, as well as

$$\sup_Q \int_0^{\delta_n} \sqrt{\log N(\varepsilon \|F_n\|_{Q,2}, \mathcal{F}_n, L_2(Q))} d\varepsilon \rightarrow 0, \quad \text{for every } \delta_n \downarrow 0.$$

Then the sequence $\{\mathbb{G}_n f_{n,t}: t \in T\}$ is asymptotically tight in $\ell^\infty(T)$ and converges in distribution to a Gaussian process provided the sequence of covariance functions $P f_{n,s} f_{n,t} - P f_{n,s} P f_{n,t}$ converges pointwise on $T \times T$.

2.11.25 Theorem. For each n , let $\mathcal{F}_n = \{f_{n,t}: t \in T\}$ be a class of measurable functions indexed by a totally bounded semimetric space (T, ρ) . Suppose (2.11.23) holds, as well as

$$\int_0^{\delta_n} \sqrt{\log N_{[]}(\varepsilon \|F_n\|_{P,2}, \mathcal{F}_n, L_2(P))} d\varepsilon \rightarrow 0, \quad \text{for every } \delta_n \downarrow 0.$$

Then the sequence $\{\mathbb{G}_n f_{n,t}: t \in T\}$ is asymptotically tight in $\ell^\infty(T)$ and converges in distribution to a tight Gaussian process provided the sequence of covariance functions $P f_{n,s} f_{n,t} - P f_{n,s} P f_{n,t}$ converges pointwise on $T \times T$.

Proofs. The random distance given in Theorem 2.11.1 reduces to

$$d_n^2(s, t) = \frac{1}{n} \sum_{i=1}^n (f_{n,s} - f_{n,t})^2(X_i) = \mathbb{P}_n(f_{n,s} - f_{n,t})^2.$$

It follows that $N(\varepsilon, T, d_n) = N(\varepsilon, \mathcal{F}_n, L_2(\mathbb{P}_n))$, for every $\varepsilon > 0$. If F_n is replaced by $F_n \vee 1$, then the conditions of the theorem still hold. Hence, assume without loss of generality that $F_n \geq 1$. Insert the bound on the covering numbers and next make a change of variables to bound the entropy integral (2.11.2) by

$$\int_0^{\delta_n} \sqrt{\log N(\varepsilon \|F_n\|_{\mathbb{P}_n,2}, \mathcal{F}_n, L_2(\mathbb{P}_n))} d\varepsilon \|F_n\|_{\mathbb{P}_n,2}.$$

This converges to zero in probability for every $\delta_n \downarrow 0$. Apply Theorem 2.11.1 to obtain the first theorem.

The second theorem is an immediate consequence of Theorem 2.11.9. ■

2.11.26 Example. The uniform entropy condition of the first theorem is certainly satisfied if, for each n , the set of functions $\mathcal{F}_n = \{f_{n,t}: t \in T\}$ is a VC-class with VC-dimension bounded by some constant independent of n .

Problems and Complements

1. The random entropy condition (2.11.2) implies that the sequence $N(\varepsilon, \mathcal{F}, d_n)$ is bounded in probability for every $\varepsilon > 0$.

[Hint: For every $\delta_n \leq \varepsilon$,

$$P\left(N(\varepsilon, \mathcal{F}, d_n) \geq M_n\right) \leq P\left(\int_0^{\delta_n} \sqrt{\log N(\varepsilon, \mathcal{F}, d_n)} d\varepsilon \geq \delta_n \sqrt{\log M_n}\right).$$

Given $M_n \rightarrow \infty$, choose $\delta_n \downarrow 0$ such that $\delta_n \sqrt{\log M_n}$ is bounded away from zero.]

2. Let $\mathcal{F} = \cup_j \mathcal{F}_{qj}$ be a sequence ($q \in \mathbb{N}$) of nested partitions of an arbitrary set $\mathcal{F}$. Take arbitrary $f_{qj} \in \mathcal{F}_{qj}$ for every partitioning set, and define $\pi_q f = f_{qj}$ and $\mathcal{F}_q f = \mathcal{F}_{qj}$ if $f \in \mathcal{F}_{qj}$. Let $\rho(f, g)$ be 2^{-q_0} for the first value q_0 (counting from 1) such that f and g do not belong to the same partitioning set at level q_0 . (Set $\rho(f, g) = 0$ if f and g are never separated.) Then ρ defines a semimetric on $\mathcal{F}$ inducing open balls

$$B(f, 2^{-q}) = \mathcal{F}_q f, \quad \text{for every } q.$$

Furthermore, if $G(f) = \sum_q 2^{-q} \xi_{q, \pi_q f}$ for i.i.d. standard normal variables $\xi_{q, f_{qj}}$, then

$$\text{var}(G(f) - G(g)) = \frac{8}{3} \rho^2(f, g),$$

for every f, g .

[Hint: We have $\rho(f, g) < 2^{-q}$ if and only if f and g are in the same partitioning set at level q if and only if $\mathcal{F}_q f = \mathcal{F}_q g$. Of course, $g \in \mathcal{F}_q g$ for every g .]

3. In the situation of the preceding problem, let the number N_q of partitioning sets $\mathcal{F}_{qj}$ at level q satisfy

$$\sum_q 2^{-q} \sqrt{\log N_q} < \infty.$$

Then the entropy numbers for the semimetric ρ satisfy

$$\int_0^\infty \sqrt{\log N(\varepsilon, \mathcal{F}, \rho)} d\varepsilon < \infty.$$

Conclude that the Gaussian process G defined in the preceding problem has a version with bounded, uniformly ρ -continuous sample paths. Hence ρ is a Gaussian semimetric.

[Hint: The convergence of the integral is immediate from the fact that $N(2^{-q}, \mathcal{F}, \rho) = N_q$. The continuity follows from Corollary 2.2.9 and Problem 2.2.17. Also see Section 2.2.6.]

4. Any semimetric d on an arbitrary set $\mathcal{F}$ such that $\int_0^\infty \sqrt{\log N(\varepsilon, \mathcal{F}, d)} d\varepsilon < \infty$ is Gaussian-dominated.

[Hint: Construct a sequence of nested partitions of $\mathcal{F} = \cup_j \mathcal{F}_{qj}$ in sets of δ -diameter at most 2^{-q} for each q . The number N_q of sets in the q th partition can be chosen such that $\sum_q 2^{-q} \sqrt{\log N_q} < \infty$. As in the preceding problems, the semimetric ρ defined from this sequence of partitions is Gaussian by the preceding problem and has the property that each ball $B_\rho(f, 2^{-q})$ equals a partitioning set; hence these balls have d -diameter at most 2^{-q} for every q . The latter implies that $d \leq 2\rho$.]

5. Any semimetric d on an arbitrary set $\mathcal{F}$ for which there exists a Borel probability measure μ with

$$\limsup_{\delta \downarrow 0} \int_{\mathcal{F}} \int_0^\delta \sqrt{\log 1/\mu(B(f, \varepsilon))} d\varepsilon = 0$$

is Gaussian-dominated. Here $B(f, \varepsilon)$ is the d -ball of radius ε .

[Hint: By Lemma 2.11.21, there exists a sequence of nested partitions of $\mathcal{F} = \cup_i \mathcal{F}_{q_i}$ in sets of δ -diameter at most 2^{-q} for each q and a Borel probability measure m such that

$$\lim_{q_0 \rightarrow \infty} \sup_f \sum_{q > q_0} 2^{-q} \sqrt{\log \frac{1}{m(\mathcal{F}_q)}} = 0.$$

As in the preceding problems, the semimetric ρ defined from this sequence of partitions has the property that each ball $B_\rho(f, 2^{-q})$ equals a partitioning set. Hence

$$\limsup_{\delta \downarrow 0} \int_{\mathcal{F}} \int_0^\delta \sqrt{\log 1/m(B_\rho(f, \varepsilon))} d\varepsilon = 0,$$

where $B_\rho(f, \varepsilon)$ is a ball with respect to ρ . By Proposition 2.2.21, the Gaussian process G defined in the first problem has a version with bounded, uniformly ρ -continuous sample paths. The inequalities $\text{diam } B(f, 2^{-q}) \leq 2^{-q}$ imply that $d \leq 2\rho$.]

6. Let $\mathcal{F}$ be a class of measurable functions such that

$$\int \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L_{2,\infty}(P))} d\varepsilon$$

is finite. Then there exists a Gaussian semimetric ρ on $\mathcal{F}$ such that

$$\left\| \sup_{f, g \in B(\varepsilon)} |f - g| \right\|_{P, 2, \infty} \leq \varepsilon,$$

for every ρ -ball $B(\varepsilon) \subset \mathcal{F}$ of radius ε .

[Hint: The finiteness of the integral implies the existence of a sequence of partitions of $\mathcal{F}$, as in the preceding problems, with the further property

$$\left\| \sup_{f, g \in \mathcal{F}_{q_j}} |f - g| \right\|_{P, 2, \infty} \leq 2^{-q}.$$

The semimetric ρ in the preceding problems is Gaussian, and every ball of radius 2^{-q} coincides with a partitioning set. Thus the inequality is valid for $\varepsilon = 2^{-q}$ and hence up to a constant 2 for every ε . Change the metric to remove the 2.]

7. Corollary 2.11.12 is a refinement of Theorem 2.5.6.
8. If the Lindeberg condition on norms in Theorems 2.11.1 and 2.11.9 is replaced by the two assumptions

$$\sup_f \left| \sum_{i=1}^{m_n} \mathbb{E} Z_{ni}(f) \{ \|Z_{ni}\|_{\mathcal{F}}^* > \eta \} \right| \rightarrow 0,$$

$$\sum_{i=1}^{m_n} \mathbb{P}^* (\|Z_{ni}\|_{\mathcal{F}} > \eta) \rightarrow 0,$$

then the conclusion that the sequence of processes $\sum_{i=1}^{m_n} (Z_{ni} - EZ_{ni})$ is asymptotically tight is still valid.

2.12

Partial-Sum Processes

The name “Donsker class of functions” was chosen in honor of Donsker’s theorem on weak convergence of the empirical distribution function. A second famous theorem by Donsker concerns the partial-sum process

$$\mathbb{Z}_n(s) = \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor ns \rfloor} Y_i = \frac{1}{\sqrt{n}} \sum_{i=1}^k Y_i, \quad \frac{k}{n} \leq s < \frac{k+1}{n},$$

for i.i.d. random variables $Y_1, \dots, Y_n$ with zero mean and variance 1. Donsker essentially proved that the sequence of processes $\{\mathbb{Z}_n(t): 0 \leq t \leq 1\}$ converges in distribution in the space $\ell^\infty[0, 1]$ to a standard Brownian motion process [Donsker (1951)].

In this chapter we discuss generalizations of this result in two directions. The first is to replace the Y_i s by processes $\{f(X_i): f \in \mathcal{F}\}$ and consider convergence in $\ell^\infty([0, 1] \times \mathcal{F})$. The second is to consider real variables on a lattice. Partial sums are then obtained by intersecting the lattice with sets in a given collection.

2.12.1 Sequential Empirical Process

Let $X_1, \dots, X_n$ be i.i.d. random elements with law P in the measurable space $(\mathcal{X}, \mathcal{A})$, and let $\mathcal{F}$ be a collection of square-integrable, measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$. The *sequential empirical process* is defined as

$$\mathbb{Z}_n(s, f) = \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor ns \rfloor} (f(X_i) - Pf) = \sqrt{\frac{\lfloor ns \rfloor}{n}} \mathbb{G}_{\lfloor ns \rfloor}(f),$$

where $\mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - P)$ is the empirical process indexed by $\mathcal{F}$. The index (s, f) ranges over $[0, 1] \times \mathcal{F}$. The covariance function of $\mathbb{Z}_n$ is given by

$$\text{cov}(\mathbb{Z}_n(s, f), \mathbb{Z}_n(t, g)) = \frac{\lfloor ns \rfloor \wedge \lfloor nt \rfloor}{n} (Pfg - PfPg).$$

By the multivariate central limit theorem, the marginals of the sequence of processes $\{\mathbb{Z}_n(s, f) : (s, f) \in [0, 1] \times \mathcal{F}\}$ converge to the marginals of a Gaussian process $\mathbb{Z}$. The latter is known as the *Kiefer-Müller process*; it has mean zero and covariance function

$$\text{cov}(\mathbb{Z}(s, f), \mathbb{Z}(t, g)) = (s \wedge t)(Pfg - PfPg).$$

The aim of this section is to show that the weak convergence $\mathbb{Z}_n \rightsquigarrow \mathbb{Z}$ is uniform with respect to the semimetric of $\ell^\infty([0, 1] \times \mathcal{F})$ for every Donsker class of functions $\mathcal{F}$. More precisely, for every such $\mathcal{F}$, the sequence $\mathbb{Z}_n$ is asymptotically tight and there exists a tight, Borel measurable version of the Kiefer-Müller process $\mathbb{Z}$.

In accordance with the general results on Gaussian processes, tightness and measurability of the limit process $\mathbb{Z}$ are equivalent to the existence of a version of with all sample paths $(s, f) \mapsto \mathbb{Z}(s, f)$ uniformly bounded and uniformly continuous with respect to the semimetric whose square is given by

$$E(\mathbb{Z}(s, f) - \mathbb{Z}(t, g))^2 = |s - t| [\rho_P^2(f)1_{s>t} + \rho_P^2(g)1_{s\leq t}] + (s \wedge t) \rho_P^2(f - g).$$

This intrinsic semimetric is somewhat complicated. However, it is up to a constant bounded (on $[0, 1] \times \mathcal{F}$) by the natural semimetric $|s - t| + \rho_P(f - g)$ if $\mathcal{F}$ is bounded for ρ_P ; in particular, if $\mathcal{F}$ is a Donsker class. By the addendum of Theorem 1.5.7, asymptotic equicontinuity with respect to the intrinsic semimetric is equivalent to asymptotic equicontinuity with respect to the natural semimetric whenever $[0, 1] \times \mathcal{F}$ is totally bounded under the natural semimetric, particularly if $\mathcal{F}$ is a Donsker class.

Call $\mathcal{F}$ a *functional Donsker class* if and only if the sequence $\mathbb{Z}_n$ converges in distribution in $\ell^\infty([0, 1] \times \mathcal{F})$ to a tight limit. Then the result can be expressed as follows.

2.12.1 Theorem. *A class of measurable functions is functionally Donsker if and only if it is Donsker.*

Proof. The empirical process $\mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - P)$ can be recovered from the sequential process as $\mathbb{G}_n(f) = \mathbb{Z}_n(1, f)$. Since a restriction map is continuous, a class is certainly Donsker if it is functionally Donsker.

For the converse, it suffices to establish the asymptotic equicontinuity of the sequence $\mathbb{Z}_n$. With the usual notation, $\mathcal{F}_\delta = \{f - g : f, g \in \mathcal{F}, \rho_P(f - g) < \delta\}$, the triangle inequality yields

$$(2.12.2) \quad \begin{aligned} & \sup_{|s-t|+\rho_P(f,g)<\delta} |\mathbb{Z}_n(s, f) - \mathbb{Z}_n(t, g)| \\ & \leq \sup_{|s-t|<\delta} \|\mathbb{Z}_n(s, f) - \mathbb{Z}_n(t, f)\|_{\mathcal{F}} + \sup_{0 \leq t \leq 1} \|\mathbb{Z}_n(t, f)\|_{\mathcal{F}_\delta}. \end{aligned}$$

In the second term on the right the parameter t may be restricted to the points k/n with k ranging over $1, 2, \dots, n$. Thus, this term is equal to $\max_{k \leq n} \|\sqrt{k/n} \mathbb{G}_k\|_{\mathcal{F}_\delta}$. By Ottaviani's inequality A.1.1,

$$\mathbf{P}^* \left(\max_{k \leq n} \sqrt{k/n} \|\mathbb{G}_k\|_{\mathcal{F}_\delta} > 2\varepsilon \right) \leq \frac{\mathbf{P}^* (\|\mathbb{G}_n\|_{\mathcal{F}_\delta} > \varepsilon)}{1 - \max_{k \leq n} \mathbf{P}^* (\sqrt{k/n} \|\mathbb{G}_k\|_{\mathcal{F}_\delta} > \varepsilon)}.$$

If the class $\mathcal{F}$ is Donsker, then the sequence $\mathbb{G}_n$ is asymptotically equicontinuous. Thus, the numerator converges to zero as $n \rightarrow \infty$ followed by $\delta \downarrow 0$. The terms of the maximum in the denominator indexed by $k \leq n_0$ can be controlled with the help of the inequality $\sqrt{k} \|\mathbb{G}_k\|_{\mathcal{F}_\delta} \leq 2 \sum_{i=1}^{n_0} F(X_i) + 2n_0 P^* F$ for an envelope function F . For sufficiently large n_0 , the terms indexed by $k > n_0$ are bounded away from 1 by the asymptotic equicontinuity of $\mathbb{G}_n$. Conclude that the denominator is bounded away from zero. Hence the second term on the right in (2.12.2) converges to zero in probability as $n \rightarrow \infty$ followed by $\delta \downarrow 0$.

To prove the same for the first term on the right in (2.12.2), it suffices to prove convergence to zero of

$$\mathbf{P}^* \left(\max_{0 \leq j\delta \leq 1} \sup_{j\delta \leq s \leq (j+1)\delta} \|\mathbb{Z}_n(s, f) - \mathbb{Z}_n(j\delta, f)\|_{\mathcal{F}} > 2\varepsilon \right).$$

By the stationarity of the increments of $\mathbb{Z}_n$ in s , the at most $\lceil 1/\delta \rceil$ terms in the maximum are identically distributed. Thus, the probability can be bounded by

$$\left\lceil \frac{1}{\delta} \right\rceil \mathbf{P}^* \left(\sup_{0 \leq s \leq \delta} \|\mathbb{Z}_n(s, f)\|_{\mathcal{F}} > 2\varepsilon \right).$$

Again replace the continuous index s by a discrete one and conclude from Ottaviani's inequality that the preceding display is not larger than

$$\left\lceil \frac{1}{\delta} \right\rceil \mathbf{P}^* \left(\max_{k \leq n\delta} \sqrt{k/n} \|\mathbb{G}_k\|_{\mathcal{F}} > 2\varepsilon \right) \leq \frac{\lceil 1/\delta \rceil \mathbf{P}^* (\sqrt{\lfloor n\delta \rfloor/n} \|\mathbb{G}_{\lfloor n\delta \rfloor}\|_{\mathcal{F}} > \varepsilon)}{1 - \max_{k \leq n\delta} \mathbf{P}^* (\sqrt{k/n} \|\mathbb{G}_k\|_{\mathcal{F}} > \varepsilon)}.$$

By the portmanteau theorem, the $\limsup$ as $n \rightarrow \infty$ of the probability in the numerator is bounded by $\mathbf{P}(\|\mathbb{G}\|_{\mathcal{F}} \geq \varepsilon/\delta^{1/2})$. Since the norm $\|\mathbb{G}\|_{\mathcal{F}}$ of a Brownian bridge has moments of all orders (cf. Proposition A.2.3), the latter probability converges to zero faster than any power of δ as $\delta \downarrow 0$. Conclude that the numerator converges to zero as $n \rightarrow \infty$ followed by $\delta \downarrow 0$. By a similar argument as before, but now also using the fact that $\mathbf{P}(\|\mathbb{G}\|_{\mathcal{F}} > \varepsilon) < 1$ for every $\varepsilon > 0$ (cf. Problem A.2.5), the denominator remains bounded away from zero. ■

2.12.2 Partial-Sum Processes on Lattices

For each positive integer n , let $Y_{n1}, \dots, Y_{nm_n}$ be independent, real-valued random variables, and let $Q_{n1}, \dots, Q_{nm_n}$ be deterministic probability measures on a measurable space $(\mathcal{X}, \mathcal{A})$. Given a collection $\mathcal{C}$ of measurable subsets of $\mathcal{X}$, consider the stochastic process

$$\mathbb{S}_n(C) = \sum_{i=1}^{m_n} Y_{ni} Q_{ni}(C).$$

Thus, $\mathbb{S}_n$ is a randomly weighted sum of the measures Q_{ni} . Both the classical partial-sum process and its smoothed version are of this type.

2.12.3 Example. Let $\mathcal{X} = [0, 1]$ be the unit interval in the real line and $\mathcal{C}$ the collection of cells $[0, s]$ with $0 \leq s \leq 1$. If Q_{ni} is the Dirac measure at the point i/n (for $1 \leq i \leq n$), then

$$\mathbb{S}_n([0, s]) = \sum_{i/n \leq s} Y_{ni} = \sum_{i=1}^k Y_{ni}, \quad \text{if } \frac{k}{n} \leq s < \frac{k+1}{n}.$$

The choice $Y_{ni} = Y_i/\sqrt{n}$ for an i.i.d. sequence $Y_1, Y_2, \dots$ gives the classical partial-sum process.

2.12.4 Example. Let $\mathcal{X} = [0, 1]$ be the unit interval in the real line and $\mathcal{C}$ the collection of cells $[0, s]$ with $0 \leq s \leq 1$. If Q_{ni} is the uniform measure on the interval $[(i-1)/n, i/n]$, then

$$\mathbb{S}_n([0, s]) = \sum_{i=1}^k Y_{ni} + \frac{s - k/n}{1/n} Y_{n,k+1}, \quad \text{if } \frac{k}{n} \leq s < \frac{k+1}{n}.$$

This is a linear interpolation of the partial-sum process in the previous example.

2.12.5 Example. If each Q_{ni} equals the Dirac measure at some point $x_{ni} \in \mathcal{X}$, then the general process reduces to

$$\mathbb{S}_n(C) = \sum_{i: x_{ni} \in C} Y_{ni}.$$

This may be visualized as random weights Y_{ni} being located at fixed points x_{ni} . Each set C is charged with the sum of the weights that it carries.

The process $\mathbb{S}_n = \sum Y_{ni} Q_{ni}$ is the sum of the independent processes $Z_{ni} = Y_{ni} Q_{ni}$. Since

$$(Z_{ni}(C) - Z_{ni}(D))^2 \leq Y_{ni}^2 Q_{ni}(C \triangle D),$$

the processes Z_{ni} are measurelike with respect to the measures $\mu_{ni} = Y_{ni}^2 Q_{ni}$. Thus for a collection $\mathcal{C}$ that satisfies the uniform entropy condition, Theorem 2.11.1 combined with Lemma 2.11.6 readily yields a central limit theorem for the sequence $\mathbb{S}_n$. This is true in particular for VC-classes of sets.

2.12.6 Theorem. For each n , let $Q_{n1}, \dots, Q_{nm_n}$ be deterministic probability measures on some measurable space, and let $Y_{n1}, \dots, Y_{nm_n}$ be independent, real-valued random variables with mean zero, satisfying

$$\sum_{i=1}^{m_n} EY_{ni}^2 = O(1),$$

$$\sum_{i=1}^{m_n} EY_{ni}^2 \{ |Y_{ni}| > \eta \} \rightarrow 0, \quad \text{for every } \eta > 0.$$

Let $\mathcal{C}$ be a class of measurable sets that satisfies the uniform entropy condition, and assume that for some probability measure Q ,

$$(2.12.7) \quad \sup_{Q(C \triangle D) < \delta_n} \sum_{i=1}^{m_n} EY_{ni}^2 (Q_{ni}(C) - Q_{ni}(D))^2 \rightarrow 0, \quad \text{for every } \delta_n \downarrow 0.$$

Finally, suppose that the covariance function $ES_n(C)S_n(D)$ converges pointwise on $\mathcal{C} \times \mathcal{C}$. Then the sequence $S_n = \sum_{i=1}^{m_n} Y_{ni}Q_{ni}$ converges weakly in $\ell^\infty(\mathcal{C})$ to a tight Gaussian process with uniformly continuous sample paths with respect to the semimetric $Q(C \triangle D)$.

Proof. Apply Theorem 2.11.1 with $Z_{ni} = Y_{ni}Q_{ni}$ and ρ the $L_1(Q)$ -semimetric. Then $\|Z_{ni}\|_{\mathcal{F}}$ can be taken equal to $|Y_{ni}|$.

The entropy condition (2.11.2) is satisfied in view of Lemma 2.11.6. Since $\mathcal{C}$ satisfies the uniform entropy condition, it is totally bounded under the $L_1(Q)$ -semimetric (for any probability measure Q).

The measurability conditions listed before Theorem 2.11.1 are satisfied, because the suprema can be replaced by countable suprema. Indeed, since $\mathcal{C}$ satisfies the uniform entropy condition, it is totally bounded and hence separable in $L_1(\sum_{i=1}^{m_n} Q_{ni} + Q)$. Thus, there exists a countable subcollection that contains, for every $C \in \mathcal{C}$, a sequence C_k with $(\sum_{i=1}^{m_n} Q_{ni} + Q)(C_k \triangle C) \rightarrow 0$ as $k \rightarrow \infty$ (and n fixed). For this sequence, $S_n(C) = \lim_{k \rightarrow \infty} S_n(C_k)$. Similar arguments applied to the multiplier processes show that the suprema that are assumed measurable in Theorem 2.11.1 can be replaced by countable suprema.

The theorem follows from Theorem 2.11.1. ■

The covariance function of the process S_n is given by

$$ES_n(C)S_n(D) = \sum_{i=1}^{m_n} EY_{ni}^2 Q_{ni}(C)Q_{ni}(D).$$

In the special case that each Q_{ni} is a Dirac measure, we have the equality $Q_{ni}(C)Q_{ni}(D) = Q_{ni}(C \cap D)$ for every pair of sets, and the covariance function reduces to $Q_n(C \cap D)$ for the measure

$$Q_n = \sum_{i=1}^{m_n} EY_{ni}^2 Q_{ni}.$$

Then pointwise convergence of Q_n to a limit on the collection of sets $\{C \cap D: C, D \in \mathcal{C}\}$ verifies convergence of the sequence of covariance functions.

For any measures Q_{ni} , the sum on the left side of (2.12.7) is bounded by $Q_n(C \triangle D)$. Thus, consideration of the measures Q_n can also be helpful for verification of the other, more involved, condition of the theorem. A simple sufficient condition for (2.12.7) is that Q_n converges uniformly to Q on the collection of sets $\{C \triangle D: C, D \in \mathcal{C}\}$. This puts further restrictions on the class $\mathcal{C}$. A simple further sufficient condition is that the sequence Q_n converges weakly to Q and, for every $\varepsilon > 0$, the collection $\mathcal{C}$ can be covered by finitely many brackets $[C_l, C^u]$ of size $Q(C^u - C_l) < \varepsilon$ and having P -continuity sets C_l and C^u as boundaries (Problem 2.12.1).

2.12.8 Example. Let $\mathcal{X} = [0, 1]^d$ be the unit cube and $\mathcal{C}$ the collection of all cells $[0, t]$ with $0 \leq t \leq 1$. Let the collection of measures Q_{ni} be the n^d Dirac measures located at nodes of the regular grid consisting of the n^d points $\{1/n, 2/n, \dots, 1\}^d$. This gives a higher-dimensional version of the classical partial-sum process. If $Y_{ni} = Y_i/n^{d/2}$ for an i.i.d. mean-zero sequence $Y_1, Y_2, \dots$ with unit variance, then the measure Q_n reduces to

$$Q_n = \frac{1}{n^d} \sum_{i=1}^{n^d} Q_{ni}.$$

This is the discrete uniform measure on the grid points. The sequence Q_n converges weakly to Lebesgue measure, and the collection of cells $[0, t]$ satisfies the bracketing condition for Lebesgue measure. Thus, (2.12.7) is satisfied for Q equal to Lebesgue measure, and the sequence of covariance functions converges. The limiting Gaussian process has covariance function $\mathbb{E}\mathbb{S}(C)\mathbb{S}(D) = \lambda(C \cap D)$ for Lebesgue measure λ (a standard *Brownian sheet*).

2.12.9 Example. Let $\mathcal{X} = [0, 1]^d$ be the unit cube and $\mathcal{C}$ a VC-class. Partition $[0, 1]^d$ into n^d cubes of volume n^{-d} , and let the collection Q_{ni} be the set of uniform probability measures on the cubes. This gives a smoothed version of the partial-sum process in higher dimensions. If each of the Y_{ni} has variance n^{-d} , then the measure Q_n reduces to Lebesgue measure. Thus (2.12.7) is satisfied for Q equal to Lebesgue measure.

If the cubes in the partition are denoted C_{ni} , then the covariance function can be written

$$\frac{1}{n^d} \sum_{i=1}^{n^d} \frac{\lambda(C_{ni} \cap C)}{\lambda(C_{ni})} \frac{\lambda(C_{ni} \cap D)}{\lambda(C_{ni})} = \mathbb{E}\mathbb{E}(1_C(U) | \mathcal{A}_n) \mathbb{E}(1_D(U) | \mathcal{A}_n),$$

where U is the identity map on $[0, 1]^d$ equipped with Lebesgue measure and $\mathcal{A}_n$ is the σ -field generated by the partition. For $n \rightarrow \infty$, the sequence $\mathbb{E}(1_C(U) | \mathcal{A}_n)$ converges in probability to $1_C(U)$ for every set C . (This follows from a martingale convergence theorem or a direct argument: it is clear that $\mathbb{E}(h | \mathcal{A}_n) \rightarrow h$ in quadratic mean for every continuous function h ; each indicator function 1_C can be approximated arbitrarily closely by a continuous function.) Thus, the sequence of covariance functions converges to the limit $\mathbb{E}1_C(U)1_D(U) = \lambda(C \cap D)$.

Problems and Complements

1. Suppose the sequence of Borel measures Q_n converges weakly to a probability measure Q . Let $\mathcal{C}$ be a collection of Borel sets that for every $\varepsilon > 0$ can be covered with finitely many brackets $[C_l, C^u]$ of size $Q(C^u - C_l) < \varepsilon$ and consisting of Q -continuity sets C_l and C^u . Then $Q_n \rightarrow Q$ uniformly in $C \in \mathcal{C} \triangle \mathcal{C}$ and $C \in \mathcal{C} \cap \mathcal{C}$.

[Hint: The following problem is helpful. Compare the proof of Theorem 2.4.1.]

2. If $\mathcal{C}$ has the bracketing property as in the preceding problem, then so do the collections $\mathcal{C} \triangle \mathcal{C}$ and $\mathcal{C} \cap \mathcal{C}$.
3. If $Z_{ni}(s, f) = f(X_i)1\{i/n \leq t\}$, then the sequential empirical process can be written $\mathbb{Z}_n = n^{-1/2}(\sum_{i=1}^n Z_{ni} - \mathbb{E}Z_{ni})$. Next the results of Chapter 2.11 can be used to deduce that the sequence $\mathbb{Z}_n$ converges in distribution. Compare the result with the present chapter.

4. **(Time reversal)** If $\mathbb{Z}$ is a Kiefer-Müller process indexed by $[1, \infty) \times \mathcal{F}$, then the process $(t, f) \mapsto t\mathbb{Z}(1/t, f)$ is a Kiefer-Müller process indexed by $(0, 1] \times \mathcal{F}$.

5. If $\mathbb{Z}_n$ is the sequential empirical process, then the sequence of processes $(t, f) \mapsto \mathbb{Z}_n(t, f)/(t \vee 1)$ converges weakly in $\ell^\infty(\mathbb{R}^+ \times \mathcal{F})$.

[Hint: The process equals $\sqrt{n}(\lfloor nt \rfloor / n(t \vee 1))(\mathbb{P}_{\lfloor nt \rfloor} - P)$. Use the reverse martingale property of $\|\mathbb{P}_n - P\|_{\mathcal{F}}^*$ given in Lemma 2.4.6.]

6. Combination of the preceding problems yields that $\sup_{m \geq n} \sqrt{n} \|\mathbb{P}_m - P\|_{\mathcal{F}}$ converges weakly to $\sup_{0 \leq t \leq 1} \|\mathbb{Z}(t, f)\|_{\mathcal{F}}$. This generalizes results of Müller (1968). See Hjort and Fenstad (1992), Section 4, for applications of this result.

7. A *standard Brownian sheet* $\mathbb{S}$ indexed by $\mathbb{R}^+ \times [0, 1]$ is a zero-mean Gaussian process with continuous sample paths and covariance function

$$\text{cov}(\mathbb{S}(s, u), \mathbb{S}(t, v)) = (s \wedge t)(u \wedge v).$$

Then the process $\mathbb{S}(s, u) - u\mathbb{S}(s, 1)$ is a classical Kiefer-Müller process (a Kiefer-Müller process for $\mathcal{X} = [0, 1]$ with uniform measure and $\mathcal{F}$ the collection of indicators of cells $[0, u]$).

8. With $\mathbb{S}$ a standard Brownian sheet, consider the processes $\mathbb{S}(s, u) - su\mathbb{S}(1, 1)$ and $\mathbb{S}(s, u) - u\mathbb{S}(s, 1) - v\mathbb{S}(1, u) + us\mathbb{S}(1, 1)$. On which sides of the unit square are these processes zero almost surely? (The second process appears in connection with tests for independence; see Chapter 3.9 and Appendix A.2.3.)

2.13

Other Donsker Classes

In this section we consider some Donsker classes of interest, that do not fit well in the framework of the preceding chapters.

2.13.1 Sequences

A countable class of functions may be shown to be Donsker by any of the criteria we have discussed so far. A very simple sufficient condition is that the sequence converges to zero sufficiently fast.

2.13.1 Theorem (Donsker Sequences). *Any sequence $\{f_i\}$ of square-integrable, measurable functions with the property $\sum_{i=1}^{\infty} P(f_i - Pf_i)^2 < \infty$ is P -Donsker.*

Proof. For a fixed natural number m , define a partition $\{f_i\} = \cup_{i=1}^{m+1} \mathcal{F}_i$ by letting $\mathcal{F}_i$ consist of the single function f_i for $i \leq m$ and $\mathcal{F}_{m+1} = \{f_{m+1}, f_{m+2}, \dots\}$. Since the variation over the first m sets in the partition is zero,

$$\begin{aligned} P\left(\sup_i \sup_{f,g \in \mathcal{F}_i} |\mathbb{G}_n(f - g)| > \varepsilon\right) &\leq P\left(\sup_{f \in \mathcal{F}_{m+1}} |\mathbb{G}_n f| > \frac{\varepsilon}{2}\right) \\ &\leq \frac{4}{\varepsilon^2} \sum_{i=m+1}^{\infty} P(f_i - Pf_i)^2 < \infty, \end{aligned}$$

by Chebyshev's inequality. For sufficiently large m , this is smaller than any prescribed $\eta > 0$. The result follows from Theorem 1.5.6. ■

2.13.2 Elliptical Classes

The preceding theorem may be combined with Theorem 2.10.3 to conclude that the class

$$\left\{ \sum_{i=1}^{\infty} c_i f_i : \sum |c_i| \leq 1, \text{ and the series converges pointwise} \right\}$$

is Donsker for any given sequence f_i with $\sum_{i=1}^{\infty} P f_i^2 < \infty$. Under the additional condition that the functions f_i are orthogonal, this can be improved.

2.13.2 Theorem (Elliptical classes). *Let $\{f_i\}$ be a sequence of measurable functions such that $P f_i f_j = 0$ for every $i \neq j$ and $\sum_{i=1}^{\infty} P f_i^2 < \infty$. Then the class of all pointwise converging series $\sum_{i=1}^{\infty} c_i f_i$, such that $\sum_{i=1}^{\infty} c_i^2 \leq 1$, is P -Donsker.*

Proof. By the condition on c , each of the series $\sum_{i=1}^{\infty} c_i f_i$ converges pointwise as well as in $L_2(P)$. The class $\mathcal{F}$ of all these series is totally bounded in $L_2(P)$, because it is bounded and can be approximated arbitrarily closely by a finite-dimensional set, since

$$P \left(\sum_{i>m} c_i f_i \right)^2 = \sum_{i>m} c_i^2 P f_i^2 \leq \max_{i>m} P f_i^2 \rightarrow 0, \quad m \rightarrow \infty.$$

It suffices to show that the sequence of empirical processes $\mathbb{G}_n$ indexed by $\mathcal{F}$ is asymptotically equicontinuous with respect to the $L_2(P)$ -seminorm. For $f = \sum c_i f_i$, $g = \sum d_i f_i$, and any natural number k ,

$$\begin{aligned} |\mathbb{G}_n(f) - \mathbb{G}_n(g)|^2 &= \left| \sum_{i=1}^{\infty} (c_i - d_i) \mathbb{G}_n(f_i) \right|^2 \\ &\leq 2 \sum_{i=1}^k (c_i - d_i)^2 P f_i^2 \sum_{i=1}^k \frac{\mathbb{G}_n^2(f_i)}{P f_i^2} + 2 \sum_{i=k+1}^{\infty} (c_i - d_i)^2 \sum_{i=k+1}^{\infty} \mathbb{G}_n^2(f_i). \end{aligned}$$

In view of the assumption that $\|c - d\|_2 \leq \|c\|_2 + \|d\|_2 \leq 2$, this expression is bounded by

$$2\|f - g\|_{P,2}^2 \sum_{i=1}^k \frac{\mathbb{G}_n^2(f_i)}{P f_i^2} + 8 \sum_{i=k+1}^{\infty} \mathbb{G}_n^2(f_i).$$

Take the supremum over all pairs of series f and g with $\|f - g\|_{P,2} < \delta$. Since $E \mathbb{G}_n^2(f_i) \leq P f_i^2$, the expectation of this supremum is bounded by $2\delta^2 k + 8 \sum_{i=k+1}^{\infty} P f_i^2$. This expression can be made arbitrarily small by first choosing k large and next δ small. ■

In terms of an orthonormal sequence $\{\psi_i\}$ in $\mathcal{L}_2(P)$, the preceding theorem can be stated in the following manner. For a given sequence of numbers b_i , the *elliptical class*

$$\mathcal{F} = \left\{ \sum_{i=1}^{\infty} c_i \psi_i : \sum \frac{c_i^2}{b_i^2} \leq 1 \text{ and the series converges pointwise} \right\}$$

is P -Donsker if $\sum_{i=1}^{\infty} b_i^2 < \infty$. The latter condition is also known to be necessary. In fact, it is already necessary for the class to be pre-Gaussian.[†] The pointwise convergence of the series forming $\mathcal{F}$ appears not to be automatic. In view of the Cauchy-Schwarz inequality, a simple sufficient condition for absolute convergence at x is that $\sum \psi_i(x)^2 b_i^2 < \infty$.

Elliptical classes are of some interest, because some well known test statistics, including the Cramér-von Mises and the Anderson-Darling statistic, arise as the (generalized) Kolmogorov-Smirnov statistic $\|\mathbb{G}_n\|_{\mathcal{F}}$ indexed by an elliptical class. This is shown in the examples ahead. The Kolmogorov-Smirnov statistics corresponding to an elliptical class can be represented as a series of uncorrelated variables. Indeed, for $\mathcal{F}$ equal to the class of functions in the previous display,

$$\|\mathbb{G}_n\|_{\mathcal{F}}^2 = \sup_f \left| \sum_{i=1}^{\infty} c_i \mathbb{G}_n(\psi_i) \right|^2 = \sum_{i=1}^{\infty} b_i^2 \mathbb{G}_n^2(\psi_i),$$

in view of the Cauchy-Schwarz inequality. If the ψ_i are uncorrelated and of unit variance, then the sequence $\|\mathbb{G}_n\|_{\mathcal{F}}^2$ is asymptotically distributed as $\sum b_i^2 Z_i^2$ for an i.i.d. sequence $Z_1, Z_2, \dots$ of standard normal variables. This follows from the series representation for $\|\mathbb{G}_n\|_{\mathcal{F}}^2$; it can also be obtained from the fact that $\|\mathbb{G}_n\|_{\mathcal{F}}^2$ converges in distribution to the square norm of a Brownian bridge $\mathbb{G}$ and a series representation for $\|\mathbb{G}\|_{\mathcal{F}}^2$, which can be obtained in an analogous manner.

2.13.3 Example (Cramér-von Mises statistic). Let $\mathbb{P}_n$ be the empirical distribution of an i.i.d. sample of size n from the uniform distribution on the unit interval $[0, 1] \subset \mathbb{R}$. Write $\mathbb{G}_n(t) = \sqrt{n}(\mathbb{P}_n - P)[0, t]$ for the classical empirical process ($t \in [0, 1]$). Let $\mathcal{F}$ be the class of functions

$$\mathcal{F} = \left\{ \sum_{j=1}^{\infty} c_j \sqrt{2} \cos \pi j t : \sum_{j=1}^{\infty} c_j^2 \pi^2 j^2 \leq 1 \right\}.$$

Since the cosines are bounded, the series defining the elements of $\mathcal{F}$ are uniformly convergent in this case. The classical Cramér-von Mises statistic equals the square of the Kolmogorov-Smirnov statistic over the elliptical class $\mathcal{F}$:

$$\int_0^1 \mathbb{G}_n^2(t) dt = \|\mathbb{G}_n\|_{\mathcal{F}}^2.$$

To see this, note that the Cramér-von Mises statistic is the square of the $L_2[0, 1]$ -norm of the function $t \mapsto \mathbb{G}_n(t)$. Since the functions $\{\sqrt{2} \sin \pi j t : j = 1, 2, \dots\}$ form an orthonormal base of $L_2[0, 1]$, Parseval's formula yields

$$\int_0^1 \mathbb{G}_n^2(t) dt = \sum_{j=1}^{\infty} \left[\int_0^1 \mathbb{G}_n(t) \sqrt{2} \sin(\pi j t) dt \right]^2 = \sum_{j=1}^{\infty} \frac{1}{\pi^2 j^2} \mathbb{G}_n^2(\sqrt{2} \cos \pi j t).$$

(Note that $-\int \mathbb{G}_n f' = \int f d\mathbb{G}_n = \mathbb{G}_n f$ by the definitions of $t \mapsto \mathbb{G}_n(t)$ and $\mathbb{G}_n f$.) The functions $\{\sqrt{2} \cos \pi j t : j = 1, 2, \dots\}$ form an orthonormal system in $L_2[0, 1]$, so that the result follows from the general series representation of $\|\mathbb{G}_n\|_{\mathcal{F}}^2$ for elliptical classes.

[†] Dudley (1967b), Proposition 6.3.

2.13.4 Example (Watson). In the situation of the preceding example, let $\mathcal{F}$ be the elliptical class

$$\left\{ \sum_{j=1}^{\infty} (c_{2j-1} \sqrt{2} \cos 2\pi j t + c_{2j} \sqrt{2} \sin 2\pi j t) : \sum (c_{2j-1}^2 + c_{2j}^2) \pi^2 j^2 \leq 1 \right\}.$$

The series are again uniformly convergent. The Watson statistic equals the square of the Kolmogorov-Smirnov statistic over the elliptical class $\mathcal{F}$:

$$\int_0^1 [\mathbb{G}_n(t) - \int \mathbb{G}_n(s) ds]^2 dt = \|\mathbb{G}_n\|_{\mathcal{F}}^2.$$

This can be seen by a similar argument. The Watson statistic is the square of the $L_2[0, 1]$ -norm of the projection of the function $t \mapsto \mathbb{G}_n(t)$ on the mean-zero functions. The functions $\{\sqrt{2} \sin 2\pi j t, \sqrt{2} \cos 2\pi j t : j = 1, 2, \dots\}$ form an orthonormal base of the mean-zero functions. Application of Parseval's formula followed by partial integration as in the preceding example yields the result.

2.13.5 Example (Anderson-Darling statistic). In the situation of the preceding example, let $\mathcal{F}$ be the elliptical class

$$\left\{ \sum_{j=1}^{\infty} c_j \sqrt{2} p_j(2t-1) : \sum c_j^2 j(j+1) \leq 1 \text{ and pointwise convergence} \right\},$$

where the functions $p_0(u) = 1/\sqrt{2}$, $p_1(u) = \sqrt{3/2}u$, $p_2(u) = \sqrt{5/8}(3u^2 - 1)$, $p_3(u) = \sqrt{7/8}(5u^3 - 3u)$, $\dots$ are the orthonormalized Legendre polynomials in $L_2[-1, 1]$. (This is the orthonormal system obtained by applying the Gram-Schmidt procedure to the functions $1, u, u^2, \dots$.) The Anderson-Darling statistic equals the square of the Kolmogorov-Smirnov statistic over the elliptical class $\mathcal{F}$:

$$\int_0^1 \frac{\mathbb{G}_n^2(t)}{t(1-t)} dt = \|\mathbb{G}_n\|_{\mathcal{F}}^2.$$

The argument is slightly more involved than in the preceding examples, but it is based on the same idea. The normalized Legendre polynomials satisfy the differential equations $p_j''(u)(1-u^2) - 2up_j'(u) = -j(j+1)p_j(u)$ for $u \in [-1, 1]$ (Problem 2.13.1). By a change of variables and partial integration, it can be deduced that

$$\begin{aligned} 2 \int_0^1 p_i'(2t-1) p_j'(2t-1) t(1-t) dt \\ = -\frac{1}{4} \int_{-1}^1 p_i(u) [p_j''(u)(1-u^2) - 2up_j'(u)] du \\ = \frac{1}{4} j(j+1) \delta_{ij}. \end{aligned}$$

It follows that the functions $2\sqrt{2}p_i'(2t-1)\sqrt{t(1-t)}/\sqrt{j(j+1)}$ with j ranging over $\{1, 2, \dots\}$ form an orthonormal base of $L_2[0, 1]$. By Parseval's formula, the Anderson-Darling statistic equals

$$\sum_{j=1}^{\infty} \left[\int_0^1 \mathbb{G}_n(t) p_j'(2t-1) dt \right]^2 \frac{8}{j(j+1)} = \sum_{j=1}^{\infty} \frac{2}{j(j+1)} \mathbb{G}_n^2(p_j(2t-1)).$$

This equals $\|\mathbb{G}_n\|_{\mathcal{F}}^2$ by the general series representation of $\|\mathbb{G}_n\|_{\mathcal{F}}^2$ for elliptical classes.

2.13.3 Classes of Sets

The preceding chapters give a wide variety of relatively easy-to-apply sufficient conditions for the Donsker property. They have little to say about necessary conditions beyond the most abstract level or in concrete cases. For classes of sets, simple necessary conditions, up to measurability, are known.

Recall from Chapter 2.6 that for a given collection $\mathcal{C}$ of sets and given points, $\Delta_n(\mathcal{C}, X_1, \dots, X_n)$ denotes the number of subsets of $\{X_1, \dots, X_n\}$ picked out by $\mathcal{C}$. Also define $K_n(\mathcal{C}, X_1, \dots, X_n)$ to be the cardinality of a maximal subset of $\{X_1, \dots, X_n\}$ shattered by $\mathcal{C}$:

$$K_n(\mathcal{C}, X_1, \dots, X_n) = \max \left\{ \#A : \Delta_n(\mathcal{C}, A) = 2^{\#A} \right\},$$

where A ranges over all possible subsets of $\{X_1, \dots, X_n\}$.

2.13.6 Theorem. *For every pointwise-separable collection of measurable sets, the following statements are equivalent:*

- (i) $\log \Delta_n(\mathcal{C}, X_1, \dots, X_n) = o_P^*(\sqrt{n})$ and $\mathcal{C}$ is P -pre-Gaussian;
- (ii) $K_n(\mathcal{C}, X_1, \dots, X_n) = o_P^*(\sqrt{n})$ and $\mathcal{C}$ is P -pre-Gaussian;
- (iii) $\mathcal{C}$ is P -Donsker;
- (iv) $\log N(\varepsilon n^{-1/2}, \mathcal{C}, L_1(\mathbb{P}_n)) = o_P^*(\sqrt{n})$ for every $\varepsilon > 0$ and $\mathcal{C}$ is P -pre-Gaussian.

Proof. Giné and Zinn (1984) and Talagrand (1988). ■

Problems and Complements

1. The Legendre polynomials on $[-1, 1]$ satisfy the differential equation

$$p_j''(u)(1 - u^2) - 2up_j'(u) = -j(j+1)p_j(u).$$

[Hint: It suffices to prove the same relationship for the polynomials $p_j(u)$ defined as u^j minus the projection of u^j on the linear space spanned by $1, u, \dots, u^{j-1}$. The left side of the differential equation is a polynomial of degree j with leading term $-j(j+1)u^j$. By definition, the right side has the same property. It suffices to show that the left side is orthogonal to the functions $1, u, \dots, u^{j-1}$. By partial integration,

$$\begin{aligned} \int p_j'(u) 2u u^k du &= 2p_j(1) \pm 2p_j(-1) - \int p_j(u) 2(k+1)u^k du \\ &= 2p_j(1) \pm 2p_j(-1), \end{aligned}$$

where $\pm$ is $+$ if $k < j$ is even and $-$ otherwise. By using partial integration twice, this can be seen to be equal to $\int p_j''(u)(1 - u^2)u^k du$.]

2. The representation of the Watson statistic as the Kolmogorov-Smirnov statistic of an elliptic class is not unique. Consider the class

$$\mathcal{F} = \left\{ \sum_{j=1}^{\infty} c_j \sqrt{2} \sin \pi j t : \sum c_j^2 \pi^2 j^2 \leq 1 \right\}.$$

Then the Watson statistic equals $\|\mathbb{G}_n\|_{\mathcal{F}}^2$. (The terms in the accompanying series representation are not asymptotically independent; hence this representation is of less interest.)

[Hint: The functions $\{\sqrt{2} \cos \pi j t : j = 1, 2, \dots\}$ form an orthonormal base of the mean-zero functions in $L_2[0, 1]$.]

2.14

Maximal Inequalities and Tail Bounds

In this chapter we derive bounds on the size of the supremum $\|\mathbb{G}_n\|_{\mathcal{F}}$ of the empirical process, also known as *maximal inequalities*.

The first two sections are mostly concerned with inequalities for the expectation $\mathbb{E}^*\|\mathbb{G}_n\|_{\mathcal{F}}$. This is bounded in terms of the uniform entropy or bracketing entropy of the class $\mathcal{F}$ of functions. For bounds in terms of the size of the envelope function of the class, it suffices that the corresponding entropy integral converges. For bounds that incorporate the sizes of the individual functions $f \in \mathcal{F}$, the exact behavior of the entropy integral near zero is important. The latter bounds are derived with a view towards statistical applications in Part 3. In particular, several of the bounds are exactly right for handling the moduli of continuity that determine the rates of convergence of M -estimators.

The uniform entropy and bracketing integrals both are stronger than the L_2 -entropy integral, which corresponds to the limiting Brownian bridge process. An alternative to using these stronger entropy integrals is to make a finite sample correction to the Gaussian integral.

In passing we derive a classical result on VC-classes, which is suboptimal in its dependence on the sample size n , but useful for moderate values of n .

Some bounds on higher moments of $\|\mathbb{G}_n\|_{\mathcal{F}}$ come for free with the bound on the first moment. Alternatively, higher moments and exponential Orlicz norms can be bounded by a combination of the first moment $\mathbb{E}^*\|\mathbb{G}_n\|_{\mathcal{F}}$ and the corresponding moment of the envelope function, by general inequalities on sums of independent random variables.

Bounds on moments imply rates of decrease of tail probabilities of the form $\mathbb{P}^*(\|\mathbb{G}_n\|_{\mathcal{F}} > t)$. For uniformly bounded classes $\mathcal{F}$, these tails are exponentially small. In Section 2.14.6, we derive exponential bounds that give the correct constant in the exponent for such classes.

All bounds in this chapter measure the size of the supremum $\|\mathbb{G}_n\|_{\mathcal{F}}$, and may be contrasted with the results presented in Chapter 2.15, which consider the deviations of this variable from its mean. Unlike the bounds in the present chapter, these deviation bounds do not depend (much) on the size of the class $\mathcal{F}$ as long as the functions in the class are uniformly bounded. They show sub-Gaussian decrease of the tails and sharp concentration around the mean. Thus tail bounds for the variable $\|\mathbb{G}_n\|_{\mathcal{F}}$ can also be viewed as consisting of two parts: a bound on deviations from its mean $E^*\|\mathbb{G}_n\|_{\mathcal{F}}$ (valid under very mild conditions, as seen in the next chapter) and bounds on the latter mean (under conditions measuring the size of the class $\mathcal{F}$ and the functions in the class, as seen in the present chapter).

Throughout the chapter, $\mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - P)$ denotes the empirical process of an i.i.d. sample $X_1, \dots, X_n$ from a probability measure P , defined as the coordinate projections of a product probability space $(\mathcal{X}^\infty, \mathcal{A}^\infty, P^\infty)$.

2.14.1 Uniform Entropy

For a class $\mathcal{F}$ of measurable functions with measurable envelope function F define the *uniform entropy integral* by

$$J(\delta, \mathcal{F} | F, L_2) = \sup_Q \int_0^\delta \sqrt{1 + \log N(\varepsilon \|F\|_{Q,2}, \mathcal{F}, L_2(Q))} d\varepsilon,$$

where the supremum is taken over all discrete probability measures Q with $\|F\|_{Q,2} > 0$. The envelope function need not be the minimal envelope function. If the envelope function is clear from the context we may also omit it from the notation and write $J(\delta, \mathcal{F}, L_2)$.

Certainly $J(1, \mathcal{F}, L_2) < \infty$ if $\mathcal{F}$ satisfies the uniform entropy condition (2.5.1). For Vapnik-Červonenkis classes $\mathcal{F}$, the function $J(\delta, \mathcal{F}, L_2)$ is of the order $O(\delta \sqrt{\log(1/\delta)})$ as $\delta \downarrow 0$.

2.14.1 Theorem. *Let $\mathcal{F}$ be a P -measurable class of measurable functions with measurable envelope function F . Then, for $p \geq 1$,*

$$\|\|\mathbb{G}_n\|_{\mathcal{F}}^*\|_{P,p} \lesssim \|J(\theta_n, \mathcal{F} | F, L_2) \|F\|_n\|_{P,p} \lesssim J(1, \mathcal{F} | F, L_2) \|F\|_{P,2\vee p}.$$

Here $\theta_n = \|\|f\|_n\|_{\mathcal{F}}^* / \|F\|_n$, where $\|\cdot\|_n$ is the $L_2(\mathbb{P}_n)$ -seminorm and the inequalities are valid up to constants depending only on the p involved in the statement.

Proof. Set $\mathbb{G}_n^o = n^{-1/2} \sum_{i=1}^n \varepsilon_i f(X_i)$ for the symmetrized empirical process. In view of the symmetrization lemma, Lemma 2.3.1, Orlicz norms of $\|\mathbb{G}_n\|_{\mathcal{F}}^*$ are bounded by two times the corresponding Orlicz norms of $\|\mathbb{G}_n^o\|_{\mathcal{F}}$.

Given $X_1, \dots, X_n$, the process $\mathbb{G}_n^o$ is sub-Gaussian for the $L_2(\mathbb{P}_n)$ -seminorm $\|\cdot\|_n$ by Hoeffding's inequality. The value $\eta_n = \|\|f\|_n\|_{\mathcal{F}}$ is an upper bound for the radius of $\mathcal{F} \cup \{0\}$ with respect to this norm. The maximal inequality Theorem 2.2.4 gives

$$\|\|\mathbb{G}_n^o\|_{\mathcal{F}}\|_{\psi_2|X} \lesssim \int_0^{\eta_n} \sqrt{1 + \log N(\varepsilon, \mathcal{F}, L_2(\mathbb{P}_n))} d\varepsilon,$$

where $\|\cdot\|_{\psi_2|X}$ is the conditional Orlicz norm for ψ_2 , given $X_1, X_2, \dots$. Make a change of variable and bound the random entropy by a supremum to see that the right side is bounded by $J(\theta_n, \mathcal{F}|F, L_2)\|F\|_n$. Every L_p -norm is bounded by a multiple of the ψ_2 -Orlicz norm. Hence

$$\mathbb{E}_\varepsilon \|\mathbb{G}_n^\circ\|_{\mathcal{F}}^p \lesssim J(\theta_n, \mathcal{F}|F, L_2)^p \|F\|_n^p.$$

Take the expectation over $X_1, \dots, X_n$ to obtain the left inequality of the theorem. Since $\theta_n \leq 1$, the right side of the preceding display is bounded above by $J(1, \mathcal{F}|F, L_2)^p \|F\|_n^p$. For $p \geq 2$, this is further bounded above by $J(1, \mathcal{F}|F, L_2)^p n^{-1} \sum F^p(X_i)$, by Jensen's inequality. This gives the inequality on the right side of the theorem. ■

Because $\|\mathbb{G}_n F\|_{P,2} = \|F\|_{P,2}$, the far right bound of the theorem can be interpreted as saying that the supremum of the empirical process has the same order of magnitude as its value on the envelope, as soon as the uniform entropy integral is finite. This is a useful result, also for many statistical applications, but has the drawback that it is insensitive to the sizes of the individual functions.

The next theorem remedies this. Its bound is not accurate when δ is very small (relative to n and the entropy integral), but has exactly the right form for applications to obtain rates of convergence (in Chapter 3.4).

2.14.2 Theorem. *Let $\mathcal{F}$ be a P -measurable class of measurable functions with envelope function $F \leq 1$ and such that $\mathcal{F}^2$ is P -measurable. If $Pf^2 < \delta^2 PF^2$, for every f and some $\delta \in (0, 1)$, then*

$$\mathbb{E}_P^* \|\mathbb{G}_n\|_{\mathcal{F}} \lesssim J(\delta, \mathcal{F}|F, L_2) \left(1 + \frac{J(\delta, \mathcal{F}|F, L_2)}{\delta^2 \sqrt{n} \|F\|_{P,2}}\right) \|F\|_{P,2}.$$

Proof. Because $\delta \mapsto J(\delta, \mathcal{F}, L_2)$ is the integral of a nonincreasing nonnegative function, it is a concave function such that the map $t \mapsto J(t)/t$, which is the average of its derivative over $[0, t]$, is nonincreasing. The concavity shows that its *perspective* $(x, t) \mapsto tJ(x/t, \mathcal{F}, L_2)$ is a concave function of its two arguments (cf. Boyd and Vandenberghe (2004), page 89). Furthermore, the “extended-value extension” of this function (which by definition is $-\infty$ if $x \leq 0$ or $t \leq 0$) is obviously nondecreasing in its first argument and was noted to be nondecreasing in its second argument. Therefore, by the vector composition rules for concave functions (Boyd and Vandenberghe (2004), pages 83–87, especially lines -2 and -1 of page 86), the function $(x, y) \mapsto H(x, y) := J(\sqrt{x/y}, \mathcal{F}, L_2) \sqrt{y}$ is concave. We have that $\mathbb{E}_P^* \mathbb{P}_n F^2 = \|F\|_{P,2}^2$. Therefore, by an application of Jensen's inequality to the first inequality in Theorem 2.14.1, we obtain, for $\sigma_n^2 = \sup_f \mathbb{P}_n f^2$,

$$(2.14.3) \quad \mathbb{E}_P^* \|\mathbb{G}_n\|_{\mathcal{F}} \lesssim J\left(\frac{\sqrt{\mathbb{E}_P^* \sigma_n^2}}{\|F\|_{P,2}}, \mathcal{F}, L_2\right) \|F\|_{P,2}.$$

The application of Jensen's inequality with outer expectations can be justified here by the monotonicity of the function H , which shows that the measurable majorant of a variable $H(U, V)$ is bounded above by $H(U^*, V^*)$, for U^* and V^* measurable

majorants of U and V . Thus $E^*H(U, V) \leq EH(U^*, V^*)$, after which Jensen's inequality can be applied in its usual (measurable) form.

The second step of the proof is to bound $E_P^* \sigma_n^2$. Because $\mathbb{P}_n f^2 = P f^2 + n^{-1/2} \mathbb{G}_n f^2$ and $P f^2 \leq \delta^2 P F^2$ for every f , we have

$$(2.14.4) \quad E_P^* \sigma_n^2 \leq \delta^2 \|F\|_{P,2}^2 + \frac{1}{\sqrt{n}} E_P^* \|\mathbb{G}_n\|_{\mathcal{F}^2}.$$

Here the empirical process in the second term can be replaced by the symmetrized empirical process $\mathbb{G}_n^o$ at the cost of adding a multiplicative factor 2, by Lemma 2.3.1. The expectation can be factorized as the expectation on the Rademacher variables ε followed by the expectation on $X_1, \dots, X_n$, and $E_\varepsilon \|\mathbb{G}_n^o\|_{\mathcal{F}^2} \leq 2E_\varepsilon \|\mathbb{G}_n^o\|_{\mathcal{F}}$ by the contraction principle for Rademacher variables, Proposition A.1.11, and the fact that $F \leq 1$ by assumption. Taking the expectation on $X_1, \dots, X_n$, we obtain that $E_P^* \|\mathbb{G}_n\|_{\mathcal{F}^2} \leq 4E_P^* \|\mathbb{G}_n^o\|_{\mathcal{F}}$, which in turn is bounded above by $8E_P^* \|\mathbb{G}_n\|_{\mathcal{F}}$ by the desymmetrization inequality, Lemma 2.3.6.

Thus $\mathcal{F}^2$ in the last term of (2.14.4) can be replaced by $\mathcal{F}$, at the cost of inserting a constant. Next we apply (2.14.3) to this term, and conclude that $z^2 := E_P^* \sigma_n^2 / \|F\|_{P,2}^2$ satisfies the inequality

$$(2.14.5) \quad z^2 \lesssim \delta^2 + \frac{J(z, \mathcal{F}, L_2)}{\sqrt{n} \|F\|_{P,2}}.$$

We apply Lemma 2.14.6 (below) with $r = 1$, $A = \delta$ and $B^2 = 1/(\sqrt{n} \|F\|_{P,2})$ to see that

$$J(z, \mathcal{F}, L_2) \lesssim J(\delta, \mathcal{F}, L_2) + \frac{J^2(\delta, \mathcal{F}, L_2)}{\delta^2 \sqrt{n} \|F\|_{P,2}}.$$

We insert this in (2.14.3) to complete the proof. ■

2.14.6 Lemma. *Let $J: (0, \infty) \rightarrow \mathbb{R}$ be a concave, nondecreasing function with $J(0) = 0$. If $z^2 \leq A^2 + B^2 J(z^r)$ for some $r \in (0, 2)$ and $A, B > 0$, then*

$$J(z) \lesssim J(A) \left[1 + J(A^r) \left(\frac{B}{A} \right)^{2r} \right]^{1/(2-r)}.$$

Proof. For $t > s > 0$ we can write s as the convex combination $s = (s/t)t + (1 - s/t)0$ of t and 0. Since $J(0) = 0$, the concavity of J gives that $J(s) \geq (s/t)J(t)$. Thus the function $t \mapsto J(t)/t$ is decreasing, which implies that $J(Ct) \leq CJ(t)$ for $C \geq 1$ and any $t > 0$.

By the monotonicity of J and the assumption on z it follows that

$$J(z^r) \leq J\left((A^2 + B^2 J(z^r))^{r/2}\right) \leq J(A^r) \left(1 + \left(\frac{B}{A}\right)^2 J(z^r)\right)^{r/2}.$$

This implies that $J(z^r)$ is bounded by a multiple of the maximum of $J(A^r)$ and $J(A^r)(B/A)^r J(z^r)^{r/2}$. If it is bounded by the second one, then $J(z^r)^{1-r/2} \lesssim J(A^r)(B/A)^r$. We conclude that

$$J(z^r) \lesssim J(A^r) + J(A^r)^{2/(2-r)} \left(\frac{B}{A}\right)^{2r/(2-r)}.$$

Next again by the monotonicity of J ,

$$\begin{aligned} J(z) &\leq J\left(\sqrt{A^2 + B^2 J(z^r)}\right) \leq J(A) \sqrt{1 + \left(\frac{B}{A}\right)^2 J(z^r)} \\ &\lesssim J(A) \left[1 + \left(\frac{B}{A}\right)^2 \left(J(A^r) + J(A^r)^{2/(2-r)} \left(\frac{B}{A}\right)^{2r/(2-r)}\right)\right]^{1/2} \\ &\lesssim J(A) \left[1 + \sqrt{J(A^r)} \left(\frac{B}{A}\right) + \left(\frac{B}{A}\right)^{2/(2-r)} J(A^r)^{1/(2-r)}\right]. \end{aligned}$$

The middle term on the right side is bounded by a multiple of the sum of the first and third terms, since $x \lesssim 1^p + x^q$ for any conjugate pair (p, q) and any $x > 0$, in particular $x = \sqrt{J(A^r)} B/A$. ■

2.14.7 Example. Suppose that $\mathcal{C}$ is a P -measurable class of measurable sets satisfying, for some constant K and every $0 < \varepsilon < K$,

$$\sup_Q N(\varepsilon, \mathcal{C}, L_1(Q)) \leq \left(\frac{K}{\varepsilon}\right)^V.$$

Then, for $a_{\mathcal{C}} = \sup_{C \in \mathcal{C}} P(C)$,

$$E_P^* \|\mathbb{G}_n\|_{\mathcal{C}} \lesssim \sqrt{a_{\mathcal{C}} V \log(K/a_{\mathcal{C}} + 1)} + \frac{V \log(K/a_{\mathcal{C}} + 1)}{\sqrt{n}}.$$

For $K \geq 1$, the constant in the inequality is universal.

Indeed, the analogue of the bound on the uniform covering numbers is true for the L_2 -norm, with the exponent $2V$ instead of V . Then the entropy integral $J(\delta, \mathcal{C}, L_2)$ is bounded above by a multiple of $\delta \sqrt{V \log(K/\delta + 1)}$. The inequality then follows from Theorem 2.14.2, with $\delta = \sqrt{a_{\mathcal{C}}} \geq a_{\mathcal{C}}$.

In the next two theorems we relax the assumption that the class $\mathcal{F}$ of functions is uniformly bounded, made in Theorem 2.14.2, to moment and exponential bounds, respectively, at the cost of a deterioration of the dependence of the bounds on δ .

2.14.8 Theorem. Let $\mathcal{F}$ be a P -measurable class of measurable functions with envelope function F such that $P F^{(4p-2)/(p-1)} < \infty$ for some $p > 1$ and such that $\mathcal{F}^2$ and $\mathcal{F}^4$ are P -measurable. If $P f^2 < \delta^2 P F^2$ for every f and some $\delta \in (0, 1)$, then

$$E_P^* \|\mathbb{G}_n\|_{\mathcal{F}} \lesssim J(\delta, \mathcal{F} | F, L_2) \left(1 + \frac{J(\delta^{1/p}, \mathcal{F} | F, L_2)}{\delta^2 \sqrt{n}} \frac{\|F\|_{P, (4p-2)/(p-1)}^{2-1/p}}{\|F\|_{P, 2}^{2-1/p}}\right)^{\frac{p}{2p-1}} \|F\|_{P, 2}.$$

2.14.9 Theorem. Let $\mathcal{F}$ be a P -measurable class of measurable functions with envelope function F such that $P \exp(F^{p+\rho}) < \infty$ for some $p, \rho > 0$ and such that $\mathcal{F}^2$ and $\mathcal{F}^4$ are P -measurable. If $P f^2 < \delta^2 P F^2$ for every f and some $\delta \in (0, 1/2)$, then for a constant c depending on $p, P F^2, P F^4$ and $P \exp(F^{p+\rho})$,

$$E_P^* \|\mathbb{G}_n\|_{\mathcal{F}} \leq c J(\delta, \mathcal{F} | F, L_2) \left(1 + \frac{J(\delta(\log(1/\delta))^{1/p}, \mathcal{F} | F, L_2)}{\delta^2 \sqrt{n}}\right).$$

Proofs. Application of the first inequality of Theorem 2.14.1 to the functions f^2 , forming the class $\mathcal{F}^2$ with envelope function F^2 , yields

$$(2.14.10) \quad \mathbb{E}_P^* \|\mathbb{G}_n\|_{\mathcal{F}^2} \lesssim \mathbb{E}_P^* J\left(\frac{\sigma_{n,4}^2}{(\mathbb{P}_n F^4)^{1/2}}, \mathcal{F}^2 | F^2, L_2\right) (\mathbb{P}_n F^4)^{1/2},$$

for $\sigma_{n,r}$ the diameter of $\mathcal{F}$ in $L_r(\mathbb{P}_n)$, i.e.

$$(2.14.11) \quad \sigma_{n,r}^r = \sup_f \mathbb{P}_n |f|^r.$$

Preservation properties of uniform entropy (see Theorem 2.10.21, applied to $\phi(f) = f^2$ with $L = 2F$) show that $J(\delta, \mathcal{F}^2 | F^2, L_2) \lesssim J(\delta, \mathcal{F} | F, L_2)$, for every $\delta > 0$. Because $\mathbb{P}_n f^2 = P f^2 + n^{-1/2} \mathbb{G}_n f^2$ and $P f^2 \leq \delta^2 P F^2$ by assumption, we find that

$$(2.14.12) \quad \mathbb{E}_P^* \sigma_{n,2}^2 \lesssim \delta^2 P F^2 + \frac{1}{\sqrt{n}} \mathbb{E}_P^* J\left(\frac{\sigma_{n,4}^2}{(\mathbb{P}_n F^4)^{1/2}}, \mathcal{F}, L_2\right) (\mathbb{P}_n F^4)^{1/2}.$$

The next step is to bound $\sigma_{n,4}$ in terms of $\sigma_{n,2}$.

By Hölder's inequality, for any conjugate pair (p, q) and any $0 < s < 4$,

$$\mathbb{P}_n f^4 \leq \mathbb{P}_n |f|^{4-s} F^s \leq (\mathbb{P}_n |f|^{(4-s)p})^{1/p} (\mathbb{P}_n F^{sq})^{1/q}.$$

Choosing s such that $(4-s)p = 2$ (and hence $sq = (4p-2)/(p-1)$), we find that

$$\sigma_{n,4}^4 \leq \sigma_{n,2}^{2/p} (\mathbb{P}_n F^{sq})^{1/q}.$$

We insert this bound in (2.14.12). The function $(x, y) \mapsto x^{1/p} y^{1/q}$ is concave, and hence the function $(x, y, z) \mapsto J(\sqrt{x^{1/p} y^{1/q}}/z, \mathcal{F}, L_2) \sqrt{z}$ can be seen to be concave by the same arguments as in the proof of Theorem 2.14.2. Therefore, we can apply Jensen's inequality to see that

$$\mathbb{E}_P^* \sigma_{n,2}^2 \lesssim \delta^2 P F^2 + \frac{1}{\sqrt{n}} J\left(\frac{(\mathbb{E}_P^* \sigma_{n,2}^2)^{1/(2p)} (P F^{sq})^{1/(2q)}}{(P F^4)^{1/2}}, \mathcal{F}, L_2\right) (P F^4)^{1/2}.$$

We conclude that $z := (\mathbb{E}_P^* \sigma_{n,2}^2)^{1/2} / \|F\|_{P,2}$ satisfies

$$\begin{aligned} z^2 &\lesssim \delta^2 + \frac{1}{\sqrt{n}} J\left(z^{1/p} \frac{(P F^2)^{1/(2p)} (P F^{sq})^{1/(2q)}}{(P F^4)^{1/2}}, \mathcal{F}, L_2\right) \frac{(P F^4)^{1/2}}{P F^2} \\ &\lesssim \delta^2 + J(z^{1/p}, \mathcal{F}, L_2) \frac{(P F^{sq})^{1/(2q)}}{\sqrt{n} (P F^2)^{1-1/(2p)}}. \end{aligned}$$

In the last step we use that $J(C\delta, \mathcal{F}, L_2) \leq C J(\delta, \mathcal{F}, L_2)$ for $C \geq 1$, and Hölder's inequality as previously to see that the present C satisfies this condition. We next apply Lemma 2.14.6 (with $r = 1/p$) to obtain a bound on $J(z, \mathcal{F}, L_2)$, and conclude the proof by substituting this bound in (2.14.3).

For the proof of Theorem 2.14.9 fix $r = 2/p$. The functions $\psi, \bar{\psi}: [0, \infty) \rightarrow [0, \infty)$ defined by

$$\psi(f) = \log^r(1+f), \quad \bar{\psi}(f) = e^{f^{1/r}} - 1,$$

are each other's inverses, and are increasing from $\psi(0) = \bar{\psi}(0) = 0$ to infinity. Thus their primitive functions $\Psi(f) = \int_0^f \psi(s) ds$ and $\bar{\Psi}(f) = \int_0^f \bar{\psi}(s) ds$ satisfy *Young's inequality* $fg \leq \Psi(f) + \bar{\Psi}(g)$, for every $f, g \geq 0$ (e.g. Boyd and Vandenberghe (2004), page 120, 3.38).

The function $t \mapsto t \log^r(1/t)$ is concave in a neighborhood of 0 (specifically: on the interval $(0, e^{1-r} \wedge 1)$), with limit from the right equal to 0 at 0, and derivative tending to infinity at this point. Therefore, there exists a *concave, increasing* function $k: (0, \infty) \rightarrow (0, \infty)$ that is identical to $t \mapsto t \log^r(1/t)$ near 0 and bounded below and above by a positive constant times the identity throughout its domain. (E.g. extend $t \mapsto t \log^r(1/t)$ linearly with slope 1 from the point where the derivative of the latter function has decreased to 1.) Write $k(t) = t \ell^r(t)$, so that ℓ^r is bounded below by a constant and $\ell(t) = \log(1/t)$ near 0. Then, for every $t > 0$,

$$(2.14.13) \quad \frac{\log(2 + t/C)}{\ell(C)} \lesssim \log(2 + t).$$

(The constant in $\lesssim$ may depend on r .) To see this, note that for $C > c$ the left side is bounded by a multiple of $\log(2 + t/c)$, whereas for small C the left side is bounded by a multiple of $[\log(2 + t) + \log(1 + 1/C)]/\ell(C) \lesssim \log(2 + t) + 1$.

From the inequality $\Psi(f) \leq f\psi(f)$, we obtain that, for $f > 0$,

$$\Psi\left(\frac{f}{\log^r(2 + f)}\right) \lesssim f.$$

Therefore, by (2.14.13) followed by Young's inequality,

$$\begin{aligned} \frac{f^4}{k(C^2)} &= \frac{f^2/C^2}{\log^r(2 + f^2/C^2)} \frac{f^2 \log^r(2 + f^2/C^2)}{\ell^r(C^2)} \\ &\lesssim \frac{f^2}{C^2} + \bar{\Psi}(F^2 \log^r(2 + F^2)). \end{aligned}$$

On integrating this with respect to the empirical measure, with $C^2 = \mathbb{P}_n f^2$, we see that, with $G = \bar{\Psi}(F^2 \log^r(2 + F^2))$,

$$\mathbb{P}_n f^4 \lesssim k(\mathbb{P}_n f^2) (1 + \mathbb{P}_n G).$$

We take the supremum over f to bound $\sigma_{n,4}^4$ as in (2.14.11) in terms of $k(\sigma_{n,2}^2)$, and next substitute this bound in (2.14.12) to find that

$$\begin{aligned} E_P^* \sigma_{n,2}^2 &\leq \delta^2 P F^2 + \frac{1}{\sqrt{n}} E_P^* J\left(\frac{\sqrt{k(\sigma_{n,2}^2)} \sqrt{1 + \mathbb{P}_n G}}{(\mathbb{P}_n F^4)^{1/2}}, \mathcal{F}, L_2\right) (\mathbb{P}_n F^4)^{1/2} \\ &\leq \delta^2 P F^2 + \frac{1}{\sqrt{n}} J\left(\frac{\sqrt{k(E_P^* \sigma_{n,2}^2)} \sqrt{1 + P G}}{(P F^4)^{1/2}}, \mathcal{F}, L_2\right) (P F^4)^{1/2}, \end{aligned}$$

where we have used the concavity of k , and the concavity of the other maps, as previously. By assumption the expected value $P G$ is finite for $r = 2/p$. It follows that $z^2 = E_P^* \sigma_{n,2}^2 / P F^2$ satisfies, for suitable constants a, b, c depending on $r, P F^2, P F^4$ and $P G$,

$$z^2 \lesssim \delta^2 + \frac{a}{\sqrt{n}} J(\sqrt{k(z^2 b)} c, \mathcal{F}, L_2).$$

By concavity and the fact that $k(0) = 0$, we have $k(Cz) \leq Ck(z)$, for $C \geq 1$ and $z > 0$. The function $z \mapsto \sqrt{k(z^2b)c}$ inherits this property. Therefore we can apply Lemma 2.14.14, with k of the lemma equal to the present function $z \mapsto \sqrt{k(z^2b)c}$, to obtain a bound on $J(z, \mathcal{F}, L_2)$ in terms of $J(\delta, \mathcal{F}, L_2)$ and $J(\sqrt{k(\delta^2b)c}, \mathcal{F}, L_2)$, which we substitute in (2.14.3). Here $k(\delta^2) = \delta^2 \log^r(1/\delta)$ for sufficiently small $\delta > 0$ and $\kappa(\delta^2) \lesssim \delta^2 \lesssim \delta^2 \log^r(1/\delta)$ for $\delta < 1/2$ and bounded away from 0. Thus we can simplify the bound to the one in the statement of the theorem, possibly after increasing the constants a, b, c to be at least 1, to complete the proof. ■

2.14.14 Lemma. *Let $J: (0, \infty) \rightarrow \mathbb{R}$ be a concave, nondecreasing function with $J(0) = 0$, and let $k: (0, \infty) \rightarrow (0, \infty)$ be nondecreasing and satisfy $k(Cz) \leq Ck(z)$ for $C \geq 1$ and $z > 0$. If $z^2 \leq A^2 + B^2J(k(z))$ for some $A, B > 0$, then*

$$J(z) \lesssim J(A) \left[1 + J(k(A)) \left(\frac{B}{A} \right)^2 \right].$$

Proof. As noted in the proof of Lemma 2.14.6 the properties of J imply that $J(Cz) \leq CJ(z)$ for $C \geq 1$ and any $z > 0$. In view of the assumed property of k and the monotonicity of J it follows that $J \circ k(Cz) \leq CJ \circ k(z)$ for every $C \geq 1$ and $z > 0$. Therefore, by the monotonicity of J and k , and the assumption on z ,

$$J \circ k(z) \leq J \circ k \left(\sqrt{A^2 + B^2J \circ k(z)} \right) \leq J \circ k(A) \sqrt{1 + (B/A)^2 J \circ k(z)}.$$

As in the proof of Lemma 2.14.6 we can solve this for $J \circ k(z)$ to find that

$$J \circ k(z) \lesssim J \circ k(A) + J \circ k(A)^2 \left(\frac{B}{A} \right)^2.$$

Next again by the monotonicity of J ,

$$\begin{aligned} J(z) &\leq J \left(\sqrt{A^2 + B^2J \circ k(z)} \right) \leq J(A) \sqrt{1 + (B/A)^2 J \circ k(z)} \\ &\lesssim J(A) \left[1 + \left(\frac{B}{A} \right) \sqrt{J \circ k(A)} + \left(\frac{B}{A} \right)^2 J \circ k(A) \right]. \end{aligned}$$

The middle term on the right side is bounded by the sum of the first and third terms. ■

2.14.15 Example (Alternative proof of the Donsker theorem). The proofs of the preceding maximal inequalities are modelled after the proofs of the uniform entropy central limit theorem, Theorem 2.5.2. The inequalities are sufficiently strong to yield short proofs of these results.

Under the uniform entropy bound (2.5.1), the uniform entropy integral $\delta \mapsto J(\delta, \mathcal{F} | F, L_2)$ is uniformly bounded and $J(\delta, \mathcal{F} | F, L_2) \rightarrow 0$ as $\delta \downarrow 0$. The entropy integral of the class $\mathcal{F} - \mathcal{F}$ (with envelope $2F$) is bounded by a multiple of $J(\delta, \mathcal{F} | F, L_2)$. Application of the L_1 -inequality of Theorem 2.14.1 followed by the Cauchy-Schwarz inequality yields, with $\theta_{n,\delta}^2 = \|\mathbb{P}_n f^2\|_{\mathcal{F}_\delta}^* / \mathbb{P}_n F^2$ and $\mathcal{F}_\delta = \{f - g: \|f - g\|_{P,2} < \delta\}$,

$$\mathbb{E} \|\mathbb{G}_n\|_{\mathcal{F}_\delta} \lesssim \mathbb{E}^* J(\theta_{n,\delta}, \mathcal{F} | F, L_2) \|F\|_n \leq \left(\mathbb{E}^* J^2(\theta_{n,\delta}, \mathcal{F} | F, L_2) P F^2 \right)^{1/2}.$$

The sequence of empirical processes indexed by $\mathcal{F}$ is asymptotically tight if the right side converges to zero as $n \rightarrow \infty$ followed by $\delta \downarrow 0$. In view of the dominated convergence theorem, this is true if $\theta_{n,\delta} \rightarrow 0$ in probability.

Without loss of generality, assume that $F \geq 1$, so that $\theta_{n,\delta}^2 \leq \|\mathbb{P}_n f^2\|_{\mathcal{F}_\delta}$. The desired conclusion follows if $\|\mathbb{P}_n f^2 - P f^2\|_{\mathcal{F}_\delta}$ converges in probability to zero, as $n \rightarrow \infty$ followed by $\delta \downarrow 0$. This is certainly the case if the class $\mathcal{G} = (\mathcal{F} - \mathcal{F})^2$ is Glivenko-Cantelli in probability. Let $\mathcal{G}_M$ be the functions $g1\{F \leq M\}$ when g ranges over $\mathcal{G}$. Since $N(4\varepsilon M\|F\|_{Q,2}, \mathcal{G}_M, L_2(Q)) \leq N(\varepsilon\|F\|_{Q,2}, \mathcal{F}, L_2(Q))^2$, the uniform entropy integral $J(\delta, \mathcal{G}_M | 4MF, L_2)$ of $\mathcal{G}_M$ with envelope $4MF$ is bounded by a multiple of $J(\delta, \mathcal{F} | F, L_2)$. A second application of Theorem 2.14.1 gives

$$E^* \|\mathbb{P}_n f^2 - P f^2\|_{\mathcal{F}-\mathcal{F}} \lesssim \frac{1}{\sqrt{n}} J(1, \mathcal{F} | F, L_2) M (P F^2)^{1/2} + P F^2 \{F > M\},$$

for every M . This converges to zero as $n \rightarrow \infty$ followed by $M \rightarrow \infty$.

2.14.2 Bracketing Entropy

Define the *bracketing integral* of a class of functions $\mathcal{F}$ with measurable envelope function F relative to a given norm as

$$J_{[]}(\delta, \mathcal{F} | F, \|\cdot\|) = \int_0^\delta \sqrt{1 + \log N_{[]}(\varepsilon\|F\|, \mathcal{F}, \|\cdot\|)} d\varepsilon.$$

The basic bracketing maximal inequality uses the $L_2(P)$ -norm.

2.14.16 Theorem. *Let $\mathcal{F}$ be a class of measurable functions with measurable envelope function F . For a given $\eta > 0$, set*

$$a(\eta) = \eta\|F\|_{P,2} / \sqrt{1 + \log N_{[]}(\eta\|F\|_{P,2}, \mathcal{F}, L_2(P))}.$$

Then for every $\eta > 0$,

$$\begin{aligned} \|\mathbb{G}_n\|_{\mathcal{F}}^* &\lesssim J_{[]}(\eta, \mathcal{F} | F, L_2(P))\|F\|_{P,2} + \sqrt{n} P F \{F > \sqrt{n} a(\eta)\} \\ &\quad + \|f\|_{P,2} \sqrt{1 + \log N_{[]}(\eta\|F\|_{P,2}, \mathcal{F}, L_2(P))}. \end{aligned}$$

Consequently, if $\|f\|_{P,2} < \delta\|F\|_{P,2}$ for every f in the class $\mathcal{F}$, then

$$\|\mathbb{G}_n\|_{\mathcal{F}}^* \lesssim J_{[]}(\delta, \mathcal{F} | F, L_2(P))\|F\|_{P,2} + \sqrt{n} P F \{F > \sqrt{n} a(\delta)\}.$$

In particular, for any class $\mathcal{F}$,

$$\|\mathbb{G}_n\|_{\mathcal{F}}^* \lesssim J_{[]} (1, \mathcal{F} | F, L_2(P))\|F\|_{P,2}.$$

The constants in the inequalities are universal.

Although the bound of the preceding theorem shows some dependence on the variances of the functions in the class, the “remainder terms” preclude easy application. The following theorems, which are bracketing analogues of Theorems 2.14.2 and 2.14.9, remedy this, at least for not too small δ .

The first theorem applies to uniformly bounded classes of functions. The second can handle more general classes, at the cost of replacing the $L_2(P)$ -norm in the bracketing integral by the “norm”

$$\|f\|_{P,B} = \left(2P(e^{|f|} - 1 - |f|)\right)^{1/2}.$$

This “Bernstein norm” neither is homogeneous nor satisfies the triangle inequality, but we shall nevertheless use it to measure the “size” of functions. It dominates the $L_2(P)$ -norm.

2.14.17 Theorem. *Let $\mathcal{F}$ be a class of measurable functions such that $Pf^2 < \delta^2 PF^2$ and $\|f\|_\infty \leq 1$ for every f in $\mathcal{F}$. Then*

$$E_P^* \|\mathbb{G}_n\|_{\mathcal{F}} \lesssim J_{[]}(\delta, \mathcal{F} | F, L_2(P)) \left(1 + \frac{J_{[]}(\delta, \mathcal{F} | F, L_2(P))}{\delta^2 \sqrt{n} \|F\|_{P,2}}\right) \|F\|_{P,2}.$$

2.14.18 Theorem. *Let $\mathcal{F}$ be a class of measurable functions such that $\|f\|_{P,B} \leq \delta \|F\|_{P,B}$ for every $f \in \mathcal{F}$. Then*

$$E_P^* \|\mathbb{G}_n\|_{\mathcal{F}} \lesssim J_{[]}(\delta, \mathcal{F} | F, \|\cdot\|_{P,B}) \left(1 + \frac{J_{[]}(\delta, \mathcal{F} | F, \|\cdot\|_{P,B})}{\delta^2 \sqrt{n} \|F\|_{P,B}}\right) \|F\|_{P,B}.$$

Since the Bernstein “norm” is not homogeneous, it may actually be beneficial to apply the second theorem to a multiple $\sigma\mathcal{F}$ of the class $\mathcal{F}$, noting that the left side equals $\sigma^{-1} E_P^* \|\mathbb{G}_n\|_{\sigma\mathcal{F}}$ for every $\sigma > 0$. Even taking $\sigma = 2$ may be helpful!

In the two preceding theorems the bracketing entropy integral is standardized relative to an envelope function. This makes the results comparable to the corresponding results without bracketing, Theorems 2.14.2–2.14.9, but the following equivalent formulations of the theorems are sometimes easier to apply. Denote the nonstandardized bracketing integral by

$$\tilde{J}_{[]}(\delta, \mathcal{F}, \|\cdot\|) = \int_0^\delta \sqrt{1 + \log N_{[]}(\varepsilon, \mathcal{F}, \|\cdot\|)} d\varepsilon.$$

2.14.17' Theorem. *Let $\mathcal{F}$ be a class of measurable functions such that $Pf^2 < \delta^2$ and $\|f\|_\infty \leq M$ for every f in $\mathcal{F}$. Then*

$$E_P^* \|\mathbb{G}_n\|_{\mathcal{F}} \lesssim \tilde{J}_{[]}(\delta, \mathcal{F}, L_2(P)) \left(1 + \frac{\tilde{J}_{[]}(\delta, \mathcal{F}, L_2(P))}{\delta^2 \sqrt{n}} M\right).$$

2.14.18' Theorem. Let $\mathcal{F}$ be a class of measurable functions such that $\|f\|_{P,B} \leq \delta$ for every $f \in \mathcal{F}$. Then

$$E_P^* \|\mathbb{G}_n\|_{\mathcal{F}} \lesssim \tilde{J}_{[]}(\delta, \mathcal{F}, \|\cdot\|_{P,B}) \left(1 + \frac{\tilde{J}_{[]}(\delta, \mathcal{F}, \|\cdot\|_{P,B})}{\delta^2 \sqrt{n}}\right).$$

The proofs of the theorems are based on the following lemma, which allows a general norm $\|\cdot\|$, or even any “monotone” measure of size of functions: it suffices that $|f| \leq |g|$ implies that $\|f\| \leq \|g\|$, for every pair of functions. In particular, the norm can be taken equal to the Bernstein “norm”. The proof is based on a chaining argument, where the mean of the supremum of the chains is bounded by the integral in the right side of the lemma and the remainders at the ends of the chains give rise to two more terms.

2.14.19 Lemma. Let $\mathcal{F}$ be an arbitrary class of measurable functions and $\|\cdot\|$ a norm that dominates the $L_2(P)$ -norm. Then, for every $\delta > 3\gamma \geq 0$, there exist deterministic functions $e_n: \mathcal{F} \rightarrow \mathbb{R}$ with $\|e_n\|_{\mathcal{F}} \leq 8\gamma\sqrt{n}$ such that

$$\begin{aligned} \left\| \sup_f (\mathbb{G}_n - e_n)^{+*} \right\|_{P,p} &\lesssim \int_{\gamma}^{\delta} \sqrt{1 + \log N_{[]}(\varepsilon, \mathcal{F}, \|\cdot\|)} d\varepsilon \\ &\quad + \left\| \sup_i |\mathbb{G}_n f_i| \right\|_{P,p} + \left\| \sup_i |\mathbb{G}_n \sup_{f \in \mathcal{F}_i} |f - f_i|^*| \right\|_{P,p}, \end{aligned}$$

for a minimal covering $\mathcal{F} = \cup_{i=1}^m \mathcal{F}_i$ by brackets of $\|\cdot\|$ -size at most δ and any choice of $f_i \in \mathcal{F}_i$. The constants in the inequalities are universal.

Proofs. For the proof of the lemma, fix integers q_0 and q_2 such that $2^{-q_0} < \delta \leq 2^{-q_0+1}$ and $2^{-q_2-2} < \gamma \leq 2^{-q_2-1}$. For $q \geq q_0$, construct a nested sequence of partitions $\mathcal{F} = \cup_{i=1}^{N_q} \mathcal{F}_{qi}$ such that the q_0 th partition is the partition given in the statement of the lemma and for, each $q > q_0$,

$$\left\| \left(\sup_{f, g \in \mathcal{F}_{qi}} |f - g| \right)^* \right\| < 2^{-q}.$$

By the choice of q_0 the number of sets in the q_0 th partition satisfies $N_{q_0} \leq N_{[]} (2^{-q_0}, \mathcal{F}, \|\cdot\|)$ and the preceding display is valid for $q = q_0$ up to a factor 2. The number N_q of subsets in the q th partition can be chosen to satisfy

$$\log N_q \leq \sum_{r=q_0}^q \log N_{[]} (2^{-r}, \mathcal{F}, \|\cdot\|).$$

Now adopt the same notation as in the proof of Theorem 2.5.6 (except define a_q with $1 + \log$ rather than $\log$) and decompose, pointwise in x (which is suppressed in the notation),

$$\begin{aligned} f - \pi_{q_0} f &= (f - \pi_{q_0} f) B_{q_0} f + \sum_{q_0+1}^{q_2} (f - \pi_q f) B_q f \\ &\quad + \sum_{q_0+1}^{q_2} (\pi_q f - \pi_{q-1} f) A_{q-1} f + (f - \pi_{q_2} f) A_{q_2} f. \end{aligned}$$

See the proof of Theorem 2.5.6 for the notation and motivation.

Apply the empirical process $\mathbb{G}_n$ to each of the three terms separately and take suprema over $f \in \mathcal{F}$. The $L_p(P)$ -norm of the first term resulting from this procedure can be bounded by

$$\left\| \left\| \mathbb{G}_n(f - \pi_{q_0} f) B_{q_0} f \right\|_{\mathcal{F}}^* \right\|_{P,p} \lesssim \left\| \left\| \mathbb{G}_n \Delta_{q_0} f \right\|_{\mathcal{F}}^* \right\|_{P,p} + \sqrt{n} \|P \Delta_{q_0} f B_{q_0} f\|_{\mathcal{F}}.$$

The first term on the right arises as the last term in the upper bound of the lemma. The second term on the right can be bounded by

$$\begin{aligned} a_{q_0}^{-1} P \Delta_{q_0}^2 f &\lesssim \delta \sqrt{1 + \log N_{[]}(\delta/2, \mathcal{F}, \|\cdot\|)} \\ &\lesssim \int_{\delta/3}^{\delta} \sqrt{1 + \log N_{[]}(\varepsilon, \mathcal{F}, \|\cdot\|)} d\varepsilon. \end{aligned}$$

This is bounded by the right side of the lemma.

The second and third terms in the decomposition can be bounded in exactly the same manner as in the proof of Theorem 2.5.6, where it may be noted that Lemma 2.2.13 bounds the ψ_1 -norm, hence every L_p -norm up to a constant. This shows that inequality (2.5.7) is also valid with the L_p -norm of $\|\mathbb{G}_n\|_{\mathcal{F}}$ on the left. Thus the second and third terms give contributions to the $L_p(P)$ -norm of $\|\mathbb{G}_n\|_{\mathcal{F}}^*$ bounded up to a constant by

$$\sum_{q_0+1}^{q_2+1} 2^{-q} \sqrt{1 + \log N_q} \lesssim \int_{2^{-q_2-1}}^{2^{-q_0}} \sqrt{1 + \log N_{[]}(\varepsilon, \mathcal{F}, \|\cdot\|)} d\varepsilon.$$

This is bounded by the right side of the lemma.

For the final term, define $e_n(f) = 2\sqrt{n}P\Delta_{q_2}f$. Then $\mathbb{G}_n(f - \pi_{q_2}f)A_{q_2}f - e_n(f)$ is bounded above by $\mathbb{G}_n\Delta_{q_2}fA_{q_2}f$, and the fourth term in the decomposition shifted by e_n yields

$$\begin{aligned} \left\| \sup_f (\mathbb{G}_n(f - \pi_{q_2}f)A_{q_2}f - e_n(f))^+ \right\|_{P,p} &\leq \left\| \left\| \mathbb{G}_n \Delta_{q_2} f A_{q_2} f \right\|_{\mathcal{F}}^* \right\|_{P,p} \\ &\lesssim a_{q_2-1} (1 + \log N_{q_2}) + 2^{-q_2} \sqrt{1 + \log N_{q_2}}. \end{aligned}$$

This is bounded by the contribution of the second and third parts of the decomposition. This concludes the proof of the lemma.

For the proof of Theorem 2.14.16, apply the preceding argument with the $L_2(P)$ -norm substituted for $\|\cdot\|$ and $q_2 = \infty$. Then the fourth term in the decomposition of $f - \pi_{q_0}f$ vanishes, and the contributions of the second and third terms to $E^*\|\mathbb{G}_n\|_{\mathcal{F}}$ can be bounded as before. The first term yields the contribution

$$E^* \left\| \mathbb{G}_n(f - \pi_{q_0}f) B_{q_0} f \right\|_{\mathcal{F}} \lesssim \sqrt{n} P^* F \{2F > \sqrt{n} a_{q_0}\}.$$

For $\delta = 8\eta\|F\|_{P,2}$, the right side is bounded by $\sqrt{n} P^* F \{F > \sqrt{n} a(\eta)\}$. Finally,

$$E^* \left\| \mathbb{G}_n \pi_{q_0} f \right\|_{\mathcal{F}} \lesssim E^* \left\| \mathbb{G}_n \pi_{q_0} f \{F \leq \sqrt{n} a(\eta)\} \right\|_{\mathcal{F}} + \sqrt{n} P^* F \{F > \sqrt{n} a(\eta)\}.$$

Each function $\pi_{q_0} f 1\{F \leq \sqrt{na}(\eta)\}$ is uniformly bounded by $\sqrt{na}(\eta)$. Inequality (2.5.7) yields a bound for the first term on the left equal to a multiple of

$$(1 + \log N_{q_0})a(\eta) + \sqrt{1 + \log N_{q_0}} \|f\|_{P,2} \|f\|_{\mathcal{F}}.$$

The first term is bounded by a multiple of $J_{[]}(\eta, \mathcal{F}, L_2(P)) \|F\|_{P,2}$. The second arises as the last term in the first inequality of the theorem.

For the second inequality of Theorem 2.14.16, it suffices to note that the integrand in the bracketing integral is decreasing so that

$$\delta \sqrt{1 + \log N_{[]}(\delta \|F\|_{P,2}, \mathcal{F}, L_2(P))} \leq J_{[]}(\delta, \mathcal{F}, L_2(P)).$$

For the final inequality of the theorem, choose $\delta = 2$ in the second. Since the class $\mathcal{F}$ fits in the single bracket $[-F, F]$, it follows that $a(2) = 2\|F\|_{P,2}$.

For the proofs of Theorems 2.14.17 and 2.14.18, apply Lemma 2.14.19 with $\gamma = 0$ and $p = 1$. It suffices to bound the second and third terms on the right appropriately.

For the proof of Theorem 2.14.17 we use Bernstein's inequality for bounded variables, as given in Example 2.2.11, and Lemma 2.2.13 (see (2.5.7)) to bound both the second and third terms by a multiple of

$$\begin{aligned} & \left(1 + \log N_{[]}(\delta \|F\|_{P,2}, \mathcal{F}, L_2(P))\right) \frac{1}{\sqrt{n}} \\ & + \sqrt{1 + \log N_{[]}(\delta \|F\|_{P,2}, \mathcal{F}, L_2(P))} \delta \|F\|_{P,2}. \end{aligned}$$

Since the entropy is decreasing in δ , the second term is smaller than the entropy integral. By the same reasoning the part in brackets of the first term is bounded by the square of the entropy integral divided by δ^2 .

Theorem 2.14.18 follows similarly, this time using the moment form of Bernstein's inequality, Lemma 2.2.10, in combination with the "norm" $\|\cdot\|_{P,B}$. ■

2.14.20 Example (Alternative proof of the Donsker theorem). The second maximal inequality given by Theorem 2.14.16 shows that $E^* \|\mathbb{G}\|_{\mathcal{F}_\delta} \rightarrow 0$ as $n \rightarrow \infty$ followed by $\delta \downarrow 0$, for $\mathcal{F}_\delta = \{f - g: \|f - g\|_{P,2} < \delta\}$. This immediately implies the simplified version of the bracketing central limit Theorem 2.5.6, which asserts that $\mathcal{F}$ is Donsker if the bracketing integral $J_{[]}(\delta, \mathcal{F}, L_2(P))$ is finite.

2.14.3 Mixed Entropy

Because the empirical process is asymptotically Gaussian, we might hope that it satisfies a maximal inequality of Gaussian form, corrected for finite sample size. Inequalities of this form can be obtained using Bernstein's inequality, Lemma 2.2.10, which similarly can be viewed as giving a Gaussian tail bound corrected for finite sample size, for a single function.

Let $\text{diam}(\mathcal{F}, \|\cdot\|)$ be the diameter of $\mathcal{F}$ relative to the norm $\|\cdot\|$.

2.14.21 Theorem. *If $\mathcal{F}$ is a separable class of measurable functions such that $Pf^2 \leq \delta^2$ for every $f \in \mathcal{F}$, then*

$$\begin{aligned} E_P \|\mathbb{G}_n\|_{\mathcal{F}} &\lesssim \int_0^\delta \sqrt{1 + \log N(\varepsilon, \mathcal{F}, L_2(P))} d\varepsilon \\ &\quad + \frac{1}{\sqrt{n}} \int_0^{\text{diam}(T, \|\cdot\|_\infty)} \left(1 + \log N(\varepsilon, \mathcal{F}, \|\cdot\|_\infty)\right) d\varepsilon. \end{aligned}$$

Proof. By Bernstein's inequality Lemma 2.2.10, as specialized in Example 2.2.11, the tail probabilities of the empirical process satisfy, for $x > 0$,

$$P(|\mathbb{G}_n f| > x) \leq 2e^{-\frac{x^2}{2Pf^2 + 2x\|f\|_\infty/(3\sqrt{n})}}.$$

This can be equivalently written as (see the discussion in Section 2.2.4), for $x > 0$,

$$P\left(|\mathbb{G}_n(f - g)| \gtrsim \sqrt{x}\|f - g\|_{P,2} + x \frac{\|f - g\|_\infty}{\sqrt{n}}\right) \leq 2e^{-x}.$$

We conclude that the empirical process satisfies the assumption of Theorem 2.2.19 with the metrics d_2 and d_1 taken equal to $\|\cdot\|_{P,2}$ and $\|\cdot\|_\infty/\sqrt{n}$. By Lemma 2.2.15 the corresponding generic chaining numbers $\gamma_1(\mathcal{F}, d_1)$ and $\gamma_2(\mathcal{F}, d_2)$ are bounded above by entropy integrals as in the statement of the theorem, where in the case of $\gamma_1(\mathcal{F}, \|\cdot\|_\infty/\sqrt{n})$, the factor $\sqrt{n}$ can be moved outside the integral by a change of variable and the fact that $N(\varepsilon/\sqrt{n}, \mathcal{F}, \|\cdot\|_\infty/\sqrt{n}) = N(\varepsilon, \mathcal{F}, \|\cdot\|_\infty)$, for every $\varepsilon > 0$. Thus the right side of the theorem dominates the expectation $E_P \sup_{f,g} |\mathbb{G}_n(f - g)|$. If the zero function belongs to $\mathcal{F}$, this dominates the left side of the theorem. It is no loss of generality to assume so. ■

The first term in the upper bound is the Gaussian entropy integral. The integral in the second term is bigger than the Gaussian integral, because it lacks the square root on the entropy and uses the uniform rather than the L_2 -norm. However, the leading multiplicative factor $1/\sqrt{n}$ compensated for this.

If the use of a stronger norm is not a problem, then alternative bounds can be obtained from the general results of Chapter 2.2 in a similar way. For instance, in view of Proposition 2.14.23, the empirical process is sub-Gaussian relative to the ψ_2 -Orlicz norm. Hence by a direct application of Corollary 2.2.9, for any separable class of measurable functions $\mathcal{F}$,

$$E_P \|\mathbb{G}_n\|_{\mathcal{F}} \lesssim \int_0^{\text{diam}(\mathcal{F}, \|\cdot\|_{P, \psi_2})} \sqrt{1 + \log N(\varepsilon, \mathcal{F}, \|\cdot\|_{P, \psi_2})} d\varepsilon.$$

If $\|f\|_{P, \psi_2} \leq \delta$, for every $f \in \mathcal{F}$, then the upper limit of the integral can be set at δ as well. Since the ψ_2 -Orlicz norm is weaker than the uniform norm, this bound remains valid if $\|\cdot\|_{P, \psi_2}$ is replaced by $\|\cdot\|_\infty$.

Alternatively, following the same route as in the preceding theorem, but now exploiting Example 2.2.12, we may derive

$$\begin{aligned} E_P \|\mathbb{G}_n\|_{\mathcal{F}} &\lesssim \text{diam}(\mathcal{F}, \|\cdot\|_{P, \psi_1}) + \int_0^\infty \sqrt{\log N(\varepsilon, \mathcal{F}, \|\cdot\|_{P, \psi_1})} d\varepsilon \\ &\quad + \frac{1}{\sqrt{n}} \int_0^\infty \log N(\varepsilon, \mathcal{F}, \|\cdot\|_{P, \psi_1}) d\varepsilon. \end{aligned}$$

The upper limits of the integrals can of course be truncated at the diameter of $\mathcal{F}$, since the entropy vanishes for ε larger than the diameter. In comparison to the bound in the preceding theorem, this bound has a bigger first integral through the use of the stronger ψ_1 -Orlicz norm, but a smaller second integral, for the opposite reason. A disadvantage of this bound is that it involves the diameter of $\mathcal{F}$ relative to the stronger ψ_1 -Orlicz norm rather than the L_2 -norm, also in the term that is not mitigated by the factor $1/\sqrt{n}$.

2.14.4 VC-Classes

Combining Theorem 2.14.1 with the bounds on covering numbers of VC-classes given by Theorem 2.6.4 or Theorem 2.6.7 shows that for every P -measurable VC-class of functions

$$E^* \|G_n\|_{\mathcal{F}} \lesssim \sqrt{V(\mathcal{F})} \|F\|_{P,2}.$$

The following classical inequality, for VC-classes of sets, adds an unnecessary logarithmic factor in the sample size to this bound, but has the benefit of having explicit and small constants.

2.14.22 Lemma (Vapnik-Červonenkis). *For any P -measurable VC-class $\mathcal{C}$ of sets and $2n \geq V$,*

$$E^* \|G_n\|_{\mathcal{C}} \leq \sqrt{2 \log(2 \max_{x \in \mathcal{X}^{2n}} \Delta_{2n}(\mathcal{C}, x))} \leq 2 \sqrt{2V(\mathcal{C}) \left(1 + \frac{\log n}{V(\mathcal{C})}\right)}.$$

Proof. To prove the first inequality, we introduce independent copies $Y_1, \dots, Y_n$ of $X_1, \dots, X_n$, and insert Rademacher variables $\varepsilon_1, \dots, \varepsilon_n$ as in the proof of Lemma 2.3.1, to see that the left side of the lemma is bounded above by $E_{X,Y} E_{\varepsilon} \|G_n(\mathcal{C})\|_{\mathcal{C}}$, where $G_n(\mathcal{C}) = n^{-1/2} \sum_{i=1}^n \varepsilon_i (1_{\mathcal{C}}(X_i) - 1_{\mathcal{C}}(Y_i))$. The coefficient vector $(1_{\mathcal{C}}(X_1) - 1_{\mathcal{C}}(Y_1), \dots, 1_{\mathcal{C}}(X_n) - 1_{\mathcal{C}}(Y_n))$ belongs to $\{-1, 0, 1\}^n$ and hence has square Euclidean norm bounded above by n . It is determined by the set of points from $X_1, \dots, X_n, Y_1, \dots, Y_n$ picked out by the set $\mathcal{C}$. Thus these vectors range over fewer than $\max_{x \in \mathcal{X}^{2n}} \Delta_{2n}(\mathcal{C}, x)$ different vectors in $\{-1, 0, 1\}^n$ as $\mathcal{C}$ ranges over $\mathcal{C}$, by the definition of the latter quantity. Consequently, the first inequality follows by a maximal inequality for Rademacher variables. (See Problem 2.2.26. Apart from the exact multiplicative constants, this is essentially Lemma 2.2.2, as the ψ_2 -norms of the $G_n(\mathcal{C})$ are bounded by $\sqrt{6}$ in view of Lemma 2.2.8.)

For the second inequality we use that $\mathcal{C}$ is a VC-class. We first bound $\max_{x \in \mathcal{X}^{2n}} \Delta_{2n}(\mathcal{C}, x)$ as in Corollary 2.6.3, and next use Problem 2.6.7 to bound this further by $(2ne/V(\mathcal{C}))^{V(\mathcal{C})}$. This gives the bound $\log(2 \max_{x \in \mathcal{X}^{2n}} \Delta_{2n}(\mathcal{C}, x)) \leq \log 2 + V(\mathcal{C}) \log(2e) + V(\mathcal{C}) \log(n/V(\mathcal{C}))$, and the bound on the right of the lemma follows as $1 \leq V(\mathcal{C})$ and $\log(4e) \leq 4$. ■

2.14.5 Bounds on Orlicz Norms

The preceding sections mostly focus on bounds on the mean $E^* \|\mathbb{G}_n\|_{\mathcal{F}}$ of the norm of the empirical process, which is a basic measure of its size. Bounds on the mean can be turned into bounds on higher moments and other Orlicz norms, or tail probabilities, by general arguments. For uniformly bounded classes of functions the deviations from the mean are exponentially small, as seen in Chapter 2.15, and integration of these exponential bounds gives bounds on higher moments or tail probabilities. Alternatively, in this section we consider bounds for more general classes of functions, using moments or Orlicz norms of the envelope function.

Bounds on Orlicz norms imply bounds on tail probabilities by Markov's inequality. For instance, a finite L_p -norm $\|X\|_p$ yields a polynomial tail bound

$$P(|X| > t) \leq \frac{1}{t^p} \|X\|_p^p.$$

Alternatively, a bound on the Orlicz norm $\|X\|_{\psi_p}$ for $\psi_p(x) = \exp x^p - 1$ and $p \geq 1$ leads to an exponential tail bound

$$P(|X| > t) \leq 2e^{-t^p / \|X\|_{\psi_p}^p}.$$

To obtain similar bounds also for p in the range $0 < p < 1$, we cannot use the functions $x \mapsto \exp x^p - 1$, as these are not convex and hence do not define Orlicz norms. However, the lack of convexity is only on the interval (c_p, ∞) , for $c_p = (1/p - 1)^{1/p}$, and can be resolved by defining $\psi_p(x)$ as $\exp x^p - 1$ for $x \geq c_p$ and to be linear and continuous on $(0, c_p)$. The resulting Orlicz norms can be used to obtain exponential bounds as before.

The following theorem shows that a general Orlicz norm of the supremum $\|\mathbb{G}_n\|_{\mathcal{F}}^*$ of the empirical process is bounded by its L_1 -norm plus the corresponding Orlicz norm of the envelope function F . In fact, Orlicz norms of sums of independent processes can in general be bounded by their L_1 -norm plus a maximum or sum of Orlicz norms of the individual terms, as noted in the proof.

2.14.23 Theorem. *Let $\mathcal{F}$ be a class of measurable functions with measurable envelope function F . Then*

$$\begin{aligned} \|\|\mathbb{G}_n\|_{\mathcal{F}}^*\|_{P,p} &\lesssim E^* \|\mathbb{G}_n\|_{\mathcal{F}} + n^{-1/2+1/p} \|F\|_{P,p} & (p \geq 2), \\ \|\|\mathbb{G}_n\|_{\mathcal{F}}^*\|_{P,\psi_p} &\lesssim E^* \|\mathbb{G}_n\|_{\mathcal{F}} + n^{-1/2} (1 + \log n)^{1/p} \|F\|_{P,\psi_p} & (0 < p \leq 1), \\ \|\|\mathbb{G}_n\|_{\mathcal{F}}^*\|_{P,\psi_p} &\lesssim E^* \|\mathbb{G}_n\|_{\mathcal{F}} + n^{-1/2+1/q} \|F\|_{P,\psi_p} & (1 < p \leq 2). \end{aligned}$$

Here $1/p + 1/q = 1$ and the constants in the inequalities $\lesssim$ depend only on the type of norm involved in the statement.

Proof. Specialization of Proposition A.1.6 to the present situation gives the inequalities

$$\begin{aligned} \|\|\mathbb{G}_n\|_{\mathcal{F}}^*\|_{P,p} &\lesssim \|\|\mathbb{G}_n\|_{\mathcal{F}}^*\|_{P,1} + n^{-1/2} \left\| \max_{1 \leq i \leq n} F(X_i) \right\|_{P,p} & (p \geq 1), \\ \|\|\mathbb{G}_n\|_{\mathcal{F}}^*\|_{P,\psi_p} &\lesssim \|\|\mathbb{G}_n\|_{\mathcal{F}}^*\|_{P,1} + n^{-1/2} \left\| \max_{1 \leq i \leq n} F(X_i) \right\|_{P,\psi_p} & (0 < p \leq 1). \end{aligned}$$

Next, Lemma 2.2.2 can be used to further bound the maxima appearing on the right. This gives the first two lines of the theorem. The third line is immediate from Proposition A.1.6. ■

To obtain tail bounds for the random variable $\|\mathbb{G}_n\|_{\mathcal{F}}^*$, the last theorem may be combined with the bounds on $E^*\|\mathbb{G}_n\|_{\mathcal{F}}$ given in the preceding sections. While Theorem 2.14.23 is valid for any class $\mathcal{F}$ of functions, the bounds on $E^*\|\mathbb{G}_n\|_{\mathcal{F}}$ rely on the finiteness of an entropy integral. If either the uniform entropy integral or the bracketing entropy integral of the class $\mathcal{F}$ is finite, then

$$E^*\|\mathbb{G}_n\|_{\mathcal{F}} \lesssim \|F\|_{P,2},$$

where the constant depends on the entropy integral. The $L_2(P)$ -norm of the envelope function can in turn be bounded by an exponential Orlicz norm. Combination with the last theorem shows that

$$\|\|\mathbb{G}_n\|_{\mathcal{F}}^*\|_{P,\psi_p} \lesssim \|F\|_{P,\psi_p} \quad (0 < p \leq 2).$$

We can conclude that $\|\mathbb{G}_n\|_{\mathcal{F}}^*$ has tails of the order $\exp(-Ct^p)$ for some $0 < p \leq 2$, whenever the envelope function F has tails of this order. Similar reasoning can be applied using the $L_p(P)$ -norms.

For large n and $0 < p < 2$, the size of the ψ_p -norm of $\|\mathbb{G}_n\|_{\mathcal{F}}^*$ is mostly determined by the entropy integral and the L_2 -norm of the envelope: according to the preceding theorem, the ψ_p -norm of the envelope enters the upper bound multiplied by a factor that converges to zero as $n \rightarrow \infty$.

To see some of the strength of the preceding theorems, it is instructive to apply them to a class $\mathcal{F}$ consisting of a single function f . Then $J(\delta, \{f\}) = \delta$ and the envelope function is $|f|$. For $\mathcal{F} = \{f\}$, Theorem 2.14.1 gives

$$\|\mathbb{G}_n f\|_{P,p} \lesssim \left\| \left(\frac{1}{n} \sum_{i=1}^n f^2(X_i) \right)^{1/2} \right\|_{P,p} \leq \|f\|_{P,p}, \quad p \geq 2.$$

This is the upper half of the *Marcinkiewicz-Zygmund inequality*, Proposition A.1.9. Theorem 2.14.23 yields a tightening of the inequality between the far left and the right sides. More interestingly, it gives bounds for the exponential Orlicz norms

$$\begin{aligned} \|\mathbb{G}_n f\|_{P,\psi_p} &\lesssim \|f\|_{P,2} + n^{-1/2}(1 + \log n)^{1/p} \|f\|_{P,\psi_p} & (0 < p \leq 1), \\ \|\mathbb{G}_n f\|_{P,\psi_p} &\lesssim \|f\|_{P,2} + n^{-1/2+1/q} \|f\|_{P,\psi_p} & (1 < p \leq 2). \end{aligned}$$

The last line for $p = q = 2$ shows that the random variable $\mathbb{G}_n f$ is sub-Gaussian whenever $\|f\|_{P,\psi_2}$ is finite. In fact, because the L_2 -norm is bounded by the ψ_2 -norm, the inequality can be simplified to

$$(2.14.24) \quad \|\mathbb{G}_n f\|_{P,\psi_2} \lesssim \|f\|_{P,\psi_2}.$$

Apart from a universal constant, this generalizes Hoeffding's inequality, which makes the same statement for uniformly bounded functions f . Hoeffding's inequality for linear combinations of Rademacher variables is given by Lemma 2.2.8 and is the essential ingredient in the proof of Theorem 2.14.1.

2.14.6 Tail Bounds

In this section we derive bounds on tail probabilities of the supremum of the empirical process $\|\mathbb{G}_n\|_{\mathcal{F}}$ for classes $\mathcal{F}$ that are uniformly bounded, under restrictions on their uniform entropy or bracketing entropy numbers. We present both Hoeffding- and Bernstein-type bounds.

Throughout the section it is assumed that $0 \leq f \leq 1$ for every $f \in \mathcal{F}$, and also that the class $\mathcal{F}$ is pointwise separable.

The prototype for the Hoeffding type bounds is the celebrated *Dvoretzky-Kiefer-Wolfowitz inequality* or *DKW-inequality*. This asserts that the classical empirical process $\sqrt{n}(\mathbb{F}_n - F)$ of an i.i.d. sample of real-valued observations from the cumulative distribution function F satisfies, for every $t > 0$,

$$\mathbf{P}(\sqrt{n}\|\mathbb{F}_n - F\|_{\infty} > t) \leq 2e^{-2t^2}.$$

Our first three theorems are generalizations of this result to the norm of the empirical process indexed by a general, uniformly bounded class of functions. The exponential factor e^{-2t^2} is retained, but additional factors enter the bound, which depend on the size of the class, as measured by covering or entropy numbers.

The first theorem is valid for classes $\mathcal{F}$ with polynomial covering numbers or polynomial bracketing numbers: for some constants $V \geq 1$ and $K \geq 1$,

$$(2.14.25) \quad \sup_Q N(\varepsilon, \mathcal{F}, L_2(Q)) \leq \left(\frac{K}{\varepsilon}\right)^V, \quad \text{for every } 0 < \varepsilon < K,$$

or

$$(2.14.26) \quad N_{[]}(\varepsilon, \mathcal{F}, L_2(P)) \leq \left(\frac{K}{\varepsilon}\right)^V, \quad \text{for every } 0 < \varepsilon < K.$$

In the first inequality, the supremum is taken over all probability measures Q . In particular, the theorem is valid for Vapnik-Červonenkis classes: according to Theorem 2.6.7, a VC-class of dimension $V(\mathcal{F})$ with envelope function $F = 1$ satisfies (2.14.25) for $V = 2V(\mathcal{F})$ and a constant K that depends on V only.

The second theorem applies to classes such that, for some constants $0 < W < 2$ and K ,

$$(2.14.27) \quad \sup_Q \log N(\varepsilon, \mathcal{F}, L_2(Q)) \leq K \left(\frac{1}{\varepsilon}\right)^W, \quad \text{for every } \varepsilon > 0.$$

Again the supremum is taken over all probability measures Q .

2.14.28 Theorem. *Let $\mathcal{F}$ be a pointwise separable class of measurable functions $f: \mathcal{X} \rightarrow [0, 1]$ that satisfies (2.14.25) or (2.14.26). Then, for every $t > 0$,*

$$\mathbf{P}(\|\mathbb{G}_n\|_{\mathcal{F}} > t) \leq \left(\frac{Dt \vee \sqrt{V}}{\sqrt{V}}\right)^V e^{-2t^2},$$

for a constant D that depends on K only.

2.14.29 Theorem. Let $\mathcal{F}$ be a pointwise separable class of measurable functions $f: \mathcal{X} \rightarrow [0, 1]$ that satisfies (2.14.27). Then, for every $\delta > 0$ and $t > 0$,

$$\mathbf{P}(\|\mathbb{G}_n\|_{\mathcal{F}} > t) \leq C e^{Dt^{U+\delta}} e^{-2t^2},$$

where $U = W(6 - W)/(2 + W)$ and the constants C and D depend on K , W , and δ only.

We note that the exponent U in the second theorem increases from 0 to 2 as W increases from 0 to 2.

While the constant 2 in the exponential e^{-2t^2} in Theorem 2.14.28 is sharp, the power of the additional term is not. For instance, the empirical distribution function on the line satisfies the exponential bound De^{-2t^2} , whereas the bound obtained from Theorem 2.14.28 is of the form $Dt^2 e^{-2t^2}$, since (2.14.25) holds with $V = 2$ in this case. We shall consider two improvements for $\mathcal{F}$ equal to a class of indicators of sets.

The exponential bounds for the suprema $\|\mathbb{G}_n\|_{\mathcal{F}}$ are based on exponential bounds for the individual variables $\mathbb{G}_n f$. For sets $\mathcal{C}$ with probability bounded away from zero and one, an improved exponential bound can be derived from the exponential bound of Talagrand for the tail of a binomial random variable

$$\mathbf{P}(|\mathbb{G}_n(\mathcal{C})| > t) \leq \frac{K}{t} e^{-2t^2}, \quad \text{for every } t > 0$$

(see Proposition A.6.4). This suggests that it might be possible to improve on Theorem 2.14.28 in the case of sets. In fact, this improvement is possible. We replace (2.14.25) and (2.14.26) by their corresponding L_1 versions. Suppose that $\mathcal{C}$ is a class of sets such that, for given constants K and V , either

$$(2.14.30) \quad \sup_Q N(\varepsilon, \mathcal{C}, L_1(Q)) \leq \left(\frac{K}{\varepsilon}\right)^V, \quad \text{for every } 0 < \varepsilon < K,$$

or

$$(2.14.31) \quad N_{[]}(\varepsilon, \mathcal{C}, L_1(P)) \leq \left(\frac{K}{\varepsilon}\right)^V, \quad \text{for every } 0 < \varepsilon < K.$$

The supremum is taken over all probability measures Q . We note that the present V corresponds to $1/2$ times the exponent V in conditions (2.14.25) and (2.14.26).

2.14.32 Theorem. Let $\mathcal{C}$ be a pointwise separable class of sets that satisfies (2.14.30) or (2.14.31) with $V \geq 1$. Then, for every $t > 0$,

$$\mathbf{P}(\|\mathbb{G}_n\|_{\mathcal{C}} > t) \leq \frac{D}{t} \left(\frac{D(t^2 \vee V)}{V}\right)^V e^{-2t^2},$$

for a constant D that depends on K only.

Considering the special case of the empirical distribution function on the line once again, we see that Theorem 2.14.32 still does not yield the known exponential bound De^{-2t^2} . Since the collection of left half-lines satisfies (2.14.30) with $V = 1$, Theorem 2.14.32 yields a bound of the form $Dt e^{-2t^2}$.

To obtain the “correct” power of t in the bounds requires consideration of the size of the sets in a neighborhood of the collection $\{C \in \mathcal{C}: P(C) = 1/2\}$ for which the variance of $\mathbb{G}_n(C)$ is maximal. To this end, define

$$\mathcal{C}_\delta = \left\{ C \in \mathcal{C}: |P(C) - 1/2| \leq \delta \right\}.$$

It is the dimensionality of such neighborhoods that really governs which power of t is possible. The following theorem gives a refinement of Theorem 2.14.32 to clarify this dependence.

2.14.33 Theorem. *Let $\mathcal{C}$ be a pointwise separable class of sets that satisfies either (2.14.30) or (2.14.31), and suppose moreover that*

$$(2.14.34) \quad N(\varepsilon, \mathcal{C}_\delta, L_1(P)) \leq K' \delta^W \varepsilon^{-V'}, \quad \text{for every } \delta \geq \varepsilon > 0,$$

for some constant K' . Then

$$P(\|\mathbb{G}_n\|_{\mathcal{C}} > t) \leq D t^{2V' - 2W} e^{-2t^2},$$

for every $t > K\sqrt{W}$ and a constant D that depends on K, K', W, V , and V' only.

For $\mathcal{C}$ equal to the collection of left half-lines in $\mathbb{R}$, inequality (2.14.34) holds with $V' = 1$ and $W = 1$ (at worst) for any probability measure P . Hence Theorem 2.14.33 gives a bound of the form $D e^{-2t^2}$ in agreement with the bound of Dvoretzky, Kiefer and Wolfowitz (1956).

When P is uniform on $[0, 1]^d$ and $\mathcal{C}$ is the collection of lower-left subrectangles of $[0, 1]^d$, (2.14.34) holds with $V' = d$, $W = 1$, and the bound given by Theorem 2.14.33 is of the form $D t^{2(d-1)} e^{-2t^2}$. However, this bound is not uniform in P . See Smith and Dudley (1992) and Adler and Brown (1986).

The preceding theorems give tail bounds in terms of entropy and envelope but do not show the dependence on the variances of the functions in $\mathcal{F}$. In many applications, especially when an entire collection $\mathcal{F}$ is replaced by a subcollection, as in the study of oscillation moduli, it is useful to introduce the maximal variance

$$(2.14.35) \quad \sigma_{\mathcal{F}}^2 = \|P(f - Pf)^2\|_{\mathcal{F}}$$

into the bound. If every f takes its values in the interval $[0, 1]$, this variance is maximally $1/4$. Thus a bound of the order $\exp(-(1/2)t^2/\sigma_{\mathcal{F}}^2)$ could be better than the bounds given by the preceding theorems. This bound is valid in the limit as $n \rightarrow \infty$, but for finite sample size a correction is necessary.

2.14.36 Theorem. *Let $\mathcal{F}$ be a pointwise separable class of measurable functions $f: \mathcal{X} \rightarrow [0, 1]$ that satisfies (2.14.25). Then for every $\sigma_{\mathcal{F}}^2 \leq \sigma^2 \leq 1$ and every $\delta > 0$,*

$$P(\|\mathbb{G}_n\|_{\mathcal{F}} > t) \leq C \left(\frac{1}{\sigma}\right)^{2V} \left(1 \vee \frac{t}{\sigma}\right)^{3V+\delta} e^{-\frac{1}{2} \frac{t^2}{\sigma^2 + (3+t)/\sqrt{n}}},$$

for every $t > 0$ and a constant C that depends on K, V , and δ only.

2.14.37 Theorem. Let $\mathcal{F}$ be a pointwise separable class of measurable functions $f: \mathcal{X} \rightarrow [0, 1]$ that satisfies (2.14.27). Then for every $\sigma_{\mathcal{F}}^2 \leq \sigma^2 \leq 1$, every $\delta > 0$, $0 < p(1-u) < 2$, and $0 < u < 1$,

$$P(\|\mathbb{G}_n\|_{\mathcal{F}} > t) \leq C e^{D(\frac{1}{\sigma})^W} \left(\frac{t}{\sigma}\right)^{p(1-u)+\delta} + 5\left(\frac{t}{\sigma}\right)^{2u} e^{-\frac{1}{2} \frac{t^2}{\sigma^2 + (3(t/\sigma)^u + t)/\sqrt{n}}},$$

for every $t > 0$, where $p = W(6-W)/(2-W)$ and the constants C and D depend on K , W , and δ only.

The exponential factor in the first theorem is roughly of Bernstein type, whereas the factor in the second theorem has an additional term $(t/\sigma)^u/\sqrt{n}$ in the exponent. This additional term, and also the leading factors in front of the exponential, are partly caused by the fact that the bounds are valid for every $t > 0$. One may note that Bernstein's inequality (Lemma 2.2.10) concerns deviations of an average from its mean. A corresponding elegant inequality for deviations of the variable $\|\mathbb{G}_n\|_{\mathcal{F}}$ from its mean is valid too, even for general classes $\mathcal{F}$, which need not satisfy entropy conditions, as shown in the next chapter (see (2.15.8)).

If we do not insist on the “sharp” factor $1/2$ in the exponent, then bounds of a simpler appearance are possible. The following results are consequences of the deviation bounds in Chapter 2.15, and will be used in the proofs of the preceding results.

2.14.38 Theorem. For every pointwise separable class $\mathcal{F}$ of measurable functions $f: \mathcal{X} \rightarrow [0, 1]$ and every $t > 0$, with $\mu_{n,\mathcal{F}} = E\|\mathbb{G}_n\|_{\mathcal{F}}$ and $\sigma_{\mathcal{F}}^2$ as in (2.14.35),

$$P(\|\mathbb{G}_n\|_{\mathcal{F}} > 4t) \leq \begin{cases} \exp\left(-\frac{t^2\sqrt{n}}{\mu_{n,\mathcal{F}} + \sqrt{n}\sigma_{\mathcal{F}}^2}\right), & \mu_{n,\mathcal{F}} \leq t \leq \mu_{n,\mathcal{F}} + \sigma_{\mathcal{F}}^2\sqrt{n}, \\ \exp\left(-t\sqrt{n} \log \frac{et}{\mu_{n,\mathcal{F}} + \sqrt{n}\sigma_{\mathcal{F}}^2}\right), & t \geq \mu_{n,\mathcal{F}} + \sigma_{\mathcal{F}}^2\sqrt{n}. \end{cases}$$

Proof. The function $h(t) = t(\log t - 1) + 1$ satisfies $h(1+t) \geq t^2/4$, for $t \leq 4$; and $h(1+t) \geq t(\log(et))/3$, for $t \geq 1$. Therefore, by Theorem 2.15.5,

$$P(\|\mathbb{G}_n\|_{\mathcal{F}} > \mu_n + s) \leq \begin{cases} e^{-s^2/(4w_n)}, & 0 \leq s \leq 4\sqrt{n}w_n, \\ e^{-\sqrt{n}s \log(es/(\sqrt{n}w_n))/3}, & s \geq \sqrt{n}w_n. \end{cases}$$

Here $w_n = \sigma^2 + 2\mu_n/\sqrt{n}$ and we drop the index $\mathcal{F}$ from $\mu_{n,\mathcal{F}}$ and $\sigma_{\mathcal{F}}$. For $s + \mu_n = 4t$, we have $s \geq 3t$ if $t \geq \mu_n$, and the two exponents can be seen to be greater than the exponents in the theorem. The range $0 \leq s \leq 4\sqrt{n}w_n$ translates into $\mu_n/4 \leq t \leq 9\mu_n/4 + \sigma^2\sqrt{n}$. The theorem follows. ■

Define a collection of functions $\varphi_s: [0, \infty) \rightarrow [0, \infty)$ by, for $s > 0$,

$$(2.14.39) \quad \varphi_s(t) = \frac{t^2}{s} 1\{t \leq s\} + t \log\left(\frac{et}{s}\right) 1\{t > s\}.$$

Every function ϕ_s is continuous and increasing, and so is $t \mapsto \phi_s(t)/t$. Theorem 2.14.38 can be formulated using the function φ_S for $S = \sqrt{n}\mu_{n,\mathcal{F}} + n\sigma_{\mathcal{F}}^2$, and asserts that

$$P(\|\mathbb{G}_n\|_{\mathcal{F}} > 4t) \leq e^{-\varphi_S(t\sqrt{n})}, \quad t \geq \mu_{n,\mathcal{F}}.$$

The functions ϕ_s correspond to a Gaussian upper bound for small values of t , and a logarithmically better than exponential bound in the far tail. They imply maximal inequalities of mixed sub-Gaussian and sub-exponential types (see Problem 2.14.10), much as Bernstein's inequality (compare Section 2.2.4).

The following lemma can be regarded as a uniform version of Kiefer's inequality for a binomial variable, Corollary A.6.3. The constant 11 does not carry special significance.

2.14.40 Lemma (Uniform small variance exponential bound). *There exists a universal constant $\sigma_0 > 0$ such that, for every pointwise separable class $\mathcal{F}$ of measurable functions $f: \mathcal{X} \rightarrow [0, 1]$ with $\sigma_{\mathcal{F}}^2 \leq \sigma_0^2$ and $\mu_{n,\mathcal{F}} \leq \sigma_0^2\sqrt{n}$,*

$$P(\|\mathbb{G}_n\|_{\mathcal{F}} > t) \leq \exp(-11t^2), \quad \text{for every } t \geq 2\mu_{n,\mathcal{F}}.$$

Proof. Since $|f(x) - Pf| \leq 1$ for all x , the probability on the left side of the above display is zero for all $t \geq \sqrt{n}$. Hence any nonnegative bound will be trivially true for $t > \sqrt{n}$, and we may restrict attention to $t \leq \sqrt{n}$.

For $t \geq 2\mu_{n,\mathcal{F}}$ the left side is bounded above by $P(\|\mathbb{G}_n\|_{\mathcal{F}} - \mu_{n,\mathcal{F}} > t/2)$. Hence the theorem follows if we can show the bound with 44 instead of 11 in the exponent for the centered norm of the empirical process.

By the assumptions $w_n = \sigma_{\mathcal{F}}^2 + 2\mu_{n,\mathcal{F}}/\sqrt{n} \leq 3\sigma_0^2$.

Therefore, for $t \leq 3\varepsilon\sqrt{n}$ we have $w_n + \frac{1}{3}t/\sqrt{n} \leq 3\sigma_0^2 + \varepsilon$, which is smaller than $1/88$ if σ_0 and ε are chosen sufficiently small. The bound then follows from the consequence (2.15.8) of Theorem 2.15.5.

Finally if $3\varepsilon\sqrt{n} \leq t \leq \sqrt{n}$, then $t/(\sqrt{n}w_n) \geq \varepsilon/\sigma_0^2$, which can be made arbitrarily large by first choosing small ε to satisfy the bound of the preceding paragraph and next choosing σ_0 to be still smaller. Because $h(t) \asymp t \log t$ as $t \rightarrow \infty$, and hence $h(t) \geq 44t$ for sufficiently large t , we then have $nw_n h(1 + t/(\sqrt{n}w_n)) \geq 44\sqrt{nt} \geq 44t^2$. In that case the bound follows from Theorem 2.15.5. ■

2.14.6.1 Proofs

We do not include proofs of all of Theorems 2.14.28–2.14.37. Theorems 2.14.29, 2.14.36, and 2.14.37 are proved completely in this subsection. For Theorem 2.14.28, we give a proof assuming (2.14.25), but with the power of t in the theorem equal to $3V + \delta$ instead of V . After developing more tools needed to prove the tighter bounds in the following subsection, we give a complete proof of Theorem 2.14.32 under the assumption (2.14.30) in Subsection 2.14.6.2. For complete proofs of Theorems 2.14.28, 2.14.32, and 2.14.33, we refer the interested reader to Talagrand (1994).

The proofs of the tail bounds stated in Theorems 2.14.29, 2.14.36, and 2.14.37 rely on two key lemmas. The first lemma bounds the tail probability of interest by a corresponding tail probability for sampling n observations without replacement from the empirical measure based on a sample of size $N = mn$ (with large m chosen dependent on t in the proof). This lemma may be considered an alternative to the symmetrization inequalities involving Rademacher variables derived previously: the present bound is more complicated, but it appears to be more economic.

Sampling without replacement can be described in terms of a random permutation. Let $(R_1, \dots, R_N)$ be uniformly distributed on the set of permutations of $\{1, \dots, N\}$, and let $X_1, \dots, X_N$ be an independent i.i.d. sample from P . Define $n' = N - n$ and

$$\tilde{\mathbb{P}}_{n,N} = \frac{1}{n} \sum_{i=1}^n \delta_{X_{R_i}}; \quad \mathbb{P}_N = \frac{1}{N} \sum_{i=1}^N \delta_{X_i}.$$

Then the tail probabilities of the empirical process of n observations can be bounded by the tail probabilities of $\tilde{\mathbb{P}}_{n,N} - \mathbb{P}_N$.

2.14.41 Lemma. *For all $0 < a < 1$, any $\sigma^2 \geq \|P(f - Pf)^2\|_{\mathcal{F}}$, and every $t > 0$,*

$$\mathbb{P}(\|\mathbb{P}_n - P\|_{\mathcal{F}} > t) \leq \left[1 - \frac{\sigma^2}{(1-a)^2 t^2 n'}\right]^{-1} \mathbb{P}\left(\|\tilde{\mathbb{P}}_{n,N} - \mathbb{P}_N\|_{\mathcal{F}} > ta \frac{n'}{N}\right).$$

Proof. See Devroye (1982) and Shorack and Wellner (1986), pages 829–830. ■

The motivation for this step is the availability of several relatively sharp bounds for the tail probabilities of the mean of a sample of size n without replacement. In the proof of the main results, these are applied conditionally on the sample $X_1, \dots, X_N$. For given real numbers $c_1, \dots, c_N$, set

$$\bar{c}_N = \frac{1}{N} \sum_{i=1}^N c_i; \quad \sigma_N^2 = \frac{1}{N} \sum_{i=1}^N (c_i - \bar{c}_N)^2; \quad \Delta_N = \max_{1 \leq i \leq N} c_i - \min_{1 \leq i \leq N} c_i.$$

The following lemma collects four exponential bounds on the tail probabilities of a mean of a sample without replacement.

2.14.42 Lemma. *Let $U_1, \dots, U_n$ be a random sample without replacement from the real numbers $\{c_1, \dots, c_N\}$. Then for every $t > 0$,*

$$\mathbb{P}(|\bar{U}_n - \bar{c}_N| > t) \leq \begin{cases} 2 \exp\left(-\frac{2nt^2}{\Delta_N^2}\right) & (\text{Hoeffding}), \\ 2 \exp\left(-\frac{2nt^2}{(1-(n-1)/N)\Delta_N^2}\right) & (\text{Serfling}), \\ 2 \exp\left(-\frac{nt^2}{2\sigma_N^2 + t\Delta_N}\right) & (\text{Hoeffding-Bernstein}), \\ 2 \exp\left(-\frac{nt^2}{m\sigma_N^2}\right) & \text{if } N = mn \text{ (Massart)}. \end{cases}$$

Proof. For a proof of the first three inequalities, see Shorack and Wellner (1986). We prove the fourth.

The sample without replacement can be generated in two steps. First, partition the set of points $\{c_1, \dots, c_N\}$ randomly into n subsets $J_1, \dots, J_n$ of m elements each. Next, randomly choose one element of each subset. Let E_2 denote expectation with respect to the second step only, treating the partition obtained in the first step as fixed. Given the partition, the variables $U_1, \dots, U_n$ are independent with means $\bar{c}_1, \dots, \bar{c}_n$ equal to the averages over the c_i in each partitioning set. The average of these averages is the grand average $\bar{c}_N$. Thus

$$E_2 e^{s(\bar{U}_n - \bar{c}_N)} = \prod_{i=1}^n E_2 e^{s(U_i - \bar{c}_i)/n} \leq \prod_{i=1}^n e^{s^2 \Delta_i^2 / 8n^2},$$

by Hoeffding's inequality (Problem 2.14.2), where the numbers $\Delta_i = \max_{j \in J_i} c_j - \min_{j \in J_i} c_j$ are the ranges of the partitioning sets. Since $\Delta_i^2 \leq 2 \sum_{j \in J_i} (c_j - \bar{c}_N)^2$, the sum of the Δ_i^2 is bounded by $2N\sigma_N^2$. Conclude that, for every s ,

$$P(\bar{U}_n - \bar{c}_N > t) \leq e^{-st} E E_2 e^{s(\bar{U}_n - \bar{c}_N)} \leq e^{-st + s^2 m \sigma_N^2 / 4n}.$$

The choice $s = 2nt/m\sigma_N^2$ yields an upper bound equal to half the upper bound given by the lemma. The probability $P(-\bar{U}_n + \bar{c}_N > t)$ can be bounded similarly. ■

Proof of Theorems 2.14.28, 2.14.29, 2.14.36, and 2.14.37. (Recall that Theorem 2.14.28 will only be proved with $3V + \delta$ instead of V .) It suffices to prove the inequalities for sufficiently large t , since their validity for small t can be ensured by choice of the constant C . Here “sufficiently large” means larger than some value depending only on K, V, W , and δ as in the statements of the theorems.

For a given $N = mn$ and a sequence of positive constants $\varepsilon_q \downarrow 0$, take a minimal ε_q -net for $\mathcal{F}$ for the $L_2(\mathbb{P}_N)$ -semimetric. For each f , let $\pi_q f$ be a closest element in this net. Fix q_0 . The series

$$(\tilde{\mathbb{P}}_{n,N} - \mathbb{P}_N) \pi_{q_0} f + \sum_{q > q_0} (\tilde{\mathbb{P}}_{n,N} - \mathbb{P}_N) (\pi_q - \pi_{q-1}) f$$

is convergent with limit $(\tilde{\mathbb{P}}_{n,N} - \mathbb{P}_N) f$. To see this, note first that the q th partial sum of the series telescopes out to $(\tilde{\mathbb{P}}_{n,N} - \mathbb{P}_N) \pi_q f$. Since $\tilde{\mathbb{P}}_{n,N} f \leq m \mathbb{P}_N f$ for every nonnegative f , we have

$$|(\tilde{\mathbb{P}}_{n,N} - \mathbb{P}_N)(f - \pi_q f)|^2 \leq 2(\tilde{\mathbb{P}}_{n,N} + \mathbb{P}_N)(f - \pi_q f)^2 \leq 2(m+1)\varepsilon_q^2 \rightarrow 0,$$

and the conclusion follows. The triangle inequality can now be applied to obtain

$$\|\tilde{\mathbb{P}}_{n,N} - \mathbb{P}_N\|_{\mathcal{F}} \leq \|(\tilde{\mathbb{P}}_{n,N} - \mathbb{P}_N) \pi_{q_0} f\|_{\mathcal{F}} + \sum_{q > q_0} \|(\tilde{\mathbb{P}}_{n,N} - \mathbb{P}_N) (\pi_q - \pi_{q-1}) f\|_{\mathcal{F}}.$$

If the ε_q -net has N_q elements, then the suprema are over at most N_{q_0} in the first term on the right and $N_q N_{q-1}$ elements in each of the terms of the series, respectively. Conclude that for any positive numbers η_q and b with $\sum_{q > q_0} \eta_q + b \leq 1$:

$$\begin{aligned} P_R(\|\tilde{\mathbb{P}}_{n,N} - \mathbb{P}_N\|_{\mathcal{F}} > t) &\leq N_{q_0} \left\| P_R(|(\tilde{\mathbb{P}}_{n,N} - \mathbb{P}_N) \pi_{q_0} f| > tb) \right\|_{\mathcal{F}} \\ (2.14.43) \quad &+ \sum_{q > q_0} N_q^2 \left\| P_R(|(\tilde{\mathbb{P}}_{n,N} - \mathbb{P}_N) (\pi_q - \pi_{q-1}) f| > t\eta_q) \right\|_{\mathcal{F}}. \end{aligned}$$

The probabilities on the right will be bounded with the help of the various exponential bounds of Lemma 2.14.42. Lemma 2.14.41 turns the resulting upper bounds into bounds on the tails of the distribution of $\|\mathbb{P}_n - P\|_{\mathcal{F}}$. For appropriate choices of tuning constants, this gives the desired results. In all four cases the second term on the right is negligible with respect to the first.

For the proofs of Theorems 2.14.28 and 2.14.29 (with $3V + \delta$ instead of V), use Hoeffding's inequality on the first term in the right of (2.14.43) and Massart's inequality on each of the terms of the series. Thus, (2.14.43) is bounded by

$$(2.14.44) \quad N_{q_0} 2 \exp(-2nt^2b^2) + \sum_{q>q_0} N_q^2 \exp\left(\frac{-nt^2\eta_q^2}{m4\varepsilon_{q-1}^2}\right).$$

Note that $\mathbb{P}_N(\pi_q f - \pi_{q-1} f)^2 \leq 2\varepsilon_q^2 + 2\varepsilon_{q-1}^2$ is bounded by $4\varepsilon_{q-1}^2$.

For the proof of Theorem 2.14.28, choose α large and

$$a = 1 - t^{-2}; \quad b = 1 - t^{-2}; \quad m = \lfloor t^2 \rfloor; \quad q_0 = 2 + \lfloor t^{2/(\alpha-1)} \rfloor; \\ \sqrt{m} \varepsilon_q = q^{-\alpha-1}; \quad \eta_q = (q-1)^{-\alpha}.$$

Combination of Lemma 2.14.41 and (2.14.25) with (2.14.43) and (2.14.44) yields as upper bound for $P(\|\mathbb{G}_n\|_{\mathcal{F}} > t)$ a constant times

$$\left(1 - \frac{n}{4(1-a)^2 t^2 n'}\right)^{-1} \left[(tq_0^{\alpha+1})^V 2 \exp(-2t^2 b^2 a^2 n'^2 / N^2) \right. \\ \left. + \sum_{q>q_0} (tq^{\alpha+1})^{2V} 2 \exp(-\frac{1}{4} t^2 (q-1)^2 a^2 n'^2 / N^2) \right].$$

For sufficiently large t , the leading multiplicative factor outside the square brackets is bounded by 2. The quotient $n'/n = 1 - 1/m$ is bounded below by $1 - 2t^{-2}$. Furthermore, the series is bounded by its q_0 th term. To see this, write it as $\sum_{q>q_0} \exp(-\psi(q))$ for $\psi(q) = (1/4)t^2(q-1)^2 a^2 n'^2 / N^2 - 2V(\alpha+1) \log q$ and apply Problem 2.14.3. Thus, for sufficiently large t , the last display is up to a constant bounded by

$$t^V \left(3 + t^{\frac{2}{\alpha-1}}\right)^{V(\alpha+1)} \exp(-2t^2(1-t^{-2})^6) \\ + t^{2V} (2+t^2)^{2V(\alpha+1)} \exp\left(-\frac{1}{4} t^2 t^{\frac{4}{\alpha-1}} (1-2t^{-2})^4\right),$$

the second term being an upper bound for the q_0 th term of the series. For large t , the second term is certainly bounded by e^{-2t^2} . For sufficiently large α , the first term is bounded by a multiple of $(1 \vee t)^{3V+\delta} e^{-2t^2}$. This concludes the proof of the first theorem.

For the proof of Theorem 2.14.29, choose α large and

$$a = 1 - t^{-2}; \quad b = 1 - t^{-r}; \quad m = \lfloor t^2 \rfloor; \quad q_0 = 2 + \lfloor t^{r/(\alpha-1)} \rfloor; \\ \sqrt{m} \varepsilon_q = q^{-\alpha-\beta}; \quad \eta_q = (q-1)^{-\alpha}; \quad \beta = \frac{W\alpha}{2-W}; \quad 2-r = W + W r \frac{\alpha+\beta}{\alpha-1}.$$

Combination of Lemma 2.14.41 and (2.14.27) with (2.14.43) and (2.14.44) yields that the probability $P(\|\mathbb{G}_n\|_{\mathcal{F}} > t)$ is bounded by a constant times

$$(2.14.45) \quad \left(1 - \frac{n}{4(1-a)^2 t^2 n'}\right)^{-1} \left[\exp K(tq_0^{\alpha+\beta})^W 2 \exp(-2t^2 a^2 b^2 n'^2 / N^2) \right. \\ \left. + \sum_{q > q_0} \exp 2K(tq^{\alpha+\beta})^W 2 \exp(-\frac{1}{4}t^2 (q-1)^{2\beta} a^2 n'^2 / N^2) \right].$$

The multiplicative factor on the left is bounded by 2 for sufficiently large t . The first term inside the square brackets is bounded by

$$2 \exp\left(Kt^W (2 + t^{r/(\alpha-1)})^{(\alpha+\beta)W} - 2t^2(1 - 2t^{-r})^6\right).$$

Since $2t^2(1 - 2t^{-r})^6 \geq 2t^2 - 24t^{2-r}$, this is further bounded by

$$2 \exp\left(Dt^{W+Wr\frac{\alpha+\beta}{\alpha-1}} + 24t^{2-r} - 2t^2\right).$$

The constant r has been chosen so that the first two terms are both of the order t^{2-r} . For $\alpha \rightarrow \infty$ (and hence $\beta \rightarrow \infty$!), the exponent $2 - r$ decreases to U as in the statement of the theorem.

The constant β has been chosen so that $2\beta = W(\alpha + \beta)$, whence the series in (2.14.45) can be written in the form $\sum_{q > q_0} e^{-\psi(q)}$ for a function of the form $\psi(q) = c(q-1)^{2\beta} - dq^{2\beta}$. For sufficiently large t , we have $c > d$ and ψ is increasing and convex for $q \geq q_0$. Then Problem 2.14.3 applies and the series is bounded by its q_0 th term. This is of much lower order than e^{-2t^2} . The second theorem is proved.

For the proofs of Theorems 2.14.36 and 2.14.37, we replace the first term of the bound (2.14.44) for (2.14.43) by a bound based on the Hoeffding-Bernstein inequality. On the set C_n where $\mathbb{P}_N(f - \mathbb{P}_N f)^2$ is bounded by $P(f - Pf)^2 + s$ for every f , the right side of (2.14.43) is bounded by

$$(2.14.46) \quad N_{q_0} 2 \exp\left(-\frac{nt^2 b^2}{2(\sigma^2 + s) + tb}\right) + \sum_{q > q_0} N_q^2 \exp\left(-\frac{nt^2 \eta_q^2}{m4\varepsilon_{q-1}^2}\right).$$

For the proof of Theorem 2.14.36, choose α large and

$$a = 1 - (t/\sigma)^{-2}; \quad b = 1 - (t/\sigma)^{-2}; \quad m = \lfloor (t/\sigma)^2 \rfloor; \\ q_0 = 2 + \lfloor (t/\sigma)^{2/(\alpha-1)} \rfloor; \\ \sqrt{m}\varepsilon_q = \sigma q^{-\alpha-1}; \quad \eta_q = (q-1)^{-\alpha}; \quad \sqrt{2N}s = 3t/\sigma.$$

Combination of Lemma 2.14.41 and (2.14.25) with (2.14.43) and (2.14.46) yields as upper bound for $P(\|\mathbb{G}_n\|_{\mathcal{F}} > t)$ a constant times

$$\left(1 - \frac{n\sigma^2}{(1-a)^2 t^2 n'}\right)^{-1} \left[\left(\frac{t}{\sigma} \frac{q_0^{\alpha+1}}{\sigma}\right)^V 2 \exp\left(-\frac{t^2 b^2 a^2 n'^2 / N^2}{2(\sigma^2 + s) + tban' / N\sqrt{n}}\right) \right. \\ \left. + \sum_{q > q_0} \left(\frac{t}{\sigma} \frac{q^{\alpha+1}}{\sigma}\right)^{2V} 2 \exp\left(-\frac{1}{4} \frac{t^2}{\sigma^2} (q-1)^2 a^2 \frac{n'^2}{N^2}\right) \right] + P^*(C_n).$$

Once again the multiplicative factor is bounded and the series is bounded by its q_0 th term. If t is sufficiently large, then so is $t/\sigma \geq t$, $m \geq (1/2)(t/\sigma)^2$, and $s \leq 3/\sqrt{n}$. The preceding expression is up to a constant bounded by

$$\begin{aligned} & \left(\frac{1}{\sigma} \frac{t}{\sigma}\right)^V \left(2 + \left(\frac{t}{\sigma}\right)^{2/(\alpha-1)}\right)^{V(\alpha+1)} \exp\left(-\frac{1}{2} \frac{t^2(1-2\sigma^2/t^2)^6}{\sigma^2 + (3+t)/\sqrt{n}}\right) \\ & + \frac{\exp(-\frac{1}{2}t^2/\sigma^2)}{\sigma^{2V}} + P^*(C_n). \end{aligned}$$

The first two terms are bounded as desired. The last term can be bounded with the help of Theorems 2.14.28 and 2.14.29. Since each $|f|$ is bounded by 1,

$$s1\{C_n\} \leq \|\mathbb{P}_N(f - \mathbb{P}_N)^2 - P(f - Pf)^2\|_{\mathcal{F}} \leq \|\mathbb{P}_N - P\|_{\mathcal{F}^2} + 2\|\mathbb{P}_N - P\|_{\mathcal{F}}.$$

The covering numbers $N(\varepsilon, \mathcal{F}^2, L_2(Q))$ of the squares are bounded above by $N(\varepsilon/2, \mathcal{F}, L_2(Q))$. Thus, under (2.14.27) the probability $P^*(C_n)$ is bounded above by $\psi(s\sqrt{N}/3) \exp(-2Ns^2/9)$ for a function ψ of the form $\psi(s) = e^{Ds^U}$ for some $U < 2$. For the present choice of s , this is bounded by a multiple of $\exp(-t^2/(2\sigma^2))$. This concludes the proof of the third theorem.

For the proof of Theorem 2.14.37, choose α large and

$$\begin{aligned} a &= 1 - (t/\sigma)^{-r}; & b &= 1 - (t/\sigma)^{-r}; & m &= \lfloor (t/\sigma)^r \rfloor; \\ q_0 &= 2 + \lfloor (t/\sigma)^{r/(\alpha-1)} \rfloor; & \beta &= \frac{W\alpha}{2-W}; & 2-r &= 2u; \\ \sqrt{m}\varepsilon_q &= \sigma q^{-\alpha-\beta}; & \eta_q &= (q-1)^{-\alpha}; & \sqrt{2N}s &= 3t/\sigma. \end{aligned}$$

Combination of Lemma 2.14.41 and (2.14.25) with (2.14.43) and (2.14.46) yields as upper bound for $P(\|\mathbb{G}_n\|_{\mathcal{F}} > t)$ a constant times

$$\begin{aligned} & \left(1 - \frac{n\sigma^2}{(1-a)^2t^2n'}\right)^{-1} \left[P^*(C_n) \right. \\ & + \exp K \left(\left(\frac{t}{\sigma}\right)^{r/2} \left(\frac{q_0^{\alpha+\beta}}{\sigma}\right)^W \right) 2 \exp\left(-\frac{t^2b^2a^2n'^2/N^2}{2(\sigma^2+s) + tban'/N\sqrt{n}}\right) \\ & + \sum_{q>q_0} \exp 2K \left(\left(\frac{t}{\sigma}\right)^{r/2} \left(\frac{q^{\alpha+\beta}}{\sigma}\right)^W \right) 2 \exp\left(-\frac{1}{4} \frac{t^2}{\sigma^2} (q-1)^{2\beta} a^2 \frac{n'^2}{N^2}\right) \Big]. \end{aligned}$$

Since $(\alpha + \beta)W = 2\beta$, the series can be written in the form $\sum_{q>q_0} e^{-\psi(q)}$ for a function of the form $\psi(q) = c(q-1)^{2\beta} - dq^{2\beta}$. The constants c and d depend on t . Elementary calculations show that ψ is convex and increasing for $q \geq q_0$ if $t^{2-Wr/2} \geq C\sigma^{2-Wr/2-W}$ for a constant C depending only on K . This is certainly the case for large t if $2-Wr/2-W \geq 0$. In that case the series can be bounded by its q_0 th term. By similar arguments as before, the preceding display can be bounded by a multiple of

$$\begin{aligned} & \exp \left(D \left(\frac{1}{\sigma}\right)^W \left(\frac{t}{\sigma}\right)^{\frac{Wr}{2} + Wr \frac{\alpha+\beta}{\alpha-1}} + 5 \left(\frac{t}{\sigma}\right)^{2-r} - \frac{\frac{1}{2}t^2}{\sigma^2 + 3((t/\sigma)^{1-r/2} + t)/\sqrt{n}} \right) \\ & + \exp \left(2D \left(\frac{1}{\sigma}\right)^W \left(\frac{t}{\sigma}\right)^{\frac{Wr}{2} + Wr \frac{\alpha+\beta}{\alpha-1}} - \frac{1}{4} \left(\frac{t}{\sigma}\right)^{2+\frac{2\beta r}{\alpha-1}} \left(1 - 2\left(\frac{\sigma}{t}\right)^r\right)^4 \right). \end{aligned}$$

For $\alpha \rightarrow \infty$, the exponent

$$\frac{Wr}{2} + Wr \frac{\alpha + \beta}{\alpha - 1} = \frac{Wr}{2} + \frac{2Wr}{2 - W} \frac{\alpha}{\alpha - 1}$$

converges to $rp/2 = (1 - u)p$. The upper bound is valid if $2 - Wr/2 - W \geq 0$, which is certainly the case if $rp/2 < 2$. ■

2.14.6.2 Proof of Theorem 2.14.32

A first step in the proof of Theorem 2.14.32 is the following “basic inequality,” which controls the supremum of the empirical process over a class $\mathcal{C}$ of small diameter consisting of sets of probability bounded away from 0 and 1.

Let σ_0^2 be the (universal) constant in Lemma 2.14.40.

2.14.47 Theorem. *There exists a universal constant K such that for any pointwise separable class $\mathcal{C}$ of measurable sets, any fixed $C_0 \in \mathcal{C}$ with $\sigma_0^2 \leq P(C_0) \leq 1 - \sigma_0^2$, any constants $a \geq \sup_{C \in \mathcal{C}} P(C \triangle C_0)$ and $\mu_n \geq E\|\mathbb{G}_n\|_{\mathcal{C} \triangle C_0}$, and any $t > 0$,*

$$P(\|\mathbb{G}_n\|_{\mathcal{C}} > t) \leq \frac{K}{t} e^{-2t^2} \exp\left(Kat^2 + K\mu_n t - \frac{t^4}{4n}\right).$$

Proof. Since $\|\mathbb{G}_n\|_{\mathcal{C}} \leq \sqrt{n}$, we may assume that $t \leq \sqrt{n}$. Abbreviate Define variables W_1, W_2 , and U through

$$\begin{aligned} W_1 &= n\mathbb{P}_n(C_0) \left\| \frac{\mathbb{P}_n(C_0 - C)}{\mathbb{P}_n(C_0)} - \frac{P(C_0 - C)}{P(C_0)} \right\|_{\mathcal{C}}, \\ W_2 &= n\mathbb{P}_n(C_0^c) \left\| \frac{\mathbb{P}_n(C - C_0)}{\mathbb{P}_n(C_0^c)} - \frac{P(C - C_0)}{P(C_0^c)} \right\|_{\mathcal{C}}, \\ U &= n|\mathbb{P}_n(C_0) - P(C_0)|. \end{aligned}$$

For $C \in \mathcal{C}$, decompose $\mathbb{P}_n(C)$ as $\mathbb{P}_n(C_0) + \mathbb{P}_n(C - C_0) - \mathbb{P}_n(C_0 - C)$, and decompose $P(C)$ similarly. This leads to a bound on the absolute difference $|\mathbb{P}_n(C) - P(C)|$ by a sum of three terms. The first term is U/n . The second term can be bounded by

$$\mathbb{P}_n(C_0^c) \left| \frac{\mathbb{P}_n(C - C_0)}{\mathbb{P}_n(C_0^c)} - \frac{P(C - C_0)}{P(C_0^c)} \right| + \mathbb{P}_n(C_0^c) \left| \frac{P(C - C_0)}{P(C_0^c)} - \frac{P(C - C_0)}{\mathbb{P}_n(C_0^c)} \right|.$$

This is bounded by $(W_2 + Ua/\sigma_0^2)/n$. The third term can be handled similarly. Together this yields the bound $\sqrt{n}\|\mathbb{G}_n\|_{\mathcal{C}} \leq U(1 + 2a/\sigma_0^2) + W_1 + W_2$. Lemma 2.14.48 below gives a bound on the joint tail probabilities of the variables U and $W = W_1 + W_2$.

If $ld \leq W < (l + 1)d$, for some given integer $l \geq 0$ and real number $d > 0$, and $U(1 + 2a/\sigma_0^2) + W \geq t\sqrt{n}$, then $U(1 + 2a/\sigma_0^2) \geq t/\sqrt{n} - (l + 1)d/n$. Conclude that, for $u(1 + 2a/\sigma_0^2) = t/\sqrt{n}$,

$$\begin{aligned} P(\|\mathbb{G}_n\|_{\mathcal{C}} > t) &\leq \sum_{l=0}^{\infty} P(U \geq u - (l + 1)d/n, W \geq ld) \\ &\leq \frac{K}{\sqrt{nu}} e^{-nh(u)} \sum_{l=0}^{\infty} e^{5u(l+1)d - \varphi(ld)}, \end{aligned}$$

by Lemma 2.14.48, where $h(u) = 2u^2 + u^4/4$. We apply this with $d \geq 1/u$ the smallest number satisfying $\varphi(d)/d \geq 11u$. This exists, as $0 < u \leq t/\sqrt{n} \leq 1$ and the function $t \mapsto \varphi(t)/t$ increases to ∞ . For $l \geq 1$ we have $\varphi(ld) \geq l\varphi(d) \geq l11ud$, by choice of d , which is greater than $5u(l+1)d + lud$. Hence $5u(l+1)d - \varphi(ld) \leq -lud$ and the right side of the preceding display is bounded above by

$$\frac{K}{\sqrt{nu}} e^{-nh(u)} \left(e^{5ud} + \sum_{l=1}^{\infty} e^{-lud} \right) \leq \frac{2K}{\sqrt{nu}} e^{-nh(u)} e^{5ud}.$$

It remains to bound ud and u in terms of t .

For sufficiently large K_1 , depending on C , we have $\phi_S(K_1 u S/C)/(K_1 u S) \geq 11u$ (see Problem 2.14.5). Since the function φ agrees with $\phi_S(\cdot/C)$ in the range $[C\sqrt{n}\mu_n, \infty)$, and is zero otherwise, and d is the minimal value with $\phi(d)/d \geq 11u$ greater than $1/u$, it follows that $d \leq \max(1/u, C\sqrt{n}\mu_n, K_1 u S)$, and hence $ud \leq 1 + C\mu_n t + K_1 t^2 S/n$. By convexity of h we have $h(u) \geq h(t/\sqrt{n}) - (t/\sqrt{n} - u)h'(t/\sqrt{n})$, where $h'(t/\sqrt{n}) \leq 5t/\sqrt{n}$, for $t \leq \sqrt{n}$, and $t/\sqrt{n} - u = 2a/\sigma_0^2/(1 + 2a/\sigma_0^2)t/\sqrt{n}$. Substituting these bounds in the preceding display yields the claimed inequality. ■

2.14.48 Lemma. *Let U and $W = W_1 + W_2$ be the variables defined in the proof of Theorem 2.14.47. Under the conditions of this theorem, we have for $S = \sqrt{n}\mu_n + na$ and $\varphi(w) = \varphi_S(w/C)1\{w \geq C\sqrt{n}\mu_n, C \triangle C_0\}$ for a sufficiently large universal constant C , for all $w \geq 0$ and $u > t > 0$,*

$$P(U \geq t, W_1 + W_2 \geq w) \leq \frac{K}{\sqrt{nu}} e^{-2nu^2 - nu^4/4 - \varphi(w) + 5nu(u-t)}.$$

Proof. For $I \subset \{1, \dots, n\}$, let Ω_I be the event that $X_i \in C_0$ for $i \in I$ and $X_i \notin C_0$ otherwise. Define P_1 on Ω_I by $P_1(A) = P(A \cap C_0)/P(C_0)$. Then for $\#I = k$, conditionally on Ω_I , W_1 has the same distribution as $\sqrt{k} \|\mathbb{G}_k^Y\|_{C_1}$ where $Y_1, \dots, Y_k$ are i.i.d. P_1 and $C_1 = \{C_0 - C: C \in \mathcal{C}\}$. Similarly, if we define P_2 on Ω_I by $P_2(A) = P(A \cap C_0^c)/P(C_0^c)$, then, conditionally on Ω_I , W_2 has the same distribution as $\sqrt{n-k} \|\mathbb{G}_{n-k}^Z\|_{C_2}$, where $Z_1, \dots, Z_{n-k}$ are i.i.d. P_2 and $C_2 = \{C - C_0: C \in \mathcal{C}\}$.

The scaled means $\sqrt{k} E\|\mathbb{G}_k^Y\|_{C_1}$ and $\sqrt{n-k} E\|\mathbb{G}_{n-k}^Z\|_{C_2}$ of these processes can be bounded by $K\sqrt{n} E\|\mathbb{G}_n\|_{C_l}$, for a constant K depending on σ_0^2 (Problem 2.14.6), and $l = 1, 2$, which are in turn bounded above by $4K\sqrt{n}\mu_n^0$ (e.g. by symmetrization and the contraction principle, Proposition A.1.11; see Problem 2.14.8). We conclude that

$$\begin{aligned} S' &:= \left[\sqrt{k} E\|\mathbb{G}_k^Y\|_{C_1} + k \|P_1(C)(1 - P_1)(C)\|_{C_1} \right] \\ &\quad \vee \left[\sqrt{n-k} E\|\mathbb{G}_{n-k}^Z\|_{C_2} + (n-k) \|P_2(C)(1 - P_2)(C)\|_{C_2} \right] \\ &\leq 4K\sqrt{n}\mu_n^0 + n \frac{a}{\sigma_0^2} \leq K'S, \end{aligned}$$

for some constant K' , depending on σ_0^2 . By Theorem 2.14.38 applied twice conditionally on Ω_I , for $w \geq 16K\sqrt{n}\mu_n^0$,

$$P(W \geq w | \Omega_I) \leq 2e^{-\varphi_{S'}(w/16)} \leq 2e^{-\varphi_S(w/C)},$$

for $C = 16K'$, in view of Problem 2.14.14. We have that $\varphi_S(w/C) = \varphi(w)$ for $w \geq C\sqrt{n}\mu_n^0$ and $\varphi(w) = 0$ otherwise. Therefore, for $u > t > 0$ and $w > 0$,

$$\begin{aligned} \mathbf{P}(U \geq t, W \geq w) &= \mathbf{E} \sum_I 1_{\Omega_I} 1_{U \geq t} \mathbf{P}(W \geq w | \Omega_I) \\ &\leq 2e^{-\varphi(w)} \mathbf{P}(U \geq t) \leq \frac{2K_2}{\sqrt{nu}} e^{-nh(u) - \varphi(w) + 5nu(u-t)}. \end{aligned}$$

The last inequality, with $h(u) = 2u^2 + u^4/4$, follows by Talagrand's inequality for the binomial tail (Proposition A.6.4) applied to the variable nU , which is the absolute deviation from the mean of a binomial $(n, P(C_0))$ variable. This concludes the proof. ■

2.14.49 Lemma (Partitioning). *For $q \in \mathbb{N}$ let $\mathcal{T}_q$ be a partition of a metric space (T, ρ) in sets of diameter $\leq 4^{-q+1}$. Then for given positive integers $p < q$ and $k_{q-1}, k_{q-2}, \dots, k_p$, there exists a sequence $\{\mathcal{T}_l\}_{p \leq l < q}$ of partitions of T with the following properties, for $p \leq l < q$:*

- (i) *each set of $\mathcal{T}_l$ has diameter less than or equal to 4^{-l+1} ;*
 - (ii) *each atom of $\mathcal{T}_l$ is the union of at most k_l atoms of $\mathcal{T}_{l+1}$;*
 - (iii) *$\#\mathcal{T}_l \leq N(4^{-l}, T, \rho) + \#\mathcal{T}_{l+1}/k_l$;*
 - (iv) *every set $T_{l,j} \in \mathcal{T}_l$ satisfies $N(4^{-m}, T_{l,j}, \rho) \leq \prod_{j=l}^m k_j$, for $l \leq m < q$.*
- In the case that $\#\mathcal{T}_q \leq (K4^q)^V$ and $N(4^{-l}, T, \rho) \leq (K4^l)^V$ and $k_l \geq 2 \cdot 4^V$ for all $p \leq l < q$, these properties imply that $\#\mathcal{T}_l \leq 2(K4^l)^V$, for $p \leq l < q$.*

Proof. The construction proceeds by decreasing induction on $l \in \{q-1, \dots, p\}$: given $\mathcal{T}_{l+1}$, we construct $\mathcal{T}_l$. For $l < q-1$ we incorporate (i)–(iii) in the induction hypothesis on $\mathcal{T}_{l+1}$; for $l = q-1$ we incorporate only (i), which is satisfied by assumption. For $N = N(4^{-l}, T, \rho)$ let $t_1, \dots, t_N$ be a 4^{-l} -net over T , and for $i = 1, \dots, N$, let A_i be the union of the sets $T_{l+1,j}$ in the partition $\mathcal{T}_{l+1}$ that have nonempty intersection with the ball $B(t_i, 4^{-l})$ around t_i . The diameter of A_i is bounded by twice the diameter of the sets $T_{l+1,j}$ plus the diameter of the ball, and hence is bounded above by $2 \cdot 4^{-l} + 2 \cdot 4^{-l} = 4^{-l+1}$. Each A_i is an aggregate of atoms of $\mathcal{T}_{l+1}$. Make the sets A_i disjoint by recursively replacing A_i by $A_i - \cup_{j < i} A_j$, for every $i = 2, \dots, N$. The new sets A_i form a partition of T that is a coarsening of $\mathcal{T}_{l+1}$, whose sets satisfy (i). To ensure (ii), partition every A_i that aggregates more than k_l elements of $\mathcal{T}_{l+1}$ into sets that unite exactly k_l elements of $\mathcal{T}_{l+1}$ and (at most) one set that unites strictly less than k_l elements of $\mathcal{T}_{l+1}$. The total number of sets then satisfies (iii), as the total number of partitioning sets with strictly fewer than k_l elements will be at most N and the total number of partitioning sets with exactly k_l elements at most $\#\mathcal{T}_{l+1}/k_l$.

Thus we obtain a sequence of partitions satisfying (i)–(iii). This satisfies also property (iv), because a set $T_{l,j}$ in $\mathcal{T}_l$ is the union of at most k_l sets $T_{l+1,i}$, and hence, by induction, the union of at most $k_l k_{l+1} \cdots k_m$ sets $T_{m+1,i}$, which each have diameter smaller than 4^{-m} by (i).

For the last assertion, note that for the given bounds on $N(4^{-l}, T, \rho)$ and k_l , we have $\#\mathcal{T}_l \leq (K4^l)^V + \#\mathcal{T}_{l+1}/(2 \cdot 4^V)$, by (iii). Next iterate (or more formally, apply decreasing induction) on l . ■

We are ready for the proof of Theorem 2.14.32. As $\|\mathbb{G}_n\|_{\mathcal{C}} \leq \sqrt{n}$, we may assume without loss of generality that $t \leq \sqrt{n}$. We may also assume that $t \gtrsim \sqrt{V}$, as we can choose the constant D in the theorem so large that its right side is larger than 1 for $t \lesssim \sqrt{V}$. Throughout we take the constant at least 2, so that $n \geq 4V$.

Let σ_0^2 be the universal constant in Lemma 2.14.40 and let $\mathcal{C}_0$ be the collection of sets $C \in \mathcal{C}$ with $P(C) \leq \sigma_0^2$ or $P(C) \geq 1 - \sigma_0^2$. Set $A = (4L\sigma_0 \vee (2L\sigma_0^{-1}) \vee \sigma_0^{-1})\sqrt{\log(K/\sigma_0^2)}$, for L the multiplicative constant in the maximal inequality of Example 2.14.7. Then $\sqrt{n} \geq t \geq A\sqrt{V}$ implies that $\sqrt{n} \geq \sigma_0^{-1}\sqrt{V \log(K/\sigma_0^2)}$ and hence, by (2.14.30) and Example 2.14.7,

$$\mathbb{E}\|\mathbb{G}_n\|_{\mathcal{C}_0} \leq L\sigma_0\sqrt{V \log(K/\sigma_0^2)} + L\frac{V \log(K/\sigma_0^2)}{\sqrt{n}} \leq 2L\sigma_0\sqrt{V \log(K/\sigma_0^2)}.$$

Consequently $\sqrt{n} \geq t \geq A\sqrt{V}$ also implies that $t \geq 2\mathbb{E}\|\mathbb{G}_n\|_{\mathcal{C}_0}$ and $\sigma_0^2\sqrt{n} \geq \sigma_0^2 t \geq \mathbb{E}\|\mathbb{G}_n\|_{\mathcal{C}_0}$, whence Lemma 2.14.40 implies that $P(\|\mathbb{G}_n\|_{\mathcal{C}_0} > t) \leq e^{-11t^2}$, which is much smaller than the bound in Theorem 2.14.32. Thus we can assume without loss of generality that $\sigma_0^2 \leq P(C) \leq 1 - \sigma_0^2$, for every $C \in \mathcal{C}$.

Fix integers $p \leq q$ such that $4^{-p+1} \leq V/t^2 \leq 4^{-p+2}$ and $3(q-p)4^q \leq n/V \leq 3(q+1-p)4^{q+1}$. Partition $\mathcal{C}$ in $N(4^{-q}, \mathcal{C}, L_1(P)) \leq (K4^q)^V$ sets of $L_1(P)$ -diameter at most 4^{-q+1} . Coarsen this partition to partitions $\mathcal{C}_{q-1}, \mathcal{C}_{q-2}, \dots, \mathcal{C}_p$ in the manner of Lemma 2.14.49, with $k_l = \lfloor 3 \cdot 4^V \rfloor \geq 2 \cdot 4^V$, for every l .

Denote the sets in the l th partition by $\mathcal{C}_{l,j}$. Clearly $\|\mathbb{G}_n\|_{\mathcal{C}} \leq \max_j \|\mathbb{G}_n\|_{\mathcal{C}_{p,j}}$, where the index j takes at most $\#\mathcal{C}_p \leq 2(K4^p)^V$ values, by the final assertion of Lemma 2.14.49. Hence, for some universal constant K_1 ,

$$\begin{aligned} P(\|\mathbb{G}_n\|_{\mathcal{C}} > t) &\leq 2(K4^p)^V \max_j P(\|\mathbb{G}_n\|_{\mathcal{C}_{p,j}} > t) \\ &\leq 2\left(\frac{16Kt^2}{V}\right)^V \frac{K_1}{t} e^{-2t^2} e^{K_1 4^{-p+1} t^2 + K_1 \mu_n^0 t - t^4/(4n)}, \end{aligned}$$

by the definition of p and Theorem 2.14.47, where

$$\mu_n^0 = \max_j \mathbb{E}\|\mathbb{G}_n\|_{\mathcal{C}_{p,j} \triangle \mathcal{C}_{0,p,j}},$$

for any fixed sets $\mathcal{C}_{0,p,j} \in \mathcal{C}_{p,j}$. The proof can be completed by showing that the exponent in the last term of the second last display is bounded above by $K_2 V$, for a constant K_2 that depends on K only.

The first term of this exponent satisfies $K_1 4^{-p+1} t^2 \leq K_1 4V$, by the definition of p . We show below that

$$(2.14.50) \quad \mu_n^0 \lesssim \frac{V}{t} + \frac{V \log(Kn/V)}{\sqrt{n}}.$$

The function $t \mapsto V \log(Kn/V)t/\sqrt{n} - t^4/(4n)$ is maximal on $(0, \infty)$ at $t = V \log(Kn/V)^{1/3} n^{1/6}$ with value $V(V/n)^{1/3} \log(Kn/V)^{4/3}$. For $4V \leq n$, which we have assumed, this is bounded by a multiple of V by the boundedness of the function $x \mapsto x^{1/3}(\log(K/x))^{4/3}$ on compact intervals.

Thus the proof will be complete once (2.14.50) has been established. Up to a multiplicative constant the expectations in the maximum are equal to $\mathbb{E}\|\mathbb{G}_n\|_{1_{\mathcal{C}} -$

$1_{C_{0,p,j}}\|_{C_{p,j}}$ (see Problem 2.14.8). A given set C in a given partitioning set $C_{p,j}$ at level p belongs to exactly one set $C_{l,i}$ in the partition $\mathcal{C}_l$ at level l , for every l . This yields a chain $C_{p,j_p}, C_{p+1,j_{p+1}}, \dots, C_{q,j_q}$ of partitioning sets to which C belongs. For every partitioning set $C_{l,j}$ at level l choose a fixed set $C_{0,l,j} \in \mathcal{C}_{l,j}$, and let $C_{0,p,j_p}, C_{0,p+1,j_{p+1}}, \dots, C_{0,q,j_q}$ be the sets connected to the partitioning sets containing C . The telescoping sum

$$1_C - 1_{C_{0,p,j_p}} = \sum_{l=p}^{q-1} (1_{C_{0,l+1,j_{l+1}}} - 1_{C_{0,l,j_l}}) + 1_C - 1_{C_{0,q,j_q}},$$

in which the sets C_{0,l,j_l} depend on C , yields the bound

$$\begin{aligned} \mathbb{E} \|\mathbb{G}_n(1_C - 1_{C_{0,p,j_p}})\|_{C_{p,j}} &\leq \sum_{l=p}^{q-1} \mathbb{E} \max_{j_l} |\mathbb{G}_n(1_{C_{0,l+1,j_{l+1}}} - 1_{C_{0,l,j_l}})| \\ &\quad + \mathbb{E} \max_{j_q} \|\mathbb{G}_n(1_C - 1_{C_{0,q,j_q}})\|_{C_{q,j_q}}. \end{aligned}$$

The maximum over j_l refers to all links attached to some $C \in C_{p,j}$, and is over at most $k_p \cdots k_l \leq (3 \cdot 4^V)^{l-p+1}$ terms. We bound the sum and the term for the partition at level q separately.

Every link $1_{C_{0,l+1,j_{l+1}}} - 1_{C_{0,l,j_l}}$ has $L_1(P)$ -size smaller than the diameter of a partitioning set in $\mathcal{C}_l$ and hence square $L_2(P)$ -norm smaller than 4^{-l+1} . From a Bernstein type maximal inequality (see (2.5.7)), it follows that the sum in the preceding display is bounded above by a multiple of

$$\sum_{l=p}^{q-1} \frac{1}{\sqrt{n}} \log(3 \cdot 4^V)^{l-p} + 2^{-l} \sqrt{\log(3 \cdot 4^V)^{l-p}} \lesssim \sum_{l=p}^{q-1} 2^{-l} \sqrt{q-p} \sqrt{V},$$

since $(l-p)V/n \leq (q-p)V/n \leq 4^{-q} \leq 4^{-l}$, by the definition of q . The right side is bounded above by $2^{-p} \sqrt{V} \leq V/t$, by the definition of p , and gives rise to the first term on the right of (2.14.50).

For the bound on the term for partition q , we employ Theorem 2.14.38 to infer that $Z_j := \|\mathbb{G}_n(1_C - 1_{C_{0,q,j_q}})\|_{C_{q,j_q}}$ satisfies $P(Z_j > 4t) \leq e^{-\phi_S(t\sqrt{n})}$, for $t \geq 2\mathbb{E}Z_j$ and $S = \sqrt{n} \max_j \mathbb{E}Z_j + n4^{-q+1}$. (The theorem is reformulated in terms of the function ϕ_S following its proof.) Then Problem 2.14.10 gives that

$$\mathbb{E} \max_j Z_j \lesssim \max_j \mathbb{E}Z_j + \frac{\sqrt{S}}{\sqrt{n}} \sqrt{\log(3 \cdot 4^V)^{q-p}} + \frac{\log(3 \cdot 4^V)^{q-p}}{\sqrt{n}}.$$

By (2.14.30) and Example 2.14.7, the first term on the right is bounded above by a multiple of

$$\begin{aligned} &\sqrt{4^{-q}V \log(K/4^{-q})} + \frac{V \log(K/4^{-q})}{\sqrt{n}} \\ &\lesssim \sqrt{V} 2^{-q} (\sqrt{\log K} + \sqrt{q}) + \frac{V(\log K + q)}{\sqrt{n}} \lesssim \frac{V \log K + Vq}{\sqrt{n}}, \end{aligned}$$

because $n4^{-q-1} \leq 3(q+1-p)V \leq 3qV$, by the definition of q , so that $2^{-q} \lesssim \sqrt{qV/n}$. The contribution of the second term is bounded above by a multiple of $\sqrt{S}\sqrt{V(q-p)}/\sqrt{n} \leq (S+Vq)/\sqrt{n}$ and hence is no bigger than the preceding display. The third term can also be seen to give no bigger contribution.

We conclude that $E \max_j Z_j$ is bounded by a multiple of the second term on the right of (2.14.50), provided $q \lesssim \log(n/V)$. By its definition $p \leq \log(16t^2/V) \leq 3 \log(n/V)$, for $4V \leq n$, as we have assumed. Also $3(q-p)4^{q-p} \leq (n/V)(V/t^2) = n/t^2$, by the definitions of q and p , which implies that $q-p \leq \log(n/t^2)$ and hence $q = q-p+p \lesssim \log(n/V)$.

Problems and Complements

1. **(Alternative definition of $\|\cdot\|_{\psi_p}$ for small p)** For $0 < p < 1$, define

$$\psi_p(x) = e^{-c_p^p} (e^{x^p} - 1); \quad \tilde{\psi}_p(x) = (\psi_p(x) - \psi_p(c_p)) 1\{x \geq c_p\}.$$

Then $\tilde{\psi}_p$ is convex and there exists a constant C_p such that $\|X\|_{\tilde{\psi}_p} \leq \|X\|_{\psi_p} \leq C_p \|X\|_{\tilde{\psi}_p}$. Thus $\|X\|_{\tilde{\psi}_p}$ defines a true Orlicz norm and can be used interchangeably with $\|X\|_{\psi_p}$ up to constants. A bound on the latter pseudonorm leads to the tail bound

$$P(|X| > t) \leq (1 + e^{c_p^p}) e^{-t^p / \|X\|_{\psi_p}^p}.$$

[Hint: Note that $\tilde{\psi}_p \leq \psi_p \leq \tilde{\psi}_p + \psi_p(c_p)$. Also $E\tilde{\psi}_p(Y/C) \leq E\tilde{\psi}_p(Y)/C$ for every $C > 1$ and Y by the convexity of $\tilde{\psi}_p$.]

2. **(Hoeffding's inequality)** If the random variable X has zero mean and takes its values in the interval $[u, v]$, then $E \exp(sX) \leq \exp(s^2(v-u)^2/8)$ for every $s \in \mathbb{R}$.
3. If $\psi: [q_0, \infty) \rightarrow \mathbb{R}$ is convex and increasing with right derivative ψ' , then

$$\sum_{q > q_0} \exp -\psi(q) \leq \psi'(q_0)^{-1} \exp -\psi(q_0).$$

[Hint: The tail series can be bounded by $\psi'(q_0)^{-1} \int_{q_0}^{\infty} \exp -\psi(x) d\psi(x)$.]

4. Let $\mathcal{F}$ be a class of measurable functions $f: \mathcal{X} \rightarrow [l, u]$ satisfying one of the following two conditions:

$$\sup_Q N(\varepsilon(u-l), \mathcal{F}, L_2(Q)) \leq \left(\frac{K}{\varepsilon}\right)^V,$$

$$\sup_Q \log N(\varepsilon(u-l), \mathcal{F}, L_2(Q)) \leq K \left(\frac{1}{\varepsilon}\right)^W,$$

for every $\varepsilon > 0$, where the supremum is taken over all discrete probability measures Q . Then the tail probabilities $P^*(\|\mathbb{G}_n\|_{\mathcal{F}} > t)$ can be bounded as in the theorems, but with t replaced by $t/(u-l)$.

5. Consider the function φ_s given in (2.14.39). For any given $C > 0$ there exists a constant $K_0 > 0$ depending only on C such that $\varphi_s(Kus/C)/Kus \geq 11u$, for every $0 < u < 1$ and $s > 0$ and $K \geq K_0$.

[Hint: The function $\varphi_s(Kus/C)/Ku^2s = KC^{-2}1_{Ku \leq C} + (uC)^{-1} \log(eKu/C)1_{Ku > C}$ is continuous and nonincreasing in $u > 0$. Its minimum over $0 < u \leq 1$ is taken at $u = 1$ and is equal to $((K/C) \wedge \log(eK/C))/C$. This is increasing in K and can be made arbitrarily large by choice of K .]

6. Let $Y_1, \dots, Y_k$ and $X_1, \dots, X_n$ be i.i.d. samples from P_1 and P , respectively, that are related by $P_1(C) = P(C \cap C_0)/P(C_0)$, for a set C_0 with $P(C_0) \geq a_0$. There exists a constant K that depends only on a_0 such that for any separable class $\mathcal{F}$ of functions and $k \leq n$,

$$\mathbb{E} \left\| \sum_{i=1}^k (f(Y_i) - P_1 f) \right\|_{\mathcal{F}} \leq K \mathbb{E} \left\| \sum_{i=1}^n (f(X_i) - P f) \right\|_{\mathcal{F}}.$$

[Hint: For K the number of X_i falling in C_0 , the left side is bounded above by twice

$$E_k := \mathbb{E} \left\| \sum_{i=1}^k \varepsilon_i f(Y_i) \right\|_{\mathcal{F}} = \mathbb{E} \left[\left\| \sum_{i=1}^n \varepsilon_i (f1_{C_0})(X_i) \right\|_{\mathcal{F}} \mid K = k \right].$$

Then $\sum_k E_k P(K = k) = \mathbb{E} \left\| \sum_{i=1}^n \varepsilon_i (f1_{C_0})(X_i) \right\|_{\mathcal{F}}$, which is smaller than $E := \mathbb{E} \left\| \sum_{i=1}^n \varepsilon_i f(X_i) \right\|_{\mathcal{F}}$, by Jensen's inequality. Combine this with the monotonicity of $k \mapsto E_k$, again by Jensen's inequality, to conclude that $E_k P(K \geq k) \leq E$, for every k , and hence $E_l \lesssim E$ for $l \asymp na_0 \leq EK$. Finally $E_n \leq (n/l + 1)E_l$, by the triangle inequality.]

7. Suppose that $x_1, \dots, x_n$ are points in a set $\mathcal{X}$ and $\varepsilon_1, \dots, \varepsilon_n$ i.i.d. Rademacher variables. If $\mathcal{C}$ is a class of subsets of $\mathcal{X}$ satisfying (2.14.30), then there exists a constant L such that, with $N = \left\| \sum_{i=1}^n 1_C(x_i) \right\|_{\mathcal{C}}$,

$$\mathbb{E} \left\| \sum_{i=1}^n \varepsilon_i 1_C(x_i) \right\|_{\mathcal{C}} \leq L \sqrt{NV \log(Kn/N)}.$$

[Hint: Use Corollary 2.2.9 with $d(C, D) = \sqrt{n \|1_C - 1_D\|_{L_1(Q_n)}}$, where Q_n is the empirical measure on the points x_i . The diameter of $\mathcal{C}$ for d is at most $\sqrt{2N}$.]

8. For a collection $\mathcal{C}$ of measurable sets and a given set C_0 define $\mathcal{C} \triangle C_0 = \{C \triangle C_0 : C \in \mathcal{C}\}$, and analogously collections $\mathcal{C} - C_0$ and $C_0 - \mathcal{C}$. The quantities $\mathbb{E}^* \|\mathbb{G}_n\|_{\mathcal{C} \triangle C_0}$, $\mathbb{E}^* \|\mathbb{G}_n(1_C - 1_{C_0})\|_{\mathcal{C}}$ and $\mathbb{E}^* \|\mathbb{G}_n\|_{\mathcal{C} - C_0} + \mathbb{E}^* \|\mathbb{G}_n\|_{C_0 - C_0}$ are equal up to multiplicative universal constants.

[Hint: By symmetrization it suffices to prove similar inequalities for the symmetrized (scaled) empirical measure $\sum_{i=1}^n \varepsilon_i \delta_{x_i}$ instead of $\mathbb{G}_n$. We can use $1_{C \triangle C_0} = 1_{C - C_0} + 1_{C_0 - C}$ and $1_C - 1_{C_0} = 1_{C - C_0} - 1_{C_0 - C}$ together with the triangle inequality to bound the first two quantities in terms of the third. We can use $1_{C - C_0} = 1_{C \triangle C_0} 1_{C_0^c}$ and $1_{C_0 - C} = 1_{C \triangle C_0} 1_{C_0}$ and the contraction principle, Proposition A.1.11, to bound the third in terms of the first, and similarly we can use $1_{C \triangle C_0} = (1_C - 1_{C_0})(1_{C_0^c} - 1_{C_0})$ to bound the first in terms of the second.]

9. Suppose that $\mathcal{C}$ is a class of sets satisfying (2.14.30) and $\sup_{C \in \mathcal{C}} P(C) \leq \sigma_0^2$. Then for some constants D and L , for all $t \geq L\sqrt{V \log K}$ and $n \geq 1$,

$$P(\|\mathbb{G}_n\|_{\mathcal{C}} > t) \leq \left(\frac{D}{t}\right) e^{-2t^2}.$$

[Hint: Use Lemma 2.14.40 and the second result proved in Exercise 2.14.7.]

10. If $Z_1, \dots, Z_m$ are random variables satisfying $P(|Z_j| \geq t) \leq \exp -\varphi_s(t/C)$, for all $t \geq D > 0$, where φ_s is the function given in (2.14.39) and $C > 0$, then

$$E \max_{1 \leq j \leq m} |Z_j| \leq D + C\sqrt{s}(\sqrt{\log m} + 2) + C(\log m + 1).$$

[Hint: Decompose $|Z_j|$ as the sum of $A_j = |Z_j|1_{|Z_j| \leq D}$, $B_j = |Z_j|1_{D < |Z_j| \leq s}$ and $C_j = |Z_j|1_{|Z_j| > s}$, and bound the expectations of the maxima over the A_j , B_j and C_j separately, by D , and multiples of $C\sqrt{s(\log m + 1)}$ and $C(\log m + 1)$, respectively, using that $P(B_j > t) \leq e^{-t^2/(C^2s)}$ and $P(C_j > t) \leq e^{-t/C}$, for every $t > 0$.]

11. Suppose that $\mathcal{C}$ is a class of subsets of a given set C_0 satisfying (2.14.31). There exists a constant K_1 such that if $P(C_0) = d \in (0, 1)$, then

$$E^* \|\mathbb{G}_n\|_{\mathcal{C}} \leq K_1 \sqrt{Vd \log \left(\frac{K}{d} \right)}.$$

[Hint: Relate $N_{[]}(\varepsilon, \mathcal{C}, L_1(P))$ to $N_{[]}(\varepsilon, \mathcal{C}, L_2(P))$, and then use Theorem 2.14.16; compare with Exercise 2.14.7.]

12. For values of δ such that $\delta \|F\|_{p_2} \ll 1/\sqrt{n}$ Theorem 2.14.2 can be improved to

$$E_p^* \|\mathbb{G}_n\|_{\mathcal{F}} \lesssim J(\delta, \mathcal{F}, L_2) \|F\|_{p_2} + J^2 \left(\frac{1}{\sqrt{n} \|F\|_{p_2}}, \mathcal{F}, L_2 \right) \sqrt{n} \|F\|_{p_2}^2.$$

[Hint: See Van der Vaart and Wellner (2011).]

13. The upper bound in Theorem 2.14.17 can (with the same proof) be slightly improved to

$$J_{[]}(\delta, \mathcal{F} | F, L_2(P)) \|F\|_{p_2} + \frac{\log N_{[]}(\delta \|F\|_{p_2}, \mathcal{F}, L_2(P))}{\sqrt{n}}.$$

A similar improvement is possible for Theorem 2.14.18.

14. The functions $t \mapsto \varphi_r(t) = (t^2/r)1_{t \leq r} + t \log(et/r)1_{t > r}$ are continuous on $(0, \infty)$; and satisfy $\varphi_s \leq \varphi_r \leq (s/r)\varphi_s$ on $(0, \infty)$ for $0 < r < s < \infty$; and satisfy $\varphi_{kr}(kt) \geq k\varphi_r(t)$, for $t > 0$ and every $k > 0$ (and hence $\varphi_{kr}(t) \geq \varphi_r(t/k)$, for $k \geq 1$).

[Hint: The lower inequality $\varphi_s \leq \varphi_r$ is obvious on $(0, r]$ and $[s, \infty)$ and next follows by concavity of $t \mapsto (\varphi_s - \varphi_r)(t)/t$ also on (r, s) . The upper inequality is obvious on $(0, r]$ with equality at r and the derivative of $t \mapsto (s/r\varphi_s - \varphi_r)(t)/t$ is nonnegative on $[s, \infty)$.]

2.15

Concentration

The location of the distribution of the norm $\|\mathbb{G}_n\|_{\mathcal{F}}$ of the empirical process, as measured for instance by its mean value $E^*\|\mathbb{G}_n\|_{\mathcal{F}}$, depends strongly on the size of the class of functions $\mathcal{F}$. Chapter 2.14 gives various methods to estimate this mean value. It turns out that the spread or “concentration” of the distribution around its center hardly depends on the complexity of the class $\mathcal{F}$.

In this chapter we present two bounds on the concentration. The first “bounded difference inequality” is an exponential inequality of Hoeffding type, and gives a bound expressed in terms of the range of the functions f . The second, deeper result is of Bennett-Bernstein type, and incorporates also the variance of the functions f . The bounds can be viewed as uniform versions of the Hoeffding and Bernstein inequalities, Lemmas 2.2.8 and 2.2.10.

A similar phenomenon of concentration is known for suprema of centered Gaussian processes, for which Borell’s inequality, Proposition A.2.1, gives a sub-Gaussian bound for probabilities of deviations from the mean that is valid without conditions.

As usual, for measurability purposes the observations $X_1, \dots, X_n$ are assumed to be defined as the coordinate projections on a product probability space.

2.15.1 Bounded Difference Inequality

By Hoeffding’s inequality the tail probabilities of the empirical process $\mathbb{G}_n f$ at a single function f with range 1 are bounded by e^{-2t^2} . The deviation probabilities of the norm of the empirical process are no bigger.

2.15.1 Theorem. *If $\mathcal{F}$ is a class of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$ such that*

$|f(x) - f(y)| \leq 1$ for every $f \in \mathcal{F}$ and every $x, y \in \mathcal{X}$, then, for all $t \geq 0$,

$$\begin{aligned} \mathbf{P}^* \left(\left| \|\mathbb{G}_n\|_{\mathcal{F}} - \mathbf{E}^* \|\mathbb{G}_n\|_{\mathcal{F}} \right| \geq t \right) &\leq 2 \exp(-2t^2), \\ \mathbf{P}^* \left(\left| \sup_f \mathbb{G}_n f - \mathbf{E}^* \sup_f \mathbb{G}_n f \right| \geq t \right) &\leq 2 \exp(-2t^2). \end{aligned}$$

This theorem is a special case of a result that applies to general statistics $T = T(X_1, \dots, X_n)$ that satisfy the *bounded difference inequality*: for all $i = 1, \dots, n$, given constants $c_1, \dots, c_n$, and every $x_1, \dots, x_n, y_i \in \mathcal{X}$,

$$(2.15.2) \quad |T(x_1, \dots, x_n) - T(x_1, \dots, x_{i-1}, y_i, x_{i+1}, \dots, x_n)| \leq c_i.$$

Both the norm $\|\mathbb{G}_n\|_{\mathcal{F}}$ and the supremum $\sup_{f \in \mathcal{F}} \mathbb{G}_n f$ satisfy this inequality, with $\sqrt{n}c_i$ the supremum over f of the range of f . (Therefore, so do their measurable cover functions almost surely under the product law P^{n+1} , if these variables are not measurable. See Problem 2.15.1.) In the application to the empirical process, the variables $X_1, \dots, X_n$ are i.i.d., but in the following theorem it suffices that they are independent. The variable Y_i (used in the statement of the theorem to relax assumption (2.15.2) to an almost sure inequality) is an independent copy of X_i .

2.15.3 Proposition (Bounded difference inequality). *If T is a measurable function of independent variables $X_1, \dots, X_n$ that satisfies (2.15.2) for almost every $(x_1, \dots, x_n, y_i)$ relative to the law of $(X_1, \dots, X_n, Y_i)$, then, for all $t > 0$,*

$$\begin{aligned} \mathbf{P}(T - \mathbf{E}T \geq t) &\leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right), \\ \mathbf{P}(T - \mathbf{E}T \leq -t) &\leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right). \end{aligned}$$

Proof. Let $\mathbf{E}(T | X_1, \dots, X_{i-1}, x)$ be an abbreviation of the conditional expectation $\mathbf{E}(T | X_1, \dots, X_{i-1}, X_i = x)$, and set, for $i = 1, \dots, n$,

$$\begin{aligned} T_i &= \mathbf{E}(T | X_1, \dots, X_i) - \mathbf{E}(T | X_1, \dots, X_{i-1}), \\ A_i &= \inf_x \left(\mathbf{E}(T | X_1, \dots, X_{i-1}, x) - \mathbf{E}(T | X_1, \dots, X_{i-1}) \right), \\ B_i &= \sup_x \left(\mathbf{E}(T | X_1, \dots, X_{i-1}, x) - \mathbf{E}(T | X_1, \dots, X_{i-1}) \right). \end{aligned}$$

(If (2.15.2) only holds almost everywhere, use the essential infimum and supremum relative to the law of X_i in the definitions of A_i and B_i instead.) With the understanding that $\mathbf{E}(T | \emptyset) = \mathbf{E}T$, we then have $T - \mathbf{E}T = \sum_{i=1}^n T_i$. Furthermore, $A_i \leq T_i \leq B_i$ and $B_i - A_i \leq c_i$ by integration of the bounded difference assumption (2.15.2) relative to $x_{i+1}, \dots, x_n$. Therefore, for all $\lambda > 0$,

$$\mathbf{E}e^{\lambda \sum_{i=1}^n T_i} = \mathbf{E}\left(e^{\lambda \sum_{i=1}^{n-1} T_i} \mathbf{E}(e^{\lambda T_n} | X_1, \dots, X_{n-1})\right) \leq \mathbf{E}e^{\lambda \sum_{i=1}^{n-1} T_i} e^{\lambda^2 c_n^2 / 8},$$

by Problem 2.15.2, where we use that $\mathbf{E}(T_n | X_1, \dots, X_{n-1}) = 0$, almost surely. By repeating the argument we see that the right side is bounded by $\exp(\lambda^2 \sum_{i=1}^n c_i^2 / 8)$.

By Markov's equality we turn this bound on the Laplace transform into the bound $e^{-\lambda t} \exp(\lambda^2 \sum_i c_i^2 / 8)$ on the probability given in the (first) left side of the theorem, valid for every $\lambda > 0$. Finally we choose $\lambda = 4t / \sum_i c_i^2$ to minimize the latter expression over λ . This completes the proof of the first inequality. The second inequality follows by applying the first to $-T$. ■

2.15.2 Talagrand's Inequality

The second bound of this chapter incorporates the maximal variance of the functions in the class $\mathcal{F}$, defined by

$$\sigma_{\mathcal{F}}^2 = \sup_{f \in \mathcal{F}} P(f - Pf)^2.$$

There are several variations of this theorem. We first state the theorem in its strongest form using the function $h: (0, \infty) \rightarrow (0, \infty)$ defined by

$$(2.15.4) \quad h(t) = t(\log t - 1) + 1,$$

or, equivalently, the function ψ defined by $\psi(t) = 2t^{-2}h(1+t)$.

2.15.5 Theorem (Bousquet–Talagrand inequality). *For any countable class $\mathcal{F}$ of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$ with $f(x) - Pf \leq 1$ for every $x \in \mathcal{X}$ and $w_n := \sigma_{\mathcal{F}}^2 + 2n^{-1/2}E \sup_f \mathbb{G}_n f$ finite, for every $t \geq 0$,*

$$\begin{aligned} P\left(\sup_f \mathbb{G}_n f - E \sup_f \mathbb{G}_n f \geq t\right) &\leq \exp\left(-nw_n h\left(1 + \frac{t}{\sqrt{nw_n}}\right)\right) \\ &= \exp\left(-\frac{t^2}{2w_n} \psi\left(\frac{t}{\sqrt{nw_n}}\right)\right). \end{aligned}$$

If, moreover, $|f(x) - Pf| \leq 1$ for every $x \in \mathcal{X}$, then this is also true with $\sup_f \mathbb{G}_n f$ replaced throughout by $\|\mathbb{G}_n\|_{\mathcal{F}}$.

Proof. The theorem is a corollary of Proposition 2.15.12 (below). It is no loss of generality to assume that $Pf = 0$ for every f . To overcome possible measurability issues in the following proof, the theorem can also be reduced first to the case of a finite collection $\mathcal{F}$.

Given $T = \sup_{f \in \mathcal{F}} \sum_i f(X_i)$ we choose

$$\begin{aligned} T_i &= \sup_{f \in \mathcal{F}} \sum_{j \neq i} f(X_j), \\ S_i &= f_i(X_i; X_j: j \neq i), \end{aligned}$$

for f_i the function in $\mathcal{F}$ for which the supremum involved in the definition of T_i is assumed, for given $(X_j: j \neq i)$. Then $S_i \leq 1$, and $E(S_i | X_j: j \neq i) = Pf_i = 0$, and $\sum_{i=1}^n E(S_i^2 | X_j: j \neq i) = \sum_{i=1}^n Pf_i^2 \leq n\sigma_{\mathcal{F}}^2$. (Here Pf_i and Pf_i^2 are the integrals of f_i and f_i^2 with respect to their first arguments, the arguments $(X_j: j \neq i)$ being

fixed.) Finally, for given $X_1, \dots, X_n$ and f_0 the function where the supremum in the definition of T is assumed,

$$(n-1)T = \sum_{i=1}^n \sum_{j \neq i} f_0(X_j) \leq \sum_{i=1}^n \sum_{j \neq i} f_i(X_j) = \sum_{i=1}^n T_i.$$

Rearranging this inequality gives that $\sum_{i=1}^n (T - T_i) \leq T$. The first assertion of the theorem follows from Proposition 2.15.12 with the choice $u = 1$.

The second assertion follows by applying the first to the class of functions $\mathcal{F} \cup (-\mathcal{F})$. ■

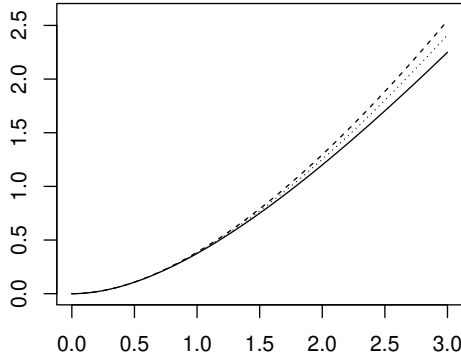


Figure 2.15.1. The three functions $t \mapsto h(1+t)$, $t \mapsto 9h_1(t/3)$ and $t \mapsto 9h_0(t/3)$ in (2.15.6).

For many applications the sequence w_n in the theorem will be bounded in n (although this is not required by the theorem). Then, for fixed or moderate t , the function $t \mapsto h(1+t)$ is mostly important in a neighborhood of 0. Two functions with a similar behavior are the functions h_0 and h_1 given by (see Figure 2.15.1)

$$h_0(t) = \frac{t^2}{2(1+t)}, \quad h_1(t) = 1 + t - \sqrt{1+2t}.$$

The three functions and their first derivatives all vanish at $t = 0$ and their second derivatives are all 1. More importantly, one can show that, for any $t \geq 0$,

$$(2.15.6) \quad h(1+t) \geq 9h_1\left(\frac{t}{3}\right) \geq 9h_0\left(\frac{t}{3}\right).$$

Thus the inequality in the theorem remains true if the function $t \mapsto h(1+t)$ is replaced by the scaled version of h_0 or h_1 given on the right.

The bound for the supremum norm resulting from h_0 can be written in the form of the following corollary.

2.15.7 Corollary. *For any countable class $\mathcal{F}$ of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$ with $|f(x) - Pf| \leq 1$ for every $x \in \mathcal{X}$ and $w_n := \sigma_{\mathcal{F}}^2 + 2n^{-1/2}E\|\mathbb{G}_n\|_{\mathcal{F}}$ finite, for every $t \geq 0$,*

$$(2.15.8) \quad P\left(\|\mathbb{G}_n\|_{\mathcal{F}} - E\|\mathbb{G}_n\|_{\mathcal{F}} \geq t\right) \leq \exp\left(-\frac{1}{2} \frac{t^2}{w_n + \frac{1}{3}t/\sqrt{n}}\right).$$

This inequality has the same form as Bernstein's inequality (Lemma 2.2.10), with w_n playing the role of the variance. If $\mathcal{F}$ consists of a single function f , then w_n is indeed the variance of f , and hence the theorem exactly recovers Bernstein's inequality. In the general case the (positive) constant $n^{-1/2}E\sup_f \mathbb{G}_n f$ must be added to the variance, increasing the bound. For a Donsker class this added term is small and tends to zero, as does the term $t/3\sqrt{n}$ in the denominator of the exponent. In the limit as $n \rightarrow \infty$ the bound takes the form $\exp(-\frac{1}{2}t^2/\sigma_{\mathcal{F}}^2)$, and hence we could interpret these additional terms as finite-sample corrections on the Gaussian bound. On the other hand, for $t \gg \sqrt{n}$ or very large classes $\mathcal{F}$ the correction terms dominate the variance term.

Replacing h by the scaled version of h_1 gives yet another bound, which has a convenient "inverse" formulation. As the inverse function of h_1 is given by $h_1^{-1}(t) = \sqrt{2t} + t$, it follows that, for every countable class $\mathcal{F}$ of measurable functions with $f - Pf \leq 1$ for every $f \in \mathcal{F}$, and for every $t > 0$,

$$(2.15.9) \quad P\left(\sup_f \mathbb{G}_n f - E\sup_f \mathbb{G}_n f \geq \sqrt{2tw_n} + \frac{t}{3\sqrt{n}}\right) \leq e^{-t}.$$

Similar bounds can be obtained for the modulus of the empirical process. In fact, these are equivalent to bounds on the supremum over the class of functions $\mathcal{F} \cup (-\mathcal{F})$. For easy use we state a simplified version (with easy but suboptimal constants) of (2.15.9) as a lemma.

2.15.10 Lemma. *Let $\mathcal{F}$ be a countable class of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$ such that $\|f\|_{\infty} \leq M$ and $Pf^2 \leq \delta^2$ for every f in $\mathcal{F}$. Then, for every $t > 0$, with probability at least $1 - e^{-t}$,*

$$\forall f \in \mathcal{F}: \quad \frac{1}{2} \mathbb{G}_n f \leq E\sup_{f \in \mathcal{F}} \mathbb{G}_n f + \frac{Mt}{\sqrt{n}} + \delta\sqrt{t}.$$

Proof. We apply (2.15.9) to the class of functions f/M , next use the inequality $\sqrt{2t(\sigma^2 + 2\mu/\sqrt{n})} \leq \sqrt{2t}\sigma + t/\sqrt{n} + \mu$, and finally bound the constants $\sqrt{2}$ and $4/3$ by 2. ■

Unlike the bounded difference inequality, Theorem 2.15.5 and its consequences (2.15.8) and (2.15.9) concern upper tail probabilities only, and do not show the size of deviations of the supremum under its mean. Although these are of the same order (up to constants), it appears that their estimation requires a separate argument.

The function h in the following theorem is as in the previous theorem and is defined in (2.15.4).

2.15.11 Theorem (Lower tail). For any countable class $\mathcal{F}$ of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$ with $|f(x) - Pf| \leq 1$ for every $x \in \mathcal{X}$ and $w_n := \sigma_{\mathcal{F}}^2 + 2n^{-1/2}E \sup_{f \in \mathcal{F}} \mathbb{G}_n f$ finite, for every $t \geq 0$,

$$P\left(\sup_{f \in \mathcal{F}} \mathbb{G}_n f - E \sup_{f \in \mathcal{F}} \mathbb{G}_n f \leq -t\right) \leq \exp\left(-\frac{nw_n}{9}h\left(1 + \frac{3t}{\sqrt{nw_n}}\right)\right).$$

The proofs of Theorems 2.15.5 and 2.15.11 rely upon the *entropy method*. The *entropy* of a nonnegative random variable Y is defined as $\text{Ent}(Y) = EY \log Y - EY \log EY$. Problems 2.15.3 and 2.15.5 derive the relevant properties of entropy.

Theorem 2.15.5 is a special case of a theorem on more general functions $T = T(X_1, \dots, X_n)$ of independent variables $X_1, \dots, X_n$. The abstract version is somewhat technical, but its proof is relatively straightforward.

2.15.12 Proposition. Let T be a measurable function of independent variables $X_1, \dots, X_n$. Suppose that there exist a constant $u \geq 0$ and random variables S_i and T_i , for $i = 1, \dots, n$, such that T_i is a measurable function of $(X_j: j \neq i)$, and

$$\begin{aligned} S_i &\leq T - T_i \leq 1, \\ E(S_i | X_j: j \neq i) &\geq 0, \\ S_i &\leq u, \\ \sum_{i=1}^n (T - T_i) &\leq T. \end{aligned}$$

If σ^2 is a constant satisfying $\sigma^2 \geq \sum_{i=1}^n E(S_i^2 | X_j: j \neq i)$, and $v = (1+u)ET + \sigma^2$, then, for all $t \geq 0$,

$$P(T - ET \geq t) \leq \exp\left(-vh\left(1 + \frac{t}{v}\right)\right).$$

Proof. Assume without loss of generality that $u \leq 1$, so that $\alpha = 1/(1+u) \in [1/2, 1]$. Throughout the proof let $\lambda > 0$.

The tensorization inequality for entropy (Problem 2.15.5) gives that

$$\text{Ent}(e^{\lambda T}) \leq E \sum_{i=1}^n \text{Ent}_i(e^{\lambda T}) = E \sum_{i=1}^n e^{\lambda T_i} \text{Ent}_i(e^{\lambda(T-T_i)}),$$

because T_i is measurable relative to $(X_j: j \neq i)$ and entropy is homogeneous (Problem 2.15.3(iii)). The variational characterization of entropy (Problem 2.15.3(ii) with $r = 1$) gives $\text{Ent}(e^Y) \leq E(1 - e^Y + Ye^Y)$, for any random variable Y . Applying this to each of the entropies $\text{Ent}_i(e^{\lambda(T-T_i)})$ conditionally given $(X_j: j \neq i)$ and substituting these bounds in the preceding display we obtain that

$$(2.15.13) \quad \text{Ent}(e^{\lambda T}) \leq E \sum_{i=1}^n \left(e^{\lambda T_i} - e^{\lambda T} + \lambda(T - T_i)e^{\lambda T} \right).$$

We shall later use the assumption $\sum_i (T - T_i) \leq T$ to bound the right side further, but first obtain an alternative upper bound using the other assumptions.

The i th term of the sum in the right side of (2.15.13) is algebraically identical to

$$g_{\lambda,\alpha}(T - T_i) \left(e^{\lambda T} - e^{\lambda T_i} - \lambda(T - T_i)e^{\lambda T_i} + \lambda\alpha(T - T_i)^2 e^{\lambda T_i} \right),$$

for the function $g_{\lambda,\alpha}: \mathbb{R} \rightarrow \mathbb{R}$ given by $g_{\lambda,\alpha}(t) = e^{\lambda t}(e^{-\lambda t} - 1 + \lambda t)/(e^{\lambda t} - 1 - \lambda t + \lambda\alpha t^2)$. Because $T - T_i \leq 1$ by assumption, and the function $g_{\lambda,\alpha}$ is increasing on $(-\infty, 1]$ (see Problem 2.15.6), and also the term in brackets is nonnegative, the expression in the preceding display increases if $g_{\lambda,\alpha}(T - T_i)$ is replaced by $g_{\lambda,\alpha}(1)$. Because $g_{\lambda,\alpha}(1) > 0$ and $-t + \alpha t^2 \leq -s + \alpha s^2$ if both $s \leq u \leq 1$ and $s \leq t \leq 1$ (remember that $\alpha = 1/(1+u)$; the parabola is symmetric about $(u+1)/2 \geq u \geq s$ and hence decreasing on $[0, (1+u)/2]$ and equal at u and 1), the preceding display is further bounded by

$$g_{\lambda,\alpha}(1) \left(e^{\lambda T} - e^{\lambda T_i} - \lambda S_i e^{\lambda T_i} + \lambda\alpha S_i^2 e^{\lambda T_i} \right).$$

We take the expected value and sum over i to obtain an upper bound on (2.15.13). Next we use that $E(-\lambda S_i e^{\lambda T_i}) \leq 0$, because $E(S_i | X_j: j \neq i) \geq 0$ and T_i is measurable relative to $(X_j: j \neq i)$. We also use that $\sum_i E S_i^2 e^{\lambda T_i} \leq \sum_i E E(S_i^2 | X_j: j \neq i) e^{\lambda T} \leq \sigma^2 E e^{\lambda T}$, where the last inequality is by assumption and the first can be argued as follows. Because $T - T_i \geq S_i$, the conditional mean of $T - T_i$ is also nonnegative, so that by Jensen's inequality applied with the exponential function $E(e^{\lambda(T-T_i)} | X_j: j \neq i) \geq 0$, or equivalently $e^{\lambda T_i} \leq E(e^{\lambda T} | X_j: j \neq i)$. Thus $E S_i^2 e^{\lambda T_i} \leq E S_i^2 E(e^{\lambda T} | X_j: j \neq i)$, where in the right side the conditioning can be transferred from $e^{\lambda T}$ to S_i^2 .

We thus have proved that

$$\frac{1}{g_{\lambda,\alpha}(1)} \text{Ent}(e^{\lambda T}) \leq E \sum_{i=1}^n \left(e^{\lambda T} - e^{\lambda T_i} \right) + \lambda\alpha\sigma^2 E e^{\lambda T}.$$

The exponentials $e^{\lambda T}$ and $e^{\lambda T_i}$ on the right side are in the reverse order relative to (2.15.13). Therefore, adding the present inequality and (2.15.13) gives

$$\text{Ent}(e^{\lambda T}) \left(\frac{1}{g_{\lambda,\alpha}(1)} + 1 \right) \leq \lambda\alpha\sigma^2 E e^{\lambda T} + E \sum_{i=1}^n \lambda(T - T_i) e^{\lambda T}.$$

The last term on the right can be further bounded by $\lambda E T e^{\lambda T}$, since $\sum_{i=1}^n (T - T_i) \leq T$ by assumption. Next we multiply across by $e^{\lambda E T}$ to rewrite the inequality as (cf. Problem 2.15.3(iii)).

$$\begin{aligned} \text{Ent}(e^{\lambda(T-ET)}) \left(\frac{1}{g_{\lambda,\alpha}(1)} + 1 \right) &\leq \lambda\alpha\sigma^2 E e^{\lambda(T-ET)} + E \lambda T e^{\lambda(T-ET)} \\ &= \lambda\alpha v E e^{\lambda(T-ET)} + E \lambda(T - ET) e^{\lambda(T-ET)}. \end{aligned}$$

The entropy $\text{Ent}(e^{\lambda(T-ET)})$ can be expressed in terms of the Laplace transform $F(\lambda) = E e^{\lambda(T-ET)}$ as $\text{Ent}(e^{\lambda(T-ET)}) = \lambda F'(\lambda) - F(\lambda) \log F(\lambda)$, and $E(T -$

$ET)e^{\lambda(T-ET)} = F'(\lambda)$. After also inserting the explicit form of $g_{\lambda,\alpha}(1)$, we can simplify the preceding inequality to

$$(\log F)'(\lambda) - \log F(\lambda) \left(\frac{e^\lambda - 1 + \alpha}{e^\lambda - 1 - \lambda + \lambda\alpha} \right) \leq \left(\frac{e^\lambda(e^{-\lambda} - 1 + \lambda)}{e^\lambda - 1 - \lambda + \lambda\alpha} \right) v\alpha.$$

Some algebra shows that the corresponding equality is solved (in $\log F$) by the function $\lambda \mapsto v\psi(\lambda)$, for $\psi(\lambda) = e^\lambda - 1 - \lambda$. Consequently, we can apply Problem 2.15.7 to see that $\log F(\lambda) \leq v\psi(\lambda)$.

Combining this with Markov's inequality, we obtain $P(T - ET \geq t) \leq e^{-\lambda t} F(\lambda) \leq e^{-(\lambda t - v\psi(\lambda))}$. The (optimal) choice $\lambda = \log(1 + t/v)$ yields the proposition. ■

A special case of Proposition 2.15.12, applicable to sums of nonnegative variables, arises if the hypotheses of Theorem 2.15.12 hold with every S_i equal to 0, so that both u and σ^2 can be chosen equal to 0. Then

$$P(T - ET \geq t) \leq \exp\left(-ET h\left(1 + \frac{t}{ET}\right)\right).$$

Boucheron, Lugosi and Massart (2000) show that under the same hypotheses, for nonnegative and bounded variables T and for $0 < t \leq ET$,

$$P(T - ET \leq -t) \leq \exp\left(-ET h\left(1 - \frac{t}{ET}\right)\right).$$

Proof of Theorem 2.15.11. It suffices to prove the theorem for a finite collection $\mathcal{F}$ of functions with $Pf = 0$. For fixed $\lambda > 0$, let $M_f(\lambda) = Ee^{-\lambda f(X_1)}$ be the Laplace transform of the variables $f(X_i)$ at $-\lambda$, and define

$$S_\lambda = \min_f \frac{e^{-\lambda \sum_{i=1}^n f(X_i)}}{M_f(\lambda)^n}.$$

The main part of the proof is to bound the expectation ES_λ . The theorem will next be derived from a bound on the Laplace transform of $T = \max_f \sum_{i=1}^n f(X_i)$ in terms of ES_λ .

By the tensorization inequality (Problem 2.15.5),

$$(2.15.14) \quad \text{Ent}(S_\lambda) \leq \sum_{i=1}^n E \text{Ent}_i(S_\lambda),$$

$$(2.15.15) \quad \leq \sum_{i=1}^n E \text{Ent}_i(S_{i,\lambda}) + \sum_{i=1}^n E(S_{i,\lambda} - S_\lambda) \log \frac{E_i S_\lambda}{S_\lambda},$$

by Problem 2.15.3(vi), for any random variables $S_{i,\lambda}$, where E_i means the expectation with respect to X_i with the other variables $(X_j; j \neq i)$ fixed. For $\hat{f}_\lambda$ a function f that attains the minimum in the definition of S_λ , we choose

$$S_{i,\lambda} = \sum_f P_i(\hat{f}_\lambda = f) \frac{e^{-\sum_{i=1}^n f(X_i)}}{M_f(\lambda)^n}.$$

Because $S_{i,\lambda}$ is a convex combination of the same variables whose minimum is S_λ , we have $S_{i,\lambda} - S_\lambda \geq 0$. Moreover, at the end of the proof we show that

$$(2.15.16) \quad \frac{E_i S_\lambda}{S_\lambda} \leq e^{\lambda \hat{f}_\lambda(X_i)} M_{\hat{f}_\lambda}(\lambda) \leq \nu(\lambda) := e^\lambda \cosh \lambda,$$

$$(2.15.17) \quad \sum_{i=1}^n E(S_{i,\lambda} - S_\lambda) \leq \nu(\lambda) \left(\sum_{i=1}^n E \text{Ent}_i(S_\lambda) - ES_\lambda \log S_\lambda \right),$$

$$(2.15.18) \quad \sum_{i=1}^n E \text{Ent}_i(S_{i,\lambda}) \leq n\nu(\lambda) ES_\lambda (\lambda L'_{\hat{f}_\lambda}(\lambda) - L_{\hat{f}_\lambda}(\lambda)).$$

Here $L_f = \log M_f$ is the cumulant generating function of the variables $f(X_i)$ at $-\lambda$. Substituting inequalities (2.15.16)–(2.15.18) in (2.15.15) and rearranging, we find that, with $\gamma = \nu \log \nu$,

$$(2.15.19) \quad \begin{aligned} & (1 - \gamma(\lambda)) \sum_{i=1}^n E \text{Ent}_i(S_\lambda), \\ & \leq n\nu(\lambda) ES_\lambda (\lambda L'_{\hat{f}_\lambda}(\lambda) - L_{\hat{f}_\lambda}(\lambda)) - \gamma(\lambda) ES_\lambda \log S_\lambda. \end{aligned}$$

For λ such that $1 - \gamma(\lambda) > 0$ (i.e. $\lambda \leq \lambda_0 \approx 0.46$), this is a bound on $(1 - \gamma(\lambda)) \text{Ent}(S_\lambda)$, by (2.15.14). The next step is to express $\text{Ent}(S_\lambda)$ and the quantities on the right side in terms of the function $F(\lambda) = ES_\lambda$ and its derivative.

Because each of the entries $\exp(-\lambda \sum f(X_i))/M_f(\lambda)^n$ in the minimum that defines S_λ is an analytic function of λ , each pair of entries is either identical or possesses graphs (as functions of λ) that agree at most on a set without limit points. It follows that $[0, \infty)$ can be partitioned into a countable number of intervals such that on each interval S_λ (as function of $\lambda \geq 0$) coincides with one of the entries. Consequently, $\lambda \mapsto S_\lambda$ is differentiable on the interior of each of these intervals, with derivative

$$S'_\lambda = \frac{\partial}{\partial \lambda} \left(\frac{e^{-\lambda \sum_{i=1}^n f(X_i)}}{M_f(\lambda)^n} \right)_{|f=\hat{f}_\lambda} = S_\lambda \left(-\sum_{i=1}^n \hat{f}_\lambda(X_i) - nL'_{\hat{f}_\lambda}(\lambda) \right).$$

The expression inside brackets on the right side can also be written as $\lambda^{-1}(\log S_\lambda + nL_{\hat{f}_\lambda}(\lambda)) - nL'_{\hat{f}_\lambda}(\lambda)$. We conclude that $F(\lambda) = ES_\lambda$ satisfies

$$\lambda F'(\lambda) = \lambda ES'_\lambda = ES_\lambda \log S_\lambda - nES_\lambda (\lambda L'_{\hat{f}_\lambda}(\lambda) - L_{\hat{f}_\lambda}(\lambda)).$$

We solve this for $ES_\lambda \log S_\lambda$, and use the definition of $\text{Ent}(S_\lambda)$ and inequalities (2.15.14) and (2.15.19) to infer that, for $\lambda \leq \lambda_0$,

$$\begin{aligned} \lambda F'(\lambda) - (1 - \gamma(\lambda)) F(\lambda) \log F(\lambda) & \leq n(\nu(\lambda) - 1) ES_\lambda (\lambda L'_{\hat{f}_\lambda}(\lambda) - L_{\hat{f}_\lambda}(\lambda)) \\ & \leq n(\nu(\lambda) - 1) F(\lambda) \max_f (\lambda L'_f(\lambda) - L_f(\lambda)). \end{aligned}$$

Here $\lambda L'_f(\lambda) - L_f(\lambda) = \text{Ent}(e^{-\lambda f(X_1)})/M_f(\lambda) \leq e^\lambda (e^{-\lambda} - 1 + \lambda) \sigma_{\mathcal{F}}^2$, by Problem 2.15.3(v) and because $M_f(\lambda) \geq 1$, by Jensen's inequality and the assumption that $Pf = 0$. We conclude that, for $\lambda \leq \lambda_0$,

$$\lambda (\log F)'(\lambda) - (1 - \gamma(\lambda)) \log F(\lambda) \leq n \frac{1}{2} (e^{2\lambda} - 1) e^\lambda (e^{-\lambda} - 1 + \lambda) \sigma_{\mathcal{F}}^2.$$

Because $M_f(\lambda) \geq 1$, we have that $F(\lambda) = \mathbb{E}S_\lambda \leq \mathbb{E}e^{-\lambda T}$, for $T = \max_f \sum_{i=1}^n f(X_i)$, and hence $\log F(\lambda)/\lambda \leq (\log \mathbb{E}e^{-\lambda T})/\lambda \rightarrow -ET$, as $\lambda \downarrow 0$. Under this side condition the differential inequality of the preceding display implies that, for $\lambda \leq \lambda_0$,

$$(2.15.20) \quad \log F(\lambda) \leq -ET\lambda e^{-I(\lambda)} + n\sigma_{\mathcal{F}}^2 \lambda e^{-I(\lambda)} J(\lambda),$$

for I and J the functions defined in Problems 2.15.8 and 2.15.9.

Because $\min_f e_f \leq \min_f (e_f/c_f) \max_f c_f$, for any positive constants c_f, e_f , we have $\mathbb{E}e^{-\lambda T} \leq \mathbb{E}S_\lambda \max_f M_f(\lambda)^n$. Combining this with Bennett's bound $\log M_f(\lambda) \leq Pf^2(e^\lambda - 1 - \lambda)$ and (2.15.20), we conclude that

$$\begin{aligned} \log \mathbb{E}e^{-\lambda(T-ET)} &\leq \lambda ET + \log F(\lambda) + n\sigma_{\mathcal{F}}^2(e^\lambda - 1 - \lambda) \\ &\leq 2ET \left(\frac{\lambda(1 - e^{-I(\lambda)})}{2} \right) + n\sigma_{\mathcal{F}}^2 \left(\lambda e^{-I(\lambda)} J(\lambda) + e^\lambda - 1 - \lambda \right) \\ &\leq w_n \frac{1}{9} (e^{3\lambda} - 1 - 3\lambda), \end{aligned}$$

by the definition of w_n , in view of Problem 2.15.9, for $\lambda \leq \lambda_0$.

For $\lambda > \lambda_0$ we use instead of (2.15.20) the trivial bound $\log F(\lambda) \leq 0$, which is valid for every λ , because the variables in the minimum defining S_λ have mean 1. It follows that $\log \mathbb{E}e^{-\lambda(T-ET)}$ is bounded by $w_n \max(\lambda/2, e^\lambda - 1 - \lambda)$, which is bounded by the right side of the preceding display for $\lambda \geq \lambda_1 \approx 0.53$. Also there exists $\lambda_2 > \lambda_1$ so that the line segment connecting the points $(\lambda_0, (e^{3\lambda_0} - 1 - 3\lambda_0)/9)$ and $(\lambda_2, \max(\lambda_2/2, e^{\lambda_2} - 1 - \lambda_2))$ is below the graph of the function $\lambda \mapsto (e^{3\lambda} - 1 - 3\lambda)/9$. By the convexity of the function $\lambda \mapsto \log \mathbb{E}e^{-\lambda(T-ET)}$, we conclude that this function is below the function $\lambda \mapsto w_n(e^{3\lambda} - 1 - 3\lambda)/9$ for every $\lambda \geq 0$. This bound on the cumulant generating function can be turned into the tail bound of the theorem using Markov's inequality.

Finally we prove the validity of inequalities (2.15.16)–(2.15.18).

For $\hat{f}_{i,\lambda}$ the minimizer of $f \mapsto e^{-\lambda \sum_{j \neq i} f(X_j)} / M_f(\lambda)^{n-1}$, which depends on $(X_j: j \neq i)$ only, the left side of (2.15.16) is bounded by

$$\frac{\mathbb{E}_i e^{-\lambda \sum_{j=1}^n \hat{f}_{i,\lambda}(X_j)} / M_{\hat{f}_{i,\lambda}}(\lambda)^n}{e^{-\lambda \sum_{j=1}^n \hat{f}_\lambda(X_j)} / M_{\hat{f}_\lambda}(\lambda)^n} = \frac{e^{-\lambda \sum_{j \neq i} \hat{f}_{i,\lambda}(X_j)} / M_{\hat{f}_{i,\lambda}}(\lambda)^{n-1}}{e^{-\lambda \sum_{j=1}^n \hat{f}_\lambda(X_j)} / M_{\hat{f}_\lambda}(\lambda)^n}.$$

By the definition of $\hat{f}_{i,\lambda}$, the right side increases if $\hat{f}_{i,\lambda}$ is replaced by $\hat{f}_\lambda$. This yields the middle inequality of (2.15.16). The second inequality follows because $e^{\lambda f} \leq e^\lambda$ for every f by the assumption that $f \leq 1$, while $M_f(\lambda) \leq \cosh \lambda$ by Problem 2.15.10.

To prove (2.15.17) we first write

$$\mathbb{E}_i S_{i,\lambda} = \sum_f \mathbb{P}_i(\hat{f}_\lambda = f) \frac{e^{-\lambda \sum_{j \neq i} f(X_j)}}{M_f(\lambda)^{n-1}} = \mathbb{E}_i S_\lambda e^{\lambda \hat{f}_\lambda(X_i)} M_{\hat{f}_\lambda}(\lambda).$$

It follows that the left side of (2.15.17) is equal to

$$\begin{aligned} & \sum_{i=1}^n \mathbb{E} S_\lambda \left(e^{\lambda \hat{f}_\lambda(X_i)} M_{\hat{f}_\lambda}(\lambda) - 1 \right) \\ &= \sum_{i=1}^n \mathbb{E} S_\lambda \left(e^{\lambda \hat{f}_\lambda(X_i)} M_{\hat{f}_\lambda}(\lambda) - 1 - \nu(\lambda) \log(e^{\lambda \hat{f}_\lambda(X_i)} M_{\hat{f}_\lambda}(\lambda)) \right) \\ & \quad - \nu(\lambda) \sum_{i=1}^n \mathbb{E} S_\lambda \log \left(\frac{e^{-\lambda \hat{f}_\lambda(X_i)}}{M_{\hat{f}_\lambda}(\lambda)} \right). \end{aligned}$$

The first term on the right increases if $e^{\lambda \hat{f}_\lambda(X_i)} M_{\hat{f}_\lambda}(\lambda)$ is replaced by $\mathbb{E}_i S_\lambda / S_\lambda$, by (2.15.16) and the fact the function $x \mapsto x - 1 - \nu \log x$ is decreasing for $x \leq \nu$. The sum in the second term on the right is equal to $\mathbb{E} S_\lambda \log S_\lambda$. It follows that the right side is bounded above by

$$\sum_{i=1}^n \mathbb{E} S_\lambda \left(\frac{\mathbb{E}_i S_\lambda}{S_\lambda} - 1 - \nu(\lambda) \log \frac{\mathbb{E}_i S_\lambda}{S_\lambda} \right) - \nu(\lambda) \mathbb{E} S_\lambda \log S_\lambda.$$

This can be seen to be identical to the right side of (2.15.17).

The convexity and homogeneity of entropy (Problem 2.15.3(iv),(iii)) yield

$$\text{Ent}_i(S_{i,\lambda}) \leq \sum_f \mathbb{P}_i(\hat{f}_\lambda = f) \frac{e^{-\lambda \sum_{j \neq i} f(X_j)}}{M_f(\lambda)^n} \text{Ent}_i(e^{-\lambda f(X_i)}).$$

Here $\text{Ent}_i(e^{-\lambda f(X_i)}) = M_f(\lambda)(\lambda L'_f(\lambda) - L_f(\lambda))$ and the order of summation and expectation can be exchanged to rewrite the right side as

$$\begin{aligned} & \mathbb{E}_i \frac{e^{-\lambda \sum_{j \neq i} \hat{f}_\lambda(X_j)}}{M_{\hat{f}_\lambda}(\lambda)^{n-1}} (\lambda L'_{\hat{f}_\lambda}(\lambda) - L_{\hat{f}_\lambda}(\lambda)) \\ &= \mathbb{E}_i S_\lambda e^{\lambda \hat{f}_\lambda(X_i)} M_{\hat{f}_\lambda}(\lambda) (\lambda L'_{\hat{f}_\lambda}(\lambda) - L_{\hat{f}_\lambda}(\lambda)). \end{aligned}$$

We bound $e^{\lambda \hat{f}_\lambda(X_i)} M_{\hat{f}_\lambda}(\lambda)$ by $\nu(\lambda)$ using (2.15.16), next take the expectation over the other variables and sum over i to arrive at (2.15.18). ■

Problems and Complements

1. Let $T: \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ be an arbitrary map defined on the product probability space $(\mathcal{X} \times \mathcal{Y}, \mathcal{A} \times \mathcal{B}, P \times Q)$ such that $|T(x, y) - T(x, z)| \leq c$ for a constant c and every $x \in \mathcal{X}$ and $y, z \in \mathcal{Y}$. Then the measurable cover $T^*: \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ for $P \times Q$ satisfies $|T^*(x, y) - T^*(x, z)| \leq c$ for $P \times Q \times Q$ almost every (x, y, z) .

[Hint: By perfectness of a coordinate projection T^* is also a version of the measurable cover for T viewed as a map $T: \mathcal{X} \times \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$ relative to $P \times Q \times Q$. Furthermore, the measurable covers of $T_1: (x, y, z) \mapsto T(x, y)$ and $T_2: (x, y, z) \mapsto T(x, z)$ relative to $P \times Q \times Q$ satisfy $|T_1^* - T_2^*| \leq |T_1 - T_2|^*$ almost surely.]

2. If X is a random variable taking values in the interval $[a, b] \subset \mathbb{R}$ with $EX = 0$, then $Ee^{\lambda X} \leq \exp(\lambda^2(b-a)^2/8)$, for all $\lambda > 0$.

[Hint: Convexity of the function $x \mapsto e^{\lambda x}$ gives, for any $x \in [a, b]$,

$$e^{\lambda x} \leq \frac{x-a}{b-a} e^{\lambda b} + \frac{b-x}{b-a} e^{\lambda a}.$$

This implies that $Ee^{\lambda X} \leq e^{\phi(\lambda(b-a))}$ for ϕ defined by $\phi(x) = -px + \log(1-p + pe^x)$ and $p = -a/(b-a)$. Then note that $\phi(0) = 0 = \phi'(0)$ while $\phi''(x) \leq 1/4$ for all p, x .]

3. **(Entropy)** The *entropy* of a nonnegative variable Y is defined by

$$\text{Ent}(Y) = E\Phi(Y) - \Phi(EY),$$

where $\Phi(y) = y \log y$ (defined by continuity at $y = 0$). It satisfies:

- (i) $\text{Ent}(Y) = \sup\{EYU: Ee^U = 1\}$, where the supremum is attained at $U = \log Y - \log EY$.
- (i') $\text{Ent}(Y) = \sup\{EY \log(V/EY): V > 0\}$, where the supremum is attained at $V = cY$.
- (ii) $\text{Ent}(Y) = \inf_{r>0} E(\Phi(Y) - (1 + \log r)Y + r) = \inf_{r>0} \{rEh(Y/r)\}$ with $h(y) = y(\log y - 1) + 1$, where the infimum is assumed at $r = EY$.
- (iii) $\text{Ent}(\gamma Y) = \gamma \text{Ent}(Y)$, for any $\gamma \in \mathbb{R}$.
- (iv) $Y \mapsto \text{Ent}(Y)$ is convex.
- (v) If $X \leq 1$, then $\text{Ent}(e^{\lambda X}) \leq EX^2 e^{\lambda} (e^{-\lambda} - 1 + \lambda)$, for every $\lambda > 0$.
- (vi) $\text{Ent}(Y) \leq \text{Ent}(Z) + E(Y - Z) \log(Y/EY)$, for every positive random variable Z .

[Hint: For (i) use that $\sup_{y>0} (yu - \Phi(y)) = e^{u-1}$ for $u \in \mathbb{R}$. For (ii) use that the function $g: (0, \infty) \rightarrow \mathbb{R}$ defined by $g(r) = E(\Phi(Y) - (1 + \log r)Y + r)$ is convex, with $g(EY) = \text{Ent}(Y)$. For (v) use (ii) with $r = 1$ and $Y = e^{\lambda X}$ to see that $\text{Ent}(e^{\lambda X}) \leq Eh(e^{\lambda X}) \leq EX^2 h(e^{\lambda})$, by the monotonicity of the function $\lambda \mapsto \lambda^{-2} h(e^{\lambda})$. Inequality (vi) follows by writing $\text{Ent}(Y)$ as $EZ \log(Y/EY) + E(Y - Z) \log(Y/EY)$ and next applying (i') to see that the first term is smaller than $\text{Ent}(Z)$.]

4. **(Herbst argument)** If a random variable X satisfies $\text{Ent}(e^{\lambda X}) \leq \frac{1}{2} \sigma^2 \lambda^2 Ee^{\lambda X}$, for every $\lambda \in \mathbb{R}$, then $X - EX$ is sub-Gaussian: $Ee^{\lambda(X-EX)} \leq e^{\sigma^2 \lambda^2 / 2}$, for every $\lambda \in \mathbb{R}$.

[Hint: The assumption entails that $\phi(\lambda) := Ee^{\lambda X}$ satisfies, for $\lambda > 0$,

$$\frac{\partial}{\partial \lambda} \frac{\log \phi(\lambda)}{\lambda} \equiv \frac{\phi'(\lambda)}{\lambda \phi(\lambda)} - \frac{\log \phi(\lambda)}{\lambda} \leq \frac{1}{2} \sigma^2.$$

For $\lambda \downarrow 0$ the function $\lambda^{-1} \log \phi(\lambda)$ can be defined by continuity as EX . Integrating gives that $\lambda^{-1} \log \phi(\lambda) - EX \leq \sigma^2 \lambda / 2$.]

5. **(Tensorization of entropy)** If T is a measurable transformation of independent variables $X_1, \dots, X_n$, then $\text{Ent}(T) \leq \sum_{i=1}^n \mathbb{E} \text{Ent}_i(T)$, for $\text{Ent}_i(T)$ the entropy of T considered as a function of X_i only with the other variables ($X_j: j \neq i$) fixed, i.e. $\text{Ent}_i(T) = \mathbb{E}(\Phi(T) | X_j: j \neq i) - \Phi(\mathbb{E}(T | X_j: j \neq i))$.

[**Hint:** Let $\mathbb{E}_I T$ denote the expectation of T relative to $(X_i: i \in I)$ given fixed values of the other variables ($X_i: i \notin I$), with $\mathbb{E}_\emptyset T = T$. Then

$$T \log T - T \log \mathbb{E} T = \sum_{i=1}^n T (\log \mathbb{E}_{1, \dots, i-1} T - \log \mathbb{E}_{1, \dots, i} T).$$

Problem 2.15.3(i') applied to T viewed as a function of X_i with $(X_j: j \neq i)$ fixed and $V = \mathbb{E}_{1, \dots, i-1} T$ for fixed $(X_j: j > i)$ gives that $\mathbb{E}_i T (\log \mathbb{E}_{1, \dots, i-1} T - \log \mathbb{E}_{1, \dots, i} T) \leq \mathbb{H}_i(T)$. Inserting this in the preceding equation and taking the expectation yields the result.]

6. The function $t \mapsto e^{\lambda t} (e^{-\lambda t} - 1 + \lambda t) / (e^{\lambda t} - 1 - \lambda t + \lambda \alpha t^2)$ is nondecreasing for $t \leq 1$, for all $\lambda \geq 0$ and all $\alpha \in [1/2, 1]$.

[**Hint:** See Lemma A.3 of Bousquet (2003).]

7. **(Differential inequality)** Let $L, L_0: [0, 1] \rightarrow \mathbb{R}$ be continuously differentiable functions that satisfy the inequality $fL' - gL \leq fL'_0 - gL_0$, for continuous, nonnegative functions $f, g: [0, 1] \rightarrow \mathbb{R}$ such that g/f is integrable. If $L(0) = L_0(0)$, then $L \leq L_0$ on $[0, 1]$.

[**Hint:** For $A(\lambda) = \exp(-\int_0^\lambda (f/g)(s) ds)$ we obtain that $(AL)' = A(L' - (g/f)L) \leq (AL_0)'$.]

8. Let $\alpha(\lambda) = (1 - \gamma)(\lambda)/\lambda$ for $\gamma = v \log v$ and $v(\lambda) = (e^{2\lambda} + 1)/2$ and let $\Lambda, \beta: [0, 1] \rightarrow \mathbb{R}$ be functions such that Λ is absolutely continuous with $\Lambda' - \alpha\Lambda \leq \beta$. Then $\Lambda(\lambda) \leq \mu \lambda e^{-I(\lambda)} + \lambda e^{-I(\lambda)} \int_0^\lambda s^{-1} e^{I(s)} \beta(s) ds$ for $\mu = \limsup_{\varepsilon \downarrow 0} \Lambda(\varepsilon)/\varepsilon$ and $I(\lambda) = \int_0^\lambda \gamma(s)/s ds$.

[**Hint:** The function I is well defined (as $\gamma(s) = O(s)$ as $s \rightarrow 0$), but the “integrating factor” $\exp(-\int_0^\lambda \alpha(s) ds)$ for the linear differential equation is not. This motivates consideration of the functions first on the interval $[\varepsilon, 1]$ for $\varepsilon \in (0, 1)$. The corresponding integrating factor is $A_\varepsilon(\lambda) = \exp(-\int_\varepsilon^\lambda \alpha(s) ds) = (\varepsilon/\lambda) e^{I(\lambda) - I(\varepsilon)}$. From the inequality $(A_\varepsilon \Lambda)' = A_\varepsilon (\Lambda' - \alpha\Lambda) \leq A_\varepsilon \beta$ we obtain that $(A_\varepsilon \Lambda)(\lambda) \leq (\Lambda A_\varepsilon)(\varepsilon) + \int_\varepsilon^\lambda (A_\varepsilon \beta)(s) ds$. Dividing by $A_\varepsilon(\lambda)$ and taking the limit as $\varepsilon \downarrow 0$ gives the desired result.]

9. The functions $I, J: [0, \infty) \rightarrow \mathbb{R}$ defined by

$$I(\lambda) = \int_0^\lambda s^{-1} \frac{1}{2} (e^{2s} + 1) \log \left(\frac{1}{2} (e^{2s} + 1) \right) ds,$$

$$J(\lambda) = \frac{1}{2} \int_0^\lambda s^{-2} e^{I(s)} (e^{2s} - 1) e^s (e^{-s} - 1 + s) ds,$$

satisfy

- (i) $\lambda e^{-I(\lambda)} J(\lambda) + e^\lambda - 1 - \lambda \leq (e^{3\lambda} - 1 - 3\lambda)/9$.
- (ii) $\lambda(1 - e^{-I(\lambda)}) \leq 2(e^{3\lambda} - 1 - 3\lambda)/9$.

10. Any random variable X with $\mathbb{E} X = 0$ and $|X| \leq 1$ satisfies $\mathbb{E} e^{\lambda X} \leq \cosh \lambda$, for $\lambda \geq 0$.

2

Notes

2.2. The maximal inequality Theorem 2.2.4 uses the special properties of the Orlicz norm only through the application of Lemma 2.2.2. Hence an identical result can be proved using other norms provided the analogue of the lemma is available. For instance, Ledoux and Talagrand (1991) show that the lemma holds for weak L_p -norms with ψ taken equal to x^p . Second, it may be remarked that often the full strength of Theorem 2.2.4 is not needed and the weakened version of this theorem, where the Orlicz norm on the left is replaced by $E \sup_{s,t} |X_s - X_t|$, suffices. This weakened version can be proved along exactly the same lines, provided a similarly weakened version of Lemma 2.2.2 is available. The latter is contained in Problem 2.2.8 for *any* Orlicz norm. In another direction Lemma 2.2.2 can be strengthened for Young functions of exponential type into a bound of the form $\| \sup_i X_i / \psi^{-1}(i) \|_\psi \lesssim \sup_i \|X_i\|_\psi$ for an arbitrary number (also countably infinite) numbers of variables; see Proposition 4.3.1, page 189, in De la Peña and Giné (1999).

The method of chaining to prove continuity or obtain bounds on suprema of stochastic processes is due to “Kolmogorov and his school”, and goes back to the 1930s and 40s. The combination of chaining with “entropy” in the study of the continuity of Gaussian processes goes back to Dudley (1967b) and Sudakov (1969). Dudley (1967b) credits V. Strassen with the idea and basic results, but see the review of Sudakov (1976) by Dudley (1978b). Strassen and Dudley (1969) apparently made the first use of entropy in connection with empirical processes. Maximal inequalities formulated as entropy bounds for general Young functions were obtained in Kono (1980) and Pisier (1983).

The sufficiency of the existence of a (continuous) majorizing measure for the boundedness (and continuity) of a Gaussian process was established by Fernique

(1974), following preliminary results of Preston (1972). The necessity was proved for stationary Gaussian processes by Fernique (1974) and for general Gaussian processes by Talagrand (1987c). For complete expositions of majorizing measures, see Adler (1990), pages 80ff., and Ledoux and Talagrand (1991), Chapters 11 and 12.

After giving simplified proofs in Talagrand (1992) and Talagrand (1996b), Talagrand (2001) introduced a more fundamental approach, which he called “generic chaining”. This approach was developed further in the books Talagrand (2005, 2014). Theorems 2.2.14 and 2.2.19 for exponential Orlicz functions are given there. Results for exponential Young functions, including the mixed ψ_1 - ψ_2 case Theorem 2.2.19, with many applications, were also obtained in Dirksen (2015). Upper bounds in terms of majorizing measures for non-Gaussian processes using general Young functions were obtained in Bednorz (2006), after initial work by Talagrand (1990) and Kono (1980). Bednorz also extended the method of generic chaining to the setting of general Young functions. Theorems 2.2.14 and 2.2.20, and Proposition 2.2.23 were obtained in Bednorz (2008, 2010, 2015). Among the rare applications of generic chaining in statistics is the paper Van de Geer (2013).

The original papers by Bernstein are apparently unavailable. Bennett (1962) discusses the history of Bernstein’s inequality, provides proofs, and compares Bernstein’s inequality to a number of other inequalities.

2.3. The method of symmetrization goes back to Kahane (1968) and Hoffmann-Jørgensen (1973, 1976). The construction of separable versions using liftings is adapted from Talagrand (1987a).

2.4. Generalizations of the classical Glivenko (1933) and Cantelli (1933) theorem for sets under a bracketing condition were first obtained by Blum (1955) and Dehard (1971). Dudley (1984) gives Theorem 2.4.1 in its present formulation. Vapnik and Červonenkis (1968, 1971, 1981) give generalizations with conditions in terms of the logarithm of the L_1 -covering numbers. The proof of the converse part of Theorem 2.4.3 is adapted from Giné and Zinn (1984), pages 981–982. Pollard (1982) formulates the reverse submartingale property of $\|\mathbb{P}_n - P\|_{\mathcal{F}}$. Measurability difficulties regarding the choice of filtration are pointed out by Strobl (1992). Theorem 2.4.7 is due to Van Handel (2013), after initial work by Adams and Nobel (2010, 2012), on extensions of the Glivenko-Cantelli property to general, not identically distributed or independent, observations. For more on Glivenko-Cantelli theorems, see Chapter 2.8, Dudley, Giné and Zinn (1991) and Theorem 3.7.14 of Giné and Nickl (2016).

2.5. The first uniform entropy central limit theorems were due to Pollard (1982) and Kolčinskij (1981) following path-breaking work by Dudley (1978a), who, among other things, established the uniform central limit theorem for the empirical process indexed by a (suitably measurable) VC-class of sets. Bracketing empirical central limit theorems were obtained by Dudley (1978a, 1984), Ossiander (1987) and Andersen, Giné, Ossiander and Zinn (1988). Ossiander (1987) obtained the result under the assumption that the $L_2(P)$ -bracketing integral is finite. The result presented in this section is more general than this, but less general than the theorems

obtained by Andersen et al. These authors relax finiteness of the two integrals to continuous majorizing measure conditions on $L_2(P)$ -balls and $L_{2,\infty}(P)$ -brackets, respectively. See Chapter 2.11 for a statement.

The chaining argument in the proof of Theorem 2.5.6 may be refined so that it is not necessary to construct a *nested* sequence of partitions. See Pollard (1989a). The present organization and notation of the chaining argument is borrowed from Arcones and Giné (1993), who chain by exponential bounds for probabilities. The use of means in combination with Lemma 2.2.13 in the proof is new.

2.6. The study of VC-classes apparently originated in Vapnik and Červonenkis (1971), who were motivated by problems in pattern recognition. See also Vapnik and Červonenkis (1981) and Vapnik (1982). Sauer's lemma and its generalizations can be found in Sauer (1972) and Frankl (1983). Theorem 2.6.4 is due to Haussler (1995), who improved an earlier result of Dudley (1978a) (see Problem 2.6.4). Our rendering of the proof owes its origins to conversations with David Pollard. Pollard (1984) showed how to obtain the result about functions from the bound for sets.

The behavior of entropies under the formation of convex hulls as described in Theorem 2.6.9 is an extension of results of Dudley (1987), who obtained the bound up to an additional δ , and Ball and Pajor (1990), who obtained the bound under the assumption that $\mathcal{F}$ is a sequence of functions decreasing at the rate $\|f_i\|_{Q,2} \leq i^{-1/V}$. The present formulation for $r = 2$ and its induction proof were new in the first edition of this work. The extensions to L_r -norms and to classes of polynomial entropy, as in Theorem 2.6.13, proved here by direct generalization of the proof for $r = 2$, were obtained in Carl, Kyrezi and Pajor (1999) and Gao (2001). See Song and Wellner (2002), Creutzig and Steinwart (2002), Mendelson (2002) and Gao (2004) for different proofs, extensions and applications. Lemma 2.6.11 is given by Pisier (1981), but is apparently due to B. Maurey.

Many results in Sections 2.6.5 and 2.6.6 are due to Assouad (1981, 1983), Pollard (1984), and Dudley (1984). Assouad (1981, 1983) gives explicit bounds for the VC-dimension of many of the classes in Lemma 2.6.19 and formulates a number of results concerning “dual density,” which we have not included here (see especially Assouad (1983), pages 245–247). Bounds on the VC-dimension of unions and intersections can be found in Van der Vaart and Wellner (2009). Lemma 2.6.18 and Example 2.6.24 improve upon an example of Dudley (1985), who shows that the class of functions $x \mapsto t^{k-1}F(x)1\{F(x) \geq t\}$ is VC-hull. We learned Exercise 2.6.21 from A. Quiroz: see Quiroz, Nakamura and Perez (1996). Lemma 2.6.14 was proved by Giné and Koltchinskii (2006).

Fat-shattering numbers were introduced by Vapnik and Červonenkis. The discussion and results in Section 2.6.7 summarizes the findings in Mendelson and Vershynin (2003) and Rudelson and Vershynin (2006).

Notable among the results not included in this section are those of Stengle and Yukich (1989) and Pisier (1984). Stengle and Yukich show how new VC classes can be generated by “semi-algebraic sets.” Pisier (1984) links up VC theory with probability in Banach spaces by showing that a class of sets is VC if and only if a certain operator is type-2. See Ledoux and Talagrand (1991) for an exposition.

2.7. Bounds on entropies for classes of smooth functions are obtained by Kolmogorov and Tikhomirov (1961), Lorentz (1966), and Birman and Solomjak (1967). In particular, Theorem 2.7.1 on Hölder classes and Theorem 2.7.16 are explicit in Kolmogorov and Tikhomirov (1961). The entropy bound on Besov classes can be obtained from Birman and Solomjak (1967) by interpolation. The proof given here is based on representation in wavelet bases as developed by Birgé and Massart (2000), Cohen, Dahmen and DeVore (2000) and Cohen, Dahmen, Daubechies and DeVore (2001). The representation by Besov bodies and the resulting entropy estimates used here follows the replicant and universal coding schemes by Kerkyacharian and Picard (2003a)-Kerkyacharian and Picard (2003b).

The literature on function spaces is substantial, and includes many reviews in the form of books, or book series. Some of these are referenced in footnotes in Section 2.7.2. Valuable introductions to Besov spaces within the context of statistics, but limited to functions on the real line, can be found in Härdle, Kerkyacharian, Picard and Tsybakov (1998) and Giné and Nickl (2016). The latter authors also give examples and counter-examples of unit balls of function spaces being Glivenko-Cantelli or Donsker. Entropies of anisotropic or weighted full domain Sobolev and Besov spaces can be found in Triebel (2006).

Sets with smooth boundaries were considered by Sun and Pyke (1982) and Dudley (1974). Dudley (1984), Chapter 7, and Dudley (1999), Chapter 8, give useful introductions to this literature. We adapted the proof of the bound for entropy with bracketing for monotone functions on $\mathbb{R}$ from Van de Geer (1991), who adapts techniques from Birman and Solomjak (1967). The results for convex sets are due to Bronštein (1976), who improved on results of Dudley (1974). These bounds were first exploited in the context of empirical processes by Bolthausen (1978). Theorem 2.7.14 was obtained by Gao and Wellner (2017), who improved on Dryanov (2009) and Guntuboyina and Sen (2013) and Guntuboyina (2016), who considered convex functions on intervals and rectangles. The restriction $r < p$ cannot be removed as for $r \geq p$, the class of convex functions is not totally bounded. Gao and Wellner (2017) also give a sharper bound for a general convex set involving the accuracy of “simplicial approximation” of the set.

Van der Vaart (1994b) gives improvements of Corollary 2.7.3 and shows that the classes of functions considered there are P -Glivenko-Cantelli if and only if $\sum M_j P(\mathcal{X}_j) < \infty$. He also shows that the bracketing integral converges if the constants $M_j P(\mathcal{X}_j)^{1/2}$ are regularly varying at infinity and $\sum M_j P(\mathcal{X}_j)^{1/2} < \infty$. Giné and Zinn (1986b) first proved that the class $C^1(\mathbb{R})_1$ is Donsker under the latter condition. Van der Vaart (1996) generalized their result to the classes of Corollary 2.7.3 in general. Nickl and Pötscher (2007) consider bracketing entropy and the Donsker property of general weighted Sobolev and Besov spaces on unbounded domains.

Bracketing numbers on Gaussian mixtures were obtained by Ghosal and Van der Vaart (2001) and Ghosal and Van der Vaart (2007), who also give slightly better bounds on ordinary covering numbers. Generalizations to mixtures of multidimensional Gaussian and other distributions can be found in Shen, Tokdar and Ghosal (2013) and Scricciolo (2011).

For other work on entropy numbers, see Carl and Stephani (1990) and Ball and Pajor (1990) and the references stated therein. These authors study the entropy of the

image of the unit ball under a (compact) operator T between Banach spaces. More precisely, they study entropy numbers denoted $\varepsilon_n(T)$ and $e_n(T)$, which as functions of n are roughly the inverses of the covering numbers and entropy numbers as in the present manuscript. For instance, $e_n(T)$ is defined as the smallest ε such that the image $T(U)$ can be covered by 2^{n-1} balls of radius ε .

2.8. Theorem 2.8.1 is due to Dudley, Giné and Zinn (1991), who give a complete treatment of the conditions under which a class of functions is universal or uniform Glivenko-Cantelli.

The development of uniform in P Glivenko-Cantelli and Donsker theorems began with the work of Dvoretzky, Kiefer and Wolfowitz (1956), Kiefer and Wolfowitz (1959), and Kiefer (1961). The exponential bounds developed in these papers yield uniform in P -Glivenko-Cantelli theorems for the classical empirical distribution function of random variables in $\mathbb{R}^d$. Moreover, these authors essentially establish the uniform Donsker property for the classical distribution function in the course of proving the asymptotic minimaxity of the empirical distribution function as an estimator of the true distribution function. Uniformity in P is implicit in the work of Vapnik and Červonenkis (1971) on the Glivenko-Cantelli theorems. The introduction of uniform entropy conditions by Pollard (1982) and Kolčinskij (1981) continued this development. Massart (1986), Theorem 5.10, page 4.11, establishes rates of convergence for weak-approximation versions of Theorem 2.8.3 under growth rate hypotheses on the uniform entropy and additional hypotheses on the envelope function of the class. (It is plausible that his rates are in fact uniform over the collection $\mathcal{P}$ for which the moment hypotheses on $\mathcal{F}$ hold uniformly.) Dudley (1987) unifies and extends results for the “universal Donsker property,” which entail that $\mathcal{F}$ is Donsker for all probability measures P on the sample space. A universal Donsker class $\mathcal{F}$ is essentially bounded: $\sup_f \text{diam } f < \infty$ (where $\text{diam } f = \sup_{x,y} (f(x) - f(y))$). For classes with this property, Giné and Zinn (1991) use Gaussian comparison methods to characterize classes that are uniformly Donsker in all possible underlying measures.

Theorem 2.8.2 is contained in Sheehy and Wellner (1992) along with other equivalences; the present proof is new.

2.9. The multiplier central limit theorem, Theorem 2.9.2, is implicit in Giné and Zinn (1984) and is stated explicitly in Giné and Zinn (1986a), who attribute the multiplier inequality to Pisier and Fernique. Alexander (1985), solving a problem posed by Hoffmann-Jørgensen, shows that no “universal multiplier moment” exists: there is no function $\psi: \mathbb{R}^2 \rightarrow \mathbb{R}$ so that ξZ satisfies the central limit theorem whenever $E\xi Z = 0$, $E\psi(|\xi|, \|Z\|) < \infty$ and Z satisfies the central limit theorem, for independent real- and Banach-space-valued random elements ξ and Z . On the other hand, Ledoux and Talagrand (1986) show that the $L_{2,1}$ -hypothesis on the multipliers ξ_i cannot be relaxed in the sense that, for every ξ with $\|\xi\|_{2,1} = \infty$, there exists a Banach space valued Z that satisfies the central limit theorem, but ξZ does not satisfy the central limit theorem. Ledoux and Talagrand (1986) and Ledoux and Talagrand (1991), Propo-

sition 10.4 on page 279, give a different proof of the basic multiplier inequality.

The almost sure conditional multiplier central limit theorem is due to Ledoux and Talagrand (1988), with contributions by J. Zinn. The present proof, based on isoperimetric methods, is adapted from Ledoux and Talagrand (1991), Theorem 10.14, on page 293. The proof using martingale difference methods originating in Yurinskii (1974), given in the “Problems and Complements” section, is taken from Ledoux and Talagrand (1988).

Lemma 2.9.11 is due to Han (2018) and first appeared in Han and Wellner (2019).

2.10. Theorem 2.10.5 is due to Van der Vaart and Wellner (2000).

Alexander (1987c) gives Theorem 2.10.1 and derives that $\mathcal{F} + \mathcal{G}$ and $\mathcal{F} \cup \mathcal{G}$ are Donsker if $\mathcal{F}$ and $\mathcal{G}$ are Donsker from his more general Proposition 2.6. Theorem 2.10.2 and Theorem 2.10.3 are due to Dudley (1985).

Versions of Theorem 2.10.8 are established by Giné and Zinn (1986a) (one-dimensional ϕ under measurability conditions) and Talagrand (1987a) (uniformly bounded classes). Lemma 2.10.16 is established by Giné and Zinn (1986a); the proof given here is new. Section 2.10.5 is based on Van der Vaart (1996) and Van der Vaart and Wellner (2000). The result of Example 2.10.28 specialized to the class $C^1(\mathbb{R})_1$ was first obtained by Giné and Zinn (1986b) by a different method. Pollard (1990), Section 5, gives a variety of results similar to those in Section 2.10.4.

2.11. The limit theory in this section has its antecedents in Koul (1970), Shorack (1973, 1979), and Van Zuijlen (1978). Also see Shorack and Beirlant (1986), Marcus and Zinn (1984), and Shorack and Wellner (1986), Section 3.3, pages 108–109.

The first subsection, based on random and uniform entropy, follows Alexander (1987b), but with a simplified proof for the main Theorem 2.11.1. See Alexander’s paper for a discussion of the minimal moment conditions on the envelope within the context of this theorem. The improvement over Theorem 2.5.2 in the i.i.d. case was already obtained by Giné and Zinn (1984). Pollard (1990) develops a combinatorial theory for the non-i.i.d. case paralleling the VC-theory of Chapter 2.6.

The bracketing central limit theorem, Theorem 2.11.11, and its corollary are due to Andersen, Giné, Ossiander and Zinn (1988), building on Ossiander (1987) and work on majorizing measures and Gaussian processes due to Fernique (1974) and Talagrand (1987c). Theorem 2.11.9 does not seem to be a corollary of their result, but appears to be the natural generalization of Ossiander’s result to the case of non-i.i.d. observations. The proof of Theorem 2.11.11 shows that the theorem remains true if the single majorizing measure, which is guaranteed to exist by the assumption of Gaussian domination, is replaced by certain sequences of majorizing measures (depending on n). The proof given here is simplified in comparison to the original proof. Lemma 2.11.21 is adapted from Talagrand (1987c) and Ledoux and Talagrand (1991), who formulate similar results in terms of ultrametrics.

Jain and Marcus proved their theorem, as given here, in Jain and Marcus (1975). Example 2.11.14 is taken from Giné and Zinn (1986a). Dudley (1985),

Section 6, gives the first satisfactory integration of the famous “Chibisov-O’Reilly theorem” treated in Example 2.11.15 with modern empirical process theory.

2.12. Partial-sum processes have a long and honorable history in probability theory and statistics, with rapid development of the classical theory occurring in the late 1940s and early 1950s. The classical work of Erdős and Kac (1946), Doob (1949), and Donsker (1951, 1952) was beautifully summarized by Billingsley (1968).

Study of the partial-sum or sequential empirical process seems to have begun with the work of Müller (1968) on the distributional invariance principle for the Glivenko-Cantelli theorem, Pyke (1968) on random-sample-size limit theorems for empirical processes, and Kiefer (1969) on embedding questions. The Hungarian school’s work on embedding theorems in the 1970s was followed by embedding theorems (explicit construction) for general empirical processes in Dudley and Philipp (1983). The present form seems to have been first stated by Sheehy and Wellner (1992), along with other equivalences of Dudley and Philipp (1983) and Dudley (1984). The present proof is analogous to the proof by Billingsley (1968) of a classical partial-sum convergence theorem: Theorem 10.1, pages 68–70.

The subsection on partial-sum processes on lattices follows Alexander and Pyke (1986), Bass and Pyke (1985), and Pyke (1983).

2.13. Theorem 2.13.1 is taken from Dudley (1984), who also shows that, in a certain sense, the condition that the sequence converges is sharp. Giné and Zinn (1986a) give further results for sequences of functions bounded in uniform norm.

Dudley (1987) notes that the square of the supremum of the empirical process over an elliptical class $\mathcal{F}$ equals a weighted sum of the squares of the empirical process acting on the basis functions defining the elliptical class, and he points out the practical importance of such classes.

The equivalence of (iii) and (iv) in Theorem 2.13.6 is due to Giné and Zinn (1984); the equivalence of (i), (ii), and (iii) was proved by Talagrand (1988).

2.14. Bounds for expectations of suprema of Gaussian processes were developed by Dudley (1967b). In the case of empirical processes, Pollard (1989b) was apparently the first to recognize the usefulness of moment bounds for suprema of the form given in Theorems 2.14.1 and 2.14.16; also see Pollard (1990) and Kim and Pollard (1990). Ledoux and Talagrand (1991) systematically developed the use of Orlicz norms; see their Notation and Chapters 6 and 11. Theorem 2.14.23 is a straightforward consequence of the general domination of Orlicz norms of sums of independent random elements by L_1 -norms plus the Orlicz norm of maximal summand proved by Talagrand (1989); see Ledoux and Talagrand (1991), Chapter 6.

Maximal inequalities that include the size of the functions in the bound are especially useful when applied to the modulus of continuity to obtain a rate of convergence of an M-estimator, as in Chapter 3.4. Theorems 2.14.17 and 2.14.18 were obtained in Chapter 3.4 of the first edition of this book. Theorems 2.14.2, 2.14.8, and 2.14.9 were obtained more recently in Van der Vaart and Wellner (2011). Theorem 2.14.21 was derived in Dirksen (2015) and Nickl and Wang (202?).

Dvoretzky, Kiefer and Wolfowitz (1956) proved the first exponential bound for the supremum distance between the empirical distribution function and the true

distribution function in the case of $\mathcal{X} = \mathbb{R}$. Massart (1990) found the best constant D multiplying the exponential for this classical case. Related bounds for the more difficult case $\mathcal{X} = \mathbb{R}^k$ were established by Kiefer (1961). Vapnik and Červonenkis (1971) proved exponential bounds of the same form as those given in Theorem 2.14.32, but with no powers of t in front and the constant D depending on (and increasing with) n . Alexander (1984) gives bounds of the same type as those in Theorems 2.14.28, 2.14.32, and 2.14.36 and also has sharper bounds than the bound given in 2.14.36.

Theorems 2.14.28, 2.14.32, and 2.14.33 are due to Talagrand (1994), while Theorems 2.14.29, 2.14.36, and 2.14.37 are from Massart (1986). For still more bounds, see Smith and Dudley (1992) and Adler and Brown (1986).

Theorem 2.14.38 is an improved version (explicit constants and a logarithm instead of a root-logarithm in the exponent) of Theorem 2.14.24 in the first edition of this book, which was there proved following Talagrand (1994). Here the proof is based on the Bousquet-Talagrand concentration bound, Theorem 2.15.5, which necessitated stronger measurability assumptions on the class of functions. The small variance bound is also due to Talagrand (1994).

2.15. The bounded difference inequality is due to McDiarmid (1989) following the method set out by Maurey (1979). Theorem 2.15.5 of Bennett-Bernstein type has its roots in Talagrand (1994), with further developments in Talagrand (1996), Ledoux (1996), Massart (2000b), Rio (2001) and Rio (2002). While the first edition of the present book presented the early results, the present chapter is based on the definitive formulation, with optimal constants, proved in Bousquet (2002, 2003). The bounds on the lower tail were obtained in Klein (2002) and Klein and Rio (2005). For a treatment of Talagrand's concentration inequality for the supremum of an empirical process and more on log-Sobolev inequalities, also see Section 3.3 of Giné and Nickl (2016). For log-Sobolev inequalities also see Ledoux (2001) and Boucheron, Lugosi and Massart (2013).



PART 3

Statistical Applications

3.1

Introduction

The empirical process methods and techniques developed in Part 2 have many applications in statistics. The present part illustrates this in some detail by applications ranging from M -estimation (limit theory and rates of convergence in infinite-dimensional applications), bootstrapping, permutation tests, tests of independence, the functional delta-method, and contiguity theory.

In agreement with its importance in statistics, a treatment of M -estimators opens this part. In Chapter 3.2 we consider estimators $\hat{\theta}_n$ defined as the value of θ maximizing (or nearly maximizing) a random criterion function $\theta \mapsto \mathbb{M}_n(\theta)$. A number of results are formulated for general, abstract criterion functions, but we are particularly interested in “empirical criterion functions” of the form $\mathbb{M}_n(\theta) = \mathbb{P}_n m_\theta$.

The limiting distribution of such M -estimators may be derived either from a continuous mapping theorem for the argmax functional or from a linearization argument. In both approaches it appears useful to split the derivation into three steps:

- establish consistency;
- establish a rate of convergence;
- derive the limiting distribution.

The most successful methods in the first step are the (generalized) method of Wald (which we do not discuss) and a method based on uniform convergence of the criterion functions. The latter method, when applied with an empirical criterion function, boils down to proving that certain classes of functions are Glivenko-Cantelli and is discussed briefly.

Rates of convergence are discussed in some detail and generality in Chapter 3.4, but in Chapter 3.2 we present a preliminary approach, which appears to be useful for the study of M -estimators of Euclidean parameters. When applied to

empirical criterion functions of the form $\theta \mapsto \mathbb{P}_n m_\theta$ this approach is based on controlling the entropy of the classes of functions $\{m_\theta - m_{\theta_0} : \|\theta - \theta_0\| < \delta\}$ (for small δ) and expresses the rate of convergence in terms of the L_2 -norms of the envelope functions of these classes.

The argmax continuous mapping theorem asserts that given a sequence of criterion functions $\mathbb{M}_n$ converging (in an appropriate distributional sense) to a limit $\mathbb{M}$, the point of maximum, or argmax, of $\mathbb{M}_n$, which is the M -estimator, converges in distribution to the argmax of the limit process $\mathbb{M}$. In obtaining a limit distribution of a sequence of M -estimators, this theorem is usually not applied with the original criterion functions $\theta \mapsto \mathbb{M}_n(\theta)$, but to (a multiple of) “localized” criterion functions of the form

$$h \mapsto \tilde{\mathbb{M}}_n(h) = \mathbb{M}_n\left(\theta_0 + \frac{h}{r_n}\right) - \mathbb{M}_n(\theta_0).$$

Here r_n is the rate of convergence and θ_0 the true parameter. These localized criterion functions usually converge functionally in distribution to a limit only if the local parameter $h = r_n(\theta - \theta_0)$ is restricted to compacta. This is the reason that it is necessary to obtain the rate of convergence first. Once it is known that $\hat{h}_n = r_n(\hat{\theta}_n - \theta_0)$ is uniformly tight, convergence in distribution of the processes $\{\tilde{\mathbb{M}}_n(h) : h \in K\}$ in the space $\ell^\infty(K)$ for each compact K suffices for successful application of the argmax theorem. Actually, this type of convergence is too strong, but it fits well within the framework of empirical processes developed in Part 2.

Using this approach, we obtain simple, but general, conditions for the asymptotic normality of maximum likelihood estimators in smooth parametric models, but we also treat a number of nonstandard problems.

An alternative approach to obtaining the limit distribution of M -estimators is to derive a characterization as the solution of a family of estimating equations of the form $\Psi_n(\hat{\theta}_n) = 0$. In Chapter 3.3 we discuss the asymptotic behavior of solutions of estimating equations, which we call Z -estimators, in general. The framework in this chapter allows an infinite-dimensional parameter space Θ and a corresponding infinite-dimensional system of estimating equations. This covers not only the classical estimating equation framework with a finite-dimensional parameter space, but also a wide range of problems in nonparametric and semiparametric inference involving estimation of parameters with values in some (Banach) space of functions.

Establishing a rate of convergence of M -estimators based on empirical criterion functions is quite amenable to empirical process methods. Chapter 3.4 develops these methods first in an abstract setting, which includes penalized criterion functions, and next specializes these to empirical criterion functions. Applications include (penalized) maximum likelihood estimation, least-squares estimation, and least-absolute-deviation estimation. In this chapter we are mostly interested in infinite-dimensional (nonparametric or semiparametric) estimation. The basic result is that a modulus of continuity of the empirical process determines the rate of convergence. Using the results of Part 2, this modulus can be bounded in terms of entropies, and the rate of convergence can be expressed in terms of entropy integrals of the parameter set. For instance, the rate of convergence r_n of a maximum

likelihood estimator can be found as the solution of the equation

$$r_n^2 \tilde{J}_{[]} \left(\frac{1}{r_n}, \mathcal{P}_n, h \right) \leq \sqrt{n}.$$

Here $\mathcal{P}_n$ is the class of densities over which the likelihood is maximized and the bracketing entropies are calculated with respect to the Hellinger distance h . If a penalty is added to a criterion, then this yields to additional considerations. The results have sufficient generality to cover a wide range of methods, including: sieved maximum contrast estimators, penalized maximum likelihood estimators, least squares regression estimators with fixed and random designs, and penalized least squares regression, including the lasso, statistical learning, and support vector machines.

Chapter 3.5 treats methods for model selection. In many current statistical applications, a model for the data is not known exactly, but instead the true state of nature is known (or can be assumed) to be in some member of a family of models. Then we can view the inference problem in terms of first choosing a “best” member of the family of models, and then making inference within that best member of the family. For example, if many predictors $\{x_j\}_{j=1}^k$ are available in a regression model with outcome or response variable Y , then model selection might be viewed as choosing the “best” subset $\{x_{\pi(j)}\}_{j=1}^{k_0}$ of the predictors to use in a regression model for the response. Making such a choice of a “best” model typical involves a trade-off between some measure of fidelity (such as least squares or log-likelihood) and some measure of model complexity (such as the number of predictors to be included in the model). The measure of model complexity is typically taken to be a non-negative function J defined on the set $\mathcal{K}$ indexing the models under consideration. Computational considerations often lead to consideration of complexity measures J that have appropriate convexity properties. In general we want to show that the resulting estimator achieves a convergence rate that corresponds to the estimator that we would have used if we know the “best” model a priori. This amounts to showing that the estimator is “adaptive”.

In applications it frequently occurs that the sample size is not fixed but is, in fact, random—perhaps even depending on the data $X_1, \dots, X_n$ in some complicated way. The effect of random sample size on the empirical processes $\mathbb{G}_n$ is investigated in Chapter 3.6. Theorem 3.6.1 asserts that if $\mathcal{F}$ is a Donsker class of functions and N_n is a sequence of random samples sizes satisfying $N_n/c_n \xrightarrow{P} \nu$ with $c_n \rightarrow \infty$, then $\mathbb{G}_{N_n} \rightsquigarrow \mathbb{G}$ in $\ell^\infty(\mathcal{F})$ provided only that $P(\nu > 0) = 1$. In this chapter we give special attention to the case that the random sample size N_n is Poisson-distributed with mean n independent of the data. In this case the *Kac empirical point process* $\mathbb{N}_n = \sum_{i=1}^{N_n} \delta_{X_i}$ is a Poisson point process with intensity measure nP , and, for $\mathcal{F}$ with $\|P\|_{\mathcal{F}} < \infty$, the *Kac empirical process* $\mathbb{Z}_n = n^{-1/2}(\mathbb{N}_n - nP)$ converges in distribution in $\ell^\infty(\mathcal{F})$ to a Brownian motion process if and only if $\mathcal{F}$ is P -Donsker.

Nonparametric bootstrap methods have become popular in statistics since their introduction by Efron (1979). This method is based on sampling from the empirical measure $\mathbb{P}_n$. Given the original sample $X_1, \dots, X_n$, let $\hat{X}_1, \dots, \hat{X}_n$ be an i.i.d. sample from $\mathbb{P}_n$. The *bootstrap empirical measure* and the *bootstrap empirical process*

are given by

$$\hat{\mathbb{P}}_n = \frac{1}{n} \sum_{i=1}^n \delta_{\hat{X}_i} \quad \text{and} \quad \hat{\mathbb{G}}_n = \sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n),$$

respectively. These are important tools for the study of a wide variety of statistical methods. Letting M_{ni} be the number of times that X_i is “redrawn” from the original sample, the bootstrap empirical measure and process can also be written as

$$(3.1.1) \quad \hat{\mathbb{P}}_n = \frac{1}{n} \sum_{i=1}^n M_{ni} \delta_{X_i} \quad \text{and} \quad \hat{\mathbb{G}}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n (M_{ni} - 1) \delta_{X_i},$$

respectively. The vector $M_n = (M_{n1}, \dots, M_{nn})$ has a multinomial distribution with n cells, n trials, and success probabilities $1/n$ for each of the n cells. Since the variables $M_{n1}, M_{n2}, \dots$ converge in distribution to a sequence of i.i.d. Poisson variables $Y_1, Y_2, \dots$ with mean 1, it is apparent that the limit theory for the bootstrap empirical process $\hat{\mathbb{G}}_n$ is closely linked to the conditional multiplier central limit theorems developed in Chapter 2.9. This is indeed the case, and our proofs of two bootstrap limit theorems due to Giné and Zinn (1990) rely on this connection. In Theorem 3.7.1 it is shown that the bootstrap empirical process satisfies $\hat{\mathbb{G}}_n \rightsquigarrow \mathbb{G}$ “in outer probability” if and only if $\mathcal{F}$ is P -Donsker, while in Theorem 3.7.2 it is shown that $\hat{\mathbb{G}}_n \rightsquigarrow \mathbb{G}$ “outer almost surely” if and only if $\mathcal{F}$ is P -Donsker and $P^* \|f - Pf\|_{\mathcal{F}}^2 < \infty$. The statistical meaning of these results is that the “distribution” of $\hat{\mathbb{G}}_n$ (which depends on the original data only) is a consistent estimator for the distribution of $\mathbb{G}_n$ (which depends on the underlying distribution P), in the sense that the difference converges to zero as $n \rightarrow \infty$. The two types of theorems yield consistency in probability and consistency in an almost sure sense, respectively. Our proofs rely heavily on the Poissonization of the bootstrap sample size. Other bootstrap methods can be based on replacing the multinomial weights in (3.1.1) by other *exchangeable* weights. For example, resampling k from the original sample of n without replacement can be treated by an appropriate choice of M_n . Conditional limit theorems for these alternative bootstrap methods are also presented.

Chapter 3.8 introduces general versions of the two-sample Kolmogorov-Smirnov statistics and shows how the corresponding two-sample tests can be implemented using either bootstrap or permutation methods. Many statistics used for testing independence can be viewed as functionals of the *independence empirical process*, which we define and study in Chapter 3.9.

One of the most important basic tools of large sample theory in statistics is the *delta-method*: if $r_n(X_n - \theta) \rightsquigarrow Z$ for some constant θ and $r_n \rightarrow \infty$, and ϕ is differentiable at θ , then $r_n(\phi(X_n) - \phi(\theta)) \rightsquigarrow \phi'(\theta)Z$. For Euclidean vectors X_n , this result is standard and $\phi'(\theta)$ is the usual derivative from calculus. Extensions to functions ϕ of infinite-dimensional statistics X_n , for instance the empirical measure $\mathbb{P}_n$, have become increasingly useful and important. The formulation of the *functional delta-method* requires careful definition of the type of derivative. A map $\phi: \mathbb{D} \rightarrow \mathbb{E}$ is called *Fréchet-differentiable* at θ if there is a continuous linear map $\phi'_\theta: \mathbb{D} \rightarrow \mathbb{E}$ such that

$$\|\phi(\theta + h) - \phi(\theta) - \phi'_\theta(h)\| = o(\|h\|), \quad \|h\| \rightarrow 0.$$

Such a map ϕ is called *Hadamard-differentiable* at θ if there is a continuous linear map $\phi'_\theta: \mathbb{D} \rightarrow \mathbb{E}$ such that

$$\frac{\phi(\theta + t_n h_n) - \phi(\theta)}{t_n} \rightarrow \phi'_\theta(h),$$

for every sequence of real numbers $t_n \rightarrow 0$ and every sequence $h_n \rightarrow h$. Hadamard differentiability is a weaker requirement than Fréchet differentiability and, fortunately, is sufficient for the functional delta-method. This asserts that if $r_n(X_n - \theta) \rightsquigarrow Z$ for a separable variable Z and ϕ is Hadamard-differentiable at θ , then

$$r_n(\phi(X_n) - \phi(\theta)) \rightsquigarrow \phi'_\theta(Z).$$

Examples treated in some detail in Chapter 3.10 include the Nelson-Aalen estimator from right-censored data, quantile and copula functions, the product integral, multivariate trimming, and M -functionals.

When specialized to a Hadamard-differentiable map ϕ on $\mathbb{D} = \ell^\infty(\mathcal{F})$ for a Donsker class $\mathcal{F}$, the functional delta-method gives

$$\sqrt{n}(\phi(\mathbb{P}_n) - \phi(P)) \rightsquigarrow \phi'_P(\mathbb{G}).$$

In this case it makes sense to ask if this continues to hold for the empirical bootstrap: is it true that

$$\sqrt{n}(\phi(\hat{\mathbb{P}}_n) - \phi(\mathbb{P}_n)) \rightsquigarrow \phi'_P(\mathbb{G})$$

in outer probability (or outer almost surely)? Several results asserting that this is true are proved in Section 3.10.3. Thus, the consistency of the bootstrap of the abstract empirical process implies the consistency of the bootstrap for many statistical functionals. This is probably the most important motivation for studying the abstract bootstrap process.

The theory of *contiguous* families of probability measures provides a notion of asymptotic absolute continuity that has proved to be of fundamental importance for asymptotic theory in statistics. Chapter 3.11 presents the basic elements of contiguity theory and also applies it to the empirical process. One result of this type concerns the empirical measure $\mathbb{P}_n$ of independent and identically distributed variables $X_{n1}, \dots, X_{nn}$ with common distribution P_n satisfying

$$\int \left[\sqrt{n}(dP_n^{1/2} - dP^{1/2}) - \frac{1}{2}h dP^{1/2} \right]^2 \rightarrow 0,$$

for some measurable function $h: \mathcal{X} \rightarrow \mathbb{R}$. If $\mathcal{F}$ is P -Donsker, then under P_n

$$\sqrt{n}(\mathbb{P}_n - P) \rightsquigarrow \mathbb{G} + s_h \quad \text{in} \quad \ell^\infty(\mathcal{F})$$

where the shift $s_h: \mathcal{F} \rightarrow \mathbb{R}$ is given by $s_h f = Pfh$.

Efficiency theory in statistics relies on lower bounds. Chapter 3.12 treats infinite-dimensional versions of the Hájek convolution and local asymptotic min-max theorems. These results apply to many statistical models, but we discuss as one example the asymptotic efficiency of the empirical distribution in the nonparametric situation.

Part 3 ends in Chapter 3.13 on “random” empirical processes. The methods developed in this chapter allow treatment of situations where $\mathbb{G}_n$ and $\mathbb{P}_n$ are indexed by a “random” class of functions $\{f_{\theta, \hat{\eta}_n} : \theta \in \Theta\}$ where $\hat{\eta}_n$ is an estimator of a (nuisance) parameter $\eta \in H$. Theory is developed for several sets of assumptions about the behavior of the estimators $\hat{\eta}_n$ and the classes $\mathcal{F} = \{f_{\theta, \eta} : \theta \in \Theta, \eta \in H\}$.

Problems and Complements

1. Suppose that for each natural number n the vector $M_n = (M_{n1}, \dots, M_{nm})$ has a multinomial distribution with n cells, n trials, and success probabilities $1/n$ for each of the n cells. Then the variables $M_{n1}, M_{n2}, \dots$ converge in distribution to a sequence of i.i.d. Poisson variables $Y_1, Y_2, \dots$ with mean 1 in $\mathbb{R}^\infty$. Furthermore, if $k(n) = o(n)$, $P_{n,k(n)}$ denotes the joint distribution of $(M_{n1}, \dots, M_{nk(n)})$ and $Q_{n,k(n)}$ denotes the joint distribution of $(Y_1, \dots, Y_{k(n)})$, then the total variation distance from $P_{n,k(n)}$ to $Q_{n,k(n)}$ converges to 0, i.e., $\sup_A |P_{n,k(n)}(A) - Q_{n,k(n)}(A)| \rightarrow 0$, for the supremum taken over the Borel sets in $\mathbb{R}^{k(n)}$.

[Hint: This is closely related to Runnenburg and Vervaat (1969) and to Wellner (1977).]

3.2

M-Estimators

The most important method of constructing statistical estimators is to choose the estimator to maximize a certain criterion function. We shall call such estimators *M-estimators* (from “maximum” or “minimum”). In the case of i.i.d. observations $X_1, \dots, X_n$, a common type of criterion function is of the form

$$\theta \mapsto \mathbb{P}_n m_\theta = \frac{1}{n} \sum_{i=1}^n m_\theta(X_i),$$

for known given functions $m_\theta: \mathcal{X} \rightarrow \mathbb{R}$ on the sample space. In particular, the method of maximum likelihood estimation corresponds to the choice $m_\theta = \log p_\theta$, where p_θ is the density of the observations.[†] The theory of empirical processes comes in naturally when studying the asymptotic properties of these estimators. In this chapter we present several results that give the asymptotic distribution of *M*-estimators. Some results are of a general nature, while others presume the set-up of i.i.d. observations.

In many situations estimators that maximize a certain map also solve a system of equations. In particular, in the case of i.i.d. observations many estimators are a zero of a map of the type

$$\theta \mapsto \mathbb{P}_n \psi_\theta,$$

for given maps ψ_θ on the sample space. We shall refer to solutions of estimating equations as *Z-estimators* (from “zero”). We note that in some of the literature the name *M*-estimator is (also) used for what we call *Z*-estimator and the distinction between the different types of estimators is not always made.

[†] For some purposes, such as for proving consistency or rates of convergence, particularly in the case of models that are convex in the parameter, it may be technically convenient to choose a different function.

The asymptotic behavior of Z -estimators is studied in Chapter 3.3. The approach given there provides an alternative to the approach in the present chapter, although generally the asymptotic properties of M -estimators are most economically studied from their characterization as a point of maximum. In this chapter we discuss two approaches that use this characterization directly.

The natural “parameter” (or functional) is denoted θ and is assumed to run through a subset Θ of a metric space; for instance, Euclidean space. It is often fruitful to separate the derivation of the limit behavior of a sequence of estimators $\hat{\theta}_n$ into subproblems, such as proving consistency, deriving the convergence rate, and finally establishing the limit distribution. In this scheme certain theorems are applied to “local” (pseudo-) parameters (such as $h = \sqrt{n}(\theta - \theta_0)$ if θ_0 is the “true” parameter). For convenience, theorems that are typically applied to local parameters are stated in terms of a parameter h .

3.2.1 Argmax Theorem

Consider finding a point of maximum as a functional called “argmax”. When applied to a random criterion function this yields an M -estimator. If the argmax functional were continuous with respect to some metric on the space of criterion functions, then convergence in distribution of the criterion functions would imply the convergence in distribution of their points of maximum, the M -estimators, to the point of maximum of the limit criterion function. This is simply a special case of the continuous mapping theorem for weak convergence.

In keeping with the main development in this book—the convergence of stochastic processes in spaces of the type $\ell^\infty(T)$, such as the empirical process—we shall develop this idea using the uniform topology on the space of criterion functions. Actually, the proofs ahead show that the (joint) convergence of suprema over certain sets is crucial for the convergence of a point of maximum, not the convergence of the processes with respect to the uniform metric. This shows that uniform convergence could be relaxed to weaker forms of convergence, such as *epi-convergence*. For simplicity, we shall not pursue such refinements, but do capture the main idea in a lemma.

Consider a sequence $\{\mathbb{M}_n(h): h \in H\}$ of stochastic processes indexed by a metric space H . For each n let the “estimator” $\hat{h}_n$ be a point of (near) maximum of the “criterion function”

$$h \mapsto \mathbb{M}_n(h).$$

Formally, $\hat{h}_n$ may be any H -valued map defined on the same probability space as the process $\mathbb{M}_n$; its (outer) distribution is evaluated accordingly. Let $\{\mathbb{M}(h): h \in H\}$ be a “limit process”, and assume that it possesses the same relation to a random point $\hat{h}$, except that $\hat{h}$ is implicitly assumed to be Borel measurable.

The argmax functional is continuous at functions $\mathbb{M}$ that have a unique, “well-separated” point of maximum: the function $h \mapsto \mathbb{M}(h)$ should be strictly smaller than $\mathbb{M}(\hat{h})$ on the complement of every neighborhood of the point $\hat{h}$. This requirement, which appears to be natural, is the first condition in the following lemma.

This lemma contains the crux of the (simple) argument; to obtain the best results, it may be necessary to tailor the underlying idea to particular examples.

3.2.1 Lemma. Let $\mathbb{M}_n, \mathbb{M}$ be stochastic processes indexed by a metric space H . Let A and B be arbitrary subsets of H . Suppose that there exists a random element $\hat{h}$ such that almost surely

$$\mathbb{M}(\hat{h}) > \sup_{h \notin G, h \in A} \mathbb{M}(h),$$

for every open set G that contains $\hat{h}$. Suppose the sequence $\hat{h}_n$ satisfies

$$\mathbb{M}_n(\hat{h}_n) \geq \sup_h \mathbb{M}_n(h) - o_P(1).$$

If $\mathbb{M}_n \rightsquigarrow \mathbb{M}$ in $\ell^\infty(A \cup B)$, then, for every closed set F ,

$$\limsup_{n \rightarrow \infty} P^*(\hat{h}_n \in F \cap A) \leq P(\hat{h} \in F \cup B^c).$$

If $\mathbb{M}_n \rightsquigarrow \mathbb{M}$ in $\ell^\infty(H)$, then the lemma could be applied with $A = B = H$. In view of the portmanteau theorem, it would yield the conclusion that $\hat{h}_n \rightsquigarrow \hat{h}$.

Unfortunately, the assumption that $\mathbb{M}_n \rightsquigarrow \mathbb{M}$ uniformly in the whole (local) parameter space is often too strong. This does not mean that the weak convergence $\hat{h}_n \rightsquigarrow \hat{h}$ is not true, since the uniform convergence of the criterion functions is much stronger than the convergence of the locations of maxima. Therefore, it may be beneficial to establish additional properties of the sequence $\hat{h}_n$ before invoking the continuous mapping theorem for the argmax functional. The following theorem requires uniform tightness, in which case uniform convergence of the criterion functions on compacta suffices.

On compacta, a unique maximum of an *upper semicontinuous* function $h \mapsto \mathbb{M}(h)$ is automatically well separated, as required by the preceding lemma. We obtain the following theorem.

3.2.2 Theorem (Argmax continuous mapping). Let $\mathbb{M}_n, \mathbb{M}$ be stochastic processes indexed by a metric space H such that $\mathbb{M}_n \rightsquigarrow \mathbb{M}$ in $\ell^\infty(K)$ for every compact $K \subset H$. Suppose that almost all sample paths $h \mapsto \mathbb{M}(h)$ are upper semicontinuous and possess a unique maximum at a (random) point $\hat{h}$, which as a random map in H is tight. If the sequence $\hat{h}_n$ is uniformly tight and satisfies $\mathbb{M}_n(\hat{h}_n) \geq \sup_h \mathbb{M}_n(h) - o_P(1)$, then $\hat{h}_n \rightsquigarrow \hat{h}$ in H .

Proofs. For the proof of the lemma, note that by the continuous mapping theorem, the sequence $\sup_{h \in F \cap A} \mathbb{M}_n(h) - \sup_{h \in B} \mathbb{M}_n(h)$ converges in distribution to the same expression, but with $\mathbb{M}$ instead of $\mathbb{M}_n$. Thus

$$\begin{aligned} \limsup_{n \rightarrow \infty} P^*(\hat{h}_n \in F \cap A) &\leq \limsup_{n \rightarrow \infty} P^*\left(\sup_{h \in F \cap A} \mathbb{M}_n(h) \geq \sup_{h \in B} \mathbb{M}_n(h) - o_P(1)\right) \\ &\leq P\left(\sup_{h \in F \cap A} \mathbb{M}(h) \geq \sup_{h \in B} \mathbb{M}(h)\right), \end{aligned}$$

by Slutsky's lemma and the portmanteau theorem. By the condition on the sample paths of $\mathbb{M}$, the event in the last probability is contained in the set $\{\hat{h} \in F\} \cup \{\hat{h} \notin B\}$, since the set $G = F^c$ is open. The lemma follows.

For the proof of the theorem, take $A = B = K$ equal to a compact set K . Then almost surely

$$\mathbb{M}(\hat{h}) > \sup_{h \notin G, h \in K} \mathbb{M}(h),$$

for every open set G around $\hat{h}$. If this were not true, then there would exist a sequence $h_m \subset G^c \cap K$ with $\mathbb{M}(h_m) \rightarrow \mathbb{M}(\hat{h})$. Since K is compact the sequence may be chosen to be convergent; by upper semicontinuity the value $\mathbb{M}(h)$ at the limit would be at least $\mathbb{M}(\hat{h})$. This contradicts the fact that $\hat{h}$ is unique, for h is contained in the closed set G^c and hence cannot equal $\hat{h}$.

Thus, the lemma may be applied and yields

$$\limsup_{n \rightarrow \infty} \mathbb{P}^*(\hat{h}_n \in F) \leq \mathbb{P}(\hat{h} \in F) + \mathbb{P}(\hat{h} \notin K) + \limsup_{n \rightarrow \infty} \mathbb{P}^*(\hat{h}_n \notin K),$$

for every closed set F . The last two terms on the right can be made arbitrarily small by the choice of K . Apply the portmanteau theorem to conclude that $\hat{h}_n \rightsquigarrow \hat{h}$. ■

The preceding results are stated in terms of the parameter h , because they are typically applied to a local parameter. However, they may also be applied to the original parameter, in which case the limit criterion function is typically nonrandom and the approach turns into a consistency proof.

3.2.3 Corollary (Consistency). *Let $\mathbb{M}_n$ be stochastic processes indexed by a metric space Θ , and let $\mathbb{M}: \Theta \rightarrow \mathbb{R}$ be a deterministic function.*

- (i) *Suppose that $\|\mathbb{M}_n - \mathbb{M}\|_{\Theta} \rightarrow 0$ in outer probability and that there exists a point θ_0 such that*

$$\mathbb{M}(\theta_0) > \sup_{\theta \notin G} \mathbb{M}(\theta),$$

for every open set G that contains θ_0 . Then any sequence $\hat{\theta}_n$, such that $\mathbb{M}_n(\hat{\theta}_n) \geq \sup_{\theta} \mathbb{M}_n(\theta) - o_P(1)$, satisfies $\hat{\theta}_n \rightarrow \theta_0$ in outer probability.

- (ii) *Suppose that $\|\mathbb{M}_n - \mathbb{M}\|_K \rightarrow 0$ in outer probability for every compact $K \subset \Theta$ and that the map $\theta \mapsto \mathbb{M}(\theta)$ is upper semicontinuous with a unique maximum at θ_0 . Then the same conclusion is true provided the sequence $\hat{\theta}_n$ is uniformly tight.*

If the estimator $\hat{\theta}_n$ maximizes the criterion function $\theta \mapsto \mathbb{M}_n(\theta)$, then the preceding theorem will typically be applied to a multiple of a “rescaled” criterion function

$$h \mapsto \mathbb{M}_n\left(\theta_0 + \frac{h}{r_n}\right) - \mathbb{M}_n(\theta_0),$$

where θ_0 is the “true” parameter and $r_n \rightarrow \infty$ is the “rate of convergence” of the estimator. We then obtain a distributional limit result for the sequence $\hat{h}_n = r_n(\hat{\theta}_n - \theta_0)$. On the other hand, the corollary will be applied to (a multiple of) the original criterion functions $\mathbb{M}_n$.

A good approach to obtain the limiting distribution of M -estimators of Euclidean parameters appears to consist of three steps:

- establish the consistency of the sequence $\hat{\theta}_n$ for a value θ_0 ;

- establish the rate of convergence r_n of the sequence $\hat{\theta}_n$ or, equivalently, establish the tightness of the sequence of local parameters $\hat{h}_n = r_n(\hat{\theta}_n - \theta_0)$;
- show that suitably rescaled versions of the criterion functions converge in distribution to a limit process $\mathbb{M}$ in the space $\ell^\infty(h: \|h\| \leq K)$ for every K .

If the sample paths $h \mapsto \mathbb{M}(h)$ of the limit process are upper semicontinuous and possess a unique maximum $\hat{h}$, then the final conclusion is that the sequence $r_n(\hat{\theta}_n - \theta_0)$ converges in distribution to $\hat{h}$.

3.2.4 Example (Parametric maximum likelihood). For the maximum likelihood estimator based on i.i.d. observations from a density p_θ we may choose $\mathbb{M}_n(\theta) = n^{-1} \sum_{i=1}^n \log p_\theta(X_i)$. If the map $\theta \mapsto p_\theta$ is sufficiently smooth, then the sequence of statistical models is locally asymptotically normal:[‡]

$$\sum_{i=1}^n \log \frac{p_{\theta+h/\sqrt{n}}(X_i)}{p_\theta} = h' \frac{1}{\sqrt{n}} \sum_{i=1}^n \dot{\ell}_\theta(X_i) - \frac{1}{2} h' I_\theta h + o_{P_\theta}(1).$$

Here $\dot{\ell}_\theta$ is the score function of the model and I_θ is the Fisher information matrix. The sequence of stochastic processes on the right (indexed by $h \in \mathbb{R}^k$) converges marginally in distribution to the Gaussian process

$$h \mapsto h' \Delta_\theta - \frac{1}{2} h' I_\theta h,$$

for a $N_k(0, I_\theta)$ -distributed random variable Δ_θ . If θ is an inner point of the parameter set, then the sequence $\hat{h}_n = \sqrt{n}(\hat{\theta}_n - \theta)$ typically converges in distribution to the maximizer $\hat{h}$ of this process when h ranges over the full Euclidean space. The classical results on asymptotic normality of maximum likelihood estimators precisely make this statement, for the maximizer is $\hat{h} = I_\theta^{-1} \Delta_\theta$ and possesses a normal $N(0, I_\theta^{-1})$ -distribution.

Asymptotic normality may be proved by means of the preceding theorem provided the marginal convergence of the processes is suitably strengthened. Simple, but already powerful, conditions are given in Example 3.2.27.

In the case of i.i.d. data and an empirical criterion function of the form $\mathbb{M}_n(\theta) = \mathbb{P}_n m_\theta$, the uniform convergence in Corollary 3.2.3 (to $\mathbb{M}(\theta) = P m_\theta$) is valid if and only if the class of functions $\{m_\theta: \theta \in \Theta\}$ is Glivenko-Cantelli. The main Glivenko-Cantelli theorems can be found in Chapter 2.4. The rescaled processes $\{\mathbb{P}_n(m_{\theta+h/r_n} - m_\theta): h \in K\}$ are up to centering at mean zero a multiple of the empirical process indexed by the classes of functions $\{m_{\theta+h/r_n} - m_\theta: h \in K\}$. Because these classes are changing with n , the convergence in distribution of these processes is somewhat more complicated than establishing a Donsker property. However, sufficient conditions for the convergence in distribution are given in Section 2.11.3.

In the next section we present a method to establish the rate of convergence.

[‡] See Le Cam (1960), or Van der Vaart (1998), Chapter 7.

3.2.2 Rate of Convergence

If θ_0 is a point of maximum of the map $\theta \mapsto \mathbb{M}(\theta)$, then the first derivative must vanish at θ_0 and the second derivative should be negative definite. Thus, it is natural to assume that for θ in a neighborhood of θ_0 ,^b

$$\mathbb{M}(\theta) - \mathbb{M}(\theta_0) \lesssim -d^2(\theta, \theta_0).$$

Consider estimators $\hat{\theta}_n$ that (nearly) maximize stochastic processes $\theta \mapsto \mathbb{M}_n(\theta)$ that approximate the functions $\mathbb{M}$. An upper bound for the rate of convergence of $\hat{\theta}_n$ can be obtained from the continuity modulus of the difference $\mathbb{M}_n - \mathbb{M}$, as can be seen intuitively as follows.

The parameter θ_0 maximizes the function $\theta \mapsto \mathbb{M}(\theta)$, but the estimator $\hat{\theta}_n$ maximizes

$$\theta \mapsto \mathbb{M}_n(\theta) = \mathbb{M}(\theta) + (\mathbb{M}_n - \mathbb{M})(\theta).$$

At $\hat{\theta}_n$ the systematic part $\mathbb{M}(\theta)$ of the criterion has fallen of the order $d^2(\hat{\theta}_n, \theta_0)$ below its maximum value, which must be compensated by a random fluctuation $(\mathbb{M}_n - \mathbb{M})(\hat{\theta}_n)$ upwards relative to its value at θ_0 of at least the same size. If the process $(\mathbb{M}_n - \mathbb{M})(\hat{\theta}_n) - (\mathbb{M}_n - \mathbb{M})(\theta_0)$ is of the order $\phi_n(d(\hat{\theta}_n, \theta_0))/\sqrt{n}$, then it follows that

$$d^2(\hat{\theta}_n, \theta_0) \asymp \frac{\phi_n(d(\hat{\theta}_n, \theta_0))}{\sqrt{n}}.$$

The following theorem makes this precise, by giving a rigorous definition to the “order” function ϕ_n . The theorem is a special case of theorems derived in Chapter 3.4, which is devoted entirely to rates of convergence.

3.2.5 Theorem (M-estimator–rate of convergence). *Let $\mathbb{M}_n$ be stochastic processes indexed by a semimetric space Θ and $\mathbb{M}: \Theta \rightarrow \mathbb{R}$ a deterministic function, such that for every θ in a neighborhood of θ_0 ,*

$$\mathbb{M}(\theta) - \mathbb{M}(\theta_0) \lesssim -d^2(\theta, \theta_0).$$

Suppose that, for every n and sufficiently small δ , the centered process $\mathbb{M}_n - \mathbb{M}$ satisfies

$$E^* \sup_{d(\theta, \theta_0) < \delta} |(\mathbb{M}_n - \mathbb{M})(\theta) - (\mathbb{M}_n - \mathbb{M})(\theta_0)| \lesssim \frac{\phi_n(\delta)}{\sqrt{n}},$$

for functions ϕ_n such that $\delta \mapsto \phi_n(\delta)/\delta^\alpha$ is decreasing for some $\alpha < 2$ (not depending on n). Let

$$r_n^2 \phi_n\left(\frac{1}{r_n}\right) \leq \sqrt{n}, \quad \text{for every } n.$$

If the sequence $\hat{\theta}_n$ satisfies $\mathbb{M}_n(\hat{\theta}_n) \geq \mathbb{M}_n(\theta_0) - O_P(r_n^{-2})$ and converges in outer probability to θ_0 , then $r_n d(\hat{\theta}_n, \theta_0) = O_P^(1)$. If the displayed conditions are valid for every θ and δ , then the condition that $\hat{\theta}_n$ is consistent is unnecessary.*

^b The notation $\lesssim$ means “is bounded above up to a universal constant”. This condition is often satisfied but is unnecessarily restrictive. Theorem 3.2.5 remains true even if the map $\theta \mapsto d^2(\theta, \theta_0)$ does not arise from a distance but is an arbitrary function from Θ to the nonnegative reals. In the last assertion “convergence of $\hat{\theta}_n$ in outer probability to θ_0 ” must then be interpreted as $d^2(\hat{\theta}_n, \theta_0) \xrightarrow{P^*} 0$.

Proof. Assume for simplicity that $\hat{\theta}_n$ truly maximizes the map $\theta \mapsto \mathbb{M}_n(\theta)$. For each n , the parameter space (minus the point θ_0) can be partitioned into the “shells” $S_{j,n} = \{\theta: 2^{j-1} < r_n d(\theta, \theta_0) \leq 2^j\}$ with j ranging over the integers. If $r_n d(\hat{\theta}_n, \theta_0)$ is larger than 2^M for a given integer M , then $\hat{\theta}_n$ is in one of the shells $S_{j,n}$ with $j > M$. In that case the supremum of the map $\theta \mapsto \mathbb{M}_n(\theta) - \mathbb{M}_n(\theta_0)$ over this shell is nonnegative by the property of $\hat{\theta}_n$. Conclude that, for every $\eta > 0$,

$$\begin{aligned} \mathbf{P}^*(r_n d(\hat{\theta}_n, \theta_0) > 2^M) &\leq \sum_{\substack{j > M \\ 2^j \leq \eta r_n}} \mathbf{P}^*\left(\sup_{\theta \in S_{j,n}} (\mathbb{M}_n(\theta) - \mathbb{M}_n(\theta_0)) \geq 0\right) \\ &\quad + \mathbf{P}^*(2d(\hat{\theta}_n, \theta_0) \geq \eta). \end{aligned}$$

If the sequence $\hat{\theta}_n$ is consistent for θ_0 , then the second probability on the right converges to 0 as $n \rightarrow \infty$ for every $\eta > 0$. Choose $\eta > 0$ small enough that the first condition of the theorem holds for every $d(\theta, \theta_0) \leq \eta$ and the second for every $\delta \leq \eta$. Then for every j involved in the sum, we have, for every $\theta \in S_{j,n}$,

$$\mathbb{M}(\theta) - \mathbb{M}(\theta_0) \lesssim -d^2(\theta, \theta_0) < \frac{-2^{2j-2}}{r_n^2}.$$

In terms of the centered process $W_n = \mathbb{M}_n - \mathbb{M}$, the series may be bounded by

$$\begin{aligned} \sum_{\substack{j > M \\ 2^j \leq \eta r_n}} \mathbf{P}^*\left(\|W_n(\theta) - W_n(\theta_0)\|_{S_{j,n}} \gtrsim \frac{2^{2j-2}}{r_n^2}\right) &\lesssim \sum_{j > M} \frac{\phi_n(2^j/r_n) r_n^2}{\sqrt{n} 2^j} \\ &\lesssim \sum_{j > M} 2^{j\alpha-2j}, \end{aligned}$$

by Markov’s inequality, the definition of r_n , and the fact that $\phi_n(c\delta) \leq c^\alpha \phi_n(\delta)$ for every $c > 1$ by the assumption on ϕ_n . This expression converges to zero for every $M = M_n \rightarrow \infty$.

The second assertion follows by a minor simplification of the preceding argument. ■

In the case of i.i.d. data and criterion functions of the form $\mathbb{M}_n(\theta) = \mathbb{P}_n m_\theta$ and $\mathbb{M}(\theta) = P m_\theta$, the centered and scaled process $\sqrt{n}(\mathbb{M}_n - \mathbb{M}) = \mathbb{G}_n m_\theta$ equals the empirical process at m_θ . The second condition of the theorem involves the suprema of the empirical process indexed by classes of functions

$$\mathcal{M}_\delta = \{m_\theta - m_{\theta_0} : d(\theta, \theta_0) < \delta\}.$$

It is not unreasonable to assume that these suprema are bounded uniformly in n . This leads to the following result.

3.2.6 Corollary. *In the i.i.d. case assume that, for every θ in a neighborhood of θ_0 ,*

$$P(m_\theta - m_{\theta_0}) \lesssim -d^2(\theta, \theta_0).$$

Furthermore, assume that there exists a function ϕ such that $\delta \mapsto \phi(\delta)/\delta^\alpha$ is decreasing for some $\alpha < 2$ and, for every n ,

$$E^* \|\mathbb{G}_n\|_{\mathcal{M}_\delta} \lesssim \phi(\delta).$$

If the sequence $\hat{\theta}_n$ satisfies $\mathbb{P}_n m_{\hat{\theta}_n} \geq \mathbb{P}_n m_{\theta_0} - O_P(r_n^{-2})$ and converges in outer probability to θ_0 , then $r_n d(\hat{\theta}_n, \theta_0) = O_P(1)$ for every sequence r_n such that $r_n^2 \phi(1/r_n) \leq \sqrt{n}$ for every n .

Thus the “continuity modulus” of the empirical process gives an upper bound on the rate. For instance, the modulus $\phi(\delta) = \delta^\alpha$ gives a rate of at least $n^{1/(4-2\alpha)}$; the “usual” rate $\sqrt{n}$ corresponds to $\phi(\delta) = \delta$.

For a Euclidean parameter space, the first condition of the theorem is satisfied if the map $\theta \mapsto Pm_\theta$ is twice continuously differentiable at the point of maximum θ_0 with nonsingular second-derivative matrix.

The preceding theorem appears to give the correct rate in fair generality, the main problem being to derive sharp bounds on the continuity modulus of the empirical process. A simple, but not necessarily efficient, method is to apply the maximal inequalities given by Theorems 2.14.1 and 2.14.16. These yield bounds in terms of the uniform entropy integral $J(1, \mathcal{M}_\delta, L_2)$ or the bracketing integral $J_{[]} (1, \mathcal{M}_\delta, L_2(P))$ of the class $\mathcal{M}_\delta$ given by

$$\begin{aligned} E_P^* \|\mathbb{G}_n\|_{\mathcal{M}_\delta} &\lesssim J(1, \mathcal{M}_\delta, L_2) (P^* M_\delta^2)^{1/2}, \\ E_P^* \|\mathbb{G}_n\|_{\mathcal{M}_\delta} &\lesssim J_{[]} (1, \mathcal{M}_\delta, L_2(P)) (P^* M_\delta^2)^{1/2}. \end{aligned}$$

Here M_δ is an envelope function of the class $\mathcal{M}_\delta$. These bounds are pessimistic in that they depend on the size of the functions $m_\theta - m_{\theta_0}$, which are likely to be small for small δ , mostly through the envelope of the class. The assumption that the entropy integrals $J(1, \mathcal{M}_\delta, L_2)$ and $J_{[]} (1, \mathcal{M}_\delta, L_2(P))$ are bounded as $\delta \downarrow 0$ appears reasonable. (This is not automatically the case, because they are defined relative to an envelope function.) In that case the preceding corollary may be applied with the upper bound $\phi^2(\delta) = P^* M_\delta^2$ and leads to a rate of convergence r_n of at least the solution of

$$(3.2.7) \quad r_n^4 P^* M_{1/r_n}^2 \asymp n.$$

In many examples involving finite-dimensional parameters this gives the correct result. In Chapter 3.4 we discuss rates of convergence in more generality.

In the remainder of this section we pursue the special case that the rate is given by (3.2.7) and present a theorem concerning the limit distribution of the sequence $r_n(\hat{\theta}_n - \theta_0)$. As is clear from the examples given in the next section, this theorem should not be viewed as the only approach, but it does give a good illustration of the combination of the argmax theorem, Theorem 3.2.2, and the rate theorem, Theorem 3.2.5.

We specialize to a Euclidean parameter set Θ and the case of i.i.d. observations. To derive the limit distribution of $r_n(\hat{\theta}_n - \theta_0)$ using the argmax theorem, we need to establish the convergence of a multiple of the processes $h \mapsto \mathbb{P}_n(m_{\theta_0+h/r_n} - m_{\theta_0})$ in $\ell^\infty(h: \|h\| \leq K)$ for every K . Theorems 2.11.24 and 2.11.25 give conditions for the convergence in distribution of the centered processes

$$\begin{aligned} h \mapsto \frac{r_n^2}{\sqrt{n}} \mathbb{G}_n(m_{\theta_0+h/r_n} - m_{\theta_0}) \\ = r_n^2 \mathbb{P}_n(m_{\theta_0+h/r_n} - m_{\theta_0}) - r_n^2 P(m_{\theta_0+h/r_n} - m_{\theta_0}). \end{aligned}$$

These are the empirical processes with indexed classes $(r_n^2/\sqrt{n})\mathcal{M}_{K/r_n}$. Theorems 2.11.24 and 2.11.25 require that the envelope functions of these classes are bounded in square expectation. In the present case this is satisfied by the definition of r_n in (3.2.7). (Of course, it was the motivation for multiplying the empirical measure by r_n^2 .) We shall also translate the other conditions to the present situation.

Assume that either the uniform entropy or bracketing integrals of the classes $\mathcal{M}_\delta$ are uniformly bounded as δ tends to 0: for some $\delta_0 > 0$

$$(3.2.8) \quad \int_0^\infty \sup_{\delta < \delta_0} \sup_Q \sqrt{\log N(\varepsilon \|M_\delta\|_{Q,2}, \mathcal{M}_\delta, L_2(Q))} d\varepsilon < \infty$$

or

$$(3.2.9) \quad \int_0^\infty \sup_{\delta < \delta_0} \sqrt{\log N_{[\cdot]}(\varepsilon \|M_\delta\|_{P,2}, \mathcal{M}_\delta, L_2(P))} d\varepsilon < \infty.$$

The first type of entropy condition can be exploited only under some measurability conditions. In the following it will be understood, that (3.2.8) includes the condition that the classes $\mathcal{M}_\delta$ satisfy the conditions of Theorem 2.14.1.

3.2.10 Theorem (M-estimator-asymptotic distribution). *For each θ in an open subset of Euclidean space, let m_θ be a measurable function such that $\theta \mapsto Pm_\theta$ is twice continuously differentiable at a point of maximum θ_0 , with nonsingular second-derivative matrix V .[‡] Let the entropy condition (3.2.8) or (3.2.9) hold. Assume that for some continuous function ϕ , such that $\phi^2(\delta) \geq P^*M_\delta^2$ and such that $\delta \mapsto \phi(\delta)/\delta^\alpha$ is decreasing for some $\alpha < 2$, and for every $\eta > 0$,*

$$\begin{aligned} \lim_{\delta \downarrow 0} \frac{P^*M_\delta^2 \{M_\delta > \eta \delta^{-2} \phi^2(\delta)\}}{\phi^2(\delta)} &= 0, \\ \lim_{\varepsilon \downarrow 0} \limsup_{\delta \downarrow 0} \sup_{\substack{\|h-g\| < \varepsilon \\ \|h\| \vee \|g\| \leq K}} \frac{P(m_{\theta_0+\delta g} - m_{\theta_0+\delta h})^2}{\phi^2(\delta)} &= 0, \\ (3.2.11) \quad \lim_{\delta \downarrow 0} \frac{P(m_{\theta_0+\delta g} - m_{\theta_0+\delta h})^2}{\phi^2(\delta)} &= E(G(g) - G(h))^2, \end{aligned}$$

[‡] It suffices that a two-term Taylor expansion $Pm_{\theta_0+h} = Pm_{\theta_0} + \frac{1}{2}h'Vh + o(\|h\|^2)$ is valid.

for all K and some zero-mean Gaussian process G such that $G(g) = G(h)$ almost surely only if $h = g$.[†] Then there exists a version of G with bounded, uniformly continuous sample paths on compacta. Define r_n as the solution of $r_n^2 \phi(1/r_n) = \sqrt{n}$. If $\hat{\theta}_n$ nearly maximizes the map $\theta \mapsto \mathbb{P}_n m_\theta$ for every n and converges in outer probability to θ_0 , then the sequence $r_n(\hat{\theta}_n - \theta_0)$ converges in distribution to the unique maximizer $\hat{h}$ of the process $h \mapsto G(h) + \frac{1}{2}h'Vh$.

Proof. By the discussion preceding the theorem and the nonsingularity of the second-derivative matrix V , the sequence $r_n(\hat{\theta}_n - \theta_0)$ is uniformly tight.

The conditions of Theorems 2.11.24 and 2.11.25 for asymptotic tightness of the sequence $(r_n/\sqrt{n})\mathbb{G}_n(m_{\theta_0+h/r_n} - m_{\theta_0})$ in the space $\ell^\infty(h: \|h\| \leq K)$ are an entropy condition and

$$\begin{aligned} \frac{r_n^4}{n} P^* M_{K/r_n}^2 &= O(1); & \frac{r_n^4}{n} P^* M_{K/r_n}^2 \{r_n^2 M_{K/r_n} > \eta n\} &= o(1), \\ \sup_{\|h-g\| < \eta_n} \frac{r_n^4}{n} P(m_{\theta_0+g/r_n} - m_{\theta_0+h/r_n})^2 &= o(1). \end{aligned}$$

These conditions are implied by the conditions of the theorem. By the two-times differentiability of the map $\theta \mapsto Pm_\theta$,

$$r_n^2 P(m_{\theta_0+h/r_n} - m_{\theta_0}) \rightarrow \frac{1}{2}h'Vh,$$

uniformly in h ranging over bounded sets. Conclude that the sequence of processes $h \mapsto r_n^2 \mathbb{P}_n(m_{\theta_0+h/r_n} - m_{\theta_0})$ is asymptotically tight in the space $\ell^\infty(h: \|h\| \leq K)$ as well. Its marginals are triangular arrays of i.i.d. random vectors that satisfy the Lindeberg condition. In view of the preceding display and (3.2.11), the mean and covariance function converge to $\frac{1}{2}h'Vh$ and $EG(g)G(h)$, respectively. Thus, the sequence of processes $h \mapsto r_n^2 \mathbb{P}_n(m_{\theta_0+h/r_n} - m_{\theta_0})$ converges in distribution in the space $\ell^\infty(h: \|h\| \leq K)$ to the Gaussian process $h \mapsto G(h) + \frac{1}{2}h'Vh$, for every K .

By Proposition A.2.25, almost all sample paths of this process attain their supremum at a unique point $\hat{h}$. Finally, apply Theorem 3.2.2. ■

The second and third lines of the display in the theorem are implied by the single condition: for every $g_\delta \rightarrow g$ and $h_\delta \rightarrow h$,

$$\lim_{\delta \downarrow 0} \frac{P(m_{\theta_0+\delta g_\delta} - m_{\theta_0+\delta h_\delta})^2}{\phi^2(\delta)} = E(G(g) - G(h))^2.$$

Instead of the variances of the differences, the condition could also be stated in terms of the covariances of m_{θ_1} and m_{θ_2} . See Problem 3.2.1 for further simplifications of the theorem.

[†] Condition (3.2.11) may be replaced by marginal convergence of the processes $h \mapsto r_n^2 \mathbb{P}_n(m_{\theta_0+h/r_n} - m_{\theta_0})$ to a process $h \mapsto G(h)$.

3.2.3 Examples

This section presents some examples that show the scope of combining the argmax theorem, Theorem 3.2.2, and the rate theorem, Theorem 3.2.5. In some examples we simply apply Theorem 3.2.10; in other examples we give the complete argument.

3.2.12 Example (Lipschitz in parameter). A simple method to verify the conditions of the preceding theorems is by a pointwise Lipschitz condition on the maps $\theta \mapsto m_\theta$. Let $X_1, \dots, X_n$ be i.i.d. random variables with common law P , and let m_θ be measurable functions such that, for every θ_1, θ_2 in a neighborhood of θ_0 ,

$$|m_{\theta_1}(x) - m_{\theta_2}(x)| \leq \dot{m}(x) \|\theta_1 - \theta_2\|^\alpha,$$

for a square-integrable function $\dot{m}$. Then the first two displayed conditions of the theorem are satisfied for $\phi(\delta) = \delta^\alpha$. The envelope function M_δ can be taken equal to $\delta^\alpha \dot{m}$, and the bracketing numbers of the class $\mathcal{M}_\delta$ are bounded by the covering numbers of Euclidean balls of radius δ for the Euclidean metric to the power α , hence polynomial. See Section 2.7.6.

Let $\hat{\theta}_n$ maximize $\theta \mapsto \mathbb{P}_n m_\theta$ for every n and be consistent for θ_0 . If in addition the map $\theta \mapsto P m_\theta$ is twice continuously differentiable at its point of maximum θ_0 with a nonsingular second-derivative matrix, then Theorem 3.2.5 gives a rate of convergence of at least $n^{1/(4-2\alpha)}$.

The limit distribution of the sequence $n^{1/(4-2\alpha)}(\hat{\theta}_n - \theta_0)$ follows by the preceding theorem under the further condition that for a zero-mean Gaussian process G ,

$$P(m_{\theta_0+\delta g} - m_{\theta_0})(m_{\theta_0+\delta h} - m_{\theta_0}) \asymp \delta^{2\alpha} \text{EG}(g)G(h).$$

In this case the sequence $n^{1/(4-2\alpha)}(\hat{\theta}_n - \theta_0)$ converges in distribution to the maximizer of the process $h \mapsto G(h) + \frac{1}{2}h'Vh$ for a version of G with continuous sample paths.

For $\alpha = 1$, it appears not unreasonable to assume that, for some vector-valued function $\dot{m}_{\theta_0}$,

$$P\left[\frac{m_{\theta_0+\delta h} - m_{\theta_0}}{\delta} - h'\dot{m}_{\theta_0}\right]^2 \rightarrow 0.$$

This yields the covariance structure $\text{EG}(g)G(h) = g'Wh$ for the matrix $W = P\dot{m}_{\theta_0}\dot{m}_{\theta_0}'$. A process G with a covariance structure of this type can be represented as $G(h) = h'\Delta$ for a $N(0, W)$ -distributed random vector Δ . The maximizer of the process $h \mapsto h'\Delta + \frac{1}{2}h'Vh$ is $\hat{h} = -V^{-1}\Delta$ and is normally distributed with mean zero and covariance matrix $V^{-1}WV^{-1}$. Thus, in this case the sequence $\sqrt{n}(\hat{\theta}_n - \theta_0)$ converges in distribution to a normal distribution.

Example 3.2.25 ahead establishes the same conclusion by a more classical linearization argument.

3.2.13 Example (Integral mode). Let $X_1, \dots, X_n$ be i.i.d., real-valued random variables with common law P with a differentiable Lebesgue density p . Define an estimator $\hat{\theta}_n$ of location as the maximizer of the function

$$\theta \mapsto \mathbb{P}_n[\theta - 1, \theta + 1].$$

Thus $\hat{\theta}_n$ is the center of an interval of length 2 that contains the largest possible fraction of the observations.

By the classical Glivenko-Cantelli theorem, the sequence $\sup_{\theta} |(\mathbb{P}_n - P)[\theta - 1, \theta + 1]|$ converges in probability to zero. Consequently, if the map $\theta \mapsto P[\theta - 1, \theta + 1]$ has a unique “well-separated” point of maximum θ_0 , then the sequence $\hat{\theta}_n$ converges in probability to θ_0 . The second derivative of $\theta \mapsto P[\theta - 1, \theta + 1]$ equals $p'(\theta + 1) - p'(\theta - 1)$, which is likely to be strictly negative at θ_0 .

The functions $m_{\theta} = 1_{[\theta-1, \theta+1]}$ are not Lipschitz in the parameter. Nevertheless, the classes of functions $\mathcal{M}_{\delta}$ satisfy the conditions of the preceding theorem. These classes are Vapnik-Červonenkis with envelope functions

$$\sup_{|\theta - \theta_0| < \delta} |1_{[\theta-1, \theta+1]} - 1_{[\theta_0-1, \theta_0+1]}| \leq 1_{[\theta_0-1-\delta, \theta_0-1+\delta]} + 1_{[\theta_0+1-\delta, \theta_0+1+\delta]}.$$

The $L_2(P)$ -norm of these functions is bounded above by a constant times $\sqrt{\delta}$. Thus, the conditions of the theorem are satisfied with $\phi(\delta) = c\sqrt{\delta}$ for a constant c , leading to a rate of convergence $n^{1/3}$. The sequence $n^{1/3}(\hat{\theta}_n - \theta_0)$ converges in distribution to the maximizer of the process $h \mapsto G(h) + \frac{1}{2}h^2(p'(\theta_0 + 1) - p'(\theta_0 - 1))$. Here G is zero-mean Gaussian with continuous sample paths and variance function

$$E(G(g) - G(h))^2 = (p(\theta_0 - 1) + p(\theta_0 + 1)) |g - h|.$$

Up to a scale factor, G is a *two-sided Brownian motion* originating from zero.

3.2.14 Example (Monotone densities). Let $X_1, \dots, X_n$ be an i.i.d. sample from a Lebesgue density f on $[0, \infty)$ that is known to be decreasing. The maximum likelihood estimator $\hat{f}_n$ of f is the step function equal to the left derivative of the least concave majorant of the empirical distribution function $\mathbb{F}_n$. For a fixed value $t > 0$, we shall derive the limit distribution of the sequence $n^{1/3}(\hat{f}_n(t) - f(t))$ under the assumption that the true density is differentiable at t with derivative $f'(t) < 0$.

Define a stochastic process $\{\hat{s}_n(a) : a > 0\}$ by

$$\hat{s}_n(a) = \operatorname{argmax}_s \{\mathbb{F}_n(s) - as\},$$

where the largest value is chosen when multiple maximizers exist. The function $\hat{s}_n$ is the inverse of the function $\hat{f}_n$ in the sense that $\hat{f}_n(t) < a$ if and only if $\hat{s}_n(a) < t$ for every t and a . This is explained in Figure 3.2.1.[‡] It follows that

$$P\left(n^{1/3}(\hat{f}_n(t) - f(t)) < x\right) = P\left(\hat{s}_n(f(t) + xn^{-1/3}) < t\right).$$

Hence the desired result can be deduced from the limiting behavior of the process of points of maximum $\hat{s}_n(f(t) + xn^{-1/3})$. By the change of variable $s \mapsto t + hn^{-1/3}$ in the definition of $\hat{s}_n$, we have

$$\begin{aligned} \hat{s}_n(f(t) + xn^{-1/3}) - t \\ = n^{-1/3} \operatorname{argmax}_h \left\{ \mathbb{F}_n(t + hn^{-1/3}) - (f(t) + xn^{-1/3})(t + hn^{-1/3}) \right\}. \end{aligned}$$

[‡] A precise proof is given in Lemma 4.1 of Van der Laan and Van der Vaart (2006). The relationship is wrong if the two strict inequalities $<$ are replaced by $\leq$.

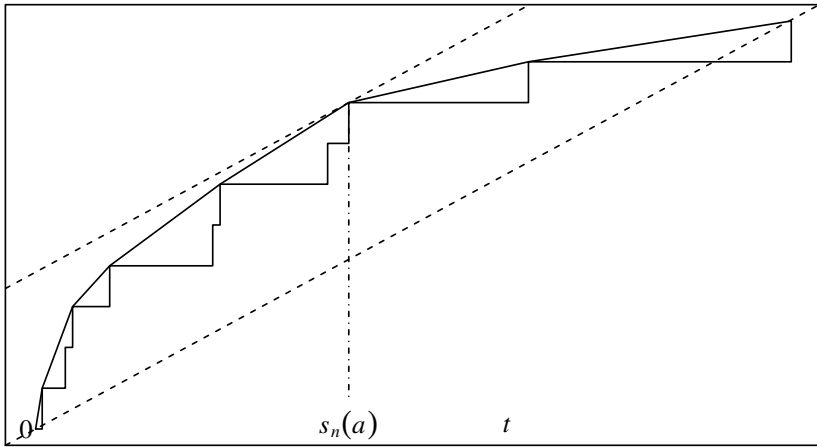


Figure 3.2.1. If $\hat{f}_n(t) \leq a$, then a line of slope a moved down vertically from $+\infty$ first hits $\mathbb{F}_n$ to the left of t . The point where the line hits is the point where $\mathbb{F}_n$ is farthest above the line of slope a through the origin.

It follows that the probability of interest is $P(\hat{h}_n < 0)$ for $\hat{h}_n$ equal to the argmax_h appearing on the right. The location of the maximum of a function does not change when the function is multiplied by a positive constant or shifted vertically. Thus, the $\operatorname{argmax}_h \hat{h}_n$ is also a point of maximum of the process

$$h \mapsto n^{2/3}(\mathbb{P}_n - P)(1_{[0, t+hn^{-1/3}]} - 1_{[0, t]}) + n^{2/3}[F(t+hn^{-1/3}) - F(t) - f(t)hn^{-1/3}] - xh.$$

By Theorem 2.11.24 or 2.11.25, this sequence of processes converges, for every K , in the space $\ell^\infty(-K, K)$ to the process

$$h \mapsto \sqrt{f(t)}\mathbb{Z}(h) + \frac{1}{2}f'(t)h^2 - xh,$$

where $\mathbb{Z}$ is a two-sided Brownian motion originating from zero: a mean-zero Gaussian process with $\mathbb{Z}(0) = 0$ and $E(\mathbb{Z}(g) - \mathbb{Z}(h))^2 = |g - h|$ for every g, h . If it can be proved that $\hat{h}_n = O_P(1)$, then the argmax continuous mapping theorem yields that the sequence $\hat{h}_n$ converges in distribution to the maximizer $\hat{h}$ of the limit process. In particular,

$$P(n^{1/3}(\hat{f}_n(t) - f(t)) < x) = P(\hat{h}_n < 0) \rightarrow P(\hat{h} < 0).$$

Elementary properties of Brownian motion allow one to rewrite the limit probability $P(\hat{h} \leq 0)$ in a simple form. See Problem 3.2.5. The final conclusion is that

$$n^{1/3}(\hat{f}_n(t) - f(t)) \rightsquigarrow |4f'(t)f(t)|^{1/3} \operatorname{argmax}_h \{\mathbb{Z}(h) - h^2\}.$$

The sequence $\hat{h}_n$ can be shown to be uniformly tight by Theorem 3.2.5 with criterion and centering function

$$\begin{aligned}\mathbb{M}_n(g) &= (\mathbb{P}_n - P)(1_{[0, t+g]} - 1_{[0, t]}) + F(t+g) - F(t) - f(t)g - xgn^{-1/3}, \\ \mathbb{M}(g) &= F(t+g) - F(t) - f(t)g.\end{aligned}$$

By its definition, $\hat{g}_n = n^{-1/3}\hat{h}_n$ maximizes $g \mapsto \mathbb{M}_n(g)$. In order to show that the sequence $n^{1/3}\hat{g}_n$ is bounded in probability, we shall apply Theorem 3.2.5. By assumption $\mathbb{M}(g) = \frac{1}{2}f'(t)g^2 + o(g^2)$ as $g \rightarrow 0$. The bracketing entropy integral of the class of functions $\{1_{[0, t+g]} - 1_{[0, t]}: |g| < \delta\}$ is bounded uniformly in δ and the envelopes satisfy $PM_\delta^2 \lesssim \delta$. It follows that we can set $\phi_n(\delta) = \sqrt{\delta} + \delta n^{1/6}$, leading to a rate of convergence $n^{1/3}$ for $\hat{g}_n$ as desired, provided $\hat{g}_n$ is consistent. The consistency can be proved by a direct argument, but also by Theorem 3.2.5 applied with $d(g, 0) = -\mathbb{M}(g)$. For this choice of “distance” the conditions of Theorem 3.2.5 are valid for every g and δ , so that by the last assertion of the theorem, $-n^{2/3}\mathbb{M}(\hat{g}_n) = O_P^*(1)$. The concavity of $\mathbb{M}$ shows that $\hat{g}_n$ converges to zero in probability.

3.2.15 Example (Current status). Let $X_1, \dots, X_n$ and $T_1, \dots, T_n$ be independent i.i.d. samples from distribution functions F and G on the nonnegative half-line, respectively. Define $\Delta_i = 1\{X_i \leq T_i\}$. Each X_i is interpreted as the (unobserved) time of onset of a disease; each T_i is a check-up time at which the patient is observed to be ill or not: $\Delta_i = 1$ or 0 . The observations consist of the pairs $(\Delta_1, T_1), \dots, (\Delta_n, T_n)$. The maximum likelihood estimator $\hat{F}_n$ for F maximizes the (partial) likelihood function

$$F \mapsto \sum_{i=1}^n \left(\Delta_i \log F(T_i) + (1 - \Delta_i) \log(1 - F(T_i)) \right).$$

This function depends on F only through the values $F(T_i)$. Hence the maximum likelihood estimator is not unique. In the following, $\hat{F}_n$ is assumed constant on the intervals $[T_{(i-1)}, T_{(i)})$.

If the observation times are initially ordered so that the observation times are increasing, then finding the values $\hat{F}_n(T_i)$ is equivalent to the “isotonic” maximization problem of maximizing the function

$$(y_1, \dots, y_n) \mapsto \sum_{i=1}^n \left(\delta_i \log y_i + (1 - \delta_i) \log(1 - y_i) \right),$$

subject to $0 \leq y_1 \leq y_2 \leq \dots \leq y_n \leq 1$. In view of Theorem 1.5.1 of Robertson, Wright and Dykstra (1988), applied with the convex function $\Phi(y) = y \log y + (1 - y) \log(1 - y)$, the solution vector $\hat{y}$ to this problem is the “isotonic regression” of the vector $(\delta_1, \dots, \delta_n)$: the vector $\hat{y}$ that minimizes $y \mapsto \sum_{i=1}^n (\delta_i - y_i)^2$ subject to the same condition as before. By Theorem 1.2.1 of the same reference, the isotonic regression $\hat{y}_i$ can be graphically represented as the slope in the interval $n^{-1}(i - 1, i)$ of the greatest convex minorant of the “cumulative-sum diagram” of the points $n^{-1}(i, \sum_{j \leq i} \delta_j)$. The cumulative-sum diagram is the function $c: [0, 1] \rightarrow \mathbb{R}$, which

equals $n^{-1} \sum_{j \leq i} \delta_i$ on the interval $n^{-1}(i-1, i]$ and equals 0 at zero. The graphical representation shows that $\hat{y}_i < a$ if and only if $\operatorname{argmin}_s \{c(s) - as\} \geq i/n$. (In case of multiple points of minimum, take the smallest.) Let $\mathbb{P}_n$ be the empirical measure of the pairs $(X_1, T_1), \dots, (X_n, T_n)$, and let $A = \{(x, t): x \leq t\}$. Define the stochastic processes

$$V_n(t) = \mathbb{P}_n 1_A 1_{\mathbb{R} \times [0, t]} = \frac{1}{n} \sum \Delta_i 1\{T_i \leq t\},$$

$$G_n(t) = \mathbb{P}_n 1_{\mathbb{R} \times [0, t]} = \frac{1}{n} \sum 1\{T_i \leq t\}.$$

Then the function $s \mapsto V_n \circ G_n^{-1}(s)$ equals the cumulative-sum diagram, whence for every observation point T_i ,

$$\hat{F}_n(T_i) < a \quad \text{if and only if} \quad \operatorname{argmin}_s \{V_n(s) - aG_n(s)\} \geq T_i.$$

Thus, the limit distribution of $\hat{F}_n(t)$ can be derived by studying the locations of the minima of the sequence of processes $s \mapsto V_n(s) - aG_n(s)$. Assume that F and G are differentiable at $t > 0$ with derivatives $f(t)$ and $g(t) > 0$. For each t , the value $\hat{F}_n(t)$ equals $\hat{F}_n(T_i)$ for some i ; hence $\hat{F}_n(t) < a$ if and only if the argmin appearing in the display is larger than the observation time T_i that is just left of t . Under the condition that $g(t) > 0$, the difference between this observation time and t is asymptotically negligible. By the change of variables $s \mapsto t + n^{-1/3}h$,

$$\begin{aligned} n^{1/3} \left(\operatorname{argmin}_s \{V_n(s) - (F(t) + xn^{-1/3})G_n(s)\} - t \right) \\ = \operatorname{argmin}_h \left\{ n^{2/3} (\mathbb{P}_n - P)(1_A - F(t))(1_{\mathbb{R} \times [0, t+hn^{-1/3}]} - 1_{\mathbb{R} \times [0, t]}) \right. \\ \left. + n^{2/3} P(1_A - F(t))(1_{\mathbb{R} \times [0, t+hn^{-1/3}]} - 1_{\mathbb{R} \times [0, t]}) \right. \\ \left. - xn^{1/3} \mathbb{P}_n(1_{\mathbb{R} \times [0, t+hn^{-1/3}]} - 1_{\mathbb{R} \times [0, t]}) \right\}. \end{aligned}$$

Let $\mathbb{Z}$ be a two-sided Brownian motion process, originating from zero. By Theorems 2.11.24 and 2.11.25, the sequence of processes inside the argmin converges, for every K , in the space $\ell^\infty(-K, K)$ to the process

$$h \mapsto \sqrt{F(1-F)g(t)} \mathbb{Z}(h) + \frac{1}{2} h^2 f(t)g(t) - xg(t)h.$$

The locations of the minima converge to the location of the minimum $\hat{h}$ of this limit process provided the sequence of locations of minima is bounded in probability. The final conclusion is

$$n^{1/3}(\hat{F}_n(t) - F(t)) \rightsquigarrow \left(\frac{4F(1-F)f}{g}(t) \right)^{1/3} \operatorname{argmin}_h \{ \mathbb{Z}(h) + h^2 \}.$$

Here we have used Problem 3.2.5 to rewrite the probability $P(\hat{h} \geq 0)$ in a simple form.

Uniform tightness of the locations of the minima can be shown with the help of Theorem 3.2.5. Define the processes

$$\begin{aligned}\mathbb{M}_n(h) &= (\mathbb{P}_n - P)(1_A - F(t))(1_{\mathbb{R} \times [0, t+h]} - 1_{\mathbb{R} \times [0, t]}) \\ &\quad + P(1_A - F(t))(1_{\mathbb{R} \times [0, t+h]} - 1_{\mathbb{R} \times [0, t]}) \\ &\quad - xn^{-1/3} \mathbb{P}_n(1_{\mathbb{R} \times [0, t+h]} - 1_{\mathbb{R} \times [0, t]}), \\ \mathbb{M}(h) &= P(1_A - F(t))(1_{\mathbb{R} \times [0, t+h]} - 1_{\mathbb{R} \times [0, t]}).\end{aligned}$$

The conditions of Theorem 3.2.5 are satisfied with $\phi_n(\delta) = \sqrt{\delta} + x\delta n^{1/6}$. Thus, the location of the maximum of $h \mapsto \mathbb{M}_n(h)$ has rate of convergence $n^{1/3}$, provided the sequence converges to zero in probability. The latter can be shown by a standard consistency proof.

3.2.4 Linearization

A (more) classical approach toward proving asymptotic normality of M -estimators is through a Taylor expansion of the criterion function. Assume that the limit criterion function $\theta \mapsto Pm_\theta$ takes its maximum at a point θ_0 ; then its first derivative must vanish at θ_0 , and the second derivative V must be negative definite. If the function $\theta \mapsto m_\theta$ is several times differentiable, then since $\mathbb{P}_n = P + n^{-1/2}\mathbb{G}_n$,

$$\begin{aligned}n\mathbb{P}_n(m_\theta - m_{\theta_0}) &= nP(m_\theta - m_{\theta_0}) + \sqrt{n}\mathbb{G}_n(m_\theta - m_{\theta_0}) \\ &\approx \frac{1}{2}\sqrt{n}(\theta - \theta_0)'V\sqrt{n}(\theta - \theta_0) + \sqrt{n}(\theta - \theta_0)\mathbb{G}_n\dot{m}_{\theta_0} \\ &\quad + n o(\|\theta - \theta_0\|^2) + \sqrt{n} o_P(\|\theta - \theta_0\|).\end{aligned}$$

If the two remainder terms are neglected, then the maximum of the right-hand side is taken for $\sqrt{n}(\theta - \theta_0) = -V^{-1}\mathbb{G}_n\dot{m}_{\theta_0}$. Thus, it may be expected that the M -estimator $\hat{\theta}_n$ that maximizes the left-hand side satisfies

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -V^{-1}\mathbb{G}_n\dot{m}_{\theta_0} + o_P(1).$$

This derivation can be made rigorous by various methods. One possibility is to impose “classical” smoothness conditions on the maps $\theta \mapsto m_\theta(x)$ (for every x) in combination with domination of derivatives. More general results can be obtained under the assumption that the criterion functions are differentiable in a stochastic sense.

The basic result does not require the i.i.d. set-up and will be stated for general criterion functions. Let $\hat{\theta}_n$ maximize the random function $\theta \mapsto \mathbb{M}_n(\theta)$ which is “centered” at the deterministic function $\theta \mapsto \mathbb{M}(\theta)$. Assume that the sequence $\hat{\theta}_n$ converges to a point of maximum θ_0 of $\theta \mapsto \mathbb{M}(\theta)$.

3.2.16 Theorem (M-estimator-asymptotic normality). *Let $\mathbb{M}_n$ be stochastic processes indexed by an open subset Θ of Euclidean space and $\mathbb{M}: \Theta \rightarrow \mathbb{R}$ a deterministic function. Assume that $\theta \mapsto \mathbb{M}(\theta)$ is twice continuously differentiable at a point of maximum $\theta_0 \in \Theta$ with nonsingular second-derivative matrix V .^b Suppose that*

$$\begin{aligned}r_n(\mathbb{M}_n - \mathbb{M})(\tilde{\theta}_n) - r_n(\mathbb{M}_n - \mathbb{M})(\theta_0) \\ = (\tilde{\theta}_n - \theta_0)'Z_n + o_P^*(\|\tilde{\theta}_n - \theta_0\| + r_n\|\tilde{\theta}_n - \theta_0\|^2 + r_n^{-1}),\end{aligned}$$

^b It suffices that a two-term Taylor expansion is valid at θ_0 .

for every random sequence $\tilde{\theta}_n = \theta_0 + o_P^*(1)$ and a uniformly tight sequence of random vectors Z_n . If the sequence $\hat{\theta}_n$ converges in outer probability to θ_0 and satisfies $\mathbb{M}_n(\hat{\theta}_n) \geq \sup_{\theta} \mathbb{M}_n(\theta) - o_P(r_n^{-2})$ for every n , then

$$r_n(\hat{\theta}_n - \theta_0) = -V^{-1}Z_n + o_P^*(1).$$

If it is known that the sequence $r_n(\hat{\theta}_n - \theta_0)$ is uniformly tight, then the displayed condition needs to be verified for sequences $\tilde{\theta}_n = \theta_0 + O_P^*(r_n^{-1})$ only.

Proof. The stochastic differentiability condition of the theorem together with the two-times differentiability of the map $\theta \mapsto \mathbb{M}(\theta)$ yields for every sequence $\tilde{h}_n = o_P^*(1)$

$$(3.2.17) \quad \begin{aligned} \mathbb{M}_n(\theta_0 + \tilde{h}_n) - \mathbb{M}_n(\theta_0) &= \frac{1}{2}\tilde{h}_n' V \tilde{h}_n + r_n^{-1}\tilde{h}_n' Z_n \\ &\quad + o_P^*(\|\tilde{h}_n\|^2 + r_n^{-1}\|\tilde{h}_n\| + r_n^{-2}). \end{aligned}$$

For $\tilde{h}_n$ chosen equal to $\hat{h}_n = \hat{\theta}_n - \theta_0$, the left side (and hence the right side) is at least $-o_P(r_n^{-2})$ by the definition of $\hat{\theta}_n$. In the right side the term $\tilde{h}_n' V \tilde{h}_n$ can be bounded above by $-c\|\tilde{h}_n\|^2$ for a positive constant c , since the matrix V is strictly negative definite. Conclude that

$$-o_P(r_n^{-2}) \leq -c\|\hat{h}_n\|^2 + r_n^{-1}\|\hat{h}_n\|O_P(1) + o_P(\|\hat{h}_n\|^2 + r_n^{-2}).$$

Complete the square to see that this implies that

$$(c + o_P(1))\left(\|\hat{h}_n\| - O_P(r_n^{-1})\right)^2 \leq O_P(r_n^{-2}).$$

This can be true only if $\|\hat{h}_n\| = O_P^*(r_n^{-1})$.

For any sequence $\tilde{h}_n$ of the order $O_P^*(r_n^{-1})$, the three parts of the remainder term in (3.2.17) are of the order $o_P(r_n^{-2})$. Apply this with the choices $\hat{h}_n$ and $-r_n^{-1}V^{-1}Z_n$ to conclude that

$$\begin{aligned} \mathbb{M}_n(\theta_0 + \hat{h}_n) - \mathbb{M}_n(\theta_0) &= \frac{1}{2}\hat{h}_n' V \hat{h}_n + r_n^{-1}\hat{h}_n' Z_n + o_P^*(r_n^{-2}), \\ \mathbb{M}_n(\theta_0 - r_n^{-1}V^{-1}Z_n) - \mathbb{M}_n(\theta_0) &= -\frac{1}{2}r_n^{-2}Z_n' V^{-1}Z_n + o_P^*(r_n^{-2}). \end{aligned}$$

The left-hand side of the first equation is larger than the second, up to an $o_P^*(r_n^{-2})$ -term. Subtract the second equation from the first to find that

$$\frac{1}{2}(\hat{h}_n + r_n^{-1}V^{-1}Z_n)'V(\hat{h}_n + r_n^{-1}V^{-1}Z_n) \geq -o_P(r_n^{-2}).$$

Since V is strictly negative definite, this yields the first assertion of the theorem.

If it is known that the sequence $\hat{\theta}_n$ is r_n -consistent, then the middle part of the preceding proof is unnecessary and we can proceed to inserting $\hat{h}_n$ and $-r_n^{-1}V^{-1}Z_n$ in (3.2.17) immediately. The latter equation is then needed for sequences $\tilde{h}_n = O_P^*(r_n^{-1})$ only. ■

The main condition of the theorem could be described as stochastic differentiability of the standardized process $r_n(\mathbb{M}_n - \mathbb{M})$, requiring a linear approximation with remainder of order

$$o_P(\|\tilde{\theta}_n - \theta_0\| + r_n\|\tilde{\theta}_n - \theta_0\|^2 + r_n^{-1}).$$

To require a remainder of the order $o_P(\|\tilde{\theta}_n - \theta_0\|)$ would be more natural, but also more stringent: if $\tilde{\theta}_n$ converges to θ_0 slower or faster than $O_P(r_n^{-1})$, then the sum in the previous display is dominated by its second and third terms, respectively. Thus, in these cases the differentiability as required by the theorem is less stringent than the “natural” condition.

The last statement of the theorem is of some interest, because it also allows one in this approach to separate establishing the rate of convergence from the derivation of the limit distribution. The rate of convergence may be derived with the help of Theorem 3.2.5. For sequences $\tilde{\theta}_n = \theta_0 + O_P(r_n^{-1})$ the stochastic differentiability condition reduces to[‡]

$$r_n(\mathbb{M}_n - \mathbb{M})(\tilde{\theta}_n) - r_n(\mathbb{M}_n - \mathbb{M})(\theta_0) = (\tilde{\theta}_n - \theta_0)'Z_n + o_P^*(r_n^{-1}).$$

This condition together with r_n -consistency of $\hat{\theta}_n$ yield the conclusion of the theorem if the map $\theta \mapsto \mathbb{M}(\theta)$ is two times differentiable with nonsingular second derivative matrix.

3.2.18 Example (Concave criterion). Consider the special case that $\Theta \subset \mathbb{R}^k$ is open and convex and all sample paths $\theta \mapsto \mathbb{M}_n(\theta)$ are concave. Suppose that for a uniformly tight sequence of random vectors Z_n and every $h \in \mathbb{R}^k$,

$$r_n^2(\mathbb{M}_n - \mathbb{M})(\theta_0 + h/r_n) - r_n^2(\mathbb{M}_n - \mathbb{M})(\theta_0) - h'Z_n \xrightarrow{P^*} 0.$$

Furthermore assume that $\theta \mapsto \mathbb{M}(\theta)$ is twice continuously differentiable at a point of maximum $\theta_0 \in \Theta$ with nonsingular second-derivative matrix V . Then any maximizers $\hat{\theta}_n$ of $\theta \mapsto \mathbb{M}_n(\theta)$ satisfy $r_n(\hat{\theta}_n - \theta_0) = -V^{-1}Z_n + o_P^*(1)$.

Thus the conclusion of Theorem 3.2.16 holds under just the stochastic expansion for deterministic perturbations. This remarkable fact arises because concavity automatically separates maxima and turns pointwise convergence into uniform convergence.

For the proof of the result we first combine the stochastic expansion and twice differentiability of $\mathbb{M}$ to see that $r_n^2(\mathbb{M}_n(\theta_0 + h/r_n) - \mathbb{M}_n(\theta_0)) - h'Z_n \xrightarrow{P^*} \frac{1}{2}h'Vh$, for every h . By concavity this convergence is then automatically uniform in h belonging to compacta (see Lemma 1.6.2). This implies that the condition of Theorem 3.2.16 is satisfied for every $\tilde{\theta}_n = \theta_0 + O_P(1/r_n)$, whence by the last assertion of the theorem it suffices to verify that $\hat{\theta}_n$ is r_n -consistent.

The function $h \mapsto h'Z_n + \frac{1}{2}h'Vh = \frac{1}{2}(h + V^{-1}Z_n)'V(h + V^{-1}Z_n) - \frac{1}{2}Z_n'V^{-1}Z_n$ is maximal at $h = -V^{-1}Z_n$. On the sphere $\{h: \|h + V^{-1}Z_n\| = 1\}$, the function is strictly bounded away from this maximum by the assumed strict

[‡] The presence of r_n^{-1} in the remainder term of the stochastic differentiability condition of the theorem ensures that one need not worry about sequences $\tilde{\theta}_n$ that converge to θ_0 too fast.

negative-definiteness of V . The difference between the function $h \mapsto r_n^2(\mathbb{M}_n(\theta_0 + h/r_n) - \mathbb{M}_n(\theta_0))$ and $h \mapsto h'Z_n + \frac{1}{2}h'Vh$ tends to zero on compacta. In view of tightness Z_n , the sphere $\{h: \|h + V^{-1}Z_n\| = 1\}$ and the point $-V^{-1}Z_n$ belong to a compact with with (arbitrarily) high probability. We conclude that the function $h \mapsto r_n^2(\mathbb{M}_n(\theta_0 + h/r_n) - \mathbb{M}_n(\theta_0))$ must also be strictly larger at $h = -V^{-1}Z_n$ than on the sphere eventually. Hence by concavity its maximum value $\hat{h}_n = r_n(\hat{\theta}_n - \theta_0)$ must be inside the sphere. This implies that the sequence $r_n(\theta_n - \theta_0)$ is uniformly tight.

3.2.4.1 I.i.d. observations

In the case of i.i.d. observations, the theorem can be applied with criterion functions $\mathbb{M}_n(\theta) = \mathbb{P}_n m_\theta$ and $\mathbb{M}(\theta) = P m_\theta$ and rate of convergence $r_n = \sqrt{n}$. In this case $\sqrt{n}(\mathbb{M}_n - \mathbb{M})(\theta) = \mathbb{G}_n m_\theta$ for the empirical process $\mathbb{G}_n$. Then the stochastic differentiability can be ascertained by empirical process methods. It is natural to require that each Z_n will be a centered and scaled sum, in which case the stochastic differentiability condition can be recast in terms of the existence of a vector-valued function $\dot{m}_{\theta_0}$ (not necessarily a pointwise partial derivative) such that, for every $\tilde{\theta}_n = \theta_0 + o_P^*(1)$,

$$(3.2.19) \quad \begin{aligned} \mathbb{G}_n(m_{\tilde{\theta}_n} - m_{\theta_0}) &= (\tilde{\theta}_n - \theta_0)' \mathbb{G}_n \dot{m}_{\theta_0} \\ &+ o_P^*(\|\tilde{\theta}_n - \theta_0\| + \sqrt{n}\|\tilde{\theta}_n - \theta_0\|^2 + n^{-1/2}). \end{aligned}$$

The following theorem gives a simple sufficient condition for this type of differentiability.

3.2.20 Theorem (M-estimator-asymptotic normality). *Suppose that there exists a vector-valued function $\dot{m}_{\theta_0}$ such that, for some $\delta > 0$,*

$$\left\{ \frac{m_\theta - m_{\theta_0} - (\theta - \theta_0)' \dot{m}_{\theta_0}}{\|\theta - \theta_0\|} : \|\theta - \theta_0\| < \delta \right\} \quad \text{is } P\text{-Donsker,}$$

$$P[m_\theta - m_{\theta_0} - (\theta - \theta_0)' \dot{m}_{\theta_0}]^2 = o(\|\theta - \theta_0\|^2).$$

Then (3.2.19) is satisfied for every sequence $\tilde{\theta}_n = \theta_0 + o_P^*(1)$. Consequently, if the map $\theta \mapsto P m_\theta$ is twice continuously differentiable at θ_0 with nonsingular second derivative matrix V ,[†] then any $\hat{\theta}_n$ such that $\mathbb{P}_n m_{\hat{\theta}_n} \geq \sup_\theta \mathbb{P}_n m_\theta - o_P(n^{-1})$ and $\hat{\theta}_n = \theta_0 + o_P^*(1)$ satisfy $\sqrt{n}(\hat{\theta}_n - \theta_0) = -V^{-1} \mathbb{G}_n \dot{m}_{\theta_0} + o_P^*(1)$.

Proof. Without loss of generality, assume that $\tilde{\theta}_n$ takes its values in $\Theta_\delta = \{\theta: \|\theta - \theta_0\| < \delta\}$. Define a function $f: \ell^\infty(\Theta_\delta) \times \Theta_\delta \rightarrow \mathbb{R}$ by $f(z, \theta) = z(\theta)$. This function is continuous at every point (z, θ_0) such that $\theta \mapsto z(\theta)$ is continuous at θ_0 .

Define a stochastic process Z_n indexed by Θ_δ by

$$Z_n(\theta) = \mathbb{G}_n \frac{m_\theta - m_{\theta_0} - (\theta - \theta_0)' \dot{m}_{\theta_0}}{\|\theta - \theta_0\|}.$$

[†] It suffices that a two-term Taylor expansion is valid at θ_0 .

By assumption, the sequence Z_n converges in $\ell^\infty(\Theta_\delta)$ to a tight Gaussian process Z . This process has continuous sample paths with respect to the semimetric ρ given by

$$\rho^2(\theta_1, \theta_2) = E(Z(\theta_1) - Z(\theta_2))^2.$$

By assumption, $\rho(\theta, \theta_0) \rightarrow 0$ if $\theta \rightarrow \theta_0$. Thus, almost all sample paths of Z are continuous at θ_0 .

By Slutsky's lemma $(Z_n, \tilde{\theta}_n) \rightsquigarrow (Z, \theta_0)$. By the continuous mapping theorem, $Z_n(\tilde{\theta}_n) = f(Z_n, \tilde{\theta}_n) \rightsquigarrow f(Z, \theta_0) = 0$. Since convergence in distribution to a constant is the same as convergence in probability, it follows that (3.2.19) holds with remainder term $o_P^*(\|\tilde{\theta}_n - \theta_0\|)$. ■

The conditions of the theorem require some type of differentiability of the map $\theta \mapsto m_\theta$ but are weak enough to allow the treatment of slightly irregular functions, such as the absolute value $x \mapsto |x - \theta|$.

The presence of the factor $\|\theta - \theta_0\|$ in the denominator appears a bit awkward. It would be unnecessary to handle this term if the rate of convergence could first be established by a different method. This may motivate separating also in this approach deriving the limit distribution from establishing the rate of convergence. If it is known that the maximizing sequence $\hat{\theta}_n$ is $\sqrt{n}$ -consistent, then (3.2.19) need hold only for sequences $\tilde{\theta}_n = \theta_0 + O_P(n^{-1/2})$ and can be replaced by

$$(3.2.21) \quad \mathbb{G}_n \sqrt{n}(m_{\tilde{\theta}_n} - m_{\theta_0}) = \sqrt{n}(\tilde{\theta}_n - \theta_0)' \mathbb{G}_n \dot{m}_{n, \theta_0} + o_P^*(1).$$

Here, for further generality, we have replaced the derivative $\dot{m}_{\theta_0}$ by a vector-valued function $\dot{m}_{n, \theta_0}$ that may depend on n . If the map $\theta \mapsto m_\theta$ is differentiable in quadratic mean at θ_0 , then the latter map may be taken equal to the derivative $\dot{m}_{\theta_0}$, but we can also, and will, use the functions with i th coordinate function equal to $\sqrt{n}(m_{\theta_0 + e_i n^{-1/2}} - m_{\theta_0})$, without a further differentiability condition.[‡]

The following theorem gives a simple sufficient condition for this type of differentiability.

3.2.22 Theorem (M-estimator-asymptotic normality). *Suppose that the classes of functions $\mathcal{M}_\delta = \{m_\theta - m_{\theta_0} : \|\theta - \theta_0\| < \delta\}$ satisfy the entropy condition (3.2.8) or (3.2.9) relative to envelope functions M_δ such that $P^* M_\delta^2 \leq \delta^2$ and $P^* M_\delta^2 \{M_\delta > \eta\} = o(\delta^2)$, for every $\eta > 0$, and assume that, for all K and some linear, zero-mean Gaussian process G ,*

$$\lim_{\varepsilon \downarrow 0} \limsup_{\delta \downarrow 0} \sup_{\substack{\|h-g\| < \varepsilon \\ \|g\| \vee \|h\| \leq K}} \frac{P(m_{\theta_0+\delta g} - m_{\theta_0+\delta h})^2}{\delta^2} = 0,$$

$$\lim_{\delta \downarrow 0} \frac{P(m_{\theta_0+\delta g} - m_{\theta_0+\delta h})^2}{\delta^2} = E(G(h) - G(g))^2.$$

[‡] If any functions $\dot{m}_{n, \theta_0}$ work, then so do the functions $\sqrt{n}(m_{\theta_0 + e_i n^{-1/2}} - m_{\theta_0})$.

Then (3.2.21) holds with $(\dot{m}_{n,\theta_0})_i = \sqrt{n}(m_{\theta_0+e_i n^{-1/2}} - m_{\theta_0})$, for every sequence $\hat{\theta}_n = \theta_0 + O_P(n^{-1/2})$. Consequently, if the map $\theta \mapsto Pm_\theta$ is twice continuously differentiable at θ_0 with nonsingular derivative matrix V , then any $\hat{\theta}_n$ such that $\mathbb{P}_n m_{\hat{\theta}_n} \geq \sup_\theta \mathbb{P}_n m_\theta - o_P(n^{-1})$ and $\hat{\theta}_n = \theta_0 + o_P(1)$, satisfy $\sqrt{n}(\hat{\theta}_n - \theta_0) = -V^{-1}\mathbb{G}_n \dot{m}_{n,\theta_0} + o_P(1)$. Moreover, the sequence $\mathbb{G}_n \dot{m}_{n,\theta_0}$ tends in distribution to the vector with i th coordinate $G(e_i)$.

Proof. To see the validity of (3.2.21), it suffices to show that the sequence of processes

$$h \mapsto \mathbb{G}_n(\sqrt{n}(m_{\theta_0+h/\sqrt{n}} - m_{\theta_0}) - h' \dot{m}_{n,\theta_0})$$

converges in probability to zero in the space $\ell^\infty(h: \|h\| \leq K)$, for every K . The sequence of processes $h \mapsto \mathbb{G}_n \sqrt{n}(m_{\theta_0+h/\sqrt{n}} - m_{\theta_0})$ can be shown to be asymptotically tight by application of Theorem 2.11.24 or 2.11.25. Taking $h = e_i$ shows that the sequence $\mathbb{G}_n \dot{m}_{n,\theta_0}$ is asymptotically tight, and hence so is the sequence of processes $h \mapsto h' \mathbb{G}_n \dot{m}_{n,\theta_0}$, for a general h . It now suffices to show that the processes in the preceding display converge marginally to zero.

By using the formula $2\langle x, y \rangle = \|x - y\|^2 - \|x\|^2 - \|y\|^2$, we can deduce from the first displayed condition that the covariance functions of the stochastic processes $h \mapsto \sqrt{n}(m_{\theta_0+h/\sqrt{n}} - m_{\theta_0})(X_1)$ converge to the covariance function of G . Then, by the definition of $\dot{m}_{n,\theta_0}$,

$$P(\sqrt{n}(m_{\theta_0+h/\sqrt{n}} - m_{\theta_0}) - h' \dot{m}_{n,\theta_0})^2 \rightarrow E(G(h) - \sum h_i G(e_i))^2.$$

The variable in the right side is zero for every h , by the linearity of G . Therefore the left side tends to zero. An application of Markov's inequality yields marginal convergence to zero of the processes in the first paragraph, and concludes the proof of (3.2.21).

For the final assertion of the theorem, note first that the rate of convergence of $\hat{\theta}_n$ is $\sqrt{n}$ by Theorem 3.2.5 (or Theorem 3.2.10). Thus, the asymptotic approximation to $\sqrt{n}(\hat{\theta}_n - \theta_0)$ follows by Theorem 3.2.16. That $(\mathbb{G}_n \dot{m}_{n,\theta_0})_i \rightsquigarrow G(e_i)$ follows by the Lindeberg-Feller theorem, the bound $|(\dot{m}_{n,\theta_0})_i| \leq \delta^{-1} M_\delta$, for $\delta = n^{-1}$, and the assumed Lindeberg condition on the envelope functions. ■

The two lines of the display in the theorem are implied by the single condition: for every $g_\delta \rightarrow g$ and $h_\delta \rightarrow h$,

$$(3.2.23) \quad \lim_{\delta \downarrow 0} \frac{1}{\delta^2} P(m_{\theta_0+\delta g_\delta} - m_{\theta_0+\delta h_\delta})^2 = E(G(g) - G(h))^2.$$

Instead of the variances of the differences, the condition could also be stated in terms of the covariances of m_{θ_1} and m_{θ_2} . In particular, the condition is implied by twice differentiability of the map $(\theta_1, \theta_2) \mapsto Pm_{\theta_1} m_{\theta_2}$ at (θ_0, θ_0) (see Problem 3.2.1).

3.2.24 Corollary. For every θ in an open subset of a Euclidean space let $x \mapsto m_\theta(x)$ be a measurable function such that $\theta \mapsto Pm_\theta$ is twice continuously differentiable at a point of maximum θ_0 with nonsingular second derivative matrix $V = W^{-1}$, and such that (3.2.23) holds for a linear, zero-mean Gaussian process G . Furthermore, suppose that the classes of functions $\mathcal{M}_\delta = \{m_\theta - m_{\theta_0} : \|\theta - \theta_0\| < \delta\}$ satisfy the entropy condition (3.2.8) or (3.2.9) relative to envelope functions M_δ such that $P^*M_\delta^2 \leq \delta^2$ and $P^*M_\delta^2\{M_\delta > \eta\} = o(\delta^2)$, for every $\eta > 0$. If $\hat{\theta}_n$ satisfies $\mathbb{P}_n m_{\hat{\theta}_n} \geq \sup_\theta \mathbb{P}_n m_\theta - o_P(n^{-1})$ and $\hat{\theta}_n = \theta_0 + o_P(1)$, then $\sqrt{n}(\hat{\theta}_n - \theta_0)$ converges in distribution to a multivariate zero-mean normal distribution with covariance matrix $(\sum_k \sum_l W_{ik} W_{jl} \text{EG}(e_k) G(e_l))$.

3.2.25 Example (Lipschitz in parameter). Let $X_1, \dots, X_n$ be i.i.d. random variables with common law P , and let m_θ be measurable functions indexed by a parameter θ ranging over an open subset of Euclidean space. Assume that, for every θ_1, θ_2 in a neighborhood of θ_0 ,

$$|m_{\theta_1}(x) - m_{\theta_2}(x)| \leq \dot{m}(x) \|\theta_1 - \theta_2\|,$$

$$P[m_\theta - m_{\theta_0} - (\theta - \theta_0)' \dot{m}_{\theta_0}]^2 = o(\|\theta - \theta_0\|^2),$$

for functions $\dot{m}$ and $\dot{m}_{\theta_0}$ with $P\dot{m}^2 < \infty$. Furthermore, assume that the map $\theta \mapsto Pm_\theta$ is twice continuously differentiable at a point of maximum θ_0 with nonsingular second-derivative matrix V . If $\hat{\theta}_n$ maximizes $\theta \mapsto \mathbb{P}_n m_\theta$ up to an $o_P(1/n)$ -term and is consistent for θ_0 , then $\sqrt{n}(\hat{\theta}_n - \theta_0) = -V^{-1} \mathbb{G}_n \dot{m}_{\theta_0} + o_P^*(1)$.

This follows from the preceding theorem. By the Lipschitz condition, the $L_2(P)$ -covering numbers of the classes $\mathcal{M}_\delta$ are polynomial and the envelopes can be taken equal to $M_\delta = \dot{m}\delta$. In view of the differentiability of $\theta \mapsto m_\theta$ in square mean, the process G can be taken as $G(h) = h' \Delta$ for normally distributed vector Δ with mean zero and covariance matrix $P\dot{m}_{\theta_0} \dot{m}'_{\theta_0}$.

It may be noted that $P\dot{m}_{\theta_0} = 0$, because this is the first derivative at θ_0 of the map $\theta \mapsto Pm_\theta$. Hence the sequence $\mathbb{G}_n \dot{m}_{\theta_0}$ is asymptotically zero-mean normal.

3.2.26 Example (Least absolute deviation regression). Given independent i.i.d. random vectors $X_1, \dots, X_n$ and $e_1, \dots, e_n$ in $\mathbb{R}^d$ and $\mathbb{R}$, respectively, let $Y_i = \theta'_0 X_i + e_i$. The least-absolute-deviation estimator $\hat{\theta}_n$ minimizes the function

$$\theta \mapsto \frac{1}{n} \sum_{i=1}^n |Y_i - \theta' X_i| = \mathbb{P}_n m_\theta,$$

where $\mathbb{P}_n$ is the empirical measure of the pairs (X_i, Y_i) and $m_\theta(x, y) = |y - \theta' x|$.

The parameter θ_0 is a point of minimum of the map $\theta \mapsto P|Y - \theta' X|$ if the distribution of the errors e_i has median zero. (Write the expectation as $E_X E_e |e - (\theta - \theta_0)' X|$ and note that the inner integral is minimized by θ_0 for every fixed value of X .) Furthermore, the maps $\theta \mapsto m_\theta$ satisfy the conditions in Example 3.2.25:

$$||y - \theta'_1 x| - |y - \theta'_2 x|| \leq \|\theta_1 - \theta_2\| \|x\|,$$

$$P\left[|Y - \theta' X| - |Y - \theta'_0 X| - (\theta - \theta_0)' X \text{ sign}(Y - \theta'_0 X)\right]^2 = o(\|\theta - \theta_0\|^2).$$

The consistency of the least-absolute-deviation estimator can be argued from the convexity of the map $\theta \mapsto |y - \theta'x|$. The map $\theta \mapsto P|Y - \theta'X|$ is twice differentiable at θ_0 if the distribution of the errors has a positive density at its median.

The final conclusion is that the sequence $\sqrt{n}(\hat{\theta}_n - \theta_0)$ is asymptotically normal with mean zero and covariance matrix $V^{-1}E(XX')V^{-1}$, for V the second-derivative matrix of $\theta \mapsto P|Y - \theta'X|$ at θ_0 .

3.2.27 Example (Maximum likelihood). Let $X_1, \dots, X_n$ be an i.i.d. sample from a density p_θ with respect to a measure μ on a measurable space $(\mathcal{X}, \mathcal{A})$. The parameter θ ranges over an open subset of Euclidean space. Assume that, for θ_1, θ_2 in a neighborhood of θ_0 ,

$$|\log p_{\theta_1}(x) - \log p_{\theta_2}(x)| \leq \ell(x) \|\theta_1 - \theta_2\|,$$

for a function ℓ with $P_{\theta_0}\ell^2 < \infty$. Furthermore, assume that the map $\theta \mapsto P_\theta \log p_\theta$ is twice continuously differentiable at θ_0 and that as $\theta \mapsto \theta_0$

$$\int \left[p_\theta^{1/2} - p_{\theta_0}^{1/2} - \frac{1}{2}(\theta - \theta_0)' \dot{\ell}_{\theta_0} p_{\theta_0}^{1/2} \right]^2 d\mu = o(\|\theta - \theta_0\|^2),$$

for some measurable vector-valued function $\dot{\ell}_{\theta_0}$. Under these conditions, if the maximum likelihood estimator $\hat{\theta}_n$ for θ_0 is consistent, then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = I_{\theta_0}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \dot{\ell}_{\theta_0}(X_i) + o_{P_{\theta_0}}^*(1),$$

provided the Fisher information matrix $I_{\theta_0} = P_{\theta_0} \dot{\ell}_{\theta_0} \dot{\ell}_{\theta_0}'$ is nonsingular.

Since the maximum likelihood estimator is an M -estimator for $m_\theta = \log p_\theta$, its asymptotic normality also follows under the conditions of Example 3.2.25 for general Lipschitz criterion functions. The present statement is better in two respects. First, the differentiability of the log density in quadratic mean is replaced by the differentiability of the root density, which has long been known to be the “right” condition for asymptotic theory.^b Second, the present statement gives the inverse of the Fisher information matrix as the asymptotic covariance matrix, rather than a matrix $V^{-1}I_{\theta_0}V^{-1}$, which would next have to be checked to be equal to $I_{\theta_0}^{-1}$ under additional conditions.

The claim can be proved as follows. Define a zero-mean stochastic process $\mathbb{Z}_n$ by

$$\mathbb{Z}_n(h) = n(\mathbb{P}_n - P_{\theta_0}) \left(\log \frac{p_{\theta_0+h/\sqrt{n}}}{p_{\theta_0}} - \frac{h' \dot{\ell}_{\theta_0}}{\sqrt{n}} \right).$$

The differentiability of the root density implies that the score is mean-zero: $P_{\theta_0} \dot{\ell}_{\theta_0} = 0$ as well as local asymptotic normality. The latter means that for every h ,

$$\begin{aligned} \mathbb{Z}_n(h) &= -\frac{1}{2} h' I_{\theta_0} h - n P_{\theta_0} \log \frac{p_{\theta_0+h/\sqrt{n}}}{p_{\theta_0}} + o_{P_{\theta_0}}(1) \\ &= -\frac{1}{2} h' I_{\theta_0} h - \frac{1}{2} h' V h + o_{P_{\theta_0}}(1). \end{aligned}$$

^b Le Cam (1960).

Here V is the second-derivative matrix of the map $\theta \mapsto P_{\theta_0} \log p_{\theta}$ at $\theta = \theta_0$. The second moment of $\mathbb{Z}_n(h)$ is equal to its variance and satisfies

$$\mathbb{E} \mathbb{Z}_n^2(h) \leq P_{\theta_0} \left(\sqrt{n} \log \frac{p_{\theta_0+h/\sqrt{n}}}{p_{\theta_0}} - h' \dot{\ell}_{\theta_0} \right)^2 \lesssim P_{\theta_0} (\dot{\ell}^2 + \|\dot{\ell}_{\theta_0}\|^2) \|h\|^2.$$

Since this is uniformly bounded in n , it follows that the sequence of means $\mathbb{E} \mathbb{Z}_n(h) = 0$ converges to the mean of the limit in probability, $-\frac{1}{2} h' I_{\theta_0} h - \frac{1}{2} h' V h$ of the sequence $\mathbb{Z}_n(h)$. Since this is true for every h , it follows that $V = -I_{\theta_0}$.

In view of Corollary 3.2.6, Theorem 3.2.10, or Example 3.2.12, the rate of the maximum likelihood estimator is $\sqrt{n}$. In view of the discussion following Theorem 3.2.16, it now suffices to verify condition (3.2.21) with $\dot{m}_{n,\theta_0} = \dot{\ell}_{\theta_0}$. In the present situation this is implied by the sequence of random variables $\sup_{\|h\| \leq K} |\mathbb{Z}_n(h)|$ converging to zero in distribution for every K . Since it was seen that the sequence of processes $\{\mathbb{Z}_n(h) : \|h\| \leq K\}$ converges marginally in probability to zero, it suffices to show that the sequence is asymptotically tight in the space of bounded functions. In view of the Lipschitz condition on the maps $\theta \mapsto \log p_{\theta}$, the bracketing numbers $N_{[]}(\varepsilon \|M_n\|_{P_{\theta_0,2}}, \mathcal{M}_n, L_2(P_{\theta_0}))$ of the classes of functions

$$\mathcal{M}_n = \left\{ \sqrt{n} \log \frac{p_{\theta_0+h/\sqrt{n}}}{p_{\theta_0}} : \|h\| \leq K \right\}, \quad M_n = K \dot{\ell},$$

are uniformly bounded by the covering numbers of a Euclidean ball. The asymptotic tightness follows from Theorem 2.11.25.

Problems and Complements

1. Assume one of the entropy conditions (3.2.8) or (3.2.9) for the class of functions $\mathcal{M}_{\delta} = \{m_{\theta} - m_{\theta_0} : \|\theta - \theta_0\| < \delta\}$ with envelope function M_{δ} . Furthermore, suppose that, for every $\eta > 0$,

$$P^* M_{\delta}^2 \leq \delta^2; \quad P^* M_{\delta}^2 \{M_{\delta} > \eta\} = o(\delta^2).$$

Finally, suppose that the maps $\theta \mapsto P m_{\theta}$ and $(\theta_1, \theta_2) \mapsto P m_{\theta_1} m_{\theta_2}$ are twice continuously differentiable at θ_0 and (θ_0, θ_0) , respectively, the first one with nonsingular second-derivative matrix V . Then a (near) maximizer $\hat{\theta}_n$ of $\theta \mapsto \mathbb{P}_n m_{\theta}$ satisfies $\sqrt{n}(\hat{\theta}_n - \theta_0) = -V^{-1} \mathbb{G}_n \dot{m}_{n,\theta_0} + o_P^*(1)$ provided it converges to θ_0 in outer probability. Here $\mathbb{G}_n \dot{m}_{n,\theta_0}$ converges in distribution to a normal distribution with mean zero and covariance matrix the derivative $\partial^2 / \partial \theta_1 \partial \theta_2 P m_{\theta_1} m_{\theta_2}$ evaluated at (θ_0, θ_0) .

2. If $M: \Theta \rightarrow \mathbb{R}$ is upper semicontinuous on a compact metric space Θ , then it achieves its maximum value. If it achieves its maximum value at a unique point θ_0 , then $M(\theta_0) > \sup_{\theta \notin G} M(\theta)$ for every open set G around θ_0 .
3. **(Consistency and Lipschitz criterion functions)** Let Θ be a measurable subset of Euclidean space and $\mathbb{M}_n$ be separable stochastic processes that converge pointwise in probability to a fixed function $\mathbb{M}$. Suppose that

$$|\mathbb{M}_n(\theta_1) - \mathbb{M}_n(\theta_2)| \leq \dot{\mathbb{M}}_n \|\theta_1 - \theta_2\|,$$

for random variables such that $\sup_n \mathbb{M}_n < \infty$ almost surely. Let $\theta \mapsto \mathbb{M}(\theta)$ be upper semicontinuous and possess a unique maximum at θ_0 . If $\hat{\theta}_n$ (nearly) maximizes $\theta \mapsto \mathbb{M}_n(\theta)$ and $\hat{\theta}_n = O_P(1)$, then $\hat{\theta}_n \rightarrow \theta_0$ in probability.

[Hint: Almost every sequence of sample paths $\theta \mapsto \mathbb{M}_n(\theta, \omega)$ is uniformly bounded and uniformly equicontinuous on compacta. By the Arzelà-Ascoli theorem almost every sequence is relatively compact for the topology of uniform convergence on compacta. Conclude that the sequence $\|\mathbb{M}_n - \mathbb{M}\|_K$ converges to zero in probability for every compact K . Apply part (ii) of Corollary 3.2.3.]

4. The argument in the last paragraph of Example 3.2.18 can be refined to yield immediately that $r_n(\hat{\theta}_n - \theta_0) = -V^{-1}Z_n$, without interception of Theorem 3.2.16.

[Hint: Use spheres of radius ε , or Problem 3.2.6.]

5. Let $\{\mathbb{Z}(h): h \in \mathbb{R}\}$ be a standard two-sided Brownian motion with $\mathbb{Z}(0) = 0$. (The process is zero-mean Gaussian and the increment $\mathbb{Z}(g) - \mathbb{Z}(h)$ has variance $|g - h|$.) Then $\operatorname{argmax}_h \{a\mathbb{Z}(h) - bh^2 - ch\}$ is equal in distribution to $(a/b)^{2/3} \operatorname{argmax}_g \{\mathbb{Z}(g) - g^2\} - \frac{1}{2}c/b$. Here a , b , and c are positive constants.

[Hint: The process $h \mapsto \mathbb{Z}(\sigma h - \mu)$ is equal in distribution to the process $h \mapsto \sqrt{\sigma} \mathbb{Z}(h) - \mathbb{Z}(\mu)$. Apply the change of variable $h \mapsto (a/b)^{2/3}g - \frac{1}{2}c/b$ and note that the location of a maximum does not change by multiplication by a positive constant or a vertical shift.]

6. (**Concave criterion**) Let $\hat{h} \in C$ be a point of maximum of the concave criterion function $h \mapsto \mathbb{M}_n(h)$ on a convex subset C of a normed space and let $\tilde{h} \in C$ and $h \mapsto M_n(h)$ be arbitrary. Then $\|\hat{h} - \tilde{h}\| > \delta$ implies that

$$2 \sup_{\|h - \tilde{h}\| \leq \delta} |(\mathbb{M}_n - M_n)(h)| \geq M_n(\tilde{h}) - \sup_{\|h - \tilde{h}\| = \delta} M_n(h).$$

Thus if M_n has a maximum at $\tilde{h}$ and $\mathbb{M}_n - M_n \xrightarrow{P^*} 0$ uniformly on balls around $\tilde{h}$, then $\hat{h} - \tilde{h} \xrightarrow{P^*} 0$.

[Hint: If $\|\hat{h} - \tilde{h}\| > \delta$, then there is $\lambda \in (0, 1)$ such that $h_\lambda = \lambda\hat{h} + (1 - \lambda)\tilde{h}$ satisfies $\|h_\lambda - \tilde{h}\| = \delta$. By concavity $\mathbb{M}_n(h_\lambda) - \mathbb{M}_n(\tilde{h}) \geq \lambda(\mathbb{M}_n(\hat{h}) - \mathbb{M}_n(\tilde{h})) \geq 0$, by the definition of $\hat{h}$. The left side of the inequality is bounded below by $M_n(h_\lambda) - M_n(\tilde{h}) + 2\Delta_n$, for $\Delta_n = \sup_{\|h - \tilde{h}\| \leq \delta} |(\mathbb{M}_n - M_n)(h)|$. Thus $2\Delta_n \geq M_n(\tilde{h}) - M_n(h_\lambda)$. (Cf. Hjort and Pollard (1993).)]

3.3

Z-Estimators

Let the parameter set Θ be a subset of a Banach space, and let

$$\Psi_n: \Theta \rightarrow \mathbb{L}, \quad \Psi: \Theta \rightarrow \mathbb{L}$$

be random maps and a deterministic map, respectively, with values in another Banach space $\mathbb{L}$. Here “random maps” means that each $\Psi_n(\theta)$ is defined on the product of Θ and some probability space. The dependence on the probability space is suppressed in the notation.

In this chapter we prove asymptotic normality of “estimators” $\hat{\theta}_n$ that (approximately) satisfy the equation

$$\Psi_n(\hat{\theta}_n) = 0.$$

Such estimators will be referred to as *Z-estimators*.

If $\mathbb{L}$ is an $\ell^\infty(\mathcal{H})$ -space, as can be assumed without loss of generality, the equation is equivalent to the collection of (real-valued) estimating equations $\Psi_n(\hat{\theta}_n)h = 0$, when h runs through $\mathcal{H}$. If θ is finite-dimensional, then the number of estimating equations is typically chosen equal to the dimension, and the space $\ell^\infty(\mathcal{H})$ can be identified with Euclidean space. In the case of an infinite-dimensional parameter, infinitely many estimating equations may be used. Letting Ψ be the “asymptotic version” of Ψ_n as n tends to infinity, we may hope that $\hat{\theta}_n$ tends to a value θ_0 satisfying

$$\Psi(\theta_0) = 0.$$

In the following, “consistency” $\hat{\theta}_n \xrightarrow{P^*} \theta_0$ is assumed from the start. We study conditions under which $\sqrt{n}(\hat{\theta}_n - \theta_0)$ converges in distribution.

The set Θ should be chosen to contain the “true value” θ_0 and the possible values of the estimators $\hat{\theta}_n$, at least asymptotically. The estimators must be defined on the same probability space as the criterion maps Ψ_n . They need not be defined

by the estimating equation and need not be measurable. Also, θ_0 need not be an interior point of the parameter set Θ and Θ need not be a linear space.

We assume that the map Ψ is Fréchet-differentiable at θ_0 . This entails that

$$\|\Psi(\theta) - \Psi(\theta_0) - \dot{\Psi}_{\theta_0}(\theta - \theta_0)\| = o(\|\theta - \theta_0\|),$$

as $\theta \rightarrow \theta_0$ within Θ , for a continuous, linear, one-to-one map $\dot{\Psi}_{\theta_0}: \text{lin } \Theta \rightarrow \mathbb{L}$. We also make the crucial assumption that the inverse $\dot{\Psi}_{\theta_0}^{-1}$ of the derivative exists and is continuous on the range of $\dot{\Psi}_{\theta_0}$. A linear operator on a finite-dimensional space is automatically continuous; in this case the assumption is simply that $\theta \mapsto \Psi(\theta)$ is differentiable at θ_0 with a nonsingular derivative matrix $\dot{\Psi}_{\theta_0}$. In the infinite-dimensional case, the assumption that the derivative $\dot{\Psi}_{\theta_0}$ is continuously invertible is harder to ascertain. If the inverse $\dot{\Psi}_{\theta_0}^{-1}$ is continuous on the range of $\dot{\Psi}_{\theta_0}$, then it has a unique continuous extension to the closure of this range. This will also be denoted $\dot{\Psi}_{\theta_0}^{-1}$ and is the inverse of the unique continuous extension of $\dot{\Psi}_{\theta_0}$ to the closure of $\text{lin } \Theta$.

3.3.1 Theorem (Z-estimator-asymptotic distribution). *Let Ψ_n and Ψ be random maps and a fixed map, respectively, from Θ into a Banach space such that*

$$(3.3.2) \quad \sqrt{n}(\Psi_n - \Psi)(\hat{\theta}_n) - \sqrt{n}(\Psi_n - \Psi)(\theta_0) = o_P^*(1 + \sqrt{n}\|\hat{\theta}_n - \theta_0\|),$$

and such that the sequence $\sqrt{n}(\Psi_n - \Psi)(\theta_0)$ converges in distribution to a tight random element Z . Let $\theta \mapsto \Psi(\theta)$ be Fréchet-differentiable at θ_0 with a continuously invertible derivative $\dot{\Psi}_{\theta_0}$. If $\Psi(\theta_0) = 0$ and $\hat{\theta}_n$ satisfies $\Psi_n(\hat{\theta}_n) = o_P^(n^{-1/2})$ and converges in outer probability to θ_0 , then*

$$\sqrt{n} \dot{\Psi}_{\theta_0}(\hat{\theta}_n - \theta_0) = -\sqrt{n}(\Psi_n - \Psi)(\theta_0) + o_P^*(1).$$

Consequently, $\sqrt{n}(\hat{\theta}_n - \theta_0) \rightsquigarrow -\dot{\Psi}_{\theta_0}^{-1} Z$. If it is known that the sequence $\sqrt{n}\|\hat{\theta}_n - \theta_0\|$ is asymptotically tight, then the first conclusion is valid without the assumption of continuous invertibility of $\dot{\Psi}_{\theta_0}$. If it is known that $\sqrt{n}(\hat{\theta}_n - \theta_0)$ is asymptotically tight, then it suffices that Ψ is Hadamard-differentiable.

Proof. By the definition of $\hat{\theta}_n$

$$(3.3.3) \quad \begin{aligned} \sqrt{n}(\Psi(\hat{\theta}_n) - \Psi(\theta_0)) &= \sqrt{n}(\Psi(\hat{\theta}_n) - \Psi_n(\hat{\theta}_n)) + o_P^*(1), \\ &= -\sqrt{n}(\Psi_n - \Psi)(\theta_0) + o_P^*(1 + \sqrt{n}\|\hat{\theta}_n - \theta_0\|), \end{aligned}$$

by the first displayed assumption of the theorem. Since the derivative $\dot{\Psi}_{\theta_0}$ is continuously invertible, there exists a positive constant c such that $\|\dot{\Psi}_{\theta_0}(\theta - \theta_0)\| \geq c\|\theta - \theta_0\|$ for every θ and θ_0 . This can be combined with the differentiability of Ψ to yield

$$\|\Psi(\theta) - \Psi(\theta_0)\| \geq c\|\theta - \theta_0\| + o(\|\theta - \theta_0\|).$$

Apply this to (3.3.3) to find that

$$\sqrt{n}\|\hat{\theta}_n - \theta_0\|(c + o_P(1)) \leq O_P(1) + o_P(1 + \sqrt{n}\|\hat{\theta}_n - \theta_0\|).$$

This implies that the sequence $\hat{\theta}_n$ is $\sqrt{n}$ -consistent for θ_0 in norm. By the differentiability of Ψ , the left side of (3.3.3) can be replaced by

$$\sqrt{n}\dot{\Psi}_{\theta_0}(\hat{\theta}_n - \theta_0) + o_P^*(\sqrt{n}\|\hat{\theta}_n - \theta_0\|).$$

The last term is $o_P(1)$ as is the remainder on the right side of (3.3.3). The second assertion of the theorem follows. Next the continuity of $\dot{\Psi}_{\theta_0}^{-1}$ and the continuous mapping theorem give the weak convergence of the sequence $\sqrt{n}(\hat{\theta}_n - \theta_0)$.

Under the additional assumptions on the sequence $\hat{\theta}_n$, the preceding proof can be simplified. ■

The preceding theorem separates the conditions for asymptotic normality into the stochastic condition (3.3.2), which requires that the remainder in a Taylor expansion is negligible, and analytical conditions on the asymptotic criterion function Ψ .

3.3.1 I.i.d. observations

In the case of i.i.d. observations in a sample space $(\mathcal{X}, \mathcal{A})$, the theorem may be applied with

$$\Psi_n(\theta)h = \mathbb{P}_n\psi_{\theta,h}, \quad \text{and} \quad \Psi(\theta)h = P\psi_{\theta,h},$$

for given measurable functions $\psi_{\theta,h}; \mathcal{X} \rightarrow \mathbb{R}$ indexed by Θ and an arbitrary index set $\mathcal{H}$. In this case $\sqrt{n}(\Psi_n - \Psi)(\theta) = \{\mathbb{G}_n\psi_{\theta,h}; h \in \mathcal{H}\}$ is the empirical process indexed by the class of functions $\{\psi_{\theta,h}; h \in \mathcal{H}\}$. Then the stochastic condition reduces to

$$(3.3.4) \quad \|\mathbb{G}_n(\psi_{\hat{\theta}_n,h} - \psi_{\theta_0,h})\|_{\mathcal{H}} = o_P^*(1 + \sqrt{n}\|\hat{\theta}_n - \theta_0\|).$$

It is a stronger condition to require that the left-hand side is $o_P^*(1)$. The following lemma gives a simple sufficient condition.

3.3.5 Lemma. *Suppose that the class of functions*

$$\left\{ \psi_{\theta,h} - \psi_{\theta_0,h}; \|\theta - \theta_0\| < \delta, h \in \mathcal{H} \right\}$$

is P -Donsker for some $\delta > 0$ and that

$$\sup_{h \in \mathcal{H}} P(\psi_{\theta,h} - \psi_{\theta_0,h})^2 \rightarrow 0, \quad \theta \rightarrow \theta_0.$$

If $\hat{\theta}_n$ converges in outer probability to θ_0 , then (3.3.4) is satisfied. In fact, the left-hand side of (3.3.4) is $o_P^(1)$.*

Proof. Without loss of generality, assume that $\hat{\theta}_n$ takes its values in $\Theta_\delta = \{\theta: \|\theta - \theta_0\| < \delta\}$. Define a function $f: \ell^\infty(\Theta_\delta \times \mathcal{H}) \times \Theta_\delta \rightarrow \ell^\infty(\mathcal{H})$ by $f(z, \theta)h = z(\theta, h)$. This function is continuous at every point (z, θ_0) such that $\|z(\theta, h) - z(\theta_0, h)\|_{\mathcal{H}} \rightarrow 0$ as $\theta \rightarrow \theta_0$.

Define a stochastic process Z_n indexed by $\Theta_\delta \times \mathcal{H}$ by

$$Z_n(\theta, h) = \mathbb{G}_n(\psi_{\theta, h} - \psi_{\theta_0, h}).$$

By assumption, the sequence Z_n converges in $\ell^\infty(\Theta_\delta \times \mathcal{H})$ to a tight Gaussian process Z . This process has continuous sample paths with respect to the semimetric ρ given by

$$\rho^2((\theta_1, h_1), (\theta_2, h_2)) = P(\psi_{\theta_1, h_1} - \psi_{\theta_0, h_1} - \psi_{\theta_2, h_2} + \psi_{\theta_0, h_2})^2.$$

By assumption, $\sup_h \rho((\theta, h), (\theta_0, h)) \rightarrow 0$ if $\theta \rightarrow \theta_0$. Conclude that the function f is continuous at almost all sample paths of Z .

By Slutsky's lemma, $(Z_n, \hat{\theta}_n) \rightsquigarrow (Z, \theta_0)$. By the continuous mapping theorem, $Z_n(\hat{\theta}_n) = f(Z_n, \hat{\theta}_n) \rightsquigarrow f(Z, \theta_0) = 0$ in $\ell^\infty(\mathcal{H})$. ■

It appears that the conditions of the preceding lemma are relatively weak. In the case of infinite-dimensional parameters, verification of the analytical conditions may require more effort. The assertion of the theorem becomes

$$(3.3.6) \quad \sqrt{n}(\hat{\theta}_n - \theta_0) = -\dot{\Psi}_{\theta_0}^{-1} \mathbb{G}_n \psi_{\theta_0} + o_P^*(1),$$

where $\dot{\Psi}_{\theta_0}$ is the derivative of the map $\theta \mapsto P\psi_\theta$. Note that $P\psi_{\theta_0} = 0$ by the “definition” of θ_0 .

3.3.7 Example (Lipschitz functions). Let $X_1, \dots, X_n$ be i.i.d. random variables with common law P , and let ψ_θ be measurable vector-valued functions indexed by a Euclidean parameter θ . (Thus, we take $\mathcal{H}$ to be a finite set, whose cardinality is the dimension of θ .) Assume that, for every θ_1, θ_2 in a neighborhood of θ_0 ,

$$\|\psi_{\theta_1}(x) - \psi_{\theta_2}(x)\| \leq \dot{\psi}(x) \|\theta_1 - \theta_2\|,$$

for some measurable function $\dot{\psi}$ with $P\dot{\psi}^2 < \infty$. Let the map $\theta \mapsto P\psi_\theta$ be differentiable at its zero θ_0 with a nonsingular derivative. Then the (near) zeros $\hat{\theta}_n$ of the map $\theta \mapsto \mathbb{P}_n \psi_\theta$ satisfy (3.3.6), provided the sequence $\hat{\theta}_n$ converges in outer probability to θ_0 .

This is a consequence of the preceding theorem. The class of functions $\{\psi_\theta - \psi_{\theta_0} : \|\theta - \theta_0\| < \delta\}$ is Donsker for sufficiently small $\delta > 0$, because its bracketing integral is finite. Furthermore, $P\|\psi_\theta - \psi_{\theta_0}\|^2 \leq P\dot{\psi}^2 \|\theta - \theta_0\|^2$ converges to zero as $\theta \rightarrow \theta_0$.

The pointwise local Lipschitz condition in the preceding example is satisfied if for each x the map $\theta \mapsto \psi_\theta(x)$ is differentiable in a neighborhood of θ_0 with derivative matrices $\dot{\psi}_\theta(x)$ such that, for some $\delta > 0$,

$$P^* \sup_{\|\theta - \theta_0\| < \delta} \|\dot{\psi}_\theta\|^2 < \infty.$$

This is comparable to the classical condition of domination of derivatives[‡], but it concerns the second moment of the first derivative rather than the first moment of the second derivative. To obtain complete analogy with the classical result, it appears to be necessary to exploit the possibility of the extra term $\sqrt{n}(\hat{\theta}_n - \theta_0)$ in the remainder of (3.3.2). See the following example.

[‡] Cf., for instance, Lehmann (1983) or Lehmann and Casella (1998).

3.3.8 Example (Classical smoothness). Let $X_1, \dots, X_n$ be i.i.d. random variables with common law P , and let ψ_θ be measurable vector-valued functions indexed by a parameter θ ranging over an open subset of Euclidean space. Assume that $P\psi_{\theta_0} = 0$ and that the map $\theta \mapsto \psi_\theta(x)$ is twice continuously differentiable with respect to θ in a neighborhood of θ_0 , for each x , with derivatives satisfying

$$P\|\dot{\psi}_{\theta_0}\|^2 < \infty; \quad P^* \sup_{\|\theta - \theta_0\| < \delta} \|\ddot{\psi}_\theta\| < \infty.$$

Then the map $\theta \mapsto P\psi_\theta$ is differentiable at θ_0 with derivative $P\dot{\psi}_{\theta_0}$ and the sequence $\sqrt{n}(\hat{\theta}_n - \theta_0)$ is asymptotically normally distributed provided $\hat{\theta}_n$ is consistent for θ_0 .

A Taylor expansion gives for each θ that is sufficiently close to θ_0

$$\mathbb{G}_n(\psi_\theta - \psi_{\theta_0}) = \mathbb{G}_n\dot{\psi}_{\theta_0}(\theta - \theta_0) + \frac{1}{2}(\theta - \theta_0)'\mathbb{G}_n\ddot{\psi}_{\tilde{\theta}}(\theta - \theta_0),$$

for a point $\tilde{\theta} = \tilde{\theta}(\theta, x)$ on the line connecting θ and θ_0 . In the quadratic term, the empirical process is applied to the function $x \mapsto \ddot{\psi}_{\tilde{\theta}(\theta, x)}(x)$. Let the function M bound $\|\ddot{\psi}_\theta\|$ for every θ in a neighborhood of θ_0 . Then $\|\mathbb{G}_n(\psi_{\hat{\theta}_n} - \psi_{\theta_0})\|$ is bounded by

$$\|\mathbb{G}_n\dot{\psi}_{\theta_0}\| \|\hat{\theta}_n - \theta_0\| + \frac{1}{2}\|\hat{\theta}_n - \theta_0\|^2 \sqrt{n}(\mathbb{P}_n + P)M = o_P(1) + o_P(1)\sqrt{n}\|\hat{\theta}_n - \theta_0\|,$$

since the sequence $(\mathbb{P}_n + P)M$ is bounded in probability. Thus (3.3.4) is satisfied.

Our emphasis on Lipschitz conditions on the criterion functions in the preceding examples may give the (wrong) impression that smoothness is necessary for the estimators to be asymptotically normal or the theory of empirical processes to be of help. This is certainly a false impression, as should be clear from Lemma 3.3.5 and Theorem 3.2.20. The point is that smoothness permits an easy general method to check the more abstract empirical process conditions, such as that the functions $\psi_{\theta, h}$ are in a Donsker class. In particular examples we can apply all methods and results of Part 2 to verify these conditions; for instance, the stability properties of Chapter 2.10. We include one example to illustrate this.

3.3.9 Example (Median and median deviation). The median and median deviation are M -estimators for the location and scale parameter (μ, σ) relative to the functions

$$\psi_{\mu, \sigma}(x) = \left(\text{sign}\left(\frac{x - \mu}{\sigma}\right), \text{sign}\left(\left|\frac{x - \mu}{\sigma}\right| - \beta\right) \right).$$

The constant β is usually chosen such that the median deviation converges to the standard deviation of some distribution of interest; for instance, it is set equal to $\Phi^{-1}(3/4)$ for the normal distribution.

The dependence $(\mu, \sigma) \mapsto \psi_{\mu, \sigma}(x)$ is not Lipschitz. However, the class of functions $\{\psi_{\mu, \sigma, 1}, \psi_{\mu, \sigma, 2}: \mu \in \mathbb{R}, \sigma > 0\}$ can easily be shown to be a Donsker class. The class of functions $x \mapsto (x - \mu)/\sigma$ belongs to a finite-dimensional vector space and hence is a VC-class. Next the stability properties of VC-classes show that the two coordinate functions of the functions $\psi_{\mu, \sigma}$ also run through VC-classes. Since they are uniformly bounded and pointwise separable, they are universally Donsker.

The preceding examples are concerned with finite-dimensional parameters or functionals. The theory of empirical processes is used to carry out the usual linearization argument under better conditions. An important advantage is that cases that require special and separate treatment in the classical approach can be dealt with in a unified manner by using the theory of empirical processes.

The next example shows that empirical processes can also be used successfully to derive properties of estimators in infinitely dimensional models. It is just one out of a number of examples that have been studied recently. These examples cannot be treated by classical methods, and the development of abstract empirical process theory seems to have occurred at the right time for the development of this part of semiparametric statistics.

3.3.10 Example (Semiparametric likelihood). Suppose one observes a sample $(X_1, Y_1), \dots, (X_n, Y_n)$ from a distribution given formally by the density

$$\int p_\theta(x|z) d\eta(z) d\eta(y).$$

Here the map $x \mapsto p_\theta(x|z)$ is for each z a probability density with respect to a measure μ on some measurable space $(\mathcal{X}, \mathcal{A})$, which is known up to a parameter θ ranging over an open subset Θ of Euclidean space. The second parameter η is a completely unknown probability distribution on a measurable space $(\mathcal{Z}, \mathcal{C})$. Write $p_\theta(x|\eta)$ for the mixture density $\int p_\theta(x|z) d\eta(z)$.

Consider the asymptotic behavior of the maximum likelihood estimator $(\hat{\theta}, \hat{\eta})$ of (θ, η) defined as the maximizer of the “likelihood”

$$(\theta, \eta) \mapsto \prod_{i=1}^n p_\theta(X_i|\eta) \prod_{i=1}^n \eta\{Y_i\}.$$

This combines the usual likelihood of the first sample, the joint density of the observations, with the “nonparametric” likelihood of the second sample, the product of the point masses $\eta\{Y_i\}$ of the observations. In this semiparametric set-up, we are mostly interested in estimating θ .

A proof of the asymptotic normality of the sequence of maximum likelihood estimators may proceed by showing that a maximum likelihood estimator solves a collection of (likelihood) equations and next apply Theorem 3.3.1. This method applies to many different kernels. Consider in particular the case that $p_\theta(x|z) = \phi((x-z)/\theta)/\theta$. Then the observations are a sample $X_1, \dots, X_n$ from the model $X = Z + \theta e$ for an (unobserved) standard normal error variable e and an (unobserved) independent variable Z with distribution η and in addition a sample $Y_1, \dots, Y_n$ without measurement error; that is with the same distribution as Z . We can think of this as observing one sample from the distribution of Z itself and an additional sample corrupted by normal measurement error. In this special example take the parameter set Θ and $\mathcal{Z}$ to be compact intervals in $(0, \infty)$.

Likelihood equations corresponding to θ can be obtained in the usual manner by partial differentiation of the log likelihood with respect to θ at $\hat{\theta}$. For simplicity, assume that θ is one-dimensional. Then we obtain

$$\mathbb{P}_n \dot{\ell}_{\hat{\theta}, \hat{\eta}} = 0,$$

where $\mathbb{P}_n = n^{-1} \sum_{i=1}^n \delta_{(X_i, Y_i)}$ is the empirical measure of the observations and $\dot{\ell}_{\theta, \eta}$ is the score function for θ of the mixture model. This can be expressed in the score functions $\dot{\ell}_{\theta}(x|z) = \partial/\partial\theta \log p_{\theta}(x|z)$ of the conditional models by

$$\dot{\ell}_{\theta, \eta}(x, y) = \frac{\int \dot{\ell}_{\theta}(x|z) p_{\theta}(x|z) d\eta(z)}{p_{\theta}(x|\eta)}.$$

Likelihood equations corresponding to the infinite-dimensional parameter η can be obtained by inserting one-dimensional submodels $t \mapsto \hat{\eta}_t$ passing through $\hat{\eta}$ in the log likelihood and differentiating with respect to t . In particular, given a bounded, measurable function h and every sufficiently small number $|t|$, we can define a probability measure $\hat{\eta}_t$ by

$$d\hat{\eta}_t = (1 + t(h - \int h d\hat{\eta})) d\hat{\eta}.$$

This leads to the likelihood equation

$$\mathbb{P}_n A_{\hat{\theta}, \hat{\eta}_t} h - P_{\hat{\theta}, \hat{\eta}_t} A_{\hat{\theta}, \hat{\eta}_t} h = 0,$$

where $A_{\theta, \eta}$ are the “score operators” given by

$$A_{\theta, \eta} h(x, y) = B_{\theta, \eta} h(x) + h(y) = \frac{\int h(z) p_{\theta}(x|z) d\eta(z)}{p_{\theta}(x|\eta)} + h(y).$$

The operators $B_{\theta, \eta}$ are the score operators for the mixture part of the model.

Let H be the set of all functions $h: \mathcal{Z} \rightarrow [0, 1]$ with $\|h\|_{lip} \leq 1$, the positive part of the unit ball of the space $C^1(\mathcal{Z})$ considered in Chapter 2.7. For each $h \in H$ the maximum likelihood estimator satisfies the equation as derived previously. Let $\Psi_n(\theta, \eta) = (\Psi_{n1}(\theta, \eta), \Psi_{n2}(\theta, \eta))$ be the element of $\mathbb{R} \times \ell^\infty(H)$ given by

$$\Psi_{n1}(\theta, \eta) = \mathbb{P}_n \dot{\ell}_{\theta, \eta}; \quad \Psi_{n2}(\theta, \eta)h = \mathbb{P}_n A_{\theta, \eta} h - P_{\theta, \eta} A_{\theta, \eta} h.$$

The map $h \mapsto \Psi_{n2}(\theta, \eta)h$ is indeed uniformly bounded on H , because the conditional expectation operator $B_{\theta, \eta}$ retains boundedness: $0 \leq B_{\theta, \eta} h \leq 1$ for every $h \in H$. The expectation of $\Psi_n(\theta, \eta)$ under the true distribution $P_0 = P_{\theta_0, \eta_0}$ is the element $\Psi(\theta, \eta) = (\Psi_1(\theta, \eta), \Psi_2(\theta, \eta))$ of $\mathbb{R} \times \ell^\infty(H)$ given by

$$\Psi_1(\theta, \eta) = P_0 \dot{\ell}_{\theta, \eta}; \quad \Psi_2(\theta, \eta)h = P_0 A_{\theta, \eta} h - P_{\theta, \eta} A_{\theta, \eta} h.$$

The maximum likelihood estimator $(\hat{\theta}_n, \hat{\eta}_n)$ and the true value (θ_0, η_0) are zeros of these maps, respectively:

$$\Psi_n(\hat{\theta}_n, \hat{\eta}_n) \equiv 0; \quad \Psi(\theta_0, \eta_0) \equiv 0.$$

We identify each probability measure η with an element of $\ell^\infty(H)$ through $\eta h = \int h d\eta$. Then both Ψ_n and Ψ can be viewed as maps from the space $\mathbb{R} \times \ell^\infty(H)$ into itself whose domain is the product of Θ and the set of probability measures in $\ell^\infty(H)$ under the given identification. The present choice of H is appropriate for our special example (and many other examples) but should be replaced by a different choice whenever convenient.

The remainder of this example is concerned with checking the conditions of Theorem 3.3.1, particularly for our special example involving a normal measurement error model.

The consistency of the sequence $(\hat{\theta}_n, \hat{\eta}_n)$ may be proved by the method of Wald or by a method using empirical processes. This requires some regularity conditions on the kernel $z \mapsto p_\theta(x|z)$. We omit a discussion.

In the present case the process $\sqrt{n}(\Psi_n - \Psi)$ takes the form

$$\sqrt{n}(\Psi_n - \Psi)(\theta, \eta) = \left(\sqrt{n}(\mathbb{P}_n - P_0)\dot{\ell}_{\theta, \eta}, \sqrt{n}(\mathbb{P}_n - P_0)A_{\theta, \eta} \right).$$

The right side is the empirical process indexed by the class of functions $\{A_{\theta, \eta}h: h \in H\} \cup \{\dot{\ell}_{\theta, \eta}\}$. In view of the discussion following the proof of Theorem 3.3.1, condition (3.3.2) of this theorem is certainly satisfied if, for some $\delta > 0$,

$$(3.3.11) \quad \{\dot{\ell}_{\theta, \eta}, A_{\theta, \eta}h: h \in H, \|\theta - \theta_0\| + \|\eta - \eta_0\| < \delta\} \quad \text{is } P_0\text{-Donsker,}$$

$$\sup_{h \in H} P_0(A_{\theta, \eta}h - A_0h)^2 + P_0(\dot{\ell}_{\theta, \eta} - \dot{\ell}_0)^2 \rightarrow 0, \quad \theta \rightarrow \theta_0, \eta \rightarrow \eta_0.$$

In view of the dominated convergence theorem, the last condition is valid if $A_{\theta, \eta}h \rightarrow A_0h$ and $\dot{\ell}_{\theta, \eta} \rightarrow \dot{\ell}_0$ pointwise, uniformly in h .

It is hard to characterize condition (3.3.11) directly in terms of the kernel $p_\theta(x|z)$. In fact, different kernels may require different approaches. Consider the case that the kernel is a smooth function on Euclidean space. Then the functions $x \mapsto B_{\theta, \eta}h(x)$ will be smooth functions as well and Theorem 2.10.27 may be appropriate. If the tail of P_0 is not too large and $\mathcal{X}$ is a subset of $\mathbb{R}^d$, then a uniform α -smoothness condition on the functions $x \mapsto B_{\theta, \eta}h(x)$ for some $\alpha > d/2$ is sufficient. Under appropriate conditions on the map $x \mapsto p_\theta(x|z)$, straightforward differentiation yields

$$\frac{\partial}{\partial x_i} B_{\theta, \eta}h(x) = \text{cov}_x \left(h(Z), \frac{\partial}{\partial x_i} \log p_\theta(x|Z) \right),$$

where for each x the covariance is computed for the random variable Z having the (conditional) density $z \mapsto p_\theta(x|z) d\eta(z)/p_\theta(x|\eta)$. Thus, for a given bounded function h ,

$$\left| \frac{\partial}{\partial x_i} B_{\theta, \eta}h(x) \right| \leq \|h\|_\infty \frac{\int \left| \frac{\partial}{\partial x_i} \log p_\theta(x|z) \right| p_\theta(x|z) d\eta(z)}{\int p_\theta(x|z) d\eta(z)}.$$

Depending on the function $\partial/\partial x_i \log p_\theta(x|z)$, this leads to a bound on the first derivative and hence on the Lipschitz norm of order 1 of the function $x \mapsto B_{\theta, \eta}h(x)$. If $\mathcal{X}$ is equal to $\mathbb{R}$, then this is sufficient for the applicability of Theorem 2.10.27. In our example we have $|\partial/\partial x \log p_\theta(x|z)| = |x - z|/\theta^2$, which is bounded by a constant times $|x|$ for θ in a neighborhood of θ_0 and η ranging over probability measures on a fixed interval. Since the tails of P_0 are Gaussian, the function $|x|$ has the required integrability properties and Theorem 2.10.27 implies that (3.3.11) is satisfied.

If $\mathcal{X}$ is a subset of a higher-dimensional Euclidean space, then the same conclusion may be reached by consideration of higher-order derivatives.

The remaining conditions of Theorem 3.3.1 are the differentiability of the map Ψ and the continuity of the inverse of the derivative. For convenience of notation, we introduce the Hilbert-space adjoint $B_{\theta,\eta}^*$ of the operator $B_{\theta,\eta}: L_2(\eta) \rightarrow L_2(p_\theta(\cdot|\eta))$ given by

$$B_{\theta,\eta}^*g(z) = \int g(x) p_\theta(x|z) d\mu(x).$$

The range of the operator $A_{\theta,\eta}$ is contained in the subset G of $L_2(p_\theta(\cdot|\eta) \times \eta)$ consisting of functions of the form $(x, y) \mapsto g_1(x) + g_2(y) + c$. The representation of a function of this type is unique if both g_1 and g_2 are taken to be zero-mean functions. The adjoint of the operator $A_{\theta,\eta}: L_2(\eta) \rightarrow G$ is given by $A_{\theta,\eta}^*(g_1 + g_2 + c) = B_{\theta,\eta}^*g_1 + g_2 + 2c$. On the set of zero-mean functions in $L_2(\eta)$, we have the identity $A_{\theta,\eta}^*A_{\theta,\eta} = I + B_{\theta,\eta}^*B_{\theta,\eta}$.

Informally, the derivative $\dot{\Psi} = (\dot{\Psi}_1, \dot{\Psi}_2)$ of the map Ψ at (θ_0, η_0) can be derived as follows. First,

$$\begin{aligned} \Psi_1(\theta, \eta) - \Psi_1(\theta_0, \eta_0) &= P_0(\dot{\ell}_{\theta,\eta} - \dot{\ell}_0) \\ &\approx (\theta - \theta_0)P_0\ddot{\ell}_0 + \int \int (\dot{\ell}_0(x|z) - \dot{\ell}_0(x)) p_0(x|z) d\mu(x) d(\eta - \eta_0)(z). \end{aligned}$$

Under regularity conditions, $P_0\ddot{\ell}_0 = -I_0$ is minus the Fisher information for θ in the situation when $\eta = \eta_0$ is known and the expectation $\int \dot{\ell}_0(x|z)p_0(x|z) d\mu(x) = 0$ for every z . Then the last line can be rewritten as

$$-I_0(\theta - \theta_0) - \int B_0^*\dot{\ell}_0 d(\eta - \eta_0).$$

The derivative of the second component of Ψ can be obtained in a similar way. Uniformly in h ,

$$\begin{aligned} \Psi_2(\theta, h)h - \Psi_2(\theta_0, \eta_0)h &= - \int A_{\theta,\eta}h d(P_{\theta,\eta} - P_0) \\ &\approx - \int A_0h d(P_{\theta,\eta} - P_0) \\ &\approx -(\theta - \theta_0) \int A_0h \dot{\ell}_0 dP_0 - \int (I + B_0^*B_0)h d(\eta - \eta_0). \end{aligned}$$

The combination of the preceding displays suggests that the derivative of Ψ at (θ_0, η_0) is given by the map

$$(\theta - \theta_0, \eta - \eta_0) \mapsto \begin{pmatrix} \dot{\Psi}_{11} & \dot{\Psi}_{12} \\ \dot{\Psi}_{21} & \dot{\Psi}_{22} \end{pmatrix} \begin{pmatrix} \theta - \theta_0 \\ \eta - \eta_0 \end{pmatrix},$$

where

$$\begin{aligned} \dot{\Psi}_{11}(\theta - \theta_0) &= -I_0(\theta - \theta_0), \\ \dot{\Psi}_{12}(\eta - \eta_0) &= - \int B_0^*\dot{\ell}_0 d(\eta - \eta_0), \\ \dot{\Psi}_{21}(\theta - \theta_0)h &= -P_0A_0h\dot{\ell}_0(\theta - \theta_0), \\ \dot{\Psi}_{22}(\eta - \eta_0)h &= - \int (I + B_0^*B_0)h d(\eta - \eta_0)(z). \end{aligned}$$

The derivation can be made rigorous under regularity conditions. We omit the somewhat tedious verification in our concrete example.

Theorem 3.3.1 requires that the derivative operator be continuously invertible on the linear span of the domain of Ψ . This can be verified by ascertaining the continuous invertibility of the two operators $\dot{\Psi}_{11}$ and $\dot{V} = \dot{\Psi}_{22} - \dot{\Psi}_{21}\dot{\Psi}_{11}^{-1}\dot{\Psi}_{12}$. In that case we have

$$\dot{\Psi}^{-1} = \begin{pmatrix} \dot{\Psi}_{11}^{-1}(\dot{\Psi}_{11} + \dot{\Psi}_{12}\dot{V}^{-1}\dot{\Psi}_{21})\dot{\Psi}_{11}^{-1} & -\dot{\Psi}_{11}^{-1}\dot{\Psi}_{12}\dot{V}^{-1} \\ -\dot{V}^{-1}\dot{\Psi}_{21}\dot{\Psi}_{11}^{-1} & \dot{V}^{-1} \end{pmatrix}.$$

The operator $\dot{\Psi}_{11}$ is continuously invertible provided the Fisher information for θ is positive. The second operator has the form

$$\dot{V}(\eta - \eta_0)h = - \int \left[(I + B_0^*B_0)h - \frac{P_0A_0h\dot{\ell}_0}{I_0}B_0^*\dot{\ell}_0 \right] d(\eta - \eta_0).$$

This operator is certainly continuously invertible if there exists a positive number ϵ such that

$$\left\{ (I + B_0^*B_0)h - \frac{P_0A_0h\dot{\ell}_0}{I_0}B_0^*\dot{\ell}_0 : h \in H \right\} \supset \epsilon H.$$

In many examples, including our concrete example, this can be ascertained by using the theory of Fredholm operators. An operator of the form $I + K$ from a Banach space in itself, where K is compact, is continuously invertible if and only if it is one-to-one.[†] In the present situation, consider the operator K given by

$$Kh = B_0^*B_0h - \frac{P_0A_0h\dot{\ell}_0}{I_0}B_0^*\dot{\ell}_0.$$

If the maps $z \mapsto p_{\theta_0}(x|z)$ are sufficiently smooth, as is the case in our example, then the operator $B_0^*: C^1(\mathcal{Z}) \rightarrow C^1(\mathcal{Z})$ can be seen to be compact by the Arzelà-Ascoli theorem. Consequently, $B_0^*B_0$ is a compact operator from $C^1(\mathcal{Z})$ into itself. The second part of K is compact, because it has a one-dimensional range. Finally, it suffices to show that $I + K$ is one-to-one. The spectrum of the self-adjoint operator $I + B_0^*B_0: L_2(\eta_0) \rightarrow L_2(\eta_0)$ is contained in $[1, \infty)$. Hence this operator is continuously invertible in the Hilbert-space sense. The equation

$$(3.3.12) \quad h + B_0^*B_0h - \frac{P_0A_0h\dot{\ell}_0}{I_0}B_0^*\dot{\ell}_0 = 0$$

can be solved to give either

$$P_0A_0h\dot{\ell}_0 = 0 \quad \text{or} \quad P_0(A_0(I + B_0^*B_0)^{-1}B_0^*\dot{\ell}_0)\dot{\ell}_0 = I_0.$$

The efficient Fisher information for θ is by definition the square length of the projection of $\dot{\ell}_0$ on the orthocomplement of the range of A_0 . This is positive in most examples, including the present one. The function $A_0(I + B_0^*B_0)^{-1}B_0^*\dot{\ell}_0 = A_0(A_0^*A_0)^{-1}A_0^*\dot{\ell}_0$ is the projection of $\dot{\ell}_0$ onto the range of A_0 . Hence the difference of the right and the left sides of the second equation in the preceding display is the

[†] See, for instance, Rudin (1973), pages 99–103.

efficient Fisher information, which is nonzero. If the first expression in the display is zero, then we find that (3.3.12) reduces to $h + B_0^* B_0 h = 0$, whence $h = 0$ almost surely under η_0 . Reinsert this into equation (3.3.12) to find that h is zero pointwise. This concludes the proof of the continuous invertibility of the operator $I + K$ and hence of the continuous invertibility of $\dot{\Psi}_0$.

Thus, all conditions of Theorem 3.3.1 have been verified. We conclude that the sequence $\sqrt{n}(\hat{\theta}_n - \theta_0, \hat{\eta}_n - \eta_0)$ is asymptotically normal in the space $\mathbb{R} \times \ell^\infty(H)$. In particular, the sequence of prime interest, $\sqrt{n}(\hat{\theta}_n - \theta_0)$, is asymptotically normal with mean zero and variance $1/P_0(\dot{\ell}_0 - A_0(A_0^* A_0)^{-1} \dot{\ell}_0)^2$.

3.4

Sieved and Penalized

M-Estimators

This chapter presents results on rates of convergence of M -estimators, including maximum likelihood estimators, least-squares estimators, empirical classifiers, and support vector machines. We first state two abstract results, which are generalizations of the theorem on rates of convergence in Chapter 3.2, and discuss general methods to establish the maximal inequalities needed for the application of these results. Next we consider the different settings in detail. Our main interest is in M -estimators of infinite-dimensional parameters, and we consider both sieved and penalized versions of such estimators.

In this introduction the parameter set is denoted Θ and may be thought of as a semimetric space Θ with semimetric d . We consider *sieved M -estimators*, sieves Θ_n being a sequence of subsets of the parameter space. Let $\mathbb{P}_n$ denote the empirical measure of observations $X_1, X_2, \dots, X_n$ taking their values in a measurable space $(\mathcal{X}, \mathcal{A})$. Given measurable maps $m_{n,\theta}: \mathcal{X} \rightarrow \mathbb{R}$, a *sieved M -estimator* $\hat{\theta}_n$ is a map defined on $\mathcal{X}^n$, that maximizes over Θ_n the criterion

$$\theta \mapsto \mathbb{P}_n m_{n,\theta}.$$

We are interested in the rate at which $\hat{\theta}_n$ approaches a target parameter θ_0 , typically a maximizer of an asymptotic version of the criterion function over the full parameter set Θ .

Actually the derivations do not use that $\hat{\theta}_n$ is a maximizer, but are based on the property that, for a given $\theta_n \in \Theta_n$,

$$\mathbb{P}_n m_{n,\hat{\theta}_n} \geq \mathbb{P}_n m_{n,\theta_n}.$$

A maximizer $\hat{\theta}_n$ verifies this inequality for *any* $\theta_n \in \Theta_n$, but the properties of $\hat{\theta}_n$ can be derived using this property for a single θ_n , typically chosen close to the target

value θ_0 . In some examples this is applied to an unobservable map $\theta \mapsto \mathbb{P}_n m_{n,\theta}$, and the inequality holds in virtue of the fact that $\hat{\theta}_n$ maximizes another, observable criterion function. The particular functions $m_{n,\theta}$ then intervene to establish the properties of $\hat{\theta}_n$, but not to define it.

For the estimators $\hat{\theta}_n$ to be consistent, the sieves Θ_n must be constructed to grow dense in Θ as $n \rightarrow \infty$. The simplest sequence of sieves with this property is the whole parameter set $\Theta_n = \Theta$. In this case, a natural choice for θ_n is the “true” parameter θ_0 . In any case the sieves must be large enough, because the rate of convergence of $d(\hat{\theta}_n, \theta_0)$ to zero is never faster than $d(\theta_n, \theta_0)$. The latter quantity may be thought of as the distance of θ_0 to the sieve Θ_n .

Maximizing a given criterion function over the whole parameter space may or may not be a good idea. A “full” maximum contrast estimator is often a good (and even asymptotically efficient) estimator, even in the case of infinite-dimensional parameter spaces. However, the supremum $\sup_{\theta} \mathbb{P}_n m_{n,\theta}$ of the criterion may be infinite and no M -estimator may be defined; or the criterion functions may have a unique point of maximum for every n , but the sequence of estimators may fail to converge to the true value, or do so at a suboptimal rate.

Alternative to restricting the maximization to a sieve, such problems can be solved by adding a penalty to the criterion function. A *penalized maximum contrast estimator* maximizes a criterion of the form

$$\theta \mapsto \mathbb{P}_n m_{n,\theta} - \hat{\lambda}_n^2 \mathcal{J}_n^2(\theta).$$

Here the *penalty function* $\mathcal{J}_n: \Theta \rightarrow [0, \infty)$ is meant to decrease the value of the criterion at parameters θ that yield a high value $\mathbb{P}_n m_{n,\theta}$, but that are not near the true parameter. The penalty may prevent *overfitting* by requiring the penalized contrast estimator to trade off the value of the empirical criterion $\mathbb{P}_n m_{n,\theta}$ versus the size of the square penalty. This trade-off is moderated by a (possibly data-dependent) *smoothing parameter* $\hat{\lambda}_n$, which will decrease to zero with n . The rate of decrease is essential for the rate of the penalized contrast estimator.

Penalized maximum (or minimum) contrast estimators can be analyzed by the same methods as sieved estimators. We show this in the second general theorem of the chapter, in Section 3.4.3, which has the same form as the first general theorem on sieved estimators, in Section 3.4.1, except that it allows for a penalty.

Both main results characterize the rate in terms of a “modulus of continuity” of the centered criterion function. We discuss the maximal inequalities that can be used to bound this modulus in Section 3.4.2. The remaining sections of the chapter apply the general results to various settings.

3.4.1 Sieved Maximum Contrast Estimators

Even though our examples involve empirical criterion functions, the scheme to establish rates of convergence applies widely. We shall formulate the theorems with arbitrary stochastic processes $\theta \mapsto \mathbb{M}_n(\theta)$ taking the place of the empirical criterion $\theta \mapsto \mathbb{P}_n m_{n,\theta}$. It is then understood that each “estimator” $\hat{\theta}_n$ is a map defined on the same probability space as $\mathbb{M}_n$, with values in the index set Θ_n of the process $\mathbb{M}_n$.

It is helpful to think of the sets Θ_n as sieves, even though in the general results they may take arbitrary form. In line with this interpretation we enlarge the index set of the processes $\mathbb{M}_n$ in the statement of the theorem with a value $\theta_{n,0}$, which may be interpreted as a “true value”. It is not assumed that $\hat{\theta}_n$ maximizes the criterion over Θ_n , but only that the criterion is no smaller at $\hat{\theta}_n$ than at some other value $\theta_n \in \Theta_n$. It is helpful to think of the latter value θ_n as an element of the sieve that is close to $\theta_{n,0}$, but again this is not an assumption.

Corresponding to the criterion functions are *centering functions* $\theta \mapsto M_n(\theta)$. Typically these are the mean functions of the criterion functions, but they can be arbitrary, even random. The value $\theta_{n,0}$ may be interpreted as a maximizer of $\theta \mapsto M_n(\theta)$, either in an absolute sense, or over the sieve (if $\theta_{n,0} \in \Theta_n$), but this is not an assumption.

In this generality we also need not assume that Θ_n is a metric space. Instead we measure the “discrepancy” or “distance” between θ and the true “value” $\theta_{n,0}$ by a map $\theta \mapsto d_n(\theta, \theta_{n,0})$. Even though this is written in a form that suggests that it derives from a distance, this may be an arbitrary map from Θ_n to $[0, \infty)$, even a stochastic process. In particular $d_n^2(\theta, \theta_{n,0})$ can be chosen equal to $M_n(\theta_{n,0}) - M_n(\theta)$ (provided this is nonnegative), in which case the first condition of the following theorem ceases to be a condition.

3.4.1 Theorem (Rate of convergence). *For each n , let $\mathbb{M}_n$ and M_n be stochastic processes indexed by a set $\Theta_n \cup \{\theta_{n,0}\}$, let $\theta \mapsto d_n(\theta, \theta_{n,0})$ be an arbitrary map from Θ_n to $[0, \infty)$, and let $\underline{\delta}_n \geq 0$. Suppose that, for every n and $\delta > \underline{\delta}_n$,*

$$\sup_{\theta \in \Theta_n: \delta/2 < d_n(\theta, \theta_{n,0}) \leq \delta} M_n(\theta) - M_n(\theta_{n,0}) \leq -\delta^2,$$

$$\mathbb{E}^* \sup_{\theta \in \Theta_n: d_n(\theta, \theta_{n,0}) \leq \delta} \sqrt{n} |(\mathbb{M}_n - M_n)(\theta) - (\mathbb{M}_n - M_n)(\theta_{n,0})| \lesssim \phi_n(\delta),$$

for increasing functions $\phi_n: [\underline{\delta}_n, \infty) \rightarrow \mathbb{R}$ such that $\delta \mapsto \phi_n(\delta)/\delta^\alpha$ is decreasing for some $\alpha < 2$. Let $\theta_n \in \Theta_n$, and let δ_n satisfy

$$\phi_n(\delta_n) \leq \sqrt{n}\delta_n^2, \quad \delta_n^2 \geq M_n(\theta_{n,0}) - M_n(\theta_n), \quad \delta_n \geq \underline{\delta}_n.$$

If the sequence $\hat{\theta}_n$ takes its values in Θ_n and satisfies $\mathbb{M}_n(\hat{\theta}_n) \geq \mathbb{M}_n(\theta_n) - O_P(\delta_n^2)$, then $d_n(\hat{\theta}_n, \theta_{n,0}) = O_P^*(\delta_n)$.

Proof. The proof is similar to that for Theorem 3.2.5. That all entities are now allowed to depend on n requires only notational changes, but the possible discrepancy between θ_n and $\theta_{n,0}$ requires some care. For $j \in \mathbb{N}$, define

$$S_{n,j} = \{\theta \in \Theta_n: 2^{j-1}\delta_n < d_n(\theta, \theta_{n,0}) \leq 2^j\delta_n\}.$$

Because the map $d_n(\theta, \theta_{n,0})$ is nonnegative, the variable $d_n(\hat{\theta}_n, \theta_{n,0})/\delta_n$ is bounded in probability if it is upper bounded in probability. Hence it suffices to show that with high probability $\hat{\theta}_n$ does not belong to an $S_{n,j}$ with large j .

If $\theta \in S_{n,j}$ and $2^j \delta_n \geq \underline{\delta}_n$, then, by the first assumption and the second lower bound on δ_n ,

$$\begin{aligned} M_n(\theta) - M_n(\theta_n) &= M_n(\theta) - M_n(\theta_{n,0}) + M_n(\theta_{n,0}) - M_n(\theta_n) \\ &\leq -2^{2j} \delta_n^2 + \delta_n^2 \lesssim -2^{2j} \delta_n^2. \end{aligned}$$

Let $G_n = \sqrt{n}(\mathbb{M}_n - M_n)$. Then the assumption on $\hat{\theta}_n$ shows that $G_n(\hat{\theta}_n) - G_n(\theta_n) \geq \sqrt{n}(M_n(\theta_n) - M_n(\hat{\theta}_n)) - O_P(\sqrt{n}\delta_n^2)$. The first term on the right is bounded below by a multiple of $\sqrt{n}2^{2j}\delta_n^2$ if $\hat{\theta}_n \in S_{n,j}$, by the preceding display. The remainder term $-O_P(\sqrt{n}\delta_n^2)$ is bounded below by a (large) multiple of $-\sqrt{n}\delta_n^2$ with high probability; for simplicity assume that this is true with probability 1. We then conclude that $G_n(\hat{\theta}_n) - G_n(\theta_n) \gtrsim \sqrt{n}2^{2j}\delta_n^2$ if $\hat{\theta}_n \in S_{n,j}$, and hence

$$P^*(\hat{\theta}_n \in S_{n,j}) \leq P^*\left(\sup_{\theta \in S_{n,j}} (G_n(\theta) - G_n(\theta_n)) \gtrsim \sqrt{n}2^{2j}\delta_n^2\right).$$

We apply Markov's inequality to the right side, where we can bound the absolute mean of $\sup_{\theta \in S_{n,j}} (G_n(\theta) - G_n(\theta_n))$ by $\phi_n(2^j \delta_n)$ plus the absolute mean of the variable $G_n(\theta_n) - G_n(\theta_{n,0})$. Because $\theta_n \in \Theta_n$, the latter is certainly bounded above by $\phi_n(d_n(\theta_n, \theta_{n,0}) \vee \underline{\delta}_n) \leq \phi_n(\delta_n) \leq \phi_n(2^j \delta_n)$, where we use that $d_n^2(\theta_n, \theta_{n,0}) \leq M_n(\theta_{n,0}) - M_n(\theta_n) \leq \delta_n^2$ if $d_n(\theta_n, \theta_{n,0}) > \underline{\delta}_n$ (where the first inequality follows from the first assumption of the theorem applied with $\delta = d_n(\theta_n, \theta_{n,0})$) and the assumption that ϕ_n is increasing.

We conclude the proof by summing over j and using the properties of ϕ_n and δ_n , as in the proof of Theorem 3.2.5. ■

The theorem can be applied with $\hat{\theta}_n$ and $\theta_{n,0}$ equal to the maximizers of $\theta \mapsto \mathbb{M}_n(\theta)$ over a sieve Θ_n and of $\theta \mapsto M_n(\theta)$ over a full parameter set Θ , respectively. The first displayed equation then requires that the centering functions fall off quadratically in the “distance” $d_n(\theta, \theta_{n,0})$ as θ moves away from the maximizing value $\theta_{n,0}$. With the choice $\underline{\delta}_n = 0$, the theorem shows that the “distance” of $\hat{\theta}_n$ to $\theta_{n,0}$ satisfies

$$(3.4.2) \quad d_n^2(\hat{\theta}_n, \theta_{n,0}) = O_P^*(\delta_n^2 + M_n(\theta_{n,0}) - M_n(\theta_n)),$$

for δ_n solving $\phi_n(\delta_n) \leq \sqrt{n}\delta_n^2$ and any $\theta_n \in \Theta_n$. Thus the rate r_n is determined by the “modulus of continuity” $\delta \mapsto \phi_n(\delta)$ of the centered processes $\sqrt{n}(\mathbb{M}_n - M_n)$ over Θ_n and the discrepancy $M_n(\theta_{n,0}) - M_n(\theta_n)$. Typically, small sieves Θ_n lead to a small modulus, hence fast δ_n in (3.4.2). On the other hand, the discrepancy $M_n(\theta_{n,0}) - M_n(\theta_n)$ of a small sieve will be large. Thus, the two terms in the right side of (3.4.2) may be loosely understood as a “variance” and a “square bias” term, which must be balanced to obtain a good rate of convergence. (In many problems an unsieved M -estimator actually performs very well, so the trade-off should not be understood too literally: it may work to reduce the “bias” to zero.)

Alternatively, the theorem may be applied with $\theta_{n,0} = \theta_n$. It is then required that the centering function decreases quadratically as θ moves away from θ_n within the sieve, at least for $d_n(\theta, \theta_n) > \underline{\delta}_n$. Even though the sieve may not contain a point of maximum, this is typically the case if θ_n is close to the point of maximum and $\underline{\delta}_n$

is a multiple of $d_n(\theta, \theta_n)$ (cf. Problem 3.4.5). The lower bound $\delta_n^2 \geq M_n(\theta_{n,0}) - M_n(\theta_n)$ on the square rate now becomes empty, but the bound $\delta_n \geq \underline{\delta}_n$ becomes active and the conclusion may be the same.

The modulus $\phi_n(\delta)$ in the theorem is defined as an expectation, over the full probability space. In the proof the expectation is used to bound a tail probability through Markov's inequality. To obtain a rate of convergence a probability bound would suffice, and it would be sufficient that this were valid on a sequence of events with probability tending to one. In the other direction, a bound on higher moments of the modulus leads to bounds on corresponding moments of $d_n(\hat{\theta}_n, \theta_{n,0})$ (see Problem 3.4.1). Another useful refinement is to shift the modulus downwards by a quadratic; this is given in the following addendum. A precise proof can follow the lines of Theorem 3.4.6.

3.4.3 Addendum. *In Theorem 3.4.1, the modulus bound can be replaced by, for events Ω_n of probability tending to one and $c \in [0, 1)$,*

$$\mathbb{E}^* 1_{\Omega_n} \sup_{\theta \in \Theta_n: d_n(\theta, \theta_{n,0}) \leq \delta} \sqrt{n} [(\mathbb{M}_n - M_n)(\theta) - (\mathbb{M}_n - M_n)(\theta_n) - c\delta^2]^+ \lesssim \phi_n(\delta).$$

3.4.2 Continuity Modulus

The main challenge for the application of the preceding theorem is to derive the maximal inequalities for the modulus of the centered processes $\sqrt{n}(\mathbb{M}_n - M_n)$. For the case that $\mathbb{M}_n(\theta) = \mathbb{P}_n m_{n,\theta}$ considered in the introduction, with centering function $M_n(\theta) = P m_{n,\theta}$, this involves the empirical process indexed by the classes of functions

$$\mathcal{M}_{n,\delta} = \left\{ m_{n,\theta} - m_{n,\theta_{n,0}} : \theta \in \Theta_n, d_n(\theta, \theta_{n,0}) \leq \delta \right\}.$$

Write $M_{n,\delta}$ for the envelope function of these classes. In several examples in Chapter 3.2 it works well to use an inequality of the type

$$\mathbb{E}_P^* \|\mathbb{G}_n\|_{\mathcal{M}_{n,\delta}} \lesssim J(1) (P^* M_{n,\delta}^2)^{1/2},$$

with $J(1)$ a bound on the uniform entropy or bracketing entropy integral of the classes $\mathcal{M}_{n,\delta}$ (cf. Theorems 2.14.1 and 2.14.16) that is uniform in (n, δ) . The rate of convergence is then driven by the sizes of the envelope functions $M_{n,\delta}$ as $\delta \downarrow 0$, and the sizes of the classes $\mathcal{M}_{n,\delta}$ are important only to guarantee a finite entropy integral. In genuinely infinite-dimensional situations, this approach appears to be less useful. It is intuitively clear that the precise entropy must make a difference for the rate of convergence in general.

Useful bounds on the modulus of continuity of the empirical process are given in Chapter 2.14 (in particular, Theorems 2.14.2, 2.14.8, 2.14.9, 2.14.17, and 2.14.18.) The general form is, for a class $\mathcal{F}$ of measurable functions such that $\|f\| < \delta$ for every f in $\mathcal{F}$,

$$(3.4.4) \quad \mathbb{E}_P^* \|\mathbb{G}_n\|_{\mathcal{F}} \lesssim J(\delta, \mathcal{F}) \left(1 + \frac{J(\delta, \mathcal{F})}{\delta^2 \sqrt{n}} M(\mathcal{F}) \right) K(\mathcal{F}).$$

Here $J(\delta, \mathcal{F})$ can be the uniform entropy integral or bracketing entropy integral relative to the given norm, and $M(\mathcal{F})$ and $K(\mathcal{F})$ are constants that depend on the envelope of the class. For uniformly bounded classes $\|\cdot\|$ can be the $L_2(P)$ -norm and L_2 -type entropy integrals suffice, but more generally stronger norms may be required, for instance the “Bernstein norm”

$$\|f\|_{P,B} = \left(2P(e^{|f|} - 1 - |f|)\right)^{1/2}.$$

This “Bernstein norm” turns out to combine well with minimum contrast problems, where the criterion is a logarithm of another, natural function, such as the log likelihood function.

If an inequality of type (3.4.4) holds for the class $\mathcal{M}_{n,\delta}$ and $\|m_{n,\theta} - m_{n,\theta_{n,0}}\| \leq d_n(\theta, \theta_{n,0})$, then the bound $\phi_n(\delta)$ on the modulus in Theorem 3.4.1 can be chosen equal to the right side of (3.4.4) with $\mathcal{F} = \mathcal{M}_{n,\delta}$. In that case the condition $\phi_n(\delta_n) \leq \sqrt{n}\delta_n^2$ on the rate δ_n is satisfied if

$$J(\delta_n, \mathcal{M}_{n,\delta}) \leq \sqrt{n}\delta_n^2.$$

Thus the rate is determined immediately by the entropy integrals of the classes $\mathcal{M}_{n,\delta}$.[‡]

We shall follow this approach in most examples, although sometimes other bounds may be available. For instance, in certain applications the empirical process indexed by the class $\mathcal{M}_{n,\delta}$ may be sub-Gaussian or satisfy other tail bounds. Then the maximal inequalities of Chapter 2.2 may be applied directly, without the additional arguments used for the empirical process in Chapter 2.14. In other examples symmetrization by Rademacher variables followed by application of sub-Gaussian maximal inequalities (which underlies the derivation of the maximal inequality Theorem 2.14.2 mentioned previously) may be applied directly.

If the bracketing entropy integral diverges at zero, then bounds of type (3.4.4) are useless, but the more general Lemma 2.14.19 can be used with a positive value of γ . The choice $\gamma = c\delta^2 \wedge \delta/3$ for a small positive constant c is appropriate, because in Theorem 3.4.1 the random process $\sqrt{n}(\mathbb{M}_n - M_n)$ could be allowed to have a quadratic drift as long as this is strictly smaller than the quadratic drift of the centering function (see Addendum 3.4.3). It suffices that the supremum of the empirical process indexed by $\mathcal{M}_{n,\delta}$ can be bounded by $Z_{n,\delta} + c\delta^2\sqrt{n}$ for $c \in [0, 1)$ and a random variable $Z_{n,\delta}$ such that

$$\mathbb{E}Z_{n,\delta}^+ \lesssim \phi_n(\delta).$$

Lemma 2.14.19 yields such a bound on the empirical processes shifted downward. This argument shows that in the following the entropy integrals can be replaced by the integrals

$$\int_{c\delta^2 \wedge \delta/3}^{\delta} \sqrt{1 + \log N_{[]}(\varepsilon, \mathcal{F}, \|\cdot\|)} d\varepsilon.$$

[‡] For δ near the origin the right side of (3.4.4) is not increasing in δ (and typically even unbounded) and hence quite inadequate as a bound on the modulus. However, the bound is of the correct order of magnitude for δ in the important interval (δ_n, ∞) , for $J(\delta_n, \mathcal{F}) \asymp \sqrt{n}\delta_n^2$, where it could also be replaced by the increasing function $J(\delta, \mathcal{F})(1 + M(\mathcal{F})K(\mathcal{F}))$.

Since this does not make a difference in most applications, this fact will not be stressed.

3.4.3 Penalized Maximum Contrast Estimators

In this section we present a variation on Theorem 3.4.1 that incorporates a penalty term. We consider a general criterion function of the form

$$\theta \mapsto \mathbb{M}_n(\theta) - \hat{\lambda}_n^2 \mathcal{J}_n^2(\theta),$$

where $\mathbb{M}_n$ is a stochastic process indexed by a set Θ_n as before, $\mathcal{J}_n: \Theta_n \rightarrow [0, \infty)$ a deterministic function and $\hat{\lambda}_n$ a nonnegative random variable defined on the same probability space as $\mathbb{M}_n$. Apart from the penalty, the set-up is the same as in Section 3.4.1.

The theorem applies to $\hat{\theta}_n$ that maximize the penalized criterion over Θ_n , and more generally to any $\hat{\theta}_n$ where the penalized criterion is bigger than at an arbitrary $\theta_n \in \Theta_n$ up to some tolerance term: it is assumed that

$$(3.4.5) \quad \mathbb{M}_n(\hat{\theta}_n) - \hat{\lambda}_n^2 \mathcal{J}_n^2(\hat{\theta}_n) \geq \mathbb{M}_n(\theta_n) - \hat{\lambda}_n^2 \mathcal{J}_n^2(\theta_n) - O_P(\delta_n^2).$$

3.4.6 Theorem (Rate of convergence). *For each n , let $\mathbb{M}_n$ and M_n be stochastic processes indexed by a set $\Theta_n \cup \{\theta_{n,0}\}$, let $\theta \mapsto d_n(\theta, \theta_{n,0})$ be an arbitrary nonnegative function on Θ_n , and let $\underline{\delta}_n \geq 0$ and $\lambda_n > 0$ be given numbers. Suppose that, for every n , $\delta > \underline{\delta}_n$, and events Ω_n of probability tending to one*

$$\sup_{\theta \in \Theta_n: \delta/2 < d_n(\theta, \theta_{n,0}) \leq \delta} M_n(\theta) - M_n(\theta_{n,0}) \leq -\delta^2,$$

$$E^* 1_{\Omega_n} \sup_{\substack{\theta \in \Theta_n: d_n(\theta, \theta_{n,0}) \leq \delta, \\ \mathcal{J}_n(\theta) < \delta/\lambda_n}} \sqrt{n} [(\mathbb{M}_n - M_n)(\theta) - (\mathbb{M}_n - M_n)(\theta_{n,0}) - c\delta^2]^+ \lesssim \phi_n(\delta),$$

for fixed $c \in [0, 1)$ and increasing functions $\phi_n: (\underline{\delta}_n, \infty) \rightarrow \mathbb{R}$ such that $\delta \mapsto \phi_n(\delta)/\delta^\alpha$ is decreasing for some $\alpha < 2$, and $\theta_n \in \Theta_n$. Let δ_n satisfy

$$\phi_n(\delta_n) \leq \sqrt{n}\delta_n^2, \quad \delta_n^2 \geq M_n(\theta_{n,0}) - M_n(\theta_n), \quad \delta_n \geq \underline{\delta}_n.$$

If $\hat{\theta}_n$ takes its values in Θ_n and satisfies (3.4.5), then on the intersection of the events $d_n(\hat{\theta}_n, \theta_{n,0}) > \underline{\delta}_n$ and $\hat{\lambda}_n \geq \lambda_n$, we have $d_n(\hat{\theta}_n, \theta_{n,0}) = O_P^*(\delta_n + \hat{\lambda}_n \mathcal{J}_n(\theta_n))$ and $\mathcal{J}_n(\hat{\theta}_n) = O_P^*(\delta_n/\hat{\lambda}_n + \mathcal{J}_n(\theta_n))$.

3.4.7 Addendum. If $\lambda_n \mathcal{J}(\theta_n) \leq \delta_n$ and $d_n(\theta_n, \theta_{n,0}) < \delta_n$, then the modulus condition may be replaced by

$$E^* \sup_{\substack{\theta \in \Theta_n: d_n(\theta, \theta_{n,0}) \leq \delta, \\ \mathcal{J}_n(\theta) < \delta/\lambda_n}} \sqrt{n} |(\mathbb{M}_n - M_n)(\theta) - (\mathbb{M}_n - M_n)(\theta_{n,0})| \lesssim \phi_n(\delta).$$

Proof. By assumption there exists a constant $k > 0$ so that the $-O_P(\delta_n^2)$ tolerance term in the defining property (3.4.5) of $\hat{\theta}_n$ is bounded below by $-k\delta_n^2$ with probability arbitrarily close to 1. It suffices to prove the assertions on the event where this

is valid, for every fixed k . It is also no loss of generality to assume that the events Ω_n are the full space.

Fix a constant $r > 1$ so that $r^2 c < 1$ for the constant $c < 1$ in the modulus condition (for instance $r = 2$ if $c = 0$). For $n \in \mathbb{N}$ and $j \in \mathbb{Z}$, and fixed $M \in \mathbb{N}$, let $S_{n,j}$ be the set

$$S_{n,j} = \left\{ (\theta, \lambda) \in \Theta_n \times [\lambda_n, \infty) : r^{2j-2} \delta_n^2 < d_n^2(\theta, \theta_{n,0}) + \lambda^2 \mathcal{J}_n^2(\theta) \leq r^{2j} \delta_n^2, \right. \\ \left. r^{2M} \lambda^2 \mathcal{J}_n^2(\theta_n) < d_n^2(\theta, \theta_{n,0}) + \lambda^2 \mathcal{J}_n^2(\theta) \right\}.$$

Then the union of the $S_{n,j}$ over $j > M$ contains all $(\theta, \lambda) \in \Theta_n \times [\lambda_n, \infty)$ with $d_n^2(\theta, \theta_{n,0}) + \lambda^2 \mathcal{J}_n^2(\theta) > r^{2M} (\delta_n^2 \vee \lambda^2 \mathcal{J}_n^2(\theta_n))$. For the theorem it suffices to prove that for sufficiently large M the limsup of the probability that $(\hat{\theta}_n, \hat{\lambda}_n)$ is in the intersection of the sets $\cup_{j>M} S_{n,j}$ and $\{(\theta, \lambda) : d_n(\theta, \theta_{n,0}) > \underline{\delta}_n \text{ or } \lambda \geq \lambda_n\}$ is arbitrarily close to 0.

The first assumption of the theorem implies that $M_n(\theta) - M_n(\theta_{n,0}) \leq -d_n^2(\theta, \theta_{n,0})$, whenever $\theta \in \Theta_n$ and $d_n(\theta, \theta_{n,0}) > \underline{\delta}_n$, and by assumption $\delta_n^2 \geq M_n(\theta_{n,0}) - M_n(\theta_n)$. Therefore, if $(\theta, \lambda) \in S_{n,j}$ and $d_n(\theta, \theta_{n,0}) > \underline{\delta}_n$, then

$$\begin{aligned} M_n(\theta) - M_n(\theta_n) - \lambda^2 \mathcal{J}_n^2(\theta) + \lambda^2 \mathcal{J}_n^2(\theta_n) \\ \leq M_n(\theta) - M_n(\theta_{n,0}) - \lambda^2 \mathcal{J}_n^2(\theta) + \lambda^2 \mathcal{J}_n^2(\theta_n) + \delta_n^2 \\ \leq -[d_n^2(\theta, \theta_{n,0}) + \lambda^2 \mathcal{J}_n^2(\theta)] + \lambda^2 \mathcal{J}_n^2(\theta_n) + \delta_n^2 \\ \leq -[d_n^2(\theta, \theta_{n,0}) + \lambda^2 \mathcal{J}_n^2(\theta)](1 - r^{-2M}) + \delta_n^2 \\ \leq -(r^{2j-2}(1 - r^{-2M}) - 1)\delta_n^2. \end{aligned}$$

By assumption $\mathbb{M}_n(\hat{\theta}_n) - \mathbb{M}_n(\theta_n) \geq \hat{\lambda}_n^2 \mathcal{J}_n^2(\hat{\theta}_n) - \hat{\lambda}_n^2 \mathcal{J}_n^2(\theta_n) - k\delta_n^2$, with probability arbitrarily close to 1. Therefore $G_n = \sqrt{n}(\mathbb{M}_n - M_n)$ satisfies

$$G_n(\hat{\theta}_n) - G_n(\theta_n) \geq \sqrt{n} [M_n(\theta_n) - M_n(\hat{\theta}_n) + \hat{\lambda}_n^2 \mathcal{J}_n^2(\hat{\theta}_n) - \hat{\lambda}_n^2 \mathcal{J}_n^2(\theta_n) - k\delta_n^2].$$

If $(\hat{\theta}_n, \hat{\lambda}_n) \in S_{n,j}$ and $d_n(\hat{\theta}_n, \theta_{n,0}) > \underline{\delta}_n$, then the right side is bounded below by $\sqrt{n}(r^{2j-2}(1 - r^{-2M}) - 1 - k)\delta_n^2$. Furthermore, by the definition of $S_{n,j}$ in this case $d_n(\hat{\theta}_n, \theta_{n,0}) \leq r^j \delta_n$ and $\mathcal{J}_n(\hat{\theta}_n) < r^j \delta_n / \hat{\lambda}_n \leq r^j \delta_n / \lambda_n$, on the event $\hat{\lambda}_n \geq \lambda_n$. It follows that $P^*((\hat{\theta}_n, \hat{\lambda}_n) \in S_{n,j}, d_n(\hat{\theta}_n, \theta_{n,0}) > \underline{\delta}_n, \hat{\lambda}_n \geq \lambda_n)$ is bounded above by

$$\begin{aligned} P^* \left(\sup_{\substack{\theta \in \Theta_n : d_n(\theta, \theta_{n,0}) \leq r^j \delta_n, \\ \mathcal{J}_n(\theta) < r^j \delta_n / \lambda_n}} (G_n(\theta) - G_n(\theta_n)) - \sqrt{n} c r^{2j} \delta_n^2 \right. \\ \left. \gtrsim \sqrt{n} (r^{2j-2}(1 - r^{-2M}) - 1 - k - c r^{2j}) \delta_n^2 \right). \end{aligned}$$

Here the number on the right of the inequality is bounded below by a multiple of $\sqrt{n} r^{2j} \delta_n^2$, for sufficiently large j and M , by the assumption that $c < 1$ and the choice of $r > 1$ so that $r^2 c < 1$. By an application of Markov's inequality and the second assumption of the theorem, the sum of the probabilities over $j > M$ is bounded above by a multiple of $\sum_{j>M} \phi_n(r^j \delta_n) / (\sqrt{n} r^{2j} \delta_n^2) \leq \sum_{j>M} r^{(\alpha-2)j}$. This tends to zero as $M \rightarrow \infty$. This completes the proof of the theorem.

For the addendum we decompose $\sqrt{n}(G_n(\theta) - G_n(\theta_n))$ into the difference of $\sqrt{n}(G_n(\theta) - G_n(\theta_{n,0}))$ and $\sqrt{n}(G_n(\theta_n) - G_n(\theta_{n,0}))$. By assumption the expectation of the supremum of the first term over the set of θ as in the preceding display is bounded by a multiple of $\phi_n(r^j \delta_n)$, as before, with c taken equal to 0. If $E\sqrt{n}|G_n(\theta_n) - G_n(\theta_{n,0})|$ is bounded by a multiple of $\phi_n(r^j \delta_n)$ also, then the proof can be finished as before. Now $\theta_n \in \Theta_n$ and $\lambda_n \mathcal{J}(\theta_n) \leq \delta_n$ and $d_n(\theta_n, \theta_{n,0}) < \delta_n$ by assumption. It follows that θ_n is an eligible value of θ in the supremum in the modulus at $\delta = \delta_n$, which implies that $E\sqrt{n}|G_n(\theta_n) - G_n(\theta_{n,0})| \lesssim \phi_n(\delta_n) \lesssim \phi_n(r^j \delta_n)$. ■

The numbers δ_n in the theorem are similar to the δ_n in the rate theorem without penalization, Theorem 3.4.1. A difference is that presently the modulus $\phi_n(\delta)$ is computed over parameters $\theta \in \Theta_n$ restricted through the values of their penalty $\mathcal{J}_n(\theta)$, rather than (only) restricted to a sieve. This gives a smaller modulus and hence a smaller solution to the inequality $\phi_n(\delta_n) \leq \sqrt{n}\delta_n^2$. This is offset by the fact that the final rate of convergence of $d_n(\hat{\theta}_n, \theta_{n,0})$ is the maximum of the rate as in Theorem 3.4.1 and the root $\hat{\lambda}_n \mathcal{J}_n(\theta_n)$ of the penalty term, which combines the smoothing parameter $\hat{\lambda}_n$ and the penalty $\mathcal{J}_n(\theta_n)$.^b

3.4.3.1 Smoothing Parameter

The smoothing parameter $\hat{\lambda}_n$ plays a crucial role for the rate of convergence. Because the rate is never faster than $\hat{\lambda}_n \mathcal{J}_n(\theta_n)$, we would like to make $\hat{\lambda}_n$ as small as possible. However, in that case λ_n , which is essentially a lower bound in probability on $\hat{\lambda}_n$, is necessarily small as well, and the modulus must be taken over the larger set of θ with $\mathcal{J}_n(\theta) < \delta/\lambda_n$, causing a large modulus and a slowly decreasing rate δ_n . The theoretical choice $\lambda_n = 0$ would reduce the modulus to the same quantity as in Theorem 3.4.1. An optimal choice of the smoothing parameter balances these effects. This will typically have $\lambda_n \rightarrow 0$ at an appropriate rate, dependent on the penalty and the set of target parameters.

For further understanding, we consider a fixed penalty $\mathcal{J}_n = \mathcal{J}$ and differentiate between the cases that the true parameter does or does not possess a finite penalty $\mathcal{J}(\theta_0)$. Finiteness of $\mathcal{J}(\theta_0)$ may seem natural, as a penalty is often motivated by a belief on the true parameter. For instance, we may believe that the true parameter possesses a finite square Sobolev norm, and next use this as a penalty. However, we shall see below that penalized estimators may also work well for true parameters for which the penalty is infinite.

In the case that $\mathcal{J}(\theta_0) < \infty$, we may apply Theorem 3.4.6 with $\theta_{n,0} = \theta_n = \theta_0$. Assume that the parameter set Θ_n consists of all θ in a linear space with $\mathcal{J}(\theta) < \infty$, and assume further that the penalty is homogeneous: $\mathcal{J}(c\theta) = c\mathcal{J}(\theta)$ for every $c > 0$ and θ . Then $\{\theta: \mathcal{J}(\theta) < \delta/\lambda_n\} = (\delta/\lambda_n)\{\theta: \mathcal{J}(\theta) < 1\}$, and the ε -covering numbers of the sets $\{\theta: \mathcal{J}(\theta) < \delta/\lambda_n\}$ are the $(\lambda_n/\delta)\varepsilon$ -covering numbers of the

^b In the main theorem the modulus also differs from the one in Theorem 3.4.1 in that it concerns the difference of the processes $G_n = \sqrt{n}(\mathbb{M}_n - M_n)$ at θ and θ_n rather than the difference at θ and $\theta_{n,0}$. The addendum gives a formulation with $\theta_{n,0}$ rather than θ_n . The difference can always be bounded above by $E|G_n(\theta_n) - G_n(\theta_{n,0})|$. If this is bounded above by $\phi_n(\delta_n)$, then the modulus in the theorem is bounded above by $\phi_n(\delta) + \phi_n(\delta_n)$, which gives the same rate δ_n .

set $\{\theta: \mathcal{J}(\theta) < 1\}$, relative to any norm. The entropy integrals of the two sets are related as

$$J(\delta, \{\theta: \mathcal{J}(\theta) < \delta/\lambda_n\}) = \frac{\delta}{\lambda_n} J(\lambda_n, \{\theta: \mathcal{J}(\theta) < 1\}).$$

If an inequality of type (3.4.4) holds for the class $\{\theta: \mathcal{J}(\theta) < 1\}$, relative to the appropriate norm, then it follows that the rate is not faster than the solution δ_n to

$$\frac{\delta_n}{\lambda_n} J(\lambda_n, \{\theta: \mathcal{J}(\theta) < 1\}) \leq \sqrt{n} \delta_n^2.$$

According to the theorem (with $\theta_{n,0} = \theta_n = \theta_0$), the rate is also not faster than $\lambda_n \mathcal{J}(\theta_0)$, leading to the rate

$$\lambda_n \vee \frac{J(\lambda_n, \{\theta: \mathcal{J}(\theta) < 1\})}{\sqrt{n} \lambda_n}.$$

The minimal solution with respect to λ_n equates the two terms and hence satisfies

$$J(\lambda_n, \{\theta: \mathcal{J}(\theta) < 1\}) \leq \sqrt{n} \lambda_n^2.$$

This is just the rate equation pertaining to using the model $\{\theta: \mathcal{J}(\theta) < 1\}$ in the situation that this covers the true parameter θ_0 .

The interpretation is that penalization with the optimal smoothing parameter results in the same rate of convergence as estimating a true parameter known to satisfy an explicit bound on $\mathcal{J}(\theta_0)$. Thus the penalization (only) has the role of lifting the necessity of knowing the value $\mathcal{J}(\theta_0)$ of the penalty of the true parameter. (In examples the penalty may be a seminorm rather than a norm and the entropy integral of $\{\theta: \mathcal{J}(\theta) < 1\}$ infinite. However, the preceding derivation can be adapted, and the interpretation can be upheld.)

In this scheme the optimal choice of the smoothing parameter λ_n is equal to the rate of convergence δ_n . The last assertion of the theorem shows that the penalty $\mathcal{J}(\hat{\theta}_n)$ of the estimator $\hat{\theta}_n$ will then be bounded in probability, matching the fact that the true parameter was assumed to possess a finite penalty value.

In the alternate case that $\mathcal{J}(\theta_0) = \infty$, Theorem 3.4.6 may still give a rate of convergence of $\hat{\theta}_n$ to θ_0 . We must now apply the theorem with $\theta_n \neq \theta_0$, as the rate of convergence is not faster than $\lambda_n \mathcal{J}(\theta_n)$, which would be infinite if $\theta_n = \theta_0$. Since the rate is also not faster than $M_n(\theta_0) - M_n(\theta_n)$, the sequence θ_n must be chosen to approximate θ_0 , which will typically make the penalty $\mathcal{J}(\theta_n)$ diverge to infinity. The smoothing parameters $\hat{\lambda}_n$ must then tend to zero faster than the convergence rate δ_n to balance the contribution of the penalty $\hat{\lambda}_n \mathcal{J}(\theta_n)$ to the other factors that determine the rate, including the modulus. Section 3.4.9 works this out for regression with a Sobolev penalty, and Section 3.4.12 for support vector machines. Of course, in practice the smoothing parameter may be set based on the data, for instance by cross validation or a Bayesian method.

The case that $\mathcal{J}(\theta_0) = \infty$ is of much interest, and even natural, in the case that the scaled criterion $n\mathbb{M}_n(\theta)$ is a log likelihood and $\theta \mapsto e^{-n\hat{\lambda}_n^2 \mathcal{J}^2(\theta)}$ can be viewed

as proportional to a prior density for the parameter. Then the penalized contrast estimator is the mode of the corresponding posterior distribution, given by

$$\hat{\Pi}_n(d\theta) \propto e^{n\mathbb{M}_n(\theta)} e^{-n\lambda_n^2 \mathcal{J}^2(\theta)}.$$

Lemma 3.4.8 below shows that a penalty equal to the norm of a *reproducing kernel Hilbert space (RKHS)*, such as a Sobolev norm, corresponds to a Gaussian prior. It is known that, in the infinite-dimensional case, such a Gaussian prior gives probability zero to its RKHS. In other words, the Gaussian prior is “supported” on parameters that have infinite penalty.[‡] For “most” parameters targeted by the prior, the penalty will then be infinite.

If a prior expresses a-priori beliefs about the parameter, then it could be natural that it does not depend on the sample size n . This corresponds to a smoothing parameter $\lambda_n = 1/\sqrt{n}$.

3.4.3.2 Posterior Mode for Gaussian priors

In this section we show that the penalized estimator for an RKHS penalty can be interpreted as a posterior mode relative to a Gaussian prior. A posterior mode is a point of maximum of the posterior density, which is proportional to the product of the likelihood and the prior density, by Bayes’s rule. For this definition to make sense the prior density should be defined relative to a dominating measure over the parameter space that is neutral, in some sense, to maximization. For a Euclidean parameter, the latter measure can be taken to be Lebesgue measure, but for an infinite-dimensional parameter there is no analog to Lebesgue measure and the notion of a posterior mode is less obvious. The most natural approach may be to lift from the level of density to probability, which has an absolute meaning, independent of the measurable space. We may then define a “ δ -posterior mode” as the center of a ball of radius δ of maximal posterior mass, and next a *posterior mode* as the limit of the δ -modes as $\delta \downarrow 0$, if this exists. In this section we carry this out for Gaussian priors on Banach spaces.

Let Π be a centered Gaussian Borel measure on a separable Banach space $(\Theta, \|\cdot\|)$ and $\mathbb{M}: \Theta \rightarrow \mathbb{R}$ a continuous function such that $0 < \int e^{\mathbb{M}(\theta)} d\Pi(\theta) < \infty$. Then we can define a Borel probability measure μ on Θ by

$$\mu(B) = \frac{\int_B e^{\mathbb{M}(\theta)} d\Pi(\theta)}{\int e^{\mathbb{M}(\theta)} d\Pi(\theta)}.$$

If $\mathbb{M}(\theta)$ is (up to an additive constant) the log likelihood of a model with parameter θ , for fixed data, then μ is the posterior distribution of θ relative to the prior distribution Π . (As the lemma below is meant to be applied for a given value of the observation, the index n has been dropped from $\mathbb{M}_n$ and the data has been suppressed from the notation.)

For given $\delta > 0$ we define the δ -mode of μ as

$$\theta^\delta := \operatorname{argmax}_{\theta \in \Theta} \mu(B(\theta, \delta)),$$

[‡] See Cox (1993), and Van der Vaart and Van Zanten (2008).

for $B(\theta, \delta)$ the ball in Θ of radius δ around θ . Let $\mathbb{H}$ be the reproducing kernel Hilbert space of Π and $\|\cdot\|_{\mathbb{H}}$ its norm, and set

$$\sigma^2(\Pi) = \sup_{b^* \in \Theta^*: \|b^*\|=1} \int (b^* \theta)^2 d\Pi(\theta).$$

The following lemma shows that in the limit as $\delta \downarrow 0$, the δ -mode is a penalized M -estimator relative to the penalty equal to the square reproducing Hilbert space norm.

3.4.8 Lemma (Posterior mode). *Suppose that $\mathbb{M}: \Theta \rightarrow \mathbb{R}$ is continuous and satisfies $\mathbb{M}(\theta) \leq M + \|\theta\|^2/(10\sigma^2(\Pi))$, for every $\theta \in \Theta$ and some constant M . Then any sequence θ^{δ_m} of δ_m -modes for $\delta_m \downarrow 0$ is relatively compact relative to the norm of Θ and any limit point is a maximizer of $\theta \mapsto \mathbb{M}(\theta) - \frac{1}{2}\|\theta\|_{\mathbb{H}}^2$.*

Proof. For ease of notation, let W be a random element with distribution Π . Set

$$\phi(\theta, \delta) = \inf_{h \in \mathbb{H}: \|h - \theta\| < \delta} \frac{1}{2} \|h\|_{\mathbb{H}}^2.$$

By the decentering inequalities for Gaussian measures[†]

$$(3.4.9) \quad \phi(\theta, \delta) \leq \log \frac{\Pi(B(0, \delta))}{\Pi(B(\theta, \delta))} \leq \frac{1}{2} \|\theta\|_{\mathbb{H}}^2.$$

The first inequality shows that $\phi(\theta, \delta) < \infty$ for every $\delta > 0$, for any θ in the support of Π . It can also be shown that in this case there exists for every $\delta > 0$ an element $h \in \mathbb{H}$ with $\|h - \theta\| \leq \delta$ such that $\phi(\theta, \delta) = \frac{1}{2} \|h\|_{\mathbb{H}}^2$. (See Problem 3.4.12. Hence the infimum is attained on the closed ball. This fact is used in the following, but is not really needed.)

If $\|w - \theta\| < \delta$, then $\|w\|^2 \leq (\|\theta\| + \delta)^2 \leq (10/9)\|\theta\|^2 + 10\delta^2$. Therefore, by the bound on $\mathbb{M}$, for $c^{-1} = E e^{\mathbb{M}(W)}$ and $\sigma = \sigma(\Pi)$,

$$\mu(B(\theta, \delta)) = c E[e^{\mathbb{M}(W)} 1_{\|W - \theta\| < \delta}] \leq c e^M e^{\|\theta\|^2/(9\sigma^2) + \delta^2/\sigma^2} \Pi(B(\theta, \delta)).$$

Also by continuity of $\mathbb{M}$ at 0, for sufficiently small $\delta > 0$,

$$\mu(B(0, \delta)) = c E[e^{\mathbb{M}(W)} 1_{\|W\| < \delta}] \geq c e^{\mathbb{M}(0)-1} \Pi(B(0, \delta)).$$

At $\theta = \theta^\delta$ the left side of the first display is bigger than the left side of the second display. The resulting inequality between the two right sides can be rearranged as

$$e^{-\|\theta^\delta\|^2/(9\sigma^2)} \frac{\Pi(B(0, \delta))}{\Pi(B(\theta^\delta, \delta))} \leq e^{\delta^2/\sigma^2 + M + 1 - \mathbb{M}(0)}.$$

We combine this with the first inequality in (3.4.9) to see that

$$-\frac{\|\theta^\delta\|^2}{9\sigma^2} + \phi(\theta^\delta, \delta) \leq \frac{\delta^2}{\sigma^2} + M + 1 - \mathbb{M}(0).$$

[†] E.g. Ghosal and Van der Vaart (2017), Lemmas I.28 and I.27.

We have $\|h\| \leq \sigma \|h\|_{\mathbb{H}}$, for every $h \in \mathbb{H}$. For $\|h - \theta\| < \delta$, we have that $\sigma \|h\|_{\mathbb{H}} \geq \|h\| \geq \|\theta\| - \delta \geq \|\theta\|/2$, if $\|\theta\| > 2\delta$. This implies that

$$\phi(\theta, \delta) \geq \frac{1}{8\sigma^2} \|\theta\|^2, \quad \text{if } \|\theta\| > 2\delta.$$

Therefore, for $\|\theta^\delta\| \geq 1$ and $\delta \rightarrow 0$, the left side of the second last display can be bounded below by $(-1/9 + 1/8)\|\theta^\delta\|^2/\sigma^2$. This shows that $\|\theta^\delta\|$ must remain bounded as $\delta \rightarrow 0$. Inserting this in the display we see that $\phi(\theta^\delta, \delta)$ must remain bounded. Hence there exists h^δ with $\|h^\delta - \theta^\delta\| \leq \delta$ such that $\frac{1}{2}\|h^\delta\|_{\mathbb{H}}^2 = \phi(\theta^\delta, \delta)$ remains bounded. By precompactness of the RKHS unit ball in Θ , it follows that the net $\{h^\delta: \delta > 0\}$ is relatively compact in Θ . Every limit point θ^0 satisfies $\|\theta^0\|_{\mathbb{H}} \leq \liminf \|h^\delta\|_{\mathbb{H}}$ (see Problem 3.4.11). In particular $\theta^0 \in \mathbb{H}$. By continuity of $\mathbb{M}$,

$$\mathbb{M}(\theta^0) - \frac{1}{2}\|\theta^0\|_{\mathbb{H}}^2 \geq \limsup_{\delta \downarrow 0} [\mathbb{M}(h^\delta) - \frac{1}{2}\|h^\delta\|_{\mathbb{H}}^2] = \limsup_{\delta \downarrow 0} [\mathbb{M}(\theta^\delta) - \phi(\theta^\delta, \delta)].$$

We finish the proof by showing that the right side is bounded below by $\mathbb{M}(\theta) - \frac{1}{2}\|\theta\|_{\mathbb{H}}^2$, for any $\theta \in \Theta$.

Also by continuity of $\mathbb{M}$, as $\delta \downarrow 0$, still with $c^{-1} = \mathbb{E}e^{\mathbb{M}(W)}$,

$$\mu(B(\theta, \delta)) = c \mathbb{E}[e^{\mathbb{M}(W)} 1_{\|W - \theta\| < \delta}] = c e^{\mathbb{M}(\theta)} \Pi(B(\theta, \delta))(1 + o(1)),$$

uniformly in θ belonging to compacta in Θ . Applying this also for $\theta = 0$, taking the quotient at θ and $\theta = 0$, we obtain

$$\log \frac{\mu(B(\theta, \delta))}{\mu(B(0, \delta))} = \mathbb{M}(\theta) - \mathbb{M}(0) + \log \frac{\Pi(B(\theta, \delta))}{\Pi(B(0, \delta))} + o(1).$$

At $\theta = \theta^\delta$ the left side is bigger than at any other θ . Comparing the resulting right hand sides yields

$$\mathbb{M}(\theta^\delta) + \log \frac{\Pi(B(\theta^\delta, \delta))}{\Pi(B(0, \delta))} \geq \mathbb{M}(\theta) + \log \frac{\Pi(B(\theta, \delta))}{\Pi(B(0, \delta))} + o(1).$$

By (3.4.9) the left side is smaller than $\mathbb{M}(\theta^\delta) - \phi(\theta^\delta, \delta)$, while the right side is bounded below by $\mathbb{M}(\theta) - \frac{1}{2}\|\theta\|_{\mathbb{H}}^2 + o(1)$. ■

3.4.4 Maximum Likelihood

Let $X_1, \dots, X_n$ be an i.i.d. sample from a density p that belongs to a set $\mathcal{P}$ of densities with respect to a measure μ on some measurable space.

The *sieved maximum likelihood estimator* $\hat{p}_n$ based on $X_1, \dots, X_n$ maximizes the log likelihood $p \mapsto \mathbb{P}_n \log p$. In this section the parameter is the density p itself and is denoted by p rather than θ . If $\hat{p}_n$ maximizes the log likelihood over a sieve $\mathcal{P}_n$ and $p_n \in \mathcal{P}_n$, then by concavity of the logarithm,

$$\mathbb{P}_n \log \frac{\hat{p}_n + p_n}{2p_n} \geq 0 = \mathbb{P}_n \log \frac{p_n + p_n}{2p_n}.$$

Thus the set-up of the introduction applies with the criterion function

$$m_{n,p} = \log \frac{p + p_n}{2p_n}.$$

(Note that it is not claimed that the maximum likelihood estimator maximizes $p \mapsto \mathbb{P}_n m_{n,p}$ over $\mathcal{P}_n$.) This choice is technically more convenient than the naive choice $\log p$, as it combines smoothly with the Hellinger distance

$$h(p, q) = \left(\int (p^{1/2} - q^{1/2})^2 d\mu \right)^{1/2}.$$

The key is the following pair of inequalities, which relate the Bernstein “norm” of the criterion functions $m_{n,p}$ to the Hellinger distance of the densities p . For densities (nonnegative functions) p, q, p_n , and p_0 such that $p_0/p_n \leq M$ and $p \leq q$, we have

$$\begin{aligned} \|m_{n,p} - m_{n,p_n}\|_{P_0, B} &\lesssim h(p, p_n), \\ \|m_{n,p} - m_{n,q}\|_{P_0, B} &\lesssim h(p + p_n, q + p_n) \leq h(p, q), \end{aligned}$$

where the constant in $\lesssim$ depends on the upper bound M only. These inequalities are proved ahead. Since the map $p \mapsto m_{n,p}$ is monotone, the second inequality shows that a bracketing partition of a class of densities p for the Hellinger distance induces a bracketing partition of the class of criterion functions $m_{n,p}$ for the Bernstein “norm” of essentially the same size. Thus, Theorem 2.14.18 becomes available for the classes of functions $\mathcal{M}_{n,\delta}$ with the entropy bounded by the Hellinger entropy of the class of densities.

Denote the (non-standardized) bracketing entropy integral by

$$\tilde{J}_{[]}(\delta, \mathcal{P}, h) = \int_0^\delta \sqrt{1 + \log N_{[]}(\varepsilon, \mathcal{P}, h)} d\varepsilon.$$

3.4.10 Lemma. *Let h denote the Hellinger distance on a class of densities $\mathcal{P}$ and set $m_{n,p} = \log \frac{1}{2}(p + p_n)/p_n$. If p_n and p_0 are probability densities with $p_0/p_n \leq M$ pointwise, then*

$$P_0(m_{n,p} - m_{n,p_n}) \lesssim -h^2(p, p_n),$$

for every probability density p such that $h(p, p_n) \geq 32M h(p_n, p_0)$. Furthermore, for the class of functions $\mathcal{M}_{n,\delta} = \{m_{n,p} - m_{n,p_n} : p \in \mathcal{P}_n, h(p, p_n) < \delta\}$,

$$E_{P_0}^* \|\mathbb{G}_n\|_{\mathcal{M}_{n,\delta}} \lesssim \sqrt{M} \tilde{J}_{[]}(\delta, \mathcal{P}_n, h) \left(1 + \frac{\sqrt{M} \tilde{J}_{[]}(\delta, \mathcal{P}_n, h)}{\delta^2 \sqrt{n}} \right).$$

In $\tilde{J}_{[]}(\delta, \mathcal{P}_n, h)$ the set $\mathcal{P}_n$ may be replaced by $\mathcal{P}_{n,\delta} = \{p \in \mathcal{P}_n : h(p, p_n) < \delta\}$ and the Hellinger distance may be replaced by the distance $h_n(p, q) = h(p + p_n, q + p_n)$.

Proof. Since $\log x \leq 2(\sqrt{x} - 1)$ for every positive x ,

$$\begin{aligned} P_0 \log \frac{q}{p_n} &\leq 2P_0 \left(\frac{q^{1/2}}{p_n^{1/2}} - 1 \right) \\ &= 2P_n \left(\frac{q^{1/2}}{p_n^{1/2}} - 1 \right) + 2 \int (q^{1/2} - p_n^{1/2}) (p_0^{1/2} - p_n^{1/2}) \frac{p_0^{1/2} + p_n^{1/2}}{p_n^{1/2}} d\mu. \end{aligned}$$

The first term on the far right equals $-h^2(q, p_n)$; the second can be bounded by the expression $2h(q, p_n)h(p_0, p_n)(\sqrt{M} + 1)$ in view of the assumption on the quotient p_0/p_n and the Cauchy-Schwarz inequality. The sum is bounded by $-\frac{1}{2}h^2(q, p_n)$ if $h(p_0, p_n)2(\sqrt{M} + 1) \leq \frac{1}{2}h(q, p_n)$.

The first statement of the theorem follows upon combining this with the inequalities

$$h(2p, p + q) \leq h(p, q) \leq 2h(2p, p + q).$$

These inequalities are valid for every pair of densities p and q and show that the Hellinger distance between p and q is equivalent to the Hellinger distance between p and $\frac{1}{2}(p + q)$. See Problem 3.4.4.

Next we establish the inequalities relating the Bernstein “norm” and the Hellinger distance given in the discussion preceding the statement of the theorem. Since $e^{|x|} - 1 - |x| \leq 4(e^{x/2} - 1)^2$ for every $x \geq -\log 2$, and $m_{n,p} \geq -\log 2$,

$$\|m_{n,p} - m_{n,p_n}\|_{P_{0,B}}^2 \lesssim P_0(e^{m_{n,p/2}} - 1)^2 = P_0\left(\frac{(p + p_n)^{1/2}}{(2p_n)^{1/2}} - 1\right)^2.$$

Since $p_0/p_n \leq M$, the right side is bounded by $\frac{1}{2}Mh^2(p + p_n, 2p_n)$. Combination with the preceding display gives the first inequality. If $p \leq q$, then $m_{n,q} - m_{n,p}$ is nonnegative. By the same inequality for $e^x - 1 - x$ as before,

$$\|m_{n,p} - m_{n,q}\|_{P_{0,B}}^2 \lesssim P_0(e^{(m_{n,q} - m_{n,p})/2} - 1)^2 = P_0\left(\frac{(q + p_n)^{1/2}}{(p + p_n)^{1/2}} - 1\right)^2.$$

This is bounded by $Mh^2(q + p_n, p + p_n)$ as before.

The maximal inequality is a consequence of Theorem 2.14.18' of the functions in $\mathcal{M}_{n,\delta}$ has Bernstein “norm” bounded by a multiple of $\sqrt{M}\delta$, while a bracket $[p^{1/2}, q^{1/2}]$ of densities of size δ leads to a bracket $[m_{n,p}, m_{n,q}]$ of Bernstein “norm” of size a multiple of $\sqrt{M}\delta$. ■

It follows that the conditions of Theorem 3.4.1 are satisfied with the Hellinger distance, $\underline{\delta}_n = 32M h(p_n, p_0)$, and

$$\phi_n(\delta) = \tilde{J}_{[]}(\delta, \mathcal{P}_n, h) \left(1 + \frac{\tilde{J}_{[]}(\delta, \mathcal{P}_n, h)}{\delta^2 \sqrt{n}}\right),$$

where $\tilde{J}_{[]}(\delta, \mathcal{P}_n, h)$ is the Hellinger bracketing integral of the sieve $\mathcal{P}_n$. (Usually this function has the property that $\phi_n(\delta)/\delta^\alpha$ is increasing for some $\alpha < 2$ as required by Theorem 3.4.1; otherwise $\tilde{J}_{[]}(\delta, \mathcal{P}_n, h)$ must be replaced by a suitable upper bound.) The condition $\phi_n(\delta_n) \lesssim \sqrt{n}\delta_n^2$ is equivalent to

$$\tilde{J}_{[]}(\delta_n, \mathcal{P}_n, h) \lesssim \sqrt{n}\delta_n^2.$$

For the unsieved maximum likelihood estimator the Hellinger integral is independent of n and any δ_n solving the preceding display gives an upper bound on the rate. Under the condition that the true density p_0 can be approximated by a sequence $p_n \in \mathcal{P}_n$ such that p_0/p_n is uniformly bounded, the sieved maximum likelihood estimator that maximizes the likelihood over $\mathcal{P}_n$ has at least the rate δ_n satisfying both

$$(3.4.11) \quad \tilde{J}_{[]}(\delta_n, \mathcal{P}_n, h) \leq \sqrt{n}\delta_n^2 \quad \text{and} \quad \delta_n \gtrsim h(p_n, p_0).$$

3.4.12 Theorem (Rate of convergence–MLE). *Given a random sample $X_1, \dots, X_n$ from a density p_0 let $\hat{p}_n$ maximize the likelihood $p \mapsto \prod_{i=1}^n p(X_i)$ over an arbitrary set of densities $\mathcal{P}_n$. Then $h(\hat{p}_n, p_0) = O_P(\delta_n)$ for any δ_n satisfying (3.4.11), where p_n can be any sequence with $p_n \in \mathcal{P}_n$ for every n and such that the functions $x \mapsto p_0/p_n(x)$ are uniformly bounded in x and n .*

Proof. The theorem is a corollary to Theorem 3.4.1, applied with $\theta_{n,0} = \theta_n = p_n$, $\underline{\delta}_n = 32M h(p_n, p_0)$, $d_n(\theta, \theta_{n,0})$ a multiple of $h(p, p_n)$, and the criterion functions $m_{n,p}$ given in Lemma 3.4.10. The conditions of the theorem and the appropriate modulus are obtained in the lemma. It was noted that the maximum likelihood estimator satisfies $\mathbb{P}_n m_{n,\hat{p}_n} \geq 0 = \mathbb{P}_n m_{n,p_n}$. ■

The last assertion of Lemma 3.4.10—that in the above the Hellinger distance may be replaced by $h(p + p_n, q + p_n)$ when computing the bracketing entropy of $\mathcal{P}_n$ —is of some interest, both because it tends to be hard to compute the entropy for the Hellinger distance and because this entropy may behave badly due to the infinite derivative at zero of the root transformation. We have

$$(3.4.13) \quad h_n^2(p, q) = h^2(p + p_n, q + p_n) \leq \int (p - q)^2 \frac{1}{p_n} d\mu.$$

Thus the distance h_n is bounded by the L_2 -distance with respect to the measure with density p_n^{-1} . If the latter measure is finite, then the usual upper estimates for the L_2 -entropy of the class of densities $\mathcal{P}_n$ are relevant and give upper bounds on the entropy of the class of densities with respect to h_n . While the assumption that p_n^{-1} is integrable is not natural, at least it allows some conclusions in situations that appear otherwise hard to handle.

3.4.14 Example (Monotone densities). Suppose the observations take their values in a compact interval $[0, T]$ in the real line and are sampled from a density that is known to be nonincreasing. According to Theorem 2.7.9, the set $\mathcal{F}$ of all nonincreasing functions $f: [0, T] \mapsto [0, 1]$ has bracketing entropy for the $L_2(\lambda)$ -norm of the order $1/\varepsilon$ for any finite measure λ on $[0, T]$, in particular Lebesgue measure. If a density p is nonincreasing, then so is its root $p^{1/2}$. Furthermore, the Hellinger distance on the densities is the $L_2(\lambda)$ -distance on the root densities. Conclude that if $\mathcal{P}$ is the set of all nonincreasing probability densities bounded by a constant C , then

$$\log N_{[]}(\varepsilon, \mathcal{P}, h) \leq \log N_{[]}(\varepsilon, \mathcal{F}, L_2(\lambda)) \lesssim \left(\frac{1}{\varepsilon}\right).$$

Thus $\tilde{J}_{[]}(\delta, \mathcal{P}, h) \lesssim \delta^{1/2}$, which yields a rate of convergence of at least $\delta_n = n^{-1/3}$ for the maximum likelihood estimator. The maximum likelihood estimator is the Grenander estimator. In Example 3.2.14 it is shown that the pointwise rate is exactly $n^{-1/3}$ if the derivative of the true density is nonzero.

3.4.15 Example (Convex densities). Suppose the observations take their values in a compact convex subset of $\mathbb{R}^d$ and are sampled from a density p that is known to be convex, uniformly bounded by a constant M , and uniformly Lipschitz:

$|p(y) - p(x)| \leq M\|y - x\|$. According to Theorem 2.7.15, the collection of all convex functions that are uniformly bounded and uniformly Lipschitz (without the restrictions that it is nonnegative and integrates to 1) has entropy of the order $\varepsilon^{-d/2}$ for the uniform norm. In view of (3.4.13), the entropy of the class of densities for the distance $h_1(p, q) = h(p + p_0, q + p_0)$ is bounded by the L_2 -entropy with respect to the measure with Lebesgue density p_0^{-1} . Thus, under the assumption that the true density satisfies $\int p_0^{-1} d\lambda < \infty$,

$$\log N_{[]}(\varepsilon, \mathcal{P}, h_1) \lesssim \left(\frac{1}{\varepsilon}\right)^{d/2}.$$

For dimension $d \leq 3$, this leads to a converging entropy integral $\tilde{J}_{[]}(\delta, \mathcal{P}, h_1) \lesssim \delta^{1-d/4}$ and yields a rate of convergence of at least $\delta_n = n^{-2/(d+4)}$. For dimension $d \geq 4$, the entropy integral diverges, but according to Addendum 3.4.3 (also see Problem 3.4.3), the relevant quantity is

$$\int_{c\delta^2 \wedge \delta/3}^{\delta} \sqrt{\log N_{[]}(\varepsilon, \mathcal{P}, h_1)} d\varepsilon \lesssim \begin{cases} \log\left(\frac{1}{\delta}\right), & \text{if } d = 4, \\ \left(\frac{1}{\delta}\right)^{d/2-2}, & \text{if } d > 4. \end{cases}$$

This leads to a rate of convergence of at least $\delta_n = (\log n)^{1/2}/n^{1/4}$ if $d = 4$ and a rate of convergence of at least $\delta_n = n^{-1/d}$ if $d > 4$.

3.4.16 Example (Gaussian mixtures). Suppose the observations are sampled from a mixture of univariate Gaussian distributions, of the form

$$p_{F,\sigma}(x) = \int \frac{1}{\sigma} \phi\left(\frac{x-z}{\sigma}\right) dF(z).$$

According to Theorem 2.7.18 the bracketing entropy for the total variation norm of the class of all mixtures $p_{F,\sigma}$ with F a probability distribution on a given compact interval and σ bounded away from 0 and ∞ is of the order $(\log(1/\varepsilon))^2$. Because the Hellinger distance is bounded by the root of the total variation distance, the Hellinger entropy is this number at ε^2 , which gives exactly the same order. Hence a multiple of the function given by $\phi_n(\delta) = \delta \log(1/\delta)$ is an upper bound on the bracketing entropy integral, and $\delta_n = \log n / \sqrt{n}$ solves $\phi_n(\delta_n) \lesssim \sqrt{n} \delta_n^2$. Thus Theorem 3.4.12 gives a rate of convergence of the maximum likelihood estimator equal to $\log n / \sqrt{n}$.

This nearly parametric rate arises, because the set of mixtures of Gaussian distributions with locations and scales restricted to compact sets in $\mathbb{R}$ and $(0, \infty)$ is nearly finite-dimensional, in terms of its entropy: $(\log(1/\varepsilon))^2$ versus $(1/\varepsilon)^d$ for some d . Theorems 2.7.18 and 2.7.19 can also be applied in various ways to obtain rates of convergence for mixtures with “bandwidth” tending to zero or with unbounded “location”, resulting in (slower) rates of convergence that depend on the bandwidth or the tail probabilities of the mixing distributions.

3.4.5 Concave Parametrizations

Assume that the parameter θ ranges over a subset of a linear space. Consider criterion functions $\theta \mapsto \mathbb{P}_n m_\theta$ for measurable functions m_θ such that the maps $\theta \mapsto \exp m_\theta(x)$ are concave. We are mostly thinking of maximum likelihood estimators based on a sample $X_1, \dots, X_n$ from a density p_θ that depends linearly on the parameter. Consequently we denote $\exp m_\theta(x)$ by $p_\theta(x)$. If $\hat{\theta}_n$ maximizes $\mathbb{P}_n \log p_\theta$ over a convex subset Θ_n , then

$$\mathbb{P}_n \log \frac{p_{\hat{\theta}_n}}{(1-t)p_{\hat{\theta}_n} + tp_{\theta_n}} \geq 0,$$

for all $0 \leq t \leq 1$ and every $\theta_n \in \Theta_n$. Differentiation at $t = 0$ yields the inequality

$$(3.4.17) \quad \mathbb{P}_n \frac{p_{\theta_n}}{p_{\hat{\theta}_n}} \leq 1, \quad \text{for every } \theta_n \in \Theta_n.$$

If $L: (0, \infty) \rightarrow \mathbb{R}$ is increasing such that the function $t \mapsto L(1/t)$ is convex, then Jensen's inequality gives

$$\mathbb{P}_n L\left(\frac{p_{\hat{\theta}_n}}{p_{\theta_n}}\right) \geq L\left(\frac{1}{\mathbb{P}_n(p_{\theta_n}/p_{\hat{\theta}_n})}\right) \geq L(1) = \mathbb{P}_n L\left(\frac{p_{\theta_n}}{p_{\hat{\theta}_n}}\right).$$

Thus the set-up of the introduction applies with

$$m_{n,\theta} = L\left(\frac{p_\theta}{p_{\hat{\theta}_n}}\right).$$

This conclusion depends essentially on the concavity of the map $\theta \mapsto p_\theta$, an assumption that is not made in the preceding section. We can take advantage of the present structure, which is not uncommon in infinite-dimensional applications, by making a clever choice of the function L . (This may even depend on n .) Some possible choices are

$$\log t; \quad \frac{\log(1+t)}{2}; \quad 1-t^{-\alpha}; \quad t^\alpha - 1; \quad \left(\frac{2t}{t+1}\right)^\alpha - 1; \quad \frac{t^\alpha - 1}{t^\alpha + 1},$$

each time with $0 < \alpha \leq 1$. The last two choices are of special interest because they are bounded functions. The first choice leads back to the original definition of $\hat{\theta}_n$, showing that (3.4.17) characterizes the M -estimator. The second choice is used in the preceding section.

3.4.6 Least-Squares Regression with Fixed Design

Given fixed “design” points $x_1, \dots, x_n$ in a set $\mathcal{X}$ and a map $\theta_0: \mathcal{X} \rightarrow \mathbb{R}$, let

$$Y_i = \theta_0(x_i) + e_i,$$

for independent and identically distributed “error” variables $e_1, \dots, e_n$. The observations consist of the pairs $(x_1, Y_1), \dots, (x_n, Y_n)$ and the unknown regression function θ_0 is known to belong to a set Θ . The sieved least-squares estimator $\hat{\theta}_n$ minimizes

$$\mathbb{P}_n(Y - \theta)^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \theta(x_i))^2$$

over a set Θ_n of regression functions. Inserting the expression for Y_i and calculating the square, we see that $\hat{\theta}_n$ maximizes

$$\theta \mapsto 2\mathbb{P}_n(\theta - \theta_0)e - \mathbb{P}_n(\theta - \theta_0)^2.$$

The latter criterion function is not observable but is of simpler character than the sum of squares. Note that the second term is assumed nonrandom, the randomness solely residing in the error terms. The set-up of Theorem 3.4.1 applies with the criterion functions $\mathbb{M}_n(\theta) = \mathbb{P}_n m_{n,\theta}$, for

$$m_{n,\theta}(x, e) = 2(\theta - \theta_0)(x) e - \mathbb{P}_n(\theta - \theta_0)^2.$$

Under the assumption that the error variables have mean zero, the mean of this expression is $M_n(\theta) = -\mathbb{P}_n(\theta - \theta_0)^2$ and can be used as a centering function. It satisfies, for every θ ,

$$M_n(\theta) - M_n(\theta_0) = -\mathbb{P}_n(\theta - \theta_0)^2.$$

Thus, Theorem 3.4.1 applies with $\theta_n = \theta_{n,0} = \theta_0$ and $d_n(\theta, \theta_0)$ equal to the $L_2(\mathbb{P}_n)$ -distance on the set of regression functions. The necessary maximal inequality takes the form

$$\mathbb{E}^* \sup_{\mathbb{P}_n(\theta - \theta_0)^2 < \delta^2, \theta \in \Theta_n} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n (\theta - \theta_0)(x_i) e_i \right| \lesssim \phi_n(\delta).$$

Since the design points are nonrandom, this involves relatively simple multiplier processes, to which the abstract maximal inequalities of Chapter 2.2 may apply directly.

In particular, if the error variables are Gaussian, then the stochastic process $\{n^{-1/2} \sum_{i=1}^n \theta(x_i) e_i : \theta \in \Theta\}$ is sub-Gaussian for the $L_2(\mathbb{P}_n)$ -semimetric on the set of regression functions. Corollary 2.2.9 shows that we may choose

$$\phi_n(\delta) = \int_0^\delta \sqrt{\log N(\varepsilon, \Theta_n, L_2(\mathbb{P}_n))} d\varepsilon.$$

This conclusion depends only on the sub-Gaussianity of the stochastic process $\{n^{-1/2} \sum_{i=1}^n \theta(x_i) e_i : \theta \in \Theta\}$ for the $L_2(\mathbb{P}_n)$ -semimetric. By Proposition A.1.6, this is more generally true if the error variables are sub-Gaussian.

For errors with heavier than Gaussian tails, Theorem 2.2.4 may be applied with a different Orlicz norm. For instance, in case of errors with sub-exponential tails, we have by Proposition A.1.6, followed by Lemma 2.2.2:

$$\begin{aligned} \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \theta(x_i) e_i \right\|_{\psi_1} &\lesssim \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \theta(x_i) e_i \right\|_1 + \left\| \max_{1 \leq i \leq n} n^{-1/2} \theta(x_i) e_i \right\|_{\psi_1} \\ &\leq \|\theta\|_{\mathbb{P}_{n,2}} \|e_1\|_2 + \frac{\log n}{\sqrt{n}} \|\theta\|_{\mathbb{P}_{n,\infty}} \|e\|_{\psi_1} \\ &\lesssim \log n \|\theta\|_{\mathbb{P}_{n,2}} \|e_1\|_{\psi_1}. \end{aligned}$$

Then Theorem 2.2.4 shows that we may choose

$$\phi_n(\delta) = \int_0^{\delta \log n} \log N(\varepsilon, \Theta_n, \log n L_2(\mathbb{P}_n)) d\varepsilon.$$

The $\log n$ terms do not affect the upper bound on the rate of convergence much. However, taking the integral of the entropy, rather than the square-root entropy, makes considerable difference. For instance, the integral does not appear to converge for monotone regression functions. Thus, this approach does not necessarily lead to a sharp upper bound on the rate. The discrepancy might be due to the fact that the present application of Theorem 2.2.4 uses the special sum structure of the process $\{n^{-1/2} \sum_{i=1}^n \theta(x_i) e_i; \theta \in \Theta\}$ only through their ψ_1 -norms. The maximal inequalities of Chapter 2.2 are designed especially for sums of independent processes and may be expected to give sharper bounds. Another explanation of the discrepancy is that too much is lost in our attempt to bound the ψ_1 -norm by the $L_2(\mathbb{P}_n)$ -norm. Unfortunately, it appears hard to use better estimates in general, since we are interested in the modulus with respect to the $L_2(\mathbb{P}_n)$ -norm.

Another approach is possible if the class of regression functions Θ_n has bounded uniform entropy. Symmetrization by Rademacher variables (Lemma 2.3.1) followed by application of the sub-Gaussian maximal inequality (Corollary 2.2.9) conditionally on the observations yields the desired maximal inequality with

$$\phi_n(\delta) = \mathbb{E}^* \int_0^{d_n} \sup_Q \sqrt{\log N(\varepsilon \|M_n\|_{Q,2}, \Theta_n, L_2(Q))} d\varepsilon (\mathbb{P}_n M_n^2 e^2)^{1/2}.$$

Here the supremum is taken over all finitely discrete measures Q , the function $M_n(x)$ is an envelope function for Θ_n , and

$$d_n^2 = \sup_{\mathbb{P}_n(\theta - \theta_0)^2 < \delta^2, \theta \in \Theta_n} \frac{\mathbb{P}_n(\theta - \theta_0)^2 e^2}{\mathbb{P}_n M_n^2 e^2} \lesssim \frac{\delta^{2/p} (\mathbb{P}_n M_n^{(1-2/p)q} e^{2q})^{1/q}}{\mathbb{P}_n M_n^2 e^2}.$$

The last step follows by Hölder's inequality for every $1/p + 1/q = 1$. Assume that the uniform entropy (the square of the integrand) is bounded by $\varepsilon^{-1/m}$ for $m > 1/2$. Then

$$\phi_n(\delta) \leq \delta^{1/p(1-1/2m)} \mathbb{E}^* \left(\mathbb{P}_n M_n^{(1-2/p)q} e^{2q} \right)^{(1-1/2m)/2q} \left(\mathbb{P}_n M_n^2 e^2 \right)^{1/4m}.$$

If the expectation on the right-hand side is bounded as $n \rightarrow \infty$, then this leads to a rate of convergence of at least

$$\delta_n = n^{-\frac{mp}{(4p-2)m+1}}.$$

For instance, if the class of regression functions is uniformly bounded, then this is a valid upper bound on the rate if $\mathbb{E}|e|^{2q} < \infty$. For every q the upper bound δ_n is slower than the rate $n^{-m/(2m+1)}$ obtained under sub-Gaussian tails, although this rate is approached arbitrarily closely as $q \rightarrow \infty$. It seems not unreasonable to expect that slower rates may pertain under just polynomial tails of the error distribution, but from the present derivation it is unclear whether the present upper bound is sharp or is the outcome of the application of suboptimal inequalities.

3.4.18 Example. If the set of regression functions is known to belong to $C^\alpha(K)_1$ for a compact convex subset K of $\mathbb{R}^d$, then the unsieved least-squares estimator attains a rate of convergence of at least

$$n^{-\frac{\alpha p}{(4p-2)\alpha+d}}$$

if the errors have finite $2q$ th moment. For errors with sub-Gaussian tails, the rate of convergence is at least $n^{-\alpha/(2\alpha+d)}$, which is the limit as $q \rightarrow \infty$ of the display.

A third approach to least-squares uses finite-dimensional sieves and can be analyzed by an elementary maximal inequality. For each n let $\psi_{1,n}, \dots, \psi_{N_n,n}$ be functions from $\mathcal{X}$ to $\mathbb{R}$ that form an orthonormal system in $L_2(\mathbb{P}_n^x)$. The number of elements $N_n \leq n$ of the system is to be determined later. Consider the sieves

$$\Theta_n = \text{lin}(\psi_{1,n}, \dots, \psi_{N_n,n}).$$

Expressing $\theta - \theta_n$ in terms of the orthonormal system and applying the Cauchy-Schwarz inequality, we obtain (if $\theta_n \in \Theta_n$)

$$\begin{aligned} \mathbb{E}^* \sup \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n \theta(x_i) e_i \right|^2 &\leq \mathbb{E}^* \left[\sup \sum_{j=1}^{N_n} \langle \theta, \psi_{j,n} \rangle \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_{j,n}(x_i) e_i \right) \right]^2, \\ &\leq \sup \sum_{j=1}^{N_n} \langle \theta, \psi_{j,n} \rangle^2 \sum_{j=1}^{N_n} \mathbb{E}^* \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_{j,n}(x_i) e_i \right)^2, \\ &\leq \delta^2 N_n \mathbb{E} e^2, \end{aligned}$$

where each time the supremum is taken over all $\theta \in \Theta_n - \theta_n$ such that $\mathbb{P}_n \theta^2 < \delta^2$. This approach gives useful results under just the assumption that the errors are mean zero with finite variance. The rate of convergence of the sieved least-squares estimator depends on the upper bound in combination with the approximation error $\|\theta_0 - \Theta_n\|_{\mathbb{P}_n, 2}$.

3.4.19 Example (Piecewise constant functions). Consider the design points $x_i = i/n$ when the set of regression functions $\theta: [0, 1] \rightarrow \mathbb{R}$ is known to be Lipschitz of order $0 < \alpha \leq 1$. The Lipschitz constants are not assumed uniformly bounded. A simple choice of the sieve is given by letting the base functions be multiples of the indicators of the cells $((i-1)/N, i/N]$. Discretization of θ_0 on the points i/N shows that

$$\|\theta_0 - \Theta_n\|_\infty \leq \|\theta_0\|_{\text{Lip}} \frac{1}{N_n^\alpha}.$$

In combination with the preceding maximal inequality, a rate of convergence for the sieved least-squares estimator in the $L_2(\mathbb{P}_n)$ -semimetric is given by the solution of

$$\delta_n \sqrt{N_n} \leq \sqrt{n} \delta_n^2; \quad \text{and} \quad \delta_n \geq \frac{1}{N_n^\alpha}.$$

The minimal choice $N_n \asymp \delta_n^{-1/\alpha}$ leads to the rate $\delta_n = n^{-\alpha/(2\alpha+1)}$. This rate is known to be optimal in this case.

3.4.20 Example (Finite-dimensional linear models). The preceding example is a very special case of the following scheme. Let Θ be a set of functions $\theta: \mathcal{X} \rightarrow \mathbb{R}$ on an arbitrary set, and let $\phi_1, \phi_2, \dots$ be a sequence of “basis” functions $\phi_j: \mathcal{X} \rightarrow \mathbb{R}$ such that

$$\|\text{lin}(\phi_1, \dots, \phi_N) - \theta\|_\infty \lesssim \left(\frac{1}{N}\right)^\alpha \|\theta\|_{\alpha|\infty},$$

for some norm $\|\theta\|_{\alpha|\infty}$ on Θ . For functions on a compact subset of Euclidean space a wavelet, spline or polynomial base are all possible, with $\|\theta\|_{\alpha|\infty}$ the appropriate Besov or Hölder norm (and α the regularity divided by the dimension). Piecewise constant functions, as in the preceding example, are possible for values of α smaller than 1 over the dimension.

Given the empirical measure $\mathbb{P}_n$ of design points $x_1, \dots, x_n$ in $\mathcal{X}$ and integers N_n to be determined, let $\psi_{1,n}, \dots, \psi_{N_n,n}$ be orthonormal in $L_2(\mathbb{P}_n)$ with the same linear span as $\phi_1, \dots, \phi_{N_n}$. For instance, these functions may be obtained by the Gram-Schmidt procedure. The error in $L_2(\mathbb{P}_n)$ when using this basis to approximate the functions in Θ is bounded above by $(1/N_n)^\alpha$, and as shown in the general discussion the appropriate modulus of continuity is bounded above by $\delta\sqrt{N_n}$. Therefore we can determine an optimal value of N_n exactly as in the preceding example, and derive a rate of convergence for $\|\hat{\theta}_n - \theta_0\|_{\mathbb{P}_n,2}$ equal to $n^{-\alpha/(2\alpha+1)}$.

3.4.7 Least-Squares Regression with Random Design

Another approach towards finding an upper bound on the rate of least-squares estimators uses the maximal inequality Theorem 2.14.18 based on the Bernstein “norm”. It is convenient to develop this approach for the situation wherein the design points are i.i.d. random variables. Thus $X_1, \dots, X_n$ and $e_1, \dots, e_n$ are independent i.i.d. samples and the observations consist of the pairs $(X_1, Y_1), \dots, (X_n, Y_n)$, where each Y_i is defined by

$$Y_i = \theta_0(X_i) + e_i.$$

As in the case of fixed design points considered in the preceding section, we consider the least-squares estimator $\hat{\theta}_n$, which minimizes the sum of squares $n^{-1} \sum_{i=1}^n (Y_i - \theta(X_i))^2$ over a sieve Θ_n . Equivalently, it maximizes

$$\theta \mapsto \mathbb{M}_n(\theta) = 2\mathbb{P}_n(\theta - \theta_0)e - \mathbb{P}_n(\theta - \theta_0)^2.$$

Under the assumption that the errors have mean zero, this leads to the centering function $M(\theta) = -P(\theta - \theta_0)^2$, which satisfies, for every θ ,

$$M(\theta) - M(\theta_0) \leq -P(\theta - \theta_0)^2.$$

Thus Theorem 3.4.1 applies with d_n equal to the L_2 -distance on the regression functions. If the class of regression functions is uniformly bounded by $M \geq 1/2$, then

$$\begin{aligned} P\left(e^{t|\theta - \theta_0||e|} - 1 - t|\theta - \theta_0||e|\right) &= \sum_{m \geq 2} \frac{P|\theta - \theta_0|^m P|te|^m}{m!} \\ &\leq P(\theta - \theta_0)^2 \mathbb{E}e^{2Mt|e|}. \end{aligned}$$

If the error variables have sub-exponential tails, then it follows that the Bernstein “norm” of a multiple of the variables $(\theta - \theta_0)(X_i)e_i$ is bounded by a multiple of the L_2 -norm of $\theta - \theta_0$. A bracket $[\theta_1, \theta_2]$ for regression functions leads to a bracket $[\theta_1 e^+ - \theta_2 e^-, \theta_2 e^+ - \theta_1 e^-]$ for the variables $\theta(X_i)e_i$. In view of Theorem 2.14.18',

$$\begin{aligned} \mathbb{E}^* \sup_{P(\theta - \theta_0)^2 < \delta^2, \theta \in \Theta_n} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n (\theta - \theta_0)(X_i) e_i \right| \\ \lesssim \tilde{J}_{[]}(\delta, \Theta_n, L_2(P)) \left(1 + \frac{\tilde{J}_{[]}(\delta, \Theta_n, L_2(P))}{\delta^2 \sqrt{n}} \right). \end{aligned}$$

This essentially gives the same rates of convergence as the approach based on Theorem 2.2.4 in Section 3.4.6. Presently, the error variables are assumed to be sub-exponential rather than sub-Gaussian. On the other hand the set of regression functions is assumed uniformly bounded.

3.4.8 Least-Absolute-Deviation Regression

The results on least-squares regression can be easily generalized to other minimization schemes. Typically, the rates of convergence of the estimators are the same, but the conditions on the error variables necessary to obtain these rates vary strongly. This is illustrated in this section by the case of least-absolute-deviation estimators.

In the set-up of Section 3.4.6 with fixed design points, let $\hat{\theta}_n$ minimize the sum

$$\frac{1}{n} \sum_{i=1}^n |Y_i - \theta(x_i)|$$

over the set of possible regression functions Θ . If the error distribution has median zero and is smooth, then the map $\mu \mapsto P|e + \mu|$ will be maximal and be twice differentiable at $\mu = 0$. Because this function increases linearly for $|\mu| \rightarrow \infty$, we cannot use a global quadratic bound. However, for every $M > 0$ there exists a constant $c_M > 0$ such that $P|e + \mu| - P|e| \leq -c_M |\mu|^2$, for $|\mu| \leq M$. Then for $\|\theta - \theta_0\|_\infty \leq M$,

$$\frac{1}{n} \sum_{i=1}^n \left(P|e_i - (\theta - \theta_0)(x_i)| - P|e| \right) \lesssim -c_M \frac{1}{n} \sum_{i=1}^n (\theta - \theta_0)^2(x_i).$$

In order to apply Theorem 3.4.1 with the empirical L_2 -semimetric of the design points, it suffices to show that $\|\hat{\theta}_n - \theta_0\|_\infty$ remains bounded in probability, and to control

$$\mathbb{E}^* \sup_{\mathbb{P}_n(\theta - \theta_0)^2 < \delta^2} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(|e_i - (\theta - \theta_0)(x_i)| - |e_i| - P|e - (\theta - \theta_0)(x_i)| + P|e| \right) \right|.$$

The centered variables within the large parentheses are bounded in absolute value by $2|\theta - \theta_0|(x_i)$. In view of Proposition A.1.6,

$$\left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(|e_i - (\theta - \theta_0)(x_i)| - |e_i| - P|e - \theta(x_i)| + P|e| \right) \right\|_{\psi_2} \lesssim (\mathbb{P}_n(\theta - \theta_0)^2)^{1/2}.$$

Thus the ψ_2 -norm of the increments is bounded by the $L_2(\mathbb{P}_n)$ -norm of the regression functions. According to Corollary 2.2.9, Theorem 3.4.1 can be applied with

$$\phi_n(\delta) = \int_0^\delta \sqrt{\log N(\varepsilon, \Theta_n, L_2(\mathbb{P}_n))} d\varepsilon.$$

Note that this is true without any tail conditions on the error distribution. It can be concluded that the method of least-absolute-deviations estimation is very robust against outliers, also in the sense of stabilizing rates of convergence in nonparametric regression problems.

3.4.9 Penalized Least-Squares Regression

As in Section 3.4.6 let the data consist of independent variables $Y_1, \dots, Y_n$ following the model

$$Y_i = \theta_0(x_i) + e_i.$$

Suppose that the “design points” $x_1, \dots, x_n$ are fixed numbers in the unit interval, and the “error” variables $e_1, \dots, e_n$ are an i.i.d. sample from a normal distribution with mean zero. As a penalty, take the Sobolev seminorm of order k , with square

$$\mathcal{J}^2(\theta) = \int_0^1 \theta^{(k)}(x)^2 dx,$$

and consider the penalized least-squares estimator $\hat{\theta}_n$ that minimizes over all functions $\theta: [0, 1] \rightarrow \mathbb{R}$ the criterion

$$\theta \mapsto \mathbb{P}_n(Y - \theta)^2 + \lambda_n^2 \mathcal{J}^2(\theta).$$

The solution to this minimization problem can be shown to be a *natural spline* function, and can be computed explicitly by matrix algebra.[‡] Here we are interested in establishing its rate of convergence.

By the same reasoning as in Section 3.4.6 we see that the penalized least squares estimator maximizes the criterion

$$\theta \mapsto 2\mathbb{P}_n(\theta - \theta_0)e - \mathbb{P}_n(\theta - \theta_0)^2 - \lambda_n^2 \mathcal{J}^2(\theta).$$

We can therefore apply Theorem 3.4.6 with the choice $\mathbb{M}_n(\theta) = \mathbb{P}_n m_{n,\theta}$, for $m_{n,\theta}(x, e) = 2(\theta - \theta_0)(x)e - \mathbb{P}_n(\theta - \theta_0)^2$ the same functions as in Section 3.4.6. As centering functions we also choose the same functions $M_n(\theta) = -\mathbb{P}_n(\theta - \theta_0)^2$, and then trivially have $M_n(\theta) - M_n(\theta_0) = -d_n^2(\theta, \theta_0)$, for d_n the $L_2(\mathbb{P}_n)$ -distance on the parameter set Θ . Also $(\mathbb{M}_n - M_n)(\theta) = 2\mathbb{P}_n(\theta - \theta_0)e$, and hence the relevant continuity modulus, for application of Theorem 3.4.6, is

$$(3.4.21) \quad \mathbb{E}^* \sup_{\mathbb{P}_n(\theta - \theta_0)^2 < \delta^2, \mathcal{J}(\theta) < \delta/\lambda_n} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n (\theta - \theta_n)(x_i) e_i \right|.$$

[‡] See Green and Silverman (1994) or Wahba (1990).

As the stochastic process in the supremum is sub-Gaussian relative to the $L_2(\mathbb{P}_n)$ -distance, by Corollary 2.2.9 the expectation is bounded above by $\|\theta_n - \theta_0\|_{\mathbb{P}_{n,2}}$ plus a multiple of

$$\phi_n(\delta) = \int_0^\delta \sqrt{1 + \log N(\varepsilon, \{\theta: \mathbb{P}_n(\theta - \theta_0)^2 < \delta^2, \mathcal{J}(\theta) < \delta/\lambda_n\}, L_2(\mathbb{P}_n))} d\varepsilon.$$

This is identical to the modulus found in Section 3.4.6, except that the Sobolev norms $\mathcal{J}(\theta)$ of the functions are restricted to be smaller than δ/λ_n . We could restrict the parameters also to sieves, but in the following will work with only this Sobolev restriction.

We shall derive that, for appropriately chosen smoothing parameters λ_n , the rate of convergence of $\hat{\theta}_n$ relative to the empirical L_2 -norm on the design points is of order $n^{-\alpha/(2\alpha+1)}$ if the true regression function is α -smooth, for every $\alpha \leq k$. (This is the minimax rate. Higher smoothness $\alpha > k$ is not exploited: for the rate the smoothness is capped at $\alpha \wedge k$.) Here we assume that the (deterministic) design points are spread in the interval $[0, 1]$ such that the smallest eigenvalue of the empirical covariance matrix of the vectors $(1, x_i, \dots, x_i^{k-1})$ is bounded away from zero as $n \rightarrow \infty$. We separate the cases that the true regression function possesses a finite penalty or not, the first being simpler.

The assumption that $\mathcal{J}(\theta_0) < \infty$ expresses the belief that the unknown regression function is Sobolev smooth of order k . For the set of parameters in the modulus, we then have $\|\theta - \theta_0\|_{\mathbb{P}_{n,2}} < \delta$ and $\mathcal{J}(\theta - \theta_0) \lesssim \delta/\lambda_n + 1$. Under the stated assumption on the design points and if $\lambda_n \lesssim 1$, this implies that $\|\theta^{(l)} - \theta_0^{(l)}\|_\infty \lesssim 1 + \delta/\lambda_n$, for $0 \leq l \leq k-1$ (cf. Problem 3.4.8). In view of Theorem 2.7.4 the entropy for the uniform norm of the set of all functions $\theta - \theta_0$ satisfying the latter bound and $\mathcal{J}(\theta - \theta_0) \lesssim 1 + \delta/\lambda_n$ is bounded above by a multiple of $((1 + \delta/\lambda_n)/\varepsilon \vee 1)^{1/k}$. Inserting this bound in the preceding display yields (note that $\delta \leq 1 + \delta/\lambda_n$ as $\lambda_n \rightarrow 0$),

$$\phi_n(\delta) \lesssim \int_0^\delta \sqrt{\left(\frac{1 + \delta/\lambda_n}{\varepsilon}\right)^{1/k}} d\varepsilon \lesssim \left(\frac{1}{\lambda_n} + \frac{1}{\delta}\right)^{1/(2k)} \delta.$$

Theorem 3.4.6, applied with $\theta_{n,0} = \theta_n = \theta_0$, $\hat{\lambda}_n = \lambda_n$ and $\delta_n = 0$, now gives a rate $\delta_n \rightarrow 0$ relative to the empirical L_2 -norm on the design points, for the solution to

$$\left(\frac{1}{\lambda_n} + \frac{1}{\delta_n}\right)^{1/(2k)} \delta_n \leq \sqrt{n} \delta_n^2, \quad \delta_n \gtrsim \lambda_n \mathcal{J}(\theta_0).$$

This is satisfied for $\lambda_n = \delta_n$ equal to a multiple of $n^{-k/(2k+1)}$. We conclude that in the case that $\mathcal{J}(\theta_0) < \infty$, the penalized least squares estimator with smoothing parameter $\lambda_n = n^{-k/(2k+1)}$ attains the rate of convergence $n^{-k/(2k+1)}$ relative to the empirical L_2 -norm on the design points. This rate is optimal for k -smooth regression functions, showing that the penalization achieves its aim of targeting functions with finite penalty.

Perhaps it is more surprising that the good performance is not limited to the case that the true parameter is smooth of the same order as the penalty. Assume now that the true regression function θ_0 belongs to the Hölder space $C^\alpha([0, 1])$, for $\alpha \leq$

k . For $\alpha = k$, this implies that $\mathcal{J}(\theta_0) < \infty$, and the preceding derivation applies, but for $\alpha < k$ it may well be that $\mathcal{J}(\theta_0) = \infty$, and the preceding derivation would not give a rate of convergence. (We should not be disappointed, as the derivation is based on the fact that the penalized criterion is smaller at $\hat{\theta}_n$ than at θ_0 , which is not informative if $\mathcal{J}(\theta_0) = \infty$.) To overcome this, we apply Theorem 3.4.6 with θ_n taken equal to a suitable approximation of θ_0 , with a finite penalty value. For every $\delta_n > 0$, there exists θ_n such that

$$\|\theta_0 - \theta_n\|_\infty \leq \delta_n, \quad \text{and} \quad \mathcal{J}(\theta_n) \lesssim (1/\delta_n)^{k/\alpha-1}.$$

The approximation θ_n can be constructed by convolving θ_0 with a kernel of a suitable bandwidth.^b As $\delta_n \downarrow 0$, the functions θ_n approximate the true regression function closely, but at the cost of an increasing penalty $\mathcal{J}(\theta_n)$.

We show below that the upper bound on the modulus (3.4.21) is still valid in this extended set-up. Since $M(\theta_0) - M(\theta_n) = \mathbb{P}_n(\theta - \theta_0)^2 \leq \delta_n^2$, by the choice of θ_n given in the preceding display, Theorem 3.4.6, applied with $\theta_{n,0} = \theta_0$, $\hat{\lambda}_n = \lambda_n$ and $\hat{\delta}_n = 0$ as before, but with θ_n equal to the approximation, gives a rate δ_n on the empirical L_2 -norm of $\hat{\theta}_n - \theta_0$ provided

$$\left(\frac{1}{\lambda_n} + \frac{1}{\delta_n}\right)^{1/(2k)} \delta_n \leq \sqrt{n} \delta_n^2, \quad \text{and} \quad \delta_n \geq \lambda_n \mathcal{J}(\theta_n) \asymp \lambda_n \left(\frac{1}{\delta_n}\right)^{k/\alpha-1}.$$

It can be checked that, still for $\alpha \leq k$, these inequalities are satisfied if $\lambda_n \asymp n^{-k/(2\alpha+1)}$ and $\delta_n \asymp n^{-\alpha/(2\alpha+1)}$. Thus the optimal rate of convergence $n^{-\alpha/(2\alpha+1)}$ for estimating the α -smooth function θ_0 is attained, even if the penalized criterion may be infinite at θ_0 .

To achieve this pleasant result it is necessary to choose the smoothing parameter λ_n dependent on α , and let it tend to zero at a faster rate for smaller α . In practice this might be achieved by cross validation or an empirical Bayesian scheme, without knowledge of α , and then the optimal rate would be obtained for any true smoothness $\alpha \leq k$.

In the special case that $\alpha = k - 1/2$, the square smoothing parameter is $\lambda_n^2 = 1/n$, which means that the penalty is fixed on the scale of the log likelihood, which is $-n\mathbb{P}_n(Y - \theta)^2/2$ up to an additive constant. One might say that the k th order Sobolev penalty corresponds to smoothness $k - 1/2$, rather than k , an interpretation that can be made rigorous from a Bayesian perspective.[‡]

We conclude with the proof that the modulus (3.4.21) does not change, for $\delta \geq \delta_n$, if $\theta_0 \in C^\alpha([0, 1])$ rather than has a finite penalty. As a first step, since $\|\theta_0 - \theta_n\|_\infty \leq \delta_n$ by construction, we see that (3.4.21) is for $\delta \geq \delta_n$ bounded above by

$$E^* \sup_{\mathbb{P}_n(\theta - \theta_n)^2 < 4\delta^2, \mathcal{J}(\theta) < \delta/\lambda_n} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n (\theta - \theta_n)(x_i) e_i \right|.$$

^b For instance, see Lemma 11.31 in Ghosal and Van der Vaart (2017), applied with $\beta \rightarrow \alpha$ and $k + 1 \rightarrow k$. The reproducing norm used there is given in Lemma 11.29 in the same reference.

[‡] Van der Vaart and Van Zanten (2007).

By sub-Gaussianity of the stochastic process inside the supremum, the expectation is bounded by a multiple of an entropy integral $\int_0^\delta \sqrt{1 + \log N(\varepsilon, \bar{\Theta}_n, L_2(\mathbb{P}_n))} d\varepsilon$ as before, but now with the set of functions taken equal to

$$\bar{\Theta}_n = \left\{ \theta: \mathbb{P}_n(\theta - \theta_n)^2 < 4\delta^2, \mathcal{J}(\theta) < \frac{\delta}{\lambda_n} \right\} \subset \left\{ \theta: \mathbb{P}_n(\theta - \bar{\theta}_n)^2 < 8\delta^2, \mathcal{J}(\theta) < \frac{\delta}{\lambda_n} \right\},$$

for any fixed $\bar{\theta}_n \in \bar{\Theta}_n$. As before, we then argue that $\mathbb{P}_n(\theta - \bar{\theta}_n)^2 < 8\delta^2$ and $\mathcal{J}(\theta - \bar{\theta}_n) < 2\delta/\lambda_n$ imply that $\|\theta - \bar{\theta}_n\|_\infty \lesssim 1 + \delta/\lambda_n$, and next use Theorem 2.7.4 to bound the entropy of $\{\theta - \bar{\theta}_n: \|\theta - \bar{\theta}_n\|_\infty \lesssim 1 + \delta/\lambda_n, \mathcal{J}(\theta - \bar{\theta}_n) < 2\delta/\lambda_n\}$. The entropy does not change if the class is shifted by the single function $\bar{\theta}_n$ and upper bounds the entropy of $\bar{\Theta}_n$.

3.4.10 Lasso

Given fixed “design points” $x_1, \dots, x_n$ in $\mathbb{R}^p$ and a true parameter $\theta_0 \in \mathbb{R}^p$, let

$$Y_i = \theta_0^t x_i + e_i, \quad i = 1, \dots, n,$$

for independent, identically distributed errors $e_1, \dots, e_n$ with a mean zero normal distribution with variance σ^2 . The *Lasso* estimator $\hat{\theta}_n$ of θ_0 minimizes the criterion, for given λ_n ,

$$\theta \mapsto \frac{1}{n} \sum_{i=1}^n (Y_i - \theta^t x_i)^2 + \lambda_n^2 \|\theta\|_1.$$

This estimator is particularly popular when p is large relative to n (so $p = p_n \rightarrow \infty$ with n and possibly even $p \gg n$), when a suitable choice of the smoothing parameter λ_n may yield *sparse* estimates $\hat{\theta}_n$, i.e. realizations of $\hat{\theta}_n$ with only few of its p coordinates different from zero. A proper choice of λ_n will under general sparsity conditions lead both to good reconstructions $\hat{\theta}_n^t x_i$ of the unknown means of the observations and to good estimates $\hat{\theta}_n$ of the parameter.

The Lasso estimator can be analyzed by simple direct methods,[†] even without the peeling device that is incorporated in the proofs of Theorems 3.4.1 and 3.4.6, but it is instructive to apply the general method of this chapter. In the obvious application of Theorem 3.4.6 the square penalty $\mathcal{J}^2(\theta)$ would be equal to $\|\theta\|_1$, but for more informative conclusions we choose

$$\mathcal{J}^2(\theta) = \|\theta_{S_0^c}\|_1.$$

Here for a subset $S \subset \{1, \dots, p\}$ of indices, the vector $\theta_S = (\theta_i: i \in S)$ is the projection of $\theta \in \mathbb{R}^p$ onto $\mathbb{R}^S$, and $S_0 = \{i: \theta_{0,i} \neq 0\}$ is the set of indices of the nonzero coordinates of the true parameter θ_0 . Next we apply Theorem 3.4.6 with criterion function given by

$$\mathbb{M}_n(\theta) = 2\mathbb{P}_n(e(\theta - \theta_0)^t x) - \mathbb{P}_n((\theta - \theta_0)^t x)^2 + \lambda_n^2 \|(\theta - \theta_0)_{S_0}\|_1.$$

[†] See e.g. pages 101–108 of Bühlmann and Van de Geer (2011).

With these definitions we have $\mathbb{M}_n(\theta_0) = \mathcal{J}(\theta_0) = 0$, and since $\|(\hat{\theta}_n - \theta_0)_{S_0}\|_1 \geq \|\theta_0\|_1 - \|\hat{\theta}_n\|_1$ by the triangle inequality, and $\|\theta\|_1 = \|\theta_S\|_1 + \|\theta_{S^c}\|_1$, for every θ and S ,

$$\begin{aligned} \mathbb{M}_n(\hat{\theta}_n) - \lambda_n^2 \mathcal{J}^2(\hat{\theta}_n) &\geq 2\mathbb{P}_n(e(\hat{\theta}_n - \theta_0)^t x) - \mathbb{P}_n((\hat{\theta}_n - \theta_0)^t x)^2 + \lambda_n^2 \|\theta_0\|_1 - \lambda_n^2 \|\hat{\theta}_n\|_1 \\ &\geq 0 = \mathbb{M}_n(\theta_0) - \lambda_n^2 \mathcal{J}^2(\theta_0), \end{aligned}$$

by the definition of $\hat{\theta}_n$ after evaluating the square in the Lasso criterion, as in Section 3.4.6. Thus Theorem 3.4.6 applies with $\theta_n = \theta_0$ to obtain a bound on the asymptotics of $\hat{\theta}_n$. As centering function M_n we do *not* choose the expectation of $\mathbb{M}_n$, but

$$M_n(\theta) = -\mathbb{P}_n((\theta - \theta_0)^t x)^2.$$

Then $M_n(\theta_0) = 0$ and the first condition of Theorem 3.4.6, with $\theta_{n,0} = \theta_0$, is clearly satisfied with

$$d_n^2(\theta, \theta_0) = \mathbb{P}_n((\theta - \theta_0)^t x)^2 = \frac{1}{n} \|X_n(\theta - \theta_0)\|_2^2,$$

where X_n is the usual design matrix: the $(n \times p)$ -matrix with rows x_i^t . The centered criterion is $\mathbb{M}_n(\theta) - M_n(\theta) = 2(\theta - \theta_0)^t \mathbb{P}_n(ex) + \lambda_n^2 \|(\theta - \theta_0)_{S_0}\|_1$, which vanishes at $\theta = \theta_0$, and hence the relevant modulus is given by

$$\phi_n(\delta) := \mathbb{E} \sup_{\substack{\frac{1}{n} \|X_n(\theta - \theta_0)\|_2^2 \leq \delta^2 \\ \|\theta_{S_0^c}\|_1 < \delta^2 / \lambda_n^2}} \sqrt{n} \left[2(\theta - \theta_0)^t \mathbb{P}_n(ex) + \lambda_n^2 \|(\theta - \theta_0)_{S_0}\|_1 - c\delta^2 \right]^+.$$

It will be important to use the refined version of the modulus, with the positive part instead of the absolute value and including a negative quadratic drift $-c\delta^2$, for a given $c < 1$. The simple bound $|(\theta - \theta_0)^t \mathbb{P}_n(ex)| \leq \|\theta - \theta_0\|_1 \|\mathbb{P}_n(ex)\|_\infty$ separates the parameter θ from the random part of the modulus. The random vector $\mathbb{P}_n(ex)$ of dimension p is Gaussian with mean zero and covariance matrix $n^{-2} X_n^t X_n$ and hence its maximum norm $Z_n := \|\mathbb{P}_n(ex)\|_\infty$ satisfies $\mathbb{E} Z_n \lesssim \sqrt{\log p} \|X_n\| / n$ by Lemma 2.2.2, where $\|X_n\|^2$ is defined as the maximum diagonal element of $X_n^t X_n$.

The supremum in the preceding display is bounded above by, with the supremum taken over the same set of θ ,

$$\sqrt{n} \sup_{\theta} \left[\|(\theta - \theta_0)_{S_0}\|_1 (2Z_n + \lambda_n^2) + \|(\theta - \theta_0)_{S_0^c}\|_1 2Z_n - c\delta^2 \right]^+.$$

For bounding this further we employ the *compatibility number* of the design matrix for the support S_0 of the true parameter, which is the biggest number $\psi(S_0)$ such that for every $\theta \in \mathbb{R}^p$

$$\|X_n\| \|\theta_{S_0}\|_1 \leq \psi(S_0)^{-1} \|X_n \theta\|_2 \sqrt{|S_0|}, \quad \text{if } \|\theta_{S_0^c}\|_1 \leq L \|\theta_{S_0}\|_1.$$

The constant L is arbitrary, but is fixed to a value strictly bigger than 1; the design matrix is assumed to be “compatible” in the sense that $\psi(S_0) > 0$. If the condition $\|\theta_{S_0^c}\|_1 \leq L \|\theta_{S_0}\|_1$ in the preceding display holds, then $\|\theta_{S_0}\|_1$ can be bounded by

a multiple of the Euclidean norm $\|X_n\theta\|_2$; otherwise it is bounded by $L^{-1}\|\theta_{S_0^c}\|_1$. Therefore, the second last display can be further bounded by

$$\begin{aligned} \sqrt{n} \sup_{\theta} \left[\frac{\|X_n(\theta - \theta_0)\|_2 \sqrt{|S_0|}}{\|X_n\|\psi(S_0)} (2Z_n + \lambda_n^2) \right. \\ \left. + \|(\theta - \theta_0)_{S_0^c}\|_1 \left[\left(1 + \frac{1}{L}\right) 2Z_n + \frac{\lambda_n^2}{L} \right] - c\delta^2 \right]^+ \\ \leq \frac{\delta n \sqrt{|S_0|}}{\|X_n\|\psi(S_0)} (2Z_n + \lambda_n^2) + \frac{\delta^2 \sqrt{n}}{\lambda_n^2} \left[\left(1 + \frac{1}{L}\right) 2Z_n + \frac{\lambda_n^2}{L} - c\lambda_n^2 \right]^+. \end{aligned}$$

We obtain a bound on the expectation of the supremum by replacing Z_n by its expectation $\mathbb{E}Z_n \lesssim \sqrt{\log p} \|X_n\|/n$ in the right side (for the choice $c = 0$). The resulting expression is a bound on the modulus $\phi_n(\delta)$. However, the second part of this bound is a quadratic function of δ , which is not allowed in the application of Theorem 3.4.6. To remedy this we restrict the expectation in the modulus to an event of the form $\Omega_n = \{Z_n \leq d\lambda_n^2\}$, and apply the refinement highlighted in Addendum 3.4.3. For λ_n^2 a big multiple of $\sqrt{\log p} \|X_n\|/n$, the probability of the complement Ω_n^c is of the order p^{-D} (for a large constant D), and hence tends to zero, as $p \rightarrow \infty$, as required by the addendum. Furthermore, for $d = (c-1/L)/(2+1/L)$, the second term on the far right side of the preceding display vanishes on the event Ω_n . Hence the restricted expectation of the modulus contains only the first part of the right side of the preceding display, which leads to the restricted modulus

$$\tilde{\phi}_n(\delta) = \frac{\delta n \sqrt{|S_0|}}{\|X_n\|\psi(S_0)} \left(\frac{\sqrt{\log p} \|X_n\|}{n} + \lambda_n^2 \right).$$

We conclude that the rate equation $\tilde{\phi}_n(\delta_n) \leq \sqrt{n}\delta_n^2$ is satisfied for

$$\delta_n \geq \frac{\sqrt{n} \sqrt{|S_0|}}{\|X_n\|\psi(S_0)} \left(\frac{\sqrt{\log p} \|X_n\|}{n} + \lambda_n^2 \right), \quad \lambda_n^2 \geq \frac{\sqrt{\log p} \|X_n\|}{n}.$$

For the minimal choice $\lambda_n^2 \asymp \sqrt{\log p} \|X_n\|/n$, this leads to the rate $\delta_n \geq \sqrt{|S_0|}/\psi(S_0) \sqrt{\log p}/\sqrt{n}$. Since $\mathcal{J}(\theta_0) = 0$, Theorem 3.4.6 yields $d_n(\hat{\theta}_n, \theta_0) = O_P(\delta_n)$, which implies

$$\|X_n(\hat{\theta}_n - \theta_0)\|_2 = O_P\left(\frac{\sqrt{\log p} \sqrt{|S_0|}}{\psi(S_0)}\right).$$

This bound on the *prediction error* can be interpreted as the “usual” rate $\sqrt{|S_0|}$, equal to the root number of nonzero parameters to be estimated in the case that the support set S_0 would be known, times the factor $\sqrt{\log p}$ as a (modest) penalty for not knowing this support. The rate deteriorates if the compatibility number tends to zero (and the assertion is vacuous if this is zero).

Theorem 3.4.6 also yields the bound $\mathcal{J}(\hat{\theta}_n) = O_P(\delta_n/\lambda_n)$, which translates into $\|(\hat{\theta}_n - \theta_0)_{S_0^c}\|_1 = O_P(\delta_n^2/\lambda_n^2)$. Under the compatibility assumption $\|(\hat{\theta}_n - \theta_0)_{S_0}\|_1$ is bounded above by the maximum of $L^{-1}\|(\hat{\theta}_n - \theta_0)_{S_0^c}\|_1$ and $\psi(S_0)^{-1}\|X_n(\hat{\theta}_n - \theta_0)\|_2|S_0|^{1/2}/\|X_n\|$. Together these bounds show that

$$\|\hat{\theta}_n - \theta_0\|_1 = O_P\left(\frac{\sqrt{\log p} |S_0|}{\psi^2(S_0) \|X_n\|}\right).$$

This has the same interpretation as the prediction bound: it is the usual rate for estimating θ when the support set is known times a logarithmic penalty for not knowing this support set.

3.4.11 Classification

In binary *classification* we obtain a random sample $(X_1, Y_1), \dots, (X_n, Y_n)$ of observations with values in $\mathcal{X} \times \{-1, 1\}$, with the interpretation that Y_i gives a class label (-1 or 1) for an individual characterized by X_i . We wish to predict the class Y corresponding to a “future” individual characterized by the observed value X , under the assumption that the vector (X, Y) follows the same distribution as the observations.

We can put this in the framework of this chapter by letting Θ be a collection of measurable functions $\theta: \mathcal{X} \rightarrow \{-1, 1\}$, and defining “loss functions” $m_\theta: \mathcal{X} \times \{-1, 1\} \rightarrow \mathbb{R}$ by

$$m_\theta(x, y) = 1_{y \neq \theta(x)}.$$

We interpret $\theta(x)$ as the predicted class label y of an individual with characteristic x , and then m_θ is a zero-one loss function for prediction error. Both a fixed function $\theta: \mathcal{X} \rightarrow \{-1, 1\}$ and a data-dependent function $\hat{\theta}: \mathcal{X} \rightarrow \{-1, 1\}$ are referred to as a *classifier*. We aim at the construction of a classifier that minimizes the probability

$$Pm_\theta = P(Y \neq \theta(X))$$

of assigning x to the wrong class.

This risk function is minimized over the set Θ of *all* measurable maps $\theta: \mathcal{X} \rightarrow \{-1, 1\}$ by the *Bayes classifier*

$$\theta_0(x) = 21_{\eta(x) > 1/2} - 1, \quad \eta(x) = P(Y = 1 | X = x).$$

This simply classifies an input x as -1 or 1 according to the biggest of the two probabilities $P(Y = -1 | X = x)$ and $P(Y = 1 | X = x)$. (The function θ_0 classifies an input as -1 if the two probabilities are equal (i.e. $\eta = 1/2$); this is just one possible choice: the minimizer of the Bayes risk is not unique on the set $\{\eta = 1/2\}$.) The Bayes classifier depends on the underlying distribution. We want to approximate it (or rather its minimal risk) as closely as possible by a classifier $\hat{\theta}(\cdot) = \hat{\theta}(\cdot; X_1, Y_1, \dots, X_n, Y_n)$ based on the observations. In this section we study the minimizer $\hat{\theta}$ of the empirical risk $\mathbb{P}_n m_\theta$ over a suitable model Θ , i.e. the element of Θ that minimizes the classification error on the “training data”. (For clarity of notation remark that η is an aspect of the distribution P of the data. As always we consider only one such distribution, and hence η is fixed throughout. The Bayes classifier θ_0 is a function of η , and receives its index 0 by virtue of minimizing the risk, for the given P or η .)

A classifier $\theta: \mathcal{X} \rightarrow \{-1, 1\}$ can be identified with the set $\{x: \theta(x) = 1\}$ of values x that it classifies as 1. Under this identification a “model” Θ is a collection of measurable subsets of $\mathcal{X}$. Alternatively, it is also customary to define classifiers as general functions $\theta: \mathcal{X} \rightarrow \mathbb{R}$ with the interpretation that x is classified according to the sign of $\theta(x)$. In this section we stick to the definition of θ as a function with values in $\{-1, 1\}$.

A natural distance between two classifiers is the probability that they give a different classification:

$$P(\theta_1(X) \neq \theta_2(X)) = \frac{1}{2} \|\theta_1 - \theta_2\|_{P,1} = \|m_{\theta_1} - m_{\theta_2}\|_{P,1}.$$

The first equality follows from the fact that the classifiers θ_i assume values in $\{-1, 1\}$, so that their distance is 2 if they differ; the second from the fact that the functions m_{θ_i} assume their values in $\{0, 1\}$. For the same reason the L_1 -distance on the far right is also equal to the square of the L_2 -distance. The following lemma relates this distance to the criterion function.

3.4.22 Lemma (Tsybakov's condition). *If $P1_{|\eta-1/2|<t} \leq Ct^\kappa$ for every $t > 0$ and some $\kappa > 0$, then, for every measurable function $\theta: \mathcal{X} \rightarrow \{-1, 1\}$,*

$$P(m_\theta - m_{\theta_0}) \geq C^{-1/\kappa} \|m_\theta - m_{\theta_0}\|_{P,1}^{1+1/\kappa}.$$

Furthermore, if η is bounded away from $1/2$, then this assertion is true with $\kappa = \infty$ and $C^{-1/\kappa}$ replaced by $\inf_x |2\eta(x) - 1|$.

Proof. A given input x is misclassified if $\theta(x) = 1$ and $y = -1$, or $\theta(x) = -1$ and $y = 1$. Therefore, conditioning on the input gives that $Pm_\theta = P((1 - \eta)1_{\theta=1} + \eta 1_{\theta=-1})$. Because θ assumes the values -1 and 1 only $1_{\theta=1} - 1_{\theta=-1} = 1_{\theta_0=-1} - 1_{\theta=-1} = \frac{1}{2}(\theta - \theta_0)$, whence $Pm_\theta - Pm_{\theta_0} = \frac{1}{2}P(1 - 2\eta)(\theta - \theta_0) = \frac{1}{2}P|1 - 2\eta||\theta - \theta_0|$, where the last equality follows because $\theta_0 = -1$ or 1 if $1 - 2\eta$ is positive or negative, respectively. For any $t > 0$,

$$P|1 - 2\eta||\theta - \theta_0| \geq tP|\theta - \theta_0|1_{|2\eta-1|<t} \geq tP|\theta - \theta_0| - tP1_{|2\eta-1|\leq t}.$$

By assumption the right side can be further bounded below by $tP|\theta - \theta_0| - C(t/2)^{1+\kappa}$. We choose $t = 2(1/2C)^{1/\kappa}(P|\theta - \theta_0|)^{1/\kappa}$ and rearrange the inequality to conclude the proof of the first assertion of the lemma. The second assertion follows by bounding the left side of the display from below by $cP|\theta - \theta_0|$ for $c = \inf_x |2\eta(x) - 1|$. ■

The condition of the lemma concerns the behavior of the function η around the *decision boundary* $\{x: \eta(x) = 1/2\}$, and is known as a *margin condition*. It can be interpreted in terms of the ease to mimic the Bayes classifier $\theta_0 = 21_{\eta>1/2} - 1$. This assigns an arbitrary input x to a class based on the position of $\eta(x)$ relative to $1/2$. Because the function η is not known, a good classifier must (implicitly or explicitly) estimate this function from the data. If $\eta(x)$ is bounded away from $1/2$, then small estimation errors do not matter for correct classification, but for inputs x with $\eta(x) \approx 1/2$ the classifier will be sensitive to deviations from the Bayes classifier. The most favorable situation is that η is bounded away from $1/2$. In the other case, the parameter κ describes the frequency of inputs for which the Bayes classifier possesses a value near $1/2$. For increasing κ the fraction of difficult inputs decreases, where the limit as $\kappa \uparrow \infty$ can be viewed as the favorable case that η is bounded away from $1/2$.

For $\kappa = \infty$ the criterion $P(m_\theta - m_{\theta_0})$ compares to the L_1 -norm $\|m_\theta - m_{\theta_0}\|_{P,1}$, which is the square of the L_2 -norm $\|m_\theta - m_{\theta_0}\|_{P,2}$. Thus in this case the criterion is quadratic as assumed in Theorem 3.4.1 with the distance d taken equal to the L_2 -distance. More generally, under the “margin condition” of the lemma the criterion function $Pm_\theta - Pm_{\theta_0}$ is quadratic in the distance

$$\bar{d}(\theta, \theta_0) := \|m_\theta - m_{\theta_0}\|_{P,1}^{(1+1/\kappa)/2} = \|m_\theta - m_{\theta_0}\|_{P,2}^{1+1/\kappa}.$$

This readily yields the following theorem.

3.4.23 Theorem (Rate of convergence–classification). *Suppose that the collection of classifiers Θ possesses finite uniform entropy integral $J(\cdot, \Theta, L_2)$ or bracketing entropy integral $J_{[]}(\cdot, \Theta, L_2(P))$. If the Bayes classifier θ_0 belongs to Θ , and $P1_{|\eta-1/2|\leq t} \leq Ct^\kappa$ for every $t > 0$ and some $\kappa > 0$, then $P(m_{\hat{\theta}} - m_{\theta_0}) = O_P(\delta_n^2)$ for $\delta_n \rightarrow 0$ satisfying $J(\delta_n^{\kappa/(\kappa+1)}) \lesssim \sqrt{n}\delta_n^2$, where J can be either of the two integrals (given suitable measurability). Consequently $\|\hat{\theta} - \theta_0\|_{P,1} = O_P(\delta_n^{2\kappa/(1+\kappa)})$.*

Proof. We apply Theorem 3.4.1 with the “square distance” in that theorem taken equal to $d^2(\theta, \theta_0) = P(m_\theta - m_{\theta_0})$. By Lemma 3.4.22 $d(\theta, \theta_0) \leq \delta$ implies that $\|m_\theta - m_{\theta_0}\|_{P,2} \leq \delta^{\kappa/(\kappa+1)}$. The relevant modulus of continuity is therefore bounded by

$$E^* \sup_{\|m_\theta - m_{\theta_0}\|_2 < \delta^{\kappa/(\kappa+1)}} |\mathbb{G}_n(m_\theta - m_{\theta_0})|.$$

This can be bounded by a multiple of $J(\delta^{\kappa/(\kappa+1)})(1 + J(\delta^{\kappa/(\kappa+1)})/(\sqrt{n}\delta^{2\kappa/(\kappa+1)}))$, using either of the two entropy integrals (at $\delta^{\kappa/(\kappa+1)}$), in view of Theorems 2.14.2 and 2.14.17. This is bounded by $\sqrt{n}\delta^2$ for $\delta \geq \delta_n$ solving the equation $J(\delta_n^{\kappa/(\kappa+1)}) \lesssim \sqrt{n}\delta_n^2$.

We conclude that the rate relative to d is δ_n . This can be translated in a rate for $\|\cdot\|_{P,1}$ by Lemma 3.4.22. ■

For instance, if $J(\delta) \asymp \delta^r$ for some $r \in (0, 2)$ and $\kappa = \infty$, then the *excess risk* $P(m_{\hat{\theta}} - m_{\theta_0})$ is of the order $(1/n)^{1/(2-r)}$. This is faster than $1/\sqrt{n}$ for any $r > 0$ and approaches $1/n$ as $r \uparrow 1$. Still under the assumption $\kappa = \infty$ these rates translate into the same rate for the discrepancy $\|\hat{\theta} - \theta_0\|_{P,1}$. The fact that the latter rate can be faster than the parametric rate $1/\sqrt{n}$ is perhaps surprising, but is merely a consequence of the fact that this discrepancy is really the square of the relevant distance.

3.4.24 Example (VC-class). For a VC-class of classifiers the uniform entropy integral is bounded above by a multiple of $\sqrt{V}\delta\sqrt{\log(1/\delta)}$, as $\delta \rightarrow 0$. This leads to an excess risk $P(m_{\hat{\theta}} - m_{\theta_0})$ of the order $(\log n/n)^{(\kappa+1)/(\kappa+2)}$, and a discrepancy $\|\hat{\theta} - \theta_0\|_{P,1}$ of the order $(\log n/n)^{\kappa/(\kappa+2)}$. For $\kappa \uparrow \infty$ these rates approach the rate $\log n/n$.

3.4.25 Example (Smooth classifiers). In many cases it is reasonable to assume that the Bayes classifier is a set with smooth boundaries. The collection $\mathcal{C}_{\alpha, d-1}$ of subgraphs (in $[0, 1]^d$) of functions in the unit ball of the Hölder space $C^\alpha([0, 1]^{d-1})$ of functions $f: [0, 1]^{d-1} \rightarrow \mathbb{R}$ considered in Corollary 2.7.22, or their intersections, has $L_2(P)$ bracketing entropy bounded above by $(1/\varepsilon)^{2(d-1)/\alpha}$ for any measure P with a bounded Lebesgue density. For $\alpha > d - 1$ this yields a bound on the entropy integral of the form $J(\delta) = \delta^{1-(d-1)/\alpha}$. The resulting rate on the excess risk is $n^{-\alpha/(\alpha+d-1)}$ in the case that $\kappa = \infty$, and slower (and expressible in κ) if $\kappa < \infty$.

3.4.26 Example (Convex classifiers). The collection of all compact, convex subsets of a bounded set in $\mathbb{R}^d$ possesses $L_2(P)$ -entropy with bracketing numbers of the order $(1/\varepsilon)^{d-1}$ (Theorem 2.7.13). Consequently, for $d = 2$ the entropy integral satisfies $J(\delta) \lesssim \sqrt{\delta}$, and we find a rate for the excess risk of the order $(1/n)^{(\kappa+1)/(3\kappa+4)}$.

For dimension $d \geq 3$ the entropy integrals diverge, and an upper bound on the rate must be obtained using the extension given by Addendum 3.4.3 (also see Problem 3.4.3).

3.4.12 Support Vector Machines

Consider the binary classification problem with input data $X_1, \dots, X_n$ belonging to a Hilbert space $(\mathbb{H}, \langle \cdot, \cdot \rangle)$, with norm $\|\cdot\|$, and corresponding class labels $Y_1, \dots, Y_n \in \{-1, 1\}$. A *support vector machine* is a hyperplane in $\mathbb{H}$ that separates the two sets of points $\{X_i: Y_i = -1\}$ and $\{X_i: Y_i = 1\}$ and that maximizes the minimum distance of the points to the hyperplane. The corresponding classifier assigns a label to a new input point x based on the half space it belongs to. We first assume that such a separating hyperplane exists, and subsequently extend the definition to more general sets of input points.

Any pair of a “normal” vector $\beta \in \mathbb{H}$ and a “displacement” $b \in \mathbb{R}$ defines a hyperplane $H_{\beta, b} = \{x \in \mathbb{H}: \langle x, \beta \rangle + b = 0\}$, and the distance of an arbitrary point $x \in \mathbb{H}$ to $H_{\beta, b}$ is

$$\|x - H_{\beta, b}\| = \frac{1}{\|\beta\|} |\langle x, \beta \rangle + b|.$$

The two half spaces defined by the hyperplane are determined by the sign of $\langle x, \beta \rangle + b$, whence the two sets of points are separated by $H_{\beta, b}$ if and only if the numbers $Y_i(\langle X_i, \beta \rangle + b)$, for $i = 1, \dots, n$, possess the same sign. It follows that the support vector machine is the hyperplane $H_{\beta, b}$ defined by a pair (β, b) that maximizes

$$\min_i \frac{1}{\|\beta\|} |\langle X_i, \beta \rangle + b|,$$

under these restrictions. Because a pair $(c\beta, cb)$ describes the same hyperplane as the pair (β, b) for any $c \in \mathbb{R}$, we can replace the restrictions by $\min_i Y_i(\langle X_i, \beta \rangle + b) = 1$. The target function to maximize then reduces to $1/\|\beta\|$; equivalently, we minimize $\|\beta\|^2$. The restriction can next be relaxed by replacing the equality by $\geq$, as a value (β, b) giving a minimum $m = \min_i Y_i(\langle X_i, \beta \rangle + b)$ strictly larger than 1 gives a larger value to the criterion than the value $(\beta/m, b/m)$. Thus the

support vector machine is the hyperplane $H_{\beta,b}$ found by minimizing $\|\beta\|^2$ over $(\beta, b) \in \mathbb{H} \times \mathbb{R}$ under the constraints $1 \leq Y_i(\langle X_i, \beta \rangle + b)$, for $i = 1, \dots, n$.

Obviously, a support vector machine only exists if the two sets of points are “linearly separable”. Moreover, even in the separable case it is fruitful to include a tuning parameter. The *soft margin support vector machine* allows the training points to be misclassified. It is defined as the hyperplane $H_{\beta,b}$ given by (β, b) obtained from minimizing, for a given nonnegative tuning parameter λ ,

$$\lambda^2 \|\beta\|^2 + \frac{1}{n} \sum_{i=1}^n \xi_i,$$

over $(\beta, b, \xi) \in \mathbb{H} \times \mathbb{R} \times \mathbb{R}^n$ under the constraints

$$\xi_i \geq 0, \quad 1 - \xi_i \leq Y_i(\langle X_i, \beta \rangle + b), \quad i = 1, \dots, n.$$

A value $\xi_i > 1$ allows the point X_i to fall on the “wrong” side of the hyperplane, and a value of $\xi_i \in (0, 1)$ is said to allow the point X_i to fall “within the margin”. This optimization problem always has a solution. Misclassification and points within the margin are discouraged by inclusion of the penalty $n^{-1} \sum_i \xi_i$ in the criterion. Lower values of the tuning constant λ increase the influence of this penalty.

The constraints in the preceding display can also be written as the set of single inequalities $\xi_i \geq (1 - Y_i(\langle X_i, \beta \rangle + b))_+$, for $i = 1, \dots, n$. We may now minimize the criterion $\lambda^2 \|\beta\|^2 + n^{-1} \sum \xi_i$ in two steps: first with respect to ξ and next with respect to (β, b) . The minimum over ξ is clearly assumed at the lower boundaries of the constraints on its coordinates ξ_i , and hence the optimization problem is equivalent to minimizing over $(\beta, b) \in \mathbb{H} \times \mathbb{R}$

$$(3.4.27) \quad \frac{1}{n} \sum_{i=1}^n \left(1 - Y_i(\langle X_i, \beta \rangle + b)\right)_+ + \lambda^2 \|\beta\|^2.$$

Thus the soft margin support vector machine is in fact a penalized empirical risk estimator.

Before using this observation to analyze rates of convergence, we generalize the support vector machine by the “kernel trick”. Suppose that inputs x in an arbitrary set $\mathcal{X}$ can be mapped into a Hilbert space H by a map $\phi: \mathcal{X} \rightarrow H$. The support vector machine can then be applied to the data $(\phi(X_i), Y_i)$ and finds a hyperplane in H parametrized by a pair $(\beta^*, b^*) \in H \times \mathbb{R}$; and a new example x is classified according to the sign of $\langle \phi(x), \beta^* \rangle_H + b^*$. The *kernel trick* is that this program can be implemented without explicitly defining the *feature map* ϕ , but solely in terms of the inner products

$$K(u, v) := \langle \phi(u), \phi(v) \rangle_H, \quad u, v \in \mathcal{X}.$$

This *kernel function* $K: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is clearly *positive definite*: the $(n \times n)$ -matrix $(K(u_i, u_j))$ is positive definite, for any $u_1, \dots, u_n \in \mathbb{H}$. Conversely any positive-definite function can be represented in the form $K(u, v) = \langle \phi(u), \phi(v) \rangle_H$, for some feature map. The following lemma gives a canonical construction. It shows that H can without loss of generality be taken equal to a *reproducing kernel Hilbert space* (RKHS) $\mathbb{H}$ (characterized by the RKHS property (ii) in the following lemma), and describes the support vector machine in terms of $\mathbb{H}$.

3.4.28 Lemma. *If $K: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a symmetric, nonnegative-definite function on the product of an arbitrary set $\mathcal{X}$ with itself, then there exists a Hilbert space $(\mathbb{H}, \langle \cdot, \cdot \rangle_{\mathbb{H}})$ of functions $f: \mathcal{X} \rightarrow \mathbb{R}$ such that:*

- (i) *the function $K(x, \cdot): \mathcal{X} \rightarrow \mathbb{R}$ is contained in $\mathbb{H}$ for every $x \in \mathcal{X}$ and the linear span of these functions is dense in $\mathbb{H}$;*
- (ii) *$h(x) = \langle K(x, \cdot), h \rangle_{\mathbb{H}}$ for every $h \in \mathbb{H}$ and $x \in \mathcal{X}$;*
- (iii) *$K(x, y) = \langle K(x, \cdot), K(y, \cdot) \rangle_{\mathbb{H}}$ for every $x, y \in \mathcal{X}$;*
- (iv) *the map $\bar{\phi}: x \mapsto K(x, \cdot)$ from $\mathcal{X}$ to $\mathbb{H}$ satisfies $\langle \bar{\phi}(x), \bar{\phi}(y) \rangle_{\mathbb{H}} = K(x, y)$.*
- (v) *the map $\bar{\phi}$ is injective if K is strictly positive definite.*

Furthermore, if $K(x, y) = \langle \phi(x), \phi(y) \rangle_H$ for some map $\phi: \mathcal{X} \rightarrow H$ into a Hilbert space H , then $\beta^* \in H$ minimizes over $\beta \in H$ the map, for arbitrary functions $\ell_i: \mathbb{R} \rightarrow \mathbb{R}$,

$$\beta \mapsto \frac{1}{n} \sum_{i=1}^n \ell_i(\langle \phi(X_i), \beta \rangle_H) + \lambda^2 \|\beta\|_H^2$$

if and only if the function f_{β^*} given by $f_{\beta^*}(x) = \langle \phi(x), \beta^* \rangle_H$ belongs to $\mathbb{H}$ and minimizes over $f \in \mathbb{H}$ the map

$$f \mapsto \frac{1}{n} \sum_{i=1}^n \ell_i(f(X_i)) + \lambda^2 \|f\|_{\mathbb{H}}^2.$$

Proof. By Kolmogorov's extension theorem, there exists a centered Gaussian process $(G_x: x \in \mathcal{X})$, defined on some probability space $(\Omega, \mathcal{U}, \mathbb{P})$, such that $K(x, y) = \mathbb{E}G_x G_y$, for every $x, y \in \mathcal{X}$. Let $\mathbb{L}$ be the closure of the linear span of the variables $(G_x: x \in \mathcal{X})$ in $L_2(\Omega, \mathcal{U}, \mathbb{P})$, and define $\mathbb{H}$ to be the set of functions

$$x \mapsto h_L(x) = \mathbb{E}G_x L, \quad L \in \mathbb{L},$$

equipped with the inner product

$$\langle h_{L_1}, h_{L_2} \rangle_{\mathbb{H}} = \mathbb{E}L_1 L_2.$$

The correspondence $L \mapsto h_L$ is one-to-one, because $h_{L_1} = h_{L_2}$ implies that $L_1 = L_2$ almost surely, by the definition of $\mathbb{L}$. Therefore, the inner product is well defined. By definition the map $L \mapsto h_L$ is an isometry between $\mathbb{L} \subset L_2(\Omega, \mathcal{U}, \mathbb{P})$ and $\mathbb{H}$, whence $\mathbb{H}$ is a Hilbert space.

Because $K(x, \cdot) = \mathbb{E}G_x G_{\cdot} = h_{G_x}(\cdot)$, we have that $K(x, \cdot) = h_{G_x} \in \mathbb{H}$. By linearity of $L \mapsto h_L$, the linear span of all functions h_{G_x} is the set of functions h_L with L in the linear span of $(G_x: x \in \mathcal{X})$, which is dense in $\mathbb{L}$, by its definition. Property (i) follows, because $L \mapsto h_L$ is an isometry. Properties (ii)-(v) follow readily from the representation $K(x, \cdot) = h_{G_x} \in \mathbb{H}$.

Consider another Hilbert space H and map $\phi: \mathcal{X} \rightarrow H$ such that $K(x, y) = \langle \phi(x), \phi(y) \rangle_H$, and define functions $f_{\beta}: \mathcal{X} \rightarrow \mathbb{R}$ by $f_{\beta}(x) = \langle \phi(x), \beta \rangle_H$, for $\beta \in H$. For $\beta \in H$ a finite linear combination of the form $\beta = \sum_i \alpha_i \phi(x_i)$, we have

$$f_{\beta}(x) = \langle \phi(x), \sum_i \alpha_i \phi(x_i) \rangle = \sum_i \alpha_i K(x, x_i) = \sum_i \alpha_i \mathbb{E}G_x G_{x_i}.$$

In other words $f_\beta = h_L$, for $L = \sum_i \alpha_i G_{x_i}$. Also, for the same β ,

$$\|\beta\|_H^2 = \sum_{i,j} \alpha_i \alpha_j \langle \phi(x_i), \phi(x_j) \rangle_H = \sum_{i,j} \alpha_i \alpha_j K(x_i, x_j) = E\left(\sum_i \alpha_i G_{x_i}\right)^2.$$

In other words, we also have $\|\beta\|_H = \|h_L\|_{\mathbb{H}} = \|f_\beta\|_{\mathbb{H}}$. It follows that the functions f_β for β in the linear span of $(\phi(x): x \in \mathcal{X}) \subset H$ are contained in $\mathbb{H}$ with $\|f_\beta\|_{\mathbb{H}} = \|\beta\|_H$. This isometry extends to the closure H_0 of the linear span of $(\phi(x): x \in \mathcal{X})$ in H , which maps to the closure of the corresponding set of functions h_L , which is $\mathbb{H}$ by definition. It follows that the two optimization problems with β restricted to H_0 and f ranging over $\mathbb{H}$ are identical, in the sense of a one-to-one correspondence.

The first given optimization problem optimizes over $\beta \in H$, which is possibly bigger than $H_0 \subset H$. It suffices to show that the solution β^* of this bigger problem is contained in H_0 .

Any $\beta \in H$ can be decomposed as $\beta = \beta_0 + (\beta - \beta_0)$ for $\beta_0 \in H_0$ and $\beta - \beta_0$ perpendicular to H_0 . Then $\langle \beta - \beta_0, \phi(x) \rangle_H = 0$ and hence the criterion $\sum_{i=1}^n \ell_i(\langle \phi(X_i), \beta \rangle_H)$ takes the same values at β and β_0 . Because $\|\beta_0\|_H \leq \|\beta\|_H$, the penalty decreases by projection. Hence $\beta^* \in H_0$. ■

If we denote the functions $f + b$ by θ , add the constants to $\mathbb{H}$ and define the seminorm $\|\theta\|_{\mathbb{H}}$ to be invariant under shifts, then we can identify a “kernelized” support vector machine with the map $\theta: \mathcal{X} \rightarrow \mathbb{R}$ that minimizes

$$\theta \mapsto \frac{1}{n} \sum_{i=1}^n (1 - Y_i \theta(X_i))_+ + \lambda^2 \|\theta\|_{\mathbb{H}}^2.$$

The corresponding classifier θ assigns an input x to the classes -1 and 1 if $\theta(x) < 0$ or $\theta(x) \geq 0$, respectively. (An input with $\theta(x) = 0$, which is “in the hyperplane”, is classified, arbitrarily, to 1 .) In other words, a support vector machine is a penalized empirical risk minimizer for the penalty $\|\theta\|_{\mathbb{H}}^2$ and the *hinge loss* function

$$m_\theta(x, y) = (1 - y\theta(x))_+.$$

Any reproducing kernel Hilbert space corresponds to a support vector machine. Although the corresponding minimization problem looks abstract in view of the reproducing kernel Hilbert penalty, it can actually be reduced to a convex finite-dimensional problem that involves the kernel function only, essentially by retracing the preceding derivation.

In Section 3.4.11 the *classification loss* is defined to be zero if the new example is correctly classified and 1 otherwise. If the classifiers are functions $\theta: \mathcal{X} \rightarrow \mathbb{R}$ and classification is according to the sign of θ , then the corresponding loss function is

$$\ell_\theta(x, y) = 1_{y\theta(x) < 0} + 1_{y=-1, \theta(x)=0}.$$

This differs from the hinge loss, and if used in the empirical criterion would lead to different classifiers. However, the corresponding prediction risks are both minimized by the Bayes classifier and their excess risks are related.

Define the functions m_θ and ℓ_θ as in the preceding two displays.

3.4.29 Lemma. Both Pm_θ and $P\ell_\theta$ are minimized over all measurable functions $\theta: \mathcal{X} \rightarrow \mathbb{R}$ by $\theta_0 = 21_{\eta > 1/2} - 1$, for $\eta(x) = P(Y = 1 | X = x)$, and

$$P\ell_\theta - P\ell_{\theta_0} \leq Pm_\theta - Pm_{\theta_0}, \quad \text{every } \theta.$$

Furthermore, if $P1_{|\eta - 1/2| \leq t} \leq Ct^\kappa$ for every $t > 0$ and some $\kappa > 0$, then, for $d^2(\theta, \theta_0) = P(m_\theta - m_{\theta_0})$,

$$\|m_\theta - m_{\theta_0}\|_{P,2} \lesssim \|\theta - \theta_0\|_\infty^{1/2} d(\theta, \theta_0) + C^{1/(2\kappa+2)} d(\theta, \theta_0)^{\kappa/(\kappa+1)}.$$

If η is bounded away from $1/2$, then this is true with $\kappa = \infty$. Furthermore, the following are true.

- (i) If η is bounded away from 0 , $1/2$ and 1 , then $P(m_\theta - m_{\theta_0}) \asymp \|\theta - \theta_0\|_{P,1}$.
- (ii) If η is bounded away from 0 and 1 , then $\|m_\theta - m_{\theta_0}\|_{P,2} \asymp \|\theta - \theta_0\|_{P,2}$.
- (iii) If θ takes its values in $[-1, 1]$, then $P(m_\theta - m_{\theta_0}) = P[|\theta - \theta_0| |1 - 2\eta|]$ and $\|m_\theta - m_{\theta_0}\|_{P,2} = \|\theta - \theta_0\|_{P,2}$.

3.4.30 Lemma. If $\Theta = B \oplus \mathbb{H}$ is the direct sum of linear spaces B and $\mathbb{H}$, then $\|b\|_{Q,2} \leq \|m_{b+h} - m_{\theta_0}\|_{P,2} + \|h\|_\infty + 1$ for every $b + h \in B \oplus \mathbb{H}$, where Q is the measure given by $dQ = (\eta \wedge (1 - \eta)) dP$.

Proofs. By conditioning on the input we can write

$$\begin{aligned} P\ell_\theta &= P(\eta 1_{\theta < 0} + (1 - \eta) 1_{\theta \geq 0}), \\ Pm_\theta &= P(\eta(1 - \theta)_+ + (1 - \eta)(1 + \theta)_+). \end{aligned}$$

The integrand in the first expression is minimized over θ by choosing θ positive or negative if η is strictly bigger or smaller than $1 - \eta$, respectively. This is equivalent to choosing $\theta \geq 0$ or < 0 if $\eta > 1/2$ or $< 1/2$ and arbitrarily if $\eta = 1/2$. The function $\theta = 21_{\eta > 1/2} - 1$ is a possible choice.

The integrand in the second expression is minimized over $\theta \in \mathbb{R}$ at every $\theta \leq -1$ if $\eta = 0$, minimal at $\theta = -1$ if $\eta \in (0, 1/2)$, minimal at every $\theta \in [-1, 1]$ if $\eta = 1/2$, minimal at $\theta = 1$ if $\eta \in (1/2, 1)$, and minimal at every $\theta \geq 1$ if $\eta = 1$. See Figure 3.4.1. Again the function $\theta = 21_{\eta > 1/2} - 1$ is a possible choice.

The minimal values of the integrands can be seen to be equal to $\eta \wedge (1 - \eta)$ and $2(\eta \wedge (1 - \eta))$, respectively. Therefore, $P\ell_\theta - P\ell_{\theta_0}$ and $Pm_\theta - Pm_{\theta_0}$ are equal to the integrals with respect to the distribution of X of the two functions

$$\begin{aligned} &\eta 1_{\theta < 0} + (1 - \eta) 1_{\theta \geq 0} - \eta \wedge (1 - \eta), \\ &\eta(1 - \theta)_+ + (1 - \eta)(1 + \theta)_+ - 2(\eta \wedge (1 - \eta)). \end{aligned}$$

It can be verified (with some effort, separately for θ in $(-\infty, -1]$, $(-1, 0)$, $\{0\}$, $(0, 1)$ and $[1, \infty)$) that the first expression is bounded above by the second for every $\theta \in \mathbb{R}$. This finishes the proof that $P\ell_\theta - P\ell_{\theta_0} \leq Pm_\theta - Pm_{\theta_0}$.

It can be shown that (use that the $+$ symbols in $(1 + \theta)_+$ and $(1 - \theta)_+$ can be deleted if $\theta \geq -1$ and $\theta \leq 1$, respectively, hence in both cases if $|\theta| \leq 1$ and always

in the case of $\theta_0 = 21_{\eta>1/2} - 1$; Figure 3.4.1 gives a graphical representation of the relevant function for the first equality)

$$\begin{aligned} P(m_\theta - m_{\theta_0}) &= P|\theta - \theta_0| [1_{\theta<-1}\eta + 1_{|\theta|\leq 1}|1 - 2\eta| + 1_{\theta>1}(1 - \eta)] \\ &\quad - P[1_{\theta<-1}1_{\eta>1/2}2(1 - \eta) + 1_{\theta>1}1_{\eta\leq 1/2}2\eta], \\ P(m_\theta - m_{\theta_0})^2 &= P|\theta - \theta_0|^2 [1_{\theta<-1}\eta + 1_{|\theta|\leq 1} + 1_{\theta>1}(1 - \eta)] \\ &\quad + P[1_{\theta<-1}1_{\eta>1/2}4(1 - \eta) + 1_{\theta>1}1_{\eta\leq 1/2}4\eta]. \end{aligned}$$

Since $1_{\eta>1/2}4(1 - \eta) \leq (\theta - \theta_0)^2\eta$ on $\theta < -1$ and $1_{\eta\leq 1/2}4\eta \leq (\theta - \theta_0)^2(1 - \eta)$ on $\theta > 1$, the second equality shows that $P(m_\theta - m_{\theta_0})^2$ is bounded above by twice $P|\theta - \theta_0|^2 [1_{\theta<-1}\eta + 1_{|\theta|\leq 1} + 1_{\theta>1}(1 - \eta)]$. The integrand in the first integral (the difference of the two parts $|\theta - \theta_0|[\cdot] - [\cdot]$) can be seen to be nonnegative and bounded below by

$$\begin{aligned} |\theta - \theta_0| &\left[(1_{\theta\leq -2\frac{1}{3}} + 1_{-2<\theta<-1}1_{\eta\leq 1/2})\eta \right. \\ &\quad + (1_{-2<\theta<-1}1_{\eta>1/2} + 1_{|\theta|\leq 1} + 1_{1<\theta<2}1_{\eta\leq 1/2})|1 - 2\eta| \\ &\quad \left. + (1_{1<\theta<2}1_{\eta>1/2} + 1_{\theta\geq 2\frac{1}{3}})(1 - \eta) \right]. \end{aligned}$$

Therefore $P(m_\theta - m_{\theta_0}) < \delta^2$ implies the three inequalities

$$\begin{aligned} P|\theta - \theta_0|(1_{\theta\leq -2} + 1_{-2<\theta<-1}1_{\eta\leq 1/2})\eta &< 3\delta^2, \\ P|\theta - \theta_0|(1_{-2<\theta<-1}1_{\eta>1/2} + 1_{|\theta|\leq 1} + 1_{1<\theta<2}1_{\eta\leq 1/2})|1 - 2\eta| &< \delta^2, \\ P|\theta - \theta_0|(1_{1<\theta<2}1_{\eta>1/2} + 1_{\theta\geq 2})(1 - \eta) &< 3\delta^2. \end{aligned}$$

We split the integral $P|\theta - \theta_0|^2 [1_{\theta<-1}\eta + 1_{|\theta|\leq 1} + 1_{\theta>1}(1 - \eta)]$ into integrals over the same three domains used in the three displayed equations. The integrals over the first and third domains are immediately seen to be bounded above by $3\|\theta - \theta_0\|_\infty \delta^2$. To complete the bound on $P|\theta - \theta_0|^2 [1_{\theta<-1}\eta + 1_{|\theta|\leq 1} + 1_{\theta>1}(1 - \eta)]$ it remains to bound $P|\theta - \theta_0|^2 G$, for $G = 1_{-2<\theta<-1}1_{\eta>1/2} + 1_{|\theta|\leq 1} + 1_{1<\theta<2}1_{\eta\leq 1/2}$. As in the proof of Lemma 3.4.22, it can be shown that the margin condition on η implies that $PH \leq (2C\|H\|_\infty)^{1/(\kappa+1)}(PH|1 - 2\eta|)^{\kappa/(\kappa+1)}$, for any nonnegative uniformly bounded variable H . We apply this with $H = |\theta - \theta_0|G$, which is uniformly bounded by 3, to see that $P|\theta - \theta_0|^2 G \leq 3P|\theta - \theta_0|G = 3PH$ is bounded above by a multiple of $C^{1/(\kappa+1)}\delta^{2\kappa/(\kappa+1)}$. This concludes the proof of the bound on $\|m_\theta - m_{\theta_0}\|_{P,2}$ in terms of $d(\theta, \theta_0)$.

If η is bounded away from $1/2$, then $PH \lesssim PH|1 - 2\eta|$ and the argument of the preceding paragraph can be improved to $\kappa = \infty$.

Assertions (i)–(iii) are clear from the expressions and bounds obtained in the preceding.

For the proof of Lemma 3.4.30, let $A = \{x: |b(x)| \leq \|h\|_\infty + 1\}$. Then $\theta = b + h$ is smaller than -1 or greater than 1 on A^c , and hence $P(m_\theta - m_{\theta_0})^2 \geq P(\theta - \theta_0)^2 1_{A^c}(\eta \wedge (1 - \eta))$, by the representation of this second moment obtained previously. In other words $\|\theta - \theta_0\|_{R,2} \leq \|m_\theta - m_{\theta_0}\|_{P,2}$, for the measure R defined by $dR = 1_{A^c}(\eta \wedge (1 - \eta)) dP$. Because $b = \theta - \theta_0 - h + \theta_0$, the triangle inequality gives $\|b\|_{R,2} \leq \|\theta - \theta_0\|_{R,2} + \|h\|_{R,2} + \|\theta_0\|_{R,2} \leq \|m_\theta - m_{\theta_0}\|_{P,2} + \|h\|_\infty + 1$.

We combine this with the inequality $Pb^2 1_A(\eta \wedge (1 - \eta)) \leq (\|h\|_\infty + 1)^2$, which is immediate from the definition of A , to see that $Pb^2(\eta \wedge (1 - \eta))$ is bounded as claimed. ■

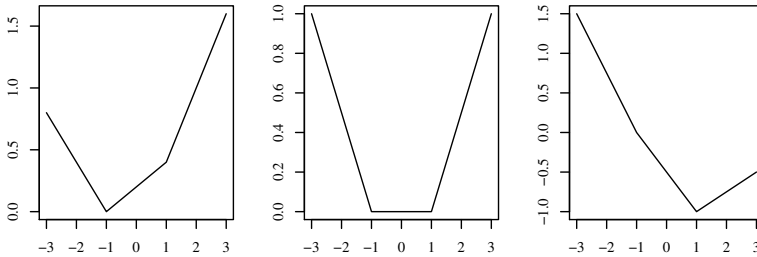


Figure 3.4.1. The function $\theta \mapsto \eta(1-\theta)_+ + (1-\eta)(1+\theta)_+ - \eta(1-\theta_0)_+ - (1-\eta)(1+\theta_0)_+$, for $\eta = 2/5, 1/2, 3/4$. In all three cases the minimum is taken at $\theta_0 = 21_{\eta>1/2} - 1$, which is equal to $-1, -1$ and 1 , respectively.

To obtain a rate of convergence of the support vector machine we may apply Theorem 3.4.6 with $\theta_{n,0} = \theta_0$ and $d^2(\theta, \theta_0) = P(m_\theta - m_{\theta_0})$. This yields a rate for $d(\hat{\theta}_n, \theta_0)$, which implies the same rate for the root of the classification error $P(\ell_{\hat{\theta}_n} - \ell_{\theta_0})$, in view of Lemma 3.4.29. The lemma also shows that the relevant modulus of continuity can be bounded by, for fixed constants C and κ in the margin condition,

$$(3.4.31) \quad \phi_n(\delta) := \mathbb{E}^* \sup_{\substack{\theta \in \mathbb{H}: \|\theta\|_{\mathbb{H}} < \delta/\lambda_n \\ \|m_\theta - m_{\theta_0}\|_{P,2} \leq \|\theta - \theta_0\|_\infty^{1/2} \delta + \delta^{\kappa/(\kappa+1)}} |\mathbb{G}_n(m_\theta - m_{\theta_n})|.$$

Under the conditions of Addendum 3.4.7, the function θ_n may be replaced by θ_0 .

We may use entropy integrals to bound this expression. Because the map $\theta \mapsto m_\theta$ is Lipschitz, the entropy of a class of functions m_θ is bounded by the entropy of the corresponding class of functions θ . Furthermore, by the reproducing formula (ii) of Lemma 3.4.28 and the Cauchy-Schwarz inequality we have $|\theta(x)| \leq \|\theta\|_{\mathbb{H}} \|K(\cdot, x)\|_{\mathbb{H}}$, so that by (iii) of Lemma 3.4.28,

$$(3.4.32) \quad \|\theta\|_\infty \leq \|\theta\|_{\mathbb{H}} \sup_x \sqrt{K(x, x)}.$$

Consequently, if the kernel is bounded on the diagonal, then in the modulus the norm $\|\theta - \theta_0\|_\infty$ may be replaced by a multiple of $\|\theta\|_{\mathbb{H}} + 1 \leq \delta/\lambda_n + 1$, and the entropy integral may be computed over the interval $(0, s_n(\delta))$, for $s_n(\delta) := \delta^{3/2}/\lambda_n^{1/2} + \delta^{\kappa/(\kappa+1)}$.

The Bayes classifier θ_0 is typically not contained in the reproducing kernel Hilbert space, whence we must apply Theorem 3.4.6 with a sequence θ_n different from θ_0 , as the rate guaranteed is not faster than $\lambda_n \mathcal{J}(\theta_n) = \lambda_n \|\theta_n\|_{\mathbb{H}}$. Theorem 3.4.6 applied with $\theta_{n,0} = \theta_0$ and arbitrary $\theta_n \in \mathbb{H}$ gives a rate of convergence not faster than the worst of the solution to $\phi_n(\delta_n) \leq \sqrt{n} \delta_n^2$, the penalty $\lambda_n \|\theta_n\|_{\mathbb{H}}$

and the root of $P(m_{\theta_n} - m_{\theta_0})$. If we choose θ_n a (near) minimizer of the map $\theta \mapsto P(m_\theta - m_{\theta_0}) + \lambda_n^2 \|\theta\|_{\mathbb{H}}^2$, then this reduces to the square root of

$$(3.4.33) \quad \delta_n^2 + \inf_{\theta} \left(P(m_\theta - m_{\theta_0}) + \lambda_n^2 \|\theta\|_{\mathbb{H}}^2 \right).$$

Here δ_n is determined by $\phi_n(\delta_n) \leq \sqrt{n}\delta_n^2$, for the modulus defined in (3.4.31). Because the Bayes classifier $\theta_0 = 21_{\eta > 1/2} - 1$ is a scaled and shifted indicator function, it appears that the second “bias” term in the display, the infimum, can seriously decelerate the convergence. (An alternative would be to apply the theorem with $\theta_{n,0}$ also an approximation of θ_0 . In the modulus we must then replace θ_0 by $\theta_{n,0}$, but as $\theta_{n,0}$ will not be the minimizing value, Lemma 3.4.29 may not be available to translate the restriction in the modulus into a readily usable form.)

Because $\theta \mapsto m_\theta$ is Lipschitz, the quantity $P(m_\theta - m_{\theta_0})$ in (3.4.33) can always be bounded above by $P|\theta - \theta_0|$. If θ takes its values in $[-1, 1]$, then it is also bounded above by $P[|\theta - \theta_0| |1 - 2\eta|]$, by (iii) of Lemma 3.4.29. Then the approximation of θ_n to θ_0 may be of smaller accuracy near the decision boundary $\{x: \eta(x) = 1/2\}$.

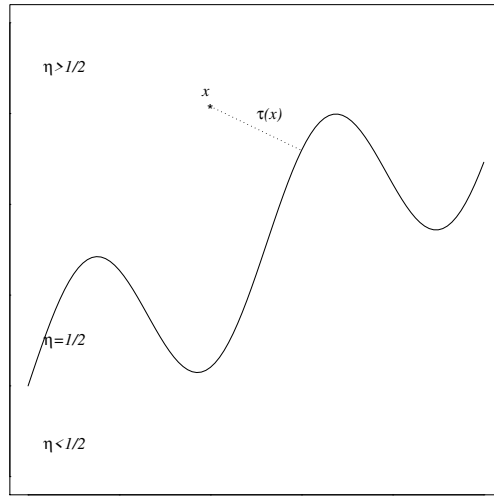


Figure 3.4.2. The function $\tau(x)$ gives the distance of the input x to the decision boundary $\{y: \eta(y) = 1/2\}$, pictured as a smooth curve in $\mathcal{X} = [0, 1]^2$. The regions above and below the curve are $\{\eta > 1/2\} = \{\theta_0 = 1\}$ and $\{\eta \leq 1/2\} = \{\theta_0 = -1\}$, for θ_0 the Bayes classifier, where by convention “below” includes the points on the curve.

For a smoothness penalty, good approximations can be constructed as convolutions $\theta_n = \theta_0 * \psi_{\sigma_n}$ of θ_0 with a smooth kernel of bandwidths $\sigma_n \downarrow 0$. The approximation errors $\theta_n(x) - \theta_0(x)$ will then be small, or even zero, for x that are in the interior of one of the regions $\{\theta_0 = 1\}$ or $\{\theta_0 = -1\}$ at distance a large multiple of σ from the decision boundary, as smoothing the constant values 1 or -1 retains the value. A nonzero error $\theta_n(x) - \theta_0(x)$ will occur only for x such that the smoothing around x involves points from both sets $\{\theta_0 = 1\}$ and $\{\theta_0 = -1\}$. Typically smaller bandwidth will give better approximation, but a larger penalty $\|\theta_n\|_{\mathbb{H}}$. A proper choice of bandwidth will be essential.

To make this quantitative, consider the map $\tau: \mathcal{X} \rightarrow [0, \infty)$, defined by

$$\tau(x) = \begin{cases} \|x - \{\eta > 1/2\}\|, & \text{if } \eta(x) \leq 1/2, \\ \|x - \{\eta \leq 1/2\}\|, & \text{if } \eta(x) > 1/2. \end{cases}$$

Since the sets $\{\eta > 1/2\}$ and $\{\eta \leq 1/2\}$ are identical to the sets $\{\theta_0 = 1\}$ and $\{\theta_0 = -1\}$, for $\theta_0 = 21_{\eta > 1/2} - 1$ the Bayes classifier, this is the distance of x to the set of inputs of the other type. Typically $\tau(x)$ can be interpreted as the distance of x to the decision boundary $\{\eta = 1/2\}$ (see Figure 3.4.2).

3.4.34 Lemma. *For any probability density ψ on $\mathbb{R}^d$, $\bar{\Psi}(s) = \int_{y: \|y\| \geq s} \psi(y) dy$, and any extension of θ_0 to a function $\theta_0: \mathbb{R}^d \rightarrow \{-1, 1\}$, with corresponding extension η ,*

$$P(m_{\theta_0 * \psi} - m_{\theta_0}) \leq 2P[\bar{\Psi}(\tau)|1 - 2\eta|].$$

Proof. Because θ_0 takes its values in $[-1, 1]$, the average $\theta_0 * \psi(x) = \int \theta_0(x - y) \psi(y) dy$ also takes its values in the interval $[-1, 1]$. Therefore the left side of the lemma is equal to $P[|\theta_0 * \psi - \theta_0| |1 - 2\eta|]$, by (iii) of Lemma 3.4.29. By the definition of τ , we have that $\theta_0(x - y) = \theta_0(x)$, for $\|y\| < \tau(x)$. Consequently, for every x , the difference $\theta_0 * \psi(x) - \theta_0(x) = \int (\theta_0(x - y) - \theta_0(x)) \psi(y) dy$ is equal to $\int_{y: \|y\| \geq \tau(x)} (\theta_0(x - y) - \theta_0(x)) \psi(y) dy$. Since θ_0 is bounded in absolute value by 1, the absolute value of the integral is bounded above by $2\bar{\Psi}(\tau(x))$. We finish by multiplying with $|1 - 2\eta(x)|$ and integrating with respect to P . ■

Applying the preceding lemma with a scaled version of ψ_σ of a given probability density ψ , we find that (3.4.33) is bounded above by, for any $\sigma > 0$,

$$\delta_n^2 + 2P[\bar{\Psi}(\tau/\sigma_n)|1 - 2\eta|] + \lambda_n^2 \|\theta_0 * \psi_{\sigma_n}\|_{\mathbb{H}}^2.$$

For inputs x with $\tau(x)$ bounded away from zero, the factor $\bar{\Psi}(\tau(x)/\sigma_n)$ in the middle term will approach zero, or be identically zero if ψ has compact support, for $\sigma_n \downarrow 0$. For x approaching the decision boundary the factor $|1 - 2\eta(x)|$ will tend to zero. The size of the middle term depends on the interaction between these two factors, which is moderated by the bandwidth σ_n . The middle term should be traded off against the reproducing norm $\|\theta_0 * \psi_{\sigma_n}\|_{\mathbb{H}}^2$, which will typically tend to infinity as $\sigma_n \downarrow 0$.

For a multi-dimensional input space, it appears to be difficult to evaluate the middle term in general. However, if x tends to the decision boundary, then typically $\tau(x) \rightarrow 0$ and $\eta(x) \rightarrow 1/2$. This suggests that $|\eta - 1/2| = \tau^p \wedge 1$, for some positive exponent p , might be an illustrative model. For this model $P[\bar{\Psi}(\tau/\sigma)|1 - 2\eta|] = 2 \int \bar{\Psi}(s/\sigma) s^p \wedge 1 g(s) ds$, for g a density of the induced distribution $P \circ \tau^{-1}$. If g is bounded and continuous at 0, this gives that $P[\bar{\Psi}(\tau/\sigma)|1 - 2\eta|] \sim \sigma^{p+1} 2g(0) \int_0^\infty \bar{\Psi}(s) s^p ds$, as $\sigma \downarrow 0$, provided the integral is finite. The latter is true for any density ψ with sufficiently small tails, for instance a normal density.

The tail function corresponding to a Gaussian density ψ takes the form $\bar{\Psi}(s) = e^{-cs^2}$. The decision boundary is said to possess a *geometric noise exponent* $\alpha > 0$ if

$$P[|1 - 2\eta|e^{-t\tau^2}] \lesssim t^{-\alpha d/2}, \quad t > 0.$$

The parameter α measures the regularity of the decision boundary, similar to the parameter κ in Tsybakov's condition.

Lemma 3.4.29 allows to handle the modulus (3.4.31) if $\|\cdot\|_{\mathbb{H}}$ is only a seminorm, for instance in the case that intercepts are added to a RKHS, as considered previously. The inequality $\|\theta\|_{\mathbb{H}} < \delta/\lambda_n$ does not restrict the intercept, so that the class of functions m_θ may have infinite entropy. The uniform bound (3.4.32) may fail for the same reason. However, by (iv) of the lemma a combination of bounds on $\|\theta\|_{\mathbb{H}}$ and $\|m_\theta - m_{\theta_0}\|_{P,2}$ does restrict the intercept to a compact interval, at least in the case that the RKHS-norm is stronger than the uniform norm. (Problem 3.4.9 allows to deal with the supremum norm of a function b .)

3.4.35 Example (Sobolev space). The restricted *Sobolev space* $W_{2,0}^k[0,1]$ of functions $\theta: [0,1] \rightarrow \mathbb{R}$ with absolutely continuous $(k-1)$ th derivative and $\theta(0) = \theta'(0) = \dots = \theta^{(k-1)}(0) = 0$ is a reproducing kernel Hilbert space relative to the square norm

$$\|\theta\|_{\mathbb{H}}^2 = \int_0^1 \theta^{(k)}(x)^2 dx.$$

The corresponding kernel is the covariance function of $(k-1)$ times *integrated Brownian motion*.[‡] The restriction $\theta(0) = 0$ would mean that the input 0 is always classified as -1 , which is odd. It is very natural to add intercepts to the functions, and also natural to add the polynomials of degree $(k-1)$ to release also the value of the derivatives at 0. Thus we let the parameter vary over all functions $\theta = b + h$ with b a polynomial of degree $(k-1)$ and $h \in W_{2,0}^k[0,1]$. Rather than make this into a RKHS (as in Lemma 11.29 of Ghosal and Van der Vaart (2017)), we equip this parameter set with the seminorm $\|\theta\|_{\mathbb{H}} := \|h\|_{\mathbb{H}}$.

The entropy of the unit ball in $W_{2,0}^k[0,1]$ for the uniform norm is of the order $(1/\varepsilon)^{1/k}$, by Theorem 2.7.4. By repeated integration using the boundary conditions $\theta(0) = \theta'(0) = \dots = \theta^{(k-1)}(0) = 0$ and the Cauchy-Schwarz inequality, it follows that $\|\theta^{(l)}\|_{\lambda,2} \leq \|\theta^{(k)}\|_{\lambda,2}$, for every integer $0 \leq l < k$. Therefore, the set of functions with $\|\theta\|_{\mathbb{H}} \leq 1$ in the restricted Sobolev space $W_{2,0}^k[0,1]$ is a subset of this unit ball and hence possesses the same entropy. However, adding all polynomials makes the entropy infinite, which complicates the situation. In view of Lemma 3.4.30, the set of functions in the supremum in the modulus (3.4.31) can be replaced by the set

$$\Theta_{n,\delta} = \left\{ \theta = b + h: \|b\|_{Q,2} \lesssim \|m_\theta - m_{\theta_0}\|_{P,2} + \|h\|_{\infty} + 1, \right. \\ \left. \|h\|_{\mathbb{H}} \leq \frac{\delta}{\lambda_n}, \|m_\theta - m_{\theta_0}\|_{P,2} \leq \|\theta - \theta_0\|_{\infty}^{1/2} \delta + \delta^{\kappa/(2\kappa+1)} \right\}.$$

Here b and h range over the polynomials of degree $k-1$ and the restricted Sobolev space $W_{2,0}^k[0,1]$, respectively. Because the uniform norm of the polynomial b is bounded by a multiple of its L_2 -norm (see Problem 3.4.9) and $\|h\|_{\infty} \lesssim \|h\|_{\mathbb{H}}$, we can first conclude that $\|b\|_{\infty} \lesssim \|\theta - \theta_0\|_{\infty}^{1/2} \delta + \delta^{\kappa/(2\kappa+1)} + \delta/\lambda_n + 1$ and next that $\|\theta - \theta_0\|_{\infty} \leq \|b\|_{\infty} + \|h\|_{\infty} + 1$ satisfies the same bound. The latter recursive

[‡] See e.g. Ghosal and Van der Vaart (2017), Section 11.4.1.

bound shows that $\|\theta - \theta_0\|_\infty \lesssim M_n(\delta) := \delta^2 + \delta/\lambda_n^{1/2} + 1$, and hence $\|b\|_\infty \leq \|\theta\|_\infty + \|h\|_\infty \lesssim M_n(\delta)$. We conclude that, for $\gamma_n(\delta) := \delta^2 + \delta^{3/2}/\lambda_n^{1/2} + \delta^{\kappa/(2\kappa+1)}$,

$$\Theta_{n,\delta} \subset \left\{ \theta = b + h : \|b\|_\infty \lesssim M_n(\delta), \|h\|_{\mathbb{H}} \lesssim \frac{\delta}{\lambda_n}, \|m_\theta - m_{\theta_0}\|_{P,2} \leq \gamma_n(\delta) \right\}.$$

Because the functions $m_\theta - m_{\theta_0}$ are uniformly bounded by a multiple of $M_n(\delta)$, we can use Theorem 2.14.17 to bound the modulus (3.4.31) (with θ_0 instead of θ_n as in Addendum 3.4.7) by

$$\phi_n(\delta) \leq \tilde{J}(\gamma_n(\delta)) \left(1 + \frac{\tilde{J}(\gamma_n(\delta))}{\sqrt{n}\gamma_n^2(\delta)} M_n(\delta) \right),$$

where $\tilde{J}(\gamma)$ is the bracketing integral of the class $\Theta_{n,\delta}$. The entropy of the class is bounded above by the sum of the entropies of the set of polynomials b and the functions h in the restricted Sobolev class. Hence the entropy integral satisfies

$$\begin{aligned} \tilde{J}(\gamma) &\leq \int_0^\gamma \sqrt{1 \vee \log\left(\frac{M_n(\delta)}{\varepsilon}\right)^k} d\varepsilon + \int_0^\gamma \sqrt{\left(\frac{\delta/\lambda_n}{\varepsilon}\right)^{1/\kappa} \vee 1} d\varepsilon \\ &\lesssim \gamma \log(M_n(\delta)/\gamma) + (\delta/\lambda_n)^{1/(2k)} \gamma^{1-1/(2\kappa)} + \gamma. \end{aligned}$$

Solving the inequality $\phi_n(\delta_n) \leq \sqrt{n}\delta_n^2$ gives an optimal value of δ_n in terms of λ_n .

We estimate the bias term in (3.4.33) with the help of Lemma 3.4.34. The k th derivative of the convolution $\theta_n = \theta_0 * \psi_\sigma$ of θ_0 with the scaling of a density ψ that is at least k times differentiable is uniformly bounded by $\sigma^{-k} \|\int \theta_0(y) \psi^{(k)}((x-y)/\sigma)/\sigma dy\|_\infty \leq \sigma^{-k} \|\psi^{(k)}\|_1$ and hence the Sobolev norm of $\|\theta_n\|_{\mathbb{H}}$ is bounded by a multiple of $1 + \sigma^{-k}$. By Lemma 3.4.34 the corresponding middle term in (3.4.33) is bounded above by $P\bar{\Psi}(\tau/\sigma)|1-2\eta|$, for $\bar{\Psi}(s) = 1 - \int_{(-s,s)} \psi(y) dy$. This will typically tend to zero as $\sigma \downarrow 0$ and must be traded off against $\lambda_n^2 \sigma^{-2k}$. For instance, if we choose the normal density ψ , and the decision boundary possesses geometric noise exponent α , then $P\bar{\Psi}(\tau/\sigma)|1-2\eta| \leq \sigma^\alpha$. This leads to minimizing $\sigma^\alpha + \lambda_n^2 \sigma^{-2k}$ over σ , giving $\sigma_n \asymp \lambda_n^{2/(\alpha+2k)}$ and value $\lambda_n^{2\alpha/(\alpha+2k)}$.

Finally we choose λ_n to minimize $\delta_n^2 + \lambda_n^{2\alpha/(\alpha+2k)}$ under the constraints $\phi_n(\delta_n) \leq \sqrt{n}\delta_n^2$ and $\lambda_n \mathcal{J}(\theta_n) \leq \delta_n$. The latter minimum is a rate for $P(m_\theta - m_{\theta_0})$ and hence also for the excess risk $P(\ell_\theta - \ell_{\theta_0})$, by Lemma 3.4.29.

3.4.36 Example (Radial basis functions). The Gaussian or *square-exponential kernel* $K(x_1, x_2) = \exp(-\|x_1 - x_2\|^2/(2\sigma^2))$ on $\mathcal{X} \times \mathcal{X}$, for $\mathcal{X}$ the unit ball of $\mathbb{R}^d$, is said to generate *radial basis functions*. The kernel is the covariance function of a stationary Gaussian process with analytic sample paths, characterized by the *spectral measure* $\mu_\sigma = N(0, \sigma^{-2}I)$. The *length scale* σ is a bandwidth parameter that must tend to zero with n for good results.

The reproducing kernel Hilbert space $\mathbb{H}$ corresponding to K is the set of real parts of the functions $\theta_\psi: \mathcal{X} \rightarrow \mathbb{C}$ given by

$$\theta_\psi: x \mapsto \int e^{i\lambda^t x} \psi(\lambda) d\mu_\sigma(\lambda),$$

when ψ runs through the complex Hilbert space $L_2(\mu_\sigma)$, with RKHS-norm equal to^b

$$\|\theta_\psi\|_{\mathbb{H}} = \|\psi\|_{\mu_\sigma, 2}.$$

These functions extend to entire functions on $\mathbb{C}^d$, and hence the entropy of the set of functions with $\|\theta_\psi\|_{\mathbb{H}} \leq \delta/\lambda_n$ can be bounded with the help of Theorem 2.7.16. (The theorem concerns functions that are analytic on a bounded neighborhood of $[0, 1]^d$, and is suboptimal for entire functions, but the difference is in logarithmic factors only.) In fact, if $\|\psi\|_{\mu_\sigma, 2} \leq 1$, then, by the Cauchy-Schwarz inequality,

$$\sup_{z \in \mathbb{C}^d: \|z\| \leq A} |\theta_\psi(z)|^2 \leq \sup_{\|z\| \leq A} \int |e^{i\lambda^t z}|^2 d\mu_\sigma(\lambda) \leq e^{2A^2/\sigma^2}.$$

Thus Theorem 2.7.16 can be applied to the functions θ_ψ divided by the uniform bound $\exp(2A^2/\sigma^2)$, for any $A > 0$, and shows that, for $\varepsilon\lambda_n/\delta \leq \exp(2A^2/\sigma^2)/2$,

$$\begin{aligned} \log N\left(\varepsilon, \{\theta_\psi: \|\theta_\psi\|_{\mathbb{H}} \leq \delta/\lambda_n\}, \|\cdot\|_\infty\right) &\lesssim \inf_A \left(\frac{1}{A}\right)^d \left(\log \frac{\exp(2A^2/\sigma^2)}{\varepsilon\lambda_n/\delta}\right)^{d+1} \\ &\lesssim \sigma^{-d} \left(\log \frac{\delta}{\varepsilon\lambda_n}\right)^{d/2+1}, \end{aligned}$$

for $\varepsilon\lambda_n/\delta < 1/2$ (choose $A^2/\sigma^2 = \log(\delta/(\varepsilon\lambda_n))$). For $\varepsilon\lambda_n/\delta \geq 1/2$, the bound for $\varepsilon\lambda_n/\delta = 1/2$ is valid, which is a multiple of σ^{-d} . Because the kernel K is bounded by 1, inequality (3.4.32) gives that $\|\theta\|_\infty \leq \|\theta\|_{\mathbb{H}}$, for every $\theta \in \mathbb{H}$, whence the functions θ_ψ in the left side of the display are uniformly bounded above by δ/λ_n . The entropy relative to the uniform norm gives brackets relative to the L_2 -norm, which allows application of the maximal inequality given by Theorem 2.14.17. With $\gamma_n(\delta) := \delta^{3/2}/\lambda_n^{1/2} + \delta + \delta^{\kappa/(\kappa+1)}$, it follows (see Problem 3.4.10) that the modulus (3.4.31) is bounded above by a multiple of

$$\phi_n(\delta) := J(\delta) \left(1 + \frac{J(\delta)}{\delta^2 \sqrt{n}} \frac{\delta}{\lambda_n}\right),$$

for

$$J(\delta) = \gamma_n(\delta) \left(\frac{1}{\sigma}\right)^{d/2} \left(\log \frac{\delta}{\gamma_n(\delta)\lambda_n} \vee 1\right)^{d/4+1/2}.$$

A fixed bandwidth σ would yield a solution δ_n to $\phi_n(\delta_n) \leq \sqrt{n}\delta_n^2$ that tends to zero at a very fast rate; for instance, the rate $(\log n)^{d/4+1/2}/\sqrt{n}$ if $\kappa = \infty$ and appropriate λ_n . However, the classification rate is a compromise between the rate obtained from the entropy and the bias term in (3.4.33). As the true classifier θ_0 will typically not be contained in the RKHS, it will be necessary to let $\sigma = \sigma_n$ tend to zero to control the bias.

To control the bias we assume that the decision boundary has geometric noise exponent α , i.e. $P[|1 - 2\eta|e^{-t\tau^2}] \lesssim t^{-\alpha d/2}$, for $t > 0$. Then the bias (3.4.33) can be estimated using Lemma 3.4.34, with ψ the centered normal density with covariance matrix $2c^2\sigma^2$. The corresponding tail function is $\bar{\Psi}(s) = P(2c^2\|Z\|^2 > s)$, for Z a standard normal vector in $\mathbb{R}^d$, which is bounded above by $2de^{-s^2/(4dc^2)}$.

^b See e.g. Van der Vaart and Van Zanten (2009), Lemma 4.1.

The expectation $P[e^{-\tau^2/(4dc^2\sigma^2)} | 1 - 2\eta]$ is bounded above by a multiple of $\sigma^{\alpha d}$, by the assumption of a geometric noise exponent. The Fourier inverse of the convolution $\theta_n = \theta_0 * \phi_{\sqrt{2}c\sigma}$ is $\lambda \mapsto \hat{\theta}_0(\lambda)e^{-c^2\|\lambda\|^2\sigma^2}$, and hence $\theta_n = \theta_\psi$, for $\psi(\lambda) = \sigma^{-d}(2\pi)^{d/2}\hat{\theta}_0(\lambda)e^{-(c^2-1/2)\|\lambda\|^2\sigma^2}$, whence

$$\|\theta_n\|_{\mathbb{H}}^2 = \|\psi\|_{\mu_{\sigma,2}}^2 = \sigma^{-d}(2\pi)^{d/2} \int |\hat{\theta}_0(\lambda)|^2 e^{-(2c^2-1/2)\|\lambda\|^2\sigma^2} d\lambda.$$

This is bounded by $\sigma^{-d}(2\pi)^{d/2}\|\hat{\theta}_0\|_2^2$, for $2c^2 \geq 1/2$. We conclude that the bias (3.4.33) is bounded above by $\delta_n^2 \geq \sigma_n^{\alpha d} + \lambda_n^2 \sigma_n^{-d}$.

We may now optimize the choices of the smoothing parameter λ_n and the bandwidth σ_n of the kernel to give the fastest rate δ_n that satisfies both the rate equation $\phi_n(\delta_n) \leq \sqrt{n}\delta_n^2$ and the bias equation $\delta_n^2 \geq \sigma_n^{\alpha d} + \lambda_n^2 \sigma_n^{-d}$. If logarithmic terms are ignored, and both parameters are optimized over powers of n , then, with some effort, it can be seen that optimal choices are given by

$$\begin{aligned} \sigma_n &\asymp n^{-\frac{2}{2d\alpha+3d}}, & \lambda_n &\asymp n^{-\frac{\alpha+1}{2\alpha+3}}, & \text{if } \alpha \leq \kappa/2 + 1/2, \\ \sigma_n &\asymp n^{-\frac{\kappa+1}{d\kappa+d+d\alpha\kappa+2d\alpha}}, & \lambda_n &\asymp n^{-\frac{(\alpha+1)(\kappa+1)}{2\kappa+2+2\alpha\kappa+4\alpha}}, & \text{if } \alpha \geq \kappa/2 + 1/2. \end{aligned}$$

This leads to the rates δ_n equal to $n^{-2\alpha/(2\alpha+3)}$ and $n^{-\alpha(\kappa+1)/(\kappa+1+\alpha\kappa+2\alpha)}$ up to logarithmic factors, in the two cases, respectively.

3.4.37 Example (Series). Given a sequence of $\psi_1, \psi_2, \dots$ of linearly independent functions $\psi_j: \mathcal{X} \rightarrow \mathbb{R}$, and numbers $\mu_j > 0$ such that $\sum_j \mu_j \psi_j^2(x) < \infty$ for every $x \in \mathcal{X}$, consider the kernel

$$K(x_1, x_2) = \sum_j \mu_j \psi_j(x_1) \psi_j(x_2).$$

The corresponding RKHS consists of all functions $\theta = \sum_j \theta_j \psi_j$ with $\sum_j \theta_j^2 / \mu_j < \infty$, with square norm $\|\theta\|_{\mathbb{H}}^2$ equal to the latter sum.[#]

In order to relate the coefficients (θ_j) to the distance $\|m_\theta - m_{\theta_0}\|_{P,2}$, we make further assumptions. First we assume that the $L_2(P)$ -norm of a function $\sum_j \theta_j \psi_j$ is equivalent to the ℓ_2 -norm $\|(\theta_j)\|_2$ of the coefficients (θ_j) ; this is true for instance if (ψ_j) is an orthonormal sequence relative to a measure μ with respect to which P has a density that is bounded away from 0 and ∞ . Second we assume that η is bounded away from 0 and 1, so that $\|m_\theta - m_{\theta_0}\|_{P,2}$ is equivalent to $\|\theta - \theta_0\|_{P,2}$, by (ii) of Lemma 3.4.29. Third, we assume that $\sup_x \sqrt{K(x, x)} < \infty$, which implies that $\|\theta\|_\infty \lesssim \|\theta\|_{\mathbb{H}}$, for every θ .

Under these conditions the supremum in the modulus (3.4.31) is taken over all $\theta = \sum \theta_j \psi_j$ with

$$\|\theta - \theta_0\|_{P,2} < \delta + \frac{\delta^{3/2}}{\sqrt{l_n}} + \delta^{\kappa/(2\kappa+1)} \quad \text{and} \quad \|\theta\|_{\mathbb{H}} < \frac{\delta}{\lambda_n}.$$

The empirical process in the modulus can be replaced by the symmetrized empirical process $\mathbb{G}_n^o$ at the cost of a factor 2. Because $\theta \mapsto m_\theta$ is Lipschitz, the contraction

[#] See e.g. Van der Vaart and Van Zanten (2008), Section 4.

principle, Proposition A.1.11, next allows to replace $\mathbb{G}_n^o(m_\theta - m_{\theta_n})$ by $\mathbb{G}_n^o(\theta - \theta_n)$, again at the cost of a factor 2. Next, for any $N \in \mathbb{N}$, by two applications of the Cauchy-Schwarz inequality,

$$|\mathbb{G}_n^o \theta|^2 = \left| \sum_j \theta_j \mathbb{G}_n^o \psi_j \right|^2 \lesssim \sum_{j \leq N} \theta_j^2 \sum_{j \leq N} |\mathbb{G}_n^o \psi_j|^2 + \sum_{j > N} \frac{\theta_j^2}{\mu_j} \sum_{j > N} \mu_j |\mathbb{G}_n^o \psi_j|^2.$$

Because $\mathbb{E}_P |\mathbb{G}_n^o \psi_j|^2 = P \psi_j^2 \lesssim 1$ for every j , it follows that, for any set Θ , and $\Pi_N \theta = \sum_{j \leq N} \theta_j \psi_j$,

$$\mathbb{E}^* \sup_{\theta \in \Theta} |\mathbb{G}_n^o(\theta - \theta_n)|^2 \lesssim N \sup_{\theta \in \Theta} \|\Pi_N(\theta - \theta_n)\|_{P,2}^2 + \sup_{\theta \in \Theta} \|\theta - \theta_n\|_{\mathbb{H}}^2 \sum_{j > N} \mu_j.$$

Therefore, because $\|\Pi_N(\theta - \theta_n)\|_{P,2} \lesssim \|\theta - \theta_0\|_{P,2} + \|\Pi_N(\theta_n - \theta_0)\|_{P,2}$, for every $N \in \mathbb{N}$, we can use

$$\phi_n(\delta) := \sqrt{N} \left[\delta + \frac{\delta^{3/2}}{\sqrt{\lambda_n}} + \delta^{\kappa/(\kappa+1)} + \|\Pi_N(\theta_n - \theta_0)\|_{P,2} \right] + \left[\frac{\delta}{\lambda_n} + \|\theta_n\|_{\mathbb{H}} \right] \sqrt{\sum_{j > N} \mu_j}.$$

This leads to the rate of convergence δ_n solving the set of inequalities, for any $N_n \in \mathbb{N}$,

$$\begin{aligned} \sqrt{N_n} \left[\delta_n + \frac{\delta_n^{3/2}}{\sqrt{\lambda_n}} + \delta_n^{\kappa/(\kappa+1)} \right] &\leq \sqrt{n} \delta_n^2, \quad \|\Pi_{N_n}(\theta - \theta_0)\|_{P,2} \leq \sqrt{n} \delta_n^2, \\ \frac{\delta_n}{\lambda_n} \sqrt{\sum_{j > N_n} \mu_j} &\leq \sqrt{n} \delta_n^2, \quad \|\theta_n\|_{\mathbb{H}} \leq \frac{\delta_n}{\lambda_n}, \quad P(m_{\theta_n} - m_{\theta_0}) \leq \delta_n^2. \end{aligned}$$

The solution depends on the approximation of θ_0 by $\theta_n \in \mathbb{H}$ through $\|\Pi_{N_n}(\theta - \theta_0)\|_{P,2}$, $\|\theta_n\|_{\mathbb{H}}$ and $P(m_{\theta_n} - m_{\theta_0})$.

If we choose the approximation $\theta_n = \sum_{j=1}^{N_n} \theta_{0,j} \psi_j$, then $\|\Pi_{N_n}(\theta - \theta_0)\|_{P,2} = 0$ and $\|\theta_n\|_{\mathbb{H}}^2 = \sum_{j > N_n} \theta_{0,j}^2 / \mu_j$, but the expectation $P(m_{\theta_n} - m_{\theta_0})$ appears to be more difficult to estimate. The bound $P(m_{\theta_n} - m_{\theta_0}) \leq P|\theta_n - \theta_0| \leq \|\theta_n - \theta_0\|_{P,2}$ implies the bound $(\sum_{j > N_n} \theta_{0,j}^2)^{1/2}$, but this loses the square nature of $P(m_{\theta_n} - m_{\theta_0})$. Moreover, the discontinuity of θ_0 may cause the coefficients $\theta_{0,j}$ to tend to zero slowly, as $j \rightarrow \infty$, depending on the basis ψ_j . While this approach would give some rate, perhaps other approximations may lead to sharper results, given specific examples of bases ψ_j and classifiers θ_0 . For θ_n a convolution of θ_0 with a kernel, the geometric noise exponent of θ_0 , or the more general bound in Lemma 3.4.34, can be used to bound $P(m_{\theta_n} - m_{\theta_0})$, but bounding the other quantities may be more difficult.

Problems and Complements

1. Under the conditions of Theorem 3.4.1 the stronger conclusion is true that $r_n d_n(\hat{\theta}_n, \theta_{n,0})$ is bounded in mean, for $r_n = 1/\delta_n$. If the second condition of the theorem is strengthened to

$$\left\| \sup_{d_n(\theta, \theta_n) \leq \delta, \theta \in \Theta_n} \sqrt{n} |(\mathbb{M}_n - M_n)(\theta) - (\mathbb{M}_n - M_n)(\theta_{n,0})| \right\|_p \lesssim \phi_n(\delta),$$

then the sequence $r_n d_n(\hat{\theta}_n, \theta_{n,0})$ is bounded in p th mean. Finally, to obtain the conclusion of the theorem (convergence in probability), it suffices that a probability version of this condition is valid.

[Hint: Bound $E^* r_n^p d_n^p(\hat{\theta}_n, \theta_{n,0})$ by the sum $\sum 2^{j/p} P^*(\hat{\theta}_n \in S_{j,n})$.]

2. For a function $M: \Theta \rightarrow \mathbb{R}$ on a metric space and $\underline{\delta} \geq 0$, consider the two conditions:

(i) $M(\theta) - M(\theta_0) \leq -d^2(\theta, \theta_0)$, for every θ with $d(\theta, \theta_0) \geq \underline{\delta}$.

(ii) $\sup_{\delta/2 < d(\theta, \theta_0) \leq \delta} (M(\theta) - M(\theta_0)) \leq -c\delta^2$, for every $\delta \geq \underline{\delta}$.

Condition (i) implies Condition (ii) with $c = 1/4$ and $d = 2$, while Condition (ii) with $c = d = 1$ implies Condition (i). (Condition (ii) without the lower bound $\delta/2$ is stronger and unnatural.)

3. Under the conditions of Theorems 2.14.17 or 2.14.18, the expectation

$$E_p^* \sup_f (\mathbb{G}_n f - \eta \sqrt{n})^+$$

is bounded by the right sides of the theorems, but with the entropy integrals computed over the interval $[\eta/8, \delta]$ rather than $[0, \delta]$.

[Hint: Use Lemma 2.14.19, as in the proofs of the theorems.]

4. For any nonnegative numbers p and q ,

$$|(2p)^{1/2} - (p+q)^{1/2}| \leq |p^{1/2} - q^{1/2}| \leq (1 + \sqrt{2}) |(2p)^{1/2} - (p+q)^{1/2}|.$$

Thus $h(p, q)$ and $h(2p, p+q)$ are equivalent up to constants.

[Hint: The lower inequality is trivial. The upper is valid with constant $\sqrt{2}$ if $q \geq p$ and with constant $1 + \sqrt{2}$ as stated if $q \leq p$.]

5. For any elements x, y , and z in a normed space and $c \geq 2$, we have $\|x - z\|^2 - \|y - z\|^2 \geq (1 - 2/c)^2 \|x - y\|^2$ whenever $\|x - y\| \geq c \|z - y\|$.

[Hint: By the triangle inequality, $\|x - y\| \leq \|x - z\| - \|y - z\| + 2\|y - z\|$. Bound $2\|z - y\|$ by $2/c$ times the left side and thus obtain

$$(1 - 2/c) \|x - y\| \leq \|x - z\| - \|y - z\|.$$

Take squares and use the fact that the right-hand side is nonnegative, since $c \geq 2$, so that one difference in the square on the right can be replaced by a sum.]

6. Consider Example 3.2.15 with the additional assumption that the distribution G of the observation times is fixed and known. Consider the class $\mathcal{P}$ of densities with respect to $\mu = G \times \nu$, where ν is the counting measure on $\{0, 1\}$, given by

$$\mathcal{P} = \{p_F(t, \delta) = F(t)^\delta (1 - F(t))^{1-\delta} : F \text{ is a d.f. on } \mathbb{R}\}.$$

Let $\hat{F}$ denote the maximum likelihood estimator as described in Example 3.2.15, given a sample $(T_1, \Delta_1), \dots, (T_n, \Delta_n)$ from p_{F_0} . Show that $h(p_{\hat{F}_n}, p_{F_0}) = O_p(n^{-1/3})$.

[Hint: See Van de Geer (1993a), pages 25 and 35.]

7. Does the result in Exercise 3.4.6 change if G is unknown?

8. Suppose that $\theta: [0, 1] \rightarrow \mathbb{R}$ satisfies $\|\theta\|_{p,2} \leq 1$ and $\|\theta^{(k)}\|_{\lambda,2} < M$, for some probability measure P on $[0, 1]$, some integer $k > 0$, and number $M \geq 1$, where λ is Lebesgue measure on $[0, 1]$. Then $\max_{0 \leq l < k} \|\theta^{(l)}\|_{\infty} \leq c_k \lambda^{-1/2} M$, for λ the smallest eigenvalue of the matrix $(Px^{i-1}x^{j-1})_{i,j=1,2,\dots,k}$ and a constant c_k that depends on k only.

[Hint: Since $|\theta^{(k-1)}(x) - \theta^{(k-1)}(0)| \leq \int_0^x |\theta^{(k)}| d\lambda \leq \|\theta^{(k)}\|_{\lambda,2}$, by the Cauchy-Schwarz inequality, the boundedness of the $L_2(\lambda)$ -norm of the k th derivative shows that $\|\theta^{(k-1)} - \theta^{(k-1)}(0)\|_{\infty} \leq M$. By repeated integration and induction it follows that the uniform distance between θ and its Taylor polynomial $\sum_{j=0}^{k-1} a_j x^j$ of degree $(k-1)$ at 0 is bounded above by M . Combined with the assumption $\|\theta\|_{p,2} \leq 1$, this shows that the $L_2(P)$ -norm of this Taylor polynomial is bounded above by $1 + M$. This implies that the absolute values of the coefficients a_j are bounded by a multiple of $1 + M$ (see Problem 3.4.9(ii)). Reinserting this in the inequalities obtained during the induction process gives the uniform boundedness of every $\theta^{(l)}$, for $l \leq k-1$.]

9. Let $\psi_j: \mathcal{X} \rightarrow \mathbb{R}$ be square-integrable functions on a measure space $(\mathcal{X}, \mathcal{A}, P)$, such that the matrix $(P\psi_i\psi_j)$ is nonsingular, and let λ denote its smallest eigenvalue.

- (i) $\|b\|_{\infty} \leq \lambda^{-1/2} \|b\|_{p,2} \|\sum_i \psi_i^2\|_{\infty}^{1/2}$, for every $b \in \text{lin}(\psi_1, \dots, \psi_k)$;
- (ii) $\|\alpha\| \leq \lambda^{-1/2} \|\alpha'\psi\|_{p,2}$, for every $\alpha \in \mathbb{R}^k$.

[Hint: For Σ the given matrix, we have $P(\alpha'\psi)^2 = \alpha'\Sigma\alpha \geq \lambda\alpha'\alpha$. By the Cauchy-Schwarz inequality $|\alpha'\psi(x)| \leq \|\alpha\| \sqrt{\sum_i \psi_i^2(x)} \leq \lambda^{-1/2} \|\alpha'\psi\|_{p,2} \|\sum_i \psi_i^2\|_{\infty}^{1/2}$.]

10. For given $r > 0$ the function $J(\delta) = \int_0^{\delta} (1 \vee \log 1/\varepsilon)^r d\varepsilon$ satisfies $J(\delta) \lesssim \delta(1 \vee \log 1/\delta)^r$, for every $\delta > 0$. Furthermore $J(\delta) \asymp \delta(\log 1/\delta)$ as $\delta \downarrow 0$, and $J(\delta) \asymp \delta$ as $\delta \uparrow \infty$, where both orders are sharp.

11. If the Hilbert space $\mathbb{H}$ is continuously embedded in the Banach space $\mathbb{B}$, and the sequence $h_n \in \mathbb{H}$ converges weakly in $\mathbb{B}$ to $b \in \mathbb{B}$, then $\|b\|_{\mathbb{H}} \leq \liminf \|h_n\|_{\mathbb{H}}$.

[Hint: We may assume that the limit of $\|h_n\|_{\mathbb{H}}$ exists and is finite. Then by Alaoglu's theorem and reflexivity of $\mathbb{H}$, there exists a subsequence $h_{n'}$ and $h \in \mathbb{H}$ with $h_{n'} \rightarrow h$ weakly in $\mathbb{H}$. Since the topology of $\mathbb{H}$ is stronger than the topology of $\mathbb{B}$ by assumption, the restriction to $\mathbb{H}$ of any b^* in the dual space $\mathbb{B}^*$ is an element of the dual space of $\mathbb{H}$. Thus $h_{n'} \rightarrow h$ also weakly in $\mathbb{B}$. Because weak limits are unique, it follows that $b = h$ and $h_{n'} \rightarrow b$ also weakly in $\mathbb{H}$. Then $\|b\|_{\mathbb{H}} = \sup_{\|g\|_{\mathbb{H}}=1} \langle g, b \rangle_{\mathbb{H}}$, where $\langle g, b \rangle_{\mathbb{H}} = \lim \langle g, h_{n'} \rangle_{\mathbb{H}} \leq \lim \|h_{n'}\|_{\mathbb{H}}$, for every g with $\|g\|_{\mathbb{H}} = 1$, and the conclusion follows.]

12. If the Hilbert space $\mathbb{H}$ is compactly embedded in the Banach space $\mathbb{B}$, then for a given open ball $B \subset \mathbb{B}$ the infimum $C = \inf\{\|h\|_{\mathbb{H}} : h \in \mathbb{H} \cap B\}$ is equal to $\bar{C} = \inf\{\|h\|_{\mathbb{H}} : h \in \mathbb{H} \cap \bar{B}\}$, for $\bar{B}$ the closure of B in $\mathbb{B}$, and the infimum in $\bar{C}$ is attained at some $h \in \mathbb{H} \cap \bar{B}$.

[Hint: It is clear that $\bar{C} \leq C$. Take any $h \in \mathbb{H} \cap \bar{B}$ and $h_n \in \mathbb{H} \cap B$ with $\|h_n\|_{\mathbb{H}} \downarrow C$. Then $g_n = \frac{1}{2}h + \frac{1}{2}h_n$ is contained in $\mathbb{H} \cap B$ and hence $C \leq \|g_n\|_{\mathbb{H}} \leq \frac{1}{2}\|h\|_{\mathbb{H}} + \frac{1}{2}\|h_n\|_{\mathbb{H}} \leq \frac{1}{2}\|h\|_{\mathbb{H}} + \frac{1}{2}(C + o(1))$. It follows that $\|h\|_{\mathbb{H}} \geq C$ and hence $\bar{C} \geq C$. Because the sequence h_n is contained in a multiple of the closed unit ball of $\mathbb{H}$, which is precompact in $\mathbb{B}$ by assumption, it has a subsequence that converges in $\mathbb{B}$. The limit h is in $\bar{B}$ and satisfies $\|h\|_{\mathbb{H}} \leq C$, by Problem 3.4.11. Thus the infimum in $\bar{C}$ is attained.]

3.5

Model Selection

The rates of convergence obtained in the preceding chapters are in some examples specific to a model of a given regularity, to which the estimation method is adapted by choice of a sieve, a penalty, or a tuning parameter. In this chapter we consider using multiple such models at once, which may come down to choosing a tuning parameter based on the data. The aim is a rate of estimation that is (nearly) equal to the rate that would have been achieved when using the best model. It is shown in some generality that this is possible by adding a penalty for model complexity to the criterion.

Many authors in statistical learning suggest that finite sample bounds for risks, with optimal constants, are crucial. However, as usual we shall not care for constants, but optimize on the simplicity and interpretability of the assertions instead. This makes the results and their proofs more transparent, but, admittedly, is not good for practical use. In particular, the implementation of the selection schemes in this chapter requires explicit forms of the penalty terms and hence of the constants involved. Minimal choices of the constants can be found in the original research papers (see the notes to this chapter).

3.5.1 General Result

For every k in an arbitrary countable set $\mathcal{K}$ let Θ_k be a subset of a metric space (Θ, d) . For each $\theta \in \Theta$ let $m_\theta: \mathcal{X} \rightarrow \mathbb{R}$ be a measurable function. We are interested in estimating

$$\theta^* = \operatorname{argmin}_{\theta \in \Theta} P m_\theta.$$

We achieve this in two steps using the “models” Θ_k . First we construct an estimator

$\hat{\theta}_k$ using the model Θ_k by minimizing the empirical contrast $\theta \mapsto \mathbb{P}_n m_\theta$ over Θ_k :

$$\hat{\theta}_k = \operatorname{argmin}_{\theta \in \Theta_k} \mathbb{P}_n m_\theta.$$

Next we choose a “best model” $\hat{k}$, resulting in the final estimator $\hat{\theta} = \hat{\theta}_{\hat{k}}$.

Because the models may differ in complexity, the best model is not necessarily the model with the smallest contrast $\inf_{\theta \in \Theta_k} \mathbb{P}_n m_\theta$. Instead we choose a best model by trading off this minimal contrast and a penalty for model complexity, given by a function $J: \mathcal{K} \rightarrow [0, \infty)$. Thus we define

$$\hat{k} = \operatorname{argmin}_{k \in \mathcal{K}} (\mathbb{P}_n m_{\hat{\theta}_k} + J(k)).$$

The resulting estimator $\hat{\theta}$ can also be written

$$\hat{\theta} = \hat{\theta}_{\hat{k}} = \operatorname{argmin}_{\theta \in \cup_k \Theta_k} \left(\mathbb{P}_n m_\theta + \sum_k J(k) 1_{\theta \in \Theta_k} \right).$$

We shall assume that the minima in the preceding are indeed assumed and the definitions well posed. (The results extend to near points of minimum, but this will encumber notation.) In case multiple points of minimum exist, we allow any consistent choice.

As shown by the preceding display, the present estimator $\hat{\theta}$ is itself an M -estimator, and hence the results of Chapter 3.4 may apply to it. For instance, Theorem 3.4.1 applied with $\mathbb{M}_n$ the empirical measure indexed by the present functions $-m_\theta - \sum_k J(k) 1_{\theta \in \Theta_k}$ with parameter set $\cup_k \Theta_k$ may yield a rate of convergence of $\hat{\theta}$. However, such a result would typically not be of the desired type in the present situation. We would like to prove a result showing that $\hat{\theta}$ achieves a rate of convergence corresponding to the model in the list that is best for the given true distribution, even though this best model is not revealed a-priori. In other words, we would like to show that the estimator *adapts* to the list of models.

Adaptive results compare the rate of convergence of the estimator $\hat{\theta}$, which is constructed without knowledge of the positioning of θ^* relative to the models Θ_k , to the rate of convergence of the estimator $\hat{\theta}_k$ in the situation that θ^* is contained in the model Θ_k . As shown by Theorem 3.4.1 the latter (inverse) rate of convergence $\delta_{n,k}$ can often be described through the modulus of continuity $\phi_{n,k}$ of the centered and scaled contrast functions by the equation

$$(3.5.1) \quad \phi_{n,k}(\delta_{n,k}) \leq D_k \sqrt{n} \delta_{n,k}^2.$$

Larger models possess a larger modulus of continuity and will yield bigger $\delta_{n,k}$. We would say that the estimator $\hat{\theta}$ *adapts* to the list of models Θ_k if it attains a rate of convergence of the order $\delta_{n,k}$ whenever $\theta^* \in \Theta_k$.

Oracle inequalities provide a different, more powerful method to formulate adaptation. They are particularly suitable to make the bias-variance trade-off visible. For each model Θ_k we may define as a best target for estimation within that model, either

$$(3.5.2) \quad \theta_k^* = \operatorname{argmin}_{\theta \in \Theta_k} P m_\theta.$$

or the projection of θ^* onto the model Θ_k relative to the distance d , given by

$$(3.5.3) \quad \bar{\theta}_k = \operatorname{argmin}_{\theta \in \Theta_k} d^2(\theta, \theta^*).$$

If we would use only model Θ_k , then we could hope to obtain a rate of convergence (cf. (3.4.2)) of the order $O_P(\delta_{n,k} + d(\bar{\theta}_k, \theta^*))$. Here the second term is the (unavoidable) bias when using model Θ_k to approximate θ^* , and the first term reflects the variance of estimation. If we would use a set of models Θ_k simultaneously, then we might hope to obtain a rate of convergence of the order

$$\inf_k (\delta_{n,k} + d(\bar{\theta}_k^*, \theta^*)).$$

However, this can only be achieved when using multiple models does not incur any penalty. This is unrealistic in general, in particular given a big list of models, as searching multiple models for good estimators will typically increase the variance of the estimator.

The *oracle inequality* given in the following theorem shows that the price to be paid can be modest, if we use the “correct” penalty for model complexity. Our seemingly unrealistic hope of doing as well as an oracle is nearly fulfilled: if the collection $(\Theta_k: k \in \mathcal{K})$ of models is not overly big, no or at most a modest loss in efficiency is incurred by using them all.

The theorem does not restrict the number of models $\#\mathcal{K}$ a priori, but measures the collection through positive constants $(x_k: k \in \mathcal{K})$, which should be large enough so that $\sum_k e^{-x_k} \leq 1$. The efficiency loss relative to using the single model Θ_k is then proportional to x_k/n , with the proportionally constant depending on the specifics of the model.

Say that the pairs of numbers (x_k, B_k) are *co-monotone* if $x_k B_{k'} \leq x_k B_k \vee x_{k'} B_{k'}$, for every $k, k' \in \mathcal{K}$.

3.5.4 Theorem (Oracle inequality). *Let $\{m_\theta: \theta \in \Theta\}$ be a class of measurable functions $m_\theta: \mathcal{X} \rightarrow \mathbb{R}$ indexed by a metric space (Θ, d) such that, for some constants B_k ,*

$$(3.5.5) \quad P(m_\theta - m_{\theta'})^2 \leq d^2(\theta, \theta'), \quad \theta, \theta' \in \Theta,$$

$$(3.5.6) \quad \|m_\theta\|_\infty \leq B_k, \quad \theta \in \Theta_k.$$

Furthermore, assume that the stochastic processes $\mathbb{G}_n m_\theta$ are separable, and for given positive constants C_k ,

$$(3.5.7) \quad C_k P(m_\theta - m_{\theta^*}) \geq d^2(\theta, \theta^*), \quad \theta \in \Theta_k,$$

$$(3.5.8) \quad \mathbb{E}^* \sup_{\theta \in \Theta_k: d(\theta, \bar{\theta}_k) < \delta} \mathbb{G}_n(m_{\bar{\theta}_k} - m_\theta) \leq \phi_{n,k}(\delta),$$

for functions $\phi_{n,k}: (0, \infty) \rightarrow \mathbb{R}$ such that $\delta \mapsto \phi_{n,k}(\delta)/\delta$ is nonincreasing, and every $\delta \geq \underline{\delta}_n$. Given $\delta_{n,k}$ and D_k satisfying (3.5.1) and B_k, C_k , let $J: \mathcal{K} \rightarrow [0, \infty)$ satisfy

$$J(k) \gtrsim \delta_{n,k}^2 C_k D_k^2 + \frac{1}{n} (B_k + C_k) (x_k + \log(B_k + C_k)).$$

for $(x_k: k \in \mathcal{K})$ numbers such that $\sum_k e^{-x_k} \leq 1$, and where the proportionality constant is universal but large enough. Assume that the x_k are chosen so that both pairs of numbers (x_k, B_k) and (x_k, C_k) are co-monotone. Then

$$E^* P(m_{\hat{\theta}} - m_{\theta^*}) \lesssim \inf_{k \in \mathcal{K}} \left(P(m_{\theta_k^*} - m_{\theta^*}) + J(k) + \frac{1}{n} \right).$$

The theorem is formulated in terms of the risk $P(m_{\hat{\theta}} - m_{\theta^*})$ rather than the square distance $d^2(\hat{\theta}, \theta^*)$, because this appears more elegant. It is also natural, given that the procedures are targeted on minimizing the empirical risk. The left side of the theorem is of course an upper bound on $E^* d^2(\hat{\theta}, \theta^*)/C_k$, but a similar replacement of the risk by the distance on the right side requires the inverse of inequality (3.5.7).

Because the risk should be interpreted as a square distance, the term $1/n$ on the far right limits the rate of convergence to be at most $1/\sqrt{n}$. This is not an important restriction. In fact in Theorem 3.4.1 the rate was restricted to this range by assumption. If the constants B_k and C_k are bounded in k , then the penalty contributes the terms $\delta_{n,k}^2$ and x_k/n to the upper bound. The first quantity we can recognize as the (inverse) rate of convergence attached to the model Θ_k , determined by its modulus through (3.5.1), which also appears in Theorem 3.4.1.

The second quantity x_k/n is the penalty incurred by using multiple models. Given finitely many models, we can choose every x_k equal to a fixed constant, rendering the penalty uniformly of order $O(1/n)$, and hence negligible. However, also given infinitely many models the term is $O(1/n)$ for any fixed model, although the proportionality constant must “increase” to infinity by the requirement $\sum_k e^{-x_k} \leq 1$. If $\mathcal{K} = \mathbb{N}$, then $x_k = k$ is (easily) possible, and hence, for instance, given a model class that contains a single model of dimension k for every $k \in \mathbb{N}$, we can choose the penalty proportional to this dimension divided by n . If the cardinality of $\mathcal{K}$ is bounded by a power n^r of the number of observations, then we can also choose $x_k \asymp \log n$ for every k , and the loss due to model selection is still only of order $\log n/n$. Thus the theorem shows that a huge number of models can be used simultaneously with the resulting procedure having risk determined by the best model.

Here “best” refers to the true distribution P of the observations. Because this distribution also determines the quantities involved in the requirement on the penalty, Theorem 3.5.4 actually provides more an abstract framework for analyzing model selection than prove the claim in the last paragraph. The quantities $\delta_{n,k}$, C_k , and D_k involved in the required lower bound on $J(k)$ refer to the true distribution P , and hence the question is whether one can actually find penalty functions that are valid for the large class of true distributions that we are interested in.

The conditions (3.5.7) and (3.5.8) in Theorem 3.5.4 are identical to the conditions in Theorem 3.4.1 on rates when using a single model. On the other hand, the remaining two conditions (3.5.5) and (3.5.6) are additional, and for this reason Theorem 3.5.4 does not recover Theorem 3.4.1. The required boundedness (3.5.6) of the criterion functions is particularly unpleasant in such applications as regression. Technically, it is used to have access to the concentration inequality for empirical processes, which is not needed to obtain rates.

The proof of the theorem is based on two lemmas. The first is Talagrand's concentration inequality, which we apply in the simplified form given by Lemma 2.15.10.

The second lemma is a version of the *peeling device* to obtain ratio theorems for the empirical process. Another version of the peeling device is present in the proof of Theorem 3.2.5. That version makes fewer assumptions (in particular it does not require the functions to be bounded), but gives a weaker conclusion.

3.5.9 Lemma. *Let $\{m_\theta: \theta \in \Theta\}$ be a class of measurable functions $m_\theta: \mathcal{X} \rightarrow [-B, B] \subset \mathbb{R}$ indexed by a metric space (Θ, d) such that the stochastic processes $\mathbb{G}_n m_\theta$ are separable, and for some function $\phi: (0, \infty) \rightarrow \mathbb{R}$ such that $\delta \mapsto \phi(\delta)/\delta$ is nonincreasing, and every $\delta \geq \underline{\delta}_n$,*

$$P(m_\theta - m_{\theta_0})^2 \leq d^2(\theta, \theta_0),$$

$$\mathbf{E}^* \sup_{\theta: d(\theta, \theta_0) < \delta} (\mathbb{G}_n(m_\theta - m_{\theta_0}))_+ \leq \phi(\delta).$$

Then for any $\delta_n \geq \underline{\delta}_n$ with $\phi(\delta_n) \leq D^2 \sqrt{n} \delta_n^2$, every $0 < \eta < D$ and every $x > 0$, with probability at least $1 - e^{-x}$,

$$\forall \theta \in \Theta: \quad \frac{1}{135\sqrt{n}} \mathbb{G}_n(m_\theta - m_{\theta_0}) \leq \eta d^2(\theta, \theta_0) + \frac{\delta_n^2 D}{\eta} + \left(B + \frac{1}{\eta}\right) \frac{x}{n}.$$

Proof. We can assume that $m_{\theta_0} = 0$, at the cost of increasing the constant B to $2B$. The assumption on ϕ implies that $\phi(c\delta) \leq c\phi(\delta)$ for every $c \geq 1$ and $\delta > 0$. Therefore, for $\delta \geq \underline{\delta}_n$,

$$\begin{aligned} \mathbf{E}^* \sup_{d(\theta, \theta_0) \geq \delta} \frac{(\mathbb{G}_n m_\theta)_+}{d^2(\theta, \theta_0) + \delta^2} &\leq \sum_{j=1}^{\infty} \mathbf{E}^* \sup_{2^{j-1}\delta \leq d(\theta, \theta_0) < 2^j \delta} \frac{(\mathbb{G}_n m_\theta)_+}{d^2(\theta, \theta_0) + \delta^2} \\ &\leq \sum_{j=1}^{\infty} \frac{\phi(2^j \delta)}{2^{2j-2} \delta^2 + \delta^2} \leq \sum_{j=1}^{\infty} \frac{2^j}{2^{2j-2}} \frac{\phi(\delta)}{\delta^2} = 4 \frac{\phi(\delta)}{\delta^2}. \end{aligned}$$

The outer expectation of the supremum of the same process over the set of θ with $d(\theta, \theta_0) < \delta$ is trivially bounded by $\phi(\delta)/\delta^2$. Hence the right side of the preceding display with the constant 4 increased to 5 is a bound on the supremum over all $\theta \in \Theta$. The functions $m_\theta/(d^2(\theta, \theta_0) + \delta^2)$ are bounded by $2B/\delta^2$ and have second moment bounded by $1/\delta^2$. Therefore, by Lemma 2.15.10, with probability at least $1 - e^{-x}$,

$$\forall \theta \in \Theta: \quad \mathbb{G}_n m_\theta \leq 15 \left(\frac{\phi(\delta)}{\delta^2} + \frac{Bx}{\delta^2 \sqrt{n}} + \frac{\sqrt{x}}{\delta} \right) (d^2(\theta, \theta_0) + \delta^2).$$

For $\delta > \delta'_n := \delta_n D/\eta > \delta_n$ the assumption on ϕ implies that $\phi(\delta)/\delta^2 \leq \phi(\delta'_n)/(\delta'_n)^2 \leq \sqrt{n}\eta$, by the definition of δ_n . We conclude the proof by choosing $\delta = \delta_n D/\eta + \sqrt{Bx/(n\eta)} + \sqrt{x/n}/\eta$ in the preceding display, which renders the three terms in the first term in brackets all smaller than $\sqrt{n}\eta$. ■

Proof of Theorem 3.5.4. The definitions of $\hat{\theta}$ and $\hat{k}$ imply that $\mathbb{P}_n m_{\hat{\theta}} + J(\hat{k}) \leq \mathbb{P}_n m_{\theta} + J(k)$, for every $\theta \in \Theta_k$. We choose $\theta = \theta_k^*$ and next rearrange the inequality as

$$(3.5.10) \quad \begin{aligned} P(m_{\hat{\theta}} - m_{\theta^*}) &\leq P(m_{\theta_k^*} - m_{\theta^*}) + J(k) - J(\hat{k}) \\ &\quad + \frac{1}{\sqrt{n}} \mathbb{G}_n(m_{\theta_k^*} - m_{\bar{\theta}_k}) + \frac{1}{\sqrt{n}} \mathbb{G}_n(m_{\bar{\theta}_k} - m_{\hat{\theta}}). \end{aligned}$$

We shall bound the two empirical process terms on the right separately.

To handle the first term we first note that for any $k, k' \in \mathcal{K}$,

$$P(m_{\theta_k^*} - m_{\bar{\theta}_{k'}})^2 \leq d^2(\theta_k^*, \bar{\theta}_{k'}) \lesssim d^2(\theta_k^*, \theta^*) + d^2(\hat{\theta}_{k'}, \theta^*),$$

by the triangle inequality and the fact that $d(\bar{\theta}_{k'}, \theta^*) \leq d(\theta, \theta^*)$ for every $\theta \in \Theta_{k'}$, by the definition of $\bar{\theta}_k$. Therefore, by Bernstein's inequality, in the form of the bound in Lemma 2.15.10 applied to the class of functions consisting of the single function $m_{\theta_k^*} - m_{\bar{\theta}_{k'}}$, with probability at least $1 - e^{-x_k - x_{k'} - \xi}$,

$$\begin{aligned} \frac{1}{\sqrt{n}} \mathbb{G}_n(m_{\theta_k^*} - m_{\bar{\theta}_{k'}}) &\lesssim (B_k \vee B_{k'}) \frac{x_k + x_{k'} + \xi}{n} \\ &\quad + (d(\theta_k^*, \theta^*) \vee d(\hat{\theta}_{k'}, \theta^*)) \sqrt{\frac{x_k + x_{k'} + \xi}{n}}. \end{aligned}$$

By the inequality $2\sqrt{ab} \leq a^2/c + b^2c$, we can bound the second term on the right by, for any $\eta > 0$,

$$\eta \frac{d^2(\theta_k^*, \theta^*) \vee d^2(\hat{\theta}_{k'}, \theta^*)}{C_k \vee C_{k'}} + \frac{1}{\eta} (C_k \vee C_{k'}) \frac{x_k + x_{k'} + \xi}{n}.$$

By the assumed co-monotonicity of (x_k, B_k, C_k) we can next separate the contributions of k and k' , and arrive at the bound

$$\frac{1}{\sqrt{n}} \mathbb{G}_n(m_{\theta_k^*} - m_{\bar{\theta}_{k'}}) \lesssim \sum_{\kappa=k, k'} \left(B_{\kappa} + \frac{C_{\kappa}}{\eta} \right) \frac{x_{\kappa} + \xi}{n} + \frac{\eta d^2(\theta_k^*, \theta^*)}{C_k} + \frac{\eta d^2(\hat{\theta}_{k'}, \theta^*)}{C_{k'}}.$$

Because $\sum_k e^{-x_k} \leq 1$ by assumption, this bound is valid simultaneously for every $k, k' \in \mathcal{K}$ with probability at least $1 - e^{-\xi}$. On the event where this is true, we can choose $k' = \hat{k}$ random, and then have the bound for every k . With this substitution the last term on the right becomes $\eta d^2(\hat{\theta}, \theta^*)/C_{\hat{k}}$.

To handle the last term on the right of (3.5.10) we apply Lemma 3.5.9 (to the functions $-m_{\theta}$; note that the supremum in the left side of (3.5.8) is automatically nonnegative because $\bar{\theta}_k \in \Theta_k$), with η of the lemma taken equal to η/C_k , to see that with probability at least $1 - e^{-x_k - \xi}$,

$$\forall \theta \in \Theta_k: \quad \frac{1}{\sqrt{n}} \mathbb{G}_n(m_{\bar{\theta}_k} - m_{\theta}) \lesssim \frac{\eta}{C_k} d^2(\theta, \bar{\theta}_k) + \frac{\delta_{n,k}^2 C_k D_k^2}{\eta} + \left(B_k + \frac{C_k}{\eta} \right) \frac{x_k + \xi}{n}.$$

In particular, the inequality is true if θ is chosen equal to $\hat{\theta}_k$. Because $d(\hat{\theta}_k, \bar{\theta}_k) \leq 2d(\hat{\theta}_k, \theta^*)$, by the triangle inequality and the definition of $\bar{\theta}_k$, the first term on the

right can next be replaced by $\eta d^2(\hat{\theta}_k, \theta^*)/C_k$, at the cost of an extra constant. With probability at least $1 - e^{-\xi}$ the resulting inequality is true for all $k \in \mathcal{K}$ simultaneously. On the event where this is true, the inequality is also valid with $\hat{k}$ replacing k . With this substitution the first term becomes $\eta d^2(\hat{\theta}, \theta^*)/C_{\hat{k}}$.

Substituting the bounds obtained in the preceding two paragraphs in the right side of (3.5.10), we obtain that with probability at least $1 - 2e^{-\xi}$, there exist a universal constant c such that, for all $k \in \mathcal{K}$,

$$\begin{aligned} cP(m_{\hat{\theta}} - m_{\theta^*}) &\leq cP(m_{\theta_k^*} - m_{\theta^*}) + cJ(k) - cJ(\hat{k}) \\ &\quad + \sum_{\kappa=k, \hat{k}} \left(B_{\kappa} + \frac{C_{\kappa}}{\eta} \right) \frac{x_{\kappa} + \xi}{n} + \frac{\eta d^2(\theta_k^*, \theta^*)}{C_k} + \frac{\eta d^2(\hat{\theta}, \theta^*)}{C_{\hat{k}}} \\ &\quad + \frac{\eta d^2(\hat{\theta}, \theta^*)}{C_{\hat{k}}} + \frac{\delta_{n, \hat{k}}^2 C_{\hat{k}} D_{\hat{k}}}{\eta} + \left(B_{\hat{k}} + \frac{C_{\hat{k}}}{\eta} \right) \frac{x_{\hat{k}} + \xi}{n}. \end{aligned}$$

The first and third terms in the third line are copies of earlier terms and can be deleted at the cost of halving the constant c . Furthermore, we can replace the scaled square distances $d^2(\theta', \theta^*)/C_k$ on the right side by $P(m_{\theta'} - m_{\theta^*})$, in view of assumption (3.5.7), and next combine them with the terms on the left and immediately following the inequality sign. The variable ξ , which now only appears in the sum, can be separated from the other terms by the inequality $B\xi \leq 2B \log B + e^{\xi/2}$, valid for $B, \xi > 0$ (see Problem 3.5.2). We thus obtain, with c increased by 2,

$$\begin{aligned} (c - \eta)P(m_{\hat{\theta}} - m_{\theta^*}) &\leq (c + \eta)P(m_{\theta_k^*} - m_{\theta^*}) + cJ(k) - cJ(\hat{k}) + \frac{\delta_{n, \hat{k}}^2 C_{\hat{k}} D_{\hat{k}}^2}{\eta} \\ &\quad + \sum_{\kappa=k, \hat{k}} \left(B_{\kappa} + \frac{C_{\kappa}}{\eta} \right) \frac{x_{\kappa}}{n} + \frac{2}{n} \sum_{\kappa=k, \hat{k}} \left(B_{\kappa} + \frac{C_{\kappa}}{\eta} \right) \log \left(B_{\kappa} + \frac{C_{\kappa}}{\eta} \right) + \frac{2e^{\xi/2}}{n}. \end{aligned}$$

We choose a fixed $\eta < c$. The penalty is chosen such that $-cJ(\hat{k})$ plus the three terms that involve $\hat{k}$ is nonpositive. Similarly, the penalty $cJ(k)$ dominates the terms involving k in the second line of the display. Therefore, we can conclude that, with probability at least $1 - 2e^{-\xi}$,

$$P(m_{\hat{\theta}} - m_{\theta^*}) \lesssim \inf_{k \in \mathcal{K}} \left(P(m_{\theta_k^*} - m_{\theta^*}) + J(k) \right) + \frac{e^{\xi/2}}{n}.$$

We integrate the implied tail bound with respect to ξ to obtain a bound on the expected value of the left side. ■

3.5.2 Statistical Learning

Theorem 3.5.4 is a natural extension of the rate theorem of Chapter 3.4, with the rate expressed in the modulus of an appropriate empirical process. Model selection was introduced in statistical learning theory as *structural risk minimization* and first analyzed without reference to the modulus. This approach is simpler and cruder, but on the positive side yields bounds with (more) explicit and simpler constants.

3.5.11 Theorem (Structural risk minimization). Let $\{m_\theta: \theta \in \Theta\}$ be a class of measurable functions $m_\theta: \mathcal{X} \rightarrow \mathbb{R}$ indexed by a metric space (Θ, d) such that $\|m_\theta\|_\infty \leq B$ for every $\theta \in \Theta_k$. Let $J: \mathcal{K} \rightarrow [0, \infty)$ be (possibly random) functions satisfying, for every $k \in \mathcal{K}$ and some $C > 0$, with probability at least $1 - e^{-x_k - \xi}$,

$$\sqrt{n} J(k) \geq E^* \sup_{\theta \in \Theta_k} \mathbb{G}_n(-m_\theta) + B\sqrt{2x_k} - C\sqrt{2\xi},$$

for given numbers x_k such that $\sum_k e^{-x_k} \leq 1$. Then

$$E^* P(m_{\hat{\theta}} - m_{\theta^*}) \leq \inf_{k \in \mathcal{K}} \left(P(m_{\theta_k^*} - m_{\theta^*}) + EJ(k) \right) + \frac{\sqrt{2\pi}(B+C)}{\sqrt{n}}.$$

Proof. By the bounded difference inequality, Theorem 2.15.1, with probability at least $1 - e^{-x_k - \xi}$,

$$\forall \theta \in \Theta_k: -\mathbb{G}_n m_\theta \leq E^* \sup_{\theta \in \Theta_k} \mathbb{G}_n(-m_\theta) + B\sqrt{2x_k + 2\xi}.$$

By the assumption on the penalty the right side is with probability at least $1 - e^{-x_k - \xi}$ smaller than $\sqrt{n} J(k) + (B+C)\sqrt{2\xi}$. Because $\sum_k e^{-x_k} \leq 1$ by assumption, the two inequalities are both true for every k on an event with probability at least $1 - 2e^{-\xi}$. On this event we can choose first $\theta = \hat{\theta}_k$ and next $k = \hat{k}$ random, so that θ in the inequality is replaced by $\hat{\theta}_{\hat{k}} = \hat{\theta}$. Thus, with probability at least $1 - 2e^{-\xi}$, for every given k ,

$$-\frac{1}{\sqrt{n}} \mathbb{G}_n m_{\hat{\theta}} \leq J(\hat{k}) + (B+C)\sqrt{\frac{2\xi}{n}}.$$

We substitute this into the bound (3.5.10) to see that, with probability at least $1 - 2e^{-\xi}$, for every given k ,

$$P(m_{\hat{\theta}} - m_{\theta^*}) \leq P(m_{\theta_k^*} - m_{\theta^*}) + J(k) + \frac{1}{\sqrt{n}} \mathbb{G}_n m_{\theta_k^*} + (B+C)\sqrt{\frac{2\xi}{n}}.$$

Integration of this bound, substituting that $E\mathbb{G}_n m_{\theta_k^*} = 0$, yields the theorem. ■

The right side of the theorem is never smaller than $O(1/\sqrt{n})$, in view of the remainder term $\sqrt{2\pi}(B+C)/\sqrt{n}$. If the left side is to be considered as a square distance, then this means that the rate given by the theorem is never better than $n^{-1/4}$, which is an important limitation.

A bigger defect is that the rate is never smaller than $n^{-1/2}$ times the expectation $E^* \sup_{\theta \in \Theta_k} \mathbb{G}_n(-m_\theta)$, which enters the upper bound through the penalty term $EJ(k)$. This expectation is an upper bound to the left side of (3.5.8), a crude one as it replaces δ in the latter inequality by ∞ . Therefore the term $\delta_{n,k}^2 C'_k D_k^2 \propto n^{-1/2} C_k D_k \phi_k(\delta_{n,k})$, which appears in the upper bound of Theorem 3.5.4 will typically be significantly smaller than the corresponding term $n^{-1/2} E^* \sup_{\theta \in \Theta_k} \mathbb{G}_n(-m_\theta)$, for every k . This renders the bound of Theorem 3.5.11 far from optimal, although it has motivated a wealth of penalization schemes.

The theorem allows the penalties $J(k)$ to be data-dependent. If they are not, then the lower bound on $\sqrt{n}J(k)$ must of course be valid with probability one, and we shall typically use the theorem with $C = \xi = 0$, giving the requirement

$$J(k) \geq \frac{1}{\sqrt{n}} E^* \sup_{\theta \in \Theta_k} \mathbb{G}_n(-m_\theta) + B \sqrt{\frac{2x_k}{n}}.$$

The right side is a measure of the complexity of the model Θ_k through the supremum of the empirical process over this model. Because the penalty must not depend on unknown quantities, we need an upper bound on this complexity that is distribution-independent.

3.5.12 Example (Rademacher complexities). The simplest estimate for the mean value $E^* \sup_{\theta \in \Theta_k} \mathbb{G}_n(-m_\theta)$ is the variable $\sup_{\theta \in \Theta_k} \mathbb{G}_n(-m_\theta)$ itself. However, this variable is dependent on the distribution of the observations through the centering of the empirical process, and hence cannot be used to construct a penalty. Symmetrization provides a way around this difficulty. By Lemma 2.3.1 the mean value of the norm of the empirical process is bounded above by twice the mean value of the norm of the corresponding symmetrized empirical process $\mathbb{G}_n^o$. Therefore, the mean value of present interest is bounded by $2E^* \|\mathbb{G}_n^o\|_{\mathcal{M}_k}$, for $\mathcal{M}_k$ the class of functions $\{m_\theta: \theta \in \Theta_k\}$. An unbiased estimate of the latter mean value is $2\|\mathbb{G}_n^o\|_{\mathcal{M}_k}$.

By the bounded difference inequality, Theorem 2.15.1, with probability at least $1 - e^{-x_k - \xi}$,

$$2E^* \|\mathbb{G}_n^o\|_{\mathcal{M}_k} \leq 2\|\mathbb{G}_n^o\|_{\mathcal{M}_k} + 2B\sqrt{2x_k + 2\xi}$$

It follows that with the same probability $2E^* \|\mathbb{G}_n^o\|_{\mathcal{M}_k} + B\sqrt{2x_k} - 2B\sqrt{2\xi}$ is bounded above by $2\|\mathbb{G}_n^o\|_{\mathcal{M}_k} + 3B\sqrt{2x_k}$. Therefore, the condition of Theorem 3.5.11 on the penalty is satisfied, with $C = 2B$, by the penalty

$$J(k) = 2 \sup_{\theta \in \mathcal{M}_k} |\mathbb{P}_n^o m_\theta| + 3B\sqrt{\frac{2x_k}{n}}.$$

This *Rademacher complexity* penalty seems not very practical, but it does work in general, within the context of Theorem 3.5.11.

3.5.13 Example (VC-class). The norm of the empirical process indexed by a VC-class can be bounded in terms of its VC-dimension. In fact, combining Theorem 2.14.1 with the bounds on covering numbers of VC-classes given by Theorem 2.6.4 or Theorem 2.6.7 shows that $E^* \|\mathbb{G}_n\|_{\mathcal{F}} \leq c\sqrt{V(\mathcal{F})} \|F\|_{P,2}$, for every P -measurable VC-class of functions, where the multiplicative constant c is universal. We conclude that

$$J(k) = c\sqrt{\frac{V_k}{n}} B + B\sqrt{\frac{x_k}{2n}}$$

gives a valid penalty within the context of Theorem 3.5.11 (with $C = 0$), if V_k is the VC-dimension of the class of functions $\mathcal{M}_k = \{m_\theta: \theta \in \Theta_k\}$.

We make no attempt to determine a minimal value for the constant c . For the case that the functions m_θ are indicators, the classical inequality due to Vapnik-Červonenkis (see Lemma 2.14.22) asserts that, for $n \geq V_k$,

$$E^* \|\mathbb{G}_n\|_{\mathcal{M}_k} \leq \sqrt{2 \log \left(2 \max_{x \in \mathcal{X}^{2n}} \Delta_{2n}(\mathcal{M}_k, x) \right)} \leq 2\sqrt{2V_k \left(1 + \log \frac{n}{V_k} \right)}.$$

This adds an unnecessary logarithmic factor to the bound, but it has the benefit of having explicit and small constants. This bound shows that the penalty in Theorem 3.5.11 can be taken equal to

$$\begin{aligned} J(k) &= \sqrt{\frac{2 \log(2 \|\Delta_{2n}(\mathcal{M}_k, \cdot)\|_\infty)}{n}} + \sqrt{\frac{x_k}{2n}}, \\ J(k) &= 2\sqrt{\frac{2V_k(1 + \log(n/V_k))}{n}} + \sqrt{\frac{x_k}{2n}}. \end{aligned}$$

In both cases the penalty is deterministic and we can choose $C = 0$. (For indicator functions m_θ the constant B can be taken equal to $1/2$ by shifting the functions down by $1/2$.)

An alternative to using the worst case picking numbers $\|\Delta_{2n}(\mathcal{M}_k, \cdot)\|_\infty$ is to evaluate these at the observed data points. It can be shown that, with probability at least $1 - e^{-\xi}$,[†]

$$2 \log \Delta_n(\mathcal{M}_k, X_1, \dots, X_n) \geq E \log \Delta_n(\mathcal{M}_k, X_1, \dots, X_n) - 2 \log 2\xi.$$

This leads to a penalty of the form

$$J(k) = 4\sqrt{\frac{\log \Delta_n(\mathcal{M}_k, X_1, \dots, X_n)}{n}} + c_1 \sqrt{\frac{x_k}{2n}},$$

where $c_1 > 1$ is a universal constant, dependent on the constant C chosen in Theorem 3.5.11. (The choices $C = 2\sqrt{\log 2}$ and $c_1 = 4.1$ work.)

[†] See Massart (2000), Section 2.1.3.

Problems and Complements

1. If $X_1, \dots, X_m$ are arbitrary random variables such that $P(X_i > x) \leq Ke^{-Cx^p}$ for every $x > 0$ and given positive constants K and C and $p \geq 1$, then $E \max_{1 \leq i \leq m} X_i \leq ((2/C) \log(1 + mK))^{1/p}$.

[Hint: Assume without loss of generality that the variables X_i are nonnegative. Because the function $x \mapsto \psi(x) = e^{x^p} - 1$ is nonnegative, convex and nondecreasing on $[0, \infty)$, by Jensen's inequality,

$$\psi\left(d^{1/p} E \max_i X_i\right) \leq E \psi\left(d^{1/p} \max_i X_i\right) = E \max_i \psi\left(d^{1/p} X_i\right) \leq \sum_{i=1}^m E \psi\left(d^{1/p} X_i\right).$$

We can compute the expectations in the right side as

$$E \psi\left(d^{1/p} X_i\right) = E \int_0^{d^{1/p} X_i} e^{x^p} p x^{p-1} dx = \int_0^\infty P(d^{1/p} X_i > x) e^{x^p} p x^{p-1} dx,$$

by Fubini's theorem. We can now insert the upper tail bound, and calculate the resulting integral as $dK/(C - d)$, provided that $d < C$. Choose $d = \frac{1}{2}C$, and next apply the function $\psi^{-1}(m) = (\log(1 + m))^{1/p}$ left and right to the inequality.]

2. **(Young inequality)** The *Legendre transform* of a convex function $f: I \rightarrow \mathbb{R}$ with domain an interval $I \subset \mathbb{R}$ is the function $g: J \rightarrow \mathbb{R}$ given by $g(y) = \sup_{x \in I} (xy - f(x))$, where the domain $J \subset \mathbb{R}$ of g is the interval where the supremum is finite. It is immediate from the definition that $xy \leq f(x) + g(y)$, for every $x \in I$ and $y \in J$. The Legendre transform of $f(x) = e^x$ on $I \subset \mathbb{R}$ is given by $g(y) = y \log y - y$ on $J = [0, \infty)$ (with $g(0) = 0$), whence $xy \leq e^x + y \log y - y$, for $x \in \mathbb{R}$ and $y > 0$. This implies that also $xy \leq e^{x/2} + 2y \log y$, for $x \in \mathbb{R}$ and $y > 0$.

3.6

Random Sample Size, Poissonization and Kac Processes

It can be argued that in practice the number of available observations is random and perhaps dependent on the same random phenomenon. In the first section it is shown in fair generality that the empirical central limit theorem is valid also for random sample size.

The Kac process is a Poisson process obtained by taking the sample size in the empirical distribution to be a Poisson distributed variable independent of the data. We study the convergence of the Kac empirical process.

3.6.1 Random Sample Size

Suppose that at “time” n a random number N_n of observations from an infinite i.i.d. sequence $X_1, X_2, \dots$ is available. Formally, N_n is any integer-valued, nonnegative map defined on the same probability space as the sequence of observations. Assume that the sequence N_n converges to infinity in the strong sense that

$$\frac{N_n}{c_n} \xrightarrow{P} \nu,$$

for a strictly positive random variable ν and a deterministic sequence $c_n \rightarrow \infty$. Let $\mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - P)$ be the empirical process as before. Then $\mathbb{G}_{N_n}$ converges in distribution whenever the sequence $\mathbb{G}_n$ does, to the same limit.

3.6.1 Theorem (Donsker random sample size). *Let $\mathcal{F}$ be a Donsker class of measurable functions. Suppose that N_n is a sequence of positive, integer-valued random variables such that $N_n/c_n \xrightarrow{P} \nu$ for a random variable ν with $P(\nu > 0) = 1$ and a deterministic sequence $c_n \rightarrow \infty$. Then the sequence $\mathbb{G}_{N_n}$ converges in distribution in $\ell^\infty(\mathcal{F})$ to a tight Brownian bridge as $n \rightarrow \infty$.*

Proof. First, suppose that $c_n = n$ and $N_n/n \leq M$ for some number M . Let $\mathbb{Z}_n(s, f) = n^{-1/2} \sum_{i=1}^{\lfloor ns \rfloor} (\delta_{X_i} - P)$ be the sequential empirical process. By a minor extension of Theorem 2.12.1, the sequence $\mathbb{Z}_n$ converges in distribution in $\ell^\infty([0, M] \times \mathcal{F})$ to a Kiefer-Müller process $\mathbb{Z}$.

The lemma below shows that the sequence $(\mathbb{Z}_n, N_n/n)$ converges weakly in the space $\ell^\infty([0, M] \times \mathcal{F}) \times \mathbb{R}$ to a pair $(\mathbb{Z}, \nu)$ of independent random elements $\mathbb{Z}$ and ν . Now

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{N_n} (\delta_{X_i} - P)f = \mathbb{Z}_n\left(\frac{N_n}{n}, f\right) = g\left(\mathbb{Z}_n, \frac{N_n}{n}\right)f,$$

where $g: \ell^\infty([0, M] \times \mathcal{F}) \times \mathbb{R} \rightarrow \ell^\infty(\mathcal{F})$ is defined by $g(z, r) = z(r, \cdot)$. The map g is continuous at every point (z, r) such that $s \mapsto z(s, \cdot)$ is continuous with respect to $\|\cdot\|_{\mathcal{F}}$. The paths of the Kiefer-Müller process $\mathbb{Z}$ possess the latter continuity property with probability 1. Thus, the continuous mapping theorem yields $g(\mathbb{Z}_n, N_n/n) \rightsquigarrow g(\mathbb{Z}, \nu) = \mathbb{Z}(\nu, \cdot)$. By the same argument,

$$\mathbb{G}_{N_n} = \frac{1}{\sqrt{N_n}} \sum_{i=1}^{N_n} (\delta_{X_i} - P) \rightsquigarrow \frac{1}{\sqrt{\nu}} \mathbb{Z}(\nu, \cdot).$$

Since ν and $\mathbb{Z}$ are independent and $\nu^{-1/2}\mathbb{Z}(\nu, \cdot)$ is distributed as a Brownian bridge for every deterministic ν , the variable on the right is distributed as a Brownian bridge. This concludes the proof under the special assumptions on the N_n .

If N_n/n is not bounded (but still $c_n = n$), define $N_{n,M} = N_n \wedge (Mn)$. By the preceding argument, $\mathbb{G}_{N_{n,M}} \rightsquigarrow \mathbb{G}$ for every M . The probability that $N_{n,M} \neq N_n$ can be made arbitrarily small by choice of M . The theorem follows.

Finally, the case that $c_n \neq n$ can be treated by relabeling the indexes. We may assume that each c_n is an integer. Furthermore, since it suffices to show that every subsequence of $\{n\}$ has a further subsequence along which $\mathbb{G}_{N_n}$ converges, it is not a loss of generality to assume that c_n is also strictly increasing. Define $N'_k = N_n$ if $c_n = k$ and define $N'_k = k\nu$ if $k \neq c_n$ for every n . Then $N'_k/k \xrightarrow{P} \nu$, hence $\mathbb{G}_{N'_k} \rightsquigarrow \mathbb{G}$. The sequence $\mathbb{G}_{N_n}$ is a subsequence. ■

3.6.2 Theorem (Mixing Donsker theorem). Let $\mathcal{F}$ be a Donsker class and ν_n a sequence of random variables with values in a Polish space $\mathbb{D}$ such that $\nu_n \xrightarrow{P} \nu$ for a random variable ν . Then the sequence of sequential empirical processes $\mathbb{Z}_n$ satisfies $(\mathbb{Z}_n, \nu_n) \rightsquigarrow (\mathbb{Z}, \nu)$ in $\ell^\infty([0, M] \times \mathcal{F}) \times \mathbb{D}$, where $\mathbb{Z}$ and ν are independent.

Proof. Let $k_n \rightarrow \infty$ slowly enough that $k_n = o(\sqrt{n})$. Set

$$\mathbb{Z}'_n(s, f) = \frac{1}{\sqrt{n}} \sum_{i=k_n+1}^{\lfloor ns \rfloor} (\delta_{X_i} - P).$$

Then the sequence $\mathbb{Z}'_n - \mathbb{Z}_n$ converges to zero in probability in $\ell^\infty([0, M] \times \mathcal{F})$. By a version of Slutsky's lemma, the sequence $(\mathbb{Z}'_n, \nu)$ has the same limit distribution as the sequence $(\mathbb{Z}_n, \nu_n)$. By Doob's martingale convergence theorem, for every

measurable set B we have $P(\nu \in B | X_1, \dots, X_k) \rightarrow P(\nu \in B | X_1, X_2, \dots)$ in mean as $k \rightarrow \infty$. Conclude that

$$\lim_{n \rightarrow \infty} P^*(\mathbb{Z}'_n \in A, \nu \in B) = \lim_{n \rightarrow \infty} E\left(1_A(\mathbb{Z}'_n)^* P(\nu \in B | X_1, \dots, X_{k_n})\right).$$

Since $\mathbb{Z}'_n$ is independent of $X_1, \dots, X_{k_n}$, the expectation in the right side can be factorized as $P^*(\mathbb{Z}'_n \in A) P(\nu \in B)$. This converges to $P(\mathbb{Z} \in A) P(\nu \in B)$ for every continuity set A . ■

The preceding theorem can be stated alternatively as the assertion that the sequence of sequential empirical processes indexed by a Donsker class *converges stably* to its limit (see Theorem 1.15.5). As the limits $\mathbb{Z}$ and ν in the lemma are independent, the stable convergence is *mixing*. Chapter 1.15 introduces these convergence concepts in general. By the continuous mapping theorem (see Problem 1.15.5), the sequence of empirical processes also converges stably and mixing.

3.6.3 Corollary. *If $\mathcal{F}$ is a Donsker class, then the sequence of empirical processes converges stably and mixing in $\ell^\infty(\mathcal{F})$ to a Brownian bridge process.*

The assumption on the sequence N_n in the preceding theorem is remarkably weak. Under the stronger assumption that $N_n/n \xrightarrow{P} 1$, the sequences $\mathbb{G}_{N_n}$ and $\mathbb{G}_n$ are not only asymptotically equal in law, but also equivalent in probability.

3.6.4 Theorem (Donsker random sample size). *Let $\mathcal{F}$ be a Donsker class of measurable functions. Suppose that N_n is a sequence of positive, integer-valued random variables such that $N_n/n \xrightarrow{P} 1$. Then the sequence $\mathbb{G}_{N_n} - \mathbb{G}_n$ converges in outer probability to zero in $\ell^\infty(\mathcal{F})$ as $n \rightarrow \infty$.*

Proof. With the same notation as in the proof of Theorem 3.6.1, the sequence $(\mathbb{Z}_n, N_n/n)$ converges in distribution to $(\mathbb{Z}, 1)$ in the space $\ell^\infty([0, 2] \times \mathcal{F}) \times \mathbb{R}$. By the continuous mapping theorem, $\mathbb{Z}_n(N_n/n, \cdot) - \mathbb{Z}_n(1, \cdot) \rightsquigarrow \mathbb{Z}(1, \cdot) - \mathbb{Z}(1, \cdot) = 0$ in $\ell^\infty(\mathcal{F})$. Convergence in distribution and in outer probability to a degenerate limit are equivalent. ■

3.6.2 Poissonization

Let the sample size N_n at “time” n be a Poisson variable with mean n and independent of the i.i.d. observations $X_1, X_2, \dots$. The random measure

$$\mathbb{N}_n = \sum_{i=1}^{N_n} \delta_{X_i}$$

is the *Kac empirical point process*. Straightforward calculations show that, for every measurable set C , the random variable $\mathbb{N}_n(C)$ is Poisson-distributed with mean $nP(C)$ (where $P(C) = P(X_i \in C)$). Furthermore, for every finite collection $C_1, \dots, C_k$ of pairwise-disjoint measurable sets, the random variables

$\mathbb{N}_n(C_1), \dots, \mathbb{N}_n(C_k)$ are independent. Thus the Kac process $\mathbb{N}_n$ is a (generalized) Poisson process on the sample space $(\mathcal{X}, \mathcal{A})$ with intensity measure nP .

Given a class $\mathcal{F}$ of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$, consider the process $\{\mathbb{N}_n(f): f \in \mathcal{F}\}$. Its mean and variance function equal $E\mathbb{N}_n(f) = nPf$ and $\text{var } \mathbb{N}_n(f) = nPf^2$. A standardized version of this process is

$$\mathbb{Z}_n = \frac{1}{\sqrt{n}}(\mathbb{N}_n - nP) = \sqrt{\frac{N_n}{n}}\mathbb{G}_{N_n} + \sqrt{n}\left(\frac{N_n}{n} - 1\right)P,$$

where $\mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - P)$ is the empirical process. The collection $\mathcal{F}$ is called *P-Kac* if the sequence $\mathbb{Z}_n$ converges in distribution to a tight limit process in $\ell^\infty(\mathcal{F})$. (The sample paths of $\mathbb{Z}_n$ are in $\ell^\infty(\mathcal{F})$ if $\mathcal{F}$ possesses a finite envelope function and $\|P\|_{\mathcal{F}} < \infty$.)

Since $N_n/n \xrightarrow{P} 1$, Theorem 3.6.4 yields that $\mathbb{G}_{N_n} - \mathbb{G}_n \xrightarrow{P} 0$ if $\mathcal{F}$ is Donsker. In that case the sequence $\mathbb{Z}_n$ is equivalent to the sequence $\mathbb{G}_n + \sqrt{n}(N_n/n - 1)P$ and converges in distribution to $\mathbb{G} + ZP$ for an independent Brownian bridge $\mathbb{G}$ and $N(0, 1)$ variable Z . Thus the limit process $\mathbb{Z} = \mathbb{G} + ZP$ in the Kac central limit theorem is a Brownian motion process and has covariance function

$$E\mathbb{Z}(f)\mathbb{Z}(g) = Pfg.$$

Each $\mathbb{Z}_n$ has the same covariance function, $E\mathbb{Z}_n(f)\mathbb{Z}_n(g) = Pfg$.

We have shown that every Donsker class with $\|P\|_{\mathcal{F}} < \infty$ is Kac. Actually, these two concepts are equivalent, which can be seen from the following concentration lemma.

Since $\mathbb{N}_n$ is a Poisson process with intensity measure nP , it can be written as the sum of n i.i.d. Poisson processes of intensity measure P . Formally, let $Y_1, Y_2, \dots$ be an i.i.d. sequence of Poisson(1) variables, and let $X_{i,j}$ be an array of i.i.d. copies of X_1 . Then the process

$$H_n = \sum_{i=1}^n \sum_{j=1}^{Y_i} (\delta_{X_{i,j}} - P)$$

is equal in distribution to $H'_n = \sum_{i=1}^{N_n} (\delta_{X_i} - P)$ (in the sense that $E^*h(H_n) = E^*h(H'_n)$ for every h if the Y_i and $X_{i,j}$ are suitably defined as coordinate projections on a product space). It follows that the random-sample central limit theorem for $\mathbb{G}_{N_n}$ is equivalent to the central limit theorem for a deterministic number of Poisson processes of the type $\sum_{j=1}^{Y_i} (\delta_{X_{i,j}} - P)$. Le Cam's lemma compares the concentration of these processes with the concentration of the empirical process.

3.6.5 Lemma (Le Cam's Poissonization lemma). *Let N_n be Poisson-distributed with mean n and be independent of the i.i.d. zero-mean stochastic processes $Z_1, \dots, Z_n$. Then for any class of functions $\mathcal{F}$*

$$\left(1 - \frac{1}{e}\right) E^* \left\| \sum_{i=1}^n Z_i \right\|_{\mathcal{F}} \leq E^* \left\| \sum_{i=1}^{N_n} Z_i \right\|_{\mathcal{F}}.$$

Proof. A Poisson variable Y_i with mean 1 satisfies $E(Y_i \wedge 1) = 1 - e^{-1}$. Let the array $Z_{i,j}$ consist of i.i.d. copies of Z_1 independent of $Y_1, \dots, Y_n$. The left side of the lemma equals

$$\begin{aligned} E_Z^* \left\| E_Y \sum_{i=1}^n (Y_i \wedge 1) Z_i \right\|_{\mathcal{F}} &\leq E^* \left\| \sum_{i=1}^n (Y_i \wedge 1) Z_i \right\|_{\mathcal{F}} \\ &\leq E_Y E_Z^* \left\| \sum_{i=1}^n \sum_{j=1}^{Y_i} Z_{i,j} \right\|_{\mathcal{F}}. \end{aligned}$$

Here the factorization of E^* as $E_Y E_Z^*$ is warranted by Lemma 1.2.7, and in the last step we replace 0 by 0 if $Y_i = 0$, Z_i by $Z_{i,1}$ if $Y_i = 1$, Z_i by $Z_{i,1} + Z_{i,2}$ if $Y_i = 2$, etc. The inequality is valid in view of the inequality $E^* \|Z_1\| = E^* \|Z_1 + E Z_2\| \leq E^* \|Z_1 + Z_2\|$ for independent processes Z_1 and Z_2 with $E Z_2 = 0$.

By the discussion preceding the lemma, the double sum on the right side is equal in distribution to $\sum_{i=1}^{N_n} Z_i$. ■

3.6.6 Theorem (Kac CLT). *A class $\mathcal{F}$ of measurable functions with $\|P\|_{\mathcal{F}} < \infty$ is Kac if and only if it is Donsker. In that case $\|\mathbb{G}_{N_n} - \mathbb{G}_n\|_{\mathcal{F}}^* = O_P(n^{-1/4})$.*

Proof. Set $\mathbb{G}'_n = n^{-1/2} \sum_{i=1}^{N_n} (\delta_{X_i} - P)$. If $N_n = n + k$, then the difference $\mathbb{G}_n - \mathbb{G}'_n$ is the sum of k terms $n^{-1/2} (\delta_{X_i} - P)$. Since $P(|N_n - n| \geq M\sqrt{n}) \leq M^{-2}$ by Chebyshev's inequality, we have for every $\varepsilon > 0$

$$\begin{aligned} P^*(\|\mathbb{G}_n - \mathbb{G}'_n\|_{\mathcal{F}} > \varepsilon) &\leq \frac{1}{M^2} + \frac{1}{\varepsilon} \sum_{|k| < M\sqrt{n}} P(N_n = n + k) E^* \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^{|k|} (\delta_{X_i} - P) \right\|_{\mathcal{F}} \\ &\leq \frac{1}{M^2} + \frac{1}{\varepsilon \sqrt{n}} E^* \left\| \sum_{i=1}^{\lfloor M\sqrt{n} \rfloor} (\delta_{X_i} - P) \right\|_{\mathcal{F}}. \end{aligned}$$

Here the last step follows, since the outer expectations $E^* \left\| \sum_{i=1}^{|k|} (\delta_{X_i} - P) \right\|_{\mathcal{F}}$ are non-decreasing in $|k|$ by Jensen's inequality.

If $\mathcal{F}$ is Donsker, then the sequence $n^{-1/4} E^* \left\| \sum_{i=1}^{\lfloor M\sqrt{n} \rfloor} (\delta_{X_i} - P) \right\|_{\mathcal{F}}$ is bounded by Lemma 2.3.11. In that case the calculation in the preceding paragraph shows that the probability $P^*(\|\mathbb{G}_n - \mathbb{G}'_n\|_{\mathcal{F}} > K n^{-1/4})$ is bounded by $M^{-2} + K^{-1} O(1)$. Its limsup as $n \rightarrow \infty$ can be made arbitrarily small by choice of M and K . Thus $\|\mathbb{G}_n - \mathbb{G}'_n\|_{\mathcal{F}}^* = O_P(n^{-1/4})$. The theorem follows, because $\|\mathbb{G}'_n - \mathbb{G}_{N_n}\|_{\mathcal{F}}^* = O_P(n^{-1/2})$.

If $\mathcal{F}$ is Kac, then $\mathbb{G}'_n = n^{-1/2} (\mathbb{N}_n - \mathbb{N}_n(1) P)$ converges in distribution to a tight limit. In view of the preceding discussion, this is equivalent to the processes $Z_i = \sum_{j=1}^{Y_i} (\delta_{X_{i,j}} - P)$ satisfying the central limit theorem. Thus the sequence of expectations $n^{-1/2} E^* \left\| \sum_{i=1}^n Z_i \right\|_{\mathcal{F}}$ is bounded by Lemma 2.3.11. By Le Cam's lemma these expectations dominate the outer expectations $n^{-1/2} E^* \left\| \sum_{i=1}^n (\delta_{X_i} - P) \right\|_{\mathcal{F}}$ up to a positive constant and the argument can be finished as before. ■

3.7

Bootstrap

In this chapter we first prove the asymptotic consistency of the empirical bootstrap estimator of the distribution of the empirical process. Next this result is generalized to more general, “exchangeable” bootstrap schemes. The results of this chapter are particularly interesting when combined with those of Section 3.10.3, which show that the consistency of the bootstrap is retained under application of the delta-method.

3.7.1 Empirical Bootstrap

Let $\mathbb{P}_n$ be the empirical measure of an i.i.d. sample $X_1, \dots, X_n$ from a probability measure P . Given the sample values, let $\hat{X}_1, \dots, \hat{X}_n$ be an i.i.d. sample from $\mathbb{P}_n$. The *bootstrap empirical distribution* is the empirical measure $\hat{\mathbb{P}}_n = n^{-1} \sum_{i=1}^n \delta_{\hat{X}_i}$, and the *bootstrap empirical process* $\hat{\mathbb{G}}_n$ is

$$\hat{\mathbb{G}}_n = \sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n) = \frac{1}{\sqrt{n}} \sum_{i=1}^n (M_{ni} - 1) \delta_{X_i},$$

where M_{ni} is the number of times that X_i is “redrawn” from the original sample. Instead of n we can also “redraw” k bootstrap values $\hat{X}_1, \dots, \hat{X}_k$. The corresponding bootstrap empirical process is

$$\hat{\mathbb{G}}_{n,k} = \sqrt{k}(\hat{\mathbb{P}}_k - \mathbb{P}_n) = \frac{1}{\sqrt{k}} \sum_{i=1}^n \left(M_{ki} - \frac{k}{n} \right) \delta_{X_i}.$$

A more precise specification of taking “bootstrap samples” $\hat{X}_1, \dots, \hat{X}_k$ should include a probability space on which these variables are defined. (This would be particularly appropriate since the results in this chapter are formulated in terms of outer

measure.) Alternatively, the definitions of the bootstrap processes can be made rigorous by simply defining them as the far right side of the preceding displays. This is more convenient for the purpose of this chapter. In the resulting definition of $\hat{\mathbb{G}}_n$ and $\hat{\mathbb{G}}_{n,k}$, it is further specified that the vector $(M_{k1}, \dots, M_{kn})$ is independent of $X_1, \dots, X_n$ and multinomially distributed with parameters k and (probabilities) $1/n, \dots, 1/n$. For computing outer expectations independence is understood in terms of a product probability space. Let $X_1, X_2, \dots$ be the coordinate projections on the first ∞ coordinates of the product space $(\mathcal{X}^\infty, \mathcal{A}^\infty, P^\infty) \times (\mathcal{Z}, \mathcal{C}, Q)$ and let the multinomial vectors M depend on the last factor only.[‡] The Poisson variables introduced later are also assumed to depend on $\mathcal{Z}$ only.

Let $\mathcal{F}$ be a class of measurable functions with a finite envelope function. Then $\hat{\mathbb{G}}_n$ considered as a process indexed by $\mathcal{F}$ can be viewed as a map in $\ell^\infty(\mathcal{F})$. The main result of this section is that the sequence $\hat{\mathbb{G}}_n$ converges in $\ell^\infty(\mathcal{F})$ conditionally in distribution to a tight Brownian bridge process given almost all sequences $X_1, X_2, \dots$ if and only if $\mathcal{F}$ is Donsker and $P^* \|f - Pf\|_{\mathcal{F}}^2 < \infty$. In that case the difference between the “conditional random law” of $\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n)$ and the “law” of $\sqrt{n}(\mathbb{P}_n - P)$ converges to zero almost surely. In statistical terms this means that the conditional law of $\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n)$ is an asymptotically consistent estimator of the law of $\sqrt{n}(\mathbb{P}_n - P)$. This result is particularly interesting in combination with the delta-method, by which the consistency result carries over to a large class of statistics which are smooth functionals of the empirical distribution (see Section 3.10.3).

The result on almost-sure weak convergence of the bootstrap empirical process is reminiscent of the multiplier central limit theorem discussed in Chapter 2.9. The difference is that in the bootstrap situation the multipliers $M_{n1} - 1, \dots, M_{nn} - 1$ are dependent. Our line of proof is to remove the dependence by Poissonization and to show that the Poissonized process and the ordinary bootstrap process are asymptotically equivalent.

Instead of n , take a Poisson number N_n of replicates, where N_n has mean n and is independent of the original observations. Then $M_{N_n,1}, \dots, M_{N_n,n}$ are i.i.d. Poisson variables with mean 1. The corresponding bootstrap empirical process can be written

$$\hat{\mathbb{G}}_{n,N_n} = \frac{1}{\sqrt{N_n}} \sum_{i=1}^n (M_{N_n,i} - 1) (\delta_{X_i} - P) - \frac{N_n - n}{\sqrt{N_n}} (\mathbb{P}_n - P).$$

Conditionally on $X_1, X_2, \dots$, the second term converges in distribution to zero almost surely if $\mathcal{F}$ is Glivenko-Cantelli. Furthermore, by the results of Chapter 2.9, the first term converges in distribution to a Brownian bridge process almost surely if and only if $\mathcal{F}$ is Donsker and $P^* \|f - Pf\|_{\mathcal{F}}^2 < \infty$. It follows under these conditions that the Poissonized bootstrap process $\hat{\mathbb{G}}_{n,N_n}$ converges to a Brownian bridge. In the proof ahead, it is shown that the difference in distribution between $\hat{\mathbb{G}}_{n,N_n}$ and $\hat{\mathbb{G}}_n = \hat{\mathbb{G}}_{n,n}$ is very small.

Conditional weak convergence is formulated in terms of the bounded Lipschitz metric, as explained in Chapters 1.12–1.13. We denote the Poisson multiplier

[‡] Given this set-up, Problem 3.7.1 gives a way of defining bootstrap values $\hat{X}_i$ on $\mathcal{X}^\infty \times \mathcal{Z}$ such that the identities for $\hat{\mathbb{G}}_n$ and $\hat{\mathbb{G}}_{n,k}$ are valid.

process by

$$\hat{Y}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n (M_{N_n, i} - 1) (\delta_{X_i} - P).$$

We write $E_M h(\hat{G}_n)$ and $E_{M, N} h(\hat{Y}_n)$ for expectations relative to the variables M and (M, N) of functions of the bootstrap empirical process and Poisson multiplier process, for fixed values of $X_1, \dots, X_n$.

As in Chapter 2.9, there are separate results for conditional weak convergence in probability and almost surely.

3.7.1 Theorem (Bootstrap). *Let $\mathcal{F}$ be a class of measurable functions with finite envelope function. The following statements are equivalent:*

- (i) $\mathcal{F}$ is Donsker;
- (ii) $\sup_{h \in BL_1} |E_{M, N} h(\hat{Y}_n) - Eh(\mathbb{G})| \xrightarrow{P^*} 0$ and $\hat{Y}_n$ is asymptotically measurable;
- (iii) $\sup_{h \in BL_1} |E_M h(\hat{G}_n) - Eh(\mathbb{G})| \xrightarrow{P^*} 0$ and $\hat{G}_n$ is asymptotically measurable.

3.7.2 Theorem (Bootstrap). *Let $\mathcal{F}$ be a class of measurable functions with finite envelope function. The following statements are equivalent:*

- (i) $\mathcal{F}$ is Donsker and $P^* \|f - Pf\|_{\mathcal{F}}^2 < \infty$;
- (ii) $\sup_{h \in BL_1} |E_{M, N} h(\hat{Y}_n) - Eh(\mathbb{G})| \xrightarrow{\text{as}^*} 0$ and the sequence $E_{M, N} h(\hat{Y}_n)^* - E_{M, N} h(\hat{Y}_n)_*$ converges almost surely to zero for every $h \in BL_1$;
- (iii) $\sup_{h \in BL_1} |E_M h(\hat{G}_n) - Eh(\mathbb{G})| \xrightarrow{\text{as}^*} 0$ and the sequence $E_M h(\hat{G}_n)^* - E_M h(\hat{G}_n)_*$ converges almost surely to zero for every $h \in BL_1$.

Here the asterisks denote the measurable cover functions with respect to M , N , and $X_1, X_2, \dots$ jointly.

Proofs. The equivalence of (i) and (ii) in both theorems is an immediate consequence of the conditional multiplier central limit theorems, Theorems 2.9.6 and 2.9.7. It suffices to prove that (i)+(ii) are equivalent to (iii). This proof will be based on a coupling of the bootstrap empirical process $\hat{G}_n$ and its (partly) Poissonized version $\hat{Y}_n$ by a special construction of multinomial variables. Let $m_n^{(1)}, m_n^{(2)}, \dots$ be i.i.d. multinomial $(1, n^{-1}, \dots, n^{-1})$ variables independent of N_n and set

$$M_n = \sum_{i=1}^n m_n^{(i)}; \quad M_{N_n} = \sum_{i=1}^{N_n} m_n^{(i)}.$$

Define $\hat{G}_n$ using M_n and $\hat{Y}_n$ using M_{N_n} . Note that $E_M h(\hat{G}_n)$ and $E_M h(\hat{G}_n)^*$ do not depend on the probability space on which M_n is defined (up to null sets). The first depends on the distribution of M_n only, in the second $h(\hat{G}_n)^*$ is equal to the measurable majorant with respect to $X_1, \dots, X_n$ only by the remarks following Lemma 1.2.7.

The absolute difference $|M_{N_n} - M_n|$ is the sum of $|N_n - n|$ of the variables $m_n^{(i)}$. Given $N_n = k$, its i th component $|M_{N_n, i} - M_{ni}|$ is binomially $(|k - n|, n^{-1})$ -distributed. For any $\varepsilon > 0$, there exists a sequence of integers ℓ_n with $\ell_n = O(\sqrt{n})$

such that $P(|N_n - n| \geq \ell_n) \leq \varepsilon$ for every n . By direct calculation,

$$P\left(\max_{1 \leq i \leq n} |M_{N_n, i} - M_{ni}| > 2\right) \leq \varepsilon + n P\left(\text{bin}(\ell_n, n^{-1}) > 2\right) \rightarrow \varepsilon.$$

It follows that for sufficiently large n all coordinates of the vector $|M_{N_n} - M_n|$ are 0, 1, or 2 with probability at least $1 - 2\varepsilon$. (The 2 is not important for the following: any bound would do.)

We can write $|M_{N_n, i} - M_{ni}| = \sum_{j=1}^{\infty} 1\{|M_{N_n, i} - M_{ni}| \geq j\}$. Alternatively, with I_n^j the set of indexes $i \in \{1, 2, \dots, n\}$ such that $|M_{N_n, i} - M_{ni}| \geq j$, we have $M_{N_n, i} - M_{ni} = \text{sign}(N_n - n) \sum_{j=1}^{\infty} 1\{i \in I_n^j\}$. Thus,

$$\begin{aligned} \hat{Y}_n - \hat{G}_n &= \frac{1}{\sqrt{n}} \sum_{i=1}^n (M_{N_n, i} - M_{ni})(\delta_{X_i} - P) \\ &= \text{sign}(N_n - n) \sum_{j=1}^{\infty} \frac{\#I_n^j}{\sqrt{n}} \left(\frac{1}{\#I_n^j} \sum_{i \in I_n^j} (\delta_{X_i} - P) \right). \end{aligned}$$

On the set where $\max_{1 \leq i \leq n} |M_{N_n, i} - M_{ni}| \leq 2$, only the first two terms of the sum over j can be nonzero. For any j we have $j(\#I_n^j) \leq |N_n - n| = O_P(\sqrt{n})$. Furthermore, the norm of the average between brackets on the far right side converges to zero outer almost surely for any j if $\mathcal{F}$ is a Glivenko-Cantelli class (cf. Lemma 3.7.16 ahead). Conclude that in that case

$$P_{M, N}(\|\hat{Y}_n - \hat{G}_n\|_{\mathcal{F}}^* > \varepsilon) \rightarrow 0,$$

as $n \rightarrow \infty$, given almost all sequences $X_1, X_2, \dots$, for every $\varepsilon > 0$. Consequently, the difference $\sup_h |E_M h(\hat{G}_n)^* - E_{M, N} h(\hat{Y}_n)^*|$ converges to zero outer almost surely. The same is true for this expression with the asterisks removed or moved to the bottom.

It follows that (i)+(ii) and (iii) are equivalent (in both theorems) provided $\mathcal{F}$ is Glivenko-Cantelli. If (i)+(ii) holds, then $\mathcal{F}$ is Donsker and certainly Glivenko-Cantelli. Thus, the proof of the theorem in the most interesting direction is complete. For the proof in the converse direction, it must be shown that (iii) implies that $\mathcal{F}$ is Glivenko-Cantelli.

It is no loss of generality to assume that $Pf = 0$ for every f in $\mathcal{F}$. Suppose that (iii) holds in probability, and assume for the moment that $P^* F^p < \infty$ for some $p > 1$. Under (iii) the sequence $n^{-1/2} \|\hat{G}_n\|_{\mathcal{F}}$ converges (unconditionally) to zero in outer probability. An application of Hölder's inequality yields

$$E^* \left(\frac{1}{\sqrt{n}} \|\hat{G}_n\|_{\mathcal{F}} \right)^p \leq E^* \frac{1}{n} \sum_{i=1}^n (M_{ni} + 1)^p F^p(X_i) = E(M_{n1} + 1)^p P^* F^p.$$

This is uniformly bounded in n . Conclude that $n^{-1/2} \|\hat{G}_n\|_{\mathcal{F}}$ converges to zero in outer mean. This implies the same for $n^{-1/2} \|\hat{G}_n\|_{\mathcal{F}}$ in view of the following in-

equalities. Since each M_{ni} is binomially (n, n^{-1}) distributed,

$$\begin{aligned}
 \left(1 - \frac{1}{n}\right)^n \mathbb{E}^* \sqrt{n} \|\mathbb{G}_n\|_{\mathcal{F}} &= \mathbb{E}_X^* \left\| \sum_{i=1}^n \mathbb{E}_M 1_{M_{ni}=0} (\delta_{X_i} - P) \right\|_{\mathcal{F}}^* \\
 &\leq \mathbb{E}^* \left\| \sum_{i=1}^n 1_{M_{ni}=0} (\delta_{X_i} - P) \right\|_{\mathcal{F}}^* \\
 &= \mathbb{E}_M \mathbb{E}_X^* \left\| \sum_{i=1}^n 1_{M_{ni}=0} (1 - M_{ni}) (\delta_{X_i} - P) \right\|_{\mathcal{F}}^* \\
 &\leq \mathbb{E}^* \left\| \sum_{i=1}^n (1 - M_{ni}) (\delta_{X_i} - P) \right\|_{\mathcal{F}}^* = \mathbb{E}^* \sqrt{n} \|\hat{\mathbb{G}}_n\|_{\mathcal{F}},
 \end{aligned}$$

where the last inequality follows from the inequality $\mathbb{E}^* \|U\| \leq \mathbb{E}^* \|U + V\|$ for independent, zero-mean processes U and V (applied conditionally given M ; also cf. Lemma 1.2.7). Conclude that $\mathcal{F}$ is Glivenko-Cantelli in outer mean. Still under the assumption that $P^* F < \infty$, a martingale argument shows that $\mathcal{F}$ is also Glivenko-Cantelli almost surely (cf. Lemma 2.4.6 and the proof of Theorem 2.4.3.)

Finally, it is shown that F satisfies $P^* F^p < \infty$, for every $p < 2$, under (iii) of the first theorem. The proof is based on the representation $\hat{\mathbb{G}}_n = n^{-1/2} \sum_{i=1}^n (\delta_{\hat{X}_i} - \mathbb{P}_n)$ of the bootstrap empirical process, where for each n given the original observations $\hat{X}_1, \dots, \hat{X}_n$ is an i.i.d. sample from $\mathbb{P}_n$. To give a precise meaning to the (conditional) outer probabilities, these bootstrap values can be defined as in Problem 3.7.1. As before, $\mathbb{P}_M$ denotes conditional probability given the original observations. Under assumption (iii), the sequence $\mathbb{P}_M(\|\hat{\mathbb{G}}_n\|_{\mathcal{F}}^* > m_n)$ converges in probability to zero for every $m_n \rightarrow \infty$. Let $\hat{Y}_1, \dots, \hat{Y}_n$ be an independent copy of $\hat{X}_1, \dots, \hat{X}_n$ given the original observations based on multinomial variables $\bar{M}$. Adapt the proof of the Lévy inequality given by Proposition A.1.2 to obtain

$$\begin{aligned}
 \mathbb{P}_{M, \bar{M}} \left(\max_{1 \leq i \leq n} \|f(\hat{X}_i) - f(\hat{Y}_i)\|_{\mathcal{F}}^* > m_n \sqrt{n} \right) \\
 \leq 2 \mathbb{P}_{M, \bar{M}} \left(\left\| \sum (f(\hat{X}_i) - f(\hat{Y}_i)) \right\|_{\mathcal{F}}^* > m_n \sqrt{n} \right) \xrightarrow{\mathbb{P}} 0.
 \end{aligned}$$

In view of Problem 2.3.2, it follows that

$$n \mathbb{P}_{M, \bar{M}} \left(\|f(\hat{X}_i) - f(\hat{Y}_i)\|_{\mathcal{F}}^* > m_n \sqrt{n} \right) \xrightarrow{\mathbb{P}} 0.$$

Since the processes $f(\hat{X}_i)$ and $f(\hat{Y}_i)$ are (conditionally) independent, the left side is an upper bound for

$$n \mathbb{P}_M \left(\|f(\hat{X}_i)\|_{\mathcal{F}}^* > 2m_n \sqrt{n} \right) \mathbb{P}_{\bar{M}} \left(\|f(\hat{Y}_i)\|_{\mathcal{F}}^* \leq m_n \sqrt{n} \right).$$

Here $\|f(\hat{X}_i)\|_{\mathcal{F}} = \sum F(X_j) 1\{m_n^{(i)} = e_j\}$, so that we can conclude that

$$n \mathbb{P}_M \left(\|f(\hat{X}_i)\|_{\mathcal{F}}^* > 2m_n \sqrt{n} \right) = \sum_{j=1}^n 1\{F^*(X_j) > 2m_n \sqrt{n}\} \xrightarrow{\mathbb{P}} 0.$$

Since the sums are binomially distributed, it follows that $n P(F^* > 2m_n \sqrt{n}) = O(1)$. ■

The preceding proof does not apply to the bootstrapped empirical process $\hat{\mathbb{G}}_{n,k}$ based on k (possibly unequal to n) replicates of the original sample $X_1, \dots, X_n$. However, the most important part of the theorem is true for arbitrary bootstrap sample sizes. If $\mathcal{F}$ is Donsker, then the “sequence” $\hat{\mathbb{G}}_{n,k}$ converges conditionally in distribution to a Brownian bridge process for every possible manner in which both $n, k \rightarrow \infty$.

As usual, let $\mathcal{F}_\delta = \{f - g : f, g \in \mathcal{F}, \rho_P(f - g) < \delta\}$.

3.7.3 Theorem (Bootstrap). *Let $\mathcal{F}$ be a Donsker class of measurable functions such that $\mathcal{F}_\delta$ is measurable for every $\delta > 0$. Then*

$$\sup_{h \in BL_1} |E_M h(\hat{\mathbb{G}}_{n,k_n}) - E h(\mathbb{G})| \xrightarrow{P^*} 0,$$

as $n \rightarrow \infty$, for any sequence $k_n \rightarrow \infty$. Furthermore, the sequence $E_M h(\hat{\mathbb{G}}_{n,k_n})^* - E_M h(\hat{\mathbb{G}}_{n,k_n})^*$ converges to zero in probability for every $h \in BL_1$. If $P^* \|f - Pf\|_{\mathcal{F}}^2 < \infty$, then the convergence is also outer almost surely.

Proof. As in the proof of the conditional multiplier theorems, Theorems 2.9.6 and 2.9.7, it suffices to establish almost-sure convergence of all the finite-dimensional marginals plus conditional convergence to zero of the sequence $\|\hat{\mathbb{G}}_{n,k}\|_{\mathcal{F}_\delta}$. Without loss of generality, assume that $Pf = 0$ for every $f \in \mathcal{F}$.

The distribution of $\hat{\mathbb{G}}_{n,k}f$ equals that of $k^{-1/2}$ times the centered sum of the i.i.d. random variables $f(\hat{X}_1), \dots, f(\hat{X}_k)$. Weak convergence of the sequence $\hat{\mathbb{G}}_{n,k}f$ can be established by the Lindeberg-Lévy theorem. By the law of large numbers for real variables,

$$\begin{aligned} E_{\hat{X}} f^2(\hat{X}_1) &= \frac{1}{n} \sum_{i=1}^n f^2(X_i) \xrightarrow{\text{as}} Pf^2, \\ E_{\hat{X}} f^2(\hat{X}_1) \{ |f(\hat{X}_1)| > \varepsilon \sqrt{k} \} &= \frac{1}{n} \sum_{i=1}^n f^2(X_i) \{ |f(X_i)| > \varepsilon \sqrt{k} \} \xrightarrow{\text{as}} 0. \end{aligned}$$

Therefore, $\hat{\mathbb{G}}_{n,k}f$ converges in distribution to a $N(0, Pf^2)$ distribution for almost every sequence $X_1, X_2, \dots$. This concludes the proof of finite-dimensional convergence.

Let $\tilde{N}_1, \tilde{N}_2, \dots$ be i.i.d. symmetrized Poisson variables with parameter $\frac{1}{2}k/n$. By Lemma 3.7.6 (ahead),

$$(3.7.4) \quad E_{\hat{X}} \|\hat{\mathbb{G}}_{n,k}\|_{\mathcal{F}_\delta} \leq 4E_{\tilde{N}} \left\| \frac{1}{\sqrt{k}} \sum_{i=1}^n \tilde{N}_i \delta_{X_i} \right\|_{\mathcal{F}_\delta}.$$

By the multiplier inequality Lemma 2.9.1, the outer expectation of the left side is for any $1 \leq n_0 \leq n$ bounded up to a constant by

$$(n_0 - 1) E \max_{1 \leq i \leq n} \frac{|\tilde{N}_i|}{\sqrt{k}} P^* F + \sqrt{\frac{n}{k}} \|\tilde{N}_i\|_{2,1} \max_{n_0 \leq j \leq n} E^* \left\| \frac{1}{\sqrt{j}} \sum_{i=n_0}^j \varepsilon_i \delta_{X_i} \right\|_{\mathcal{F}_\delta}.$$

In view of Problem 3.7.4, the first term is bounded by $(n_0 - 1)2\sqrt{2} (n \wedge k)^{-1/4} P^* F$. It converges to zero as $n, k \rightarrow \infty$ for every fixed $n_0 > 1$. By Problem 3.7.3, $\sqrt{n/k} \|\tilde{N}_i\|_{2,1} \leq 2\sqrt{2}$ for all n, k . Hence the second term converges to 0 as $n, k \rightarrow \infty$ followed by $n_0 \rightarrow \infty$ and $\delta \downarrow 0$, in view of Lemma 2.3.11. It follows that $E^* \|\hat{G}_{n,k}\|_{\mathcal{F}_\delta} \rightarrow 0$ as $n, k \rightarrow \infty$ followed by $\delta \downarrow 0$. This concludes the proof of the first statement of the theorem.

Assume that $P^* F^2 < \infty$. If $\varepsilon_1, \varepsilon_2, \dots$ are i.i.d. Rademacher variables independent of $\tilde{N}_1, \tilde{N}_2, \dots$ and $X_1, X_2, \dots$, then $\tilde{N}_i$ in (3.7.4) can be replaced by $\varepsilon_i |\tilde{N}_i|$. Next, Lemma 3.7.7 applied with $Z_i = \varepsilon_i \delta_{X_i}$ and X_i fixed yields

$$\begin{aligned} E_{\hat{X}} \|\hat{G}_{n,k}\|_{\mathcal{F}_\delta}^* &\leq (n_0 - 1) E \max_{1 \leq i \leq n} \frac{|\tilde{N}_i|}{\sqrt{k}} E_\varepsilon \frac{1}{n} \sum_{i=1}^n \|\varepsilon_i \delta_{X_i}\|_{\mathcal{F}_\delta}^* \\ &\quad + \sqrt{\frac{n}{k}} \|\tilde{N}_i\|_{2,1} \max_{n_0 \leq j \leq n} E_{\varepsilon,R} \left\| \frac{1}{\sqrt{j}} \sum_{i=n_0}^j \varepsilon_i \delta_{X_{R_i}} \right\|_{\mathcal{F}_\delta}^*. \end{aligned}$$

The first term on the right is bounded by $(n_0 - 1) 2\sqrt{2} (n \wedge k)^{-1/4} 2\mathbb{P}_n F$. Since $\mathcal{F}$ is Glivenko-Cantelli, the first term converges to zero almost surely for every fixed n_0 . To conclude the proof, it suffices to show that

$$(3.7.5) \quad \max_{n_0 \leq j \leq n} E_{\varepsilon,R} \left\| \frac{1}{\sqrt{j}} \sum_{i=n_0}^j \varepsilon_i \delta_{X_{R_i}} \right\|_{\mathcal{F}_\delta}^* \xrightarrow{\text{as}} 0,$$

as $n, k \rightarrow \infty$ followed by $n_0 \rightarrow \infty$ and $\delta \downarrow 0$. By Jensen's inequality, the expression is bounded by

$$\max_{n_0 \leq j \leq n} E_{\varepsilon,R} \left\| \frac{1}{\sqrt{j}} \sum_{i=1}^j \varepsilon_i \delta_{X_{R_i}} \right\|_{\mathcal{F}_\delta}^* = \max_{n_0 \leq j \leq n} E(U_j | \mathcal{S}_n) \leq E\left(\max_{n_0 \leq j} U_j | \mathcal{S}_n\right),$$

where $U_j = E_\varepsilon \left\| j^{-1/2} \sum_{i=1}^j \varepsilon_i \delta_{X_i} \right\|_{\mathcal{F}_\delta}^*$ and $\mathcal{S}_n$ is the σ -field generated by all functions $f: \mathcal{X}^\infty \rightarrow \mathbb{R}$ that are symmetric in their first n coordinates. By Corollary 2.9.9, the variable $\max_j U_j$ is integrable. The sequence $\mathcal{S}_n$ decreases to the "symmetric" σ -field $\mathcal{S}$, which consists of sets of probability 0 or 1 only by the Hewitt-Savage zero-one law.^b It follows that, as $n \rightarrow \infty$,

$$E\left(\max_{n_0 \leq j} U_j | \mathcal{S}_n\right) \xrightarrow{\text{as}} E\left(\max_{n_0 \leq j} U_j | \mathcal{S}\right) = E \max_{n_0 \leq j} U_j.$$

By Lemma 2.9.8, we have that $\limsup_{j \rightarrow \infty} U_j \leq K E\|\mathbb{G}\|_{\mathcal{F}_\delta}$ almost surely. Hence $\max_{n_0 \leq j} U_j \rightarrow 0$ almost surely as $n_0 \rightarrow \infty$ followed by $\delta \downarrow 0$. Its expectation converges to zero by dominated convergence. This proves (3.7.5). ■

^b Cf. Dudley (1993).

3.7.6 Lemma. For fixed elements $x_1, \dots, x_n$ of a set $\mathcal{X}$, let $\hat{X}_1, \dots, \hat{X}_k$ be an i.i.d. sample from $\mathbb{P}_n = n^{-1} \sum_{i=1}^n \delta_{x_i}$. Then, for every class $\mathcal{F}$ of functions $f: \mathcal{X} \rightarrow \mathbb{R}$,

$$\mathbb{E}_{\hat{X}} \left\| \sum_{j=1}^k (\delta_{\hat{X}_j} - \mathbb{P}_n) \right\|_{\mathcal{F}} \leq 4 \mathbb{E}_{N, N'} \left\| \sum_{i=1}^n (N_i - N'_i) \delta_{x_i} \right\|_{\mathcal{F}},$$

for i.i.d. Poisson variables $N_1, N'_1, \dots, N_n, N'_n$ with mean $\frac{1}{2}k/n$.

Proof. Let ε_j be i.i.d. Rademacher variables, independent of $\hat{X}_1, \dots, \hat{X}_k$. By the symmetrization lemma, Lemma 2.3.1, and Le Cam's Poissonization lemma, Lemma 3.6.5,

$$\mathbb{E} \left\| \sum_{j=1}^k (\delta_{\hat{X}_j} - \mathbb{P}_n) \right\|_{\mathcal{F}} \leq 2 \mathbb{E} \left\| \sum_{j=1}^k \varepsilon_j \delta_{\hat{X}_j} \right\|_{\mathcal{F}} \leq 4 \mathbb{E} \left\| \sum_{j=1}^N \varepsilon_j \delta_{\hat{X}_j} \right\|_{\mathcal{F}},$$

where N is Poisson(k)-distributed, independent of the i.i.d. sequence $\hat{X}_1, \hat{X}_2, \dots$ and the Rademacher variables ε_j . Set

$$N_i = \#(j \leq N: \hat{X}_j = x_i, \varepsilon_j = 1); \quad N'_i = \#(j \leq N: \hat{X}_j = x_i, \varepsilon_j = -1).$$

Given $N = m$, the vector $(N_1, N'_1, \dots, N_n, N'_n)$ is multinomially distributed with parameters $(m, 1/2n, \dots, 1/2n)$. Hence, unconditionally the coordinates of this vector are i.i.d. Poisson-distributed with mean $k/2n$. The lemma follows, since $\sum_{j=1}^N \varepsilon_j \delta_{\hat{X}_j} = \sum_{i=1}^n (N_i - N'_i) \delta_{x_i}$. ■

3.7.7 Lemma (Multiplier inequalities). For $n \in \mathbb{N}$ and arbitrary stochastic processes $Z_1, \dots, Z_n$, every exchangeable random vector $(\xi_1, \dots, \xi_n)$ that is independent of $Z_1, \dots, Z_n$, and any $1 \leq n_0 \leq n$,

$$\begin{aligned} \mathbb{E}_{\xi} \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}}^* &\leq 2(n_0 - 1) \frac{1}{n} \sum_{i=1}^n \|Z_i\|_{\mathcal{F}}^* \mathbb{E}_{\xi} \max_{1 \leq i \leq n} \frac{|\xi_i|}{\sqrt{n}} \\ &\quad + 2 \|\xi_1\|_{2,1} \max_{n_0 \leq k \leq n} \mathbb{E}_R \left\| \frac{1}{\sqrt{k}} \sum_{i=n_0}^k Z_{R_i} \right\|_{\mathcal{F}}^*. \end{aligned}$$

Here $(R_1, \dots, R_n)$ is uniformly distributed on the set of all permutations of $\{1, 2, \dots, n\}$ and independent of $Z_1, \dots, Z_n$ and the asterisks denote measurable covers with respect to all variables jointly. Furthermore,

$$\begin{aligned} \mathbb{E} \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i Z_i \right\|_{\mathcal{F}}^* &\leq 2(n_0 - 1) \frac{1}{n} \sum_{i=1}^n \mathbb{E} \|Z_i\|_{\mathcal{F}}^* \mathbb{E}_{\xi} \max_{1 \leq i \leq n} \frac{|\xi_i|}{\sqrt{n}} \\ &\quad + 2 \|\xi_1\|_{2,1} \max_{n_0 \leq k \leq n} \mathbb{E} \left\| \frac{1}{\sqrt{k}} \sum_{i=n_0}^k Z_{R_i} \right\|_{\mathcal{F}}^*. \end{aligned}$$

For nonnegative variables ξ_i , the two constants 2 on the right can be deleted.

Proof. Since the ξ_i can be split into their positive and negative parts, it suffices to consider the case that they are nonnegative.

Choose the vector $R = (R_1, \dots, R_n)$ independent of $\xi_1, \dots, \xi_n$. By exchangeability of $\xi_1, \dots, \xi_n$, the left side times $\sqrt{n}$ equals

$$E_{\xi} \left\| \sum_{i=1}^n \xi_{R_i} Z_i \right\|_{\mathcal{F}}^* = E_{\xi, R} \left\| \sum_{i=1}^n \xi_i Z_{R_i} \right\|_{\mathcal{F}}^* = E_{\xi, R} \left\| \sum_{i=1}^n \xi_{(i)} Z_{R_{S_i}} \right\|_{\mathcal{F}}^*,$$

where $\xi_{(1)} \geq \dots \geq \xi_{(n)}$ are the reverse order statistics of $\xi_1, \dots, \xi_n$ and $S = (S_1, \dots, S_n)$ is a (random) permutation such that $\xi_{(i)} = \xi_{S_i}$. To determine S uniquely also in the presence of ties, we can require that $S_i < S_{i+1}$ whenever $\xi_{S_i} = \xi_{S_{i+1}}$. Then the vector $R \circ S$ (with i th component R_{S_i}) is distributed as R and is independent of S and $\xi_1, \dots, \xi_n$. By the triangle inequality, the right-hand side of the last display is bounded by

$$E_{\xi, R} \left\| \sum_{i=1}^{n_0-1} \xi_{(i)} Z_{R_{S_i}} \right\|_{\mathcal{F}}^* + E_{\xi, R} \left\| \sum_{j=n_0}^n \xi_{(j)} Z_{R_{S_j}} \right\|_{\mathcal{F}}^*.$$

The first term is bounded by

$$E_{\xi, R} \max_{1 \leq i \leq n} \xi_i \sum_{i=1}^{n_0-1} \|Z_{R_{S_i}}\|_{\mathcal{F}}^* \leq E_{\xi} \max_{1 \leq i \leq n} \xi_i \frac{n_0-1}{n} \sum_{i=1}^n \|Z_i\|_{\mathcal{F}}^*.$$

The second term can be bounded by the same method as in the proof of Lemma 2.9.1.

The second inequality is proved in the same manner. ■

3.7.2 Exchangeable Bootstrap

For each n , let $(W_{n1}, \dots, W_{nn})$ be an exchangeable, nonnegative random vector. Consider the *weighted bootstrap empirical measure*

$$\hat{\mathbb{P}}_n = \frac{1}{n} \sum_{i=1}^n W_{ni} \delta_{X_i}.$$

The corresponding weighted bootstrap empirical process is defined as

$$\hat{\mathbb{G}}_n = \sqrt{n}(\hat{\mathbb{P}}_n - \bar{W}_n \mathbb{P}_n) = \frac{1}{\sqrt{n}} \sum_{i=1}^n (W_{ni} - \bar{W}_n) \delta_{X_i} = \frac{1}{\sqrt{n}} \sum_{i=1}^n W_{ni} (\delta_{X_i} - \mathbb{P}_n).$$

This corresponds to resampling W_{ni} times the variable X_i . However, it is not required that the weights W_{ni} be integer-valued. In the case that in total n variables are resampled, the average $\bar{W}_n$ is equal to one and the weighted bootstrap process is simply $\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n)$. It is assumed that

$$\begin{aligned} \sup_n \|W_{n1} - \bar{W}_n\|_{2,1} &< \infty, \\ n^{-1/2} E \max_{1 \leq i \leq n} |W_{ni} - \bar{W}_n| &\rightarrow 0, \\ n^{-1} \sum_{i=1}^n (W_{ni} - \bar{W}_n)^2 &\xrightarrow{P} c^2 > 0. \end{aligned} \tag{3.7.8}$$

Sufficient for the second condition is that the variables $W_{ni} - \bar{W}_n$ are uniformly weak- L_2 (Problem 2.3.3). Both the first and second conditions are valid if $\|W_{n1}\|_{2+\varepsilon}$ is uniformly bounded for some $\varepsilon > 0$.

The multinomial weights considered in the preceding section satisfy these conditions. The following theorem shows that the conditions (3.7.8) alone imply the conditional weak convergence of the bootstrap process. This gives a large number of different ways to obtain bootstrap estimators for the distribution of the empirical process.

3.7.9 Example (Renormalized i.i.d.). If $Y_1, \dots, Y_n$ are i.i.d. positive random variables with $\|Y_1\|_{2,1} < \infty$, then the weights $W_{ni} = Y_i/\bar{Y}_n$ satisfy (3.7.8) with $c = \sigma(Y_1)/EY_1$.

If the variables Y_i are exponentially distributed with mean 1, then the weight vector $(W_{n1}, \dots, W_{nn})$ follows the n -dimensional Dirichlet distribution with parameter vector $(1/n, \dots, 1/n)$. The resulting scheme is known as the *Bayesian bootstrap*, as it is the weak limit of the posterior distributions given a sequence of Dirichlet priors on the law P of the observations $X_1, \dots, X_n$ with base measures tending to zero.[#]

3.7.10 Example (Multinomial). Multinomial vectors $(W_{n1}, \dots, W_{nn})$ with parameters n and (probabilities) $(1/n, \dots, 1/n)$ satisfy (3.7.8) with $c = 1$.

3.7.11 Example (Multinomial subsampling). The vectors $(W_{n1}, \dots, W_{nn})$ equal to $\sqrt{n/k}$ times multinomial vectors with parameters k and (success probabilities) $(1/n, \dots, 1/n)$ satisfy (3.7.8) with $c = 1$ provided $k \rightarrow \infty$. Thus, the theorem in this section contains the consistency of the empirical bootstrap based on k replicates.

3.7.12 Example (Wild bootstrap). If $W_{n1}, \dots, W_{nn}$ are i.i.d. vectors with finite $L_{2,1}$ -norm, then the conditions (3.7.8) are satisfied with $c^2 = \text{var } W_{1,1}$. In this case the total mass $\bar{W}_n$ of $\mathbb{P}_n$ is a random variable. This choice of weights $\{W_{ni}\}$ has been called the *wild bootstrap* by some authors.

3.7.13 Theorem (Bootstrap exchangeable). Let $\mathcal{F}$ be a Donsker class of measurable functions such that $\mathcal{F}_\delta$ is measurable for every $\delta > 0$. For each n let $(W_{n1}, \dots, W_{nn})$ be an exchangeable, nonnegative random vector independent of $X_1, X_2, \dots$ such that the conditions (3.7.8) are satisfied. Then as $n \rightarrow \infty$,

$$\sup_{h \in BL_1} |E_W h(\hat{\mathbb{G}}_n) - E h(c \mathbb{G})| \xrightarrow{P^*} 0.$$

Furthermore, the sequence $E_W h(\hat{\mathbb{G}}_n)^* - E_W h(\hat{\mathbb{G}}_n)_*$ converges to zero in outer probability. If $P^* \|f - Pf\|_{\mathcal{F}}^2 < \infty$, then the convergence is also outer almost surely.

[#] See e.g. Section 4.7 in Ghosal and Van der Vaart (2017).

Proof. Without loss of generality, assume that $\bar{W}_n = 0$ and that $Pf = 0$ for every n and f . As before, it suffices to prove the conditional almost-sure weak convergence of every marginal plus the conditional asymptotic equicontinuity in probability or almost surely.

The variables W_{ni} satisfy the conditions of Lemma 3.7.15 (ahead). Thus this lemma with $a_{ni} = f(X_i) - \mathbb{P}_n f$ implies that conditionally on the sequence $X_1, X_2, \dots$ the sequence $\hat{\mathbb{G}}_n f$ is asymptotically normal $N(0, c^2 P f^2)$ given every sequence $X_1, X_2, \dots$ such that, as $n \rightarrow \infty$ followed by $M \rightarrow \infty$,

$$\mathbb{P}_n(f - \mathbb{P}_n f)^2 \rightarrow P f^2 \quad \text{and} \quad \mathbb{P}_n(f - \mathbb{P}_n f)^2 \{ |f - \mathbb{P}_n f| > M \} \rightarrow 0.$$

Both are valid almost surely, since $P f^2 < \infty$. Combined with the Cramér-Wold device, this establishes finite-dimensional convergence.

Let $\mathcal{F}_\delta$ be the set of differences $f - g$ of elements $f, g \in \mathcal{F}$ with $P(f - g)^2 < \delta^2$, as usual. By the multiplier inequalities given by Lemma 3.7.7 (applied with $Z_i = \delta_{X_i} - \mathbb{P}_n$ fixed), the conditional expectation $E_W \|\hat{\mathbb{G}}_n\|_{\mathcal{F}_\delta}$ equals

$$\begin{aligned} E_W \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n W_{ni} (\delta_{X_i} - \mathbb{P}_n) \right\|_{\mathcal{F}_\delta}^* &\lesssim \frac{n_0 - 1}{n} \sum_{i=1}^n \|\delta_{X_i} - \mathbb{P}_n\|_{\mathcal{F}_\delta}^* E \max_{1 \leq i \leq n} \frac{|W_{ni}|}{\sqrt{n}} \\ &\quad + \|W_{n1}\|_{2,1} \max_{n_0 \leq k \leq n} E_R \left\| \frac{1}{\sqrt{k}} \sum_{i=n_0}^k (\delta_{X_{R_i}} - \mathbb{P}_n) \right\|_{\mathcal{F}_\delta}^*. \end{aligned}$$

The first term on the right converges to zero outer almost surely for every fixed n_0 . In the second term the vector $(X_{R_1}, \dots, X_{R_n})$ is a random sample without replacement from $X_1, \dots, X_n$. By Hoeffding's inequality (Lemma A.1.10), this term increases if the vector is replaced by a sample with replacement $\hat{X}_1, \dots, \hat{X}_n$. Next, the sum can be extended to the range from 1 to k by Jensen's inequality. Conclude that the second term is bounded by

$$\sup_n \|W_{n1}\|_{2,1} \max_{n_0 \leq k \leq n} E_{\hat{X}} \|\hat{\mathbb{G}}_{n,k}\|_{\mathcal{F}_\delta}^*.$$

Here $\hat{\mathbb{G}}_{n,k} = k^{-1/2} \sum_{i=1}^k (\delta_{\hat{X}_i} - \mathbb{P}_n)$ is the multinomial bootstrap process encountered in the previous section. By (the proof of) Theorem 3.7.3, this expression converges to zero in outer probability as $n_0, n \rightarrow \infty$ followed by $\delta \downarrow 0$. Under the additional condition on the envelope function, the convergence is outer almost surely. ■

3.7.14 Example (Bootstrap without replacement). The bootstrap without replacement is based on resampling $k < n$ observations from $X_1, \dots, X_n$ without replacement. This can be incorporated in the scheme of the theorem by letting $(W_{n1}, \dots, W_{nn})$ be a row of k times the number $n(n-k)^{-1/2} k^{-1/2}$ and $n-k$ times the number 0, ordered at random, independent of the X s. Then the conditions (3.7.8) on the weights are satisfied for $c = 1$, provided both $k \rightarrow \infty$ and $n-k \rightarrow \infty$. (For this choice the sample standard deviation is even identically equal to $c = 1$.)

In this case the assertion of the theorem can be phrased in terms of the empirical measures

$$\tilde{\mathbb{P}}_{k,n} = \frac{1}{k} \sum_{i=1}^k \delta_{X_{R_{ni}}}; \quad \tilde{\mathbb{Q}}_{n-k,n} = \frac{1}{n-k} \sum_{i=k+1}^n \delta_{X_{R_{ni}}},$$

where $(R_{n1}, \dots, R_{nn})$ is a random permutation of the numbers $1, 2, \dots, n$. If both $k \rightarrow \infty$ and $n - k \rightarrow \infty$, then the sequence

$$\sqrt{\frac{nk}{n-k}}(\tilde{\mathbb{P}}_{k,n} - \mathbb{P}_n) = \sqrt{\frac{k(n-k)}{n}}(\tilde{\mathbb{P}}_{k,n} - \tilde{\mathbb{Q}}_{n-k,n})$$

converges conditionally in distribution to a tight Brownian bridge.

3.7.15 Lemma. For each n , let $(a_{n1}, \dots, a_{nn})$ and $(W_{n1}, \dots, W_{nn})$ be a vector of numbers and an exchangeable random vector such that

$$\begin{aligned} \bar{a}_n &= 0; \quad \frac{1}{n} \sum_{i=1}^n a_{ni}^2 \rightarrow \sigma^2 > 0; \quad \lim_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n a_{ni}^2 \{ |a_{ni}| > M \} = 0; \\ \bar{W}_n &= 0; \quad \frac{1}{n} \sum_{i=1}^n W_{ni}^2 \xrightarrow{P} \tau^2 > 0; \quad \frac{1}{n} \max_{1 \leq i \leq n} W_{ni}^2 \xrightarrow{P} 0. \end{aligned}$$

Then the sequence $n^{-1/2} \sum_{i=1}^n a_{ni} W_{ni}$ converges weakly to a mean zero normal distribution with variance $\sigma^2 \tau^2$.

Proof. Without loss of generality, assume that both $n^{-1} \sum a_{ni}^2$ and $n^{-1} \sum_{i=1}^n W_{ni}^2$ are equal to 1 for every n . Then

$$\sum_i \sum_j \frac{a_{ni}^2 W_{nj}^2}{n n} \{ |a_{ni} W_{nj}| > \varepsilon \sqrt{n} \} \leq \frac{1}{n} \sum_i a_{ni}^2 \{ |a_{ni}| \max_j |W_{nj}| > \varepsilon \sqrt{n} \}.$$

Since $\max_j |W_{nj}| / \sqrt{n}$, converges to zero in probability and the array a_{ni} is “uniformly integrable”, this expression converges to zero in probability as $n \rightarrow \infty$, for every $\varepsilon > 0$.

Combination with the assumptions shows that every subsequence of $\{n\}$ has a further subsequence along which, for every $\varepsilon > 0$,

$$\max_{1 \leq j \leq n} \frac{|W_{nj}|}{\sqrt{n}} \xrightarrow{\text{as}} 0; \quad \sum_i \sum_j \frac{a_{ni}^2 W_{nj}^2}{n n} \{ |a_{ni} W_{nj}| > \varepsilon \sqrt{n} \} \xrightarrow{\text{as}} 0.$$

This means that for almost every realization of the W_{nj} , the conditions of Hájek’s rank central limit theorem (Proposition A.5.3) are satisfied along the subsequence. Hence conditionally on the W_{nj} , the subsequence of rank statistics $n^{-1/2} \sum_{i=1}^n a_{ni} W_{n, R_{ni}}$ is asymptotically standard normally distributed.

This argument implies that every subsequence of $\{n\}$ has a further subsequence along which

$$\sup_{h \in BL_1} \left| E_R h \left(\frac{1}{\sqrt{n}} \sum a_{ni} W_{n, R_{ni}} \right) - E h(\xi) \right| \rightarrow 0,$$

almost surely, for a standard normal variable ξ . Conclude that this bounded Lipschitz distance converges to zero in probability along the whole sequence $\{n\}$. Take the expectation with respect to the W_{ni} to see that the sequence $n^{-1/2} \sum a_{ni} W_{n, R_{ni}}$ is unconditionally asymptotically normal as well. By the exchangeability of the W_{nj} , this sequence is equal in distribution to the sequence $n^{-1/2} \sum a_{ni} W_{ni}$. ■

The following lemma can be called a “multiplier Glivenko-Cantelli theorem” with exchangeable multipliers. A special case of the lemma is used in the proof of Theorem 3.7.2.

3.7.16 Lemma. *Let $\mathcal{F}$ be a Glivenko-Cantelli class of measurable functions. For each n , let $(W_{n1}, \dots, W_{nn})$ be an exchangeable nonnegative random vector independent of $X_1, X_2, \dots$ such that $\sum_{i=1}^n W_{ni} = 1$ and $\max_{1 \leq i \leq n} |W_{ni}|$ converges to zero in probability. Then, for every $\varepsilon > 0$, as $n \rightarrow \infty$,*

$$P_W \left(\left\| \sum_{i=1}^n W_{ni} (\delta_{X_i} - P) \right\|_{\mathcal{F}}^* > \varepsilon \right) \xrightarrow{\text{as}} 0.$$

Proof. By a modification of the multiplier inequality given by Lemma 3.7.7 (the L_r form with $r < 1$ with Glivenko-Cantelli normalization instead of the L_1 form with Donsker normalization in the bound of the second term), with $Z_i = \delta_{X_i} - P$,

$$\begin{aligned} E_W \left\| \sum_{i=1}^n W_{ni} Z_i \right\|_{\mathcal{F}}^{*r} &\leq (n_0 - 1) E \max_{1 \leq i \leq n} W_{ni}^r \frac{1}{n} \sum_{j=1}^n \|Z_j\|_{\mathcal{F}}^{*r} \\ &\quad + (n E W_{n1})^r \max_{n_0 \leq k \leq n} E_R \left\| \frac{1}{k} \sum_{j=n_0}^k Z_{R_j} \right\|_{\mathcal{F}}^{*r}. \end{aligned}$$

In the first term on the right, the average $n^{-1} \sum_{i=1}^n \|Z_i\|_{\mathcal{F}}^{*r}$ is bounded by $\mathbb{P}_n F^{*r} + P^* F^r$, which converges almost surely to $2P^* F^r$. Since the variables W_{ni} take their values in $[0, 1]$, the sequence $\max_{1 \leq i \leq n} W_{ni}^r$ converges to zero in mean by the dominated convergence theorem. Thus, the first term on the right converges almost surely to zero for every fixed n_0 .

The factor $n E W_{n1}$ in the second term equals 1 by exchangeability and the fact that the sum of the W_{ni} is 1. By the triangle inequality, the second term is bounded by

$$\begin{aligned} 2 \max_{n_0-1 \leq k \leq n} E_R \left\| \frac{1}{k} \sum_{j=1}^k Z_{R_j} \right\|_{\mathcal{F}}^{*r} &= 2 \max_{n_0-1 \leq k \leq n} E(U_k^r | \mathcal{S}_n) \\ &\leq 2 E \left(\max_{n_0-1 \leq k} U_k^r | \mathcal{S}_n \right), \end{aligned}$$

where $U_k = \|k^{-1} \sum_{j=1}^k Z_j\|_{\mathcal{F}}^*$, and $\mathcal{S}_n$ is the σ -field generated by all functions $f: \mathcal{X}^\infty \rightarrow \mathbb{R}$ that are symmetric in their first n coordinates. Since $\mathcal{F}$ is Glivenko-Cantelli, we have $E \max_k U_k^r < \infty$ by Problem 2.3.6 (also see Corollary A.1.8) and the sequence U_k converges almost surely to zero. The sequence $\mathcal{S}_n$ decreases to the symmetric σ -field $\mathcal{S}$, which consists of sets of probability 0 or 1 by the Hewitt-Savage zero-one law. It follows that as $n \rightarrow \infty$

$$E \left(\max_{n_0-1 \leq j} U_j^r | \mathcal{S}_n \right) \xrightarrow{\text{as}} E \left(\max_{n_0-1 \leq j} U_j^r | \mathcal{S} \right) = E \max_{n_0-1 \leq j} U_j^r.$$

The right side converges to zero as $n_0 \rightarrow \infty$. Conclude that the left side converges to zero almost surely when $n \rightarrow \infty$ followed by $n_0 \rightarrow \infty$.

Apply Markov's inequality to complete the proof. ■

Problems and Complements

1. **(Formally defining bootstrap samples)** Let $m_n^{(1)}, m_n^{(2)}, \dots$ be i.i.d. n -dimensional multinomial $(1, 1/n, \dots, 1/n)$ variables defined on some probability space $(\mathcal{Z}, \mathcal{C}, Q)$. Let $X_1, X_2, \dots$ be the coordinate projections of $(\mathcal{X}^\infty, \mathcal{A}^\infty, P^\infty)$. Then bootstrap values can be defined on $\mathcal{X}^\infty \times \mathcal{Z}$ by $\hat{X}_i = X_j$ if $m_n^{(i)}$ equals the j th unit vector in $\mathbb{R}^n$. Then

$$P^\infty \times Q((\hat{X}_1, \dots, \hat{X}_n) = (X_{j_1}, \dots, X_{j_n}) | X_1, \dots, X_n) = \left(\frac{1}{n}\right)^n$$

for every vector $j \in \{1, 2, \dots, n\}^n$, and taking conditional expectation given $X_1, \dots, X_n$ is the same as taking expectation with respect to $m_n^{(1)}, m_n^{(2)}, \dots$ only.

2. For each n , let $Z_{n1}, \dots, Z_{nn}$ be i.i.d. stochastic processes. If $\left\|n^{-1/2} \sum_{i=1}^n (Z_{ni} - EZ_{ni})\right\|_{\mathcal{F}}^* = O_P(1)$ and $P^*(\|Z_{n1}\|_{\mathcal{F}} > \sqrt{n}) \rightarrow 0$ as $n \rightarrow \infty$, then $\limsup_{n \rightarrow \infty} n P^*(\|Z_{n1}\|_{\mathcal{F}} > t\sqrt{n})$ converges to zero as $t \rightarrow \infty$.

[Hint: Let $\tilde{Z}_{n1}, \dots, \tilde{Z}_{nn}$ be an i.i.d. copy of $Z_{n1}, \dots, Z_{nn}$. Fix $0 < \varepsilon < 1/4$. For any sufficiently large t ,

$$\limsup P^*\left(\left\|\sum_{i=1}^n (Z_{ni} - \tilde{Z}_{ni})\right\|_{\mathcal{F}} > t\sqrt{n}\right) < \varepsilon.$$

By a Lévy inequality followed by Problem 2.3.2, there exists an N_t such that, for all $n \geq N_t$,

$$n P^*(\|Z_{n1} - \tilde{Z}_{n1}\|_{\mathcal{F}}^* > t\sqrt{n}) < 4\varepsilon.$$

Independent random processes Z and $\tilde{Z}$ satisfy for every t the inequality $P^*(\|Z - \tilde{Z}\|_{\mathcal{F}} > t) \geq P^*(\|Z\|_{\mathcal{F}} > 2t)P^*(\|\tilde{Z}\|_{\mathcal{F}} \leq t)$.

3. A symmetrized Poisson variable $\tilde{N}$ with parameter λ satisfies $\|\tilde{N}/\sqrt{\lambda}\|_{2,1} \leq 4$.

[Hint: $E\tilde{N}^4 \leq 12\lambda^2 + 2\lambda$.]

4. The maximum of i.i.d. symmetrized Poisson variables $\tilde{N}_1, \dots, \tilde{N}_n$ with parameter λ satisfies $E \max_{1 \leq i \leq n} |\tilde{N}_i| \leq n^{1/4} (12\lambda^2 + 2\lambda)^{1/4}$.

[Hint: This bound is based on Lemma 2.2.2 with the L_4 -norm and can be greatly improved.]

5. If $\tilde{N}_\lambda$ is a symmetrized Poisson variable with parameter λ , then the family of variables $\{\tilde{N}_\lambda/\sqrt{\lambda}: \lambda > 0\}$ is not uniformly square integrable (even though the family is uniformly bounded in the $L_{2,1}$ -norm).
6. It appears that for the consistency of the bootstrap without replacement with $n - k = o(n)$, the fact that $\mathcal{F}$ is Donsker is not used as much in the proof, as one would expect.

3.8

Two-Sample Problem

Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independent random samples from distributions P and Q on a measurable space $(\mathcal{X}, \mathcal{A})$. We wish to test the null hypothesis $H_0: P = Q$ versus the alternative $H_1: P \neq Q$.

One possible test statistic is the *two-sample Kolmogorov-Smirnov*. Given a class $\mathcal{F}$ of measurable functions $f: \mathcal{X} \rightarrow \mathbb{R}$, set

$$D_{m,n} = \sqrt{\frac{mn}{m+n}} \|\mathbb{P}_m - \mathbb{Q}_n\|_{\mathcal{F}},$$

where $\mathbb{P}_m$ and $\mathbb{Q}_n$ are the empirical measures of the X s and Y s, respectively. For real-valued observations and $\mathcal{F}$ equal to the set of indicators of cells $(-\infty, t]$, this reduces to the classical Kolmogorov-Smirnov statistic, which is the supremum distance between the two empirical distribution functions. If the underlying distributions P and Q are a priori assumed to be atomless, then the classical Kolmogorov-Smirnov statistic is distribution-free under the null hypothesis and both its finite sample and asymptotic null distribution are well known.

In contrast, for more general sample spaces and indexing classes $\mathcal{F}$, even the asymptotic null distribution of the “sequence” $D_{m,n}$ may depend on the common underlying measure $P = Q$. In this chapter we discuss two ways of setting critical values for the test and show that the test is asymptotically consistent against every alternative (P, Q) with $\|P - Q\|_{\mathcal{F}} > 0$, if $\mathcal{F}$ is a Donsker class with respect to both P and Q .

If $\mathcal{F}$ is a Donsker class under both P and Q , then the sequences $\mathbb{G}_m = \sqrt{m}(\mathbb{P}_m - P)$ and $\mathbb{G}'_n = \sqrt{n}(\mathbb{Q}_n - Q)$ converge jointly in distribution to independent Brownian bridges $\mathbb{G}_P$ and $\mathbb{G}_Q$. Set $N = m + n$ and note that

$$D_{m,n} = \left\| \sqrt{\frac{n}{N}} \mathbb{G}_m - \sqrt{\frac{m}{N}} \mathbb{G}'_n + \sqrt{\frac{mn}{N}} (P - Q) \right\|_{\mathcal{F}}.$$

If $\|P - Q\|_{\mathcal{F}} > 0$, then $D_{m,n} \rightarrow \infty$ in probability as $m, n \rightarrow \infty$. On the other hand, under the null hypothesis $P = Q$ the sequence $D_{m,n}$ converges in distribution to the random variable $\|\sqrt{1 - \lambda} \mathbb{G}_P - \sqrt{\lambda} \mathbb{G}_Q\|_{\mathcal{F}}$ if $m, n \rightarrow \infty$ such that $m/N \rightarrow \lambda$. For $P = Q$, the limit variable $\sqrt{1 - \lambda} \mathbb{G}_P - \sqrt{\lambda} \mathbb{G}_Q$ possesses the same distribution as $\mathbb{G}_P$. It follows that the test that rejects the null hypothesis if $D_{m,n} > c_{m,n}$ has asymptotic level α if the critical values $c_{m,n}$ can be chosen such that

$$c_{m,n} \rightarrow c_P = \inf \left\{ t: \mathbb{P}(\|\mathbb{G}_P\|_{\mathcal{F}} > t) \leq \alpha \right\}.$$

Since $c_P < \infty$, the test is then consistent in the sense that $\mathbb{P}(D_{m,n} > c_{m,n}) \rightarrow 1$ against any alternative with $\|P - Q\|_{\mathcal{F}} > 0$.

In general, the upper α -quantile, c_P , of the limit distribution depends on the underlying measure, P . Then it is impossible to find numbers $c_{m,n}$ with the given convergence property for every P in the null hypothesis. Instead we use data-dependent “critical points” $\tilde{c}_{m,n} = c_{m,n}(X_1, \dots, X_m, Y_1, \dots, Y_n)$, and properly speaking the test statistic is $D_{m,n} - \tilde{c}_{m,n}$ rather than $D_{m,n}$. Similar reasoning as before shows that the sequence of tests that reject the null hypothesis if $D_{m,n} > \tilde{c}_{m,n}$ is asymptotically consistent and of level α if the sequence $\tilde{c}_{m,n}$ behaves well in probability. More precisely, the asymptotic level is α in the sense that $\limsup \mathbb{P}(D_{m,n} > \tilde{c}_{m,n}) \leq \alpha$ for $P = Q$ if[†]

$$\tilde{c}_{m,n} \xrightarrow{P} c_P.$$

Furthermore, the sequence of tests is consistent against any alternative (P, Q) for which $\tilde{c}_{m,n}$ is bounded in probability and $\|P - Q\|_{\mathcal{F}} > 0$.

“Critical values” $\tilde{c}_{m,n}$ with these properties can be determined by a permutation or a bootstrap approach. These procedures amount to sampling without and with replacement, respectively, from the pooled data $(Z_{N1}, \dots, Z_{NN}) = (X_1, \dots, X_m, Y_1, \dots, Y_n)$. Let $\lambda_N = m/N$, and consider the *pooled empirical measure*

$$\mathbb{H}_N = \frac{1}{N} \sum_{i=1}^N \delta_{Z_{Ni}} = \lambda_N \mathbb{P}_m + (1 - \lambda_N) \mathbb{Q}_n.$$

Since $\mathbb{P}_m - \mathbb{H}_N = (1 - \lambda_N)(\mathbb{P}_m - \mathbb{Q}_n)$, the Kolmogorov-Smirnov test is equivalent to a test based on the discrepancy between $\mathbb{P}_m$ and the pooled empirical measure.

3.8.1 Permutation Empirical Processes

Sampling without replacement from the pooled empirical measure $\mathbb{H}_N$ can be represented in terms of a random permutation. Let the random vector $R = (R_1, \dots, R_N)$ be uniformly distributed on the set of all permutations of $\{1, 2, \dots, N\}$ and be independent from $X_1, \dots, X_m, Y_1, \dots, Y_n$. (As usual, when outer expectations are

[†] Since H_0 is composite, it is perhaps more proper to call the sequence of tests asymptotically of level α if $\sup_{P=Q} \mathbb{P}(D_{m,n} > \tilde{c}_{m,n})$ is asymptotically bounded by α . We do not consider this stronger concept. In view of the results of Chapter 2.8 on uniform in P Donsker classes, the present test is asymptotically of level α also in this stronger sense for many Donsker classes $\mathcal{F}$.

involved, “independence” is understood in terms of a product probability space.) The *two-sample permutation empirical measures* are

$$\tilde{\mathbb{P}}_{m,N} = \frac{1}{m} \sum_{i=1}^m \delta_{Z_{NR_i}}; \quad \tilde{\mathbb{Q}}_{n,N} = \frac{1}{n} \sum_{i=m+1}^N \delta_{Z_{NR_i}}.$$

The *permutation empirical process* corresponding to the first sample is $\sqrt{m}(\tilde{\mathbb{P}}_{m,N} - \mathbb{H}_N)$. The following theorems establish the almost-sure and in-probability limiting behavior of this process.

3.8.1 Theorem. *Let $\mathcal{F}$ be a class of measurable functions that is Donsker under both P and Q and satisfies both $\|P\|_{\mathcal{F}} < \infty$ and $\|Q\|_{\mathcal{F}} < \infty$. If $m, n \rightarrow \infty$ such that $m/N \rightarrow \lambda \in (0, 1)$, then $\sqrt{m}(\tilde{\mathbb{P}}_{m,N} - \mathbb{H}_N) \rightsquigarrow \sqrt{1-\lambda} \mathbb{G}_H$ given $X_1, X_2, \dots, Y_1, Y_2, \dots$ in probability. Here $\mathbb{G}_H$ is a tight Brownian bridge process corresponding to the measure $H = \lambda P + (1-\lambda)Q$.*

3.8.2 Theorem. *If in addition $\mathcal{F}$ possesses an envelope function F with both $P^*F^2 < \infty$ and $Q^*F^2 < \infty$, then also $\sqrt{m}(\tilde{\mathbb{P}}_{m,N} - \mathbb{H}_N) \rightsquigarrow \sqrt{1-\lambda} \mathbb{G}_H$ given almost every sequence $X_1, X_2, \dots, Y_1, Y_2, \dots$.*

Proof. Without loss of generality, assume that $Pf = 0$ for every f . Otherwise, replace f by $f^o = f - Pf$ for every f , in which case we still have $\|Q\|_{\mathcal{F}^o} < \infty$. Given fixed values of the original pooled sample, let $\hat{Z}_{N1}, \dots, \hat{Z}_{NN}$ be an i.i.d. sample from $\mathbb{H}_N$.

For marginal convergence it suffices by the Cramér-Wold device to show that, for every f with $Hf = 0$ and $Hf^2 < \infty$,

$$\frac{1}{\sqrt{m}} \sum_{i=1}^m (f(Z_{NR_i}) - \mathbb{H}_N f) \rightsquigarrow N(0, (1-\lambda) Hf^2),$$

given almost every sequence $X_1, X_2, \dots, Y_1, Y_2, \dots$. This follows from the central limit theorem for two-sample rank statistics, Proposition A.5.3, applied with $a_{N,i} = f(Z_{Ni})$ and $b_{N,i}$ equal to 1 or 0 if $i \leq m$ or $i > m$.

Set $\mathcal{F}_\delta = \{f - g, f, g \in \mathcal{F}, H(f - g)^2 < \delta^2\}$. The assumptions imply that $\mathcal{F}$ is totally bounded in both $L_2(P)$ and $L_2(Q)$. Then $\mathcal{F}$ is also totally bounded in $L_2(H)$. By Hoeffding’s inequality, Proposition A.1.10,

$$\mathbb{E}_R \left\| \frac{1}{\sqrt{m}} \sum_{i=1}^m (\delta_{Z_{NR_i}} - \mathbb{H}_N) \right\|_{\mathcal{F}_\delta} \leq \mathbb{E}_{\hat{Z}} \left\| \frac{1}{\sqrt{m}} \sum_{i=1}^m (\delta_{\hat{Z}_{Ni}} - \mathbb{H}_N) \right\|_{\mathcal{F}_\delta}.$$

Let $\tilde{N}_1, \dots, \tilde{N}_N$ be i.i.d. symmetrized Poisson($m/2N$) variables. By the Poissonization inequality, Lemma 3.7.6, the right side is bounded by 4 times

$$\mathbb{E}_{\tilde{N}} \left\| \frac{1}{\sqrt{m}} \sum_{i=1}^N \tilde{N}_i \delta_{Z_{Ni}} \right\|_{\mathcal{F}_\delta} \leq \mathbb{E}_{\tilde{N}} \left\| \frac{1}{\sqrt{m}} \sum_{i=1}^m \tilde{N}_i \delta_{X_i} \right\|_{\mathcal{F}_\delta} + \mathbb{E}_{\tilde{N}} \left\| \frac{1}{\sqrt{m}} \sum_{i=1}^n \tilde{N}_i \delta_{Y_i} \right\|_{\mathcal{F}_\delta}.$$

By the closedness of the symmetrized Poisson family under convolution and Jensen’s inequality, the parameter of the Poisson variables can be increased to $1/2$.

According to the unconditional multiplier theorem, Theorem 2.9.2, the sequences of processes $m^{-1/2} \sum_{i=1}^m \tilde{N}_i \delta_{X_i}$ and $n^{-1/2} \sum_{i=1}^n \tilde{N}_i \delta_{Y_i}$ converge in distribution to tight Gaussian processes Z_P and Z_Q , respectively.

By Lemma 2.3.11, the outer expectation with respect to the original observations $X_1, X_2, \dots, Y_1, Y_2, \dots$ of the right side of the last display is asymptotically bounded by a multiple of the expression $E\|Z_P\|_{\mathcal{F}_\delta} + \sqrt{(1-\lambda)/\lambda} E\|Z_Q\|_{\mathcal{F}_\delta}$ as $m, n \rightarrow \infty$. This converges to 0 as $\delta \downarrow 0$ since both processes have uniformly continuous sample paths with respect to the $L_2(H)$ -semimetric. This concludes the proof of the first theorem.

If $\mathcal{F}$ possesses an envelope function that is square integrable under both P and Q , then the limsup as $m, n \rightarrow \infty$ of the last display is bounded by a constant times $E\|Z_P\|_{\mathcal{F}_\delta} + \sqrt{(1-\lambda)/\lambda} E\|Z_Q\|_{\mathcal{F}_\delta}$, for almost every $X_1, X_2, \dots, Y_1, Y_2, \dots$, by Corollary 2.9.8 combined with Lemma 2.3.11. Again, this upper bound converges to 0 as $\delta \rightarrow 0$. ■

The preceding theorem implies that the sequence

$$\tilde{D}_{m,n} = \sqrt{\frac{mn}{m+n}} \|\tilde{\mathbb{P}}_{m,N} - \tilde{\mathbb{Q}}_{n,N}\|_{\mathcal{F}} = \frac{1}{\sqrt{1-\lambda_N}} \sqrt{m} \|\tilde{\mathbb{P}}_{m,N} - \mathbb{H}_N\|_{\mathcal{F}}$$

converges in distribution to $\|\mathbb{G}_H\|_{\mathcal{F}}$ given the original observations. Consequently, the upper α -quantiles

$$(3.8.3) \quad \tilde{c}_{m,n} = \inf\left\{t: P_R(\tilde{D}_{m,n} > t) \leq \alpha\right\}$$

of the conditional distribution of $\tilde{D}_{m,n}$ can be used as “critical values” for the Kolmogorov-Smirnov test. The distribution of the norm $\mathbb{G}_H$ of a tight Brownian bridge is known to be absolutely continuous with a positive density.[‡] Thus, under the conditions of the preceding theorems,

$$\tilde{c}_{m,n} \rightarrow c_H = \inf\left\{t: P(\|\mathbb{G}_H\|_{\mathcal{F}} > t) \leq \alpha\right\}$$

in probability and almost surely, respectively.

It follows that the sequence of tests that reject the null hypothesis $H_0: P = Q$ if $D_{m,n} > \tilde{c}_{m,n}$, is asymptotically of level α for any $P = Q$ for which the index set $\mathcal{F}$ is Donsker. Furthermore, the sequence of tests is consistent against any alternative (P, Q) for which $\|P - Q\|_{\mathcal{F}} > 0$ and for which $\mathcal{F}$ is both P - and Q -Donsker. In most cases there exist simple index classes $\mathcal{F}$ that satisfy these requirements for every pair $P = Q$ and $P \neq Q$ on the sample space, respectively.

3.8.4 Example. In Euclidean space any suitably measurable, uniformly bounded VC-class of functions is universally Donsker. The collection of all indicators of cells $(-\infty, t]$ can be added to ensure that $\|P - Q\|_{\mathcal{F}} > 0$ for every $P \neq Q$. The resulting class satisfies the two requirements, and the sequence of Kolmogorov-Smirnov tests is asymptotically consistent against every possible alternative.

[‡] See Beran and Millar (1986), Proposition 2, page 442, or Corollary A.2.10 for a weaker result.

While every reasonable VC-class will thus ensure consistency against every possible alternative, the particular choice will determine the power of the resulting test. See Chapter 3.11 for a study of empirical processes under contiguous alternatives.

3.8.5 Example. If the σ -field in the sample space is countably generated, then there exists a sequence of measurable sets A_i such that $P \neq Q$ if and only if $P(A_i) \neq Q(A_i)$ for at least one i . (Replace a given countable generator by the collection of all finite intersections. This is a countable generator that is intersection-stable and hence a measure determining class.) By Theorem 2.13.1, the sequence $\{c_i 1_{A_i}\}$ is a universal Donsker class for every sequence c_i that decreases to zero sufficiently fast.

Thus, any sample space with a countably generated σ -field supports a class $\mathcal{F}$ for which the Kolmogorov-Smirnov test is asymptotically consistent against every possible alternative. (The Donsker class in the preceding paragraph proves existence but is not necessarily recommended to obtain power against “reasonable” alternatives.)

The test that rejects the null hypothesis if $D_{m,n} > \tilde{c}_{m,n}$ with $\tilde{c}_{m,n}$ determined by (3.8.3) is exactly a classical *permutation test*. There are $N!$ possible orderings of the pooled sample $X_1, \dots, X_m, Y_1, \dots, Y_n$ and, counting ties by multiplicity, equally many corresponding values of the test statistic $D_{m,n}$. Given the values in the pooled sample, each possible value of $D_{m,n}$ is equally probable under the null hypothesis (still counting ties by multiplicity). The definition (3.8.3) of $\tilde{c}_{m,n}$ is such that the null hypothesis is rejected if the observed value $D_{m,n}(x_1, \dots, x_m, y_1, \dots, y_n)$ is among the α -fraction of largest possible values of $D_{m,n}$.

It follows that given the values of the pooled sample the probability of an error of the first kind is smaller than α . This being true for every m, n and every pooled sample also yields the conclusion that the sequence of tests is asymptotically of level α . The abstract approach given previously shows in addition that the sequence of tests is consistent and that the critical values $\tilde{c}_{m,n}$ converge (in probability or almost surely) to a deterministic value. It can also be used to study the power of the sequence of tests under contiguous alternatives.

3.8.2 Two-Sample Bootstrap

Instead of sampling values without replacement from the pooled sample, we can sample with replacement. This leads to *two-sample bootstrap empirical measures*

$$\hat{\mathbb{P}}_{m,N} = \frac{1}{m} \sum_{i=1}^m \delta_{\hat{Z}_{Ni}}; \quad \hat{\mathbb{Q}}_{n,N} = \frac{1}{n} \sum_{i=1}^n \delta_{\hat{Z}_{N,m+i}},$$

where $\hat{Z}_{N1}, \dots, \hat{Z}_{NN}$ is an i.i.d. sample from the pooled empirical measure $\mathbb{H}_N$. Unlike the permutation empirical measures, the bootstrap empirical measures corresponding to the two samples are independent.

3.8.6 Theorem (Bootstrap). *Let $\mathcal{F}$ be a class of measurable functions that is Donsker under both P and Q and satisfies both $\|P\|_{\mathcal{F}} < \infty$ and $\|Q\|_{\mathcal{F}} < \infty$. If $m, n \rightarrow \infty$ such that $m/N \rightarrow \lambda \in (0, 1)$, then $\sqrt{m}(\hat{\mathbb{P}}_{m,N} - \mathbb{H}_N) \rightsquigarrow \mathbb{G}_H$ given $X_1, X_2, \dots, Y_1, Y_2, \dots$ in probability. Here $\mathbb{G}_H$ is a tight Brownian bridge process corresponding to the measure $H = \lambda P + (1 - \lambda)Q$.*

3.8.7 Theorem. *If in addition $\mathcal{F}$ possesses an envelope function F with both $P^*F^2 < \infty$ and $Q^*F^2 < \infty$, then also $\sqrt{m}(\hat{\mathbb{P}}_{m,N} - \mathbb{H}_N) \rightsquigarrow \mathbb{G}_H$ given almost every sequence $X_1, X_2, \dots, Y_1, Y_2, \dots$.*

The theorems are proved by the same arguments as used in the proofs of the permutation theorems. In fact, the proofs are simpler, because Hoeffding's inequality is no longer needed and the finite-dimensional convergence follows from the Lindeberg theorem.

The theorems combined with the analogous results for the bootstrap process corresponding to the second sample imply that the sequence

$$\begin{aligned} \hat{D}_{m,n} &= \sqrt{\frac{mn}{N}} \|\hat{\mathbb{P}}_{m,N} - \hat{\mathbb{Q}}_{n,N}\|_{\mathcal{F}} \\ &= \left\| \sqrt{m(1 - \lambda_N)} (\hat{\mathbb{P}}_{m,N} - \mathbb{H}_N) - \sqrt{n\lambda_N} (\hat{\mathbb{Q}}_{n,N} - \mathbb{H}_N) \right\|_{\mathcal{F}} \end{aligned}$$

converges in distribution to $\|\sqrt{1 - \lambda}\mathbb{G}_H - \sqrt{\lambda}\mathbb{G}'_H\|_{\mathcal{F}}$ for independent Brownian bridges $\mathbb{G}_H$ and $\mathbb{G}'_H$. The process $\sqrt{1 - \lambda}\mathbb{G}_H - \sqrt{\lambda}\mathbb{G}'_H$ is a version of an H -Brownian bridge itself. Consequently, the upper α -quantiles

$$\hat{c}_{m,n} = \inf \left\{ t: \mathbf{P}_{\hat{Z}}(\hat{D}_{m,n} > t) \leq \alpha \right\}$$

of the conditional distribution of $\hat{D}_{m,n}$ can be used as “critical values” for the Kolmogorov-Smirnov test. Under the conditions of the preceding theorems,

$$\hat{c}_{m,n} \rightarrow c_H = \inf \left\{ t: \mathbf{P}(\|\mathbb{G}_H\|_{\mathcal{F}} > t) \leq \alpha \right\},$$

in probability and almost surely, respectively. The critical values set by the permutation method in the preceding section possess exactly the same behavior. Hence at this level of analysis the two methods are of equal accuracy.

Problems and Complements

1. If $\mathcal{F}$ is P -Donsker and $\mathbb{P}_m$ and $\mathbb{Q}_n$ are the empirical measures of independent i.i.d. samples of sizes m and n from P , then $\sqrt{mn/N}(\mathbb{P}_m - \mathbb{Q}_n)$ converges in distribution to a tight Brownian bridge if $m, n \rightarrow \infty$ (also if $m/n \rightarrow 0$ or does not converge).

2. (**k-sample problem**) For each $j = 1, \dots, k$, let $X_{j1}, \dots, X_{jn_j}$ be an i.i.d. sample from a probability measure P_j . Consider testing the null hypothesis $H_0: P_1 = \dots = P_k$. Let $\mathbb{P}_{j,n_j}$ be the empirical measure of the j th sample, and let $\mathbb{H}_N$ be the empirical measure of the pooled sample. If $\mathcal{F}$ is a P -Donsker class and $\|P\|_{\mathcal{F}} < \infty$ for every P , then under the null hypothesis

$$K = \sum_{j=1}^k n_j (\mathbb{P}_{j,n_j} - \mathbb{H}_N)^2 \rightsquigarrow \sum_{j=1}^{k-1} \mathbb{G}_j^2,$$

in $\ell^\infty(\mathcal{F})$, where $\mathbb{G}_1, \dots, \mathbb{G}_{k-1}$ are i.i.d. tight P -Brownian bridge processes. Critical values for a test based on $\|K\|_{\mathcal{F}}$ can be set both by a permutation and a bootstrap procedure.

3. In the setting of the preceding problem, consider tests based on the statistics defined as $\sum_{i \neq j} \sqrt{n_i n_j / N} \|\mathbb{P}_{i,n_i} - \mathbb{P}_{j,n_j}\|_{\mathcal{F}}$ and $\sup_{i \neq j} \sqrt{n_i n_j / N} \|\mathbb{P}_{i,n_i} - \mathbb{P}_{j,n_j}\|_{\mathcal{F}}$.
4. (**Deficient two-sample bootstrap**) Let $\hat{\mathbb{P}}_m$ and $\hat{\mathbb{Q}}_n$ be the bootstrap empirical measures based on independent samples of sizes m and n from $\mathbb{P}_m$ and $\mathbb{Q}_n$, respectively. If $\mathcal{F}$ is P -Donsker, then the sequence $\hat{D}_{m,n} = \sqrt{mn/N} \|\hat{\mathbb{P}}_m - \hat{\mathbb{Q}}_n\|_{\mathcal{F}}$ converges under the null hypothesis $P = Q$ in distribution to $\|\mathbb{G}_P\|_{\mathcal{F}}$ conditionally on the original observations (in probability or almost surely). Thus the upper α -quantile of the distribution of $\hat{D}_{m,n}$ can be used as critical values for the test based on $D_{m,n}$. Is the resulting test consistent for every $\|P - Q\|_{\mathcal{F}} > 0$?

[**Hint:** The variables $\hat{D}_{m,n}$ defined this way mimic the variables $\hat{D}_{m,n}$ under the alternative hypothesis as well as under the null hypothesis in the sense that $D_{m,n} / \sqrt{mn/N} \xrightarrow{P} \|P - Q\|_{\mathcal{F}}$ and $\hat{D}_{m,n} / \sqrt{mn/N} \xrightarrow{P} \|P - Q\|_{\mathcal{F}}$. Hence $\hat{c}_\alpha \xrightarrow{P} \infty$.]

3.9

Independence

Empirical Processes

Let H be a probability measure on the measurable space $(\mathcal{X} \times \mathcal{Y}, \mathcal{A} \times \mathcal{B})$ with marginal laws P and Q on $(\mathcal{X}, \mathcal{A})$ and $(\mathcal{Y}, \mathcal{B})$, respectively. Given a sample $(X_1, Y_1), \dots, (X_n, Y_n)$ of independently and identically distributed vectors from H , we want to test the null hypothesis of independence $H_0: H = P \times Q$ versus the alternative hypothesis $H_1: H \neq P \times Q$. Let $\mathbb{H}_n$ be the empirical measure of the observations, and let $\mathbb{P}_n$ and $\mathbb{Q}_n$ be its marginals. The latter are the empirical measures of the X_i s and Y_i s, respectively.

Given classes $\mathcal{F}$ and $\mathcal{G}$ of real-valued measurable functions defined on $\mathcal{X}$ and $\mathcal{Y}$, respectively, let $\mathcal{F} \times \mathcal{G}$ be the class of all functions $f \times g$ from $\mathcal{X} \times \mathcal{Y}$ to $\mathbb{R}$ defined as

$$(f \times g)(x, y) = f(x)g(y),$$

when f and g range over $\mathcal{F}$ and $\mathcal{G}$, respectively. Consider the test statistics

$$K_n = \sqrt{n} \|\mathbb{H}_n - \mathbb{P}_n \times \mathbb{Q}_n\|_{\mathcal{F} \times \mathcal{G}}.$$

The *independence empirical process* $\mathbb{Z}_n$ is the process indexed by $\mathcal{F} \times \mathcal{G}$ given by

$$\begin{aligned} \mathbb{Z}_n(f, g) &= \sqrt{n} [(\mathbb{H}_n - \mathbb{P}_n \times \mathbb{Q}_n)(f \times g) - (H - P \times Q)(f \times g)] \\ &= \sqrt{n}(\mathbb{H}_n - H)(f \times g) - \sqrt{n}(\mathbb{P}_n - P)f \mathbb{Q}_n g - P f \sqrt{n}(\mathbb{Q}_n - Q)g \\ &= \sqrt{n}(\mathbb{H}_n - H)((f - Pf) \times (g - Qg)) - \sqrt{n}(\mathbb{P}_n - P)f(\mathbb{Q}_n - Q)g. \end{aligned}$$

Under the null hypothesis $H = P \times Q$ the test statistic can be written as $K_n = \|\mathbb{Z}_n\|_{\mathcal{F} \times \mathcal{G}}$, while under the alternative hypothesis $K_n = \|\mathbb{Z}_n + \sqrt{n}(H - P \times Q)\|_{\mathcal{F} \times \mathcal{G}}$.

The second term in the third representation of $\mathbb{Z}_n$ is asymptotically negligible. The following result follows from Slutsky's lemma.

3.9.1 Theorem. Let $\mathcal{F}$ and $\mathcal{G}$ be classes of measurable functions on measurable spaces $(\mathcal{X}, \mathcal{A})$ and $(\mathcal{Y}, \mathcal{B})$, respectively. If $\mathcal{F} \times \mathcal{G}$, $\mathcal{F}$, and $\mathcal{G}$ are H -Donsker and $\|P\|_{\mathcal{F}} < \infty$ and $\|Q\|_{\mathcal{G}} < \infty$, then the sequence of independence processes $\mathbb{Z}_n$ converges in distribution in $\ell^\infty(\mathcal{F} \times \mathcal{G})$ to the Gaussian process $\mathbb{Z}_H(f \times g) = \mathbb{G}_H((f - Pf) \times (g - Pg))$ for a tight H -Brownian bridge $\mathbb{G}_H$.

Under the null hypothesis $H = P \times Q$, the covariance function of the limit process can be calculated as

$$\text{cov}(\mathbb{Z}_H(f_1 \times g_1), \mathbb{Z}_H(f_2 \times g_2)) = (Pf_1f_2 - Pf_1Pf_2)(Qg_1g_2 - Qg_1Qg_2).$$

Thus, the limiting process has the same mean and covariance function as the product $\mathbb{G}_P(f)\mathbb{G}_Q(g)$ of two independent Brownian bridges (which is not Gaussian).

3.9.2 Example (Completely tucked Brownian sheet). When $\mathcal{X} \times \mathcal{Y}$ is the unit square in the Euclidean plane with uniform measure and both $\mathcal{F}$ and $\mathcal{G}$ are the indicators of the cells $\{[0, t]: 0 \leq t \leq 1\}$, the limit process is the *completely tucked Brownian sheet*. The class $\mathcal{F} \times \mathcal{G}$ is the set of all indicators of cells $[0, t]$ in the unit square, and the limit process can be identified with a process $\mathbb{Z}(s, t)$ indexed by the unit square. This is a zero-mean Gaussian process with covariance function

$$\mathbb{E}\mathbb{Z}(s_1, t_1)\mathbb{Z}(s_2, t_2) = (s_1 \wedge s_2 - s_1s_2)(t_1 \wedge t_2 - t_1t_2).$$

A “completely tucked Brownian sheet” derives its name from the fact that $\mathbb{Z}(s, t) = 0$ almost surely for all (s, t) in the boundary of the unit square. The distribution of the norm $\|\mathbb{Z}\|_{[0,1]^2}$ appears to be unknown in closed form.

The same limiting distribution for the sequence K_n arises when $\mathcal{X} \times \mathcal{Y} = \mathbb{R}^2$ and $\mathcal{F}$ and $\mathcal{G}$ are equal to the class of indicators of cells $\{(-\infty, t]: t \in \mathbb{R}\}$. Under the null hypothesis of independence, the weak limit of the sequence $\mathbb{Z}_n$ can be represented as $\mathbb{Z}(P(s), Q(t))$, where $\mathbb{Z}$ is a standard, completely tucked Brownian sheet. Under the assumption that the marginal distribution functions $P(s)$ and $Q(t)$ are continuous, the variable $\|\mathbb{Z}(P(s), Q(t))\|_{\mathbb{R}^2}$ is distributed as the norm of a completely tucked Brownian sheet.

The conclusion of the preceding theorem immediately yields that under the null hypothesis the sequence of test statistics $K_n = \sqrt{n}\|\mathbb{H}_n - \mathbb{P}_n \times \mathbb{Q}_n\|_{\mathcal{F} \times \mathcal{G}}$ converges in distribution to the norm $\|\mathbb{Z}_H\|_{\mathcal{F} \times \mathcal{G}}$ of the limit independence process. Furthermore, since the test statistics can be written

$$K_n = \|\mathbb{Z}_n + \sqrt{n}(H - P \times Q)\|_{\mathcal{F} \times \mathcal{G}},$$

the sequence of test statistics converges in probability to $+\infty$ under every H in the alternative hypothesis such that $\|H - P \times Q\|_{\mathcal{F} \times \mathcal{G}} > 0$.

To implement a test based on K_n , it remains to approximate the critical points of its distribution. Since the limit distribution of the sequence K_n depends on the unknown marginal distributions P and Q even under the null hypothesis, it is typically not practical to obtain critical points from the asymptotic distribution. However, critical points can be set by a bootstrap approach.

Under the null hypothesis, a natural estimate for the underlying distribution H of the observations is the product $\mathbb{P}_n \times \mathbb{Q}_n$ of the empirical measures of the two samples. For the bootstrap test of independence, let $(\hat{X}_1, \hat{Y}_1), \dots, (\hat{X}_n, \hat{Y}_n)$ be an i.i.d. sample from the measure $\mathbb{P}_n \times \mathbb{Q}_n$, and let $\hat{\mathbb{H}}_n = n^{-1} \sum_{i=1}^n \delta_{(\hat{X}_i, \hat{Y}_i)}$ be the corresponding bootstrap empirical measure. The *bootstrap independence process* is defined by

$$\hat{\mathbb{Z}}_n = \sqrt{n}(\hat{\mathbb{H}}_n - \hat{\mathbb{P}}_n \times \hat{\mathbb{Q}}_n).$$

Note that, given the original observations, the variable $\hat{\mathbb{Z}}_n(f \times g)$ is centered at mean zero since the bootstrap sample is taken from $\mathbb{P}_n \times \mathbb{Q}_n$, not $\mathbb{H}_n$. Thus, there is a symmetry in the definitions of $\hat{\mathbb{Z}}_n$ and $\mathbb{Z}_n$ only if the latter is considered under the null hypothesis $H = P \times Q$. We shall still be interested in the asymptotic behavior of $\hat{\mathbb{Z}}_n$ conditionally on the original observations, under both null and alternative hypotheses.

With $\hat{K}_n = \|\hat{\mathbb{Z}}_n\|_{\mathcal{F} \times \mathcal{G}}$, define a critical point

$$\hat{c}_n = \inf\{t > 0: P_{\hat{X}, \hat{Y}}(\hat{K}_n > t) \leq \alpha\}.$$

Here $P_{\hat{X}, \hat{Y}}$ denotes the conditional law given $(X_1, Y_1), \dots, (X_n, Y_n)$. The bootstrap test of independence rejects the null hypothesis for values of K_n exceeding $\hat{c}_n$. This procedure is a *model-based* bootstrap in that the bootstrap resampling is done from the estimated model under the null hypothesis.

To derive the asymptotic properties of this test, the convergence of the process $\hat{\mathbb{Z}}_n$ must be established under both the null and alternative hypotheses. Although theorems as general as those of Chapter 3.8 are lacking for this process, a number of conclusions are possible using the methods developed in Sections 2.8 and 2.9. These methods apply to a wide range of problems involving model-based bootstrapping.

We apply the results of Section 2.8.3 with P_n of that section taken to be $\mathbb{P}_n \times \mathbb{Q}_n$ for given values of the original observations.

3.9.3 Theorem (Bootstrap). *Let $\mathcal{F}$ and $\mathcal{G}$ be separable classes of measurable functions on measurable spaces $(\mathcal{X}, \mathcal{A})$ and $(\mathcal{Y}, \mathcal{B})$, respectively, such that $\mathcal{F} \times \mathcal{G}$ satisfies the uniform entropy condition for envelope functions F, G , and $F \times G$ that are H -square integrable. Then*

$$\hat{\mathbb{G}}_n = \sqrt{n}(\hat{\mathbb{H}}_n - \mathbb{P}_n \times \mathbb{Q}_n) \rightsquigarrow \mathbb{G}_{P \times Q}, \quad \text{in } \ell^\infty(\mathcal{F} \times \mathcal{G}),$$

and

$$\hat{\mathbb{Z}}_n = \sqrt{n}(\hat{\mathbb{H}}_n - \hat{\mathbb{P}}_n \times \hat{\mathbb{Q}}_n) \rightsquigarrow \mathbb{Z}_{P \times Q}, \quad \text{in } \ell^\infty(\mathcal{F} \times \mathcal{G}),$$

given H^∞ -almost every sequence $(X_1, Y_1), (X_2, Y_2), \dots$. Here $\mathbb{G}_{P \times Q}$ is a $P \times Q$ -Brownian bridge.

Proof. The bootstrap independence process can be rewritten as

$$\begin{aligned} \hat{\mathbb{Z}}_n(f, g) &= \sqrt{n}(\hat{\mathbb{H}}_n - \mathbb{P}_n \times \mathbb{Q}_n)(f \times g) - \sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n)f \mathbb{Q}_n g \\ &\quad - \mathbb{P}_n f \sqrt{n}(\hat{\mathbb{Q}}_n - \mathbb{Q}_n)g - \sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n)f(\hat{\mathbb{Q}}_n - \mathbb{Q}_n)g. \end{aligned}$$

The assertions now follow from an empirical central limit theorem for a triangular array of observations. In the present case the n th row of the array is the sample

$(\hat{X}_1, \hat{Y}_1), \dots, (\hat{X}_n, \hat{Y}_n)$ from $\mathbb{P}_n \times \mathbb{Q}_n$. Since $\mathcal{F} \times \mathcal{G}$ satisfies the uniform entropy condition, Theorem 2.8.9 shows that $\hat{\mathbb{G}}_n$ converges in distribution to $\mathbb{G}_{P \times Q}$ for every sequence of original observations for which

$$\begin{aligned} (\mathbb{P}_n \times \mathbb{Q}_n)(F \times G)^2 &= O(1), \\ (\mathbb{P}_n \times \mathbb{Q}_n)(F \times G)^2 \{ |F \times G| \geq \varepsilon \sqrt{n} \} &\rightarrow 0, \quad \text{every } \varepsilon > 0, \\ \sup_{h_1, h_2 \in \mathcal{F} \times \mathcal{G}} |\rho_{\mathbb{P}_n \times \mathbb{Q}_n}(h_1, h_2) - \rho_{P \times Q}(h_1, h_2)| &\rightarrow 0. \end{aligned}$$

The quantities in this display are two-sample U-statistics (for dependent samples if H is not a product measure), but in view of the special structure of the kernels can be handled by the strong law of large numbers for i.i.d. variables. For instance, the second line of the display can be written as

$$\frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n F^2(X_i) G^2(Y_j) \{ |F(X_i) G(Y_j)| \geq \varepsilon \sqrt{n} \}.$$

For every $M^2 \leq \varepsilon \sqrt{n}$ this is bounded above by

$$\mathbb{P}_n F^2 \{ F \geq M \} \mathbb{Q}_n G^2 + \mathbb{P}_n F^2 \mathbb{Q}_n G^2 \{ G \geq M \}.$$

For $n \rightarrow \infty$ this converges almost surely to a fixed value, which can be made arbitrarily small by choosing M large. This concludes the proof of the weak convergence of $\hat{\mathbb{G}}_n$.

By Slutsky's lemma and the continuous mapping theorem, the sequence $\hat{\mathbb{Z}}_n$ converges in distribution to $\mathbb{G}_{P \times Q}(f \times g) - \mathbb{G}_{P \times Q}(f \times 1)Qg - Pf\mathbb{G}_{P \times Q}(1 \times g)$, which is a mean zero Gaussian process with the same covariance function as $\mathbb{Z}_{P \times Q}$ (Cf. the second representation of $\mathbb{Z}_n$ in the introduction). ■

The preceding theorem implies that the sequence $\hat{K}_n = \|\hat{\mathbb{Z}}_n\|_{\mathcal{F} \times \mathcal{G}}$ converges almost surely conditionally in distribution to $\|\mathbb{Z}_{P \times Q}\|_{\mathcal{F} \times \mathcal{G}}$ under both null and alternative hypotheses. Thus, it has the same limiting distribution as the test statistic K_n under the null hypothesis $H = P \times Q$. Hence the upper α -quantiles $\hat{c}_n$ of its (conditional) distribution satisfy

$$\hat{c}_n \rightarrow \inf \{ t > 0 : P(\|\mathbb{Z}_{P \times Q}\|_{\mathcal{F} \times \mathcal{G}} > t) \leq \alpha \}, \quad \text{a.s.}$$

It follows that the sequence of tests that reject the null hypothesis $H_0: H = P \times Q$ if $K_n > \hat{c}_n$ is asymptotically of level α for any $H = P \times Q$ for which the index set $\mathcal{F} \times \mathcal{G}$ satisfies the conditions of the preceding theorem. Furthermore, the sequence of tests is consistent against any alternative H for which $\|H - P \times Q\|_{\mathcal{F} \times \mathcal{G}} > 0$.

An alternative to using the bootstrap is to set critical points by a permutation procedure. For a permutation test of independence let $R = (R_1, \dots, R_n)$ be uniformly distributed on the set of permutations of $\{1, \dots, n\}$ and independent of the original observations. Then $\tilde{\mathbb{H}}_n = n^{-1} \sum_{i=1}^n \delta_{(X_i, Y_{R_i})}$ is the permutation empirical measure, and the process

$$\tilde{\mathbb{Z}}_n = \sqrt{n}(\tilde{\mathbb{H}}_n - \mathbb{P}_n \times \mathbb{Q}_n)$$

is called the *permutation independence process*. Now with $\tilde{K}_n \equiv \|\tilde{Z}_n\|_{\mathcal{F} \times \mathcal{G}}$ define as critical point

$$\tilde{c}_n = \inf\{t > 0: \mathbf{P}_R(\tilde{K}_n > t) < \alpha\}.$$

The permutation test of independence rejects the null hypothesis for values of K_n larger than $\tilde{c}_n$. The conclusions of Theorem 3.9.3 for the bootstrap independence process $\hat{Z}_n$ carry over to the permutation independence process $\tilde{Z}_n$, under the same conditions, and hence the permutation test of independence is also consistent.^b

Problems and Complements

1. Suppose that $\mathcal{F} \times \mathcal{G}$ is H -Donsker where H is a distribution on $\mathcal{X} \times \mathcal{Y}$. Then the finite-dimensional distributions of the permutation independence process $\tilde{Z}_n$ converge in distribution to those of $\mathbb{Z}_{P \times Q}$, given H^∞ -almost every sequence $(X_1, Y_1), (X_2, Y_2), \dots$

[Hint: Use the rank central limit theorem, Proposition A.5.3.]

^b Dümbgen (1994).

3.10

Delta-Method

After giving the general principle of the delta-method, we consider the special case of Gaussian limits and the “conditional” delta-method, which applies to the bootstrap. The chapter closes with a large number of examples.

3.10.1 Main Result

Suppose X_n is a sequence of real-valued maps with $\sqrt{n}(X_n - \theta) \rightsquigarrow X$ for some constant θ . If $\phi: \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at θ , then by Slutsky’s theorem

$$\sqrt{n}(\phi(X_n) - \phi(\theta)) = \frac{\phi(X_n) - \phi(\theta)}{X_n - \theta} \sqrt{n}(X_n - \theta) \rightsquigarrow \phi'(\theta) X.$$

This is the simplest form of the *delta-method*.

A more general form of the delta-method is valid for maps $\phi: \mathbb{D} \rightarrow \mathbb{E}$ between normed spaces $\mathbb{D}$ and $\mathbb{E}$ or, even more generally: metrizable, topological vector spaces.[‡] The appropriate form of differentiability of ϕ is Hadamard differentiability. A map $\phi: \mathbb{D}_\phi \subset \mathbb{D} \rightarrow \mathbb{E}$ is called *Hadamard-differentiable* at $\theta \in \mathbb{D}_\phi$ if there is a continuous linear map $\phi'_\theta: \mathbb{D} \rightarrow \mathbb{E}$ such that

$$\frac{\phi(\theta + t_n h_n) - \phi(\theta)}{t_n} \rightarrow \phi'_\theta(h), \quad n \rightarrow \infty,$$

for all converging sequences $t_n \rightarrow 0$ and $h_n \rightarrow h$ such that $\theta + t_n h_n \in \mathbb{D}_\phi$ for every n . An equivalent definition is that (for normed spaces $\mathbb{E}$)

$$(3.10.1) \quad \sup_{h \in K, \theta + th \in \mathbb{D}_\phi} \left\| \frac{\phi(\theta + th) - \phi(\theta)}{t} - \phi'_\theta(h) \right\| \rightarrow 0, \quad t \rightarrow 0,$$

[‡] A *topological vector space* is a vector space for which addition and scalar multiplication are continuous operations: if $x_n \rightarrow x$, $y_n \rightarrow y$, and $c_n \rightarrow c$, then $x_n + y_n \rightarrow x + y$ and $c_n x_n \rightarrow cx$. Every normed space is a topological vector space; another example is $\mathbb{R}^\infty$ with product topology.

for every compact $K \subset \mathbb{D}$. The first definition may be refined to *Hadamard-differentiable tangentially* to a set $\mathbb{D}_0 \subset \mathbb{D}$ by requiring that every $h_n \rightarrow h$ in the definition has $h \in \mathbb{D}_0$; the derivative need then be defined on $\mathbb{D}_0$ only. The second definition more or less shows why Hadamard differentiability is the appropriate type of differentiability for our purposes: the compacts K match up with the asymptotic tightness of weakly convergent sequences.

The domain $\mathbb{D}_\phi$ is allowed to be an arbitrary subset of $\mathbb{D}$. In particular, it is not necessary that it is all of $\mathbb{D}$ and it need not be an open subset. This is often important for applications, for instance when considering functions of distribution functions viewed as element of a Skorokhod space.

3.10.2 Example (Hadamard and Fréchet differentiability). For maps $\phi: \mathbb{D}_\phi \subset \mathbb{R}^k \rightarrow \mathbb{R}^m$, Hadamard differentiability is equivalent to the usual differentiability of calculus, and the derivative $\phi'_\theta: \mathbb{R}^k \rightarrow \mathbb{R}^m$ is given by the matrix of partial derivatives $(\partial\phi_i/\partial\theta_j)$ evaluated at θ . In general, a map $\phi: \mathbb{D}_\phi \rightarrow \mathbb{E}$ is called *Fréchet-differentiable* if there exists a continuous, linear map $\phi'_\theta: \mathbb{D} \rightarrow \mathbb{E}$ such that (3.10.1) holds uniformly in h in bounded subsets of $\mathbb{D}_\phi$. For normed spaces, this is equivalent to

$$\|\phi(\theta + h) - \phi(\theta) - \phi'_\theta(h)\| = o(\|h\|), \quad \|h\| \rightarrow 0.$$

Since a compact set is bounded, Fréchet differentiability is a stronger requirement than Hadamard differentiability. For normed spaces, they are equivalent if and only if the unit ball is compact, that is, $\mathbb{D}$ is finite-dimensional.

The chain rule asserts that the composition of two differentiable maps is differentiable. It makes possible a calculus of differentiable maps, in which complicated maps can be decomposed into more basic maps. We shall use it frequently when dealing with examples.

3.10.3 Lemma (Chain rule). *If $\phi: \mathbb{D}_\phi \subset \mathbb{D} \rightarrow \mathbb{E}_\psi$ is Hadamard-differentiable at $\theta \in \mathbb{D}_\phi$ tangentially to $\mathbb{D}_0$ and $\psi: \mathbb{E}_\psi \rightarrow \mathbb{F}$ is Hadamard-differentiable at $\phi(\theta)$ tangentially to $\phi'_\theta(\mathbb{D}_0)$, then $\psi \circ \phi: \mathbb{D}_\phi \rightarrow \mathbb{F}$ is Hadamard-differentiable at θ tangentially to $\mathbb{D}_0$ with derivative $\psi'_{\phi(\theta)} \circ \phi'_\theta$.*

Proof. Rewrite the difference $\psi \circ \phi(\theta + th_t) - \psi \circ \phi(\theta)$ as $\psi(\phi(\theta) + tk_t) - \psi(\phi(\theta))$, with k_t given by $(\phi(\theta + th_t) - \phi(\theta))/t$. First apply Hadamard differentiability of ϕ to see that k_t converges to $\phi'_\theta(h)$, and next Hadamard differentiability of ψ . ■

For measurable maps X_n , the delta-method is an almost immediate consequence of the almost-sure representation theorem. If $r_n \rightarrow \infty$, $r_n(X_n - \theta) \rightsquigarrow X$ in $\mathbb{D}$, and every X_n is Borel measurable, then there are Borel measurable $\tilde{X}_n$ that are equal in law to X_n , take their values in the range of X_n , and satisfy $r_n(\tilde{X}_n - \theta) \xrightarrow{\text{as}} \tilde{X}$. (Apply the representation theorem to $r_n(X_n - \theta)$ and transform.) If ϕ is Hadamard-differentiable at θ , then

$$r_n(\phi(\tilde{X}_n) - \phi(\theta)) = \frac{\phi(\theta + r_n^{-1} r_n(\tilde{X}_n - \theta)) - \phi(\theta)}{r_n^{-1}} \xrightarrow{\text{as}} \phi'_\theta(\tilde{X}).$$

If ϕ is measurable the left side is equal in distribution to $r_n(\phi(X_n) - \phi(\theta))$. Moreover, the almost sure convergence implies convergence in law. Thus, we obtain that the sequence $r_n(\phi(X_n) - \phi(\theta))$ converges in distribution to $\phi'_\theta(\tilde{X})$.

Unfortunately, this argument breaks down if the measurability assumptions fail. It can be fixed by using the (refined) continuous mapping theorem for outer almost-sure convergence, or by a direct argument. The latter approach appears preferable. In any case, the delta-method “works” without the need of explicit measurability assumptions.

3.10.4 Theorem (Delta-method). *Let $\mathbb{D}$ and $\mathbb{E}$ be metrizable topological vector spaces. Let $\phi: \mathbb{D}_\phi \subset \mathbb{D} \rightarrow \mathbb{E}$ be Hadamard-differentiable at θ tangentially to $\mathbb{D}_0$. Let $X_n: \Omega_n \rightarrow \mathbb{D}_\phi$ be maps with $r_n(X_n - \theta) \rightsquigarrow X$ for some sequence of constants $r_n \rightarrow \infty$, where X is separable and takes its values in $\mathbb{D}_0$. Then $r_n(\phi(X_n) - \phi(\theta)) \rightsquigarrow \phi'_\theta(X)$. If ϕ'_θ is defined and continuous on the whole of $\mathbb{D}$, then the sequence $r_n(\phi(X_n) - \phi(\theta)) - \phi'_\theta(r_n(X_n - \theta))$ converges to zero in outer probability.*

Proof. The map $g_n(h) = r_n(\phi(\theta + r_n^{-1}h) - \phi(\theta))$ is defined on the domain $\mathbb{D}_n = \{h: \theta + r_n^{-1}h \in \mathbb{D}_\phi\}$ and satisfies $g_n(h_n) \rightarrow \phi'_\theta(h)$ for every $h_n \rightarrow h \in \mathbb{D}_0$. By the extended continuous mapping theorem, Theorem 1.11.1, one has $g_n(r_n(X_n - \theta)) \rightsquigarrow \phi'_\theta(X)$. This is the first assertion of the theorem.

The second assertion follows upon applying the first to the map $\psi: \mathbb{D}_\phi \rightarrow \mathbb{E} \times \mathbb{E}$ defined by $\psi(d) = (\phi(d), \phi'_\theta(d))$. Since this map is Hadamard-differentiable at (θ, θ) with derivative $(\phi'_\theta, \phi'_\theta)$, it follows that

$$(r_n(\phi(X_n) - \phi(\theta)), r_n(\phi'_\theta(X_n) - \phi'_\theta(\theta))) \rightsquigarrow (\phi'_\theta(X), \phi'_\theta(X)).$$

By the continuous mapping theorem, the difference of the coordinates of these vectors converges weakly to $\phi'_\theta(X) - \phi'_\theta(X) = 0$. ■

A remarkable fact about the previous theorem is that it suffices that ϕ is differentiable at just the single point θ . This is certainly convenient for applications to abstract-valued random elements, such as the empirical process, when continuous differentiability can easily fail.

A stronger form of differentiability may be needed for a “uniform” delta-method. Suppose $\theta_n \rightarrow \theta$ and $r_n(X_n - \theta_n) \rightsquigarrow X$. Under what condition does it follow that $r_n(\phi(X_n) - \phi(\theta_n)) \rightsquigarrow \phi'_\theta(X)$? Consider two cases.

First, if θ_n converges sufficiently fast to θ —specifically if the sequence $r_n(\theta_n - \theta)$ is relatively compact—then the conditions of the previous theorem apply. Simply write the object of interest as the sum $r_n(\phi(X_n) - \phi(\theta)) + r_n(\phi(\theta) - \phi(\theta_n))$ and apply the previous theorem to the first term and the definition of Hadamard differentiability to the second. If $r_n(\theta_n - \theta) \rightarrow h$ (along a subsequence), this yields $r_n(\phi(X_n) - \phi(\theta_n)) \rightsquigarrow \phi'_\theta(X + h) + \phi'_\theta(-h) = \phi'_\theta(X)$.

Second, for general θ_n , it suffices that $\phi: \mathbb{D}_\phi \rightarrow \mathbb{E}$ be uniformly differentiable.

3.10.5 Theorem (Delta-method). Let $\mathbb{D}$ and $\mathbb{E}$ be metrizable topological vector spaces and r_n constants with $r_n \rightarrow \infty$. Let $\phi: \mathbb{D}_\phi \subset \mathbb{D} \rightarrow \mathbb{E}$ satisfy

$$r_n(\phi(\theta_n + r_n^{-1}h_n) - \phi(\theta_n)) \rightarrow \phi'_\theta(h),$$

for every converging sequence h_n with $\theta_n + r_n^{-1}h_n \in \mathbb{D}_\phi$ for all n and $h_n \rightarrow h \in \mathbb{D}_0$ and some arbitrary map ϕ'_θ on $\mathbb{D}_0$. If $X_n: \Omega_n \rightarrow \mathbb{D}_\phi$ are maps with $r_n(X_n - \theta_n) \rightsquigarrow X$, where X is separable and takes its values in $\mathbb{D}_0$, then $r_n(\phi(X_n) - \phi(\theta_n)) \rightsquigarrow \phi'_\theta(X)$. If ϕ'_θ is defined, linear, and continuous on the whole of $\mathbb{D}$, then the sequence $r_n(\phi(X_n) - \phi(\theta_n)) - \phi'_\theta(r_n(X_n - \theta_n))$ converges to zero in outer probability.

This theorem is proved in exactly the same manner as the previous theorem. A sufficient condition for uniform differentiability is continuous differentiability in a neighborhood of θ . For convex domains $\mathbb{D}_\phi$, a weak form of this is that $\phi: \mathbb{D}_\phi \subset \mathbb{D} \rightarrow \mathbb{E}$ be differentiable at every $\vartheta \in \mathbb{D}_\phi$ that is sufficiently close to θ , where the derivatives ϕ'_ϑ satisfy

$$(3.10.6) \quad \begin{aligned} \lim_{\substack{\eta \rightarrow \vartheta \\ \eta \in \mathbb{D}_\phi}} \phi'_\eta(h) &= \phi'_\vartheta(h), & \text{for every } h \in \text{lin } \mathbb{D}_\phi, \\ \lim_{\substack{\vartheta \rightarrow \theta \\ \vartheta \in \mathbb{D}_\phi}} \phi'_\vartheta(h) &= \phi'_\theta(h), & \text{uniformly in } h \in K, \end{aligned}$$

for every totally bounded subset K of $\text{lin } \mathbb{D}_\phi$.

3.10.7 Lemma. Let $\mathbb{D}$ and $\mathbb{E}$ be normed spaces and $\mathbb{D}_\phi \subset \mathbb{D}$ convex. Let $\phi: \mathbb{D}_\phi \rightarrow \mathbb{E}$ be Hadamard-differentiable at every $\vartheta \in \mathbb{D}_\phi$ in a neighborhood around θ , tangentially to $\mathbb{D}_0$, where the derivatives satisfy (3.10.6). Then ϕ is uniformly differentiable along every sequence $\theta_n \rightarrow \theta$ in $\mathbb{D}_\phi$, tangentially to $\mathbb{D}_0$.

Proof. Let $t_n \downarrow 0$ be given. Fix sufficiently large n and h such that $\theta_n + t_n h \in \mathbb{D}_\phi$, and define a map $f_n(t) = \phi(\theta_n + t h)$ on the interval $0 \leq t \leq t_n$. Since $\mathbb{D}_\phi$ is convex, this is well defined, and by Hadamard differentiability of ϕ ,

$$\frac{f_n(t+g) - f_n(t)}{g} \rightarrow \phi'_{\theta_n + t h}(h), \quad \text{as } g \rightarrow 0,$$

for every t . Thus f_n is differentiable on the interval $[0, t_n]$. By assumption its derivative is continuous in t . Hence for every element e^* from the dual space of $\mathbb{E}$, the fundamental theorem of calculus allows one to write $e^*(f_n(t_n) - f_n(0))$ as the integral of its derivative. In other words

$$e^*(\phi(\theta_n + t_n h) - \phi(\theta_n)) = t_n \int_0^1 e^* \phi'_{\theta_n + t t_n h}(h) dt.$$

For every $h_n \rightarrow h$ such that $\theta_n + t_n h_n \in \mathbb{D}_\phi$, we have by assumption that

$$\int_0^1 |e^* \phi'_{\theta_n + t t_n h_n}(h_n) - e^* \phi'_\theta(h_n)| dt \rightarrow 0,$$

uniformly in $\|e^*\| \leq 1$. The combination of the last two equations shows that the sequence $t_n^{-1}(\phi(\theta_n + t_n h_n) - \phi(\theta_n))$ has the same limit as the sequence $\phi'_\theta(h_n)$. ■

3.10.2 Gaussian Limits

The delta-method yields the conclusion that the transformed sequence of variables $r_n(\phi(X_n) - \phi(\theta))$ converges to a limit variable $\phi'_\theta(X)$ for a certain continuous, linear map $\phi'_\theta: \mathbb{D} \rightarrow \mathbb{E}$. In the case that $\mathbb{D} = \mathbb{R}^k$ and $\mathbb{E} = \mathbb{R}^m$, the map ϕ'_θ is an $(m \times k)$ -matrix. If the limit X of the original sequence is normally $N_k(\mu, \Sigma)$ -distributed, then $\phi'_\theta(X)$ is $N_m(\phi'_\theta\mu, \phi'_\theta\Sigma(\phi'_\theta)^t)$ -distributed. In particular, the asymptotic normality is retained. In this section it is noted that the same conclusion is true in the infinite-dimensional situation.

By definition, a Borel measurable random element X in a Banach space $\mathbb{D}$ is normally distributed (or Gaussian) if the real-valued random variable d^*X is normally distributed for every element d^* of the dual space (the collection of continuous, linear real maps on $\mathbb{D}$). From this definition it is immediate that $\phi'_\theta(X)$ is normally distributed in $\mathbb{E}$ for every continuous, linear map $\phi'_\theta: \mathbb{D} \rightarrow \mathbb{E}$, whenever X is normally distributed in $\mathbb{D}$. It suffices to note that $e^* \circ \phi'_\theta$ is an element of the dual space of $\mathbb{D}$ for every element e^* of the dual space of $\mathbb{E}$.

Many applications concern stochastic processes with bounded sample paths. By definition, a stochastic process $\{X_t: t \in T\}$ is Gaussian if every one of its finite-dimensional marginals $(X_{t_1}, \dots, X_{t_k})$ is multivariately normally distributed. If the stochastic process is also defined as a Borel measurable map into $\ell^\infty(T)$, then the definition of Gaussianity of a Banach-valued random element given in the preceding paragraph may be applied as well. Thus, X is Gaussian if d^*X is normally distributed for every element d^* of the dual space of $\ell^\infty(T)$. It is not immediately clear that these two definitions are equivalent. The second definition appears to be more stringent, as the first definition requires that d^*X is normally distributed for every linear combination $d^* = \sum \alpha_i \pi_{t_i}$ of coordinate projections, but not necessarily for every element of the dual space of $\ell^\infty(T)$. The following lemma shows that the two definitions are equivalent if X is tight.

3.10.8 Lemma. *Let X be a tight, Borel measurable map into $\ell^\infty(T)$ such that the vector $(X_{t_1}, \dots, X_{t_k})$ is multivariately normally distributed for every finite set $t_1, \dots, t_k$ in T . Then $\phi(X)$ is normally distributed for every continuous, linear map $\phi: \ell^\infty(T) \rightarrow \mathbb{E}$ into a Banach space $\mathbb{E}$.*

Proof. It suffices to consider the case that $\mathbb{E}$ is the real line. By Lemma 1.5.9 there exists a semimetric ρ on T that makes T totally bounded and such that almost all sample paths $t \mapsto X_t$ are uniformly continuous. Every bounded, uniformly continuous function $z: T \rightarrow \mathbb{R}$ has a unique extension $\bar{z}$ to a continuous function on the completion $\bar{T}$ of T . This completion is compact for (the extension of) ρ . By a minor modification of the Riesz representation theorem,[†] there exists a signed Borel measure μ on the completion such that

$$\phi(z) = \int_{\bar{T}} \bar{z}(t) d\mu(t).$$

By discretization we can construct a sequence of maps of the form $\phi_m = \sum \alpha_{mi} \pi_{t_{mi}}$, with every $t_{mi} \in T$, that converges pointwise to ϕ on $UC(T, \rho)$. In

[†] E.g., Rudin (1966).

particular, $\phi_m(X) \rightarrow \phi(X)$ almost surely. Since $\phi_m(X)$ is normally distributed by assumption, so is $\phi(X)$. ■

It is clear from the preceding proof that in applications of the delta-method to maps $\phi: \ell^\infty(T) \rightarrow \mathbb{R}$, the resulting limit variable $\phi'_\theta(X)$ can be written as $\int_{\bar{T}} \bar{X}(t) d\mu(t)$, whenever X is tight. The preceding lemma shows that $\phi'_\theta(X)$ is normally distributed. With the help of Fubini's theorem, its mean and variance can be computed as

$$\int_{\bar{T}} \mathbb{E} \bar{X}(t) d\mu(t); \quad \int_{\bar{T}} \int_{\bar{T}} \mathbb{E} \bar{X}(s) \bar{X}(t) d\mu(s) d\mu(t),$$

respectively. Often mean and variance can be computed more easily from the asymptotic linear expansion guaranteed by the second assertion of Theorem 3.10.4.

3.10.3 Conditional Delta-Method

The various bootstrap processes discussed in Chapter 3.7 were shown to converge conditionally in distribution given the “original” observations. In this case, application of the delta-method should lead to a statement concerning the conditional weak convergence of the transformed processes. Proper care of measurability issues requires a separate discussion. The results are not specific to bootstrap processes, but apply more generally to processes that converge conditionally in distribution given certain random variables, in probability or almost surely.

Consider random elements $\mathbb{P}_n = \mathbb{P}_n(X_n)$ and $\hat{\mathbb{P}}_n = \hat{\mathbb{P}}_n(X_n, M_n)$ in a normed space $\mathbb{D}$ such that the sequence $\sqrt{n}(\mathbb{P}_n - P)$ converges unconditionally and the sequence $\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n)$ converges conditionally given X_n in distribution to a tight random element $\mathbb{G}$. A precise formulation of the second is that

$$(3.10.9) \quad \sup_{h \in BL_1(\mathbb{D})} |\mathbb{E}_M h(\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n)) - \mathbb{E} h(\mathbb{G})| \rightarrow 0, \\ \mathbb{E}_M h(\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n))^* - \mathbb{E}_M h(\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n))_* \rightarrow 0,$$

in outer probability or outer almost surely, with h ranging over the bounded Lipschitz functions. Given a map ϕ , we wish to show that

$$(3.10.10) \quad \sup_{h \in BL_1(\mathbb{E})} |\mathbb{E}_M h(\sqrt{n}(\phi(\hat{\mathbb{P}}_n) - \phi(\mathbb{P}_n))) - \mathbb{E} h(\phi'_P(\mathbb{G}))| \rightarrow 0, \\ \mathbb{E}_M h(\sqrt{n}(\phi(\hat{\mathbb{P}}_n) - \phi(\mathbb{P}_n)))^* - \mathbb{E}_M h(\sqrt{n}(\phi(\hat{\mathbb{P}}_n) - \phi(\mathbb{P}_n)))_* \rightarrow 0,$$

in outer probability or outer almost surely.

For statistical purposes, consistency in probability appears to be sufficient. This is retained under just Hadamard differentiability at the single point P .

3.10.11 Theorem (Conditional delta-method in probability). *Let $\mathbb{D}$ and $\mathbb{E}$ be normed spaces. Let $\phi: \mathbb{D}_\phi \subset \mathbb{D} \rightarrow \mathbb{E}$ be Hadamard-differentiable at P tangentially to a subspace $\mathbb{D}_0$. Let $\mathbb{P}_n$ and $\hat{\mathbb{P}}_n$ be maps as indicated previously with values in $\mathbb{D}_\phi$ such that $\sqrt{n}(\mathbb{P}_n - P) \rightsquigarrow \mathbb{G}$ and (3.10.9) holds in outer probability, where $\mathbb{G}$ is separable and takes its values in $\mathbb{D}_0$. Then (3.10.10) holds in outer probability.*

In particular, the weak consistency of the bootstrap estimators considered in Chapter 3.7 carries over to any Hadamard-differentiable functional: the sequence of “conditional random laws” (given $X_1, X_2, \dots$) of $\sqrt{n}(\phi(\hat{\mathbb{P}}_n) - \phi(\mathbb{P}_n))$ is asymptotically consistent in probability for estimating the laws of the random elements $\sqrt{n}(\phi(\mathbb{P}_n) - \phi(P))$.

For almost-sure consistency, we need a stronger differentiability property of ϕ . The next two results impose uniform Hadamard differentiability and Fréchet differentiability with a rate, respectively. First, suppose that

$$\sqrt{n}(\phi(\mathbb{P}_n + n^{-1/2}h_n) - \phi(\mathbb{P}_n)) \rightarrow \phi'_P(h)$$

for almost every sequence $\mathbb{P}_n$ and every converging sequence $h_n \rightarrow h \in \mathbb{D}_0$. Then Theorem 3.10.5 yields

$$\sqrt{n}(\phi(\hat{\mathbb{P}}_n) - \phi(\mathbb{P}_n)) \rightsquigarrow \phi'_P(\mathbb{G}),$$

given almost every sequence $\mathbb{P}_n$ for which $\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n) \rightsquigarrow \mathbb{G}$. Thus, without further effort we have obtained an almost-sure version of the preceding theorem. However, the result implies at best that the first expression in (3.10.10) converges almost surely to zero, not necessarily outer almost surely. (In particular, it does not imply the in-outer-probability statement of the preceding theorem.) In a given application the measurability of the transformed bootstrap process could be argued directly. Alternatively, outer-almost-sure convergence in (3.10.10) can be obtained directly under a stronger type of differentiability. Assume that

$$(3.10.12) \quad \frac{\phi(P_n + t_n h_n) - \phi(P_n)}{t_n} \rightarrow \phi'_P(h),$$

for all sequences $t_n \downarrow 0$, $P_n \rightarrow P$, $h_n \rightarrow h \in \mathbb{D}_0$ such that P_n and $P_n + t_n h_n \in \mathbb{D}_\phi$ for every n and a continuous, linear map $\phi'_P: \mathbb{D}_0 \rightarrow \mathbb{E}$.

3.10.13 Theorem (Conditional delta-method almost surely). *Let $\mathbb{D}$ and $\mathbb{E}$ be normed spaces. Let $\phi: \mathbb{D}_\phi \subset \mathbb{D} \rightarrow \mathbb{E}$ satisfy (3.10.12) for a given measurable subspace $\mathbb{D}_0$. Let $\mathbb{P}_n$ and $\hat{\mathbb{P}}_n$ be maps as indicated previously with values in $\mathbb{D}_\phi$ such that $\sqrt{n}(\mathbb{P}_n - P) \rightsquigarrow \mathbb{G}$, $\|\mathbb{P}_n - P\| \rightarrow 0$ outer almost surely and (3.10.9) holds outer almost surely, where $\mathbb{G}$ is tight and takes its values in $\mathbb{D}_0$. Then (3.10.10) holds outer almost surely.*

The uniform Hadamard differentiability needed for the preceding theorem is generally satisfied by functionals on finite-dimensional spaces, but can easily fail in infinite dimensions. An alternative is to use Fréchet differentiability. It suffices to

consider Fréchet differentiability at a fixed point, but we need control over the order of the remainder term. Assume that

$$\phi(P + h) - \phi(P) = \phi'_P(h) + O(q(\|h\|)), \quad h \rightarrow 0,$$

for a continuous, linear map $\phi'_P: \mathbb{D} \rightarrow \mathbb{E}$ and a given monotone function q with (at least) $q(t) = o(t)$. The extra control over the remainder term can be exploited if there is also additional information on the almost-sure behavior of $\|\mathbb{P}_n - P\|$. Assume that

$$(3.10.14) \quad \limsup_{n \rightarrow \infty} \frac{\sqrt{n} \|\mathbb{P}_n - P\|^*}{b_n} \leq 1 \quad \text{a.s.,}$$

for some sequence $b_n \rightarrow \infty$. For $\mathbb{P}_n$ equal to the empirical distribution function and $b_n = \sqrt{2 \log \log n} \|P(f - Pf)^2\|_{\mathcal{F}}^{1/2}$, this follows from the law of the iterated logarithm, which asserts equality in the preceding display. This is well known to be valid for a Donsker class $\mathcal{F}$ with square-integrable envelope function.[‡] The following theorem requires that $b_n \rightarrow \infty$ sufficiently slowly that $\sqrt{n} q(cb_n/\sqrt{n}) \rightarrow 0$ for some $c > 1$. For $b_n \asymp \sqrt{\log \log n}$, this condition is satisfied already for $q(t) = t/(\log \log(1/t))^r$ and $r > 1/2$.

3.10.15 Theorem (Conditional delta-method almost surely). *Let $\mathbb{D}$ and $\mathbb{E}$ be normed spaces. Suppose $\phi: \mathbb{D}_\phi \subset \mathbb{D} \rightarrow \mathbb{E}$ is Fréchet-differentiable at P with rate function q . Let $\mathbb{P}_n$ and $\hat{\mathbb{P}}_n$ be maps as indicated previously with values in $\mathbb{D}_\phi$ such that $\sqrt{n}(\mathbb{P}_n - P) \rightsquigarrow \mathbb{G}$, and (3.10.9) holds outer almost surely, where $\mathbb{G}$ takes its values in a separable subspace $\mathbb{D}_0$. Furthermore let (3.10.14) hold for a sequence b_n such that $b_n = o(\sqrt{n})$ and $\sqrt{n} q(cb_n/\sqrt{n}) \rightarrow 0$ for some $c > 1$. Then (3.10.10) holds outer almost surely.*

Proofs. Without loss of generality assume, that the derivative $\phi'_P: \mathbb{D} \rightarrow \mathbb{E}$ is defined and continuous on the whole space. (By assumption it is defined and continuous on a subspace $\mathbb{D}_0$ of $\mathbb{D}$. We can always embed $\mathbb{E}$ isometrically into another Banach space $\tilde{\mathbb{E}}$ for which an extension $\phi'_P: \mathbb{D} \rightarrow \tilde{\mathbb{E}}$ exists, as shown in Problem 3.10.18. Every bounded Lipschitz function $h \in BL_1(\mathbb{E})$ has an extension to a function $h: \tilde{\mathbb{E}} \rightarrow \mathbb{R}$ of the same bounded Lipschitz norm,^b and so the result with ϕ and ϕ'_P considered maps into $\tilde{\mathbb{E}}$ implies the result for $\mathbb{E}$.) For every $h \in BL_1(\mathbb{E})$, the function $h \circ \phi'_P$ is contained in $BL_{\|\phi'_P\|}(\mathbb{D})$. Thus (3.10.9) implies

$$\sup_{h \in BL_1(\mathbb{E})} \left| E_M h \left(\phi'_P(\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n)) \right) - E h(\phi'_P(\mathbb{G})) \right| \rightarrow 0,$$

in outer probability or outer almost surely, corresponding to which of the two assumptions is made in (3.10.9). Next

$$(3.10.16) \quad \begin{aligned} & \sup_{h \in BL_1(\mathbb{E})} \left| E_M h \left(\sqrt{n}(\phi(\hat{\mathbb{P}}_n) - \phi(\mathbb{P}_n)) \right) - E_M h \left(\phi'_P(\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n)) \right) \right| \\ & \leq \varepsilon + 2P_M \left(\left\| \sqrt{n}(\phi(\hat{\mathbb{P}}_n) - \phi(\mathbb{P}_n)) - \phi'_P(\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n)) \right\|^* > \varepsilon \right). \end{aligned}$$

[‡] Cf. Dudley and Philipp (1983), Theorem 1.3, together with Kuelbs (1976); see also Ledoux and Talagrand (1991).

^b See Dudley (1993), Proposition 11.2.3.

The three theorems are proved once it has been shown that the conditional probability on the right converges to zero in outer probability (Theorem 3.10.11) or outer almost surely (Theorem 3.10.13 and 3.10.15).

Both sequences $\sqrt{n}(\mathbb{P}_n - P)$ and $\sqrt{n}(\hat{\mathbb{P}}_n - P)$ converge (unconditionally) in distribution to separable random elements that concentrate on the space $\mathbb{D}_0$. By Theorem 3.10.4,

$$\begin{aligned}\sqrt{n}(\phi(\hat{\mathbb{P}}_n) - \phi(P)) &= \phi'_P(\sqrt{n}(\hat{\mathbb{P}}_n - P)) + o_P^*(1), \\ \sqrt{n}(\phi(\mathbb{P}_n) - \phi(P)) &= \phi'_P(\sqrt{n}(\mathbb{P}_n - P)) + o_P^*(1).\end{aligned}$$

Subtract these equations to conclude that the sequence $\sqrt{n}(\phi(\hat{\mathbb{P}}_n) - \phi(\mathbb{P}_n)) - \phi'_P(\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n))$ converges (unconditionally) to zero in outer probability. Thus, the conditional probability on the right in (3.10.16) converges to zero in outer mean. This concludes the proof of the first theorem.

For the proof of the second theorem, fix $\varepsilon > 0$ and choose a compact set $K \subset \mathbb{D}_0$ such that $P(\mathbb{G} \notin K) < \varepsilon$. By the uniform Hadamard differentiability of ϕ , there exist $\delta, \eta > 0$ such that for every $P' \in \mathbb{D}_\phi$, $\|P' - P\| < \eta$, $t < \eta$, $P' + th \in \mathbb{D}_\phi$, and $h \in K^\delta$:

$$\left\| \frac{\phi(P' + th) - \phi(P')}{t} - \phi'_P(h) \right\| < \varepsilon.$$

Consequently, for $n \geq 1/\eta^2$, the right side of (3.10.16) is bounded by

$$\varepsilon + 2E_M 1_{\mathbb{D}-K^\delta}(\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n))^* + \{\|\mathbb{P}_n - P\| \geq \eta\}^*.$$

The last term converges to zero almost surely by assumption. The function $h(z) = \delta^{-1}(d(z, K) \wedge \delta)$ is bounded and Lipschitz and satisfies $1_{\mathbb{D}-K^\delta} \leq h \leq 1_{\mathbb{D}-K}$. Hence the conditional expectation in the middle term is bounded by

$$E_M h(\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n))^* \xrightarrow{\text{as}} Eh(\mathbb{G}) \leq P(\mathbb{G} \notin K) < \varepsilon.$$

Thus, the conditional probability in the right side of (3.10.16) converges to zero almost surely. This concludes the proof of the second theorem.

If $\text{rem}(h) = \phi(P + h) - \phi(P) - \phi'_P(h)$, then the norm in the right side of (3.10.16) can be written $\sqrt{n}\|\text{rem}(\hat{\mathbb{P}}_n - P) - \text{rem}(\mathbb{P}_n - P)\|$ and can be bounded above by the triangle inequality. By assumption, there exist $K, \eta > 0$ such that $\text{rem}(h) \leq Kq(\|h\|)$ for every $\|h\| \leq \eta$. Thus

$$\begin{aligned}P_M\left(\sqrt{n}\|\text{rem}(\mathbb{P}_n - P)\| > \varepsilon\right)^* \\ \leq \{\|\mathbb{P}_n - P\|^* > \eta\} + \left\{\sqrt{n}q(\|\mathbb{P}_n - P\|) > \frac{\varepsilon}{K}\right\}^*.\end{aligned}$$

This converges to zero almost surely by assumption. Next, for $\eta_n/\sqrt{n} + \eta/2 < \eta$, the triangle inequality and monotonicity of q yield

$$\begin{aligned}P_M\left(\sqrt{n}\|\text{rem}(\hat{\mathbb{P}}_n - P)\|^* > \varepsilon\right) &\leq P_M\left(\sqrt{n}\|\hat{\mathbb{P}}_n - \mathbb{P}_n\|^* > \eta_n\right) \\ &+ \left\{\|\mathbb{P}_n - P\|^* > \frac{\eta}{2}\right\} + \left\{\sqrt{n}q(\eta_n/\sqrt{n} + \|\mathbb{P}_n - P\|) > \frac{\varepsilon}{K}\right\}^*.\end{aligned}$$

The first two terms on the right converge to zero almost surely for every $\eta_n \rightarrow \infty$ and $\eta > 0$. The third term converges to zero for $\eta_n = o(b_n)$. ■

3.10.4 Random Centering

The conditional delta-method concerned variables $\sqrt{n}(\phi(\hat{\mathbb{P}}_n) - \phi(\mathbb{P}_n))$ centered at random variables $\phi(\mathbb{P}_n)$, but where the centering variables tend (rapidly) to deterministic limits. If the centering variables remain stochastic, the weak convergence of the sequence $\sqrt{n}(\hat{\mathbb{P}}_n - \mathbb{P}_n)$ is not enough for a delta-method. The following theorem shows that stable convergence, as defined in Chapter 1.15, does work.

Suppose we wish to prove the weak convergence of the sequence $\sqrt{n}(\phi(X_n) - \phi(\Theta))$, for random variables Θ defined on the same, common probability space as the X_n .

3.10.17 Theorem (Stable delta-method). *For separable normed spaces $\mathbb{D}$ and $\mathbb{E}$ and constants $r_n \rightarrow \infty$, let $\phi: \mathbb{D}_\phi \subset \mathbb{D} \rightarrow \mathbb{E}$ be maps with domain a linear subspace $\mathbb{D}_\phi$ such that*

$$r_n(\phi(\theta_n + r_n^{-1}h_n) - \phi(\theta_n)) \rightarrow \phi'_\theta(h),$$

for every sequence $h_n \rightarrow h \in \mathbb{D}_0$ and $\theta_n \rightarrow \theta \in \mathbb{D}_\phi$. If X_n and Θ are random elements in $\mathbb{D}_\phi$ defined on the same probability space such that $r_n(X_n - \Theta)$ converges $\mathcal{A}_0$ -stably to X and Θ is $\mathcal{A}_0$ -measurable, then the sequence $r_n(\phi(X_n) - \phi(\Theta))$ converges $\mathcal{A}_0$ -stably to $\phi'_\Theta(X)$ in $\mathbb{E}$.

Proof. The maps defined by $g_n(h, \theta) = r_n(\phi(\theta + r_n^{-1}h) - \phi(\theta))$ satisfy $g_n(h_n, \theta_n) \rightarrow \phi'_\theta(h)$, for every $h_n \rightarrow h \in \mathbb{D}_0$ and every $\theta_n \rightarrow \theta \in \mathbb{D}_\phi$. By Theorem 1.15.5 and stable convergence, the sequence $(\sqrt{n}(X_n - \Theta), \Theta)$ converges in distribution to the variable (X, Θ) , which takes its values in $\mathbb{D}_0 \times \mathbb{D}_\phi$, by assumption. Thus $g_n(\sqrt{n}(X_n - \Theta), \Theta)$ converges in distribution to $\phi'_\Theta(X)$, by the refined continuous mapping theorem, Theorem 1.11.1. ■

3.10.5 Examples

Throughout this section, $D[a, b]$ is the Banach space of all cadlag functions $z: [a, b] \rightarrow \mathbb{R}$ on an interval $[a, b] \subset \bar{\mathbb{R}}$ equipped with the uniform norm. The notation $\int |dA|$ is used for the total variation of a function A , and $BV_M[a, b]$ is the set of all cadlag functions of total variation bounded by M . Product spaces, such as $D[a, b] \times D[a, b]$, are always equipped with a product norm. Given an arbitrary set $\mathcal{X}$ and Banach space $\mathcal{Y}$, the Banach space $\ell^\infty(\mathcal{X}, \mathcal{Y})$ is the set of all maps $z: \mathcal{X} \rightarrow \mathcal{Y}$ that are uniformly norm-bounded equipped with the norm $\|z\| = \sup_x \|z(x)\|_{\mathcal{Y}}$.

3.10.5.1 Wilcoxon Statistic

Given a cadlag function A and a function of bounded variation B on an interval $[a, b] \subset \bar{\mathbb{R}}$, define

$$\phi(A, B) = \int_{(a, b]} A dB \quad \text{and} \quad \psi(A, B)(t) = \int_{(a, t]} A dB.$$

These maps are Hadamard-differentiable if the domain is restricted to pairs (A, B) such that B is of total variation bounded by some fixed constant.

3.10.18 Lemma. For each fixed M , the maps $\phi: D[a, b] \times BV_M[a, b] \rightarrow \mathbb{R}$ and $\psi: D[a, b] \times BV_M[a, b] \rightarrow D[a, b]$ are Hadamard-differentiable at each $(A, B) \in D_\phi$ such that $\int |dA| < \infty$. The derivatives are given by

$$\phi'_{A,B}(\alpha, \beta) = \int A d\beta + \int \alpha dB; \quad \psi'_{A,B}(\alpha, \beta)(t) = \int_{(a,t]} A d\beta + \int_{(a,t]} \alpha dB,$$

where $\int A d\beta$ is defined via integration by parts[‡] if β is not of bounded variation.

Proof. For $\alpha_t \rightarrow \alpha$ and $\beta_t \rightarrow \beta$, define $A_t = A + t\alpha_t$ and $B_t = B + t\beta_t$. By convention, we consider only perturbations such that (A_t, B_t) is contained in the given domain. In particular, the variation of B_t is bounded by M . Write

$$\frac{\int A_t dB_t - \int A dB}{t} - \phi'_{A,B}(\alpha_t, \beta_t) = \int \alpha d(B_t - B) + \int (\alpha_t - \alpha) d(B_t - B).$$

Since A is of bounded variation, the derivative map as stated is continuous (with respect to the uniform norm). It suffices to show that the expression in the display converges to zero. The second term on the right is bounded in absolute value by $\|\alpha_t - \alpha\|_\infty 2M$ and hence converges to zero. Since α is cadlag on $[a, b]$, there exists a partition $a = t_0 < t_1 < \dots < t_m = b$ such that α varies less than ε on each interval $[t_{i-1}, t_i)$. Let $\tilde{\alpha}$ be the discretization that is constant and equal to $\alpha(t_{i-1})$ on $[t_{i-1}, t_i)$. Then

$$\begin{aligned} \left| \int \alpha d(B_t - B) \right| &\leq \|\alpha - \tilde{\alpha}\|_\infty 2M + \sum_{i=1}^m |\alpha(t_{i-1})| |(B_t - B)[t_{i-1}, t_i)| \\ &\quad + |\alpha(b)| |(B_t - B)\{b\}|. \end{aligned}$$

The first term on the right can be made arbitrarily small by the choice of ε . The second term is bounded by $(2m+1)\|B_t - B\|_\infty \|\alpha\|_\infty$ and hence converges to zero for every fixed partition.

The proof for ψ is basically the same as that for ϕ . ■

The preceding lemma has many corollaries. Here we discuss two simple statistical consequences.

3.10.19 Example (Wilcoxon statistic). Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be independent samples from distribution functions F and G on the real line, respectively. If $\mathbb{F}_m$ and $\mathbb{G}_n$ are the empirical distribution functions of the two samples, then

$$\phi(\mathbb{F}_m, \mathbb{G}_n) = \int \mathbb{F}_m d\mathbb{G}_n$$

is the Mann-Whitney form of the Wilcoxon statistic. It is an estimator of $\phi(F, G) = \int F dG = P(X \leq Y)$. As a consequence of the differentiability of ϕ , if $m/(m +$

[‡] Thus $\int A d\beta = (A\beta)(b) - (A\beta)(a) - \int (\beta-) dA$.

$n) \rightarrow \lambda$, then

$$\begin{aligned} \sqrt{\frac{mn}{m+n}} \left(\int \mathbb{F}_m d\mathbb{G}_n - \int F dG \right) &\rightsquigarrow \phi'_{F,G}(\sqrt{1-\lambda}\mathbb{G}_F, \sqrt{\lambda}\mathbb{G}_G) \\ &= \sqrt{\lambda} \int F d\mathbb{G}_G + \sqrt{1-\lambda} \int \mathbb{G}_F dG, \end{aligned}$$

where $\mathbb{G}_F$ and $\mathbb{G}_G$ are independent tight, F - and G -Brownian bridge processes. The limit variable is zero-mean normally distributed with variance $\lambda \text{var } F(Y) + (1-\lambda) \text{var } G(X)$. An elementary way to see this is to use the stronger assertion of Theorem 3.10.4, which gives the weak representation

$$\begin{aligned} \sqrt{\frac{mn}{m+n}} \left(\int \mathbb{F}_m d\mathbb{G}_n - \int F dG \right) \\ = \sqrt{\lambda} \int F d\mathbb{G}_{n,G} + \sqrt{1-\lambda} \int \mathbb{G}_{m,F} dG + o_P(1). \end{aligned}$$

The variable on the right is asymptotically normal by the central limit theorem and Slutsky's lemma. (Also see Problems 3.10.2 and 3.10.3.)

3.10.20 Example (Nelson–Aalen). The Nelson-Aalen estimator of a cumulative hazard function based on censored data is a Hadamard-differentiable map of the empirical distribution function of the observations. Let $X_1, \dots, X_n$ be i.i.d. “failure times” distributed according to the distribution function F and let $C_1, \dots, C_n$ be i.i.d. “censoring times” distributed according to the distribution function G . Failure times and censoring times are assumed independent. Observed are the pairs $(Z_1, \Delta_1), \dots, (Z_n, \Delta_n)$, where $Z_i = X_i \wedge C_i$, and $\Delta_i = 1\{X_i \leq C_i\}$ indicates whether a failure time is censored or not. By definition, the cumulative hazard function of the failure times is

$$\Lambda(t) = \int_{[0,t]} \frac{1}{\bar{F}} dF = \int_{[0,t]} \frac{1}{\bar{H}} dH^{uc},$$

where $\bar{F}(t) = P(X \geq t)$ and $\bar{H}(t) = P(Z \geq t)$ are (left-continuous) survival distributions, and $H^{uc}(t) = P(Z \leq t, \Delta = 1)$ is the subdistribution function of the uncensored observations. The *Nelson-Aalen estimator* is given by

$$\Lambda_n(t) = \int_{[0,t]} \frac{1}{\bar{\mathbb{H}}_n} d\mathbb{H}_n^{uc},$$

where

$$\mathbb{H}_n^{uc}(t) = \frac{1}{n} \sum_{i=1}^n \Delta_i 1\{Z_i \leq t\} \quad \text{and} \quad \bar{\mathbb{H}}_n(t) = \frac{1}{n} \sum_{i=1}^n 1\{Z_i \geq t\}$$

are the empirical subdistribution functions of the uncensored failure times and survival function of the observation times, respectively. The Nelson-Aalen estimator depends on the pair $(\mathbb{H}_n^{uc}, \bar{\mathbb{H}}_n)$ through the two maps

$$(A, B) \mapsto \left(A, \frac{1}{B}\right) \mapsto \int_{[0,t]} \frac{1}{B} dA.$$

For given $M, \varepsilon > 0$, the composition map is Hadamard-differentiable on the domain $\{(A, B): \int |dA| \leq M, B \geq \varepsilon\}$ at every point (A, B) such that $1/B$ is of bounded variation. If t is restricted to an interval $[0, \tau]$ such that $H(\tau) < 1$, then the pair $(\mathbb{H}_n^{uc}, \bar{\mathbb{H}}_n)$ is contained in this domain with probability tending to 1 for $M \geq 1$ and sufficiently small ε . The derivative map is given by $(\alpha, \beta) \mapsto \int (1/B) d\alpha - \int (\beta/B^2) dA$.

The pair $(\mathbb{H}_n^{uc}, \bar{\mathbb{H}}_n)$ can be identified with the empirical distribution of the observations indexed by the functions $\delta 1\{z \leq t\}$ and $1\{z > t\}$, when t ranges over $\mathbb{R}$. Any of the main empirical central limit theorems yields

$$\sqrt{n}(\mathbb{H}_n^{uc} - H^{uc}, \bar{\mathbb{H}}_n - \bar{H}) \rightsquigarrow (\mathbb{G}^{uc}, \bar{\mathbb{G}}), \quad \text{in } (D[0, \tau])^2,$$

where $(\mathbb{G}^{uc}, \bar{\mathbb{G}})$ is a tight, zero-mean Gaussian process with covariance structure

$$\begin{aligned} E\mathbb{G}^{uc}(s)\mathbb{G}^{uc}(t) &= H^{uc}(s \wedge t) - H^{uc}(s)H^{uc}(t), \\ E\bar{\mathbb{G}}(s)\bar{\mathbb{G}}(t) &= \bar{H}(s \vee t) - \bar{H}(s)\bar{H}(t), \\ E\mathbb{G}^{uc}(s)\bar{\mathbb{G}}(t) &= (H^{uc}(s) - H^{uc}(t-))1\{t \leq s\} - H^{uc}(s)\bar{H}(t). \end{aligned}$$

By the delta-method we conclude that

$$\sqrt{n}(\Lambda_n - \Lambda) \rightsquigarrow \int_{[0, t]} \frac{1}{\bar{H}} d\mathbb{G}^{uc} - \int_{[0, t]} \frac{\bar{\mathbb{G}}}{\bar{H}^2} dH^{uc},$$

where the first term on the right is to be understood via partial integration. The covariance function of the Gaussian limit may be evaluated by martingale calculus.[†]

The process $\mathbb{M}^{uc}(t) = \mathbb{G}^{uc}(t) - \int_{[0, t]} \bar{\mathbb{G}} d\Lambda$ is a zero-mean Gaussian martingale with covariance function

$$E\mathbb{M}^{uc}(s)\mathbb{M}^{uc}(t) = \int_{[0, s \wedge t]} \bar{H}(1 - \Delta\Lambda) d\Lambda.$$

The limit variable can be expressed as the stochastic integral $\int_{[0, t]} 1/\bar{H} d\mathbb{M}^{uc}$ and therefore is a martingale. It is distributed as $\mathbb{Z}(C)$ for a standard Brownian motion $\mathbb{Z}$ and the function C given by

$$C(t) = \int_{[0, t]} \frac{(1 - \Delta\Lambda)}{\bar{H}} d\Lambda.$$

The final conclusion is that the sequence $\sqrt{n}(\Lambda_n - \Lambda)$ converges in distribution to $\mathbb{Z}(C)$ in $D[0, \tau]$ for every τ such that $H(\tau) < 1$.

[†] See, for instance, Revuz and Yor (1994), Chapter 4, Section 2, page 138.

3.10.5.2 Inverse Map

For a nondecreasing function $A \in D[a, b]$ and fixed $p \in \mathbb{R}$, let $\phi(A) \in [a, b]$ be an arbitrary point in $[a, b]$ such that

$$A(\phi(A)-) \leq p \leq A(\phi(A)).$$

The natural domain $\mathbb{D}_\phi$ of the resulting map ϕ is the set of all nondecreasing A such that there exists a solution to the pair of inequalities. If there exists more than one solution, then the precise choice of $\phi(A)$ is irrelevant. For instance, $\phi(A)$ may be taken equal to $\inf\{t: A(t) \geq p\}$.

3.10.21 Lemma. *Let $A \in \mathbb{D}_\phi$ be differentiable at a point $\xi_p \in (a, b)$ such that $A(\xi_p) = p$, with strictly positive derivative. Then $\phi: \mathbb{D}_\phi \subset D[a, b] \rightarrow \mathbb{R}$ is Hadamard-differentiable at A tangentially to the set of functions $\alpha \in D[a, b]$ that are continuous at ξ_p . The derivative is given by $\phi'_A(\alpha) = -\alpha(\xi_p)/A'(\xi_p)$.*

Proof. Let $\alpha_t \rightarrow \alpha$ uniformly on $[a, b]$ for a function α that is continuous at ξ_p . Write ξ_{pt} for $\phi(A + t\alpha_t)$. By the definition of ϕ , we have

$$(A + t\alpha_t)(\xi_{pt} - \varepsilon_t) \leq p \leq (A + t\alpha_t)(\xi_{pt}),$$

for every $\varepsilon_t > 0$. Choose ε_t positive and such that $\varepsilon_t = o(t)$. Since α_t converges uniformly to a bounded function, the sequence α_t is uniformly bounded. Conclude that $A(\xi_{pt} - \varepsilon_t) + O(t) \leq p \leq A(\xi_{pt}) + O(t)$. By assumption, the function A is monotone and bounded away from p outside any interval $(\xi_p - \varepsilon, \xi_p + \varepsilon)$ around ξ_p . To satisfy the preceding inequalities, the numbers ξ_{pt} must be to the right of $\xi_p - \varepsilon$ eventually, and the numbers $\xi_{pt} - \varepsilon_t$ must be to the left of $\xi_p + \varepsilon$ eventually. In other words, $\xi_{pt} \rightarrow \xi_p$.

By the uniform convergence of α_t and the continuity of the limit, $\alpha_t(\xi_{pt} - \varepsilon_t) \rightarrow \alpha(\xi_p)$ for every $\varepsilon_t \rightarrow 0$. Using this and Taylor's formula on the preceding display yields

$$\begin{aligned} p + (\xi_{pt} - \xi_p)A'(\xi_p) - o(\xi_{pt} - \xi_p) + O(\varepsilon_t) + t\alpha(\xi_p) - o(t) \\ \leq p \leq p + (\xi_{pt} - \xi_p)A'(\xi_p) + o(\xi_{pt} - \xi_p) + O(\varepsilon_t) + t\alpha(\xi_p) + o(t). \end{aligned}$$

Conclude first that $\xi_{pt} - \xi_p = O(t)$. Next, use this to replace the $o(\xi_{pt} - \xi_p)$ in the display by $o(t)$ -terms and conclude that $(\xi_{pt} - \xi_p)/t \rightarrow -(\alpha/A')(\xi_p)$. ■

3.10.22 Example (Empirical quantiles). Fix $0 < p < 1$ and consider the map that assigns to each cumulative distribution function its p th quantile $F^{-1}(p) = \inf\{x: F(x) \geq p\}$. This map is Hadamard-differentiable at every cumulative distribution F that is differentiable at $F^{-1}(p)$ with strictly positive derivative $f(F^{-1}(p))$, tangentially to the set of functions that are continuous at $F^{-1}(p)$.

If $\mathbb{F}_n$ is the empirical distribution function of an i.i.d. sample of size n from F , then the sequence $\sqrt{n}(\mathbb{F}_n - F)$ converges in $D(\mathbb{R})$ to the process $\mathbb{G} \circ F$, where $\mathbb{G}$ is a standard Brownian bridge. Since F is continuous at $F^{-1}(p)$, almost all the sample

paths of this process are continuous at the point $F^{-1}(p)$. Thus, the delta-method yields

$$\sqrt{n}(\mathbb{F}_n^{-1}(p) - F^{-1}(p)) \rightsquigarrow \phi'_F(\mathbb{G} \circ F) = -\frac{\mathbb{G}(p)}{f(F^{-1}(p))}.$$

This variable on the right is zero-mean normally distributed with its variance given by $p(1-p)/f^2(F^{-1}(p))$.

3.10.23 Example. While the delta-method yields an elegant proof of the asymptotic normality of the empirical quantile function, this result can easily be obtained directly. The real attraction of the delta-method is that it separates the analysis and the probability and can be applied without much work in a variety of situations. In the preceding derivation, the empirical distribution can be replaced by any estimator $\tilde{F}_n$ such that the sequence $\sqrt{n}(\tilde{F}_n - F)$ converges weakly in $D[a, b]$ for an interval $[a, b]$ that contains $F^{-1}(p)$ as an interior point. The estimator $\tilde{F}_n$ need not be based on an i.i.d. sample, and F may be an arbitrary centering function.

Two concrete applications are the inverse of the Nelson-Aalen estimator and the quantiles of the Kaplan-Meier estimator (see Examples 3.10.20 and 3.10.33).

Next consider the inverse function $\phi(F) = F^{-1}$ as a map from the set of distribution functions into the space $\ell^\infty(p, q)$ for given $0 < p < q < 1$. For definiteness, let F^{-1} denote the left-continuous inverse $F^{-1}(p) = \inf\{t: F(t) \geq p\}$. Given an interval $[a, b] \subset \mathbb{R}$ let $\mathbb{D}_1$ be the set of all restrictions of distribution functions on $\mathbb{R}$ to $[a, b]$ and let $\mathbb{D}_2$ be the subset of $\mathbb{D}_1$ of distribution functions of measures that concentrate on $(a, b]$.

3.10.24 Lemma.

- (i) Let $0 < p < q < 1$, and let F be continuously differentiable on the interval $[a, b] = [F^{-1}(p) - \varepsilon, F^{-1}(q) + \varepsilon]$ for some $\varepsilon > 0$, with strictly positive derivative f . Then the inverse map $G \mapsto G^{-1}$ as a map $\mathbb{D}_1 \subset D[a, b] \rightarrow \ell^\infty[p, q]$ is Hadamard-differentiable at F tangentially to $C[a, b]$.
- (ii) Let F have compact support $[a, b]$ and be continuously differentiable on its support with strictly positive derivative f . Then the inverse map $G \mapsto G^{-1}$ as a map $\mathbb{D}_2 \subset D[a, b] \rightarrow \ell^\infty(0, 1)$ is Hadamard-differentiable at F tangentially to $C[a, b]$.

In both cases the derivative is the map $\alpha \mapsto -(\alpha/f) \circ F^{-1}$.

Proof. The proof follows the same pattern as the proof for the preceding lemma, but with care to keep the needed uniformity in p . We give the proof of (ii) only.

Let $\alpha_t \rightarrow \alpha$ uniformly in $D[a, b]$, where α is continuous and $F + t\alpha_t$ is contained in $\mathbb{D}_2$ for all t . Abbreviate $F^{-1}(p)$ and $(F + t\alpha_t)^{-1}(p)$ to ξ_p and ξ_{pt} , respectively. Since F and $F + t\alpha_t$ are concentrated on $(a, b]$ (by assumption), we have $a < \xi_{pt}, \xi_p \leq b$ for all $0 < p < 1$. Thus the numbers $\varepsilon_{pt} = t^2 \wedge (\xi_{pt} - a)$ are positive, whence by definition

$$(F + t\alpha_t)(\xi_{pt} - \varepsilon_{pt}) \leq p \leq (F + t\alpha_t)(\xi_{pt}).$$

By the smoothness of F we have $F(\xi_p) = p$ and $F(\xi_{pt} - \varepsilon_{pt}) = F(\xi_{pt}) + O(\varepsilon_{pt})$, uniformly in $0 < p < 1$. It follows that

$$-t\alpha(\xi_{pt}) + o(t) \leq F(\xi_{pt}) - F(\xi_p) \leq -t\alpha(\xi_{pt} - \varepsilon_{pt}) + o(t).$$

The $o(t)$ -terms are uniform in $0 < p < 1$. The far left side and the far right side are $O(t)$, while the middle is bounded above and below by a constant times $|\xi_{pt} - \xi_p|$. Conclude that $|\xi_{pt} - \xi_p| = O(t)$, uniformly in p . Next, the lemma follows by the uniform differentiability of F . ■

3.10.25 Example (Empirical quantile process). Suppose that F is a distribution function with continuous and positive derivative f on the interval $[F^{-1}(p) - \varepsilon, F^{-1}(q) + \varepsilon]$ for some $\varepsilon > 0$. The empirical distribution function $\mathbb{F}_n$ of an i.i.d. sample of size n from F satisfies $\sqrt{n}(\mathbb{F}_n - F) \rightsquigarrow \mathbb{G} \circ F$ in $D(\mathbb{R})$ for a standard Brownian bridge $\mathbb{G}$. Almost all sample paths of the limit process are continuous on the interval $[F^{-1}(p) - \varepsilon, F^{-1}(q) + \varepsilon]$. Since the inverse map is Hadamard-differentiable at F tangentially to the subspace of functions that are continuous on this interval, the delta-method yields

$$\sqrt{n}(\mathbb{F}_n^{-1} - F^{-1}) \rightsquigarrow -\frac{\mathbb{G} \circ F(F^{-1})}{f(F^{-1})}, \quad \text{in } \ell^\infty[p, q].$$

The limit process is Gaussian with zero-mean and covariance function

$$\frac{s \wedge t - st}{f(F^{-1}(s))f(F^{-1}(t))}, \quad s, t \in \mathbb{R}.$$

The second part of the preceding lemma may be used to obtain uniform convergence of the whole quantile process for distributions with compact support and strictly positive, continuous density, such as the uniform distribution.

3.10.5.3 Composition

Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a fixed map. Given an arbitrary set $\mathcal{X}$, consider the map $\phi: \ell^\infty(\mathcal{X}) \rightarrow \ell^\infty(\mathcal{X})$ given by $\phi(A)(x) = g(A(x))$. The natural domain of this map is the set of elements of $\ell^\infty(\mathcal{X})$ that take their values in the domain of g .

3.10.26 Lemma. *Let $g: (a, b) \subset \mathbb{R} \rightarrow \mathbb{R}$ be differentiable with uniformly continuous and bounded derivative, and let $\mathbb{D}_\phi = \{A \in \ell^\infty(\mathcal{X}): a < A < b\}$. Then the map $A \mapsto g \circ A$ is Hadamard-differentiable as a map $\mathbb{D}_\phi \subset \ell^\infty(\mathcal{X}) \rightarrow \ell^\infty(\mathcal{X})$ at every $A \in \mathbb{D}_\phi$. The derivative is given by $\phi'_A(\alpha) = g'(A(x))\alpha(x)$.*

3.10.27 Example. The map $A \mapsto 1/A$ is differentiable on the domain of functions that are bounded away from zero.

The proof of the preceding lemma is easy. Consider the more complicated composition map into which the function g , which is fixed in the preceding result,

is a second variable. Given maps $A: \mathcal{X} \rightarrow \mathcal{Y}$ and $B: \mathcal{Y} \rightarrow \mathcal{Z}$ define the composition map $\phi(A, B): \mathcal{X} \rightarrow \mathcal{Z}$, by

$$\phi(A, B)(x) = B \circ A(x) = B(A(x)).$$

We shall assume that $\mathcal{Y}$ and $\mathcal{Z}$ are subsets of normed spaces. If B is a uniformly norm-bounded map from $\mathcal{Y} \rightarrow \mathcal{Z}$, then $\phi(A, B)$ is a uniformly norm-bounded map from $\mathcal{X} \rightarrow \mathcal{Z}$. Consider ϕ as a map with domain $\ell^\infty(\mathcal{X}, \mathcal{Y}) \times \ell^\infty(\mathcal{Y}, \mathcal{Z})$ equipped with the norm $\|(A, B)\|_\infty = \sup_x \|A(x)\|_{\mathcal{Y}} \vee \sup_y \|B(y)\|_{\mathcal{Z}}$.

3.10.28 Lemma. *Suppose that $B: \mathcal{Y} \rightarrow \mathcal{Z}$ is Fréchet-differentiable uniformly in y in the range of A with derivatives B'_y such that $y \mapsto B'_y$ is uniformly norm-bounded.[‡] Then the composition map $\phi: \ell^\infty(\mathcal{X}, \mathcal{Y}) \times \ell^\infty(\mathcal{Y}, \mathcal{Z}) \rightarrow \ell^\infty(\mathcal{X}, \mathcal{Z})$ is Hadamard-differentiable at (A, B) tangentially to the set $\ell^\infty(\mathcal{X}, \mathcal{Y}) \times UC(\mathcal{Y}, \mathcal{Z})$. The derivative is given by*

$$\phi'_{A,B}(\alpha, \beta)(x) = \beta \circ A(x) + B'_{A(x)}(\alpha(x)), \quad x \in \mathcal{X}.$$

Proof. Let $\alpha_t \rightarrow \alpha$ and $\beta_t \rightarrow \beta$ in $\ell^\infty(\mathcal{X}, \mathcal{Y})$ and $\ell^\infty(\mathcal{Y}, \mathcal{Z})$, respectively. Write

$$\begin{aligned} & \frac{(B + t\beta_t) \circ (A + t\alpha_t) - B \circ A}{t} - \beta \circ A - B'_A(\alpha) \\ &= (\beta_t - \beta) \circ (A + t\alpha_t) + \beta(A + t\alpha_t) - \beta(A) \\ & \quad + \frac{B(A + t\alpha_t) - B(A)}{t} - B'_A(\alpha_t) + B'_A(\alpha_t - \alpha). \end{aligned}$$

The first term converges to zero in $\ell^\infty(\mathcal{X}, \mathcal{Z})$, since $\beta_t \rightarrow \beta$ uniformly in y . The second term converges to zero for every $\beta \in UC(\mathcal{Y}, \mathcal{Z})$. The third term converges to zero by uniform Fréchet differentiability. The fourth term converges to zero since the map $y \mapsto B'_y$ is norm-bounded. ■

3.10.5.4 Copula Function

Denote the marginals of a bivariate distribution function H by $F(x) = H(x, \infty)$ and $G(y) = H(\infty, y)$, and consider the map ϕ from bivariate distribution functions H on $\mathbb{R}^2$ to functions on $[0, 1]^2$ defined by

$$(3.10.29) \quad \phi(H)(u, v) = H(F^{-1}(u), G^{-1}(v)), \quad (u, v) \in [0, 1]^2.$$

Here F^{-1} and G^{-1} are the left-continuous quantile functions corresponding to F and G , respectively. We shall refer to the function $\phi(H)$ as the *pseudo copula function* corresponding to H .

By definition a *copula function* is a cumulative distribution function of a probability distribution on $[0, 1]^2$ with uniform marginals. By *Sklar's theorem* every cumulative distribution function H on $\mathbb{R}^2$ can be written as $H = C(F, G)$, for some

[‡] Differentiability at each y in a convex set plus the uniform norm continuity of the derivatives $y \mapsto B'_y$ implies uniform Fréchet differentiability. See Problem 3.10.1.

copula C . This copula is unique if and only if F and G are continuous, and in that case given by $\phi(H)$ as in (3.10.29).^b The nonuniqueness in the case of distributions with jumps arises by the necessity of smoothing out point masses to meet the requirement of uniform marginals, which in two dimensions may be achieved with different types of dependency between the coordinates. For instance, the empirical distribution $\mathbb{H}_n$ of a sample of bivariate observations is supported on a grid of points, and its copulas are created by smoothing the masses over the rectangles in the grid, besides mapping the distribution to $[0, 1]^2$ by the quantile transformations. Any dependence structure may be used for this smoothing; in fact different copulas could be used for different rectangles. While this leads to multiple copulas, the function $\phi(\mathbb{H}_n) = \mathbb{H}_n(F_n^{-1}, G_n^{-1})$ is customarily referred to as *the empirical copula*, even though it is not a copula. Consideration of this “pseudo copula” is justifiable both by the simplicity of the definition, and the nonuniqueness of the true empirical copula.

In the following lemma we assume that the true marginal distributions are continuous, so that the target of estimation, a copula, is uniquely defined. In this case various versions of estimators, including the pseudo copula (3.10.29), will typically have the same asymptotic behavior, notwithstanding the fact that in general the pseudo copula $\phi(H)$ is not even a cumulative distribution function.

The following lemma gives Hadamard differentiability on the full domain, tangentially to the set of functions that are *constant on the flats of the true marginals*. We say that a function h satisfies the latter condition if $h = h(F_+^{-1} \circ F, G_+^{-1} \circ G)$ everywhere, where $F_+^{-1}(u) = \inf\{x: F(x) > u\}$ is the right-continuous quantile function of a cumulative distribution function F on $\mathbb{R}$. These tangential restrictions seem harmless in applications, and allow significantly weaker conditions on the true copula. (In particular, strictly increasing F and G do not have flats and the corresponding restriction is empty.)

3.10.30 Lemma. *Suppose that H is a distribution function on $\mathbb{R}^2$ with continuous marginal distribution functions F and G , and copula $C: [0, 1]^2 \rightarrow [0, 1]$ whose partial derivatives $\dot{C}_1$ and $\dot{C}_2$ exist and are continuous on the sets $(0, 1) \times [0, 1]$ and $[0, 1] \times (0, 1)$, respectively. Then the map $\phi: \mathbb{D}_\phi \subset D(\mathbb{R}^2) \rightarrow \ell^\infty((0, 1)^2)$ defined by (3.10.29) on the domain $\mathbb{D}_\phi$ of all cumulative distribution functions on $\mathbb{R}^2$ whose marginals give probability one to an open interval inside the convex hulls of the supports of F and G is Hadamard-differentiable at H tangentially to the set $\mathbb{D}_0$ of all functions $h \in C(\bar{\mathbb{R}}^2)$ that are constant on the flats of F and G and satisfy $h(F^{-1}(1), G^{-1}(1)) = 0$. Its derivative in the direction h is given by*

$$\begin{aligned} \phi'_H(h)(u, v) &= h(F^{-1}(u), G^{-1}(v)) - \dot{C}_1(u, v) h(F^{-1}(u), G^{-1}(1)) \\ &\quad - \dot{C}_2(u, v) h(F^{-1}(1), G^{-1}(v)). \end{aligned}$$

Proof. First we reduce the distribution functions to the unit square by the mapping $H_t \mapsto H_t(F_+^{-1}, G_+^{-1})$, where F_+^{-1} and G_+^{-1} are the right-continuous quantile functions of the true marginals F and G . The ordinary quantile function $F^{-1}(u)$ and the right-continuous version $F_+^{-1}(u)$ differ only if F has a flat at height $F \circ F^{-1}(u)$,

^b See Nelsen (1999), Theorem 2.3.3, or Problem 3.10.13.

where the former is the left endpoint and the latter the right endpoint of the flat. Because F is assumed continuous, we have $F \circ F^{-1}(u) = u = F \circ F_+^{-1}(u)$. This and the analogous properties of G show that the true copula satisfies $C = H(F^{-1}, G^{-1}) = H(F_+^{-1}, G_+^{-1})$. In particular, at the true cumulative distribution function the reduction results in the true copula C , and this is identical to $\phi(C)$. Since $H_t(F_+^{-1}, G_+^{-1})$ has marginals $F_t \circ F_+^{-1}$ and $G_t \circ G_+^{-1}$, the pseudo copula (3.10.29) at the reduction of a general cumulative distribution function H_t is

$$\phi(H_t(F_+^{-1}, G_+^{-1})) = H_t(F_+^{-1} \circ (F_t \circ F_+^{-1})^{-1}, G_+^{-1} \circ (G_t \circ G_+^{-1})^{-1}).$$

If we can show that this is equal to $\phi(H_t) + o(t)$ if $h_t := (H_t - H)/t \rightarrow h \in \mathbb{D}_0$, then $\phi(H_t) - \phi(H) = \phi(H_t(F_+^{-1}, G_+^{-1})) - \phi(C) + o(t)$, and hence it next suffices to prove the theorem for $H = C$, where we use that $h_t(F_+^{-1}, G_+^{-1}) \rightarrow h(F_+^{-1}, G_+^{-1}) = h(F^{-1}, G^{-1})$ to find the transformed derivative. We also note that the marginals of the reduced distributions $H_t \mapsto H_t(F_+^{-1}, G_+^{-1})$ do not put mass at 0, for any $H_t \in \mathbb{D}_\phi$, by the definition of the latter set.

To see that the preceding display is equal to $\phi(H_t) + o(t)$, using the continuity of F , first rewrite the right side of the display as

$$H(F_+^{-1} \circ F \circ F_t^{-1}, G_+^{-1} \circ G \circ G_t^{-1}) + t h(F_+^{-1} \circ F \circ F_t^{-1}, G_+^{-1} \circ G \circ G_t^{-1}) + o(t).$$

The map $F_+^{-1} \circ F$ pushes a point in a flat of F to the right end of the flat. As F does not put mass on its flats, the first term does not change if $F_+^{-1} \circ F$ is replaced by the identity, which is a lower bound. Similarly $G_+^{-1} \circ G$ can be replaced by the identity, so that the first term is identical to $H(F_t^{-1}, G_t^{-1})$. For $h \in \mathbb{D}_0$ the same substitutions can be made in the second term without changing its value, by the definition of $\mathbb{D}_0$, whence the whole expression is $(H + th)(F_t^{-1}, G_t^{-1}) + o(t) = \phi(H_t) + o(t)$.

Finally we prove that ϕ on the domain $\mathbb{D}_\phi$ of cumulative distribution functions on $[0, 1]^2$ with marginals that give zero mass to the point 0 is Hadamard differentiable at C , tangentially to the set of functions $h \in C([0, 1]^2)$ with $h(1, 1) = 0$. Let U denote the uniform marginals of C . The map ϕ can be decomposed as

$$H \mapsto (H, F, G) \mapsto (H, F^{-1}, G^{-1}) \mapsto H(F^{-1}, G^{-1}).$$

- (i) The first map $\phi_1: D([0, 1]^2) \rightarrow D([0, 1]^2) \times D[0, 1] \times D[0, 1]$ is linear and continuous, and hence is Hadamard differentiable at C .
- (ii) The second map $\phi_2: D([0, 1]^2) \times \mathbb{D}_2 \times \mathbb{D}_2 \rightarrow D([0, 1]^2) \times \mathbb{D}_{-2} \times \mathbb{D}_{-2}$, for $\mathbb{D}_2$ the cumulative distribution functions on $[0, 1]$ without point mass at 0 and $\mathbb{D}_{-2}$ its image, given by $(H, F_1, F_2) \mapsto (H, F_1^{-1}, F_2^{-1})$, is Hadamard differentiable at (C, U, U) tangentially to $D([0, 1]^2) \times C[0, 1] \times C[0, 1]$ by Lemma 3.10.24(ii), with derivative $\phi'_{2,(C,U,U)}(h, g_1, g_2) = (h, -g_1, -g_2)$.
- (iii) The third map $\phi_3: \mathbb{D}_\phi \times \mathbb{D}_{-2} \times \mathbb{D}_{-2} \rightarrow \ell^\infty((0, 1)^2)$, given by $(H, Q_1, Q_2) \mapsto H(Q_1, Q_2)$ is Hadamard differentiable at (C, U, U) tangentially to $\mathbb{D}_0 \times \mathbb{E}_0 \times \mathbb{E}_0$, for $\mathbb{E}_0 = \{h \in C[0, 1]: h(0) = h(1) = 0\}$, with derivative $\phi'_{3,(C,U,U)}(h, g_1, g_2)(u, v) = \dot{C}_1(u, v)g_1(u) + \dot{C}_2(u, v)g_2(v) + h(u, v)$. This is shown in the next paragraph. (Under stronger conditions this also follows from the result of the preceding section.)

The conclusion of the lemma next follows from the chain rule, where we note that if $(H_t - C)/t \rightarrow h$, for distributions H_t with marginals without point mass at zero, implies that h vanishes on the sets $[0, 1] \times \{0\}$ and $\{0\} \times [0, 1]$. If also $h(1, 1) = 0$, then $\phi'_{2,(C,U,U)} \circ \phi'_{1,C}(h) \in D([0, 1]^2) \times \mathbb{E}_0 \times \mathbb{E}_0$, so that the tangent spaces match up correctly.

To prove the differentiability (iii), let $(h_t, g_{1,t}, g_{2,t}) \rightarrow (h, g_1, g_2) \in \mathbb{D}_0 \times \mathbb{E}_0^2$ relative to the uniform norm (and such that $(H + th_t, U + tg_{1,t}$ and $G + tg_{2,t})$ are a bivariate distribution function and quantile functions; in particular so that they assume values in $[0, 1]$). By the mean value theorem there exists $t^* = t^*(u, v) \in [0, t]$ such that

$$\begin{aligned} & t^{-1} \left((C + th_t)(u + tg_{1,t}(u), v + tg_{2,t}(v)) - C(u, v) \right) \\ &= \sum_{j=1}^2 \dot{C}_j(u + t^*g_{1,t}(u), v + t^*g_{2,t}(v)) g_{j,t}(u_j) + h_t(u + tg_{1,t}(u), v + tg_{2,t}(v)), \end{aligned}$$

where $u_1 = u$ and $u_2 = v$. By the uniform convergence of h_t to the uniformly continuous function h , the last term on the right tends to $h(u, v)$. By the assumed continuity of $\dot{C}_1$ the term $\dot{C}_1(u + t^*g_{1,t}(u), v + t^*g_{2,t}(v))$ tends to $\dot{C}_1(u, v)$ uniformly in (u, v) with $\delta \leq u \leq 1 - \delta$, while $\dot{C}_1(u + t^*g_{1,t}(u), v + t^*g_{2,t}(v))|g_{1,t}(u)|$ becomes uniformly small as $u \rightarrow 0$ or $u \rightarrow 1$, as $g_{1,t}$ tends uniformly to $g_1 \in \mathbb{E}_0$, and $\dot{C}_1(u, v)$ is uniformly bounded by 1 (as follows from the fact that C has uniform marginals). This and the analogous argument for $\dot{C}_2$, shows that the first term on the right side of the preceding display tends to $\sum_{j=1}^2 \dot{C}_j(u, v)g_j(u_j)$. ■

For a smooth distribution H the partial derivatives of the copula are given by

$$\dot{C}_1(u, v) = \frac{H_1(F^{-1}(u), G^{-1}(v))}{f(F^{-1}(u))}, \quad \dot{C}_2(u, v) = \frac{H_2(F^{-1}(u), G^{-1}(v))}{g(G^{-1}(v))}.$$

The presence of the densities in the denominator suggests ill behavior in the tails of the distribution, but in many cases a small denominator is balanced by the numerator, at least on the sets $(0, 1) \times [0, 1]$ and $[0, 1] \times (0, 1)$, where this matters (cf. Problem 3.10.15). An explanation is that a copula function is uniformly Lipschitz, due to its uniform marginals. Alternatively, one may formulate a theorem on Hadamard differentiability on a smaller domain.

3.10.31 Example (Empirical copula function). Suppose that $(X_1, Y_1), \dots, (X_n, Y_n)$ are i.i.d. vectors from a distribution function H , with continuous marginals F and G . The empirical estimator for the copula function $C = H(F^{-1}, G^{-1})$ is

$$\mathbb{C}_n(u, v) = \mathbb{H}_n(\mathbb{F}_n^{-1}(u), \mathbb{G}_n^{-1}(v)),$$

where $\mathbb{H}_n$, $\mathbb{F}_n$, and $\mathbb{G}_n$ are the joint and marginal empirical distribution functions of the observations. If C satisfies the hypotheses of the preceding lemma, then the sequence $\sqrt{n}(\mathbb{C}_n - C)$ converges in distribution in $\ell^\infty((0, 1)^2)$ to $\phi'_C(\mathbb{G}_C)$ for a tight Brownian bridge $\mathbb{G}_C$.

We may obtain this result by applying the preceding lemma directly to H and $\mathbb{H}_n$, and next simplifying the distribution of the limit variable $\phi'_H(\mathbb{G}_H)$ to $\phi'_C(\mathbb{G}_C)$. Alternatively, we may first reduce the observations (X_i, Y_i) by the probability integral transformation to the pairs $U_i = F(X_i)$ and $V_i = G(Y_i)$, and next apply the lemma to the empirical distribution of the sample $(U_1, V_1), \dots, (U_n, V_n)$ from C . The empirical pseudo copula of the latter variables can be seen to be identical to $\mathbb{C}_n$.

The limit variable can also be represented as $\mathbb{G}_C(\dot{\phi}_{u,v})$ for the functions $\dot{\phi}_{u,v}$ given by

$$\dot{\phi}_{u,v}(x, y) = 1\{x \leq u, y \leq v\} - \dot{C}_1(u, v)1\{x \leq u\} - \dot{C}_2(u, v)1\{y \leq v\}.$$

If $C(u, v) = uv$ is the uniform distribution function on the unit square, then the limit process is a “completely tucked” Brownian sheet with covariance function

$$\text{cov}(\mathbb{Z}(s, t), \mathbb{Z}(u, v)) = (s \wedge u - su)(t \wedge v - tv).$$

In general, the covariance function has a more complicated structure.

3.10.5.5 The Product Integral

For a function $A \in D(0, b]$, let $\Delta A(t) = A(t) - A(t-)$ and $A^c(t) \equiv A(t) - \sum_{0 < s \leq t} \Delta A(s)$ be the jump part and the continuous part, respectively. The *product integral* is defined as

$$\phi(A)(t) = \prod_{0 < s \leq t} (1 + dA(s)) = \prod_{0 < s \leq t} (1 + \Delta A(s)) \exp(A^c(t)).$$

The middle expression is simply notation, which is motivated by the fact that

$$\phi(A)(t) = \lim_{\max_i |t_i - t_{i-1}| \rightarrow 0} \prod_i \left(1 + (A(t_i) - A(t_{i-1}))\right)$$

if the limit is taken over partitions $0 = t_0 < t_1 < \dots < t_n = t$ with mesh-width decreasing to zero. Alternatively, the product integral can be defined in terms of a Peano series or as the unique solution of a Volterra integral equation (see Problem 3.10.5). As additional notation, we shall use

$$\phi(A)(s, t] = \prod_{s < u \leq t} (1 + dA(u)) = \frac{\phi(A)(t)}{\phi(A)(s)}, \quad s < t.$$

Here both the first and second expressions are defined by the right-hand side, although their definition in terms of the jump and continuous parts of A restricted to the interval $(s, t]$ is clear.

Some basic properties of product integrals, such as the “forward” and “backward” equation and the *Duhamel equation*, are given in the Problems and Complements section (Problems 3.10.6 and 3.10.7). The Duhamel equation is the key to proving the Hadamard differentiability of the product integral. It asserts that

$$(\phi(B) - \phi(A))(t) = \int_{(0, t]} \phi(A)(0, u) \phi(B)(u, t] (B - A)(du).$$

Integration by parts shows that this difference is bounded by a constant times $\|A - B\|_\infty$ (also see Problem 3.10.8). Thus, product integration is uniformly continuous.

3.10.32 Lemma. For fixed, finite, positive constants b and M , the product integral map $\phi: BV_M[0, b] \subset D[0, b] \rightarrow D[0, b]$ is Hadamard-differentiable with derivative

$$\phi'_A(\alpha)(t) = \int_{(0,t]} \phi(A)(0, u) \phi(A)(u, t] d\alpha(u).$$

Here the integral with respect to α is defined by integration by parts (as given in Problem 3.10.8) if α is of unbounded variation.

Proof. Set $A_n = A + t_n \alpha_n$ for a sequence $\alpha_n \rightarrow \alpha$. In view of the Duhamel equation, it suffices to show that

$$\int_{(0,t]} \phi(A)(0, u) \phi(A_n)(u, t] d\alpha_n(u) \rightarrow \int_{(0,t]} \phi(A)(0, u) \phi(A)(u, t] d\alpha(u),$$

uniformly in $0 \leq t \leq b$. In each side the error if α_n or α is replaced by a function $\tilde{\alpha}$ is bounded by a constant times $\|\alpha_n - \tilde{\alpha}\|_\infty$ or $\|\alpha - \tilde{\alpha}\|_\infty$, respectively. This follows by integration by parts. Choose a function $\tilde{\alpha}$ of bounded variation that is close to α in the uniform norm. Then it is also close to α_n for sufficiently large n , and the error is small on both sides. Conclude that it suffices to show that

$$\int_{(0,t]} \phi(A)(0, u) \phi(A_n)(u, t] d\tilde{\alpha}(u) \rightarrow \int_{(0,t]} \phi(A)(0, u) \phi(A)(u, t] d\tilde{\alpha}(u),$$

for every function of bounded variation $\tilde{\alpha}$. This follows since the product integrals $\phi(A_n)$ converge uniformly to their limit $\phi(A)$, by a second application of the Duhamel equation. ■

The product integral is important in statistics because it transforms cumulative hazard functions into the corresponding survival functions. This can be seen most conveniently from the fact that the product integral is the unique solution of the Volterra equation defined by the negative of the cumulative hazard function. The cumulative hazard function corresponding to the distribution function F is defined as

$$\Lambda(t) = \int_{[0,t]} \frac{1}{1 - F(s-)} dF(s).$$

This immediately yields

$$1 - \int_{[0,t]} (1 - F(s-)) d\Lambda(s) = 1 - F(t).$$

Thus the survival function $S = 1 - F$ solves the Volterra equation $B = 1 + \int B(s-) d(-\Lambda)(s)$ induced by $-\Lambda$. According to Problem 3.10.5,

$$1 - F(t) = \prod_{s \leq t} (1 - d\Lambda(s)).$$

The Hadamard differentiability of the product integral now implies that the asymptotic normality of estimators for the cumulative hazard function carries over to the asymptotic normality of the corresponding estimators of the survival function.

Let the estimator $\hat{\Lambda}_n$ have the property that the sequence $\sqrt{n}(\hat{\Lambda}_n - \Lambda)$ converges in distribution in $D[0, \tau]$ to a limit $\mathbb{Z}$. Let $1 - \mathbb{F}_n = \hat{\phi}(-\hat{\Lambda}_n)$ be the corresponding estimator of $1 - F = \phi(\Lambda)$. If the estimators $\hat{\Lambda}_n$ are of uniformly bounded total variation (with probability tending to 1), then the delta-method, gives in $D[0, \tau]$,

$$\begin{aligned} \sqrt{n}(\mathbb{F}_n - F) &\rightsquigarrow -\phi'_{-\Lambda}(-\mathbb{Z}) = \int_{(0,t]} S(u-) \frac{S(t)}{S(u)} d\mathbb{Z}(u) \\ &= S(t) \int_{(0,t]} \frac{1}{1 - \Delta\Lambda(u)} d\mathbb{Z}(u). \end{aligned}$$

3.10.33 Example (Kaplan–Meier). The Nelson–Aalen estimator has limit distribution $\mathbb{Z} = \mathbb{B} \circ C$ for a standard Brownian motion $\mathbb{B}$ and the function C as given in Example 3.10.20. In that case, the zero-mean Gaussian process $\phi'_{-\Lambda}(\mathbb{Z})$ in the preceding display has covariance function

$$S(s)S(t) \int_{[0, s \wedge t]} \frac{1}{(1 - \Delta\Lambda)\overline{H}} d\Lambda, \quad 0 \leq s, t \leq \tau.$$

The corresponding estimator for the survival function is the *Kaplan–Meier estimator*.

3.10.5.6 Multivariate Trimming

Throughout this section fix a number $0 < \alpha < 1/2$ and let $\mathcal{H}$ denote the collection of all closed half-spaces in $\mathbb{R}^d$. For a given probability distribution P on $\mathbb{R}^d$, define a compact, convex set by

$$K_P = \cap \{H \in \mathcal{H}: P(H) \geq 1 - \alpha\}.$$

It will be assumed that K_P contains a neighborhood of the origin. (If K_P has a nonempty interior, this can always be achieved by translation of P .) We shall be interested in the distance of the origin to the boundary of K_P . The distance in the direction $u \in S^{d-1}$ equals

$$R_P(u) = \inf\{r \geq 0: ru \notin K\}.$$

Define a functional ϕ from the probability measures $\mathcal{M}$ on $\mathbb{R}^d$ to $\ell^\infty(S^{d-1})$ by setting $\phi(P)(u)$ equal to $R_P(u)$. Under suitable conditions, this map is Hadamard-differentiable.

The set K_P at the true P is assumed to be regular in the sense that it has a unique supporting hyperplane at each of its boundary points $R_P(u)u$. Let $V_P(u)$ be the outward unit-normal vectors of these supporting hyperplanes.

3.10.34 Lemma. Let P be a probability distribution on $\mathbb{R}^d$ such that K_P is regular and the maps $u \mapsto R_P(u)$ and $u \mapsto V_P(u)$ are continuous. Furthermore, assume that $u'X$ has a uniformly bounded density $p(\cdot; u)$ such that $p(z; V_P(u))$ is positive and continuous in $u \in S^{d-1}$ and z near $R_P(u)u'V_P(u)$. Then $\phi: \mathcal{M} \subset \ell^\infty(\mathcal{H}) \rightarrow \ell^\infty(S^{d-1})$ is Hadamard-differentiable at P tangentially to $UC(\mathcal{H}, \rho_P)$ with derivative given by

$$\phi'_P(h)(u) = -\frac{h(H_P(u))}{u'V_P(u)p(u'V_P(u)R_P(u); V_P(u))},$$

for $h \in UC(\mathcal{H}, \rho_P)$ and $u \in S^{d-1}$. Here $H_P(u)$ is the half-space containing K_P whose boundary is equal to the supporting hyperplane at $R_P(u)u$.

Proof. Write $H(u, r)$ for the half-space $\{x: u'x \leq r\}$. Let $P_t = P + th_t$ be a sequence of probability measures viewed as elements of $\ell^\infty(\mathcal{H})$ such that $h_t \rightarrow h \in UC(\mathcal{H}, \rho_P)$. Let K_t be the corresponding compact, convex sets, and define for every $u \in S^{d-1}$

$$\begin{aligned} R_t(u) &= \inf\{r \geq 0: ru \notin K_t\}, \\ Q_t(u) &= \inf\{r \geq 0: P_t H(u, r) \geq 1 - \alpha\}. \end{aligned}$$

Furthermore, let $V_t(u)$ be the outward unit-normal of some supporting hyperplane to K_t at the point $R_t(u)u$. Thus

$$K_t \subset \{x: V_t(u)'x \leq R_t(u)V_t(u)'u\} = H(V_t(u), R_t(u)V_t(u)'u).$$

For $t = 0$, the supporting hyperplanes are unique by assumption. If there are more candidates, then any one will do in the following argument. For simplicity of notation, we drop the index t if $t = 0$.

The half-space $H(v, Q_t(v))$ is the minimal half-space in the direction v that contains K_t . Combining the fact that it contains K_t with the fact that $R_t(u)u \in K_t$ immediately gives

$$R_t(u)v'u \leq Q_t(v), \quad \text{for every } u, v \in S^{d-1}.$$

For the normal $v = V_t(u)$ of a supporting hyperplane at $R_t(u)u$, this inequality becomes the equality

$$R_t(u)V_t(u)'u = Q_t(V_t(u)), \quad \text{for every } u \in S^{d-1}.$$

Otherwise, the half-space $H(v, Q_t(v))$ would not be minimal, in view of the fact that $K_t \subset H(V_t(u), R_t(u)V_t(u)'u)$ by the definition of a supporting hyperplane.

The combination of the two preceding displays shows that

$$\frac{Q_t(V_t(u)) - Q(V_t(u))}{u'V_t(u)} \leq R_t(u) - R(u) \leq \frac{Q_t(V(u)) - Q(V(u))}{u'V(u)}.$$

The left and right sides divided by t will be shown to converge to the same limit.

First we show consistency of Q_t and V_t . The numbers $Q_t(u)$ are the $(1 - \alpha)$ -quantiles of the random variables $u'X_t$ if X_t is distributed according to P_t . The

assumption $P_t \rightarrow P$ in $\ell^\infty(\mathcal{H})$ shows that the distribution functions of these variables converge uniformly, uniformly in u . A standard argument shows that $Q_t \rightarrow Q$ in $\ell^\infty(S^{d-1})$ (Problem 3.10.10).

Since the density $p(z; V(u))$ of $u'X$ is bounded away from zero for z close to $Q(u)$, uniformly in u , it follows that $\sup_u PH(u, \varepsilon) < 1 - \alpha$ for $\varepsilon < \inf_u Q(u)$. Here $\inf_u Q(u)$ is positive, because K contains an interior point. Conclude that for sufficiently small t we have $\sup_u P_t H(u, \varepsilon) < 1 - \alpha$, whence K_t contains the ball of radius ε . According to Problem 3.10.11, $u'V_t(u)$ is bounded away from zero uniformly in u , for sufficiently small t . Conclude that the denominators in the preceding display are bounded away from zero and next that $R_t \rightarrow R$ uniformly in u .

For the proof that $V_t \rightarrow V$ in $\ell^\infty(S^{d-1}, \mathbb{R}^d)$, consider arbitrary sequences u_n and $t_n \rightarrow 0$. By compactness of S^{d-1} , there exists a subsequence along which $u_n \rightarrow u$ and $V_{t_n}(u_n) \rightarrow v$ for some u and v in S^{d-1} . By the definition of R and R_t , we have for every n

$$\left(\inf_u \frac{R_{t_n}(u)}{R(u)} \right) K \subset K_{t_n} \subset H(V_{t_n}(u_n), R_{t_n}(u_n) V_{t_n}(u_n)' u_n).$$

The left side approaches K and the right side $H(v, R(u)v'u)$. Conclude that $K \subset H(v, R(u)v'u)$. By regularity, it follows that $v = V(u)$, and next by continuity of V , we have $V_{t_n}(u_n) - V(u_n) \rightarrow 0$ along the subsequence.

The absolute value of the difference $P_t H(v, Q_t(v)) - (1 - \alpha)$ is bounded by the jump size of the distribution function of $v'X_t$ at the point $Q_t(v)$. By assumption, $v'X$ has no atoms and, since $v'X$ has a bounded density and h is uniformly ρ_P -continuous, $r \mapsto h(H(v, r))$ is continuous. Since $P_t = P + th_t$ we can conclude that the difference is $o(t)$. Thus, uniformly in $v \in S^{d-1}$,

$$\begin{aligned} o(t) &= P_t H(v, Q_t(v)) - PH(v, Q(v)) \\ &= PH(v, Q_t(v)) - PH(v, Q(v)) + th_t(H(v, Q_t(v))) \\ &= (p(Q(v); v) + o(1))(Q_t(v) - Q(v)) + th(H(v, Q(v))) + o(t). \end{aligned}$$

This may be rewritten as

$$\frac{Q_t(v) - Q(v)}{t u'v} = \frac{-h(H(v, Q(v))) + o(1)}{u'v(p(Q(v); v) + o(1))}.$$

Apply this with $v = V_t(u)$ and $v = V(u)$ to complete the proof. ■

3.10.35 Example. Let $X_1, \dots, X_n$ be an i.i.d. sample from the distribution P on $\mathbb{R}^d$ with empirical distribution $\mathbb{P}_n$. The collection of half-spaces is a separable Vapnik-Červonenkis class and hence a Donsker class. Thus, the empirical process $\sqrt{n}(\mathbb{P}_n - P)$ converges in distribution in $\ell^\infty(\mathcal{H})$ to a tight Brownian $\mathbb{G}_P$. Define a random compact, convex set $\mathbb{K}_n$ by

$$\mathbb{K}_n = \cap \{H \in \mathcal{H} : \mathbb{P}_n(H) \geq 1 - \alpha\}.$$

If P satisfies the hypotheses of the preceding theorem, then

$$\sqrt{n}(R_{\mathbb{P}_n} - R_P) \rightsquigarrow - \frac{\mathbb{G}_P(H_P(u))}{u'V_P(u)p(u'V_P(u)R_P(u); V_P(u))}, \quad \text{in } \ell^\infty(S^{d-1}).$$

The limit process is zero-mean Gaussian with covariance function

$$\frac{PH_P(u) \cap H_P(v) - (1 - \alpha)^2}{g(u)g(v)}, \quad u, v \in S^{d-1},$$

where $g(u) = u'V_P(u)f(R(u)u'V(u); V(u))$.

A *multivariate trimmed mean* of a probability measure P could be defined as

$$T(P) = \frac{1}{\lambda(K_P)} \int_{K_P} x \, d\lambda(x).$$

The empirical trimmed mean is obtained by replacing P by the empirical distribution $\mathbb{P}_n$. We can prove the asymptotic normality of this estimator, and more generally the asymptotic normality of the trimmed mean of estimators that converge in $\ell^\infty(\mathcal{H})$, by application of the preceding lemma. For this we write the trimmed mean in the form

$$T(P) = \frac{\int_{S^{d-1}} \int_0^{R_P(u)} ru \, J(u, r) \, dr \, du}{\int_{S^{d-1}} \int_0^{R_P(u)} J(u, r) \, dr \, du},$$

where $J(u, r)$ is the Jacobian of the transformation $x \mapsto (u, r)$ and S^{d-1} can be identified with an interval in $\mathbb{R}^{d-1}$. For instance, for $d = 2$, we can use polar coordinates and obtain

$$T(P) = \frac{\int_0^{2\pi} R_P(u)^3 / 3 \, du}{\int_0^{2\pi} R_P(u)^2 / 2 \, du}.$$

It is easy to see that the map $R_P \mapsto T(P)$ is Hadamard-differentiable. Combining this with Lemma 3.10.34 and the chain rule, we see that the map $P \mapsto T(P)$ is Hadamard-differentiable as well.

3.10.5.7 Z-Functionals

Estimators defined by equations, Z -estimators, are considered in Chapter 3.2, and their asymptotic distribution may be established by Theorem 3.3.1. Solving an estimating equation is the same as assigning a zero to a random function. In this section it is shown that the functional that assigns a zero is Hadamard-differentiable. Thus, the delta-method may be used for deriving the asymptotic distribution of Z -estimators. This route is not necessarily recommended, however; while the packaging of the argument by Hadamard differentiability is elegant, it seems to require conditions that are stronger than those of the more direct Theorem 3.3.1.

Given an arbitrary subset Θ of a Banach space and another Banach space $\mathbb{L}$, let $\ell^\infty(\Theta, \mathbb{L})$ be the Banach space of all uniformly norm-bounded functions $z: \Theta \rightarrow \mathbb{L}$. Let $Z(\Theta, \mathbb{L})$ be the subset consisting of all maps with at least one zero. Let $\phi: Z(\Theta, \mathbb{L}) \rightarrow \Theta$ be a map that assigns one of its zeros $\phi(z)$ to each element $z \in Z(\Theta, \mathbb{L})$. In case of multiple zeros the precise choice of a zero is irrelevant.

3.10.36 Lemma. Assume $\Psi: \Theta \rightarrow \mathbb{L}$ is uniformly norm-bounded, is one-to-one, possesses a θ_0 and has an inverse (with domain $\Psi(\Theta)$) that is continuous at 0. Let Ψ be Fréchet-differentiable at θ_0 with derivative $\dot{\Psi}_{\theta_0}$, which is one-to-one and continuously invertible on $\text{lin } \Theta$. Then the map $\phi: Z(\Theta, \mathbb{L}) \subset \ell^\infty(\Theta, \mathbb{L}) \rightarrow \Theta$ is Hadamard-differentiable at Ψ tangentially to the set of $z \in \ell^\infty(\Theta, \mathbb{L})$ that are continuous at θ_0 . The derivative is given by $\phi'_\Psi(z) = -\dot{\Psi}_{\theta_0}^{-1}(z(\theta_0))$.

Proof. Let $z_t \rightarrow z$ uniformly on Θ for a map $z: \Theta \rightarrow \mathbb{L}$ that is continuous at θ_0 . By definition the element $\theta_t = \phi(\Psi + tz_t)$ satisfies $\Psi(\theta_t) + tz_t(\theta_t) = 0$. Conclude that $\Psi(\theta_t) = O(t)$. Since Ψ is one-to-one and has an inverse that is continuous at zero, it follows that $\theta_t = \Psi^{-1}(\Psi(\theta_t)) \rightarrow \Psi^{-1}(0) = \theta_0$. By the Fréchet differentiability of Ψ ,

$$\liminf_{t \downarrow 0} \frac{\|\Psi(\theta_t) - \Psi(\theta_0)\|}{\|\theta_t - \theta_0\|} \geq \liminf_{t \downarrow 0} \frac{\|\dot{\Psi}_{\theta_0}(\theta_t - \theta_0)\|}{\|\theta_t - \theta_0\|} \geq \inf_{\|h\|=1} \|\dot{\Psi}_{\theta_0}(h)\|,$$

where h ranges over $\text{lin } \Theta$. Since the inverse of $\dot{\Psi}_{\theta_0}$ is continuous, the right side is positive. Conclude that there exists a positive constant c such that $\|\theta_t - \theta_0\| < c\|\Psi(\theta_t) - \Psi(\theta_0)\| = c\|tz_t(\theta_t)\|$ for all sufficiently small $t > 0$. Conclude that $\|\theta_t - \theta_0\| = O(t)$. By Fréchet differentiability,

$$\Psi(\theta_t) - \Psi(\theta_0) = \dot{\Psi}_{\theta_0}(\theta_t - \theta_0) + o(\|\theta_t - \theta_0\|).$$

The remainder term is $o(t)$. Conclude that

$$\frac{\theta_t - \theta_0}{t} = \dot{\Psi}_{\theta_0}^{-1}\left(\frac{\Psi(\theta_t) - \Psi(\theta_0)}{t} + o(1)\right) \rightarrow \dot{\Psi}_{\theta_0}^{-1}(-z(\theta_0)),$$

since $t^{-1}(\Psi(\theta_t) - \Psi(\theta_0)) = -z_t(\theta_t) \rightarrow z(\theta_0)$. ■

3.10.37 Example (Z-estimators). Let $\hat{\theta}_n$ be “estimators” solving the estimating equation $\Psi_n(\theta) = 0$ for given random criterion functions $\Psi_n: \Theta \rightarrow \mathbb{L}$. Assume that the sequence $\sqrt{n}(\Psi_n - \Psi)$ converges in distribution to a process $\mathbb{Z}$ in the space $\ell^\infty(\Theta, \mathbb{L})$. If Ψ satisfies the conditions of the preceding lemma and almost all sample paths $\theta \mapsto \mathbb{Z}(\theta)$ are continuous at θ_0 , then the delta-method yields

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \rightsquigarrow \phi'_\Psi(\mathbb{Z}) = -\dot{\Psi}_{\theta_0}^{-1}(\mathbb{Z}(\theta_0)).$$

This is the same conclusion as given by Theorem 3.3.1.

It should be noted that the uniformity in convergence with respect to Θ together with the requirement that the limit process is continuous at θ_0 implies condition (3.3.2). Thus, Theorem 3.3.1 gives the same result under weaker conditions, except that consistency of $\hat{\theta}_n$ is part of the assumptions.

For finite-dimensional θ , the maps Ψ_n will usually take their values in Euclidean space. The weak convergence (in $\ell^\infty(\Theta, \mathbb{L})$) needed for the preceding argument is then the weak convergence of each coordinate function in $\ell^\infty(\Theta)$. For $\mathbb{L} = \ell^\infty(\mathcal{F})$ for some set $\mathcal{F}$, the space $\ell^\infty(\Theta, \mathbb{L})$ can be isometrically identified with $\ell^\infty(\Theta \times \mathcal{F})$ through $z(\theta)f \leftrightarrow z(\theta, f)$, and the weak convergence can be interpreted accordingly.

Problems and Complements

1. **(Uniform Fréchet differentiability)** Let $\mathbb{D}$ and $\mathbb{E}$ be normed spaces and $\mathbb{D}_\phi \subset \mathbb{D}$ convex. Let $\phi: \mathbb{D}_\phi \rightarrow \mathbb{E}$ be Fréchet-differentiable at every $\theta \in \mathbb{D}_\phi$, where the derivatives $\theta \mapsto \phi'_\theta$ are uniformly norm-continuous. Then ϕ is uniformly Fréchet-differentiable on $\mathbb{D}_\phi$.

[Hint: The first part of the argument is the same as the argument in the proof of Lemma 3.10.7.]

2. Arbitrary distribution functions F and G on the real line satisfy the identity

$$\int \int [G(s \wedge t) - G(s)G(t)] dF(s)dF(t) = \int F^2 dG - \left(\int F dG \right)^2.$$

3. If $\mathbb{G}_G$ is a G -Brownian bridge process indexed by the half-lines $(-\infty, t]$, then

$$-\int \mathbb{G}_G dF \sim \mathbb{G}_G(F),$$

where the right side denotes a G -Brownian bridge process indexed by the single function F .

[Hint: Use the previous exercise.]

4. **(Quantile-quantile)** The *quantile-quantile transformation* $\phi(F, G) = F \circ G^{-1}$ for distribution functions F and G on $\mathbb{R}$ is Hadamard-differentiable tangentially to $D[a, b] \times C[a, b]$ as a map $D(\mathbb{R})^2 \rightarrow D[a, b]$ is Hadamard-differentiable tangentially to $D[a, b] \times C[a, b]$. This may be used to derive the limit distribution of $\mathbb{F}_n \circ \mathbb{G}_n^{-1}$ for the empirical distribution functions $\mathbb{F}_n$ and $\mathbb{G}_n$ based on independent samples from F and G . [Dudley (1992) establishes Fréchet differentiability of this map ϕ with other norms on the domain and range spaces.]

5. **(Volterra equation and Peano series)** The product integral $B = \phi(A)$ is the unique solution of the *Volterra equation*

$$B(t) = 1 + \int_{(0,t]} B(s-) dA(s).$$

This may be used to write the product integral as the *Peano series*

$$B(t) = 1 + \sum_{n=1}^{\infty} \int \dots \int_{0 < t_1 < \dots < t_n \leq t} dA(t_1) \dots dA(t_n).$$

6. **(Forward and backward equation)** The product integral $B = \phi(A)$ satisfies the forward equation

$$B(s, t] - 1 = \int_{(s,t]} B(s, u) dA(u)$$

and the backward equation

$$B(s, t] - 1 = \int_{(s,t]} dA(u) B(u, t].$$

- 7. (Duhamel equation)** Product integrals $B_1 = \phi(A_1)$ and $B_2 = \phi(A_2)$ satisfy the *Duhamel equation*

$$B_2(s, t] - B_1(s, t] = \int_{(s, t]} B_1(s, u) B_2(u, t] (A_2 - A_1)(du).$$

[Hint: Define a Borel measure $A_{1,2}$ on $\mathbb{R}_+^{m+n}$ by

$$A_{1,2}(C_1 \times \dots \times C_m \times D_1 \times \dots \times D_n) = A_1(C_1) \cdots A_1(C_m) A_2(D_1) \cdots A_2(D_n).$$

Then integrate the indicator of the set $\{s < u_1 < \dots < u_{m+n} \leq t\}$ with respect to $A_{1,2}$, apply Fubini's theorem, and sum both sides of the resulting equality over $m \geq 1$ and $n \geq 1$.]

- 8. (Integration by parts for the Duhamel equation)** Suppose that $A, B \in D(\mathbb{R}^+)$ and that $h \in D(\mathbb{R}^+)$ is of bounded variation. (When h is of unbounded variation, the left side is defined by the right side!) Then

$$\begin{aligned} & \int_{(0, t]} \prod (1 + dB)(0, u) dh(u) \prod (1 + dA)(u, t] \\ &= h(t) + \int_{(0, t]} \prod (1 + dB)(0, s) dB(s) [h(t) - h(s)] \\ & \quad + \int_{(0, t]} h(r-) dA(r) \prod (1 + A)(r, t] \\ & \quad + \int \int_{0 < s < r \leq t} \prod (1 + dB)(0, s) dB(s) [h(r-) - h(s)] \\ & \quad \times \prod (1 + A)(r, t] dA(r). \end{aligned}$$

[Hint: Use the forward and backward equations to replace product integrals on the left side, and then use Fubini's theorem.]

- 9. (Existence of zeros)** Lemma 3.10.36 takes the domain of the zero-functional equal to the set of maps that have at least one zero. This avoids some technical problems. In many situations it can be shown that the domain is open by the following result. Let $\Psi: \Theta \rightarrow \mathbb{R}^p$ be a homeomorphism of a neighborhood of $\theta_0 \in \mathbb{R}^p$ onto a neighborhood of $0 \in \mathbb{R}^p$. Then every continuous $z: \Theta \rightarrow \mathbb{R}^p$ for which $\|z - \Psi\|_\infty$ is sufficiently small has at least one zero.

[Hint: Without loss of generality the neighborhood of 0 can be chosen to be the ball $B(0, r)$. Fix $z \in C(\Theta)$ with $\|z - \Psi\|_\infty < r$. The function $x \mapsto x - z \circ \Psi^{-1}(x)$ maps the ball $B(0, r)$ into itself, since

$$\|z \circ \Psi^{-1}(x) - x\| \leq \|z - \Psi\|_\infty.$$

If z is continuous, then so is the map $x \mapsto x - z \circ \Psi^{-1}(x)$. By Brouwer's fixed-point theorem, it has a fixed point x_z . This satisfies $z(\Psi^{-1}(x_z)) = 0$.]

- 10.** Let $x \mapsto F_n(x; u)$ be distribution functions that converge uniformly in x and u to distribution functions $x \mapsto F(x; u)$. If there exist points ξ_{pu} such that for every $\delta > 0$

$$\sup_u F(\xi_{pu} - \delta; u) < p < \inf_u F(\xi_{pu} + \delta; u),$$

then $F_n^{-1}(p; u) \rightarrow F^{-1}(p; u)$ uniformly in u . The condition is satisfied if every $F(\cdot; u)$ has a density that is uniformly bounded away from zero in a neighborhood of its p th quantile ξ_{pu} .

11. For a compact, convex set $K \subset \mathbb{R}^d$, let $V(u)$ be the outward normal of a supporting hyperplane at the point $R(u)u$, where $R(u) = \inf\{r: ru \notin K\}$. If K contains the closed ball of radius ε around the origin, then $u'V(u) \geq \varepsilon / \text{diam } K$ for every $u \in S^{d-1}$.

[Hint: The convex hull of the ball of radius ε and the point $R(u)u$ is contained in K . Hence, by the definition of a supporting hyperplanes $V(u)'(\frac{1}{2}R(u)u + \frac{1}{2}\varepsilon V(u)) \leq V(u)'R(u)u$, since $\varepsilon V(u)$ is contained in the ball of radius ε .]

12. In Lemma 3.10.34 the distance ρ_P may be replaced by $d(H(u, r), H(v, s)) = \|u - v\| + |r - s|$.
13. **(Sklar's theorem)** Let H be a bivariate cumulative distribution function with marginals F and G . For (X, Y) distributed according to H and A and B possibly dependent variables such that A given X and B given Y is uniformly distributed, define $U = F(X-) + AF\{X\}$ and $V = G(Y-) + BG\{Y\}$.

- (i) The distribution function C of (U, V) is a copula of H : the variables U and V are uniformly distributed and $H = C(F, G)$.
- (ii) Any copula of H arises in this manner.
- (iii) Consequently, if F and G are continuous, then the law of $(F(X), G(Y))$ is a unique copula of H .

[Hint: Verify that $F^{-1}(F(x-) + aF\{x\}) = x$, for every $x \in \mathbb{R}$ and $a \in (0, 1]$, and conclude that $F^{-1}(U) = X$, almost surely. Similarly $F^{-1}(V) = Y$. Also, by the quantile transformation $C(F(x), G(y)) = P(F^{-1}(U) \leq x, F^{-1}(V) \leq y)$. For (ii) start with $(U, V) \sim C$ and define $X = F^{-1}(U)$, $Y = F^{-1}(V)$, and $A = (U - F(X-))/F\{X\}$ and $B = (V - G(Y-))/G\{Y\}$, and arbitrary uniform if $F\{X\} = 0$ or $G\{Y\} = 0$. It suffices to show that A given X is uniform and B given Y is uniform.]

14. Let H be a bivariate cumulative distribution function with marginals F and G , and let $U = F(X)$ and $V = G(Y)$, for (X, Y) distributed according to H .
- (i) The cumulative distribution function of (U, V) is

$$\begin{cases} H(F_+^{-1}(u), G_+^{-1}(v)), & \text{if } u \in \text{ran}(F) \text{ and } v \in \text{ran}(G), \\ H(F^{-1}(u)-, G^{-1}(v)-), & \text{if } u \notin \text{ran}(F) \text{ and } v \notin \text{ran}(G), \\ H(F^{-1}(u)-, G_+^{-1}(v)), & \text{if } u \notin \text{ran}(F) \text{ and } v \in \text{ran}(G), \\ H(F_+^{-1}(u), G^{-1}(v)-), & \text{if } u \in \text{ran}(F) \text{ and } v \notin \text{ran}(G). \end{cases}$$

- (ii) If F and G are continuous, then only the first case is relevant. In that case the expression is identical to $H(F^{-1}(u), G^{-1}(v))$.

[Hint: The left-continuous cdf has the simple expression $P(F(X) < U, G(Y) < v) = P(X < F^{-1}(u), Y < G^{-1}(v))$.]

15. **(Bivariate normal copula)** The copula of the bivariate normal distribution function on $\mathbb{R}^2$ with mean 0, variances 1, and correlation ρ satisfies the condition of Lemma 3.10.30, but is not continuously differentiable on the full unit square.

16. **(Archimedian copula)** The copula $C(u, v) = \psi^{-1}(\psi(u) + \psi(v))$, where $\psi(t) = (t^{-\theta} - 1)/\theta$, for $0 < t \leq 1$ and $\theta > 0$, satisfies the condition of Lemma 3.10.30.

17. Lemma 3.10.30 can be extended to dimension > 2 .

18. **(Extension of linear map)** Let $T: \mathbb{D}_0 \rightarrow \mathbb{E}$ be a continuous linear map defined on a linear subspace $\mathbb{D}_0 \subset \mathbb{D}$ with values in a Banach space $\mathbb{E}$. Then there exists a Banach space $\tilde{\mathbb{E}}$, an isometry $i: \mathbb{E} \rightarrow \tilde{\mathbb{E}}$ onto a subspace of $\tilde{\mathbb{E}}$ and a continuous linear map $\tilde{T}: \mathbb{D} \rightarrow \tilde{\mathbb{E}}$ of the same norm as T such that $\tilde{T}y = iTy$, for every $y \in \mathbb{D}_0$. (It is often not possible to choose $\tilde{\mathbb{E}} = \mathbb{E}$.)

[Hint: Identify $\mathbb{E}$ isometrically with a subspace of its second dual $\mathbb{E}^{**}$ through the canonical embedding $e \mapsto e^{**}$ (where $e^{**}(e^*) = e^*(e)$), and next, for $\mathbb{E}_1^*$ the unit ball of $\mathbb{E}^*$, identify $\mathbb{E}^{**}$ isometrically with a subspace of $\tilde{\mathbb{E}} := \ell^\infty(\mathbb{E}_1^*)$ by restriction of domain. This defines the map $i: \mathbb{E} \rightarrow \tilde{\mathbb{E}}$. Every map $e^*T: \mathbb{D}_0 \rightarrow \mathbb{R}$ can be extended to a continuous linear map $\overline{e^*T}: \mathbb{D} \rightarrow \mathbb{R}$ of the same norm by the Hahn-Banach theorem. Define a map $\tilde{T}: \mathbb{D} \rightarrow \tilde{\mathbb{E}}$ through $\tilde{T}y(e^*) = \overline{e^*T}(y)$. It can be verified that $\tilde{T}$ extends $iT: \mathbb{D}_0 \rightarrow \tilde{\mathbb{E}}$ and has the same norm. (Although $\tilde{T}y = iTd \in i\mathbb{E}$ for $y \in \mathbb{D}_0$, this is not necessarily true for every $y \in \mathbb{D}$.)]

3.11

Contiguity

Let P and Q be probability measures on a measurable space $(\Omega, \mathcal{A})$. If Q is absolutely continuous with respect to P , then the Q -law of a measurable map $X: \Omega \rightarrow \mathbb{D}$ can be calculated from the P -law of the pair $(X, dQ/dP)$ through the formula

$$\mathbb{E}_Q f(X) = \mathbb{E}_P f(X) \frac{dQ}{dP}.$$

With M equal to the induced P -law of the pair $(X, dQ/dP)$, this identity can also be expressed as

$$Q(X \in B) = \int_{B \times \mathbb{R}} v \, dM(x, v).$$

The validity of these formulas depends essentially on the absolute continuity of Q with respect to P . Indeed, it is clear that no P -law contains information about the part of Q that is singular with respect to P .

Consider an asymptotic version of the problem. Let $(\Omega_\alpha, \mathcal{A}_\alpha)$ be measurable spaces, each with a pair of probability measures P_α and Q_α on it. Under what conditions can a Q_α -limit law of maps $X_\alpha: \Omega_\alpha \rightarrow \mathbb{D}$ be obtained from suitable P_α -limit laws? In view of the above, it is necessary that Q_α is asymptotically absolutely continuous with respect to P_α in a suitable sense.

3.11.1 Definition. Let P_α and Q_α be nets of probability measures on measurable spaces $(\Omega_\alpha, \mathcal{A}_\alpha)$. Then Q_α is *contiguous* with respect to P_α if $P_\alpha(A_\alpha) \rightarrow 0$ along a subnet implies $Q_\alpha(A_\alpha) \rightarrow 0$ along this subnet for every choice of measurable sets A_α (and every subnet). This is denoted $Q_\alpha \triangleleft P_\alpha$. The nets P_α and Q_α are *two-sided contiguous* if both $P_\alpha \triangleleft Q_\alpha$ and $Q_\alpha \triangleleft P_\alpha$. This is denoted $P_\alpha \triangleleft\!\!\!\triangleright Q_\alpha$.

Contiguity can be characterized by the asymptotic behavior of the *likelihood ratios* of P_α and Q_α . Given two measures P and Q , write dQ/dP for the Radon-Nikodym density of the part of Q that is P -absolutely continuous. This function is only P -a.s. uniquely defined, but since only its P -law will be studied here, it is not necessary to remove the ambiguity. One way of forming dQ/dP is to take densities p and q of P and Q with respect to some measure for which there exists such densities (for instance, a σ -finite measure that dominates both, such as $P + Q$) and set $dQ/dP = q/p$.

Log likelihood ratios are always nonnegative and integrable. In particular, for nets of probability measures P_α and Q_α ,

$$\mathbb{E}_{Q_\alpha} \frac{dP_\alpha}{dQ_\alpha} \leq 1 \quad \text{and} \quad \mathbb{E}_{P_\alpha} \frac{dQ_\alpha}{dP_\alpha} \leq 1.$$

Thus, the nets of likelihood ratios dP_α/dQ_α and dQ_α/dP_α are uniformly tight as maps into $[0, \infty)$ under Q_α and P_α , respectively. The properties of their limit points determine contiguity.

3.11.2 Theorem (Contiguity). *Let P_α and Q_α be nets of probability measures on measurable spaces $(\Omega_\alpha, \mathcal{A}_\alpha)$. Then the following statements are equivalent:*

- (i) $Q_\alpha \triangleleft P_\alpha$;
- (ii) if $dP_\alpha/dQ_\alpha \overset{Q_\alpha}{\rightsquigarrow} L$ along a subnet, then $L(0, \infty) = 1$;
- (iii) if $dQ_\alpha/dP_\alpha \overset{P_\alpha}{\rightsquigarrow} V$ along a subnet, then $\mathbb{E}V = 1$;
- (iv) for any $T_\alpha: \Omega_\alpha \rightarrow \mathbb{R}$, if $T_\alpha \overset{P_\alpha}{\rightsquigarrow} 0$ along a subnet, then $T_\alpha \overset{Q_\alpha}{\rightsquigarrow} 0$ along this subnet.

Proof. The equivalence of (i) and (iv) is an easy exercise.

(i) $\Rightarrow$ (ii). Suppose, without loss of generality, that $dP_\alpha/dQ_\alpha \overset{Q_\alpha}{\rightsquigarrow} L$. Then by the portmanteau theorem, $\liminf Q_\alpha(dP_\alpha/dQ_\alpha < \varepsilon) \geq L[0, \varepsilon)$ for every $\varepsilon > 0$. For $i \in \mathbb{N}$, take α_i such that, for all $\alpha \geq \alpha_i$,

$$(3.11.3) \quad Q_\alpha \left(\frac{dP_\alpha}{dQ_\alpha} < \frac{1}{i} \right) \geq L[0, \frac{1}{2i}] - \frac{1}{i}.$$

Next, define for any α a number $\varepsilon_\alpha = \inf\{1/i: \alpha \geq \alpha_i\}$. (Let the infimum over the empty set be 2.) Then $\varepsilon_\alpha \leq 1/j$ for $\alpha \geq \alpha_j$; hence the net ε_α satisfies $\varepsilon_\alpha \rightarrow 0$. Moreover, for α sufficiently large (already for $\alpha \geq \alpha_1$), either $\varepsilon_\alpha = 1/i$ for some i or $\varepsilon_\alpha = 0$. In the first case, $\alpha \geq \alpha_i$, so that by (3.11.3),

$$(3.11.4) \quad Q_\alpha \left(\frac{dP_\alpha}{dQ_\alpha} \leq \varepsilon_\alpha \right) \geq L[0, \frac{1}{2}\varepsilon_\alpha] - \varepsilon_\alpha.$$

In the second case, (3.11.3) holds for infinitely many i ; again conclude that (3.11.4) holds. Now

$$P_\alpha \left(\frac{dP_\alpha}{dQ_\alpha} \leq \varepsilon_\alpha \wedge dQ_\alpha > 0 \right) = \int_{dP_\alpha/dQ_\alpha \leq \varepsilon_\alpha} \frac{dP_\alpha}{dQ_\alpha} dQ_\alpha \leq \int \varepsilon_\alpha dQ_\alpha \rightarrow 0.$$

If Q_α is contiguous with respect to P_α , then the Q_α -probability of the set on the left goes to zero too. Combination with (3.11.4) shows that $L\{0\} = 0$.

(iii) $\Rightarrow$ (i). Suppose $P_\alpha(A_\alpha) \rightarrow 0$ along some subnet. By Prohorov's theorem, every further subnet has a subnet along which $(dQ_\alpha/dP_\alpha, 1_{\Omega_\alpha - A_\alpha}) \rightsquigarrow (V, 1)$ under P_α for some weak limit V . Along the subnet, by the portmanteau theorem,

$$\liminf Q_\alpha(\Omega_\alpha - A_\alpha) \geq \liminf \int_{\Omega_\alpha - A_\alpha} \frac{dQ_\alpha}{dP_\alpha} dP_\alpha \geq E1 \cdot V,$$

since the continuous function $(v, t) \mapsto vt$ is bounded from below on $[0, \infty) \times \{0, 1\}$. If (iii) holds, then $EV = 1$, so $Q_\alpha(A_\alpha) \rightarrow 0$ along the subnet.

(ii) $\Leftrightarrow$ (iii). In view of Prohorov's theorem, given any subnet there is a further subnet such that along it

$$\frac{dP_\alpha}{dQ_\alpha} \xrightarrow{Q_\alpha} L \quad ; \quad \frac{dQ_\alpha}{dP_\alpha} \xrightarrow{P_\alpha} V \quad ; \quad \frac{dP_\alpha}{dP_\alpha + dQ_\alpha} \xrightarrow{P_\alpha + Q_\alpha} W,$$

for some weak limits L, V , and W , where W is a random element of total mass 2 with $EW = 1$. Assume for simplicity that this statement is true for the whole net. It must be shown that $L\{0\} = 0$ and $EV = 1$ are equivalent. By the last of the three convergences and the continuous mapping theorem,

$$\left(\frac{dP_\alpha}{dQ_\alpha}, \frac{dQ_\alpha}{dP_\alpha}, \frac{dP_\alpha}{dP_\alpha + dQ_\alpha} \right) \xrightarrow{P_\alpha + Q_\alpha} \left(\frac{W}{1 - W}, \frac{1 - W}{W}, W \right),$$

in $[0, \infty] \times [0, \infty] \times [0, 1]$ (with $1/0 = \infty$; with this convention the functions $w \mapsto w/(1 - w)$ and $w \mapsto (1 - w)/w$ are continuous from $[0, 1]$ into $[0, \infty]$). Let $f: [0, \infty] \rightarrow \mathbb{R}$ be bounded and continuous. Then so is the function $(x, v) \mapsto f(x)v$, defined on $[0, \infty] \times [0, 1]$. Consequently,

$$E_{Q_\alpha} f \left(\frac{dP_\alpha}{dQ_\alpha} \right) = E_{P_\alpha + Q_\alpha} f \left(\frac{dP_\alpha}{dQ_\alpha} \right) \frac{dQ_\alpha}{dP_\alpha + dQ_\alpha} \rightarrow Ef \left(\frac{W}{1 - W} \right) (1 - W).$$

Conclude that $L(A) = E1_A(W/(1 - W))(1 - W)$; so $L\{0\} = P(W = 0)$. By a similar argument, $Ef(V) = Ef((1 - W)/W)W$; so $EV = E((1 - W)/W)W = E(1 - W)1_{W>0} = 1 - P(W = 0)$. Hence the three statements $L\{0\} = 0$, $EV = 1$, and $P(W = 0) = 0$ are equivalent. ■

Since subnets of sequences are not necessarily subsequences, the preceding theorem is unnecessarily complicated if the directed set is equal to the natural numbers. For sequences the theorem can be simplified to the following theorem.

3.11.5 Theorem (Contiguity). *Let P_n and Q_n be sequences of probability measures on measurable spaces $(\Omega_n, \mathcal{A}_n)$. Then the following statements are equivalent:*

- (i) $Q_n \triangleleft P_n$;
- (ii) if $dP_n/dQ_n \xrightarrow{Q_n} L$ along a subsequence, then $L(0, \infty) = 1$;
- (iii) if $dQ_n/dP_n \xrightarrow{P_n} V$ along a subsequence, then $EV = 1$;
- (iv) for any $T_n: \Omega_n \rightarrow \mathbb{R}$, if $T_n \xrightarrow{P_n} 0$, then $T_n \xrightarrow{Q_n} 0$;
- (v) for any A_n in $\mathcal{A}_n$: if $P_n(A_n) \rightarrow 0$, then $Q_n(A_n) \rightarrow 0$.

Proof. The equivalence of (ii) through (v) can be shown by the same arguments as before. Furthermore, in every metric space a limit point of a sequence (possibly along a subnet) is always also the limit of a subsequence. ■

3.11.6 Example (Log-normality). Let P_α and Q_α be probability measures on arbitrary measurable spaces. Suppose

$$\frac{dP_\alpha}{dQ_\alpha} \stackrel{Q_\alpha}{\rightsquigarrow} e^{N(\mu, \sigma^2)}.$$

Then $Q_\alpha \triangleleft P_\alpha$ and $Q_\alpha \ntriangleleft P_\alpha$ if and only if $\mu = -\frac{1}{2}\sigma^2$. For the last it suffices to note that $E \exp\{N(\mu, \sigma^2)\} = 1$ if and only if $\mu = -\frac{1}{2}\sigma^2$.

The following theorem solves the problem posed in the introduction. It is a version of *Le Cam's third lemma*.

3.11.7 Theorem (Le Cam's third lemma). Let P_α and Q_α be nets of probability measures on measurable spaces $(\Omega_\alpha, \mathcal{A}_\alpha)$ and $X_\alpha: \Omega_\alpha \rightarrow \mathbb{D}$ maps with values in a metric space. Let $Q_\alpha \triangleleft P_\alpha$ and

$$\left(X_\alpha, \frac{dQ_\alpha}{dP_\alpha}\right) \stackrel{P_\alpha}{\rightsquigarrow} (X, V).$$

Then $L(B) = E 1_B(X) V$ defines a probability measure and $X_\alpha \stackrel{Q_\alpha}{\rightsquigarrow} L$. If X is tight or separable, then so is L .

Proof. Clearly $V \geq 0$ and by contiguity $EV = 1$; so L is a probability measure. It is immediate from the definition of L that $\int f dL = Ef(X) V$ for every indicator function or nonnegative f . Conclude, in steps, that the same is true for every simple function f and any integrable measurable function.

If $f: \mathbb{D} \rightarrow \mathbb{R}$ is lower semicontinuous and nonnegative, then so is the function $(t, v) \mapsto f(t) v$ on $\mathbb{D} \times [0, \infty)$. Thus by the portmanteau theorem,

$$\liminf E_{Q_\alpha, *} f(X_\alpha) \geq \liminf \int f(X_\alpha) * \frac{dQ_\alpha}{dP_\alpha} dP_\alpha \geq Ef(X) V.$$

Finally, apply the portmanteau theorem a second time, in the converse direction.

The last statement of the theorem follows immediately from the representation of L : if X concentrates on S , so does L . ■

3.11.8 Example (Le Cam's third lemma). The name *Le Cam's third lemma* is often reserved for the following special case of the previous theorem. Suppose

$$\left(X_\alpha, \log \frac{dQ_\alpha}{dP_\alpha}\right) \stackrel{P_\alpha}{\rightsquigarrow} N_{k+1} \left(\begin{pmatrix} \mu \\ -\frac{1}{2}\sigma^2 \end{pmatrix}, \begin{pmatrix} \Sigma & \tau \\ \tau' & \sigma^2 \end{pmatrix} \right).$$

Then $X_\alpha \stackrel{Q_\alpha}{\rightsquigarrow} N_k(\mu + \tau, \Sigma)$. To see this, let (X, W) have the given $(k+1)$ -dimensional normal distribution. Then $\int e^{it'x} dL(x) = E e^{it'X} e^W$ equals the characteristic function of this normal distribution at the point $(t, -i)$, that is,

$$\int e^{it'x} dL(x) = e^{it'\mu - \frac{1}{2}\sigma^2 - \frac{1}{2}(t', -i) \begin{pmatrix} \Sigma & \tau \\ \tau' & \sigma^2 \end{pmatrix} \begin{pmatrix} t \\ -i \end{pmatrix}} = e^{it'(\mu + \tau) - \frac{1}{2}t'\Sigma t}.$$

Two-sided contiguity can of course be characterized by “doubling” any of the characterizations of Theorem 3.11.2, but it is customary to write a characterization in terms of the logarithms of the likelihood ratios. If it is understood that $\log 0 = -\infty$, then

$$\log \frac{dP_\alpha}{dQ_\alpha} : \mathcal{X}_\alpha \rightarrow \bar{\mathbb{R}}$$

is a well-defined map, although it is only Q_α -a.s. unique. Similarly,

$$\log \frac{dQ_\alpha}{dP_\alpha} : \mathcal{X}_\alpha \rightarrow \bar{\mathbb{R}}$$

is P_α -a.s. uniquely defined. We are interested only in the “laws” of these maps under Q_α and P_α , respectively, so for now the ambiguity of definition need not be removed. The extended logarithm $x \mapsto \log x$ is a continuous bijection from $[0, \infty)$ onto $[-\infty, \infty)$, with a continuous inverse. Consequently, a likelihood ratio converges weakly in $[0, \infty)$ if and only if a log likelihood ratio converges weakly in $[-\infty, \infty)$ and the limits are related through taking logarithms and exponentials. This leads to the following reformulation of Theorem 3.11.2.

3.11.9 Theorem. *Let P_α and Q_α be nets of probability measures on measurable spaces $(\Omega_\alpha, \mathcal{A}_\alpha)$. Then the following statements are equivalent:*

- (i) $Q_\alpha \triangleleft P_\alpha$;
- (ii) if $\log dP_\alpha/dQ_\alpha \overset{Q_\alpha}{\rightsquigarrow} L$ and $\log dQ_\alpha/dP_\alpha \overset{P_\alpha}{\rightsquigarrow} M$ in $\bar{\mathbb{R}}$ along a subnet, then $L(\mathbb{R}) = M(\mathbb{R}) = 1$;
- (iii) if $\log dP_\alpha/dQ_\alpha \overset{Q_\alpha}{\rightsquigarrow} L$ in $\bar{\mathbb{R}}$ along a subnet, then $L(\mathbb{R}) = 1$ and $\int e^x dL(x) = 1$.

Proof. Use characterization (ii) of Theorem 3.11.2 to get equivalence of the three statements

$$\begin{aligned} & Q_\alpha \triangleleft P_\alpha, \\ & \text{if } dP_\alpha/dQ_\alpha \overset{Q_\alpha}{\rightsquigarrow} V \text{ along a subnet, then } P(V \in (0, \infty)) = 1, \\ & \text{if } \log dP_\alpha/dQ_\alpha \overset{Q_\alpha}{\rightsquigarrow} W, \text{ then } P(W \in \mathbb{R}) = 1. \end{aligned}$$

Next use characterization (iii) to obtain the equivalence of

$$\begin{aligned} & P_\alpha \triangleleft Q_\alpha, \\ & \text{if } dP_\alpha/dQ_\alpha \overset{Q_\alpha}{\rightsquigarrow} V \text{ along a subnet, then } EV = 1, \\ & \text{if } \log dP_\alpha/dQ_\alpha \overset{Q_\alpha}{\rightsquigarrow} W, \text{ then } Ee^W = 1. \end{aligned}$$

Together this gives the equivalence of (i) and (iii) of the present theorem.

To obtain the equivalence of (i) and (ii), apply characterization (ii) of Theorem 3.11.2 twice, to both $P_\alpha \triangleleft Q_\alpha$ and $Q_\alpha \triangleleft P_\alpha$. ■

3.11.1 Empirical Process

For each n , let $X_{n1}, \dots, X_{nn}$ be i.i.d. random elements in a measurable space $(\mathcal{X}, \mathcal{A})$. Under the “null hypothesis”, the common law is a fixed measure P , whereas under the “alternative hypothesis”, the common law is P_n . Set $\mathbb{P}_n = n^{-1} \sum_{i=1}^n \delta_{X_{ni}}$. In many studies of the asymptotic efficiency of estimators and tests, it is of interest

to study the behavior of statistics based on $X_{n1}, \dots, X_{nn}$ under the assumption that P_n converges to P in the sense that

$$(3.11.10) \quad \int \left[\sqrt{n}(dP_n^{1/2} - dP^{1/2}) - \frac{1}{2}h dP^{1/2} \right]^2 \rightarrow 0,$$

for some measurable function $h: \mathcal{X} \rightarrow \mathbb{R}$. In this section we derive the asymptotic distribution of the empirical process $\sqrt{n}(\mathbb{P}_n - P)$ under both the null and alternative hypotheses.

Sequences P_n as above are often referred to as “contiguous alternatives” to P . A more precise statement is that the sequence of distributions P_n^n and P^n (of $(X_{n1}, \dots, X_{nn})$ on $\mathcal{X}^n$) are contiguous. In fact, the sequence of log likelihood ratios allows a linear expansion, and contiguity follows from the central limit theorem and Example 3.11.6.

3.11.11 Lemma. *Let the sequence of probability measures P_n satisfy (3.11.10). Then necessarily $Ph = 0$ and $Ph^2 < \infty$ and*

$$\sum_{i=1}^n \log \frac{dP_n}{dP}(X_{ni}) = \frac{1}{\sqrt{n}} \sum_{i=1}^n h(X_{ni}) - \frac{1}{2}Ph^2 + R_n,$$

where the sequence R_n converges to zero in probability under both P^n and P_n^n .

Proof. See, e.g., Hájek and Šidák (1967), Chapter VI, Section 1, pages 201–209; Bickel, Klaassen, Ritov and Wellner (1993), Appendix A.9, page 498–513; or Van der Vaart (1988), Proposition A.8, page 182. ■

Let $\overset{P}{\rightsquigarrow}$ denote weak convergence under the assumption that $(X_{n1}, \dots, X_{nn})$ are distributed according to P^n . The linear expansion of the log likelihood ratio $\Lambda_n(P_n, P) = \log dP_n^n/dP^n$ given by the preceding lemma combined with Slutsky’s lemma and the multivariate central limit theorem gives, for any f with $Pf^2 < \infty$,

$$\left(\sqrt{n}(\mathbb{P}_n - P)f, \Lambda_n(P_n, P) \right) \overset{P}{\rightsquigarrow} N \left(\begin{pmatrix} 0 \\ -\frac{1}{2}Ph^2 \end{pmatrix}, \begin{pmatrix} P(f - Pf)^2 & Pfh \\ Pfh & Ph^2 \end{pmatrix} \right).$$

According to Le Cam’s third lemma, this implies that

$$\sqrt{n}(\mathbb{P}_n - P)f \overset{P_n}{\rightsquigarrow} N(Pfh, P(f - Pf)^2).$$

It follows that the naturally centered empirical process $\sqrt{n}(\mathbb{P}_n - P_n)f$ converges marginally under P_n if and only if the deterministic sequence $\sqrt{n}(P_n - P)f$ converges to a limit. This is not necessarily true under (3.11.10), but for instance if $\|P_n f^2\|_{\mathcal{F}} = O(1)$, then the sequence $\sqrt{n}(P_n - P)f$ converges to Pfh . In that case $\sqrt{n}(\mathbb{P}_n - P_n)f$ converges under P_n in distribution to a $N(0, P(f - Pf)^2)$ -distribution, the same limit distribution as $\sqrt{n}(\mathbb{P}_n - P)f$ under P , as expected. The following theorem records the uniform version of this result.

3.11.12 Theorem. Let $\mathcal{F}$ be a P -Donsker class of measurable functions with $\|P\|_{\mathcal{F}} < \infty$. If the sequence of probability measures P_n satisfies (3.11.10), then the sequence $\sqrt{n}(\mathbb{P}_n - P)$ converges under P_n in distribution in $\ell^\infty(\mathcal{F})$ to the process $f \mapsto \mathbb{G}(f) + Pf h$, where $\mathbb{G}$ is a tight Brownian bridge. Moreover, if $\|P_n f^2\|_{\mathcal{F}} = O(1)$, then $\|\sqrt{n}(P_n - P)f - Pf h\|_{\mathcal{F}} \rightarrow 0$ and the sequence $\sqrt{n}(\mathbb{P}_n - P_n)$ converges under P_n in distribution to $\mathbb{G}$.

Proof. According to the discussion preceding the theorem, the sequence $\sqrt{n}(\mathbb{P}_n - P)$ converges under P_n marginally to the process $f \mapsto \mathbb{G}(f) + Pf h$. Since the two marginal sequences of $(\sqrt{n}(\mathbb{P}_n - P), \Lambda_n(P_n, P))$ are asymptotically tight under P in $\ell^\infty(\mathcal{F})$ and $\mathbb{R}$, respectively, this sequence is jointly asymptotically tight in $\ell^\infty(\mathcal{F}) \times \mathbb{R}$ under P . By Le Cam's third lemma and the converse part of Prohorov's theorem, the sequence $\sqrt{n}(\mathbb{P}_n - P)$ is asymptotically tight in $\ell^\infty(\mathcal{F})$ under P_n . Conclude that $\sqrt{n}(\mathbb{P}_n - P)$ converges under P_n in distribution to the process $f \mapsto \mathbb{G}(f) + Pf h$.

The proof is complete if it is shown that the sequence $\sqrt{n}(P_n - P)f - Pf h$ converges to zero uniformly in f . For any f ,

$$\begin{aligned} \sqrt{n}(P_n - P)f - Pf h &= \frac{1}{2} \int f h dP^{1/2} (dP_n^{1/2} - dP^{1/2}) \\ &\quad + \int f \left[\sqrt{n}(dP_n^{1/2} - dP^{1/2}) - \frac{1}{2} h dP^{1/2} \right] [dP_n^{1/2} + dP^{1/2}]. \end{aligned}$$

By the Cauchy-Schwarz inequality, the second term on the right converges to zero uniformly in f , because $\sup_f \int f^2 [dP_n^{1/2} + dP^{1/2}]^2 = O(1)$ by assumption. The first term is bounded by

$$\frac{1}{2} M \int h dP^{1/2} |dP_n^{1/2} - dP^{1/2}| + \frac{1}{2} \left(\int_{F>M} h^2 dP \right)^{1/2} \left(\int f^2 (dP_n + dP) \right)^{1/2}.$$

This converges to zero for $M = M_n \rightarrow \infty$ sufficiently slowly. ■

3.11.13 Example (Power Kolmogorov-Smirnov test). A test of the null hypothesis $H_0: P = P_0$, that $X_1, \dots, X_n$ are i.i.d. according to a fixed probability measure P_0 , can be based on the (generalized) Kolmogorov-Smirnov statistic $\|\mathbb{P}_n - P_0\|_{\mathcal{F}}$ for a given P_0 -Donsker class $\mathcal{F}$. If c_α is the upper α -quantile of the distribution of the norm $\|\mathbb{G}\|_{\mathcal{F}}$ of a tight Brownian bridge $\mathbb{G}$, then the test that rejects the null hypothesis if $\sqrt{n}\|\mathbb{P}_n - P_0\|_{\mathcal{F}} > c_\alpha$ has asymptotic level α . The sequence of tests is consistent against every alternative P such that $\mathcal{F}$ is P -Donsker and $\|P - P_0\|_{\mathcal{F}} > 0$. The power against a sequence of alternatives P_n satisfying (3.11.10) equals

$$P_{P_n}^* (\sqrt{n}\|\mathbb{P}_n - P_0\|_{\mathcal{F}} > c_\alpha) \rightarrow P(\|\mathbb{G}(f) + P_0 f h\|_{\mathcal{F}} > c_\alpha).$$

Unfortunately, the last expression can rarely be evaluated explicitly.

3.11.2 Change-Point Alternatives

For each n , let $X_{n1}, \dots, X_{nn}$ be independent random elements in a measurable space $(\mathcal{X}, \mathcal{A})$. Under the “null hypothesis”, the random elements are i.i.d. with common law P , whereas under the alternative hypothesis, X_{ni} has law P_{ni} . Suppose that the marginal laws P_{ni} converge to the measure P in the following sense:

$$(3.11.14) \quad n^{-1} \sum_{i=1}^n \int \left[\sqrt{n} (dP_{ni}^{1/2} - dP^{1/2}) - \frac{1}{2} h\left(\cdot, \frac{i}{n}\right) dP^{1/2} \right]^2 \rightarrow 0,$$

for a function $h \in L_2(\mathcal{X} \times [0, 1], P \times \lambda)$ such that $\int h(x, t) dP(x) = 0$ for every $t \in [0, 1]$. Assume that the functions $t \mapsto h(x, t)$ are continuous in the sense that the sequence of discretizations $h_n(x, t) = \sum_{i=1}^n h(x, i/n) 1_{((i-1)/n, i/n]}(t)$ converges in second mean to h :

$$(3.11.15) \quad (P \times \lambda)(h_n - h)^2 \rightarrow 0.$$

Under these assumptions, the following extension of the local asymptotic normality lemma, Lemma 3.11.11, holds.

3.11.16 Lemma. *Let the sequence of probability measures $P_n = P_{n1} \times \dots \times P_{nn}$ satisfy (3.11.14) and (3.11.15). Then*

$$\sum_{i=1}^n \log \frac{dP_{ni}}{dP}(X_{ni}) = \frac{1}{\sqrt{n}} \sum_{i=1}^n h\left(X_{ni}, \frac{i}{n}\right) - \frac{1}{2} (P \times \lambda) h^2 + R_n,$$

where the sequence R_n converges to zero in probability under both P^n and P_n and the sequence $n^{-1/2} \sum_{i=1}^n h(X_{ni}, i/n)$ converges under P^n in distribution to a normal distribution with mean zero and variance $(P \times \lambda) h^2$.

Proof. Condition (3.11.15) implies the convergence of variances as well as the Lindeberg condition for the tangent vectors $h(\cdot, i/n)$:

$$(P \times \lambda)(h_n^2) = \frac{1}{n} \sum_{i=1}^n P h^2\left(\cdot, \frac{i}{n}\right) \rightarrow (P \times \lambda) h^2$$

$$\frac{1}{n} \sum_{i=1}^n P h^2\left(\cdot, \frac{i}{n}\right) \left\{ \left| h\left(\cdot, \frac{i}{n}\right) \right| \geq \varepsilon \sqrt{n} \right\} \rightarrow 0, \quad \text{for every } \varepsilon > 0.$$

Next the lemma follows from, for instance, Proposition A.8, page 182, Van der Vaart (1988); or Hájek and Šidák (1967), Chapter VI, Section 1, pages 201–209; or Bickel, Klaassen, Ritov and Wellner (1993), Appendix A.9, page 498–513. ■

Let $\mathbb{Z}_n$ be the sequential empirical process $n^{-1/2} \sum_{i=1}^{\lfloor ns \rfloor} (f(X_{ni}) - Pf)$ under the null hypothesis. The linear expansion of the log likelihood ratio $\Lambda_n(P_n, P^n) = \log dP_n/dP^n$ given by the preceding lemma combined with Slutsky’s lemma and the Lindeberg-Feller central limit theorem gives, for any s and any f with $Pf^2 < \infty$,

$$\left(\mathbb{Z}_n(s, f), \Lambda_n(P_n, P^n) \right) \overset{P}{\rightsquigarrow} N \left(\begin{pmatrix} 0 \\ -\frac{1}{2} \sigma^2 \end{pmatrix}, \begin{pmatrix} sP(f - Pf)^2 & c(s, f) \\ c(s, f) & (P \times \lambda) h^2 \end{pmatrix} \right).$$

Here $c(s, f)$ is the asymptotic covariance of $\mathbb{Z}_n$ and the log likelihood ratios and is given by

$$c(s, f) = (P \times \lambda)fh[0, s] = \int_0^s \int_{\mathcal{X}} f(x)h(x, t) dP(x) dt.$$

In view of Le Cam's third lemma, this implies that under the alternative hypothesis, for every (s, f) in $[0, 1] \times \mathcal{F}$,

$$\mathbb{Z}_n(s, f) \overset{P_n}{\rightsquigarrow} N(c(s, f), sP(f - Pf)^2).$$

The following theorem yields the corresponding conclusion for the whole process $\mathbb{Z}_n$ in $\ell^\infty([0, 1] \times \mathcal{F})$.

3.11.17 Theorem. *Let $\mathcal{F}$ be a P -Donsker class of measurable functions with $\|P\|_{\mathcal{F}} < \infty$. If the triangular array of probability measures P_{ni} satisfies (3.11.14) and (3.11.15), then the sequence $\mathbb{Z}_n$ converges weakly in $\ell^\infty([0, 1] \times \mathcal{F})$ to the process*

$$(s, f) \mapsto \mathbb{Z}(s, f) + (P \times \lambda)fh[0, s],$$

where $\mathbb{Z}$ is a tight P -Kiefer-Müller process. If, in addition, the sequence of average measures $\bar{P}_n = n^{-1} \sum_{i=1}^n P_{ni}$ satisfies $\|\bar{P}_n f^2\|_{\mathcal{F}} = O(1)$, then the sequence of processes $n^{-1/2} \sum_{i=1}^{\lfloor ns \rfloor} (\delta_{X_{ni}} - P_{ni})f$ converges in distribution under $P_n = P_{n1} \times \dots \times P_{nn}$ in $\ell^\infty([0, 1] \times \mathcal{F})$ to $\mathbb{Z}$.

Proof. The first assertion follows by combining Prohorov's theorem and Le Cam's third lemma as in the proof of Theorem 3.11.12.

For the second assertion it suffices to show that

$$\left\| \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor ns \rfloor} (P_{ni} - P)f - (P \times \lambda)fh[0, s] \right\|_{[0, 1] \times \mathcal{F}} \rightarrow 0.$$

By extension of the argument in the proof of Theorem 3.11.12 the sequence $n^{-1} \sum_{i=1}^{\lfloor ns \rfloor} [\sqrt{n}(P_{ni} - P)f - Pf h(\cdot, i/n)]$ converges to zero uniformly in (s, f) . Furthermore, since $|\int_0^s h_n(x, t) dt - n^{-1} \sum_{i=1}^{\lfloor ns \rfloor} h(x, i/n)|$ is bounded above by $n^{-1}|h(x, (\lfloor ns \rfloor + 1)/n)|$, it follows that for every (s, f)

$$\begin{aligned} & \left| \frac{1}{n} \sum_{i=1}^{\lfloor ns \rfloor} Pf h\left(\cdot, \frac{i}{n}\right) - (P \times \lambda)fh1_{[0, s]} \right| \\ & \leq \frac{1}{n} P \sup_i \left| h\left(\cdot, \frac{i}{n}\right) f \right| + (P \times \lambda) |fh_n - fh|. \end{aligned}$$

In view of condition (3.11.15), the sequence $n^{-1} \sum_{i=1}^n Ph^2(\cdot, i/n) = (P \times \lambda)h_n^2$ is bounded. Two applications of the Cauchy-Schwarz inequality show that the expressions in the last display converge to zero uniformly in (s, f) . ■

3.11.18 Example (Change-point alternatives). Let $\{P_n\}$ be a given sequence that satisfies (3.11.10), and let $t \in (0, 1)$ be fixed. Define P_{ni} to be P for $1 \leq i \leq \lfloor nt \rfloor$ and define it to be P_n for $\lfloor nt \rfloor < i \leq n$. Thus the distribution governing the observations changes from P to P_n between $i = \lfloor nt \rfloor$ and $i = \lfloor nt \rfloor + 1$.

These “change-point alternatives” satisfy the conditions of the preceding theorem, with $h(x, s) = h(x)1\{s > t\}$.

Problems and Complements

- 1. (Contiguity and limit experiments)** Let P_α and Q_α be nets of probability measures on measurable spaces $(\mathcal{X}_\alpha, \mathcal{B}_\alpha)$. There exist probability measures P and Q on a measurable space $(\mathcal{X}, \mathcal{B})$ and a subnet such that $dP_\alpha/dQ_\alpha \rightsquigarrow dP/dQ$ under Q_α and $dQ_\alpha/dP_\alpha \rightsquigarrow dQ/dP$ under P_α along the subnet. (Here the laws of the limits are computed under Q and P , respectively.) Contiguity $Q_\alpha \triangleleft P_\alpha$ holds if and only if for every such subnet and pairs P and Q one has $Q \ll P$. For sequences this is already true with subsequence replacing subnet.

[Hint: Let W be a weak limit point of the maps $dP_\alpha/(dP_\alpha + dQ_\alpha): \mathcal{X}_\alpha \rightarrow [0, 1]$ under $P_\alpha + Q_\alpha$. (W has total mass 2.) By the continuous mapping theorem applied in $\mathbb{R}$, one has $dP_\alpha/dQ_\alpha \rightsquigarrow W/(1 - W)$ and $dQ_\alpha/dP_\alpha \rightsquigarrow (1 - W)/W$ (take $1/0 = \infty$). On $\mathcal{X} = [0, 1]$ with Borel sets, define $P(A) = E1_A(W)W$ and $Q(A) = E1_A(W)(1 - W)$. Then $(P + Q)(A) = E1_A(W)$, so that $dP/dQ(w) = w/(1 - w)$ and $dQ/dP(w) = (1 - w)/w$. The second assertion follows from the first, the characterization of contiguity in terms of likelihood ratios and the equivalence of the three statements: $Q \ll P$; $Q(dP/dQ = 0) = 0$; and $E_P dQ/dP = 1$.]

- 2.** Let P_α and Q_α be nets of probability measures on measurable spaces $(\Omega_\alpha, \mathcal{A}_\alpha)$ such that $P_\alpha(A_\alpha) \rightarrow 0$ implies $Q_\alpha(A_\alpha) \rightarrow 0$ for every choice of measurable sets A_α . Then not necessarily $Q_\alpha \triangleleft P_\alpha$.

[Hint: Subnets are tricky. See Le Cam (1986), page 87.]

- 3. (Conditions that imply the Lindeberg-Feller condition)**

Let $A_n(s) = n^{-1} \sum_{i=1}^{\lfloor ns \rfloor} h^2(X_{ni}, i/n)$ and $A(s) = \int_0^s Ph^2(\cdot, t)dt$.

- (i) If $A_n(1) \xrightarrow{P} A(1)$ and $h_n \rightarrow h$ in $P \times \lambda$ -measure, then the Lindeberg condition holds.
(ii) If $A_n(s) \xrightarrow{P} A(s)$ for each $0 \leq s \leq 1$ and $h_n \rightarrow h$ in $L_2(P \times \lambda)$, then the Lindeberg condition holds.

[Hint: Greenwood and Shiryayev (1985), remark 1, page 116.]

- 4.** Use Theorem 3.11.12 to investigate the local asymptotic power of the bootstrap and permutation independence tests defined in Chapter 3.9.
5. Suppose that $X_1, \dots, X_n$ are i.i.d. with distribution function F on $[0, 1]$. Let F_0 be the Uniform $[0, 1]$ distribution function. Consider testing the null hypothesis $F = F_0$ versus the alternative that $F(x) \geq F_0(x)$ for all x with strict inequality for some x , based on the statistics

$$A_n^+ = \int_0^1 \sqrt{n} [\mathbb{F}_n(t) - F_0(t)]^+ dt \quad \text{and} \quad B_n = \int_0^1 \sqrt{n} [\mathbb{F}_n(t) - F_0(t)] dt.$$

Investigate the local asymptotic power of these two tests.

3.12

Convolution and Minimax Theorems

Let H be a linear subspace of a Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$. For each $n \in \mathbb{N}$ and $h \in H$, let $P_{n,h}$ be a probability measure on a measurable space $(\mathcal{X}_n, \mathcal{A}_n)$. Consider the problem of estimating a “parameter” $\kappa_n(h)$ given an “observation” X_n with law $P_{n,h}$. The convolution theorem and the minimax theorem give a lower bound on how well $\kappa_n(h)$ can be estimated asymptotically as $n \rightarrow \infty$. Suppose the sequence of statistical experiments $(\mathcal{X}_n, \mathcal{A}_n, P_{n,h}: h \in H)$ is “asymptotically normal” and the sequence of parameters $\kappa_n(h)$ is “regular” then the limit distribution of every “regular” estimator sequence is the convolution of a certain Gaussian distribution and a noise factor. Furthermore, the maximum risk of any estimator sequence is bounded below by the “risk” of this Gaussian distribution. These concepts are defined as follows.

For ease of notation, let $\{\Delta_h: h \in H\}$ be the *iso-Gaussian process*: the Gaussian process with zero mean and covariance function $E\Delta_{h_1}\Delta_{h_2} = \langle h_1, h_2 \rangle$. The sequence of experiments $(\mathcal{X}_n, \mathcal{A}_n, P_{n,h}: h \in H)$ is called *asymptotically (shift) normal* if

$$\log \frac{dP_{n,h}}{dP_{n,0}} = \Delta_{n,h} - \frac{1}{2}\|h\|^2,$$

for stochastic processes $\{\Delta_{n,h}: h \in H\}$ such that

$$\Delta_{n,h} \xrightarrow{0} \Delta_h \quad \text{marginally.}$$

Here $\xrightarrow{h}$ denotes weak convergence under $P_{n,h}$. This terminology arises from the theory of limiting experiments due to Le Cam, which, however, we shall not use in this chapter.

The sequence of parameters $\kappa_n(h)$ is assumed to belong to a Banach space $\mathbf{B}$. It should be regular in the sense that

$$r_n(\kappa_n(h) - \kappa_n(0)) \rightarrow \dot{\kappa}(h), \quad \text{for every } h \in H,$$

for a continuous, linear map $\kappa: H \rightarrow \mathbf{B}$ and certain linear maps $r_n: \mathbf{B} \rightarrow \mathbf{B}$ (“norming operators”). Any maps $T_n: \mathcal{X}_n \rightarrow \mathbf{B}$ are considered estimators of the parameter. A sequence of estimators T_n is *regular* with respect to the norming operators r_n if

$$r_n(T_n - \kappa_n(h)) \overset{h}{\rightsquigarrow} L, \quad \text{for every } h \in H,$$

for a fixed, tight, Borel probability measure L on $\mathbf{B}$.

This set-up covers many examples. In the simplest and most common examples, the observation at time n is a sample of n independent observations from a fixed measure P .

3.12.1 Example (I.i.d. observations). Let $X_1, \dots, X_n$ be an i.i.d. sample from a probability measure P on the measurable space $(\mathcal{X}, \mathcal{A})$. The common law P is known to belong to a class $\mathcal{P}$ of probability measures. It is required to estimate the parameter $\kappa(P)$.

Sequences of asymptotically normal experiments arise as “localized” or “rescaled” experiments. Fix some $P \in \mathcal{P}$ and set $P_{n,0} = P^n$. Consider one-dimensional submodels $t \mapsto P_t$ (maps from $[-1, 1] \subset \mathbb{R}$ to $\mathcal{P}$) such that

$$\int \left[\frac{dP_t^{1/2} - dP^{1/2}}{t} - \frac{1}{2}h dP^{1/2} \right]^2 \rightarrow 0, \quad t \rightarrow 0,$$

for some measurable function $h: \mathcal{X} \rightarrow \mathbb{R}$. Single out those paths such that also

$$\frac{\kappa(P_t) - \kappa(P)}{t} \rightarrow \dot{\kappa}(h),$$

for some continuous linear map $\dot{\kappa}: L_2(P) \rightarrow \mathbf{B}$. A function $h \in L_2(P)$ for which there exists a path $t \mapsto P_t$ with the first property is called a *score function*, and the collection of all score functions is called the *tangent set* at P (relative to $\mathcal{P}$). Let H be a linear subspace of the tangent set such that for every $h \in H$ there exists a path $t \mapsto P_t$ with both properties. For each h , set $P_{n,h} = P_{1/\sqrt{n}}$ for a corresponding path $t \mapsto P_t$. Then Lemma 3.11.11 implies that the sequence of experiments $(\mathcal{X}^n, \mathcal{A}^n, P_{n,h}: h \in H)$ is asymptotically normal. (Since the experiments are “local” the sequence is often called *locally asymptotically normal* (LAN).) Furthermore, the sequence of parameters $\kappa(P_{n,h})$ is regular with respect to the norming operators $\sqrt{n}$ (the maps $b \mapsto \sqrt{n}b$).

A sequence of estimators $T_n = T_n(X_1, \dots, X_n)$ is regular if

$$\sqrt{n}(T_n - \kappa(P_{n,h})) \overset{h}{\rightsquigarrow} L, \quad \text{for every } h \in H,$$

for a some fixed, tight Borel measure L . The weak convergence in this display refers to a sequence of laws $P_{1/\sqrt{n}}$ that converges to P . Thus, in this situation regularity comes down to a certain uniformity in reaching the limit measure. In particular, regularity is weaker than weak convergence uniformly in P' ranging through a “neighborhood” of P combined with continuity of the limiting measures as P' approaches P .

The continuous, linear map $\dot{\kappa}: H \rightarrow \mathbf{B}$ has an adjoint map $\dot{\kappa}^*: \mathbf{B}^* \rightarrow \bar{H}$, which maps the dual space of $\mathbf{B}$ into the completion of H . This is determined by $\langle \dot{\kappa}^*b^*, h \rangle = b^*\dot{\kappa}(h)$.

3.12.2 Theorem (Convolution theorem). Assume that the sequence of statistical experiments $(\mathcal{X}_n, \mathcal{A}_n, P_{n,h}; h \in H)$ is asymptotically normal and let the sequences of parameters $\kappa_n(h)$ and estimators T_n be regular. Then the limit distribution L of the sequence $r_n(T_n - \kappa_n(0))$ equals the distribution of a sum $G + W$ of independent, tight, Borel measurable random elements in $\mathbf{B}$ such that

$$b^*G \sim N(0, \|\kappa^*b^*\|^2), \quad \text{for every } b^* \in \mathbf{B}^*.$$

Here $\kappa^*: \mathbf{B}^* \rightarrow \bar{H}$ is the adjoint of κ . The law of G concentrates on the closure of $\kappa(H)$.

Proof. (a) Assume first that the dimension of H is finite. Let $h_1, \dots, h_k$ be an orthonormal base and set

$$\begin{aligned} Z_{n,a} &= r_n(T_n - \kappa_n(\sum a_i h_i)), \\ \Lambda_n(a) &= \log \frac{dP_{n,h}}{dP_{n,0}} = \Delta_{n,h} - \frac{1}{2}\|h\|^2, \quad \text{for } h = \sum a_i h_i. \end{aligned}$$

By assumption, the sequence $Z_{n,0}$ and each sequence $\Delta_{n,h}$ converge in distribution in $\mathbf{B}$ and $\bar{\mathbb{R}}$, respectively. By Prohorov's theorem, there exists a subsequence of $\{n\}$ such that

$$(Z_{n',0}, \Delta_{n',h_1}, \dots, \Delta_{n',h_k}) \xrightarrow{0} (Z, \Delta_{h_1}, \dots, \Delta_{h_k}),$$

in $\mathbf{B} \times \mathbb{R}^k$. By assumption, Z has (marginal) distribution equal to L .

The random variable $\Delta_{\sum a_i h_i} - \sum a_i \Delta_{h_i}$ has second moment zero. Hence this variable is zero almost surely. Conclude that the sequence $\Delta_{n, \sum a_i h_i} - \sum a_i \Delta_{n, h_i}$ converges to zero in probability under $P_{n,0}$. Combination with the preceding displays and the regularity of the sequence of parameters yield, for every $a \in \mathbb{R}^k$,

$$(Z_{n',a}, \Lambda_{n'}(a)) \xrightarrow{0} \left(Z - \sum a_i \kappa(h_i), \sum a_i \Delta_{h_i} - \frac{1}{2}\|a\|^2 \right).$$

Apply Le Cam's third lemma, Theorem 3.11.7, to see that $Z_{n',a} \xrightarrow{a} Z_a$, where $\xrightarrow{a}$ denotes weak convergence under $P_{n,h}$ for $h = \sum a_i h_i$, and Z_a is distributed according to

$$(3.12.3) \quad P(Z_a \in B) = E 1_B \left(Z - \sum a_i \kappa(h_i) \right) e^{\sum a_i \Delta_{h_i} - \frac{1}{2}\|a\|^2}.$$

Since the sequence T_n is regular, the left side equals $L(B)$ for each a . Average the left side and the right side over a with respect to a $N_k(0, \lambda^{-1}I)$ weight function. Straightforward calculations yield

$$L(B) = \int E 1_B \left(Z - \frac{\sum \Delta_{h_i} \kappa(h_i)}{1 + \lambda} - \frac{\sum a_i \kappa(h_i)}{(1 + \lambda)^{1/2}} \right) c_\lambda(\Delta) dN_k(0, I)(a),$$

where $c_\lambda(\Delta) = (1 + \lambda^{-1})^{k/2} \exp(\frac{1}{2}(1 + \lambda)^{-1} \sum \Delta_{h_i}^2)$. Conclude that L can be written as the law of the sum $G_\lambda + W_\lambda$ of independent random elements $-G_\lambda$ and W_λ , where $G_\lambda = -\sum A_i \kappa(h_i)/(1 + \lambda)^{1/2}$ for a $N_k(0, I)$ -distributed vector $(A_1, \dots, A_k)$ and W_λ is distributed according to

$$P(W_\lambda \in B) = E 1_B \left(Z - \frac{\sum \Delta_{h_i} \kappa(h_i)}{1 + \lambda} \right) c_\lambda(\Delta).$$

As $\lambda \downarrow 0$, we have $G_\lambda \rightsquigarrow G = \sum A_i \dot{\kappa}(h_i)$. The variable $b^*G = \sum A_i b^* \dot{\kappa}(h_i)$ is normally distributed with zero mean and variance

$$Eb^*G^2 = \sum (b^* \dot{\kappa}(h_i))^2 = \|\dot{\kappa}^* b^*\|^2.$$

By the converse part of Prohorov's theorem, the variables $\{G_\lambda: 0 < \lambda < 1\}$ are uniformly tight. Combined with the tightness of L and the convolution statement, it follows that the variables $\{W_\lambda: 0 < \lambda < 1\}$ are uniformly tight (Problem 3.12.4). If $W_{\lambda_m} \rightsquigarrow W$ for a sequence $\lambda_m \downarrow 0$, then $(G_{\lambda_m}, W_{\lambda_m}) \rightsquigarrow (G, W)$, where G and W are independent and $G + W$ is distributed according to L . This concludes the proof of the theorem for finite-dimensional H .

(b) Let H be arbitrary. For any finite orthonormal set $h_1, \dots, h_k$, the previous argument yields tight independent processes G_k and W_k such that $G_k + W_k$ is distributed according to L and G_k is zero-mean Gaussian with

$$Eb^*G_k^2 = \sum_{i=1}^k \langle \dot{\kappa}^* b^*, h_i \rangle^2.$$

The set of all variables G_k and W_k so obtained is uniformly tight. Indeed, by tightness of L , there exists for any given $\varepsilon > 0$ a compact set K such that $L(K) = \int \mathbf{P}(G_k \in K - x) dP^{W_k}(x) > 1 - \varepsilon$. Thus there exists x_0 with $\mathbf{P}(G_k \in K - x_0) > 1 - \varepsilon$. By symmetry, $\mathbf{P}(G_k \in x_0 - K) > 1 - \varepsilon$, whence $\mathbf{P}(G_k \in \frac{1}{2}(K - K)) > 1 - 2\varepsilon$. Next, the uniform tightness of L and the collection G_k imply the uniform tightness of the collection W_k .

Direct the finite-dimensional subspaces of H by inclusion, and construct variables (G_k, W_k) for every subspace. Every weak limit point (G, W) of the net of laws $(G_{k'}, W_{k'})$ satisfies the requirements of the theorem. ■

In the convolution theorem the distribution of the Gaussian variable G is interpreted as the optimal limit law. The distribution of G is completely determined by the model, through the properties of $\dot{\kappa}$ and H . The variable W is interpreted as a noise factor that is zero for optimal estimator sequences.

The convolution of a zero-mean Gaussian variable with an arbitrary second variable leads to a loss of concentration. This is most easily understood in terms of variances: convolution increases variance. The following lemma makes the statement precise in terms of general subconvex risk functions. A nonnegative map $\ell: \mathbf{B} \rightarrow \mathbb{R}$ is *subconvex* if the set $\{y: \ell(y) \leq c\}$ is closed, convex, and symmetric for every c .

3.12.4 Lemma (Anderson's lemma). *Let $\ell: \mathbf{B} \rightarrow \mathbb{R}$ be subconvex. Let G be a tight, zero-mean, Borel measurable Gaussian variable and W be an arbitrary, tight, Borel measurable variable independent of G . Then*

$$E\ell(G + W) \geq E\ell(G).$$

In particular, $\mathbf{P}(\|G + W\| \leq c) \leq \mathbf{P}(\|G\| \leq c)$ for every c .

Proof. For a finite-dimensional Banach space, this assertion is a special case of Anderson's lemma. The general case can be proved by the approximation of ℓ from below by cylinder functions. See step (b) of the proof of the minimax theorem, Theorem 3.12.5. ■

A subconvex function is certainly lower semicontinuous. Therefore, the portmanteau theorem, the lemma, and the convolution theorem can be combined to yield, for any regular estimator sequence T_n and subconvex loss function ℓ ,

$$\liminf_{n \rightarrow \infty} E_{0*} \ell(r_n(T_n - \kappa_n(0))) \geq E\ell(G).$$

This inequality may fail for non-regular estimator sequences. However, according to the minimax theorem, the maximum risk $\sup_h E_{h*} \ell(r_n(T_n - \kappa_n(h)))$ can never asymptotically fall below $E\ell(G)$ under just some measurability conditions on the estimator sequence T_n . The following theorem gives a slightly stronger result.

A little (asymptotic) measurability is the only requirement on T_n , but measurability can be restrictive, so we need to be careful about it. Let $\mathbf{B}'$ be a given subspace of $\mathbf{B}^*$ that separates points of $\mathbf{B}$, and let $\tau(\mathbf{B}')$ be the weak topology induced on $\mathbf{B}$ by the maps $b': \mathbf{B} \rightarrow \mathbb{R}$ when b' ranges over $\mathbf{B}'$. A map $\ell: \mathbf{B} \rightarrow \mathbb{R}$ is called $\tau(\mathbf{B}')$ -subconvex if the sets $\{y: \ell(y) \leq c\}$ are $\tau(\mathbf{B}')$ -closed, convex, and symmetric for every c . An estimator sequence T_n is asymptotically $\mathbf{B}'$ -measurable if

$$E_0^* f(r_n(T_n - \kappa_n(0))) - E_{0*} f(r_n(T_n - \kappa_n(0))) \rightarrow 0,$$

for every function f of the form $f(y) = g(b'_1 y, \dots, b'_k y)$ with $b'_1, \dots, b'_k$ in $\mathbf{B}'$ and $g: \mathbb{R}^k \rightarrow \mathbb{R}$ continuous and bounded. Clearly, every (asymptotically) measurable sequence is asymptotically $\mathbf{B}^*$ -measurable. A sequence of stochastic processes in a function space is $\mathbf{B}'$ -measurable for $\mathbf{B}'$ equal to the set of coordinate projections.

3.12.5 Theorem (Minimax theorem). Assume that the sequence of experiments $(\mathcal{X}_n, \mathcal{A}_n, P_{n,h}; h \in H)$ is asymptotically normal and let the sequence of parameters $\kappa_n(h)$ be regular. Suppose a tight, Borel measurable Gaussian element G , as in the statement of the convolution theorem, exists. Then for every asymptotically $\mathbf{B}'$ -measurable estimator sequence T_n and $\tau(\mathbf{B}')$ -subconvex function ℓ ,

$$\sup_{I \subset H} \liminf_{n \rightarrow \infty} \sup_{h \in I} E_{h*} \ell(r_n(T_n - \kappa_n(h))) \geq E\ell(G).$$

Here the first supremum is taken over all finite subsets I of H .

Proof. (a). Assume first that the loss function can be written in the special form $\ell(y) = \sum_{i=1}^r 1_{K_i^c}(b'_{i,1}y, \dots, b'_{i,p_i}y)$ for compact, convex, symmetric subsets $K_i \subset \mathbb{R}^{p_i}$ and arbitrary elements $b'_{i,j}$ of $\mathbf{B}'$. Fix an arbitrary orthonormal set $h_1, \dots, h_k$ in H , and set

$$Z_{n,a}^i = (b'_{i,1}, \dots, b'_{i,p_i}) \circ r_n(T_n - \kappa_n(\sum a_i h_i)), \quad 1 \leq i \leq r.$$

Considered as maps into the one-point compactification of $\mathbb{R}^{p_i}$, the sequences $Z_{n,a}^i$ are certainly asymptotically tight. The sequences are asymptotically measurable by assumption.

Direct the finite subsets of H by inclusion. There exists a subnet $\{n_I: I \subset H, \text{ finite}\}$ such that the left side of the statement of the theorem equals

$$\text{minimax risk} = \limsup_I \sup_{h \in I} E_{h*} \ell(r_{n_I}(T_{n_I} - \kappa_{n_I}(h))).$$

By the same arguments as in the proof of the convolution theorem there is a further subnet $\{n'\} \subset \{n_I\}$ such that $Z_{n',a}^i \xrightarrow{a} Z_a^i$ in the one-point compactifications, for every $a \in \mathbb{R}^k$ and every i . Here the limiting processes satisfy, for each i ,

$$\int \mathcal{L}(Z_a^i) dN_k(0, \lambda^{-1}I) \sim G_\lambda^i + W_\lambda^i,$$

for independent elements G_λ^i and W_λ^i such that

$$G_\lambda^i = (b'_{i,1}, \dots, b'_{i,p_i}) \circ G_\lambda = (b'_{i,1}, \dots, b'_{i,p_i}) \circ \frac{\sum_j A_j k(h_j)}{(1+\lambda)^{1/2}},$$

for a $N_k(0, I)$ -distributed vector $(A_1, \dots, A_k)$. By the portmanteau theorem,

$$\text{minimax risk} \geq \liminf_{n'} \sum_{i=1}^r P_{a*}(Z_{n',a}^i \notin K_i) \geq \sum_{i=1}^r P(Z_a^i \notin K_i).$$

Since this is true for every a , the left side is also bounded below by the average of the right side. Combination with Lemma 3.12.4 (only the finite-dimensional case is needed here) yields

$$\text{minimax risk} \geq \sum_{i=1}^r P(G_\lambda^i + W_\lambda^i \notin K_i) \geq \sum_{i=1}^r P(G_\lambda^i \notin K_i) = E\ell(G_\lambda).$$

Finish the proof for this special form of loss function by letting $\lambda \downarrow 0$ followed by taking the limit along finite-dimensional subspaces of H .

(b). An arbitrary subconvex ℓ can be approximated from below by a sequence of functions ℓ_r of the type as in (a). More precisely, the sequence ℓ_r can be constructed such that $0 \leq \ell_r \leq \ell$ everywhere and $\ell_r \uparrow \ell$ G -almost surely. It follows that the minimax risk decreases when ℓ is replaced by ℓ_r . Combination with (a) shows that the minimax risk is bounded below by $E\ell_r(G)$ for every r . The theorem follows by letting $r \rightarrow \infty$.

For the construction of ℓ_r , note first that

$$0 \leq 2^{-r} \sum_{i=1}^{2^{2r}} 1\{y: \ell(y) > i2^{-r}\} \uparrow \ell(y), \quad \text{for every } y.$$

Each of the sets $\{y: \ell(y) \leq i/r\}$ is convex, $\tau(\mathbf{B}')$ -closed, and symmetric. Thus, it suffices to approximate functions ℓ of the type 1_{C^c} for a convex, $\tau(\mathbf{B}')$ -closed, and symmetric set C .

By the Hahn-Banach theorem, any such set C can be written

$$C = \bigcap_{b' \in \mathbf{B}'} \{y: |b'y| \leq c_{b'}\}.$$

Thus the complement of C intersected with the support S of the limit variable G is the union of the sets $\{y \in S: |b'y| > c_{b'}\}$. These sets are relatively open in S and S is separable. Since a separable set is Lindelöf, the possibly uncountable union can be replaced by a countable log likelihood. Thus there exists a sequence b'_i in $\mathbf{B}'$ and numbers c_i such that $C^c \cap S = \bigcup_{i=1}^{\infty} \{y \in S: |b'_i y| > c_i\}$. This implies that

$$1_{C^c \cap S} = \sup_r 1_{K_r^c}(b'_1 y, \dots, b'_r y),$$

for the subsets of $\mathbb{R}^r$ defined by $K_r = \bigcap_{i=1}^r \{x \in \mathbb{R}^r: |x_i| \leq c_i\}$. ■

3.12.6 Example (Separable Banach space). A convex set in a Banach space is closed with respect to the norm if and only if it is weakly closed. This means that “ $\tau(\mathbf{B}^*)$ -subconvex” is identical to “subconvex”.

We conclude that the minimax theorem is valid for Borel measurable estimator sequences T_n and every subconvex loss function. The restriction that each T_n be Borel measurable is reasonable in separable Banach spaces, but less so in nonseparable spaces.

3.12.7 Example (Skorokhod space). Consider the Skorokhod space $D[a, b]$ for a given interval $[a, b] \subset \mathbb{R}$, equipped with the uniform norm. The dual space consists of maps of the form

$$d^*(z) = \int z(u) d\mu(u) + \sum_{i=1}^{\infty} \alpha_i (z(u_i) - z(u_i-)),$$

for a finite signed measure μ on $[a, b]$, an arbitrary sequence u_i in $(a, b]$, and a sequence α_i with $\sum |\alpha_i| < \infty$.[‡] Each such d^* is the pointwise limit of a sequence of linear combinations of coordinate projections. Thus, the σ -field generated by the dual space equals the σ -field generated by the coordinate projections.

It follows that an estimator sequence is $D[a, b]^*$ -measurable if and only if it is a stochastic process. Since “ $\tau(D[a, b]^*)$ -subconvex” is identical to “subconvex with respect to the norm” (Problem 3.12.3), the minimax theorem is valid for any sequence of stochastic processes T_n and subconvex loss function ℓ .

Examples of subconvex loss functions include

$$\begin{aligned} z &\mapsto \ell_0(\|z\|_{\infty}), \\ z &\mapsto \int |z|^p(t) d\mu(t), \end{aligned}$$

for a nondecreasing, left-continuous function $\ell_0: \mathbb{R} \rightarrow \mathbb{R}$, a finite Borel measure μ , and $p \geq 1$.

[‡] Van der Vaart (1988), pages 81–85.

3.12.8 Example (Bounded functions). On the space $\ell^\infty(\mathcal{F})$, functions of the type

$$z \mapsto \ell_0\left(\left\|\frac{z}{q}\right\|_{\mathcal{F}}\right),$$

for a nondecreasing, left-continuous function $\ell_0: \mathbb{R} \rightarrow \mathbb{R}$ and an arbitrary map $q: \mathcal{F} \rightarrow \mathbb{R}$ are subconvex with respect to the linear space spanned by the coordinate projections $z \mapsto z(f)$. Indeed, for any c there exists d such that

$$\left\{z: \ell_0\left(\left\|\frac{z}{q}\right\|_{\mathcal{F}}\right) \leq c\right\} = \left\{z: \left\|\frac{z}{q}\right\|_{\mathcal{F}} \leq d\right\} = \bigcap_{f \in \mathcal{F}} \left\{z: |z(f)| \leq dq(f)\right\}.$$

Thus, the minimax theorem is valid for any estimator sequence T_n that is coordinatewise measurable and any loss function of this type.

For general loss functions that are subconvex with respect to the norm, the preceding minimax theorem applies only under strong measurability conditions on the estimator sequences. It is of interest that these measurability conditions are satisfied by sequences T_n such that $T_n(f)$ is measurable for every f and such that the sequence $r_n(T_n - \kappa_n(0))$ is asymptotically tight under $P_{n,0}$. Indeed, according to Lemma 1.5.2, such sequences are asymptotically $\tau(\ell^\infty(\mathcal{F})^*)$ -measurable. It follows that, given any subconvex loss function, the minimax theorem may be used to designate optimal estimator sequences among the asymptotically tight sequences.

3.12.1 Efficiency of the Empirical Distribution

Let $\mathcal{F}$ be a P -Donsker class of functions on a measurable space $(\mathcal{X}, \mathcal{A})$. In this section it is shown that the empirical distribution $\mathbb{P}_n$ of a sample of size n from a measure P is an asymptotically efficient estimator for P , when both are (and can be) viewed as elements $f \mapsto \mathbb{P}_n f$ and $f \mapsto Pf$ in $\ell^\infty(\mathcal{F})$ and when P is considered completely unknown.

The efficiency here is understood in the sense of best (locally) regular at P and locally asymptotically minimax at P within the context of Example 3.12.1. For local efficiency at all P in a collection $\mathcal{P}$, we would assume that $\mathcal{F}$ is P -Donsker with $\|P\|_{\mathcal{F}} < \infty$ for every $P \in \mathcal{P}$. At the end of this section we also briefly discuss global minimaxity over a class of underlying measures $\mathcal{P}$.

We use the set-up of Example 3.12.1 with the “model” $\mathcal{P}$ equal to the class of all probability measures on $(\mathcal{X}, \mathcal{A})$. Then the tangent set can be shown to be equal to all square-integrable measurable functions h such that $Ph = 0$. We do not really need a result as strong as this (and would actually not be able to use it either, because the parameter might be irregular with respect to this large set). Instead we note that for every bounded, measurable function $h: \mathcal{X} \rightarrow \mathbb{R}$, the measures with densities $dP_t(x) = (1 + th(x)) dP(x)$ are well defined for sufficiently small $|t|$ and define probability measures if $Ph = 0$. These paths can be seen to be differentiable in quadratic mean by elementary arguments. We define the set H to be equal to all bounded functions with mean zero and let $P_{n,h}$ be the measure $P_{1/\sqrt{n}}$.

The parameter to be estimated is the map $f \mapsto Pf$, which is assumed to belong to $\ell^\infty(\mathcal{F})$. Since $P_t f = Pf + tPf h$ for every f , the derivative of this parameter is

the map $\dot{\kappa}: h \rightarrow (f \mapsto Pf h)$. Since $\mathcal{F}$ is a Donsker class, it is bounded in $L_2(P)$, which implies that the derivative $\dot{\kappa}$ is continuous and the parameter regular with respect to our choice of H .

Since $\mathcal{F}$ is assumed a Donsker class, the sequence $\sqrt{n}(\mathbb{P}_n - P)$ converges in distribution under P to a Brownian bridge process $\mathbb{G}_P$. By Theorem 3.11.12, the sequence $\sqrt{n}(\mathbb{P}_n - P_{n,h})$ has the same limit distribution if the observations are sampled from $P_{n,h}$. Thus, the sequence of empirical distributions is regular (at P). Furthermore, for every bounded, continuous function $\ell: \ell^\infty(\mathcal{F}) \rightarrow [0, \infty)$,

$$(3.12.9) \quad \sup_{I \subset H} \limsup_{n \rightarrow \infty} \sup_{h \in I} E_{h*} \ell(\sqrt{n}(\mathbb{P}_n - P_{n,h})) = E\ell(\mathbb{G}_P).$$

Thus, we may conclude that the empirical distribution is asymptotically efficient in terms of both the convolution and the minimax theorems, (for certain loss functions) if the tight Gaussian element G in these theorems equals the Brownian bridge. For our choice of H , this is indeed the case and easy and to prove.

It suffices to calculate the covariance function of G in Theorem 3.12.2. The projection $\pi_f: \ell^\infty(\mathcal{F}) \rightarrow \mathbb{R}$ on the f -coordinate, given by $z \mapsto z(f)$, is an element of the dual space of $\ell^\infty(\mathcal{F})$. By definition of $\dot{\kappa}^*$, we have

$$P(\dot{\kappa}^* \pi_f)h = \pi_f(\dot{\kappa}h) = Pf h = P(f - Pf)h, \quad \text{for every } h \in H.$$

Since the completion of H consists of all zero-mean functions and $\dot{\kappa}^*$ is required to map $\ell^\infty(\mathcal{F})$ into $\tilde{H}$, this means that $\dot{\kappa}^* \pi_f$ is equal to $f - Pf$. Conclude that $G(f)$ in Theorem 3.12.2 has variance $P(f - Pf)^2$. A similar argument applied to linear combinations of projections shows that the covariance function of G is equal to the covariance of a Brownian bridge $\mathbb{G}_P$.

The condition that ℓ is bounded and continuous can be relaxed in many situations. In view of Theorem 1.11.3, equation (3.12.9) is valid for every function ℓ that is continuous almost surely under the distribution of $\mathbb{G}_P$ and such that the sequence $\ell(\sqrt{n}(\mathbb{P}_n - P_{n,h}))$ is asymptotically equi-integrable under $P_{n,h}$.

The result (3.12.9) asserts that the empirical distribution is asymptotically minimax in a local sense. If $\mathcal{F}$ is uniformly Donsker in a class $\mathcal{P}$ of underlying measures and ℓ is bounded and Lipschitz-continuous, then

$$\lim_{n \rightarrow \infty} \sup_{P \in \mathcal{P}} E_P \ell(\sqrt{n}(\mathbb{P}_n - P)) = \sup_{P \in \mathcal{P}} E\ell(\mathbb{G}_P).$$

From consideration of the lower bounds for the local minimax risks, the right-hand side is the minimum value attainable for any (asymptotically measurable) estimator sequence if the local submodels $P_{n,h}$ corresponding to the measure(s) P for which the right-hand side is (nearly) maximal also belong to $\mathcal{P}$. This is particularly the case if $\mathcal{P}$ is the collection of all probability measures. In this situation the empirical measure is “globally asymptotically minimax over $\mathcal{P}$ ” with respect to the loss ℓ .

In the classical situation of cells $(-\infty, t]$ in Euclidean space, the global asymptotic minimax character has been shown to be valid also for loss functions that are not continuous or bounded. (See the Notes to this chapter.) In more general situations this question appears not to have been investigated.

Problems and Complements

1. Both the convolution and the minimax theorem remain valid if the set H is not a linear space but does contain a convex cone that has the same closed linear span as H .

[Hint: Van der Vaart (1988).]

2. Suppose that $\mathcal{P} = \{P_\theta: \theta \in \Theta\}$ is a parametric model indexed by an open subset of Euclidean space such that

$$\int \left[dP_\theta^{1/2} - dP_{\theta_0}^{1/2} - \frac{1}{2}(\theta - \theta_0)' \dot{\ell}_{\theta_0} dP_{\theta_0}^{1/2} \right]^2 d\mu = o(\|\theta - \theta_0\|^2),$$

for some measurable, vector-valued function $\dot{\ell}_{\theta_0}$. Derive the optimal variance for estimating a parameter $q(\theta)$ given a differentiable map $q: \Theta \rightarrow \mathbb{R}$. Is this parameter always regular?

3. A convex set in a Banach space is closed with respect to the norm if and only if it is weakly closed. Thus, “ $\tau(\mathbf{B}^*)$ -subconvex” is identical to “subconvex”.

[Hint: Apply the Hahn-Banach theorem, or see Rudin (1973).]

4. Suppose L_n and M_n are uniformly tight sequences of Borel probability measures on a metric space. If $L_n = M_n * N_n$ for every n , then the sequence N_n is uniformly tight.

3.13

Random Empirical Processes

In this chapter we consider the empirical process indexed by “random functions”. Given a collection $\{f_{\theta,\eta}: \theta \in \Theta, \eta \in H\}$ of measurable functions $f_{\theta,\eta}: \mathcal{X} \rightarrow \mathbb{R}$ on a measurable space $(\mathcal{X}, \mathcal{A})$, indexed by sets Θ and H , and “estimators” $\hat{\eta}_n$, we consider the stochastic process

$$(3.13.1) \quad \mathbb{P}_n f_{\theta, \hat{\eta}_n} = \frac{1}{n} \sum_{i=1}^n f_{\theta, \hat{\eta}_n}(X_i).$$

The purpose is to provide conditions that ensure that, as $n \rightarrow \infty$,

$$(3.13.2) \quad \sup_{\theta \in \Theta} |\mathbb{G}_n(f_{\theta, \hat{\eta}_n} - f_{\theta, \eta_0})| \xrightarrow{\mathbb{P}^*} 0.$$

Here an “estimator” $\hat{\eta}_n$ is a random element with values in H defined on the same probability space as the variables $X_1, \dots, X_n$ that define the empirical measure $\mathbb{P}_n$, and $\eta_0 \in H$ is a fixed element, which is typically a limit in probability of the sequence $\hat{\eta}_n$.

The result (3.13.2) helps to derive the limit behavior of the stochastic process

$$\sqrt{n}(\mathbb{P}_n f_{\theta, \hat{\eta}_n} - P f_{\theta, \eta_0}) = \mathbb{G}_n(f_{\theta, \hat{\eta}_n} - f_{\theta, \eta_0}) + \mathbb{G}_n f_{\theta, \eta_0} + \sqrt{n}P(f_{\theta, \hat{\eta}_n} - f_{\theta, \eta_0}).$$

If (3.13.2) holds, then the first term on the right converges to zero in probability. If the functions f_{θ, η_0} are square-integrable, the second term on the right will converge to a Gaussian process, in $\ell^\infty(\Theta)$ if the functions form a Donsker class. The behavior of the third term depends on the estimators $\hat{\eta}_n$, and would typically follow from an application of the delta-method, applied to the map $\eta \mapsto (P f_{\theta, \eta}: \theta \in \Theta)$. In Section 3.13.4 we give several concrete examples of this set-up.

We present two different approaches to establishing (3.13.2). The first is a straightforward application of techniques from the preceding chapters, and is useful

when little is known about the estimators $\hat{\eta}_n$, beyond possibly that they are consistent for some value η_0 . In the second approach it is assumed that the standardized sequence of estimators $\sqrt{n}(\hat{\eta}_n - \eta_0)$ converges in distribution.

3.13.1 Uniform Function Classes

The consistency of the estimators assumed in the first approach is understood in the sense that

$$(3.13.3) \quad \sup_{\theta \in \Theta} P(f_{\theta, \hat{\eta}_n} - f_{\theta, \eta_0})^2 \xrightarrow{P^*} 0.$$

Define entropy integrals $J_{[]}(\delta, \mathcal{F} | F, L_2(P))$ and $J(\delta, \mathcal{F} | F, L_2)$ of a class of functions $\mathcal{F}$ relative to an envelope function F as in Chapter 2.14. The *Lindeberg condition* on a sequence of envelope functions is understood to mean that $PF_n^2 = O(1)$ and $PF_n^2 1_{F_n \geq \varepsilon \sqrt{n}} \rightarrow 0$, for every $\varepsilon > 0$.

3.13.4 Theorem. Suppose that H_0 is a fixed subset of H such that $P(\hat{\eta}_n \in H_0) \rightarrow 1$ and suppose that the class of functions $\{f_{\theta, \eta}; \theta \in \Theta, \eta \in H_0\}$ is P -Donsker. If (3.13.3) holds, then (3.13.2) is valid.

3.13.5 Theorem. Suppose that H_n are subsets of H such that $P(\hat{\eta}_n \in H_n) \rightarrow 1$ and such that the classes of functions $\mathcal{F}_n = \{f_{\theta, \eta}; \theta \in \Theta, \eta \in H_n\}$ satisfy either $J_{[]}(\delta_n, \mathcal{F}_n | F_n, L_2(P)) \rightarrow 0$, or $J(\delta_n, \mathcal{F}_n | F_n, L_2) \rightarrow 0$, for every sequence $\delta_n \rightarrow 0$, relative to measurable envelope functions F_n that satisfy the Lindeberg condition. In the second case also assume that the classes of functions $\mathcal{F}_{n, \delta}$ defined in the proof (for $\delta > 0$) and $\mathcal{F}_n^2$ are P -measurable. If (3.13.3) holds, then (3.13.2) is valid.

Proofs. The first theorem can be proved along the lines of the proof of Lemma 3.3.5. For the proof of the second theorem we may assume without of generality that $f_{\theta, \eta_0} = 0$; otherwise, we replace the functions in $\mathcal{F}_n$ by the functions $f_{\theta, \eta} - f_{\theta, \eta_0}$, giving classes of functions with similar entropy integrals relative to the envelopes $2F_n$. Because the entropy integrals decrease if the envelope functions are replaced by bigger functions, we may and shall assume that the envelope functions F_n are bounded below by 1.

For $\delta > 0$, let $\mathcal{F}_{n, \delta}$ be the set of functions $\{f \in \mathcal{F}_n: Pf^2 \leq \delta^2\}$. Then by (3.13.3) and the assumption that $P(\hat{\eta}_n \in H_n) \rightarrow 1$, we have $\sup_{\theta \in \Theta} |\mathbb{G}_n f_{\theta, \hat{\eta}_n}| \leq \sup_{f \in \mathcal{F}_{n, \delta}} |\mathbb{G}_n f|$ on a set of probability tending to one, for every $\delta > 0$. It suffices to show that the variable on the right tends to zero in probability, as $n \rightarrow \infty$ followed by $\delta \downarrow 0$.

The envelope function F_n of the class $\mathcal{F}_n$ is also an envelope function of $\mathcal{F}_{n, \delta}$, and the bracketing numbers of the latter class are smaller than those of $\mathcal{F}_n$. By Theorem 2.14.16,

$$E^* \sup_{f \in \mathcal{F}_{n, \delta}} |\mathbb{G}_n f| \lesssim J_{[]}(\delta, \mathcal{F}_n | F_n, L_2(P)) \|F_n\|_{P, 2} + \frac{PF_n^2 1\{F_n > a_{n, \delta} \sqrt{n}\}}{a_{n, \delta}},$$

where $a_{n, \delta} = \delta \|F_n\|_{P, 2} / (1 + \log N_{[]}(\delta \|F_n\|_{P, 2}, \mathcal{F}_n, L_2(P)))^{1/2}$. Under the bracketing and Lindeberg assumptions of the second theorem, the first term on the

right side tends to zero for every $\delta = \delta_n \downarrow 0$. This implies that it must be bounded at every fixed $\delta > 0$ (see Problem 3.13.3), which in turn implies that $\log N_{[]}(\delta \|F_n\|_{P,2}, \mathcal{F}_n, L_2(P))$ must be bounded and hence that $a_{n,\delta}$ is bounded away from zero, for sufficiently small fixed δ , as $n \rightarrow \infty$. By the Lindeberg condition we conclude that $a_{n,\delta}^{-1} P F_n^2 1\{F_n > a_{n,\delta} \sqrt{n}\} \rightarrow 0$ as $n \rightarrow \infty$, for every fixed $\delta > 0$. This concludes the proof of the second theorem under the bracketing assumption.

By Theorem 2.14.1 and the Cauchy-Schwarz inequality,

$$(3.13.6) \quad \mathbb{E}^* \sup_{f \in \mathcal{F}_{n,\delta}} |\mathbb{G}_n f| \lesssim \left[\mathbb{E}^* J^2(\theta_{n,\delta}, \mathcal{F}_{n,\delta} | F_n, L_2) \mathbb{E} \mathbb{P}_n F_n^2 \right]^{1/2},$$

for

$$\theta_{n,\delta}^2 = \frac{\|\mathbb{P}_n f^2\|_{\mathcal{F}_{n,\delta}}}{\mathbb{P}_n F_n^2} \leq \|\mathbb{P}_n f^2\|_{\mathcal{F}_{n,\delta}} \wedge 1.$$

The covering numbers of $\mathcal{F}_{n,\delta}$ are bounded by the covering numbers of $\mathcal{F}_n$ and hence the uniform entropy integral of $\mathcal{F}_{n,\delta}$ is bounded by the uniform entropy integral of $\mathcal{F}_n$. Thus for every $\varepsilon > 0$ the square of the right side of (3.13.6) is bounded above by a multiple of

$$\left[J^2(1, \mathcal{F}_n | F_n, L_2) \mathbb{P}^*(\theta_{n,\delta} \geq \varepsilon) + J^2(\varepsilon, \mathcal{F}_n | F_n, L_2) \right] P F_n^2.$$

The theorem is proved once it is shown that $\theta_{n,\delta} \rightarrow 0$ in outer probability as $n \rightarrow \infty$, followed by $\delta \rightarrow 0$.

Because $P f^2 \leq \delta^2$ for every $f \in \mathcal{F}_{n,\delta}$, by the definition of the class $\mathcal{F}_{n,\delta}$ and $\mathcal{F}_{n,\delta} \subset \mathcal{F}_n$, this is certainly the case if $\|\mathbb{P}_n - P\|_{\mathcal{F}_n^2}$ tends to 0 in outer probability. By the Cauchy-Schwarz inequality $|\mathbb{P}_n f^2 - P f^2| \leq \|f - g\|_{\mathbb{P}_{n,2}} \|2F_n\|_{\mathbb{P}_{n,2}}$, so that $N(2\varepsilon \|F_n\|_{\mathbb{P}_{n,2}}, \mathcal{F}_n^2, L_1(\mathbb{P}_n)) \leq N(\varepsilon, \mathcal{F}_n, L_2(\mathbb{P}_n))$. Boundedness of the uniform entropy integral shows that the latter numbers are bounded, for every sufficiently small $\varepsilon > 0$. The convergence $\|\mathbb{P}_n - P\|_{\mathcal{F}_n^2} \xrightarrow{P^*} 0$ then follows from a straightforward extension of the Glivenko-Cantelli theorem, Theorem 2.4.3, to classes of functions that depend on n . ■

The preceding theorems do not take the rate of approach of $\hat{\eta}_n$ to η_0 into account. One might hope that the sequence δ_n in the theorems could be restricted to this rate, thus giving a considerable strengthening of their assertion. This appears to be false hope in general, but it is possible to go in this direction by application of more dedicated maximal inequalities, as illustrated in the following theorem, which applies to uniformly bounded functions. Instead of (3.13.3) assume that, for some $\delta_n \downarrow 0$,

$$(3.13.7) \quad \sup_{\theta \in \Theta} P(f_{\theta, \hat{\eta}_n} - f_{\theta, \eta_0})^2 = O_P^*(\delta_n).$$

3.13.8 Theorem. Suppose that H_n are subsets of H such that $P(\hat{\eta}_n \in H_n) \rightarrow 1$ and such that the classes of functions $\mathcal{F}_n = \{f_{\theta, \eta}; \theta \in \Theta, \eta \in H_n\}$ are uniformly bounded and satisfy either $J_{[]}(\delta_n, \mathcal{F}_n | 1, L_2(P)) \rightarrow 0$ and $J_{[]}(\delta_n, \mathcal{F}_n | 1, L_2(P)) \lesssim \sqrt{n}\delta_n^2$ or $J(\delta_n, \mathcal{F}_n | 1, L_2) \rightarrow 0$ and $J(\delta_n, \mathcal{F}_n | 1, L_2) \lesssim \sqrt{n}\delta_n^2$. In the second case also assume that the classes $\mathcal{F}_{n, \delta}$ and $\mathcal{F}_{n, \delta}^2$ defined in the proof of Theorem 3.13.5 are P -measurable. If (3.13.7) holds, then (3.13.2) is valid.

Proof. This follows the same lines as the first part of the proof of Theorem 3.13.5, but we apply the maximal inequality given by Theorem 2.14.17 instead of the one of Theorem 2.14.16, or the maximal inequality Theorem 2.14.2 in the case uniform entropy integrals are employed, to the class of functions $\mathcal{F}_{n, M\delta_n}$, for a (every) fixed large constant M . ■

Given the many techniques to verify that a given class of functions is Donsker, or to compute bounds on its entropy, the preceding three theorems give quick results, if they apply. Furthermore, they appear to be close to best possible unless more information about the estimators $\hat{\eta}_n$ can be brought in, possibly about the dependence between $\hat{\eta}_n$ and the empirical measure $\mathbb{P}_n$. Information that the estimators $\hat{\eta}_n$ converge at a certain rate and/or possess certain regularity properties (such as having uniformly bounded derivatives), cannot be exploited in Theorem 3.13.4, but could be used for the choice of the sets H_n or is explicit in Theorems 3.13.5 and 3.13.8.

3.13.2 Weakly Converging Estimators

An alternative approach assumes that the estimators $\hat{\eta}_n$ converge in distribution, after centering and scaling. Let H be a Banach space, and suppose that the sequence $\sqrt{n}(\hat{\eta}_n - \eta_0)$ converges in distribution to a tight, Borel-measurable random element in H . The tight limit of the sequence $\sqrt{n}(\hat{\eta}_n - \eta_0)$ takes its values in a σ -compact subset $H_0 \subset H$. For $\theta \in \Theta$, $h_0 \in H_0$, and $\delta > 0$, let $F_{n, \theta, h_0, \delta}$ be an arbitrary measurable envelope function for the class of functions

$$\mathcal{F}_{n, \theta, h_0, \delta} = \left\{ f_{\theta, \eta_0 + n^{-1/2}h} - f_{\theta, \eta_0} : h \in H, \|h - h_0\| < \delta \right\}.$$

3.13.9 Theorem. Suppose that the sequence $\sqrt{n}(\hat{\eta}_n - \eta_0)$ converges in distribution to a tight, random element with values in a given σ -compact subset H_0 of H . Then (3.13.2) is valid if (i)–(iii) hold, where:

- (i) $\sup_{\theta \in \Theta} |\mathbb{G}_n(f_{\theta, \eta_0 + n^{-1/2}h_0} - f_{\theta, \eta_0})| \xrightarrow{P^*} 0$, for every $h_0 \in H_0$.
- (ii) $\sup_{\theta \in \Theta} |\mathbb{G}_n F_{n, \theta, h_0, \delta}| \xrightarrow{P^*} 0$, for every $\delta > 0$ and every $h_0 \in H_0$;
- (iii) $\sup_{\theta \in \Theta} \sup_{h_0 \in K} \sqrt{n} P F_{n, \theta, h_0, \delta_n} \rightarrow 0$, for every $\delta_n \rightarrow 0$ and every compact $K \subset H_0$.

3.13.10 Addendum. The set H_0 may be reduced to the union $H_0 = \cup_m K_m$ of any sequence of compact sets $K_m \subset H$ such that the weak limit of $\sqrt{n}(\hat{\eta}_n - \eta_0)$ gives probability one to $\cup_m K_m^\delta$, for every $\delta > 0$, and the compact sets K in (iii) may be restricted to the sets K_m .

Proof. Suppose that $\sqrt{n}(\hat{\eta}_n - \eta_0) \rightsquigarrow Z$ and let $\varepsilon > 0$ be fixed. There exists a compact set $K \subset H_0$ with $P(Z \in K) > 1 - \varepsilon$ and hence for every $\delta > 0$, with K^δ the set of all points at distance less than δ to K ,

$$(3.13.11) \quad \liminf_{n \rightarrow \infty} P_*(\sqrt{n}(\hat{\eta}_n - \eta_0) \in K^{\delta/2}) > 1 - \varepsilon.$$

In view of the compactness of K there exist finitely many elements $h_1, \dots, h_{p(\delta)} \in K \subset H_0$ such that the balls of radius $\delta/2$ around these points cover K . Then $K^{\delta/2}$ is contained in the union of the balls of radius δ , by the triangle inequality. Thus, with $B(h, \delta)$ denoting the ball of radius δ around h in the space H ,

$$\left\{ \sqrt{n}(\hat{\eta}_n - \eta_0) \in K^{\delta/2} \right\} \subset \bigcup_{i=1}^{p(\delta)} \left\{ \hat{\eta}_n \in B(\eta_0 + n^{-1/2}h_i, n^{-1/2}\delta) \right\}.$$

It follows that with inner probability at least $1 - \varepsilon$, as $n \rightarrow \infty$,

$$\begin{aligned} \sup_{\theta} |\mathbb{G}_n(f_{\theta, \hat{\eta}_n} - f_{\theta, \eta_0})| &\leq \sup_{\theta} \max_i \sup_{\|h - h_i\| < \delta} |\mathbb{G}_n(f_{\theta, \eta_0 + n^{-1/2}h} - f_{\theta, \eta_0})| \\ &\leq \sup_{\theta} \max_i \sup_{\|h - h_i\| < \delta} \left[|\mathbb{G}_n(f_{\theta, \eta_0 + n^{-1/2}h} - f_{\theta, \eta_0 + n^{-1/2}h_i})| \right. \\ &\quad \left. + |\mathbb{G}_n(f_{\theta, \eta_0 + n^{-1/2}h_i} - f_{\theta, \eta_0})| \right] \\ &\leq \sup_{\theta} \max_i |\mathbb{G}_n F_{n, \theta, h_i, \delta}| + 2 \sup_{\theta} \sup_{h_0 \in K} \sqrt{n} P F_{n, \theta, h_0, \delta} \\ &\quad + \sup_{\theta} \max_i |\mathbb{G}_n(f_{\theta, \eta_0 + n^{-1/2}h_i} - f_{\theta, \eta_0})|, \end{aligned}$$

where in the last step we use the inequality $|\mathbb{G}_n f| \leq 2\sqrt{n}PF + |\mathbb{G}_n F|$, valid for any functions f and F with $|f| \leq F$. The maxima in the display are over the finite set $i = 1, \dots, p(\delta)$, and the elements $h_1, \dots, h_{p(\delta)} \in K$ depend on δ . By assumptions (ii) and (i) the first and third terms converge to zero as $n \rightarrow \infty$, for every fixed δ . It follows that there exists $\delta_n \downarrow 0$ such that these terms with δ_n substituted for δ converge to 0. For this δ_n , all three terms converge to zero in probability as $n \rightarrow \infty$, in view of (iii).

The addendum follows, because the compact sets K in the preceding proof need only satisfy (3.13.11), and this is also true for the sets K_m and sufficiently large m . ■

The two conditions (i) and (ii) of Theorem 3.13.9 concern the empirical process indexed by the classes of functions

$$\begin{aligned} &\{f_{\theta, \eta_0 + n^{-1/2}h_0} - f_{\theta, \eta_0} : \theta \in \Theta\}, \\ &\{F_{n, \theta, h_0, \delta} : \theta \in \Theta\}. \end{aligned}$$

These classes are indexed by Θ only, and hence Theorem 3.13.9 avoids conditions that involve the complexity of the class of functions $\{f_{\theta, \eta} : \theta \in \Theta, \eta \in H\}$ due to the parameter $\eta \in H$. The following two lemmas, which are in the spirit of Theorems 3.13.4 and 3.13.5, provide easy means to verify the conditions (i) and (ii).

3.13.12 Lemma. Suppose that the class of functions $\cup_n \mathcal{F}_n$ is P -Donsker. If $\sup_{f \in \mathcal{F}_n} P f^2 \rightarrow 0$, then $\sup_{f \in \mathcal{F}_n} |\mathbb{G}_n f| \xrightarrow{P^*} 0$.

3.13.13 Lemma. Suppose that for measurable envelope functions F_n that satisfy the Lindeberg condition either $J_{[\cdot]}(\delta_n, \mathcal{F}_n | F_n, L_2(P)) \rightarrow 0$ or $J(\delta_n, \mathcal{F}_n | F_n, L_2) \rightarrow 0$, for all $\delta_n \downarrow 0$. In the second case also assume that each class $\mathcal{F}_n$ is suitably measurable. If $\sup_{f \in \mathcal{F}_n} P f^2 \rightarrow 0$, then $\sup_{f \in \mathcal{F}_n} |\mathbb{G}_n f| \xrightarrow{P^*} 0$.

Condition (iii) of Theorem 3.13.9 is an analytical condition, involving the means of the envelope functions $F_{n,\theta,h_0,\delta}$ and not the empirical process. Any envelope functions may be used, as long as the same envelopes are used in both (ii) and (iii). For the minimal envelope functions (if measurable) the condition takes the form: for any $\delta_n \downarrow 0$,

$$\sup_{\theta} \sup_{h_0 \in K} \sqrt{n} P \sup_{\|h-h_0\| < \delta_n} |f_{\theta, \eta_0+n^{-1/2}h} - f_{\theta, \eta_0+n^{-1/2}h_0}| \rightarrow 0.$$

This is an “integrated uniform Lipschitz assumption” on the map $\eta \mapsto f_{\theta,\eta}$ “locally at η_0 ”. The set K in the preceding display, or (iii), is a compact set in the support of the limit distribution of the sequence $\sqrt{n}(\hat{\eta}_n - \eta_0)$. The display, or (iii), may be valid for any compact $K \subset H$, but sometimes more precise information about the limit process must be exploited. For instance, if the weak limit is a Gaussian process in $\ell^\infty(T)$, then K may be taken to be a set of functions $z: T \rightarrow \mathbb{R}$ that is uniformly bounded and uniformly equicontinuous relative to the standard deviation metric of the limit process, and T will be totally bounded (see Example 1.5.10).

The addendum helps to make this more precise. For instance, it shows that if the weak limit is zero-mean Gaussian, then K may be restricted to multiples of the unit ball in the reproducing kernel Hilbert space of the limit variable (see Problem 3.13.1), and hence H_0 may be taken equal to the reproducing Hilbert space. For instance, if the limit is Brownian motion or bridge on the unit interval, the functions can be assumed to be elements of the Cameron-Martin space, i.e. absolutely continuous with square-integrable derivative.

3.13.3 Composition

An interesting particular case is that the functions $f_{\theta,\eta}$ take the form

$$f_{\theta,\eta}(x) = \theta(\eta(x)),$$

for maps $\theta: \mathbb{R}^d \rightarrow \mathbb{R}$ and $\eta: \mathcal{X} \rightarrow \mathbb{R}^d$. The realizations of the estimators $\hat{\eta}_n$ are then functions $x \mapsto \hat{\eta}_n(x)$ on the sample space $\mathcal{X}$ and can be evaluated at the observations to obtain the *pseudo observations* $\hat{\eta}_n(X_1), \dots, \hat{\eta}_n(X_n)$, which are (dependent) random vectors in $\mathbb{R}^d$. The random empirical process $\{\mathbb{P}_n f_{\theta,\hat{\eta}_n}: \theta \in \Theta\}$ is the empirical measure

$$\mathbb{P}_n \theta \circ \hat{\eta}_n = \frac{1}{n} \sum_{i=1}^n \theta(\hat{\eta}_n(X_i))$$

of these pseudo observations, indexed by the functions θ . Note that we have suppressed the randomness in the notation $\hat{\eta}_n$; the pseudo observations would typically depend on the data “twice”, in the form $\hat{\eta}_n(X_i) = \hat{\eta}_n(X_i; X_1, \dots, X_n)$.

We first give a general set of sufficient conditions for the validity of condition (i) of Theorem 3.13.9, and next consider separately the cases that the functions θ are smooth or of bounded variation, respectively. The second case is more involved and requires more stringent conditions, but includes the classical empirical process of the pseudo observations, when the functions θ are chosen indicator functions of cells. Variations of the results are easily possible; for instance, $\mathbb{R}^d$ could be replaced by a more general Banach space, and the Banach space H could be chosen with a stronger norm to make the Hadamard differentiability work, as long as the sequence $\sqrt{n}(\hat{\eta}_n - \eta)$ converges in H . Weighted uniform norms are alternatives to the norm of $\ell^\infty(\mathcal{X}, \mathbb{R}^d)$.

3.13.14 Lemma. *Suppose that Θ is a suitably measurable class of functions with measurable envelope function Θ_{env} such that $J(1, \Theta | \Theta_{env}, L_2) < \infty$ and $P(\Theta_{env} \circ \eta_0)^2 < \infty$, and that the functions $\Theta_{env} \circ (\eta_0 + n^{-1/2}h_0)$ satisfy the Lindeberg condition in $L_2(P)$, for every $h_0 \in H_0$. If, for every $h_0 \in H_0$,*

$$\sup_{\theta \in \Theta} P(\theta \circ (\eta_0 + n^{-1/2}h_0) - \theta \circ \eta_0)^2 \rightarrow 0,$$

then condition (i) of Theorem 3.13.9 is satisfied for the functions of the form $f_{\theta, \eta} = \theta \circ \eta$.

Proof. For $f_{\theta, \eta} = \theta \circ \eta$, condition (i) of Theorem 3.13.9 takes the form $\sup_{f \in \mathcal{F}_n} |\mathbb{G}_{n, Q} f| \xrightarrow{P^*} 0$, for $\mathbb{G}_{n, Q}$ the empirical process of a random sample from the distribution $Q = P \circ (\eta_0, h_0)^{-1}$ of the pair $(\eta_0(X_1), h_0(X_1))$, and $\mathcal{F}_n$ the class of functions

$$\mathcal{F}_n = \{(y, z) \mapsto \theta(y + n^{-1/2}z) - \theta(y) : \theta \in \Theta\}.$$

The class $\mathcal{F}_n$ is contained in the difference of the classes of functions

$$\begin{aligned} \mathcal{F}'_n &= \{(y, z) \mapsto \theta(y + n^{-1/2}z) : \theta \in \Theta\}, \\ \mathcal{F}'' &= \{(y, z) \mapsto \theta(y) : \theta \in \Theta\}. \end{aligned}$$

These classes possess envelope functions F'_n and F'' defined by

$$\begin{aligned} F'_n(y, z) &= \Theta_{env}(y + n^{-1/2}z), \\ F''(y, z) &= \Theta_{env}(y). \end{aligned}$$

The uniform entropy of $\mathcal{F}''$ relative to F'' is finite by assumption. The uniform entropy of $\mathcal{F}'_n$ relative to F'_n is exactly the same, as the law of $Y + n^{-1/2}Z$ runs through all possible laws on $\mathbb{R}$ if the law of (Y, Z) runs through all possible laws on $\mathbb{R} \times \mathbb{R}$. The uniform entropy of $\mathcal{F}_n$ relative to $F_n = F'_n + F''$ is bounded by the sum of the uniform entropies of $\mathcal{F}'_n$ and $\mathcal{F}''$, in view of Theorem 2.10.22. The lemma follows by an application of Lemma 3.13.13. ■

3.13.3.1 Smooth Functions

For differentiable functions θ , the conditions for convergence of the processes $\theta \mapsto \sqrt{n}(\mathbb{P}_n \theta \circ \hat{\eta}_n - \theta \circ \eta_0)$ are mild. The following two results separately give sufficient conditions for handling the remainder term (3.13.2) and the drift term in the decomposition. If the conditions of both lemmas are satisfied and the class Θ is $P \circ \eta_0^{-1}$ -Donsker, then the given sequence will be tight in $\ell^\infty(\Theta)$, and typically converge in distribution, depending on $\hat{\eta}_n$.

3.13.15 Lemma. *Suppose that Θ is a suitably measurable collection of uniformly bounded, uniformly Lipschitz functions $\theta: \mathbb{R}^d \rightarrow \mathbb{R}$ of which the restrictions Θ_K to any compact $K \subset \mathbb{R}^d$ satisfy $J(1, \Theta_K | 1, L_2) < \infty$. If the sequence $\sqrt{n}(\hat{\eta}_n - \eta_0)$ converges weakly in $\ell^\infty(\mathcal{X}, \mathbb{R}^d)$ to a tight random element, for some $\eta_0 \in \ell^\infty(\mathcal{X}, \mathbb{R}^d)$, then $\sup_{\theta \in \Theta} |\mathbb{G}_n(\theta(\hat{\eta}_n) - \theta(\eta_0))| \xrightarrow{P^*} 0$.*

3.13.16 Lemma. *If Θ is a collection of continuously differentiable functions $\theta: \mathbb{R}^d \rightarrow \mathbb{R}$ in Θ with derivatives $\dot{\theta}$ such that $\|\dot{\theta}(u)\| \leq 1$ for every $u \in \mathbb{R}^d$, then the map $\eta \mapsto (P\theta \circ \eta; \theta \in \Theta)$ from $L_1(\mathcal{X}, \mathcal{A}, P)^d$ to $\ell^\infty(\Theta)$ is Hadamard differentiable at η_0 with derivative $h \mapsto (P[\dot{\theta}(\eta_0)h]; \theta \in \Theta)$.*

Proofs. The assumption that the functions θ are uniformly Lipschitz, implies that, for every $x \in \mathcal{X}$,

$$\left| \theta(\eta_0(x) + n^{-1/2}h(x)) - \theta(\eta_0(x) + n^{-1/2}h_0(x)) \right| \leq \frac{\|h(x) - h_0(x)\|}{\sqrt{n}}.$$

The norm in the right side is bounded by the supremum norm $\|h - h_0\|_\infty$ on $\ell^\infty(\mathcal{X}, \mathbb{R}^d)$. It follows that the envelope functions in Theorem 3.13.9 can be taken equal to $F_{n,\theta,h_0,\delta} = \delta/\sqrt{n}$. Then condition (ii) of the theorem is trivially satisfied because the empirical process vanishes on constants, and the validity of condition (iii) is immediate.

For the verification of condition (i) of Theorem 3.13.9 it suffices to consider the functions θ on the union of the ranges of the functions $\eta_0 + h_0/\sqrt{n}$ for a fixed h_0 in the support of the limit distribution of the sequence $\sqrt{n}(\hat{\eta}_n - \eta_0)$. This union is contained in a fixed compact set $K \subset \mathbb{R}^d$ and hence we may restrict the domain of the functions θ to K . By assumption $J(1, \Theta_K | 1, L_2) < \infty$, and by the preceding display with $h = 0$, the displayed condition of Lemma 3.13.14 is (easily) satisfied. Thus by the latter lemma condition (i) of Theorem 3.13.9 is also satisfied, whence the first lemma is a corollary of the latter theorem.

For the proof of the second lemma, given a sequence of functions $h_t: \mathcal{X} \rightarrow \mathbb{R}^d$ with $P\|h_t - h\| \rightarrow 0$, for some $h \in L_1(\mathcal{X}, \mathcal{A}, P)^d$ we can write, by Fubini's theorem,

$$\begin{aligned} \left| \frac{P(\theta(\eta_0 + th_t) - \theta(\eta_0))}{t} - P\dot{\theta}(\eta_0)h \right| &= \left| \int_0^1 P[\dot{\theta}(\eta_0 + sth_t)h_t - \dot{\theta}(\eta_0)h] ds \right| \\ &\leq \int_0^1 P\|\dot{\theta}(\eta_0 + sth_t) - \dot{\theta}(\eta_0)\| \|h_t\| ds + P\|\dot{\theta}(\eta_0)\| \|h_t - h\|. \end{aligned}$$

The second term on the right is bounded above by $P\|h_t - h\|$ and converges to

zero by assumption. The first term on the right converges to zero by the dominated convergence theorem. ■

If $d = 1$, then any collection of uniformly bounded, Lipschitz functions $\theta: K \rightarrow \mathbb{R}$ on a compact interval K satisfies $J(1, \Theta_K | 1, L_2) < \infty$. Thus in that case the assertion of the preceding theorem is true for any collection of uniformly bounded Lipschitz functions. If $d > 1$ further restrictions on the class Θ are necessary, for instance Hölder smoothness of order $\alpha > d/2$, or VC-type conditions.

Lemma 3.13.16 gives Hadamard differentiability of the drift term $\eta \mapsto P\theta \circ \eta$ relative to the $L_1(P)^d$ -norm on $\ell_m^\infty(\mathcal{X}, \mathbb{R}^d) \subset L_1(\mathcal{X}, \mathcal{A}, P)^d$, and hence certainly relative to the uniform norm on $\ell_m^\infty(\mathcal{X}, \mathbb{R}^d)$.

3.13.3.2 Functions of Bounded Variation

For functions θ that are not smooth, the conditions for convergence (3.13.2) are less mild. We consider the case that every function $\theta: \mathbb{R}^d \rightarrow \mathbb{R}$ in Θ is the survival function $\theta(x) = \int 1_{[x, \infty)} d\theta$ of a subprobability measure on $\mathbb{R}^d$.

The following lemmas require that the variable $\eta_0(X_1)$ possesses a continuous density, and that for every h_0 in the support of the limit of the sequence $\sqrt{n}(\hat{\eta}_n - \eta_0)$ the conditional distribution of $h_0(X_1)$ given $\eta_0(X_1) = s$ depends continuously on the value of s . The second condition is clearly unpleasant, but appears to be natural in the present situation.

3.13.17 Lemma. *Let Θ be a collection of survival functions $\theta: \mathbb{R}^d \rightarrow [0, 1]$ of subprobability measures supported on an interval $(a, b) \subset \mathbb{R}^d$. If the sequence $\sqrt{n}(\hat{\eta}_n - \eta_0)$ converges in distribution in $\ell^\infty(\mathcal{X}, \mathbb{R}^d)$ to a tight Borel measure concentrating on the σ -compact set H_0 , and for every $h_0 \in H_0$ there exists a version of the conditional distribution of $h_0(X_1)$ given $\eta_0(X_1) = s$ that is weakly continuous in s , then $\sup_{\theta \in \Theta} |\mathbb{G}_n(\theta(\hat{\eta}_n) - \theta(\eta_0))| \xrightarrow{P^*} 0$.*

3.13.18 Lemma. *Let Θ be a collection of survival functions $\theta: \mathbb{R} \rightarrow [0, 1]$ of subprobability measures supported on a compact interval $[a, b] \subset \mathbb{R}$. If the variable $\eta_0(X_1)$ possesses a continuous Lebesgue density f , then the map $\eta \mapsto (P\theta \circ \eta: \theta \in \Theta)$ from $\ell^\infty(\mathcal{X})$ to $\ell^\infty(\Theta)$ is Hadamard differentiable at η_0 tangentially to the set of all h_0 such that there exists a version of the conditional distribution of $h_0(X_1)$ given $\eta_0(X_1) = s$ that is weakly continuous in s . The derivative is the map*

$$h \mapsto \left(- \int \mathbb{E}(h(X_1) | \eta(X_1) = s) f(s) d\theta(s): \theta \in \Theta \right).$$

Proofs. Because every θ is nonincreasing in each of its arguments, we have for every h with $\|h - h_0\|_\infty < \delta$,

$$\begin{aligned} & |\theta(\eta_0 + n^{-1/2}h) - \theta(\eta_0 + n^{-1/2}h_0)| \\ & \leq \theta(\eta_0 + n^{-1/2}h_0 - n^{-1/2}\delta) - \theta(\eta_0 + n^{-1/2}h_0 + n^{-1/2}\delta) \\ & = \int 1\{s - n^{-1/2}\delta \leq \eta_0 + n^{-1/2}h_0 \leq s + n^{-1/2}\delta\} d\theta(s), \end{aligned}$$

where δ in the right side is the vector $(\delta, \dots, \delta)$ (and $d\theta$ refers to the nonnegative measure defined by θ). It follows that the envelope function $F_{n, \theta, h_0, \delta}$ in The-

orem 3.13.9 can be taken equal to the function on the right of the display. The expectation of the integrand with h_0 replaced by a converging sequence $h_n \rightarrow h_0$ and δ replaced by $\delta_n \rightarrow 0$ tends to zero, by Lemma 3.13.19 (below) with the variables (X, Y, Y_t) taken equal to $(\eta_0(X_1), h_0(X_1), h_n(X_1))$, $t = 1/\sqrt{n}$, $g = 1$, $x_t = s - n^{-1/2}\delta$, $a_t = 1$ and $b_t = 2\delta_n \rightarrow 0 = b$. Therefore, by Fubini's theorem, $\sqrt{n}PF_{n,\theta,h_0,\delta_n} \rightarrow 0$, for every $\delta_n \downarrow 0$, uniformly in h_0 contained in a compact set $K \subset \ell^\infty(\mathcal{X}, \mathbb{R}^d)$ and uniformly in θ , because the total variation norms $\|\theta\|$ are uniformly bounded. This verifies condition (iii) of Theorem 3.13.9. Because the functions θ take values in $[0, 1]$, squaring the left side of the display makes it smaller and hence $P(\theta(\eta_0) - \theta(\eta_0 + n^{-1/2}h_0))^2$ is bounded above by the expectation of the right side of the display with $h = 0$ (and δ taken equal to $\|h_0\|_\infty$) and tends to 0 (even at $\sqrt{n}$ -rate), uniformly in θ , for every fixed h_0 . Because the survival functions of subprobability measures are in the convex hull of the set of indicator functions $1_{[t,\infty)}$ and the zero function, which is a VC-class, $J(1, \Theta | 1, L_2) < \infty$, by Corollary 2.6.12. We can now see that (i)–(ii) of Theorem 3.13.9 are satisfied with the help of Lemma 3.13.14. This concludes the proof of the first lemma.

For given $s \in \mathbb{R}$ and $u > 0$ let $\chi_{s,u}$ be the continuous function that takes the value 0 on $(-\infty, s - u]$, takes the value 1 on $[s, \infty)$ and is linear on the interval $[s - u, s]$. Then $1_{[s,\infty)} \leq \chi_{s,u}$, and hence

$$P(1_{s \leq \eta_0 + th_t} - 1_{s \leq \eta_0}) \leq P(\chi_{s,u}(\eta_0 + th_t) - \chi_{s,u}(\eta_0)) + P(\chi_{s,u}(\eta_0) - 1_{[s,\infty)}(\eta_0)).$$

Because $\eta_0(X_1)$ possesses a Lebesgue density that is bounded on compact sets and $1_{[s,\infty)} - \chi_{s,u}$ vanishes off the set $(s - u, s)$, the second term on the right is bounded in absolute value by a multiple of u , uniformly in s ranging through compacta. By choosing $u = \delta t$ this term divided by t can be made arbitrarily small by choice of δ . Because $\chi_{s,u}$ is absolutely continuous with derivative $1/u$ on $(s - u, s)$ and 0 elsewhere, the first term on the right divided by $1/t$ can be written in the form

$$\int_0^1 \frac{1}{u} P(h_t 1_{s-u < \eta_0 + vth_t \leq s}) dv.$$

For $u = \delta t$ and $\|h_t - h_0\|_\infty \rightarrow 0$, this converges to $E(h_0(X_1) | \eta_0(X_1) = s)f(s)$, by Lemma 3.13.19 (with $x_t = s - \delta t$, $X = \eta_0(X_1)$, $g(Y_t) = Y_t = h_t(X_1)$, $a_t = v$ and $b_t = \delta$), uniformly in s ranging over compacta. It follows that, uniformly in s ranging over compacta

$$\limsup_{t \downarrow 0} \frac{1}{t} P(1_{s \leq \eta_0 + th_t} - 1_{s \leq \eta_0}) \leq E(h_0(X_1) | \eta_0(X_1) = s)f(s).$$

A similar argument using the functions $\chi_{s+u,u}$ instead of $\chi_{s,u}$ gives a corresponding lower bound, whence the quotient on the left side converges to the right side, uniformly in s ranging through compacta.

For a subsurvival function θ we can write $P(\theta(\eta_0 + th_t) - \theta(\eta_0)) = -\int P(1_{s \leq \eta_0 + th_t} - 1_{s \leq \eta_0}) d\theta(s)$. By the preceding paragraph the right side times t^{-1} tends to $-\int E(h_0(X_1) | \eta_0(X_1) = s)f(s) d\theta(s)$, uniformly in subsurvival functions $\theta \in \Theta$ that are supported on a compact interval with total variation bounded by 1. ■

3.13.19 Lemma. Suppose that X, Y, Y_t (for $t > 0$) are real-valued random variables on a common probability space such that:

- (i) X possesses a continuous Lebesgue density f ;
- (ii) $\|Y_t - Y\|_\infty \rightarrow 0$ and $\|Y\|_\infty < \infty$;
- (iii) a regular version of $\mathcal{L}(Y|X = x)$ is weakly continuous in x .

Then for every continuous function $g: \mathbb{R} \rightarrow \mathbb{R}$ and every converging sequences $x_t \rightarrow x$, $a_t \rightarrow a$ and $0 \leq b_t \rightarrow b$, as $t \rightarrow 0$,

$$\frac{1}{t} \mathbb{E}[g(Y_t) 1_{x_t < X + ta_t Y_t \leq x_t + tb_t}] \rightarrow b \int g(y) \mathbb{E}(g(Y)|X = x) f(x).$$

Proof. First consider the case that $Y_t = Y$ for every t . For f as in (i) and $K(x, \cdot)$ the weakly continuous regular version of $\mathcal{L}(Y|X = x)$ in (iii),

$$\begin{aligned} & \frac{1}{t} \mathbb{E}[g(Y) 1_{x_t < X + ta_t Y < x_t + tb_t}] \\ &= \frac{1}{t} \int \mathbb{E}[g(Y) 1_{t^{-1}(x_t - s) < a_t Y \leq t^{-1}(x_t - s) + b_t} | X = s] f(s) ds \\ &= \int \int g(y) 1_{u < a_t y \leq u + b_t} K(x_t - ut, dy) f(x_t - ut) du. \end{aligned}$$

The inner integral is equal to $\mathbb{E}g(Z_t) 1_{u < a_t Z_t \leq u + b_t}$ for Z_t possessing the law $K(x_t - ut, \cdot)$. By assumption $a_t Z_t$ converges in distribution to the law of aY for Y possessing the law $K(x, \cdot)$. It follows that the inner integral converges to $\int g(y) 1_{u < ay \leq u + b} K(x, dy)$ for any (u, b) such that u and $u + b$ are not among the atoms of the law of aY . Because this includes almost every u for every fixed b , and the range of u in the double integral is bounded by the assumption that $\|Y\|_\infty$ is finite, we can conclude by the dominated convergence theorem that the right side of the display converges to

$$\iint g(y) 1_{u < ay \leq u + b} K(x, dy) f(x) du.$$

This reduces to $b \int g(y) K(x, dy) f(x)$ by Fubini's theorem. This concludes the proof of the lemma for the case that $Y_t = Y$.

Because g is continuous, Y possesses bounded range and $\|Y_t - Y\|_\infty \rightarrow 0$, we have that $\|g(Y_t) - g(Y)\|_\infty \rightarrow 0$. Therefore

$$\begin{aligned} & \frac{1}{t} \mathbb{E}|g(Y_t) - g(Y)| 1_{x_t < X + ta_t Y_t < x_t + tb_t} \\ & \leq o(1) \frac{1}{t} \mathbb{P}(x_t - ta_t \|Y_t\|_\infty < X < x_t + tb - t + ta_t \|Y_t\|_\infty). \end{aligned}$$

This converges to zero as X has a density that is bounded on bounded intervals.

Finally the difference $1_{x_t < X + ta_t Y_t < x_t + tb_t} - 1_{x_t < X + ta_t Y < x_t + tb_t}$ is nonzero only if $X + ta_t Y$ is in the union of two (nonrandom) intervals of total length bounded by $ta_t \|Y_t - Y\|_\infty$ in a neighborhood of x . By the lemma with $Y_t = Y$ and $g = 1$, which is already proved, $t^{-1} \mathbb{P}(x_t < X + ta_t Y < x_t + tb_t) \rightarrow cf(x)$ for any

sequences $x_t \rightarrow x$ and $c_t \rightarrow c$. Hence this probability is $o(t)$ for $c_t = ta_t \|Y_t - Y\|_\infty$. We conclude from this that

$$\begin{aligned} & \frac{1}{t} |\mathbb{E}g(Y)(1_{x_t < X + ta_t Y_t < x_T + tb_t} - 1_{x_t < X + ta_t Y < x_t + tb_t})| \\ & \leq \|g(Y)\|_\infty \frac{1}{t} \mathbb{E}|1_{x_t < X + ta_t Y_t < x_T + tb_t} - 1_{x_t < X + ta_t Y < x_t + tb_t}| \rightarrow 0. \end{aligned}$$

The proof of the lemma is complete upon combining the preceding paragraphs. ■

3.13.4 Examples

In this section we apply the preceding results to several examples.

3.13.4.1 Regression Residuals

Suppose that $(X_1, Y_1), \dots, (X_n, Y_n)$ are a random sample distributed according to the regression model $Y = g_\eta(X) + e$. For given estimators $\hat{\eta}_n$ we form the residuals $\hat{e}_i = Y_i - g_{\hat{\eta}_n}(X_i)$ and are interested in the empirical process corresponding to $\hat{e}_1, \dots, \hat{e}_n$, i.e. the process, for a given set Θ of functions $\theta: \mathbb{R} \rightarrow \mathbb{R}$,

$$\theta \mapsto \frac{1}{n} \sum_{i=1}^n \theta(\hat{e}_i) = \frac{1}{n} \sum_{i=1}^n \theta(Y_i - g_{\hat{\eta}_n}(X_i)).$$

This is of the form (3.13.1) for the functions $f_{\theta, \eta}$ defined as $f_{\theta, \eta}(x, y) = \theta(y - g_\eta(x))$.

In many cases it will be possible to apply the simplest of the preceding theorems, Theorem 3.13.4. As a particular example, consider that g_η is a polynomial in $x \in \mathbb{R}^d$, and that Θ is the class of indicator functions $1_{(-\infty, \theta]}$, for $\theta \in \mathbb{R}$. Then the functions $f_{\theta, \eta}$ are the indicator functions of the sets $\{(x, y): y - g_\eta(x) - \theta \leq 0\}$. Because the set of functions $(x, y) \mapsto y - g_\eta(x) - \theta$ is contained in a finite-dimensional vector space, it is a VC-class, and hence so is the collection of its negativity sets (see Lemmas 2.6.16 and 2.6.20). Thus the class of functions $f_{\theta, \eta}$ is Donsker, and Theorem 3.13.4 yields (3.13.2).

To obtain the weak convergence of the residual empirical process in $\ell^\infty(\Theta)$, we need to combine this with the differentiability of the map $\eta \mapsto Pf_{\theta, \eta} = \mathbb{E}\theta[e + (g_{\eta_0} - g_\eta)(X)]$ at η_0 , uniformly in $\theta \in \Theta$. For smooth functions θ this will typically be true without further conditions, whereas for indicators $1_{(-\infty, \theta]}$ differentiability will depend on sufficient smoothness of the error distribution.

3.13.4.2 Kendall's Process

For $\hat{\eta}_n$ the empirical distribution function of a random sample $X_1, \dots, X_n$ from a distribution η_0 on $\mathbb{R}^d$, consider the process

$$\mathbb{K}_n(\theta) = \frac{1}{n} \sum_{i=1}^n 1_{\hat{\eta}_n(X_i) \leq \theta}, \quad \theta \in \mathbb{R}.$$

The standardized version $\sqrt{n}(\mathbb{K}_n - K_0)$, for $K_0(\theta) = P(\eta_0(X_1) \leq \theta)$, is known as *Kendall's process*. The asymptotic normality of this process may be proved using the preceding results, applied with the composition map $f_{\theta, \eta} = 1_{(-\infty, \theta]} \circ \eta$.

An attempt to apply Theorem 3.13.4 to the classical Kendall process would lead to the consideration of the class of all indicator functions of sets of the form $\{x \in \mathbb{R}^d: \eta(x) \leq \theta\}$, for η ranging over the cumulative distribution functions on $\mathbb{R}^d$ and $\theta \in \mathbb{R}$. This class is similar to the collection of all “lower layers” in $\mathbb{R}^d$, and, unfortunately, fails to be Donsker for most distributions.[†] It appears to be necessary to exploit the limit behavior of the sequence $\sqrt{n}(\hat{\eta}_n - \eta_0)$, which is possible through an application of Theorem 3.13.9. Of course, this sequence of empirical processes tends in distribution in $\ell^\infty(\mathbb{R}^d)$ to a d -dimensional η_0 -Brownian sheet.

If $d = 1$, then the Brownian sheet is a Brownian bridge and can be represented as $B \circ \eta_0$ for B a standard Brownian bridge on the unit interval. A typical function in the support of the limit distribution of the sequence $\sqrt{n}(\hat{\eta}_n - \eta_0)$ can be represented as $h_0 = h \circ \eta_0$ for some function $h: [0, 1] \rightarrow \mathbb{R}$. The conditional law of the variable $h_0(X_1)$ given $\eta_0(X_1) = s$ is the Dirac measure at $h(s)$. Because the standard Brownian bridge is continuous, the corresponding Markov kernels $K(z, \cdot) = \delta_z(\cdot)$ are weakly continuous in z , as required in Lemmas 3.13.17 and 3.13.18.

In the more interesting case that $d > 1$, the conditioning event $\eta_0(X_1) = s$ will typically restrict X_1 to a one-dimensional curve in $\mathbb{R}^d$. Under sufficient smoothness of η_0 , this curve will vary continuously with s , and under further smoothness conditions on the law of X_1 , the conditional distribution of $h_0(X_1)$ given $\eta_0(X_1) = s$ for a continuous function h_0 will vary continuously as well, and the following corollary applies.

3.13.20 Corollary. *Suppose that for a given interval $[a, b] \subset (0, 1)$:*

- (i) *the distribution K_0 of the variable $\eta_0(X_1)$ possesses a Lebesgue density k_0 that is continuous on a neighborhood of $[a, b]$.*
- (ii) *the conditional distribution of X_1 given $\eta_0(X_1) = s$, has a regular version that is weakly continuous in $s \in [a, b]$.*

Then the sequence of processes $\sqrt{n}(\mathbb{K}_n - K_0)$ tends in $\ell^\infty[a, b]$ in distribution to the process $(\mathbb{G}_{\eta_0} f_\theta: \theta \in [a, b])$, for $\mathbb{G}_{\eta_0}$ an η_0 -Brownian bridge process and $f_\theta: [0, 1]^d \rightarrow \mathbb{R}$ defined by $f_\theta(x) = 1_{\eta_0(x) \leq \theta} - k_0(\theta) \mathbb{E}[1_{x \leq X_1} | \eta_0(X_1) = \theta]$.

Proof. The corollary is a consequence of Lemmas 3.13.17 and 3.13.18.

To ascertain the given form of the limit process, note first that the derivative in Lemma 3.13.18 under the identification $\theta = 1_{(-\infty, \theta]}$ is given by $L(h)(\theta) = -\mathbb{E}[h(X_1) | \eta_0(X_1) = \theta] k_0(\theta)$. This is, of course, linear in h , so that the approximation to the drift term $\sqrt{n}(P(1_{(-\infty, \theta]} \circ \hat{\eta}_n - 1_{(-\infty, \theta]} \circ \eta_0))$ satisfies

$$L(\sqrt{n}(\hat{\eta}_n - \eta_0)) = \frac{1}{\sqrt{n}} \sum_{i=1}^n L(1_{X_i \leq \cdot} - \eta_0(\cdot)) = \mathbb{G}_n L(1_{x \leq \cdot}).$$

In these expressions $1_{X_i \leq \cdot}$ is the function $s \mapsto 1_{X_i \leq s}$, L works on this function for fixed X_i , and the empirical process works on the function $x \mapsto L(1_{x \leq \cdot})$ as a function of x , for fixed $\cdot$. ■

[†] See Dudley (2014), pages 300, 400.

Alternatively, we may consider the Kendall process indexed by functions $\sqrt{n}(\mathbb{P}_n \theta \circ \hat{\eta}_n - P\theta \circ \eta_0)$, for a general class of functions $\theta: \mathbb{R} \rightarrow \mathbb{R}$. Under the condition that the functions θ are continuously differentiable with derivatives bounded by 1, Lemmas 3.13.15 and 3.13.16 give without further conditions that this process converges in distribution in $\ell^\infty(\Theta)$ to the process $(\mathbb{G}_{\eta_0} f_\theta: \theta \in \Theta)$, for $\mathbb{G}_{\eta_0}$ an η_0 -Brownian bridge process in $\ell^\infty(\Theta)$ and $f_\theta: \mathbb{R}^d \rightarrow \mathbb{R}$ defined as $f_\theta(x) = \theta \circ \eta_0(x) - \int_{[x, \infty)} \dot{\theta} \circ \eta_0 d\eta_0$.

3.13.4.3 Copula Function

The *copula function* of a distribution function η on $\mathbb{R}^d$ is the function

$$C(u) = \eta(\eta_1^{-1}(u_1), \dots, \eta_d^{-1}(u_d)), \quad u = (u_1, \dots, u_d) \in [0, 1]^d.$$

Here η_j is the j th marginal distribution function of η , and η_j^{-1} its quantile function. Given an i.i.d. sample $X_1, \dots, X_n$ from a distribution η_0 , we can form the empirical distribution function $\hat{\eta}_n(x) = n^{-1} \sum_{i=1}^n 1_{X_i \leq x}$, extract its marginal empirical distribution functions $\hat{\eta}_{n,j}$, for $j = 1, \dots, d$, form the “pseudo-observations”

$$\hat{e}_{n,i} = (\hat{\eta}_{n,1}(X_{i,1}), \dots, \hat{\eta}_{n,d}(X_{i,d}))^t,$$

and construct an estimator of the true copula function C_0 by

$$\hat{C}_n(u) = \frac{1}{n} \sum_{i=1}^n 1_{\hat{e}_{n,i} \leq u}, \quad u \in [0, 1]^d.$$

This is close to the multivariate version $\hat{\eta}_n(\hat{\eta}_{n,1}^{-1}, \dots, \hat{\eta}_{n,d}^{-1})$ of the *empirical copula function*, discussed in Section 3.10.5.4, the difference being in the definitions of the inverses. The convergence in distribution of the sequence $\sqrt{n}(\hat{C}_n - C_0)$ in $\ell^\infty((0, 1)^2)$ can be established through Hadamard differentiability exactly as for the empirical copula function, as in Example 3.10.31. Alternatively, the convergence may be established through the results in the present chapter.

A straightforward application of Lemmas 3.13.17 and 3.13.18 to this aim, with $f_{\theta, \eta}(x) = 1_{\eta_1(x_1) \leq \theta_1, \dots, \eta_d(x_d) \leq \theta_d}$, is possible if the true copula function is continuously differentiable on its domain $[0, 1]^d$. Unfortunately, the latter restriction excludes copulas that are less well-behaved near the boundary of the unit cube, including the Gaussian copula. If the true copula is continuous differentiable on the open cube, then it is still straightforward to deduce the convergence of the sequence $\sqrt{n}(\hat{C}_n - C_0)$ on a closed subinterval of $[0, 1]^d$. Alternatively, the arguments could be refined by using different norms and more elaborate analytical developments.

As it is shown in Section 3.10.5.4, that the assumption of continuous differentiability on the full domain $[0, 1]^d$ can be relaxed, we omit a discussion of the refinement, but instead discuss convergence of the copula process indexed by smooth functions. For a function $\theta: [0, 1]^d \rightarrow \mathbb{R}$, set

$$\hat{C}_n \theta = \frac{1}{n} \sum_{i=1}^n \theta(\hat{e}_{n,i}) = \mathbb{P}_n f_{\theta, \hat{\eta}_n},$$

for

$$f_{\theta, \eta}(x) = \theta(\eta_1(x_1), \dots, \eta_d(x_d)), \quad x = (x_1, \dots, x_d).$$

Then $\hat{C}_n \theta$ is an estimator of $\eta_0 f_{\theta, \eta}$, which is equal to $C_0 \theta = \int \theta dC_0$, if η_0 has

continuous marginals, as $(\eta_{0,1}(X_1), \dots, \eta_{0,d}(X_d))$ is distributed according to C_0 in that case.

3.13.21 Corollary. *For every suitably measurable collection Θ of continuously differentiable functions $\theta: [0, 1]^d \rightarrow \mathbb{R}$ with derivatives $\dot{\theta}$ such that $\sup_u \|\dot{\theta}(u)\|_\infty \leq 1$ satisfying $J(1, \Theta | 1, L_2) < \infty$, the sequence of processes $(\sqrt{n}(\hat{C}_n - C_0)\theta: \theta \in \Theta)$ converges in distribution in $\ell^\infty(\Theta)$. If η_0 has continuous marginal distributions, the limiting process can be represented as $(\mathbb{G}_{C_0} f_\theta: \theta \in \Theta)$, where $\mathbb{G}_{C_0}$ is a C_0 -Brownian bridge process in $\ell^\infty(\Theta)$ and $f_\theta: [0, 1]^d \rightarrow \mathbb{R}$ is the function defined by $f_\theta(u) = \theta(u) + \int \dot{\theta}(s)(1_{u_1 \leq s_1}, \dots, 1_{u_d \leq s_d})_{j=1,d}^t dC_0(s)$.*

Proof. We apply Lemmas 3.13.15 and 3.13.16, with the estimators $(\hat{\eta}_{n,1}, \dots, \hat{\eta}_{n,d})$ and with the functions $f_{\theta,\eta}$ given in the display before the corollary, which are the composition of θ with the functions $x \mapsto (\eta_1(x_1), \dots, \eta_d(x_d))$. By Lemma 3.13.16 the approximation to the drift term $P_0(f_{\theta,\hat{\eta}_n} - f_{\theta,\eta_0})$ takes the form

$$\begin{aligned} & L(\sqrt{n}(\hat{\eta}_n - \eta_0)) \\ &= \int \dot{\theta}(\eta_{0,1}(x_1), \dots, \eta_{0,d}(x_d)) (\sqrt{n}(\hat{\eta}_{n,j}(x_j) - \eta_{0,j}(x_j)))_{j=1,d}^t dP_0(x) \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \int \dot{\theta}(\eta_{0,1}(x_1), \dots, \eta_{0,d}(x_d)) (1_{X_{i,j} \leq x_j} - \eta_{0,j}(x_j))_{j=1,d}^t dP_0(x). \end{aligned}$$

The weak limit process is the process $\mathbb{G}_{\eta_0}[f_{\theta,\eta_0} + g_{\theta,\eta_0}]$, where g_{θ,η_0} is any function such that $g_{\theta,\eta_0}(X_i) - P_0 g_{\theta,\eta_0}$ is the i th term of the sum on the right side, and $\mathbb{G}_{\eta_0}$ is the η_0 -Brownian motion (and $\eta_0 = P_0$).

If the marginals $\eta_{0,i}$ are continuous, we can change variables in the integral defining g_{θ,η_0} , writing this in the form

$$g_{\theta,\eta_0}(x) = \int \dot{\theta}(u)(1_{\eta_{0,j}(x_j) \leq u_j})_{j=1,d}^t dC_0(u).$$

We can further use that the process $\theta \mapsto \mathbb{G}_{\eta_0} \theta \circ (\eta_{0,1}, \dots, \eta_{0,d})$ is equal in distribution to the Brownian bridge $\theta \mapsto \mathbb{G}_{C_0} \theta$. ■

Problems and Complements

1. A zero-mean Gaussian random element in a separable Banach space gives probability one to $\cup_{m \in \mathbb{N}} K_m^\delta$, for every $\delta > 0$, for K_m equal to m times the unit ball of its reproducing kernel Hilbert space.

[Hint: This is a consequence of the geometric version of Borell's inequality; see Borell (1975) or Proposition A.2.13.]

2. If C_0 is continuously differentiable on $[0, 1]^d$ with gradient ∇C_0 , the sequence of copula processes $\sqrt{n}(C_n - C_0)$ given in Section 3.13.4.3 can be shown to converge in distribution in $\ell^\infty([0, 1]^d)$ to the process $(\mathbb{G}_{\eta_0} f_\theta: \theta \in [0, 1]^d)$, where $\mathbb{G}_{\eta_0}$ is an η_0 -Brownian bridge process, and $f_\theta(u) = 1_{u \leq \theta} - \nabla C_0(\theta)^t (1_{u_1 \leq s_1}, \dots, 1_{u_d \leq s_d})$.

3. Suppose $J_n(\delta) = \int_0^\delta j_n(\varepsilon) d\varepsilon$ for a nonincreasing function $j_n: (0, \infty) \rightarrow (0, \infty)$. If $J_n(\delta_n) \rightarrow 0$ for every $\delta_n \downarrow 0$, then $J_n(\delta)$ and $j_n(\delta)$ are bounded for every fixed $\delta > 0$.

[Hint: If $J_n(\delta)$ is bounded for some $\delta > 0$, then it is clearly bounded for all smaller values of δ , but also for bigger values as $J_n(k\delta) \leq kJ_n(\delta)$. If there were no such δ , then for $\delta = 1, 1/2, 1/3, \dots$ we could find $n_1 < n_2 < n_3 < \dots$ with $J_{n_i}(1/i) > i$. Thus for $\delta_{n_i} = 1/i$ we have $J_n(\delta_n) \rightarrow \infty$ along the subsequence $\{n_i\}$. This contradicts the assumption. For the second assertion we use $\delta j_n(\delta) \leq J_n(\delta)$.]

4. **(Mean absolute deviation)** For $X_1, \dots, X_n$ i.i.d. with distribution P on $\mathbb{R}^d$, set $D_n = n^{-1} \sum_{i=1}^n \|X_i - \bar{X}_n\|_p$.

- (i) If $d = p = 1$ and $E|X_1| < \infty$, then $D_n \rightarrow P|X - \mu|$, almost surely.
- (ii) If $d = p = 1$ and $EX_1^2 < \infty$, then $\sqrt{n}(D_n - d) \rightarrow_d N(0, v)$, for some d, v under appropriate additional assumptions.
- (iii) Similar results hold for general $d, p \in \mathbb{N}$.

[Hint: Use elementary arguments, or the results of this chapter.]

5. In the setting of Lemma 3.13.16, suppose that $P\|\eta_0\|^p < \infty$ and $P \sup_\eta \|\dot{\theta}(\eta)\|^q \lesssim 1$, for every θ , for $q \geq 1$ such that $1/p + 1/q = 1$. Then the map $\eta \mapsto P\theta \circ \eta$ is Hadamard differentiable at η_0 tangentially to $L_p(\mathcal{X}, \mathcal{A}, P)^d$. The supremum over η can be restricted to an $L_p(\mathcal{X}, \mathcal{A}, P)$ -neighborhood of η_0 . (Contributed by Thomas Nagler.)

3

Notes

3.2. Versions of the continuous mapping theorem for the argmax functional have already been used by Prakasa Rao (1969), Le Cam (1970a), and Ibragimov and Has'minskii (1981). An explicit statement of a slightly more special theorem than the theorem presented here occurs in Kim and Pollard (1990). It is clear from the proof that uniform weak convergence of the criterion functions can be relaxed to the weak convergence of pairs of maxima of the form $(\sup_{h \in A} \mathbb{M}_n(h), \sup_{h \in B} \mathbb{M}_n(h))$. This is slightly weaker, but in general still much stronger than needed. If the limit process possesses multiple points of maximum, then the limit behavior of the points of maximum $\hat{h}_n$, even if these are unique, will depend on lower order terms in the processes $\mathbb{M}_n$. Under the assumption that the points of maximum of the limit process can be ordered, Ferger (2004) shows that the points $\hat{h}_n$ will asymptotically be distributed between the smallest and the largest point of maximum.

For general notes on results on rates of convergence, see the notes for Chapter 3.4. Theorem 3.2.10 is essentially due to Kim and Pollard (1990). The maximum likelihood estimator for a monotone decreasing density (or concave distribution function) is known as the Grenander estimator. Its limit behavior was derived by Prakasa Rao (1969). In his study of the concave majorant of Brownian motion Groeneboom (1983) introduced an “inverse process” similar to $\hat{s}_n$ and the corresponding basic switching identity which is the basis of our proof (with the inequalities, strict or not, slightly corrected as in Van der Laan and Van der Vaart (2006)). Groeneboom (1985) goes on to initiate the study of global functionals of $\hat{f}_n$, in particular the L_1 distance from $\hat{f}_n$ to f using the structure of the inverse process $\hat{s}_n$. The limit distribution of the maximum likelihood estimator based on interval censored observations was first obtained by Groeneboom (1987). The present ex-

ample is an adaptation of the presentation in Part 2 of Groeneboom and Wellner (1992). Groeneboom (1988) gave a systematic study of the “inverse process” associated with Brownian motion minus a parabola, including a determination of its infinitesimal generator. As a corollary of this deep and detailed analysis, Groeneboom characterized the distribution of the point of maximum of Brownian motion minus a parabola, and also its joint distribution with the maximum itself.

Rigorous proofs of the asymptotic normality of the maximum likelihood estimator were first given in the 1930s and 1940s. See Stigler (2007) for more on the early history. For a direct proof under the classical smoothness conditions, see, for instance, Cramér (1946). General M -estimators were studied systematically by Huber (1967), who also introduced bracketing conditions to ensure their asymptotic consistency and normality. Pollard (1984, 1985) shows the gains of applying results from empirical processes, as in this monograph, to this problem. Theorems 3.2.16 and 3.3.1 are generalizations of results in this work. Hoffmann-Jørgensen (1994) also aims at applying techniques from probability in Banach spaces to maximum likelihood estimators (in mostly parametric models).

Example 3.2.18 can be found in Hjort and Pollard (1993). Example 3.2.12 follows Huang (1993). Example 3.2.27 is taken from lecture notes by the first author. The Lipschitz condition on $\theta \mapsto \log p_\theta$ can of course be relaxed, but it seems to cover all the standard examples. For one-dimensional parameters, a better approach is possible based on the maximal inequality given by Theorem 2.2.4, applied with $\psi(\varepsilon) = \varepsilon^2$ and the Hellinger distance. Then the pointwise-Lipschitz condition can be replaced by the condition $\int [p_{\theta_1}^{1/2} - p_{\theta_2}^{1/2}]^2 \leq |\theta_1 - \theta_2|^2$. See Le Cam (1970a) and Ibragimov and Has'minskii (1981) for results in this direction.

3.3. Estimators satisfying estimating equations were studied systematically by Huber (1967). Pollard (1984, 1985) shows the benefits of applying results from empirical processes, as in this monograph, to this problem. Theorems 3.3.1 is a generalization of results in his work to infinite dimensions. Also, see Bickel, Klaassen, Ritov and Wellner (1993) or Van der Vaart (1996) for results of this type. Extensions to estimating equations with nuisance parameters can be found in Chapter 3.13.

Theorem 3.3.1 is used by Van der Vaart (1994a, 1994c, 1995) and Murphy (1995) to prove the asymptotic efficiency of the maximum likelihood estimator in some semiparametric models. Similar results can be found in Bickel, Klaassen, Ritov and Wellner (1993). Example 3.3.10 is based on Van der Vaart (1994a, 1994c). For a general treatment of infinite-dimensional estimating equations for semiparametric models, see Chapter 25 of Van der Vaart (1998). For other applications of empirical processes in obtaining the limit distribution of estimators in semiparametric models, see Van der Laan (1993), Huang (1996), and Van der Vaart (1996, 1998).

3.4. This chapter is a thorough revision and extension of the corresponding chapter in the first edition, of which a major part was inspired on the exposition by Van de Geer (1993b). The modulus of continuity, present in this paper and in the original chapter, was rediscovered in several papers in machine learning, where it is sometimes referred to as “local complexities”. The extension Theorem 3.4.6 to penalized

criteria goes back to at least Murphy and Van der Vaart (1997) and Van der Vaart (1998). The connection of the scaling to priors in Bayesian estimation is made in Wahba (1990), Van der Vaart and Van Zanten (2007, 2008b) and Szabó, Van der Vaart and Van Zanten (2013). A version of Lemma 3.4.8 was obtained in Dashti, Law, Stuart and Voss (2013), although already present in a different form in earlier papers (see Wahba (1990)).

Van de Geer (1990) studies least-squares and least-absolute-deviation estimators for regression functions. In the case of least-squares estimators, she uses chaining for sub-Gaussian processes, under the assumption that the error distribution is sub-Gaussian. The results on least-squares with random design were obtained by Birgé and Massart (1993). Van de Geer (1993b) considers maximum likelihood estimators for densities under the assumption that the model is convex or that the log likelihood ratios are uniformly bounded. Her arguments are based on chaining by bracketing.

Birgé and Massart (1993) suggest using the functions $\log(p + p_0)/2p_0$ as criterion functions rather than the log likelihood ratios. They also introduce the use of the refined version of Bernstein's inequality in the bracketing-chaining argument, both for regression and maximum likelihood estimators. Using this refined inequality, Wong and Shen (1995), removed unnecessary boundedness conditions and obtain a version of Theorem 3.4.12. Other references in this area are Wong and Severini (1991) and Shen and Wong (1994).

The maximum likelihood estimator for a monotone density was introduced by Grenander (1956). The rate at a point for the maximum likelihood estimator of a convex regression function is $n^{-2/5}$, as shown by Mammen (1991) and Wang (1993). Jongbloed (1995) obtains the same pointwise rate for the maximum likelihood estimator of a convex density. Limiting distributions were obtained for convex regression and the maximum likelihood estimator of a convex decreasing density by Groeneboom, Jongbloed and Wellner (2001); see also Groeneboom and Jongbloed (2014), Chapter 11. Gaussian mixtures were considered in Ghosal and Van der Vaart (2001) and Ghosal and Van der Vaart (2007).

Penalized least squares and their relationship to splines were studied Kimeldorf and Wahba (1971), Wahba (1978) and Wahba (1990), among others. The treatment given here borrows from Van der Vaart (1998). The Bayesian connection is developed in Van der Vaart and Van Zanten (2007) and Szabó, Van der Vaart and Van Zanten (2013).

Support vector machine were introduced by Aizerman, Braverman and Rozonoer (1964), Vapnik (1982) and Cortes and Vapnik (1995). Some authors call empirical risk estimators with hinge loss also support vector machines, but it seems the name ought to be connected to the RKHS penalty and kernel methods more than to the hinge loss. Our treatment of rates based on Theorem 3.4.6 on general minimum contrast estimators gives similar, but sometimes sharper, results as in Blanchard, Bousquet and Massart (2008) and Steinwart and Scovel (2007). The latter paper introduced the “geometric noise exponent”. The calculations in Example 3.4.36 are based on Jankova (2012) and Jankova and Van der Vaart (2013).

3.5. The main theorem of this chapter was adapted from Blanchard, Bousquet and

Massart (2008); also see Massart and Nédélec (2006) and Massart (2007). It combines neatly with the results on rates of convergence in Chapter 3.4.

The subsection on statistical learning goes back further. The idea of “structural risk minimization” was set forth in Vapnik and Červonenkis (1974) and was developed further in, among others, Vapnik (1982, 1995, 1998, 2000), Devroye, Györfi and Lugosi (1996), Anthony and Bartlett (1999) and Koltchinskii (2006, 2011).

3.6. The basic method of Poissonization was used by Kolmogorov (1933) and Donsker (1952) in their classic papers; see, e.g., Shorack and Wellner (1986), Chapter 8. Poissonization techniques in the study of Banach-space-valued random variables began in Le Cam (1970b) and were developed further in Araujo and Giné (1980). Kac (or Poissonized) empirical processes have also played an important role in obtaining rates of convergence for limit theorems and invariance principles; see, e.g., Massart (1989), in particular Lemma 2, and Dudley (1984), Section 8.3, where Poissonization is used to study rates of convergence for classes of sets too large to satisfy the central limit theorem. Durst and Dudley (1981) prove a partial result in the direction of Theorem 3.1 for classes of sets. In the classical one-dimensional case $\mathcal{X} = \mathbb{R}$, there is a substantial literature concerned with finite sample and asymptotic distributions of statistics connected with the Kac empirical process; see Csörgő-S. (1981) and the references therein.

Pyke (1968) proves Theorem 3.6.1 for the one-dimensional distribution function under the assumption that ν equals 1 almost surely. Billingsley (1968) and Csörgő-S. (1974), following Blum, Hanson and Rosenblatt (1963), show that the result continues to hold under the present conditions, which allow a general positive limit random variable ν . These results were extended by Wichura (1968) and Fernandez (1970) to allow, for instance, the q -metrics mentioned by Pyke (1968) in his closing remark. Theorem 3.6.1 contains their results as well as many others. Lemma 3.6.5 and its proof come from Giné and Zinn (1990), who only consider the symmetric case; this lemma also follows from the proof of a result of Le Cam (1970b), Theoreme 3, on accompanying Poisson laws.

Part of Theorem 3.6.6 is contained in Theorem 3.4.9 of Araujo and Giné (1980); see also their Exercise 2 on page 122, and note that their $X_{n,j}$ are assumed symmetric. Evarist Giné has shown us a proof of Theorem 3.6.6 via symmetrization and desymmetrization. The present statement of Theorem 3.6.6 is from Klaassen and Wellner (1992).

3.7. The bootstrap central limit theorems, Theorems 3.7.1 and 3.7.2, follow essentially from the bootstrap results of Giné and Zinn (1990), who prove the equivalence of (i) and (iii) under measurability assumptions on the class $\mathcal{F}$, the equivalence of (i) and (ii) being immediate from the multiplier central limit theorems. A new detail in the present statement is the absence of measurability assumptions.

The proofs of Giné and Zinn use symmetrization by Rademacher variables. If we write s(i), s(ii), and s(iii) for the corresponding symmetrized parts of Theorem 3.7.1 and s(ii)u for the unconditional symmetrized version of (ii), then the proof of Giné and Zinn (1990) consists of the steps $s(i) \Rightarrow s(ii)u \Rightarrow s(iii) \Rightarrow s(i)$. Here the first implication follows from Lemma 2.9 of Giné and Zinn (1984); the

second follows from the almost-sure multiplier central limit theorem of Ledoux and Talagrand (1988) and Proposition 2.2 of Giné and Zinn (1990); the last implication comes from Proposition 2.2 again. Giné and Zinn (1990) next show that the s 's can be removed from the string of implications. (They do not have $s(ii)$ or $s(ii)u$ in the statement of their main result, but only in their Lemma 2.1 and Proposition 2.2.)

The finite-dimensional convergence of the exchangeable bootstrap empirical process is proved by Mason and Newton (1990), who also establish Theorem 3.7.13 for the special case that $\mathcal{F}$ is the class of indicators of cells in the real line. The general version of Theorem 3.7.13 (under somewhat stronger assumptions, including that $\sum_{i=1}^n W_{ni} = n$) is from Praestgaard and Wellner (1993), as is the almost-sure part of Theorem 3.7.3. The in-probability part of Theorem 3.7.3 was noted by Arcones and Giné (1992).

3.9. For observations in the unit square, an alternative to the Kolmogorov-Smirnov statistic is

$$B_n = \iint n(\mathbb{H}_n(s, t) - \mathbb{P}_n(s)\mathbb{Q}_n(t))^2 d\mathbb{H}_n(s, t).$$

Blum, Kiefer and Rosenblatt (1961) show that under independence and continuity of the marginal distributions P and Q

$$B_n \rightsquigarrow \iint \mathbb{Z}_{P \times Q}^2(s, t) dP(s)dQ(t) \sim \int_0^1 \int_0^1 \mathbb{Z}_{[0,1]^2}^2(u, v) dudv.$$

They also compute the characteristic function of the limit variable and use this to compute and tabulate its distribution. For higher-dimensional generalizations and extensions, see Dugue (1975) and Deheuvels (1981). The sequence B_n is asymptotically equivalent to a statistic proposed by Hoeffding (1948).

For material on distributions related to the completely tucked Brownian sheet, see, e.g., Adler (1990), Section V.5.

3.10. The use of Hadamard instead of Fréchet differentiability to derive limiting distributions of transformed statistics was developed systematically by Reeds (1975), although there are earlier applications in work by Le Cam.

A number of the examples can be found in Gill (1989). Comprehensive reviews of the product integral can be found in Gill and Johansen (1990) and Gill (1994). We borrowed the symbol for the product integral from the latter author. They work in the more general framework of $p \times p$ matrices of elements of $D(\mathbb{R}^+)$, in which case the definition given here is appropriate only in the “commutative case.” The limit behavior of estimators connected to multivariate trimming is studied by Nolan (1992). We have reformulated her result in terms of Hadamard differentiability, under slightly simpler conditions. The result on the Hadamard differentiability of M -functionals is taken from Heesterman and Gill (1992) and Van der Vaart (1995), although this is also one of the main examples in Reeds (1975). For still more applications of the delta-method, see, e.g., Grübel and Pitts (1992, 1993) and Pitts (1994).

The weak convergence of the sample quantile process was proved by Bickel (1967). Substantial refinements, which involve further interplay between analysis and probability, are due to Shorack (1972) and Csörgő-M. and Révész (1978); see

Shorack and Wellner (1986), Chapter 18. See also Shorack and Wellner (1986), Chapter 15, for connections with the well known Bahadur-Kiefer theorems. For further applications of the approach taken here, see Doss and Gill (1992).

In the first edition of this work differentiability of the copula map was limited to intervals strictly within the supports, which does not allow to obtain the original result on empirical copulas due to Gaenssler and Stute (1987). This was noted by Fermanian, Radulović and Wegkamp (2004), Segers (2012), and Bücher and Volgushev (2013), who also introduced refinements. The present formulation gives another slight improvement.

The use of Hadamard differentiability does not allow conclusions about the speed of convergence. On the other hand, Fréchet differentiability, with a rate on the remainder term plus information on the speed of convergence of the original sequence, gives a rate on the linear approximation for the transformed statistics. In a series of papers and books Dudley (1990, 1992, 1994) and Dudley and Norvaiša (2009, 2011) explore Fréchet differentiability of some of the standard functionals with respect to nonstandard norms, such as p -variation on distribution functions.

3.11. Contiguity was introduced and studied by Le Cam. See Le Cam (1960, 1969). Also see Le Cam (1986), pages 85–90, for connections with limiting experiments and different characterizations. Differentiability in quadratic mean was introduced by Le Cam as well. Le Cam’s third lemma derives its name from being listed as the third in a list of basic lemmas. For the other lemmas, see the original papers by Le Cam or see Hájek and Šidák (1967).

3.12. The convolution and minimax theorems for locally asymptotically normal parametric models were established by Hájek (1970, 1972). Le Cam (1972) shows the connection with approximation of statistical experiments and states similar theorems for non-Gaussian limit experiments. The proofs for the Gaussian case in the present chapter have the benefit of being short and easy, but they lack the insight provided by Le Cam’s method, which is to show the convergence of a sequence of experiments to a limit in a general sense first and next obtain concrete results (such as a convolution theorem) from an analysis of the limit experiment. The convolution theorem for infinite-dimensional parameters was proved in increasing generality by Levit (1978), Millar (1983, 1985), Van der Vaart (1988), and Van der Vaart and Wellner (1989), among others. The assumption of asymptotic tightness and measurability can be relaxed considerably, as well as the assumption that the tangent space is linear, see Van der Vaart (1988). For many examples of tangent spaces in semiparametric models, see Bickel, Klaassen, Ritov and Wellner (1993).

Anderson (1955) proves a concentration lemma for symmetric distributions on Euclidean space. Lemma 3.12.4, taken from Millar (1983), extends his result to infinite-dimensional Gaussian variables.

Kiefer and Wolfowitz (1959) shows the global asymptotic minimax character of the empirical distribution function in Euclidean space. Here “global” means that they took the supremum over all probability measures, rather than over shrinking neighborhoods as in the present chapter. This is better in terms of attainment of

the bound, but less interesting for the lower bound. Millar (1979) shows that the empirical distribution is also asymptotically globally minimax in certain models not consisting of all probability distributions on the sample space. His results can be further refined to local minimaxity, for which the key property is that the tangent space is sufficiently large. See Problem 3.12.1.

3.13. This chapter is based on Van der Vaart and Wellner (2007). Theorem 3.13.5 is taken from Van der Vaart (2002).

The notion of “pseudo-observations” was coined in the papers Ghoudi and Rémillard (1998, 2004), who studied empirical processes of such pseudo-observations in some detail. Kendall’s process was studied earlier in Barbe, Genest, Ghoudi and Rémillard (1996). The latter paper (pages 202–208) gives a number of explicit examples, where the density function k_0 is unbounded at 0 and hence not continuous on $[0, 1]$ (which include the case in which η_0 is the uniform distribution on $[0, 1]^d$). For such distributions Corollary 3.13.20 does not yield convergence of Kendall’s process in the space $\ell^\infty([0, 1])$, even though this may still be valid. Under the growth condition $k_0(t) = o(t^{-1/2}(\log(1/t))^{-1/2-\varepsilon})$, as $t \downarrow 0$, for some $\varepsilon > 0$, Barbe, Genest, Ghoudi and Rémillard (1996) prove convergence on the full domain using weighted empirical processes.

The empirical copula function was first studied by Stute (1984) and Gaenssler and Stute (1987). More recent work on this subject includes Fermanian, Radulović and Wegkamp (2004), Segers (2012), Bücher and Volgushev (2013), Ghoudi and Rémillard (1998) and Genest, Nešlehová and Rémillard (2017). See also Section 3.10.5.4.

Bücher, Segers and Volgushev (2014) discuss alternative metrics and modes of convergence related to the theme of this chapter.

PART A

Appendix

A.1

Inequalities

This section presents several inequalities for sums of independent stochastic processes. For a stochastic process $\{X_t: t \in T\}$ indexed by some arbitrary index set, the notation $\|X\|$ is an abbreviation for the supremum $\sup_t |X_t|$.

The stochastic processes need not be measurable maps into a Banach space. Independence of the stochastic processes $X_1, X_2, \dots$ is understood in the sense that each of the processes is defined on a product probability space $\prod_{i=1}^{\infty} (\Omega_i, \mathcal{U}_i, P_i)$ with X_i depending on the i th coordinate of $(\omega_1, \omega_2, \dots)$ only. The process X_i is called symmetric if X_i and $-X_i$ have the same distribution. In the case of nonmeasurability, the symmetry of independent processes $X_1, X_2, \dots$ may be understood in the sense that outer probabilities remain the same if one or more X_i are replaced by $-X_i$.[†]

Throughout the chapter S_n equals the partial sum $X_1 + \dots + X_n$ of given stochastic processes $X_1, X_2, \dots$.

A.1.1 Proposition (Ottaviani's inequality). *Let $X_1, \dots, X_n$ be independent stochastic processes indexed by an arbitrary set. Then for $\lambda, \mu > 0$,*

$$P^* \left(\max_{k \leq n} \|S_k\| > \lambda + \mu \right) \leq \frac{P^* (\|S_n\| > \lambda)}{1 - \max_{k \leq n} P^* (\|S_n - S_k\| > \mu)}.$$

Proof. Let A_k be the event that $\|S_k\|^*$ is the first $\|S_j\|^*$ that is strictly greater than $\lambda + \mu$. The event on the left is the disjoint union of $A_1, \dots, A_n$. Since $\|S_n - S_k\|^*$

[†] This is the case, for instance, if each $(\Omega_i, \mathcal{U}_i, P_i)$ is a product $(\mathcal{X}_i, \mathcal{A}_i, P_i)^2$ of two identical probability spaces and $X_i(x_1, x_2) = Z_i(x_1) - Z_i(x_2)$ for some stochastic process Z_i .

is independent of $\|S_1\|^*, \dots, \|S_k\|^*$,

$$\begin{aligned} P(A_k) \min_{j \leq n} P(\|S_n - S_j\|^* \leq \mu) &\leq P(A_k, \|S_n - S_k\|^* \leq \mu) \\ &\leq P(A_k, \|S_n\|^* > \lambda), \end{aligned}$$

since $\|S_k\|^* > \lambda + \mu$ on A_k . Sum up over k to obtain the result. ■

A.1.2 Proposition (Lévy's inequalities). *Let $X_1, \dots, X_n$ be independent, symmetric stochastic processes indexed by an arbitrary set. Then for every $\lambda > 0$ we have the inequalities*

$$\begin{aligned} P^*\left(\max_{k \leq n} \|S_k\| > \lambda\right) &\leq 2P^*(\|S_n\| > \lambda), \\ P^*\left(\max_{k \leq n} \|X_k\| > \lambda\right) &\leq 2P^*(\|S_n\| > \lambda). \end{aligned}$$

Proof. Let A_k be the event that $\|S_k\|^*$ is the first $\|S_j\|^*$ that is strictly greater than λ . The event on the left in the first inequality is the disjoint union of $A_1, \dots, A_n$. Write T_n for the sum of the sequence $X_1, \dots, X_k, -X_{k+1}, \dots, -X_n$. By the triangle inequality, $2\|S_k\|^* \leq \|S_n\|^* + \|T_n\|^*$. It follows that

$$P(A_k) \leq P(A_k, \|S_n\|^* > \lambda) + P(A_k, \|T_n\|^* > \lambda) = 2P(A_k, \|S_n\|^* > \lambda),$$

since $X_1, \dots, X_n$ are symmetric. Sum up over k to obtain the first inequality.

For the second inequality, let A_k be the event that $\|X_k\|^*$ is the first $\|X_j\|^*$ that is strictly greater than λ . Write T_n for the sum of the variables $-X_1, \dots, -X_{k-1}, X_k, -X_{k+1}, \dots, -X_n$. By the triangle inequality, $2\|X_k\|^* \leq \|S_n\|^* + \|T_n\|^*$. Proceed as before. ■

An interesting consequence of Lévy's inequalities is the following theorem concerning the convergence of random series of independent processes.

A.1.3 Proposition (Lévy–Itô–Nisio). *Let $X_1, X_2, \dots$ be independent stochastic processes with sample paths in $\ell^\infty(T)$. Then the following statements are equivalent:*

- (i) $\{S_n\}$ converges outer almost surely to a tight limit;
- (ii) $\{S_n\}$ converges in outer probability to a tight limit;
- (iii) $\{S_n\}$ converges weakly to a tight limit.

Proof. The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) are true for general random sequences. We must prove the implications in the converse direction.

(iii) $\Rightarrow$ (ii). Since any Cauchy sequence is convergent, it suffices to show that $S_{n_{k+1}} - S_{n_k}$ converges in outer probability to zero as $k \rightarrow \infty$ for any sequence $n_1 < n_2 < \dots$. The sequence $(S_{n_{k+1}}, S_{n_k})$ is asymptotically tight and asymptotically measurable. By Prohorov's theorem, every subsequence has a further subsequence along which $Y_k = S_{n_{k+1}} - S_{n_k}$ converges in distribution to a tight limit Y . Then $Y_n(t) \rightsquigarrow Y(t)$ for every t . For every real number s at which the characteristic

function of the weak limit of $S_n(t)$ is nonzero, in particular for every s sufficiently close to zero,

$$\mathbb{E}e^{isY_k(t)} = \frac{\mathbb{E}e^{isS_{n_{k+1}}(t)}}{\mathbb{E}e^{isS_{n_k}(t)}} \rightarrow 1.$$

Thus the characteristic function of $Y(t)$ is 1 in a neighborhood of zero. Conclude that $Y = 0$ almost surely.

(ii) $\Rightarrow$ (i). Write S for the limit in probability of S_n . First assume that the processes X_j are symmetric. There exists a subsequence $n_1 < n_2 < \dots$ such that $\mathbb{P}^*(\|S_{n_k} - S\| > 2^{-k}) < 2^{-k}$ for every k . By the Borel-Cantelli lemma, $S_{n_k} \xrightarrow{\text{as*}} S$ as $k \rightarrow \infty$. By a Lévy inequality,

$$\mathbb{P}^*\left(\max_{n_k < n \leq n_{k+1}} \|S_n - S_{n_k}\| > 2^{-k+1}\right) \leq 2\mathbb{P}^*\left(\|S_{n_{k+1}} - S_{n_k}\| > 2^{-k+1}\right).$$

The right side is smaller than a multiple of 2^{-k} . Hence $\max_{n_k < n \leq n_{k+1}} \|S_n - S_{n_k}\|^*$ converges almost surely to zero by the Borel-Cantelli lemma. This concludes the proof that S_n converges outer almost surely for symmetric X_i .

Given general processes, construct an independent copy $Y_1, Y_2, \dots$ defined on a copy of the original probability space $(\Omega, \mathcal{U}, \mathbb{P})$, and let T_n be the corresponding partial sums. Then the elements of the sequence $S_n - T_n$ are the partial sums of the symmetric variables $X_i - Y_i$ and converge in outer probability. (It is formally defined on $(\Omega, \mathcal{U}, \mathbb{P}) \times (\Omega, \mathcal{U}, \mathbb{P})$.) By the preceding paragraph, $S_n - T_n$ converges outer almost surely. By Fubini's theorem there exists ω such that $S_n - T_n(\omega)$ converges outer almost surely. Then it converges also in distribution. Since S_n converges in distribution as well, it follows that the sequence $T_n(\omega)$ is convergent. ■

A.1.4 Proposition (Hoffmann-Jørgensen inequalities). *Let $X_1, \dots, X_n$ be independent stochastic processes indexed by an arbitrary set. Then for any $\lambda, \eta > 0$,*

$$\mathbb{P}^*\left(\max_{k \leq n} \|S_k\| > 3\lambda + \eta\right) \leq \mathbb{P}^*\left(\max_{k \leq n} \|S_k\| > \lambda\right)^2 + \mathbb{P}^*\left(\max_{k \leq n} \|X_k\| > \eta\right).$$

If $X_1, \dots, X_n$ are independent and symmetric, then also

$$\mathbb{P}^*(\|S_n\| > 2\lambda + \eta) \leq 4\mathbb{P}^*(\|S_n\| > \lambda)^2 + \mathbb{P}^*\left(\max_{k \leq n} \|X_k\| > \eta\right).$$

Proof. Let A_k be the event that $\|S_k\|^*$ is the first $\|S_j\|^*$ that is strictly greater than λ . The (disjoint) union of $A_1, \dots, A_n$ is the event that $\max_{k \leq n} \|S_k\|^*$ is greater than λ . By the triangle inequality, $\|S_j\|^* \leq \|S_{k-1}\|^* + \|X_k\|^* + \|S_j - S_k\|^*$ for every $j \geq k$. On A_k the first term on the right is bounded by λ . Conclude that on A_k

$$\max_{j \geq k} \|S_j\|^* \leq \lambda + \max_{k \leq n} \|X_k\|^* + \max_{j > k} \|S_j - S_k\|^*.$$

On A_k this remains true if the maximum on the left is taken over all $\|S_j\|^*$. Since the processes are independent, we obtain for every k

$$\begin{aligned} & \mathbb{P}\left(A_k, \max_{k \leq n} \|S_k\|^* > 3\lambda + \eta\right) \\ & \leq \mathbb{P}\left(A_k, \max_{k \leq n} \|X_k\|^* > \eta\right) + \mathbb{P}(A_k)\mathbb{P}\left(\max_{j > k} \|S_j - S_k\|^* > 2\lambda\right). \end{aligned}$$

In the probability on the far right the variable $\max_{j>k} \|S_j - S_k\|^*$ can be bounded by $2 \max_{k \leq n} \|S_k\|^*$. Next sum over k to obtain the first inequality of the proposition.

To prove the second inequality, first establish by the same method that

$$P(A_k, \|S_n\|^* > 2\lambda + \eta) \leq P(A_k, \max_{k \leq n} \|X_k\|^* > \eta) + P(A_k)P(\|S_n - S_k\|^* > \lambda).$$

In the probability on the far right, bound the variable $\|S_n - S_k\|^*$ by $\max_{k \leq n} \|S_n - S_k\|^*$. Next sum over k to obtain

$$\begin{aligned} P(\|S_n\|^* > 2\lambda + \eta) \\ \leq P\left(\max_{k \leq n} \|X_k\|^* > \eta\right) + P\left(\max_{k \leq n} \|S_k\|^* > \lambda\right)P\left(\max_{k \leq n} \|S_n - S_k\|^* > \lambda\right). \end{aligned}$$

The processes S_k and $S_n - S_k$ are the partial sums of the symmetric processes $X_1, \dots, X_n$ and $X_n, \dots, X_2$, respectively. Apply Lévy's inequality to both probabilities on the far right to conclude the proof. ■

A.1.5 Proposition (Hoffmann–Jørgensen inequalities for moments). *Let $0 < p < \infty$ and suppose that $X_1, \dots, X_n$ are independent stochastic processes indexed by an arbitrary index set T . Then there exist constants C_p and $0 < u_p < 1$ such that*

$$E^* \max_{k \leq n} \|S_k\|^p \leq C_p \left(E^* \max_{k \leq n} \|X_k\|^p + F^{-1}(u_p)^p \right),$$

where F^{-1} is the quantile function of the random variable $\max_{k \leq n} \|S_k\|^*$. Furthermore, if $X_1, \dots, X_n$ are symmetric, then there exist constants K_p and $0 < v_p < 1$ such that

$$E^* \|S_n\|^p \leq K_p \left(E^* \max_{k \leq n} \|X_k\|^p + G^{-1}(v_p)^p \right),$$

where G^{-1} is the quantile function of the random variable $\|S_n\|^*$. For $p \geq 1$, the last inequality is also valid for mean-zero processes (with different constants).

Proof. Take $\lambda = \eta = t$ in the first inequality of the preceding proposition to find that, for any $\tau > 0$,

$$\begin{aligned} E^* \max_{k \leq n} \|S_k\|^p &= 4^p \int P\left(\max_{k \leq n} \|S_k\|^* > 4t\right) d(t^p) \\ &\leq (4\tau)^p + 4^p \int_{\tau}^{\infty} P\left(\max_{k \leq n} \|S_k\|^* > t\right)^2 d(t^p) \\ &\quad + 4^p \int_{\tau}^{\infty} P\left(\max_{k \leq n} \|X_k\|^* > t\right) d(t^p) \\ &\leq (4\tau)^p + 4^p P\left(\max_{k \leq n} \|S_k\|^* > \tau\right) E^* \max_{k \leq n} \|S_k\|^p + 4^p E^* \max_{k \leq n} \|X_k\|^p. \end{aligned}$$

Choosing τ such that $4^p P(\max_{k \leq n} \|S_k\|^* > \tau)$ is bounded by $\frac{1}{2}$, and rearranging terms, yield the claimed inequality. The second inequality can be proved in a similar manner, this time using the second inequality of the preceding proposition.

The inequality for mean-zero processes follows from the inequality for symmetric processes by symmetrization and desymmetrization: by Jensen's inequality, $E^* \|S_n\|^p$ is bounded by $E^* \|S_n - T_n\|^p$ if T_n is the sum of n independent copies of $X_1, \dots, X_n$. ■

Hoffmann-Jørgensen's inequality strengthens probability statements about sums to statements concerning expectations or p -moments, provided the corresponding moment of the maximum of the individual terms can be controlled. Thus it can be used to "invert" the consequences of Markov-type inequalities under an additional condition. A typical application is to a sequence of sums $\sum_{i=1}^n X_{ni}$. Boundedness in probability $\|\sum_{i=1}^n X_{ni}\|^* = O_P(1)$ implies that $G_n^{-1}(u) = O(1)$ for the sequence of quantile functions of the variables $\|\sum_{i=1}^n X_{ni}\|^*$. Conclude that the sequence of expectations $E^*\|\sum_{i=1}^n X_{ni}\|^p$ is $O(1)$ if the same is true for the sequence $E^* \max_{1 \leq i \leq n} \|X_{ni}\|^p$.

Hoffmann-Jørgensen's inequality may also be used to bound higher moments of sums in terms of a first moment plus a higher moment of the maximum of the individual terms. The first part of the following proposition is in this spirit, although its constant is much smaller than the constant obtainable from Hoffmann-Jørgensen's inequality.

A.1.6 Proposition. *Let $X_1, \dots, X_n$ be independent, mean-zero stochastic processes indexed by an arbitrary index set T . Then*

$$\begin{aligned} \|\|S_n\|^*\|_{P,p} &\leq K \frac{p}{\log p} \left[\|\|S_n\|^*\|_{P,1} + \|\max_{1 \leq i \leq n} \|X_i\|^*\|_{P,p} \right], \quad (p > 1), \\ \|\|S_n\|^*\|_{\psi_p} &\leq K_p \left[\|\|S_n\|^*\|_{P,1} + \|\max_{1 \leq i \leq n} \|X_i\|^*\|_{\psi_p} \right], \quad (0 < p \leq 1), \\ \|\|S_n\|^*\|_{\psi_p} &\leq K_p \left[\|\|S_n\|^*\|_{P,1} + \left(\sum_{i=1}^n \|\|X_i\|^*\|_{\psi_p}^q \right)^{1/q} \right], \quad (1 < p \leq 2). \end{aligned}$$

Here $1/p + 1/q = 1$, and K and K_p are a universal constant and a constant depending on p only, respectively.

Proof. The first part with the inferior constant $24(2^{1/p})(3^p)$ follows from the preceding Hoffmann-Jørgensen inequality by noting that $(1-v)G^{-1}(v) \leq \int_v^1 G^{-1}(s) ds \leq E^*\|S_n\|$ for every v , and then substitution of $v_p = 1 - 3^{-p}/8$ and $K_p = 2(3^p)$. The proofs of the first part with the improved constant $p/\log p$ and of the second and third parts are long and rely on the isoperimetric methods developed by Talagrand (1989). See Ledoux and Talagrand (1991), pages 172–175. ■

Another consequence of Hoffmann-Jørgensen's inequality is the following equivalence of moments of the supremum of sums of independent stochastic processes and the same moment of the maximal summand.

A.1.7 Proposition. *Let $0 < p < \infty$ and suppose that $X_1, X_2, \dots$ are independent stochastic processes indexed by an arbitrary set T . If $\sup_{n \geq 1} \|S_n\|^* < \infty$ almost surely, then $E^* \sup_{n \geq 1} \|S_n\|^p < \infty$ if and only if $E^* \sup_{n \geq 1} \|X_n\|^p < \infty$.*

Proof. Since $\|X_n\| \leq \|S_n\| + \|S_{n-1}\|$, finiteness of the first expectation clearly implies finiteness of the second expectation. For any fixed n , it follows from Proposition A.1.5 that

$$E^* \max_{k \leq n} \|S_k\|^p \leq C_p \left(E^* \max_{k \leq n} \|X_k\|^p + F^{-1}(u_p)^p \right),$$

where F^{-1} is the quantile function of the random variable $\max_{k \leq n} \|S_k\|^*$, and $0 < u_p < 1$. Since $\sup_{n \geq 1} \|S_n\|$ is assumed finite almost surely, there is an M so that $F^{-1}(u_p) \leq M$ for all n . Hence letting $n \rightarrow \infty$ in the last display shows that finiteness of the second expectation implies finiteness of the first expectation. ■

An interesting corollary of this proposition for normalized sums of independent stochastic processes is as follows.

A.1.8 Corollary. *Let $0 < p < \infty$ and let $\{a_n\}$ be a sequence of positive numbers that increases to infinity. Let $X_1, X_2, \dots$ be independent stochastic processes indexed by an arbitrary set T . If $\sup_{n \geq 1} (\|S_n\|/a_n)^* < \infty$ almost surely, then $E^* \sup_{n \geq 1} \|S_n\|^p / a_n^p < \infty$ if and only if $E^* \sup_{n \geq 1} \|X_n\|^p / a_n^p < \infty$.*

Proof. See Ledoux and Talagrand (1991), page 159. ■

Verification of the second statement in the preceding corollary can be carried out by use of Problem 2.3.5.

A.1.9 Proposition (Marcinkiewicz inequalities). *Let $X_1, X_2, \dots$ be independent, mean zero stochastic variables. Then for $p \geq 1$ there exist constants A_p and B_p such that*

$$A_p E \left(\sum_{i=1}^n X_i^2 \right)^{p/2} \leq E \left| \sum_{i=1}^n X_i \right|^p \leq B_p E \left(\sum_{i=1}^n X_i^2 \right)^{p/2}.$$

Proof. This can be proved using symmetrization followed by Khinchine's inequality, Proposition A.6.10. Alternatively see Chow and Teicher (1978), page 356, or De la Peña and Giné (1999), Lemma 1.4.13. ■

A.1.10 Proposition (Hoeffding's inequality). *Let $\{c_1, \dots, c_N\}$ be elements of an arbitrary vector space $\mathbb{V}$, and let $U_1, \dots, U_n$ and $V_1, \dots, V_n$ denote samples of size $n \leq N$ drawn without and with replacement, respectively, from $\{c_1, \dots, c_N\}$. Then, for every convex function $\phi: \mathbb{V} \rightarrow \mathbb{R}$,*

$$E \phi \left(\sum_{j=1}^n U_j \right) \leq E \phi \left(\sum_{j=1}^n V_j \right).$$

Proof. See Hoeffding (1963) and Marshall and Olkin (1979), Corollary A.2.e, page 339. ■

A.1.11 Proposition (Contraction principle). *Let $X_1, \dots, X_n$ be arbitrary stochastic processes and $\gamma_1, \dots, \gamma_n$ arbitrary, real-valued (measurable) random variables with $0 \leq \gamma_i \leq 1$. Let $\xi_1, \dots, \xi_n$ be independent, real random variables with zero means independent of $(X_1, \dots, X_n, \gamma_1, \dots, \gamma_n)$. Then*

$$E^* \left\| \sum_{i=1}^n \xi_i \gamma_i X_i \right\| \leq E^* \left\| \sum_{i=1}^n \xi_i X_i \right\|.$$

Proof. Since the γ_i can be taken out one at a time, it suffices to show that

$$E^* \|\xi \gamma X + Y\| \leq E^* \|\xi X + Y\|$$

whenever ξ has zero mean and is independent of (γ, X, Y) . By the triangle inequality, the left side is bounded by

$$E^* \|(\xi X + Y)\gamma\| + E^* \|Y(1 - \gamma)\| = E^* \|(\xi X + Y)\| \gamma + E^* \|Y\| (1 - \gamma).$$

The second term on the right can be written as

$$E_{Y,\gamma}^* \|Y + (E_\xi \xi)X\| (1 - \gamma).$$

By Jensen's inequality and Fubini's theorem, it is bounded by $E^* \|Y + \xi X\| (1 - \gamma)$. The result follows since γ is measurable. ■

Problems and Complements

1. In the first inequality of Proposition A.1.5, the constants $C_p = 2(4^p)$ and $u_p = 1 - 4^{-p}/2$ will do. The second inequality is true for symmetric processes with $K_p = 2(3^p)$ and $v_p = 1 - 3^{-p}/8$; for mean-zero processes with $K_p = 4(6^p)$ and $v_p = 1 - 3^{-p}/16$.

A.2

Gaussian Processes

Since Gaussian processes arise as the limits in distribution of empirical processes, their properties are of importance for many of the developments in Parts 2 and 3. Inequalities for Gaussian processes have also been used in a number of proofs involving multiplier processes with Gaussian multipliers. In this chapter we discuss a number of important results, most of them with full proofs.

A.2.1 Maximal Inequalities and Comparison

A stochastic process $X = (X_t: t \in T)$, indexed by a set T , is *Gaussian* if all finite linear combinations $\sum_i a_i X_{t_i}$ are normally distributed, and *mean-zero* if $EX_t = 0$ for all $t \in T$. The customary definition of a “separable” process requires that T is a semimetric space (see Section 2.3.3). In the present chapter the Gaussian process may be understood to be *separable* if there is a countable subset $T_0 \subset T$ so that the supremum of the process or its increments, or absolute values of these, is equal to the corresponding supremum restricted to T_0 . It is no real loss to assume that T is countable itself; extension to an uncountable set is always by taking limits (and trivially so).

For a separable process we write $\|X\|$ for the supremum $\sup_{t \in T} |X_t|$, and define the median $M(X)$ of $\|X\|$ by the two requirements

$$P(\|X\| \leq M(X)) \geq \frac{1}{2}, \quad P(\|X\| \geq M(X)) \geq \frac{1}{2}.$$

It can be shown that this pair of inequalities define a unique number $M(X)$ (see the proof of the following theorem). Furthermore, we define

$$\sigma^2(X) = \sup_{t \in T} \text{var } X_t.$$

The following proposition shows that the distribution of the supremum of a zero-mean Gaussian process has sub-Gaussian tails, whenever it is finite.

A.2.1 Proposition (Borell's inequality). *Let X be a mean-zero, separable Gaussian process X with finite median $M(X)$. Then $\sigma^2(X)$ and $E\|X\|^2$ are finite, and for every $\lambda > 0$,*

$$\begin{aligned} P(\|X\| - M(X) \geq \lambda) &\leq \exp\left(-\frac{\lambda^2}{2\sigma^2(X)}\right), \\ P(\|X\| - E\|X\| \geq \lambda) &\leq 2\exp\left(-\frac{\lambda^2}{2\sigma^2(X)}\right), \\ P(\|X\| \geq \lambda) &\leq 2\exp\left(-\frac{\lambda^2}{8E\|X\|^2}\right). \end{aligned}$$

Proof. The proof is based on finite-dimensional approximation and the concentration inequalities for the finite-dimensional, standard normal distribution given in the lemma ahead.

First, consider the case that the index set T consists of finitely many points $t_1, \dots, t_k$. Then the process X can be represented as AZ , for Z a standard normal vector and A the symmetric square root of the covariance matrix of X . By the Cauchy-Schwarz inequality,

$$\max_{1 \leq i \leq k} |Az|_i \leq \max_{1 \leq i \leq k} \|A_{i\cdot}\| \|z\| = \max_{1 \leq i \leq k} (A^2)_{ii}^{1/2} \|z\|.$$

This implies that the function $f(z) = \max_i |Az|_i$ is Lipschitz of norm bounded by $\sup_t \sigma(X_t) = \sigma(X)$. Apply Lemma A.2.2 to both f and $-f$, and combine the results to obtain the theorem for the case where the index set is finite.

Since X is separable by assumption, the supremum $\|X\|$ is the almost-sure, monotone limit of a sequence of finite suprema. The supremum M of the sequence of medians can be seen to be a median (the smallest) of $\|X\|$. Approximate $\|X\| - M(X)$ by a sequence of similar objects for finite suprema to obtain the first inequality of the theorem with $M = M(X)$. The proof of this inequality is complete if it is shown that M is the only median of $\|X\|$.

Since the median of $|X_t|$ is bounded above by the median of $\|X\|$, it follows that $P(|X_t| \leq M(X)) \geq \frac{1}{2}$ for every t . Taking into account the normal distribution of X_t , we obtain that $\sigma(X_t)$ is bounded above by $M(X)/\Phi^{-1}(3/4)$. Therefore, $\sigma(X)$ is finite and the exponential inequality obtained previously is nontrivial. The argument actually shows that both the left-tail probability $P(\|X\| \leq M - \lambda)$ and the right-tail probability $P(\|X\| \geq M + \lambda)$ are bounded by $1/2$ times the exponential upper bound. This means that these probabilities are strictly less than $1/2$ for $\lambda > 0$, whence M is a unique median of $\|X\|$.

To obtain the second inequality of the theorem, we note first that $E\|X\|$ is finite in view of the exponential tail bound for $\|X\| - M(X)$. Next we use Lemma A.2.2 and take limits along finite subsets as before.

The third inequality is trivially satisfied if $0 \leq \lambda < 2E\|X\|$, because in that case the exponential is larger than $\exp(-\frac{1}{2}) \geq 0.6$. For $\lambda > 2E\|X\|$, the probability is bounded by $P(\|X\| > E\|X\| + \lambda/2)$, which can be further bounded by the second inequality. ■

A.2.2 Lemma. Let Z be a d -dimensional random vector with the standard normal distribution. Then for every Lipschitz function $f: \mathbb{R}^d \rightarrow \mathbb{R}$ with $\|f\|_{Lip} \leq 1$,

$$\begin{aligned} \mathbf{P}(f(Z) - \text{med } f(Z) > \lambda) &\leq \frac{1}{2} \exp(-\tfrac{1}{2}\lambda^2), \\ \mathbf{P}(f(Z) - \mathbf{E}f(Z) > \lambda) &\leq \exp(-\tfrac{1}{2}\lambda^2). \end{aligned}$$

Proof. For the first inequality consider, the set $A = \{z: f(z) \leq \text{med } f(Z)\}$. Since f has Lipschitz norm bounded by 1, the set A^λ of points at distance at most λ from A is contained in the set $\{z: f(z) \leq \text{med } f(Z) + \lambda\}$. It follows that

$$\mathbf{P}(f(Z) \leq \text{med } f(Z) + \lambda) \geq \mathbf{P}(Z \in A^\lambda).$$

By definition of the median, the set A has probability at least $1/2$ under the standard normal distribution. According to the isoperimetric inequality for the normal distribution, Corollary A.2.12, a half-space H with boundary at the origin is an extreme set in the sense that $\mathbf{P}(Z \in A^\lambda) \geq \mathbf{P}(Z \in H^\lambda)$ for every $\lambda > 0$, for any other measurable set A with probability at least $1/2$. The proof of the first inequality of the lemma is complete upon noting that $\mathbf{P}(Z \in H^\lambda) = \Phi(\lambda)$ and $1 - \Phi(\lambda) \leq \frac{1}{2} \exp(-\frac{1}{2}\lambda^2)$.

For the proof of the second inequality, assume without loss of generality that $\mathbf{E}f(Z) = 0$. An arbitrary Lipschitz function f can be approximated by arbitrarily smooth functions with compact supports and of no larger Lipschitz norms. Therefore, it is no loss of generality to assume that f is sufficiently regular. In particular assume without loss of generality that f is differentiable with gradient ∇f uniformly norm bounded by 1.

Let Z_t be normally distributed with mean zero and covariance matrix $(1 - e^{-2t})I$. For $t \geq 0$, define functions $P_t f$ by

$$P_t f(x) = \mathbf{E}f(e^{-t}x + Z_t).$$

The operators P_t form a semigroup ($P_0 = I$ and $P_s P_t = P_{s+t}$) on a domain that includes all sufficiently regular functions: the Ornstein-Uhlenbeck or Hermite semigroup defined via Mehler's formula. The *generator* of the group is the derivative $A = d/dt P_t|_{t=0}$ and has the property that $d/dt P_t = A P_t$. We shall need the integration-by-parts formula

$$-\mathbf{E}f(Z)Ag(Z) = \mathbf{E}\langle \nabla f(Z), \nabla g(Z) \rangle.$$

This is valid for sufficiently regular functions f and g .

For fixed $r > 0$, define $G(t) = \mathbf{E} \exp(r P_t f(Z))$. Then $G(0) = \mathbf{E} \exp(r f(Z))$ is the moment-generating function of $f(Z)$ and $G(\infty) = 1$. Differentiating under the expectation and using the integration-by-parts formula, we obtain

$$\begin{aligned} -G'(t) &= -r \mathbf{E} e^{r P_t f(Z)} A P_t f(Z) \\ &= r \mathbf{E} \langle \nabla e^{r P_t f(Z)}, \nabla P_t f(Z) \rangle = r^2 \mathbf{E} e^{r P_t f(Z)} \|\nabla P_t f(Z)\|^2. \end{aligned}$$

Differentiating under the expectation in the definition of $P_t f$, we see that the gradient $\nabla P_t f(x)$ is equal to $\mathbf{E} \nabla f(e^{-t}x + Z_t) e^{-t}$ and its norm is bounded above

by $\sup_x \|\nabla f(x)\| e^{-t}$. Substitution in the preceding display yields the inequality $-G'(t) \leq r^2 e^{-2t} G(t)$. Equivalently,

$$(\log G)'(t) \geq \left(\frac{1}{2} r^2 e^{-2t}\right)'.$$

Since the functions $\log G$ and $\frac{1}{2} r^2 e^{-2t}$ are both zero at ∞ and hence equal, it follows that $\log G$ is smaller than $\frac{1}{2} r^2 e^0$ at zero. This concludes the proof that $E \exp r f(Z) = G(0) \leq \exp(\frac{1}{2} r^2)$.

By Markov's inequality, it follows that $P(f(Z) \geq \lambda) \leq \exp(\frac{1}{2} r^2 - r\lambda)$ for every $r > 0$. Optimize to conclude the proof of the lemma. ■

The process X having bounded sample paths is the same as $\|X\|$ being a finite random variable. In that case the median $M(X)$ is certainly finite and $\sigma(X)$ is finite by the argument in the proof of the preceding proposition. Next, the inequalities in the preceding proposition show that $\|X\|$ has moments of all orders. In fact, we have the following proposition.

A.2.3 Proposition. *Let X be a mean-zero, separable Gaussian process such that $\|X\|$ is finite almost surely. Then $\sigma^2(X) < \infty$ and*

$$E \exp(\alpha \|X\|^2) < \infty \quad \text{if and only if} \quad \alpha 2\sigma^2(X) < 1.$$

Proof. It is shown in the preceding proof that $\sigma(X) \leq M(X)/\Phi^{-1}(3/4)$. The sufficiency of the condition $\alpha 2\sigma^2(X) < 1$ follows by integrating Borell's inequality. The necessity may be proved by considering the individual X_t , whose normal distributions can be handled explicitly. ■

Thus, for Gaussian processes, finiteness of lower moments implies finiteness of higher moments. Higher moments can even be bounded by lower ones up to universal constants. The following proposition reverses the usual Liapunov inequality for moments.

A.2.4 Proposition. *There exist constants $K_{p,q}$ depending on $0 < p \leq q < \infty$ only such that*

$$(E\|X\|^q)^{1/q} \leq K_{p,q} (E\|X\|^p)^{1/p},$$

for any separable Gaussian process X for which $\|X\|$ is finite almost surely.

The distribution of a mean-zero Gaussian process is completely determined by its covariance function. According to Anderson's lemma, Lemma 3.12.4, a smaller covariance function indicates that the process is more concentrated near zero. The following theorem is in the same spirit and shows that smaller variances of differences imply a stochastically smaller maximum value.

A.2.5 Proposition (Slepian–Fernique–Marcus–Shepp inequalities). *Let X and Y be separable, mean-zero Gaussian processes indexed by a common index set T such that*

$$\mathbb{E}(X_s - X_t)^2 \leq \mathbb{E}(Y_s - Y_t)^2, \quad \text{for all } s, t \in T.$$

Then $\mathbb{E} \sup_{t \in T} X_t \leq \mathbb{E} \sup_{t \in T} Y_t$, and, for every convex, increasing $\Phi: [0, \infty) \rightarrow [0, \infty)$,

$$\mathbb{E} \Phi\left(\sup_{s,t} |X_s - X_t|\right) \leq \mathbb{E} \Phi\left(\sup_{s,t} |Y_s - Y_t|\right).$$

If, moreover, $\mathbb{E} X_t^2 = \mathbb{E} Y_t^2$ for every $t \in T$, then $\mathbb{E} \|X\| \leq 2 \mathbb{E} \|Y\|$ and

$$\mathbb{P}\left(\sup_{t \in T} X_t \geq \lambda\right) \leq \mathbb{P}\left(\sup_{t \in T} Y_t \geq \lambda\right), \quad \text{for all } \lambda > 0.$$

Finally, if T is a compact metric space and Y has a version with continuous sample paths, then so does X .

Proof. A proof of the inequality comparing the suprema of the deviations $|X_s - X_t|$ and $|Y_s - Y_t|$ can be found in Fernique (1974). (The choice $\Phi(x) = x$ gives a comparison of the mean suprema of deviations, of which a proof is also given in Marcus and Rosen (2006), Lemma 5.5.3. Also see Theorem 2.9 in Adler (1990).) Since $\sup_{s,t} |X_s - X_t| = \sup_s X_s + \sup_t (-X_t)$, and $-X$ is distributed as X , we have $\mathbb{E} \sup_{s,t} |X_s - X_t| = 2 \mathbb{E} \sup_t X_t$. Combined with the same equality for Y , this gives the inequality between the expected suprema of the X_t and Y_t .[‡]

If the second moments are equal, then the extended stochastic processes $\bar{X}$ and $\bar{Y}$ to which a zero variable is added, say $X_\infty = 0$ and $Y_\infty = 0$ for $\infty \notin T$, satisfy the increment inequality $\mathbb{E}(\bar{X}_s - \bar{X}_t)^2 \leq \mathbb{E}(\bar{Y}_s - \bar{Y}_t)^2$, for all $s, t \in \bar{T} = T \cup \{\infty\}$. Then $\mathbb{E} \sup_t |X_t| \leq \mathbb{E} \sup_{s,t \in \bar{T}} |\bar{X}_s - \bar{X}_t| \leq \mathbb{E} \sup_{s,t \in \bar{T}} |\bar{Y}_s - \bar{Y}_t|$, by the first inequality, which is bounded above by $2 \mathbb{E} \sup_t |\bar{Y}_t|$, by the triangle inequality and the fact that $|Y_\infty| = 0$.

The probability bound for the suprema of X and Y , also under the additional condition of equal second moments, can be found in Corollary 3.12 of Ledoux and Talagrand (1991). ■

For a centered Gaussian process $\{X_t: t \in T\}$, let ρ denote its standard deviation semimetric on T defined by

$$\rho(s, t) = \sigma(X_s - X_t), \quad s, t \in T.$$

Let $N(\varepsilon, T, \rho)$ be the covering number of T with respect to ρ .

By Corollary 2.2.9, the expectation $\mathbb{E} \|X\|$ can be bounded above by an entropy integral with respect to ρ . Sudakov's inequality gives a bound in the opposite direction.

[‡] Simpler proofs with a factor 2 or under the assumption of equal second moments can be found in many references.

A.2.6 Proposition (Sudakov's inequality). *For every mean-zero, separable Gaussian process X ,*

$$\sup_{\varepsilon > 0} \varepsilon \sqrt{\log N(\varepsilon, T, \rho)} \leq \sqrt{2\pi \log 2} \operatorname{E} \sup_{t \in T} X_t.$$

Moreover, if almost all sample paths of X are bounded and uniformly continuous on (T, ρ) , then $\varepsilon \sqrt{\log N(\varepsilon, T, \rho)} \rightarrow 0$ as $\varepsilon \rightarrow 0$.

Proof. Let $t_1, \dots, t_N$ be a maximal set of points in T with $\rho(t_i, t_j) \geq \varepsilon$, for all $i \neq j$. Then the balls of radius ε around the t_i cover T , and hence $N(\varepsilon, T, \rho) \leq N$. If $Z_1, \dots, Z_N$ are i.i.d. standard normal random variables, then $\operatorname{E}(\varepsilon Z_i / \sqrt{2} - \varepsilon Z_j / \sqrt{2})^2 = \varepsilon^2 \leq \operatorname{E}(X_{t_i} - X_{t_j})^2$, for $1 \leq i \neq j \leq N$, and hence Slepian's inequality, Proposition A.2.5, gives that $\operatorname{E} \sup_{t \in T} X_t \geq \operatorname{E} \max_{1 \leq i \leq N} \varepsilon Z_i / \sqrt{2}$.

Thus the problem is reduced to proving that $\operatorname{E} \max_{1 \leq i \leq N} Z_i \geq c_N := (\pi \log 2)^{-1/2} (\log N)^{1/2}$. For $N = 1$ this is clear, while for $N = 2$ the inequality can be verified (with equality) by direct calculation. By the last assertion of Corollary A.2.9 applied to the convex function $z \mapsto \max_i z_i$, the expected value of $\max_{1 \leq i \leq N} Z_i$ is bounded below by its median, which can be directly calculated to be equal to $\Phi^{-1}(2^{-1/N})$. Thus the proof can be completed by verifying that $\Phi^{-1}(2^{-1/N}) \geq c_N$, or equivalently $\operatorname{P}(Z_1 \geq c_N) > 1 - 2^{-1/N}$, for all $N \geq 3$.

For the final assertion, see Lemma 2.10.17. ■

A.2.2 Geometric Inequalities

A *Gaussian random element* in a separable Banach space $\mathbb{B}$ is a Borel measurable map X from some probability space into $\mathbb{B}$ such that b^*X possesses a univariate Gaussian distribution, for every element b^* of the dual space of $\mathbb{B}$. It is zero-mean if the latter distributions are centered. The results in this section are more conveniently stated for Gaussian random elements than for stochastic processes, although some also have a stochastic process version. (Tightness of such a random element plays a role; the results extend to Radon maps into more general locally convex Hausdorff spaces, but need modification for stochastic processes.)

The moment inequalities and tail bounds from the preceding section are also true for the norm $\|X\|$ of such a Gaussian random element, with $\sigma^2(X) = \sup_{\|b^*\| \leq 1} \operatorname{var} b^*X$ playing the role of maximal marginal variance. Furthermore, each such random element possesses an expectation $\operatorname{E}X$: an element of $\mathbb{B}$ such that $b^*(\operatorname{E}X) = \operatorname{E}b^*X$, for every element of the dual space, the *Pettis integral* of X .

A.2.7 Proposition (Ehrhard–Latała–Borell inequality). *For every Gaussian random element X in a separable Banach space $\mathbb{B}$, any Borel sets $A, B \subset \mathbb{B}$ and any $\mu \in (0, 1)$,^{b‡}*

$$\Phi^{-1}[\operatorname{P}(X \in \mu A + (1 - \mu)B)] \geq \mu \Phi^{-1}[\operatorname{P}(X \in A)] + (1 - \mu) \Phi^{-1}[\operatorname{P}(X \in B)].$$

^b The sets $\mu A + (1 - \mu)B$, although not necessarily Borel, are analytic, being a continuous image of the Borel sets $A \times B$ in $\mathbb{B} \times \mathbb{B}$. Therefore they are measurable in the completion of the Borel sets under any measure, including the law of X , making the left side well defined, although it might be more correct to write P_* in the left side. (See Dudley (1989), Theorems 13.2.1 and 13.2.6.)

[‡] The inequality is also valid if one of $\operatorname{P}(X \in A)$ or $\operatorname{P}(X \in B)$ is 0 or 1, with $\Phi^{-1}(0) = -\infty$ and $\Phi^{-1}(1) = \infty$, as long as the right side makes sense by the usual conventions: $\infty + a = a + \infty = \infty$, if $a > -\infty$, and $-\infty + a = a - \infty = -\infty$.

Proof. The inequality for a general Gaussian random element can be obtained from the inequality for a standard normal vector in Euclidean space, through centering and a spectral decomposition, as in the proof of Proposition A.2.13. We concentrate on the latter special case, denoting by $X = Z$ a d -dimensional standard normal vector.

Because the probabilities $P(Z \in A)$ and $P(Z \in B)$ can be approximated arbitrarily closely from below by probabilities $P(Z \in K)$ for compact sets K within A and B , it is not a loss of generality to assume that A and B are compact. This allows to approximate the indicators of these sets and their convex combination by C^∞ -functions $f, g, h: \mathbb{R}^d \rightarrow (0, 1)$ satisfying, for given $0 < \delta < \eta < \alpha < 1$,

$$f = \begin{cases} \delta & \text{on } (A^\varepsilon)^c \\ \alpha & \text{on } A, \end{cases} \quad g = \begin{cases} \delta & \text{on } (B^\varepsilon)^c \\ \alpha & \text{on } B, \end{cases} \quad h = \begin{cases} \eta & \text{on } ((\mu A^\varepsilon + (1 - \mu)B^\varepsilon)^\varepsilon)^c \\ \alpha & \text{on } \mu A^\varepsilon + (1 - \mu)B^\varepsilon, \end{cases}$$

and assuming intermediate values for the remaining arguments. (Later we shall choose $\eta = \eta(\delta, \alpha)$.) As $\delta, \eta \rightarrow 0$ followed by $\alpha \uparrow 1$ and $\varepsilon \downarrow 0$, these three functions tend pointwise to the indicator functions of A , B and $\mu A + (1 - \mu)B$. (We can use compactness of A and B to see this for the third.) Therefore it suffices to prove the inequality with $Ef(Z)$, $Eg(Z)$ and $Eh(Z)$ taking the places of $P(Z \in A)$, $P(Z \in B)$ and $P(Z \in \mu A + (1 - \mu)B)$.

Consider the function C given by

$$C(t, x, y) = \Phi^{-1} [Eh(\mu x + (1 - \mu)y + \sqrt{t}Z)] \\ - \mu \Phi^{-1} [Ef(x + \sqrt{t}Z)] - (1 - \mu) \Phi^{-1} [Eg(y + \sqrt{t}Z)].$$

For $t = 1$ and $x = y = 0$, the nonnegativity $C(1, 0, 0) \geq 0$ is the desired inequality on the functions f, g, h . For $t = 0$ the inequality $C(0, x, y) \geq 0$ reduces to

$$\Phi^{-1} \circ h(\mu x + (1 - \mu)y) \geq \mu \Phi^{-1} \circ f(x) + (1 - \mu) \Phi^{-1} \circ g(y).$$

This inequality is true in the two cases $x \in A^\varepsilon, y \in B^\varepsilon$ and $x \notin A^\varepsilon, y \notin B^\varepsilon$, as the left side takes the (maximal) value $\Phi^{-1}(\alpha)$, or the right side takes the (minimal) value $\Phi^{-1}(\delta)$, in these cases, as $\delta < \eta < \alpha$. The inequality is also true if $x \in A^\varepsilon, y \notin B^\varepsilon$ or $x \notin A^\varepsilon, y \in B^\varepsilon$ provided we choose η such that $\Phi^{-1}(\eta) \geq \max(\mu \Phi^{-1}(\alpha) + (1 - \mu) \Phi^{-1}(\delta), \mu \Phi^{-1}(\delta) + (1 - \mu) \Phi^{-1}(\alpha))$. Since $0 < \mu < 1$, the minimal value η satisfying this inequality satisfies $\eta \rightarrow 0$ as $\delta \rightarrow 0$, for fixed α . Thus it is possible too let $\delta \alpha 0, \eta \downarrow 0, \alpha \uparrow 1$ and $\varepsilon \downarrow 0$ so that the inequality is valid in all cases, i.e. $C(0, x, y) \geq 0$, for all $x, y \in \mathbb{R}^d$.

The remainder of the proof is to show that C satisfies a partial differential equation that preserves the nonnegativity $C(0, x, y) \geq 0$ at its initial time $t = 0$ at all future times.

For a function $u: \mathbb{R}^d \rightarrow \mathbb{R}$ define $\underline{u}(t, \xi) = Eu(\xi + \sqrt{t}Z)$ and $U = \Phi^{-1} \circ \underline{u}$. It can be checked (by partial integration or the Feynman-Kac formula) that $\underline{u}$ satisfies the heat equation $\underline{u}_t = \frac{1}{2} \underline{u}_{\xi\xi}$, where $\underline{u}_{\xi\xi} = \Delta \underline{u}$ is the Laplacian for the space variable

if $a < \infty$. In particular, if one of $P(X \in A)$ or $P(X \in B)$ is 1 and the other positive, then the inequality is true with ∞ on both sides. With the added convention that $-\infty + \infty = \infty + (-\infty) = -\infty$, the inequality is always true, but in this additional case it is trivial, because its right side is $-\infty$. The inequality extends to $\mu \in \{0, 1\}$, but is also trivial in these cases.

ξ . Substituting $\underline{u} = \Phi \circ U$, we infer that $U_t = \frac{1}{2}(U_{\xi\xi} - U\|U_{\xi}\|^2)$, for $U_{\xi} = \nabla U$ the space gradient of U . Next, for F, G and H corresponding to f, g and h as U to u , we can derive for $C(t, x, y) = H(t, \mu x + (1 - \mu)y) - \mu F(t, x) - (1 - \mu)G(t, y)$ that

$$C_t = \mathcal{A}C + b_1 \cdot C_x + b_2 \cdot C_y - aC,$$

for $\mathcal{A}$ the elliptic second order differential operator

$$\mathcal{A} = \frac{1}{2} \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2} + \frac{1}{2} \sum_{i=1}^d \frac{\partial^2}{\partial y_i^2} + \sum_{i=1}^d \frac{\partial^2}{\partial x_i \partial y_i},$$

and $b_1, b_2: [0, 1] \times \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $a: [0, 1] \times \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ the functions given by

$$\begin{aligned} b_1(t, x, y) &= -\frac{1}{2}F(t, x)[H_{\xi}(t, \mu x + (1 - \mu)y) + F_{\xi}(t, x)], \\ b_2(t, x, y) &= -\frac{1}{2}G(t, x)[H_{\xi}(t, \mu x + (1 - \mu)y) + G_{\xi}(t, x)], \\ a(t, x, y) &= \frac{1}{2}\|H_{\xi}(t, \mu x + (1 - \mu)y)\|^2. \end{aligned}$$

If C is somewhere negative on $R = \{(t, x, y): 0 \leq t \leq 1, x, y \in \mathbb{R}^d\}$, then for sufficiently small $\beta > 0$ the function $D: (t, x, y) \mapsto \beta t + C(t, x, y)$ is negative somewhere on R as well. Since $\liminf_{0 \leq t \leq 1} C(t, x, y) \geq 0$ as $|x| \rightarrow \infty$ or $|y| \rightarrow \infty$ (by choice of $\eta > \delta$), the function D must then have a point of (negative) minimum somewhere in R . This point cannot be on the boundary $t = 0$, as there $C \geq 0$. The point must be stationary relative to (x, y) , even if were on the boundary $t = 1$. Since C and D differ only by a function of t , their derivatives with respect to (x, y) agree, and hence at the point of negative minimum we must have $C < 0$, $C_x = C_y = 0$ and the Hessian of C must be nonnegative definite. Because $\mathcal{A}$ is elliptic, the latter implies that $\mathcal{A}C \geq 0$ (see Problem A.2.12). Since also $a \geq 0$, the partial differential equation implies that $C_t \geq 0$ at the point of minimum and hence $D_t \geq \beta > 0$. This contradicts that the point is a point of minimum of D . We conclude that $C \geq 0$, everywhere in R , in particular at $(1, 0, 0)$. ■

A.2.8 Corollary (Log-concavity). *For every Gaussian random element X in a separable Banach space $\mathbb{B}$, any Borel sets $A, B \subset \mathbb{B}$ and any $\mu \in (0, 1)$,*

$$\log P(X \in \mu A + (1 - \mu)B) \geq \mu \log P(X \in A) + (1 - \mu) \log P(X \in B).$$

Proof. Apply the increasing function $x \mapsto \log \circ \Phi(x)$ across the inequality in Proposition A.2.7, and use its concavity on the right side (cf. Problem A.2.8). ■

A.2.9 Corollary. *For every Gaussian random element X in a separable Banach space $\mathbb{B}$, and convex measurable function $f: \mathbb{B} \rightarrow [-\infty, \infty)$, the function $h: \mathbb{R} \rightarrow \mathbb{R}$ defined by $h(\lambda) = \Phi^{-1}(P(f(X) \leq \lambda))$ is concave. Unless the variable $f(X)$ is degenerate, the function $\lambda \mapsto P(f(X) \leq \lambda)$ is strictly increasing on the support of $f(X)$, which is an interval, and has at most a single jump, at the left endpoint of the support of $f(X)$. Also the median of the variable $f(X)$ is smaller than its mean $Ef(X)$, if the latter exists.*

Proof. By convexity of f the sets $C(t) = \{x: f(x) \leq t\}$ satisfy $C(\mu s + (1 - \mu)t) \supset \mu C(s) + (1 - \mu)C(t)$, for any $s, t \in \mathbb{R}$ and $\mu \in [0, 1]$. The given function is $h(\lambda) = \Phi^{-1}(\mathbf{P}(X \in C(\lambda)))$, and hence $h(\mu s + (1 - \mu)t) \geq \Phi^{-1}(\mathbf{P}(X \in \mu C(s) + (1 - \mu)C(t)))$. This is lower bounded by $\mu h(s) + (1 - \mu)h(t)$, by Proposition A.2.7, applied with the sets $A = C(s)$ and $B = C(t)$.

If $h(s) > -\infty$ and $h(t) = \infty$, for some $s < t$, then by concavity $h(\lambda) = \infty$, for every $\lambda \in (s, t)$ (see the footnote to Proposition A.2.7), but then $h(s) = \infty$, by right-continuity. Thus $h(t) = \infty$ for some t implies that there exist $m \geq -\infty$ such that $h(\lambda) = -\infty$, for $\lambda < m$ and $h(\lambda) = \infty$ otherwise, whence $f(X)$ is degenerate at m . In the other case, that $h(\lambda) < \infty$ for all λ , we have $h(\lambda) \uparrow \infty$ as $\lambda \uparrow \infty$, since $f(X)$ is a finite random variable, and the concavity shows that h is continuous on the interval where it is finite and cannot be constant on any nontrivial interval.

The median m of $f(X)$ satisfies $\mathbf{P}(f(X) < m) \leq 1/2 \leq \mathbf{P}(f(X) \leq m)$, which implies that $h(m-) \leq 0 \leq h(m)$. If $h(m-) = -\infty$, then $\mathbf{P}(f(X) < m) = 0$ and hence certainly $\mathbf{E}f(X) \geq m$. If $h(m-) > -\infty$, then h is finite and continuous in an interval around m , whence $h(m) = 0$, and there must exist $b > 0$ such that $h(\lambda) \leq b(\lambda - m)$, for every $\lambda \in \mathbb{R}$, as h is concave and increasing. Then $\mathbf{P}(f(X) \leq \lambda) = \Phi(h(\lambda)) \leq \Phi(b(\lambda - m))$, for every λ , whence $f(X)$ is stochastically larger than a Gaussian variable with mean m . In particular $\mathbf{E}f(X) \geq m$. ■

A.2.10 Corollary. *For every non-degenerate Gaussian random element X in a separable Banach space $\mathbb{B}$ whose support contains 0 the distribution function of the norm $\|X\|$ is continuous on $\mathbb{R}$ and strictly increasing on $\mathbb{R}^+$. In particular this is true if X is zero-mean and non-zero.*

Proof. Since $x \mapsto \|x\|$ is convex, the distribution function H of $\|X\|$ is strictly increasing and continuous on its support, which is an interval, by Corollary A.2.9. We have $H(0) = 0$, and $H(\lambda) < 1$, for every $\lambda > 0$, since $\|X\| \geq |b^*X|$ for every b^* in the unit ball of the dual space and at least one of the variables b^*X is Gaussian and nonzero by the assumed nondegeneracy of X . That 0 is contained in the support of X is equivalent to $H(\lambda) > 0$, for every $\lambda > 0$. Thus the support of $\|X\|$ is $[0, \infty)$.

The support of a centered Gaussian random element in a separable Banach space contains 0. Otherwise, there would exist $\lambda > 0$ so that the ball of radius λ would have probability 0, which would imply that every ball of radius λ had probability zero, as $\mathbf{P}(\|X - x\| \leq \lambda)$ is maximal for $x = 0$ by Anderson's lemma. This would contradict that the full space can be covered by countably many balls of any positive radius, by the assumed separability of the Banach space. ■

A.2.11 Corollary (Anderson's inequality). *For every zero-mean Gaussian random element X in a separable Banach space $\mathbb{B}$, convex, symmetric, measurable subset $C \subset \mathbb{B}$ and $a \in \mathbb{B}$, we have $\mathbf{P}(X \in C + a) \leq \mathbf{P}(X \in C)$.*

Proof. Since $\frac{1}{2}(C + a) + \frac{1}{2}(C - a) = C$ by convexity of C , Proposition A.2.7 gives that $\Phi^{-1}(\mathbf{P}(X \in C)) \geq \frac{1}{2}\Phi^{-1}(\mathbf{P}(X \in C + a)) + \frac{1}{2}\Phi^{-1}(\mathbf{P}(X \in C - a))$. The

symmetry of C and X give that $P(X \in C - a) = P(X \in C + a)$, so that the right side is equal to $\Phi^{-1}(P(X \in C + a))$. The corollary follows upon applying Φ left and right of the inequality. ■

For $A \subset \mathbb{R}^d$ and $\lambda > 0$, let $A^\lambda = \{x \in \mathbb{R}^d: \|x - A\| < \lambda\}$ be the λ -enlargement of A . The Gaussian isoperimetric inequality says that half-spaces minimize among sets of equal measure the growth of their probability under the standard normal distribution when enlarging the set.

A half-space is understood to be a set of the form $\{x \in \mathbb{R}^d: \langle x, u \rangle \leq r\}$, for some $u \in \mathbb{R}^d$ and $r \in \mathbb{R}$.

A.2.12 Corollary (Isoperimetric inequality). *If Z is a standard normal vector in $\mathbb{R}^d$, then for any Borel set $A \subset \mathbb{R}^d$ and half-space $H \subset \mathbb{R}^d$, the inequality $P(Z \in A) \geq P(Z \in H)$ implies that $P(Z \in A^\lambda) \geq P(Z \in H^\lambda)$, for any $\lambda > 0$. In particular, if $P(Z \in A) \geq 1/2$, then $P(Z \in A^\lambda) \geq \Phi(\lambda)$, for any $\lambda > 0$.*

Proof. Let B be the unit ball in $\mathbb{R}^d$. Since $\lambda B = (1 - \mu)(\lambda B / (1 - \mu))$ and $\lambda B / (1 - \mu)$ is convex, Proposition A.2.7 gives, for $\mu \in (0, 1)$ and $\lambda > 0$,

$$\Phi^{-1}[P(Z \in \mu A + \lambda B)] \geq \mu \Phi^{-1}[P(Z \in A)] + (1 - \mu) \Phi^{-1}[P(Z \in \lambda B / (1 - \mu))].$$

By continuity of the Gaussian scale family Z/μ relative to the total variation norm, we have $P(Z \in \mu A + \lambda B) = P(Z \in A + \lambda B/\mu) + o(1)$, as $\mu \uparrow 1$, which tends to $P(\|Z - A\| \leq \lambda)$. Thus the left side of the display tends to $\Phi^{-1}[P(\|Z - A\| \leq \lambda)]$. The first term on the right of the display clearly increases to $\Phi^{-1}[P(Z \in A)]$, while the second term on the right is equal to $(1 - \mu) \Phi^{-1} \circ F(\lambda/(1 - \mu))$, for F the cumulative distribution of $\|Z\|$. Because $\Phi^{-1} \circ F(u) \sim u$, as $u \rightarrow \infty$ (cf. Problem A.2.9), this term tends to λ , as $\mu \uparrow 1$. Thus, letting $\mu \uparrow 1$ in the preceding display, and applying Φ to the limits on the left and right, we conclude that $P(\|Z - A\| \leq \lambda) \geq \Phi[\Phi^{-1}(P(Z \in A)) + \lambda]$. Because the right side is continuous, this remains true if the left side is replaced by $P(\|Z - A\| < \lambda) = P(Z \in A^\lambda)$. We have proved that $P(Z \in A^\lambda) \geq \Phi[\Phi^{-1}(P(Z \in A)) + \lambda]$.

By the rotational symmetry of the standard normal distribution it is not a loss of generality to assume that the half-space in the assumption takes the form $H = \{z \in \mathbb{R}^d: z_1 \leq r\}$. In that case $P(Z \in H) = \Phi(r)$, and $H^\lambda = \{z \in \mathbb{R}^d: z_1 < r + \lambda\}$, so that $P(Z \in H^\lambda) = \Phi(r + \lambda)$. If $P(Z \in A) \geq P(Z \in H)$, then $\Phi^{-1}(P(Z \in A)) \geq r$ and the right side of the final equality in the preceding paragraph is no smaller than $\Phi(r + \lambda) = P(Z \in H^\lambda)$. This gives the first assertion of the corollary.

The final statement of the corollary follows from the inequality $P(Z \in A^\lambda) \geq \Phi(r + \lambda)$, where we can choose $r \geq 0$ since $P(Z \in A) \geq 1/2$, so that $\Phi(r + \lambda) \geq \Phi(\lambda)$. ■

The isoperimetric inequality extends to Gaussian random elements with values in a general separable Banach space. To every such a Gaussian random element corresponds a so-called *reproducing kernel Hilbert space*, which may be considered embedded in the Banach space. Let $\mathbb{H}_1$ be the open unit ball of this reproducing kernel Hilbert space.

A.2.13 Proposition (Geometric Borell's inequality). *For any zero-mean Gaussian random element X in a separable Banach space $\mathbb{B}$ and any measurable set $A \subset \mathbb{B}$, we have $P(X \in A + \lambda \mathbb{H}_1) \geq \Phi(\Phi^{-1}(P(X \in A)) + \lambda)$, for any $\lambda > 0$.*

Proof. We can represent the Gaussian variable as $X = \sum_{j=1}^{\infty} Z_j h_j$, for i.i.d. standard normal variables $Z_1, Z_2, \dots$ and an orthonormal basis $h_1, h_2, \dots$ of the reproducing kernel Hilbert space, where the series converges in $\mathbb{B}$, almost surely (see e.g. Van der Vaart and Van Zanten (2008), Section 4.), or if X is finite-dimensional as a finite such sum. In the remainder of the proof we address the infinite-dimensional case.

The reproducing kernel Hilbert space is the set of all series $\sum_{j=1}^{\infty} \xi_j h_j$, with $\xi = (\xi_j) \in \ell_2$, normed by the ℓ_2 -norm $\|\xi\|_2 = \|\sum_{j=1}^{\infty} \xi_j h_j\|_{\mathbb{H}}$. The almost sure convergence of $X = \sum_{j=1}^{\infty} Z_j h_j$ can alternatively be expressed by saying that $Z = (Z_j)$ belongs with probability one to the set $B_0 \subset \mathbb{R}^{\infty}$ of all $x \in \mathbb{R}^{\infty}$ such that $\sum_{j=1}^{\infty} x_j h_j$ converges in $\mathbb{B}$. Let $A_0 \subset B_0$ be the set of coefficients x of such converging series with limit belonging to A . From the measurability of the (continuous) maps $x \mapsto \sum_{j=m}^n x_j h_j$ from $\mathbb{R}^{\infty}$ to $\mathbb{B}$ for every $n \geq m \in \mathbb{N}$, it can be ascertained that B_0 is measurable in $\mathbb{R}^{\infty}$, and next that $x \mapsto \sum_{j=1}^{\infty} x_j h_j$ is measurable from B_0 to $\mathbb{B}$, so that A_0 is measurable as well. Clearly $P(X \in A) = P(Z \in A_0)$, and also $P(X \in A + \lambda \mathbb{H}_1) = P(\|Z - A_0\|_2 < \lambda)$. Thus it suffices to show that $P(\|Z - A_0\|_2 < \lambda) \geq \Phi(\Phi^{-1}(P(Z \in A_0)) + \lambda)$, for any $\lambda > 0$. Since the right side is continuous in λ , it also suffices to prove this with the strict inequality in the event on the left side replaced by smaller or equal.

Let $P_n z = (z_1, \dots, z_n, 0, 0, \dots)$ denote the projection of $\mathbb{R}^{\infty}$ onto the first n coordinates. If A_0 is compact in $\mathbb{R}^{\infty}$, then $P_n A_0$ is compact and hence measurable, and $\|P_n z - P_n A_0\|_2 \uparrow \|z - A_0\|_2$, as $n \rightarrow \infty$, for any $z \in \mathbb{R}^{\infty}$. To see the latter, note first that $\|P_n z - P_n A_0\|_2 \leq \|z - A_0\|_2$, for every n , since $\|P_n z - P_n a\|_2 \leq \|z - a\|_2$, for every a . A sequence $a_n \in A_0$ that minimizes $a \mapsto \|P_n z - P_n a\|_2$ over A_0 for every n , has a converging subsequence $a_{n'} \rightarrow a \in A_0$ in $\mathbb{R}^{\infty}$ (so coordinatewise), and then $\|P_m z - P_m a\|_2 = \lim_{n'} \|P_m z - P_m a_{n'}\|_2 \leq \lim_{n'} \|P_{n'} z - P_{n'} A_0\|_2$, for every fixed m . Letting $m \rightarrow \infty$, we see that also $\|z - a\|_2$ is smaller than the right side and hence so is $\|z - A_0\|_2$.

It follows that, for compact $A_0 \subset \mathbb{R}^{\infty}$, by the portmanteau theorem,

$$P(\|Z - A_0\|_2 \leq \lambda) \geq \limsup_{n \rightarrow \infty} P(\|P_n Z - P_n A_0\|_2 \leq \lambda).$$

Since $P_n Z$ is essentially a standard normal vector in $\mathbb{R}^n$, the right side is bounded below by $\Phi(\Phi^{-1}(P(P_n Z \in P_n A_0)) + \lambda)$, by Corollary A.2.12, for any $\lambda > 0$. Since the event $\{P_n Z \in P_n A_0\}$ contains the event $\{Z \in A_0\}$, this is further bounded below by $\Phi(\Phi^{-1}(P(Z \in A_0)) + \lambda)$, for any $\lambda > 0$.

This concludes the proof when A_0 is compact. Because every Borel measure on $\mathbb{R}^{\infty}$ is inner regular, for a general A_0 there exists a sequence of compact sets $K_n \subset A_0$ with $P(Z \in K_n) \uparrow P(Z \in A_0)$. The argument of the preceding paragraph applied with K_n instead of A_0 gives that $P(\|Z - K_n\|_2 \leq \lambda) \geq \Phi(\Phi^{-1}(P(Z \in K_n)) + \lambda)$. Since $K_n \subset A_0$, the left side is smaller than $P(\|Z - A_0\|_2 \leq \lambda)$, for every n , while the right side increases to $\Phi(\Phi^{-1}(P(Z \in A_0)) + \lambda)$, as $n \rightarrow \infty$. ■

A.2.14 Proposition (Gaussian correlation inequality). *For any zero-mean Gaussian vector X in $\mathbb{R}^d$ and any closed, symmetric sets $C, D \subset \mathbb{R}^d$, we have $P(X \in C \cap D) \geq P(X \in C)P(X \in D)$.*

Proof. See Royen (2014) or Latała and Matlak (2017). ■

A.2.3 Exponential Bounds

Exponential bounds for $\|X\| = \sup_{t \in T} |X_t|$ without the centering by $M(X)$ or $E\|X\|$ as in Proposition A.2.1, require hypotheses that give control of $E\|X\|$ (or equivalently of $M(X)$). The following theorems are examples of results of this type. They are closely related to Theorems 2.14.28, 2.14.32 and 2.14.33 for empirical processes.

Recall that $\rho(s, t) = \sigma(X_s - X_t)$ is the intrinsic semimetric of the given Gaussian process X , and $\sigma(X)$ is the supremum of the standard deviations $\sigma(X_t)$.

A.2.15 Proposition. *Let X be a separable, mean-zero Gaussian process such that for some $K > \sigma(X)$, some $V > 0$, and some $0 < \varepsilon_0 \leq \sigma(X)$,*

$$N(\varepsilon, T, \rho) \leq \left(\frac{K}{\varepsilon}\right)^V, \quad 0 < \varepsilon < \varepsilon_0.$$

Then there exists a universal constant D such that, for all $\lambda \geq \sigma^2(X)(1 + \sqrt{V})/\varepsilon_0$,

$$P\left(\sup_{t \in T} X_t \geq \lambda\right) \leq \left(\frac{DK\lambda}{\sqrt{V}\sigma^2(X)}\right)^V (1 - \Phi)\left(\frac{\lambda}{\sigma(X)}\right).$$

A.2.16 Proposition. *Let X be a separable, zero-mean Gaussian process, and for $\delta > 0$, let $T_\delta = \{t \in T: EX_t^2 \geq \sigma^2(X) - \delta^2\}$, and for $V \geq W \geq 1$, suppose that, for all $0 < \delta^2 \leq \sigma^2(X)$ and all $0 < \epsilon \leq \delta(1 + \sqrt{V})/\sqrt{W}$,*

$$(A.2.17) \quad N(\epsilon, T_\delta, \rho) \leq K\delta^W \epsilon^{-V}.$$

Then there exists a universal constant D such that, for $\lambda \geq 2\sigma(X)\sqrt{W}$,

$$P\left(\sup_{t \in T} X_t \geq \lambda\right) \leq K \frac{W^{W/2}}{V^{V/2}} D^{V+W} \left(\frac{\lambda}{\sigma^2(X)}\right)^{V-W} (1 - \Phi)\left(\frac{\lambda}{\sigma(X)}\right).$$

The preceding propositions bound the tail probabilities $P(\sup_{t \in T} X_t \geq \lambda)$ of the suprema of general Gaussian processes, using entropy bounds only. For particular Gaussian processes, both the renewal theoretic methods of Siegmund (1988, 1992), and the “Poisson clump heuristic” methods of Aldous (1989), yield results on the asymptotic behavior as $\lambda \rightarrow \infty$, that often seem to provide accurate approximations for λ on the order of $1.5\sigma(X)$. The following are some particular examples.

A.2.18 Example (Brownian sheet on $[0, 1]^2$). The standard Brownian sheet B is a zero-mean Gaussian process indexed by $T = [0, 1]^2$, with covariance function

$$\text{cov}(B(s), B(t)) = (s_1 \wedge t_1)(s_2 \wedge t_2).$$

Then $\sigma^2(B) = 1$, and (A.2.17) holds with $V = 4$ and $W = 4$. Hence, for some constant M , and $\lambda > 0$,

$$\mathbb{P}\left(\sup_{t \in T} B_t \geq \lambda\right) \leq M\lambda^{-1} \exp\left(-\frac{\lambda^2}{2}\right).$$

In fact, it was shown by Goodman (1976) that, for all $\lambda > 0$,

$$4 \int_{\lambda}^{\infty} z(1 - \Phi)(z) dz \leq \mathbb{P}\left(\sup_{t \in T} B_t \geq \lambda\right) \leq 4(1 - \Phi)(\lambda).$$

These inequalities imply that, for $\lambda \rightarrow \infty$,

$$\mathbb{P}\left(\sup_{t \in T} B_t \geq \lambda\right) \sim 4(1 - \Phi)(\lambda) = 4\mathbb{P}(B(1, 1) \geq \lambda) \sim \sqrt{\frac{8}{\pi}} \lambda^{-1} \exp\left(-\frac{\lambda^2}{2}\right).$$

A.2.19 Example (Brownian bridge on $[0, 1]^2$). The Brownian sheet B^0 pinned down to 0 at $t = (1, 1)$ is a zero-mean Gaussian process indexed by $[0, 1]^2$, with covariance function

$$\text{cov}(B^0(s), B^0(t)) = (s_1 \wedge t_1)(s_2 \wedge t_2) - s_1 s_2 t_1 t_2.$$

Alternatively, B^0 can be defined from the Brownian sheet of Example A.2.18, by setting

$$B^0(t_1, t_2) = B(t_1, t_2) - t_1 t_2 B(1, 1).$$

This is just the P -Brownian bridge process $\mathbb{G}_P$ indexed by the collection of sets $\{[0, t]: t \in [0, 1]\}$, with P equal to the uniform distribution (Lebesgue measure) on $[0, 1]^2$. Then $\sigma^2(B^0) = 1/4$, and this supremum is achieved for every $t \in [0, 1]^2$ with $t_1 t_2 = 1/2$. It can be shown that (A.2.17) holds with $V = 4$ and $W = 1$. Therefore, for some constant M ,

$$\mathbb{P}\left(\sup_{t \in T} B_t^0 \geq \lambda\right) \leq M\lambda^2 \exp(-2\lambda^2).$$

It has been shown by Hogan and Siegmund (1986) and also by Aldous (1989), page 202, that as $\lambda \rightarrow \infty$,

$$\mathbb{P}\left(\sup_{t \in T} B_t^0 \geq \lambda\right) \sim (4 \log 2) \lambda^2 \exp(-2\lambda^2).$$

In this case Goodman (1976) established the lower bound

$$\mathbb{P}\left(\sup_{t \in T} B_t^0 \geq \lambda\right) \geq (1 + 2\lambda^2) \exp(-2\lambda^2), \quad \lambda > 0.$$

The asymptotic formula of Hogan and Siegmund becomes greater than Goodman's lower bound for $\lambda \geq (2(\log 4 - 1))^{-1/2} = 1.138 \dots$. For this and some numerical comparisons, see Adler (1990), Chapter V.

A.2.20 Example (Tucked Brownian sheet). The tucked Brownian sheet is the zero-mean Gaussian process indexed by $[0, 1]^2$, with covariance function

$$\text{cov}(B^{00}(s), B^{00}(t)) = (s_1 \wedge t_1 - s_1 t_1)(s_2 \wedge t_2 - s_2 t_2).$$

This can be obtained from the Brownian sheet given in Example A.2.18, by pinning it down to 0 on the entire boundary of $[0, 1]^2$, i.e.

$$B^{00}(t_1, t_2) = B(t_1, t_2) - t_1 B(1, t_2) - t_2 B(t_1, 1) + t_1 t_2 B(1, 1).$$

This process arises in testing independence and in connection with the estimation of copula functions; see Chapters 3.9 and 3.10. In this case $\sigma^2(B^{00}) = 1/16$, and this supremum is uniquely achieved for $t = (1/2, 1/2)$. It can be shown that (A.2.17) holds with $V = 4$ and $W = 2$. Therefore, for some constant M ,

$$\mathbb{P}\left(\sup_{t \in T} B_t^{00} \geq \lambda\right) \leq M \lambda \exp(-8\lambda^2).$$

It follows from the methods of Siegmund (1988, 1992), or alternatively from Aldous (1989), that as $\lambda \rightarrow \infty$,

$$\mathbb{P}\left(\sup_{t \in T} B_t^{00} \geq \lambda\right) \sim (4\sqrt{2\pi}) \lambda \exp(-8\lambda^2).$$

A.2.21 Example (Kiefer-Müller process on $[0, 1]^2$). The Kiefer-Müller process Z on $[0, 1]^2$ can be obtained from a Brownian sheet B by way of

$$Z(t_1, t_2) = B(t_1, t_2) - t_2 B(t_1, 1).$$

Then Z has covariance function

$$\text{cov}(Z(s_1, t_1), Z(s_2, t_2)) = (s_1 \wedge s_2)(t_1 \wedge t_2 - t_1 t_2).$$

This process is a special case of the P -Kiefer process in Chapter 2.12 with P the uniform distribution on $[0, 1]$ and $\mathcal{F} = \{1_{[0, t]}: t \in [0, 1]\}$. In this case $\sigma^2(Z) = 1/4$, and this is uniquely achieved for $(s, t) = (1, 1/2)$. It can be shown that (A.2.17) holds with $V = 4$ and $W = 3$. Therefore, for some constant M ,

$$\mathbb{P}\left(\sup_{t \in T} Z_t \geq \lambda\right) \leq M \exp(-2\lambda^2).$$

It follows from the methods of Siegmund (1988, 1992), or alternatively from Aldous (1989), that as $\lambda \rightarrow \infty$,

$$\mathbb{P}\left(\sup_{t \in T} Z_t \geq \lambda\right) \sim 2 \exp(-2\lambda^2).$$

A.2.22 Example (Brownian bridge indexed by convex subsets of $[0, 1]^2$). Consider the P -Brownian bridge process with P uniform on $[0, 1]^2$ indexed by the collection $\mathcal{C}$ of all convex subsets of $[0, 1]^2$. It follows from Corollary 2.7.13 (with $d = r = 2$) that

$$(A.2.23) \quad \log N_{[]}(\varepsilon, \mathcal{C}, L_2(P)) \leq K \left(\frac{1}{\varepsilon} \right),$$

and hence $\mathcal{C}$ is P -preGaussian. Although neither Theorem A.2.15 nor A.2.16 apply in this case, the methods of Samorodnitsky (1991) apply, and show that for some constants C and D ,

$$(A.2.24) \quad \mathbb{P} \left(\sup_{C \in \mathcal{C}} \mathbb{G}_P(C) \geq \lambda \right) \leq C \exp(D\lambda^{2/3}) \exp(-2\lambda^2).$$

More generally, the methods of Samorodnitsky (1991) show that if the power 1 in (A.2.23) is replaced by V , then the power $2/3$ in (A.2.24) must be replaced by $2V/(V + 2)$.

A.2.4 Argmax

Many M -estimators are asymptotically distributed as the point of maximum of a Gaussian process. This point of maximum is not always easy to evaluate. The following proposition shows that a point of absolute maximum is unique, if it exists.

A.2.25 Proposition. *Let $\{X_t: t \in T\}$ be a Gaussian process with continuous sample paths, indexed by a separable metric space T . If $\text{var}(X_s - X_t) \neq 0$ for every $s \neq t$, then almost every sample path achieves its supremum at most at one point.*

Proof. If a given sample path $t \mapsto X_t$ attains its supremum at both t_0 and t_1 , then the suprema $\sup_{t \in G_0} X_t$ and $\sup_{t \in G_1} X_t$ are equal (and equal to the overall supremum) for all neighborhoods G_0 and G_1 of t_0 and t_1 . Thus it suffices to prove

$$\mathbb{P} \left(\forall t_0 \neq t_1 \in T, \exists G_0 \ni t_0, G_1 \ni t_1: \sup_{t \in G_0} X_t \neq \sup_{t \in G_1} X_t \right) = 1.$$

Here we may restrict the neighborhoods to a given countable base of the open sets in T , and therewith reduce the problem to showing that

$$\forall t_0 \neq t_1 \in T, \exists G_0 \ni t_0, G_1 \ni t_1: \quad \mathbb{P} \left(\sup_{t \in G_0} X_t \neq \sup_{t \in G_1} X_t \right) = 1.$$

Fix a pair t_0 and t_1 for the remainder of the proof.

The covariance $\text{cov}(X_{t_0}, X_{t_1})$ must be unequal to at least one of the variances $\text{var} X_{t_0}$ or $\text{var} X_{t_1}$, as otherwise $\text{var}(X_{t_0} - X_{t_1}) = 0$; this variance can then not be zero. For definiteness, let this be true for t_0 , and consider the process $Y_t = X_t - \beta(t)X_{t_0}$, for $\beta(t) = \text{cov}(X_t, X_{t_0}) / \text{var} X_{t_0}$ so that Y is stochastically independent of X_{t_0} . The function β is continuous by assumption, and by construction $1 = \beta(t_0) \neq \beta(t_1)$. It follows that for sufficiently small neighborhoods $G_0 \ni t_0$ and $G_1 \ni t_1$, we have either $\sup_{t \in G_0} \beta(t) < \inf_{t \in G_1} \beta(t)$

or $\sup_{t \in G_1} \beta(t) < \inf_{t \in G_0} \beta(t)$, depending on the sign of $\beta(t_1) - \beta(t_0)$. Since $X_t = Y_t + \beta(t)X_{t_0}$, the probability in the preceding display can be written

$$E_Y P\left(\sup_{t \in G_0} (Y_t + \beta(t)X_{t_0}) \neq \sup_{t \in G_1} (Y_t + \beta(t)X_{t_0}) \mid Y\right).$$

By the independence of X_{t_0} from Y , the inner probability refers to X_{t_0} only, and for a given sample path $\alpha(t) := Y_t$ is equal to the probability that X_{t_0} does not fall in the set of x such that $\sup_{t \in G_0} (\alpha(t) + \beta(t)x) = \sup_{t \in G_1} (\alpha(t) + \beta(t)x)$. The two functions in this equality are convex with slopes (of their supporting lines) that are strictly separated. Therefore they can be equal at most at one point and hence the probability that X_{t_0} falls in the set is zero. ■

Problems and Complements

1. For every separable Gaussian process, $\sigma^2(X) \leq M^2(X)/\Phi^{-1}(3/4)^2$.

[Hint: See the proof of Proposition A.2.1.]

2. For every separable Gaussian process $E\|X\|^p \leq K_p M(X)^p$ for a constant depending on p only.

[Hint: Integrate Borell's inequality to bound $E\|X - M(X)\|^p$. Next, use the preceding problem.]

3. For every separable Gaussian process, $|E\|X\| - M(X)| \leq \sqrt{\pi/2} \sigma(X)$.
4. Suppose that $\mathcal{F}$ and $\mathcal{G}$ are pre-Gaussian classes of measurable functions. Use majorizing measures to show that $\mathcal{F} \cup \mathcal{G}$ is pre-Gaussian.
5. Every tight, Borel measurable centered, Gaussian variable X in $\ell^\infty(T)$ satisfies $P(\|X\| < \varepsilon) > 0$ for every $\varepsilon > 0$.

[Hint: By Anderson's lemma, $P(\|X - x\| \leq \varepsilon) \leq P(\|X\| \leq \varepsilon)$ for every x . The support of X can be covered by countably many balls.]

6. The Brownian sheet B satisfies the tail bound, for every $\lambda > 0$,

$$P\left(\sup_{0 \leq t_1, t_2 \leq 1} B(t_1, t_2) \geq \lambda\right) \leq 4(1 - \Phi(\lambda)).$$

[Hint: Use Lévy's inequality twice to see $2P(\sup_{0 \leq t_2 \leq 1} B(1, t_2) \geq \lambda) \leq 4P(B(1, 1) \geq \lambda)$. (The reflection principle for Brownian motion implies that the second inequality actually holds with equality.)]

7. The Kiefer-Müller process Z satisfies the tail bound, for every $\lambda > 0$,

$$P\left(\sup_{0 \leq t_1, t_2 \leq 1} Z(t_1, t_2) \geq \lambda\right) \leq 2 \exp(-2\lambda^2).$$

[Hint: Use Lévy's inequality to bound the left side by $2P(\sup_{0 \leq t_2 \leq 1} Z(1, t_2) \geq \lambda)$. Since the process $\{Z(1, t_2): 0 \leq t_2 \leq 1\}$ is a standard Brownian bridge process on $[0, 1]$, this can be further bounded using the known form of the distribution of the one-sided supremum (see Doob (1949), or Shorack and Wellner (1986), page 34).]

8. The function $x \mapsto \log \circ \Phi(x)$ is concave on $\mathbb{R}$.

[Hint: Use *Mills's inequality* $1 - \Phi(x) \leq \phi(x)/x$, for $x > 0$, to see that the second derivative is negative.]

9. If Φ^{-1} is the quantile function of the standard normal distribution and F is the cumulative distribution function of the root of a chisquare variable, then $\Phi^{-1}(F(u))/u \rightarrow 1$, as $u \rightarrow \infty$.

[Hint: The cumulative distribution function G of a Gamma $\Gamma(k, 1)$ -variable satisfies $1 - G(u) \sim u^{k-1}e^{-u}/\Gamma(k)$, as $u \rightarrow \infty$, and $F(u) = G(u^2/2)$, for k the degrees of freedom divided by 2. Also by Mills's ratio $\Phi^{-1}(v) \sim (-2 \log(1 - v))^{1/2}$, as $v \uparrow 1$.]

10. The distribution function of $f(X)$ for a convex function f and a Gaussian random element X may have a jump of any size at the left endpoint of its support (cf. Corollary A.2.9).

[Hint: Try X^+ for X univariate normal.]

11. The support of $\|X\|$ for a non-zero Gaussian random element X may be strictly contained in $(0, \infty)$.

[Hint: Try $X = (Z, 1)$, for a standard normal variable Z .]

12. If the twice continuously differentiable function $C: \mathbb{R}^d \rightarrow \mathbb{R}$ possesses a local minimum at x_0 , then its Hessian matrix C evaluated at x_0 satisfies $\sum_i \sum_j A_{ij} C_{ij} \geq 0$, for every nonnegative-definite symmetric matrix A .

[Hint: In terms of the spectral decomposition $A = \sum_k \lambda_k a_k a_k'$ the double sum is $\sum_k \lambda_k a_k' C a_k$.]

13. If $X_1, \dots, X_n$ are jointly normal variables with mean zero, then $P(\max_i |X_i| \leq x) \geq \prod_i P(|X_i| \leq x)$, for every x .

[Hint: Use the Gaussian correlation inequality.]

14. If (X_1, X_2) is bivariate normal with mean zero and $E(X_1 - X_2)^2 = \rho^2$, then $EX_1 \vee X_2 = |\rho|/\sqrt{2\pi}$.

[Hint: Write $X_1 \vee X_2 = (X_1 + X_2 + |X_1 - X_2|)/2$.]

A.3

Rademacher Processes

Let $\varepsilon_1, \dots, \varepsilon_n$ be Rademacher variables and $x_1, \dots, x_n$ arbitrary real functions on some “index” set T . The *Rademacher process* $\{\sum_{i=1}^n \varepsilon_i x_i(t) : t \in T\}$ shares several properties with Gaussian processes. In this chapter we restrict ourselves to two propositions that have been used in Part 2.[†]

Let $\|x\|$ be the supremum norm $\sup_{t \in T} |x(t)|$ for a given function $x: T \rightarrow \mathbb{R}$. Define

$$\sigma^2 = \sup_{t \in T} \text{var} \sum_{i=1}^n \varepsilon_i x_i(t) = \left\| \sum_{i=1}^n x_i^2 \right\|.$$

The tail probabilities of the norm of a Rademacher process are comparable to those of Gaussian processes. The following inequalities are similar to the Borell inequalities, although it appears likely that the constants can be improved.

A.3.1 Proposition. *Let M be a median of the norm of the Rademacher process $\sum_{i=1}^n \varepsilon_i x_i$. Then there exists a universal constant C such that, for every $\lambda > 0$,*

$$\begin{aligned} \mathbb{P}\left(\left|\sum \varepsilon_i x_i\right| - M > \lambda\right) &\leq 4 \exp\left(-\frac{\lambda^2}{8\sigma^2}\right), \\ \mathbb{P}\left(\left|\sum \varepsilon_i x_i\right| - \mathbb{E}\left\|\sum \varepsilon_i x_i\right\| > \lambda\right) &\leq C \exp\left(-\frac{\lambda^2}{9\sigma^2}\right). \end{aligned}$$

Proof. For the first inequality, see Ledoux and Talagrand (1991), Theorem 4.7, on page 100.

To obtain the second inequality set $\mu = \mathbb{E}\|\sum \varepsilon_i x_i\|$. Integrate over the first inequality to obtain $|\mu - M| \leq 4\sqrt{2\pi}\sigma$. For $c\lambda > |\mu - M|$, the event in the

[†] See, for instance, Ledoux and Talagrand (1991) for a further discussion.

second inequality is contained in the event $\{|\|\sum \varepsilon_i x_i\| - M| > (1 - c)\lambda\}$. Choose $c > 0$ sufficiently small to bound the probability of this event by $4 \exp(-\lambda^2/9\sigma^2)$. For $c\lambda \leq |\mu - M|$, the right side in the second inequality is never smaller than $C \exp(-(\mu - M)^2/9c^2\sigma^2)$, which is at least 1 for sufficiently large C . ■

A.3.2 Proposition (Contraction principle). *For each i , let x_i take its values in the interval $[-1, 1]$, and let $\phi_i: [-1, 1] \rightarrow \mathbb{R}$ be Lipschitz of norm 1 with $\phi_i(0) = 0$. Then*

$$\mathbb{E} \left\| \sum \varepsilon_i \phi_i(x_i) \right\| \leq 2 \mathbb{E} \left\| \sum \varepsilon_i x_i \right\|.$$

Proof. See Ledoux and Talagrand (1991), Theorem 4.12, on page 112. ■

The last proposition with the choices $\phi_i(x) = \frac{1}{2}x^2$ is applied in Chapter 2.14.6 to obtain $\mathbb{E} \left\| \sum \varepsilon_i x_i^2 \right\| \leq 4 \mathbb{E} \left\| \sum \varepsilon_i x_i \right\|$.

A.4

Isoperimetric Inequalities for Product Measures

Let $(\mathcal{X}^n, \mathcal{A}^n)$ be the product of n copies of a measurable space $(\mathcal{X}, \mathcal{A})$. Given points $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$, say that y controls a subset of coordinates $\{x_i : i \in I\}$ of x if the corresponding coordinates of y are the same: $y_i = x_i$ for every $i \in I$. Given vectors $y^1, \dots, y^q$, let $f(y^1, \dots, y^q, x)$ be the number of coordinates of x that are not controlled by any y^j . Given subsets $A_1, \dots, A_q$ of $\mathcal{X}^n$, define $f(A_1, \dots, A_q, x)$ to be the minimum of the numbers $f(y^1, \dots, y^q, x)$ when the y^j range over A_j . Thus $n - f(A_1, \dots, A_q, x)$ is the maximal number of coordinates of x that can be controlled by the clever choice of y^j from A_j . Also, if $f(A_1, \dots, A_q, x) = k$, then there exist points $y^j \in A_j$ such that the number of indices $i \in \{1, \dots, n\}$ such that $x_i \notin \{y_i^1, \dots, y_i^q\}$ is k (and a choice of y^j giving more control is impossible).

A basic inequality giving an upper bound on the size of the control functions $f(A_1, \dots, A_q, x)$ is as follows.

A.4.1 Proposition. *If the coordinates of $X = (X_1, \dots, X_n)$ are i.i.d. random elements in $\mathcal{X}$, then for any measurable sets $A_1, \dots, A_q$ in $\mathcal{A}^n$ and any integer $q \geq 1$*

$$\mathbf{E}^* q^{f(A_1, \dots, A_q, X)} \leq \left(\prod_{l=1}^q \mathbf{P}(X \in A_l) \right)^{-1}.$$

Consequently, for every set $A \in \mathcal{A}^n$ with $\mathbf{P}(X \in A) \geq 1/2$

$$\mathbf{P}^*(f(A, \dots, A, X) \geq k) \leq \frac{2^q}{q^k} \leq \left(\frac{2}{q} \right)^k, \quad k \geq q.$$

Here q copies of A appear in the left side.

Proof. It will first be shown that, for any random variables $U_1, \dots, U_q$ taking values in $[0, 1]$,

$$\mathbb{E}\left(q \wedge \min_{1 \leq l \leq q} U_l^{-1}\right) \prod_{l=1}^q \mathbb{E}U_l \leq 1.$$

Here $1/0$ is understood to be infinite. Indeed, the random variable U defined by $U^{-1} = q \wedge \min_{1 \leq l \leq q} U_l^{-1}$ satisfies $1/q \leq U \leq 1$. Since $U_l \leq U$ for every l , the left side is bounded above by $\mathbb{E}U^{-1}(\mathbb{E}U)^q$. Since $\log x \leq x - 1$, the logarithm of this expression is bounded by $\mathbb{E}U^{-1} - 1 + q\mathbb{E}U - q$. This is bounded above by zero, because on the interval $[1/q, 1]$ the function $z \mapsto z^{-1} + qz$ attains a maximum value of $1 + q$ (at the boundary points).

The proof of the theorem proceeds by induction on n . For $n = 1$ the control number $f(A_1, \dots, A_q, X)$ equals 0 if X is contained in some A_l and 1 otherwise. Thus

$$q^{f(A_1, \dots, A_q, X)} = q \wedge \min_{1 \leq l \leq q} 1_{A_l}(X)^{-1}.$$

The inequality in the first paragraph yields the result.

Suppose the theorem is true for n , and consider a vector X and subsets A_l in $\mathcal{X}^{n+1}$. Write Y and Z for the first n and the last coordinate of $X = (Y, Z)$, respectively. Furthermore, for a subset A of $\mathcal{X}^{n+1}$, define the sections and projection relative to the first n coordinates by

$$\begin{aligned} A(Z) &= \{Y \in \mathcal{X}^n : (Y, Z) \in A\}, \\ B &= \{Y \in \mathcal{X}^n : (Y, Z) \in A, \text{ for some } Z\}. \end{aligned}$$

If $y^1, \dots, y^q$ control all except k coordinates of Y , then $(y^1, z^1), \dots, (y^q, z^q)$ control all except at most $k + 1$ coordinates of (Y, Z) ; they control all except k coordinates of (Y, Z) if one z^j is chosen equal to Z . Therefore, it follows that, for every $l \leq q$,

$$\begin{aligned} f(A_1, \dots, A_q, X) &\leq 1 + f(B_1, \dots, B_q, Y) \\ f(A_1, \dots, A_q, X) &\leq f(B_1, \dots, B_{l-1}, A_l(Z), B_{l+1}, \dots, B_q, Y). \end{aligned}$$

Write $\mathbb{E}_Y$ for the expectation taken with respect to the first n coordinates only and let P be the distribution of Y . The first inequality of the display yields

$$\mathbb{E}_Y q^{f(A_1, \dots, A_q, X)} \leq q \mathbb{E}_Y q^{f(B_1, \dots, B_q, Y)} \leq q \left(\prod_{l=1}^q P(B_l) \right)^{-1},$$

by the induction hypothesis. Similarly, the second inequality together with the induction hypothesis implies that, for every $1 \leq l \leq q$,

$$\mathbb{E}_Y q^{f(A_1, \dots, A_q, X)} \leq \left(\prod_{j \neq l} P(B_j) P(A_l(Z)) \right)^{-1}.$$

Combine the last two displays to see that their common left side is bounded above by $q \wedge \min_{1 \leq l \leq q} Z_l^{-1}$ times $\left(\prod_{l=1}^q P(B_l) \right)^{-1}$ for the random variables $Z_l = P(A_l(Z))/P(B_l)$. An application of the inequality in the first paragraph completes the proof.

Actually the preceding proof ignores issues of measurability and is false in general, because maps of the type $X \mapsto f(A_1, \dots, A_q, X)$ need not be measurable, as required for the application of Fubini's theorem.

With some effort it can be seen that this problem does not arise in the case that the sample space is Polish with Borel σ -field and compact sets $A_1, \dots, A_q$. This follows from the identity

$$\{x: n - f(A_1, \dots, A_q, x) \leq k\} = \bigcap_{|\cup I_i| > k} \bigcup_{i=1}^q \{x: \pi_{I_i} x \notin \pi_{I_i} A_i\},$$

where the intersection is taken over all collections of q subsets $I_1, \dots, I_q$ of $\{1, 2, \dots, n\}$ and π_I is the projection $\pi_I: \mathcal{X}^n \rightarrow \mathcal{X}^I$ on the I th coordinates. (Note that $\pi_I x \in \pi_I A$ if and only if there exists $y \in A$ that controls $\{x_i: i \in I\}$.) Projections of compact sets are compact, hence measurable. Thus, in this special situation the proof is correct.

Next, the case of general Borel sets in a Polish (product) space can be obtained by approximation from within by compact sets. By replacing every set A_i by a compact $K_i \subset A_i$, the control numbers increase (there is less control), whence the left side of the theorem increases. Next, apply the theorem and note that the resulting upper bound decreases to the right side of the theorem if the probabilities $P(A_i - K_i)$ are made to decrease to zero. The latter is possible by the regularity of Borel measures on Polish spaces.

The case of Polish spaces is sufficient for most applications but can be extended to arbitrary sample spaces. We sketch the extension.

Since any event in $\mathcal{A}^n$ is contained in the completion of the product of sub- σ -fields of $\mathcal{A}$ that are countably generated, it is no loss of generality to assume that $\mathcal{A}$ is countably generated, say with generator the sets $A_1, A_2, \dots$. Then the map $\phi: \mathcal{X} \rightarrow \{0, 1\}^\infty$ given by

$$\phi(x) = (1_{A_1}(x), 1_{A_2}(x), \dots)$$

is Borel measurable into the Polish space $\{0, 1\}^\infty \equiv R$ and $\mathcal{A} = \phi^{-1}(\mathcal{B})$ for the Borel sets $\mathcal{B}$. Let $\phi_n: \mathcal{X}^n \rightarrow R^n$ be the map $(x_1, \dots, x_n) \mapsto (\phi(x_1), \dots, \phi(x_n))$, and write P for the underlying measure on $\mathcal{X}$.

By construction there exists for every $A \in \mathcal{A}^n$ a Borel set $B \in \mathcal{B}^n$ with $A = \phi_n^{-1}(B)$. Suppose that there exists a Borel set $B' \subset B$ of the same measure under $(P \circ \phi^{-1})^n$ with the property that for every $z' \in B'$, there exists $z \in B \cap \phi(\mathcal{X})^n$ such that $z_i = z'_i$ for all i such that $z'_i \in \phi(\mathcal{X})$. Thus, the coordinates of z' that are not in $\phi(\mathcal{X})$ can be changed, meanwhile leaving the other coordinates the same, so as to obtain a point in B with all coordinates in $\phi(\mathcal{X})$. A vector in B with all coordinates in $\phi(\mathcal{X})$ is the image of some element in A and it can be seen that

$$\{x: f(B'_1, \dots, B'_q, \phi_n(x)) \leq k\} \subset \{x: f(A_1, \dots, A_q, x) \leq k\}.$$

Consequently,

$$(P^n)^*(f(A_1, \dots, A_q, x) > k) \leq ((P \circ \phi^{-1})^n)^*(z: f(B'_1, \dots, B'_q, z) > k).$$

Since $(P \circ \phi^{-1})^n(B') = P^n(A)$, the problem has been reduced to the case of a Polish sample space.

The existence of the sets B' can be argued as follows. For every partition $\{1, \dots, n\} = I \cup J$, define

$$B_{I,J} = \left\{ x \in R^n : (P \circ \phi^{-1})^J(B(\pi_I x)) = 0 \right\}.$$

Here $B(\pi_I x)$ is the section of B at $\pi_I x$, which is contained in R^J . By Fubini's theorem, each set $B_{I,J}$ is a $(P \circ \phi^{-1})^n$ null set. Thus, $B' = B - \cup_{I,J} B_{I,J}$ has the same measure as B . If $z' \in B'$, then

$$(P \circ \phi^{-1})^J(B(\pi_I z')) > 0; \quad ((P \circ \phi^{-1})^J)^*(\phi(\mathcal{X})^J) = 1.$$

This means that the set $B(\pi_I z') \cap \phi(\mathcal{X})^J$ cannot be empty. If the coordinates $\{z'_i : i \in J\}$ are not contained in $\phi(\mathcal{X})$, then they can be changed in the desired manner. ■

A typical application of the control numbers $f(A_1, \dots, A_q, X)$ is as follows. For natural numbers n , let $S_n : \mathcal{X}^n \rightarrow \mathbb{R}$ be permutation-symmetric functions that satisfy

$$(A.4.2) \quad \begin{aligned} S_n(x) &\leq S_{n+m}(x, y), & \text{for every } x \in \mathcal{X}^n, y \in \mathcal{X}^m, \\ S_{n+m}(x, y) &\leq S_n(x) + S_m(y), & \text{for every } x \in \mathcal{X}^n, y \in \mathcal{X}^m. \end{aligned}$$

An example is the functions $S_n = E_\varepsilon \left\| \sum_{i=1}^n \varepsilon_i f(x_i) \right\|_{\mathcal{F}}$ for Rademacher random variables ε_i . Given sets $A_1, \dots, A_q$ in $\mathcal{X}^n$, the elements of a random sample $X_1, \dots, X_n$ in $\mathcal{X}$ can be split into $q+1$ sets, the first one consisting of the $k = f(A_1, \dots, A_q, X)$ variables not controlled by the sets A_l and the other q sets consisting of variables that also occur in some $Y^l \in A_l$ for $l = 1, \dots, q$. By the “triangle inequality” in (A.4.2), the number $S_n(X_1, \dots, X_n)$ can be bounded by the sum of $q+1$ terms. By the first inequality in (A.4.2), the last q terms can be bounded by $S_n(Y^l)$ for $l = 1, \dots, q$. It follows that on the set $f(A_1, \dots, A_q, X) = k$,

$$S_n(X) \leq \sup_{x \in \mathcal{X}^k} S_k(x) + \sum_{l=1}^q \sup_{y \in A_l} S_n(y).$$

The first term on the right should be small if k is small, while the size of the second term can be influenced by the choice of the control sets A_l .[‡]

A.4.3 Lemma. *Let $S_n : \mathcal{X}^n \rightarrow [0, n]$ be permutation-symmetric functions satisfying (A.4.2). Then, for every $t > 0$ and n ,*

$$P^*(S_n \geq t) \leq \exp\left(-\frac{1}{2}t \log \frac{t}{12(E^* S_n \vee 1)}\right).$$

This is true for any $S_n = S_n(X)$ and any vector $X = (X_1, \dots, X_n)$ with i.i.d. coordinates.

[‡] The full force of the definition of $f(A_1, \dots, A_q, X)$ is not used for this application. The control numbers could be reduced to the minimal k such that there exist $y^l \in A_l$ with at most k of the X_i not occurring in the set $\{y^l : l = 1, \dots, q, i = 1, \dots, n\}$.

Proof. Set $\mu = E^* S_n$. By Markov's inequality, the set $A = \{S_n \leq 2\mu\}_*$ has probability at least $\frac{1}{2}$. Since $S_n \leq n$ by assumption, the argument preceding the lemma shows that $S_n(X) \leq f(A, \dots, A, X) + q2\mu$ for any integer $q \geq 1$. Thus, the probability in the lemma is bounded by

$$P^* \left(f(A, \dots, A, X) \geq t - 2q\mu \right) \leq \frac{2^q}{q^{t-2q\mu}} \leq e^{q \log q(2\mu + 1) - t \log q},$$

for $q \geq 2$ by Theorem A.4.1. For $t \geq 12(\mu \vee 1)$ and $q = \lfloor t/6(\mu \vee 1) \rfloor$, the exponent is bounded above by $-\frac{1}{2}t \log \lfloor t/6(\mu \vee 1) \rfloor$. The lemma follows since $\lfloor x \rfloor \geq \frac{1}{2}x$ for $x \geq 1$. For $t \leq 12(\mu \vee 1)$ the lemma is trivially true, because the right side is larger than 1. ■

A.5

Some Limit Theorems

In this chapter we present the strong law of large numbers and the central limit theorem uniformly in the underlying measure, as well as the rank central limit theorem.

Let $\bar{X}_n$ be the average of the first n variables from a sequence $X_1, X_2, \dots$ of independent and identically distributed random vectors in $\mathbb{R}^d$. Let $\mathcal{P}$ be a class of underlying probability measures. For instance, let X_i be the i th coordinate projection of $(\mathbb{R}^d)^\infty$, and let $\mathcal{P}$ consist of Borel probability measures on $\mathbb{R}^d$.

A.5.1 Proposition (Uniform law of large numbers). *Let $X_1, X_2, \dots$ be i.i.d. random variables with distribution $P \in \mathcal{P}$ such that*

$$\lim_{M \rightarrow \infty} \sup_{P \in \mathcal{P}} \mathbb{E}_P |X_1| \{ |X_1| \geq M \} = 0.$$

Then the strong law of large numbers holds uniformly in $P \in \mathcal{P}$ in the sense that, for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \sup_{P \in \mathcal{P}} \mathbb{P}_P \left(\sup_{m \geq n} |\bar{X}_m - \mathbb{E}_P X_1| \geq \varepsilon \right) = 0.$$

Proof. See Chung (1951). ■

The *Prohorov distance* π between probability laws on $\mathbb{R}^d$ dominates twice the bounded Lipschitz distance and generates the weak topology. The following theorem gives an explicit upper bound on the Prohorov distance between the distribution of $\sqrt{n}\bar{X}_n$ and the limiting normal distribution. This can easily be converted into a central limit theorem uniformly in P ranging over $\mathcal{P}$. Write $\pi(X, Y)$ for the Prohorov distance between the distributions of random vectors X and Y .

A.5.2 Proposition (Uniform central limit theorem). Let $X_1, \dots, X_n$ be i.i.d. random vectors with mean zero and finite covariance matrix Σ . Then for every $\varepsilon > 0$,

$$\pi(\sqrt{n}\bar{X}_n, N(0, \Sigma)) \leq 2\varepsilon^{-2}g(\varepsilon\sqrt{n}) \vee \varepsilon + 4^{1/3}g(\varepsilon\sqrt{n})^{1/3} \\ + C(d\varepsilon g(0))^{1/4} \left(1 + \left|\log \frac{\varepsilon g(0)}{d}\right|^{1/2}\right).$$

Here C is an absolute constant and $g(\varepsilon) = \mathbb{E}\|X\|^2 \{\|X\| \geq \varepsilon\}$.

Proof. For a given $\varepsilon \in (0, 1)$ and $1 \leq i \leq n$, let Y_i be the truncated variable $X_i 1\{|X_i| \leq \varepsilon\sqrt{n}\}$ and Z_i the centered variable $Y_i - \mathbb{E}Y_i$. By Strassen's theorem (see Dudley (1989), Theorem 11.6.2), it follows that

$$\pi(\sqrt{n}\bar{X}_n, \sqrt{n}\bar{Y}_n) \vee \pi(\sqrt{n}\bar{Y}_n, \sqrt{n}\bar{Z}_n) \leq (\varepsilon^{-2} \vee \varepsilon^{-1})\mathbb{E}\|X\|^2 \{\|X\| \geq \varepsilon\sqrt{n}\} \vee \varepsilon.$$

Furthermore, $\mathbb{E}\|Z_i\|^3 \leq 8\mathbb{E}\|Y_i\|^3 \leq 8\varepsilon\sqrt{n}\mathbb{E}\|X_i\|^2$. Thus, by Theorem 1 of Yurinskii (1977), as corrected in Theorem B on page 395 of Dehling (1983),

$$\pi(\sqrt{n}\bar{Z}_n, N(0, \Sigma_Z)) \lesssim (d\varepsilon g(0))^{1/4} \left(1 + \left|\log \frac{8\varepsilon g(0)}{d}\right|^{1/2}\right),$$

where Σ_Z is the covariance matrix of Z_1 . Finally,

$$\pi(N(0, \Sigma_Z), N(0, \Sigma))^3 \leq 4\mathbb{E}\|X\|^2 \{\|X\| \geq \varepsilon\sqrt{n}\},$$

by Lemma 2.1, page 402, of Dehling (1983). ■

For each n , let $a_{n1}, \dots, a_{nn}$ and $b_{n1}, \dots, b_{nn}$ be real numbers, and let $(R_{n1}, \dots, R_{nn})$ be a random vector that is uniformly distributed on the $n!$ permutations of $\{1, \dots, n\}$. Consider the rank statistic

$$S_n = \sum_{i=1}^n b_{ni} a_{n, R_{ni}}.$$

The mean and variance of S_n are equal to $\mathbb{E}S_n = n\bar{a}_n\bar{b}_n$ and $\text{var } S_n = A_n^2 B_n^2 / (n-1)$, where A_n^2 and B_n^2 are the sums of squared deviations from the mean of the numbers $a_{n1}, \dots, a_{nn}$ and $b_{n1}, \dots, b_{nn}$, respectively. Thus $A_n^2 = \sum_{i=1}^n (a_{ni} - \bar{a}_n)^2$.

A.5.3 Proposition (Rank central limit theorem). Suppose that

$$\max_{1 \leq i \leq n} \frac{|a_{ni} - \bar{a}_n|}{A_n} \rightarrow 0; \quad \max_{1 \leq i \leq n} \frac{|b_{ni} - \bar{b}_n|}{B_n} \rightarrow 0.$$

Then the sequence $(S_n - \mathbb{E}S_n) / \sigma(S_n)$ converges in distribution to a standard normal distribution if and only if

$$\sum \sum_{(i,j): \sqrt{n}|a_{ni} - \bar{a}_n| |b_{nj} - \bar{b}_n| > \varepsilon A_n B_n} \frac{|a_{ni} - \bar{a}_n|^2 |b_{nj} - \bar{b}_n|^2}{A_n^2 B_n^2} \rightarrow 0, \quad \text{for every } \varepsilon > 0.$$

Proof. See Hájek (1961). ■

A.6

More Inequalities

This chapter collects well known and less known inequalities for binomial, multinomial, and Rademacher random variables. Some of the results are related to the inequalities obtained by other means in Chapters 2.2 and 2.14.

A.6.1 Binomial Random Variables

Throughout this subsection $\bar{Y}_n$ is the average of independent Bernoulli variables $Y_1, \dots, Y_n$ with success probability p .

One of the simplest inequalities for the tails of sums of bounded random variables—in particular, for the binomial distribution—is that of Hoeffding (1963).

A.6.1 Proposition (Hoeffding's inequality). *Let $X_1, \dots, X_n$ be independent random variables taking values in $[0, 1]$. Let $\mu = E\bar{X}_n$. Then for $0 < t < 1 - \mu$,*

$$\begin{aligned} P(\bar{X}_n - \mu \geq t) &\leq \left(\left(\frac{\mu}{\mu + t} \right)^{\mu+t} \left(\frac{1 - \mu}{1 - \mu - t} \right)^{1-\mu-t} \right)^n \\ &\leq \exp -nt^2 g(\mu) \\ &\leq \exp -2nt^2. \end{aligned}$$

Here $g(\mu)$ is equal to $(1 - 2\mu)^{-1} \log(1/\mu - 1)$, for $0 < \mu < 1/2$, and $(2\mu(1 - \mu))^{-1}$, for $1/2 \leq \mu < 1$.

Proof. See Hoeffding (1963). ■

Applied to Bernoulli variables, Hoeffding's inequality yields

$$P(\sqrt{n}|\bar{Y}_n - p| \geq \lambda) \leq 2 \exp -2\lambda^2, \quad \lambda > 0.$$

This inequality is valid for any n and p . For p close to zero or one, it is possible to improve on the factor 2 in the exponent considerably.

A.6.2 Proposition (Bennett's inequality). For all n and $0 < p < 1$,

$$P(\sqrt{n}|\bar{Y}_n - p| \geq \lambda) \leq 2 \exp - \frac{\lambda^2}{2p} \psi\left(\frac{\lambda}{\sqrt{np}}\right), \quad \lambda > 0.$$

Here $\psi(x) = 2h(1+x)/x^2$, for $h(x) = x(\log x - 1) + 1$.

Proof. See Bennett (1962), Hoeffding (1963), or Shorack and Wellner (1986), page 440. ■

A.6.3 Corollary (Kiefer's inequality). For all n and $0 < p < e^{-1}$,

$$P(\sqrt{n}|\bar{Y}_n - p| \geq \lambda) \leq 2 \exp - \lambda^2 (\log(1/p) - 1).$$

Proof. Since the probability on the left side is zero if $\lambda > \sqrt{n}$, we may assume that $\lambda \leq \sqrt{n}$. Then $\psi(\lambda/\sqrt{np}) \geq \psi(1/p)$, because ψ is decreasing. Now apply Bennett's inequality and finish the proof by noting that $\psi(1/p)/(2p) \geq \log(1/p) - 1$ for $0 < p < e^{-1}$. ■

For $p \downarrow 0$, the function $\log(1/p) - 1$ increases to infinity. Thus Kiefer's inequality gives bounds of the type $2 \exp - C\lambda^2$ for large constants C for sufficiently small p . For example, $2 \exp - 11\lambda^2$ for $p \leq e^{-12}$.

For p bounded away from zero and one, Hoeffding's inequality can be improved as well—at least for large values of λ .

A.6.4 Proposition (Talagrand's inequality). There exist constants K_1 and K_2 depending only on p_0 , such that for $p_0 \leq p \leq 1 - p_0$ and all n ,

$$\begin{aligned} P(\sqrt{n}(\bar{Y}_n - p) = \lambda) &\leq \frac{K_1}{\sqrt{n}} \exp - \left(2\lambda^2 + \frac{\lambda^4}{4n}\right), \quad \lambda > 0, \\ P(\sqrt{n}(\bar{Y}_n - p) \geq t) &\leq \frac{K_2}{\lambda} \exp - \left(2\lambda^2 + \frac{\lambda^4}{4n}\right) \exp 5\lambda(\lambda - t), \quad 0 < t < \lambda, \\ P(\sqrt{n}(\bar{Y}_n - p) \geq \lambda) &\leq \frac{K_2}{\lambda} \exp - \left(2\lambda^2 + \frac{\lambda^4}{4n}\right), \quad \lambda > 0. \end{aligned}$$

Proof. Stirling's formula with bounds asserts that

$$\sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{1/(12n+1)} \leq n! \leq \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{1/(12n)}.$$

Hence it follows, that for $1 \leq k \leq n - 1$,

$$(A.6.5) \quad \binom{n}{k} p^k (1-p)^{n-k} \leq \frac{1}{\sqrt{2\pi}} \left(\frac{n}{k(n-k)}\right)^{1/2} e^{-n\Psi(u,p)}.$$

Here set $u = k/n - p$ and define

$$\Psi(u, p) = (u + p) \log\left(\frac{u + p}{p}\right) + (1 - (u + p)) \log\left(\frac{1 - (u + p)}{1 - p}\right).$$

The function Ψ satisfies $\Psi(0, p) = 0$, $\partial/\partial u \Psi(0, p) = 0$, and

$$\frac{\partial^2}{\partial u^2} \Psi(u, p) = \frac{4}{1 - 4(u - (\frac{1}{2} - p))^2} \geq 4 \left(1 + 4(u - (\frac{1}{2} - p))^2 \right).$$

Integrate this inequality to obtain

$$\frac{\partial}{\partial u} \Psi(u, p) \geq 4u + \frac{16}{3} \left((u - (\frac{1}{2} - p))^3 + (\frac{1}{2} - p)^3 \right) \geq 4u + \frac{4}{3} u^3.$$

In the last step, use the fact that the function $\beta \mapsto (u - \beta)^3 + \beta^3$ takes its minimal value at $\beta = u/2$. Integrate again to obtain the inequality $\Psi(u, p) \geq 2u^2 + u^4/3$.

For the proof of the first part of the proposition, put $k = \sqrt{n}\lambda + np$. Then $k \geq np_0$. Furthermore, for $k \leq n(1 - p_0/2)$, we have $n - k \geq np_0/2$, and the right-hand side of (A.6.5) is bounded above by

$$\frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{np_0^2/2}} e^{-n(2u^2 + u^4/3)}.$$

For $n(1 - p_0/2) \leq k < n$, we have $u = k/n - p \geq p_0/2$, and the right-hand side of (A.6.5) is bounded by

$$\frac{1}{\sqrt{2\pi}(1 - p_0/2)} e^{-n(2u^2 + u^4/3)} \leq \frac{K}{\sqrt{n}} e^{-n(2u^2 + u^4/4)} \sup_{n \geq 1} \sqrt{n} e^{-n(p_0/2)^4/12}.$$

Finally, for $k = n$, the inequality of the proposition can be reduced to

$$-\log(1/p) + 2(1 - p)^2 + \frac{1}{4}(1 - p)^4 \leq \frac{1}{n} \log \left(\frac{K_1}{\sqrt{n}} \right).$$

The function $h(p)$ on the left side of this last inequality is negative for $0 < p < 1$, with $h(0) = -\infty$ and $h(1) = 0$, and has a local maximum of $-0.160144\dots$ at $0.344214\dots$ and a local minimum of $-0.184428\dots$ at $0.598428\dots$. It follows that the supremum of the left side over any interval $p_0 \leq p \leq 1 - p_0$ is strictly less than zero. On the other hand, the infimum of the right side over n is bounded below by $-1/(2eK_1^2)$. Thus, the inequality is valid for sufficiently large K_1 .

For the proof of the second part of the proposition, consider the function $h(x) = 2x^2 + x^4/4$. The derivative $h'(x) = 4x + x^3$ is bounded below by $4x$ and is bounded above by $5x$ for $x \leq 1$. Since h is convex, $h(x) \geq h(u) + (x - u)h'(u)$ for all x . Apply the first inequality of the proposition to find that, with $k_0 = \lceil np + \sqrt{nt} \rceil$,

$$\begin{aligned} \mathbf{P}(\sqrt{n}(\bar{Y}_n - p) \geq t) &\leq \sum_{k \geq k_0} \frac{K_1}{\sqrt{n}} e^{-n(h(u) + (k/n - p - u)h'(u))} \\ &\leq \frac{K_1}{\sqrt{n}} e^{-nh(u)} \frac{e^{(nu - (k_0 - np))h'(u)}}{1 - e^{-h'(u)}} \leq \frac{K_1}{\sqrt{n}} e^{-nh(u)} \frac{1}{4u} e^{5nu(u - t/\sqrt{n})}. \end{aligned}$$

This concludes the proof of the proposition. ■

A.6.2 Multinomial Random Vectors

In this section $(N_1, \dots, N_k)$ is a multinomially distributed vector with parameters n and $(p_1, \dots, p_k)$. It is helpful to relate this to empirical processes. Let $\{C_i\}_{i=1}^k$ be a partition of a set $\mathcal{X}$ and P the probability measure on the σ -field $\mathcal{C}$ generated by the partition such that $P(C_i) = p_i$. If $\mathbb{P}_n$ is the empirical measure corresponding to a sample of size n from P , then the vector $(n\mathbb{P}_n(C_1), \dots, n\mathbb{P}_n(C_k))$ has the same distribution as $(N_1, \dots, N_k)$.

The first two propositions concern the L_1 -distance $\sum_{i=1}^k |N_i - np_i|$. In the representation using the empirical process, this is equivalent to the Kolmogorov-Smirnov distance

$$\|\mathbb{P}_n - P\|_{\mathcal{C}} = \sup_{C \in \mathcal{C}} (\mathbb{P}_n - P)(C) = \frac{1}{2} \sum_{i=1}^k |\mathbb{P}_n(C_i) - P(C_i)|.$$

A.6.6 Proposition (Bretagnolle–Huber–Carol inequality). *If the random vector $(N_1, \dots, N_k)$ is multinomially distributed with parameters n and $(p_1, \dots, p_k)$, then*

$$P\left(\sum_{i=1}^k |N_i - np_i| \geq 2\sqrt{n}\lambda\right) \leq 2^k \exp -2\lambda^2, \quad \lambda > 0.$$

Proof. In terms of the Kolmogorov-Smirnov statistic, the left side is equal to $P(\|\mathbb{G}_n\|_{\mathcal{C}} \geq \lambda)$. Each of the probabilities $P((\mathbb{P}_n - P)(C) > \lambda)$ can be bounded by $\exp(-2\lambda^2)$ by a one-sided version of Hoeffding's inequality. There are 2^k sets in $\mathcal{C}$. Alternatively, first reduce $\mathcal{C}$ by dropping all sets with probability more than $1/2$. This does not change $\|\mathbb{P}_n - P\|_{\mathcal{C}}$. Now the two-sided version of Hoeffding's inequality can be applied to the 2^{k-1} remaining sets. ■

The following inequality results from combining Kiefer's inequality, Corollary A.6.3, and Talagrand's inequality, Proposition A.6.4.

A.6.7 Proposition. *If the random vector $(N_1, \dots, N_k)$ is multinomially distributed with parameters n and $(p_1, \dots, p_k)$, then there is a universal constant K such that*

$$P\left(\sum_{i=1}^k |N_i - np_i| \geq 2\sqrt{n}\lambda\right) \leq \frac{K}{\lambda} 2^k \exp -2\lambda^2, \quad \lambda > 0.$$

Proof. The proof is slightly easier in the language of empirical processes as introduced above. Let $\mathcal{C}_0$ be the class of sets $C \in \mathcal{C}$ such that $P(C) < e^{-12}$, and let $\mathcal{C}_1$ be the class of sets with $e^{-12} \leq P(C) \leq 1/2$. Then $\|\mathbb{G}_n\|_{\mathcal{C}} = \|\mathbb{G}_n\|_{\mathcal{C}_0} \vee \|\mathbb{G}_n\|_{\mathcal{C}_1}$. By the remark following Kiefer's inequality, Corollary A.6.3, applied to $\|\mathbb{G}_n\|_{\mathcal{C}_0}$ and Talagrand's inequality, Proposition A.6.4, applied to $\|\mathbb{G}_n\|_{\mathcal{C}_1}$, there exists a universal constant K_2 such that

$$P(\|\mathbb{G}_n\|_{\mathcal{C}} \geq \lambda) \leq 2^k 2 \exp -11\lambda^2 + 2^k \frac{K_2}{\lambda} \exp -2\lambda^2.$$

The proposition follows easily. ■

The last inequality for multinomial random vectors concerns a slight modification of the classical chi-square statistic (or weighted L_2 -distance).

A.6.8 Proposition (Mason–van-Zwet inequality). Let $(N_1, \dots, N_k)$ be multinomially distributed with parameters n and $(p_1, \dots, p_k)$ such that $p_i > 0$ for $i < k$. Then for every $C > 0$ and $\delta > 0$ there exist constants $a, b, c > 0$, such that for all $n \geq 1$ and $\lambda, p_1, \dots, p_{k-1}$ satisfying $\lambda \leq Cn \min\{p_i: 1 \leq i \leq k-1\}$ and $\sum_{i=1}^{k-1} p_i \leq 1 - \delta$,

$$P\left(\sum_{i=1}^{k-1} \frac{(N_i - np_i)^2}{np_i} > \lambda\right) \leq a \exp(bk - c\lambda).$$

We do not give a full derivation of this inequality here, but note that the chi-square statistic is an example of a supremum over an elliptical class of functions. Therefore, the preceding inequality can be deduced from Talagrand's general empirical process inequality, Theorem 2.14.38.

A.6.3 Rademacher Sums

The following inequality improves Hoeffding's inequality for Rademacher sums, Lemma 2.2.8, in much the same way that Talagrand's inequality improves on Hoeffding's inequality. Let $\varepsilon_1, \dots, \varepsilon_n$ denote independent Rademacher variables.

A.6.9 Proposition (Pinelis–Bentkus–Dzindzalieta inequality). For any numbers $a_1, \dots, a_n$ with $\sum_{i=1}^n a_i^2 = 1$, for $c_* = 4^{-1}(1 - \Phi(\sqrt{2}))^{-1} \approx 2.378$,

$$P\left(\sum_{i=1}^n a_i \varepsilon_i \geq \lambda\right) \leq c_* (1 - \Phi(\lambda)) \leq c_* \frac{\phi(\lambda)}{\lambda}, \quad \lambda > 0.$$

A.6.10 Proposition (Khinchine's inequality). For any real numbers $a_1, \dots, a_n$ and any $p \geq 1$, there exist constants A_p and B_p such that

$$A_p \|a\|_2 \leq \left\| \sum_{i=1}^n a_i \varepsilon_i \right\|_p \leq B_p \|a\|_2.$$

Problems and Complements

1. The function $g(p) = \psi(1/p)/(2p)$ is bounded below by $\log(1/p) - 1$ for $0 < p < e^{-1}$. It is not bounded below by $\log(1/p) - c$ for any $c < 1$.

[Hint: $\psi(1/p)/(2p) - (\log(1/p) - 1) \rightarrow 0$ as $p \rightarrow 0$.]

2. **(Bennett's inequality plus Jensen equals Bernstein's inequality)** Show that the function ψ in Proposition A.6.2 is decreasing, $\psi(0) = 1$, $x\psi(x)$ is increasing, and $\psi(x) \geq 1/(1+x/3)$.

[Hint: Let $f(x) = h(1+x) = (x+1)\log(x+1) - x$, and write

$$f(x) = \int_0^1 x f'(xt) dt = \int_0^1 x^2 \int_0^1 1\{0 \leq s \leq t \leq 1\} f''(xs) ds dt,$$

where $f'(x) = \log(1+x)$ and $f''(x) = 1/(1+x)$. Hence

$$\psi(x) = 2 \int_0^1 \int_0^1 1\{0 \leq s \leq t \leq 1\} f''(xs) ds dt.$$

To prove the last inequality, interpret the right side of this identity as an expectation.]

3. The probability in Kiefer's inequality is bounded below by $\exp[-\lambda^2(1-p)^{-2} \log(1/p)]$ when $\lambda = \sqrt{n}(1-p)$.
4. The constant K resulting from the proof of Proposition A.6.7 may be minimized by choosing an optimal cut-off point instead of e^{-12} . This involves understanding the dependence of the constant K_2 in Proposition A.6.4 on p_0 .
5. Independent binomial $(1, p_i)$ variables $X_1, \dots, X_n$ satisfy

$$P\left(\sum_{i=1}^n X_i > k\right) \leq \exp\left(-n\bar{p}_n h\left(\frac{k}{n\bar{p}_n}\right)\right) < \left(\frac{en\bar{p}_n}{k}\right)^k,$$

for the function $h(x) = x(\log x - 1) + 1$ and $\bar{p}_n$ the average of the success probabilities.

A

Notes

A.1. Ottaviani (1939) proves his inequality for the case of real-valued random variables. Hoffmann-Jørgensen (1974) proves his inequalities for Banach-valued random variables. For both cases Dudley (1984) gives careful proofs for random elements in nonseparable Banach spaces.

Proposition A.1.6 is due to Talagrand (1989).

A.2. The first inequality of Proposition A.2.1 is a consequence of the finer isoperimetric results obtained independently by Borell (1975) and Sudakov and Tsirel'son (1978) (see Corollary A.2.12 and Proposition A.2.13). The second inequality was found by Ibragimov, Sudakov and Tsirel'son (1976) and rediscovered by Maurey and Pisier (see Pisier (1986)). The present proof is based on the exposition by Ledoux (1996). See Ledoux and Talagrand (1991), page 59, for the third inequality of Proposition A.2.1 as well as other interesting results.

The moment inequality Proposition A.2.4 is given by Pisier (1986).

Adler (1990) gives a good summary of many of the Gaussian comparison results, including Sudakov's minorization inequality, Dudley's majorization inequality, and the more recent work on majorizing measures. The inequalities of Proposition A.2.5 are due to Fernique after earlier work by Slepian (1962), Marcus and Shepp (1971) and Landau and Shepp (1970). The last assertion of Proposition A.2.5 was noted by Marcus and Shepp and is stated explicitly in Jain and Marcus (1978).

Proposition A.2.7 was first proved by Ehrhard (1993) in the case that both A and B are convex, next by Łatała (1996) when one set is convex, and finally for general Borel sets, as given here with its remarkable and short proof, by Borell (2003). Related to this, Borell (2008) studied the maximal set of coefficients α_i giving an analogous inequality for a (possibly nonconvex) linear combination $\sum_i \alpha_i A_i$

of (possibly more than two) sets, and found that these sets are different for convex or nonconvex A_i . The log-concavity given in Corollary A.2.8 was first obtained by Borell (1974). See Davydov, Lifshits and Smorodina (1998) for generalizations, examples and alternative proofs, or Giné and Nickl (2016), Section 2.4.1, for a direct proof. The application to the continuity of the distribution function of the norm is also made explicit in Gaenssler, Molnár and Rost (2007). The isoperimetric inequality and the geometric form of Borell's inequality were proved by Sudakov and Tsirel'son (1978) and Borell (1975), for general locally convex Hausdorff spaces.

Anderson's inequality was proved for a more general class of symmetric unimodal distributions in Anderson (1955). The Gaussian correlation inequality was long thought to be true, but proved only relatively recently by Royen (2014); Latała and Matlak (2017) give a nice exposition.

The section on exponential bounds for Gaussian processes is based mostly on Talagrand (1994). The interested reader should also consult Samorodnitsky (1991), Adler and Brown (1986), Adler and Samorodnitsky (1987), Adler (1990), and Goodman (1976). The renewal theoretic large deviation methods of Hogan and Siegmund (1986), Siegmund (1988, 1992) give precise results on the asymptotic tail behavior for the suprema of particular Gaussian fields, especially those satisfying what Aldous (1989) calls the "uncorrelated orthogonal increments" property. See Aldous (1989), Chapter J, pages 190–219 for a variety of fascinating asymptotic formulas. Computation of the asymptotic behavior in Example A.2.20 was carried out by David Siegmund (personal communication).

Lemma A.2.25 and its proof are due to Kim and Pollard (1990).

A.4. The basic inequality in this chapter was first proved by Talagrand (1989) and Ledoux and Talagrand (1991). The present elegant proof by induction is taken from Talagrand (1995), who discusses many extensions and applications. The name "isoperimetric inequality" appears not entirely appropriate. It resulted from the analogy with exponential inequalities for Gaussian variables, which can be derived from isoperimetric inequalities for the standard normal distribution. This asserts that, among all sets A with $P(A)$ equal to a fixed number, the probability $P(A^\varepsilon)$ increases the least as ε increases from zero for A of the form $\{x: |x| > c\}$. See Ledoux and Talagrand (1991), Ledoux (1996) and Ledoux (2001) for further references.

A.5. In connection to Proposition A.5.2, consult the papers by Yurinskii (1977) and Dehling (1983).

A.6. Bennett's inequality was proved by Bennett (1962); the function ψ was introduced by Shorack (1980). For Kiefer's inequality, see Kiefer (1961); the present proof from Bennett's inequality is new. For Talagrand's inequality, see Talagrand (1994), and for the L_1 -inequality for multinomial variables, see Bretagnolle and Huber (1978). Greene and Wellner (2017) establish inequalities for sampling without replacement analogous to Talagrand's inequality. Mason and Van Zwet (1987) prove inequality A.6.8 and use it to obtain an interesting refinement of the Komlos-Major-Tusnady inequality. Proposition A.6.9 with the (optimal) constant c_* is due to Bentkus and Dzindzalieta (2015); Pinelis (1994) and Pinelis (2007) proved the

inequality with slightly worse constants; the results improve on conjectures of Eaton (1974). Problem A.6.2 was communicated to us by David Pollard.

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List of Symbols

$\mathcal{A}(\mathcal{X})$	analytic functions $f: \mathcal{X} \rightarrow \mathbb{R}$, 245
$\mathbf{B}^*$	dual space of the Banach space $\mathbf{B}$, 560
$BL(\mathbb{D})$	the set of bounded, Lipschitz functions on $\mathbb{D}$, 72
BL_1	subset of $BL(\mathbb{D})$ of norm bounded by 1, 72
$B_{p,q}^\alpha(\mathcal{X})$	Besov space of functions $f: \mathcal{X} \rightarrow \mathbb{R}$ of smoothness α , 229
$\mathbb{B}_p^\delta$	unit ball of coefficients of a Besov body of δ , 236
$\mathbb{C}$	complex numbers, 28
$C_b(\mathbb{D})$	bounded, continuous, real functions on $\mathbb{D}$, 15
$C(T)$	continuous functions from T to $\mathbb{R}$, 33
$C^\alpha(\mathcal{X})$	Hölder space of functions $f: \mathcal{X} \rightarrow \mathbb{R}$ of smoothness α , 223
$C^\alpha(\mathcal{X})_M$	subset of $C^\alpha(\mathcal{X})$ of functions with $\ f\ _\alpha \leq M$, 223
$\text{conv } \mathcal{F}$	convex hull of $\mathcal{F}$, 207
$\text{cov } X$	covariance matrix of a random vector X , 270
d_u, d_∞	Skorokhod metrics, 93, 93
∂B	boundary of the set B , 17
$\text{diam}(T, d)$	diameter of the set T for metric d , 62
δ_x	point mass at x or Dirac measure, 46
$\Delta_h f, \Delta_h^r f$	first and r th difference of function f over vector h , 229
$D(\varepsilon, \mathcal{F}, d)$	packing number of $\mathcal{F}$ relative to metric d , 147
$(\mathbb{D}, d), (\mathbb{E}, e)$	metric spaces, 2
$D[a, b]$	cadlag functions on $[a, b]$, 46, 82
$D[0, \infty)$	cadlag functions, Skorokhod space, on $[0, \infty)$, 82
ΔX	jump process $X(t) - X(t-)$ of cadlag process X , 104
$\mathbb{E}^*$	outer integral, 4
$\mathcal{F}, \mathcal{G}, \mathcal{H}$	collections of functions, 129, 136
$\mathbb{F}_n$	empirical distribution function, 131

- $\mathbb{G}$ Brownian bridge process, 130
- $\mathbb{G}_n$ empirical process, 129
- $\text{Ent}(X)$ entropy of random variable X , 369
- $H^\alpha(\mathcal{X})$ Sobolev space $W_2^\alpha(\mathcal{X})$ of smoothness α , 228
- $J(\delta, \mathcal{F} | F, L_2)$ uniform entropy integral of $\mathcal{F}$ relative to envelope F , 330
- $J_{[]}(\delta, \mathcal{F} | F, \|\cdot\|)$ uniform entropy integral of $\mathcal{F}$ relative to envelope F , 337
- $\ell^\infty(T)$ set of all uniformly bounded real functions on T , 33
- $\ell^\infty(T_1, T_2, \dots)$ set of all locally bounded real functions on $T = \cup_{i=1}^\infty T_i$, 41
- $\mathcal{L}_r$ r -integrable functions, 133
- L_r equivalence classes of r -integrable functions, 133
- $\mathcal{L}_\infty$ essentially bounded functions, 133, 181
- L_∞ equivalence classes of essentially bounded functions, 133, 181
- $\text{Lip}_p^\alpha(\mathcal{X})$ Lipschitz functions $f: \mathcal{X} \rightarrow \mathbb{R}$ of smoothness α , 230
- $\mathcal{L}(\cdot)$ law or distribution of a random variable, 15
- λ Lebesgue measure, 3, 206
- lin linear span, 50
- $^2\log m$ logarithm base 2, 253
- $\langle M \rangle$ predictable variation of martingale M , 103
- $\mathbb{N}$ set of natural numbers, 21
- $\mathbb{N}_0$ set of natural numbers and 0, 222
- $N(\varepsilon, \mathcal{F}, d)$ covering number of $\mathcal{F}$ relative to metric d , 147
- $N_{[]}(\varepsilon, \mathcal{F}, d)$ bracketing number of $\mathcal{F}$ relative to metric d , 132
- $N(\mu, \sigma^2)$ normal (Gaussian) distribution on $\mathbb{R}$, 129
- $N_k(\mu, \mathcal{S})$ normal (Gaussian) distribution on $\mathbb{R}^k$, 130
- P, Q, H probability measures on underlying sample space $\mathcal{X}$, 2, 506
- $\mathbb{P}_n, \mathbb{Q}_n$ empirical measures, 129, 506
- P^* outer probability, 6
- $\mathcal{P}$ collection of probability measures, 9
- $\mathbb{Q}$ set of rational numbers, 16
- $\mathbb{R}$ real numbers, 4
- $\bar{\mathbb{R}}$ extended real numbers, $[-\infty, \infty]$, 4
- $\mathbb{R}^k$ k -dimensional Euclidean space, 18
- ρ, ρ_0 semimetrics on a set T or $\mathcal{F}$, 36, 37
- ρ a lifting, 181
- ρ_P variance semimetric, 138
- $\text{sconv } \mathcal{F}$ symmetric convex hull of $\mathcal{F}$, 207, 280
- T index set for a stochastic process, 33
- $(\bar{T}, \rho)$ ρ -completion of T , 38
- T^* minimal measurable majorant of the map/function T , or measurable cover function of T , 4
- $UC(T, \rho)$ uniformly continuous functions from (T, ρ) to $\mathbb{R}$, 40
- $\text{var } X$ variance of a real random variable X , 153
- $V(\mathcal{C}), V(\mathcal{F})$ VC-index of class of sets $\mathcal{C}$ or functions $\mathcal{F}$, 200, 206
- $V(\varepsilon, \mathcal{F})$ fat-shattering index, 216
- $w(z, \delta)$ Skorokhod modulus of function z , 85, 95
- $w_{\infty, T}(z, \delta)$ Skorokhod modulus of function z , 85, 95

$W_p^\alpha(\mathcal{X})$	Sobolev space of functions $f: \mathcal{X} \rightarrow \mathbb{R}$ of smoothness α , 228
$\mathbb{Z}$	set of integers, 99
$\sigma(X)$	standard deviation of a real random variable X , 151
$\Omega, \Omega_n, \Omega_\alpha$	sample spaces, 6, 2, 16

Symbols not connected to Greek or Roman letters

$\underline{x}$	greatest integer strictly less than x , 222
$\#A, A $	cardinality of the (finite) set A , 129, 148
κ^*	adjoint of the linear map κ , 560
A^t	transpose of the matrix A , 80
a', a^t	transpose of the vector a , 19
$\xrightarrow{\text{as}}$	convergence almost surely, 52
$\xrightarrow{\text{au}}$	convergence almost uniformly, 52
$\xrightarrow{P^*}$	convergence in outer probability, 52, 57
$\xrightarrow{P}$	convergence in probability, 53
$\xrightarrow{\text{as}^*}$	convergence outer almost surely, 52
$\rightsquigarrow$	weak convergence, 2
$\overset{P}{\rightsquigarrow}$	weak convergence under P , 554
$\rightarrow$	convergence, 2
$\rightarrow, \mapsto$	function specifier, 4
$\hookrightarrow$	continuous embedding, 231
■	end-of-proof symbol, 7
$\ f\ _{Q,r}$	$L_r(Q)$ norm of function f , 133
$\ f\ _{Q,2,\infty}$	$L_{2,\infty}(Q)$ norm of function f , 195
$\ \xi\ _{p,1}$	$L_{p,1}$ -norm of the random variable ξ , 137, 267
$\ X\ _\psi$	Orlicz norm of the random variable X , 139
$\ f\ _\infty$	supremum or uniform norm of function f , 21
$\ z\ _{\mathcal{F}}$	supremum or uniform norm of map $z: \mathcal{F} \rightarrow \mathbb{R}$, 130
$\ L\ _{BL}$	bounded Lipschitz norm of signed measure L , 73
$\ f\ _{BL}$	bounded Lipschitz norm of function f , 72
$\ f\ _\alpha$	Hölder norm of smoothness α of function f , 222
$\ f\ _{\alpha p}$	Sobolev norm of smoothness α of function f , 228
$\ f\ _{\alpha;p}$	Lipschitz norm of smoothness α of function f , 230
$\ f\ _{\alpha p,q}$	Besov norm of smoothness α of function f , 229, 233
$\ f\ _{\alpha;p,q}$	Lizorkin norm of smoothness α of function f , 255
$\equiv$	equals by definition, 136
$\approx$	approximately equals, 407
$\lesssim$	left side bounded by a constant times the right side, 154
$\asymp$	left side bounded above and below by constants times the right side, 195
$\sim$	quotient of left and right sides is asymptotic to 1, 170
$\times$	Cartesian product, 10
$\sim$	equal in distribution to, 79
$\perp\!\!\!\perp$	independence of random variables, 79
$\ll$	absolutely continuous with respect to, 9
$\triangleleft$	contiguous with respect to, 549

- $\Leftrightarrow$ two-sided (or mutually) contiguous, 549
- $\vee$ maximum, 7, 23, 136
- $\wedge$ minimum, 7, 23, 136
- $\sqcap$ pairwise intersections of two classes of sets, 213
- $\sqcup$ pairwise unions of two classes of sets, 213
- $\triangle$ symmetric difference of sets, 218
- $-$ difference of sets, 356
- $\emptyset$ empty set, 14
- 1_A indicator function of the set A , 53